5 Methods for Protein Synthesis

A. M. Felix

An ultimate goal of many peptide chemists is to develop methods for the synthesis of proteins. The development of such methods continues to be a formidable and challenging undertaking. Thorough evaluation and utilization of the strategies outlined in the earlier chapters of this treatise can provide a useful background for undertaking protein synthesis. The synthesis of insulin, achieved nearly simultaneously by three independent groups, may be considered to have been the first synthetic protein.[1–3] These early syntheses of insulin (Section 5.1.4), as well as the subsequent solid-phase synthesis of insulin (Section 5.1.5), represent landmark achievements by peptide chemists.

The preparation of Ribonuclease A, by solid-phase methods,[4] and Ribonuclease S-protein, by solution methods,[5] represented the first syntheses of enzymes (Section 5.1.6). A general protocol for protein synthesis in solution utilizes the combination of Boc/Bzl/Pac groups (Section 5.2.1) and a careful choice of fully protected segments (Section 5.2.2). Since many of these segments exhibit solubility problems, it was necessary to develop new solvent systems containing mixtures of trifluoroethanol or hexafluoroisopropanol with chloroform or dichloromethane.[6] The new solvent systems were used for the synthesis of Human Midkine (Section 5.2.4), a protein containing 121 amino acid residues consisting of an N-terminal domain (containing three disulfide bonds) and a C-terminal domain (containing two disulfide bonds). This technology has been extended to the synthesis of the 238-residue Green Fluorescent Protein using a total of 26 fully protected fragments.[7]

The development of other procedures to improve solubility of protected peptide segments by incorporating backbone protection can overcome difficulties in peptide synthesis. The inclusion of a tertiary amide bond every five or six amino acid residues increases the solubility of these segments. Incorporation of acetylated 2-hydroxy-4-methoxybenzyl amino acids provides segments that are soluble in nonpolar solvents. This procedure has been used for a novel solid-phase condensation strategy for the synthesis of Ribonuclease A using eight segments (Section 5.3.1).

Chemical ligation methods for peptide synthesis using thioester chemistry in solution have been previously documented (see Vol. E 22a, Section 4.1.5). Generalized procedures for solid-phase ligation have been developed that simplify the overall procedure. One method uses a safety-catch acid labile linker at the C-terminus and was used for the synthesis of a 71-amino acid chemokine, vMIP I (Section 5.3.2.1). Another procedure uses a selectively cleavable glycolate ester linkage (Section 5.3.2.2).

Semisynthesis involves the combination of synthetic peptides (including fragments of proteins) with a large protein segment (which may be derived from natural sources, total synthesis, or recombinant methods). The conversion of porcine insulin into human insulin represents one of the earliest semisynthetic proteins. An important advantage of semisynthesis is that proteins can be prepared that incorporate unnatural amino acids or nonpeptidic linkages. The ligation of unprotected proteins (or their fragments) can be achieved through their amino and carboxy termini via peptide bonds (Section 5.4.1). The use of protein splicing units (intein) which generate a thioester at the C-terminus of a recombinant protein have recently been reported.[8,9] The ligation of unprotected proteins (or their fragments) can also be achieved resulting in nonpeptide bonds. Examples include ligation with hydrazides, aminoxy functions, and thiols, which are reactive with aldehydes (Section 5.4.2).

for references see p 2

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5.1 History of Protein Synthesis

R. B. Merrifield

5.1.1 Origins of Protein Synthesis

For nearly one hundred years, chemists have been interested in the total synthesis of a protein. The idea was first expressed by Emil Fischer in his elegant 1906 quotation:[1]

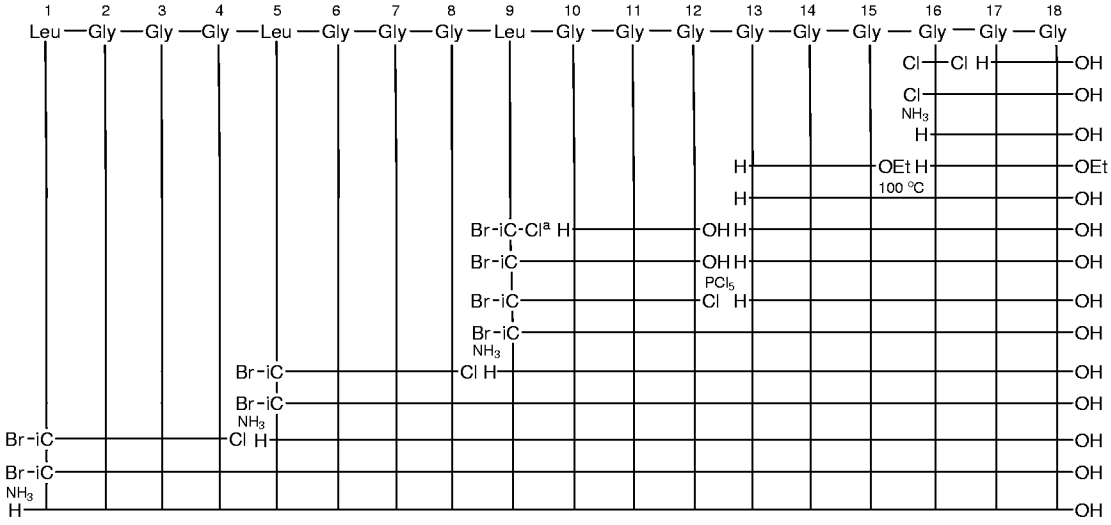
"Whereas professional colleagues fear that a rational study of this class of compounds, because of their complicated structure and their highly inconvenient physical characteristics, would today still uncover insurmountable difficulties, other optimistically endowed observers, among which I will count myself, are inclined to the view that an attempt should at least be made to besiege this virgin fortress with all the expedients of the present; because only through this hazardous affair can the limitations of the ability of our methods be ascertained."

This expresses the feelings of others in this field, and even today most syntheses have been undertaken mainly to examine the new methodology at hand and to test its limits. The second objective then becomes the application of the technology to answer important chemical and biological questions through the chemical synthesis of peptides and proteins.

It, of course, was not possible for Fischer to make a naturally occurring protein because at that time no structure of a protein was known. In fact, many believed that proteins were not even discrete single large molecules, but were actually colloidal mixtures of smaller compounds. It was, however, possible for Fischer to apply his new methods to the synthesis of large peptides that he felt would simulate the properties of natural proteins.

Fischer succeeded in preparing an 18-residue peptide composed of glycine and L-leucine (Scheme 1).[2] He used a forerunner of what eventually became the classical fragment synthesis strategy in which amino acids are combined to give small peptides that can be coupled together to produce larger peptides. In his case, there was no reversible α-amine protecting group and he was forced to resort to several methods to circumvent the problem, including the α-halo acid halide method, peptide ester condensation, and hydrolyses of piperazinediones.

Scheme 1 Synthesis of the Octadecapeptide H-L-Leu-$(Gly)_3$-L-Leu-$(Gly)_3$-L-Leu-$(Gly)_8$-Gly-OH[2]



[a] The precursor of Leu is an isocaproic acid derivative (iC).

for references see p 38

Fischer began by hydrolyzing glycine piperazinedione to give glycylglycine. Chloroacetyl chloride was then added to produce the α-chloro tripeptide, which could be aminated with ammonia to the tripeptide. Triglycine ethyl ester was prepared and condensed with itself at 100 °C to give mainly hexaglycine ester. α-Bromoisocaproyl-triglycine was prepared by reaction of triglycine with α-bromoisocaproyl chloride (2-bromo-4-methylpentanoyl chloride), and coupled to the hexapeptide to give the decapeptide after ammonolysis. The peptide was then extended using similar methodologies to give the tetradeca- and octadecapeptides. This was a major extension of what had already been achieved and it did establish the reasonable limits of what could be done at that time. Due to its size, this peptide has some of the properties of a protein, i.e. it can be salted out of aqueous solution and it gives a positive biuret test. Thus, Fischer had accomplished his goal of testing the limits of his methods, but he realized that he was still far away from his true objectives.

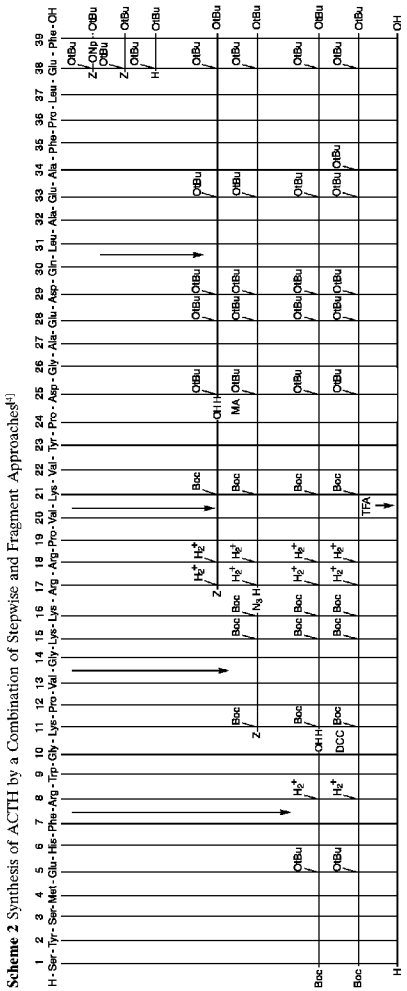
There are now several general strategies for the assembly of peptides and proteins. Initially the techniques of classical organic chemistry were used. The monomer units were amino acids or compounds that could be converted into amino acids; they could be condensed with one another through amide bonds (peptide bonds) one at a time in a stepwise manner, or small peptides could be coupled together to produce larger peptides by the fragment (segment) condensation approach. The latter is often referred to as a convergent synthesis. Much later it became possible to apply these two strategies to the elongation of peptides that were attached to solid supports and were, therefore, not in homogeneous solution, but in a highly solvated system. Many different tactics, i.e. detailed chemical procedures and techniques, can be applied to each of these general approaches.

5.1.2 Reversible N^{α}-Protecting Groups

After the Fischer period (see Section 5.1.1), the next step was the search for improved methodology, particularly for a reversible α-amine protecting group. The major discovery came 25 years later by Fischer's former student Max Bergmann and a colleague Leonidas Zervas[3] when they invented the carbobenzoxy group. This benzyloxycarbonyl protecting group, which could be selectively removed without damage to the peptide chain, quickly became the key to the synthesis of small peptides and, later, of large peptides and small proteins. Other parts of the synthesis problem involved the development of better carboxy-activating groups and appropriate protecting groups for the various side-chain functional groups. All of these reagents have been described in detail (see Vol. E22a).

5.1.3 Representative Peptides Synthesized in Solution

The availability of the new chemistry (see Section 5.1.2) soon led to the synthesis of many small peptides and to several larger ones containing any of the naturally occurring amino acids in virtually any desired sequence. Among these were oxytocin, angiotensin, α-MSH, β-MSH, calcitonin, gastrin, glucagon, and ACTH. The latter 39-residue peptide was obtained via a particularly elegant synthesis[4] employing both stepwise assembly of fragments using the nitrophenyl ester technique and coupling of these fragments using the early azide method of Curtius or by the mixed anhydride and *N,N*-dicyclohexylcarbodiimide methods, as outlined in Scheme 2. The fragments were synthesized stepwise with Z-(amino acid)-ONp esters and deprotected using H_2 with Pd/C. This was the largest peptide synthesized through 1963.



Scheme 2 Synthesis of ACTH by a Combination of Stepwise and Fragment Approaches[41]

for references see p 38

5.1.4 Early Syntheses of Proteins

The first molecule that had been sequenced and could be considered a true protein was insulin.[5] It consists of two peptide chains of 21 and 30 residues held together by one intrachain and two interchain disulfide bonds. The molecule was originally thought to be a protein of about 12000 Da molecular weight, but was convincingly shown[6] to be composed of only these two chains (Mwt 5733 Da). It could be reduced and separated into the two chains, and the chains could be recombined to give an active hormone. Thus, the stage was set for an all-out synthetic attack on a protein.

5.1.4.1 Fragment Synthesis of Insulin

The synthesis of insulin was achieved by three groups working independently in three countries virtually simultaneously. Pannoyatis Katsoyannis, working in Pittsburgh, PA, first reported his synthesis in June, 1963; however, that paper was not in print until January, 1964.[7] A German group under the direction of Helmut Zahn published their synthesis in December, 1963,[8] while a large Chinese group, working in Shanghai and Beijing, published their complete synthesis of crystalline bovine insulin in 1965.[9]

All these syntheses involved the separate preparation of the 21-residue A chain and the 30-residue B chain by fragment methods, followed by a combination of the two purified chains by oxidation of the six cysteines to give the three disulfide bonds. The general strategies were the same, but each group used different chemistry and different connecting points for the various fragments. The conditions for refolding the chains and oxidizing the thiol groups were also different. It is difficult to calculate overall yields, but each research group was able to prepare purified A and B chains and to combine synthetic A with natural B, synthetic B with natural A, and synthetic A with synthetic B to obtain biologically active insulin.

The synthesis of the A chain of sheep insulin by Katsoyannis et al.,[11] in which the final coupling was between residues 1–9 and 10–21, is illustrated in Scheme 3.

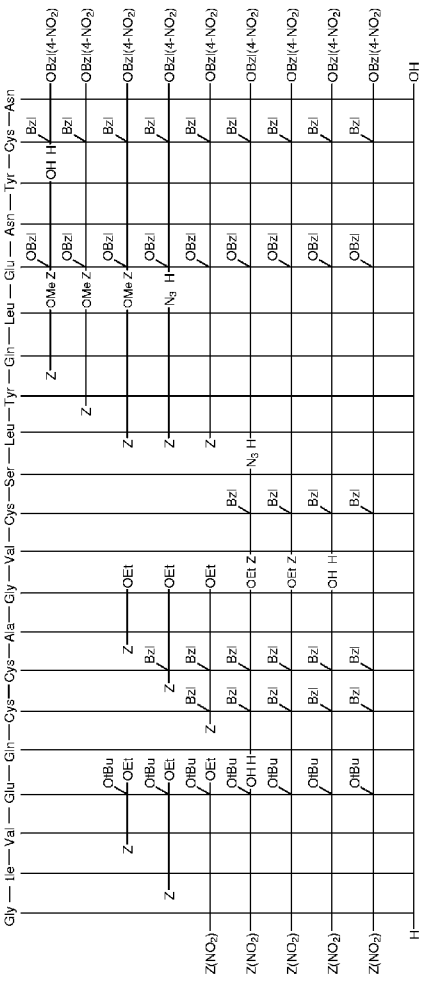
The A-chain synthesis by Meienhofer et al.[8] used the same 1–9 and 10–21 fragments, but they employed azide or anhydride activation for the various condensations. Likewise, Kung et al.[9] used the 1–9 and 10–21 condensation at the last step of their synthesis.

The synthesis of the insulin B chain can be illustrated by the scheme of Meienhofer and Schnabel (Scheme 4).[12] They used 9–20 and 21–30, and then 1–8 fragment couplings using the azide method or 4-nitrophenyl esters. The Chinese group[9] selected fragments 9–16 and 17–30, and then 1–8, while the Katsoyannis group[7] used fragments 10–13 and 14–30, and then 1–9. In all three syntheses, the side-chain protection and coupling methods differed in a variety of ways.

Combination of the A and B chains[8,11] involved deprotection of Cys(Bzl), Lys(Tos), and Arg(Tos) or Arg(NO_2) by sodium in liquid ammonia and slow air oxidation to form the intrachain A_6–A_{11} disulfide and the two interchain A_7-B_7 and A_{20}-B_{19} disulfides. In some experiments, the *S*-sulfonates were utilized for combination of the chains. All three syntheses produced material with insulin activity in the mouse convulsion assay or the glucose oxidation assay.

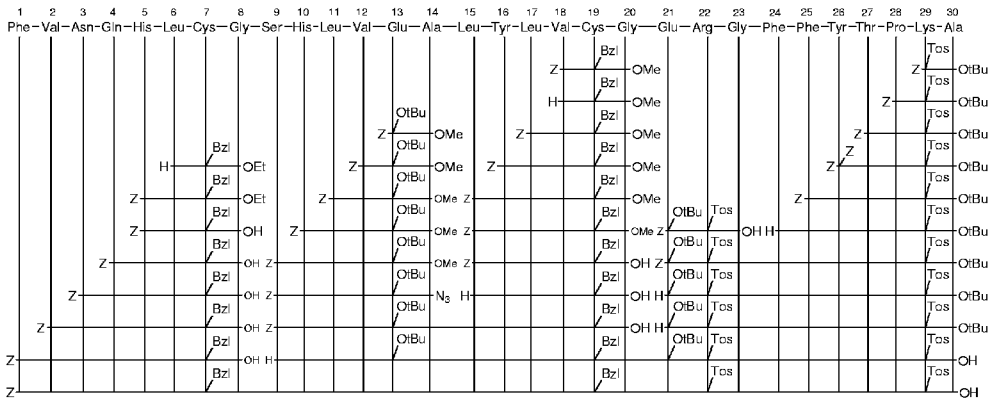
These methods initially led to low recovery of insulin from its natural chains, but the synthetic chains together, or in combination with a natural chain, gave comparable activity (in the 0.5 to 1% range). The Chinese group used the method of Du et al.[10] for the combination of the A and B chains. At a ratio of 1.5:1, the specific activity was approximately 60% based on the A chain. These were remarkable, labor-intensive projects, which represent the first syntheses of a natural protein. They provided new evidence that the structure of insulin proposed by Sanger[5] was correct.

Scheme 3 Synthesis of the Sheep Insulin A Chain[111]



for references see p 38

Scheme 4 Synthesis of the Insulin B Chain[12]


1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
Phe–Val–Asn–Gln–His–Leu–Cys–Gly–Ser–His–Leu–Val–Glu–Ala–Leu–Tyr–Leu–Val–Cys–Gly–Glu–Arg–Gly–Phe–Phe–Tyr–Thr–Pro–Lys–Ala
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5.1.4.2 Insulin by Controlled Disulfide Formation

In 1977, Sieber et al.[13] carried out an elegant controlled synthesis of human insulin in which the three disulfide bonds were formed separately using pairs of orthogonally protected cysteines. This eliminated the very difficult and low-yielding random oxidation used previously. The protecting groups were *S*-Trt for A_6 and A_{11}, *S*-Acm for A_7 and B_7, and also *S*-Acm for A_{20} and B_{19}. The following steps were then carried out in the order indicated in Scheme 5: (1) The two Trt cysteines in fragment A_{6-13} were oxidized to cystine A_6–A_{11} by treatment with iodine in TFE, without any influence on the Acm group at A_7. (2) The fragment A_{1-5} was coupled by the azide method to A_{6-13}. (3) The disulfide between A_{20} and B_{19} was formed by treatment of the Cys residues in A_{20-21} and B_{17-20} with iodine, and fragment coupling was continued to give Bpoc-A_{14-21}-(A_{20}–B_{19})-Trt-B_{17-30}. (4) The Trt group was selectively removed, in acid, from Leu^{17} of the B chain, leaving Bpoc-Tyr^{14} of the A chain intact. Then, B_{1-16} was coupled to complete the B chain. (5) The Bpoc group was removed and A_{1-13} was coupled using DCC to complete the A chain. (6) Finally, the third disulfide bond between A_7 and B_7 was formed using iodine, either before or after all the acid-labile protecting groups were removed in 95% TFA. Following purification, crystalline human insulin was obtained in high yield with full biological activity.

5.1.4.3 Proinsulin

When proinsulin was discovered by Steiner et al.,[14] the way was open for a total synthesis of this insulin precursor, which consists of a single chain composed of B-chain-(connecting peptide)-A chain (86 residues). This native molecule can be reduced and then reoxidized in high yield to give the same three disulfide bonds found in native insulin. Furthermore, there are two exopeptides in pancreatic β-cell granules that, together with carboxypeptidase H, will remove Arg^{64}-Arg^{65} and Lys^{31}-Arg^{32} selectively to give insulin.

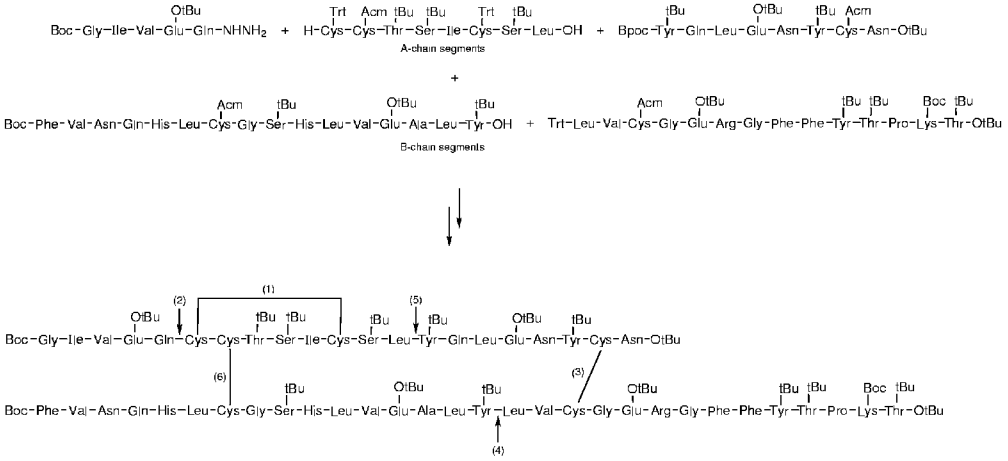
Human proinsulin has been synthesized in homogeneous solution from 11 protected fragments using azide coupling.[15] The difficulties with insoluble intermediates were sufficiently overcome to allow the 86-residue peptide to be synthesized. The product was deprotected, converted into the *S*-sulfonate, and then reduced and reoxidized to form the three disulfide bonds. The product was extensively purified and analyzed, and shown to be pure proinsulin. This product could then be converted into insulin by the use of endopeptidases I and II from pancreatic β-cell granules, together with carboxypeptidase H, which removed the four basic residues 31, 32, 64, and 65, and split out C-peptide.[16]

5.1.5 Solid-Phase Peptide Synthesis

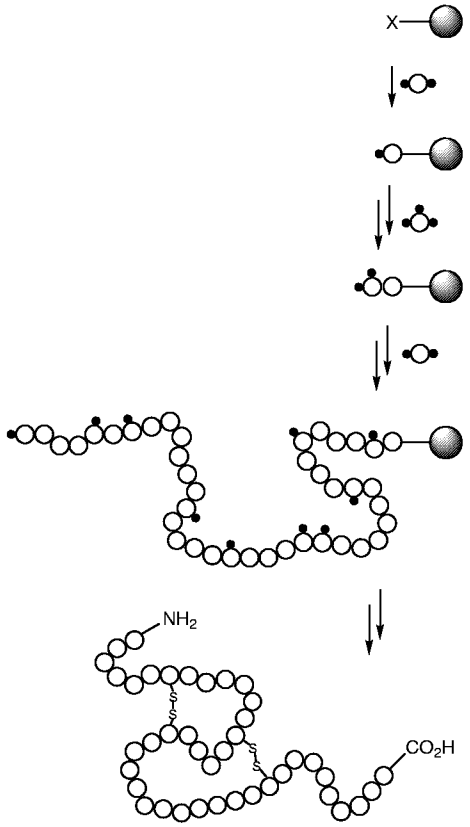
In 1962, solid-phase peptide synthesis was introduced, and described in detail in 1963.[17] This new synthetic principle was developed with the hope that it would simplify and accelerate the synthesis of peptides. The idea was to covalently anchor the C-terminal residue of a peptide to an insoluble support and then to assemble the remaining amino acids in a stepwise manner with activated amino acids while the peptide was in the insoluble solid phase, and finally to cleave the peptide from the solid support and liberate it into solution. The general methodology is shown in Scheme 6.

for references see p 38

Scheme 5 Total Synthesis of Human Insulin by Selective Formation of Disulfides at Different Stages of a Fragment Condensation Scheme[1,3]



Scheme 6 The Basic Idea of Solid-Phase Peptide Synthesis[17]

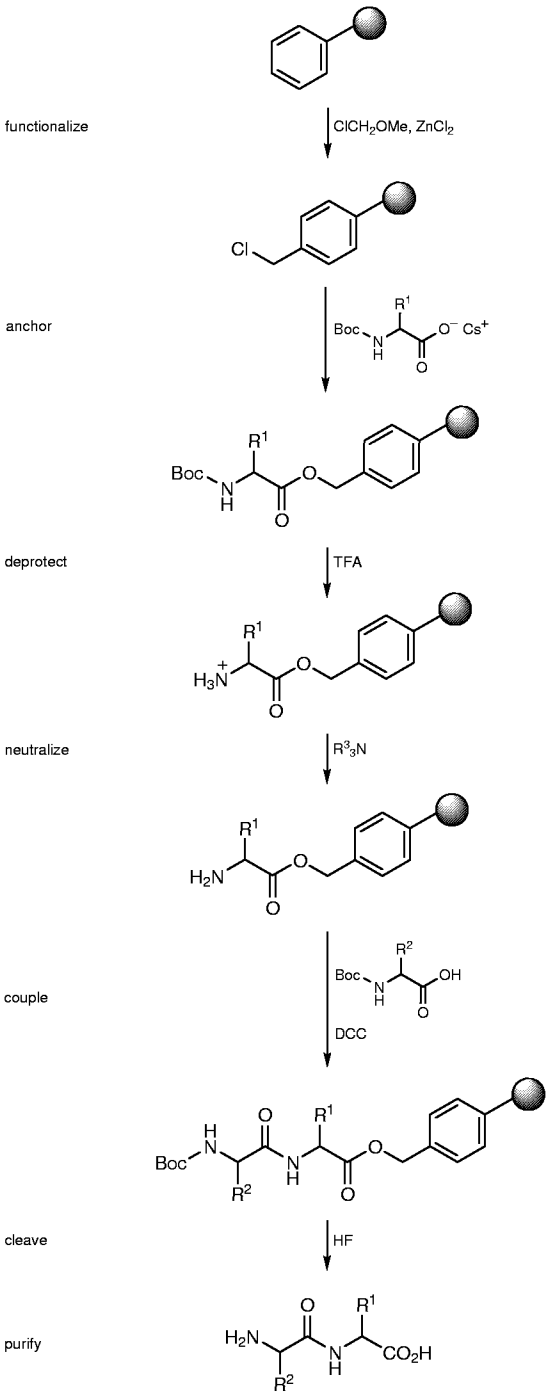


The first useful support was a small (100-μm diameter) bead of copoly(styrene-1%-divinylbenzene). It was not soluble in any common solvent, but was highly solvated and swollen in aromatic and chlorinated hydrocarbons, which allowed the solvents and reagents to readily enter the beads and to react very rapidly with the growing peptide chain. The aromatic rings of the polymer were chloromethylated and the first N^{α}-protected amino acid was attached as a substituted benzyl ester. The N^{α}-carbobenzoxy group was initially examined, but the more acid-labile Boc group was found to be more suitable. After each coupling it was removed with anhydrous HCl in dioxane or with TFA. Activation of the carboxy group was possible utilizing many of the procedures already developed for classical synthesis, but the DCC method was the most useful. The incoming protected amino acid in CH_2Cl_2 was added to the highly swollen beads in this solvent, followed by DCC. The activation was rapid and the coupling was also rapid; kinetic studies showed complete coupling in two minutes or less. Alternatively, preformed symmetrical anhydrides could be used, and many new methods are now available for the coupling.

Finally, the completed peptide was deprotected and cleaved by strong acid, either HBr in AcOH or in TFA, or by anhydrous HF. Many variations of the support and of each step of the chemistry have been developed over the succeeding years.[18] An early synthetic scheme is shown in Scheme 7.

for references see p 38

Scheme 7 A Synthetic Scheme for the Preparation of Peptides on a Solid Support[19]



5.1.5.1 Early Examples of Small Peptides Using Solid-Phase Synthesis

The solid-phase method was tested with the synthesis of many small peptides,[19] including bradykinin, angiotensin, desaminooxytocin, and the 20-residue S-peptide of ribonuclease A, and was found to be useful and efficient.

5.1.5.2 Solid-Phase Synthesis of Insulin

Two years after the first syntheses of insulin had been published, it was shown that the A and B chains of insulin could be prepared by solid-phase synthesis;[20] the two chains were partially deprotected and cleaved from the resin support in good yields. Final removal of the *S*-benzyl groups was with sodium in liquid ammonia, and the A and B chains were each converted into their *S*-sulfonates with sodium sulfite and sodium tetrathionate in 37 and 21% overall yield, respectively, calculated from the first amino acid of each chain. These derivatives were highly purified by countercurrent distribution. The $A(SSO_3)_4$ and $B(SSO_3)_2$ chains were mixed, reduced with thioglycolic acid, and oxidized in air at pH 10. The oxidation mixture was extracted using the method of Du et al.[21] and lyophilized. The electrophoretic mobility of the insulin preparation was the same as that of standard insulin. The product had 8% of the mouse convulsion activity of standard insulin, which is comparable to that obtained initially by the fragment method.[7–9]

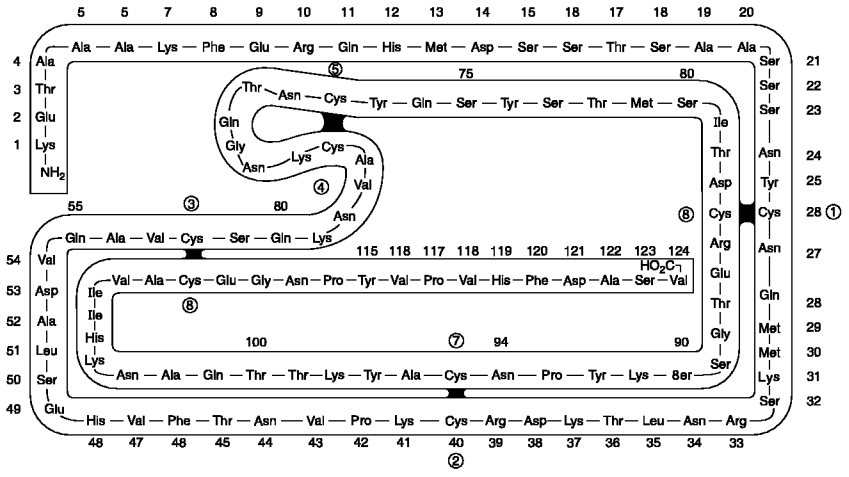
5.1.6 Synthesis of Enzymes

In 1968, it was decided that it was time to apply our solid-phase methods to the synthesis of an enzyme. Much as Fischer had done over 50 years earlier, we wanted to test the limits of the new technique and to answer some important chemical and biological questions.

5.1.6.1 Solid-Phase Synthesis of Ribonuclease A

Ribonuclease A (RNase A) was selected as the target enzyme for solid-phase synthesis because its sequence was known (Scheme 8),[22,23] and an X-ray structure had been deduced.[24] Importantly, it had been shown that this 124-residue protein could be reduced and unfolded and then reoxidized to re-form the four disulfide bonds with recovery of full enzymatic activity.[25]

Scheme 8 The Two-Dimensional Sequence of Ribonuclease A[23][a]



[a] Adapted from Smyth; Stein; Moore,[23] with permission.

for references see p 38

The synthesis[26,27] made use of the standard C-to-N stepwise solid-phase assembly of the peptide chain, which was covalently anchored through a benzyl ester linkage to co-poly(styrene-1%-divinylbenzene) beads. N^{α}-Boc amino acids were activated by DCC or, in the case of Asn and Gln, by 4-nitrophenyl esters. The esters were coupled in DMF, while all other couplings were performed in CH_2Cl_2. The maximum side-chain protection strategy was adopted, using benzyl esters for Asp and Glu, benzyl ethers for Ser, Thr, and Tyr, benzyl sulfide for Cys, the Z group for Lys, the NO_2 group for N^{γ}-Arg, and the sulfoxide for Met. Histidine was unprotected.

The Boc group was removed at each step with 50% TFA in CH_2Cl_2, under conditions where the other protecting groups were largely stable. Following neutralization of the α-amino trifluoroacetate with TEA at each step, the H_2N-peptide-resin was ready for the next coupling. The synthesis was conducted in an automated peptide synthesizer[28] at the rate of four residues per day. Numerous analytical samples were collected during the synthesis and larger samples were saved after the formation of RNase(26–124)-resin and RNase(21–124)-resin for later cleavage and workup. A total of 47% of the starting resin was removed during the synthesis. Approximately 1.4% of the anchoring benzyl ester was lost at each deprotection step, resulting in a decrease from 0.21 to 0.036 mmol of peptide per gram of polystyrene (17% peptide assembly yield).

The completed RNase(1–124)-resin was cleaved in three ways: (1) 60% Anhydrous HF plus 20% TFA and 20% anisole at 0 °C for one hour (yield: 41%). (2) 80% HF plus 20% anisole (cleavage yield: 70%). (3) HBr bubbled through a mixture of 10 mL of TFA plus 10 mL of CH_2Cl_2 and 2 mL of anisole for 90 minutes. After evaporation, anhydrous HF containing 10% anisole at 0 °C was used to remove the *S*-benzyl groups (yield: ca. 41%). Any N→O shift was reversed in NH_4HCO_3 after each treatment. Material from cleavage method 2 was passed through Sephadex G-75 to give 3 peaks. The center peak contained 45% of the total, with a specific activity of 5% against RNA substrate, while aggregates were at the exclusion volume and small peptides appeared late. The peptide from the main center peak was reduced, reoxidized, and fractionated on IRC-50. Three peaks were obtained, but only peak 1 (60%) had RNase activity (13% specific activity) and a satisfactory amino acid analysis. This product was run on a CM-cellulose column and eluted at the same position as natural RNase (50 ± 1 mL).

Native RNase is quite resistant to digestion with trypsin, even at a w/w ratio of 1:20, but small or unfolded fragments would be expected to be digested. When the synthetic enzyme was treated with trypsin, a 70% recovery of protein with a specific activity of 61% was obtained. Treatment of this material with saturated ammonium sulfate (diluted 16:26), pH 4.6, gave 33% of amorphous precipitate and 66% of soluble RNase A. The overall yield from the first Val residue was only 3%, but the specific activity was quite high at 78%. This is as far as the purification was carried out at that time.

Peptide maps of tryptic digests of oxidized synthetic RNase A showed the expected 14 components plus a spot at the position of lysine.

The substrate specificity of the synthetic enzyme was also studied. The enzyme was active against RNA, but not DNA, and it was active against cyclic cytidine phosphate, but not cyclic guanosine phosphate, 5′-(3′-guanylyl)cytidylic acid, or adenylyladenylic acid. Thus, it was active against both large and small substrates and it required ribose, not deoxyribose. It was effective against pyrimidine substrates, but was inactive against purine substrates or cytidine in the 3′-position of a dinucleotide. This is the substrate specificity expected for RNase A. Finally, a rabbit antibody to crystalline RNase A neutralized equivalent amounts of natural and synthetic RNase A.

From this first synthesis of an enzyme, two general conclusions were drawn. Firstly, it is possible to begin with free amino acids and to assemble them in the laboratory to give a real protein that possesses true enzymatic activity; secondly, a new and independent kind of evidence was provided for the view that the primary structure of a protein determines its tertiary structure.

It was known from the work of Richards that RNase A could be cleaved between residues 20 and 21 by digestion with subtilisin.[29] The N-terminal 20-residue peptide (S-peptide) could be separated from the 104-residue 21–124 protein (S-protein), and each was enzymatically inactive; however, when they were mixed in a 1:1 ratio, full RNase activity was regenerated even though the covalent bond between residues 20 and 21 was not re-formed. This finding was the basis for a great many structure/function studies by the synthesis of analogues of the S-peptide and their noncovalent recombination with natural S-protein.[30,31]

5.1.6.1.1 Ribonuclease S and des(21–25) Ribonuclease S

The synthesis of RNase A (see Section 5.1.6.1) allowed us to examine changes in the S-protein (104-residue 21–124 protein) by removing samples after coupling Cys^{26} and again after coupling Ser^{21}.[27] These two proteins, RNase(26–124) and RNase(21–124) were cleaved and isolated, combined with synthetic or natural S-peptide (the N-terminal 20-residue peptide), and oxidized to give RNase S and des(21–25) RNase S. All four products showed the same enzymatic activity, demonstrating that the five residues Ser^{21}, Ser^{22}, Ser^{23}, Asn^{24}, and Tyr^{25} are not necessary for RNase activity, even though residues 24 and 25 had been predicted to be important.

5.1.6.1.2 The Role of C-Terminal Residues of Ribonuclease

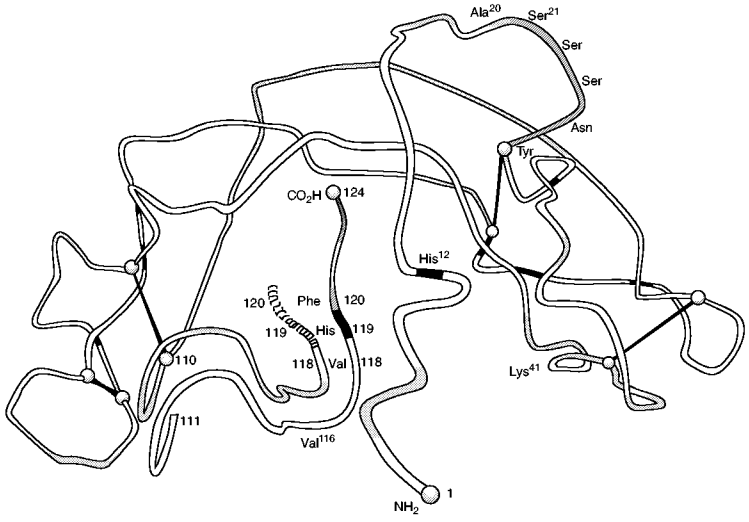
It was possible to carry out a semisynthesis study at the C-terminus of RNase because it had been shown[32] that enzymatic removal of 4, 5, or 6 residues from the C-terminus of RNase A greatly decreases the stability of the protein and eliminates the enzymatic activity. The synthetic tetradecapeptide RNase(111–124) can associate noncovalently with natural RNase(1–118) and regenerate 80% of the enzymatic activity of RNase.[32] This was quite similar to the S-peptide/S-protein system (see Section 5.1.6.1). By the synthesis of shorter peptides or of replacement analogues, the roles of individual residues were examined in this semisynthesis system. It was found that replacement of Phe^{120} by Tyr^{120} gave a complex 98% as active toward 2′,3′-cyclic cytidylic acid as the parent Phe^{120}.[33] In contrast, when 2′,3′-uridylic acid was the substrate, the Tyr^{120} complex was 190% as active. A semisynthetic analogue that is more active than the native sequence was a novel finding. Even more extreme, the complex between RNase(1–118) and [Ala^{123}]-RNase(111–124) was 19 times as active toward cytidine-containing substrates as it was toward uridine-containing substrates. These relationships, illustrated in Scheme 9, suggest that a semisynthetic enzyme could be designed with high substrate specificity.

5.1.6.1.3 An Active Enzyme by a Noncovalently Bound Three-Component System

Finally, for the first time it was shown that a three-component system could lead to an active protein complex.[32] Equimolar amounts of RNase(1–20) containing His^{12}, RNase(111–124) containing His^{119}, and RNase(21–118) containing Lys^{41} were combined noncovalently and 30% enzymatic activity was generated. Thus, the three known catalytic residues of RNase in three separate chains must have become precisely aligned to give a catalytic site very similar to that found in active RNase A (see Scheme 9).

for references see p 38

Scheme 9 A Three-Dimensional Representation of RNase A, Showing the Relationship of the S-Peptide (1–20), C-Peptide (111–124), S-Protein (21–124), and Shortened S-Protein Derivatives[24][a]



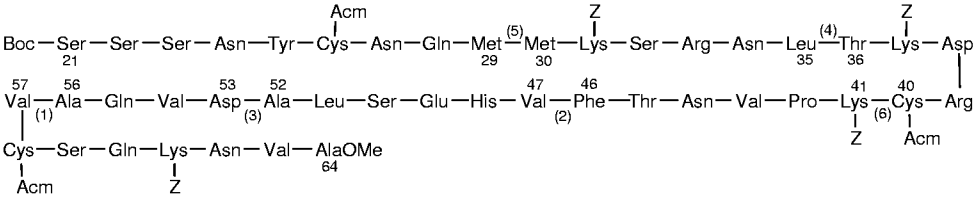
[a] Reprinted from [Kartha; Bello; Harker, *Nature (London)*, Vol. 213], Copyright (1967), p 862, with permission from Macmillan Publishers Ltd.

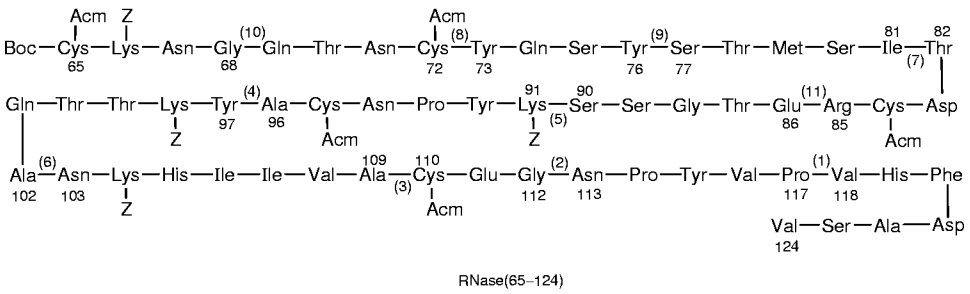
5.1.6.2 Solution-Phase Syntheses of Ribonucleases

5.1.6.2.1 Ribonuclease S

Simultaneously with the RNase A synthesis by Gutte and Merrifield,[26] a large group at Merck, Sharp, and Dohme under Denkewalter and Hirschmann published on their fragment synthesis of S-protein.[34] Since it was known that noncovalent combination of the 20-residue S-peptide with the 104-residue S-protein will regenerate full enzymatic activity, it was reasoned that synthesis of RNase(21–124) would constitute the synthesis of an enzyme. Thus, a series of fragments was prepared, using in part a new, controlled *N*-carboxyanhydride method, and combined in solution to give protected RNase(21–64) and RNase(65–124) (Scheme 10). It is difficult to estimate the overall synthetic yield, but it appears that the 19 fragments were combined one at a time, in the numerical order shown in brackets in Scheme 10, with individual yields of 35 to 60% at each coupling. Finally, these two large 44- and 60-residue fragments were combined by the azide method in HMPA, which after workup produced S-protein [RNase(21–124)]. When 58 mg of the HF-deprotected S-protein was combined with natural S-peptide, it generated RNase S that had 2 μg of ribonuclease activity.

Scheme 10 Fragment Syntheses of RNase(21–64) and RNase(65–124)[34]





5.1.6.2.2 Ribonuclease A

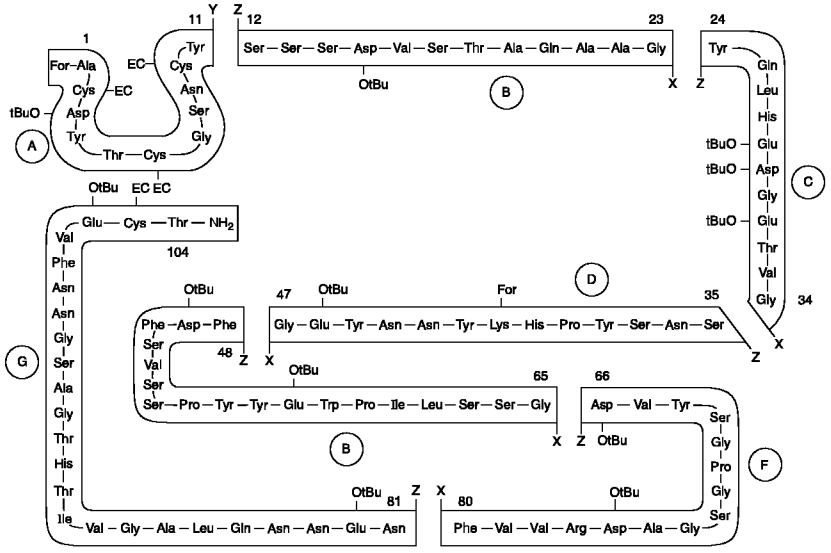
Some 10 years later, Yajima and Fujii completed a fragment synthesis of RNase A using partial protection of side chains and TFOH for deprotection.[35] The yields were quite high and the secret of success was the use of many small peptides (only 3–12 residues) in a stepwise design with azide coupling, so the inefficient condensation of two very large peptides could be avoided. A mixed solvent composed of DMF, DMSO, and HMPA was used. The product was oxidized in air at pH 8.0 in the presence of reduced and oxidized glutathione and gave a protein that was 13% active relative to natural RNase. It is interesting that this is exactly the same activity as reported earlier for the solid-phase synthesis (see Section 5.1.6.1).[26] Merrifield et al.'s further purification by controlled tryptic digestion and ammonium sulfate fractionation raised the specific activity to 78%.[27] Yajima and Fujii went on to further purify their product by affinity chromatography, which was now available to them, and then on CM-cellulose, and obtained a protein that could be crystallized and was fully active in the RNase assay!

5.1.6.2.3 Ribonuclease T1

Following the original publications on RNase A and RNase S-protein, progress on the synthesis of ribonuclease T1 was reported.[36] The plan followed a typical fragment scheme (Scheme 11) in which a total of seven peptides were prepared by way of numerous small di- or tripeptide intermediates. Further stepwise elongation of these with additional Boc-protected dipeptides was by azide coupling. A minimal protection strategy was employed in which only Lys(For), Glu(OtBu), Asp(OtBu), and Cys(EC), together with N^{α}-For and C^{α}-NHNH-Boc, were used. The seven fragments A(1–11), B(12–23), C(24–34), D(35–47), E(48–65), F(66–80), and G(81–104) were purified to apparent homogeneity and carefully analyzed. Fragments B, C, and D were coupled and then fragment A was added by the azide method to give residues 1–47.[37] In addition, E and F were combined to give residues 48–80. Throughout the synthesis low solubility was encountered, requiring the use of mixtures of DMF and DMSO. For purification, 45% HCO_2H was sometimes useful and in other cases 8 M urea plus TEA was preferred. Condensation of these two large fragments followed by coupling with amide fragment G(81–104) was apparently unsuccessful, presumably because of extreme insolubility problems.

for references see p 38

Scheme 11 Plan for the Solution Fragment Synthesis of Ribonuclease T1[36][a]



[a] Reprinted from (Yanaihara; Yanaihara; Dupuis; Beacham; Camble; Hofmann, *Journal of the American Chemical Society*, Vol. 91), Copyright (1969), p 2185, with permission from The American Chemical Society.

The fragments were then assembled in a different order which provided RNase T1(66–104);[38] however, the further addition of residues 1–65 was not reported and is assumed to have been unsuccessful as a result of solubility problems.

5.1.6.3 Solution-Phase Synthesis of Lysozyme

Following the ribonuclease T1 study (Section 5.1.6.2.3), the syntheses of a number of long peptides and small proteins were undertaken in several laboratories using the solution-phase approach. These included cytochrome c, ferridoxin, trypsin inhibitor, and a 129-residue analogue of lysozyme.[39] In the latter synthesis, the complete sequence was assembled through a great effort by many people, but the complete deprotection of the eight *S*-Acm groups was not possible. Apparently, during the treatment with mercury(II) acetate the intermediates became insoluble and prevented further reaction. The reason for this synthetic undertaking and the details of the chemistry were beautifully presented in George Kenner's Bakerian Lecture before the Royal Society in 1976.[39] His first aim was to demonstrate how a protein can be synthesized using all the methods available at the time, and the second was to "open doors to the understanding of biological processes, including the interaction between small peptides and protein receptors". A molecule differing from hen egg lysozyme (129 residues) at 28 residues was selected to reduce the expected synthetic difficulties. Thus, several Arg, His, and Met residues were deleted or replaced by Gly. The molecule was divided into 12 segments for subsequent condensation and the eventual construction of two large protected peptides containing 75 and 54 residues. A maximum protection scheme was adopted. It included many acid-labile side-chain groups, and the Acm group for cysteine. α-Nitrogen was temporarily blocked by the Z or, where hydrogenolysis was not possible, by the biphenylisopropyloxycarbonyl group. Temporary carboxy protection was achieved with a phenyl ester method. The many potential side reactions and other problems were very carefully considered. Couplings were performed with mixed anhydrides, active aryl esters, or DCC plus HOSu. Solubility problems were frequently encountered with the hydrophobic intermediates; the DMF/HMPA solvent system was helpful in overcoming this problem. All

fragments were purified chromatographically. Finally, the two segments 1–75 and 76–129 were successfully coupled to give the fully protected lysozyme analogue 1–129.

In these and other fragment syntheses one of the main problems continued to be solubility of intermediates, which caused slow or incomplete coupling reactions or gave no product at all. Several solutions to this problem have been proposed with varying success. The stepwise addition of much smaller fragments is one of the best ways,[35] and the development of better solvent mixtures possessing higher solvating power is another. The recent advances by Sakakibara and his colleagues (Section 5.1.7) are remarkably good and mark the beginning of a new era in the solution synthesis of proteins.

5.1.7 The Search for a General Protocol for Protein Solution Synthesis

Following the experiences and difficulties illustrated with the solution synthesis of proteins (*vide supra*), Sakakibara and his team at the Protein Research Institute in Osaka, Japan, began a search for a general protocol that would minimize the existing problems. In 1981, the essential features of such a plan were outlined as being (1) maximal side-chain protection with the use of benzyl-type groups where possible, (2) use of a water-soluble carbodiimide plus HOBt for fragment condensations, (3) cleavage of all protecting groups simultaneously with HF, and (4) detection and removal of byproducts using HPLC.[40] Although Sakakibara was fully aware of the solubility problems frequently reported in the literature and from the experiences of his own group, this was not listed as one of the four main considerations of his plan. However, this later became a foremost consideration and led to the development of powerful new solvent systems. Maximum protection, benzyl ester protection, a carbodiimide activating reagent, cleavage and deprotection with HF, and purification using HPLC were standard practice in solid-phase peptide synthesis at that time.

The great advantage of solution synthesis was well recognized by everyone in the field to lie in the ability to purify small intermediates. The disadvantages were solubility problems, slow and incomplete coupling reactions, and the large amounts of time and labor required. The Sakakibara plan overcame most of the problems and the new strategy for the synthesis of large peptides in solution was demonstrated by the synthesis of [Asn^{76}]-human parathyroid hormone [hPTH(1–84)] from 13 smaller fragments, each protected with a C-terminal phenacyl ester and an N^{α}-Boc group.[41]

5.1.7.1 Further Development of Solvent Systems

The Sakakibara group, armed with their new strategy (Section 5.1.7), went on to synthesize numerous large peptides and small proteins, including angiogenin (123 residues).[42] This was assembled from 15 small segments and only one segment, 100–106, presented solubility problems. The final coupling was carried out in a mixture of NMP/DMSO rather than in either alone. In the assembly of a large number of peptide segments, 95% were soluble in DMF or in a mixture of DMF and NMP. Of the insoluble peptides, 3% became soluble in NMP plus DMSO. The remaining 15 peptides (1%) were insoluble in these solvents or in pure DMSO. Therefore, the search for even better solvent systems was continued.[43] The insolubility problems were more dependent on sequence than on size, and were considered to be due to β-sheet aggregation resulting from intra- or interchain hydrogen bonds.

The most powerful new solvent systems utilized were mixtures of TFE or HFIP and $CHCl_3$ or CH_2Cl_2. For water-soluble carbodiimide plus HOOBt couplings of fully protected Boc fragments with other sparingly soluble peptide esters, TFE/$CHCl_3$ or TFE/CH_2Cl_2 was the best solvent system, and reactions proceeded without racemization or trifluoroethyl ester

for references see p 38

formation. A second effective solvent system was a mixture of $CHCl_3$/PhOH (3:1). Thus, several of the larger fragments that were not soluble in TFE/$CHCl_3$ were soluble in $CHCl_3$/PhOH and this solvent system was used for those coupling reactions. The slight formation of phenyl esters was not considered to be a serious problem. These solvent mixtures are much more effective than chaotropic agents such as LiCl.

5.1.7.2 Examples of Solution Syntheses of Larger Proteins

The new solvent systems (Section 5.1.7.1) were an important advance and opened the way for the solution segment synthesis of even larger and more complex proteins. For example, human midkine (121 residues) containing 5 disulfide bonds was assembled from Boc-(1–59)-OH and H-(60–121)-OBzl, which were not soluble in DMSO, but were readily soluble in $CHCl_3$/TFE (3:1) (Scheme 12).[44] The coupling was undertaken with a water-soluble carbodiimide containing HOOBt and was complete in one hour.

The protected protein was deprotected with HF and then with mercury(II) acetate in 50% AcOH to remove the Acm groups. Oxidative folding gave the desired peptide. A tryptic digest was separated on HPLC and the seven identified peaks confirmed the correct disulfide bonds in the product.

5.1.7.2.1 The 238-Residue Green Fluorescent Protein

The Sakakibara group (*vide supra*) has extended the range of accessible proteins via fragment coupling in solution by a factor of 2. Their new synthetic strategy was applied to the preparation of a linear 238-residue protein, the precursor of the *Aequorea* green fluorescent protein.[45] A total of 26 fragments were combined stepwise from the C-termini to give four large fragments containing 51 to 65 residues each. These were then combined into two very large protected peptides, Boc-(1–116)-OPac and Boc-(117–238)-OBzl and finally these two were coupled in $CHCl_3$/TFE to give the fully protected protein (Mwt 27037.2 by mass spectrometry). It was deprotected as usual and air oxidized to form the Cys^{48}/Cys^{70} disulfide, passed through a Sephadex G-25 column, and further purified by HPLC. It was shown to be of high purity and gave the expected analytical data.

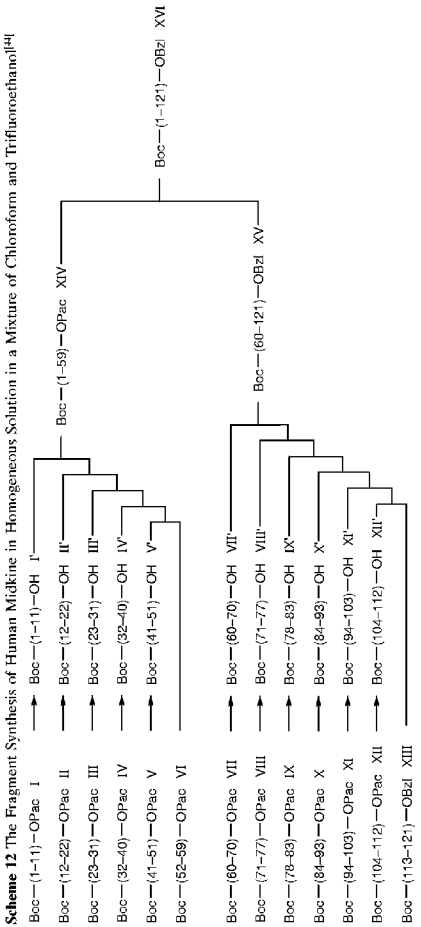
It is of some interest to note that 24 of the starting fragments were prepared by solid-phase methods in order to simplify and accelerate the synthesis. This confirms what we have believed for a long time, i.e. that the various approaches (chemical, enzymatic, and biological) to the synthesis of peptides and proteins are not mutually exclusive but, rather, they can complement one another when judiciously selected for the particular problem at hand.

5.1.8 Continuing Improvements in Solid-Phase Synthesis

Just as improvements have been made in classical solution-phase protein synthesis (see Section 5.1.7), so have important and effective improvements been made in solid-phase synthetic methodology. These improvements have enhanced the ease and speed of synthesis, the quality and yields of products, and the size of proteins accessible to chemical synthesis.[18]

Numerous laboratories have studied each of the steps in the solid-phase process and the details of the changes are described in detail in other sections of this volume (see Vol. E 22a, Section 4.3). Briefly, the following modifications have been forthcoming:

Scheme 12 The Fragment Synthesis of Human Midkine in Homogeneous Solution in a Mixture of Chloroform and Trifluoroethanol[44]



for references see p 38

1. New polymeric solid supports have been devised, which include macroporous styrene-divinylbenzene containing large fixed pores, porous glass beads, insoluble carbohydrate polymers, poly(ethylene oxide), cross-linked derivatives of polyacrylamide resins, and graft copolymers of polystyrene and poly(ethylene oxide). The last two have been the most effective and widely used and have competed well with the original copoly(styrene-divinylbenzene) beads.

2. Many new functional groups have been attached to these supports either directly or via various handles. They were designed to provide anchoring bonds for the peptides that could be cleaved at the end of a synthesis by a wide range of acidic reagents, from AcOH, through TFA, to the strong TFOH and HF. Other anchoring bonds are stable to acid but cleaved by basic reagents, while some are broken by neutral reagents or photolysis.

3. Initially, the Boc group was used for reversible α-amine protection and most side-chain functional groups were protected as benzyl derivatives, which are stable to Boc removal in HCl or TFA. This Boc/Bzl strategy is still frequently used and is the method of choice in several laboratories. It has, however, been replaced in many laboratories by the base-labile Fmoc group, which allows weak acid deprotection of *tert*-butyl groups from the side chains. For short- and moderate-sized peptides both systems are effective. For protein synthesis, the relative merits have not yet been fully established.

4. Some of the most important and useful changes in the chemistry have been in the development of new carboxy-activating reagents. The original DCC reagent continues to be effective, but it has been supplemented by additives, especially HOSu, HOBt, or HODhbt, or replaced by water-soluble carbodiimides. Preformed active esters (4-nitrophenyl, pentachlorophenyl, benzotriazol-1-ol) continue to be used, especially for coupling Asn and Gln to avoid nitrile formation. Other important activating reagents include symmetrical anhydrides, acid fluorides, and urethane-protected *N*-carboxyanhydrides. The most frequently used new reagents are based on phosphonium derivatives, such as BOP,[46] or uronium derivatives, such as HBTU.[47]

5. Just as they are in solution syntheses, solvent systems and coupling conditions are very important in solid-phase synthesis, although for different reasons. Extreme insolubility of intermediate peptide fragments is often encountered in solution syntheses and powerful solvent systems such as TFE/$CHCl_3$ are needed (see Section 5.1.7.1). This is not encountered in solid-phase work because of the strong peptide-solvating effects of the resin support.[48] There are, however, certain peptides that self-aggregate and do not couple efficiently under the usual conditions. This can be partially overcome by solvent changes, decreased peptide concentration, increased temperature, chaotropic agents, pH adjustment, and reversible peptide-bond derivatives. The usual solvents are CH_2Cl_2, DMF, NMP, and DMSO.

When solid-phase peptide synthesis was initially being developed, the question of whether or not a separate neutralization step is necessary was considered. Since it was known from the work of others that the chloride ion promotes racemization during the coupling step in classical peptide synthesis, and since we were deprotecting the Boc group with HCl, it seemed advisable to neutralize the hydrochloride by treatment with TEA and to remove chloride by filtration and washing. This short, additional step was simple and convenient and became the standard protocol. Subsequently, we became aware of three other reasons why neutralization was desirable: (1) to avoid weak acid catalysis of piperazine-2,5-dione formation,[49] (2) to avoid acid-catalyzed formation of pyroglutamic acid (5-oxopyrrolidine-2-carboxylic acid),[50] and (3) to avoid amidine formation between DCC and protonated peptide-resin. The latter does not occur with the free amine.

What was not known by us at that time was that some peptide sequences have a tendency to change conformation when the α-amine hydrochloride or trifluoroacetate is neutralized. In these cases, neutralization in the presence of the incoming activated amino acid reduces the time during which the free α-amine is present, and appears to minimize the conformational change to a β-sheet structure that can cause aggregation and lead to in-

complete coupling. The idea of simultaneous neutralization and coupling was introduced by Castro et al.,[51,52] and several excellent syntheses were reported which used BOP as the activating reagent, even when the coupling time was reduced from 1 or 2 hours to 30 minutes. This procedure was adopted as part of an improved protocol in which the activating reagent was changed from DCC or BOP to HBTU.[53] Rapid-flow washes were introduced and, in addition, the time of the coupling reaction was further reduced to 10 minutes, so that the overall time of the synthetic cycle was only 15 to 19 minutes; the synthesis of a large peptide or small protein could be completed much more quickly, at a rate of 75 residues per day. This protocol using HBTU and in situ neutralization was compared with procedures involving a separate neutralization step followed by addition of a Boc amino acid activated with HOBt plus DIC or with preformed HOBt esters. In situ neutralization was found to give better coupling yields and fewer side reactions, especially with respect to the formation of 5-oxo-pyrrolidine-2-carboxylic acid from glutamine residues. HIV protease synthesis and a number of other examples demonstrated that this was an improved procedure. The initial concern about racemization by the chloride ion apparently is not a problem when the trifluoroacetate ion is present during coupling reactions.

Opinions differ as to which support, which anchoring bond, which side-chain protection, which α-amine protecting groups, and which activation reagent is best. Most of the changes in chemistry have certain advantages, but also certain disadvantages. Choices will depend on individual laboratories and especially on the problem at hand.

5.1.8.1 Representative Large Peptides and Small Proteins Using Solid-Phase Synthesis

Representative proteins that have been synthesized using solid-phase methods are presented in Table 1. The chemical system employed is also indicated to give an idea of how the choices have changed over the past 30 years. Very few comparative studies have been made, however, so it remains difficult to conclude which approach is actually the best one. It was clear from the beginning that the crude synthetic product is not entirely homogeneous, that purification steps are necessary, and that very good analytical procedures are required to assess the identity and purity of the product. Many researchers have been unwilling to agree that good products can be produced or that useful information can be obtained by this approach to the synthesis of peptides. My rationale was expressed in the following quote from 1976:[84] "The pessimist says that a pure product cannot be isolated and that, even if it were, its purity cannot be demonstrated. Since this attitude produces no progress, we prefer the pragmatic approach of being aware of the problem of purity and simply using the best methods currently available during synthesis, isolation, and characterization of the product. Improvements in separation techniques are appearing regularly; what cannot be achieved today may seem simple tomorrow. In the meantime much useful information can be gained using peptides purified by the available procedures." Many of these improvements have come to pass, and in the meantime much experience and valuable information has been gained.

for references see p 38

Table 1 Large Peptides and Small Proteins Synthesized Using Solid-Phase Methods

No. of Residues	No. of SH	Name of Peptide/ Protein	Solid Support	Machine Type	Protection of N^{α}-Side Chains	Coupling	Date	Ref
64	0	pre-procecropin A	PAM-PSty	batch	Boc, Bzl	DCC	1989	[54]
70	6	insulin-like growth factor	PAM-PSty	batch	Boc, Bzl	SA, DCC	1990	[55]
70	2	uteroglobin	PEG-PSty	fragment on resin	Fmoc, tBu	HATU, PyBOP	1996	[56]
72	4	interleukin-8 (Human)	PAM-PSty	batch	Boc, Bzl	SA, DCC	1991	[57]
74	2	β-chorionic gonadotropin	polyamide (Pepsyn K)	flow	Fmoc, tBu	SA	1989	[58]
76	0	ubiquitin	Wang	batch	Fmoc, tBu	SA, HOBt, DIC	1994	[59]
77	4	interleukin-8 (endothelial)	PAM-PSty	batch	Boc, Bzl	TBTU	1992	[60]
79	0	P9 cell regulator	polyamide	flow	Fmoc, tBu	TBTU, HOBt	1992	[61]
80	0	SRY (DNA binder)	Pepsyn KA	flow	Fmoc, tBu	TBTU, HOBt	1992	[62]
84	0	parathyroid hormone	PAM-PSty	batch	Boc, Bzl	DCC	1983	[63]
86	0	HIV-1 gag p24	PAM-PSty	batch	Boc, Bzl	DCC, HOBt	1990	[64]
86	7	HIV-1 Tat protein	PAM-PSty	batch	Fmoc, tBu	DCC, HOBt	1996	[65]
88	2	cardiodilatin	PAM-PSty	batch	Boc, Bzl	DCC, HOBt	1988	[66]
92	3	pre-progonadotropin releasing hormone	PAM-PSty	manual	Boc, Bzl	DCC, HOBt	1992	[67]
97	0	Gro ES chaperonin	PAM-PSty	batch	Boc, Bzl	DCC	1990	[68]
99	2	HIV-1 protease	PAM-PSty	batch	Boc, Bzl	DCC, HOBt, SA	1988	[69]
99	2	HIV-1 protease	PAM-PSty	batch	Boc, Bzl	DCC, SA, DMF	1988	[70]
99	2	SIV protease	PAM-PSty	batch	Boc, Bzl	SA or HOBt ester	1992	[71]
100	0	chaperonin 10	PAM-PSty	batch	Boc, Bzl	HBTU, HOBt	1996	[72]
104	2	ribonuclease T1	S-DVB	manual	Boc, Bzl	DCC	1974	[73]
109	0	prothymosin α	Trt(2-Cl)	batch fragment	Fmoc, tBu	DCC, HOBt	1991	[74]
115	2	McPC $V_H(1-115)$	PAM-PSty	batch	Boc, Bzl	DCC	1993	[75]
115	2	migration inhibitor factor	TentaGel	batch	Fmoc, tBu	TBTU, HOBt	1996	[76]
124	8	ribonuclease A	S-DVB	batch	Boc, Bzl	DCC	1971	[27]
124	8	ribonuclease A	MBHA	batch, fragment	Boc, Bzl	BOP subtiligase	1994	[77]
126	0	leukemia virus protease	PAM-PSty	batch	Boc, Bzl	DCC	1992	[78]

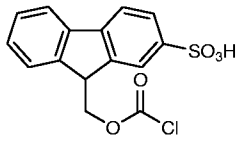
Table 1 (cont.)

No. of Residues	No. of SH	Name of Peptide/ Protein	Solid Support	Machine Type	Protection of N^{α}-Side Chains	Coupling	Date	Ref
126		integrin α 11b-β 3	PAM-PSty	batch, ligate	Boc, Bzl	HBTU, HOBt	1994	[79]
127	4	colony stimulating factor	PAM-PSty	batch	Boc, Bzl	SA, DCC	1988	[80]
134	0	p24(146–279)	PAM-PSty	batch	Boc, Bzl	DCC	1993	[81]
140	4	interleukin-3	PAM-PSty	batch	Boc, Bzl	SA (DCC), DMF	1986	[82]
166	5	interferon α_1	PAM-PSty	batch	Boc, Bzl	DCC	1983	[83]

Among the purification techniques now in use are gel filtration, ion-exchange chromatography, countercurrent distribution, isoelectric focusing, high performance reversed phase chromatography, affinity chromatography, antibody columns, and crystallization, while the analytical techniques include amino acid analysis, peptide sequencing, fingerprinting of enzyme digests, capillary electrophoresis, gel electrophoresis, analytical HPLC, NMR spectroscopy, and mass spectrometry.

A purification procedure for synthetic peptides was devised,[85] which uses the sulfobenzoic anhydride reagent of Wieland et al.[86] to terminate unreacted amino groups at each synthetic step and Sulfmoc chloride[87] (**1**, Scheme 13) to derivatize the deprotected full-length peptide at the end of the synthesis.

Scheme 13 Structure of Sulfmoc Chloride



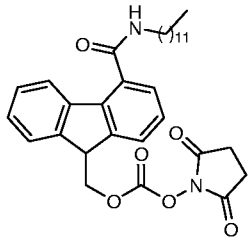
1

Following cleavage with hydrogen fluoride, the various classes of peptides were separated in a one-step purification procedure on a tertiary or quaternary amine column. After removal of the Sulfmoc group with 5% TEA, homogeneous Leu-Ala-Gly-Val, for example, was obtained. The Sulfmoc procedure was also very effective for purification of synthetic thymosin α_1 (28 residues).[87] This was the first use of an Fmoc derivative for selective and reversible orthogonal peptide purification.

A particularly nice application of a related purification procedure and analytical technique was published in 1996 for the Boc/Bzl-based synthesis of the 100-residue chaperonin 10,[72] which is a more demanding target than thymosin α_1. The procedure also incorporates a termination of unreacted amino groups at each cycle using *N*-(2-chlorobenzyloxycarbonyloxy)succinimide and the addition of a reversible Fmoc-based hydrophobic group after deprotection to give a distinctive position on HPLC. This reagent was *N*-[4-(dodecylaminocarbonyl)fluoren-9-ylmethoxycarbonyloxy]succinimide (**2**, Scheme 14).

for references see p 38

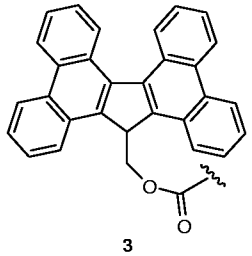
Scheme 14 Structure of the Fmoc-Based Derivatization Reagent for Chaperonin 10[72]



2

Following reversed phase HPLC purification, the highly homogeneous protein was obtained with good analytical data, including mass spectrometry and capillary electrophoresis. The protein aggregated to the expected heptameric structure and was active as a chaperone in the folding of enolase. These experiments showed that another chaperone was not necessary for folding of this rat chaperone 10. Similarly, a hydrophobic affinity-based N^{α}-protecting group was used by Brown et al.[88] to purify the 76-residue monocyte chemotactic protein. For this purpose, the TbFmoc group (**3**, Scheme 15) was developed.

Scheme 15 Structure of the TbFmoc Group



3

Tomasselli et al.[71] applied an interesting purification strategy in their preparation of simian immunodeficiency virus protease (SIV, 99 residues). Using a standard solid-phase procedure, they assembled residues 99–1, and continued with the next eight residues that occur in the SIV gag/pol precursor, and finally added a biotin residue. This construction gave an Ala^{-1}-Pro^{1} linkage, which was a substrate for the mature protease. After cleavage from the resin and deprotection, the biotin-linker-protease was purified by avidin chromatography. Upon refolding, the protease became active and the product became a substrate for autocatalysis, thus producing the desired purified SIV 1–99.

Of course, the goal of every synthetic organic chemist is to obtain crystalline products, and a few cases of crystalline synthetic proteins have been reported. These include the ribonuclease A synthesized in solution,[35] which crystallized after chemical workup and affinity chromatography (see Section 5.1.6.2.2). Further examples include an HIV protease analogue,[70] a ubiquitin analogue,[59] and monellin,[89] which were each prepared by solid-phase methods and purified by HPLC.

Monellin provides a nice example of a protein that has been studied utilizing chemical synthesis to learn about the role of various residues in determining receptor binding and biological activity. It is a very sweet protein consisting of two noncovalently associated peptide chains, A and B, containing 44 and 50 residues, respectively. It has been synthesized using Fmoc solid-phase methods and has been crystallized by a vapor-diffusion method. A number of replacement analogues have been synthesized[89,90] and it has been concluded that Asp^{7} of the B chain is at the active site, whereas several specific residues are not involved in receptor binding. The A and B chains separately are not sweet, but some analogues of the

combined chains are as much as 7500 times as sweet as sucrose on a weight basis (>200000 times on a molar basis).

To illustrate the diversity of strategies available for solid-phase synthesis, several fairly recent protein syntheses can be cited. SRY, an 80-residue DNA binding protein, was synthesized[62] on a Pepsyn support (polyacrylamide gel beads) in a continuous flow machine using Fmoc/tBu protection and TBTU/HOBt activation. HPLC gave a homogeneous product with molecular weight 10051 Da (calculated 10033 Da). It bound to DNA, as expected.

Kaiser and Voelter[76] undertook a comparative study of various procedures for the synthesis of the 115-residue macrophage migration inhibitor factor. They began with the solid-phase stepwise approach both by batch and by continuous flow systems using Fmoc/tBu maximum protection tactics and various polystyrene resins. In all cases, the synthesis in the C-terminal region was unsatisfactory. They then turned to an Fmoc convergent strategy using 2-chlorotrityl resin for preparation of the fragments and for fragment assembly on the resin. Once again, these couplings were not satisfactory and the full synthesis of the protein could not be completed. Finally, a successful, stepwise approach was found which used a graft copolymer of ethylene oxide and styrene-divinylbenzene (TentaGel R trityl resin), and TBTU/HOBt/DIPEA activation of Fmoc amino acids in DMF. A two-step cleavage and deprotection with 20% TFA/CH_2Cl_2 and then with TFA/EDT/thioanisole/TIS/H_2O/phenol (80:8:2:1:1:3) gave a high yield of crude peptide that was 57% homogeneous before final purification on HPLC. The theoretical mass of 12503 was found. This protein has a high percentage of β-sheet, which may be responsible for the synthetic difficulties.[18]

In contrast, the uteroglobin monomer (8122 Da) was readily synthesized using a PEG-(S-DVB)-resin-based convergent synthesis from five synthetic fragments that had been prepared on an MBHA resin.[56] HATU and PyBOP were used for the condensations of the fragments; the protein was purified by HPLC and showed a good mass spectrum. Thus, in this case, a convergent synthesis on a support was successful.

Human interferon α_1 (hIFN α_1, 166 residues) is one of the largest of the proteins to be synthesized using solid-phase methods.[83] It was assembled manually on PAM resin using Boc/Bzl protection and DCC activation. Each coupling was monitored by the quantitative ninhydrin reaction and no major deletions were detected, although with a molecule of that size the sensitivity is not as good as for smaller peptides. After cleavage by low/high HF and reduction in 6 M guanidine hydrochloride and β-ME, the product was passed through Sephadex G-200 in 6 M guanidine hydrochloride, and the 18000 Da fraction was fractionated on Sephadex G-25 in Tris buffer containing GSH and GSSG. The oxidized product was purified on an IFN α antibody column in pH 2 glycine. The overall yield was, as expected, quite low at this point. [Ser^1]-IFN α, and IFN α_2, containing 10 fewer residues, as well as its Ser^1 analogue, were also synthesized. The proteins were characterized by ion-exchange and reversed phase chromatography, as well as isoelectric points, and were the same as the corresponding natural IFNs. They bound reversibly to polyclonal anti IFN α antibodies. Broad-spectrum antiviral specific activity was equivalent to the natural IFNs, in the 10^8 units per mg range. Growth inhibition of K562 cells was also similar. Finally, killer cell activity was measured. It was found that the Cys^1 and Ser^1 analogues counteract one another. [Cys^1]-IFN α_2 increases activation like the natural protein and [Ser^1]-IFN α_2 decreases it. This IFN project, a joint effort of Elizabeth Merrifield, Matilda Krum, Emri Mecs, Virender Sarin, and Floyd Fox, was a severe test of the synthetic methodology and represented an upper size limit at that time.

for references see p 38

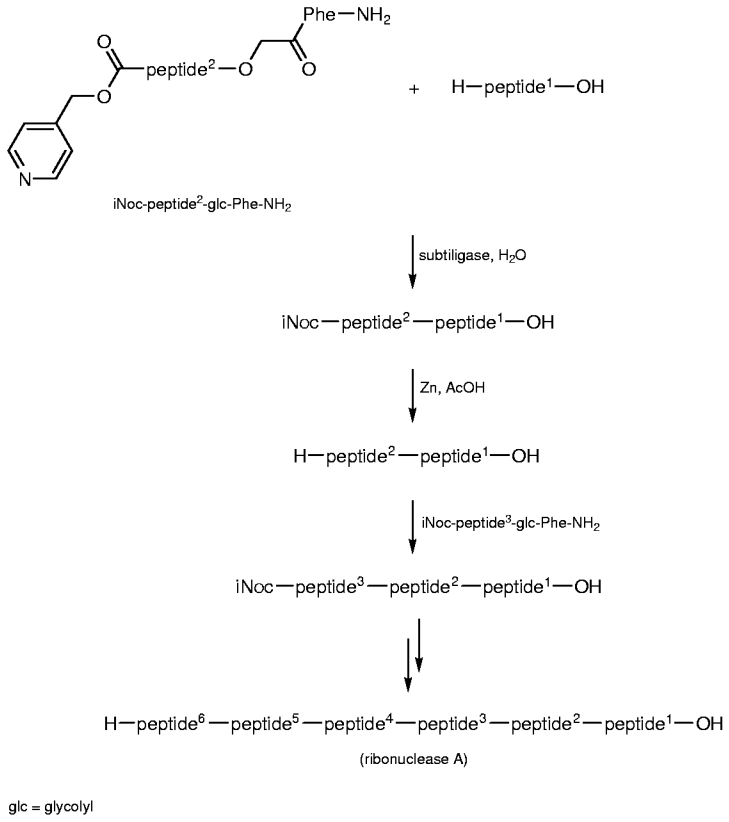
5.1.9 Enzyme-Catalyzed Synthesis of Peptides and Proteins

An alternative to the synthesis of proteins by classical fragment synthesis in solution or by solid-phase synthesis on a support is the use of enzyme-catalyzed condensation of amino acids or peptides. This possibility was first demonstrated in 1938[91] with the synthesis of poorly soluble benzoyl-leucyl-leucine anilide by papain catalysis. After many years, this approach was extended to the preparation of peptide hormones such as Leu-enkephalin[92] and dynorphin(1–8).[93] This was made possible by the use of highly purified enzymes and by careful control of reaction conditions. The basic principles of protease-catalyzed peptide bond formation have been discussed.[94]

Enzyme-catalyzed condensation reactions can be carried out in organic solvents, such as octane containing only traces of water, with good amide bond formation but with greatly reduced peptide bond hydrolysis.[95] Human growth hormone releasing factor GRF(1–44)-NH_2 was synthesized by trypsin-catalyzed coupling of Leu-NH_2 to Arg^{43} of GRF(1–43)-OH in 76% DMA.[126]

Since 1991, there has been a marked advance in the enzyme-catalyzed ligation of peptide fragments.[77] Thus, the serine protease subtilisin was engineered to produce "subtiligase" in which Ser^{221} is replaced by Cys, and Pro^{225} is replaced by Ala.[96] The modified enzyme is many times more selective for the formation of peptide bonds from a protected peptide ester and a peptide bearing a free α-amine than for the hydrolysis of peptide bonds. The enzyme has been applied to the synthesis of ribonuclease A[78] (Scheme 16). Six peptide esters were prepared by standard solid-phase Boc/Bzl chemistry and BOP coupling, and then the α-amine was blocked with an acid-stable iNoc group. The purified peptides (HPLC) were coupled one at a time starting with the C-terminal peptide fragment (116–124) of RNase, containing a free α-amine, by subtiligase catalysis in aqueous solution. These steps went well with an average yield of 66% based on the limiting C-terminal peptide. The iNoc group was removed with zinc in acetic acid in 86% average yield, giving an overall yield of ~5.8% based on the first peptide. Each of the purified peptides was present in three- to fivefold excess for the ligations, giving ~1.5% yield based on all the purified peptides. If each of these had been synthesized and purified in a 40% yield, based on the first amino acid (not on excess of amino acid derivatives), then the overall yield of this synthesis would appear to be about 0.6%. Nevertheless, 10-mg quantities of ribonuclease A were prepared in this way. Also synthesized was RNase A containing 4-fluorohistidine at positions 12, 119, and 12 plus 119, and it was shown that there was little change in k_{cat} despite a change in pK_a from 6.8 to 3.5 for each residue. This indicates that the imidazoles function as general acids and bases, but that the proton-transfer steps are not rate limiting. This is a remarkable advance in enzyme-catalyzed peptide synthesis.

Scheme 16 Enzyme-Catalyzed Fragment Synthesis of Ribonuclease A in Aqueous Solution Using a Modified Subtiligase[77]



glc = glycolyl

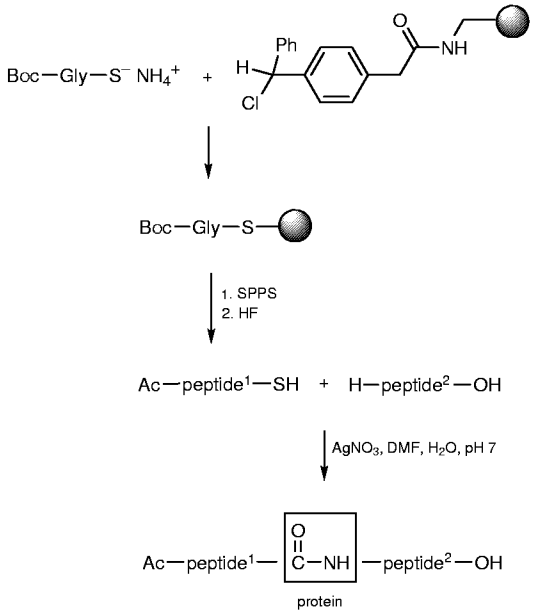
5.1.10 Peptide Ligation

5.1.10.1 Highly Selective Second-Order Intermolecular Reactions

As early as 1951,[97] it was shown that amino acid phenyl thioesters are highly activated and suitable for the formation of a peptide bond. The use of thioacids and thioesters to form larger peptides has evolved in an interesting way. The goal has been to activate one or both components with functional groups that can selectively react with one another. Peptide thioacids have been made[98] that, after activation with silver ions, react in aqueous solution with the α-amine of a peptide blocked only on the lysine ε-amino and the cysteine thiol groups (Scheme 17). Although this is a second-order ligation reaction and is expected to proceed slowly, the high selectivity results in an avoidance of competing side reactions. With this procedure, apocytochrome c (104 residues) was synthesized from four segments.

for references see p 38

Scheme 17 Silver Ion Activated Coupling between a Thiocarboxy Group and the α-Amine of Another Peptide Segment[98]

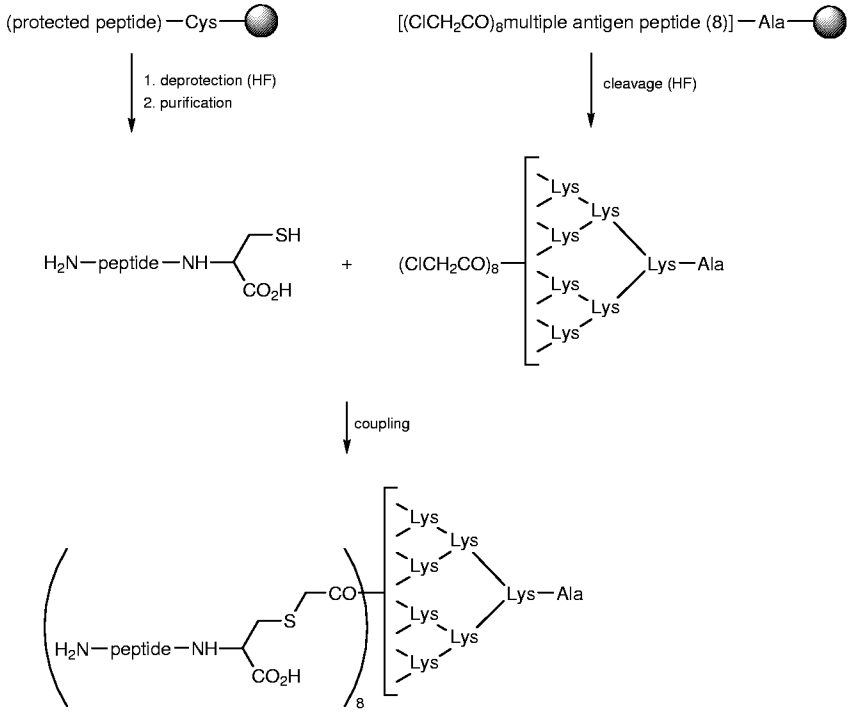


Several years later, a ligation method was developed that coupled N- or C-terminal cysteine peptides with a chloroacetyl peptide.[99] This reaction was so specific that, for the first time, fully unprotected peptides could be utilized for coupling reactions (Scheme 18). The reaction produced a sulfide at the ligation site rather than an amide; however, the purpose was to prepare branched multiple antigen peptides rather than linear proteins. The multiple antigen peptides were highly immunogenic and the sulfide bond did not interfere with their antigenic properties. Molecules of 72 and 96 or more residues could be assembled by adding 4 or 8 cysteinyl peptides of 9 or 24 residues to a branched lysine scaffold, and thus approached the size of a protein.

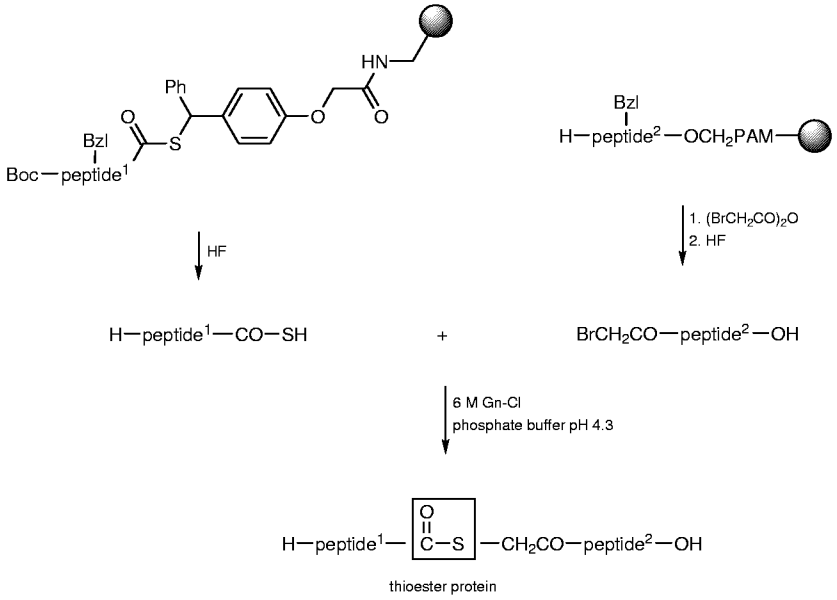
The following year, a variation[100] of the earlier technique[98] was developed in which the C^{α}-thioacid was coupled with a bromoacetyl peptide. The reaction was carried out in aqueous solution and with no side-chain protection because of its high selectivity. This chemistry also does not produce a normal amide bond at the site of ligation, rather a thioester (Scheme 19). For HIV-1 protease (99 residues) this was not a problem because fully active enzyme was obtained in high yield and purity. This technique was also used to show that the geometry of the β-turn at residues 15–18 in the two subunits of an HIV-1 protease is not critical for activity. This conclusion was reached from data on a constrained nonpeptidic type II β-turn replacement analogue.[101] This work was extended with the synthesis of two 99-residue HIV-1 derivatives, each containing a thioester and a free thiol group, which were then coupled via a 2,2′-dipyridyl disulfide to produce a 198-residue protein held together at the center by a disulfide bond.[102]

The technique has been used for the total synthesis of truncated forms of the basic/helix-loop-helix/zipper domains of the nuclear protein transcription regulation factors cMyc and Max. These were covalently joined through an oxime bond to form a single protein-like entity with two amino termini, i.e. cMyc-Max heterodimer and Max-Max homodimer, each 172 residues, which bind to DNA and activate or inhibit mRNA transcription.[103]

Scheme 18 Synthesis of a 72-Residue, 8-Chain Multiple Antigen Peptide[99]



Scheme 19 Segment Synthesis by Reaction of a Peptide Thioacid with a Bromoacetyl Peptide[100]



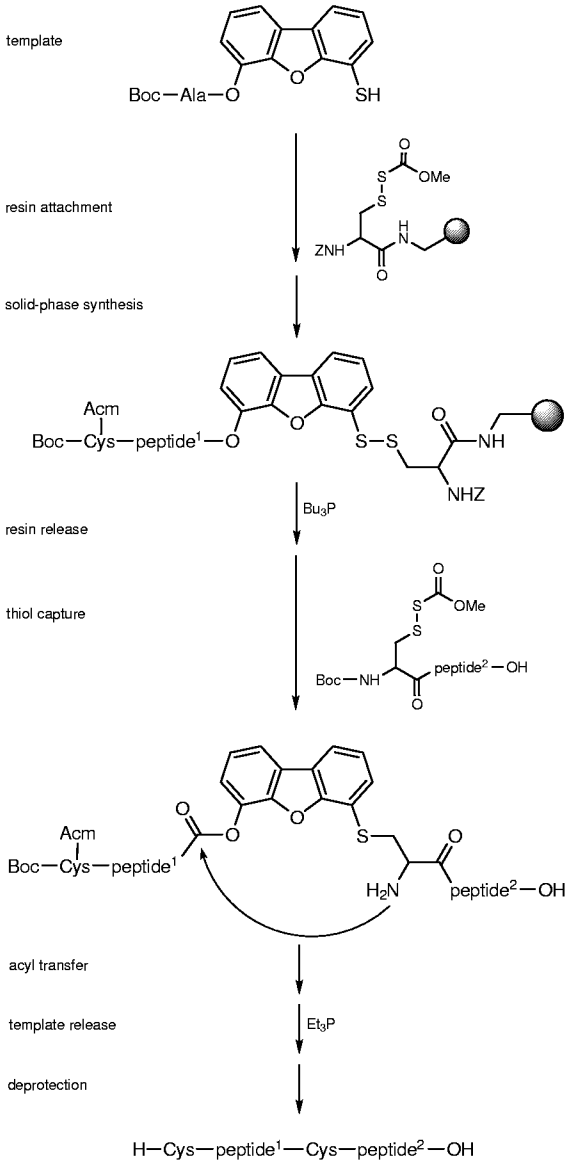
5.1.10.2 First-Order Intramolecular Ligation

Workers in three laboratories have studied ligation via cyclic transition-state intermediates. The couplings described in Section 5.1.10.1 to form the amide or thioester bond involve second-order *intermolecular* reactions of two peptide components, which are necessarily

for references see p 38

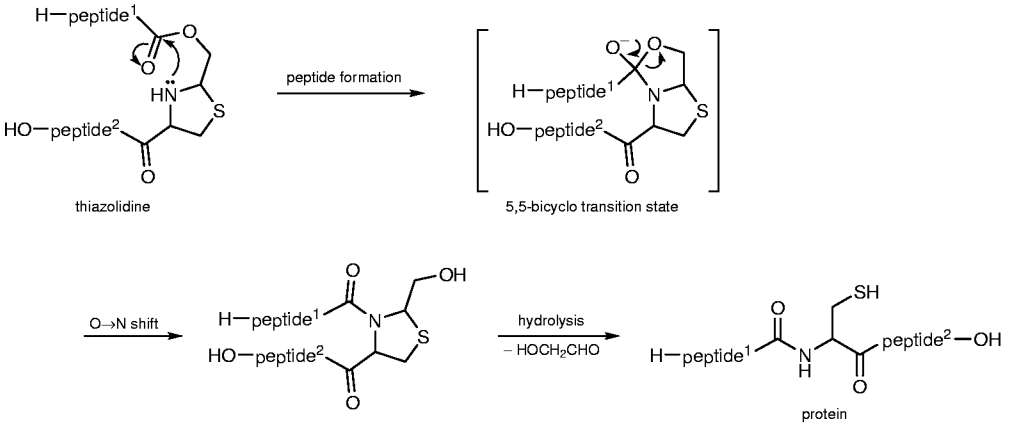
present in low concentrations and, unless highly activated, have a low potential for effective reaction. However, new methods are being developed to overcome this defect. Kemp et al.[104] have devised techniques leading to *intramolecular* amide bond formation that can occur at very high effective concentration. From kinetic data it was deduced that the amide bond forming reaction occurred at a concentration equivalent to 10 M! The synthesis starts with a template containing properly spaced phenolic and thiophenolic groups, as shown in Scheme 20. The template is reacted with Boc-Ala and a functionalized cysteine attached to a resin. Peptide[1] is synthesized stepwise, followed by release from the resin. A "thiol capture" is carried out with another cysteinyl-peptide[2] derivative, followed by the intramolecular acyl-transfer reaction and finally the template release. The product can be recycled to give an expanded fragment synthesis. The value of this elegant method was demonstrated by the synthesis of a 29-residue segment of trypsin inhibitor.

Scheme 20 Segment Synthesis by Thiol Capture[104]



An intramolecular ligation strategy has also been developed that does not depend on a "thiol capture", but on an "aldehyde capture" (Scheme 21).[105] In this procedure, the carboxy component of a synthetic peptide is converted into a glycol aldehyde ester. The amine component is a synthetic peptide containing an N-terminal serine, threonine, or cysteine. Both peptides are completely unprotected. In aqueous solution at pH 4, the α-amine is captured by the aldehyde to form an oxazolidine or thiazolidine. The ring nitrogen is then in close proximity to the C-terminal carbonyl of the other peptide segment and forms a 3.3.0-fused ring system in the transition state of an O-to-N acyl-transfer reaction. At pH 6 to 9 in an aqueous solution, this reaction also occurs at very high effective concentration. A hydroxymethyl-α-thioproline is formed at the ligation site. The method was evaluated by the synthesis of a 50-residue analogue of the transforming growth factor. If an aldehyde reaction occurs between other unprotected amino groups, it is nonproductive and is reversible and does not interfere with the synthesis.

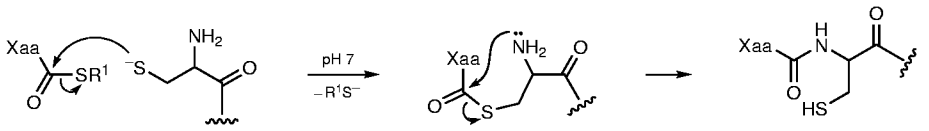
Scheme 21 Segment Synthesis by Aldehyde Capture[105]



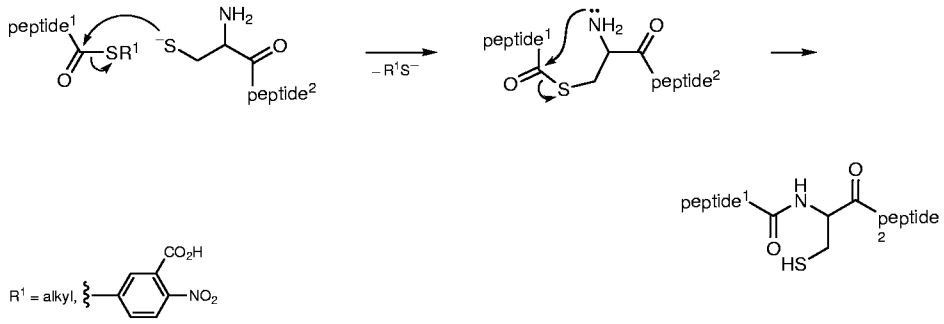
It is interesting to recall that Wieland[106] in 1953 showed that an amino acid thioester of cysteine would undergo an intramolecular transacylation to form an N^{α}-acyl cysteine containing a peptide bond (Scheme 22). This principle has subsequently been used for ligation of two peptide segments to give a protein (Scheme 23).[107] This represents an improvement over the previous ligation method between a peptide thioacid and a bromoacetyl peptide,[100] which results in a thioester at the ligation site (see Section 5.1.10.1). The new method gave a normal peptide bond and it was carried out with fully unprotected peptides in aqueous solution. The value of the technique was demonstrated by the synthesis of [Ala^{33}]-interleukin-8(1–72) ([Ala^{33}]-IL-8) in good yield and high purity of correct molecular weight. Air oxidation gave [Ala^{33}]-IL-8 containing the two disulfide bonds that are essential for activity. The starting peptide thioester was prepared by solid-phase peptide synthesis on a thioester resin according to the Blake,[98] and Yamashiro and Li[108] procedures. After hydrogen fluoride cleavage from the resin, the resulting peptide thioacid was converted in a second step into the 3-carboxy-4-nitrophenyl thioester by treatment with 5,5-disulfanediylbis(2-nitrobenzoic acid) (Ellman's reagent) in 8 M urea. The resulting α-derivatized IL-8(1–33) was then used for the ligation with a Cys-peptide. IL-8(34–72) was also made by solid-phase techniques using a PAM resin, N^{α}-Boc protection, and side-chain benzyl protection. Following HF cleavage from the resin, the fully unprotected 34–72 peptide, containing an N-terminal Cys and a free α-carboxy group, was purified. The two peptides were reacted in aqueous solution to give [Ala^{33}]-IL-8(1–72) (Scheme 23). Numerous other proteins have been synthesized using this new methodology, including the SH3 domain of cCrk,[109] barnase analogues (110 residues),[110] and human secretory PLA_2 107 (124 residues).[111]

for references see p 38

Scheme 22 Intramolecular Transacylation of Thioesters To Form Dipeptides[106]



Scheme 23 Intramolecular Transacylation of Thioesters for the Synthesis of a Protein[107,112]



Essentially simultaneously, a closely related intramolecular ligation method using fully unprotected peptide segments was developed independently.[112] A peptide thioester on a resin support was also prepared, but then converted in only a single step into the desired alkyl thioester by a thiol–thioester exchange under reducing conditions at pH 7.6 in the presence of an alkylphosphine and an alkylthiol.[113,114] Ligation of this alkyl thioester with the amino peptide component containing an N-terminal Cys gave the two peptides linked by a Cys thioester. The resulting thioester, containing a free α-amine in close proximity, spontaneously underwent an intramolecular S-to-N acyl transfer via a five-membered-ring intermediate to produce the peptide bond between the two peptide segments (Scheme 23). The mechanism of the various reactions was studied in considerable detail[112] and conditions were developed that greatly reduced side reactions and gave good yields. The method was tested on peptides containing up to 54 residues, with yields of 60 to 88% based on the limiting component, with the smaller size component in 1.2- to 2.5-fold excess. Peptides containing internal unprotected cysteine or lysine were not a problem because their reactions were reversible and nonproductive.

This first-order intramolecular ligation method is, to quote Kent et al.,[115] "a robust practical way of making proteins. It has been used in a wide variety of target systems. It is simple, generally applicable, proceeds cleanly with minimal side reactions and gives high yields. Virtually any amino acid, even β-branched ones, can be used as the Xaa component at the Xaa-Cys ligation site. Furthermore, the reaction is regioselective, that is, it gives specific quantitative ligation at a single site even in the presence of other Cys residues in either or both peptide segments". The technique is facilitated by the solid-phase synthesis of the peptide fragments that are to be ligated.

5.1.11 Semisynthesis

There are many instances where a problem in protein chemistry or biology can be studied efficiently and effectively by semisynthesis. For this section, "semisynthesis" means the combination of a synthetic peptide with a large protein segment derived from natural sources. The combination can be by a covalent peptide bond or, in certain cases, by a specific

noncovalent binding of the two components that leads to an active product. For the present purpose, examples in which a protein is modified by acylation or other reactions on side chains are considered to be chemically modified proteins. A synthesis in which multiple amino acids are coupled stepwise to a natural protein fragment represents a semisynthesis. With the semisynthesis approach, many peptide analogues can be designed and synthesized readily, and combined with a protein to produce further analogues of interest, without resorting to the more difficult synthesis of the total protein each time. The early work on protein semisynthesis has been reviewed.[116]

5.1.11.1 Association of Synthetic Peptides and Natural Protein Segments

An example of a noncovalent system has already been discussed in the case of ribonuclease A (see Section 5.1.6).[29–32] Thus, much information was gained from the relatively easy synthesis of many peptide analogues with sequences derived from either end of the 124-residue protein, and their combination with the protein segments RNase(21–124), RNase(1–118), or RNase(21–118) produced by selective enzymatic digestions.

Other examples of noncovalent peptide-protein systems include cytochrome c,[117,118] somatotropin,[119] and staphylococcal nuclease.[120] A limited tryptic digest of the nuclease from *Staphylococcus aureus* produced three fragments. Peptide P_2 (6–48) and P_3 (49–149) in water formed a noncovalent complex with significant enzyme activity. Many analogues of P_2 were synthesized and their complexes with P_3 assayed. Residues $Asp^{19,21,40}$, Glu^{43}, and Arg^{35} were found to have an influence on activity, but His^{46} was not important for enzyme activity. Other tryptic cleavages have given fragments 1–126 and 99–149,[121] and a series of analogues with replacements near the carboxy terminus was prepared.

In the case of somatotropin,[119] a synthetic analogue of residues 145–191 of growth hormone was combined with the natural fragment 1–134, which regenerated full growth-promoting activity.[124]

In the experiments on cytochrome c,[117,118] and staphylococcal nuclease,[120] as well as on myoglobin,[128] there were overlapping regions with retention of activity, just as in the case of RNase(1–118) plus RNase(111–124) where eight residues are in common; for the somatotropin experiment, 10 residues (135–144) could be missing in the complex without loss of activity.

5.1.11.2 Covalent Combination of Peptides with Protein Segments

There are many examples in which a chain of amino acids has been assembled stepwise on the amino terminus of a natural protein or on a chain derived by partial enzymatic degradation of a protein. Special precautions are needed to prevent substitution at more than one site. A large series of semisynthesis studies has been conducted by Slotboom et al. on phospholipase A.[122,123] For example, the 130-residue precursor of the enzyme was acetimidylated on the lysines and then selectively cleaved by a protease at Arg^{7}; one, two, or three residues were then removed by Edman degradation. Different Boc amino acid active esters could then be added to give a series of replacement analogues. Some of the replacement sites were in the region of the enzyme that recognizes the presence of a lipid/water interface; this provided considerable new information about the required structure including the pK value of the α-amino group. Similar studies on the stepwise elongation of protein segments have been carried out with trypsin, ferridoxin, insulin, myoglobin, and other proteins.

In contrast to the stepwise addition of amino acids to a protein, much of the semi-synthesis of proteins has been concerned with the covalent combination of synthetic peptides with natural proteins or their partially degraded fragments.

For example, it has been shown that a covalent bond between Ala^{20} and Ser^{21} of RNase S can be formed in 50% yield by treatment with subtilisin in 90% glycerol in water.[125] This

for references see p 38

system should be useful for preparing semisynthetic RNase A (see also Section 5.1.6) and certain other proteins.

Cytochrome c(1–65) and -(66–104), derived by cyanogen bromide cleavage, form a stable complex in which the 65–66 peptide bond spontaneously slowly re-forms to give a protein with full biological activity.[124]

Construction of protein analogues by site-specific condensation of unprotected fragments has been reported.[127] In this application, analogues of human granulocyte colony stimulating factor were prepared. A recombinant analogue (75–174) containing a single Lys^{75} residue adjacent to serine was digested with a lysine protease, and the resulting N-terminal serine was oxidized with periodate to form an aldehyde. The aldehyde reacted with another protein fragment (1–62) containing a C-terminal hydrazide to give the modified protein containing an internal hydrazone.

There are three main requirements for a semisynthesis of a protein: (1) the synthesis of the appropriately protected and activated peptide, (2) the cleavage or partial degradation of the protein to be studied, and (3) the proper protection of the protein component. The first two are fairly well managed, but the third is more difficult and not always adequately selective.

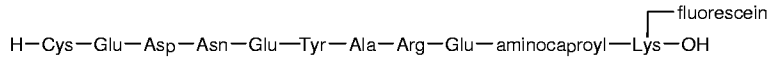
A semisynthesis of myoglobin has been carried out.[128] The lysines were blocked by acetimidylation and then the protein was cleaved at Trp^{14}. The 15–153 fragment was coupled with N^{α}-Boc-N^{in}-Trp-OSu; the 14–153 protein then was coupled with synthetic 6–13 peptide, and finally with peptide 1–5 to give the complete myoglobin(1–153). This nice piece of work was preparatory for the semisynthesis of suitable myoglobin analogues.

Semisynthesis using these methods is somewhat restricted because the cleavages are usually dependent on the natural positions of Met, Trp, Lys, and Arg in the protein, although recombinant genetic engineering can introduce these groups at almost any position. However, it would appear to be simpler to make the necessary fragment directly if this technique is to be applied.

Significant advances in protein chemistry evolved in the late 1990s with the combination of ligation methods with semisynthesis. Thus, instead of intermolecular reactions of amino acids or peptides with larger, protected, natural protein segments, the new intramolecular ligation reactions have been introduced utilizing unprotected components that will react in aqueous solutions. Two examples can be given to illustrate the technique.

In the first, Muir et al.[129] have introduced a protein engineering term they call "expressed protein ligation", which means ligating a synthetic peptide to a recombinant protein to give a new semisynthetic protein. In one example, they began with the synthesis of a small peptide (Scheme 24) by solid-phase methods using Boc chemistry, and the corresponding phosphotyrosine peptide was made using Fmoc chemistry. Fluorescein *N*-hydroxysuccinimide ester was added to the lysine ε-amino group of each, and the peptides were purified by HPLC.

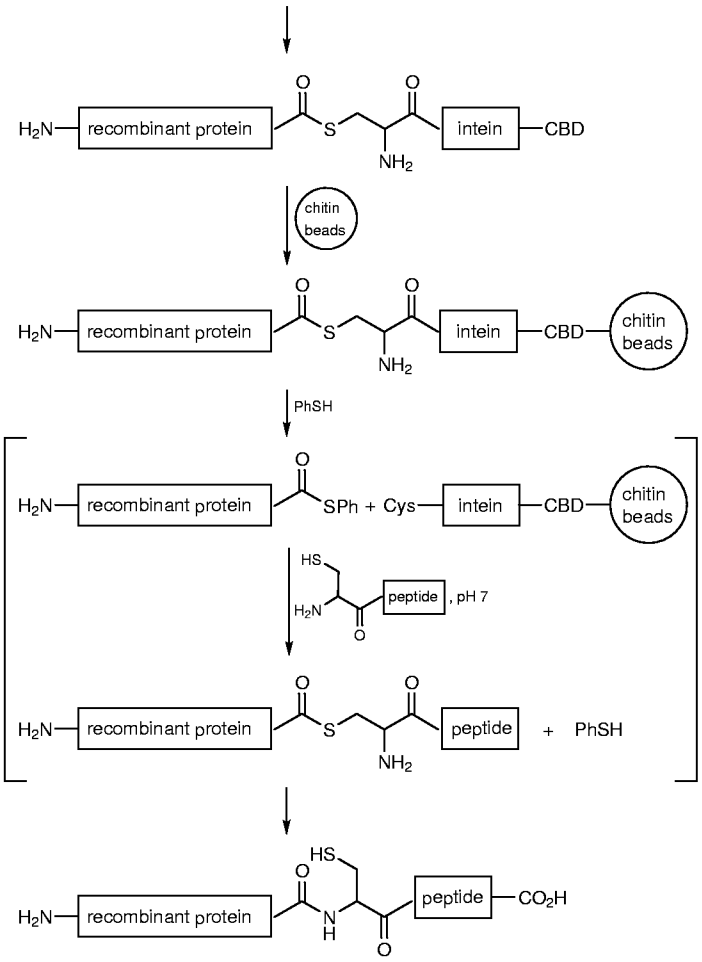
Scheme 24 Synthetic Peptide Precursor for Semisynthesis[129]



A full-length human csk DNA that codes for Csk, a 50-kDa protein that catalyzes the phosphorylation of a tyrosine within the C-terminal tail of Srk, was inserted in a plasmid and in-frame with a modified intein-chitin binding domain (CBD) encoding sequence, where intein is a protein-splicing element.[130] The expressed protein underwent a conversion into a thioester of the N-terminal cysteine of the intein, but the normal second step of an intein splicing did not occur. The protein was then bound to a chitin resin and washed. The resin containing the bound protein was treated with 2% benzenethiol to cleave the thioester and give the free phenyl thioester, which was immediately treated with one of the synthetic

peptides (2 mM) in aqueous pH 7 buffer for 24 hours, filtered, and washed. This was repeated with the other peptide. Dialysis and phosphotyrosine affinity purification gave homogeneous products, with SDS-PAGE and mass spectrometric data on the two peptides of 52547 Da and 52629 Da, in excellent agreement with the calculated values. The synthetic scheme is outlined in Scheme 25. The time course of phosphorylation of Lck could then be studied. The Csk peptide was much more active than the phosphorylated Csk peptide.

Scheme 25 Semisynthesis of a Protein from a Recombinant Protein and a Synthetic Peptide via a Thioester[129]



A closely related study[131] involved the 566 amino acid residue protein of the σ^{70} subunit of *Escherichia coli* RNA polymerase and the location of its binding site for the bacteriophage T4 anti-sigma protein (Asi A). The σ^{70} was fused with intein-chitin binding domain and expressed as described above. A 34-residue peptide corresponding to residues 568–600 of σ^{70}, but with an additional Cys at the N-terminus, was synthesized using solid-phase methods and Boc-chemistry. After ligation and purification of the resulting semisynthetic protein, it was homogeneous, had the correct molecular weight, and was biologically active. This study provided evidence that Asi A inhibits transcription directly by occluding the DNA binding surface in the 568–600 residue region of σ^{70}. The semisynthetic method promises to contribute substantially to the understanding of other protein-protein and protein-nucleic acid contacts that govern transcription activation and promoter recognition.

for references see p 38

5.1.12 Conclusions

It can be fairly concluded that there have been large and important advances in the chemical synthesis of proteins in the one hundred years since the pioneering work of Emil Fischer, beginning with the first syntheses of insulin in the early 1960s and the first syntheses of an enzyme in the late 1960s. Advances in peptide chemistry, purification, and analysis have all played major roles, and advances in biology have created a need for, and an understanding of, new active peptide and protein target molecules.

There have been two major developments in the past decade that make the total synthesis of proteins a practical reality; the first developed and refined by people trained in classical solution chemistry and the second developed and refined by people trained in solid-phase chemistry. There are fundamental differences in the two approaches, but it is most interesting to see how they have converged into a unified approach to a common problem.

The first development has been led most recently by the Sakakibara group (see Section 5.1.7) and has culminated in the synthesis of a protein of 238 amino acid residues. This approach begins with the preparation of a number of peptides by solid-phase synthesis followed by coupling of the purified building blocks into larger and larger peptides by solution methods in new and highly effective solvent systems. These are bimolecular reactions of protected peptide components that proceed in good yields, and the proteins are deprotected in the final stage of the synthesis.

The second approach has been led most recently by the groups of Kent and of Tam (see Section 5.1.10.2) and has culminated in the synthesis of proteins in the 100–200 residue range. It begins with the solid-phase synthesis and purification of large peptide components, followed by their complete deprotection, and finally their intramolecular ligation in aqueous solution to give the target protein. Thus, there is a combined use of solid-phase synthesis and homogeneous solution synthesis leading to the same overall goal. There is good reason to believe that larger and more complex proteins will be accessible with these methods of total synthesis.

Corresponding improvements in the semisynthesis of proteins have been developed (see Section 5.1.11). They, too, have involved intermolecular and intramolecular reactions to form normal peptide bonds between small synthetic peptides and large protein components derived from natural sources or produced by recombinant techniques. In this case, there is the potential to produce derivatives of much larger proteins than can currently be prepared by total chemical synthesis. The total synthesis and semisynthesis approaches are useful for somewhat different purposes, but can be considered complementary; together, they greatly broaden the field of protein chemistry and biology.

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5.2 Protein Synthesis in Solution

S. SAKAKIBARA

Among procedures thus far reported for the synthesis of proteins in solution,[1–3] the strategy of using a combination of Boc/Bzl/Pac groups[3,4] is potentially the most widely applicable for the synthesis of a variety of proteins. In this section, the principle of the Boc/Bzl/Pac strategy is described in detail using as examples the syntheses of dendrotoxin-I, a 60 residue peptide,[5] and human midkine, a 121 residue protein.[6]

5.2.1 Principle of the *tert*-Butoxycarbonyl/Benzyl/Phenacyl Strategy

The entire structure of a target protein is initially divided into small segments, each of the size of about 10 residues, which is a reasonable size for confirming the homogeneity of a segment by currently available analytical procedures, even if the segment is in a fully protected form. Each protected segment is arranged in the form of Boc-peptide-OPac, using ordinary Boc amino acids developed for standard solid-phase synthesis as building blocks, except for the segment at the C-terminus of the target protein. If the C-terminus of the target protein is in the free carboxylic acid form, the C-terminal segment is synthesized in the form of Boc-peptide-OBzl. If the C-terminus of the target protein has the carboxamide structure, the terminal segment is synthesized in the form of Boc-peptide-NH_2. The Boc group can be removed selectively by treatment with TFA, leaving all of the other functional groups protected. The Pac group can be removed selectively by treatment with zinc powder in acetic acid, leaving all of the other functional groups protected. The carboxy and amino components thus formed can be coupled together to obtain a larger sized intermediate segment using EDC as the coupling reagent in appropriate organic solvents in the presence of HOBt or HODhbt as an additive.[7] Generally speaking, these fully protected segments are soluble in either DMF, NMP, or DMSO. However, if some segments are insoluble in these solvents, they can be dissolved in a mixture of $CHCl_3$/TFE (3:1).[8] This solvent mixture is powerful in terms of dissolving fully protected segments, and it can also be used as the solvent for coupling peptide segments with EDC, in which neither TFE ester formation nor epimerization at the activated amino acid residues is observed if HODhbt is used as an additive.[8] A mixture of $CHCl_3$/phenol or HFIP can also be used as an alternative solvent for the segment elongation reactions. However, risk of ester formation with phenol or HFIP is higher than in the case of $CHCl_3$/TFE (3:1) as the solvent. After completion of the formation of the full sequence of the target protein in a fully protected form, the product is treated with HF in the presence of appropriate scavengers to remove protecting groups, and the target protein can be isolated by HPLC and subjected to appropriate folding reactions.

5.2.2 Design and Synthesis of Segments

To apply this strategy successfully, the segments should be designed according to the following principles:

(1) The C-terminal amino acid residue of a segment, which is to be coupled with the amino group of another segment, should be selected from group A or B in Table 1, but not from group C.

(2) There are no special requirements in the selection of an N-terminal amino acid residue in a segment with which the carboxy component of a segment is to be coupled, unless a highly hindered amino acid or a secondary amino acid is selected. If a Pro or Hyp residue is located at the N-terminus of the segment, monitoring of the coupling reaction with ninhydrin or fluorescamine is extremely difficult.

(3) The size of a segment should be limited to less than 15 residues.

(4) Each fully protected segment must be soluble in DMF, NMP, DMSO, or $CHCl_3$/TFE (3:1). If the solubility of a protected segment is poor in any of these solvents, a change in the segment structure must be considered, or the use of $CHCl_3$/phenol or $CHCl_3$/HFIP as the coupling solvent must be considered at the risk of ester formation with phenol or HFIP.

(5) In order to obtain a reliable product, each fully protected segment must be carefully synthesized in solution. However, if homogeneity of the synthetic segments can be controlled properly, the synthesis can also be carried out by the solid-phase method.

(6) Side-chain protecting groups suitable for the present strategy can be selected from the list given in Table 2. However, if Bom is used for protecting the His group, the use of 10 to 50 equivalents of Cys per Bom is recommended as a scavenger during the HF deprotection reaction.[9] Similarly, in the case where Asn(Xan) or Gln(Xan) is used, TIS must be used as a scavenger during the step involving the removal of the Xan group with TFA.

Table 1 C-Terminal Amino Acid Residues Acceptable for Segment Condensation Reactions

Group A Suitable	Group B Acceptable	Group C Not Suitable
Gly, Ala, Leu, Pro, Hyp(Bzl)	Asn, Gln, Lys[Z(2-Cl)], Gly(OCy), Asp(OCy), Trp(Hoc), Trp(For), Ser(Bzl)	Thr(Bzl), Ile, Val, Phe, Tyr[Z(2-Br)], His(Bom), Cys(Acm), Arg(Tos), Tyr(3-Pn)

Table 2 Side-Chain Protecting Groups Suitable for the Boc/Bzl/Pac Strategy

Amino Acid	Protecting Groups Suitable for Solution Synthesis	Protecting Groups Suitable for Solid-Phase Synthesis on HMFS Resin
Asp, Glu	Cy	Cy
Arg	Tos	Tos
Tyr	Z(2-Br), 3-Pn	3-Pn
Lys	Z(2-Cl)	Z(2-Cl)
His	Bom	Bom
Trp	Hoc, For	Hoc
Cys	Acm	Acm
Asn, Gln		Xan

5.2.3 Synthesis of Dendrotoxin-I

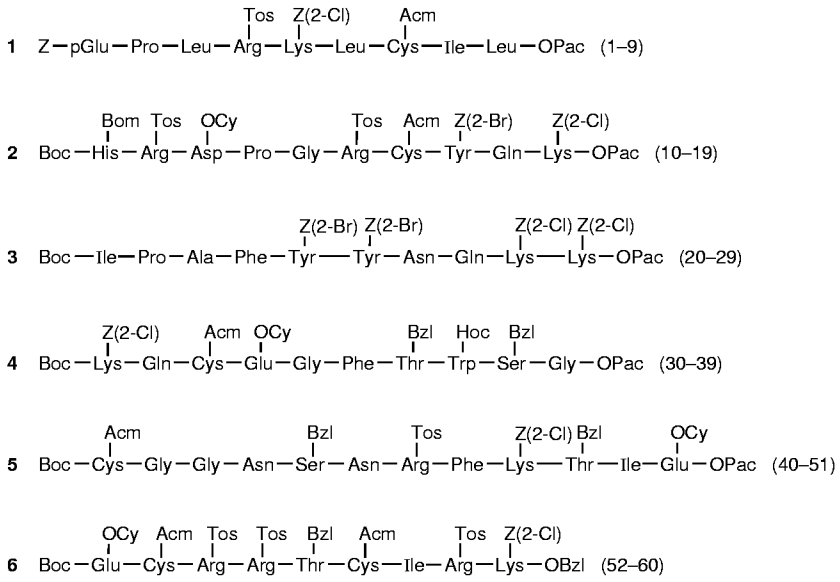
The primary structure of dendrotoxin-I is shown in Scheme 1. A design for assembling the segments in the synthesis of dendrotoxin-I is shown in Scheme 2.[5]

Scheme 1 Primary Structure of Dendrotoxin-I



for references see p 62

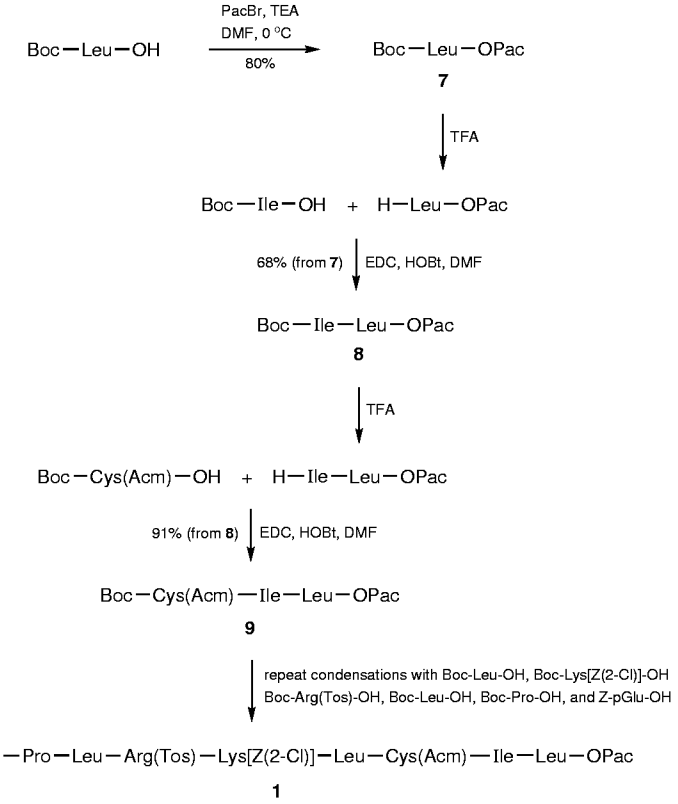
Scheme 2 Structure of Segments Designed for the Synthesis of Dendrotoxin-I[5]



5.2.3.1 Synthesis of Boc-Peptide-OPac in Solution

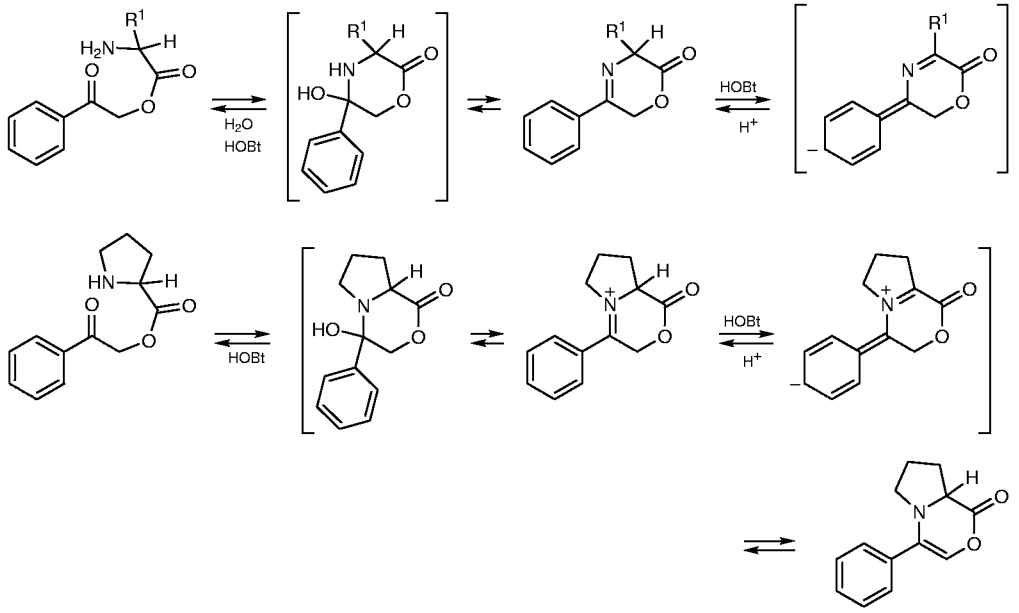
In this section, the synthesis of the first segment, compound **1** in Scheme 2, is presented as an example. Elongation of the segment is performed stepwise from the C-terminal amino acid Pac ester as shown in Scheme 3.[5]

Scheme 3 Synthesis of Segment **1**[5,10,11]



Amino acid Pac esters can be synthesized in general as described in the following experimental section, but several precautions must be observed for the synthesis of dipeptides having a structural unit of Pro-OPac, Hyp(Bzl)-OPac, or Gly-OPac at the C-terminus because the Pac ester contains an active carbonyl group. When Boc-Ser(Bzl)-OH is coupled with H-Gly-OPac in DMF, for example, using EDC and HOBt as the coupling agents, the yield of the product, Boc-Ser(Bzl)-Gly-OPac, is relatively low because H-Gly-OPac has a tendency to cyclize to form a Schiff base or a carbinolamine derivative as shown in Scheme 4. This type of side reaction is also a serious problem in the coupling reaction with H-Pro-OPac or H-Hyp(Bzl)-OPac under the same reaction conditions.[12]

Scheme 4 Proposed Mechanism of Side-Product Formation during the Coupling of Gly-OPac or Pro-OPac with Boc Amino Acid



As a consequence, the product must be purified by repeated crystallization. If the yield is too low, it is recommended that another route be taken such as starting with Boc-Ser(Bzl)-Gly-OH and converting it into the Pac ester using PacBr, as in the case of the synthesis of Boc-Xaa-OPac. When Boc-Leu-OH was coupled with H-Pro-OPac in DMF using EDC as the coupling agent in the presence of HOBt, the yield of the L-L form of Boc-Leu-Pro-OPac was 54% with the product contaminated by the formation of the L-D isomer in 16% yield. This type of epimerization occurs when H-Pro-OPac or H-Hyp(Bzl)-OPac is coupled with other Boc amino acids in DMF in the presence of HOBt, particularly when a trace of water is present as a contaminant in the DMF used.[12] Thus, after the synthesis of Boc-dipeptide-OPac by coupling a Boc amino acid with an amino acid Pac ester, the product must be purified by repeated crystallization, and the homogeneity of the product must be carefully checked by HPLC. It is fortunate, however, that almost all Boc-dipeptide phenacyl esters are highly crystallizable compounds. When using an ordinary L-amino acid Pac ester as an amino component, the yield of the Boc-dipeptide-OPac is 70–80%, and formation of the epimerized side product is less than 1% when EDC (in the presence of HOBt) is used as the coupling agent and dry DMF is used as the solvent.

After the removal of the Boc group by TFA treatment, excess TFA must be carefully removed from the reaction product, otherwise the coupling reaction which follows does not

for references see p 62

proceed stoichiometrically. To remove excess TFA from the treated peptides, replacement of the TFA salt with HCl is effective. For this purpose, the TFA-treated product is triturated with ca. 1.5 equivalents of 5 M HCl in dioxane, and the product is precipitated as the HCl salt by addition of Et_2O and subsequent washing with Et_2O until the pH of the wash becomes 6. Use of the free base EDC (an oily compound; 1 mole of EDC is equal to 183 mL) is convenient for coupling the carboxy component with the HCl salt of an amino component. During the coupling reaction, the pH of the mixture must be kept between 5 and 6, which can be checked simply by applying a tiny volume of the reaction mixture to wet pH-test paper. If crystalline EDC·HCl is used as the coupling reagent, it is essential that an additional 1 equivalent of TEA be used to adjust the pH of the reaction mixture to 5 or 6.

Boc-Leu-OPac (7):[10,11]
TEA (17.6 mL, 126 mmol) was added slowly to a stirred soln of Boc-Leu-OH·H_2O (30 g, 120 mmol) and PacBr (25 g, 126 mmol) in DMF (150 mL) at 0 °C. After 3 h, completion of the reaction was confirmed by TLC, and the product was precipitated by addition of excess H_2O. The product was extracted with EtOAc, and the extract was washed successively with H_2O, 1 M HCl, H_2O, and sat. brine. The washed extract was dried ($MgSO_4$), concentrated, and crystallized by triturating with hexane. The product was further purified by recrystallization (EtOAc/hexane); yield: 35 g (80%).

Boc-Ile-Leu-OPac (8):[5]
Compound **7** (24.5 g, 70 mmol) was dissolved in TFA (200 mL) at 0 °C with stirring for 10 min and the mixture was allowed to react for an additional 50 min at rt. TFA was removed by evaporation in vacuo, the residue was triturated with 5.7 M HCl (14.7 mL, 1.2 equiv) in dioxane, and the product was precipitated as the HCl salt of H-Leu-OPac by addition of Et_2O, and dried. The dried HCl salt was dissolved in DMF (200 mL) together with Boc-Ile-OH·0.5H_2O (17.2 g, 72 mmol, 1.02 equiv) and HOBt (10.4 g, 77 mmol, 1.1 equiv). Then, EDC (free-base form, 14.1 mL, 77 mmol, 1.1 equiv) was added slowly to the soln with stirring at −10 °C, and the stirring was continued for an additional 3 h at rt. After confirming the completion of the coupling reaction by TLC using the ninhydrin test as well as the fluorescamine test, the product was precipitated as an oil by addition of excess H_2O (up to 7 times the volume of DMF used). The oil was extracted with EtOAc and the extract was washed successively with 5% aq $NaHCO_3$, H_2O, 1 M HCl, H_2O, and sat. brine, and then dried (Na_2SO_4). The dried soln was concentrated in vacuo and the residue was crystallized by triturating with hexane. The crystals were collected and further purified by recrystallization (EtOAc/hexane); yield: 22 g (68%).

Boc-Cys(Acm)-Ile-Leu-OPac (9):[5]
Compound **8** (12 g, 26 mmol) was dissolved in TFA (150 mL) at 0 °C with stirring for 10 min and the soln allowed to react for an additional 50 min at rt. The resulting soln was concentrated by evaporation in vacuo, the residue was dissolved in 5.7 M HCl in dioxane (5.5 mL), and the product was precipitated by addition of Et_2O as HCl·H-Ile-Leu-OPac. EDC (free base, 5.3 mL, 29 mmol, 1.1 equiv) was added dropwise to a soln of the HCl salt, Boc-Cys(Acm) (8.4 g, 29 mmol, 1.1 equiv), and HOBt (3.8 g, 29 mmol, 1.1 equiv) in DMF (100 mL) at −10 °C, and the mixture was allowed to react for an additional 3 h at rt. After confirming the completion of the reaction by TLC, the product was precipitated as an oily residue by addition of excess H_2O (ca. 7 times the volume of DMF used), and extracted with EtOAc. The extract was washed successively with 5% aq $NaHCO_3$, H_2O, 1 M HCl, H_2O, and sat. brine, and then dried (Na_2SO_4). The dried soln was concentrated in vacuo, the residue was crystallized by triturating with hexane, and further purified by recrystallization (EtOAc/hexane); yield: 15 g (91%). Homogeneity of the product was confirmed by HPLC.

Z-pGlu-Pro-Leu-Arg(Tos)-Lys[Z(2-Cl)]-Leu-Cys(Acm)-Ile-Leu-OPac (1):[5]
The peptide, Boc-Pro-Leu-Arg(Tos)-Lys[Z(2-Cl)]-Leu-Cys(Acm)-Ile-Leu-OPac (3.85 g, 2.46 mmol), synthesized stepwise as in the case of **9**, was dissolved slowly in TFA (80 mL) at 0 °C over 10 min and the soln was kept for an additional 50 min at rt. TFA was removed by evaporation in vacuo, the residue was triturated with 4.1 M HCl in dioxane (0.72 mL, 1.2 equiv), precipitated by addition of excess Et_2O, and dried in a vacuum desiccator over NaOH pellets. The dried product was dissolved in DMF (30 mL), together with Z-pGlu-OH (680 mg, 2.58 mmol, 1.05 equiv) and HOBt (367 mg, 2.71 mmol, 1.1 equiv). EDC (free base, 496 µL, 2.71 mmol, 1.1 equiv) was added slowly to the DMF soln at −10 °C with stirring over 1 h. The mixture was then allowed to react for an additional 4 h at rt. After confirming the completion of the reaction by TLC, the mixture was poured into excess H_2O with vigorous stirring to precipitate the product. The precipitate was collected by filtration, thoroughly washed with H_2O, triturated twice with MeCN, washed with Et_2O, and dried over P_2O_5 in vacuo; yield: 3.5 g (83%).

Synthesis of Other Protected Segments:[5]
Compounds **2**, **3**, **4**, **5**, and **6** can be synthesized by the same procedure described for **1**. Analytical data for these compounds are shown in Table 3.

Table 3 Analytical Data for the Protected Segments Used for the Synthesis of Dendrotoxin-I[a]

Compound	R_f[b]	t_R[c] (min)	PD-MS $[M + H]^+$ (Calcd)
1	0.58 (1)	20.8 (A)	1713.0 (1713.5)
2	0.45 (2)	19.8 (A)	2442.9 (2442.0)
3	0.64 (3)	23.7 (B)	2276.5 (2276.0)[d]
4	0.64 (2)	21.5 (B)	2012.4 (2011.1)[d]
5	0.54 (3)	16.4 (B)	2201.6 (2201.0)
6	0.59 (1)	21.9 (A)	2302.2 (2301.2)

[a] This table is assembled from data available in ref [5].
[b] TLC was carried out on Kieselgel 60F-254 (Merck). Developing solvent: (1) $CHCl_3$/MeOH/AcOH (85:10:5); (2) $CHCl_3$/MeOH/AcOH (85:15:5); (3) $CHCl_3$/80% AcOH/TFE (6:1:1).
[c] RP-HPLC (YMC-Pack ODS, 4.6 × 150 mm); eluent: (A) 30–90% MeCN/0.1% TFA for 25 min at 40 °C; (B) 45–95% MeCN/0.1% TFA for 25 min at 40 °C.
[d] This value was detected as $[M + Na]^+$.

5.2.3.2 Synthesis of Boc-Peptide-OH in Solution

Pac esters can be removed from the fully protected peptide segments by zinc reduction in AcOH at ca. 40 °C.[11] This reaction is so mild that other protecting groups used in the present strategy are kept intact except when $CHCl_3$ is used as a cosolvent. Under these reduction conditions, $CHCl_3$ is also reduced to form HCl, which attacks the Boc group at 40 °C in AcOH. If Met- or Cys-containing peptides are treated with zinc powder under these reduction conditions, the sulfur group is catalytically oxidized to sulfoxide by oxygen (from the air) on the surface of the zinc. Thus, the reaction mixture must be covered with inert gas such as argon or oxygen-free nitrogen during the reaction. If the fully protected peptide Pac ester is not highly soluble in AcOH, the peptide is dissolved in HFIP, and the solution is diluted with AcOH to 20–50% concentration. The Pac ester can then be removed by treatment with zinc powder as in the case of the reduction in AcOH.

Removal of Pac Esters from Compounds 1 and 2:[5]
Powdered Zn (5.4 g, 50 equiv) was added with stirring to a soln of **1** (2.8 g, 1.64 mmol) in glacial AcOH (20 mL) under argon at 40 °C, and stirring was continued for 90 min at the same temperature. After confirming the removal of the Pac group by TLC, the remaining Zn was removed by filtration, the filtrate was concentrated under reduced pressure, and the residue washed successively with chilled 0.1 M HCl, H_2O, and Et_2O, and then dried in a desiccator. The dried material was further purified by dissolving it in a mixture of $CHCl_3$ and MeOH followed by precipitation with Et_2O; yield of **13**: 2.3 g (88%). Yield of **14** from 4 g of **2**: 3.5 g (92%).

Removal of Pac Esters from Compounds 4 and 5:[5]
Compound **4** (15 g, 7.46 mmol) was dissolved in HFIP (150 mL), and the soln diluted with AcOH (150 mL). Zn powder (25 g, ca. 50 equiv) was added to the soln with stirring at 40 °C under argon. After 3 h of stirring, the mixture was treated as in the case of the synthesis of **13**; yield of **12**: 10.2 g (72%). Compound **5** (8.9 g, 4.05 mmol) was also treated as described above to obtain **15**; yield: 7.6 g (90%).

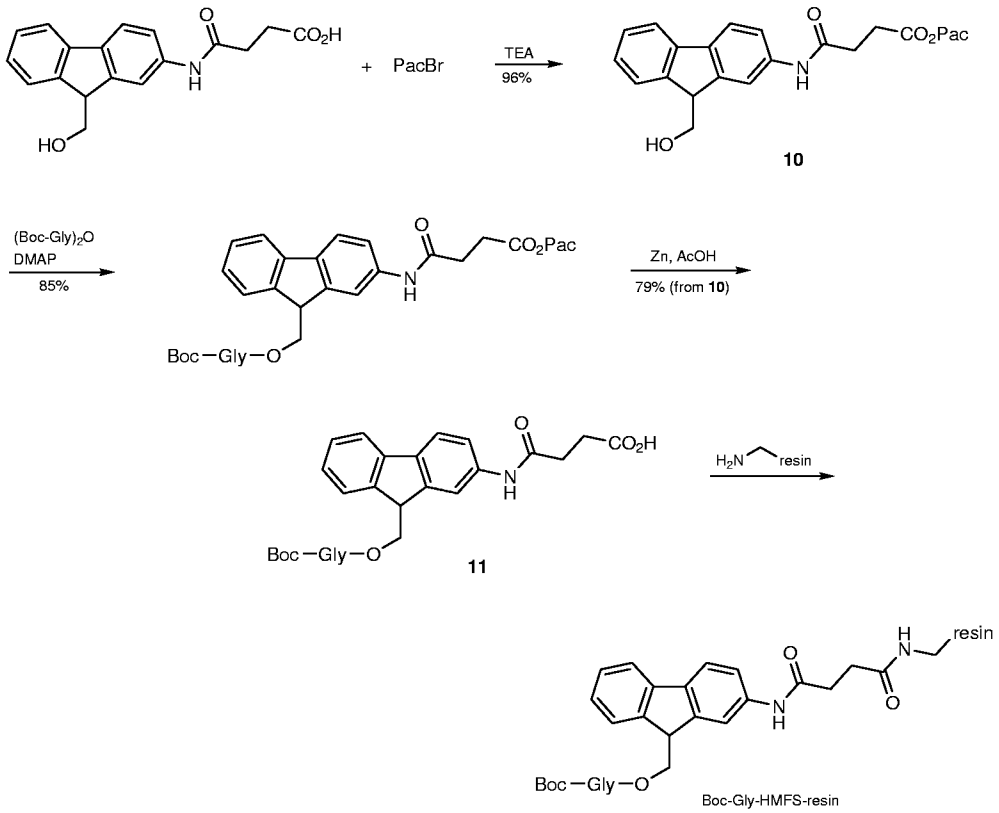
5.2.3.3 Synthesis of Boc-Peptide-OH on *N*-[9-(Hydroxymethyl)fluoren-2-yl]-succinamic Acid Resin

Although Boc-peptide-OH can be synthesized in solution following the procedure as described above, the same fully protected segment can be synthesized by the solid-phase method using the Boc strategy on an *N*-[9-(hydroxymethyl)-2-fluorenyl]succinamic acid

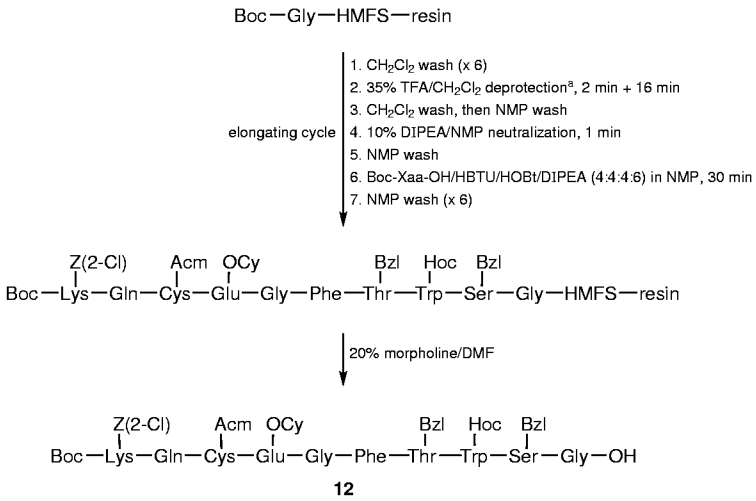
for references see p 62

(HMFS) resin.[13,14] The principle is based on the fact that the 9-fluorenylmethyl ester group is stable to TFA, which is an essential reagent for elongating peptides by the Boc strategy, and is susceptible to a secondary base such as morpholine or piperidine, which is essential for cleaving the protected peptides from the resin while keeping all the protecting groups intact. Thus, the protected peptide segments, Boc-peptide-OH, can be conveniently synthesized by the solid-phase procedure as shown in Schemes 5 and 6. In order to synthesize protected segments successfully on a HMFS resin, all of the side-chain protecting groups must be stable under the treatment conditions with a secondary base. Among protecting groups listed in Table 2, the Z(2-Br) used for protecting Tyr and For for protecting Trp are too unstable in the presence of a secondary base. Thus, Hoc[15] and 3-Pn[16] must be used for protecting Trp and Tyr residues, respectively. By applying the solid-phase method on the HMFS resin, at least one or two protected decapeptides can be synthesized within a week using a semi-automatic peptide synthesizer, such as ACT Model 90 (Advanced ChemTech, Louisville, Kentucky). For elongating each Boc amino acid within a short period of time, the use of HBTU is recommended as the coupling reagent. However, during the coupling reaction with HBTU, significant dehydration of the side-chain amide groups of Asn and Gln is observed; to suppress the side reaction, protection of the amide groups with Xan[17] is also recommended. Xan can be easily removed with TFA. However, in the course of cleaving the Xan group with TFA, addition of 1% TIS in TFA is effective in eliminating unnecessary side reactions with the formed xanthenyl cation.

Scheme 5 Synthesis of HMFS Resin and Attachment of the First Boc Amino Acid[13,14]



Scheme 6 Elongation of Component Amino Acids on the HMFS Resin[13,14]



[a] 1% TIS must be added to the TFA/CH_2Cl_2 soln when Boc-Asn(Xan)- or Boc-Gln(Xan)-peptide-HMFS-resin is treated.

Synthesis of Pac-Protected HMFS 10:[14]

TEA (9.9 mL, 70 mmol) was added with stirring to a soln of HMFS[13] (20 g, 64.2 mmol) and PacBr (14.0 g, 70 mmol) in DMF (100 mL) at rt. After 3 h of stirring, the mixture was partitioned between EtOAc (300 mL) and H_2O (2 L), and the organic layer was washed successively with 1 M HCl, H_2O, and sat. brine, and then dried ($MgSO_4$). The dried soln was concentrated and the residue purified by recrystallization (EtOAc/hexane); yield: 26.5 g (96%).

Synthesis of *N*-[9-(Boc-Gly-OCH$_2$)fluoren-2-yl]succinamic Acid (11):[13,14]

DCC (4.95 g, 24 mmol) was added with stirring to a soln of Boc-Gly-OH (8.4 g, 48 mmol) in CH_2Cl_2 (50 mL) at rt. After stirring for 30 min, the DCU formed was removed by filtration, and the filtrate was added to a soln of **10** (8.59 g, 20 mmol) in DMF (30 mL) at 0 °C. DMAP (2.4 g, 2.0 mmol) was added to the entire mixture at once at 4 °C and the mixture was stirred continuously for an additional 2 h at rt. The resulting soln was concentrated under reduced pressure, the residue was partitioned between EtOAc (300 mL) and 1 M HCl (150 mL), the organic layer was washed successively with H_2O, sat. aq $NaHCO_3$, H_2O, and sat. brine, and then dried ($MgSO_4$). Removal of the solvent gave a residue, which was crystallized (EtOAc/hexane); yield: 10 g (85%). The product was dissolved in AcOH (100 mL), and the soln treated with Zn powder (20 g) at 40 °C with vigorous stirring for 30 min. Excess Zn was removed by filtration, the filtrate was concentrated under reduced pressure, and the residue triturated with 1 M HCl, washed thoroughly with H_2O, and dried ($MgSO_4$). The dried material was purified by recrystallization (EtOAc/hexane) to give **11**; yield: 7.4 g (79% from **10**).

Attachment of Compound 11 onto an Aminomethylated Polystyrene Resin:[13,14]

A 1% cross-linked aminomethylated polystyrene resin (0.83 mmol of amino group per gram of resin, 2.41 g, 2 mmol) was placed in an ATC Model 90 reaction vessel (capacity 200 mL) and the resin washed successively with NMP, 10% DIPEA in NMP, and NMP. Then, **11** (1.41 g, 3 mmol, 1.5 equiv), HBTU (1.14 g, 3 mmol), and HOBt (0.41 g, 3 mmol) were dissolved in NMP (20 mL) in a flask, and DIPEA (0.78 mL, 4.5 mmol) was added to the soln immediately. The mixture was introduced into the reaction vessel containing the resin within 2 min after addition of DIPEA, and the entire mixture was agitated for 1 h at rt. The soln was drained from the reaction vessel and the resin washed well with NMP. The completion of the coupling reaction was confirmed by a ninhydrin test.

On confirming that the ninhydrin test is negative, the resin is subjected to the segment elongation reaction. If the ninhydrin test is positive, the same coupling procedure should be repeated until the ninhydrin test becomes negative.

Elongation of Protected Segments on the HMFS Resin; Synthesis of 12 on the Resin:[13,14]

Segment **12** was elongated from the Boc-Gly-HMFS-resin (2 mmol scale) in the same reaction vessel following the coupling cycles as shown in Scheme 6. Coupling was carried out with 8 mmol each of Boc-Ser(Bzl)-OH, Boc-Trp(Hoc)-OH, Boc-Thr(Bzl)-OH, Boc-Phe-OH, Boc-Gly-OH, Boc-Glu(OCy)-OH, Boc-Cys(Acm)-OH, Boc-Gln(Xan)-OH, and Boc-Lys[Z(2-Cl)]-OH, in this order, using HBTU/HOBt/

for references see p 62

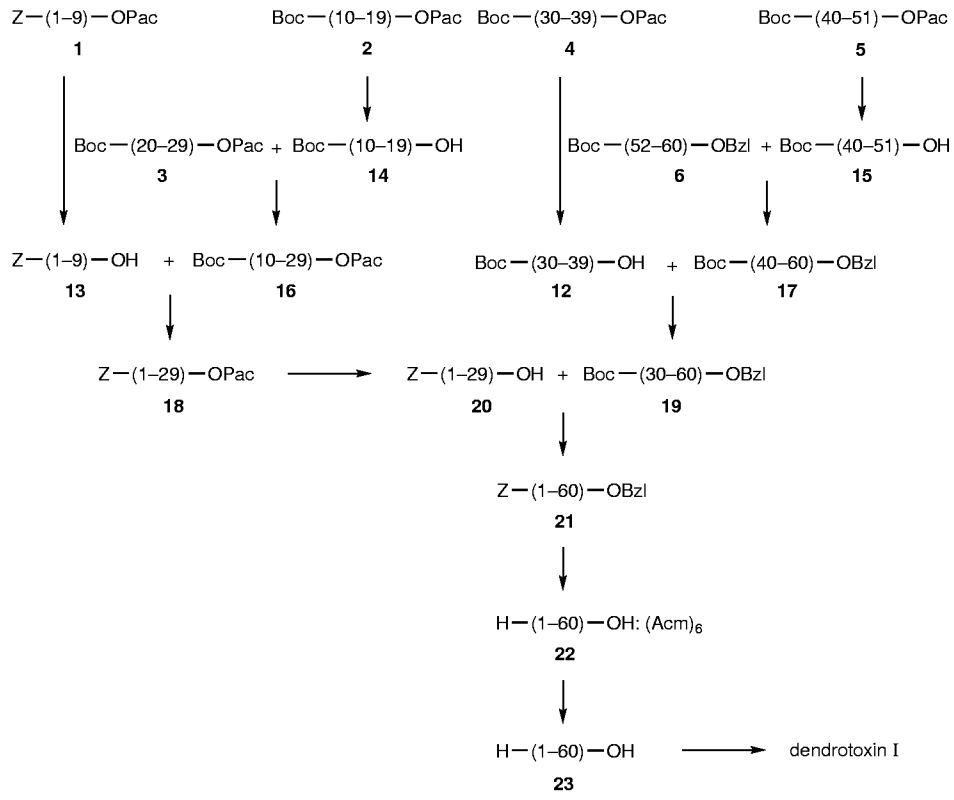
DIPEA (8:8:12 mmol each) as the coupling reagents. Completion of each coupling reaction was monitored by the ninhydrin test. Finally, the fully protected peptide-HMFS-resin was collected by filtration on a glass filter, washed well with MeOH and dried; wt of the resin: 5.97 g.

Isolation of 12 from the Resin:[13,14]
The fully protected peptide-HMFS-resin (5.97 g) was placed in a vessel (200 mL) equipped with a glass filter at the bottom, commonly designed for the manual solid-phase synthesis of peptides, and the resin was soaked in DMF for several min until swollen, after which 20% morpholine in DMF (80 mL) was added over the resin. The vessel was then rotated gently for 30 min at rt to cleave off the protected segment from the resin. The DMF soln was collected by filtration, the resin remaining in the vessel was treated again with 20% morpholine in DMF (80 mL) for 30 min, and the DMF soln was collected by filtration. The DMF solns were combined and concentrated under reduced pressure. The residue was crystallized by trituration with 0.1 M HCl and with H_2O, the crystals collected by filtration, washed successively with H_2O and then Et_2O, and dried. The product was purified by reprecipitation from DMF with MeOH, washed with MeOH and Et_2O, and dried to give **12**; yield: 2.44 g (69% calcd from Boc-Gly-HMFS-resin); TLC (Merck Kieselgel 60F-254): R_f 0.45 ($CHCl_3$/MeOH/AcOH 85:10:5); HPLC (YMC-ODS 4.6 × 150 mm): t_R 19.5 min (45–95% MeCN/0.1% TFA in 25 min at 40 °C); MALDI-TOF-MS (*m/z*): $[M + Na]^+$ calcd, 1893.55; found, 1893.53. These values were completely identical to those observed for **12** synthesized by the solution procedure.

5.2.3.4 Assembly of Segments into the Full Sequence of Dendrotoxin-I

In order to couple two fully protected segments, the Pac ester group must be removed from the carboxy component and the Boc group must be removed from the amino component, as has been described in Section 5.2.1. These protected segments thus synthesized are assembled into the full sequence of the target protein, dendrotoxin-I, as shown in Scheme 7.[5]

Scheme 7 Assembly of Synthetic Segments to the Fully Protected Dendrotoxin-I in Solution[5]



When carrying out the elongation reactions with fully protected segments, it is important to compare the difference in solubility between the amino and carboxy components. If the difference in solubility is significant, the more soluble component must be used in slight excess, up to 1.1 equivalents, versus the less soluble component. This process is particularly effective in making purification of the product easier.

Boc-(10–29)-OPac (16):[5]
Compound **3** (1.45 g, 0.645 mmol) was dissolved in TFA (30 mL) at 0 °C for 10 min. After allowing to react for an additional 50 min at rt, the soln was concentrated under reduced pressure. The residue was triturated with 5 M HCl in dioxane (0.19 mL, 1.5 equiv), the product was precipitated by addition of excess Et_2O, and dried over NaOH pellets in a vacuum desiccator. The dried product and **14** (1.5 g, 0.67 mmol) were dissolved together in DMF (50 mL), and HODhbt (0.11 g, 0.67 mmol) was added to the soln. After cooling to −10 °C, EDC (free base, 0.12 mL, 0.67 mmol) was added slowly into the soln with stirring, and the entire mixture was then allowed to react for 16 h at rt. After confirming the completion of the coupling reaction by TLC, the product was precipitated by adding an excess of chilled H_2O, the precipitate collected by filtration, and washed successively with H_2O and Et_2O. The product was further purified by reprecipitation from MeOH with Et_2O, and dried; yield: 2.5 g (89%). Formation of the desired product was confirmed by amino acid analysis.

Z-(1–29)-OPac (18):[5]
Compound **16** (2.1 g, 0.48 mmol) was treated with TFA (30 mL) as described above and the product was converted into the HCl salt, as shown in the synthesis of **1** or **9**. The HCl salt and **13** (0.79 g, 0.50 mmol) were both dissolved in DMF (50 mL). After adding HODhbt (81.6 mg, 0.50 mmol), the soln was chilled to −10 °C, and EDC (free base, 91.5 μL, 0.50 mmol) was added slowly with stirring. The entire mixture was allowed to react for 16 h at rt. After confirming the completion of the reaction by TLC, the product was precipitated by addition of excess chilled H_2O, the precipitates were collected by filtration, and washed successively with H_2O and Et_2O. The product was further purified by reprecipitation from MeOH with Et_2O, and dried; yield: 2.6 g (92%). Formation of the desired product was confirmed by amino acid analysis.

Boc-(40–60)-OBzl (17):[5]
Compound **6** (7.5 g, 3.26 mmol) was dissolved in TFA (50 mL) at −10 °C, and the soln was allowed to react for 40 min at rt. After the removal of excess TFA in vacuo, the residue was triturated with 4.1 M HCl (0.96 mL, 3.94 mmol) in dioxane, the product precipitated with Et_2O, and dried over NaOH pellets in a vacuum desiccator. The dried product and **15** (7.1 g, 3.41 mmol) were dissolved together in NMP (100 mL). After addition of HODhbt (0.56 g, 3.43 mmol), the soln was kept at −10 °C, and EDC (free base, 0.63 mL, 3.44 mmol) was added slowly with stirring. At this point, the pH of the mixture was checked with pH-test paper and was kept between 5 and 6. The entire mixture was stirred overnight at rt, during which period the progress of the reaction was monitored by TLC using the ninhydrin, as well as the fluorescamine test. After confirming the completion of the coupling reaction, the product was precipitated by addition of excess of chilled H_2O, the precipitates collected by filtration, and washed successively with H_2O and MeCN. The product was purified by reprecipitation from DMF with EtOAc to give **17**; yield: 13.0 g (93.5%). Formation of the desired compound was confirmed by amino acid analysis.

Boc-(30–60)-OBzl (19):[5]
Compound **17** (7.0 g, 1.64 mmol) was dissolved in TFA (50 mL) at 0 °C to remove the Boc group as in the case of the synthesis of **18**, the product was obtained as the HCl salt, and the product was dissolved in NMP (200 mL) together with **12** (3.2 g, 1.71 mmol) and HOBt (0.23 g, 1.72 mmol). After cooling the soln to −10 °C, EDC (free base, 0.32 mL, 1.75 mmol) was added to the soln with stirring, and the entire mixture was allowed to react for 16 h at rt. During this period, progress of the coupling reaction was monitored by TLC. The product was precipitated by addition of excess MeCN, the precipitate was collected by filtration, and washed well with MeCN and then with Et_2O. The washed product was purified by reprecipitation from NMP with MeCN, and dried; yield: 9.0 g (91%). Formation of the desired compound was confirmed by amino acid analysis.

Z-(1–29)-OH (20):[5]
Compound **18** (2.36 g, 0.4. mmol) was treated with Zn powder (1.17 g, 18 mmol) in AcOH (50 mL) under argon at 45 °C for 1 h. The remaining Zn powder was removed by filtration, washed with a little AcOH, and the filtrate concentrated to dryness under reduced pressure. The residue was washed successively with 1 M aq HCl, H_2O, MeCN, and Et_2O, and then dried; yield: 2.0 g (87%).

for references see p 62

Z-(1–60)-OBzl (21):[5]
Compound **19** (1.34 g, 0.22 mmol) was treated with TFA (30 mL), as in the case of the synthesis of **17**, and the product was converted into the HCl salt. The HCl salt, compound **20** (1.4 g, 0.24 mmol) and HODhbt (40.8 mg, 0.25 mmol) were dissolved together in a mixture of DMF/NMP/DMSO (1:2:1, 100 mL), and EDC (free base, 45.8 μL, 0.25 mmol) was slowly added to the soln with stirring at −10 °C. The entire mixture was allowed to react for an additional 16 h at rt. The product was precipitated by addition of excess MeCN, the precipitate collected by filtration, and washed successively with MeCN and Et_2O. The washed material was purified by trituration with MeCN and Et_2O, and dried; yield: 2.2 g (87%). At this stage, formation of the desired product was confirmed by amino acid analysis.

5.2.3.5 Final Deprotection and Folding to Dendrotoxin-I

The final deprotection reaction is performed in HF in the presence of an appropriate scavenger using a properly designed HF reaction apparatus.[18] As a scavenger, anisole is most practical and efficient since it is easily soluble in HF. In general, more than 5 equivalents of anisole must be added compared to the number of protecting groups to be cleaved by HF, and the volume of HF used must be 5- to 10-fold greater than the volume of anisole. *p*-Cresol can also be used as a scavenger, but it is less efficient in mixing with peptides in HF unless the reaction mixture is agitated vigorously. The temperature of the HF reaction mixture is also an important factor and affects the tendency to form side products during the deprotection reactions in HF since the Friedel–Crafts reaction is temperature dependent. Thus, all of the HF deprotection reactions must be carried out at −5 °C, during which time the HF reaction vessel must be immersed in an ice–salt bath at a temperature lower than −5 °C, and after the reaction time is over, excess HF must be removed as rapidly as possible at a temperature lower than −5 °C. For evacuating the HF reaction vessel efficiently, the use of a mechanical pump with a liquid nitrogen trap made of Teflon or Kel-F, is recommended.

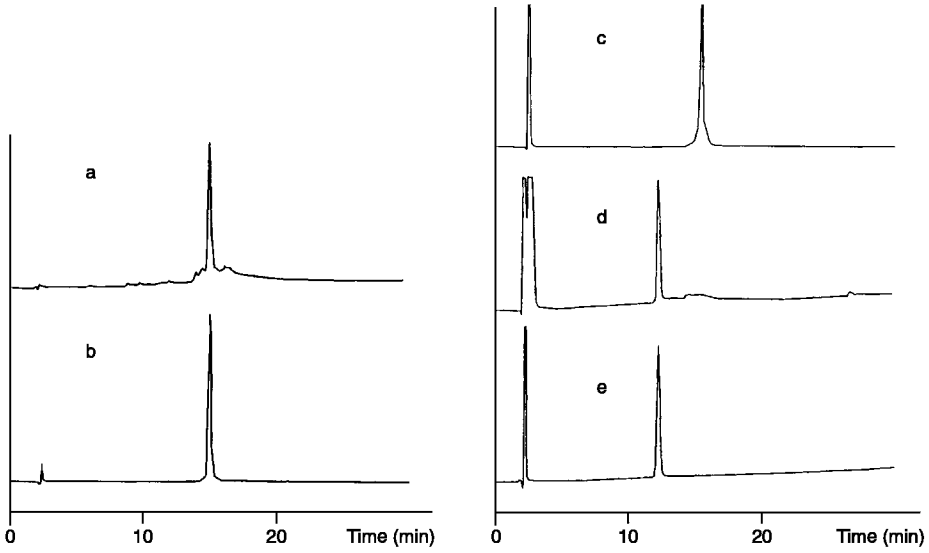
The For group for protecting Trp is stable in HF. Thus, after completion of the normal HF reaction with anisole as the scavenger and removal of HF as well as anisole from the reaction mixture, fresh HF is again introduced into the same reaction vessel after adding butanedithiol into the vessel as a new scavenger. On the other hand, if the Hoc group[15] is used for protecting the Trp residue, it is removed simultaneously with all other protecting groups. For this reason, the use of Hoc is advantageous compared to For. Acm,[19] used for protecting Cys, is also stable in HF. In the case of the fully protected peptide containing Cys(Acm) groups, all protecting groups other than Acm groups are removed by a single HF reaction with anisole. The product can then be thoroughly purified by HPLC and the Acm groups removed by $Hg(OAc)_2$, thus, isolating the completely deprotected product. For the deprotection reaction with $Hg(OAc)_2$, 50% AcOH must be used as the solvent.[20] This condition is important in order to minimize the attachment of β-ME to the indole moiety if the peptide contains Trp residues (β-ME is an essential reagent for removing Hg ions from the liberated peptide by gel filtration). Removal of the Acm group by the two-step deprotection reaction is extremely effective in making it easier for the subsequent folding reaction to take place.

Removal of Protecting Groups from 21 in HF:[5]
Compound **21** (1 g, 86 μmol) and anisole (4.4 mL) were placed in a Kel-F HF reaction vessel (capacity 200 mL) and HF (~40 mL) was introduced into the vessel using a HF reaction apparatus, which was cooled to −80 °C using an acetone/dry ice bath. The dry ice bath was replaced with a $NaCl/H_2O$/ice bath at −5 °C and the HF-containing soln was stirred vigorously for 1 h using a Teflon magnetic stirrer. Excess HF was then removed as rapidly as possible by distillation under reduced pressure using a mechanical pump, and the residue was completely dried. The residue was triturated with Et_2O, collected by filtration, washed well with Et_2O, and dried; yield of the crude product: 800 mg. The product was purified by preparative RP-HPLC (YMC-Pac ODS column, 30 × 250 mm, linear gradient 20–30% MeCN/0.1% aq TFA for 60 min) to give 130 mg of purified **22**. The HPLC chromatograms are shown in Figure 1 (a) and (b).

Removal of Acm Groups from 22:[5]
To a soln of **22** (155 mg, 20.5 μmol) in 50% aq AcOH (20 mL), $Hg(OAc)_2$ (43 mg, 135.2 μmol) was added with stirring under argon, and the mixture allowed to react for 2 h at rt in the dark. After adding β-ME (0.32 mL), the mixture was further stirred for an additional 2 h at rt under argon and the mixture was applied onto a Sephadex G-25 column (20 × 100 mm, 1 M AcOH). The fractions containing the principal product were collected and lyophilized to give 130 mg of crude product **23**. The product was dissolved in a soln of DTT (22 mg, 0.14 mmol) in 0.1 M NH_4OAc buffer, pH 7.8, the soln kept for 10 min at rt, and then purified by preparative HPLC after addition of TFA to adjust the soln to an acidic pH. The peptide was developed on the HPLC column (gradient 20–40% MeCN/0.1% aq TFA for 60 min) to give 110 mg of purified **23**. The HPLC chromatogram is shown in Figure 1 (c).

Oxidative Folding of 23:[5]
Compound **23** (100 mg, 14.0 μmol) was dissolved in 0.1 M Tris buffer pH 8.7 (1 L) containing 1 mM EDTA, and reduced (430 mg) and oxidized (86 mg) glutathione, and the soln gently stirred for 20 h at rt. The mixture was subjected to purification by preparative HPLC (linear gradient 10–40% MeCN/0.1% aq TFA for 60 min) to give 60 mg of the target peptide, dendrotoxin-I. The HPLC profiles of the folding soln and the purified material are shown in Figure 1 (d) and (e). Amino acid analysis after 6 M HCl hydrolysis: Asp (4), 3.98; Thr (3), 2.73; Ser (2), 1.83; Glu (7), 7.01; Gly (5), 4.90; Ala (1), 1.05; $1/2(Cys)_2$ (6), 5.68; Ile (4), 3.82; Leu (3), 3.01; Tyr (3), 2.93; Phe (3), 3.00; His (1) 0.99; Lys (7), 6.93; Trp (1), 0.97; Arg (7), 6.98; Pro (3), 2.86; MALDI-TOF-MS (*m/z*): $[M + H]^+$ calcd, 7134.3; found, 7134.3.

Figure 1 HPLC Profile of Synthetic Dendrotoxin-I Intermediates in the Deprotection and Folding Process. (a) Crude Product Obtained by the HF Deprotection Reaction. (b) Purified Peptide (6 Acm). (c) Completely Deprotected Peptide. (d) After Folding Reaction. (e) Final, Purified Product. Column: YMC-Pac ODS A-302 (4.6 × 150 mm); Eluent: 20–40% MeCN/0.1% TFA; 25 min at 40 °C; Flow Rate: 1.0 mL·min^{-1}; Detection: 220 nm. Individual Chromatographs from this Figure have Previously Appeared in Ref[5]

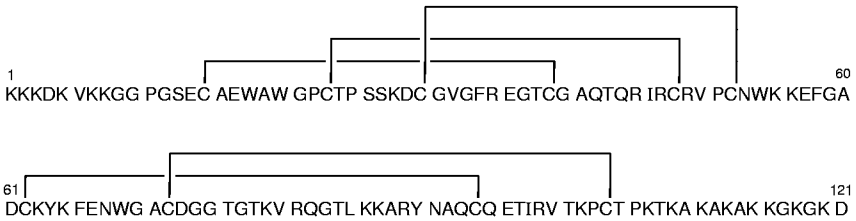


5.2.4 Synthesis of Human Midkine

Human midkine is a linear protein having 121 amino acid residues consisting of two domains, the structure of which is stabilized by five intramolecular disulfide bonds. The N-terminal half domain, containing three disulfide bonds, is linked to a C-terminal half domain, containing two disulfide bonds, as shown in Scheme 8. Thus, synthesis of the entire molecule is achieved by coupling two large N- and C-half segments.[6] The N-half segment is assembled from six segments and the C-half from seven segments, as shown in Scheme 9. The basic principles for the synthesis of each segment are the same as shown in Section 5.2.2, but in this case the For group is used for protecting the Trp residues. Analytical data for these fully protected segments thus synthesized are shown in Table 4.

for references see p 62

Scheme 8 Primary Structure of Human Midkine



Scheme 9 Structure of Segments Designed for the Synthesis of Human Midkine[6]

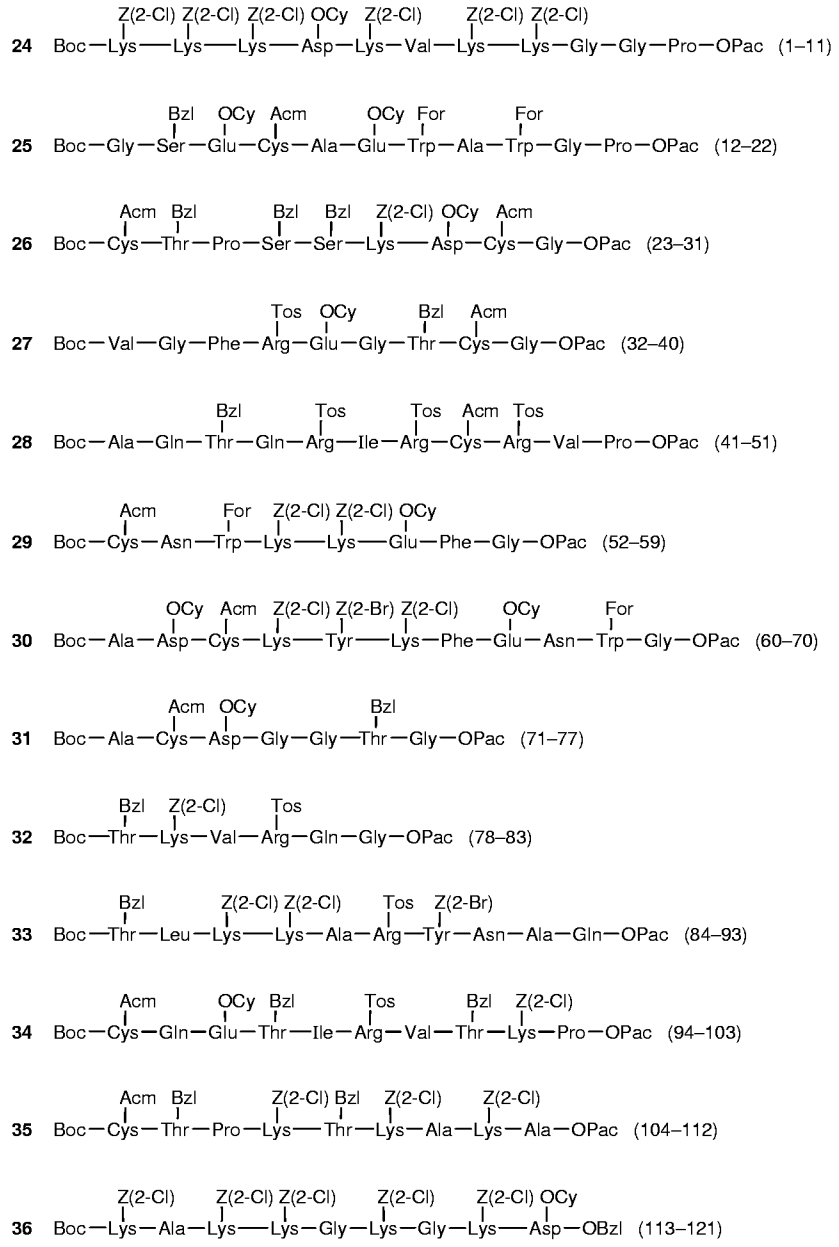


Table 4 Analytical Data for the Protected Segments Used for the Synthesis of Human Midkine[a]

Compound	R_f[b]	t_R[c] (min)
24	0.51 (1)	20.8 (A)
25	0.53 (3)	18.2 (A)
26	0.49 (1)	19.8 (A)
27	0.54 (3)	23.7 (B)
28	0.31 (1)	21.5 (B)
29	0.50 (1)	16.4 (B)
30	0.60 (3)	23.1 (B)
31	0.62 (1)	11.2 (B)
32	0.64 (3)	16.1 (B)
33	0.59 (2)	21.2 (A)
34	0.70 (3)	17.2 (A)
35	0.61 (1)	19.2 (A)
36	0.71 (3)	25.1 (A)

[a] This table is assembled from data available in ref[6].
[b] TLC was carried out on Kieselgel 60F-254 (Merck). Developing solvent: (1) $CHCl_3$/MeOH/AcOH (85:10:5); (2) $CHCl_3$/MeOH/AcOH (85:15:5); (3) $CHCl_3$/80% AcOH/TFE (6:1:1).
[c] RP-HPLC (YMC-Pack ODS, 4.6 × 150 mm); eluent: (A) 30–90% MeCN/0.1% TFA for 25 min at 40 °C; (B) 45–95% MeCN/0.1% TFA for 25 min at 40 °C.

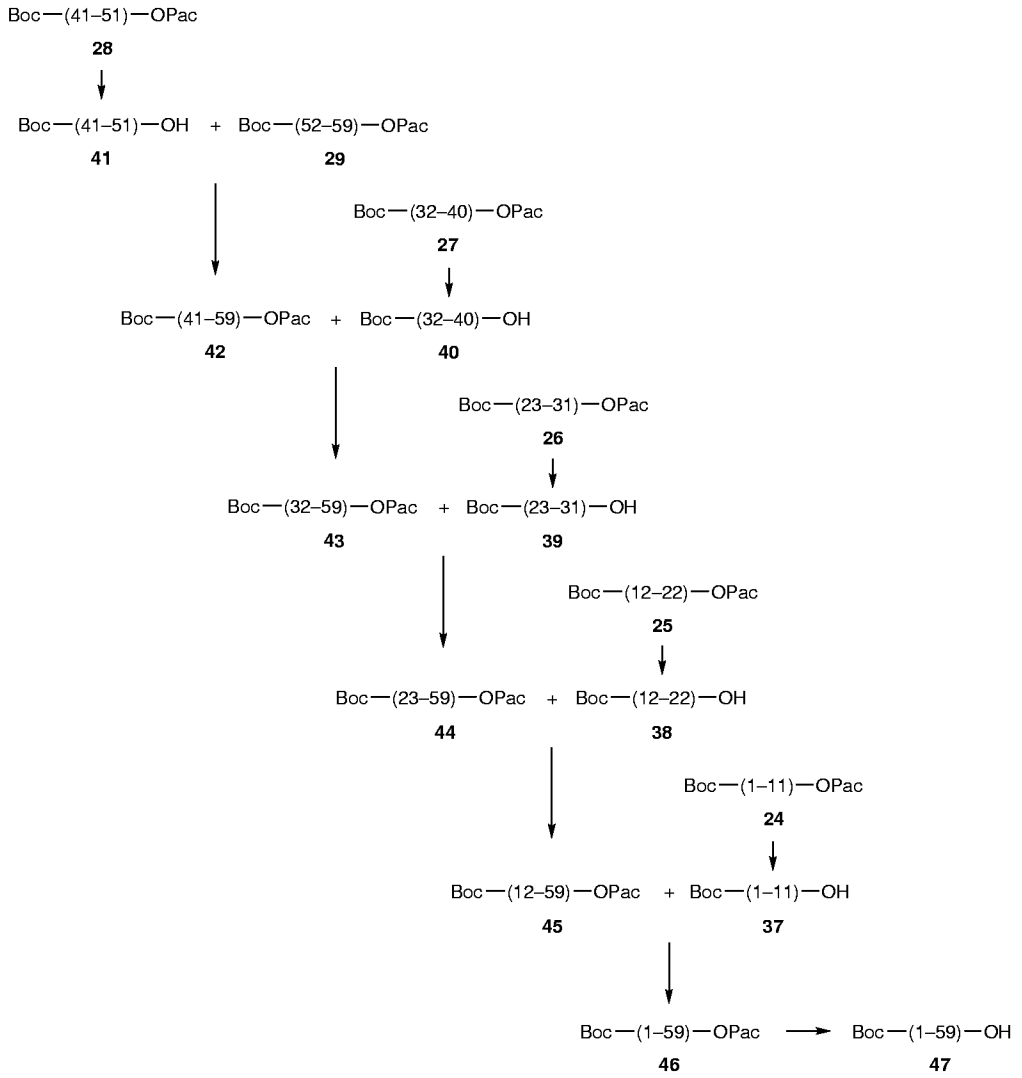
Compounds **26**, **27**, **31**, **32**, **34**, and **35** are freely soluble in AcOH. Consequently, their Pac esters can be removed in AcOH, as described in Section 5.2.3.2 for the synthesis of compounds **1** and **2**. Compounds **24**, **25**, **28**, **30**, and **33** are insoluble in AcOH. Thus, they are dissolved in HFIP and the solutions are treated with zinc powder after diluting with the same volume of AcOH as discussed in Section 5.2.3.2 for the synthesis of compounds **4** and **5**.

5.2.4.1 Assembly of Segments into the Full Sequence of Human Midkine

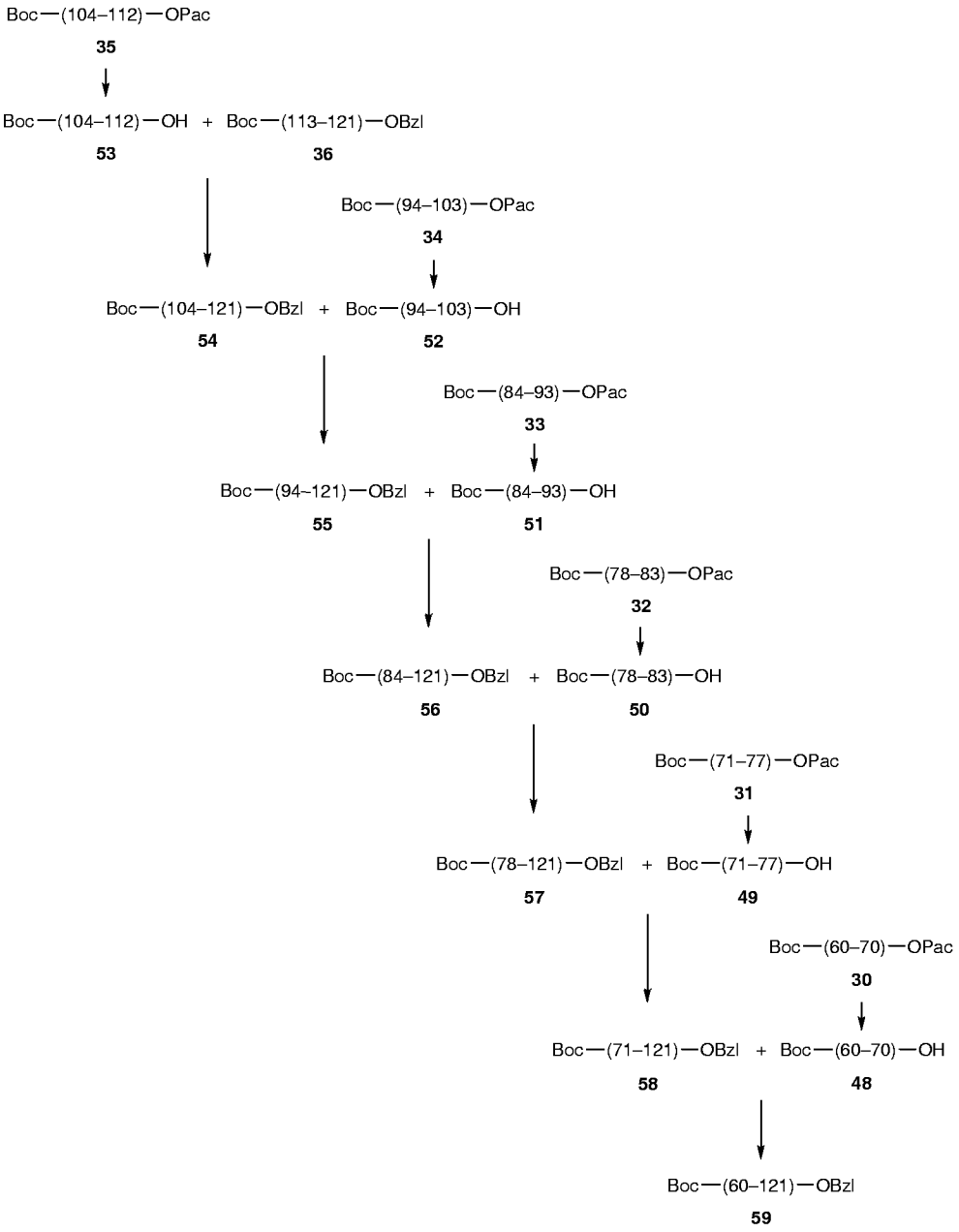
The protected segments synthesized above were assembled as shown in Schemes 10, 11, and 12.[6]

for references see p 62

Scheme 10 Assembly of Synthetic Segments to the Fully Protected Midkine N-Half Domain in Solution[6]

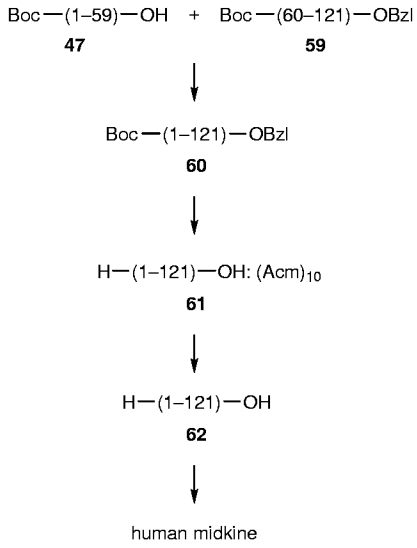


Scheme 11 Assembly of Synthetic Segments to the Fully Protected Midkine C-Half Domain in Solution[6]



for references see p 62

Scheme 12 Final Steps for the Synthesis of Human Midkine[6]



Boc-(41–59)-OPac (42):[6]
Compound **29** (2.44 g, 1.4 mmol) was dissolved in TFA (50 mL) at −10 °C and the soln allowed to react for 1 h at rt. The soln was concentrated to a residue, which was then triturated with 5.2 M HCl in dioxane (0.4 mL, 2.1 mmol) and the residue precipitated with Et_2O to give a powder. The product was washed with Et_2O and dried. The entire product and **41** (3.0 g, 1.47 mmol) were dissolved in DMF (100 mL), and HODhbt (0.24 g, 1.47 mmol) was then added. EDC (free base, 0.27 mL, 1.47 mmol) was added to the soln with stirring at −10 °C, and the mixture allowed to react at rt for 8 h. The product was precipitated by adding an excess of H_2O, the precipitate collected by filtration, washed successively with MeOH, EtOAc, and hexane, and then dried; yield: 4.5 g (87%); TLC (Merck Kieselgel 60F-254): R_f 0.41 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **42** was confirmed by amino acid analysis.

Boc-(32–59)-OPac (43):[6]
Compound **42** (4.48 g, 1.22 mmol) was treated with TFA (50 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **40** (1.92 g, 1.28 mmol) were dissolved in a mixture of DMF (70 mL) and NMP (100 mL), and HOBt (0.17 g, 1.28 mmol) was then dissolved in the soln. EDC (free base, 0.23 mL, 1.28 mmol) was added to the soln with stirring at −10 °C, and the mixture allowed to react at rt for 16 h. The product was precipitated by adding an excess of H_2O, the precipitate collected by filtration, washed successively with MeOH, EtOAc, and hexane, and then dried; yield: 5.33 g (88%); TLC (Merck Kieselgel 60F-254): R_f 0.60 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **43** was confirmed by amino acid analysis.

Boc-(23–59)-OPac (44):[6]
Compound **43** (5.3 g, 1.06 mmol) was treated with TFA (50 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **39** (1.85 g, 1.11 mmol) were dissolved in a mixture of DMF (50 mL) and NMP (100 mL), and HOBt (0.15 g, 1.11 mmol) was then dissolved in the soln. EDC (free base, 0.20 mL, 1.11 mmol) was added to the soln with stirring at −10 °C, and the mixture allowed to react at rt for 16 h. The product was precipitated by adding an excess of H_2O, the precipitate collected by filtration, washed with MeOH, and purified by dissolving in a small volume of a mixture of $CHCl_3$/TFE (3:1) followed by precipitation with EtOAc; yield: 6.18 g (90%); TLC (Merck Kieselgel 60F-254): R_f 0.61 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **44** was confirmed by amino acid analysis.

Boc-(12–59)-OPac (45):[6]
Compound **44** (6.11 g, 0.95 mmol) was treated with TFA (50 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **38** (1.78 g, 1.05 mmol) were dissolved in a mixture of $CHCl_3$/TFE (3:1, 100 mL), and HODhbt (0.17 g, 1.05 mmol) was then dissolved in the soln. EDC (free base, 0.19 mL, 1.05 mmol) was added dropwise to the soln with stirring at −10 °C, and the mixture allowed to react for 4 h at rt. The product was precipitated by adding an excess of MeCN, the precipitate collected by filtration, washed successively with H_2O and MeOH, and purified by dissolving

in a small volume of a mixture of $CHCl_3$ and TFE (3:1) followed by precipitation with MeCN; yield: 6.48 g (85%); TLC (Merck Kieselgel 60F-254): R_f 0.50 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **45** was confirmed by amino acid analysis.

Boc-(1–59)-OPac (46):[6]
Compound **45** (6.45 g, 0.81 mmol) was treated TFA (50 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **37** (2.12 g, 0.88 mmol) were dissolved in a mixture of $CHCl_3$/TFE (3:1, 150 mL), and HODhbt (0.14 g, 0.88 mmol) was then dissolved in the soln. EDC (free base, 0.16 mL, 0.88 mmol) was added dropwise to the soln with stirring at −10 °C, and the mixture allowed to react for 3 h at rt. The product was precipitated by adding an excess of MeCN, the precipitate collected by filtration, washed successively with H_2O and MeOH, and purified by dissolving in a small volume of $CHCl_3$/TFE (3:1) followed by precipitation with EtOAc; yield: 7.88 g (96%); TLC (Merck Kieselgel 60F-254): R_f 0.41 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **46** was confirmed by amino acid analysis.

Boc-(1–59)-OH (47):[6]
Compound **46** (7.85 g, 0.77 mmol) was dissolved in HFIP (100 mL), and the soln diluted with AcOH (100 mL). Powdered Zn (2.7 g) was added to the soln with stirring at 40 °C under argon. After stirring for 2 h, excess Zn was removed by filtration, the filtrate concentrated to a residue under reduced pressure, the residue triturated with ice-cold 0.1 M HCl, and collected by filtration. The filtrate was washed successively with H_2O, MeOH, EtOAc, and hexane, and then dried; yield: 7.1 g (93%); TLC (Merck Kieselgel 60F-254): R_f 0.41 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **47** was confirmed by amino acid analysis.

Boc-(104–121)-OBzl (54):[6]
Compound **36** (7.8 g, 3.76 mmol) was treated with TFA (50 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **53** (7.2 g, 3.95 mmol) were dissolved in DMF (60 mL), and HODhbt (0.64 g, 3.95 mmol) was then dissolved in the soln. EDC (free base, 0.72 mL, 3.95 mmol) was added dropwise to the soln with stirring at −10 °C, and the mixture allowed to react at rt for 16 h. The product was precipitated by adding an excess of chilled H_2O, the precipitate collected by filtration, washed successively with H_2O and MeOH, and purified by reprecipitation from DMF with EtOAc; yield: 13.7 g (97%); TLC (Merck Kieselgel 60F-254): R_f 0.75 ($CHCl_3$/MeOH/AcOH 85:10:5). Formation of the desired product **54** was confirmed by amino acid analysis.

Boc-(94–121)-OBzl (55):[6]
Compound **54** (11.3 g, 3.00 mmol) was treated with TFA (60 mL), as described in the synthesis of **42**, and the product was converted into the HCl salt. The HCl salt and **52** (6.08 g, 3.15 mmol) were dissolved in a mixture of DMF (70 mL) and NMP (60 mL), and HOBt (0.43 g, 3.15 mmol) was then dissolved in the soln. EDC (free base, 0.58 mL, 3.15 mmol) was added dropwise to the soln with stirring at −10 °C, and the mixture allowed to react for 16 h at rt. The product was precipitated by adding an excess of chilled H_2O, the precipitate collected by filtration, washed well with MeOH, and purified by reprecipitation from $CHCl_3$/TFE with MeCN; yield: 16.1 g (96%); TLC (Merck Kieselgel 60F-254): R_f 0.35 ($CHCl_3$/MeOH/AcOH 85:10:5). Formation of the desired product **55** was confirmed by amino acid analysis.

Boc-(84–121)-OBzl (56):[6]
Compound **55** (13.9 g, 2.5 mmol) was treated with TFA (60 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **51** (5.7 g, 2.75 mmol) were dissolved in $CHCl_3$/TFE (3:1, 150 mL), and HODhbt (0.45 g, 2.75 mmol) was then dissolved in the soln. EDC (free base, 0.50 mL, 2.75 mmol) was added dropwise in the soln with stirring at −10 °C, and the mixture allowed to react for 4 h at rt. After evaporation of the solvents, the product was precipitated by adding an excess of chilled H_2O, the precipitate collected by filtration, washed with MeOH, and purified by reprecipitation from $CHCl_3$/TFE with EtOAc; yield: 17.9 g (95%); TLC (Merck Kieselgel 60F-254): R_f 0.35 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **56** was confirmed by amino acid analysis.

Boc-(78–121)-OBzl (57):[6]
Compound **56** (13.2 g, 1.75 mmol) was treated with TFA (70 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **50** (2.3 g, 1.93 mmol) were dissolved in $CHCl_3$/TFE (3:1, 150 mL), and HODhbt (0.32 g, 1.93 mmol) was then dissolved in the soln. EDC (free base, 0.35 mL, 1.93 mmol) was added dropwise to the soln with stirring at −10 °C, and the mixture allowed to react for 8 h at rt. The product was precipitated by adding an excess of MeCN, the precipitate collected by filtration, washed successively with H_2O and MeOH, and purified by reprecipitation from $CHCl_3$/TFE with EtOAc; yield: 14.7 g (97.5%); TLC (Merck Kieselgel 60F-254): R_f 0.49 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **57** was confirmed by amino acid analysis.

for references see p 62

Boc-(71–121)-OBzl (58):[6]
Compound **57** (14.7 g, 1.7 mmol) was treated with TFA (70 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **49** (1.7 g, 1.87 mmol) were dissolved in $CHCl_3$/TFE (3:1, 200 mL), and HODhbt (0.31 g, 1.87 mmol) was then dissolved in the soln. EDC (free base, 0.34 mL, 1.93 mmol) was added dropwise to the soln with stirring at −10 °C, and the mixture allowed to react for 8 h at rt. The product was precipitated by adding an excess of MeCN, the precipitate collected by filtration, washed successively with H_2O and MeOH, and purified by reprecipitation from $CHCl_3$/TFE with MeCN; yield: 15 g (94%); TLC (Merck Kieselgel 60F-254): R_f 0.43 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **58** was confirmed by amino acid analysis.

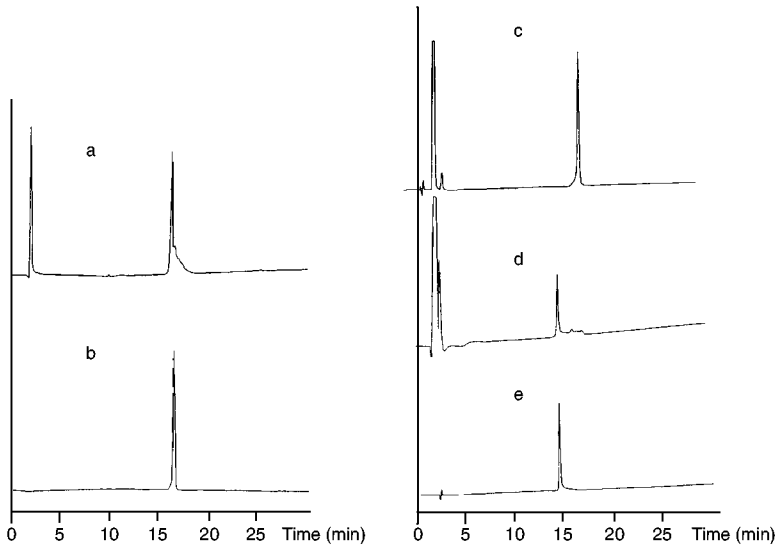
Boc-(60–121)-OBzl (59):[6]
Compound **58** (15 g, 1.6 mmol) was treated with TFA (80 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **48** (4.0 g, 1.76 mmol) were dissolved in $CHCl_3$/TFE (3:1, 200 mL), and HODhbt (0.29 g, 1.76 mmol) was then dissolved in the soln. EDC (free base, 0.32 mL, 1.76 mmol) was added dropwise to the soln with stirring at −10 °C, and the mixture allowed to react for 2 h at rt. The product was precipitated by adding an excess of MeCN, the precipitate collected by filtration, washed successively with H_2O, MeOH and DMF, and purified by reprecipitation from $CHCl_3$/TFE with MeCN; yield: 17.0 g (92%); TLC (Merck Kieselgel 60F-254): R_f 0.53 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **59** was confirmed by amino acid analysis.

Boc-(1–121)-OBzl (60):[6]
Compound **59** (6.03 g, 0.52 mmol) was treated with TFA (50 mL), as described in the synthesis of **42**, and the product converted into the HCl salt. The HCl salt and **47** (4.87 g, 0.57 mmol) were dissolved in $CHCl_3$/TFE (3:1, 200 mL), and HODhbt (0.09 g, 0.57 mmol) was then dissolved in the soln. EDC (free base, 0.10 mL, 0.57 mmol) was added dropwise to the soln with stirring at −10 °C, and the mixture allowed to react for 1 h at rt. During this period, completion of the coupling reaction was confirmed by TLC. The product was precipitated by adding an excess of MeCN, the precipitate collected by filtration, washed successively with H_2O, MeOH, EtOAc, and hexane, and purified by reprecipitation from $CHCl_3$/TFE with MeCN; yield: 11.0 g (97%); TLC (Merck Kieselgel 60F-254): R_f 0.40 ($CHCl_3$/80% aq AcOH/TFE 6:1:1). Formation of the desired product **60** was confirmed by amino acid analysis.

5.2.4.2 Final Deprotection and Folding to Human Midkine

In this synthesis of human midkine, the For group was used for protecting the Trp residues. Thus, reaction with HF to remove the protecting groups has to be carried out in two separate steps; one for all protecting groups other than For and Acm groups, and the other for the For groups. The Boc group attached to the N-terminus of the fully protected midkine molecule can be removed by HF, but during the cleavage reaction with HF at −5 °C the *tert*-butyl cation is formed, which will attach to the Met or Cys residues. Thus, it is recommended that the Boc group is cleaved by TFA prior to the final HF reaction. The product containing Cys(Acm) groups thus obtained can be thoroughly purified by RP-HPLC. Following confirmation of the homogeneity, it can then be subjected to the folding reaction after removal of the Acm groups. Homogeneity of these intermediate products and progress of the folding reaction are monitored by HPLC; the profiles are shown in Figure 2.

Figure 2 HPLC Profile of Synthetic Midkine Intermediates in the Deprotection and Folding Process. (a) Crude Product Obtained by the HF Deprotection Reaction. (b) Purified Peptide (10 Acm). (c) Completely Deprotected Peptide. (d) After Folding Reaction. (e) Finally Purified Product. Column: YMC-Pack ODS A-302 (4.6 × 150 mm); Eluent: 10–40% MeCN/0.1% TFA, 25 min at 40 °C; Flow Rate: 1.0 mL·min^{-1}; Detection: 220 nm. Individual Chromatographs from this Figure have Previously Appeared in Ref[6]



Removal of Protecting Groups from Boc-(1–121)-OBzl (60):[6]
Compound **60** (2 g, 0.09 mmol) was dissolved in TFA (50 mL) at −10 °C, and the soln was allowed to react for 1 h at rt. The soln was concentrated to a residue, precipitated as a powder by triturating with Et_2O, and dried. The dried product and anisole (8 mL) were placed in a HF reaction vessel (200-mL capacity), the vessel attached to a HF reaction apparatus,[18] and HF (70 mL) introduced into the vessel using an acetone/dry ice bath at −80 °C. The mixture was allowed to react for 1 h at −5 °C, as in the case of the synthesis of dendrotoxin-I (see Section 5.2.3.4). HF was removed as rapidly as possible by distillation, the residue washed with Et_2O by decantation three times, then butanedithiol (12 mL) was added into the reaction vessel, and HF (30 mL) was introduced again in the cylinder as described before. After allowing it to react for 30 min at −5 °C, the HF was evaporated off completely, the residue triturated with Et_2O, and collected by filtration; yield of the crude product: 1.30 g. The crude product was purified by RP-HPLC on a YMC-Pack ODS column using 0.1% aq TFA as buffer A and 0.1% TFA in MeCN as buffer B. The conditions and elution profile of the HPLC are shown in Figure 2. The HPLC-purified product was further purified by CM-cellulose column chromatography using a linear gradient of 0.2 M (pH 5.0) to 0.7 M NH_4OAc buffer (pH 6.0) in 3 M urea followed by RP-HPLC to give 100 mg of the purified product **61**.

Removal of Acm Groups from H-(1–121)-OH: $(Acm)_{10}$:[6]
To a soln of **61** (100 mg, 7.16 μmol) in 50% aq AcOH (14.3 mL), $Hg(OAc)_2$ (25.1 mg, 78.8 μmol) was added with stirring under argon, and the mixture allowed to react for 2 h in the dark at rt. After adding β-ME (0.17 mL), the soln was further stirred for 2 h at rt under argon, the mixture applied onto a Sephadex G-25 column (20 × 700 mm, 1 M AcOH). The fractions containing the principal product were collected and lyophilized to give 77 mg of **62**. The HPLC chromatogram is shown in Figure 2 (c).

Oxidative Folding of Compound 62 to Human Midkine:[6]
Compound **62** (75 mg, 5.66 μmol) was dissolved in 50 mM acetate buffer pH 7.7 (566 mL) containing 1 mM EDTA, 2 M $(NH_4)_2SO_4$, and reduced (174 mg) and oxidized (35 mg) glutathione, and the soln gently stirred for 48 h at 5 °C. The mixture was then acidified to pH 3 by adding TFA [the HPLC profile of the product at this point is shown in Figure 2 (d)] and subjected to purification by preparative HPLC using a linear gradient (16–30% MeCN/0.1% aq TFA for 60 min) to obtain 48 mg of the target protein. The folding process was also monitored by HPLC [HPLC profile of the finally purified product is shown in Figure 2 (e)]. Amino acid analysis after 6 M HCl hydrolysis: Asp (8), 7.52; Thr (10), 9.51; Ser (3), 2.86; Glu (11), 11.12; Gly (16), 16.00; Ala (10), 10.73; 1/2$(Cys)_2$ (10), 8.41; Val (5), 4.87; Ile (2), 1.83; Leu (1), 1.02; Tyr (2), 2.01; Phe (3), 3.10; Lys (23), 23.32; Trp (4), 2.88; Arg (7), 6.90; Pro (6), 5.98; ES-MS (*m/z*): $[M + H]^+$ calcd, 13 240.2; found, 13 240.9.

for references see p 62

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5.3 Examples of Protein Synthesis on Solid Support

P. E. Dawson and J. L. Offer

Solid-phase synthesis has transformed the synthesis of biopolymers and more recently, the field of organic chemistry as a whole. Attachment to a swollen resin support solubilizes protected peptide intermediates, allowing the use of high concentrations of reagent and their straightforward handling and washing.[1] Stepwise solid-phase peptide synthesis (SPPS) has given rapid synthetic access to large quantities of moderate sized peptides (20–50 amino acids). However, the synthesis of the polypeptide chains that compose proteins has proven extremely challenging using the stepwise SPPS procedure. Consequently, larger polypeptides (50–200 amino acids) have been synthesized by coupling fully[2] or minimally[3] side-chain protected smaller peptides (20–50 amino acids) in solution (fragment condensation) or fully unprotected peptides in solution (chemical ligation).[4]

It has long been recognized that fragment condensation on a solid phase would be highly desirable.[1,5,6] However, the solid-phase fragment coupling approach has been limited by the notoriously poor solubility properties of fully protected peptides that make them difficult to purify and characterize.

The problem of peptide solubility has been addressed by two conceptually opposite approaches, backbone protection and chemical ligation. Protection of the polypeptide backbone has been demonstrated as an effective approach for improving the solubility of fully protected polypeptides in both organic and mixed aqueous solvents. Consequently, these protected peptides can be purified by HPLC with acetonitrile/aqueous-based solvents and easily coupled at high peptide concentration in solvents such as dichloromethane that minimize epimerization during segment activation and coupling. The application of this method is described in Section 5.3.1. In contrast, the chemical ligation approach takes advantage of fully unprotected peptides that are easily purified and characterized and are highly soluble in denaturing aqueous buffers such as 6 M guanidine hydrochloride (Gn-Cl). Peptide coupling takes place using highly chemoselective reactions that form a native amide bond at the site of ligation. Two approaches for carrying out native chemical ligation on a solid support are described in Section 5.3.2.

5.3.1 Solid-Phase Segment Condensation – The Application of Backbone Protection to Solid-Phase Protein Synthesis

Poor peptide solubility has been identified as a major obstacle to efficient segment assembly of peptides on a solid phase particularly for fully protected strategies.[1,5,6] Backbone protection directly addresses this problem and is the basis for a general route to the synthesis of small proteins.[7,8] However, the inclusion of backbone protection in a synthesis is not yet automated or routine and requires careful planning.

The peptide bond has a central role in determining peptide conformation and secondary structure. A tertiary amide bond, present for example as either proline or sarcosine, increases peptide solubility in organic and aqueous based solvents by preventing the formation of β-sheet packing and conferring an unstructured fully solvated state.[9,10] This effect has been exploited to prepare fully protected peptide segments of typically 15–25 amino acid residues that are extraordinarily soluble, up to $500\,mg \cdot mL^{-1}$ in DMF or even dichloromethane.[7,8] The optimum spacing of the tertiary amide for solubility has been identified as every five or six amino acid residues for both fully protected peptides in organic solvents and deprotected peptides in aqueous solution.[10,11] Improved peptide solubility al-

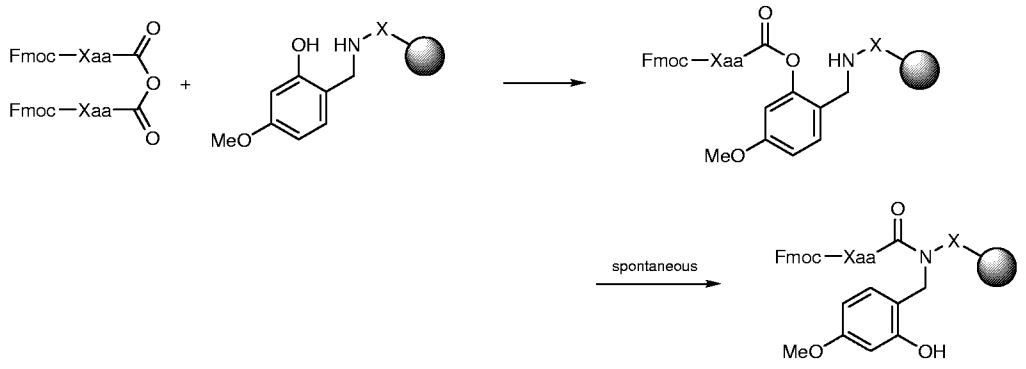
for references see p 80

lowed the routine purification of fully protected peptide fragments by preparative HPLC for the first time.[8]

The insolubility of peptides is not limited to the solution phase and can occur during peptide synthesis on the solid phase where it manifests itself as the difficult sequence problem. On-resin peptide aggregation and poor peptide segment solubility are due to the formation of β-sheet-like structures. Difficult sequences contain predominantly hydrophobic residues that consistently show incomplete acylation and a characteristic shrinkage of the peptide-resin during synthesis.[1] Reversible backbone protection was introduced into Fmoc/tBu SPPS after the study of synthesis failure during the assembly of difficult sequences[12] and is now routinely used to circumvent synthesis failure in SPPS.

The dramatic improvement in peptide solubility accompanying amide bond substitution was first investigated by Weygand et al. using dimethoxybenzyl-substituted (Dmb) peptides prepared by solution methods.[13,14] Narita et al. systematically incorporated reversible Dmb substitution into several sequences at regularly spaced intervals for maximum solubility.[10] However, N-substitution of amino acids other than glycine disfavored subsequent acylation. To overcome nonquantitative N-acylation of the substituted amino acid *N*-(2-hydroxy-4-methoxybenzyl)-substituted (Hmb) amino acids were introduced and investigated on the solid phase. The presence of an ortho-hydroxy site for acylation, more sterically accessible than the secondary amine, facilitated quantitative acylation by a highly activated amino acid, except Hmb-Gly where routine couplings were adequate.[15] Acylation initially formed an acyl phenyl ester, suitably positioned for subsequent intramolecular O-to-N acyl migration (Scheme 1). This migration was anticipated to be favored as it occurred through a six-membered intermediate; however, steric crowding with the more sterically demanding amino acids slowed it considerably. The acyl rearrangement from O-to-N in the 2-hydroxybenzylamine system had previously been studied in detail by Kemp who had described this anomalously large steric effect.[16] In the case of coupling β-branched amino acids, Val, Ile, and Thr, as their symmetrical anhydrides, onto Hmb-substituted amino acids, with the exception of Hmb-Gly, the coupling time required was prohibitively long for routine synthesis (greater than 24 hours with repeat couplings).[15] Coupling of the Hmb amino acids themselves to the growing peptide chain is straightforward when introduced as their *N,O*-bis-Fmoc-*N*-(2-hydroxy-4-methoxybenzyl)amino acid pentafluorophenyl esters. The protection of the hydroxy group of Hmb with Fmoc is necessary to prevent formation of the intramolecular phenyl ester between the activated carboxy group and the hydroxy group during amino acid activation and coupling.[17] The introduction of 2-hydroxy-4-methoxybenzaldehyde by reductive amination to the end of the growing peptide chain is a potentially useful preparative alternative.[18]

Scheme 1 Acylation of a 2-Hydroxy-4-methoxybenzyl-Substituted Peptide[15]



X = side-chain protected peptide

O-Acylated Hmb is stable to acidolysis by TFA giving a useful additional feature to Hmb-containing peptides. Side-chain protection and the N^{α}-protection, if it is first replaced with Boc before Hmb O-acetylation, can be removed and backbone protection retained. After deacetylation of Hmb with aqueous hydrazine the peptide can be further purified before removal of the Hmb groups, useful when the product is poorly soluble. This feature is also useful for the purification of large polypeptides as the presence of backbone protection on the side-chain deprotected peptide prevents the formation of relatively stable folded structures that can complicate HPLC purification.[11]

Segment Condensation

The concept of segment condensation on a solid phase is accepted to have many advantages over stepwise SPPS whilst conserving many of its desirable features.[1,5,6,19] The use of rigorously purified peptide segments allows synthetic access to larger peptide targets, >60 amino acids, unavailable from stepwise SPPS. Deletion sequences from segment condensation differ by the length of a whole segment and, therefore, are simple to purify by preparative HPLC, in contrast to single amino acid deletions in SPPS which present a more challenging purification task and introduce unacceptable microheterogeneity to the product. Segment condensation reduces the number of chemical cycles the growing chain is exposed to and therefore decreases side-product formation. Lastly, it facilitates the rapid synthesis of analogues through changes in a single segment.

Although impressive targets have been synthesized by solid-phase segment condensation[19] (SPSC) it has not, until recently, been accepted as a general method. There have been two main interrelated obstacles to its implementation. Firstly, fully protected peptides are unpredictably and often poorly soluble and consequently they are difficult to handle, purify, and they give poor coupling kinetics. The second problem is that the stereogenic integrity of the C-terminal amino acid is no longer protected during activation and coupling by urethane protection as in the stepwise case. The epimerization problem is related to the solubility problem as the use of nonpolar solvents can suppress epimerization but poor solubility prevents their use,[8] in addition, poor solubility exacerbates the first-order process of epimerization by prolonging acylation time. The use of Hmb to increase solubility, therefore, addresses both of these problems. The Hmb-substituted peptide segment can be dissolved in dichloromethane at high concentration giving rapid coupling with low epimerization. The Hmb-backbone protection makes the peptide segments so soluble that often segment solubility ceases to dictate the volume of solvent used during a segment coupling. The resins

for references see p 80

used so far have had a loading of around 50 μmol·g^{-1}, but higher resin loading should be tolerated allowing higher concentrations of coupling peptide segment.

Planning a Synthesis

In the segment assembly method the target sequence is subdivided into smaller segments. The first consideration in choosing the segments is the stereogenic integrity of the C-terminal amino acid, classically limited to glycine, which does not possess a chiral center, and proline. However, this constraint cannot always be accommodated within the synthetic scheme. The improved solubility of Hmb-containing peptides confers solubility in nonpolar solvents such as dichloromethane. Activation and coupling in dichloromethane proceeds with minimal epimerization and gives greater flexibility to the choice of C-terminal amino acid and greater freedom in choosing segment coupling sites.[8]

The following guidelines were used during the planning of routine stepwise syntheses and apply equally to the synthesis of fully protected peptides. The application of these rules has so far been unproblematic. Previously, many protein syntheses required several attempts at synthesis with frequent redesigning and tedious resynthesis and purification of segments.[6] When these rules were adhered to no redesigning has been required. Indeed the requirements for spacing of backbone protection have proved to be more relaxed than these rules indicate:

1. During Fmoc-based syntheses the aspartyl side chain can undergo intramolecular cyclization during repetitive base treatment, especially Asp-Asn, Asp-Gly, Asp-Gln, and Asp-Ser. Backbone protection prevents aspartimide formation and is introduced wherever prudent.[20]

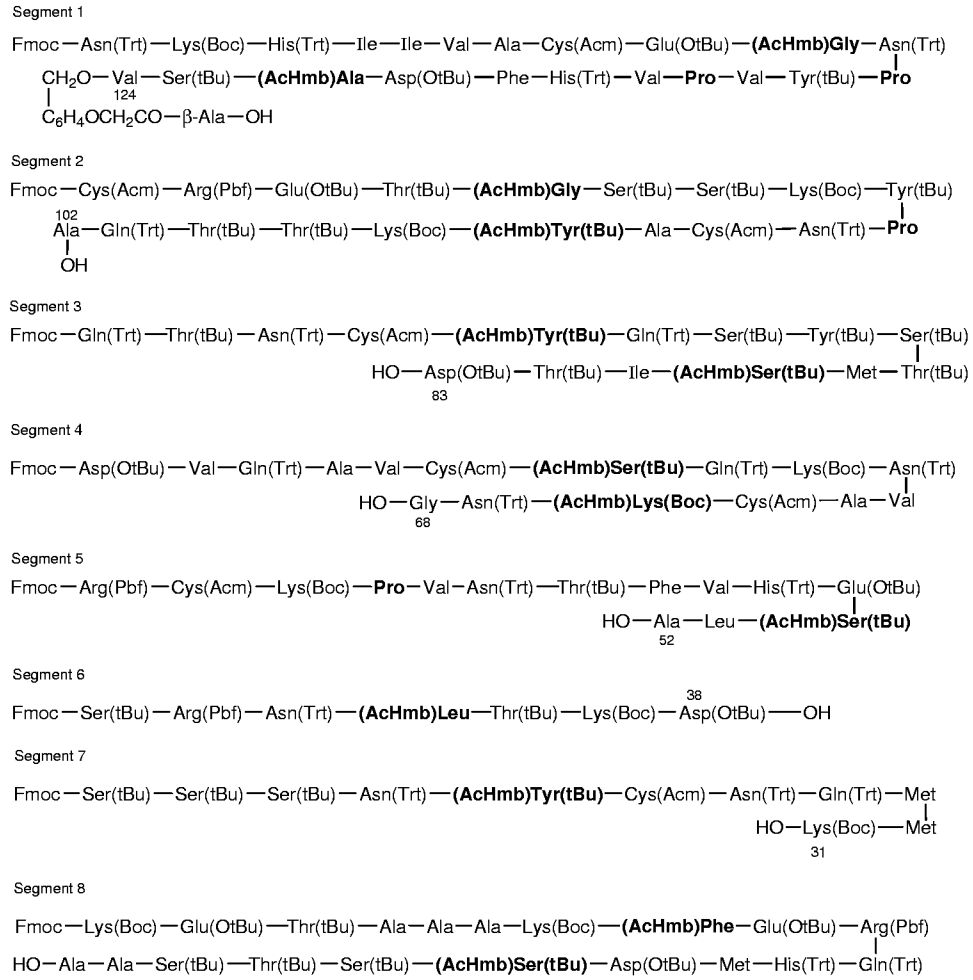
2. During a difficult synthesis, the onset of collapse of acylation and deprotection efficiency usually occurs between 8–12 residues into the synthesis.[1] Proline is the most effective residue for preventing synthesis failure; however, arginine(Pmc), histidine(Trt) and to a lesser extent glutamine(Trt) and asparagine have been shown to prevent a synthesis from failing.[21,22] As a general rule, Hmb substitution is included at positions 6, 12, 18, and 24 in the absence of the aforementioned residues. When the chemistry is not compatible with these positions the substitution should be within the first six residues and then at intervals of every sixth residue or less from the substituted position. The positioning of these substitutions relaxes with increasing chain length. The synthesis will rarely collapse after 25 residues.

3. If the terminal Hmb-substituted amino acid is glycine then the subsequent residue, even if β-branched, may be coupled through its pentafluorophenyl ester or similar.[15] For any other Hmb amino acid the subsequent residue must be coupled as its symmetric anhydride in dichloromethane. The only limitation to the inclusion of Hmb into a sequence is that β-branched amino acids will not quantitatively N-acylate a terminal Hmb residue, except glycine, on a practical timescale.

4. Hmb cannot be removed near residues that are easily protonated, such as the N-terminal amide bond, the N-terminus, and to a much lesser extent close to histidine residues.[15] As chain length increases Hmb cleavage from amide bonds around basic regions becomes easier.

An example of the practical application of these rules is shown in Scheme 2, where all residues with tertiary amide bonds are highlighted.

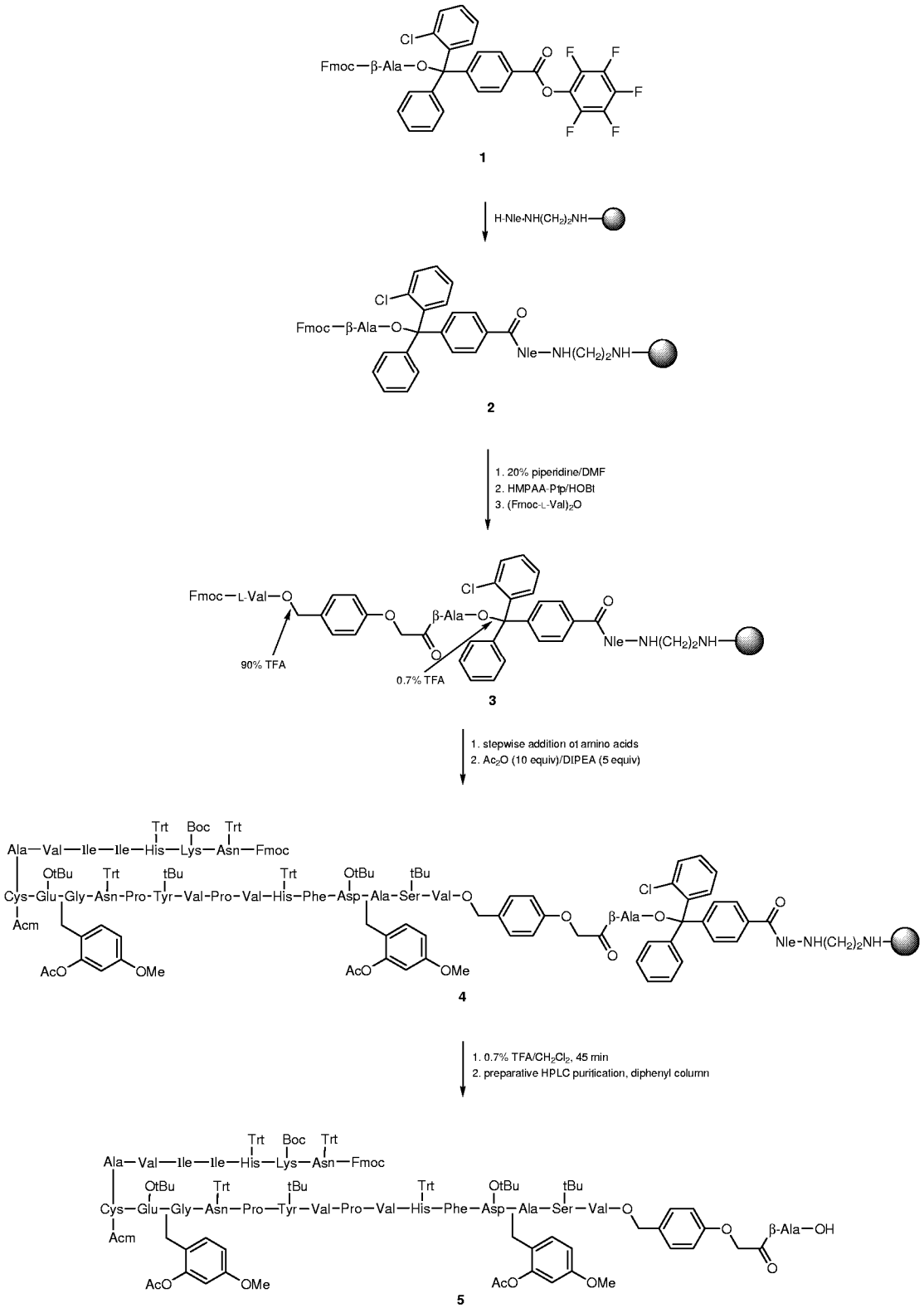
Scheme 2 Subdivision of Ribonuclease A into Smaller Segments



A requirement of a segment approach is that the fully protected peptide must be cleanly removed from the solid support while retaining α-amino and side-chain protecting groups. Several linkers are available for the synthesis of fully protected segments, the most attractive being the 2-chlorotrityl linker, developed by Barlos et al.[23] The segment can be cleaved while fully protected under weakly acidic conditions (0.7% TFA in CH_2Cl_2). Although small amounts of the trityl side-chain protection of histidine are observed to be lost during cleavage and purification, these losses are very minor. The use of 2-chlorotrityl has a further advantage since segments with C-terminal proline and glycine can be synthesized without loss of material because of piperazine-2,5-dione (DKP) formation. The synthesis of the C-terminal peptide, segment 1, used a double linker strategy, allowing purification and re-attachment of the first segment whilst preserving the chiral integrity of C-terminal valine (Scheme 3).[5,6]

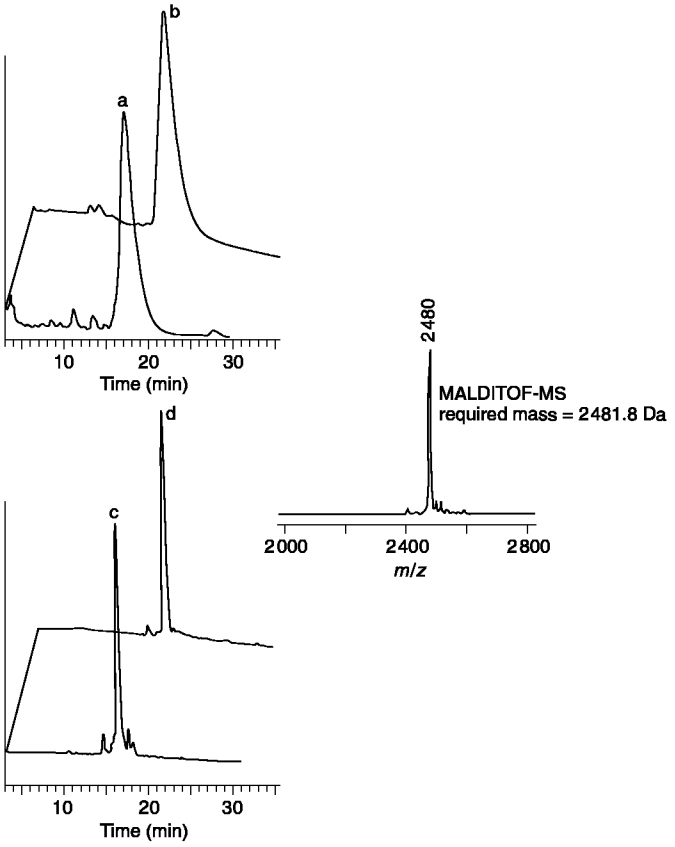
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Scheme 3 Assembly of C-Terminal Segment 1 of Ribonuclease A[8]



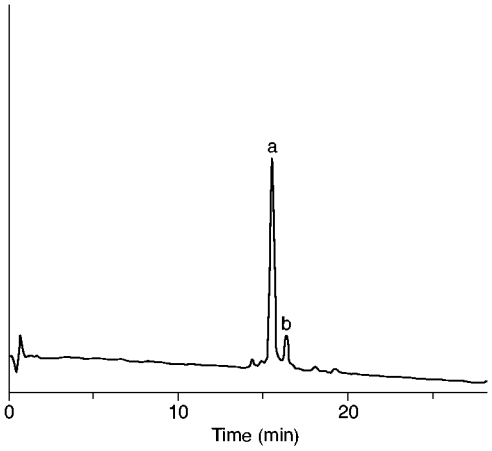
One of the most important consequences of the application of backbone protection to solid-phase segment condensation has been the ability to purify fully protected segments using standard chromatographic purification procedures. As demonstrated in the example, peptide segments are soluble in a range of organic and aqueous organic mixtures (MeCN/0.1% aq TFA). Segments were routinely purified by multiple injections on a Vydac diphenyl 22 × 250 mm column. Before purification analytical HPLC (diphenyl) was carried out on the crude protected segments (Figure 1). A sample of each segment was globally deprotected to give the base peptide which was analyzed by analytical HPLC (C8) and MS. The same exercise was repeated after purification of the peptide. Segment 1 contains 23 amino acids with side-chain protection, a handle, and N^{α}-protection. Preparative HPLC purification of the fully protected segment as is shown in Figure 2 can improve the purity of the base peptide. The limit of protected peptide size has not yet been systematically investigated.

Figure 1 HPLC Profiles of Crude (a) and Purified (b) Fully Protected Segment 1. Column: Vydac Diphenyl (250 × 4.6 mm); Eluant 80–90% Solution B in Solution A; Linear Gradient for 25 min. HPLC Profiles of *H*-Asn[103]-Val[124]-*OH* Cys(Acm)[110] (c) and (d) Obtained from Deprotection of Crude (a) and Purified (b) Fully Protected Segment 1, Respectively. Column: Vydac C_8 (250 × 4.6 mm); Eluant: 20–40% Solution B in Solution A Linear Gradient 25 min (Solution A: 0.1% aq TFA and Solution B: 90% MeCN/10% Solution A); Detection: 215 nm



for references see p 80

Figure 2 HPLC Profile of Crude Base Peptide (a) H-Cys(Acm)^{84}-Val^{124} Cys(Acm)^{95,110}-OH from the Addition of Segments 1 and 2. Peak (b) was Identified as the Base Peptide + Hmb. Column: Vydac C_8 (250 × 4.6 mm); Eluant: 15–40% Solution B in Solution A; Linear Gradient for 25 min, Solvents as Figure 1; Detection: 215 nm



After completion of the synthesis of the segments on the 2-chlorotrityl resin they are O-acylated to prevent irreversible modification of the phenolic group during segment activation.[7] The growing peptide-resin as well as the coupling segment should be fully solvated throughout the synthesis.[23] However, unlike the incoming segment, the Hmb groups do not have to be acylated as the risk of irreversible modification seems to be limited to the coupling segment. A major advantage of the use of backbone-protected peptides is that only a single coupling with two equivalents of coupling segment is required, which is a considerable improvement over comparable methods.[5] Coupling procedures were limited to the use of BOP/HOBt/DIPEA in DMF when the chiral integrity of the C-terminal amino acid was not a consideration and DIC/HOBt in dichloromethane when it was an issue. The new generation of coupling agents available such as HATU suggest that similar efficient couplings can be anticipated with even less coupling segment.

Another important consideration was the choice of resin. The use of Hmb was developed using resins composed of polydimethylacrylamide polymerized within the pores of a solid, macroporous support, either Pepsyn or polyhipe, both are exemplified below. The use of Hmb involves the switching of solvents from DMF to dichloromethane; the resins mentioned have excellent swelling properties in both solvents. However, some commercial polystyrene supports have given poor results because of the need to switch solvents, which can cause problems due to resin shrinkage.[1]

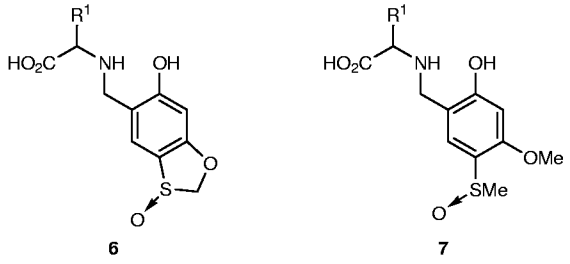
Limitations

A major drawback of this approach is that the fully protected peptides cannot be directly characterized. The segments have to be deprotected before they can be analyzed by mass spectroscopy. The free phenol group of Hmb must be acetylated after completion of the peptide assembly as it can be irreversibly modified during activation and coupling.[7] However, acetylation causes small losses of the segment from the resin as the presence of acetic acid gradually cleaves the segment from the hyper acid labile resin. Moreover, incomplete acetylation gives rise to additional peaks, complicating purification and reducing the yield of purified material. In common with all protected segment approaches, the removal of side-chain protection requires exposure, typically two hours, to acidic conditions (95% TFA) that degrades some proteins and generates deprotection artifacts.[6] This is probably the single

major disadvantage of this approach in comparison to ligation as deprotection artifacts are a major source of impurities.

The major shortcoming of using Hmb-substituted amino acids is that they cannot easily be included in a sequence where the subsequent coupling amino is β-branched. In addition automation is problematic due to the change from standard coupling conditions to the use of a symmetric anhydride in dichloromethane for N-acylation of the terminal Hmb residue. In an attempt to address both of these problems by improving the kinetics of the O-to-N acyl transfer the *N*-(6-hydroxy-3-oxido-1,3-benzoxathiol-5-yl)methyl backbone protection **6** was developed (Scheme 4).[17] An electron-withdrawing sulfoxide group was positioned para to the free phenolic group and acylation of this phenolic group under standard coupling conditions produced an activated ester. Acid lability of this backbone protection was reintroduced using the quantitative sulfoxide-to-sulfide safety catch of proven compatibility with SPPS. Automated acylation of the N-terminal safety-catch backbone-protected residue of the peptide-resin was demonstrated to be quantitative under standard conditions using a simple variant *N*-(2-hydroxy-4-methoxy-6-methylsulfinylbenzyl) backbone protection **7** (Scheme 4).[24] Coupling of β-branched amino acids is now potentially possible. However, the application of safety-catch backbone protection to the synthesis of fully protected peptide fragments has yet to be investigated and the existence of sulfoxide diastereomers could potentially complicate segment purification.

Scheme 4 Safety-Catch Backbone Protection Groups[17,24]



R^1 = amino acid residue

Solid-phase segment condensation of two segments of RNase is illustrated in the experimental section. RNase A has frequently been targeted by chemists as a model small protein for synthesis.[25] It has attracted interest because of its size, 124 amino acids, ease of folding, and assaying. Although it is abundant from natural sources, it is a notoriously challenging synthetic target, containing difficult sequences, eight cysteine residues, and aspartimide prone positions and, in the case of its glycosylated variant RNase B, a single glycosylation site.

Pentafluorophenyl 4-[(2-Chlorophenyl)(Fmoc-β-alanyloxy)phenylmethyl]benzoate (1):[7]
2-Chloro-4′-carboxytriphenylmethanol pentafluorophenyl ester (345 mg, 0.68 mmol) was dissolved in dry CH_2Cl_2 (10 mL), AcBr (2 mL) was added and the mixture left at rt for 2 h, concentrated, and redissolved in CH_2Cl_2 (5 mL). Fmoc-β-Ala (212 mg, 0.68 mmol) dissolved in CH_2Cl_2 (5 mL) and DIPEA (264 mg, 2.0 mmol) was stirred for 5 min and then added to the crude bromide. The mixture was left standing for 2 h at rt, diluted with CH_2Cl_2 (35 mL), washed with sat. aq $NaHCO_3$, dried ($MgSO_4$), and the solvent removed under reduced pressure to give a white gum; yield: 580 mg. This material was used crude for direct coupling to the resin.

Preparation of Fmoc-β-Ala-Trt(2-Cl)-CO-Nle-HN$(CH_2)_2$NH-SU 500 Resin (2):[7]
To Nle-HN$(CH_2)_2$NH-SU 500 resin (1 g) was added the ester **1** (560 mg, 0.68 mmol) and HOBt (103 mg, 0.67 mmol) in DMF (4 mL). The mixture was left until the resin gave a negative ninhydrin response (12 h). The derivatized resin **2** was thoroughly washed and acetylated with Ac_2O (204 mg) and DIPEA (103 mg) premixed in DMF (4 mL) to cap any underivatized trityl-resin.

for references see p 80

Preparation of Fmoc-Val-$OCH_2C_6H_4OCH_2CO$-β-Ala-Trt(2-Cl)-CO-Nle-$NH(CH_2)_2NH$-SU 500 Resin (3):[7]

The Fmoc group was removed from peptide-resin **2** and coupled with pentafluorophenyl 4-hydroxymethylphenoxyacetate (313 mg, 0.9 mmol) and HOBt (138 mg, 0.9 mmol) in DMF until ninhydrin negative (2 h). Fmoc-L-Val (1018 mg, 3 mmol) was dissolved in CH_2Cl_2 (15 mL) at 0 °C. DIC (187 mg, 1.48 mmol) was added and stirred at 0 °C for 1 h and concentrated. The resulting symmetric anhydride was dissolved in DMF (6 mL) and added to the peptide-resin. After 2 min, DMAP (37 mg, 0.3 mmol) in DMF (1 mL) was added and the mixture left at rt for 2 h. The peptide-resin was thoroughly washed with DMF, Et_2O and dried; yield: 1.35 g.

Preparation of Fully Protected Segment 1 (4):

Peptide-resin **3** was coupled with the appropriate Fmoc amino acid pentafluorophenyl ester (1.25 mmol) on an automated synthesizer programmed to carry out acylation reactions in DMF for 45 min and Fmoc deprotection reactions in 20% piperidine/DMF for 10 min. For the incorporation of Hmb-Ala, the peptide-resin was removed from the automated synthesizer and coupled with *N,O*-bis-Fmoc-*N*-(2-hydroxy-4-methoxybenzyl)alanine pentafluorophenyl ester (3 equiv)/HOBt (3 equiv) in DMF, until ninhydrin negative. The peptide-resin was washed and returned to the automated synthesizer for Fmoc deprotection, removed, washed, and dried. The symmetric anhydride of Fmoc-Asp(OtBu)-OH was prepared using the amino acid (2.5 mmol) and DIC (1.25 mmol) in CH_2Cl_2 at 0 °C. The peptide-resin was added to the mixture and left overnight, thoroughly washed with MeOH, CH_2Cl_2, DMF, and Et_2O, dried, and returned to the synthesizer. An indication of the extent of coupling was obtained by analyzing a small peptide-resin sample (2.5 mg) for quantitative UV absorbance at 290 nm and comparing with standards. For the incorporation of Hmb-Gly the peptide-resin was removed from the synthesizer and coupled with *N,O*-bis-Fmoc-*N*-(2-hydroxy-4-methoxybenzyl) glycine pentafluorophenyl ester (3 equiv). Hmb-Gly does not require strongly activated amino acid esters to acylate it[15] so the peptide-resin was returned to the automated synthesizer and the synthesis continued under routine coupling conditions. After the final amino acid was added the peptide-resin was acetylated with Ac_2O (255 mg)/DIPEA (160 mg) for 1 h. The acetylation was carried out on the synthesizer to enable any cleavage of peptide from the resin by AcOH to be monitored by UV and minimized. The fully assembled peptide-resin **4** was washed and dried.

Cleavage and Purification of Fully Protected Segment 1:

Fully protected peptide-resin **4** was treated with 0.7% TFA/CH_2Cl_2 (50 mL) for 45 min. The mixture was filtered, the resin washed with CH_2Cl_2 and the filtrate treated with *N*-methylmorpholine (350 µL), washed with sat. NaCl (2 × 500 mL), dried ($MgSO_4$), and the solvent removed under reduced pressure. The crude segment (535 mg) was dissolved in TFE (15 mL)/H_2O(5 mL) and purified by injection of aliquots onto a Vydac diphenyl 22 × 250 mm preparative HPLC column using a high % MeCN/0.1% aq TFA gradient. Fractions were pooled, diluted with H_2O, neutralized with Na_2CO_3, MeCN was removed using a rotary evaporator, and the aqueous residue lyophilized. The product was redissolved in CH_2Cl_2 (100 mL.) washed with sat. NaCl (3 × 500 mL), dried ($MgSO_4$), and the solvent removed under reduced pressure. The residue was dissolved in TFE (10 mL) transferred to a falcon tube, concentrated to ca. 2 mL, diluted with H_2O (15 mL), and lyophilized to give the purified segment 1 (**5**; 270 mg) which was stored at rt under vacuum. To deprotect either crude or purified protected peptides for analytical HPLC and MS analysis the same procedure was carried out. Protected peptide (2 mg) was treated with 20% piperidine/DMF (approximately 50 µL) for 30 min in a 15-mL falcon tube, precipitated with ice-cold Et_2O (10 mL), and left in a CO_2/acetone bath for 5 min then centrifuged for 7 min. The Et_2O was decanted off and the wash repeated. The resultant pellet was dried in vacuo then treated with TFA/EDT/TES/H_2O (85:10:3:2, 1 mL) for 2 h. The soln was sparged with N_2 and washed with cold Et_2O as before. Base peptide MALDI-TOF-MS at 2480.0 theoretical mass = 2481.8 Da.

Preparation and Purification of Fully Protected Segment 2:

The ester **1** was prepared and loaded onto Nle-$HN(CH_2)_2NH$-SU 500 resin as described above. The peptide was then elaborated as described for segment 1, adding Hmb-Tyr (3 equiv) manually to economize on Hmb amino acid. Fmoc-Ala was added as the symmetric anhydride in CH_2Cl_2. Gly^{88} was added as an Hmb amino acid and the following residues were all added routinely on the automated synthesizer. The protected peptide was cleaved as described for segment 1 to yield crude segment 2 (430 mg, 102 µmol). The product was dissolved in TFE (15 mL)/H_2O(5 mL) and purified by 20 × 1 mL injections onto a Vydac diphenyl 22 × 250 mm preparative HPLC column using 85–95% soln B in soln A; eluting for 25 min where soln A was 0.1% aq TFA and soln B was 90% MeCN/10% soln A. Workup as for segment 1 gave pure segment 2; yield: 182 mg. MALDI-TOF-MS at 2253.0 theoretical mass = 2251.5 Da.

Loading of Segment 1 onto Resin:
To purified segment 1 (189 mg, 41.2 μmol) in a 15-mL falcon tube was added BOP (18.2 mg), HOBt (6.3 mg, 41 μmol), and DIPEA (7.2 μL, 81 μmol) in dry DMF (1 mL). The mixture was stirred for 5 min for activation. H-Nle-NovaSyn-KD-resin (340 mg, approximate loading 0.1 mmol · g^{-1}) was added, followed by the minimum of DMF required to barely cover the peptide-resin, the mixture gently agitated, and left overnight. The peptide-resin was washed, acetylated, and dried, yielding 460 mg of peptide-resin; loading approximately 47.4 μmol · g^{-1} assessed by quantitative UV analysis. The peptide-resin was treated with 20% piperidine-DMF for 3 and 7 min, thoroughly washed, and dried.

Addition of Segment 2:
Purified segment 2 (182 mg, 43 μmol) and HOBt (6.6 mg, 43 μmol) were placed in dry CH_2Cl_2 (1.5 mL) and stirred at 0 °C in a 15-mL falcon tube. DIC (5.44 mg, 43 μmol) was added and the soln stirred for 1 h. at 0 °C. HOBt, which is insoluble in CH_2Cl_2 forms the soluble peptide HOBt ester over 1 h. The segment 1 peptide-resin (420 mg) was added and the coupling reaction left overnight. Routine workup yielded peptide-resin (490 mg). An aliquot (2.0 mg) of peptide-resin (Fmoc-deprotected) was cleaved with a premixed soln of TFA/EDT/TES/H_2O (90:5:3:2, 500 μL) for 4 h to give deprotected peptide (Figure 2).

5.3.2 Solid-Phase Chemical Ligation

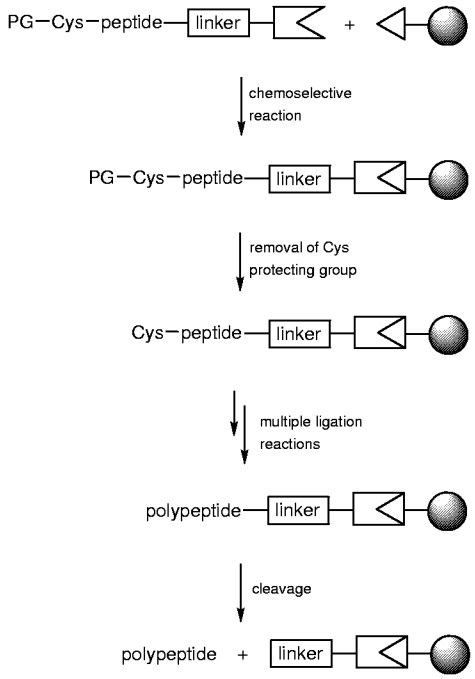
Native chemical ligation[26] has been used extensively for the synthesis of large polypeptides that correspond to folded proteins and domains.[4] Chemical ligation addresses the peptide solubility problem by utilizing fully unprotected peptides that have generally good solubility properties in denaturing aqueous solutions such as 6 M Gn-Cl. Additional advantages of using unprotected peptides are that the peptides can be characterized by electrospray and MALDI mass spectrometry during the purification process, the final product does not require an additional deprotection step to give the target polypeptide, and only a single equivalent of each peptide is required.

Typical ligation reactions proceed with high efficiency (>90%) and a single ligation step produces the desired polypeptide of 70–100 amino acids in high purity and reasonable yields (40–60% recovered). Unfortunately, carrying out multiple ligation reactions in solution requires repeated HPLC separations. These purification steps result in considerable handling losses when synthesizing polypeptides from several segments. As a result, recovered yields of <15% for a three segment ligation[27] and <10% for a four segment ligation[28] have been recently reported. Carrying out ligation reactions on a solid phase would eliminate the need for chromatographic separations after each ligation and consequently increase the yield and reduce the time required for the assembly of the target protein.[4]

Despite the advantages of the solid-phase chemical ligation method, the general applicability of the method has been restrained by the complexity of selecting a compatible set of ligation chemistries, resin linker, and protecting groups. The solid-phase ligation approach requires that a variety of specifications be met. The solid support must be chemically stable and have good swelling characteristics for all ligation, deprotection, and cleavage conditions; a chemoselective reaction to link the first peptide to the solid support; a cleavable linker to remove the polypeptide from the solid support after completion of the peptide assembly; and an N-terminal protecting group to prevent cyclization/polymerization reactions with the internal (N-terminal Cys and thioester-containing) segments during the assembly. These properties are summarized in Scheme 5.

for references see p 80

Scheme 5 Generalized Solid-Phase Ligation[32]



The primary limitation to this approach is that the cleavable linker and the N-terminal protecting group have to be stable to TFA and anhydrous HF during the Boc/Bzl SPPS assembly and cleavage and also to thiol nucleophiles under the native chemical ligation conditions (6 M Gn-Cl, 2% PhSH, 2% Bzl-SH, pH 7.0).[28,29] In addition, the linker must be stable to the removal of the N-terminal Cys protecting group. Finally, all reaction conditions must keep the unprotected peptide/solid-support network fully solvated.[30]

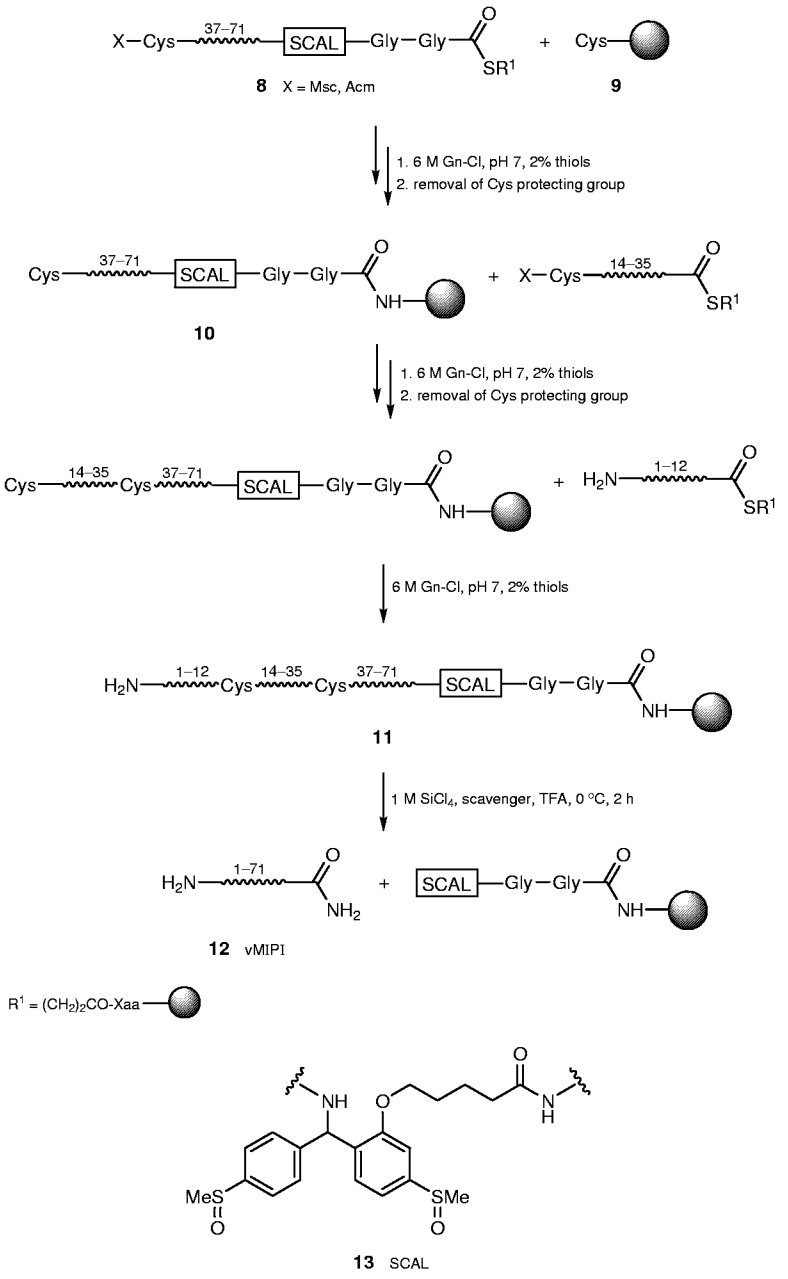
Two approaches for solid-phase chemical ligation have been described. Canne et al. have developed an elegant system that utilizes an oxime forming ligation to attach the first peptide to the resin, a selectively cleavable ester link to remove the peptide from the resin as a C-terminal carboxylic acid, and the Acm group to protect the N-terminal cysteine residue.[31] A complementary approach has been developed by Brik et al. that utilizes native chemical ligation to attach the first peptide to the solid support, a safety-catch acid labile linker to remove the final polypeptide from the support as a C-terminal amide and either Acm or Msc group for N-terminal cysteine protection.[32]

5.3.2.1 Using the Safety-Catch Acid Labile Linker

A robust and synthetically flexible strategy for chemical ligation on a solid support is illustrated in Scheme 6. This C-to-N assembly terminus approach began with a sepharose solid support bearing an N-terminal cysteine residue **9**. The initial peptide segment **8** contains a thioester C-terminus[28,33,34] to allow straightforward attachment to the solid support by native chemical ligation. In addition, a safety-catch acid labile (SCAL) linker **13** was incorporated into the C-terminal peptide. The SCAL linker is stable to both SPPS and ligation conditions, but was selectively cleaved by TFA following reduction of the SCAL linker sulfoxide groups.[35] Finally, The N-terminal cysteine residue is protected on either the side-chain thiol and/or the terminal amine using two different protecting groups; N^{α}-Msc[36] and S^{β}-Acm.[37] The utility of the methodology is demonstrated by the synthesis of vMIP I protein from three

peptide segments.[32] The crude polypeptide product was recovered quantitatively from the solid support and purified in 20% recovered yield.

Scheme 6 Strategy for Solid-Phase Synthesis of vMIP I Protein Using a Safety-Catch Linker[32]



for references see p 80

Solid Support

The support for solid-phase ligation must remain swollen in a variety of solvents, and be stable to a broad range of pH values and to both oxidizing and reducing conditions. Sepharose resin consisting of 4% cross-linked agarose functionalized with ca. 20 μmol primary amine · mL^{-1} of gel has proven to be an excellent solid support. Commercially available "Immobilized Diaminodipropylamine Gel" is functionalized with free amines at 16–20 μmol · mL^{-1}. This support can be acylated using standard solid-phase synthesis procedures.[38]

The sepharose support is compatible with basic conditions, pH 13, 20 min combined reaction times, and is stable for over two hours in TFA. The support also remains highly solvated throughout the synthesis, even when it is derivatized with long polypeptide chains. As observed during standard SPPS on polystyrene resins[30] the sepharose solid support swells significantly, greater than twofold in volume, as the polypeptide length increases. This mutual solvation of peptide and resin allows the attached peptide to be soluble at high (ca. 10 mM) concentrations and should allow for large, >200 amino acids, polypeptides to be assembled.

Attachment of the C-Terminal Peptide to the Solid Support

Chemical ligation uses side-chain unprotected peptides, consequently, a highly chemoselective reaction must be used to link the first peptide to the solid support. Many ligation chemistries have been developed that form thioester,[39] sulfide,[40] oxime/hydrazone,[41,42] thiazolidine,[43] and amide bonds[43] at the C-terminus. The amide bond forming "native chemical ligation"[26] was selected to allow other chemoselective reactions to be utilized later in the synthesis. Another advantage of using native ligation to attach the C-terminal peptide to the solid support is that the same chemical reaction is used for the subsequent assembly of the target polypeptide from its segments. Attachment of the first segment to the solid support proceeds rapidly in denaturing aqueous solution with only a small excess (1.1 equiv) of peptide. If desired, a limiting amount of peptide could be used, followed by capping the remaining active groups on the resin with small molecules bearing a thioester group.

Safety-Catch Amide Linker

The safety-catch amide linker (SCAL) **13** developed by Lebl et al.[35,44] possesses ideal properties for solid-phase chemical ligation (Scheme 6). In the sulfoxide form, SCAL is stable to a wide range of chemical conditions. It is stable to both TFA and HF used during Boc-chemistry SPPS, and is compatible with Fmoc and Alloc protection strategies. The linker is also compatible with aqueous PhSH, reducing agents such as TCEP and DTT, basic conditions (pH 13), and also with mercury(II) acetate. However, following reduction of the aryl sulfoxide to the corresponding aryl sulfide, the SCAL linker becomes labile to TFA. These properties of the SCAL linker make it compatible with the variety of subsequent chemical manipulations in the course of the polypeptide synthesis.

The SCAL linker has been used for the synthesis of many short peptides and peptide libraries, although several procedures for SCAL cleavage have been reported[35,44] the use of silicon tetrachloride/TFA gave the cleanest results with large peptides on the sepharose support. The cleavage reaction could be visually monitored by the blue color of the benzhydrylium ion that is slowly trapped by scavengers such as ethanedithiol, thioanisole, and *m*-cresol.[44] The blue color disappeared after one hour at which point the cleavage was found to be >90%.

N-Terminal Cysteine Protecting Groups

Multiple component syntheses using native chemical ligation require protection of the N-terminal cysteine residue for all middle segments in order to prevent polymerization or cyclization. Since native chemical ligation requires the unique arrangement of amine and thiol groups, protection of either functionality is sufficient to prevent reaction. The N-terminal cysteine may be protected at either the amine group using Msc or at the thiol side-chain using Acm. The Msc group has been used in the synthesis of several multiple segment proteins with great success despite the necessity for brief (5 min) treatments at pH 13.[28] The addition of the Acm group to the methodology was particularly advantageous for the synthesis of proteins with base-sensitive groups. The Acm group is removed by mercury(II) acetate at pH 4.

To demonstrate the applicability of the solid-phase ligation approach to protein synthesis, vMIP I , a 71 amino acid chemokine was synthesized.[35] This target was chosen since it contains all 20 natural amino acids and has four cysteine residues that form two disulfide bonds upon oxidative folding. As shown in Scheme 6, the sequence can be divided into three segments, 1–12, 13–35, and 36–71. The resulting Cys^{12}-Cys^{13} and Ala^{35}-Cys^{36} ligation sites are fully compatible with the native chemical ligation approach.

Synthesis of vMIP I Protein (12):[35]
Preparation of Polymer Support 9:
Commercially available Immobilized Diaminodipropylamine Gel (4% cross-linked agarose, 16–20 μmol available amino group · mL^{-1} of gel) was used as a polymer support. The resin was allowed to swell in DMF, followed by normal coupling of Boc-Cys(Trt)-OH (6 equiv) for 30 min. The modified resin was then treated with 1 M NaOH for 5 min in order to remove any esters of the hydroxy group of the agarose with cysteine. The Boc and the Trt groups were removed by treating the modified resin with a mixture of diisopropylsilane/H_2O/TFA (2.5:2.5:95) for 1 min, followed by DMF, CH_2Cl_2 flow washes. The resin **9** was allowed to swell in 6 M Gn-Cl pH 7, and was ready for the first ligation.

SCAL Linker:
The Fmoc protecting group of commercial Fmoc-SCAL linker (*N*-Fmoc-{5-[2-{amino[4-(methylsulfinyl)phenyl]methyl}-5-(methylsulfinyl)phenoxy]}pentanoic acid) was replaced by the Boc protecting group. Removal of the Fmoc was achieved using DBU (1.2 equiv) in $CHCl_3$. The free amine was protected as a Boc derivative using Boc_2O (2 equiv) and DIPEA (2 equiv) in $CHCl_3$ to give Boc-SCAL in >95% yield for the two steps.

Synthesis of the Peptide-Thioester 8 Incorporating the SCAL Linker:
The C-terminus of vMIP I (36–71) **8** was synthesized on a thioester resin, incorporating the SCAL linker and a Gly-Gly spacer between the SCAL linker and the thioester (Scheme 6). This peptide was synthesized on a TAMPAL thioester resin (0.4 mmol). The method for synthesis of the TAMPAL resin was fully described by Hackeng et al.[28] Following deprotection of the trityl group protecting the thiol of 2-(tritylsulfanyl)propanoylalanyl-MBHA-resin, Boc-Gly-OH was coupled using standard in situ neutralization protocols for Boc-SPPS. Boc-Gly-OH (2.2 mmol) was dissolved in 0.5 M HBTU in DMF (4 mL, 2.0 mmol) and DIPEA (1.0 mL) and allowed to react for 2 min. The activated amino acid was added to the resin and allowed to couple for 1 h to ensure complete formation of the thioester. The Boc group was deprotected with 100% TFA (2 × 1 min) and flow-washed with DMF for 40 s. The second Gly was coupled as described above except for only 15 min. Following removal of the Boc group and neutralization of the resin with 10% DIPEA in DMF, the SCAL linker was incorporated by dissolving Boc-SCAL (315 mg, 0.6 mmol, 1.5 equiv) in 0.5 M HBTU (1.2 mL, 0.6 mmol) and DIPEA (0.6 mmol), activating for 2 min, and adding the soln to the resin for 20 min (99.9% coupling).
The synthesis of the C-terminal peptide segment bearing the SCAL linker **8** (Scheme 6) was completed using standard in situ neutralization protocols for Boc SPPS and proved to be straightforward. HPLC purification gave a homogeneous product by HPLC and ES-MS (30% from resin loading). The resulting peptide was comparable in purity to a similar peptide (vMIP I 36–71) synthesized on traditional PAM resin. This synthesis demonstrates the compatibility of the SCAL linker with Boc chemistry and thioester linkages.

for references see p 80

Attachment of 8 to the Resin 9:
The modified resin **9** (0.4 μmol, 25 μl) was swollen in 6 M Gn-Cl pH 7. The C-terminal segment of vMIP I **8** [Msc-(36–71)SCAL-GG-SR[1]] (0.4 μmol, 2mg) was dissolved in ligation buffer (60 μl, 6 mM peptide in 6M Gn-Cl, pH 8.5 containing PhSH (2% v/v), and Bzl-SH (2% v/v), and added to the modified resin. The ligation was carried out at rt and mixed thoroughly periodically to equilibrate the thiol additives. The reaction was monitored using HPLC. A sample (1 μL) of ligation mixture was removed at 2.5 h intervals and diluted with 6 M Gn-Cl (9 μL) and injected into an HPLC apparatus to follow the consumption of the peptide. Following completion of the ligation (8–10 h) the Msc-protecting group was removed by exposing the ligation mixture to pH 13 for 5 min by adding 1 M NaOH, followed by a 6 M Gn-Cl flow-wash to yield **10**.

Assembly of the vMIP I Polypeptide 11:
The same steps were repeated for the other vMIP I segments using 1.5 equiv peptides relative to the Cys-modified gel. In the case where N-terminal Cys of vMIP I segments were protected with the Acm group, the resin was washed with $AcOH/H_2O/MeCN$ (10/45/45) followed by treatment for 30 min with a soln of $Hg(OAc)_2$ (15 mg·mL^{-1}) in 10% AcOH soln at pH 4. After thorough washing of the resin with the AcOH soln followed by 6 M Gn-Cl, 0.1 M Na_3PO_4, pH 7, the resin was treated with 20% β-ME in the same 0.1 M Na_3PO_4 pH 7.0 buffer for 30 min to yield **11**.

Cleavage of vMIP I Polypeptide 12 from the Resin:
The fully assembled polypeptide-resin **11** was washed with 6 M Gn-Cl, 100 mM Na_3PO_4, pH 7.0, DMF, and CH_2Cl_2. A cleavage cocktail of TFA (315 μL), thioanisole (50 μL), *m*-cresol (50 μL), EDT (25 μL), and $SiCl_4$ (60 μL) was added to the resin and left at 0 °C for 2 h. The TFA mixture was sparged and precipitated with cold Et_2O (5 mL), and the precipitated product isolated and dissolved in $MeCN/H_2O$ (1:1) containing tris(carboxyethyl)phosphine to reduce any oxidized product. The polypeptide **12** was purified using an analytical RP-HPLC column (C8).

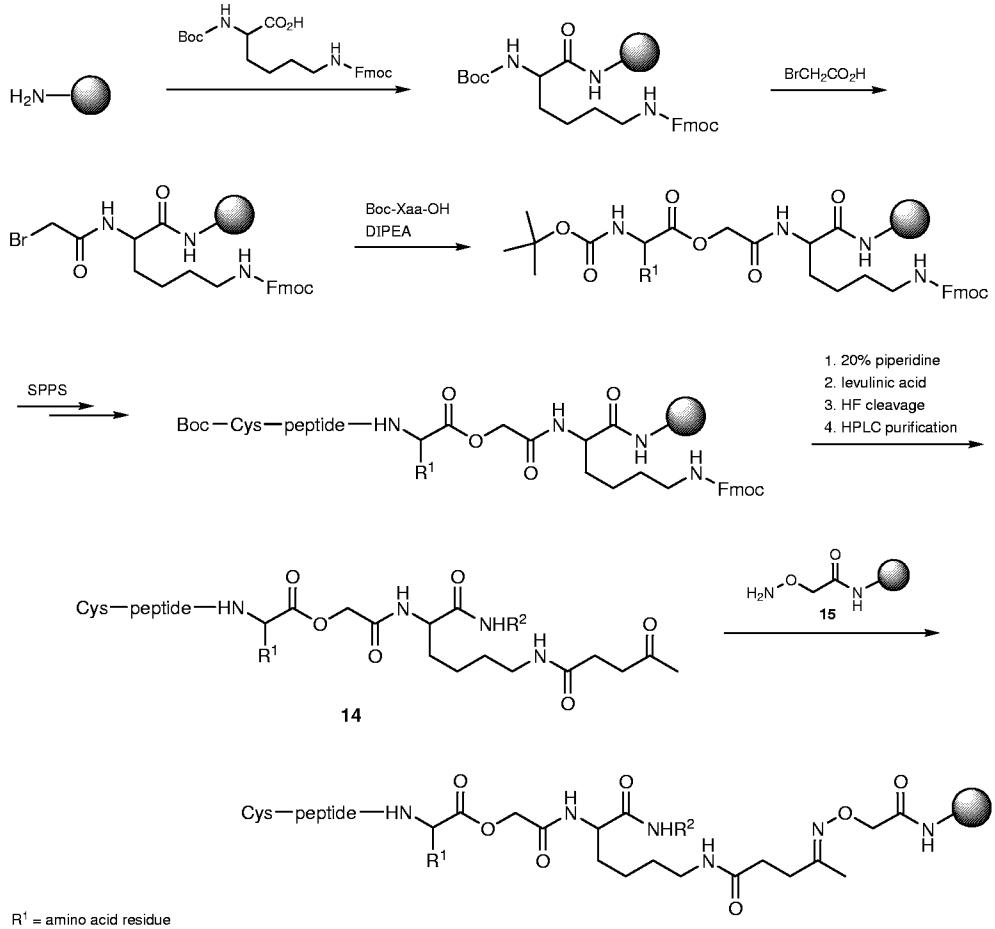
5.3.2.2 Using a Glycolate Ester Linker

A second approach for the assembly of unprotected peptides on a solid support was developed by Canne et al.[31] This approach differs from the previously described method in several aspects. (1) The C-terminal peptide segment was attached to the resin using an oxime-forming ligation reaction. (2) The C-terminal segment contained a base-labile glycolate (hydroxyacetate) ester group that was hydrolyzed after assembly of the polypeptide to generate a C-terminal carboxylate. (3) An amino Spherilose affinity resin was used as the polymer support.

The key element of the approach was the C-terminal peptide which was synthesized as described in Scheme 7.

Assembly of the protein consisted of attachment of C-terminal peptide to the solid support by an oxime-forming ligation between the ketone on the peptide and the aminooxy group on the support at pH 4.6. An Acm group was utilized to protect the N-terminal Cys residue and the polypeptide was assembled as previously described (Section 5.3.2.1). The required polypeptide was then cleaved at the glycolate residue by hydrolysis using a brief treatment with NaOH to yield the protein with a C-terminal carboxylate. Care was taken when removing the Acm group to avoid hydrolysis of the glycolate ester linker during the β-ME treatment.

Scheme 7 Application of the Glycolate Ester Linking Group[31]



Aminooxy Solid Support 15:[31]
Boc-aminooxyacetic succinimide ester in DMF (10 equiv, 0.5 M) was added to commercial amino Spherilose affinity resin (1 equiv) preswollen in DMF for 3 h. The resin was washed with DMF and the Boc group removed by treatment with neat TFA (2 min). The resin was washed again with DMF and neutralized with 10% DIPEA/DMF, and washed again with DMF. After thorough washing with CH_2Cl_2, the derivatized solid support **15** was dried.

Coupling of Modified C-Terminal Peptide 14 to Aminooxy-Resin:[31]
The ketone-containing peptide (ca. 5 mM) was dissolved in 6 M Gn-Cl, 0.1 M NaOAc, and 0.15 M Met, pH 4.6. This mixture was added to dry aminooxy-Spherilose and allowed to stand for 18 h.

Native Chemical Ligation:[31]
The peptide segment to be ligated to the polymer supported bound peptide was dissolved in 6 M Gn-Cl, 0.1 M Na_3PO_4 (pH 7.0), 0.1 M Met, and 1% PhSH (10–50 mM peptide), added to the N-terminal Cys-peptide-resin, and allowed to react at rt. Ligation reactions were complete in 2–24 h.

Cys(Acm) Deprotection:[31]
The N-terminal Cys(Acm)-peptide-resin was washed with 3% aq AcOH followed by treatment with $Hg(OAc)_2$ (15 mg·mL^{-1}) in 3% aq AcOH for 30 min. The resin was washed with 3% aq AcOH and 6 M Gn-Cl, and 0.1 M Na_3PO_4 (pH 7.0). The resin was left in 20% β-ME in pH 7.0 buffer for 30 min to remove the remaining mercury.

Cleavage of Glycolate Ester Linker:[31]
The peptide-resin was washed with 8 M urea and 0.1 M Na_3PO_4 (pH 7.0), and treated with 0.25 M NaOH in the same urea buffer (pH 12–14) for 2 min, washed with an equal volume of 0.25 M HCl in urea buffer

for references see p 80

(pH 1–2) and one volume of the pH 7 urea buffer. Eluants were combined and purified by semi-preparative HPLC.

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5.4 Semisynthesis

H. F. Gaertner and R. E. Offord

Semisynthesis can be defined as the use of fragments of proteins, or intact proteins themselves, as ready-made intermediates in the chemical synthesis of proteins. Many modifications to the native structure of a protein can be expected to have profound effects, and produce materials of the greatest academic or practical value, but only involve a change of one or a few atoms out of the thousands that often go to make up the complete molecule. If a semisynthetic method can be found to incorporate, at the correct site, a small synthetic fragment carrying the wanted change, there is no need to go to the trouble of constructing the rest of the molecule when it has already been preformed for us biosynthetically.

As originally conceived, the technique used natural proteins as the source of intermediates, but as recombinantly derived proteins became available these were immediately brought into use. Even totally synthetic proteins could be used in this way, in order to add to them structural features that had not occurred to those who carried out the synthesis, or that could best be put in place not during the total synthesis but afterwards, using a route that would traditionally have been called semisynthetic.

There is a tendency to reserve semisynthetic and totally synthetic methods for the introduction of bonds and residues that cannot be specified by the genetic code. The present chapter will concentrate on these aspects. However, semisynthesis can have a role to play even when building structures that are completely accessible to the genetic code. The first industrial challenge for the emerging technologies of total chemical synthesis, recombinant protein expression, and semisynthesis was the economic production of human insulin in pharmaceutically usable quantity and quality. The semisynthetic human insulin that was made from porcine insulin proved exceptionally convenient to produce, and was the first introduction to human insulin for very many patients.

Semisynthetic proteins can involve the combination of a synthetic with a biosynthetic portion by either noncovalent or covalent bonds. Among covalent bonds, there are available for use disulfide bonds, peptide bonds, and other, completely unnatural, bonds such as oximes and hydrazones.

A recent example of the noncovalent approach is that of Roy and Imperiali.[1] Earlier work on noncovalent and disulfide-bridge semisynthesis is cited in a review[2] and we limit ourselves in this chapter to the covalent preparation of semisynthetic proteins, using peptide bonds, oximes, hydrazones, and thiazolidines.

Originally, the natural fragments were linked by the same coupling reactions as used in chemical synthesis, i.e. carbodiimide reactions, azide coupling, etc. This made it necessary to distinguish chemically between, on the one hand, the single α-amino group in a given fragment and the large number of ε-amino groups potentially present, and, on the other hand, between the single α-carboxy groups and all the β- and γ-carboxy ones. If this was not done, the product of the coupling of the two fragments could be wildly heterogeneous, with the wanted component, having a contiguous backbone chain formed only through the α-functional groups, representing only a small minority in the product mixture. Although it was not at first sight easy to see how to protect side-chain groups on a native peptide or protein while leaving the α-groups free, methods were found (see, for example, ref[3]). It is rare for such methods to be needed today, since ways have been developed of coupling, using the α-groups, proteins, and peptides that are essentially unprotected, without any significant production of molecules linked through the side chains.

As a result, protein molecules can be constructed in quantities well beyond those accessible in cell-free recombinant expression systems, which incorporate residues or linkages that cannot be specified by the genetic code. Very large chimeric fusion proteins can be made, and since the reaction conditions are generally quite mild, this can be done under circumstances where the individual protein modules retain their native conformation at all

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times. In this way one avoids the problem occasionally encountered in recombinant fusion technology that the different modules, not being designed to fold up in the presence of each other, fail to do so, with very low expression yields in consequence. Finally, linear macromolecules other than proteins can be grafted as contiguous extensions of the polypeptide chain just as easily as proteins or peptides can.

We now describe and exemplify some of the principal methods of attaining these objectives.

5.4.1 Ligation of Unprotected Proteins or Their Fragments through Their Amino and Carboxy Termini Using Peptide Bonds

When synthesizing proteins it is natural to think first in terms of the creation of a backbone consisting only of peptide bonds. Two approaches are used in semisynthesis: native chemical ligation and reverse proteolysis.

Native Ligation

Native chemical ligation, which is described in Section 5.3, has proven to be a generally useful route to the total synthesis of proteins of up to medium size. A recent adaptation which utilizes a protein splicing element, an intein, to generate a reactive thioester at the C-terminus of a recombinant protein provides an interesting variant of this technique.[4,5]

Reverse Proteolysis

Proteolytic enzymes, particularly Ser proteases, have been used extensively in peptide synthesis because of their selectivity, their ability to function under mild reaction conditions, and the minimal requirement for protecting groups. Most recent work has been devoted to the development of engineered enzymes where the aminolysis of the acyl-enzyme intermediate is particularly strongly favored over hydrolysis. An illustration of this approach is the semisynthesis of human growth hormone analogues containing an N-terminal reporter group,[6] and an RNase A synthesis in good yield and high purity using an engineered subtilisin subtiligase, which was used to ligate six peptide fragments ranging in length from 12 to 30 residues.[7] It would be desirable if this interesting enzyme were to be more readily available.

Applications of reverse proteolysis in the modular assembly of new proteins are rather scarce. This is due to the fact that high-efficiency ligations of two large fragments call for some conformational assistance from prior noncovalent association between them, or some sort of conformational trapping of the product.

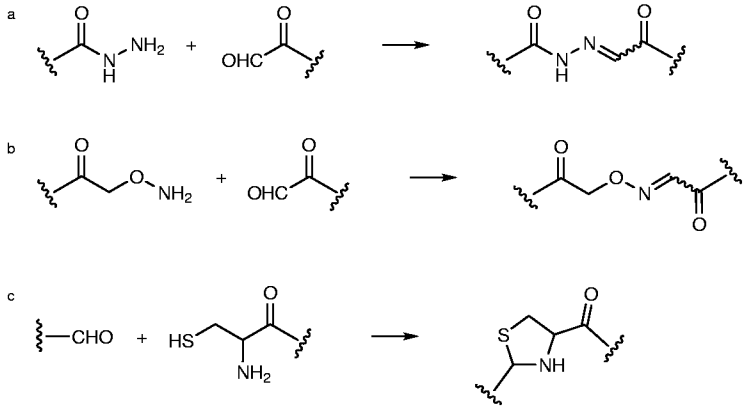
Reverse proteolysis is nevertheless an attractive approach as the first step in some of the ligation schemes developed in Section 5.4.2, in the sense that it is a powerful means to introduce the appropriate functionality at the C-terminus of a protein fragment. In this case, the reverse proteolysis only has to introduce a small molecule, which can usually be present in high concentrations, and so conformational assistance is not needed.

5.4.2 Ligation of Unprotected Proteins or Their Fragments through Their Amino and Carboxy Termini Using other than Peptide Bonds

We have seen that a key requirement in the ligation of unprotected protein fragments is chemoselectivity, so that only the intended sites react with each other in the presence of numerous side-chain functional groups. This requirement can be met by a general approach using carbonyl chemistry in which a weak base is ligated with an aldehyde under acidic conditions (Scheme 1). Under acidic conditions, the nucleophiles in basic side chains are

protonated and so are disfavored in the reaction. On the other hand, weak bases, including hydrazides, aminooxy groups, and thiols, are still good nucleophiles under these conditions, and are highly reactive toward aldehydes.

Scheme 1 Examples of Chemoselective Ligation Strategies Resulting in the Formation of Nonpeptidic Bonds: (a) Hydrazone, (b) Oxime, (c) Thiazolidine



As illustrated in Scheme 1, hydrazide and aminooxy groups readily form stable hydrazone and oxime bonds, respectively, with aldehydes at pH values between 3 and 5.5. The Cys 1-amino-2-thiol moiety forms a stable thiazolidine ring with the aldehyde between pH 4 and 5, whereas at these pH values an amino group, if it reacts at all, gives a readily reversible Schiff base.

The limiting step in the synthesis of a protein via chemical ligation is generally not the ligation step itself, which is relatively straightforward to perform, but rather the preparation of the peptide segments possessing the appropriate reactive end groups. The most convenient method of introducing an aldehydic or, rather, carbonyl group into a peptide or a protein is by periodate oxidation of a 1-amino-2-alcohol. This grouping is found in polypeptides, at the N-terminus, but only when Ser or Thr are in that position. This is admittedly a restriction, but a number of important natural sequences conform to this requirement, and others can readily be produced by site-directed mutation or total chemical synthesis to serve as intermediates in the production of semisynthetic structures which neither of those two methods could reach on their own. A Ser or a Thr residue can often be conveniently revealed by limited exopeptidase digestion. This has been done, for example, with the human cytokine GCSF where the N-terminal Met was easily removed with an aminopeptidase to reveal a Thr.[8] A prolyl-specific endopeptidase has been used to remove the N-terminal Met-Arg-Pro sequence of the interleukin-1 soluble receptor antagonist, also known as Antril, to reveal an N-terminal Ser without loss of biological activity.[21]

It is possible that N-terminal Cys would react toward periodate in the same way, but this seems not to have been tested in practice.

If ligation through a chosen side chain is desired, it can be introduced in the form of N^{ε}-seryl-Lys, particularly if the fragment in question is to be made by total chemical synthesis. The periodate oxidation of a Ser that has a free N-terminus is so simple and satisfactory that we regard such a residue simply as a protected aldehyde.

The sensitivity to periodate of Ser and Thr is thousands of times greater than that of the better known periodate oxidation of 1,2-*cis*-diols, for example in the sugars of glycoproteins.[9] This means that the concentration of reagent can be kept very low, in only small excess over the 1-amino-2-alcohol, and the reaction time kept short (3 to 10 min). It is, thus, usually possible to avoid oxidative attack of the side chains of Met, Trp, and His, but some risk of

for references see p 91

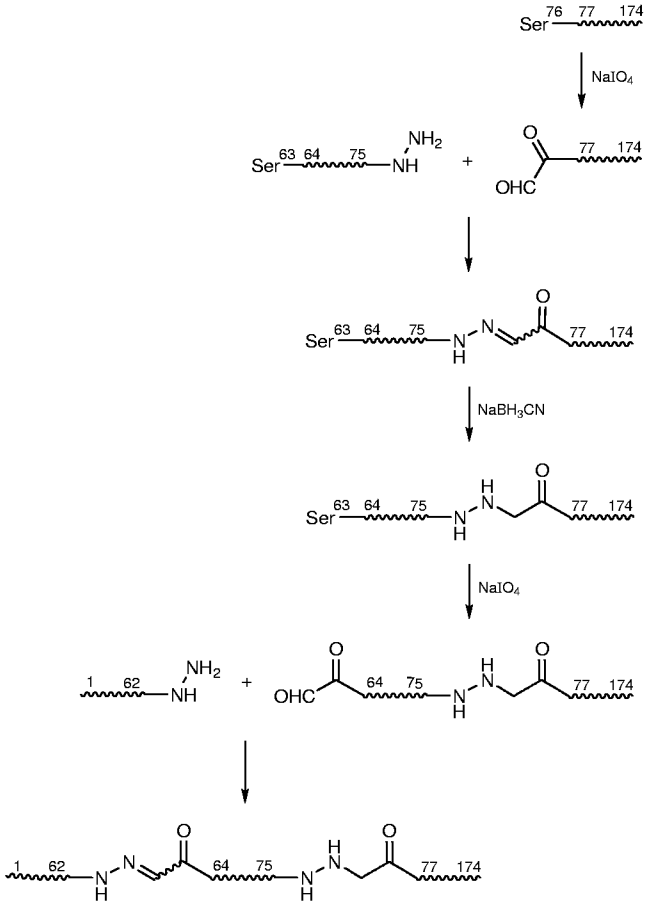
partial conversion of Met into Met sulfoxide remains, and should be guarded against by the inclusion of a scavenger, as described in Section 5.4.2.2 (Periodate Oxidation of Interleukin-8).

Peptides containing an N-terminal aminooxy group, a Cys residue, or even a C-terminal hydrazide can be generated readily by standard solid-phase peptide synthesis on commercially available supports, and can be used as partners in one or other of the above-mentioned reactions. In contrast, introduction of the appropriate reactive end at the C-terminus of a large peptide or a protein is much more difficult, but is illustrated in a few examples below.

5.4.2.1 Ligation within the Backbone Structure of a Single Protein Molecule

This operation is illustrated by work on the human cytokine GCSF, a protein of 174 residues. The disulfide-linked loop (64–74) of this protein, thought to be involved in binding to the receptor, was replaced by various non-coded synthetic structures.[10] In this case recombinant methods were used to produce a variant of the parent protein having the appropriate cleavage sites for the production of the wanted fragments.

Scheme 2 Construction of GCSF Analogue (1–62)-NHN=CHCO-(64–75)-NHNHCH$_2$CO-(77–174)[10]



The N-terminal fragment 1–62 and the C-terminal fragment 76–174 of two recombinant analogues of GCSF were isolated after enzymatic cleavage with the Lys-specific protease from *Achromobacter lyticus*. In one fragment, the only Lys in the protein was at position 62, and was immediately followed by Ser. In the other fragment, the only Lys, also followed by a

Ser, was at position 75. The strategy for the construction is shown in Scheme 2 and relies on the generation of a glyoxylyl group at the N-terminus and of a hydrazide at the C-terminus of the fragments to be ligated. Key steps that will be described here are the hydrazide formation by reverse proteolysis, ligation through hydrazone formation, and reduction of the hydrazone linkage. We describe the oxidation of N-terminal Ser, another key part of the scheme, in Section 5.4.2.2.1.1.

It can be seen from the scheme that virtually any linking structure, peptide or not, can be incorporated in the place of residues 63–75.

The final protein product has two hydrazone bonds. If the middle section (residues 63–75) is synthesized according to the natural sequence, the reassembled protein with two unreduced hydrazone bonds has a biological activity relative to the starting cytokine of roughly 1/1000. With one of the two bonds reduced the activity is 1/100 or 1/50 of the native sequence, depending on which bond is involved. With both bonds reduced the activity is a little less than that of the starting cytokine, but not to a statistically significant extent. This improvement of activity on reduction is doubtless due to the release of the rotation constraint around the N–C bond.

Hydrazide Formation by Reverse Proteolysis:[22]
A soln of 1 M $BocNHNH_2$ was prepared in DMSO/0.1 M NaOAc (1:1), pH 5.5, and the apparent pH at the glass electrode was adjusted to 5.5 with glacial AcOH. The polypeptide fragment (8 mg) was dissolved in the hydrazide soln at 20 $mg \cdot mL^{-1}$. Achromobacter protease was added as a freshly prepared soln in H_2O (10 $mg \cdot mL^{-1}$) at an enzyme/substrate ratio of 1:50 (w/w) and the sample incubated at rt for 15 h. Coupling was almost quantitative as determined by analytical HPLC, so that adsorption on a C-18 Sep-Pak, which separated the excess hydrazide and the enzyme from the protein fragment, gave material sufficiently pure for the following step. The mixture was diluted with 5 volumes of glacial AcOH and 20 volumes of 0.1% aq TFA and applied on the C18 Sep-Pak cartridge previously washed with MeOH and then equilibrated with 0.1% TFA. After washing with 0.1% TFA (5 mL) containing 30% MeCN, the sample was eluted with 0.1% TFA (5 mL) containing 90% MeCN. Solvent was removed by rotary evaporation and the hydrazide derivative lyophilized. The Boc group was cleaved by dissolving the product in TFA (1 mL) for 45 min at rt. TFA was evaporated under a flow of N_2, the material taken up in H_2O, freeze dried and characterized by ESI-MS; yield: 7 mg (85%); MS *m/z*: calcd for fragment (1–62)-NH-NH_2, 6948.1; found, 6948.1±1.4.

Formation of the Hydrazone Linkage:[10]
A twofold molar excess of the hydrazide fragment (2.5 mM in H_2O) was added to a soln of the oxidized C-terminal fragment (2 $mg \cdot mL^{-1}$) in 0.1 M NaOAc, pH 5.3, in 0.1% Sarcosyl detergent. The coupling reaction was followed by analytical HPLC. Although the coupling reaction was definitely faster at lower pH values, the reaction was carried out at pH 5.3, which corresponds to the lower limit at which Sarcosyl is usefully soluble. The hydrazone-linked conjugate could be isolated by HPLC after 15 h incubation at rt, or reduced in situ as described below.

Reduction of the Hydrazone Linkage:[10]
To 1 volume of the hydrazone mixture there was added, with mixing, 2 volumes of 0.6 M $NaBH_3CN$ in 0.1 M NaOAc/MeCN (1:1); the apparent pH was previously adjusted (glass electrode) to 4.6 with glacial AcOH. The apparent pH was readjusted to an apparent value of 4.6 with glacial AcOH, the soln incubated for 4 d at rt in the dark, and the reduced hydrazone finally isolated by HPLC. The same conditions could be used for both the ligation steps shown in Scheme 2.

5.4.2.2 Ligation at the N-Terminus of the Polypeptide Chain

As suggested above, the presence of an N-terminal Ser or Thr in a protein can be used to generate a reactive carbonyl group. This will be at a unique site, as there is only one N-terminus per polypeptide chain. Furthermore, that site is exactly the one required for there to be contiguity of the backbone of the chimeric molecule under construction.

In terms of chimeric macromolecules, the backbone of the oxidized protein or peptide can be extended by means of a mono-aminooxy functionalized protein, or a similar derivative of a nonprotein polymer (macromolecular adducts, see below). As an exercise in site-

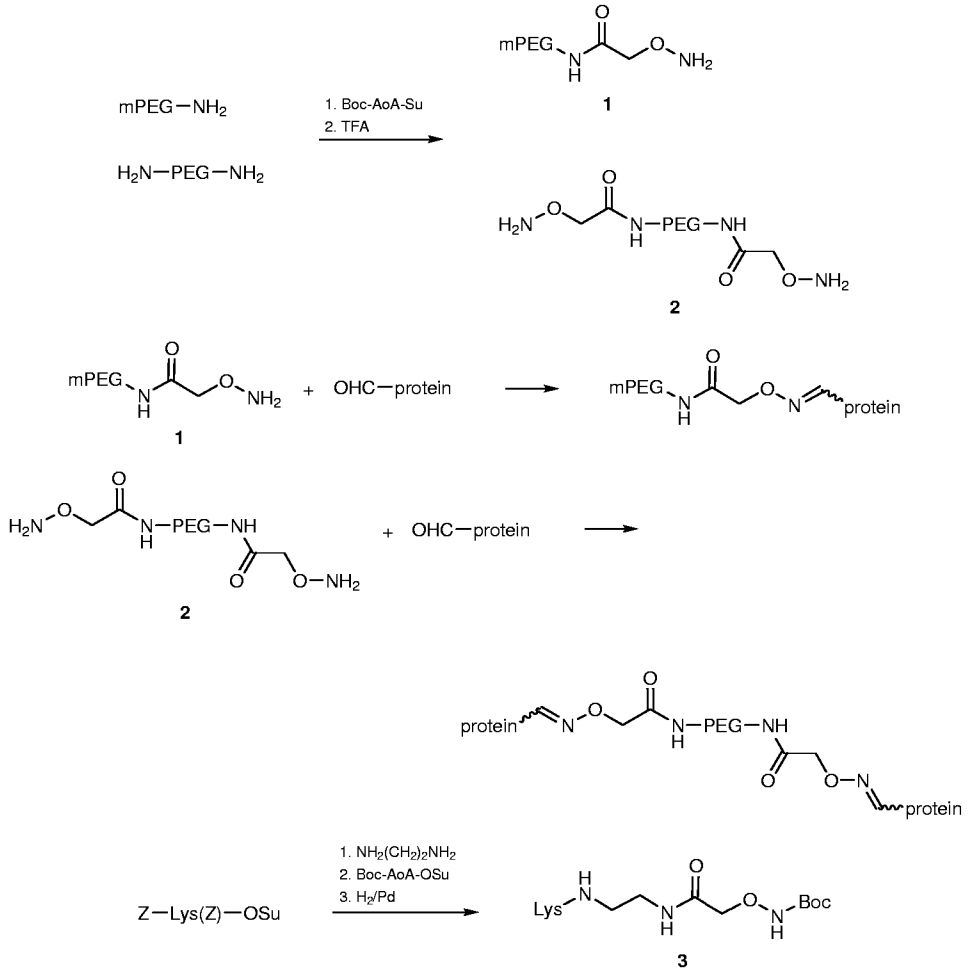
for references see p 91

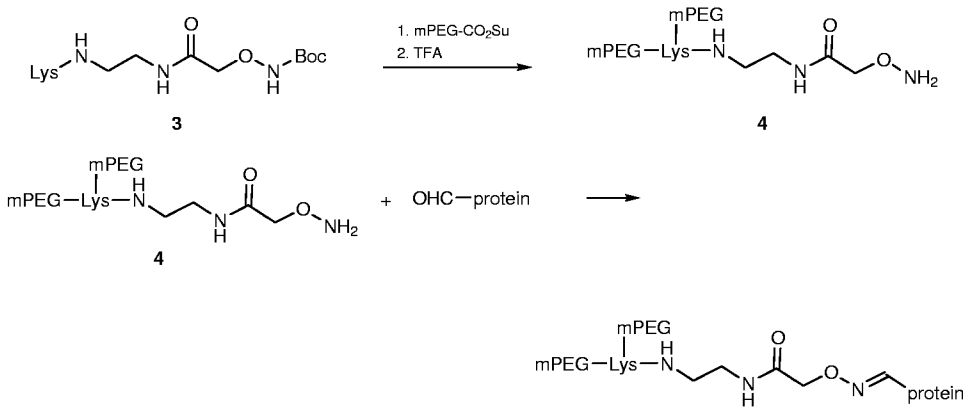
specific labeling, the unique carbonyl can become the point of attachment to which a radioactive label, reporter group, imaging agent, or drug can be grafted in an entirely site-specific manner (see Section 5.4.2.2.2).

5.4.2.2.1 Macromolecular Adducts

The N-terminal oxidation approach has been exemplified in several ways, for example in the semisynthesis of a fusion protein by coupling periodate-oxidized β-lactamase (the N-terminal residue is Thr) to a specially modified single-sulfanyl (i.e., thiol) murine F(ab′) by means of an oxime bond.[11] Here, in order to emphasize that the fusion constructions can involve nonprotein segments of backbone, we describe the extension of the polypeptide backbone of the chemokine interleukin-8 by PEG chains of various types (linear or branched). This operation was carried out so as to modify the pharmacokinetics of the protein. Scheme 3 gives some examples of the different possibilities involved.

Scheme 3 Presentation of the Different Aminooxy Derivatives of PEG and Conjugation to the Reactive Carbonyl Group Generated at the N-Terminus of the Protein[8]





5.4.2.2.1.1 Site-Specific Polyethylene Glycolated Interleukin-8

Periodate Oxidation of Interleukin-8:[12]
Interleukin-8 (2 mg) was dissolved in 1% NH_4HCO_3 (pH 8.3; 1 mL) and 0.2 M Met (60 µL) in H_2O added as a scavenger in 50-fold excess. Then, 50 mM $NaIO_4$ (48 µL) in H_2O (10-fold excess) was added, the soln incubated for 10 min in the dark and the reaction stopped by the addition of 10 M ethylene glycol soln (250 µL) in H_2O (10 000-fold excess). The soln was further incubated for 15 min and the product separated from the reagents on a PD10 gel-filtration column equilibrated in 0.1 M NaOAc at pH 4.6, or by dialysis in the same buffer. The approximate concentration of the soln of oxidized protein was determined by UV spectroscopy. The soln was stored frozen if not used immediately. Oxidation of the amino-terminal Ser was shown to be almost quantitative by ESI-MS, where a mass of 8350±1.4 Da was obtained, corresponding to a loss of 31 Da by the native structure to form the glyoxylyl derivative (calcd *m/z* 8351.3). A dehydrated form of this compound could also be detected (calcd *m/z* 8333.1; found *m/z* 8333.3±1.2).

Oxidized protein can be stored at −20 °C as a frozen soln for 6 months, and after desalting and lyophilization for up to 12 months.

Preparation of a Linear (Aminooxy)acetyl-Functionalized PEG 1:[8]
Commercially obtained aminomethoxypolyethylene glycol (NH_2-mPEG) was acylated with Boc-$NHOCH_2CO_2Su$ (see Section 5.4.2.2.2.1) under the following conditions. The active ester (144 mg, 0.5 mmol) was added to NH_2-mPEG (1 g) of M_r 5000 (0.2 mmol) dissolved in DMSO (4 mL). The apparent pH (externally indicated) was adjusted to 8.0 with NMM. After overnight stirring of the mixture at rt, the degree of acylation was checked by a quantitative ninhydrin procedure on an aliquot (10 µL) of the mixture. The acylated mPEG was then recovered by dilution with 5 volumes of H_2O, followed by extensive dialysis against H_2O, and lyophilization. The Boc group was removed by dissolving the product in anhydrous TFA (10 mL) for 1 h at rt. After removal of the acid by rotary evaporation, the material was taken up in H_2O, extensively dialyzed against H_2O, and finally lyophilized to give aminooxy-functionalized PEG polymer **1**; yield: 0.93 g (93%). The same procedure was used to prepare the aminooxyacetyl compounds AoA-mPEG$_{10\,kDa}$ and AoA-mPEG$_{20\,kDa}$. Branched polymers could be prepared from linear ones, as shown in Scheme 3, by, for example, reacting mPEG-CO_2Su with Lys-$NH(CH_2)_2NHCOCH_2ONHBoc$ (**3**) to give (mPEG$_{5\,kDa}$)$_2$Lys-$NH(CH_2)_2NHCOCH_2ONH_2$ (**4**).[8]

Conjugation with Aminooxy-Functionalized PEG Polymers:[8]
A 10-fold molar excess of a 10 mM aq soln of aminooxy-PEG polymer **1** was added to the oxidized protein (~ 0.25 mM), the pH adjusted to 3.6 with glacial AcOH, and the mixture incubated for 20 h at rt. The conjugation product was purified from reactants by ion-exchange chromatography on a Pharmacia MonoS column equilibrated in 25 mM NaOAc buffer, pH 4.7, using a linear gradient from 0 to 2 M NaCl over 20 min (to eliminate the excess of polymer). This step was followed by RP-HPLC (see ref[8] for full details of the column and gradient) to separate unreacted interleukin-8. The HPLC profiles allowed an estimate of the coupling yield of about 80% in the case of AoA-mPEG$_{5\,kDa}$ and AoA-mPEG$_{10\,kDa}$. The lower coupling yields observed with AoA-mPEG$_{20\,kDa}$ and (mPEG$_{5\,kDa}$)$_2$Lys-$NH(CH_2)_2NHCOCH_2ONH_2$ after 20 h of incubation (~30%) could be significantly increased by prolonging the reaction period to 2–3 d.

The use of the bifunctional PEG **2** resulted in the formation of a protein dimer linked by a PEG chain. It is worth stressing that this dimer construction was obtained in a single coupling step, even with a

for references see p 91

5- to 10-fold excess of the bifunctional polymer over the oxidized protein. This result is probably related to the fact that the protein already exists as a dimer in solution. Under such circumstances, it is quite likely that a bifunctional PEG compound of appropriate chain length will cross-link the two N-termini of the dimer components.

The chemoselectivity of the ligation reaction was such as to leave the protein with significant biological activity. While interleukin-8 (IL-8) had an EC_{50} of 2 nM in the neutrophil chemotaxis assay (concentration corresponding to 50% of the maximal chemoattractant activity), $mPEG_{5\,kDa}$ and $mPEG_{10\,kDa}$ derivatives were shown to have an EC_{50} of 10 nM, and for $mPEG_{20\,kDa}$, $(mPEG_{20\,kDa})_2$Lys-IL-8, and IL-8-$PEG_{20\,kDa}$-IL-8 the EC_{50} ranged between 20 and 40 nM.

5.4.2.2.2 Small Molecule Adducts

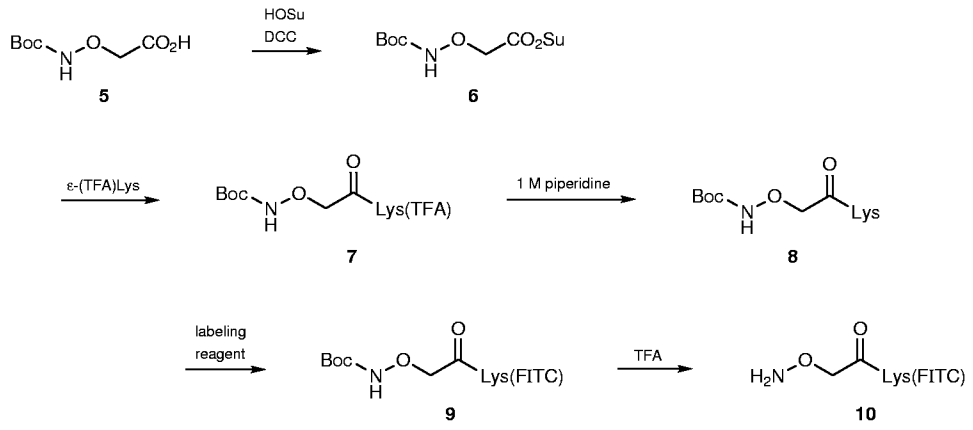
5.4.2.2.2.1 Site-Specific Fluorescein Labeling of Interleukin-8 through an Oxime Bond

The site-specific fluorescent labeling of chemokines is of growing interest as an aid to following receptor behavior and the direct evaluation of chemokine binding. The products lend themselves to such methods as confocal fluorescence microscopy, fluorescence-activated cell-sorting analysis, and fluorescence correlation spectroscopy. Chemokines can often be labeled at the N-terminus via an oxime bond,[12] since many have, or can easily be given, an N-terminal Ser. This can be converted into a carbonyl group, and the fluorophore can be introduced as an oxime via an aminooxy derivative. Alternatively, as already implied above, a specific oximation site can be introduced at any point in a sequence of a synthetic chemokine by introducing an N^{ε}-seryl-Lys. In such a case, if the chemokine has N-terminal Ser it is better to use N^{ε}-levulinyl-Lys, which is ready for oximation without oxidation.[13]

The following method describes the preparation of an aminooxy fluorophore and its conjugation to an oxidized protein. The fluorescent molecule is built up on the Lys skeleton, with the fluorophore on the ε-amino group, and the AoA group on the α-amino group. In the description below, the fluorophore is the fluorescein thiocarbamyl group, but ref[12] describes the introduction of other fluorophores, such as the NBD group, CY5 (cyanine 5 fluorescent dye), and Texas Red; ref[14] describes the introduction of a rhodamine-based fluorophore.

The different steps of the construction are shown in Scheme 4.[15]

Scheme 4 Construction of the Aminooxy-Functionalized Fluorescein[12]



Preparation of an Aminooxy Fluorophore:[12]

Boc-aminooxyacetyl Hydroxysuccinimido Ester (6):

To Boc-aminooxyacetic acid (**5**; 1.91 g, 10 mmol) (Novabiochem, Laüfelfingen, Switzerland) in EtOAc (30 mL) was added a soln of HOSu (1.15 g, 10 mmol) in EtOAc (30 mL), and then at rt, under constant mixing, was added a soln of DCC (2.06 g, 10 mmol) dissolved in EtOAc (3 mL). After agitation for 5 h at 20 °C, there was an abundant precipitate of the dicyclohexylurea. The complete disappearance of the

carbodiimide was verified by TLC (silica gel) of 5 µL samples ($CHCl_3$/MeOH 1:1) developed with a spray of 1,3-dimethylbarbituric acid (500 mg in 9 mL pyridine plus 1 mL H_2O).[16] A 5 µL spot of carbodiimide (30 mg in 1 mL of EtOAc) was run alongside for comparison and gave a blue-violet color. If some carbodiimide remained, further **5** (1.45 mg, 5 mmol) was added and the reaction continued until no carbodiimide remained. Once all the carbodiimide had been consumed, the mixture was filtered, and the filtrate was partially dried to obtain an oily soln. The material was then crystallized by the slow addition of an equal volume of Et_2O and by cooling for a few hours in an ice bath. The crystallized material was filtered, washed with previously cooled Et_2O, and finally dried in a desiccator under oil-pump vacuum; yield: 1.3 g (45%). Material remaining in the filtrate was dried down (1.2 g) and resolubilized in EtOAc for another crystallization.

N^{α}-(Boc-aminooxyacetyl),N^{ε}-TFA-Lys (7):[12]
Compound **6** (570 mg, 2 mmol) was added to N^{ε}-TFA-Lys (300 mg, 1.2 mmol) (Novabiochem, Laüfelfingen, Switzerland) suspended in DMSO (3 mL). NMM (~150 µL) was added until the externally indicated pH (moist narrow-range pH paper) was between approximately 8 and 9. By this time the mixture was a clear soln. After 15 h at 20 °C and verification that the externally indicated pH was still alkaline, H_2O (5 mL) was added, the external apparent pH was brought up if necessary to its original value with DIPEA, and the mixture left for 1 h at 37 °C to hydrolyze any remaining active ester. The soln was then cooled to 0 °C (ice bath) and the apparent pH (glass electrode, uncorrected) was brought to 3.0 by careful addition of glacial AcOH under constant agitation. Cold H_2O (13 mL) was then added and the product was isolated on a preparative-scale HPLC. One half of the mixture could be loaded at a time on a preparative HPLC equipped with a PrepLC 25 mm Module (Waters) containing two PrepPak 25 × 100-mm cartridges filled with Nova-pak HR C18 6 µm, 60 Å resin and a 25 × 10-mm guard Pak cartridge. The flow rate was 20 mL·min^{-1}, with a 30-min gradient between 20 and 50% of MeCN/TFA/H_2O (900:1:100), the balance being 0.1% aq TFA; yield: 380 mg (76%).

N^{α}-(Boc-aminooxyacetyl)-Lys (8):[12]
Dried **7** was dissolved in H_2O (4 mL) and piperidine (0.44 mL) was added (final concentration 1 M). After 4 h at 20 °C the mixture was adjusted to pH 3.0 at 0 °C as before and then diluted with 0.1% TFA (10 mL). The deprotected material was then isolated by preparative HPLC using a 0–30% gradient (same eluants as above) over 30 min with a flow rate of 20 mL·min^{-1} to give the trifluoroacetyl salt; yield: 370 mg (94%).

N^{α}-(Boc-aminooxyacetyl),N^{ε}-(Fluorescein Thiocarbamoyl)-Lys (9):[12]
Compound **8** (40 mg) was dissolved in DMF (300 µL), FITC (40 mg) was added, and the mixture adjusted to pH 8.0 with NMM. After incubation for 15 h in the dark, the product was purified by chromatography (silica) on a Kieselgel 60 1.5 × 20-cm column (Fluka Chemie, Buchs, Switzerland) equilibrated in MeOH/$CHCl_3$ (1:1). The excess FITC eluted in the flow-through fraction, and a second yellow fraction containing the expected product was eluted with MeOH/$CHCl_3$ (4:1). The solvent was removed by rotary evaporation, and the product was characterized by ESI-MS; *m/z*: $[M+H]^+$ calcd, 709.6; found, 709.4. Further purification was by semipreparative RP-HPLC on a Nucleosil-C8 column (10 × 250 mm) at a flow rate of 4 mL·min^{-1} using a 30–60% gradient (same components as above) over 30 min.

N^{α}-(Aminooxyacetyl),N^{ε}-(Fluorescein Thiocarbamoyl)-Lys (10):[12]
The dried product **9** (40 mg) was dissolved in TFA to a concentration not exceeding 20 mg·mL^{-1} and left for 45 min at 20 °C. The majority of the TFA was then evaporated in a stream of air and drying was completed by lyophilization. If **9** had not been purified by RP-HPLC as described above, then purification of **10** was equally possible on the semipreparative column, using a gradient of 25–40% B over 15 min; yield: 18 mg (32%); ESI-MS *m/z*: $[M+H]^+$ calcd, 609.6; found, 609.6. The FITC derivatives normally elute as doublets on low-loading analytical HPLC because of the use of the commercial mixture of fluorescein 5- and 6-isothiocyanate for the coupling to **8**.

Oximation of N-Terminally Oxidized Interleukin-8:[12]
A threefold excess of a 10 mM soln of H_2NOCH_2CO-Lys(FITC)-OH (**10**) in H_2O containing 20% MeCN was added to the oxidized interleukin-8 soln (2 mg·mL^{-1} in 0.1 M NaOAc buffer, pH 4.6). MeCN was added to a final concentration of 10% to ensure solubilization of the reagent, and the pH of the soln was adjusted to 4.6 with glacial AcOH. After incubation for 15 h at rt in the dark, the product was purified from reactants by RP-HPLC on a C8 column. The coupling yield was estimated at 80% from the recovery of coupled protein. The conjugate had the expected molecular mass; *m/z*: calcd, 8941.6; found, 8940.6±1.0.

for references see p 91

5.4.2.2.2.2 Biotinylated Human Parathyroid Hormone through Thiazolidine Ring Formation

Human parathyroid hormone, a 34-residue peptide, has an N-terminal Ser suitable for conversion into the N^{α}-glyoxyl moiety, thus, permitting a subsequent thiazolidine conjugation with 1-amino-2-thiol derivatized biotin.[17]

Cys-NH(CH_2)$_6$NH-biotin:[17]
BocNH(CH_2)$_6$$NH_2$ · HCl (1.15 g, 4.4 mmol) and NMM (0.49 mL, 4.4 mmol) were added to biotinylimidazolide[18] (1.18 g, 4 mmol) obtained by reaction of equimolar amounts (4.1 mmol) of D-biotin and CDI in dry DMF (20 mL). The soln was stirred overnight at rt. DMF was removed under vacuum and the residue recrystallized (iPrOH) to give biotin-NH(CH_2)$_6$NHBoc; yield: 1.5 g (83%). The product has an R_f value of 0.81 on TLC (silica, $CHCl_3$/MeOH/AcOH 85:10:5). Biotin-NH(CH_2)$_6$NHBoc (0.89 g, 2 mmol) was dissolved in 50% TFA/CH_2Cl_2 (10 mL). After 30 min, TFA and CH_2Cl_2 were removed and the product was precipitated with dry Et_2O. The resultant solid was washed with dry Et_2O (4 × 10 mL) and then taken up in dry DMF (10 mL). After the addition of NMM (0.44 mL, 4 mol) and Boc-Cys(Trt)-OH (0.92 g, 2 mmol), the mixture was cooled to 4 °C and HBTU (0.76 g, 2 mmol) was added to the soln. The soln was slowly warmed to rt and stirred for an additional 18 h. The solvent was removed under vacuum and the residue taken up in EtOAc (100 mL). The EtOAc phase was washed with sat. NaCl soln (2 × 15 mL), 5% $KHSO_4$ (2 × 15 mL), 5% Na_2CO_3 (3 × 15 mL), and H_2O (2 × 15 mL). The organic phase was dried (Na_2SO_4) and concentrated under vacuum to yield an oily residue, which solidified on addition of hexane. The white solid was then treated with TFA/thioanisole (10 mL, 8.5:1.5) for 4 h. After removal of the solvent the product was precipitated with dry Et_2O. The white powder was dissolved in glacial AcOH and precipitated again with dry Et_2O. After repeating this procedure several times, the product was taken up in H_2O and lyophilized; yield: 0.67 g; MALDI-MS *m/z*: $[M+H]^+$ calcd, 446.6; found, 446.5±1.0.

Oxidation of Human Parathyroid Hormone and Biotinylation:[17]
Human parathyroid hormone (1 mg, 0.24 μmol) was dissolved in 0.1 M citric acid/0.2 M Na_2HPO_4 buffer (1 mL) containing Met (1.2 μmol), pH 7.0, and treated with $NaIO_4$ (1 equiv, 52 μg, 0.24 μmol). Oxidation was allowed to proceed at 4 °C for 4 min. The oxidized material was purified on a semipreparative C8 column with a gradient of 6 to 48% MeCN in 0.05% TFA over 60 min. The purified N^{α}-glyoxylyl parathyroid hormone was characterized by MALDI-MS, giving the expected molecular mass; *m/z*: $[M+H]^+$ calcd, 4087.7; found, 4087.9±1.0.

N^{α}-Glyoxylyl parathyroid hormone (0.2 mg, 0.05 μmol) was dissolved in 0.2 M NaOAc buffer (200 μL) containing 0.01 M EDTA, pH 5.2. To this was added functionalized biotin (100 μL of 1 mM in the same buffer used for the aldehyde). The conjugation was monitored by HPLC. After 7 h, N^{α}-glyoxylyl parathyroid hormone disappeared completely with the formation of a new peak. The product was then isolated by semipreparative HPLC; yield: ~95%; *m/z*: $[M+H]^+$ calcd, 4516.3; found, 4515.6±1.0.

5.4.2.3 Ligation at the C-Terminus of the Polypeptide Chain

In order to ligate at the C-terminus without the risk of unwanted conjugation through side-chain carboxy groups, it is usually best to direct reactivity to this site, and to this site only, using the specificity for the α-carboxy shown by many proteases. This is best done by using reverse proteolysis to substitute the α-carboxy with a chemically activated group.

5.4.2.3.1 Macromolecular Adducts

The general methodology described in this chapter can be used to produce head-to-tail or tail-to-tail chimerae between two proteins. For example, a site-specific linked immunoconjugate was prepared between an F(ab′)-like fragment of the monoclonal anti-CEA murine IG1 A5B7 and a mutant of the dimeric enzyme carboxypeptidase G2. The enzyme normally has an Ala at the N-terminus, but it was found to be relatively simple to express a recombinant form with N-terminal Thr.[19] As previously demonstrated, an aldehyde was introduced at the N-terminus of the enzyme by mild periodate oxidation. A carbohydrazide group (-NHNHCONHNH$_2$) was specifically introduced at the C-terminus of the truncated heavy chain of the F(ab′)$_2$-like fragment by reverse proteolysis. The two proteins were then

conjugated by the formation of a hydrazone bond. It was as though the Fc portion of the IgG molecule had been removed and directly replaced by the enzyme.

The final construction had $M_r = 153$ kDa. In size, and homogeneity of structure, this illustrates the advantages of this general methodology for the preparation of constructions not readily accessible by any other method. The subsequent biological characterization showed, as hoped, that the immunoreactivity of the construction was significantly better than one made by random chemical coupling between side chains.

5.4.2.3.1.1 An F(ab′)$_2$-Enzyme Conjugate

Carbohydrazide-Substituted F(ab′)$_2$:
F(ab′)$_2$ was obtained by digestion of a 5 mg·mL^{-1} antibody soln in 50 mM Tris·HCl, pH 8.4, with lysyl endopeptidase at a 3% w/w enzyme/substrate ratio over 20 h at 37 °C. The F(ab′)$_2$ was then purified by cation-exchange chromatography on an SP trisacryl column, desalted, and concentrated again to yield 35–40% of the theoretical maximum for F(ab′)$_2$. Solid carbohydrazide was added to the purified F(ab′)$_2$, typically about 5 mg protein in 10 mM Tris·HCl (700 µL), pH 8.0, to get a final carbohydrazide concentration of 2.5 M. The pH was then lowered to 5.5 by addition of glacial AcOH prior to addition of lysyl endopeptidase (10 mg·mL^{-1} in H_2O, enzyme/F(ab′)$_2$ ratio of 5% w/w). The mixture was allowed to stand at rt for 3 h, whereupon the reaction was stopped by addition of the pancreatic trypsin inhibitor (Trasylol, 100 mg·mL^{-1} in H_2O, 30-fold mass excess over the lysyl endopeptidase). The mixture was then gel filtered on a Superose 12 column in 0.1 M NaOAc, pH 4.6, at a flow rate of 0.6 mL·min^{-1}. The peak corresponding to F(ab′)$_2$ was collected in a tube containing Trasylol (10-fold mass excess over the quantity of peptidase used for the reverse proteolysis), concentrated, and made 10 mM in urea by adding 1 M urea in H_2O. After 1 h at rt, the sample was desalted on a fast desalting column equilibrated in the same NaOAc buffer. The protein recovery was quantitative.

It is important to keep the F(ab′)$_2$-carbohydrazide, once removed from the 2.5 M carbohydrazide soln, in the presence of a large excess of Trasylol. If this is not done, the coupling is rapidly reversed by remaining traces of lysyl endopeptidase.

Typically, the incorporation of carbohydrazide obtained with antibody A5B7 was about 0.5 molecules of carbohydrazide per F(ab′)$_2$ as determined by incorporation of ^{55}Fe-labeled aldehyde.[20] ESI-MS after reduction of disulfide bridges by dialysis in 10 mM DTT confirmed that the light chain was not modified by this procedure. On the other hand, the heavy-chain mass was 26 489 Da in the F(ab′)$_2$-carbohydrazide and only 26 416 in the F(ab′)$_2$. The 73 Da difference between the two heavy chains is in agreement with the 72 Da increment expected by the incorporation of one carbohydrazide group.

Conjugation of the Two Proteins:
The F(ab′)$_2$-carbohydrazide in the NaOAc buffer was mixed with aldehydic CPG2 which had been prepared according to Section 5.4.2.2.1.1 (Periodate Oxidation of Interleukin-8) (0.8 mg of CPG2 for 1 mg of F(ab′)$_2$-carbohydrazide, corresponding to about four aldehyde groups per hydrazide). The mixture was then concentrated in a Centripep 10 concentrator to a final volume of about 700 µL and was allowed to stand for about 60 h, after which time it was gel filtered on Superose 12 in phosphate-buffered saline. Fractions containing the conjugate were pooled and concentrated; yield: ~30%. The overall yield of the conjugate preparation was 10–15% based on starting IgG, and 30% based on starting CPG2.

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6 Specific Methods

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The structural diversity of peptides and proteins from ribosomal biosynthesis derives from the combination of 20 (if we include selenocysteine, 21) amino acids. The diversity is exponentially increased by nature via complex co- and posttranslational processing to the functionally mature forms. This involves folding with the help of specialized foldases for concomitant aminoacyl-proline *cis/trans* isomerization and disulfide formation, limited proteolysis of the folded precursor proteins, and transpeptidation reactions (peptide bond splicing with peptide deletions and/or permutations). However, the most significant enzyme-assisted chemical modifications occur on single residues of the polypeptide chain. As a consequence, the actual number of amino acids found in proteins is much closer to 200 than to the 20 specified by the genetic code. Moreover, even chiral inversion from L- to D-amino acid residues has been recognized as a true posttranslational process (for recent reviews on the subject, see refs[1,2]).

With the advent of proteomics unambiguous identification of these posttranslational modifications, using advanced mass spectrometric techniques,[3] has become of fundamental importance in transport, compartmentalization, and function. Similarly, synthetic replicates are increasingly required to facilitate the understanding of the biological function of these structural motifs at molecular level. Attempts to copy nature in this structural diversity by synthesis has been the challenge of peptide chemistry since its inception. Although significant progress has been achieved in the chemistry involved, many of these posttranslational modifications are still awaiting efficient synthetic methodology.

The modifications are often discovered by ingenuity and serendipity, but also by alert observation and elegant experimentation. Reactions that occur posttranslationally as a consequence of environmental conditions[4] are also observed as side products in peptide synthesis. These include peptide bond isomerization involving Asp or Asn, and to a minor extent Glu and Gln with concomitant racemization[5–7] as well as transpeptidation at Ser or Cys residues.[8] The latter structure-dependent spontaneous peptide bond isomerization represents the basic reaction in autocatalytic protein splicing,[8] which has been exploited for the development of the native chemical ligation methodology (see Vol. E 22a, Section 4.1.4). Additional nonenzymatic transformations are α-oxo acid[9] and pyroglutamic acid[28,29] formation and also oxidation of Met[146] and Cys disulfides.[94] These chemical modifications that occur more or less spontaneously dependent upon the environment and aging, serve for both functional maturation of peptides and proteins, but also for their inactivation. In particular cases, e.g. Met(O), repair enzymes are available such as Met(O)-peptide reductase[146] to reverse such modifications.

Most of the other posttranslational modifications involving the N- or C-terminus (Table 1) as well as the side-chain functionalities (Table 2) of the polypeptide chains occur under the control of enzymes that also dictate the regioselectivity of such chemical transformations. This regioselectivity is difficult to attain by synthetic procedures. Sophisticated protection schemes are required when additional chemistry must be performed on preassembled peptides, unless enzymatic methods can be used to supplement the synthetic strategies. As a consequence, the use of suitably modified amino acids as synthons is generally the preferred approach as will be discussed in the following sections.

for references see p 96

Table 1 Natural N- and C-Terminal Modifications[a]

N-Terminus	C-Terminus
N-acyl (C1;[10–12] C2;[13–16] C4, C6, C8, C10;[17] C12;[18] C14;[19–21]) (Section 6.4.1.1)	amide[34,35] (Vol. E 22a, Section 2.3.1)
N-tetradeca(mono and di)enoyl (C14:1; C14:2)[18]	*O*-(ADP-ribosyl)[36,37]
N-aminoacyl[22,23]	*O*-methyl[38,39]
N-α-oxoacyl[9]	glycosylphosphatidylinositol (GPI anchors)[40–42] (Section 6.4)
N-methyl[24–27]	N^{α}-Tyr[43,44]
N-pyroglutamyl[28,29] (Section 6.7)	
N-glucuronyl[30,31] (Section 6.3)	
N-glycosyl[32,33] (Section 6.3.)	

[a]This list of posttranslational modifications was compiled from refs[1,2] and sections of Vol. E22 are reported in which related synthetic procedures are described.

Table 2 Natural Side-Chain Modifications[a]

Amino Acid	Side-Chain Modification
Arg	N^{ω}-(ADP-ribosyl)[45,46] N^{ω}-methyl[47] N^{ω}-dimethyl[47] $N^{\omega},N^{\omega'}$-dimethyl[47] N^{ω}-phosphoryl[50] ornithine[48] citrulline[49]
Asn	*N*-glycosyl[51–53] (Section 6.3.3) *N*-methyl[54] N^{ε}-(β-aspartyl)lysine[55] *erythro*-β-hydroxy[56] *N*-(ADP-ribosyl)[57] Asp and D-Asp (desamidation and α→β transpeptidation)[5–7,58–60] (Vol. E 22a, Section 2.3.3)
Asp	α→β transpeptidation[5–7] (Vol. E 22a, Section 2.2.2) D-Asp (racemization)[6,7,61] β-carboxy[62] β-methylsulfanyl[63] *O*-phosphoryl[64] *O*-methyl[65] (Vol. E 22a, Section 2.2.2.1) *erythro*-β-hydroxy[66,67]
Cys	disulfide formation[68,69] (Section 6.1) disulfide *S*-oxides, sulfenic acid, sulfonic acid[94] polysulfides[95–97] (Section 6.2.1) *S*-γ-glutamyl[70,71] *S*-(2-histidyl)[72] *S*-(3-tyrosyl)[73] *S*-(*sn*-1-glyceryl)[74] *S*-(*sn*-1-diacylglyceryl)[75] (Section 6.4.1.4) *S*-(*sn*-1-{2,3-di-*O*-[3′,7′,11′,15′-tetramethylhexadecyl]}glyceryl)[76] *S*-palmitoyl[77–79] (Section 6.4.1.3) *S*-farnesyl[38,80–82] (Section 6.4.1.2) *S*-geranylgeranyl[80,81,83] (Section 6.4.1.2) *S*-heme[84] *S*-phycocyanobilin[85] *S*-*p*-coumaroyl[87] *S*-(6-flavin)[86] *S*-(8α-flavin)[86] *S*-(coenzyme A)[88] *S*-(ADP-ribosyl)[89] *S*-glycosyl[90] (Section 6.3.5) dehydroalanine[91,92] (Vol. E 22c, Section 11.1)

Table 2 (cont.)

Amino Acid	Side-Chain Modification
	lysinoalanine[93] lanthionine[93] (Section 6.2.2)
Gln	Glu (desamidation)[58,98] N^{ε}-(γ-glutamyl)lysine[99,100] *N*-(γ-glutamyl)ornithine[101] *N*-(γ-glutamyl)polyamine[102,103] *N*,*N*-(bis-γ-glutamyl)polyamine[102,103] N^{5}-methyl[104]
Glu	*O*-(ADP-ribosyl)[46,105,106] γ-carboxy[107–109] *O*-methyl[65,110] N^{α}-(γ-glutamyl)-$[Glu]_{1-5}$[111] N^{α}-(γ-glutamyl)-$[Gly]_{3-34}$[112] *N*-(γ-glutamyl)ethanolaminephosphate[113]
His	diphthamide[114–116] N^{τ}-(ADP-ribosyl)diphthamide[114,115,117,118] *N*-phosphoryl[119,120] (Section 6.5.3) N^{π}-methyl[121,122] 4-iodo- and diiodo[123] N^{τ},N^{π}-(8α-flavin)[86] N^{π}-(8α-flavin)[124]
Lys	N^{ε}-acetyl[125] N^{ε}-(N^{α}-monomethylalanyl)[126] N^{ε}-murein (peptidoglycan)[127] N^{ε}-lipoyl[128] N^{ε}-biotinyl[129] N^{ε}-ubiquitinyl[130,131] N^{ε}-phosphoryl[120] N^{ε}-phophopyridoxyl[132] N^{ε}-retinyl[133] N^{ε}-glycosyl[134] N^{ε}-mono-, di-, trimethyl[47,122] hypusine: N^{ε}-(4-amino-2-hydroxybutyl)[135,136] allysine[137,138] Δ-hydroxy[139,140] Δ-hydroxyallysine[139] cross-links (desmosines, syndesines, pyridinolysines)[138,141–144] Δ-glycosyloxy[145]
Met	sulfoxide[146] (Vol. E 22a, Section 2.6.5)
Phe	β-glycosyloxy[31]
Pro	3-hydroxy[147] (Vol. E 22c, Section 9.2.3.1) 4-hydroxy[140] (Vol. E 22c, Section 9.2.3.1) 3,4-dihydroxy[148] 4-glycosyloxy[149] (Section 6.3.2.5)
Ser	selenocysteine[150] (Section 6.2.3) *O*-phosphate[151,152] (Section 6.5.1) *O*-pantetheinephosphoryl[153] *O*-(GlcNAc-1-phosphoryl)[154] *O*-(glycerol-1-phosphoryl)[155] alanino(π- and τ-histidine)[156] lanthionine[157] (Section 6.2.2) *O*-acetyl[158] (Vol. E 22a, Section 2.6.4.1.2) *O*-fattyacyl[77] (Section 6.4.1.3) *O*-methyl[159] *O*-glycosyl[160] (Section 6.3.2.4)

for references see p 96

Table 2 (cont.)

Amino Acid	Side-Chain Modification
Thr	*O*-phosphate[151,152] (Section 6.5.1) *O*-glycosyl[160] (Section 6.3.2.4) *O*-fattyacyl[77] (Section 6.4.1.3) *O*-methyl[159] β-methyllanthionine[157] (Section 6.2.2)
Trp	2,4′-bis-Trp-6′,7′-dione[161,162] C2-mannopyranosyl[163,164] (Section 6.3.4) 6-bromo[165]
Tyr	3-iodo- (bromo-, chloro-)[166] 3,5-diiodo- (bromo-, chloro-)[166] thyronine[166,167] 3,4-dihydroxy-Phe (DOPA)[168] 3,4,6-trihydroxy-Phe (TOPA)[162] *O*-phosphate[169] (Section 6.5.2) *O*-sulfate[170] (Section 6.6) *O*-adenylyl[171] *O*-uridylyl[171] *O*-(8α-flavin)[86] 3,3′-bis(Tyr)[172] 3,3′;5,5′-tris(Tyr)[173] isodi-Tyr[174] β-glycosyloxy[30] (Section 6.3.2.6) 6-(N^{ε}-Lys)-3,4-dihydroxy-Phe[175]

[a]This list of posttranslational modifications was compiled from refs[1,2] and sections of Vol. E22 are reported in which related synthetic procedures are described.

In addition to the direct modifications of single or multiple amino acid side chains in functionally mature peptides and proteins, enzyme-assisted posttranslational modifications generate cyclic and polycyclic structural elements of differing ring sizes via side-chain disulfide[68,69] (see Section 6.1) or polysulfide[95–97] (Section 6.2.1) as well as lanthionine[157] bridges (Section 6.2.2), which are all responsible for the induction and stabilization of bioactive architectures. Similarly, small side-chain-to-backbone ring closures to heterocycles such as oxazoles and thiazoles[176,177] (Section 6.8.5.2.2) serve for local conformational restrictions. An even larger diversity of homodetic and heterodetic cyclic peptide structures is produced in microorganisms by the action of nonribosomal peptide synthetases, which serve as Nature's lesson to develop artificial conformational restraints in synthetic peptides for the modulation of biological functions (Section 6.8).

In view of the large diversity of structures listed in Tables 1 and 2, only the synthetic methods developed for the most common and biologically most important peptide derivatives will be discussed in the following sections. Additional attention will be paid to the often-ingenious design of artificial modifications capable of mimicking the native peptide transformations in an efficient and biologically competent manner.

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6.1 Synthesis of Cystine Peptides

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Disulfide bonds are found in practically all classes of extracellular peptide and protein molecules, including hormones, growth factors, toxins, and enzyme inhibitors. Intramolecular disulfides serve to covalently cross-link portions of the polypeptide chain that are apart in the linear sequence, but are close in three dimensions, and intermolecular disulfides link two or more separate polypeptide chains. From multiple-cysteine-containing peptides various cystine networks can be formed in a molecule, which may correspond to the theoretically predictable isomeric forms, but also to polymeric products. These networks become even more complicated when the number of cysteine residues increases and, particularly, when both intra- and intermolecular disulfides are to be generated in the molecule. Thus, chemical control over the specificity of the cysteine pairings is essential to assure the formation of correct intra- and/or interchain disulfide bridges from the precursor cysteine peptides. This is unless sequence-encoded information suffices for the thermodynamically controlled correct oxidative refolding of the synthetic cysteine peptides. Despite the great progress achieved in the field,[1–6] the chemistry of regioselective disulfide formation still remains the key to the successful syntheses of many biologically active cystine peptides or proteins and therefore a significant challenge for peptide chemistry. However, low yields are frequently observed due to problems of intermolecular dimerization/oligomerization, of scrambling of preformed disulfides, as well as modification of sensitive amino acid side chains.

The practical aspects of applying oxidative refolding procedures as the ultimate step in the synthesis of multiple-cystine-containing peptides are covered in Section 6.1.5, while sections 6.1.1–6.1.4.3 describe the chemistry available for the regioselective synthesis of homo- (symmetrical cystine peptides) and heterodisulfides (unsymmetrical cystine peptides) from the thiol-free and/or thiol-protected cysteine peptide precursors. This chemistry is strongly related to the thiol protecting groups available for cysteine residues, their differentiated stabilities, and more or less orthogonal deprotections; particularly if multiple disulfide bridges have to be generated in regioselective manner. For the chemistry of these S-protected cysteine derivatives and related references to original literature, see Vol. E 22a, Section 2.6.6.

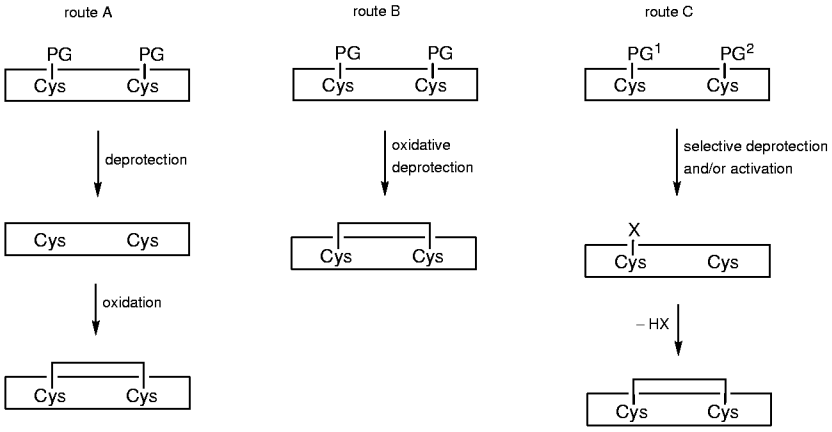
6.1.1 Intrachain-Bridged Mono-disulfide Peptides

There are three main approaches to intramolecular mono-disulfide formation in synthetic cysteine peptides as shown in Scheme 1. Route A relies on deprotection of a pair of cysteine residues, followed by oxidation with molecular oxygen or another appropriate reagent to provide the intrachain disulfide. In route B, the cysteine protecting groups are removed oxidatively by electrophilic reagents such as iodine, to produce the disulfide bond directly, whereas route C requires two different cysteine protecting groups that are selectively removed and/or activated to allow a subsequent site-directed disulfide formation. All three synthetic routes suffer from side reactions due to the reactivity of sensitive amino acids present in the peptide chain, and from intermolecular dimerization and oligomerization. The latter side reactions can be controlled to varying extents by the use of high dilution in solution, or of pseudo-dilution in polymer-supported reactions. While the synthetic routes A and B are applied to the formation of intramolecular single or multiple disulfide bridges, the synthetic approach C which is also for the generation of intramolecular disulfides, represents the sole unambiguous procedure for regioselective intermolecular disulfide bridging of two or more polypeptide chains. In all these synthetic approaches conformational preferences

for references see p 137

can be of great help in preventing scrambling or disproportionation that may occur under strongly acidic or even weakly basic conditions.[4] In this context, double-chain unsymmetrical cystine peptides are particularly prone to disproportionation into the symmetrical cystine peptides, and thus exposure to conditions that favor such side reactions should be minimized.

Scheme 1 General Approaches for the Production of Disulfide Bonds in Cysteine Peptides



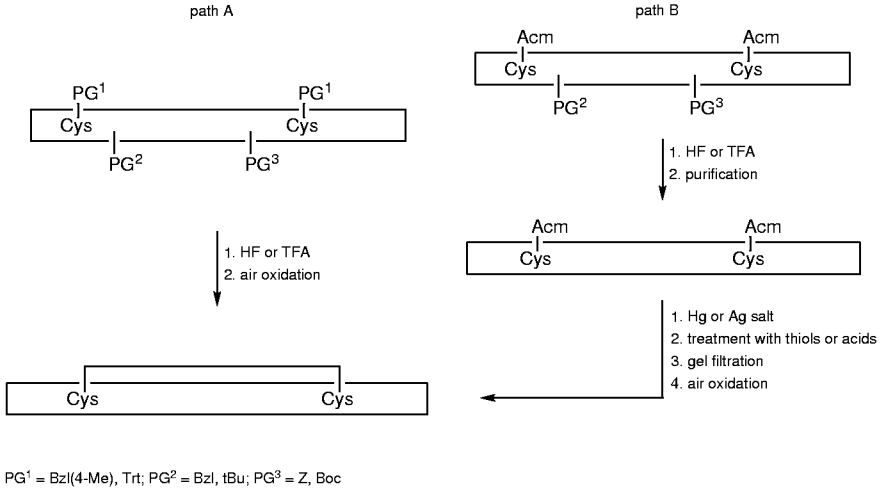
6.1.1.1 Oxidation of S-Unprotected Bis-cysteine Peptides

The conceptually most simple approach for the synthesis of intrachain mono-disulfide bridged peptides involves full deprotection of bis-cysteine peptides followed by careful oxidation of the resultant molecules. The clear advantage of this procedure is that it requires only one type of S-protecting group. Among the many basic reactions that have been developed for the formation of a single intramolecular disulfide bond, air oxidation is one of the most established methods. In this simple procedure, oxidation of the bis-cysteine peptide is carried out in the presence of atmospheric oxygen in aqueous or aqueous/organic media under slightly alkaline conditions, pH 7.5–8.5. High dilutions (10–100 μM) are used in order to prevent intermolecular disulfide formation as well as thiol/disulfide exchange reactions, particularly in cases where intramolecular cyclization is conformationally and thus thermodynamically less favored. Occasionally moderate levels of denaturants such as 0.5–3 M urea or 0.1–1.5 M guanidine hydrochloride are added to avoid aggregation.[7,8] Since the reaction relies on the concept that a linear peptide will appropriately fold into the most stable conformer to yield the correct disulfide pairings, at least in the case of naturally occurring single- and multiple-cystine peptides,[9] the air oxidation approach still remains the first choice for the oxidative refolding of the related synthetic cysteine precursors (see Section 6.1.5). However, even for simple mono-disulfide peptides careful attention has to be paid to optimize the reaction conditions (pH, peptide concentration, temperature, ionic strength, etc.) in view of possible dimerizations and oligomerizations that are responsible for low yields.

Oxidation by molecular oxygen most likely occurs via a radical mechanism[10–13] and the reaction rates are generally slow unless traces of metal ions are present which are known to drastically affect the reaction rate,[14] thus making control of the air-oxidation reaction rather difficult. The rates can also be significantly enhanced by adding charcoal to induce a surface-assisted catalysis of the intramolecular disulfide bond formation.[15] Nonetheless the difficult control of this oxidation procedure can lead to partial oxidation of Met and Trp residues when peptides are exposed for longer periods of time to air oxygen.[16]

The general synthetic procedure is shown in Scheme 2. For relatively short peptides, after full deprotection, oxidation is carried out without intermediate purification of the linear precursor (path A), since this operation may lead to partial oxidation in the chromatographic and workup steps, thus requiring an additional reduction step prior to the final oxidation to the cyclic intrachain disulfide-bridged peptides. In the case of larger peptides, purification of the linear S-protected precursor is advisable to improve the overall yields (path B). For this purpose, the acid-stable *S*-Acm protection, which is compatible with both the Boc and Fmoc strategies (see Vol. E 22a, Section 2.6.6), as well as the *S*-tBu group, which is compatible with the Fmoc strategy (see Vol. E 22a, Section 2.6.6), are generally employed. These protecting groups allow the acidolytic removal of the residual acid-labile protecting groups and the cleavage of the peptide from the resin, if required, in the thiol-protected form. Following the purification of such crude products, the Acm groups are then removed by mercury(II)[17] or silver(I) salts[18] and the *S*-tBu protection by trialkylphosphines.[19–21] In the case of sulfides the heavy metal ions have to be removed by treatment with excess of thiols or acids. Then the bis-cysteine peptides are exposed to oxidation by atmospheric oxygen, a process that is easily monitored by HPLC or spectroscopically with the Grassetti[22] and Ellman's reagents.[23] Since quantitative oxidation is difficult to achieve, traces of thiol compounds may lead in the subsequent workup of the highly diluted solutions to disulfide scrambling via thiol/disulfide exchange reactions and thus to oligomerization. Acid quenching of the solutions[24] or trapping of the free thiol groups with highly reactive soluble or polymer-bound reagents[25] is highly advisable. Despite these shortcomings of the air oxidation procedure, a large variety of mono-disulfide bridged cyclic bioactive peptides have been successfully prepared by this approach even on a large scale, for example oxytocin,[26–28] vasopressin,[29,30] somatostatin,[31–35] calcitonin,[36] and vasopressin antagonist.[37]

Scheme 2 Intramolecular Disulfide Formation by Atmospheric Oxygen



PG^1 = Bzl(4-Me), Trt; PG^2 = Bzl, tBu; PG^3 = Z, Boc

Besides molecular oxygen, numerous relatively mild oxidizing agents are known that convert thiols into disulfides and also considerably increase the reaction rates (Table 1). Among these potassium ferricyanide has been particularly useful for the oxidation of small-size water-soluble bis-cysteine peptides.[27,29,34,38] Potassium ferricyanide was also used in the selective formation of the first disulfide bond in several peptides containing two disulfide bridges.[39,40] However, side products may appear due to the oxidation of Met and Trp residues,[41] and an additional step is required to remove the oxidizing reagent. Early studies used hydrogen peroxide in aqueous solution,[24] but a comparative analysis clearly confirmed

for references see p 137

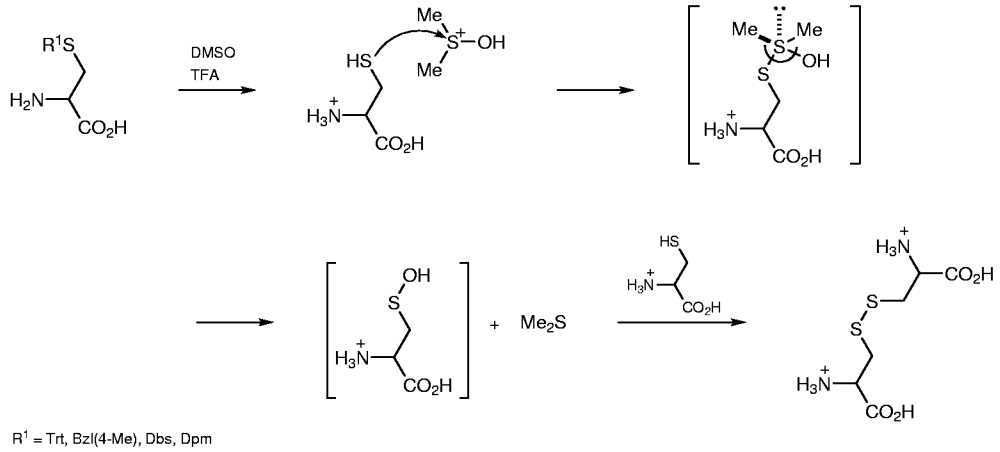
the high rates of oxidation of sensitive amino acids.[42] Such oxidation side products are also observed when the disulfide bonds are produced with iodine or iodoethane[24] in aqueous or aqueous/organic media.[38,42–45]

Table 1 Mild Reagents Used for the Oxidation of Bis-cysteine Peptides[2,9,24,27,29,34,38,42–45,47–51]

Reagent	Solvent	Ref
air oxygen	aq or aq/organic high dilution, pH 7.5–8.5	[2,9,24]
potassium ferricyanide	aq	[27,29,34,38]
hydrogen peroxide	aq	[24,42]
iodine	aq or aq/organic	[24,42–45]
iodoethane	aq or aq/organic	[24,42–45]
DMSO	aq, pH 1–8	[47–51]

Alternatively, DMSO-promoted oxidation of cysteine peptides[46] has found more frequent application as it can be applied over an extended pH range of 1 to 8.[47–51] Generally, faster oxidation rates are obtained with DMSO in acidic media and the proposed reaction mechanism is shown in Scheme 3. Since DMSO is known to disrupt aggregates, higher DMSO concentrations may serve to solubilize peptides. Deprotection of *S*-Acm peptide derivatives with silver(I) trifluoroacetate[18] followed by oxidation with 10% or even better 50% DMSO in 1 M HCl leads to disulfide formation in high yields.[52] The aqueous HCl serves to liberate the thiol functions from the silver(I) sulfides by precipitation of silver chloride[53] and additionally, it offers the acidity for the fast oxidation of the dithiol peptides. Modifications at sensitive amino acids such as Met, Trp, and Tyr are not observed by this oxidation method. However, problems can arise in removing larger amounts of DMSO from the reaction mixture, and adverse effects were noted in RP-HPLC.

Scheme 3 Intramolecular Disulfide Formation by Dimethyl Sulfoxide/Acid[46]

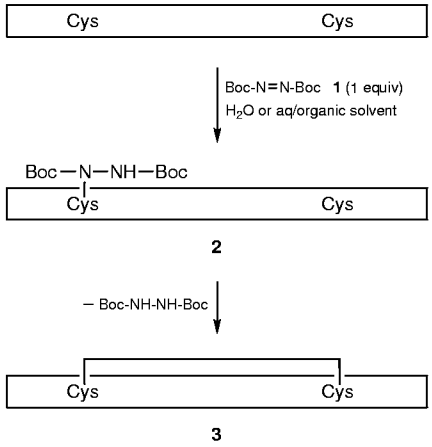


R^1 = Trt, Bzl(4-Me), Dbs, Dpm

The general conclusion to be drawn from the experience reported with these various oxidative methods is that with increasing reactivity of the oxidizing agents more side reactions are to be expected at sensitive amino acid residues, in particular at Met, Trp, and Tyr. In this context, the azodicarboxylic acid derivatives could represent a valid alternative[54,55] since these reagents are devoid of any side reaction at these sensitive residues.[56] The reaction of di-*tert*-butyl azodicarboxylate (**1**) with a peptide cysteine thiol leads to the intermediate formation of a sulfenohydrazide adduct **2** that reacts with the second thiol to generate the

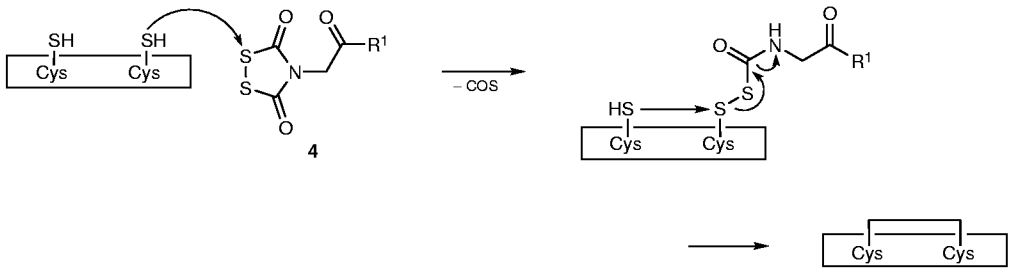
disulfide **3** as shown in Scheme 4.[57] Hence, 1 equivalent of reagent suffices for the oxidation of a bis-cysteine peptide. With the possibility of using the water-soluble azodicarboxylic acid dimorpholide or the ethyl and *tert*-butyl esters which are soluble in organic solvents,[58] oxidations can be performed in either aqueous or organic solvents.

Scheme 4 Intramolecular Disulfide Formation with Azodicarboxylic Acid Derivatives[57]



Similarly mild, but highly efficient oxidizing reagents are 2,2′-dithiodipyridine[22] and 5,5′-dithiobis(2-nitrobenzoic acid)[23] which were originally used for analytical quantification of thiol groups in peptides and proteins. While these type of reagents were used only sporadically for the synthesis of mono-disulfide peptides,[24] extensive use of this method is made in the site-directed synthesis of interchain disulfide bridges (see Section 6.1.3.2). This principle of readily reduced disulfides led to the introduction of bis(1-methyl-1*H*-tetrazol-5-yl) disulfide[59] and more recently dithiasuccinoylglycine or its polymer-bound derivative **4** as efficient oxidizing reagents for disulfide formation.[60] In the latter case the reduced species decomposes and the reaction becomes irreversible, although involving a thiol/disulfide exchange-type reaction (Scheme 5). A similar principle operates with the polymer-supported Ellman's reagent in which 5,5′-dithiobis(2-nitrobenzoic acid) is linked via the carboxy groups to a solid support.[61] Such polymer-bound reagents are highly promising, since they may allow the exploitation of all the advantages of solid-phase methodologies, but their practical usefulness is not yet sufficiently documented.

Scheme 5 Intramolecular Disulfide Formation with Dithiasuccinoylglycine[60]



R^1 = OH, (Dts-Gly-OH), Gly-βAla-Gly-PAL-PEG-PS

for references see p 137

Air Oxidation:

Synthesis of Somatostatin Using Path A (Scheme 2):[19]
Boc-Ala-Gly-Cys(StBu)-Lys(Adoc)-Asn-Phe-Phe-Trp-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Cys-(StBu)-OtBu (5.0 g, 2.01 mmol) was dissolved in ice-cold TFA (200 mL) containing 2-methylindole (5.4 g, 41 mmol) and anisole (20 mL). After 90 min, the soln was poured into ice-cold Et_2O. The precipitate was collected, washed with Et_2O, and dissolved in argon-sat. 80% AcOH (300 mL) and reacted with Ph_3P (10.5 g, 40 mmol). After 6 d stirring in the dark, the product was precipitated with EtOAc and dissolved in 0.01 M NH_4OAc buffer (34 L, pH 6.8). To the soln was added toluene (few mL) and air oxidation was allowed to proceed with gentle stirring at rt in the dark. After 7 d the soln was concentrated in batches to 1/10 of the volume and then lyophilized. The crude product was chromatographed on LH-20 with 0.1 M NH_4OAc buffer (pH 6.5) as eluent. Fractions containing the desired product were collected and lyophilized; yield: 1.67 g (46%); the product was characterized by HPLC and amino acid analysis.

Synthesis of Porcine Brain Natriuretic Peptide (pBNP-32) Using Path B (Scheme 2):[16]
The purified S-protected pBNP-32 [H-Ser-Pro-Lys-Thr-Met-Arg-Asp-Ser-Gly-Cys(Tacm)-Phe-Gly-Arg-Arg-Leu-Asp-Arg-Ile-Gly-Ser-Leu-Ser-Gly-Leu-Gly-Cys(Tacm)-Asn-Val-Leu-Arg-Arg-Tyr-OH] (28 mg, 7.4 μmol) in ice-cold TFA (4 mL) was treated with $AgBF_4$ (56 mg, 0.29 mmol) in the presence of anisole (32 μL) for 60 min, this was followed by the addition of dry Et_2O. The resulting powder was dissolved in 1 M AcOH (4 mL) and DTT (90 mg) was added. The soln was stirred at 25 °C for 3 h and the supernatant was gel-filtered on Sephadex G-25 with 1 M AcOH as eluent. The desired fractions were diluted with H_2O (125 mL) and the pH of this soln was adjusted to 7.5 with 5% NH_4OH. The soln was kept at 25 °C and the progress of air oxidation was monitored by HPLC. After 3 d, the mixture was lyophilized and the crude product was purified by preparative HPLC. The product was isolated by lyophilization; yield: 11% (based on the fully protected pBNP-32); $[\alpha]_D^{26}$ −60 (*c* 0.1, 1 M AcOH); the product was characterized by FAB-MS and amino acid analysis.

Dimethylsulfoxide/Acid Oxidation:

Synthesis of Oxytocin:[52]
[Cys(Acm)^{1,6}]-oxytocin [H-Cys(Acm)-Tyr-Ile-Gln-Asn-Cys(Acm)-Pro-Leu-Gly-NH_2] (0.79 μmol) was treated with AgOTf (8 mg, 40 equiv) in TFA (200 μL) in the presence of anisole (3 μL) at 4 °C for 1.5 h. Dry Et_2O (1 mL) was added to the mixture affording a powder. After being washed with dry Et_2O (3×3 mL) the product was treated with 50% DMSO/1 M HCl (1.3 mL) at rt for 7 h. After removal of the AgCl precipitate by filtration, the filtrate was diluted with H_2O (10 mL). The crude peptide was purified by HPLC; yield: 0.57 μmol (70%, based on [Cys(Acm)^{1,6}]-oxytocin); the product was characterized by FAB-MS, amino acid analysis, and HPLC.

6.1.1.2 Oxidation of S-Protected Bis-cysteine Peptides

The advantages of this approach are the opportunity to purify the intermediate S-protected derivatives, the introduction of the disulfide bonds in the early stages of the synthesis,[62] and also a "one-pot" deprotection/resin cleavage/oxidation method.[63–65] The main reagents used for this purpose are summarized in Table 2.

Table 2 Reagents for Oxidative Cleavage of S-Protecting Groups[45,48,66,67,72,89,90,93–96]

Reagent	Thiol Protecting Group	Solvent	Ref
I_2	Acm, Trt	organic (AcOH, MeOH, $CHCl_3$)	[45,66,67,72]
DMSO/H^+	Trt, Mob	organic (TFA)	[48,93]
Tl(III)	Mob, Acm, Ada	organic (TFA)	[89,90]
$ClSiH_3$	Acm, Mob, tBu	organic (TFA)	[94–96]

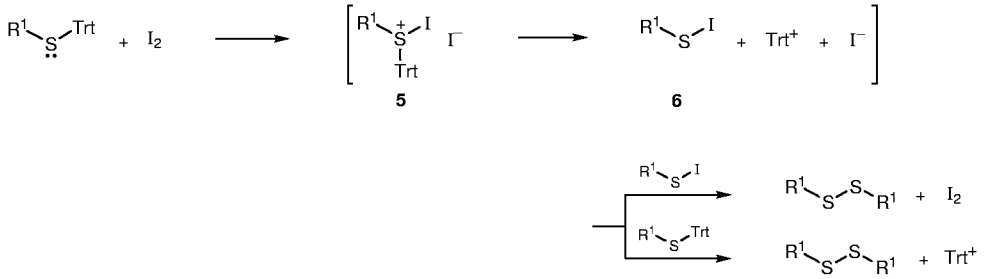
6.1.1.2.1 Intramolecular Disulfide Formation with Iodine

The prototype of oxidizing agents that convert S-protected cysteine peptides directly into cystine peptides (Scheme 1, route B) is iodine. This procedure was initially introduced by Kamber and Rittel for the oxidation of *S*-Trt cysteine peptides[66] and has since been extended to the *S*-Acm,[45,67] *S*-Tacm,[68] *S*-Phacm,[69–71] and *S*-Tmb derivatives.[64] It can also be

applied to other less common Acm- and alkyl-type S-protecting groups. Suitable solvents are aqueous or aqueous/organic media as well as neat MeOH, trifluoroethanol, dioxane, $CHCl_3$, and DMF. Aqueous or neat acetic acid and trifluoroacetic acid are also used. Since the protecting groups usually applied to other functionalities are stable to iodine,[62,66,72] this procedure can be used for disulfide formation in fully protected intermediate fragments as elegantly demonstrated in the first regioselective total synthesis of human insulin.[68] Iodine oxidation is now mainly employed as one step in the regioselective stepwise multiple-disulfide formation in solution or on solid support (see Sections 6.1.2 and 6.1.4).

A reasonable mechanism for the iodine oxidation of *S*-Trt cysteine peptides is given in Scheme 6.[45] Reaction of iodine with the divalent sulfur atom leads to the iodosulfonium ion **5** which is then transformed to the sulfenyl iodide **6** and the trityl cation. Sulfenyl iodides are also postulated as intermediates in the iodine oxidation of thiols to disulfides. The disulfide bond is then formed by disproportionation of two sulfenyl iodides or by reaction between the electrophilic sulfur atom of R^1-S-I and the nucleophilic S-atom of a second R^1-S-Trt molecule. The proposed mechanism suggests that any sulfur substitution (i.e., thiol protecting group) capable of forming a stabilized species on cleavage, such as the trityl cation, can be oxidatively cleaved by iodine.

Scheme 6 Intramolecular Disulfide Formation by Iodine[45]



The *S*-Trt and *S*-Acm cysteine derivatives are oxidatively cleaved by iodine with different reaction rates that strongly depend on the solvent employed (Table 3).[45] In $CHCl_3$, CH_2Cl_2, or trifluoroethanol the Trt group is cleaved within seconds, while for the Acm group the reaction takes hours to complete. Both S-protecting groups are, however, oxidatively cleaved with nearly identical rates when the reaction is carried out in MeOH, AcOH, or dioxane. By exploiting this particular feature, regioselective disulfide formation from *S*-Trt cysteine derivatives in the presence of *S*-Acm protection can be carried out, as efficiently exploited in the synthesis of a suitably protected disulfide-bridged fragment[45] required for the total synthesis of human insulin.[68] These differential reactivities can also be applied to the stepwise regioselective iodine-mediated formation of two disulfides.[73,74] Conversely, direct application of the iodine oxidation to a single-step formation of the multiple disulfide results in a complex mixture of products.[75]

Table 3 The Solvent Effect on $t_{1/2}$ Values for the Iodine Oxidation of *S*-Trt- and *S*-Acm-Protected Peptide Derivatives[45]

Solvent	$t_{1/2}$ for *S*-Trt (s)	$t_{1/2}$ for *S*-Acm (h)	Ref
MeOH	3–5	0.02	[45]
AcOH	70–80	0.67–0.75	[45]
dioxane	60	1.5–2	[45]
$CHCl_3$, CH_2Cl_2	1–2	1.5–2	[45]
HFIP/$CHCl_3$ (1:1)	1–2	> 2	[45]
TFE/$CHCl_3$ (1:1)	5–6	> 2	[45]

for references see p 137

Iodine oxidation carries some serious limitations due to the high reactivity toward sensitive amino acid residues, such as Tyr, Met, Trp, present in peptides; although, the extent of these side reactions can be largely controlled by the appropriate choice of solvents and reaction conditions.[45] Nevertheless, in view of the strongly sequence-dependent reactivity of Trp, formation of β-3-oxyindolylalanine was reported in the synthesis of human endothelin-1[76] and bombyxin-II,[77] in the latter case accompanied by Trp-Xaa peptide bond cleavage. In the case of human endothelin-1 oxidation of the Trp residue could be suppressed by addition of a large excess of Trp derivatives as scavengers.[76] Moreover, in the presence of Trp residues an additional side reaction may involve the indole side-chain group with generation of indole-2-thioether derivatives. This reaction is expected to occur via the intermediately formed sulfenyl iodide, since the less-reactive sulfenyl chlorides are well established reagents for conversion of Trp residues into the related 2-thioether derivatives.[78] The identical reaction was used preparatively for the Cys/Trp side-chain to side-chain cyclization of phalloidin peptides.[79] Model studies have shown that this serious side reaction may occur whenever conditions are applied which lead to a markedly retarded rate of disulfide bond formation and, particularly, if the ring size of the resulting thioether compound is favoring such cyclization.[80] An additional side reaction discovered more recently, involves the S→O transfer of the acetamido methyl group onto unprotected hydroxy groups of serine, threonine,[81,82] and hydroxyproline residues.[83] In the synthesis of 99-residue HIV-protease, removal of the Acm group with iodine was so slow that partial oxidation of methionine occurred.[84]

The iodine oxidation method has also been applied in one-pot synthetic strategies to mono-disulfide peptides in which the bis-cysteine peptide is cleaved from the resin, deprotected, and oxidized in a single step.[63,64] A comparative analysis of the iodine procedure (TFA/10 equiv I_2) with other TFA/oxidant mixtures using oxytocin as the target model peptide clearly revealed formation of side products to a significant extent.[85]

Besides molecular iodine, *N*-iodosuccinimide,[86] cyanogen iodide,[87] and diacetoxyphenyliodine[88] have been proposed as alternative sources of the iodonium ion.

Boc-Ala-Gly-Cys-Lys(Boc)-Asn-Phe-Phe-Trp-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Cys-OtBu:[45]

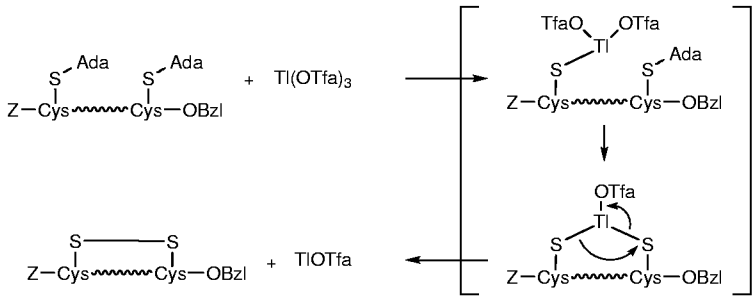
CAUTION: *Iodine is a skin and mucous membrane irritant. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

A soln of Boc-Ala-Gly-Cys(Acm)-Lys(Boc)-Asn-Phe-Phe-Trp-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser-(tBu)-Cys(Acm)-OtBu (24.8 g, 0.01 mol) in DMF (1L) was poured in one portion (requiring about 30 s) into a well-stirred soln of I_2 (30.5 g, 0.12 mol) in MeOH (9 L). After 5 min, a soln of ascorbic acid (23 g) in citrate buffer (2 L, pH 5) was added, producing within about 15 s a colorless and slightly turbid soln. This soln was concentrated to about 1.5 L. The resultant suspension was then diluted with H_2O (2 L) and stirred at 0 °C for 1 h yielding a white precipitate. This was collected, washed with Et_2O (3×150 mL), and then purified by counter-current distribution; yield: 16.8 g (78%).

6.1.1.2.2 Intramolecular Disulfide Formation with Thallium(III) Trifluoroacetate

As a useful alternative to iodine, thallium(III) trifluoroacetate represents a relatively mild oxidant that can be used to cleave various S-protecting groups with spontaneous formation of the disulfide group.[89] The reaction of thallium(III) trifluoroacetate with *S*-Ada protected bis-cysteine peptide derivatives is thought to proceed according to the mechanism shown in Scheme 7. The reagent thallium(III) trifluoroacetate acts first as a soft Lewis acid to cleave the Ada group, then as a mild oxidant to form the disulfide bond. During the reaction thallium(III) is reduced to thallium(I) and the S-protecting group is liberated as a cation which is trapped with appropriate scavengers.

Scheme 7 Intramolecular Disulfide Formation with Thallium(III) Trifluoroacetate[89]



This cleavage/oxidation reaction is mainly carried out in TFA, since this solvent represents the optimal choice to solubilize both fully protected and deprotected peptides. Among the S-protecting groups generally used in peptide chemistry, the *S*-Acm, *S*-Trt, *S*-Mob, *S*-tBu, *S*-Ada, and *S*-Dbs derivatives are deprotected smoothly by this procedure to yield the related cystine peptides. The usefulness of this procedure is demonstrated by the successful syntheses of oxytocin, urotensin II, and human calcitonin gene-related peptides.[90,91] It may well find wider application in the synthesis of both single- and multiple-disulfide peptides, in the latter case as one of the regioselective steps. This oxidative deprotection procedure in TFA lends itself to the one-pot strategy for peptides synthesized on solid support,[63,64] but in a comparative analysis with other methods it proved to be remarkably less efficient than other oxidants.[85]

When cystine peptides are exposed to the thallium(III) reagent in aqueous solution, oxidation to cysteic acid occurs. Therefore, precipitation of the cystine peptide from the reaction with diethyl ether is required prior to workup. Moreover, Trp residues undergo modifications with formation of several unidentified side products and Met residues are partially oxidized to the corresponding sulfoxide.[92] Trp needs to be protected during the disulfide bond formation, and thus, additional synthetic steps are required after disulfide formation for the removal of this protecting group as well as for the reduction of the Met(O) by procedures not affecting the disulfide group.[91] Other sensitive amino acid residues such as His and Tyr are fully resistant on exposure to the thallium(III) reagent. The practical limitation of this reagent is, however, its high toxicity. Also thallium, as other metals, can be difficult to remove quantitatively from the cystine peptides.

Synthesis of Oxytocin:[91]

CAUTION: *Thallium salts are very toxic when inhaled, ingested, or absorbed through the skin. Appropriate safety precautions and procedures should be taken when handling and disposing of thallium compounds.*

To an ice-cold soln of Z(OMe)-Cys(Mob)-Tyr-Ile-Gln-Asn-Cys(Mob)-Pro-Leu-Gly-NH_2 (37 mg, 26.8 μmol) in TFA (1 mL) and anisole (0.1 mL), thallium(III) trifluoroacetate (15.5 mg, 28.8 μmol) was added. After 60 min the product was precipitated with hexane, the supernatant was removed, and the residue treated with dry Et_2O. The resulting solid was dissolved in 1 M AcOH and chromatographed on Sephadex G-15 with the same solvent. The product was further purified by preparative HPLC; yield: 7.2 mg (27%); $[\alpha]_D^{20}$ −25.2 (*c* 0.2, H_2O); the product was characterized by amino acid analysis.

6.1.1.2.3 Intramolecular Disulfide Formation with Dimethyl Sulfoxide/Acid

The use of DMSO as an oxidant[46] produces water and dimethyl sulfide as a nonreactive byproduct (Scheme 3). This method has been used in aqueous and aqueous/organic media in the pH range from 1 to 8 (see Section 6.1.1.1), and also in neat TFA.[48] Under the latter

for references see p 137

conditions the *S*-Trt and *S*-Mob cysteine derivatives are deprotected with in situ conversion of the resulting bis-cysteine peptides into the cystine peptides in good yields.[93] This procedure has been adopted for the one-pot strategy where it proved to be highly superior to the TFA/I_2 and TFA/thallium(III) trifluoroacetate methods since quantitative oxidation was observed without detectable side products.[85] The full stability of the *S*-Acm cysteine derivatives under these conditions opens other promising approaches to the synthesis of multiple-cystine peptides.

Synthesis of Oxytocin:[85]
H-Cys(Trt)-Tyr(tBu)-Ile-Gln(Trt)-Asn(Trt)-Cys(Trt)-Pro-Leu-Gly-Rink-Linker-PEGA-resin (60 mg, 0.09 mmol · g^{-1}, 5.4 μmol) was suspended in 95% aq TFA (10 mL) for 1 h at rt, followed by the addition of DMSO (2 mL) at 0 °C. After 30 min at 0 °C and 1 h at rt, the resin was rinsed with 95% aq AcOH. The solvent was removed under reduced pressure and the product precipitated with cold Et_2O and purified by HPLC; yield: 3.0 mg (56%). The product was characterized by MALDI-MS, amino acid analysis, and HPLC.

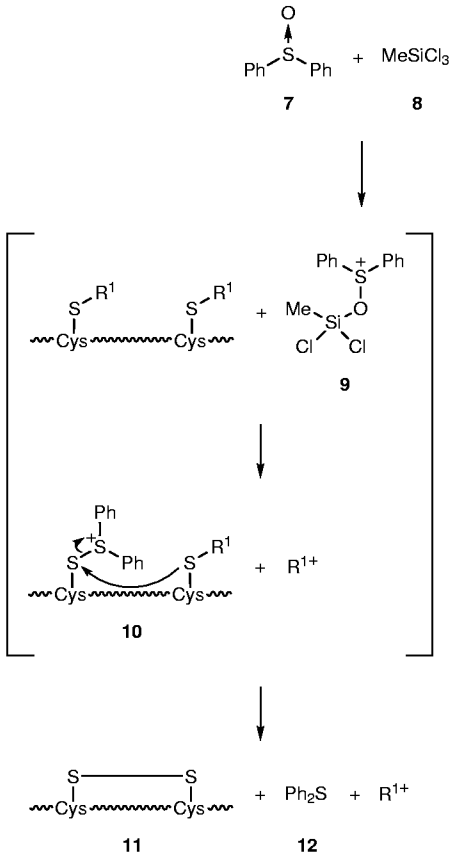
Synthesis of α-Human Calcitonin Gene-Related Peptide (α-hCGRP):[93]
Protected α-hCGRP [Boc-Ala-Cys(Ada)-Asp-Thr-Ala-Thr-Cys(Ada)-Val-Thr-His-Arg(Mts)-Leu-Ala-Gly-Leu-Leu-Ser(Bzl)-Arg(Mts)-Ser(Bzl)-Gly-Gly-Val-Val-Lys(Z)-Asn-Asn-Phe-Val-Pro-Thr-Asn-Val-Gly-Ser(Bzl)-Lys(Z)-Ala-Phe-NH_2] (20.5 mg, 4.05 μmol) was treated with a mixture of *m*-cresol (244 μl), thioanisole (600 μl), TFA (3.45 mL), and TMSOTf (970 μl) in an ice bath for 2 h, then cold Et_2O was added. The precipitate was treated with 10% (v/v) DMSO in TFA containing 1% anisole (5.0 mL) at rt for 1 h. The crude product was precipitated with Et_2O and collected by centrifugation, then washed thoroughly with Et_2O and purified by HPLC; yield: 2.7 mg (17%, based on the protected peptide); the product was characterized by FAB-MS, amino acid analysis, and HPLC.

6.1.1.2.4 Intramolecular Disulfide Formation with Chlorosilane/Sulfoxide

Disulfide formation using a mixture of chlorosilanes and sulfoxides is a novel oxidation procedure developed independently by Akaji et al.[94,95] and Koide et al.[96] The reagents oxidatively cleave almost all types of S-protecting groups including those which are not oxidatively cleaved by iodine. Therefore, this procedure offers an additional dimension to the strategies of regioselective multiple-disulfide formation (see Section 6.1.4.2).

The proposed mechanism of the oxidative cleavage of S-protecting groups by the chlorosilane/sulfoxide procedure is outlined in Scheme 8.[95] The first reaction is considered to be formation of the sulfonium cation **9** from diphenyl sulfoxide (**7**) and the oxygenophilic silyl compound **8**. The formation of a sulfonium ion of this type is known and has been utilized for the reduction of sulfoxides.[97] Subsequent electrophilic attack of **9** on the sulfur atom of the S-protected cysteine residue leads to the formation of intermediate **10**, whereby the nature of the silyl chloride employed should be the main factor that influences the electrophilicity of **9**. The postulated intermediate **10** may then act as the electrophile and react with another S-protected cysteine residue to generate the disulfide **11** and the inert byproduct diphenyl sulfide (**12**). This final step is analogous to the reaction of a sulfenyl iodide as discussed in Section 6.1.1.2.1.

Scheme 8 Intramolecular Disulfide Formation with Chlorosilane/Sulfoxide[95,96]



The cleavage/oxidation reaction is fast and no disulfide exchange is observed when applied to the synthesis of multiple-disulfide peptides (Section 6.1.2). Again TFA is usually used as the solvent for solubility reasons. The thiol-protecting groups of the Acm-type as well as the *S*-Mob, *S*-[Bzl(4-Me)], and *S*-tBu derivatives are cleaved oxidatively by this procedure within 10–60 minutes, while the *S*-iPr and *S*-[Bzl(4-NO_2)] protection is unaffected. A combination of methyltrichlorosilane and diphenyl sulfoxide is most commonly used, although certain chlorosilane/sulfoxide combinations are more effective than others, depending upon the S-protecting groups.[96] So far, this new procedure has been applied successfully to the synthesis of various cystine peptides such as oxytocin,[95] human brain natriuretic peptide (hBNP),[95] and somatostatin.[95] Due to the fast reaction in TFA, the chlorosilane/sulfoxide method is also well suited for the one-pot strategy.[65]

The chlorosilane/sulfoxide method is compatible with most amino acid side-chains including the sensitive His, Tyr, and Met residues. The Trp residue, however, has to be protected at the indole side-chain group as the N^{in}-formyl derivative to avoid its quantitative chlorination.[95] The indole ring acts as an excellent scavenger to push the reaction to a simple reduction of the sulfoxide. The N^{in}-formyl group is later removed by rapid base treatment without seriously affecting the disulfides.

Synthesis of Human Brain Natriuretic Peptide (hBNP):[95]

CAUTION: *Methyltrichlorosilane is highly flammable and the vapor irritates eyes. It is also corrosive to the respiratory tract. Appropriate safety precautions and procedures should be adopted when handling this substance.*

for references see p 137

To a soln of [Cys(Tacm)^{10,26}]-hBNP (8.4 mg, 2.5 μmol), prepared by Fmoc/tBu solid-phase synthesis followed by FPLC purification, in TFA (8.4 mL) $MeSiCl_3$ (27 μL; 250 μmol), PhS(O)Ph (4.6 mg, 25 μmol), and anisole (25 μL) were added at 25 °C. After 10 min, dry Et_2O (50 mL) and 1 M AcOH (10 mL) were added to the mixture. The crude product in the aqueous phase was purified by FPLC to give homogeneous hBNP; yield: 3.9 mg (50%, based on [Cys(Tacm)^{10,26}]-hBNP); $[\alpha]_D^{30}$ −57.1 (*c* 0.2, 1 M AcOH); the product was characterized by amino acid analysis and HPLC.

6.1.1.3 Disulfide Formation on Solid Support

The principle of oxidation of bis-cysteine peptides on resin consists of deprotection and oxidation of a cysteine pair in a single step or in two separate steps to form the disulfide bond while the peptide chain remains attached to the solid support. The strategy is especially effective for the formation of intramolecular disulfides since the solid support simulates high dilution via the pseudo-dilution effect, thus, largely preventing intermolecular reactions.[63,98–100]

6.1.1.3.1 Oxidation of Resin-Bound S-Unprotected Bis-cysteine Peptides

Some of the methods of disulfide formation in solution can be transferred to the solid-phase method, wherein particular attention has to be paid to the solvents that effectively swell the peptide-resin. In early examples air oxygen or 1,2-diiodoethane were used as oxidizing agents,[39,101–103] and also 1 M aqueous potassium ferricyanide/DMF (1:10).[104] Besides the *S*-tBu thiol protection which is compatible with the Fmoc chemistry and allows orthogonal reductive cleavage on the resin,[104] the *S*-Fm protection, which is compatible with the Boc-chemistry, has been successfully used.[63] The selective removal of the Fm group with piperidine/DMF/β-ME followed by oxidation with Ellman's reagent in DMF or the direct deprotection/oxidation by piperidine/DMF (1:1) leads to yields that are superior to those obtained with the identical chemistry in solution. Similar results have been obtained by this solid-phase chemistry with 2-(2,4-dinitrophenyl)ethyl thiol protection.[105]

The Tmb thiol protection has also been exploited for the peptide dithiol formation on a resin by using dilute TFA.[64] The use of the mild oxidant CCl_4 in NMP[106] for the oxidation of a resin-bound dithiol peptide is an interesting variation to this synthetic strategy.[64]

Synthesis of Oxytocin:[63]
Boc-Cys-Tyr[Z(Br)]-Ile-Gln-Asn-Cys-Pro-Leu-Gly-MBHA-resin [prepared from the corresponding *S*-Fm-protected resin-linked peptide (50–60 mg, 0.35 mmol/g, 19 μmol) by treatment with piperidine/DMF/β-ME (10:10:0.7) at 25 °C for 3 h] was suspended in pH 8 buffered DMF (3 mL) in an open vessel at 25 °C for 1 h, washed, deblocked with TFA/CH_2Cl_2, and cleaved with HF/anisole (9:1) at 0 °C for 1 h to provide the crude peptide; yield 16 μmol (88%). This material was purified by MPLC on a Lobar RP-8 column; yield: 8.5 μmol (53%); the product was characterized by amino acid analysis and HPLC.

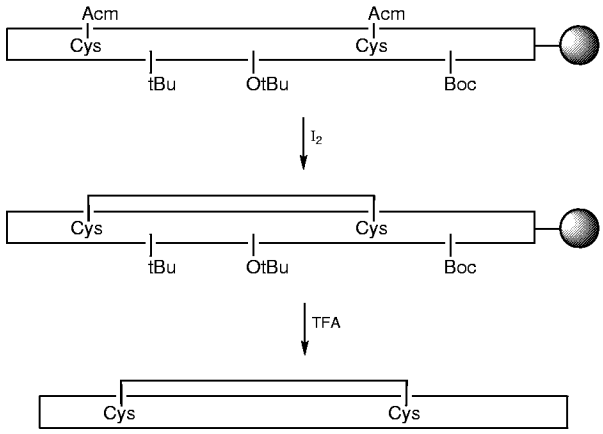
6.1.1.3.2 Oxidative Deprotection of Resin-Bound S-Protected Cysteine Peptides

Much of the experience gained in the field of solution-phase oxidative deprotection of cysteine peptides can be applied to solid-phase chemistry. For this purpose the iodine, thallium(III), and chlorosilane/sulfoxide procedures have been usually employed,[63,107–112] generally, the results obtained with the thallium(III) reagent are better than with iodine.

A general procedure for the iodine reaction in the solid phase is shown in Scheme 9. Both Boc and Fmoc chemistry can be used to assemble the linear S-protected bis-cysteine peptides and for thiol protection the well-established Acm and Trt groups are usually used. Suitable solvents for the thiol-deprotection/oxidation step by iodine to form the disulfide are CH_2Cl_2, DMF, or aqueous AcOH. The final deblocking and cleavage from the resin is carried out under standard conditions. Modification at sensitive amino acid residues caused

by iodine can be suppressed, since the reaction is carried out on the fully protected peptide-resin. However, the deblocking/cleavage reaction has to be carefully controlled to prevent scrambling of the disulfide already formed, hence, the addition of thiol scavengers or prolonged reaction times have to be avoided.

Scheme 9 Iodine-Mediated Disulfide Formation on Solid Support[107]



Disulfide formation in resin-bound peptides is particularly advisable in the case of small-size cystine peptides with a high tendency to intermolecular disulfide formation when oxidation is carried out in solution.[113] Additionally, removal of the reagents is facilitated in comparison to the solution methods. This is particularly useful in the case of the thallium(III) procedure which has been applied successfully in the solid-phase mode using DMF as solvent in the oxidative cleavage of the Acm groups, although generally higher yields are obtained with TFA as solvent.[63] Interestingly, in DMF no reaction is observed with *S*-Trt derivatives.[63] The thallium(III)-mediated oxidation has also been applied to the solid-phase disulfide formation of oxytocin using the acid-labile Xan group for cysteine protection.[114] In a different approach to solid-phase disulfide formation, simultaneous cysteine oxidation and release of a protected disulfide-bridged peptide from 2-chlorotrityl resin by using iodine in AcOH/TFE/CH_2Cl_2 has been investigated. This allows the preparation of suitably protected peptide fragments for subsequent synthetic steps.[115] To obtain moderate to good yields by this solid-phase approach, preliminary model studies on the reaction conditions such as solvent, resin substitution level, support characteristics (polarity, extent of cross-linking), and monitoring methods are recommended.

H-Cys-Arg-Gly-Asp-Phe-Cys-NH_2:[113]

The peptide resin Boc-Cys(Acm)-Arg(Tos)-Gly-Asp(OCy)-Phe-Cys(Acm)-MBHA-resin (0.5 mmol) was swollen with CH_2Cl_2 (10 mL) and suspended in DMF (10 mL). A soln of I_2 (381 mg, 1.5 mmol) in MeOH (10 mL) was added and the mixture was stirred for 3 h at rt. The resin was washed repeatedly with DMF and CH_2Cl_2/MeOH (1:1) and dried. It was then treated with HF/anisole (95:5) at 0 °C for 1 h. After removal of the HF, the residue was washed repeatedly with Et_2O and extracted with 5% AcOH, followed by lyophilization. The product was purified by HPLC and converted into its hydrochloride by repeated lyophilization from HCl-containing H_2O; yield: 51 mg (16%); $[\alpha]_D^{28}$ −68.8 (*c* 1.0, H_2O); the product was characterized by amino acid and MS analysis.

When H-Cys(Acm)-Arg-Gly-Asp-Phe-Cys(Acm)-NH_2, prepared from the same protected peptide resin by HF treatment, was subjected to I_2 oxidation in soln, the peptide and other reagents could not be separated, and the desired product was not obtained.

for references see p 137

6.1.2 Intrachain-Bridged Multiple-Disulfide Peptides

Intramolecular multiple disulfide bonds are formed either by simple air oxidation using a single-type S-protecting group or by regioselective methods based on an orthogonal thiol-protecting scheme. The air oxidation method is a basic procedure that depends on the thermodynamic stability of the target peptide (see Section 6.1.5). For the synthesis of naturally occurring peptides and proteins this oxidative refolding strategy is usually applied. However, even for natural peptides the correct cystine network is not formed in all cases. In such examples, as well as for the synthesis of nonnatural isomers or of de novo designed constructs, efficient chemical control of the production of multiple disulfide bonds is required.

The regioselective methodology discussed in this section is a rather complex multiple-step procedure which has to be highly efficient to guarantee the formation of the target disulfide bonds including disulfide bridges of thermodynamically less stable nonnatural cystine networks. In this approach, the multiple disulfides are formed successively step-by-step and in regioselective manner using a combination of the oxidation methods described in Sections 6.1.1–6.1.1.3.2 for mono-disulfide formation. For practical convenience the synthetic steps can be carried out in solution or on solid-phase or by combining both procedures.

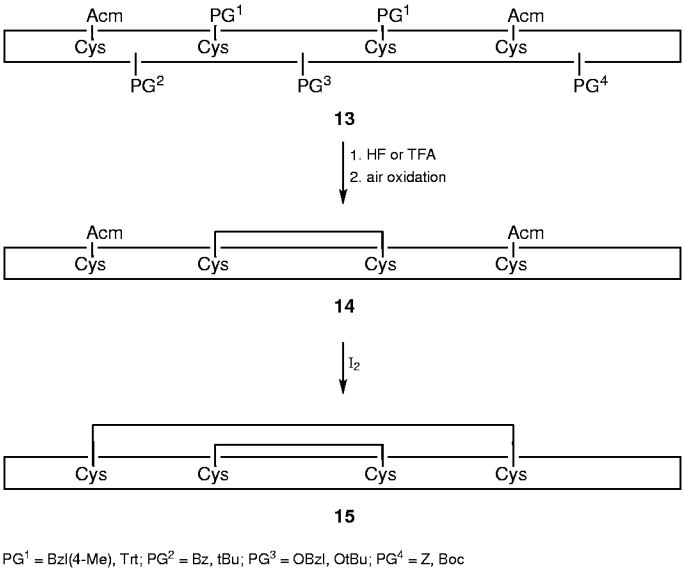
6.1.2.1 Regioselective Bis-disulfide Formation in Solution

The general strategy is based on selective deprotection and oxidation of pairwise cysteine residues, as specified by the orthogonal protection scheme selected for this purpose. Since multiple disulfide bonds are formed step-by-step, reaction conditions are required that prevent breaking or scrambling of the disulfide bonds already formed. Therefore, throughout the synthesis, exposure to thiols or alkaline conditions as well as lengthy reaction times that may cause disulfide disproportionation must be avoided.

6.1.2.1.1 Disulfide Formation by a Combination of Air Oxygen and Iodine

This combination of oxidation methods is the most general procedure for regioselective formation of two intramolecular disulfide bonds, although many variations regarding the thiol protection strategy can be found in the literature. This strategy is outlined in Scheme 10 which takes into account the compatibility of the thiol protecting groups with Boc and Fmoc chemistry. For this procedure two pairs of S-protecting groups are essential that are cleaved in a fully orthogonal manner by different mechanisms. Generally, a combination of acid-labile S-protecting groups and acid-stable, but iodine-cleavable S-protecting groups are employed in this synthetic route. The acid-labile thiol-protecting groups *S*-Trt or *S*-Tmb derivatives are used in Fmoc-based synthesis and the *S*-[Bzl(4-Me)] derivatives in Boc chemistry. The second pair of cysteine residues is usually protected with the Acm group which is stable to acidolysis, but smoothly cleaved by iodine.

Scheme 10 Bis-disulfide Formation with Air Oxygen and Iodine



PG^1 = Bzl(4-Me), Trt; PG^2 = Bz, tBu; PG^3 = OBzl, OtBu; PG^4 = Z, Boc

For the formation of the first disulfide bond, the acid-labile thiol protecting groups were removed in the acidolytic deprotection/cleavage of the protected peptide (or peptide-resin) **13** by treatment with TFA (Fmoc chemistry) or HF (Boc chemistry). Air oxidation was then employed to generate the first disulfide bond yielding **14**. To improve the reaction rates and to bypass solubility problems with the target peptide in aqueous buffer as an alternative to air oxygen, potassium ferricyanide, and DMSO may be used as oxidants. The second disulfide bridge is then generated by oxidative cleavage of the two Acm groups with iodine to give **15**. The reaction conditions of this synthetic step have to be optimized to avoid overoxidation as well as to minimize modification of sensitive residues. The sulfenyl iodides which are formed as reactive intermediates as well as iodine itself can theoretically attack the already formed disulfide with consequent scrambling. However, such side reactions have been observed only sporadically and when occurring, these have been attributed to the thermodynamic stability of the side products.[83,116,117] It is further confirmed that this predictable side reaction is largely prevented by the fast consumption of the sulfenyl iodide via reaction with the second *S*-Acm derivative by the successful regioselective formation of the two disulfide bridges of human endothelin-1.[118] Human endothelin-1 shows a relatively high stability for a second isomer, since on oxidative refolding besides the natural isomer a second isomer is formed in 25% (see Section 6.1.5).

This regioselective cysteine pairing procedure was used for the synthesis of conotoxin GI[119] and mast cell degranulating peptide[120,121] with a combination of the Bzl(4-Me) and Acm thiol protection. Other acid-labile thiol protecting groups well suited for this strategy are the Trt, Tmb, and Xan groups, since these are fully orthogonal with the *S*-Acm protection. For the synthesis of conotoxin GI[122] an alternative protection scheme was used that is based on the *S*-tBu and *S*-Acm cysteine derivatives. This combination allows the reductive cleavage of the two *S*-tBu groups for air oxidation prior to the iodine oxidation step of the Acm-protected cysteine pair. Moreover, the *S*-Fm protection that is compatible with Boc chemistry and oxidatively cleaved with piperidine/DMF[121] could serve for the generation of the first disulfide bond to allow iodine-mediated production of the second disulfide bond from the pair of Acm-protected cysteine residues. The use of the Phacm group for protecting one pair of cysteine residues can be envisaged as it is selectively removed by the action of the enzyme penicillin amidohydrolase even in presence of the Acm group.[69–71]

for references see p 137

If Trp residues are present in the molecule, it is advisable to protect the indole function in order to suppress possible side reactions. For this purpose usually the N^{in}-formyl derivative is used which is removed by rapid base treatment without affecting the disulfide bonds already formed.[118]

The unambiguous formation of disulfide bonds by the synthetic strategy discussed in this section is also well suited for determining the cystine networks of newly isolated natural peptides or isomers.[123]

Synthesis of Human Endothelin-1:[118]
Following N^{α}-deprotection of Boc-Cys(Acm)-Ser(Bzl)-Cys[Bzl(4-Me)]-Ser(Bzl)-Ser(Bzl)-Leu-Met-Asp(OCy)-Lys[Z(Cl)]-Glu(OCy)-Cys[Bzl(4-Me)]-Val-Tyr[Z(Br)]-Phe-Cys(Acm)-His-Leu-Asp(OCy)-Ile-Ile-Trp(For)-OBzl (4.3 g, 1.08 mmol) with TFA (40 mL) the resulting product was treated with HF/*p*-cresol (8:1; 100 mL) at −2 °C for 1 h. The crude product (3 g) was dissolved in a 0.1 M NH_4OAc buffer containing 8 M urea (2 L, pH 7.2), to which a soln of $K_3Fe(CN)_6$ (527 mg, 1.6 mmol) in H_2O (16 mL) was added slowly over 30 min. The oxidation reaction was quenched with AcOH and the mixture was passed through a column of Dowex 1 X2 (acetate form). The product was adsorbed onto a column of Diaion HP-20, which was washed with H_2O and eluted with MeCN/0.1% TFA (8:1). The eluate was concentrated and the residue (3 g) was dissolved in 80% aq MeOH (350 mL) containing HCl (15 equiv). This soln was added in one portion to a soln of 0.1 M I_2 in MeOH (15 equiv, 160 mL). After 30 min, the mixture was quenched with an excess of ascorbic acid in citrate buffer (pH 5). The product was isolated using a column of Diaion HP-20 as described above. The eluant was concentrated, the residue was suspended in H_2O (250 mL), and poured into 0.1 M NaOH (25 mL) at 2 °C. After stirring for 2 min, the mixture was neutralized with 1 M HCl (2.5 mL), then the product was isolated using a column of Diaion HP-20 as described above. The crude product was purified by ion-exchange chromatography on DEAE-cellulose, followed by preparative HPLC on Vydac 30 C18 (57×300 mm) and Asahipak ODP-90 (21.5×300 mm); yield: 0.5 g (20%); $[\alpha]_D^{25}$ −122 (*c* 0.27, 1% AcOH); the product was characterized by amino acid analysis.

Synthesis of a Fragment of Bovine Pituitary Peptide:[121]
H-Leu-Gly-Arg(Tos)-Thr(Bzl)-Gly-Ser(Bzl)-Cys(Acm)-Gln-Asn-Cys(Acm)-Trp(For)-Leu-Cys[Bzl(4-Me)]-Ser(Bzl)-Gln-Asn-Cys[Bzl(4-Me)]-Ala-Gly-Val-OCH_2-PAM-Phe-resin (0.52 g) was treated with HF/*p*-cresol (9:1; 5 mL) at 0 °C for 1 h. The product was purified by preparative MPLC. Fractions containing the purified major product were combined and added slowly dropwise to a 0.05 M Tris·HCl buffer, (pH 8, 500 mL) with vigorous stirring in a flat crystallization dish. When the addition was complete, the pH was adjusted to 8 with 0.1 M Tris·HCl buffer (pH 8) and the oxidation was allowed to proceed for 36 h at rt. The mixture was loaded onto a column of Diaion HP-20 and after exhaustive column washing with H_2O, the adsorbed peptide was eluted with MeCN/H_2O 0.1% TFA (8:1) and lyophilized. The residue was dissolved with H_2O/MeOH (3:1) in the presence of HCl (15 equiv) and the soln was added in one portion into a vigorously stirred soln of 0.1 M I_2 in MeOH (16.5 mL). The oxidation was stopped after 40 min by addition of a sat. soln of ascorbic acid in 0.05 M sodium citrate buffer, pH 5, until total decoloration. After adsorption-elution on Diaion HP-20 as above and lyophilization, the salt-free peptide was treated with 0.1 M NaOH at 2 °C in a UV-cell for 2 min. Fast UV scans of the soln showed immediate disappearance of the 300-nm absorption due to the formyl group. The reaction was quenched by addition of 10% AcOH, the peptide material was once again isolated by means of the Diaion HP-20 column and purified by MPLC; yield: 3 mg (6%); the product was characterized by TOF-MS and amino acid analysis.

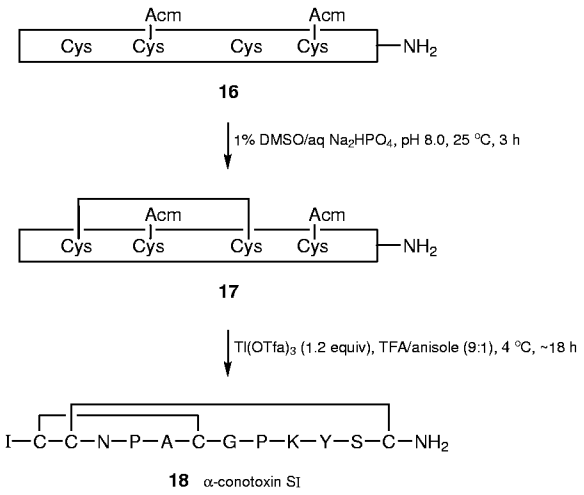
6.1.2.1.2 Disulfide Formation by a Combination of Dimethyl Sulfoxide/Acid and Thallium(III)

In this synthetic approach, thallium(III) trifluoroacetate (see Section 6.1.1.2.2) is employed for the regioselective disulfide generation in the second step. Although the thallium(III) reagent represents a valid alternative to iodine, only a few examples have been reported.

The basic procedure is outlined in Scheme 11.[107] The sequence of synthetic steps is analogous to that described in Section 6.1.2.1.1. The two disulfide bonds are formed stepwise by successive oxidation with air oxygen or aqueous DMSO followed by the thallium(III) reagent. Again for the formation of the first disulfide, acid-labile thiol protecting groups such as the Tmb group are used, whereas the Acm group is employed for the formation of the

second disulfide bridge by oxidative cleavage with the thallium(III) reagent. No disulfide exchange reaction is observed in the second synthetic step. A detailed examination of the efficiency of this procedure has been carried out in the regioselective synthesis of α-conotoxin SI.[107]

Scheme 11 Bis-disulfide Formation with Dimethyl Sulfoxide and Thallium(III) Trifluoroacetate[107]



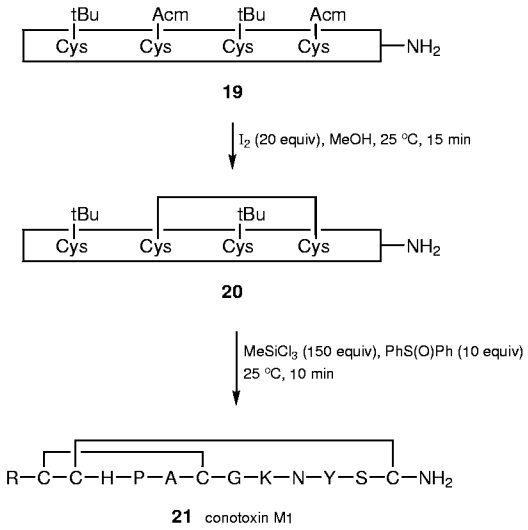
Synthesis of α-Conotoxin SI (18):[107]
The peptide-resin H-Ile-Cys(Mob)-Cys(Acm)-Asn-Pro-Ala-Cys(Tmb)-Gly-Lys(Boc)-Tyr(tBu)-Ser-(tBu)-Cys(Acm)-NH-PAL-resin (25 mg, 4.5 μmol) was swollen in CH_2Cl_2 (2 mL) for 5 min, filtered, and treated with TFA/CH_2Cl_2/Et_3SiH/H_2O/anisole (95:4:0.5:0.5:0.5) to provide the crude [bis-thiol,bis-Acm]-intermediate **16**. This crude product (6.1 mg) was dissolved in 0.01 M phosphate buffer (pH 8.0, 4.0 mL) and DMSO (40 μL) was added. After 3 h at 25 °C, the reaction was quenched by lyophilization to provide the [mono-disulfide, bis-Acm]-intermediate **17**; yield: 4.2 mg. This intermediate (~2.8 μmol) was dissolved in TFA/anisole (19:1, 3.0 mL), cooled in an ice bath and treated with thallium(III) trifluoroacetate (1.8 mg, 1.2 equiv). After 18 h at 4 °C, the mixture was concentrated, the peptide precipitated with Et_2O (10 mL), and centrifuged. The trituration/centrifugation cycle was repeated twice to ensure removal of the Tl salt. After HPLC purification the homogeneous product **18** was obtained; yield: 1.6 mg (27%); the product was characterized by FAB-MS.

6.1.2.1.3 Disulfide Formation by a Combination of Iodine and Chlorosilane/Sulfoxide

By combining the iodine with the chlorosilane/sulfoxide method for the stepwise formation of two disulfide bonds the time-consuming air oxidation step is efficiently bypassed. The typical procedure of this synthetic strategy is illustrated in Scheme 12. Two acid-stable thiol protecting groups are employed, the tBu and the Acm group, both of which are known to be resistant to TFA and even HF, although the stability of the *S*-tBu protection to HF depends on the scavengers employed. In view of this choice of S-protecting groups, the linear precursor peptide can be constructed either by Boc or Fmoc chemistry, and after the acidic deprotection (and cleavage from the resin) the peptide can be purified in the thiol-protected form prior to the disulfide bridging steps.

for references see p 137

Scheme 12 Bis-disulfide Formation with Iodine and Chlorosilane/Sulfoxide[124]



The combination of the Acm and tBu groups for thiol protection is based on their different stability toward iodine.[124] While the *S*-Acm derivatives are oxidatively cleaved with iodine,[67] the *S*-tBu derivatives are resistant to these conditions.[125] Consequently, the first cysteine pairing is achieved with iodine exploiting the reactivity of the Acm groups. The second disulfide is then generated by the chlorosilane/sulfoxide procedure. Although the *S*-tBu derivatives are known to be cleaved by sulfenyl chlorides,[126,127] the intermediate sulfenyl iodide reacts with the second *S*-Acm cysteine derivative at sufficiently high rates to allow selective pairing of the two Acm-protected cysteine residues. Similarly, the sulfonium ion formed in the second step does not induce detectable scrambling of the disulfide already formed. Again, if Trp residues are present in the molecule, the indole group has to be protected in view of the possible side reactions related to the iodine treatment and particularly, to the second step where chlorination of the indole side chain by chlorosilanes/sulfoxide occurs preferentially over the desired disulfide formation. Examples of successful application of this procedure are the syntheses of conotoxin M1[124] and of the Trp-containing human endothelin-1.[128]

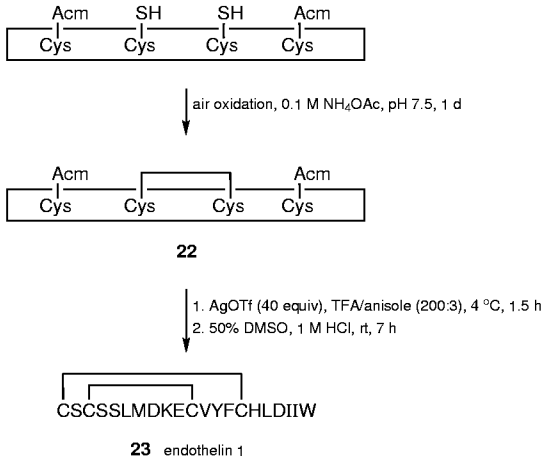
Synthesis of Conotoxin M1 (21):[124]
To a soln of H-Gly-Arg-Cys(tBu)-Cys(Acm)-His-Pro-Ala-Cys(tBu)-Gly-Lys-Asn-Tyr-Ser-Cys(Acm)-NH_2 (**19**; 20 mg, 11 μmol) in 80% aq MeOH (11.4 mL) containing HCl (2 equiv), was added in one portion a 20% soln of I_2 in MeOH (0.29 mL). After 15 min at 25 °C, the excess I_2 was reduced with 1 M ascorbic acid in H_2O. Diaion HP-20 (ca. 10 g) in 0.1% aq TFA (100 mL) was added, and the mixture was stirred for 15 min at 25 °C. The Diaion HP was separated and the absorbed peptide was eluted with MeCN/0.1% aq TFA (8:2). The solvent was removed by lyophilization to give **20**; yield: 18.3 mg. This intermediate (17.3 mg, 11 μmol) was then reacted in TFA (10.8 mL) with $MeSiCl_3$ (180 μL) in the presence of PhS(O)Ph (21.9 mg, 0.12 mmol) and anisole (117 μL) at 25 °C for 10 min. Dry Et_2O (100 mL) and 1 M AcOH (100 mL) were added to the mixture. The aqueous phase was separated, concentrated, and the residue chromatographed by FPLC to give **21**; yield: 6.5 mg (41%); the product was characterized by amino acid analysis and FAB-MS.

6.1.2.1.4 Disulfide Formation by a Combination of Air Oxygen and Dimethyl Sulfoxide/Acid

By a combination of acid-labile S-protecting groups, such as Trt or Mob, with the acid-stable *S*-Acm protection consecutive formation of two disulfide bonds can be achieved by two very mild oxidation procedures namely air oxidation followed by acid-catalyzed DMSO oxidation

as shown in Scheme 13. After the first intramolecular disulfide bridging, deprotection of the *S*-Acm cysteine derivatives with silver(I) trifluoroacetate[18] should proceed without affecting the disulfide group. However, in the synthesis of endothelin-1, a second isomer is formed besides the desired compound although only at a 1:8 ratio.[52] The second disulfide bond is produced in high yields by oxidation with 10% or even better 50% DMSO in 1 M HCl.[52] The aqueous HCl serves to liberate the thiol group from the silver(I) sulfide by precipitation of silver chloride[53] and also provides the acidity to prevent thiol/disulfide exchange reactions on the first disulfide.

Scheme 13 Bis-disulfide Formation by Air Oxygen and Dimethyl Sulfoxide/Acid[52]



Human Endothelin-I (23):[52]
[Cys(Acm)^{1,15}]-endothelin I [H-Cys(Acm)-Ser-Cys-Ser-Ser-Leu-Met-Asp-Lys-Glu-Cys-Val-Tyr-Phe-Cys(Acm)-His-Leu-Asp-Ile-Ile-Trp] (**22**; 0.41 μmol) was treated with silver(I) trifluoroacetate/DMSO/1 M HCl in an analogous manner to that described in Section 6.1.1.1. The HPLC elution pattern of the crude peptide showed two peaks, with retention times identical to those of endothelin-I and an isomer in a ratio of 8:1. Each product was purified by HPLC to yield endothelin-I (**23**) and its isomer (7.5%; yield 62%); these products were characterized by HPLC, FAB-MS, and amino acid analyses.

6.1.2.2 Regioselective Bis-disulfide Formation on Solid Support

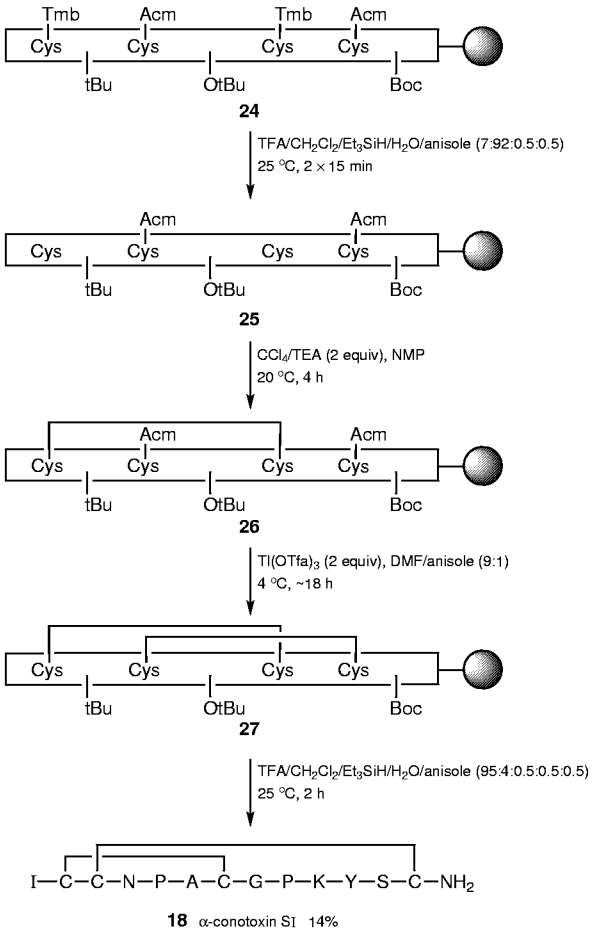
So far, this methodology has been applied only sporadically although offering the advantage of the pseudo-dilution effect. It has been used by combining both the solid-phase method for the first disulfide bond formation via iodine or *N*-iodosuccinimide on the resin-bound peptide, followed by cleavage from the polymeric support and air oxidation of the second cysteine pair.[129] An alternative was reported for the synthesis of a fragment of bovine pituitary peptide, a Trp-containing 21-residue cystine peptide.[121] In this synthesis, the base-labile Fm thiol protection was employed for the first disulfide formation on the resin via piperidine/DMF-mediated cleavage/oxidation. After the acid treatment, the second disulfide was formed in solution again by air oxidation at high dilution. In both approaches thiol/disulfide exchange reactions were expected to occur at the disulfide bond formed in the first step despite the high dilution. In the case of the pituitary peptide fragment random oxidation was found to produce the correct cystine connectivity and the two nonnatural isomers in a ratio of 70:18:12.[121] Thus, the high preference for the correct natural cystine network sensibly reduces disulfide scrambling by thiol/disulfide exchange reactions in the air-oxidation step. Generally however, such a strategy is not recommended. A combination of the *S*-Fm

for references see p 137

and *S*-Acm protection that would allow in the second step iodine- or thallium(III)-mediated disulfide formation would fulfill the requirements of regioselectivity.

An interesting approach based entirely on the solid-phase route has been reported for the synthesis of α-conotoxin SI (Scheme 14) where a combination of acid-labile and acid-stable thiol protecting groups are employed for the required orthogonality.[107]

Scheme 14 Solid-Phase Bis-disulfide Formation with Carbon Tetrachloride and Thallium(III)[107]



Using the PAL linker the first pair of Tmb-protected cysteine residues in **24** were converted with mild acid treatment into the bis-cysteine form **25** which was oxidized by the mild CCl_4 method to **26**.[106] The second disulfide bond was then generated from the Acm-protected cysteine pair with the thallium(III) reagent to yield **27**. Finally, α-conotoxin SI (**18**) was freed from the solid support by acid deprotection/cleavage. The yield obtained by this solid-phase method was lower than that of the corresponding synthesis in solution due to the following limitations: (i) the first pair of thiol protecting groups had to be generated by acid treatment while keeping the other protecting groups and particularly the linkage to the solid support intact, (ii) conventional air oxidation even with DMSO is difficult to effect on solid supported peptides, and (iii) a pronounced formation of oligomers or polymers was observed.

Synthesis of α-Conotoxin SI:[107]

The peptide-resin H-Ile-Cys(Tmb)-Cys(Acm)-Asn-Pro-Ala-Cys(Tmb)-Gly-Pro-Lys(Boc)-Tyr(tBu)-Ser(tBu)-Cys(Acm)-NH-PAL-resin (**24**; 25 mg, 4.5 μmol) was swollen in CH_2Cl_2 (2 mL) for 5 min, fil-

tered and treated with TFA/CH_2Cl_2/Et_3SiH/H_2O/anisole (7:92:0.5:0.5:0.5, 2 mL) to selectively cleave the Tmb groups. The resin-bound [bis-thiol, bis-*S*-Acm]-intermediate **25** was washed with CH_2Cl_2, DMF, and NMP (each for 5×2 min, 2 mL), swollen further in NMP (2 mL) for 5 min and then treated with 35 mM CCl_4 (0.9 µL) and Et_3N (1 µL) in NMP (0.4 mL). After 4 h at 20 °C, the excess oxidizing agent was removed by filtration and washing with DMF (5×2 min; 2 mL) to yield **26**. The peptide-resin **26** was swollen in DMF (2 mL) for 5 min, filtered and then treated with 25 mM thallium(III) trifluoroacetate (4.9 mg) in DMF/anisole (19:1, 0.35 mL). After ~18 h at 4 °C, the Tl reagent was removed with DMF (5×2 min, 2 mL) and the peptide-resin washed further with CH_2Cl_2 (5×2 min; 2 mL) to give **27**. Cleavage with TFA/CH_2Cl_2/Et_3SiH/H_2O/anisole (95:4:0.5:0.5:0.5, 2 mL) provided crude α-conotoxin SI (**18**) in ca. 14%. On a preparative scale, the peptide-resin (90 mg) was subjected to the same series of transformations as described above (scaled up proportionally in all aspects) and the crude product was purified by HPLC to give the homogeneous product; yield: 2.2 mg (10%); this was characterized by amino acid analysis and FAB-MS.

6.1.2.3 Regioselective Multiple-Disulfide Formation

In principle, formation of three disulfide bonds by the combination of two different oxidative procedures such as iodine and air oxidation, can also be envisaged, although in this case separation of isomers or additional synthetic procedures are inevitable because of the incomplete regioselectivity in the two disulfide bridging steps.[130] The orthogonality of the S-protecting groups as well as the methods developed for their oxidative deprotection theoretically allow the regioselective formation of three and even four disulfide bonds, particularly if the enzymatic cleavage of the Phacm thiol protecting group is also exploited.[2] To our knowledge, regioselective formation of three disulfide bonds has been reported only for insulin and related double-stranded peptides (Section 6.1.4.2), but not for single-chain cystine peptides. This may also be due to the fact that such highly cross-bridged relatively short polypeptide chains, mainly belonging to the toxin family of peptides, show a protein-like high tendency for correct oxidative refolding (see Section 6.1.5).

6.1.3 Interchain Mono-disulfide Peptides

For the synthesis of double-stranded symmetrical and unsymmetrical monocystine peptides the formation of an intermolecular disulfide bridge is required. For homodimerization of cysteine peptides all the methods discussed in Section 6.1.1 can be applied taking into account the reactivity of the different oxidative agents toward sensitive amino acid residues present in the peptide sequences. Synthetic approaches based on the direct use of suitable cystine derivatives can be envisaged, at least for small-size peptides since disproportionation would in all cases retain the homodimeric structure.[24]

For the synthesis of heterodimeric cystine peptides where two different peptide chains are cross-linked by a disulfide bridge random co-oxidation of the two chains besides producing the heterodimer leads in the optimal case to the additional two homodimers in statistical distribution. Therefore, chemical control of the disulfide bridging via site-directed disulfide formation techniques is required since a thermodynamic control for generation of heterodimers is extremely difficult to achieve (see Section 6.1.5).

In order to obtain heterodimers a cysteine residue of one peptide has to be specifically activated for subsequent reaction with the cysteine residue of the second peptide. Thiols are activated by introducing an electron-withdrawing substituent that increases the responsiveness of the sulfur to nucleophilic substitution. This topic has been reviewed.[4,131] Of the various older methods known in sulfur chemistry for the preparation of unsymmetrical disulfides, i.e. thiosulfates,[132] *S*-alkylthiosulfonates,[133] *S*-alkylthiosulfinates,[134] sulfanylamines,[135] sulfanylimides,[136] sulfanyl thiocyanates and sulfenyl chlorides,[137,138] sulfan-

for references see p 137

ylthiocarbonate,[139] and sulfanyl(diethyl)hydrazodicarboxylate[55] only the last three procedures were successfully converted into more or less efficient methods for site-directed disulfide formation in peptides.

Adaptation of the sulfenyl chloride method was attempted by exploiting the solvent-dependent rates of iodolysis of the Acm and Trt thiol protecting groups. However, iodine treatment of an equimolar mixture of *S*-Acm and *S*-Trt cysteine peptides leads only to the preferred formation of the heterodimer,[45] thus, making this method of limited use.[140] Similarly, the pioneering studies of Hiskey et al.[141,142] with thiocyanogen for selective activation of one cysteine peptide to form, via thiolysis by or with a second cysteine peptide, the desired heterodimers, led to the observation that sulfanylthiocyanates are thermally unstable and that fast reactions occur with basic nitrogen atoms,[143] again limiting this approach.

These early studies clearly revealed the inherent problems of the chemistry for site-directed formation of unsymmetrical disulfides that has to avoid formation of homodimers. These can result from (i) slow rates of activation of the cysteine peptide, (ii) disproportionation of the activated cysteine species, (iii) weak activation and thus slow thiolysis by the second cysteine component and thus its oxidation to the homodimer as well as thiol/disulfide exchange reactions on the unsymmetrical disulfide present in the reaction media. The latter side reactions are partly controlled by operating in degassed argon-saturated buffers or in organic solvents and preferably under acidic conditions where thiol/disulfide exchange reactions on the nonactivated disulfides, i.e. on the target unsymmetrical cystine peptide occurs at slow rates.

As mentioned above, various older methods were not meeting these stringent requirements,[24] thus limiting advances in the field for the demanding synthesis of biologically important double-stranded peptides such as human insulin and related peptides. The most efficient methods that emerged from the research outlined above are listed in Table 4. The activation of thiols via mixed disulfides is best achieved when the leaving group is deactivated and unable to induce thiol/disulfide exchange reactions. In this context, the thiol-induced fragmentation of sulfanylcarbonates releases unreactive COS, alcohols,[139,144,153] and sulfanylpyridines[145,146] which immediately tautomerize to the inactive thiopyridone. Similarly, thiolysis of sulfenohydrazides produces the inactive hydrazide as a byproduct,[55] while in the Cys(O)-based procedure inactive alcohols are formed.[147–150] Despite the efficiency of these methods the development of new reagents or application of the existing methodology to new synthetic strategies still remains an important issue in the synthesis of complex cystine peptides.

Table 4 Cysteine Derivatives Suitable for Interchain Disulfide Formation[56,145,147–150,152–155,163,164,166]

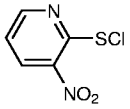
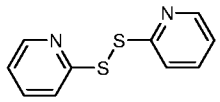
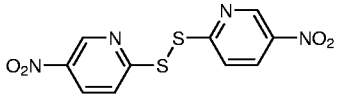
Cysteine derivative	Activating reagent	Solvent	Ref
Cys(SCO_2Me)	CH_3O-CO-S-Cl [MeO_2CSCl]	organic ($CHCl_3$, MeOH)	[152–154]
Cys(SC(O)NPhMe)	$Ph(CH_3)$N-CO-S-Cl [MePhN(O)CSCl]	organic ($CHCl_3$)	[155]
Cys(R^1)(O)	S-protected cysteine sulfoxide	organic (TFA)	[147,148] [149,150]
Cys(Npys)	 [Npys-Cl]	aq (pH 3~5)	[163,164]

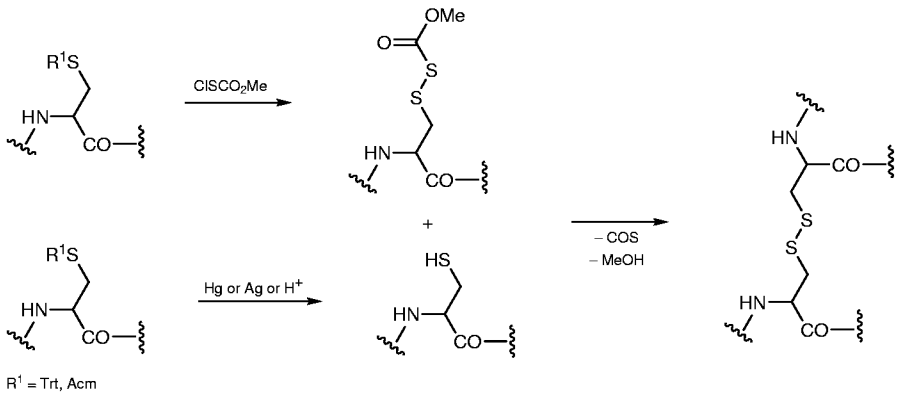
Table 4 (cont.)

Cysteine derivative	Activating reagent	Solvent	Ref
Cys(Py)	 [di(2-pyridyl)disulfide]	aq (pH 5~7)	[145]
Cys(pNpys)	 {di[5-nitro(2-pyridyl)]disulfide}	aq (pH 3~5)	[166]
Cys(BocNNHBoc)	Boc-N=N-Boc	organic (DMF)	[56]

6.1.3.1 Site-Directed Interchain Disulfide Formation with Alkoxycarbonylsulfenyl Chloride

Reports concerning the quantitative displacement of various alkyl-, aryl-, and Acm-type S-protecting groups with sulfenyl chlorides[126,151] and the facile thiolysis of sulfanylcarbonates[144] led to the development of methoxycarbonylsulfenyl chloride[152] as a suitable reagent for conversion of S-protected or free-thiol cysteine peptides into the related S-activated sulfanylcarbonates that readily undergo thiolysis by a second S-unprotected cysteine peptide to form disulfide-bridged heterodimers (see Scheme 15).[153,154] The unreactive byproducts COS and methanol are formed. The *S*-methoxycarbonylsulfanyl cysteine peptide derivatives are readily obtained by reacting the *S*-Trt, *S*-Acm, *S*-Xan, *S*-Tmb, and *S*-tBu precursors with methoxycarbonylsulfenyl chloride in organic solvents under neutral or preferably weak acidic conditions.[70,153] In the case of S-protected or unprotected cysteine peptides that contain amino groups, the reaction has to be carried out in aqueous/organic media under acidic conditions in order to prevent their sulfanylation. Taking into account that the *S*-methoxycarbonylsulfanyl derivatives are stable to even strong acid treatment such as HF or TfOH but unstable to bases, methoxycarbonylsulfanyl-protected cysteine peptides are also accessible by classical Z[154] or solid-phase Boc chemistry[155] with due precautions. Thus, the methoxycarbonylsulfanyl group can act not only as an activating group for thiolysis, but also as an acid-stable thiol protecting group. Similar properties are shown by the *S*-ethyloxycarbonylsulfanyl cysteine peptide derivatives.[156] Better stabilities, however, were reported

Scheme 15 Site-Directed Interchain Disulfide Formation with *S*-Alkoxycarbonylsulfanyl Cysteine Derivatives[153,154]

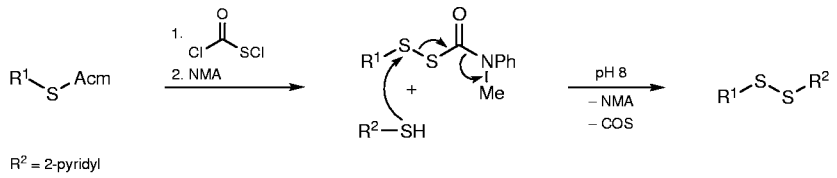


for references see p 137

for the *S*-benzyloxycarbonylsulfanyl cysteine derivative,[157] although this type of protecting/activating group has not found extensive use.

A serious limitation of this thiol-activation method is the presence of Trp residues in the peptide chains, since sulfenyl chlorides are known to react with the indole group to form tryptophan-2-thioethers at high rates even under neutral or acidic conditions[78] as required for the preparation and handling of the base-labile *S*-methoxycarbonylsulfanyl derivative. A bypass to this problem is the assembly of the peptide with suitable methoxycarbonylsulfanyl-protected cysteine derivatives. Besides the lability of the *S*-methoxycarbonylsulfanyl derivatives to bases and therefore incompatibility with the Fmoc chemistry, these exhibit a high tendency to S→N migration when the cysteine residue is located in the N-terminal position.[155] To avoid this side reaction, the (*N*-methyl-*N*-phenylcarbamoyl)sulfanyl group was developed for thiol protection and activation.[155] The *S*-(*N*-methyl-*N*-phenylcarbamoyl)sulfanyl derivatives of cysteine compounds and cysteine peptides are obtained from the related *S*-Acm derivatives in two steps, as shown in Scheme 16, and they are easier to handle in the synthesis of peptides. The *S*-(*N*-methyl-*N*-phenylcarbamoyl)sulfanyl derivatives are as stable to strong acids as the *S*-methoxycarbonylsulfanyl derivatives. Due to their acid stability, such sulfanylcarbonates should be well suited for a one-pot intermolecular disulfide bridging technique.

Scheme 16 Conversion of *S*-Acm Cysteine Derivatives into Activated *S*-Sulfanyl Cysteine Derivatives[155]



Boc-Cys(SCO$_2$Me)NPhMe)-Asn-OtBu:[153]

To a soln of Boc-Cys(Trt)-Asn-OtBu (1.585 g, 2.5 mmol) in $CHCl_3$/MeOH (2:1, 15 mL) was added methoxycarbonylsulfenyl chloride (0.44 mL, 4.9 mmol) at 0 °C. After stirring at 0 °C for 50 min, 1 M aq DEA (5.5 mL, 5.5 mmol) and $CHCl_3$ (100 mL) were added. The soln was washed with 1 M citric acid and H_2O, and then dried. The soln was taken to dryness and the residue crystallized ($CHCl_3$/petroleum ether); yield: 0.95 g (79%); mp 102–104 °C; $[\alpha]_D^{20}$ –19 (*c* 1.1, $CHCl_3$). The same dipeptide ester was obtained by a similar procedure from Boc-Cys(Acm)-Asn-OtBu in 81% yield.

Boc-Cys-Asn-OtBu

Boc-Val-Cys-Gly-Glu(OtBu)-OtBu:[153]

CAUTION: *Mercury(II) acetate [Hg(OAc)$_2$] is highly toxic and may be fatal if ingested, inhaled, or swallowed. It is moderately toxic by skin contact. Appropriate safety precautions and procedures should be taken during all stages of its handling and disposal.*

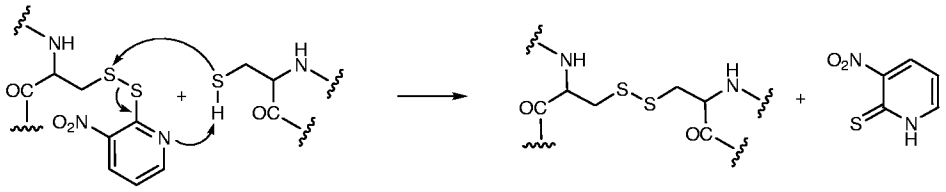
A soln of Boc-Val-Cys(Trt)-Gly-Glu(OtBu)-OtBu (8.61 g, 10 mmol) and $Hg(OAc)_2$ (3.5 g, 11 mmol) in EtOAc/MeOH (2:1, 90 mL) was kept at 20 °C for 4 h. H_2S was passed through the soln for 10 min and the resulting mixture was filtered. The filtrate was taken to dryness and the residue was washed with petroleum ether giving Boc-Val-Cys(H)-Gly-Glu(OtBu)-OtBu as an oil; yield: 4.42 g. Boc-Val-Cys(H)-Gly-Glu(OtBu)-OtBu (1.33 g, 2 mmol) and Boc-Cys(SCO$_2$Me)-Asn-OtBu (0.963 g, 2 mmol) were dissolved in CH_2Cl_2 (20 mL) and the mixture was allowed to react at 20 °C for 90 min. The soln was taken to dryness and the residue was washed with petroleum ether; yield: 1.95 g (97%).

6.1.3.2 Site-Directed Interchain Disulfide Formation with Pyridylsulfenyl Chloride Derivatives

Peptides containing cysteine residues in the free thiol form are efficiently oxidized by di(2-pyridyl)disulfide, di(4-pyridyl)disulfide and di[5-nitro-2-pyridyl]disulfide which are reagents used for the spectroscopic quantification of thiols in peptides and proteins.[145,146] Since thiols are oxidized to disulfides by these reagents with the generation of unreactive pyridine-2(1*H*)-thione derivatives their use for the synthesis of cystine peptide homodimers,[158,159] the preparation of mixed disulfides, and their use for disulfide cross-bridging in proteins has been reported.[160,161] This chemistry has subsequently been exploited for the preparation of activated mixed disulfides of cysteine peptides which readily undergo a thiol/disulfide exchange reaction with a second cysteine component to form the interchain heterodisulfides, as shown in Scheme 17.[162]

The reaction rates of the thiolytic fragmentation increase with increasing pH, but concomitantly the rate of disproportionation is enhanced, thus leading to the undesired accumulation of symmetrical disulfides. Since formation of the disulfide bond is driven by the low pK_a of the aromatic thiol, the reaction takes place even under slightly acidic conditions which minimize thiol/disulfide exchange reactions once the desired disulfide bond is formed. While thiolysis of the *S*-2-nitrophenylsulfanyl cysteine derivatives[126] is slow under these conditions, the reaction is relatively fast in the case of the 2-pyridylsulfanyl (SPy) derivatives.[162] To further enhance the rates of disulfide formation under acidic conditions the 3-nitro-2-pyridylsulfanyl (Npys)[163–165] and 5-nitro-2-pyridylsulfanyl[166] were investigated. These groups exhibit sufficiently high reaction rates even in the pH range of 3.5 to 6 in comparison to 5 to 7 in the case of the *S*-SPy derivatives.

Scheme 17 Site-Directed Interchain Disulfide Formation with Pyridylsulfanyl Cysteine Derivatives[162–166]



The activated mixed disulfides are prepared directly from the unprotected cysteine peptides in aqueous acidic media such as aqueous acetic acid, by reaction with an excess of di(2-pyridyl)disulfide[162] or di[5-nitro(2-pyridyl)]disulfide[166] to yield the *S*-SPy and *S*-5-nitro-2-pyridylsulfanyl derivatives, respectively. The progress of the reaction can be monitored spectroscopically. Similarly to the *S*-Nps[126] and *S*-methoxycarbonylsulfanyl derivatives,[153] the *S*-SPy[162] and *S*-Npys derivatives[163] are readily obtained from *S*-alkyl, *S*-aryl, and *S*-Acm cysteine or cysteine peptide derivatives by reaction with SPy-Cl[162] or Npys-Cl.[167,168] Since the sulfenyl halides suffer from extreme sensitivity toward hydrolysis from atmospheric moisture, reactions have to be carried out with the exclusion of water, and exact quantities can only be achieved with freshly prepared reagents. As in the case of the *S*-methoxycarbonylsulfanyl derivative (see Section 6.1.3.1), displacement of the S-protecting groups with the sulfenyl chlorides is carried out in organic solvents, preferably in presence of acetic acid to suppress any disproportionation. In the case of S-protected cysteine peptides which are deprotected at all other functions the reaction is run in mixtures of acetic and trifluoroacetic acid to prevent sulfanylation of amino groups.[169] Again this procedure cannot be applied to tryptophan-containing peptides. Therefore, for such peptides S-deprotection of the cysteine residues is required and conversion of the free thiol to the *S*-pyridinylsulfanyl derivatives is achieved with di(2-pyridyl)disulfide[49,170,171] or particularly with di[5-nitro(2-

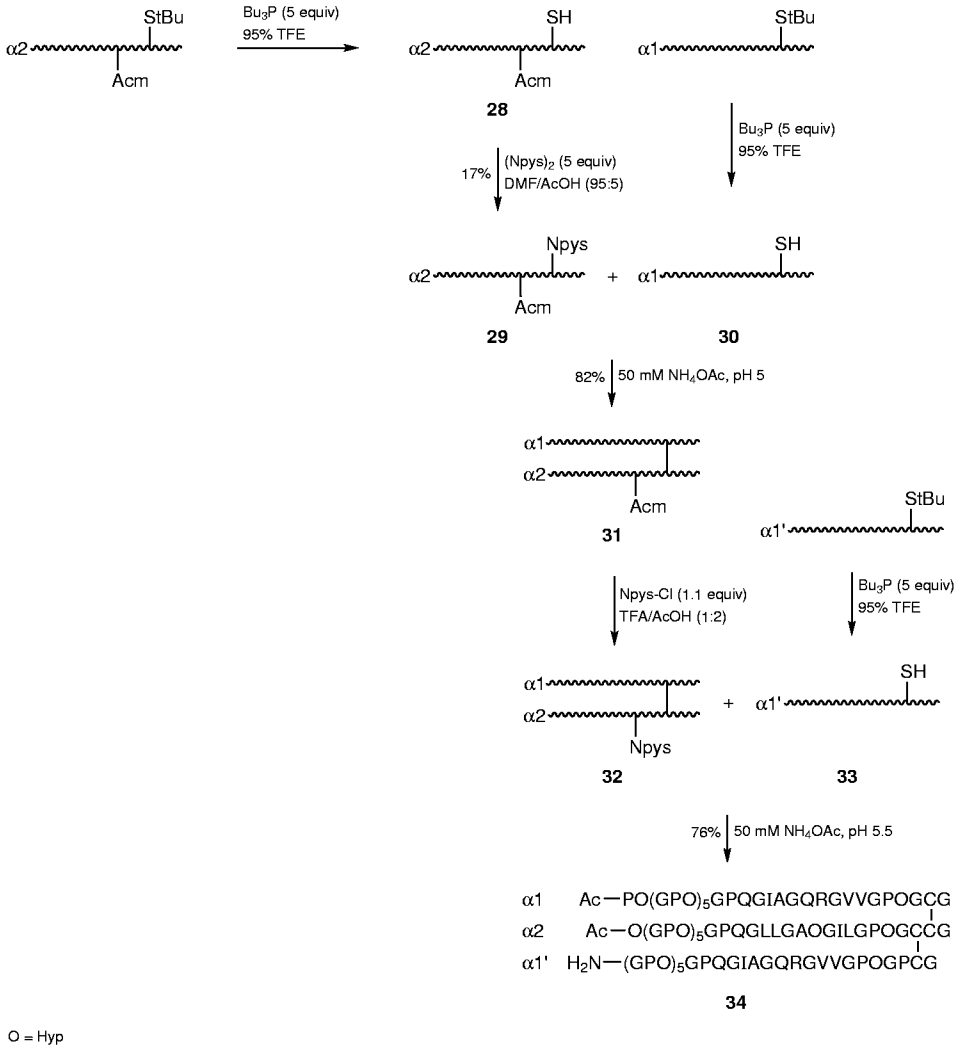
for references see p 137

pyridyl)]disulfide.[166] With the latter reagent the S-activation step is faster under the relatively acidic conditions required to prevent homodimer formation (Table 4, Section 6.1.3).

Similarly to the methoxycarbonylsulfanyl-protected cysteine derivatives, the *S*-Npys compounds are stable toward strong acid treatment[164,165,172] and thus, they can be exploited in the dual purpose of S-protection and S-activation. The Npys group has been used with considerable success in combination with Boc chemistry for the preparation of peptides.[173–175] This S-protecting/activating group, however, is labile to piperidine treatment and can, therefore, not be used in Fmoc-based peptide synthesis.

Besides being presently the most efficient method for site-directed interchain disulfide bridging of two different cysteine peptides, this procedure is also recommended for a controlled peptide–protein conjugation by reacting Npys-protected cysteine peptides with properly thiol-functionalized protein carriers.[176] In this conjugation procedure the amount of peptide grafted to the protein is quantitatively determined by measuring spectrophotometrically at 430 nm the amount of 3-nitropyridine-2(1*H*)-thione liberated in the reaction. An example of this approach is outlined in Scheme 18.

Scheme 18 Interchain Disulfide Formation Using *S*-(3-Nitro-2-pyridylsulfanyl)cysteine[169]



O = Hyp

Synthesis of Disulfide-Bridged Collagen Peptide 34:[169]
Synthesis of the Monomer 29:
The 4 mM soln of the crude peptide [H-(Gly-Pro-Hyp)$_5$-Gly-Pro-Gln-Gly-Leu-Leu-Gly-Ala-Hyp-Gly-Ile-Leu-Gly-Cys(Acm)-Cys-Gly-Gly-OH] **28** in degassed argon-sat. DMF/AcOH (95:5) was added dropwise to a 100 mM soln of di[5-nitro(2-pyridyl)]disulfide (5 equiv) in DMF/AcOH (95:5) with exclusion of air oxygen. The reaction was monitored spectroscopically at 430 nm, and after completion (1 to 2 h), the solvent was removed under reduced pressure. The resulting residue was dissolved in H_2O and the excess reagent was filtered off. The H_2O was removed under reduced pressure and the residue was reprecipitated from TFE with methyl *tert*-butyl ether and purified by preparative HPLC to give **29**; yield: 13%; the product was characterized by MALDI-TOF-MS, HPLC, and amino acid analysis.

Synthesis of the Heterodimer 31:
The peptide [H-(Gly-Pro-Hyp)$_5$-Gly-Pro-Gln-Gly-Ile-Ala-Gly-Gln-Arg-Gly-Val-Val-Gly-Pro-Hyp-Gly-Pro-Cys-Gly-OH] **30** was dissolved at 4 mM concentration in degassed and argon-sat. 50 mM NH_4OAc buffer (pH 5.5). The resulting soln was added dropwise to a 10 mM soln of peptide **29** in 50 mM NH_4OAc buffer (pH 5.5) with exclusion of air oxygen. The reaction was monitored spectroscopically at 430 nm. After 2–3 h the mixture was concentrated and the heterodimer **31** isolated by gel chromatography or HPLC; yield: 68%; the product was characterized by HPLC and MALDI-TOF-MS.

Synthesis of the Heterodimer 32:
To a 50 mM soln of heterodimer **31** in TFA/AcOH (1:2) freshly prepared Npys-Cl (1.1 equiv) was added. After 30 min the bulk of the solvent was removed under reduced pressure and the residue was purified by gel chromatography to give **32**; yield: 77%; the product was characterized by HPLC and MALDI-TOF-MS.

Synthesis of the Heterodimer 34:
A soln of peptide 33 in degassed argon-sat. 50 mM NH_4OAc buffer (pH 5.5) at 4 mM concentration was added dropwise to a 10 mM soln of 1 equiv of heterodimer 32 in 50 mM NH_4OAc (pH 5.5) under exclusion of air oxygen. The reaction was monitored spectroscopically at 430 nm and by HPLC. After 2–3 h the solvent was removed and the product isolated by HPLC; yield: 35%; the product was characterized by ESI-MS, HPLC, and amino acid analysis.

Synthesis of a Disulfide Heterodimer:[165]
To Ac-Cys(Npys)-Gly-Glu(OBzl)-Gln-His(Tos)-His(Tos)-Pro-Gly-Gly-Gly-Ala-Lys[Z(Cl)]-Gln-Ala-NH_2 (16 mg, ca. 10 μmol) and Ac-Gly-Glu(OBzl)-Gln-His(Tos)-His(Tos)-Pro-Gly-Gly-Gly-Ala-Lys-[Z(Cl)]-Gln-Cys-NH_2 (14 mg, ca. 10 μmol) was added 1 M AcOH (2.0 mL) and the resulting mixture was allowed to react for 48 h at rt. The progress of the reaction was monitored by analytical HPLC and judged to be >90% after 21 h. The mixture was purified by preparative HPLC; yield: 12 mg (40%).

6.1.3.3 Site-Directed Interchain Disulfide Formation with Azodicarboxylic Acid Derivatives

This method is based on the activation of one cysteine peptide as a sulfenohydrazide derivative by the reaction of the free thiol group of the cysteine peptide with azodicarboxylic acid as a dialkyl ester or dimorpholide depending on the solubility requirements.[56] Following the isolation and characterization of the S-activated intermediate, reaction with a second free thiol cysteine component via thiolysis leads to the desired intermolecular disulfide bond, as shown in Scheme 4.

Serious side reactions at the sensitive amino acid residues including Trp are not observed to occur under the usual operating conditions of this thiol activating procedure. However, neutral or slightly basic conditions are required for this method both in the activation and thiolysis step; therefore, conditions have to be applied that prevent homodimerization of the unprotected cysteine peptides by careful exclusion of air oxygen and preferential use of organic solvents.[166,177,178] In aqueous or aqueous/organic media quantitative activation of the cysteine peptide is difficult to achieve and the thiolysis step can be insufficiently fast to prevent both homodimerization of the second cysteine peptide or thiol/disulfide exchange reactions on the unsymmetrical disulfide group present in the reaction media.[83]

for references see p 137

H-Thr(tBu)-Cys[N(Boc)-NH(Boc)]-Pro-Pro-Cys(Acm)-Pro-Ala-Pro-OH:[166]
A soln of H-Thr(tBu)-Cys-Pro-Pro-Cys(Acm)-Pro-Ala-Pro-OH (1.53 g, 1.68 mmol) in argon-sat. DMF (50 mL) was added dropwise to a soln of azodicarboxylic acid di-*tert*-butyl ester (1.16 g, 5.04 mmol) in DMF (20 mL). The mixture was stirred at rt for an additional 4 h until the Ellman's test was negative. The soln was then taken to dryness and the residue triturated with EtOAc/Et_2O (1:2) to give the title product; yield: 1.7 g (84%); $[\alpha]_D^{20}$ −136.2 (*c* 1.0, MeOH).

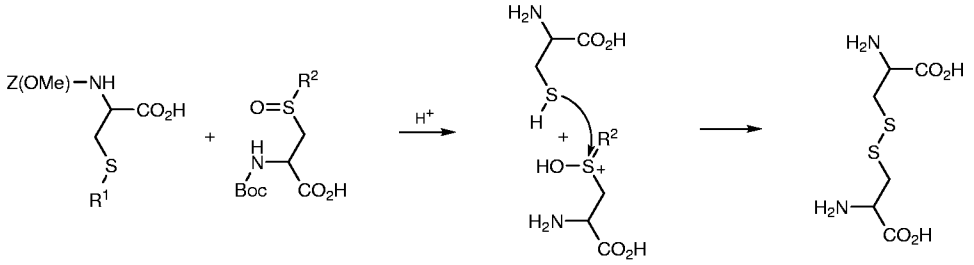
H-Thr(tBu)-Cys-Pro-Pro-Cys(Acm)-Pro-Ala-Pro-OH

HO-Pro-Ala-Pro-Cys-Pro-Pro-Cys(Acm)-Thr(tBu)-H:[166]
H-Thr(tBu)-Cys(Acm)-Pro-Pro-Cys-Pro-Ala-Pro-OH (0.313 g, 0.33 mmol), prepared by treatment of the corresponding *S*-tBu peptide with Bu_3P, was dissolved in argon-sat. DMF (15 mL) and then added dropwise to a soln of the above sulfenohydrazide peptide (0.48 g, 0.40 mmol) in argon-sat. DMF (15 mL). After 5 h (Ellman's test was negative) the solvent was removed under reduced pressure. The residue was purified by silica gel chromatography followed by reprecipitation from MeOH/iPr_2O to give the title product; yield: 0.6 g (85%); $[\alpha]_D^{20}$ −161.5 (*c* 1.0, MeOH).

6.1.3.4 Site-Directed Interchain Disulfide Formation with Cysteine-Sulfoxide by Acid Catalysis

A further development of the DMSO/H^+ method for oxidation of cysteine peptides led to the cysteine-sulfoxide acid-catalyzed intermolecular disulfide formation with a second S-unprotected or acid-labile protected cysteine component as shown in Scheme 19.[147] The protonation of the sulfoxide by TfOH in the case of *S*(*O*)-Mob or TFA in the case of *S*(*O*)-Acm derivatives provides electrophilicity to the sulfur atom to allow attack by the second S-unprotected cysteine component (formed by the fast deprotection of the *S*-Mob group with TfOH in presence of dimethylsulfide) to generate in a site-directed manner the interchain disulfide bond. Although extensive experience with this method has not been accumulated for interchain disulfide bridging, it has been successfully applied for intrachain site-directed disulfide bond formation in chicken calcitonin-gene-related peptide.[179]

Scheme 19 Site-Directed Interchain Disulfide Formation by the Cysteine-Sulfoxide/Acid Method[147]



Synthesis of Protected Cys(O)(Acm) and Cys(O)(Mob) Derivatives; Typical Procedures:[149]

Z(OMe)-Cys(Mob)(O)-OH:[149]
To a stirred soln of Z(OMe)-Cys(Mob)-OH (4.0 g, 9.8 mmol) in EtOAc (40 mL) was added $NaBO_3 \cdot 4H_2O$ (1.65 g, 1.1 equiv) in H_2O (20 mL) and the mixture was stirred overnight at rt. The organic layer was washed with 5% citric acid, brine, and dried (Na_2SO_4). The solvent was removed under reduced pressure and the residue was recrystallized from THF with Et_2O; yield: 3.43 g (84%); mp 148–151 °C; $[\alpha]_D^{22}$ −72.8 (*c* 1.1, DMF).

Boc-Cys(Acm)(O)-OH:[149]
To a stirred soln of Boc-Cys(Acm)-OH (4.0 g, 13.7 mmol) in EtOAc (30 mL) was added $NaBO_3 \cdot 4H_2O$ (2.32 g, 1.1 equiv) in H_2O (10 mL) and the mixture was stirred overnight at rt. The aqueous layer was washed with EtOAc and then acidified with citric acid. After standing for 2 d, the precipitated product was recrystallized (MeOH with EtOH); yield: 1.79 g (42%); mp 159.5–161 °C; $[\alpha]_D^{18}$ −46.3 (*c* 1.1, MeOH).

H-Cys-Ala-NH_2/H-Cys-Gly-OBzl:[148]
An equimolar mixture of Z(OMe)-Cys(Mob)-Gly-OBzl (0.1 mmol) and Z(OMe)-Cys(Mob)(O)-Ala-NH_2 (0.1 mmol) was treated with 1 M TfOH/TFA (3 mL) containing Me_2S (0.5 M) in an ice-bath for 1 h, then hexane was added. The resulting precipitate was dissolved in H_2O and the aqueous phase was washed with Et_2O. The soln was treated with Amberlite CG-4B (acetate form) for 30 min then the resin was removed by filtration. The filtrate was applied to a column of Sephadex G15 which was eluted with 1 M AcOH, the fractions corresponding to the front main peak were combined and lyophilized. The residue was purified by HPLC to give the title product; yield: 87%; $[\alpha]_D^{22}$ −60.9 (*c* 0.2, H_2O). The product was further characterized by FAB-MS and amino acid analysis.

6.1.4 Peptides with Multiple Inter- and Intrachain Disulfides

Following the first interchain cross-bridging of two identical or two different cysteine peptides to produce homo- or heterodimers by the methods described in Section 6.1.3, any additional inter- or intrachain disulfide bridging of a pair of cysteine residues corresponds to an intramolecular reaction that requires all the precautions and selective chemistry discussed in Section 6.1.2 in order to avoid scrambling of existing disulfides and dimerization/oligomerization via intermolecular reactions. Therefore, multiple inter- and intrachain disulfide bonds have to be built-up step-by-step in a regioselective manner by combining the above mentioned oxidation/activation methods. From suitable combinations of these methods that are compatible with each other and with the orthogonality of the presently available thiol protecting groups several strategies have been developed that allow the regioselective formation of up to three or more disulfide bonds.

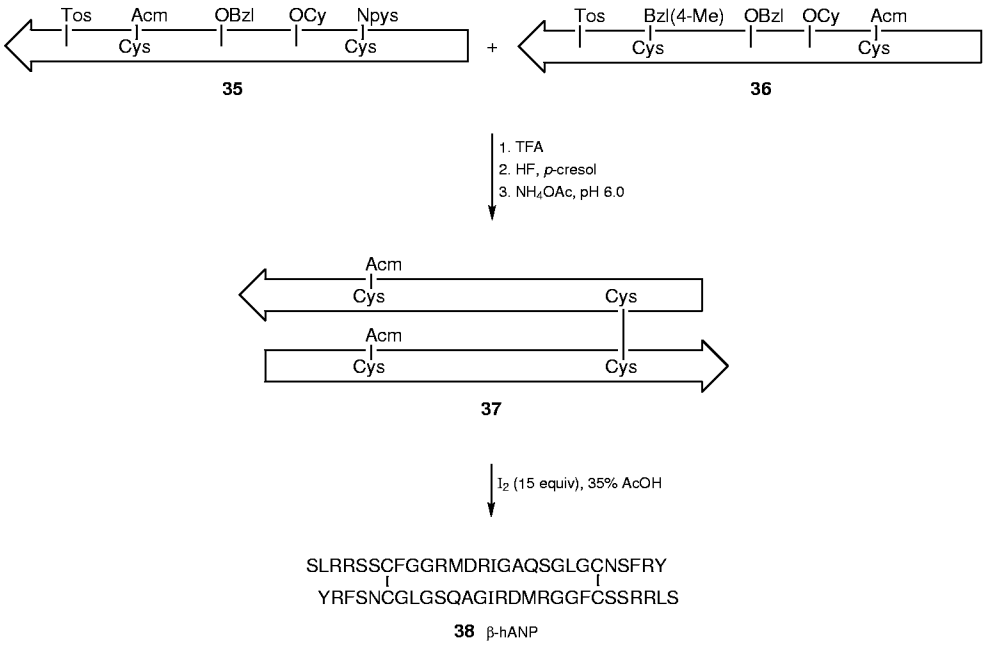
6.1.4.1 Peptides with Multiple Interchain Disulfide Bonds

The simplest approach is offered by parallel homodimers which require only two orthogonal protecting groups. After deprotection of one thiol group mild oxidation by one of the methods discussed in Section 6.1.1.1 or following direct selective oxidative cleavage of one S-protected cysteine residue by the methods described in Section 6.1.1.2, the second disulfide bond has to be produced by selective procedures compatible with the presence of a disulfide as described in Section 6.1.2.

For the synthesis of antiparallel homodimers or parallel and antiparallel heterodimers the first interchain cross-bridging requires site-directed disulfide formation by one of the methods described in Section 6.1.3. Among these, the most widely applied procedure relies on the use of Npys-protection/activation of one cysteine peptide for reaction with the unprotected cysteine residue of the second chain (Scheme 20). The second pair of cysteine residues are protected by the acid-stable Acm group which allows the second disulfide formation mainly via the iodine-mediated oxidation. This scheme allows several variations in terms of protecting groups for the first as well as for the second disulfide bond formation. Scheme 20 illustrates an example of this approach with one bis-cysteine peptide **35** protected as the *S*-Acm and *S*-Npys derivative synthesized by Boc chemistry taking advantage of the acid-stability of the Npys group, and the second bis-cysteine peptide **36** protected as *S*-[Bzl(4-Me)] and *S*-Acm derivative. By one-pot acid treatment of both peptide chains in a 1:1 molar ratio the free thiol group generated by cleavage of the acid-labile *S*-[Bzl(4-Me)] derivative reacts directly with the Npys-activated cysteine residue upon adjustment of the pH to 3.5–6.5 to form the first interchain disulfide bridge, compound **37**.[173] The second interchain disulfide bond is then generated by oxidative cleavage of the two Acm-protected cysteine residues with iodine to give **38**. Both parallel and antiparallel dimers are obtained by this procedure by preselected positioning of the three types of S-protecting groups.[171,180]

for references see p 137

Scheme 20 Synthesis of Biscystine Dimers by the *S*-Nitropyridylsulfanyl and Iodine Procedure[173]



Alternatively, both peptide chains could be protected at one cysteine residue as a *S*-Acm derivative and at the second cysteine residue by an acid-labile [Trt, Mob, Xan, or Bzl(4-Me)], base-labile (Fm), or reduction-labile (*S*-tBu) group. Both peptide chains may then be separately converted into the free thiol/Acm-protected form for selective activation of one chain as *S*-SPy or *S*-Npys derivatives by reaction with di(2-pyridyl)disulfide or di[5-nitro(2-pyridyl)]disulfide, or as a sulfenohydrazide derivative by reaction with azodicarbocylic acid derivatives for formation of the first interchain disulfide bridge.

At the second iodine-mediated disulfide formation step careful control of the reaction conditions is required in order to suppress side reactions at sensitive amino acid residues. Moreover, depending on the thermodynamic stability of the product, isomerization by a disulfide-exchange reaction caused by iodine can also take place, as observed in the synthesis of IGg I hinge peptide.[116] Besides iodine-mediated oxidation, application of the other methods described in Section 6.1.2 can be envisaged. This general approach has been successfully applied to the synthesis of β-human atrial natriuretic peptide,[173] of the uteroglobin cavities,[180] and of a human IGg-I fragment,[116] all in parallel and antiparallel alignment as well as of artificially designed dimers[181] and of IGg-I/gastrin constructs.[182,183]

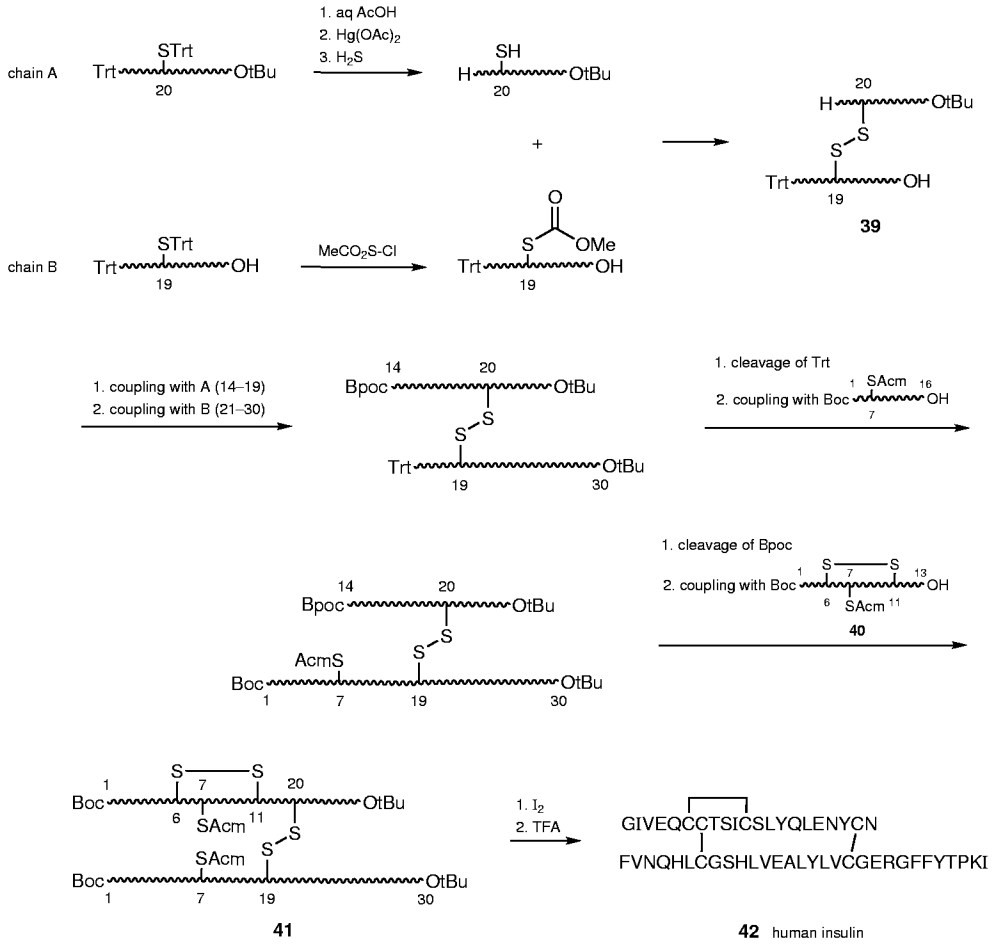
Synthesis of β-hANP (38):[173]

The peptide derivatives **35** (R^1 = Acm; R^2 = Npys) and **36** [R^1 = Bzl(4-Me); R^2 = Acm] of Boc-Ser(Bzl)-Leu-Arg(Tos)-Arg(Tos)-Ser(Bzl)-Ser(Bzl)-Cys(R^1)-Phe-Gly-Gly-Arg(Tos)-Met-Asp(OCy)-Arg(Tos)-Ile-Gly-Ala-Gln-Ser(Bzl)-Gly-Leu-Gly-Cys(R^2)-Asn-Ser(Bzl)-Phe-Arg(Tos)-Tyr[Z(Br)]-OBzl were synthesized by conventional Boc chemistry in soln. Equivalent amounts of peptides **35** and **36** (0.2 mmol scale) were exposed to TFA treatment to remove the Boc groups and then to liq HF in the presence of *p*-cresol to remove all protecting groups except Acm and Npys. The residue was dissolved in 0.1 M degassed NH_4OAc buffer (pH 6.0) and kept for 30 min at rt. After purification by CM-cellulose chromatography by elution with 0.1 to 1 M gradient of NH_4OAc buffer, the product **37** (570 mg, 50%) was treated with I_2 (15 equiv) in 0.2 M HCl in 35% aq AcOH (peptide concentration; 10^{-3} mol); conversion yield according to HPLC: 63%. The crude product was purified by preparative HPLC to homogeneity to yield **38**; the product was characterized by HPLC and amino acid analysis.

6.1.4.2 Peptides with Intra- and Interchain Disulfide Bonds

The classical example of peptides containing both inter- and intrachain disulfide bonds is insulin which has been object of extensive studies of oxidative refolding as discussed in Section 6.1.5. Due to the low yields obtained in these experiments the total synthesis of insulin has been one of the main targets of peptide chemistry for decades. A first total synthesis of human insulin with regioselective formation of both the inter- and intrachain disulfides was achieved by Sieber et al.[184,185] with a strategy based on fragment condensation procedures in solution as outlined in Scheme 21. Use was made of an unsymmetrical double-chain fragment **39** prepared by the site-directed disulfide bridging with the *S*-methoxycarbonylsulfanyl activation method.[153] With selective N- and C-terminal elongation of this unsymmetrical cystine peptide and using a fragment of the A-chain **40** with preformed intramolecular disulfide bridge[186] the final double-chain synthetic intermediate **41** was obtained. The third disulfide bond linkage was produced by iodine-oxidation of the bis-*S*-Acm precursor **41** to produce, after final deprotection with acid, human insulin (**42**) both with the natural and nonnatural cystine networks.[41] Attempts to apply a similar strategy in the solid-phase synthesis of insulin was unsuccessful.[187]

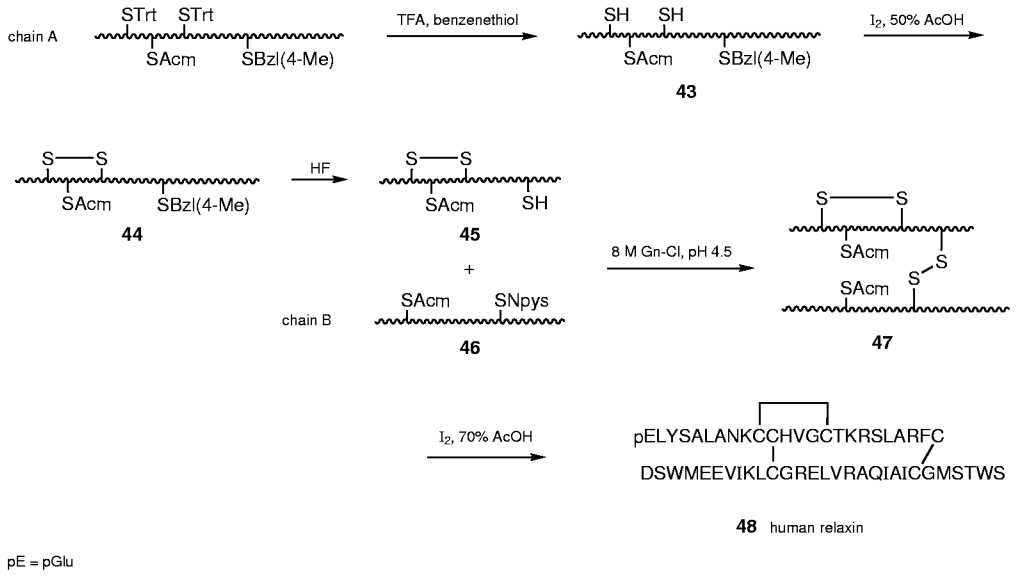
Scheme 21 Synthesis of Human Insulin by the Fragment Condensation Strategy in Solution[184,185]



for references see p 137

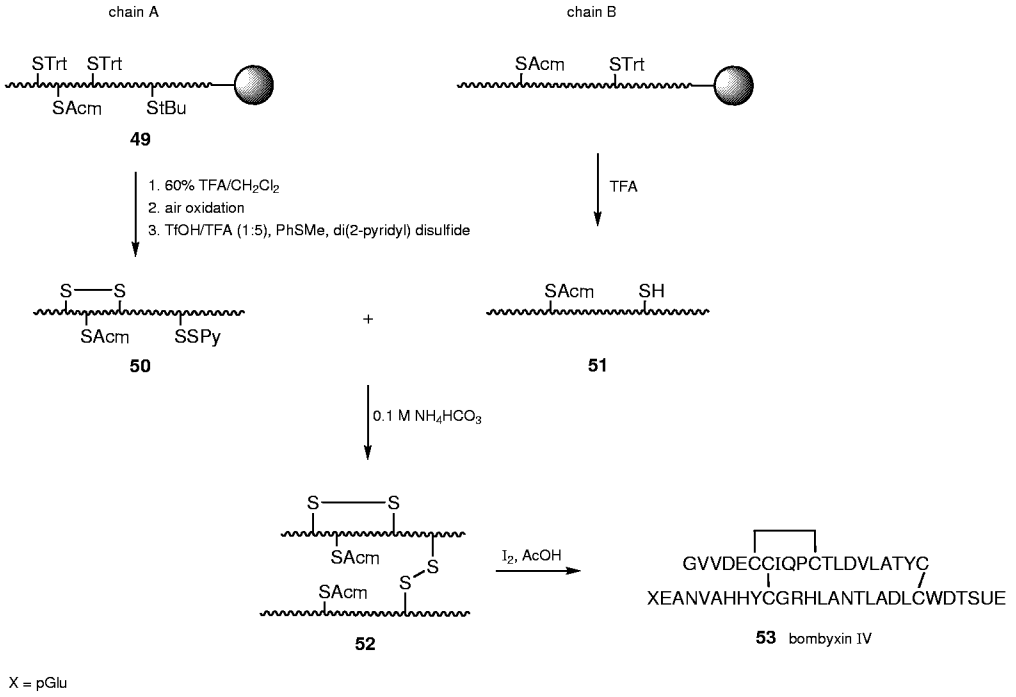
The new developments in the cysteine protection strategy have been successfully applied in direct disulfide cross-linking of two peptide chains for the synthesis of human relaxin which exhibits the identical cystine network as insulin, but contains as additional difficulties two Met and one Trp residue.[188] The synthetic route is shown in Scheme 22. Again for the A-chain the first intrachain disulfide bridge was obtained from **43** by titration with iodine in aqueous acetic acid to give **44** without affecting the *S*-Acm group. The subsequent *S*-Bzl(4-Me) deprotection of the A-chain provided the intermediate **45** for site-directed intrachain cross-linking with the B-chain **46** bearing the suitably positioned activated *S*-Npys cysteine residue to yield **47**. Finally following the procedure of Sieber et al.[184] the third disulfide was obtained by iodine oxidation to give human relaxin (**48**). Particularly problematic in this synthesis was the reduction of the Met(O) residues incorporated to prevent their partial oxidation in the iodine-treatment steps. Since soft bases (e.g., thioethers in strong acids[189]) tend to scramble unsymmetrical disulfides,[175] reduction with iodide in acid[190,191] was successfully achieved using ammonium iodide in TFA.

Scheme 22 Synthesis of Human Relaxin[188]



For the synthesis of bombyxin IV (**53**)[192] regioselective formation of three disulfide bonds was achieved by exploiting the differentiated acid-stability of the *S*-Trt vs the *S*-tBu protection in **49** that allowed the air-mediated intramolecular disulfide formation in the A-chain (Scheme 23). For subsequent activation of the third cysteine residue from the precursor *S*-tBu derivative the rather drastic conditions of TfOH/TFA (1%) were applied in presence of di(2-pyridyl) disulfide which despite the strong acidity allowed a concomitant deprotection/activation of this residue to give **50**. Subsequent reaction of **50** with the B-chain derivative **51** established the intrachain heterodisulfide cross-link in **52** which on oxidation gave bombyxin IV (**53**).

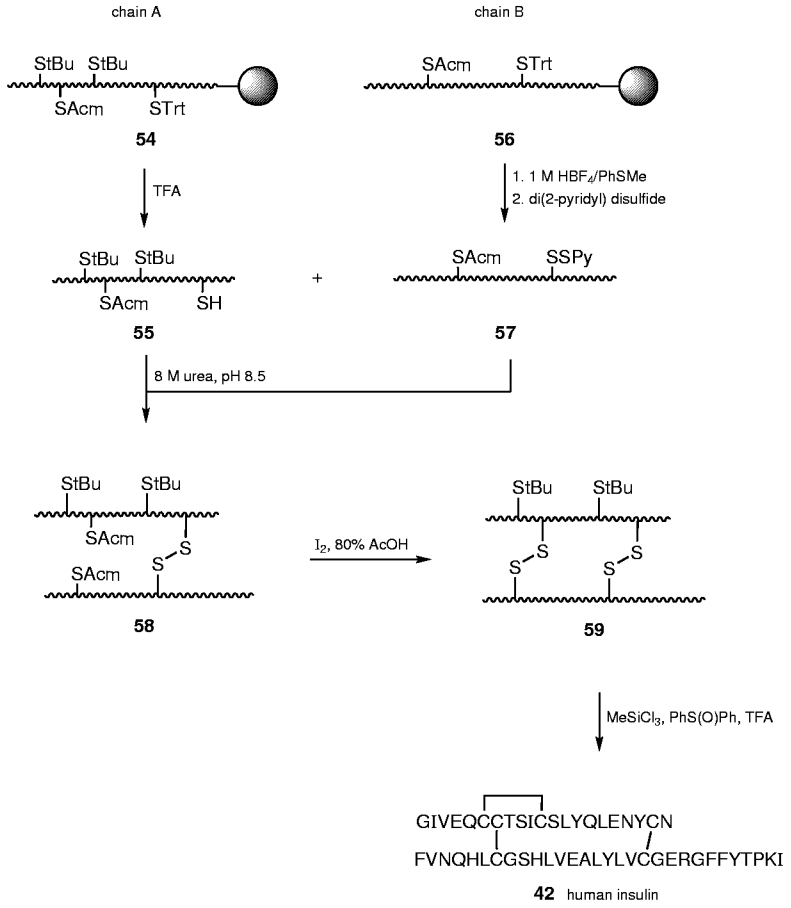
Scheme 23 Synthesis of Bombyxin IV[192]



The use of the *S*-pyridinesulfanyl activation procedure for site-directed disulfide bridging of two peptide chains followed by two regioselective disulfide bridging steps based on iodine and chlorosilane/sulfoxide oxidation methods was the first practical procedure for regioselective formation of three disulfide bonds without exposure of preformed disulfides to strong acid treatment. This strategy was used by Akaji et al.[193] for the synthesis of human insulin from the two suitably protected A and B chains as shown in Scheme 24. According to this synthetic route the three disulfide bonds were generated in three separate steps by exploiting three orthogonal S-protecting groups, i.e. Trt, Acm, and tBu. The Trt groups in **54** and **56** were selectively cleaved with acids in the presence of the *S*-Acm group and the *S*-tBu derivative, and the *S*-Acm group in **58** was oxidatively cleaved with iodine leaving the two tBu-protected cysteine residues intact yielding **59**.

for references see p 137

Scheme 24 Synthesis of Human Insulin by Regioselective Disulfide Bridging of the A- and B-Chains[193]



Cleavage of the Trt group of one chain **54** with a weak acid to give **55** and its subsequent thiolysis of the *S*-SPy derivative of the second chain **57** directs the formation of the first interchain disulfide bond in **58**. The second interchain disulfide bridge is formed between the two Acm-protected cysteine residues of the [bis(Acm), bis(tBu), mono-disulfide]-heterodimer **58** by treatment with iodine. Finally, treatment of **59** with chlorosilane/sulfoxide produces the third disulfide bond between the two tBu-protected cysteine residues yielding human insulin (**42**).

[Cys(Acm)^{A7},Cys(tBu)^{A6,11}]/[Cys(Acm)^{B7}]-Human Insulin (58):[193]
The [Cys(Acm)^{B7},Cys^{B19}]-B chain (**56**; 37 mg, 10.7 μmol) in iPrOH /2 M AcOH (1:1, 8 mL) was treated with di(2-pyridyl)disulfide (24 mg, 10 equiv) in iPrOH/2 M AcOH (1:1, 1 mL) at 25 °C for 40 min. The product was gel-filtered to give [Cys(Acm)^{B7},Cys(SPy)^{B19}]-B chain **57**; yield: 31 mg (80%). The product **57** (7.7 mg, 2.2 μmol) and [Cys(Acm)^{A7},Cys(tBu)^{A6,11},Cys^{A20}]-A chain (**55**; 11 mg, 4.3 μmol) were dissolved in 8 M urea buffer (pH 8.5, 5 mL) and the mixture was stirred at 25 °C for 50 min. The pH of the mixture was adjusted to 4 with AcOH. After gel-filtration on Sephadex G-25, the heterodimer **58** was purified by semipreparative HPLC; yield: 2.0 mg (15%); the compound was characterized by amino acid analysis and FAB-MS.

[Cys(tBu)^{A6,11}]-Human Insulin (59):[193]
A 20% soln of I_2 in MeOH (120 μL) was added in one portion to a soln of the mono-disulfide bridged human insulin (**58**; 5.8 mg, 0.96 μmol) in 80% aq AcOH (5 mL) and the mixture was stirred at 25 °C for 60 min. The excess I_2 was reduced with 1 M ascorbic acid in H_2O and the soln was gel-filtered on

Sephadex G-25. The product **59** was further purified by semipreparative HPLC; yield: 2.3 mg (42%); the compound was characterized by amino acid analysis and FAB-MS.

Human Insulin (42):[193]
The bis-disulfide bridged human insulin **59** (1.0 mg, 0.17 μmol) in TFA (0.6 mL) was treated with $MeSiCl_3$ (5 μL; 250 equiv) in the presence of PhS(O)Ph (0.7 mg, 20 equiv) at 25 °C for 15 min. NH_4F (3 mg) was added to the mixture, and the solvent was removed in under reduced pressure. The residue was dissolved in 50% AcOH (1 mL) and the soln was gel-filtered on Sephadex G-25. The product **42** was further purified by semipreparative HPLC; yield: 0.6 mg (61%); the synthetic human insulin was characterized by amino acid analysis and FAB-MS, it exhibited identical chromatographic and biological properties as a reference compound.

6.1.4.3 Multiple-Chain Cystine Peptides

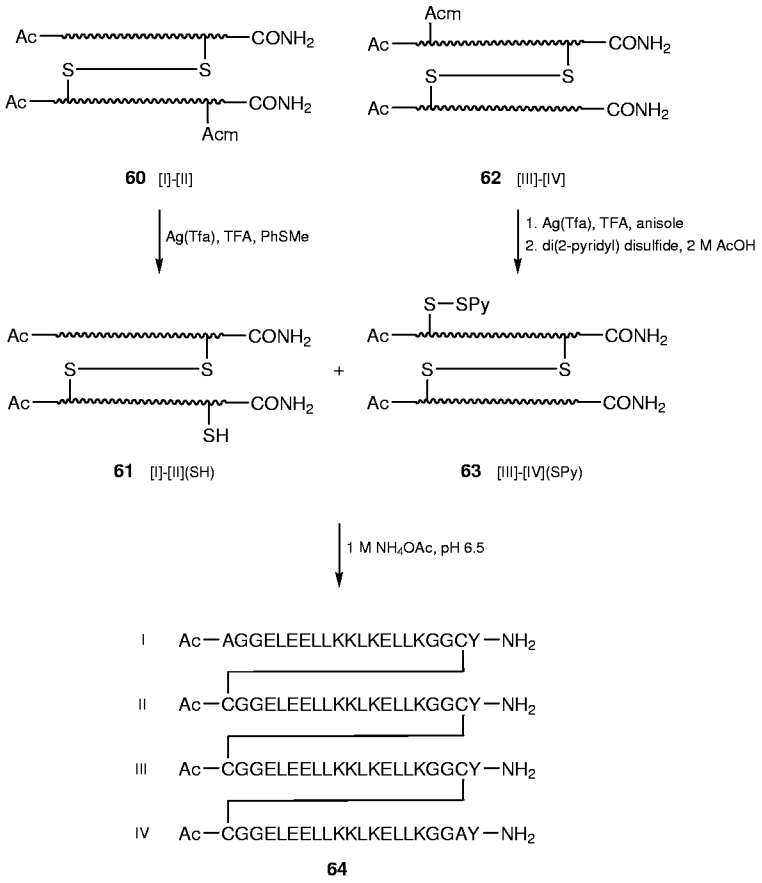
Nature often uses cystine knots of variable complexity to assemble polypeptide chains in the correct order and to stabilize the structure of the resulting proteins. Classical examples are proteins of the extracellular matrix with collagens and gelatines as the main components. This simple cystine-knot approach has been applied to synthetic constructs to ligate in a desired order helix bundles.[166,170,194–196] It has also been used to mimic natural multiple-chain containing proteins like collagen.[83,169,197]

In principle all the methods described in Section 6.1.3 for site-directed interchain disulfide formation can be applied to the stepwise interchain cross-linking of peptide chains by artificial cystine knots into homomeric or heteromeric constructs. As shown in Scheme 18 for the synthesis of heterotrimeric collagen peptides,[169,197] a combination of the two orthogonal cysteine protecting groups Acm and StBu was selected. This choice allows the selective conversion of the StBu-protected cysteine residues by reduction with phosphines into the S-unprotected cysteine peptides **28** and **30**, one of which **28** can be activated in acidic media with di[5-nitro(2-pyridyl)] disulfide to the *S*-Npys derivative for thiolysis by the second S-unprotected cysteine peptide to form the heterodimer **31** (see Section 6.1.3.2 for the Experimental Procedure). Reaction of the Acm-protected cysteine residue of the heterodimer **31** with Npys-Cl serves to activate it as *S*-Npys derivative for reaction with a third cysteine peptide **33** to form the heterotrimer **34**. In principle this cycle of reactions can be repeated allowing the preparation of peptide ladders cross-linked by disulfide bridges in a defined order with the sole limitation that peptide chains to be incorporated may not contain tryptophan residues. This residue terminates the reaction cycle due to its reactivity toward the activating reagent Npys-Cl, unless *S*-Acm deprotection is carried out with silver(I) trifluoroacetate in TFA as recommended by Fujii et al.[18] for peptides containing disulfides, followed by liberation of the thiol group via precipitation of the metal ion as silver chloride. This approach has been utilized by Futaki et al.[170,194,195] for the synthesis of a four-helix bundle as shown in Scheme 25.

In this approach an orthogonal S-protection consisting of the acid-labile Mob and the Acm group was selected. Free thiol peptides were prepared by acid treatment of the precursors and activated with di(2-pyridyl) disulfide to afford the *S*-SPy derivatives which served for the preparation of the two heterodimers **60** and **62** via thiolysis with a second cysteine peptide. In both heterodimers **60** and **62** the Acm group was cleaved with silver(I) trifluoroacetate in TFA to generate the mercaptide. Following gel filtration in acidic media the free thiol from **62** was activated as the *S*-SPy derivative **63** and subsequent thiolysis with the second heterodimer **61** in the free thiol form produced the heterotetramer **64**.

for references see p 137

Scheme 25 Synthetic Route for a Heterotetrameric Four-Helix-Bundle Cross-linked by Disulfides[170]



Synthesis of a Four-Helix-Bundle Construct:[170]

[Cys-I], [Cys,Cys(Acm-II], [Cys(Acm),Cys-III], and [Cys-IV]; Typical Procedure:[170]
The monomeric peptides [Cys-I], [Cys,Cys(Acm-II], [Cys(Acm),Cys-III], and [Cys-IV] were synthesized by Fmoc chemistry on Rink-amide resin as *S*-Mob derivatives and cleaved/deprotected with 1 M TMSBr/thioanisole in TFA in the presence of *m*-cresol and 1,2-ethanedithiol as scavengers. Following gel filtration on Sephadex G-10 with 1 M AcOH as solvent and HPLC purification the peptides were obtained in 30–40% yield. Each product was characterized by LSIMS, HPLC, and amino acid analysis.

[Cys(SPy),Cys(Acm)-II]:[170]
Di(2-pyridyl)disulfide (3.3 mg, 15 μmol) in propan-2-ol (0.3 mL) was added to a soln of [Cys, Cys(Acm)-II] (12.2 mg, 5 μmol) in 2 M AcOH (1 mL). The mixture was stirred at rt for 1 h. The soln was applied to a Sephadex G-10 column and the fractions corresponding to the front main peak were collected and lyophilized. The resulting powder (8.7 mg) was further purified by HPLC to give the product as a white fluffy powder; yield: 3.8 mg (30%); the product was characterized by LSIMS and HPLC.

[Cys(SPy)-IV]:[170]
Di(2-pyridyl)disulfide (30.8 mg, 60 μmol) in propan-2-ol (1.2 mL) was added to a soln of [Cys-IV] (46.7 mg, 20 μmol) in 2M AcOH (4 mL). The mixture was stirred at rt for 1 h and the product was purified as described above; yield: 18.4 mg (38%); the product was characterized by LSIMS and HPLC.

Disulfide-Cross-linked [I]/[II] (60):
[Cys-I] (7.4 mg, 3.2 μmol) in 0.1 M NH_4OAc buffer (pH 6.5, 3.2 mL) was mixed with the soln of [Cys-(SPy),Cys(Acm)-II] (8.1 mg, 3.2 μmol) in 0.1 M NH_4OAc buffer (pH 6.5, 3.2 mL). The mixture was stirred for 15 min at rt and the product (**60**) was purified as described above; yield: 11.0 mg (74%); the

product was characterized by LSIMS, HPLC, and amino acid analysis. The dimer **62** was prepared and characterized in a similar manner.

Synthesis of the Heterotetramer [I]/[II]/[III]/[IV] (64):[170]
Heterodimer **62** (7.9 mg, 1.7 μmol) was treated with Ag(Tfa) (43 mg, 0.17 mmol) in TFA (0.8 mL) in the presence of anisole (20 μL) in an ice bath for 1.5 h. Et_2O (40 mL) was added and the resulting precipitate was collected by centrifugation and dried. The resulting powder was dissolved in 50% AcOH (2 mL) and gel-filtered on Sephadex G-10 using 1 M AcOH as solvent. To the resulting heterodimer (6.4 mg, 1.4 μmol) in 2 M AcOH (650 μL) di(2-pyridyl)disulfide (0.91 mg, 4.1 μmol) in iPrOH (90 μL) was added. The mixture was stirred at rt for 1 h and the product was purified by gel filtration to give **63**; yield: 5.6 mg (72%).
Heterodimer **61** was obtained from **60** (10.7 mg, 2.3 μmol) by treatment with Ag(Tfa) (59 mg, 0.23 mmol) in TFA/anisole (1.1 mL/20 μL) followed by gel filtration as described above to give the desired product **61**; yield: 9.0.
The heterodimer **61** (2.31 mg, 0.5 μmol) was added to a soln of **63** (2.36 mg, 0.5 μmol) in 0.1 M NH_4OAc buffer containing 6 M urea buffer (pH 6.5, 1 mL). After 2 h at rt, the product **64** was purified by preparative HPLC; yield: 5.9 mg (48%); the product was characterized by amino acid analysis and LSIMS.

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6.1.5 Oxidative Refolding of Multiple-Cystine Peptides

T. KIMURA

Disulfides constitute essential structural elements in a large variety of bioactive peptides such as hormones, neurotransmitters, growth factors, inhibitors, and toxins, which are generated by proteolytic processing of prefolded protein precursors and then released from the cells in the bioactive form into the extracellular medium to exert their physiological function. In these naturally occurring bioactive peptides intra- and/or intermolecular disulfide bridges form single or multiple cyclic heterodetic structures which impart considerable conformational restraints and thus are important in inducing and maintaining the bioactive folds.

With Anfinsen's classic in vitro experiments on the oxidative refolding of ribonuclease A[1,2] the central principle of self-assembly of proteins was established. Correspondingly, most proteins and even single domains of proteins contain within their amino acid sequence sufficient information to direct folding into the correct functional spatial structure, although more recently it has been recognized that the folding efficiency in vitro and particularly in vivo is supported by a wide variety of molecular chaperons and folding catalysts. These do not determine the final conformation of the polypeptide chain, but rather increase the efficiency of the folding process by inhibiting off-pathway aggregation phenomena[3,4] or by catalyzing reshuffling of disulfides[5] and rate-limiting isomerization steps.[6] Extensive studies on disulfide containing proteins, including bovine pancreatic trypsin inhibitor,[7–9] ribonuclease A,[10–12] and ribonuclease T1[13] have clearly demonstrated that folding and disulfide bond formation are thermodynamically coupled processes, and that the specificity of disulfide formation is determined by the extent to which noncovalent interactions favor conformations suitable for the correct cysteine pairings.[14,15]

In the case of bioactive cystine-peptides that are products of posttranslational processing of protein precursors, the essential structural information for correct oxidative refolding may have been partially or totally lost. Thus, a drastically decreased efficiency is expected for such refolding processes that should lead to poor yields of the correctly folded products or even to a total failure. For peptides containing two cysteine residues only one intramolecular disulfide bond can be formed if oligomerization is avoided by the use of high dilution methods; however, with increasing numbers of cysteine residues the possible isomers formed in random oxidation processes increase according to the equation:

$$N_{2n}^{j} = \frac{(2n)!}{2^j(2n-2j)!j!}$$

where N is the number of isomers resulting from random oxidation of $2n$ thiol groups to form j disulfide bonds.[16] This means that with four, six, or eight cysteine residues in principle 3, 15, and 105 different intramolecular disulfide isomers can be formed, and the complexity of possible isomers increases enormously when two or more polypeptide chains are cross-linked by interchain disulfide bridges as in the classical case of insulin.[17] For this double-stranded polypeptide the theoretically expected yield from random oxidation should therefore be close to zero; nevertheless under optimized conditions insulin is obtained in yields of up to 25–30%[18,19] suggesting that the native insulin represents the most stable among the possible isomers.[20]

Since an increased number of disulfide bridges in relatively short polypeptide chains leads to compact globular structures with the disulfides mainly buried in the nonpolar core, such excised protein fragments should represent, even in the precursor molecules, stable subdomains. Therefore, sufficient structural information can be retained for a correct refolding at least to some extent, if appropriate experimental conditions are applied in terms of peptide concentration, redox reagents, temperature and/or reaction buffers. A great deal of

knowledge in this context has been acquired with model systems, for example with bovine pancreatic trypsin inhibitor.[7] However, formation of larger amounts of disulfide isomers are still often observed.

The inherent drawbacks of the oxidative refolding approach for synthetic polypeptides containing multiple cysteine residues is the individual behavior of each peptide that derives from the encoded sequence, more or less pronounced structural information which prevents general procedures to be elaborated and proposed. Nevertheless, this synthetic approach remains attractive because of its simplicity compared to the synthetic strategies for regioselective disulfide bond formation (Section 6.1.1–6.1.4), and it is certainly indispensable if the number of cysteine residues exceeds the presently available chemistry for site-directed cysteine pairings.

6.1.5.1 Oxidative Refolding of Peptides with Two Intramolecular Disulfide Bonds

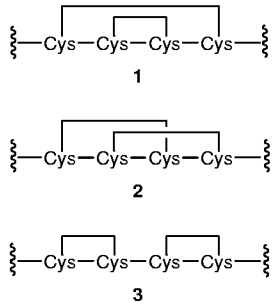
The simplest approach to disulfide bond formation of chemically synthesized peptides and proteins involves (i) complete deprotection including that of the cysteine thiol functions, and (ii) mild oxidation of the thiol groups to form the folded product with the native cystine connectivity. With careful attention to experimental conditions, the native-type folding of the peptide can be accomplished; however, misfolded disulfide isomers are often produced as the main products in spite of efforts to optimize the reaction conditions.

In principle, peptides containing two cysteine residues on oxidation should form only one disulfide-bridged compound if high dilution is used. It was, however, shown that the readiness to form the disulfide loop depends on the number of intervening residues m between the two cysteines which is more important than the type of residues at least up to $m = 6$.[21] The case of $m = 0$ as present in malformin A_1[110] is very rare as shown by a statistical evaluation of peptides and proteins with known disulfide connectivities.[111] The adjacent cysteine residues can form an intramolecular disulfide only when the peptide bond connecting them is in the *cis* configuration.[22] Due to the enthalpic cost of the *trans*→*cis* isomerization, besides the cyclic monomer, high proportions of parallel and antiparallel dimer are formed[23] and thus, the small-size loop is difficult to obtain by air oxidation.[112] Therefore formation of such disulfide bonds is highly improbable in multiple-cysteine peptides. It has also been demonstrated with model peptides that with $m > 0$, disulfide loop formation is favored with even-numbered m values and unfavored with odd-numbered m values.[21] Correspondingly, intramolecular disulfide bridging with $m = 1$ or 3 is highly disfavored. Peptides with $m = 1$ have been synthesized and the conformational restriction is strong allowing the ring structure to adopt only a γ-turn centered on the intervening amino acid residue.[24,25] The amplitude of the oscillating K_{ox} values diminishes with increasing m values to degenerate into a general dependence from the reduced probability of random collision between two cysteines when they are separated by longer distances in the linear sequence, where the K_{ox} becomes proportional to $m^{-3/2}$.[26,27] With the results derived from small-sized disulfide loops in model peptides good estimates for the most favored isomers, at least in relatively short peptides containing two disulfide bridges, are possible.[28]

When peptides containing four cysteine residues in the molecule are randomly oxidized, three different disulfide isomers can be formed (Scheme 1), isomer **1** with the disulfides aligned in parallel manner, isomer **2** with crossed disulfides and isomer **3** with sequential disulfides; these isomers are also named in a more descriptive manner as the ribbon, globule and string-of-beads isomers.[28] The preferred formation of one of the isomers strongly depends upon the sequential topology of the cysteine residues, but also upon the sequence composition that may favor more or less globular structures with hydrophobic cores.

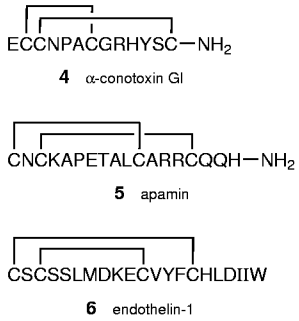
for references see p 159

Scheme 1 The Three Possible Disulfide Isomers of Bis-cystinyl Peptides



Of this family of peptides containing two intramolecular disulfide bonds the most studied in terms of oxidative refolding are α-conotoxins with two adjacent cysteine residues, i.e. with $m = 0$, the bee venom toxins, for example apamin and mast cell degranulating peptide, and snake venom toxins, exemplified by sarafotoxins, and endothelins, mammalian peptide hormones with the characteristic Cys-$(Xaa)_1$-Cys/Cys-$(Xaa)_3$-Cys motif (Scheme 2). With $m = 0$ or 1 all these peptides are expected to show a weak tendency to form the isomer **3** with a disulfide bond between two proximal cysteine residues. This was fully confirmed by oxidative refolding experiments.

Scheme 2 Structure of α-Conotoxin GI, Apamin, and Endothelin-1



6.1.5.1.1 Air Oxidation in Aqueous Media

Air oxidation of fully reduced α-conotoxin GI (**4**) led to an isomer **1/2/3** distribution of 2:7:1 with the native isomer **2** highly predominant,[29,30] whereas in the case of the reduced apamin **5** the native isomer **2** was formed exclusively.[31,32] Air oxidation of the fully reduced endothelin-1 (**6**) generated the isomers **1** and **2** in the ratio of 75:25 and the amount of isomer **3** was negligible, even if the reaction conditions were varied.[33–35] Thus, again the most readily formed isomer is the native one. Due to solubility problems, the reduced **6** has to be dissolved in 0.1% aqueous ammonia for the oxidation experiments. Other authors have reported the use of iPrOH / H_2O mixtures (70:30) at pH 8.2.[36]

Since the introduction of the redox pair GSH/GSSG as an auxiliary oxidative refolding agent for proteins by Saxena and Wetlaufer,[37] this redox reagent is routinely used to allow, via thiol/disulfide exchange reactions, the thermodynamically most stable oxidation product to be formed. By applying this redox agent in the case of α-conotoxin GI the native-type disulfide isomer **2** is obtained almost quantitatively, whereas for **6** the formation of the native-type disulfide isomer **1** cannot be improved. Interestingly, the identical isomer **1/2** ratio of 75:25 is obtained on dissolving separately the isomer **1** and **2** of **6** in the GSH/GSSG redox buffer, clearly indicating that this ratio represents the thermodynamically favored product distribution.

Air Oxidation of Reduced Endothelin-1 (6):[33]

CAUTION: *Hydrofluoric acid (HF) is highly toxic and corrosive, and severely irritating to the respiratory system and eyes. All operations should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

A fully-protected peptide-resin (550 mg) with four Cys[Bzl(4-Me)] residues was treated at −2 °C with HF (10 mL) in the presence of 1,4-butanedithiol (2 mL) and *p*-cresol (0.6 mL). The fully deprotected peptide (50 mg) was dissolved in 0.1% aq NH_3 (100 mL) containing 8 M urea and the soln was kept at rt with gentle stirring in contact with air-oxygen. The progress of the oxidation reaction was monitored by RP-HPLC. After 3 h the soln was acidified to pH 2 with TFA, desalted and purified by RP-HPLC using MeCN containing 0.1% TFA as eluant; yield: 8 mg (16%). The product was characterized by amino acid analysis and elemental analysis, and the cystine connectivities were confirmed by comparison with the product synthesized by the technique of regioselective disulfide bond formation.

6.1.5.1.2 Air Oxidation in Aqueous Organic Media

The three-dimensional solution structure of endothelin-1 (**6**), as determined by NMR conformational analysis,[38–40] consists of an α-helix spanning the Cys-$(Xaa)_3$-Cys portion of the molecule on which a hairpin loop is disulfide-bridged to the N-terminal Cys-$(Xaa)_1$-Cys part aligned in an extended conformation and in an antiparallel orientation to the C-terminus. A very similar structure is assumed by apamin (**5**) with its isomer type **2** disulfide connectivity.[41–43] Again the Cys-$(Xaa)_3$-Cys portion is folded into an α-helical stretch which is crosslinked by the disulfide bonds to the N-terminus aligned in parallel orientation upon a hairpin loop followed by an additional β-turn. Due to the presence of a common α-helix, this type of structure has been named the cystine-stabilized α-helix motif.[38,44,45] The high preference of **5** for the isomer type **2** cystine framework has been attributed to the presence of the proline in position 6 that favors the more globular fold in comparison to the ribbon-type fold of **6** as disulfide isomer **1**.[46] Conversely for α-conotoxin GI (**4**), which according to NMR conformational analysis is folded into a tight globular structure,[47,48] the high preference for the type **2** cystine connectivity has been attributed to the highly favored cystine loop with $m = 4$ vs a cystine loop with $m = 3$ as type **1** isomer.[28]

Studies of the effect of α-helix inducing solvents such as trifluoroethanol (TFE) on the product distribution of oxidative folding of endothelin-1 (**6**) (0.01 mM peptide concentration in 80% TFE, pH 9.5) clearly revealed that induction of the α-helical conformation favors the onset of the cystine-stabilized α-helix structural motif with concomitant enhancement of isomer **1** (88:12 vs 75:25) (Table 1).[49] A similar enhancement of the correctly folded **6** by the addition of TFE has also been observed under other conditions.[50]

Table 1 The Effects of TFE on the Oxidative Folding of Endothelin-1[49]

Concentration of TFE %	Ratio (Isomer **1**/Isomer **2**)	
	Without GSH/GSSG	With GSH/GSSG
0	75:25	71:29
80	88:12	92:8

Whilst the dichroic properties of fully reduced **6** in aqueous buffer (pH 9.5) are typical of a random coil structure, addition of TFE leads to a CD spectrum that indicates an increased content of α-helical structure as its pattern becomes similar to the CD spectrum of **6** in 50% TFE at pH 7.0.[45] These results confirm that induction of α-helical conformation by addition of TFE is responsible for the observed increased content of the native isomer **1**.

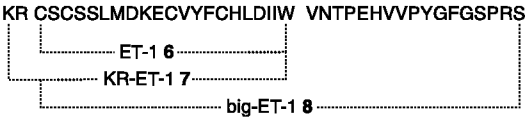
for references see p 159

6.1.5.1.3 Effect of the Prosequence on the Oxidative Folding of Endothelin-1

In general, folding of peptides and proteins in vivo are considered to occur in their precursor forms before processing. Therefore, the prosequence is assumed to participate in the correct folding pathways. For example, the prosequence of subtilisin, α-lytic protease, and of carboxypeptidase Y have been reported to act as intramolecular chaperons during the folding.[51–54] For bovine pancreatic trypsin inhibitor the single cysteine residue in its prosequence has been reported to accelerate the disulfide bond formation and to rearrange the mismatched disulfide bonds to the native ones and, thus, to act as an intramolecular thiol reagent.[55]

In the case of endothelin (**6**), the initial gene product consisting of 212 amino acid residues[56] is first converted into the so-called big-ET-1 (**8**) by cleavage at the Lys-Arg sites, followed by scission of the Trp-Val bond between position 21 and 22 (Scheme 3). Since folding processes in vivo produce exclusively the biologically active compounds, addition of a part of the prosequence of **6** was expected to possibly favor the correct oxidative folding even when performed in vitro. The C-terminal extension of **6** by the 17-residue prosequence to the big-ET-1 form (**8**) was without any effect, since isomers **1** and **2** are produced in the same ratio as in the case of **6** (Scheme 3).[57]

Scheme 3 Primary Structures of Endothelin-1, KR-ET-1 and big-ET-1



Conversely, when **6** was extended N-terminally by a small portion of the prosequence, that is by the Lys-Arg dipeptide to give KR-ET-1 (**7**), and subjected to oxidation in 0.1 M NH_4OAc buffer (pH 9.5) at 25 °C for 24 hours the ratio of native to nonnative-type disulfide isomer increases remarkably (88:12 vs 75:25 for **6**), whilst isomer **3** is not detectable. In the presence of GSH/GSSG an additional increase to almost quantitative formation of the native isomer was observed (Table 2).[58] This improvement was completely abolished by substituting Asn for Asp at position 8 (D8N-KR-ET-1), whereas most of the increase was maintained with similar carboxamide analogues in positions 10 and 18 (Table 2).

Table 2 Ratio of Isomers **1** and **2** in the Oxidative Folding Reaction of Reduced Endothelin-1, KR-ET-1 and Carboxamide Analogues of KR-ET-1 in the Presence or Absence of Redox Reagents[58]

Peptide	Ratio (Isomer **1**/Isomer **2**)	
	Without GSH/GSSG	With GSH/GSSG
6	75:25	71:29
7	88:12	96:4
8	75:25	71:29
D8N-KR-ET-1	66:34	68:32
E10Q-KR-ET-1	88:12	94:6
D12N-KR-ET-1	80:20	90:10

CD and NMR structural analysis of **7** clearly revealed an extended α-helical stretch in the C-terminal portion of the molecule which is stabilized by a particular salt-bridge between the Arg^{-1} guanido group and the Asp^{8} carboxylate which is abolished in the D8N-KR-ET-1 analogue with correspondingly lower α-helical content.[59] These results imply that similarly to the TFE effect, a stabilization of the α-helix via the Arg^{-1}/Asp^{8} salt-bridge favors remarkably the correct oxidative folding of the short endothelin proform **7**.

Double basic residues such as Lys-Arg in precursor proteins are known to be a recognition sequence of the processing enzyme(s). Since the above results demonstrate a correlation between oxidative folding efficiency in vitro and structure stabilization of **6** by the N-terminal extension with the Lys-Arg sequence, such double basic residues could possibly favor the correct oxidative folding of other multiple-cysteine peptides.

6.1.5.2 Oxidative Refolding of Peptides with Three or More Intramolecular Disulfide Bonds

Oxidative refolding of ribonuclease A, fully reduced under denaturing conditions, regenerates in high yields the native four disulfide bond pairings, conformation, and activity on removal of the denaturant and reductant, although statistically 105 different disulfide isomers could be formed.[1] The easy correct intramolecular disulfide bond formation in the case of this enzyme derives predominantly from conformational parameters.[10–12] Oxidative folding of synthetic peptides containing three or more disulfide bonds in the molecule would be of practical use, if similarly to the ribonuclease A, the polypeptide chains in their reduced form would fold preferentially into native-like conformations coupled to the correct cystine connectivities. However, nonnative isomers may easily form because of misfolding or even as obligatory intermediates in the folding pathways. In general, the more complex peptides or proteins are obtained in relatively low yields, even after optimization of experimental protocols for synthesis, purification and oxidative folding. Therefore, great attention must always be paid to experimental conditions such as pH, buffer, peptide concentration, redox reagents, temperature, additives, denaturants, and organic solvents to generate the native-type product in satisfactory yields. Of course, the success of the oxidative folding greatly depends also on the purity of the starting reduced peptide.

6.1.5.2.1 Air Oxidation in the Absence of Redox Reagents

ω-Conotoxins from marine snails of the genus *Conus* are characterized by their ability to bind to and inhibit presynaptic Ca^{2+} channels and are relatively short peptides (25–30 residues) with three disulfide bonds. Due to the highly conserved precursor sequences, a role of these prosequences in folding of ω-conotoxins with their low level of sequence homology was postulated,[60] but experimentally not observed, since air oxidation of synthetic peptides corresponding to their mature forms leads to the native disulfide-coupled folds with rather high efficiency.[61] A detailed study of the effect of the prosequences on the folding of ω-conotoxins clearly revealed that in the precursor form generation of the native fold is not favored thermodynamically or kinetically.[62] Thus, an efficient burial of the nonpolar disulfides as revealed by the NMR conformational analysis of this family of toxins,[63] may be an important determinant of the folding specificity in these molecules.

Air oxidation of synthetic fully reduced ω-conotoxin GVIA (**9**) (Scheme 4), a 27-residue peptide,[64] generated a main product that possessed the identical biological activities as the natural product whose cystine framework was determined with the synthetic compound.[65] The choice of buffer system and peptide concentration can, however, dramatically affect the folding efficiency when the DMSO-mediated oxidation method was applied.[66]

Scheme 4 ω-Conotoxin GVIA[64,65]



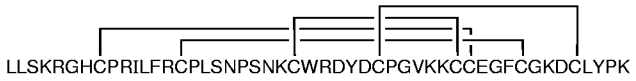
9

O = Hyp

for references see p 159

Results similar to those reported for ω-conotoxin GVIA were obtained with the Na^+, K^+-ATPase inhibitor-I (**10**),[67] a 49 amino acid residue peptide containing four intramolecular disulfide bonds (Scheme 5).[68] Even in this case the addition of redox reagents to the oxidation buffer had a minor effect on the product distribution.

Scheme 5 Na^+, K^+-ATPase Inhibitor-I[64]



10

Oxidative Folding of ω-Conotoxin GVIA (9):[65]
The 27-residue peptide was synthesized in fully protected form by soln procedures. All of the protecting groups except the six Acm thiol protecting groups were removed with liq HF at 0 °C and the resulting intermediate was purified by CM-cellulose chromatography followed by RP-HPLC. The purified peptide was treated with $Hg(OAc)_2$ (1.1 equiv/Acm) in 50% AcOH and the resulting crude product was purified on Sephadex G-25 after adding β-ME. The reduced peptide (130 mg) at 0.05 mM concentration in 0.05 M sodium acetate buffer (800 mL, pH 7.5) was then exposed to air oxidation for 3 d at 4 °C. The progress of the reaction was monitored with Ellman reagent and by HPLC. The product was isolated by preparative HPLC; yield: 16%. The product was characterized by MALDI-TOF-MS and amino acid analysis; the cystine connectivities were determined by enzymatic digestion and amino acid analysis in combination with synthetic approaches.

Oxidative Folding of Na^+, K^+-ATPase Inhibitor-I (10):[67]
The reduced form of Na^+, K^+-ATPase inhibitor-I (**10**) was obtained by treatment of the protected peptide synthesized by the soln procedure with HF, followed by reaction with $Hg(OAc)_2$. After purification of the crude product on Sephadex G-25, the reduced peptide (110 mg) was dissolved in 0.1 M NH_4OAc buffer (1 L, pH 7.8) at a peptide concentration of 0.018 mM and then stirred at rt. After 24 h, the major peak in the HPLC, which coeluted with the natural product, corresponded to 55% of the product distribution. The mixture was acidified to pH 3 with AcOH and **10** was purified by RP-HPLC. When the oxidation was carried out in the presence redox reagents at a peptide/GSH/GSSG ratio of 1:100:10, after 24 h the major oxidation product increased to 69%. The mixture was acidified with AcOH and the product (**10**) isolated by preparative HPLC; yield: 20%. The product was characterized by MALDI-TOF-MS and amino acid analysis; a combination of enzymatic peptide mapping and synthetic approaches were applied to assign the cystine connectivities.

6.1.5.2.2 Air Oxidation in the Presence of Redox Reagents

Many scorpion toxins, insect defensins, and enzyme inhibitors are cystine-rich polypeptides containing three to four disulfide bonds. In a large number of these toxins, two cystines are involved in the consensus Cys-$(Xaa)_1$-Cys/Cys-$(Xaa)_3$-Cys framework which is responsible for the common characteristic fold consisting of an α-helix and a two- or three-stranded antiparallel β-sheet (αββ-fold or βαββ-fold). For a review see ref[69]. The overall compact globular structures of these cystine-rich peptides contain the cystine stabilized α-helix motif (Section 6.1.5.1.2) which is further stabilized by a third disulfide bond between the N-terminus and the β-strand adjacent to the helix and in some cases by an additional fourth disulfide bridge. Due to the presence of the cystine stabilized α-helix motif, a preferred initial formation of this motif followed by its stabilization via the additional disulfides was expected. However, in contrast to what was observed for the cystine peptides containing only the cystine stabilized α-helix motif, simple air oxidation is not successful.

As exemplarily shown in the case of charybdotoxin, a 37-residue peptide with three intramolecular disulfide bonds,[70] operating in redox buffer was crucial for efficient formation of the correct disulfide bonds.[71] When the reduced peptide was oxidized in 0.1 M NH_4OAc buffer (pH 8.0) at 0.11 mM concentration in the presence of redox reagents (peptide/GSH/GSSG ratio of 1:60:6), the main product was the native peptide contaminated

only by small amounts of additional oxidation products; in the absence of redox reagents, these contaminants were formed in similar amounts as the desired native isomer. On addition of GSH/GSSG in the same proportion as above, the reaction mixture resulting from direct air-oxidation re-equilibrated to the product distribution obtained by oxidation in redox buffer. Apparently, reduced charybdotoxin folds into a number of disulfide isomers or intermediates in the absence of redox reagents, whereas with GSH/GSSG a reshuffling is taking place via thiol/disulfide exchange reactions that generate in high yield the thermodynamically most stable native cystine connectivity. A similar crucial effect of the redox buffer has been reported for oxidative folding of calciseptine (**11**) (Scheme 6),[72] dendrotoxin I,[73] and ω-conotoxin GS.[74] The GSH/GSSG ratios had to be optimized for the single peptides and kept at overall concentrations similar to those of glutathione in the secretory organelles. Alternatively, under optimized conditions even Cys/$(Cys)_2$ was found to act as an efficient redox pair.[75]

Scheme 6 Calciseptine[72]



11

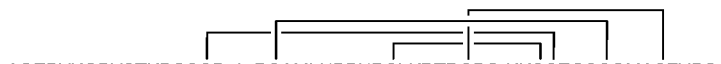
Oxidative Folding of Calciseptine (11):[72]
The reduced form of calciseptine (148 mg, 21 μmol) was dissolved at 0.11 mM concentration in 4.2 L of 0.1 M NH_4OAc buffer (pH 7.8) containing GSH (645 mg) and GSSG (129 mg) at a peptide/GSH/GSSG ratio of 1:100:10, and the oxidative folding was carried out with gentle stirring for 48 h at 25 °C. The mixture was acidified to pH 2.5 by adding TFA, and the folded peptide **11** was purified by preparative RP-HPLC; yield: 61 mg (42%). The product was characterized by amino acid analysis and MALDI-TOF-MS; the cystine connectivities were assessed by enzymatic digestion, sequence analyses, and MS.[76]

6.1.5.2.3 Air Oxidation in the Presence of Denaturants and Organic Solvents

Oxidative folding of peptides in aqueous solution needs to be carried out at high dilution in order to avoid aggregation and intermolecular oligomerization. However, the solubility of the peptides in both reduced and oxidized forms can often raise serious problems. Occasionally, denaturants such as urea or guanidine hydrochloride (Gn-Cl) are added to prevent aggregation. Similarly, the addition of organic solvents is effective in some cases not only to avoid the solubility problem, but also to accelerate the folding reaction. DMSO, known to be itself an oxidizing reagent, is used as a denaturing cosolvent which improves the solubility of peptides in the oxidative folding reactions.[77] However, limited use of DMSO has been made so far, particularly in the oxidative refolding of peptides containing more than three disulfide bonds. The beneficial effects of added denaturants or organic solvent to the redox buffer is well exemplified with the oxidation of elafin (**12**),[78] the presynaptic calcium channel blocker PLTX-II (**13**),[79] and ω-agatoxin IVA (**14**).[80]

Elafin (**12**), an elastase-specific inhibitor isolated from human skin, is a 57-residue peptide containing four intramolecular disulfide bonds (Scheme 7).[81,82]

Scheme 7 Elafin[78]

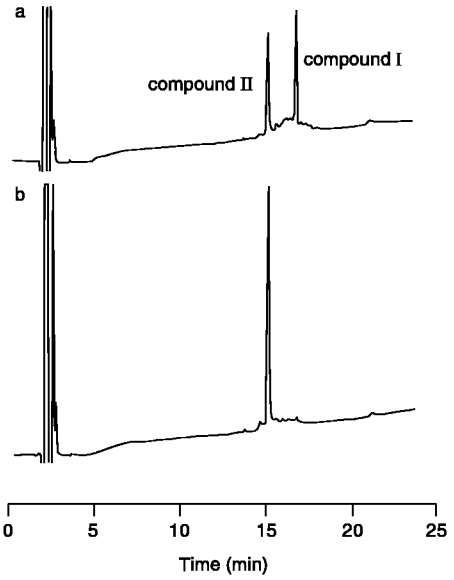


12

for references see p 159

For dissolving the reduced peptide at a concentration of 0.017 mM, addition of Gn-Cl as a denaturant was necessary. Air oxidation of the peptide in 0.1 M NH_4OAc (pH 7.8) containing 2 M Gn-Cl in absence of GSH/GSSG led to two major products, compounds I and II in a 1:1 ratio (Figure 1a). Bioassays confirmed that compound II corresponded to the native isomer **12**. The ratio of the two oxidation products varies with the concentrations of Gn-Cl and redox reagents in the opposite manner. An increased concentration of Gn-Cl leads to a decreased amount of compound II, whereas the amount of compound I increases. The opposite effect is obtained by varying the concentration of redox reagent, and as shown in Figure 1b, with 0.1 M Gn-Cl and at a peptide/GSH/GSSG (1:100:10) ratio the native-type disulfide bonds are formed exclusively.

Figure 1 HPLC Profile of Oxidative Folding of Elafin; (a) After Folding Reaction in NH_4OAc Buffer Containing 2 M Gn-Cl, without Redox Reagents. (b) After Folding Reaction in NH_4OAc Buffer Containing 2 M Gn-Cl with Redox Reagents (100:10).[78] [a]



[a] Column: YMC-Pak ODS (4.6×150 mm); eluent: 10–60% MeCN/0.1% TFA in 25 min at 50 °C; flow rate 1.0 mL min^{-1}; detection: 220 nm.

A similar optimization of reaction conditions was required for oxidation of PLTX-II (**13**) (Scheme 8). This presynaptic calcium channel blocker is a 44-residue peptide containing five intramolecular disulfide bonds and an O-palmitoylated Thr amide at the carboxy-terminus.[83]

Scheme 8 Presynaptic Calcium Channel Blocker PLTX-II[79]

ADCSATGDTCDHTKKCCDDCYTCRCGTPWGANCRCDYYKARCDT(Pal)—NH_2

13

In order to investigate the factors that affected yields of the correct oxidative folding of PLTX-II (**13**) the following experiments were carried out: to the aqueous solution of the reduced peptide (NH_4OAc, pH 7.7 and higher) containing GSH/GSSG were added organic solvents and inorganic salts, and the reaction temperature, pH, and peptide concentrations were varied.[79] Addition of organic solvents such as MeOH, EtOH, glycerol, ethyleneglycol,

or MeCN did not improve the solubility. Good solvents for the peptide were 50% iPrOH and 50% TFE, but with the latter solvent the oxidation reaction led to a complex mixture of unidentified products. The peptide precipitated from a buffer containing 50% iPrOH at peptide concentrations higher than 0.05 mM. The oxidation reaction was very slow (3–4 days) and the reaction rate decreased significantly at 5 °C. The folding reaction was accelerated by SO_4^{2-} and HPO_4^{2-} anions, and suppressed by the ClO_4^- anion. The initial rate of the folding reaction was increased by increasing pH, but at pH > 9 the folding reaction gave again a complex mixture of unidentified products. From these results optimal conditions could be derived (see following experimental procedures).

For ω-agatoxin IVA (**14**), a toxic 48-residue peptide with four intramolecular disulfide bonds (Scheme 9),[84] the main beneficial effects were obtained by adding 20% glycerol to a 0.01 mM peptide solution in 0.2 M NH_4OAc (pH 7.8) containing 1 mM EDTA in the presence of redox reagents at a peptide/GSH/GSSG ratio of 1:100:10, and by operating at 4 °C.[80]

Scheme 9 Structure of ω-Agatoxin IVA



14

Oxidative Folding of Elafin (12):[78]
The reduced peptide (150 mg) was air-oxidized in 0.1 M NH_4OAc buffer (1.5 L) (pH 7.8) at 0.017 mM concentration in the presence of 0.1 M Gn-Cl and of glutathione at a peptide/GSH/GSSG ratio of 1:100:10 for 22 h at 25 °C. The folding reaction was monitored by RP-HPLC after which the mixture was acidified to pH 4.3 with AcOH and desalted by preparative RP-HPLC. The product was further purified by CM-cellulose chromatography using a linear gradient of 0.02 M NH_4OAc (pH 5.8) to 0.35 M NH_4OAc as eluent followed by preparative RP-HPLC; yield: 53%. The product was characterized by ESI-MS and amino acid analysis; the cystine connectivities were determined by enzymatic digestion, amino acid analysis, and MS.

Oxidative Folding of the Presynaptic Calcium Channel Blocker PLTX II (13):[79]
The reduced peptide (52 mg) was dissolved at 0.01 mM concentration in 25 mM NH_4OAc (pH 8.5) containing 50% iPrOH, 25 mM Na_2HPO_4 and 1 mM EDTA in the presence of glutathione at a peptide/GSH/GSSG ratio 1:100:10 and the soln was gently stirred at 30 °C for 72 h. The mixture was acidified to pH 2.0 with TFA, desalted, and purified by RP-HPLC using MeCN containing 0.1% TFA as eluant; yield: 17 mg (33%). The product was characterized by amino acid analysis and FAB-MS; the cystine connectivities have not been determined due to the resistance of PLTX II to almost all enzymes.

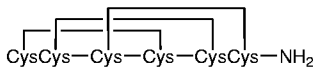
6.1.5.2.4 Effect of Peptide Concentration

Although it is a general rule to operate at high dilution (10^{-4}–10^{-5} M) in oxidative folding reactions to avoid formation of intermolecular disulfide-bonded species, μ-conotoxin GIIIB offers an opposite example (Scheme 10). This toxin belongs to a family of conotoxins from marine snails consisting of 22 amino acid residues with three intramolecular disulfide bonds.[85]

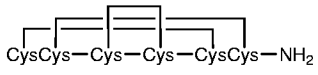
When air oxidation of the reduced μ-conotoxin GIIIB (**18**) was carried out in 0.1 M NH_4OAc buffer (pH 7.5) at 0.01 mM peptide concentration and at 10 °C, three major products, isomers **15**, **16**, and **17** were produced after 40 hours in a ratio of 1:4:3 (Figure 2).[86] The disulfide structures of each isomer were determined by enzymatic digestion followed by amino acid analyses, mass spectrometry, sequence analyses, as well as by the synthetic approach (Scheme 10).

for references see p 159

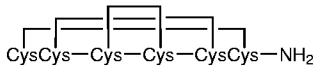
Scheme 10 Disulfide Connectivities of Isomers of μ-Conotoxin GIIIB[86]



15



16



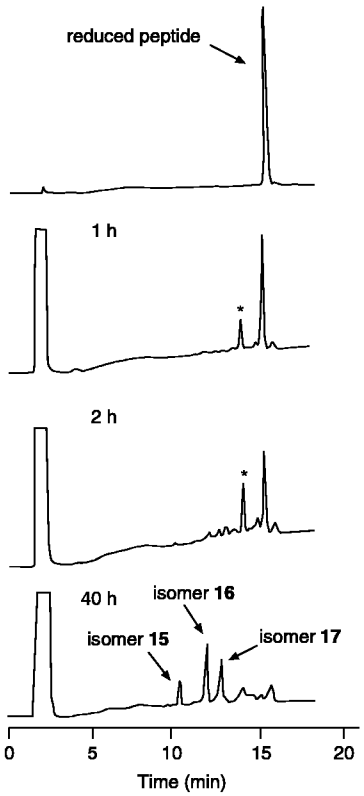
17

RDCCTOORKCKDRRCKOMKCCA—NH_2

18

O = Hyp

Figure 2 HPLC Profile of Oxidative Folding of Reduced μ-Conotoxin GIIIB; Peak Labelled * was Identified as the Cys^{10}/Cys^{15} Disulfide Bonded Intermediate[86] a, b



[a] Column: YMC-Pak ODS (4.6×150 mm); eluent: 7–16% aq MeCN/0.1% TFA; detection: 210 nm.

[b] Reprinted from (Kubo; Chino; Kimura; Sakakibara, *Peptide Research*, Vol. 6), Copyright (1993), p 66, with permission from Eaton Publishing.

The toxicity of each product was measured and only isomer **15**, the minor component of the oxidation mixture, showed the same toxicity as that of the natural product; its identity was further confirmed to be analogous to natural μ-conotoxin GIIIB by coelution experiments on RP-HPLC. The sequential topology of the cysteine residues in μ-conotoxin GIIIB indicates the presence of two adjacent cysteines in the N-terminus and two in the C-terminus of the sequence with the remaining two cysteine residues located in the central part with four intervening amino acid residues (Scheme 10). Since disulfide bridging between adjacent residues is extremely unfavored and a disulfide loop formation with $m = 4$ highly favored, the preferred generation of isomers **16** and **17** was to be expected. This was fully confirmed by monitoring the time course of the reaction (Figure 2) which allowed the identification after 1–2 hours of the formation of a folding intermediate (indicated with an asterisk). This intermediate was reacted with 4-vinylpyridine to convert unreacted thiol functions into the stable 4-pyridylethyl thioether derivatives. By this procedure the intermediate was identified as the Cys^{10}/Cys^{15} disulfide-bonded species, which as the most favored was also formed most rapidly during the oxidation reaction. Moreover, it leads to a chain reversal which facilitates collision of the N- and C-terminal cysteines to produce the remaining two disulfide bonds resulting in almost statistical distribution between isomers **16** and **17** in the product mixture. This implies that the preferentially formed Cys^{10}/Cys^{15}disulfide bond has to be rearranged in order to generate the native isomer **15**.

Changes in salt concentration and pH of the reaction medium, reaction temperature, and/or addition of Gn-Cl were without any effect. However, enhancing the peptide concentration or addition of redox reagents (GSH/GSSG) strongly affected the product distribution with the isomer **15** becoming the predominant species. An increase of the peptide concentration from 0.01 mM to 0.5 mM led mainly to isomer **15**, whereas a further increase provokes formation of other oxidation products, most probably intermolecular oligomers. Even at 0.5 mM peptide concentrations the Cys^{10}/Cys^{15} disulfide-bridged intermediate was formed initially; however, the high density of SH groups in the medium allowed for thiol/disulfide exchange reactions to occur with rearrangement of the mismatched Cys^{10}/Cys^{15} disulfide bond. A similar effect was observed by addition of GSH/GSSG at a ratio of 100:10, which also accelerated the generation of isomer **15**, even at the low 0.01 mM peptide concentration. Thus, in appropriate cases higher reduced peptide concentrations can mimic the effect of redox reagents.

Oxidative Folding of μ-Conotoxin GIIIB (15):[86]
A 0.5 mM soln of the reduced docosapeptide **18** in 1 M NH_4OAc buffer (pH 8.0) was gently stirred at 25 °C for 2 d. After acidification to pH 4.0 with TFA, the mixture was chromatographed on a preparative YMC ODS column and the product was eluted with aq MeCN containing 0.1% TFA as eluant. The pooled fractions were further purified by ion-exchange column chromatography on a CM-Toyopearl 650M column using a gradient elution of NaCl in 20 mM sodium phosphate buffer (pH 6.0), followed by RP-HPLC; yield: 60%. The product was characterized by amino acid analysis and MALDI-TOF-MS; the cystine connectivities as **15** were deduced by enzymatic digestion, sequence analyses, MS, and synthetic approaches.

6.1.5.2.5 Effect of Temperature and Salts

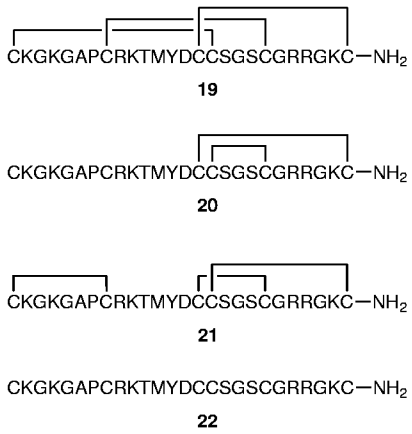
All the disulfide-folding pathways have the common feature that disulfide formation is initially random and not energetically favored due to the unfolded state of the reduced polypeptide chain. Disulfide formation becomes less random as certain disulfides are favored energetically as a result of the acquisition of nonrandom local conformations. In turn, the disulfides stabilize such conformations, so that disulfide formation and folding become cooperative. Thereby the sequence encoded structural information may fully suffice to direct the correct oxidative refolding as shown for ω-conotoxin GVIA (**9**) (Section 6.1.5.2.1) which assumes the native-type cystine connectivities directly even in absence of redox reagents and

for references see p 159

thus, without the need of reshuffling folding intermediates. Conversely for μ-conotoxin GIIIB (**15**), rearrangement of initially formed mismatched disulfide bonds by a moderate thiol concentration (0.5 mM to 1 mM) of either the reduced peptide or redox reagents was required for formation of the native disulfide isomer. However, in cases where only weak structural information is encoded in the sequence, local conformations that facilitate the correct folding have to be additionally stabilized, as well exemplified in the case of ω-conotoxin MVIIC (**19**). Detailed folding studies to optimize the production of the correct disulfide isomer of this toxin again indicate the strong effect of the individual character of each class of multiple-cystine peptides.

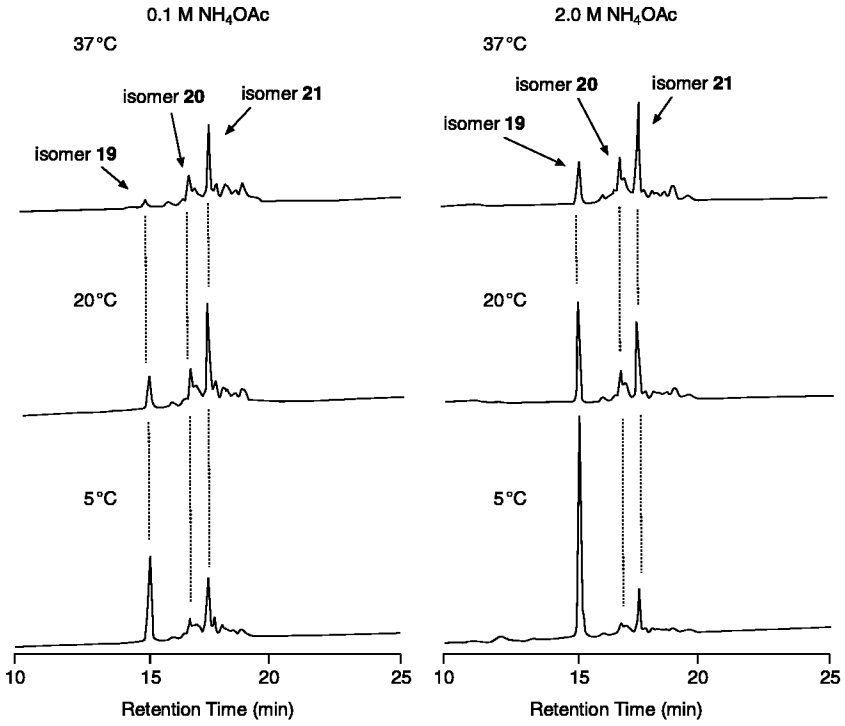
ω-Conotoxin MVIIC (**19**), a specific blocker of the P- and Q-type calcium channels, is a 26-residue peptide with three intramolecular disulfide bonds.[87,88] Due to four Lys and three Arg residues, the molecule is a highly basic peptide. When the reduced peptide **22** was oxidized under standard conditions in 0.1 M NH_4OAc buffer (pH 7.7) at a peptide concentration of 0.01 mM in the absence or presence of redox reagents (GSH/GSSG) at 21 °C for 24 hours, three major components, the isomers **19**, **20**, and **21**, were formed in yields of 4, 17, and 38% and 11, 15, and 25%, respectively.[89] The disulfide connectivities of the three isomers were determined and are shown in Scheme 11. Comparative biological assays and chromatographic analysis revealed that compound **19**, the minor oxidation product, was the native disulfide isomer of ω-conotoxin MVIIC. This result would suggest that at room temperature a well expressed energy minimum is not characterizing the native fold.

Scheme 11 Disulfide Connectivities of Isomers of ω-Conotoxin MVIIC[89]



Isomers **19**, **20**, and **21** are stable in 0.1 M NH_4OAc buffer (pH 7.7) in the absence of redox reagents and when GSH/GSSG are added to reach the thermodynamically most favored product distribution again a ratio of 11:15:25 was obtained for isomers **19/20/21**. As is well known, preferred local conformations can be stabilized by lowering the temperature. Whilst almost no isomer **19** was detectable on increasing the temperature, by lowering it to 5 °C the isomer **19** became the predominant oxidation product (Figure 3). Similarly, by increasing the ionic strength to 2 M NH_4OAc and thus by lowering electrostatic repulsions in the highly positively charged molecule, a shift of the product distribution in favor of the native fold was observed (Figure 3). This became even more pronounced by the salting out effect of 2 M $(NH_4)_2SO_4$ which may favor a burial of the disulfides, as it should occur with the disulfide connectivities of isomer **19**. Under such optimized conditions, 2 M $(NH_4)_2SO_4$ at 5 °C, ω-conotoxin MVIIC was formed in >80% yield (Figure 4).

Figure 3 HPLC Profile of the Effects of Reaction Temperature and NH_4OAc Concentration on the Folding of Reduced ω-Conotoxin MVIIC in the Presence of Redox Reagents; Buffer: NH_4OAc pH 7.7; Peptide/GSH/GSSG 1:100:10; Time 24 h[89] a, b



[a] Column: YMC-Pak ODS (4.6×150 mm); eluent: 1–25% aq MeCN/0.1% TFA.

[b] Reprinted from (Kubo; Chino; Kimura; Sakakibara, *Biopolymers*, Vol. 38), Copyright (1996), p 773, with permission from John Wiley and Sons, Inc.

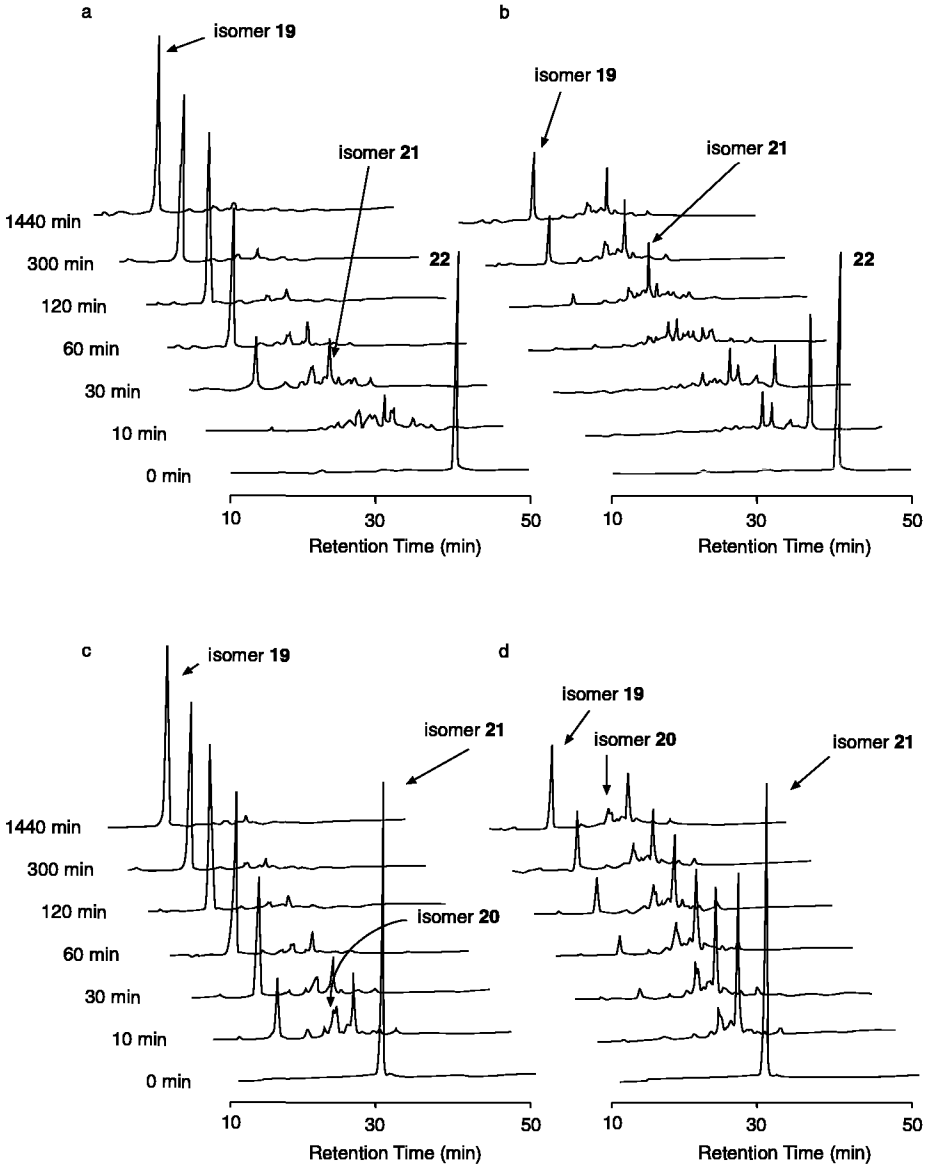
Information on the folding pathway of ω-conotoxin MVIIC was obtained by monitoring the progress of the oxidation reaction of the reduced peptide **22** as well as of the disulfide rearrangement reaction of isomer **21** in 50 mM NH_4OAc buffer or buffer containing 2 M $(NH_4)_2SO_4$ (both at pH 7.7) at 5 °C in the presence of redox reagents (Figure 4).[89] In buffer containing 2 M $(NH_4)_2SO_4$ (Figure 4a), the reduced peptide disappeared within 10 minutes and a complex pattern of peaks was detected on the chromatogram. After 30 minutes, both isomers **19** and **21** were formed, however, the yield of isomer **21** gradually decreased with concomitant increase of isomer **19**. After 24 hours, a ratio of isomer **19**/**21** of 84:4 was reached. Similarly, when isomer **21** was dissolved in the identical redox buffer (Figure 4c), a fast conversion of isomer **21** into isomer **19** was observed. A similar, but less pronounced and preferred formation of isomer **19** was observed in 50 mM NH_4OAc buffer (Figure 4b and d).

Isomer **20** contains a disulfide bridge that spans an intervening sequence with $m = 3$ which is not a favored ring size, while in isomer **21** a highly favored disulfide loop with $m = 4$ is present, which agrees with its preferred formation in the absence of structure stabilizing effects and as mismatched folding intermediate even under optimized conditions. The presence of productive folding intermediates with disulfide connectivities not present in the final native structure has been observed in various folding pathways, for example hirudin.[90] However, for a rational explanation of the functional role of isomer **21** in the oxidative folding pathway of ω-conotoxin MVIIC (**22**), a detailed structural comparison of isomers **19** and **21** would be required. Most probably the main reason for rearranging the disulfide Cys^{15}/Cys^{20} of isomer **21** to form three large disulfide loops is to bury all three disulfides in the

for references see p 159

inner core of the molecule with the highly charged surface exposed to the bulk solvent. CD conformational analysis of the reduced ω-conotoxin MVIIC (**22**) and its isomers **19** and **21** at different temperatures and salt concentrations confirmed the presence of more ordered local conformations in the native-type disulfide isomer of ω-conotoxin MVIIC which are stabilized at lower temperature and higher ionic strength.[89]

Figure 4 HPLC Profile of the Progress of the Oxidative Folding of Reduced ω-Conotoxin MVIIC and Stability of Isomer **21** in the Presence of Redox Reagents in 50 mM NH_4OAc Buffer Containing 2 M $(NH_4)_2SO_4$ at pH 7.7 (a, c) and in 50 mM NH_4OAc Buffer at pH 7.7 (b, d) at 5 °C; Aliquots were Analyzed at the Time Shown in Each Panel[89] a, b



[a] Column: YMC-Pak ODS (4.6×150 mm); eluent: 1–25% aq MeCN/0.1% TFA in 25 min; detection: 220 nm.
[b] Reprinted from (Kubo; Chino; Kimura; Sakakibara, *Biopolymers*, Vol. 38), Copyright (1996), p 773, with permission from John Wiley and Sons, Inc.

6.1.5.3 Oxidative Refolding of Double-Stranded Cystine Peptides

As discussed in the preceding sections, oxidative folding of single-stranded multiple-cysteine peptides generally requires detailed optimization studies if satisfactory yields are the goal. In the case of double-stranded multiple-cystine peptides the problems encountered are exponentially increased as well demonstrated by the classical example of insulin that has been the object of extensive studies in the last four decades. Insulin is the product of post-translational processing of proinsulin which as a one-chain precursor folds almost quantitatively to the correct cystine connectivities under appropriate oxidative conditions.[91,92] Cleavage of the C-peptide linker sequence, leads to a double-stranded molecule crosslinked by two interstrand disulfide bonds and containing one additional intrachain disulfide bridge with $m = 4$. Since suitable reversible crosslinking of the two chains (A- and B-chain of insulin)[93,94] allows oxidative refolding under carefully controlled conditions in >90% yield, the role of the C-peptide is mainly to bring and keep the two chains together as to favor formation of the native disulfide bonds.[18,20] Indeed, the two chains contain sufficient information to form under optimized oxidative conditions insulin in yields far superior to that expected from a statistical product distribution.[18,19]

This concept of reversible chemical crosslinking of the chains to drastically decrease the enthalpic penalty of the folding process and to exploit the highly favored disulfide loop formation in the A-chain with $m = 4$, has been further developed into artificial peptide linker chains that can be excised selectively by enzymatic processing, to allow for bioexpression of the "artificial proinsulins" on an industrial scale.[95,96] However, to apply this approach rationally to other double-stranded cystine peptides, knowledge about their three-dimensional structure is essential.

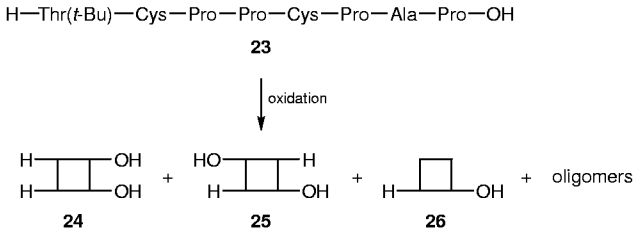
6.1.5.3.1 Air Oxidation of Bis-cysteine Peptides to Homodimers

Oxidation of mono-cysteine peptides to the dimer is a straightforward reaction that can produce only the desired product. In the case of bis-cysteine peptides statistically the oxidation leads to the homodimers in parallel and antiparallel orientation as well as to the disulfide-bridged monomer and oligomers. When the two cysteine residues are placed in the adjacent position formation of homodimers is highly favored over the cyclic monomer (Section 6.1.5.1) and the product distribution depends strongly on the peptide concentration. Such a type of intermolecular disulfide bridging is present in bovine seminal ribonuclease, where an antiparallel alignment occurs at the interface of the dimer.[97]

Dimerization of two protein domains via two proximal cysteine residues occurs in the hinge region of the antibody family of proteins. In human IgG1 this hinge region is composed of the parallel dimer of Cys-Pro-Pro-Cys which acts like a pivot for the functional movements of the Fab arms and Fc portion.[98] Air-oxidation of N- and C-terminally extended hinge peptide **23** at 3 mM concentration in aqueous solution, where thiol/disulfide exchange reactions were allowed to produce the thermodynamically most favored product distribution, led to the compounds **24**, **25**, and **26** in a ratio of 90:8:2 (Scheme 12).[99,100] The antiparallel hinge dimer **25** was converted into the identical product distribution when incubated with the bis-cysteine peptide **23**. Formation of the correct disulfide pattern in the human IgG1 hinge-peptide 225–232/225′–232′ was expected to occur as a consequence of a thermodynamically highly favored conformation. This was confirmed by CD[101] and NMR conformational analysis,[102] which indicated for the excised fragment the identical fold, that is two parallel poly(Pro)-II helices connected by disulfide bridges of identical chirality as in the X-ray structure of the Kol-protein.[103] The protein subdomain character of the hinge peptide was retained even after N- or C-terminal incorporation of IgG1 unrelated peptide sequences.[104,105]

for references see p 159

Scheme 12 Expected Products from Random Air Oxidation of the Bis-cysteine-Octapeptide 225–232 of Human IgG1 Protein[99,100]



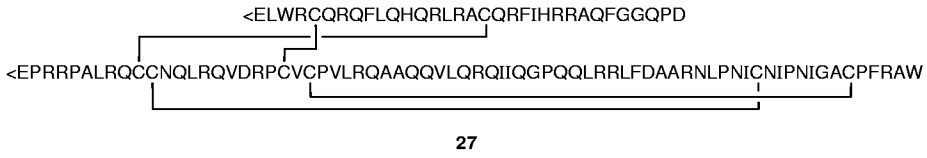
A detailed oxidation study on an artificially designed palindromic peptide with a predicted high tendency to form antiparallel α-helices confirmed that stabilization of the α-helix by increasing amounts of TFE allowed the control of the product distribution in favor of the antiparallel dimer to a certain degree, although increasing amounts of cyclic trimers were formed.[106] In the absence of or at low levels of TFE, formation of the cyclic monomer was predominant.

(H-Thr(tBu)-Cys-Pro-Pro-Cys-Pro-Ala-Pro-OH)$_2$, Parallel Dimer (24):[107]
A soln of H-Thr(tBu)-Cys-Pro-Pro-Cys-Pro-Ala-Pro-OH (**23**; 0.36 g, 0.42 mmol) in 0.1 M NH_4OAc (pH 6.8) (140 mL) was gently stirred at rt for 11 d. The soln was lyophilized and the residue purified by silica gel chromatography followed by gel filtration on LH 20. The product was precipitated from MeOH with MeCN to give **24**; yield: 0.37 g (85%); peptide content by amino acid analysis: 81%.

6.1.5.3.2 Oxidative Refolding of Peptides with Multiple Inter- and Intrastrand Disulfide Bonds

Besides insulin discussed above, the sweet protein mabinlin II (**27**) is a heterodimer consisting of a 33-residue A-chain and a 72-residue B-chain crosslinked by two interstrand disulfide bonds and containing two additional disulfide loops, as shown in Scheme 13.[108] Interestingly, the cysteine topology in the B-chain shows two consecutive ($m = 0$) and two proximal cysteine residues ($m = 1$) both characterized by a low tendency to form small-sized intramolecular disulfide loops. Air oxidation of a mixture of the fully reduced A- and B-chains produced mabinlin II (**27**), but also significant amounts of A-chain and B-chain dimers.[109] When oxidation was carried out in the presence of GSH/GSSG mainly glutathione adducts were formed, whereas in the presence cysteine/cystine the correct product was obtained in the surprisingly high yield of 47.4%. The disulfide connectivities in the synthetic material were confirmed by enzymatic digestion combined with mass spectrometry and sequencing.

Scheme 13 Structure of Mabinlin II[108]



Oxidative Folding of Mabinlin II (27):[109]
To a solution of the A-chain (8.06 mg; 1.95 μmol), B-chain (16.16 mg, 1.95 μmol), and cysteine (23.6 mg, 0.195 mmol) in 0.1 M NH_4OAc, pH 8 (12.12 mL), cystine (4.68 mg, 0.0195 mmol) was added and the mixture was kept at rt for 7 d. The mixture was purified in portions on Pharmacia Resource S column by elution with a linear gradient from aq 0.1% TFA to 60% MeCN containing 0.1% TFA; the fractions were collected and lyophilized to give **27**; yield: 11.47 mg (47.4%); ESI-MS; amino acid analysis.

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6.1.6 Determination of Disulfide Connectivities in Multiple-Cystine Peptides

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When multiple-cysteine peptides or proteins are subjected to oxidative folding, their theoretical intra- and/or intermolecular disulfide bond connectivities are dependent on the number of cysteine residues in the molecule (see Section 6.1.5). Nevertheless, the disulfide bonding pattern is unique for each particular peptide and protein in the native state due to the structural information encoded in the sequence.[1] There is, however, no simple method that allows the prediction of the native disulfide topology of multiple-cystine peptides and proteins merely by inspection of their sequence composition, even taking into account statistical analyses based on known disulfide patterns of peptides and proteins.[2–5] This difficulty is well illustrated by the family of peptides related to endothelin and sarafotoxin which exhibit the identical distribution of the cysteine residues along their sequences and which, despite distinct amino acid composition, are characterized by an identical disulfide topology, that is Cys^{1}/Cys^{15} and Cys^{3}/Cys^{11}, the ribbon isomer. Conversely, in another homologous peptide, apamin, the equivalent cysteine residues do not pair in the same manner, but form Cys^{1}/Cys^{11} and Cys^{3}/Cys^{15} disulfide bridges, the globule isomer. For a definition of the isomers see Section 6.1.5.1.

As discussed in Section 6.1.5.1, oxidative refolding of cysteine peptides in vitro often does not result exclusively in the generation of the native isomers. For example, the oxidation of endothelins and sarafotoxins yielded two disulfide isomers corresponding to the native ribbon and nonnative globule isomer in a ratio of 75:25, respectively,[6] while in the case of apamin, the isomer with the native globule-type disulfide connectivity was formed exclusively.[7] Significant formation of a second disulfide isomer was also observed in the oxidative folding of insulin-like growth factor-1; besides the native isomer with Cys^{6}/Cys^{48}, Cys^{18}/Cys^{61}, and Cys^{47}/Cys^{52} disulfide bonds, the misfolded disulfide isomer with Cys^{6}/Cys^{47}, Cys^{18}/Cys^{61}, and Cys^{48}/Cys^{52} was also formed.[8] In other cases, for example ω-conotoxin MVIIC, predominant formation of nonnative disulfide isomers is observed unless conditions are optimized (see Section 6.1.5.2.5).[9] These observations clearly underline the necessity of an unambiguous assignment of the disulfide connectivities of such oxidation products and of a comparative analysis with the natural products. One way of confirming the correct oxidative refolding of synthetic products is to analyze their chromatographic behavior and their bioactivities. Identical potencies in the biological activity and/or identical retention times in chromatographic elution profiles as those of the natural peptides strongly support the correct disulfide bond patterns, since both biopotencies and chromatographic properties are closely related to the specific tertiary structure which in turn is defined by the native disulfide connectivity. On the other hand, the disulfide connectivities have to be determined for both the native isomer and sometimes even for nonnative isomers if knowledge about the nature of misfolded isomers allows the identification and thus optimization of the oxidative folding pathway. In this section the main methods are discussed which are currently used to determine disulfide topologies of both natural and synthetic multiple-cystine peptides.

6.1.6.1 Direct Analysis of Disulfide Topologies of Multiple-Cystine Peptides

When the disulfide connectivity of a short peptide, consisting of about 20 amino acid residues or less with two intrachain disulfide bonds, has to be determined, the task is generally carried by synthesizing the three possible disulfide isomers (globule/ribbon/beads) and subsequent chromatographic comparison with the natural peptides. The isomers are synthesized by the two-step selective disulfide forming methods based on orthogonal cysteine

protection schemes as discussed in Section 6.1.2. The three synthetic disulfide isomers are generally well separated as single entities on RP-HPLC as well documented in various cases such as α-conotoxin GI[10] and endothelin-1.[6] Thus, an unambiguous assignment of the disulfide pattern is possible by comparative chromatographic analysis with the natural peptide. It is worth noting that the elution order of three isomers may vary from peptide to peptide; for example, the isomers of α-conotoxin GI elute in the order of ribbon, beads, and globule isomer, whereas those of endothelin-1 in the order of globule, ribbon, and beads isomer. If a peptide contains more than two intramolecular disulfide bonds, this synthetic approach is theoretically still feasible, but very time consuming in view of the large number of possible isomers. In these cases, direct assignment of the disulfide topology is obtained in the straightforward manner by X-ray crystallographic analysis, if crystals can be grown of the natural or synthetic multiple-cystine peptide, or by NMR structural analysis, if signal overlapping does not seriously hamper assignments of the half-cystine C^{α} and C^{β} proton resonances.

6.1.6.1.1 NMR Spectrometric Analysis

Some of the multiple-cystine peptides are resistant to enzymatic and/or chemical cleavage, and thus the more simple methods described in Section 6.1.6.2 cannot be applied. In these cases assignment of their disulfide connectivities is generally attempted by NMR structural analysis. Thereby, if the two half-cystine residues are located on opposite sides of the NMR-derived 3D structure, a disulfide bond between them is improbable.

An essential requirement for assigning disulfide bonds by NMR conformational analysis is the observation of local nuclear Overhauser enhancements (NOEs) between protons in the half-cystine residues, especially those between $C^{\beta}H/C^{\beta'}H$, $C^{\alpha}H/C^{\beta'}H$, and $C^{\alpha'}H/C^{\beta}H$.[11] Similarly, the deviation of the average S^{γ}-S^{γ} distance from the ideal disulfide length may be useful for the identification of favored and unfavored disulfide connectivities in the NMR-derived 3D-structure of the peptides. This approach was successfully applied to echistatin,[12] iberiotoxin,[13] and scyllatoxin.[14] Calculation with ambiguous distance restraints[15] was applied to an automatic assignment of the disulfide connectivities of the mamba intestinal toxin 1 as Cys^{7}/Cys^{19}, Cys^{13}/Cys^{31}, Cys^{18}/Cys^{59}, Cys^{41}/Cys^{67}, and Cys^{61}/Cys^{77}.[16] It is interesting to note that some of the disulfide connectivities determined for this toxin (Cys^{18}/Cys^{19}, Cys^{31}/Cys^{41}, and Cys^{7}/Cys^{77}) by the classical enzymatic fragmentation procedure (Section 6.1.6.2) were not identical to those obtained by NMR analysis. The reason for this discrepancy was the presence of a minor contaminant in the toxin, which was cleaved by enzymes. On RP-HPLC fractionation of the digest, the fragments were used for the topological assignment of the disulfides. The toxin itself, however, was fully resistant to the enzymatic digestion and the origin of the contaminant was unknown. This fact clearly underlines the fundamental importance of a quantitative analysis for an unambiguous determination of disulfide connectivities in natural and synthetic multiple-cystine peptides.

6.1.6.2 Analysis of Disulfide Topologies of Multiple-Cystine Peptides after Fragmentation

The most common method used for determining the disulfide patterns in peptides and proteins is based on enzymatic and chemical fragmentation of the polypeptide chain whilst maintaining the native disulfide bridges intact. The assignment will be successful if all the peptide bonds between the half cystines are cleaved so that each generated fragment contains only one disulfide bond. For this purpose the target molecules have sometimes to be subjected to repetitive and/or combined cleavage methods. In this context it is important to note that disulfide bonds in peptides might not be perfectly stable under the experimental

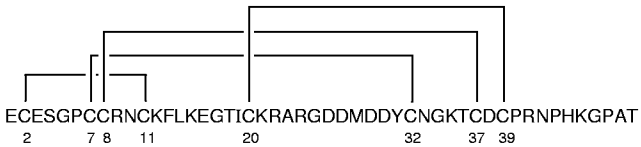
for references see p 175

conditions applied, since they exhibit a more or less pronounced tendency to isomerization and scrambling particularly at extreme pH values. In the pH range of 2.0 to 6.5 they are generally stable.[17] Accordingly, appropriate conditions have to be chosen for the fragmentation reactions to avoid undesirable scrambling of the disulfides.

6.1.6.2.1 Chemical Fragmentation

Particular peptide bonds are cleaved by acids under milder conditions than those required for complete hydrolysis. Applying this observation in the early 1950s the disulfide connectivities of insulin were determined utilizing fragments resulting from such partial acid hydrolysis. In this case, 5 M H_2SO_4 was used for hydrolyzing the chymotryptic fragment at 100 °C for 45 minutes and fragments were generated which allowed the correct disulfide assignment.[18] Currently, dilute oxalic acid is the most commonly used acid,[19–22] and partial hydrolysis is generally carried out with 0.25 M oxalic acid at 100 °C for several hours. This method was applied to the elucidation of the disulfide connectivities of echistatin **1** (Scheme 1).

Scheme 1 Echistatin[20]



1

Peptide bonds are cleaved in a nonselective, but not in a completely random manner. Based on anchimeric side-chain assistance, steric factors, and bond strains, acid-labile peptide bonds are predicted to include sites containing Asp, Glu, Ser, Thr, Asn, Gln, Gly, and Pro.[22] The disulfide topologies of circulin B and cyclopsychotride, backbone-cyclized peptides with three disulfide bonds, were determined by partial hydrolysis for 5 hours.[22] Occasionally, the bond between adjacent half-cystine residues is cleaved due to the nonselective nature of the mechanism of partial acid hydrolysis.[21] By this procedure, in all cases, a complex mixture of peptide fragments is produced which requires careful chromatographic separation by RP-HPLC for subsequent analysis by mass spectrometry (see Section 6.1.6.2.7).

Chemical cleavage at selective amino acid residues is an alternative method for scission of peptide bonds in target molecules. Cleavage at the Met-Xaa bond is achieved by CNBr, at the Trp-Xaa bond by the tryptophan-directed reagent 3-bromo-3-methyl-2-[(2-nitrophenyl)sulfanyl]-3*H*-indole (BNPS-skatole) and at the Asp-Xaa bond by 2% formic acid.[23,24] All these reactions are carried out at acidic pH under standard conditions for each reagent.

Disulfide Connectivities of Echistatin (1):[20]
Echistatin (0.1 mg), a 49 amino acid peptide containing 8 Cys residues, was dissolved in degassed and N_2-purged 250 mM oxalic acid (100 µl). The soln was sealed in a glass capillary tube and hydrolyzed at 100 °C for 4 h. The fragments were isolated by RP-HPLC and analyzed by ES-MS. Twelve peaks were associated with fragments containing the Cys^2/Cys^{11} and Cys^{20}/Cys^{39} disulfide bonds, respectively. Four peaks were assigned to fragments containing two disulfide bonds, the connectivities of which were determined by MS/MS to be Cys^7/Cys^{32} and Cys^8/Cys^{37}. The disulfide topology of echistatin (**1**) derived from this analysis is shown in Scheme 1.

6.1.6.2.2 Enzymatic Fragmentation

Proteolysis of peptides and proteins by enzymes occurs in a selective or nonselective manner. Chymotrypsin, trypsin, lysyl endopeptidase, *Staphylococcus aureus* V8 protease, and endopeptidase Asp-N are frequently listed as selective enzymes, whereas thermolysin, pepsin, subtilisin, and elastase belong to the nonselective enzymes, although thermolysin preferentially cleaves peptide bonds before hydrophobic residues.

Trypsin was used to distinguish between the parallel and antiparallel alignment of the two chains of β-human atrial natriuretic peptide, which contains two interchain disulfide bridges. Only a fragment containing a single cystine residue was obtained by this proteolytic digestion, which allowed the definite assignment of the antiparallel alignment of the two chains of this peptide with $Cys^{7}/Cys^{23'}$ and $Cys^{7'}/Cys^{23}$ disulfide bonds.[25] Thermolytic fragments of the topological isomers of human uroguanylin, which are well separated by RP-HPLC, revealed the identical disulfide connectivities of Cys^{4}/Cys^{12} and Cys^{7}/Cys^{15} for both isomers, clearly showing that the two forms do not correspond to disulfide isomers.[26]

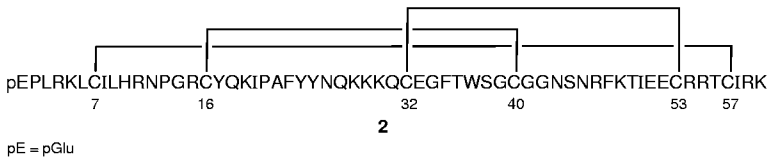
In some cases a combination of enzymes is used simultaneously and in other cases successively to produce the desired fragments, which in the ideal case should contain only a single cystine residue. For example, all peptide bonds between six half-cystine residues of charybdotoxin were cleaved by a mixture of trypsin and chymotrypsin, allowing an easy determination of the disulfide connectivities as Cys^{7}/Cys^{28}, Cys^{13}/Cys^{33}, and Cys^{17}/Cys^{35}.[27] Successive cleavage with different enzymes followed by chemical fragmentation is also often used to generate adequate fragments for the assignment of disulfide topologies.[28]

Since retainment of the native disulfide bonds in a peptide during fragmentation represents an essential prerequisite, conditions of enzymatic digestion have to be chosen that prevent disulfide scrambling. The most important variable is the pH of the reaction medium which has to be between 2.0 and 6.5.[17] Although the optimum pH values for peptide digestion by trypsin, chymotrypsin, and some other enzymes are within a slightly alkaline range, proteolysis by these enzymes has to be carried out at pH 6.5 or even lower to suppress disulfide exchange reactions. In this context, pepsin is a good example since cleavage by this particular enzyme occurs in the low pH range, even in 5% formic acid.[29] Alternatively, the enzymatic reactions are carried in the presence of thiol-trapping reagents such as iodoacetic acid in order to scavenge any nascent thiolates. Note that enzymatic reactions at pH 6.5 may induce transpeptidation at sensitive sites, e.g. Asp-Xaa, as a minor side reaction.[30]

The unexpected behavior of peptides toward enzymatic digestions has been observed in which a peptide was resistant to proteolysis by one enzyme, but digested by a second. For example elafin was resistant to trypsin or lysyl endopeptidase, but was cleaved by thermolysin.[31] Complete resistance to the classical enzymes has also been described,[32] which may be associated with the tight 3D structure resulting from the multiple disulfide cross-linking of the polypeptide chain. In such cases partial denaturation of the peptides represents a possible bypass strategy, since some enzymes retain their proteolytic activity even in the presence of a certain amount of denaturants such as urea, guanidine hydrochloride, or sodium dodecyl sulfate (SDS).[33–35] As an alternative, chemical cleavage prior to enzymatic digestion may prove to be useful.[36] Moreover, although endopeptidases recognize specific amino acid residues for cleavage of the peptide bond, these may not in all cases be scissile and extra bonds are also cleaved.[37–39] For example trypsin is reported to cleave the His-Asn bond of agelenin and prolyl endopeptidase the Ala-Xaa bond of hen egg white lysozyme which had been pretreated with CNBr and trypsin. However, these complex cleavage patterns do not hamper the evaluation of peptide mappings, since the molecular weight determination is greatly facilitated by the presently available sophisticated MS techniques (see Section 6.1.6.2.7). The disulfide connectivities of dendrotoxin I (**2**) were determined by this method (Scheme 2).

for references see p 175

Scheme 2 Dendrotoxin I[39]



Adjacent half-cystine residues are present in many peptides and proteins. In most of the cases they form two disulfide bonds with other cysteines in the molecule, unless the peptide bond between them is *cis* (see Section 6.1.5.1).[40,41] Specific enzymes that cleave the Cys-Xaa bond have not yet been discovered, although there are a few reports of cleavage of the Cys-Cys bond by enzymes such as elastase and pepsin.[42,43] For peptides with Cys-Cys bonds the cleavage method in Section 6.1.6.2.4 is recommended.

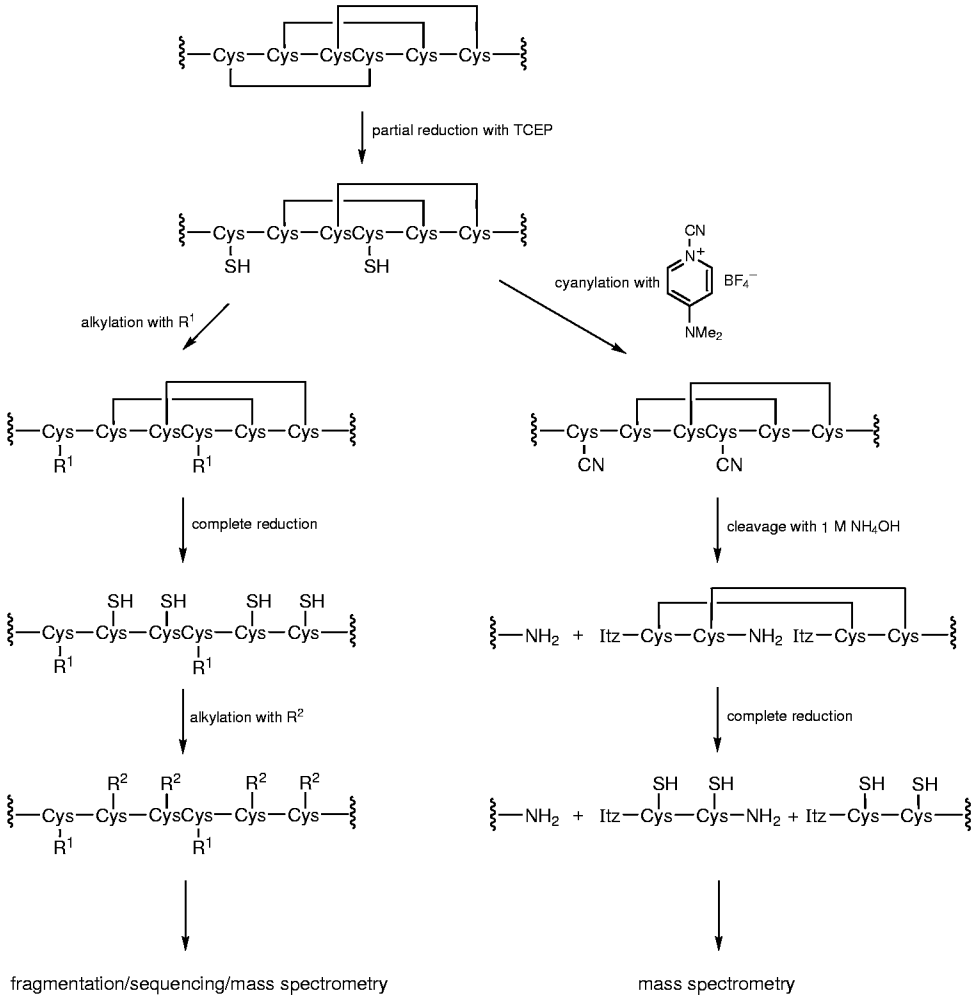
Disulfide Connectivities of Dendrotoxin I (2):[39]
Dendrotoxin I (0.16 mg), a 60 amino acid peptide containing 6 Cys residues, was digested with lysyl endopeptidase (enzyme/substrate ratio of 1:5) in Tris·HCl buffer containing 1 M guanidine hydrochloride at 37 °C for 4 h. The fragment composed of three peptide chains cross-linked by 3 disulfide bonds, was isolated by RP-HPLC and subdigested with *N*-tosyl-L-phenylalanine chloromethylketone-treated trypsin (enzyme/substrate ratio of 1:5) in the same buffer as above at 37 °C for 3 h. All the Arg-Xaa bonds and unexpectedly even the Trp-Ser bond were cleaved, resulting in three fragments, all of which contained a single disulfide bond. By amino acid analysis, MS and sequence analysis of each fragment the disulfide topology of dendrotoxin I (**2**) was determined as shown in Scheme 2.

6.1.6.2.3 Partial Reduction of Multiple-Cystine Peptides Followed by Alkylation

Complete reduction of all the disulfide bonds in peptides and proteins is generally achieved by treatment with excesses of dithiothreitol (DTT) under denaturing conditions. Due to the highly differentiated redox potentials of the disulfides, selective reduction of single disulfide bonds can also be achieved. Subsequent alkylation of the thiol groups generated by this partial reduction leads to stable derivatives, which are useful in the analysis of disulfide connectivities.

Partial reduction of disulfide bonds is carried out at both alkaline[44] and acidic pH. As mentioned previously (Section 6.1.6.2), under acidic conditions thiol/disulfide exchange reactions are suppressed allowing for a better retainment of the disulfide integrity. Tris(2-carboxyethyl)phosphine (TCEP) is an effective reductant even at acidic pH.[45] Thus, partial reduction with TCEP has now become one of the most powerful methods for determining disulfide connectivities in peptides.[46] In principle, this method relies on five successive steps as shown in Scheme 3: (1) partial reduction with TCEP at acidic pH; (2) alkylation of the generated pair of thiols at acidic pH; (3) separation of the modified peptide by RP-HPLC; (4) complete reduction of the remaining disulfide bond(s); and, (5) alkylation of the generated thiols by a reagent different from that used in step 2. Since the disulfide bonds, as mentioned previously, differ more or less significantly in their redox potentials, in some cases selective partial reduction may be difficult and in other cases disulfide bonds may be fully resistant toward TCEP. Accordingly, conditions for partial reduction with TCEP have to be carefully optimized for each individual peptide by varying the amount of reagent, the pH of the reaction medium as well as temperature and time. Generally, in a first trial, the reactions are carried out in 10 mM TCEP at pH 3 for 2 minutes at room temperature.[46]

Scheme 3 Schematic Route for Determination of Disulfide Topologies by Differentiated Alkylation of Partially Reduced Disulfides and/or Peptide Bond Cleavage by the Cyanylation Method[46,56]

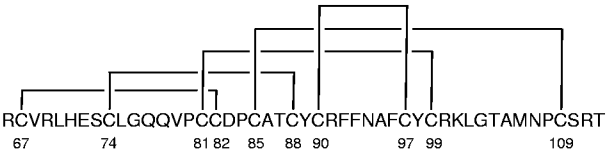


Immediate alkylation after partial reduction is crucial for the successful application of this method. Thiol/disulfide exchange reactions or reoxidation to the starting compound may occur if the pH is raised to slightly alkaline conditions (pH 8.0) as required for efficient alkylation of the thiols with alkyl groups such as carboxymethyl (CM) or aminocarbonylmethyl (Cam). To prevent this side reaction as much as possible, alkylation at pH 8.0 is carried out under forcing conditions using a supersaturated solution of the reagent such as iodoacetamide.[46] Alternatively, the alkylation is run under acidic conditions with 4-vinylpyridine at pH 3.0[46] or 6.0,[47] with *N*-(hydroxymethyl)benzamide in TFA,[48] or N-ethylmaleimide (NEM) at pH 6.0–6.5.[49,50] The resulting alkylated peptides are then subjected to complete reduction with either excesses of TCEP or DTT, followed by the second alkylation step with reagents different from those used in the first alkylation step. 4-Pyridylethyl,[51] carboxymethyl, or aminocarbonylmethyl derivatives are often used in this second step. Pairwise determination of disulfide connectivities is accomplished using the fragments obtained by the enzymatic digestion and/or chemical cleavage of the modified peptides as well as by direct sequence analysis without fragmentation (Section 6.1.6.2.6). The

for references see p 175

disulfide connectivities of human agouti-related protein (66–112) (**3**) were elucidated by this method (Scheme 4).[50]

Scheme 4 Human Agouti-Related Protein (66–112)[50]



3

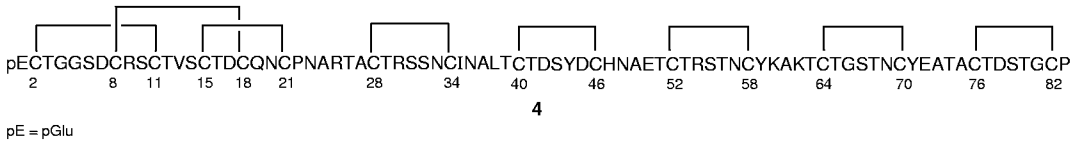
Cleavage of the polypeptide chain prior to partial reduction is an efficient way of controlling the reaction. For example, ω-agatoxin IVB was completely reduced with TCEP, but upon treatment with trypsin, partially reduced species were detected under the same conditions.[48] In contrast to these all or none characteristics in reduction, the four disulfide bonds of echistatin (**1**) can be reduced in a stepwise manner, which even allowed the elucidation of its reductive unfolding pathway.[52]

Disulfide Connectivities of Human Agouti-Related Protein (66–112) (3):[50]
Human agouti-related protein (66–112), a 47 amino acid peptide containing 10 half-cystine residues, was dissolved in 0.1% TFA (100 µl; pH 2.0) and reacted with 10 mM TCEP at 45 °C for 10–30 min. To the mixture was added 20 mM NEM and then an equal volume of 8 M Gn-Cl. The pH of the solution was adjusted to 6.0–6.5 with 1 M Tris base for alkylation of the generated thiols by NEM at rt for 60 min. Partially reduced and alkylated peptides was separated by RP-HPLC and subjected to complete reduction with 10 mM DTT in 0.2 M Tris · HCl/6 M Gn-Cl (pH 8.0) at 45 °C for 60 min. Free thiols were carboxymethylated with 15 mM iodoacetic acid at rt for 30 min in the dark. The differentially alkylated cysteine peptide was analyzed by sequencing and MS of both the intact peptide and the related thermolytic fragments. The elucidated disulfide connectivity of the protein fragment **3** is shown in Scheme 4.

6.1.6.2.4 Partial Reduction of Multiple-Cystine Peptides and Peptide Backbone Cleavage by Cyanylation

Cyanylation of thiols generated by partial reduction with TCEP is an interesting alternative to the alkylation method. By this procedure cleavage of the peptide bonds preceding the modified cysteine residues is achieved under alkaline conditions.[53] This type of fragmentation at the Xaa-Cys bond after cyanylation represents a selective method since it occurs only at modified cysteine residues. Cyanylation is carried out with 2-nitro-5-thiocyanobenzoic acid at pH 8.0.[54] However, using 1-cyano-4-(dimethylamino)pyridinium tetrafluoroborate[55] the reaction can be carried out at acidic pH. Accordingly, partial reduction and subsequent cyanylation of nascent thiols can be achieved simultaneously at pH 3.0.[56] The Xaa-Cys(CN) bond is cleaved with 0.01 M NaOH or 1 M NH_4OH at pH 12, where a possible dehydroalanine formation is largely suppressed.[56–59] Due to easy removal under reduced pressure, 1 M NH_4OH is frequently used as cleaving reagent which yields the related N-terminal fragment as the amide and the C-terminal fragment as the 2-iminothiazolidine-4-carbonyl (Itz) derivative as shown in Scheme 3. The bond cleaved in the peptide by this procedure is determined before and after complete reduction even if it is a Cys-Cys bond.[60] This approach was used to determine the disulfide connectivities of antifreeze protein 1 (**4**) from *Dendroides canadensis* (Scheme 5).

Scheme 5 Antifreeze Protein 1[59]



Although dehydroalanine formation represents a minor side reaction, the related byproduct may serve to confirm further the assigned disulfide pattern since the Xaa-dehydroalanine bond is stable toward the base treatment. Progress of the oxidative folding of human epidermal growth factor was analyzed by trapping the intermediates via cyanylation followed by their disulfide assignments.[61]

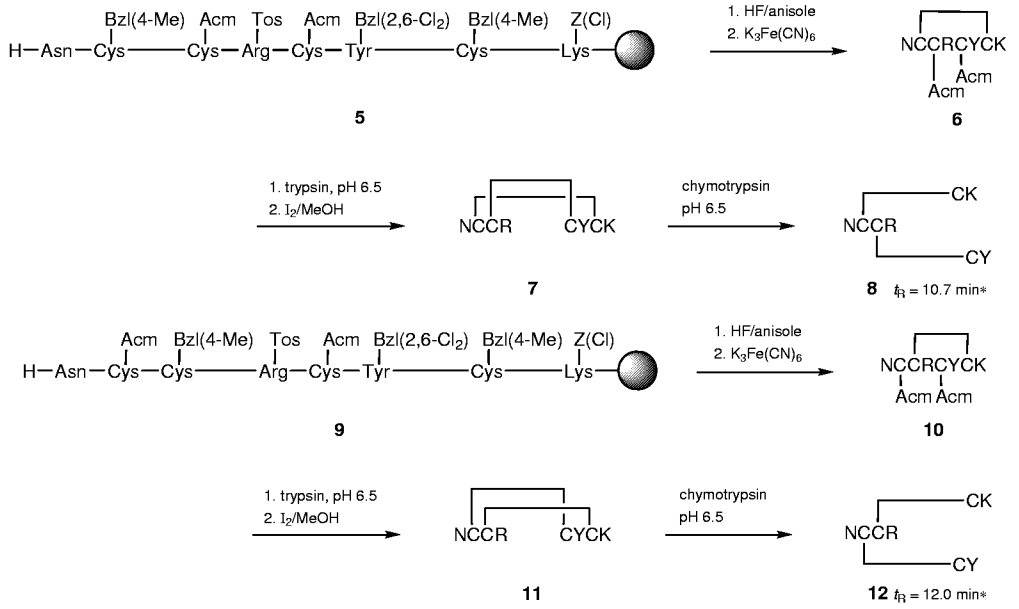
Disulfide Connectivities of Antifreeze Protein 1 (4) from *Dendroides canadensis*:[59]
Antifreeze protein-1 (3 mg · mL^{-1}), an 83 amino acid peptide including 16 half-cystines, was digested with pepsin (enzyme/substrate ratio of 1:20) in aq HCl (pH 2.0) at 37 °C for 28 h. Five disulfide connectivities were determined by analyzing the resulting peptide fragments, leaving three in the N-terminal sequence region undetermined. This N-terminal fragment (1–24) with three disulfide bridges was dissolved in 0.1 M citrate buffer (10 µl; pH 3.0), and 1 equiv TCEP for each disulfide group was added and allowed to react at rt for 15 min. To the mixture a 20-fold molar excess of 1-cyano-4-(dimethylamino)pyridinium tetrafluoroborate over the total cysteine content was added and the mixture was incubated at rt for another 15 min. Single disulfide reduced and cyanylated peptides were isolated by RP-HPLC, dried, and incubated in 1 M NH_4OH (100 µl; pH 12.0) at rt for 1 h. After removal of excess NH_4OH by vacuum centrifugation, the remaining disulfide bonds were fully reduced with 0.15 mM TCEP (100 µl) at 37 °C and pH 3.0 for 30 min. All the disulfide connectivities of antifreeze protein 1 (**4**) were determined from the results obtained using the peptide fragments and by analyzing the completely reduced fragments by MS.

6.1.6.2.5 Comparative Analysis with Synthetic Fragments

After enzymatic and/or chemical fragmentation of peptides containing adjacent half-cystine residues, the generated fragments contain two interchain disulfide bonds, unless the Cys-Cys bond was cleaved. The connectivities of these disulfide bonds, which cross-link the three component segments, can be determined by synthetic approaches. Two interchain disulfide bonds are most probably present at the Cys-Cys sequence due to the lower probability of an intramolecular Cys-Cys connectivity (see Section 6.1.5.1), so that only two synthetic disulfide isomers are usually required for comparative analysis. The design and synthesis of these two isomers with two interchain disulfides are generally as follows: (1) design of a linear peptide by introducing two artificial peptide bonds between the three component segments; (2) pairwise selection of two orthogonal protecting groups for the Cys residues according to the desired final interchain disulfide bridges; (3) synthesis of the peptides with the same sequence but with the exchanged pairwise combination of Cys protecting groups; (4) formation of the first intramolecular disulfide bond; (5) formation of the second intramolecular disulfide bond with or without concomitant cleavage of the peptide backbone; and, (6) selective cleavage of the artificial peptide bond(s) to produce the two types of disulfide isomers with two distinct interchain linkages. This approach is shown in Scheme 6 for the determination of the disulfide connectivities of ω-conotoxin GVIA (**13**) (Scheme 7).[62]

for references see p 175

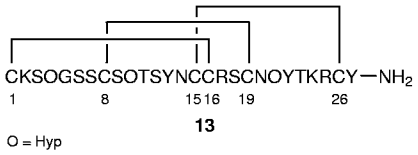
Scheme 6 Formation of Two Interchain Disulfide Bridges with Different Connectivities Using Two Kinds of Protecting Groups for the Cysteine Residues. Synthetic Schemes for the Two Isomers of the Tryptic/Chymotryptic Fragment of ω-Conotoxin-GVIA[62]



*The retention times (t_R) of the two isomers **8** and **12** were determined by RP-HPLC on Nucleosil 5C18 under isocratic elution with 4% MeCN/0.1 M phosphate buffer (pH 6.5).

The two types of target isomers **8** and **12** were well separated by RP-HPLC[9,37,62,63] as well as by ion-exchange chromatography.[62] By comparative analysis of the synthetic isomers with the fragments resulting from enzymatic and/or chemical fragmentation of the multiple-cystine peptide, the disulfide connectivities of two adjacent half-cystine residues in the target peptide were definitely determined. It is important to note that the sequence of the starting peptides **5** and **9** is not the natural one, but an artificial precursor designed to facilitate the synthesis.

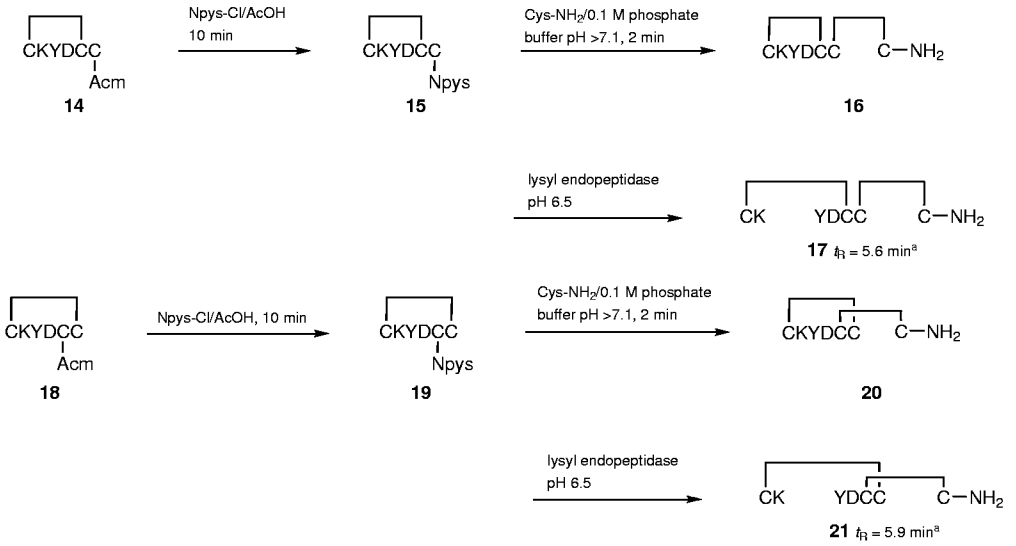
Scheme 7 ω-Conotoxin GVIA[62]



An alternative strategy for the synthesis of isomers with two interchain disulfides is similar to that described above up to the formation of the first disulfide bond. However, the subsequent activation of one Cys residue followed by the formation of an interchain (intermolecular) disulfide bond by procedures described in Section 6.1.3 gave a second disulfide bridge. As an example the syntheses of peptides **17** and **21** with the different possible disulfide connectivities of ω-conotoxin MVIIC (**22**) are illustrated in Scheme 8.[9] Two peptides **14** and **18** containing three Cys residues were designed by inserting a peptide bond between the two constituent peptide segments. The peptides **14** and **18** were prepared by standard resin-based methods followed by the disulfide bond formation between the thiol-liberated Cys residues. The Acm group on the remaining cysteines of **14** and **18** were then

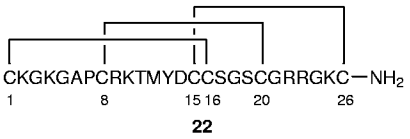
replaced by the thiol activating 3-nitro-2-pyridylsulfanyl (Npys) group[64] in acetic acid[65] yielding **15** and **19**. The subsequent intermolecular disulfide bond formation with the free thiol-containing peptide was carried out in aqueous buffer at as low pH as possible (such as <6.5) to suppress thiol/disulfide exchange reactions yielding **16** and **20** (see Section 6.1.3.2).[66,67] Treatment of **16** and **20** with lysyl endopeptidase gave the target peptides **17** and **21**. The retention time of **21** was identical to that derived from ω-conotoxin MVIIC allowing the disulfide connectivity of ω-conotoxin MVIIC to be determined as **22** (Scheme 9).

Scheme 8 Formation of Two Interchain Disulfide Bonds with Different Connectivities Using Two Kinds of Protecting Groups for the Cysteine Residues and Activation via the Npys Derivative. Synthetic Schemes for the Two Isomers of the ω-Conotoxin MVIIC Fragment Obtained by Digestion with Lysyl Endopeptidase and Thermolysin[9]



[a]The retention times (t_R) of the two isomers **17** and **21** were determined by RP-HPLC on YMC-ODS under isocratic elution with 1% MeCN/0.1% TFA.

Scheme 9 ω-Conotoxin MVIIC[9]



A tryptic fragment with three interchain disulfide bonds between four peptide chains was synthesized by repetitive application of this Cys activation/disulfide forming method, and by this procedure the disulfide connectivities of synthetic porcine C5a-anaphylatoxin was confirmed to be identical to those for the natural peptide.[67]

Disulfide Connectivities of ω-Conotoxin GVIA (13):[62]
ω-Conotoxin GVIA, a 27 amino acid peptide containing 6 half-cystines, was subjected to simultaneous digestion with trypsin and chymotrypsin in phosphate buffer (pH 6.5) for 36 h. By amino acid analysis of each peak on the HPLC peptide mapping the Cys^{8}/Cys^{19} disulfide pairing was identified and a clustering of the other four half-cystine residues ($Cys^{1,15,16,26}$) in the CT1 fragment which consisted of the sequence portions (1–2), (14–17), and (26–27) cross-linked by two interchain disulfide bridges. A peptide **8** with the Cys^{1}/Cys^{15} and Cys^{16}/Cys^{26} connectivities and a peptide **12** with Cys^{1}/Cys^{16} and Cys^{15}/Cys^{26} pairings were designed to be synthesized from the peptide obtained by connecting the constituent peptide chains in the order (14–17), (26–27), and (1–2) (Scheme 6). The two required pairings of disulfide bridges were

for references see p 175

realized by starting from the protected peptides **5** and **9** which were prepared on resin by standard techniques. Upon HF cleavage of the peptides from the resin with concomitant deprotection of the Bzl(4-Me) groups on Cys residues, the first intramolecular disulfide bond was formed by oxidation with $K_3Fe(CN)_6$ to give **6** and **10**. Tryptic digestion at pH 6.5 cleaved the bond between Arg^{17} and Cys^{26}, the second disulfide bond was generated between the Cys(Acm) residues by oxidation with I_2 in MeOH to yield **7** and **11**. After cleavage of the bond between Tyr^{27} and Cys^{1} with chymotrypsin at pH 6.5, RP-HPLC, and ion-exchange chromatography on TSK-gel CM-2SW clearly revealed distinct retention times of the resulting two peptide isomers **8** and **12**. Comparative chromatographic analysis of CT1 with **8** and **12** showed coelution of CT1 with peptide **12**. Thus, for ω-conotoxin GVIA the disulfide topology was determined as **13** (Scheme 7).

Disulfide Connectivities of ω-Conotoxin MVIIC (22):[9]
ω-Conotoxin MVIIC, a 26 amino acid peptide containing 6 half-cystines, was digested with lysyl endopeptidase in 0.1 M NH_4OAc buffer (pH 6.5) at 37 °C for 2 h. After isolation of the fragment with three disulfide bonds, it was subdigested with thermolysin in 0.1 M HCO_2NH_4 buffer containing 1 mM $CaCl_2$ (pH 6.5) at 37 °C for 24 h. Analysis of the digest by peptide mapping on RP-HPLC followed by amino acid and MS analysis as well as sequencing revealed a Cys^{8}/Cys^{20} connectivity; the remaining four half-cystine residues were located in the C1T1 fragment composed of the sequence portions (1–2), (13–16), and Cys-NH_2. As shown in Scheme 8 the intermediates of the synthetic disulfide isomers **17** and **21**, one of which related to the disulfide connectivity of the C1T1 fragment, were synthesized as reported above for ω-conotoxin GVIA. A peptide with a bond between Lys^{2} and Tyr^{13} was synthesized and the first disulfide bond formed by oxidation with $K_3Fe(CN)_6$ to yield **14** and **18**. Each of these peptides with one disulfide bond and one Cys(Acm) residue was reacted in AcOH with 1 equiv Npys-Cl at rt for 10 min to give **15** and **19**. After isolation of these Npys-peptide derivatives by RP-HPLC and subsequent concentration by vacuum centrifugation, the residues were reacted in 0.1 M phosphate buffer at pH 7.1 with H-Cys-NH_2 (approximately 1 equiv) at rt for 2 min. The products **16** and **20** were isolated by RP-HPLC and digested with lysyl endopeptidase in 0.1 M Tris · HCl buffer (pH 6.5) at 37 °C for 2 h giving the target isomers **17** and **21**. A comparison of the retention times on RP-HPLC of **17** and **21** and C1T1 indicated that **21** was identical to C1T1, thus, establishing the disulfide connectivities of ω-conotoxin MVIIC as **22** (Scheme 9).

6.1.6.2.6 Sequence Analysis

Sequence analysis is one of the most useful tools for determining disulfide connectivities by detecting the phenylthiohydantoins (PTH) of either cystine or cysteine derivatives. Sequencing can be carried out on the intact peptide, although most frequently disulfide-bridged fragments are subjected to this type of analysis.

Edman degradation of a cystine-containing peptide yields diPTH-cystine [$(PTH\text{-}Cys)_2$] as a major product and the DTT adduct of PTH-dehydroalanine as a minor one. The latter compound derives from DTT added as stabilizer to the TFA reagent. $(PTH\text{-}Cys)_2$ is detected at the cycle where the second half-cystine is located. The recovery is dependent on the sequence position of the second half-cystine, generally the yield is 40% in the early cycles, but decreases to about 8% after 10 degradation cycles. Similarly the ratio of the two products, $(PTH\text{-}Cys)_2$ and the dehydroalanine derivative, changes from 10:1 in the early cycles to approximately 1:1 after 10 Edman reactions.[28] For an unambiguous assignment of half-cystine pairs it is therefore recommended to shorten the length of the N-terminal sequence preceding the second half-cystine by suitable fragmentation methods. It is also important to know that the recovery of $(PTH\text{-}Cys)_2$ is as high as 20% when the disulfide-linked half-cystines in different chains are located at the same position from the N-terminus in both chains; in the opposite case the recovery is lowered to almost 1/10.[68]

$(PTH\text{-}Cys)_2$ is detectable in significant amounts only when the Edman degradation cycles reach the second half-cystine because the generated PTH-half cystine (PTH-Cys) is attached to the second half-cystine via a disulfide linkage. However, small amounts of $(PTH\text{-}Cys)_2$ are detectable after derivatization of the first half-cystine which may arise from disulfide exchange reactions occurring during the Edman cycles and are also responsible for the relatively low recovery of $(PTH\text{-}Cys)_2$ after repetitive Edman degradations. Analysis of the reduced and then S-pyridylethylated or S-carboxymethylated peptides is carried out in

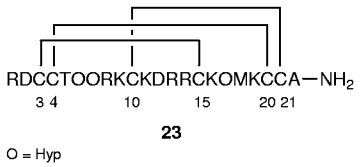
addition to that of the disulfide-intact peptide to unambiguously identify the disulfide topology of the target molecule.[69,70]

Two operating modes, gas-phase sequencing and pulsed-liquid sequencing, are used in automated sequencers. In the former method formation of the DTT adduct of PTH-dehydroalanine occurs to much lower extents than in the latter method due to the gaseous delivery of TFA which contains DTT as stabilizer.[71,72] The conversion-flask temperature in the Edman degradation reaction also affects the recovery of $(PTH\text{-}Cys)_2$ which is higher at lower temperatures (in the case of ABI sequencer 50 °C vs the standard 64 °C) by preventing the decomposition of $(PTH\text{-}Cys)_2$.[73] The retention time of $(PTH\text{-}Cys)_2$ is almost the same as that of PTH-Tyr, and the DTT adduct of PTH-dehydroalanine elutes just before $(PTH\text{-}Cys)_2$ under normal conditions on the ABI sequencer.[28,68] It is important to remember that the latter compound is also produced in the Ser cycles.

Disulfide connectivities of a partially reduced and alkylated peptide can be identified by the pairwise recognition of the specifically derivatized PTH-Cys on sequencing.[46] PTH-Cys(Cam) and PTH-Cys(CM) are reported to elute on the ABI sequencer with PTH-Glu and PTH-Ser, respectively,[46] and PTH-Cys(NEM) after PTH-Pro.[49] Therefore, it is important to know the primary structure of the target molecule in advance and to focus on the determination of its disulfide arrangement during sequence analysis in order to avoid misassignment of the amino acids at the cycles of the modified cysteine residues.

Alternatively, the Edman degradation reaction may be used as a cleavage method at the Cys-Cys bond of multichain peptides. After repeated Edman degradations up to the Cys-Cys bond in the sequence, an additional cycle releases from the parent fragment a part of the fragment with a pair of half-cystines. Disulfide connectivities are then easily determined by amino acid and/or MS analysis of the generated fragments. For example, the disulfide connectivities of bovine neutrophil β-defensin-12 were determined by a combination of trypsin digestion and one-step Edman degradation[74] and those of cardiotoxin I by the successive digestions with trypsin and endopeptidase Asp-N, followed by three steps of Edman degradation.[75] The disulfide connectivities of μ-conotoxin GIIIB (**23**) were also determined using these techniques (Scheme 10).[72]

Scheme 10 μ-Conotoxin GIIIB[72]



23

O = Hyp

Disulfide Connectivities of μ-Conotoxin GIIIB:[72]

μ-Conotoxin GIIIB, a 22 amino acid peptide containing 6 half-cystines, was digested with lysyl endopeptidase (enzyme/substrate ratio of 1:10) in 0.1 M NH_4OAc buffer (pH 6.5) at 37 °C for 30 h. The resulting cystine-containing fragment was composed of the four segments (1–9), (10–11), (12–16) and (20–22) which were cross-linked by three interchain disulfide bridges. Gas-phase sequence analysis of the fragment was carried out under modified conditions where DTT was removed from TFA used for the cleavage reaction. The yield of $(PTH\text{-}Cys)_2$, quantified by summing up $(PTH\text{-}Cys)_2$ and the DTT-adduct of PTH-dehydroalanine, revealed that cystine was detected at the second and the fourth cycle in a ratio of 1:2. Such a pattern is consistent with one of the eight possible disulfide connectivities expected from the components of the lysyl endopeptidase fragment (Table 1). From this result the disulfide topology **23** was determined for μ-conotoxin GIIIB (Scheme 10).

for references see p 175

Table 1 Eight Possible Lysyl Endopeptidase Fragments of μ-Conotoxin GIIIB with Different Disulfide Connectivities and Ratios of $(PTH\text{-}Cys)_2$ during Their Sequencing[72][a]

	Cycle				Ref		Cycle				Ref
	1	2	3	4			1	2	3	4	
DRRCK RDCCTOORK CCA—NH_2 CK	0	1	0	2	[72]	CK RDCCTOORK CCA—NH_2 DRRCK	0	0	1	2	[72]
DRRCK RDCCTOORK CCA—NH_2 CK	0	1	1	1	[72]	CK RDCCTOORK CCA—NH_2 DRRCK	0	0	1	2	[72]
DRRCK RDCCTOORK CCA—NH_2 CK	1	0	0	2	[72]	CK RDCCTOORK CCA—NH_2 DRRCK	0	0	1	2	[72]
DRRCK RDCCTOORK CCA—NH_2 CK	1	0	1	1	[72]	CK RDCCTOORK CCA—NH_2 DRRCK	0	0	1	2	[72]

[a] O = hydroxyproline, Hyp.

6.1.6.2.7 Mass Spectrometric Analysis

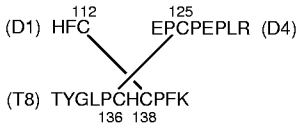
Molecular weight analysis by MS is an essential analytical tool for the determination of disulfide connectivities of multiple-cystine peptides. MS analysis is applied in all the methods described above. Three types of MS analyses are presently used, FAB-MS, MALDI-MS, and ES-MS. The related methodology is developing rapidly and thus continuously increasing the analytical power of these techniques.

Fragmentation of multiple-cystine peptides by the proper enzymatic and/or chemical methods without disulfide exchange and the subsequent mapping of the generated fragments on RP-HPLC usually affords multiple peaks in the chromatogram which are analyzed and characterized by their molecular weight by MS regardless of the homogeneity of the peak components. When the fragments, upon extensive fragmentation, contain single disulfide bonds, their molecular weight analysis allows unambiguous assignment of the disulfide connectivities in the parent molecule. Similarly MS analysis is indispensable to identify the paired half-cystines in fragments obtained by partial acid hydrolysis (Section 6.1.6.2.1) or by the partial reduction (Section 6.1.6.2.3) and the cyanylation cleavage method (Section 6.1.6.2.4), independently of whether the disulfide bonds are intact or reduced and alkylated. A very particular method is based on the synthetic incorporation of ^2H-labelled Cys residues at defined sites which upon oxidative folding of the multiple-cysteine peptide and its complete hydrolysis, allows the identification of cystine connectivities by FAB-MS analysis. In practice, analysis is carried out after Fmoc-derivatization of the hydrolysate as shown in the case of heat-stable enterotoxin.[76]

When the enzymatic and/or chemical fragments still contain multiple disulfide bonds, MS fragmentation is an efficient tool for the assignment of these disulfide connectivities. During MS fragmentation under low energy (i.e., under collision-induced dissociation conditions), disulfide bonds are not preferentially cleaved and disulfide connectivities are assigned by exploiting scissile backbone ion peaks between the half-cystine residues, which is well documented by the analysis of the three disulfide isomers (globule/ribbon/beads) of

human guanylin(99–115).[77] MALDI-postsource decay analysis is also a useful fragmentation procedure, although by this technique some disulfide bonds are cleaved or remain intact.[78,79] This technique was applied to elucidating the disulfide connectivities of a GM2 activator protein fragment **24** (Scheme 11).

Scheme 11 GM2 Activator Protein Fragment[79]



24

Alternatively, MS fragmentation at the disulfide bonds without peptide backbone cleavage by MALDI-ion-trap-MS can be applied for the analysis of enzymatic and/or chemical fragments consisting of multiple peptide chains cross-linked by multiple disulfide bonds. Series of fragment ion peaks are generated by random cleavage of disulfide bonds, thus allowing the identification of the disulfide connectivities between the component chains.[80,81]

Disulfide Connectivities of a GM2 Activator Protein Fragment (24) Prepared by Digestion with Trypsin, *S. aureus* V8 Protease and Endopeptidase Asp-N:[79]

The enzymatic fragment composed of the three segments D1(110–112), D4(123–130) and T8(131–141), where the half-cystine residues were located at positions 112, 125, 136, and 138, was analyzed by MALDI-MS which revealed signals at *m/z* 2606.9 (D1+D4+T8), 2203.6 (D4+T8), 1670.0 (D1+T8), and 940.2 (D4). The ion at 2606.9 was subjected to MALDI-postsource decay analysis which resulted in signals with the combined molecular weights of the T8-derived C-terminal peptides (137–141) or (138–141) and D1 (1034.2 and 897.4, respectively), indicating a disulfide connectivity of Cys^{112}/Cys^{138}. The N-terminal part of T8 containing Cys^{136} was observed in the two signals corresponding to the calculated values of [(131–137)+(124–125)] and [(131–137)+D4] (972.5 and 1711.2, respectively). From these results, disulfide connectivities of this particular enzymatic fragment of GM2 activator protein are as shown in Scheme 11.

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6.2 Synthesis of Cystine Peptide Analogues

L. Moroder

The disulfide bridge is one of the conspicuous structural components of many naturally occurring peptides that favors and stabilizes bioactive conformations and, thus, plays a crucial role in triggering their biological functions. The conformational restraints imparted by the disulfide cross-linked ring structure are additionally modulated by nature via ring expansion with polysulfide groups[1] (Section 6.2.1) or ring restriction with lanthionine,[2] which is the monosulfide analogue of cystine consisting of two alanine-like residues that are bridged side chain to side chain by a sulfide group (Section 6.2.2). This natural variation offers the additional advantage of high chemical and metabolic stability in contrast to the cystine disulfide that is unstable in reducing environments as well as toward nucleophiles and bases. In this context, even the selenocystine with its strongly reducing redox potential of −381 mV[3] yields a diselenide-bridged ring structure which is isosteric to the disulfide, but of significantly enhanced stability (Section 6.2.3). With increasing frequency selenocysteine and selenocystine are encountered in natural proteins.[4] To bypass the chemical lability of the cystine disulfide group, artificial isosteric cystine mimetics have been developed in which one sulfur atom is replaced by a methylene group (monocarba-analogue) or both sulfur atoms by an ethylene bridge (dicarba-analogue). As a result, extended and shortened ring sizes were additionally built into peptides to develop more stable analogues and also to study the topochemical contribution of the disulfide bridge to the biological activity (Section 6.2.4). Of the naturally occurring diaminodicarboxylic acids that can be considered as carba-analogues of cystine, only *meso*-2,6-diaminopimelic acid is found as a key cross-linking amino acid in the peptidoglycan of the cell wall layer of bacteria.[5,6] However, cystathionine, an intermediate in the transsulfurization pathway for the biosynthesis of cysteine,[7] and djenkolic acid[8] have not yet been detected as constituents of peptides or proteins.

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6.2.1 Polysulfide Cystine Analogues

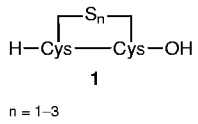
H.-J. Musiol, G. Loidl, and L. Moroder

The only thermodynamically stable form of sulfur at room temperature is an eight-atom cyclic molecule crystallizing orthorhombically (α-S_8). All other sulfur molecules and modifications are converted into this crown-shaped cyclooctasulfur form with half-lives between 10^{-3} seconds and one year.[1] With regard to the relatively high stability of multiple S–S bonds in α-S_8 it is not surprising that various natural products have been found to contain higher-order sulfides such as trisulfides. The interesting biological properties of trisulfide-containing natural products, such as esperamicins[2] and calicheamicins[3] as a new class of antitumor antibiotics, or the pentasulfides diallylpentasulfide[4] and Varacin[5] with antitumor, antifungal, and cytotoxic properties, have generated significant interest. Moreover, a trisulfide moiety has also been identified in a biosynthetic human growth hormone produced in *E. coli*.[6] Whereas growth hormone isolated from human pituitaries contains two disulfide bridges between Cys^{53} and Cys^{165}, and Cys^{182} and Cys^{189},[7] respectively, a single trisulfide bridge was identified in the biosynthetic human growth hormone both at the first[8] or at the latter position.[6,9]

Stability of Polysulfides in Peptides

Trisulfide-bridged peptides **1** (n = 3) (Scheme 1) are highly stabile when stored as lyophilisates at 25 °C, with 70% of the trisulfide remaining intact after five weeks.[10] In solution the stability is strongly dependent on pH. The trisulfide bond is stable for more than two weeks at pH 2, while at pH 7 degradation is already detectable after three days, and at pH 9 degradation of the disulfide occurs within three days.[11] Due to this instability under basic conditions, the trisulfide groups are incompatible with a variety of reagents commonly used for Fmoc cleavage.[10] Tetra- and pentasulfides are remarkably stable, but the latter decompose slowly to the corresponding tetrasulfides in aqueous buffer at neutral pH.[12] Treatment with dithiothreitol (DTT) reduces polysulfides to the related free dithiols within minutes.[12,13]

Scheme 1 Structure of Polysulfide Cystine Peptides



1

n = 1–3

Analytical Characterization of Polysulfide-Bridged Peptides

Cystine peptides containing polysulfides can be separated from the corresponding disulfides by RP-HPLC under acidic conditions. The retention time increases with the number of sulfur atoms between the cysteine residues, in correlation with the hydrophobicity of the compounds.[12,14,15] In combination with HPLC, mass spectrometry based on a variety of ionization sources has been extensively applied.[10–12] The ^{1}H-NMR signals of trisulfide-bridged peptides compared to their disulfide analogues are very similar with only minor shifts of the signals for the β-protons of the half-cystine residues flanking the trisulfide bridge.[6,16] To distinguish polysulfides from disulfides, peptides containing polysulfides are treated with DTT or cysteine and the liberated hydrogen sulfide is detected by the blackening of filter paper soaked in lead acetate solution.[6,13,14]

for references see p 183

6.2.1.1 Polysulfides as Side Products and Intermediates in the Synthesis of Cystine Peptides

Mass spectrometry has gained widespread application for the characterization of synthetic peptides and has led, in combination with HPLC, to the identification of accompanying byproducts such as polysulfides.[17]

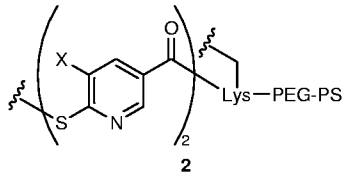
6.2.1.1.1 Side Products in Disulfide Formation

The controlled formation of disulfide bridges in peptide synthesis has been studied extensively in the past, but only recently have polysulfides been identified as unexpected side products in these reactions.

6.2.1.1.1.1 Side Products in Disulfide Formation with Ellman's Type Reagents

A family of solid-phase reagents, derived from Ellman's reagent, has been introduced for the formation of disulfide bridges in peptides.[18] Two new reagents of this type (**2**, X = H or NO_2; Scheme 2) have been proposed where both amino groups of the lysine residue are acylated by the Ellman's reagents and cross-linked by the related disulfide.[19] The nitro derivative displays higher efficiency as an oxidation reagent, but under certain reaction conditions besides producing the expected disulfide it leads to byproducts which based on LC-MS analysis were assigned to the related tri- and tetrasulfide peptide derivatives.

Scheme 2 New Reagents for Disulfide Formation[19]



2

X = H, NO_2

6.2.1.1.1.2 Side Products in Disulfide Formation with *N*-(Dithiasuccinoyl)glycine

The *N*-dithiasuccinoyl group was originally proposed for amino protection as it can be removed by thiolysis.[20] By exploiting the reaction mechanism involved in the deprotection of this group *N*-(dithiasuccinoyl)glycine was proposed as a mild oxidant for disulfide formation.[15] With the reduced nonapeptide hormon oxytocin (OXT) and deamino-oxytocin as model peptides, the oxidation with *N*-(dithiasuccinoyl)glycine has been investigated in detail both in solution and/or on a solid support.[15] Under suboptimal conditions, such as high pH and/or a high *N*-(dithiasuccinoyl)glycine to peptide ratio, the formation of byproducts including dimers as well as up to 30% of the corresponding trisulfides has been observed.

6.2.1.1.2 Side Products in Peptide Synthesis with Cystine Derivatives

The standard approach to the synthesis of peptides containing disulfide bridges is to use S-protected cysteine derivatives for chain elongation and then subsequently form the disulfide by oxidation of the final S-protected or S-unprotected cysteine peptides (Section 6.1.1). The alternative approach, based on the use of suitably protected symmetrical or unsymmetrical cystine derivatives for the synthesis of disulfide bridged homo-, or heterodimers, besides exhibiting the difficulties discussed in Section 6.1.3, was recently found to generate under

particular conditions trisulfide analogues to appreciable extents.[16] In fact, by coupling *N,N'*-bis(benzyloxycarbonyl)cystine with glycine(3-dimethylamino)propylamide by the DCC/HOBt/DIPEA procedure, up to 10% of the trisulfide is formed as side product depending on the reaction time. While formation of the trisulfide was not detectable in DMF/DIPEA alone, 15–20% of the trisulfide is formed with BOP/HOBt or HBTU/HOBt as coupling reagents after 24 hours. With addition of the radical quenching agent hydroquinone trisulfide, formation of the trisulfide is prevented; nonetheless, the mechanism by which the trisulfides are generated remains elusive.

6.2.1.1.3 Polysulfides as Intermediates in Peptide Ligation

The development of peptide fragment coupling strategies exploiting the proximity-driven intramolecular acylation for peptide bond formation has been an important advance in peptide synthesis.[21] Compared to intermolecular acylation reactions, intramolecular peptide bond-forming processes are advantageous for segment condensation due to a reduced entropic barrier resulting in high regiospecificity without the need for protecting groups.

An orthogonal coupling approach through an acyl disulfide mediated intramolecular acylation step has been proposed for the ligation of unprotected peptide fragments and applied to the synthesis of a 32-residue model peptide.[14] The key step of this method consists of a nucleophilic attack of the C^{α}-thiocarboxylic acid moiety of the acyl segment at the Npys-activated thiol group of the N-terminal cysteine residue of the amino segment. The resulting acyl disulfide undergoes an intramolecular *S,N*-acyl transfer yielding a hydrodisulfide, and after thiolytic reduction with DTT the native cysteine residue adjacent to the ligation site is regenerated. The intermediate hydrodisulfide is unstable and undergoes disulfide exchange and oxidation reactions before DTT treatment to give various tri- and tetrasulfide products, including the trisulfide product formed between the hydrodisulfide and the Npys-activated amino segment as well as tri- and tetrasulfides formed between two hydrodisulfide molecules. These intermediates are all converted into the desired cysteine product upon treatment with DTT and, therefore, have no effect on the yield of the ligation reaction or on the purity of the final product.

6.2.1.2 Synthesis of Polysulfide-Bridged Cystine Peptides

The chemistry of diorgano polysulfides has been extensively investigated for decades and a variety of methods has been described for their preparation.[1,22–24] However, only three major methods have been applied to the synthesis of polysulfides in cystine peptides thus far.

By reacting a cystine peptide with an excess of elemental sulfur at pH 10 in EtOH/$CHCl_3$/CS_2/NH_4OH (45:5:2:2) at room temperature,[16] a mixture of the starting material and the corresponding tri-, tetra-, and pentasulfides are obtained, which can be resolved by RP-HPLC. However, when this protocol was applied to the synthesis of the vasopressin trisulfide analogue, attempts to purify the target compound from the resulting mixture of polysulfides failed.[11] Similarly, oxidation of bis(cysteine) peptides in the presence of bis(tetrabutylammonium) hexasulfide in aqueous solution at pH 7 resulted in a mixture of products, which could be resolved by RP-HPLC.[12] With equimolar amounts of the hexasulfide salt, peptides containing two to five sulfur atoms were prevalently formed, whereas with a five-fold excess of peptide only the di- and trisulfides were detected.

An alternative method for formation of trisulfides[10] has been derived from well-known procedures for the formation of disulfides that involve sulfenyl thiocarbonates as protecting/activating groups for cysteine.[25] In analogy to the chemistry for disulfides, cysteine side chains were protected and activated as extra-sulfur analogues, i.e. as *S*-(methoxycarbonyl)disulfanyl or *S*-(*N*-methyl-*N*-phenylcarbamoyl)disulfanyl derivatives.[10] Preferably,

for references see p 183

these cysteine derivatives were introduced as N-terminal amino acids, and upon intramolecular thiolytic displacement of the disulfanyl groups by a second cysteine thiol group, the cyclized peptide trisulfides were obtained. The observed formation of di- and tetrasulfides during the cyclization step was attributed to disproportionation reactions. Appearance of these side products in varying amounts makes this method of little application for preparative purposes.

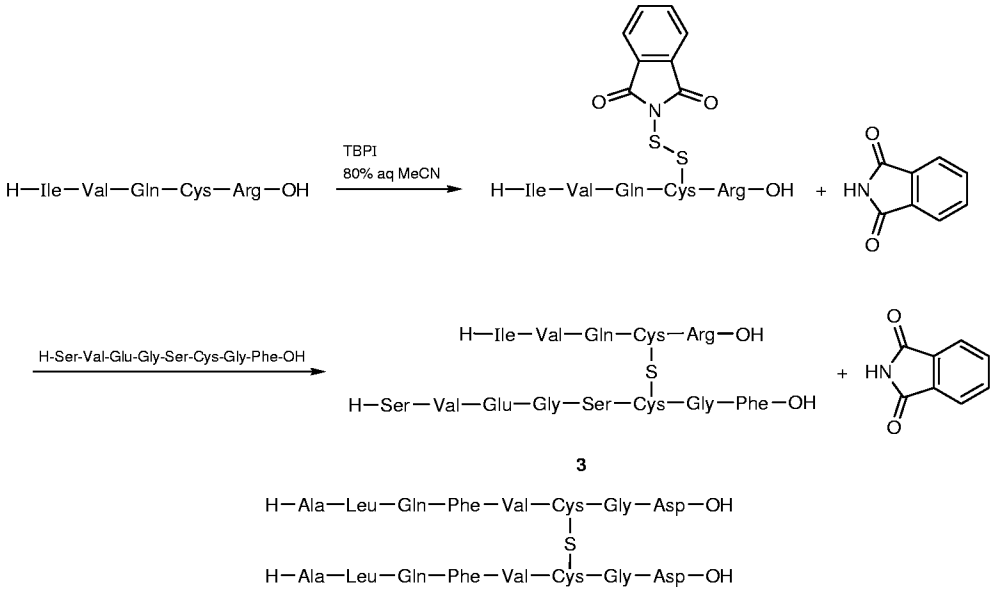
6.2.1.2.1 Synthesis of Trisulfides with *N,N'*-Thiobisphthalimide

The only approach for the selective synthesis of peptides containing a trisulfide bridge known so far involves *N,N'*-thiobisphthalimide (TBPI), a reagent that is largely used for sulfur transfer in the preparation of nonpeptidic trisulfides.[26–29] This reagent has been used for the synthesis of interchain[13] as well as intrachain trisulfide peptides[11,13] from the corresponding S-unprotected cysteine peptides.

6.2.1.2.1.1 Interchain Trisulfide-Bridged Peptides

The preparation of a trisulfide bridge between two different linear peptides is outlined in Scheme 3. In a first step, the free thiol of one peptide forms an activated mixed disulfide by reaction with *N,N'*-thiobisphthalimide, and upon the subsequent addition of the second cysteine peptide, nucleophilic displacement of the phthalimide moiety by the free thiol yields the desired interchain trisulfide bridged peptide **3**.[13] Homodimeric analogues, e.g. **4**, are prepared similarly.[13]

Scheme 3 Synthesis of Trisulfide-Bridged Cystine Peptide Analogues[13]



Trisulfide-Bridged Cystine Peptide 3 **(Heterodimer):**[13]
To a stirred soln of TBPI in 80% aq MeCN (0.75 mM) was added at rt H-Ile-Val-Gln-Cys-Arg-OH (1 equiv). After 6 h, H-Ser-Val-Glu-Gly-Ser-Cys-Gly-Phe-OH (1 equiv) was added and the soln was kept at rt overnight. Upon completion of the reaction as monitored by RP-FPLC, the MeCN was removed under reduced pressure and the precipitated phthalimide was removed by filtration. The resulting

filtrates were lyophilized and the residue was purified by gel-permeation chromatography; yield: 55%. The product was characterized by amino acid analysis and FAB-MS.

Trisulfide-Bridged Cystine Peptide 4 (Homodimer):[13]
To a stirred soln of TBPI in 80% aq MeCN (0.75 mM) was added at rt H-Ala-Leu-Gln-Phe-Val-Cys-Gly-Asp-OH (2 equiv) and the reaction was stirred overnight. The mixture was worked up as described for the heterodimer; yield: 54%.

6.2.1.2.1.2 Intrachain Trisulfide-Bridged Peptides

To avoid the formation of dimers, high-dilution conditions have been used for the synthesis of intrachain trisulfides derivatives of vasopressin[11] and a peptide related to the C-terminal region of human growth hormone.[13] Typically, a 0.07 mM solution of a peptide containing two thiol groups is reacted with one equivalent of *N,N'*-thiobisphthalimide under the conditions described in the experimental procedure of Section 6.2.1.2.1.1; workup, purification procedures, and yield are identical.

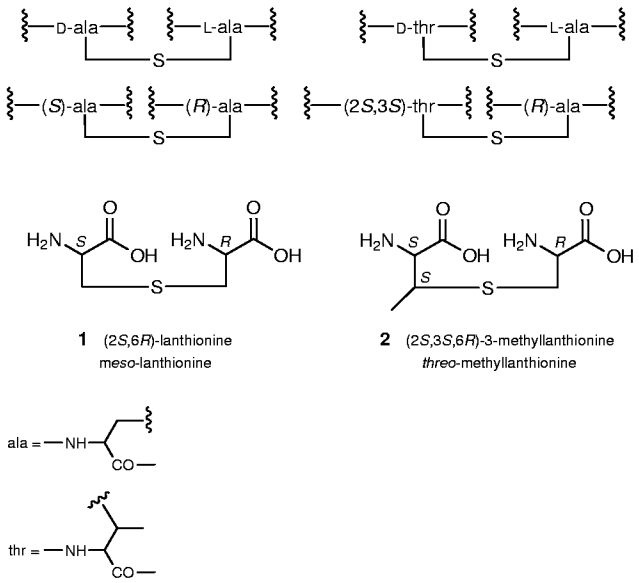
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6.2.2 Lanthionine Peptides

R. D. Süssmuth, R. W. Jack, and G. Jung

Lanthionine (Lan) (**1**) is a naturally-occurring, but relatively rare dicarboxylic diamino acid, which until recently, was largely ignored. The amino acid was initially identified as an artifact in amino acid analysis of wool hydrolyzates[1] which later turned out to be a result of the alkaline hydrolysis procedure used. Subsequently, it was discovered that a novel class of bacterial-derived peptide antibiotics also contained lanthionine and the closely related amino acid 3-methyllanthionine (MeLan) (**2**) (Scheme 1).

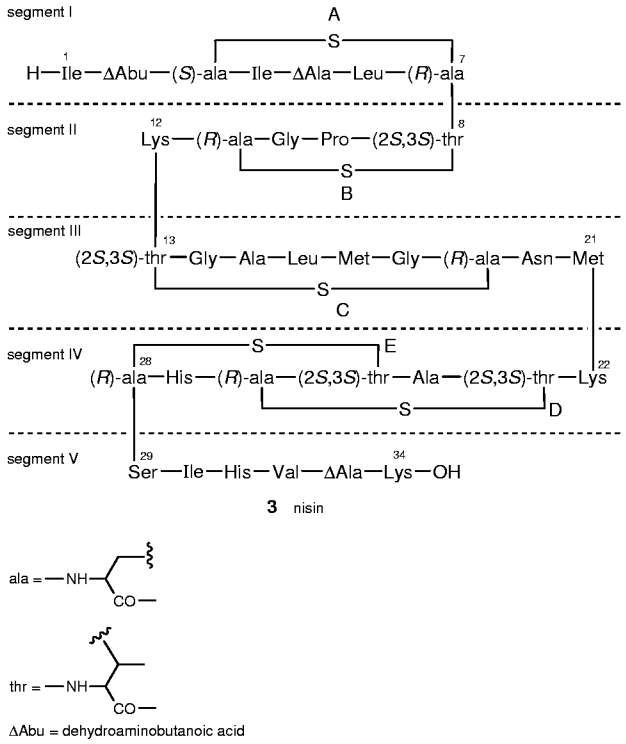
Scheme 1 The Structures of the Naturally Occurring Thioether Amino Acids *meso*-Lanthionine and *threo*-Methyllanthionine and Common Abbreviations within the Peptide Chain, Alanine and Threonine Denote the Amino Acid Core Structures Linked to the Sulfur Atom through the Side Chain



The rising interest in these antimicrobial peptides, the class of which was named lantibiotics[2] to denote that they are Lan-containing and antibiotic in nature, led to a significant burst of research into their chemical nature and biosynthesis, the three-dimensional structure of the peptides, and their mechanism(s) of killing action.[74] (For recent books and reviews on the subject, see refs[3–9].) In the bacterial cells producing lantibiotics, it is now well-established that the Lan and MeLan residues arise from a two-step process, the first step of which involves the dehydration of specific Ser and Thr residues, respectively, in the nascent peptide chain. In the subsequent step, the thiol group of a neighboring cysteine residue forms the second half of the Lan/MeLan through a stereospecific Michael-type addition reaction.

In the bacterial production of lantibiotics, enzymes carry out these reactions in a highly organized and regulated manner. Within the last 15 years, a considerable increase in the number of such peptides being isolated and characterized has led to a vastly increased understanding of these peptides. One of the most intensely studied lantibiotics is nisin (**3**) (Scheme 2), which was probably first identified in 1928, before the discovery of penicillin.[10,11]

Scheme 2 Structure of the Pentacyclic 34-Residue Peptide Antibiotic Nisin and Strategy Used for Segment Condensation[13]



However, it was not until 1971, when Gross and Morrell reported the structure elucidation of the purified nisin peptide,[12] that it became clear that nisin contained Lan and MeLan residues. Since nisin is an important peptide antibiotic, regularly used in the food production industry as a preservative (preservative EC 234), it has attracted a great deal of attention. Nisin is produced by fermentation, however, partly because of its intriguing structure, it also became the target for the only complete synthesis of a lantibiotic reported to date in the literature. This work has been reviewed.[13,14]

Stereochemistry

Lanthionine is a dicarboxylic diamino acid possessing two stereocenters. However, because of the symmetry of the molecule there exist only three stereoisomers. Thus, with respect to lanthionine stereochemistry, one must differentiate between the 2*R*,6*R*-lanthionine (L-lanthionine), 2*S*,6*S*-lanthionine (D-lanthionine) and 2*S*,6*R*-lanthionine (*meso*-lanthionine, D,L-lanthionine) (**1**); the latter is the naturally occurring stereoisomer of lanthionine isolated from lantibiotics. However, care must be taken when determining and assigning the stereochemistry of lanthionine residues present within peptide sequences. For naturally occurring lanthionine peptides the sulfide amino acid Lan within the peptide is no longer symmetrical and, in the majority of cases, it has been shown to be present as 2*S*,6*R*-lanthionine.

Considering the stereochemistry and nomenclature of lanthionines, the variety may become somewhat confusing. Here, we have endeavored to restrict ourselves to the D,L-nomenclature; however, because of the diversity of nomenclatures found in the literature, this is not always possible. In this context, it has to be noted, that L-cysteine is (2*R*)-cysteine

for references see p 211

due to the higher priority of the sulfur atom, in contrast to all other proteinogenic L-amino acids, which are (2*S*)-amino acids. Furthermore, the problem of stereochemistry and nomenclature[15] is significantly exacerbated by the formation of an additional stereocenter in the 3-methyl derivative of Lan, MeLan (**2**). Thus, from the stereochemical point of view, eight different 3-methyllanthionines are conceivable; however, in nature (2*S*,3*S*,6*R*)-3-methyllanthionine, also known as *threo*-3-methyllanthionine, appears to predominate.

Stability and Analytical Characterization of Lanthionine Peptides

Lanthionine peptides are considerably more stable to reduction than their disulfide counterparts; indeed, the stability of lanthionine rings has been one of the reasons behind the interest in synthesis of peptides in which disulfide rings have been exchanged with lanthionines. Furthermore, because of the stability of the sulfide bridge in acids, hydrolysis for 16 hours at 110 °C in 6 M HCl followed by amino acid composition analysis has often been used for identification and analysis of lanthionine in peptides.[16] However, in contrast to total hydrolysis of most amino acids, lanthionines suffer considerably from epimerization during acid hydrolysis. For example, heating above 96 °C in acidic solutions (6 M HCl or HCl/AcOH) for several hours has been reported to lead to racemization to a mixture of D-, L-, and *meso*-lanthionine.[12,17] It has also been reported that complete racemization occurs under basic conditions.[12]

6.2.2.1 Synthesis of Monolanthionine Peptides

Shortly after the detection of lanthionine in naturally occurring proteins and peptides, attempts to synthesize lanthionine began. Historically, the synthesis of lanthionine was achieved using a number of different approaches, including the reaction of cysteine with α-amino-β-chloropropanoic acid,[18] of sodium phthalimidomalonic ester with bis(chloromethyl)sulfide,[19] and of cysteine with *N*-acetylamino acrylic acid.[20] In particular the first and last of these methods have proven to be rather useful and they have been successfully employed in more recent preparations of lanthionine peptides. Ignoring differences in yield and the relative difficulty in obtaining particular precursors, the currently employed methods for the preparation of lanthionine peptides can be broadly divided into different subgroups. Each of these subgroups can be distinguished by its transferability to solid-phase synthesis and/or its stereochemical outcome. Each is discussed in turn below.

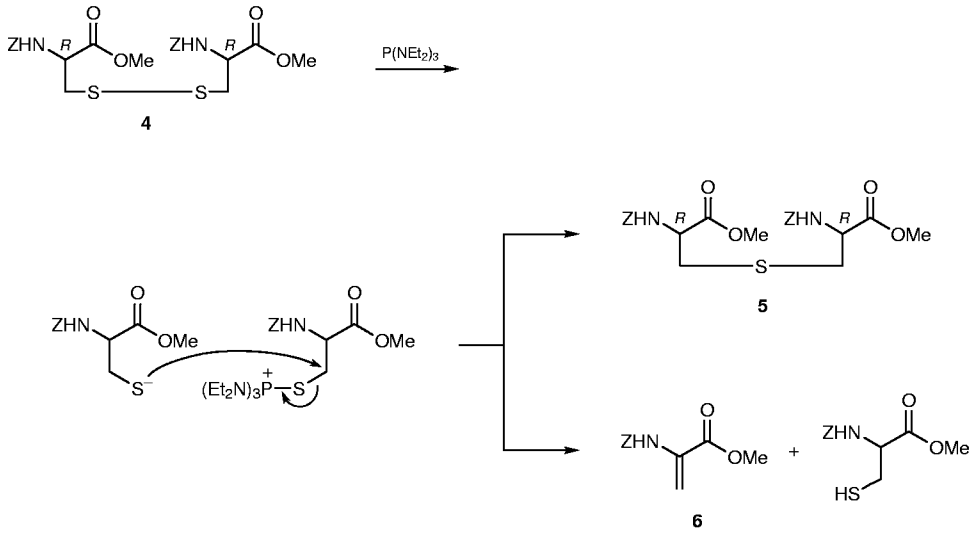
6.2.2.1.1 Synthesis of Lanthionine Derivatives by Desulfurization of Cystine Compounds

One of the most versatile and advanced approaches in lanthionine synthesis involves desulfurization with aminophosphines.[21,22] In the reactions reported, tris(diethylamino)phosphine [$P(NEt_2)_3$] is used as the sulfur acceptor, converting the disulfide **4** into the respective sulfide **5** (Scheme 3). The sulfur extrusion has been found to proceed most effectively using $P(NEt_2)_3$ in an 25-fold excess,[23] in comparison to other phosphines such as Bu_3P, Ph_3P, $P(OPh)_3$, and $P(NMe_2)_3$. Reaction times were about 12 hours when using DMF as a solvent.

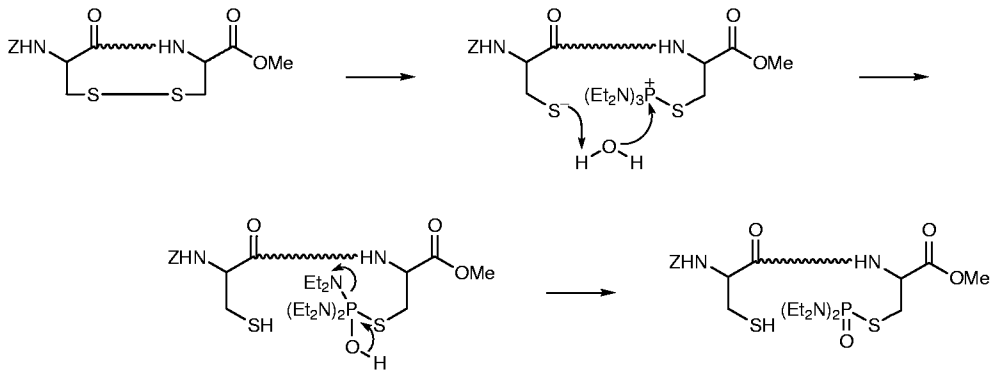
Investigations with regard to the solvent dependence of the sulfur extrusion procedure have been carried out using *N*,*N*′-bis(benzyloxycarbonyl)cystine dimethyl ester (**4**) as a model compound.[24] Amongst the organic solvents tested, DMF was found to be the most suitable, whilst others such as dioxane, THF, benzene, $CHCl_3$, EtOAc, and MeCN appeared to support the formation of dehydroalanine **6** as a major byproduct. In a further contribution from the same authors,[25] the desulfurization has been investigated with regard to the combination of solvent dependence, reaction temperature, and amount of phosphine. Here it was found that benzene is also a suitable solvent for desulfurization using the example of a

tetrapeptide in which sulfur extrusion resulted in the formation of a 3-methyllanthionine ring. These two studies are important as they provide an insight into a potential problem associated with the sulfur extrusion method when used in the synthesis of lanthionine peptides. Thus, if solvents which have not been carefully dried are used for the desulfurization of peptides, byproduct formation through reaction with the $P(NEt_2)_3$ is likely to occur (Schemes 3 and 4).[24]

Scheme 3 Desulfurization Reaction and Proposed Reaction Mechanism Leading to Protected (2*R*,6*R*)-Lanthionine and Byproducts[24]



Scheme 4 Proposed Mechanism of Byproduct Formation During Desulfurization in Solvents which are Not Water Free[24]



6.2.2.1.1.1 Desulfurization of Symmetrical Cystine Derivatives

This reaction was first demonstrated using the desulfurization of *N*,*N*′-bis(trifluoroacetyl)-L-cystine dimethyl ester which proceeded in 96% yield.[21] However, mild alkaline hydrolysis to obtain the free acid resulted in significant product losses to yield only 64% of L-lanthionine (2*R*,6*R*-lanthionine). Alternative protecting strategies have been used in order to raise the versatility of the reaction for peptide synthesis, thus, *N*,*N*′-bis(benzyloxycarbonyl)-L-cystine diethyl ester could be converted into L-lanthionine in 86% yield.

for references see p 211

***N,N'*-Bis(benzyloxycarbonyl)-L-lanthionine Diethyl Ester:**[21]
N,N'-Bis(benzyloxycarbonyl)-L-cystine diethyl ester (2.26 g, 4.0 mmol) was suspended in dry benzene (10 mL) and $P(NEt_2)_3$ (1.2 g, 4.8 mmol) was added slowly to the suspension. The mixture was stirred for 1 h, during which time the soln cleared. The solvent was removed and the residue chromatographed on silica gel. The phosphine sulfide was first eluted with hexane/EtOAc (9:1) then the title product with hexane/EtOAc (1:1); yield: 1.83 g (86%); mp 63–67 °C; $[\alpha]_D^{25}$ −15.9 (*c* 1.1, MeOH).

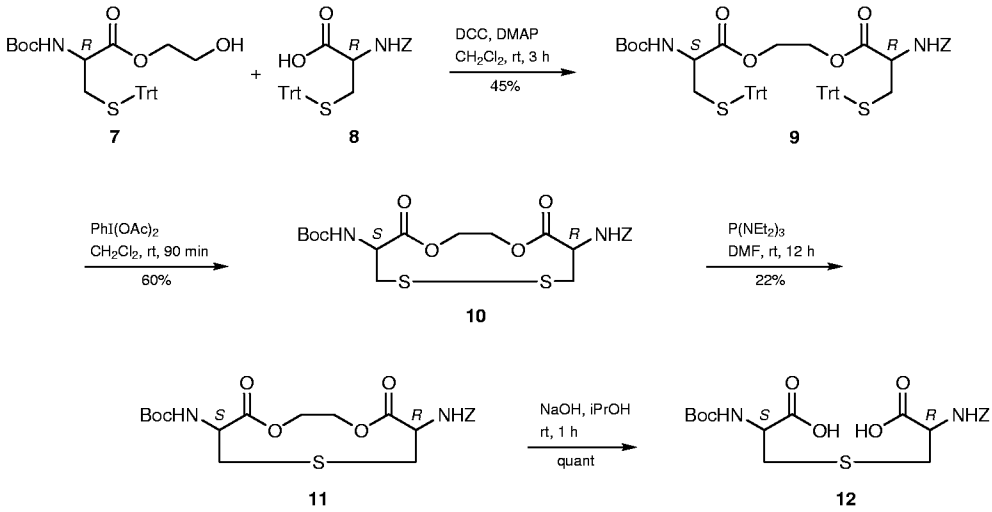
6.2.2.1.1.2 Desulfurization of Unsymmetrical Cystine Derivatives

Critical appraisal of the method,[23,26] using attempts to synthesize nonsymmetrically substituted lanthionines, resulted in rearrangement of the products, presumably due to phosphine-catalyzed disproportionation of the unsymmetrical disulfides. This reaction should proceed more rapidly than the desulfurization process and is thought to occur because sulfur extrusion takes place via a reversible reaction by recombination of ionic intermediates (Scheme 3).[21,22,24] Thus, the reaction of nonsymmetrical cystine derivatives results in the formation of a mixture of three different lanthionines.

From a stereochemical viewpoint, the sulfur extrusion methods[21,22] using $P(NEt_2)_3$ seem to have certain advantages, because chiral centers at alpha carbon atoms are not involved in the lanthionine formation. However, due to the possible formation of ionic intermediates prior to formation of the sulfide bridge, undesired symmetrically substituted or also stereochemically undesired lanthionines may inadvertently be formed. Possible solutions to circumvent these problems have also been reported.[23,26]

In order to bypass problems in the synthesis of nonsymmetrical lanthionines, a slightly modified approach using sulfur extrusion to generate *meso*-lanthionine has been reported.[23] Firstly, the synthesis of a cyclic cystine derivative **10** was carried out (Scheme 5) from the cysteines **7** and **8** (L or D, respectively) in which the amino groups were protected with the Boc and Z group, respectively. Coupling of **7** and **8** gave **9** in which the carboxy groups of the cysteine derivatives were linked through an ethylene glycol diester. The trityl groups were removed and the disulfide **10** was formed by oxidation with diacetoxyphenyl iodide in yields of 55–90%. The subsequent sulfur extrusion from **10** to give **11** was found to proceed most effectively using $P(NEt_2)_3$ in a 25-fold excess. To obtain the orthogonally protected free acid **12**, the lanthionine diester **11** was hydrolyzed with 0.1 M NaOH. Whilst the yield was only about 22%, no epimerization was reported to take place during the sulfur extrusion step. However, due to the strongly basic nature of the phosphine, base-labile protecting groups, (e.g., Fmoc), cannot be used in the sulfur extrusion step. This principal approach is very similar to the sulfur extrusion of peptides reported later in this review (Section 6.2.2.1.4), with the exception that the ester bonds in the backbone can be reversibly removed after lanthionine formation. However, one disadvantage of this method is that after alkaline hydrolysis of the ethylene glycol ester, the carboxy groups cannot be selectively protected for further use in peptide synthesis.

Scheme 5 Synthesis of Unsymmetrically Substituted Lanthionines Using the Desulfurization Reaction of Ethylene Glycol Bridged Cystines[23]



A further approach for the synthesis of nonsymmetrically protected lanthionines is the conversion of thiosulfinates of symmetrically protected cystine derivatives into nonsymmetrically protected cystines via a reaction with a cysteine derivative and subsequently the conversion of the resulting unsymmetrically protected cystine into the nonsymmetrically protected lanthionines with a tris(dialkylamino)phosphine.[26] The oxidation of the symmetrically protected cystine, e.g. *N*,*N*′-bis(benzyloxycarbonyl)-L-cystine diethyl ester, of one stereochemical configuration to the thiosulfinate with *m*-chloroperoxybenzoic acid is essentially quantitative. The nonsymmetrical cystine is then formed in a subsequent step by the addition of the *N*-*tert*-butoxycarbonyl-L-cysteine *tert*-butyl ester derivative to give *N*-Z-*N*′-Boc-L-cystine ethyl *tert*-butyl diester. The desired 2*R*,6*R*-lanthionine is then formed in the presence of $P(NEt_2)_3$ in yields of >50%.

The one-pot synthesis of lanthionines starting from thiosulfinates and cysteines and subsequent addition of tris(diethylamino)phosphine was not successful. Moreover, with this method based on L-cystine derivatives only L-lanthionines have been synthesized. In this context it is worth noting, that alternative and possibly more efficient methods can be applied for the synthesis of the unsymmetrically protected cystines as intermediates (Section 6.1.3).

***N*-Benzyloxycarbonyl-*N*′-*tert*-butoxycarbonyl-L,L-lanthionine *tert*-Butyl Ethyl Diester:**[26]
To a soln of *N*-Z-*N*′-Boc-L-cystine *tert*-butyl ethyl diester (0.96 g, 1.74 mmol) in benzene (25 mL) was added $P(NEt_2)_3$ (0.47 g, 1.91 mmol) dropwise, with stirring under N_2. After stirring at rt for 5 h, the solvent was removed under reduced pressure and the oily residue purified by MPLC (hexane) to yield the title compound; yield: 0.47 g (52%); $[\alpha]_D^{25}$ +2.3 (*c* 7.9, $CHCl_3$).

6.2.2.1.2 Synthesis of Lanthionine Derivatives from Precursor Molecules

In contrast to the syntheses described above, which all start from cystine derivatives to form lanthionine, the lanthionine syntheses in this section all start from a protected cysteine as the nucleophilic precursor, which is then allowed to react with any of a variety of different substrates. These subsequent reactions are the Michael addition with dehydroalanine, the nucleophilic substitution of halo amino acids, or the ring-opening reaction of serine β-lactones and aziridines, respectively. However, it must be emphasized that the Michael

for references see p 211

addition to form lanthionine is a reaction without diastereospecificity, except in those cases described later for the synthesis of lanthionine peptides.[27]

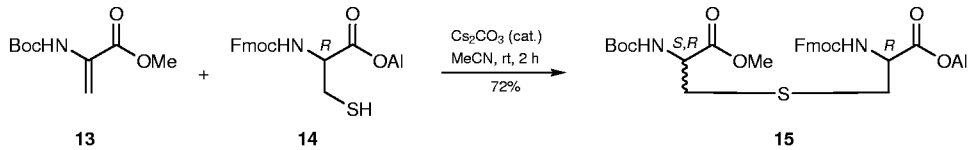
The methods presented below are all suitable for the synthesis of lanthionine and its protected derivatives. However, the Michael addition of cysteine to dehydroalanine results in the formation of two diastereomers. This is particularly problematic when attempting to synthesize fragments of naturally occurring lanthionine-containing peptides, since these generally possess Lan in positionally defined stereochemical configurations. Methods involving the nucleophilic substitution of β-halogenated alanines as well as those based on ring-opening reactions of serine β-lactones[28,29] or aziridines[30] with cysteines frequently result in optically pure lanthionine derivatives. However, in the case of the nucleophilic substitution of haloalanines, strongly basic conditions have to be prevented in order to suppress dehydroalanine formation which in turn leads to diastereomeric product formation. Whilst, in the case of the aziridine ring-opening reaction, it has to be mentioned that, although optically pure compounds can be obtained, yields are very low. Moreover, even experimental procedures for the ring opening of serine β-lactones are not particularly satisfactory.[28]

6.2.2.1.2.1 Dehydroalanine/Cysteine Addition

The synthesis of lanthionines via dehydroalanines has been carried out using an *O,S*-dibenzoylated Z-L-Cys-L-Ser-OMe.[31] Using β-elimination under basic conditions in sodium methoxide and subsequent intramolecular Michael addition, the corresponding L-lanthionine derivative was formed. However, as a final step the cyclolanthionyl dipeptide would have to be cleaved at the amide bond to obtain lanthionine. The same authors reported the use of the O-tosylated serine dipeptide which was subjected to β-elimination, forming the desired dehydroalanine derivative. However, this synthesis also yielded cyclic L-lanthionine dimers as a result of an intermolecular Michael addition. The amounts of cyclolanthionine and lanthionine dimer derivatives depended on the initial concentration of the dipeptides. In addition, the directed synthesis of cyclic lanthionine dimers, which were produced from *N*-benzyloxycarbonyl-*N'*-tritylanthionine monomethyl ester, has been reported.[32]

An alternative approach using Michael addition for the synthesis of orthogonally protected lanthionines has been described, which seems to be of more practicable use.[33] It is based on the reaction of Fmoc-L-Cys-OAl (**14**) with Boc-ΔAla-OMe (**13**) in MeCN in the presence of catalytic amounts of Cs_2CO_3 and the product **15** was obtained in good yield (Scheme 6). The resultant lanthionine is designed such that the protecting groups are suitable for solid-phase synthesis. However, the disadvantage of the Michael addition is that the reaction is regio-, but not stereospecific, so that the resulting diastereomers have to be separated by RP-HPLC. As the authors note, the synthesis of stereochemically pure lanthionine using the ring-opening reaction of serine β-lactones[28] (Section 6.2.2.1.2.3) was not practicable with an allyl ester as a protecting group.

Scheme 6 Protected Partially Racemic Lanthionine Formation via Michael Addition of L-Cysteine to Dehydroalanine Derivatives[33]



13 **14** **15**

In order to generate the dehydroalanine required in another lanthionine synthesis, a number of different reactions have been used.[75] One method is the β-elimination of O-tosylated or O-diphenylphosphorylated serines and threonines;[34] however, in this reaction aziridines are formed as side products on treatment with TEA. Hofmann degradation has also been used, by starting with α,β-diaminopropanoic acid and subsequent permethylation with iodomethane in $KHCO_3$ to generate dehydroalanine in 60–85% yield.[35] Some less common methods such as the reaction of serines and threonines with DIC/CuCl to produce dehydroalanine derivatives via the formation of isoureas,[36] or refluxing of Z-Ser-OMe in MeOH/HCl to yield the dehydroalanine derivative in 82% yield[37] have also been successfully applied. However, possibly the best method for the synthesis of dehydroalanine is the dehydration of serine with *N*,*N*′-disuccinimidyl carbonate (DSC).[38]

***N*′-*tert*-Butyloxycarbonyl-*N*-fluorenylmethoxycarbonyllanthionine *O*-Allyl *O*′-Methyl Diester (15):**[33] Fmoc-L-Cys-OAl (825 mg, 2.15 mmol) and a catalytic amount of Cs_2CO_3 (ca. 20 mg) were added to a soln of Boc-ΔAla-OMe (430 mg, 2.14 mmol) in MeCN (10 mL). The mixture was stirred for 2 h, then diluted with H_2O (10 mL) and extracted with EtOAc (2×20 mL). The EtOAc layers were combined, washed with H_2O (2×20 mL), brine (20 mL), dried ($MgSO_4$), and the solvent removed under reduced pressure. The residue was purified by column chromatography (silica gel, petroleum ether/EtOAc/AcOH 75:25:1); yield: 0.9 g (72%); the diastereomers were separated by HPLC.

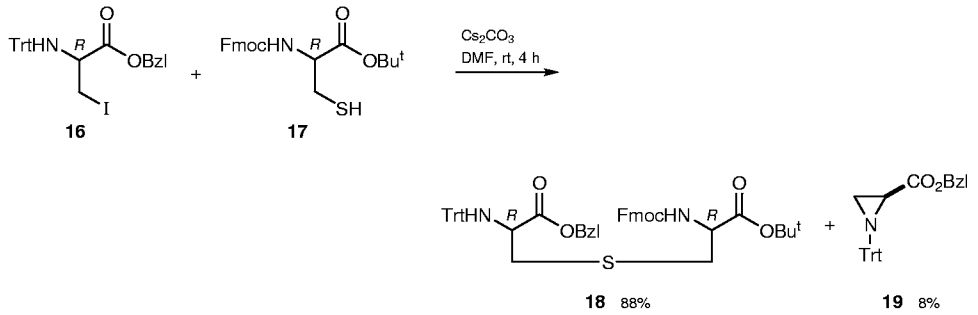
6.2.2.1.2.2 3-Haloalanine/Cysteine Addition

A widely used method for the synthesis of lanthionines is the reaction of 3-chloroalanine with the nucleophilic sulfur, either starting with cystine[39] or cysteine.[17] However, the manner in which the reaction is carried out significantly influences the yield; reactions in aqueous alkaline solutions result in moderate yields (34%),[39] but phase-transfer catalysts in water/diethyl ether considerably increase the usefulness of this procedure giving 68% yield.[17] Whilst the authors report a high enantiomeric excess with the above mentioned syntheses, other reports[23,40] have claimed that the strongly basic conditions lead to β-elimination. Thus, diastereomeric mixtures should be obtained. There are various methods for producing 3-chloroalanine, such as reacting serine with phosphorus pentachloride (yield 82%),[41] or cystine with chlorine (0 °C, 1 h) in yields of 68%.[42]

With regard to stereochemical considerations, a further very interesting approach is the synthesis of lanthionines from variously protected cystine derivatives and *N*-trityl-3-iodoalanine benzyl ester (**16**) (Scheme 7).[40] The cystine derivative is reduced to **17** with tributylphosphine prior to the addition of the 3-iodoalanine derivative **16**. After the reaction, equimolar amounts of Cs_2CO_3 are added to quench the iodide. The relatively mild reaction conditions reduce the possibility of β-elimination and yields are very good, varying from 79–88%. As a side reaction, the formation of the corresponding aziridine-2-carboxylates **19** in yields of 7–17% has been observed; however, others report up to 35% aziridine formation.[76] An enantiomeric excess of >98% was determined by degradation to alanine and subsequent precolumn derivatization with a chiral reagent prior to HPLC analysis. The 3-iodoalanine derivative **16** was synthesized from Trt-Ser-OBzl via mesylate formation and nucleophilic substitution with sodium iodide.

for references see p 211

Scheme 7 (2*R*,6*R*)-Lanthionine Formation Proceeds with Retention of Configurations via Nucleophilic Substitution with 3-Haloalanines[40]

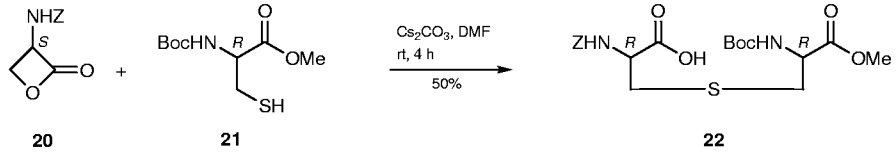


(2*R*,6*R*)-*N*-Fluorenylmethoxycarbonyl-*N'*-trityllanthionine *O'*-Benzyl *O*-*tert*-Butyl Diester (18):[40]
N,*N'*-Bis(fluorenylmethoxycarbonyl)cystine di-*tert*-butyl ester (399 mg, 0.5 mmol) was stirred with Bu_3P (125 µl, 0.5 mmol) in THF for 15 min at rt. The solvent was removed under reduced pressure, the residue was redissolved in EtOAc and washed with citric acid (10%) and brine. The solvent was again removed under reduced pressure to yield **17** which was used without purification. To a soln of **17** and *N*-trityl-3-iodoalanine benzyl ester (547 mg, 1 mmol) in dry DMF was added Cs_2CO_3 (326 mg; 1 mmol) and the reaction was stirred for 4 h at rt. After dilution with iPr_2O, the soln was washed with citric acid (10%) and the aqueous layer was then extracted with iPr_2O. The combined organic layers was washed with citric acid (10%) and brine. The solvent was removed under reduced pressure and the product was purified by flash chromatography (silica gel, EtOAc/hexane 15:85); yield: 719 mg (88%); $[\alpha]_D^{20}$ −6 (*c* 0.5, EtOAc).

6.2.2.1.2.3 Ring Opening of Serine β-Lactone Derivatives

An alternative approach to the synthesis of orthogonally protected lanthionines **22** involves the regioselective ring opening of *N*-benzyloxycarbonylserine β-lactones **20** by the respective thiolate anions of cysteine derivatives **21** (Scheme 8).[28,29,43,77] A priori there are two conceivable possibilities on opening of the serine β-lactone ring. The *O*-acyl fission resulting in thioester formation and the *O*-alkyl fission yielding the desired lanthionine. Reaction conditions have been optimized to obtain the desired lanthionine in satisfactory yields of >50%, dependent on the β-lactone used. To promote lanthionine formation, cesium is used as a counter ion for the thiol group of the cysteine. Thus, the reaction is performed using Cs_2CO_3 or $CsHCO_3$ in DMF at room temperature (Scheme 8). Despite the limitations imposed by the moderate yields obtained for unsubstituted lanthionines, the optical purity of the compounds of any desired stereoisomer makes this method very attractive. The serine β-lactones required for the synthesis are formed using modified Mitsunobu conditions (Ph_3P/dimethyl azodicarboxylate) without racemization and in yields of 60–70%.[29,77]

Scheme 8 Lanthionines with High Optical Purity via Ring Opening of Serine β-Lactones with Cysteines[29]



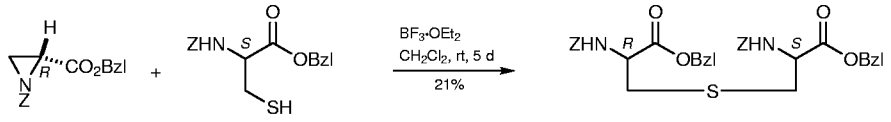
(2*R*,6*R*)-*N'*-Benzyloxycarbonyl-*N*-*tert*-butyloxycarbonyllanthionine *O*-Methyl Ester (22):[28]
To a stirred soln of Z-L-Ser-β-lactone (**20**; 1.0 g, 4.5 mmol) and Boc-L-Cys-OMe (**21**; 1.0 g, 4.3 mmol) in dried DMF (15 mL) under N_2 was added Cs_2CO_3 (0.7 g; 2.2 mmol). The mixture was stirred for 4 h. After removal of the solvent under reduced pressure, the residue was dissolved in EtOAc (100 mL), washed with aq $NaHSO_4$ (pH 4), and brine. The organic layer was dried ($MgSO_4$), concentrated under reduced

pressure, and the product **22** was obtained by chromatography (silica gel, $CHCl_3$/AcOH 100:0 to 95:5) to give **22** as an oil; yield: 0.97 g (50%); $[\alpha]_D^{25}$ −20.8 (*c* 0.48, MeOH).

6.2.2.1.2.4 Ring Opening of Aziridines

A further ring-opening reaction which results in the formation of lanthionines occurs via nucleophilic attack of the cysteine sulfur on aziridines in the presence of $BF_3 \cdot OEt_2$ (Scheme 9).[30] This method was originally developed for a variety of different alkyl thiols, resulting in yields of 50–90% for the synthesis of methionine analogues. However, the use of this reaction in the formation of symmetrically Z-protected lanthionines results in low yields varying from 12–37% and reaction times are in the range of 2–5 days. The aziridines are prepared from the corresponding serine and threonine derivatives[30] via intramolecular nucleophilic substitution (Section 9.2.1).[44–46] The ring opening of aziridines in phosphate buffer has also been reported.[47] However, yields were also rather low (<30%), due to the preferred formation of 2-amino-3-[(2-amino-1-carboxypropyl)thio]propanoic acid.

Scheme 9 Ring-Opening Reaction of Aziridines with Cysteines Yielding Lanthionines[30]



23 **24** **25**

(2*S*,6*R*)-*N*,*N*′-Bis(benzyloxycarbonyl)lanthionine *O*,*O*′-Dibenzyl Ester (25):[30]
Z-L-Cys-OBzl (**24**; 345 mg, 1 mmol) and $BF_3 \cdot OEt_2$ (75 µl) was added to a soln of benzyl *N*-benzyloxycarbonyl-D-aziridine-2-carboxylate (**23**; 311 mg, 1 mmol) in CH_2Cl_2 (5 mL) at rt. After 5 d, the solvent was removed and the residue dissolved in EtOAc. The organic phase was washed with 1 M $NaHCO_3$, dried (Na_2SO_4), and the solvent removed under reduced pressure. The crude product was purified by chromatography (silica gel, $CHCl_3$) and recrystallized (EtOAc/Et_2O/hexane) to give **25**; yield: 149 mg (21%); mp 86–87 °C.

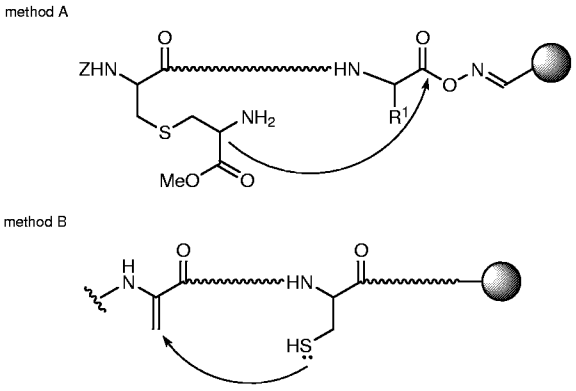
6.2.2.1.3 Synthesis of Lanthionine Peptides with Lanthionine Derivatives

The methods presented in the sections above are all suitable for the synthesis of lanthionines as amino acid derivatives. However, a very important synthetic goal remains the synthesis of lanthionine-containing peptides, for example the synthesis of lantibiotics and other peptide analogues where lanthionine has replaced cystine. In the following Sections some syntheses of lanthionine-containing peptides comprising analogues of cystine peptides, such as enkephalin, somatostatin, and oxytocin are described.

Due to the vast numbers and rapidity of novel developments in solid-phase synthesis over the past ten years, a number of reports currently found in the literature deal with solid-phase syntheses of lanthionine peptides. There are at least two different approaches to synthesize lanthionine peptides in which the sulfide bond links amino acid halves that are not direct neighbors within the peptide chain (Scheme 10). One obvious approach, method A, is based on the coupling of a preformed, orthogonally protected lanthionine monomer to the N-terminus of a peptide oxime resin.[48] This is then followed by acid-catalyzed cyclization and simultaneous release from the resin during amide bond formation with the C-terminal carboxy group via the peptide cyclization method on oxime resin (see Section 6.7.3.2.2). The alternative approach is lanthionine formation after peptide synthesis from amino acid derivatives, such as serine and cysteine (method B).

for references see p 211

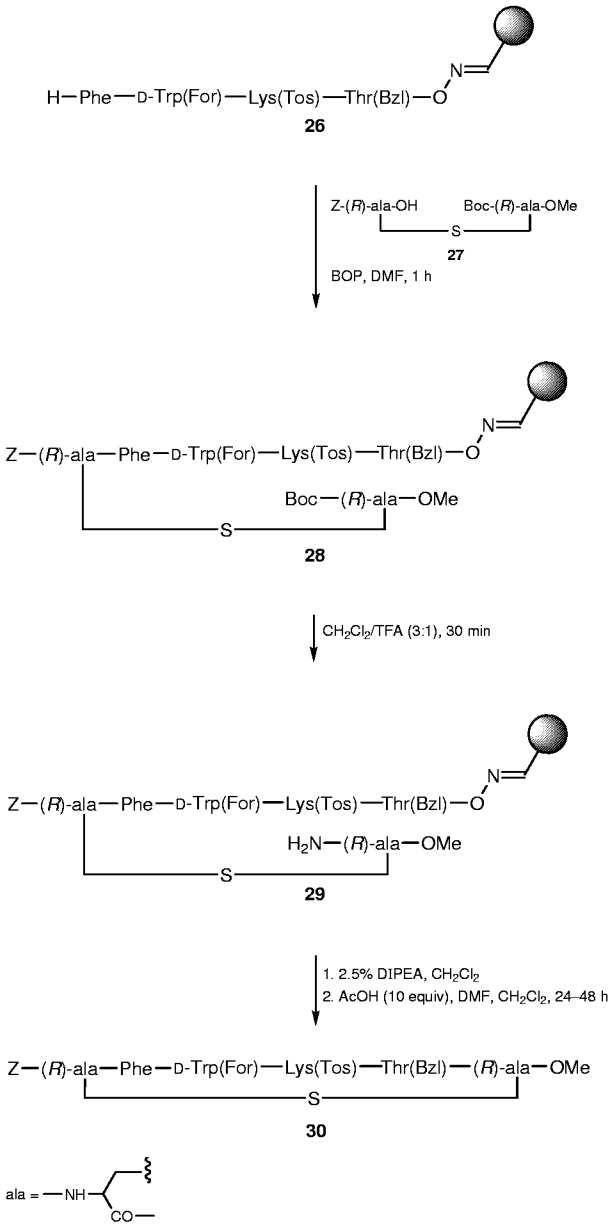
Scheme 10 Solid-Phase Macrocyclization Approaches for Lanthionine Peptide Synthesis; (A) Macrocyclization from Preformed Lanthionine Derivatives; (B) Macrocyclization from Precursors



6.2.2.1.3.1 Macrocyclization with Selectively Protected Lanthionine Derivatives

The synthetic approach for the synthesis of lanthionine peptides using preformed, selectively protected lanthionines has been pursued both on solid phase[48] and in solution, as in the case of the nisin ring A synthesis and of sandostatin analogues.[49,78] The synthesis of lanthionine peptides utilising peptide cyclization and simultaneous cleavage from the Kaiser oxime resin[50,51] was demonstrated with peptides containing 2–6 amino acid residues and using a Boc-based protection strategy.[48] *N*-Z-*N*′-Boc-lanthionine monomethyl ester was used as a building block in the synthesis, with the cyclization taking place in the last step (Scheme 11). In a further report this work was extended to the synthesis of a somatostatin analogue (a tetradecapeptide hormone), as a means of improving the peptide's biological half-life.[52] The peptide segment 3–14 was synthesized on a solid support with coupling of *N*-Boc-*N*′-Fmoc-lanthionine monomethyl ester as the last step. After cyclizing cleavage from the oxime resin, a 2+12 segment condensation was carried out in solution to obtain, on deprotection, the 14-mer Lan-somatostatin analogue in yields of 41% with respect of the resin loading and an excellent purity of 94%. The orthogonally protected lanthionine was synthesized from a suitably protected serine β-lactone.[28,78] The oxime resin mediated cyclization method has also been used for the synthesis of Lan-sandostatin analogues;[53] however, here epimerization occurs, perhaps due to the strongly basic deprotection conditions used to obtain the free peptide.

Scheme 11 Synthesis of a Lanthionine Peptide via Solid-Phase Peptide Synthesis and Final Cleavage from Kaiser Oxime Resin[48]



Lanthionine Peptide 30:[48]

The peptide H-Phe-D-Trp(For)-Lys(Tos)-Thr(Bzl)-OH was synthesized on an oxime-polystyrol resin (resin loading: 0.2–0.3 mmol·g^{-1}) using Boc-based peptide synthesis with BOP coupling and TFA/CH_2Cl_2 cleavage to yield **26**. *N*-Z-*N'*-Boc-lanthionine monomethyl ester (**27**) was then coupled to the resin (3 equiv BOP/DMF, 1 h) to give **28**. The Boc group was cleaved with TFA/CH_2Cl_2 (1:3, 30 min) followed by extensive washing with DMF. The cyclization of **29** was carried out in DMF/CH_2Cl_2 (2:1) with DIPEA (2.5% v/v) for neutralization and an addition of AcOH (10 equiv based on the resin loading). After 24 h, the mixture was filtered and the resin was washed with DMF. The combined filtrates were concentrated and the product **30** precipitated from DMF with H_2O; yield: 65%.

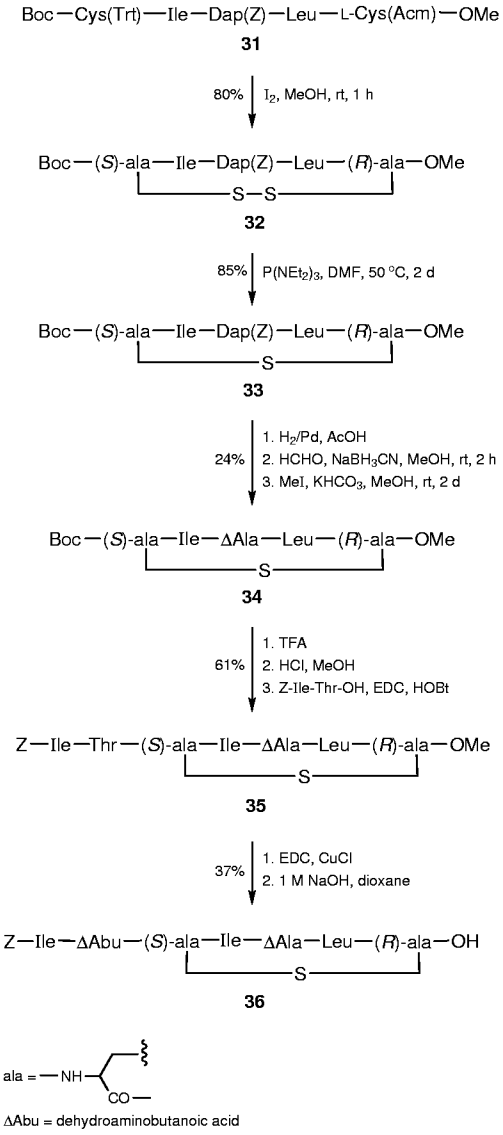
for references see p 211

6.2.2.1.4 Synthesis of Lanthionine Peptides by Desulfurization of Cystine Peptides

The synthesis of lanthionine peptides by the desulfurization of cystine peptides is exemplified by the synthesis of the nisin ring A (Scheme 2).[24] Due to the presence of two dehydroamino acids, the synthesis of ring A, together with the synthesis of rings D and E, was a considerable achievement. These syntheses were carried out in solution. A solid-phase based synthesis of methyllanthionine peptides using the desulfurization reaction has also been achieved.[54]

In the optimized synthesis reported for ring A fragment,[24] the protected pentapeptide Boc-D-Cys(Trt)-Ile-Dap(Z)-Leu-L-Cys(Acm)-OMe **31** was synthesized using *N*-hydroxysuccinimide ester couplings (Scheme 12). The protected cyclic disulfide-bridged peptide methyl ester **32** was prepared by applying the iodine-induced oxidative Trt/Acm-deprotection of the cysteine residues, the N-terminal residue of which was the D-isomer. Desulfurization with $P(NEt_2)_3$ leads to the *meso*-lanthionine containing peptide. In the final steps, two dehydroamino acids had to be introduced. The first dehydroalanine residue was obtained from the diaminopropanoic acid (Dap) residue, which was deprotected and converted into ΔAla by Hofmann degradation yielding **34**. Subsequent removal of the N^{α}-Boc protection and condensation with Z-Ile-Thr-OH produced the N^{α}-Z-protected N-terminal dipeptide **35**.[68] Generation of the second dehydrated residue (ΔAbu) could then proceed and resulted from EDC/CuCl-mediated dehydration of the Thr residue at position 2 of the peptide. In the final step, the methyl ester was saponified to yield the protected peptide acid **36** serving as segment I for the fragment condensation steps in the total synthesis of nisin (**3**) (Section 6.2.2.3).[68]

Scheme 12 Synthesis of Ring A Segment of Nisin which Contains Dehydroamino Acids and *meso*-Lanthionine[24,68]



Nisin Segment I, Ring A (33):[24]
$P(NEt_2)_3$ (1.26 g, 5.1 mmol) in anhyd DMF (20 mL) was added to a soln of the disulfide-bridged peptide **32** (200 mg, 0.255 mmol) in anhyd DMF (80 mL). The mixture was stirred for 2 d at 50 °C and then concentrated. The residue was dissolved in EtOAc and washed with citric acid (10%) and brine. The organic layer was dried ($MgSO_4$) and the solvent removed under reduced pressure. The oily residue was crystallized by trituration with hexane to give **33**; yield: 164 mg (85%); mp 226.5–227 °C; $[\alpha]_D^{26}$ –47.6 (*c* 1.0, DMF).

6.2.2.1.5 Synthesis of Lanthionine Peptides by Macrocyclization of Precursors

The principal disadvantage of the intramolecular cyclization approach using Michael addition of a cysteine-based thiol to dehydroalanine is its nonstereoselectivity. Similarly, even when using 3-bromoalanine derivatives in a halide-based substitution reaction with cysteine-

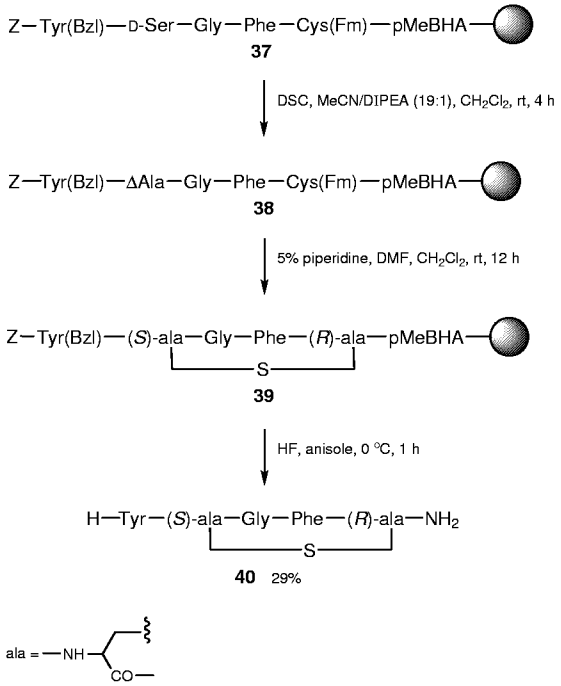
for references see p 211

derived thiols, the formation of dehydroalanine prior to the subsequent cyclization step cannot be fully excluded.[55] Perhaps somewhat surprisingly then, the syntheses presented below which follow these synthetic approaches are reported to succeed in yielding the desired stereochemistry of the lanthionine, presumably due to sequence-dependent steric factors. In contrast, the oxime resin mediated cyclization circumvents the stereochemical problem by using a preformed and orthogonally protected lanthionine derivative of the desired final stereochemistry. One major advantage of the macrocyclization on solid support is the prevention of potential dimerization and cyclodimerization. A disadvantage of the oxime resin mediated cyclization using the *N*-Z-*N'*-Boc-lanthionine monomethyl ester is that, at least in some cases, exocyclic residues have to be attached in a classical solution-phase coupling strategy that adds an additional purification step to the synthesis. In any case, regardless of the synthetic approach chosen, cyclization on a solid support appears to provide peptides in high purity.

6.2.2.1.5.1 Macrocyclization by the Dehydroalanine/Cysteine Method

Using the Michael addition approach, the solid-phase macrocyclization of a cyclic lanthionine-containing 5mer-enkephalin analogue **38** prepared from Z-Tyr(Bzl)-D-Ser-Gly-Phe-Cys(Fm)-pMeBHA-resin **37** has been achieved as shown in Scheme 13.[27] After synthesis of the linear peptide **37** using Boc-based solid-phase peptide synthesis, the unprotected serine was converted into 2,3-dehydroalanine using *N,N'*-disuccinimidyl carbonate to give **38**.[38] The subsequent cyclization to **39** occurred upon deprotection of the cysteine residue, and cleavage from solid support and purification by HPLC gave the desired product **40** in 29% yield. The relatively low overall yield results from the incomplete conversion of serine into dehydroalanine (48%). Although Michael additions are not stereoselective, only one diastereomer was obtained (D,L-Lan), which may be an exception, possibly favored by steric effects and dependent on the peptide sequence. In addition, the high stereoselectivity of Michael addition type reactions was reported for the synthesis of an epidermin B-ring analogue[79] and subtilin rings A, B, and E.[80] From two possible regioisomers of the subtilin A-ring product, the naturally occurring regioisomer is formed. However, formation of diastereomers is observed in a 3:1 ratio favoring the desired diastereomer. These results show that in many cases, even for small hexa- or heptapeptides, the naturally occurring stereoisomers of lanthionine are formed and that solution-phase or solid-phase strategies render the same results. As a consequence, it has to be noted, that in terms of diastereoselectivity the Michael addition approach might also fail in several cases.[76] More recently, *Se*-phenylselenocysteine residues were shown to present most useful precursors of dehydroalanine for the synthesis of lanthionine peptides[81,82] (see Section 6.8.3.4).

Scheme 13 Solid-Phase Peptide Synthesis of an Enkephalin Analogue by Macrocyclization of Cysteine and Dehydroalanine[27]



Enkephalin Analogue 40:[27]
The resin-bound peptide **37** (3 g, 0.36 mmol·g^{-1}) was suspended and swollen in CH_2Cl_2 (20 mL). A soln of DSC (387 mg, 1.51 mmol) in MeCN (10 mL) was added followed by a soln of DIPEA in CH_2Cl_2 (5.22 mL, 1.5 mmol). After 4 h shaking at rt under N_2, the mixture was filtered and the resin **38** was washed with CH_2Cl_2 (4 ×). The resin was treated with a soln of piperidine/DMF (1:4, 20 mL, 2 × 50 min) and shaken in piperidine/CH_2Cl_2/DMF (0.1:1:1, 40 mL) for 12 h. The resin **39** was filtered off, washed with DMF (1 ×), CH_2Cl_2 (2 ×), and EtOH (2 ×), and dried; yield: 3.7 g. For cleavage of the product from the resin, the peptidyl resin **39** (3.7 g) was treated with anhyd HF (74 mL) at 0 °C in the presence of anisole (3.7 mL) in a Teflon HF apparatus. After removal of volatile components, the remaining material was washed with EtOAc (2 × 20 mL) and the product was extracted with AcOH followed by extraction with AcOH/H_2O (1:9). The combined fractions were lyophilized and purified by preparative HPLC (H_2O/MeCN/TFA) and gel-permeation chromatography (Sephadex G-15) to give **40**; yield: 155 mg (29.1%).

6.2.2.1.5.2 Macrocyclization by the 3-Bromoalanine/Cysteine Method

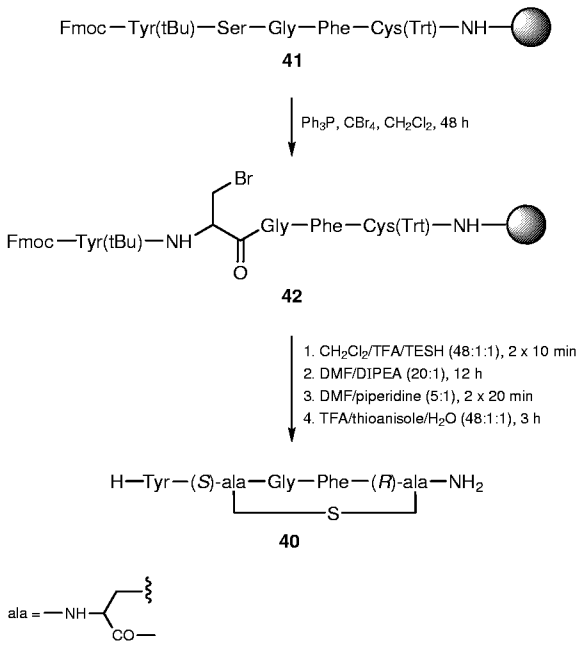
Another elegant macrocyclization synthesis of lanthionine-containing 5-mer and 8-mer peptides has been reported using Fmoc-based solid-phase synthesis on Rink amide resin as shown in Scheme 14.[55] In this case, intramolecular thioether formation could be achieved using 3-bromoalanine and cysteine. In the first step, the linear peptide sequence was assembled on the solid support to give **41**. Then the serine residue, which had been incorporated in the unprotected form during peptide synthesis, was converted into 3-bromoalanine with Ph_3P/CBr_4 to yield **42**. The side-chain protecting group of cysteine (Trt) was selectively removed with TFA/TESH and subsequent cyclization was carried out under basic conditions (DIPEA/DMF). Finally, the N-terminal Fmoc-protecting group was removed with piperidine and the peptide **40** was cleaved from the solid support in high purity (>95%). However, the mechanism of the thioether formation remains unclear; it is thought that under the chosen conditions the reaction proceeds via β-elimination prior to the cyclization step, as opposed to direct substitution of the halide. This was demonstrated with the use of D- and L-

for references see p 211

serine in the synthesis, both of which gave stereochemically identical lanthionine peptides with the *S*-configuration of the N-terminal lanthionine α-carbon. This route yields the same target compound as Scheme 13.

The use of brominated amino acids for cyclization with cysteine derivatives leads to so called cystathione or monocarba analogues of peptides, in which the disulfide bridge is replaced by an isosteric -CH_2–S- linkage (Section 6.2.4.2.1). This has been demonstrated by the solid-phase synthesis of a 9-residue peptide monocarba analogue of oxytocin, which was synthesized via macrocyclization of γ-bromo-α-aminobutanoic acid with cysteine.[56] The peptide was obtained in 45% yield, with respect to the initial loading of the resin. The bromo derivative was synthesized from Fmoc-protected L-homoserine lactone with HBr/AcOH (rt, 1 h) in high yields (70%).[57]

Scheme 14 Solid-Phase Peptide Synthesis of an Enkephalin Analogue by Macrocyclization of Cysteine and 3-Bromoalanine[55]



6.2.2.2 Synthesis of 3-Methyllanthionine Peptides

With respect to synthetic methods, the most thoroughly investigated lanthionine from the group of alkylated lanthionines is *threo*-3-methyllanthionine (**2**), a lanthionine which is also found in a number of natural peptides such as nisin[12] and epidermin.[4] In this section, some of the synthetic methods are compared and discussed with respect to their relative applicability and their possible extension to other alkylated lanthionines.

6.2.2.2.1 Synthesis of 3-Methyllanthionine

Whilst lanthionine is commercially available from a number of sources, 3-methyllanthionine is generally difficult to obtain; indeed, literature for the synthesis of unprotected MeLan is also scarce. One such procedure is the synthesis of MeLan from L-cysteine and (4*Z*)-4-ethylidene-2-phenyl-1,3-oxazol-5(4*H*)-one.[58]

The synthesis of selectively protected monoalkyl lanthionines, or otherwise alkylated and/or nonsymmetrically substituted lanthionines is made all the more difficult due to the introduction of additional stereocenters. For example, for the monoalkylated derivative 3-methyllanthionine, eight different stereoisomers are possible; obviously, similar problems arise for the stereochemistry of other alkylated lanthionines.

Despite this problem, the synthesis of 3-methyllanthionines can be achieved by different methods, most of which are analogous to the methods for the synthesis of lanthionine presented in previous sections. Using threonine instead of serine as a precursor the stereochemical aspects become analogous to those of the above mentioned lanthionine syntheses. However, limitations in the method of synthesis may well arise from the limited availability of potential precursors. Various 2,3-dehydroamino acid derivatives suitable for the synthesis of diversely substituted methyllanthionines via Michael addition have also been reported.[59] The respective 2,3-dehydroamino acids were synthesized from side-chain alkylated cysteine analogues in yields of 80–90%. The reagent methyl fluorosulfonate in $CHCl_3$ was used to generate the sulfonium ion which then undergoes elimination via base catalysis.

(2*RS*, 3*RS*, 6*R*)-3-Methyllanthionine:[58]
A suspension of L-cysteine (12 g, 0.1 mol) and (4*Z*)-4-ethylidene-2-phenyl-1,3-oxazol-5(4*H*)-one (22 g, 0.12 mol) in THF (100 mL) and H_2O (200 mL) was shaken under N_2 until the disappearance of any thiol reactivity. Following removal of THF, concd HCl (220 mL) was added and the mixture was refluxed for 6 h. The mixture was cooled, the benzoic acid filtered off, and the filtrate concentrated under reduced pressure to dryness. The residue was dissolved in MeOH (50 mL) and the 3-methyllanthionine was precipitated by the addition of a mixture of pyridine (15 mL) and $CHCl_3$ (30 mL). The title compound was filtered off and washed with $CHCl_3$; yield: 12 g (55%); mp 224 °C (dec 205 °C); $[\alpha]_D^{20}$ +27.4 (*c* 0.475, 1M HCl).

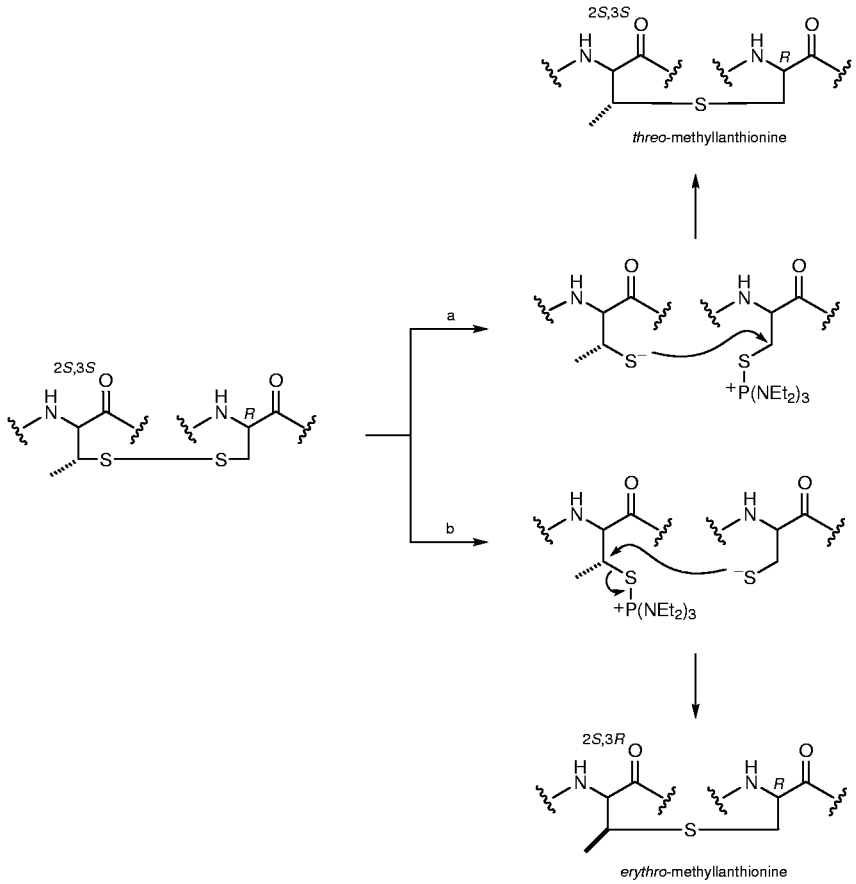
6.2.2.2.2 Desulfurization of 3-Methylcystine Peptides

The sulfur extrusion method[21] has been extensively investigated for the synthesis of *threo*-3-methyllanthionine;[25] however, in this case desulfurization was not used to generate the amino acid (either in free or protected form), but rather to form different sulfide-bridged cyclic peptides as precursors for the total synthesis of nisin (**3**). Development of synthetic methods in this area was particularly important in respect to the synthesis of nisin, since this lantibiotic contains four *threo*-3-methyllanthionine residues. The synthesis proceeded from a protected *threo*-3-methyl-D-cysteine in the N-terminal position and L-cysteine in the C-terminal position.

It is interesting to note that the desulfurization reaction of the disulfide intermediate formed using these building blocks could theoretically lead to the generation of two different diastereomers (Scheme 15). However, during solution-phase synthesis of various nisin rings, generation of the undesired *erythro*-3-methyllanthionine configuration has not been observed. Although the reasons for this stereospecificity are not fully understood, it has been suggested that it is probably due to steric reasons, and/or to an electron-donating effect of the methyl group which stabilizes the thiolate anion of 3-methyl-D-cysteine in favor of cysteine.[14,25] As mentioned in Section 6.2.2.1.1 the formation of byproducts is observed when the synthesis is carried out in less dry solvents.[24]

for references see p 211

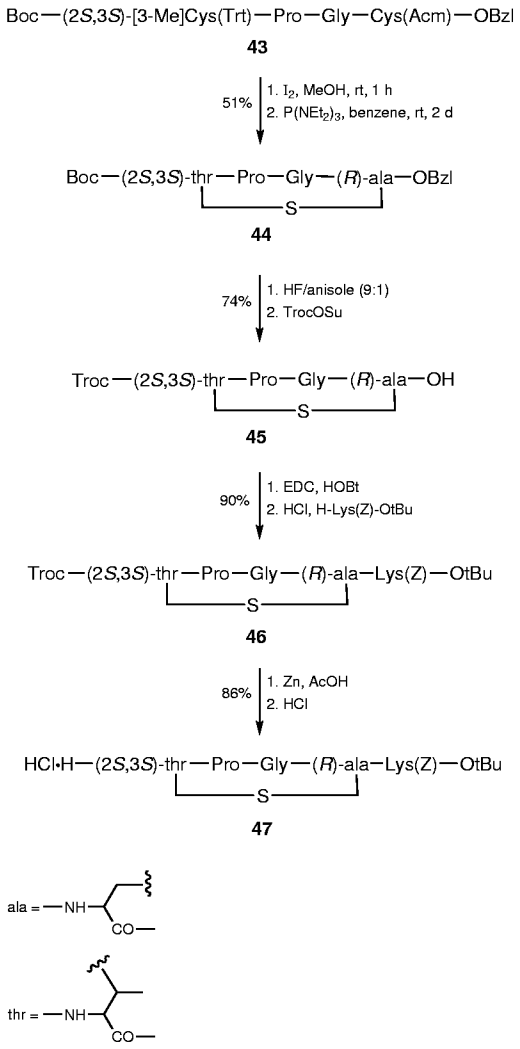
Scheme 15 Methyllanthionine Formation via Desulfurization with Tris(diethylamino)phosphine[25]a



[a]Only the *threo*-isomer was isolated after this reaction.

The synthesis of rings B and C of nisin (segments II and III, Scheme 2) are examples of the synthesis of methyllanthionine peptides by desulfurization. For ring B, the peptide Boc-D-[3-Me]Cys(Trt)-Pro-Gly-Cys(Acm)-OMe (**43**) was prepared with *threo*-D-3-methylcysteine ([3-Me]Cys) and Cys flanking the termini of the peptide (Scheme 16).[25] Oxidation to the disulfide was carried out using iodine and then desulfurization gave **44**. Following exchange of the N^{α}-Boc protection in the tetrapeptide fragment **44** for Troc, H-Lys(Z)-OtBu was coupled to give **46**. Removal of the N^{α}-Troc group completed the preparation of the protected pentapeptide ester of segment II (**47**), suitable for further segment condensation.[68]

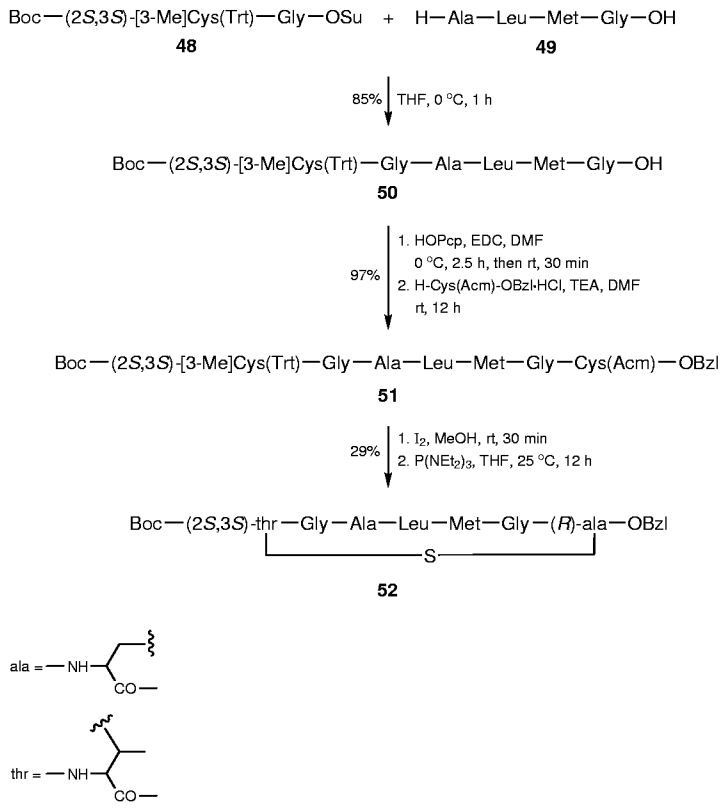
Scheme 16 Synthesis of Segment II of Nisin[25]



As with ring B, the synthesis of the ring-C-containing heptapeptide (segment III, Scheme 2) was dependent on the formation of a MeLan residue, the starting amino acid for which was Boc-*threo*-methylcysteine.[60] As shown in Scheme 17, the heptapeptide **51** was assembled in two stages, first by fragment condensation of a tetrapeptide **49** with the dipeptide **48** to yield **50** followed by installation of the C-terminally positioned H-Cys(Acm)-OBzl. The protected, disulfide-bridged peptide was again formed through oxidative deprotection of **51** with iodine and the respective thioether **52** obtained via desulfurization with $P(NEt_2)_3$ in yields of 29%. To serve in the final segment condensation in the total synthesis of nisin, **52** was debenzylated, the N-terminal Boc protection exchanged for Troc, and the resulting acid condensed with H-Asn(Mbh)-Met-OtBu followed by removal of the Troc protection.[68]

for references see p 211

Scheme 17 Synthesis of Ring C Precursor of Segment III of Nisin[60]



One problem encountered in the solution-phase synthesis of the C ring of nisin is that, during desulfurization, a large amount of dimeric compound is formed, even with a large excess of $P(NEt_2)_3$.[61] This problem can be alleviated through desulfurization of the fragment on solid phase, although such procedures may lead to partial formation of the undesired diastereomer, a problem not encountered during solution-phase synthesis.[60] In order to synthesize this fragment on solid phase, the C-terminal 4-oxymethylphenylacetic acid ester of **51** was assembled (Scheme 17).[61] Following oxidative deprotection of *threo*-D-[3-Me]Cys and L-Cys residues to form the disulfide, the cyclic heptapeptide was attached to aminomethylpolystyrene resin. The desulfurization reaction was then carried out using $P(NEt_2)_3$ in tetrahydrofuran.

The preparation of the required *threo*-3-methylcysteine component in this synthesis, used the stereoselective ring opening of aziridines.[62,63] It was shown to be possible to maintain stereospecificity of the reaction using the aziridine ring as an intermediate.

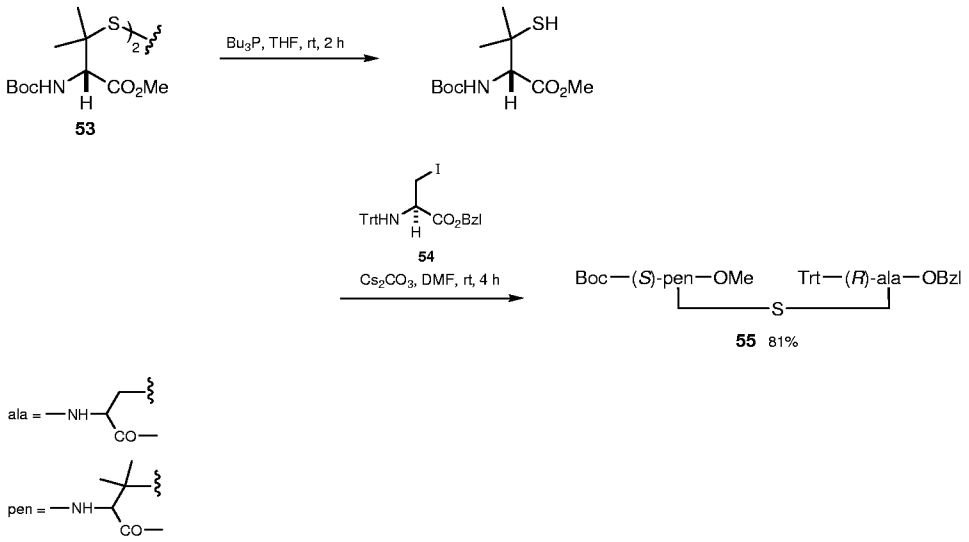
6.2.2.2.3 Nucleophilic Substitution of Triphenylmethyl-Protected 3-Iodoalanine

The synthesis of four out of eight possible stereoisomers of 3-methyllanthionine [(2*S*,3*S*,6*R*), (2*S*,3*R*,6*R*), (2*S*,3*S*,6*S*), (2*S*,3*R*,6*S*)] has been achieved using the reaction of Z-protected 3-methyl-D-cysteine with D- or L-3-chloroalanine in yields of 35–53%.[64] The methyl-D-cysteine stereoisomers were obtained by two routes. Firstly from (2*S*,3*S*)-threonine via O-tosylation and subsequent inversion of configuration by nucleophilic attack with thiobenzoic acid. The resulting derivative was debenzoylated and oxidized to the respective cystine derivative prior to the reduction with Zn/HCl to give the *erythro*-3-methyl-D-cysteine

(2*S*,3*R*). Secondly, the *threo*-3-methyl-D-cysteine (2*S*,3*S*) was prepared using the previously described addition of thiocarboxylic acid to threonine-derived aziridines.[62]

The use of Trt-protected 3-iodoalanines for lanthionine synthesis is also a highly promising method for the synthesis of 3-methyl- and 3,3-dimethyllanthionines.[40] This method is based on the use of *N*-trityl-3-iodoalanine benzyl ester (**54**) and the symmetrically protected bis(Boc)-cystine-derivative dimethyl esters derived from L-*threo*-3-methylcysteine and D-penicillamine. Yields of the respective lanthionine derivatives are >80%; however, enantiomeric excesses have not been determined for the 3-substituted lanthionines (Scheme 18).

Scheme 18 Synthesis of a Substituted Lanthionine by Nucleophilic Substitution of Trt-Protected 3-Iodoalanine[40]



3,3′-Dimethyllanthionine Derivative 55:[40]

N,*N*′-Bis(Boc)-penicillamine disulfide dimethyl ester (**53**; 262 mg, 0.5 mmol) was stirred with Bu_3P (125 µl, 0.5 mmol) in THF for 15 min at rt. The solvent was removed under reduced pressure, the residue was redissolved in EtOAc, and washed with citric acid (10%) and brine. The solvent was again removed under reduced pressure to yield the thiol. To a soln of the thiol and *N*-trityl-3-iodoalanine benzyl ester (**54**; 547 mg, 1 mmol) in dry DMF was added Cs_2CO_3 (326 mg, 1 mmol) and the mixture was stirred for 4 h at rt. After dilution with iPr_2O, the soln was washed with citric acid (10%) and the aqueous layer was then extracted with iPr_2O. The organic layers were combined and washed with citric acid (10%) and brine. After removal of the solvent, the residue was purified by flash chromatography (silica gel, EtOAc/hexane 1:9) to give **55**; yield: 548 mg (81%); $[\alpha]_D^{20}$ +7 (*c* 1.0, EtOAc).

6.2.2.2.4 Ring Opening of Serine β-Lactone with Penicillamine Derivatives

Although, the synthesis of 3-methyllanthionine has not been achieved by ring opening of Z-protected serine β-lactones, the synthesis of dialkylated derivatives has been carried out with *N*-Boc-penicillamine methyl ester and *N*-Boc-β,β-pentamethylene cysteine methyl ester.[28] Of the two possibilities for the ring-opening reaction, (i.e., thioether or thiol ester formation), that can result during the synthesis of lanthionine analogues, thioether formation is clearly favored in the case of the 3-alkylated analogues. This becomes apparent from the yields generally obtained for the penicillamine derivative, 78 and 84%, or 92% for the pentamethylene cysteine derivative compared to unsubstituted lanthionines (50% yield). Steric effects have been suggested to explain the high regioselectivity of the sulfide-bridge

for references see p 211

forming reaction when using β-substituted cysteines. In the model developed, the access of the tertiary thiolate anion to the carbonyl carbon of the β-lactone is blocked. β,β-Dimethylcyclolanthionine dipeptides[65] have also been synthesized using the previously described methods developed for peptide cyclization on oxime resin.[48]

6.2.2.2.5 Ring Opening of Methylaziridines

Analogously to the synthesis of lanthionines by the ring opening of aziridines (Section 6.2.2.1.2.4), this method has also been investigated for the synthesis of 3-methyllanthionines.[30,44] The synthesis involves ring opening of threonine-derived methyl aziridines by Z-L-Cys-OBzl in the presence of $BF_3 \cdot OEt_2$, but yields were extremely low (12 and 22%). The precursor *threo*-3-methyl-D-cysteine was synthesized via the ring opening of (2*R*,3*R*)-1-*tert*-butoxycarbonyl-3-methylaziridinecarboxamide derived from D-threonine in a three-step process and which retained the appropriate stereochemistry.[62,63]

6.2.2.3 Synthesis of Multiple Lanthionine and 3-Methyllanthionine Peptides

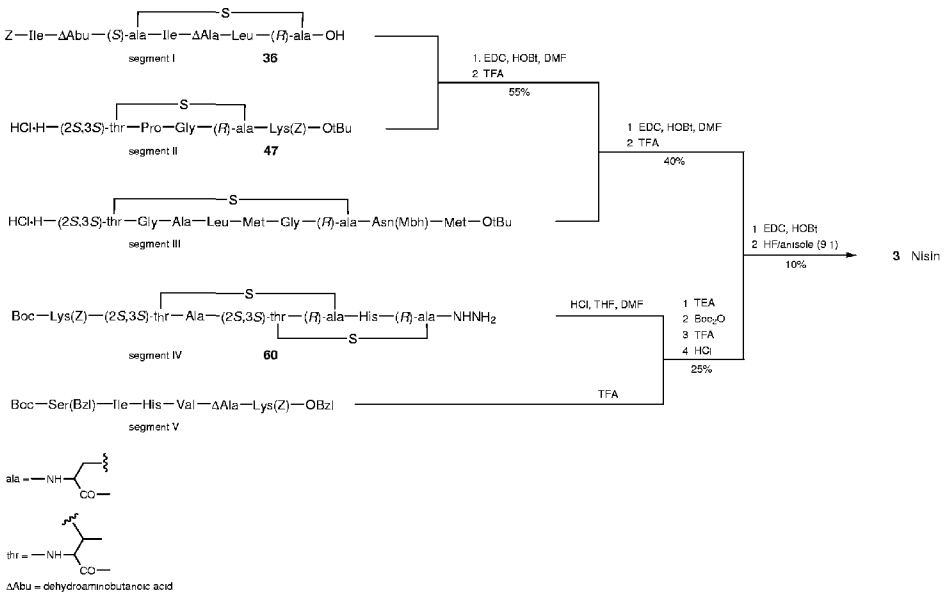
Nisin, because of its continued use as a preservative for more than 30 years,[66] became an obvious target for a total synthesis.[13] To date, it remains the only naturally occurring lanthionine peptide for which this feat has been achieved. In contrast to the lanthionine peptide syntheses of most other approaches already discussed, the total synthesis of nisin is based on a solution-synthesis strategy. Whilst, the total synthesis of the pentacyclic, 34-residue peptide nisin also demonstrated that such syntheses are not a commercially viable option for large amounts of lantibiotics, they have opened the way to useful synthetic methods for the synthesis of novel conformationally constrained peptides and peptidomimetic compounds.

Nisin (**3**) consists of five rings, three single rings A, B, C, and two overlapping rings D and E, and in addition to Lan and MeLan several α,β-unsaturated amino acids are present arising from dehydration of the corresponding 3-hydroxy α-amino acids together with a single Ser residue that is not dehydrated (Scheme 2). The synthesis of lanthionine-containing cyclic structures as precursors for the total synthesis of a lantibiotic is dependent on at least two different factors. The first is the synthesis of the cyclic structures Lan and MeLan, whilst the second is the synthesis of additional modified residues, for example dehydroalanine or dehydroaminobutanoic acid, which must not be allowed to react further during the additional reactions required for the final assembly of the peptide.

Despite the early work by Photaki et al.[49] suggesting that ring A can be synthesized using selectively protected lanthionine derivatives, Shiba et al.[13,14] have found that the use of the sulfur extrusion of preformed disulfides with $P(NEt_2)_3$[21,22] provides a facile method to access the thioether cycles. Desulfurizations were performed at high dilution (0.62 mM) to prevent intermolecular dimerization reactions. Hofmann degradation of diaminopropanoic acid[35] is a suitable method for the generation of the α,β-unsaturated amino acid ΔAla (ring A synthesis). Dehydroaminobutanoic acid (ΔAbu) has already been successfully introduced into peptides by the DCC/CuCl-induced dehydration of threonine residues.[36]

Thus, the solution-based synthesis of the complete molecule required development of a number of different synthetic and protecting group strategies for the peptide fragments (Scheme 2), which were then combined via segment condensation to provide the final product. General coupling procedures for peptide synthesis, *N*-hydroxysuccinimide esters, DCC/EDC, or symmetrical anhydride couplings were used. From the four segments I–IV and the additional linear segment V, the total synthesis of nisin proceeded as shown in Scheme 19.[13] Conventional *tert*-butyl and benzyl protection was used throughout, since the dehydroamino acids proved to be sufficiently stable in TFA and HF under anhydrous conditions.

Scheme 19 Total Synthesis of Nisin via Condensation of Five Preformed Segments To Give Biologically Active Nisin[68]



for references see p 211

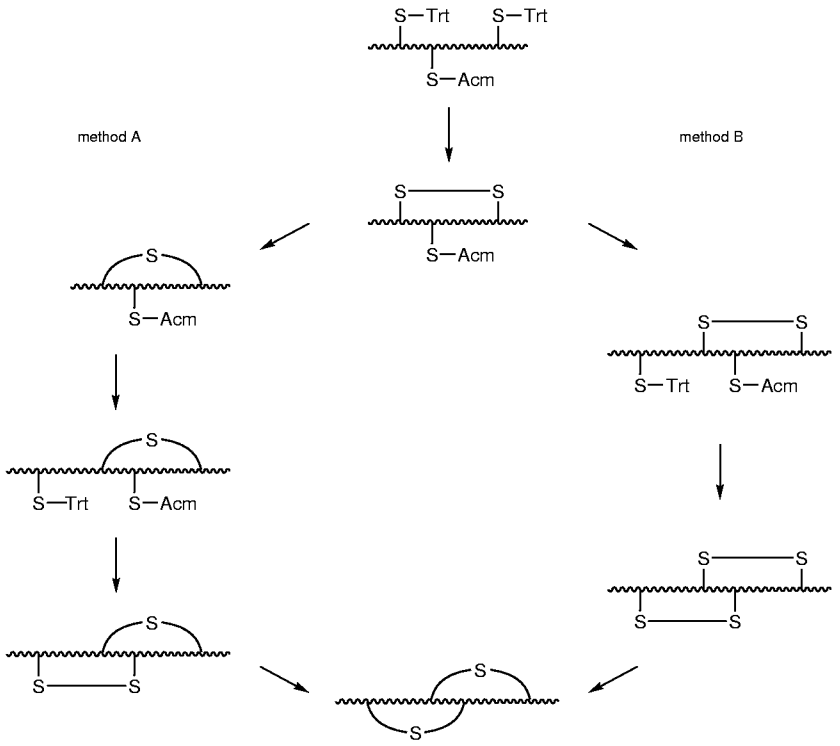
Initially segments I, II, and III were successively condensed, as were segments IV and the linear segment V. The result was two fragments which are identical to the naturally occurring BrCN cleavage products of nisin. The final condensation of these two products then afforded nisin.

The synthetic pentacyclic nisin was compared by analytical techniques including amino acid analysis, mass spectrometry, and NMR and was shown to be identical to natural nisin in all respects. Furthermore, the biological activity of the synthetic nisin exhibited the same activities as native nisin confirming the success of the synthesis.[13,14,67–69]

In addition to the total synthesis of nisin, a large number of related fragments, in some cases with various labels or other functional groups attached, have also been synthesized.[13,14,67,70] Similarly, fragments related to other similar lanthionine-containing peptide antibiotics have been synthesized, many of which may provide substrates for assaying the enzymes isolated from lantibiotic-producing bacteria.[2–8,70–72] In conjunction with natural nisin, these chemically synthesized fragments and precursors have proved invaluable in elucidating both the three-dimensional structure and biological activities of nisin and a number of other lantibiotics. For further information on these aspects, the reader is directed to the host of excellent reviews and books available on this subject.[2–8]

At the end of this section dealing with desulfurization of multiple lanthionine containing peptides, the different synthetic strategies for overlapping rings D and E of nisin (Scheme 2 and 20) are discussed in detail. Since rings D and E form a bicyclic structure composed of two residues of MeLan, the first, and perhaps most important step in the synthesis of the overlapping rings D and E was the organization of ring closures so that the thioether bonds connect the correct amino acid positions (thioether 1: amino acids 23 and 26; thioether 2: amino acids 25 and 28, segment IV, Scheme 2). Two different approaches were pursued,[73] the first (method A) a two-step desulfurization and the second (method B) a one-step desulfurization resulting in simultaneous formation of the two thioethers (Scheme 20). Both peptides, regardless of the procedure used, were identical with regard to mass spectra, HPLC, NMR, and amino acid analyses. However, method B turned out to be the more favorable one, since it involved fewer reaction steps, ease of peptide formation, and provided a better yield after desulfurization.

Scheme 20 Desulfurization Strategies for the Synthesis of Rings D and E (Segment IV) of Nisin

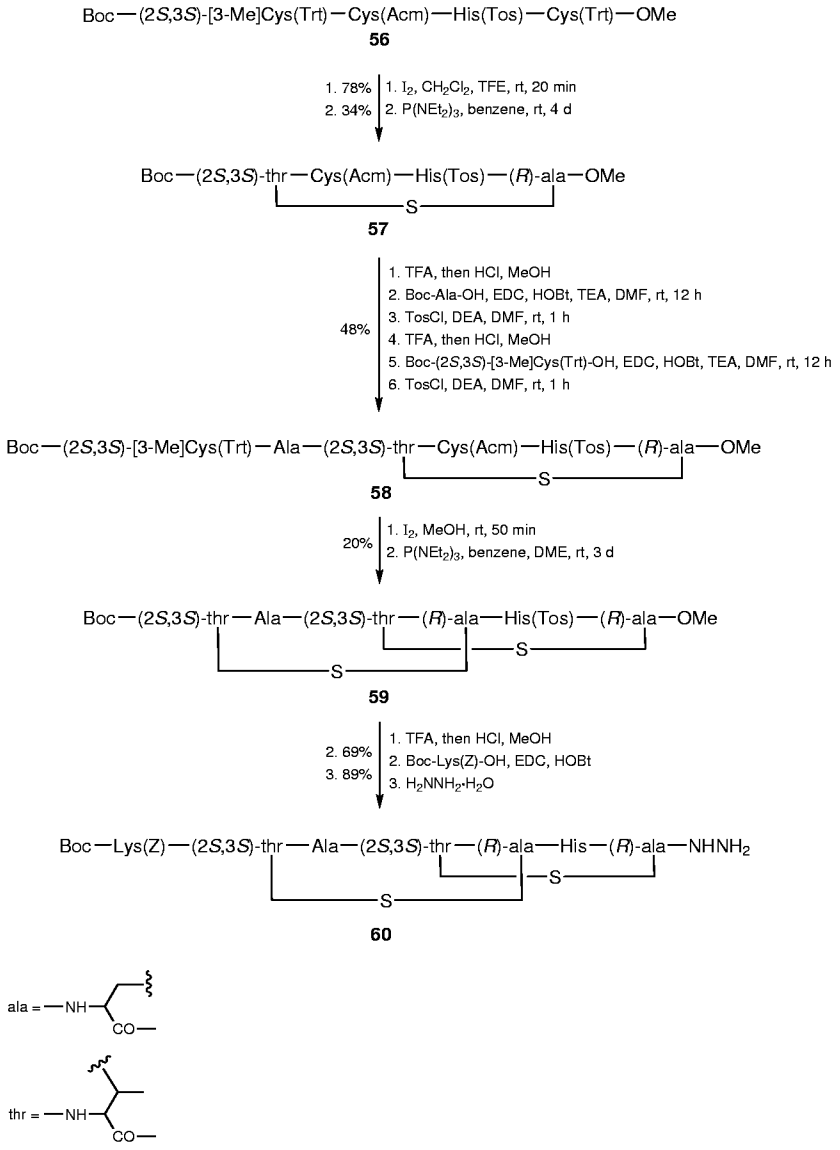


6.2.2.3.1 Stepwise Lanthionine Formation

Obviously for a sequential approach to lanthionine-ring closure, an orthogonal protection strategy is required. Thus, the fully protected tetrapeptide **56** containing one residue each of *S*-Trt-protected *threo*-3-methyl-D-cysteine and L-cysteine, as well as an Acm-protected L-cysteine in the second position, was prepared by standard methods (Scheme 21).[73] The trityl protecting groups were selectively removed and simultaneously oxidized with iodine to afford the first cystine peptide which was then desulfurized to form the lanthionine cycle **57**. The tetrapeptide **57** was then N-terminally elongated with Boc-Ala-OH and Boc-D-[3-Me]Cys(Trt)-OH to yield **58** and oxidation and desulfurization was carried out for a second time to obtain the second lanthionine in the peptide **59**. Following N^{α}-Boc-deprotection and elongation to the heptapeptide with Boc-Lys(Z)-OH, the C-terminal methyl ester was converted into the hydrazide **60** for azide coupling in the segment condensation step.

for references see p 211

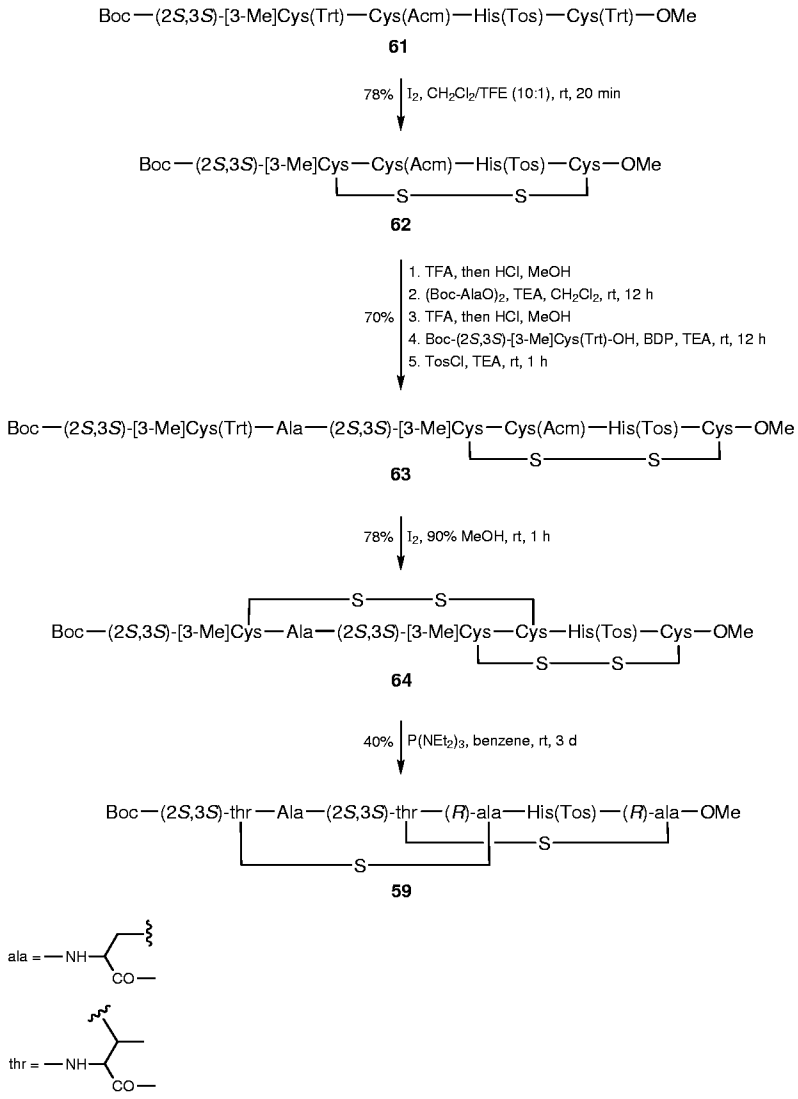
Scheme 21 Stepwise Lanthionine Formation in the Synthesis of Rings D and E Precursor of Segment IV of Nisin[73]



6.2.2.3.2 Desulfurization of Multiple Cystine Peptides

In contrast to the stepwise desulfurization process, the trityl groups of the peptide **61** were selectively removed and simultaneously oxidized using iodine to afford the first cystine **62** (Scheme 22).[73] The tetrapeptide **62** was N-terminally elongated via coupling of Boc-Ala-OH and Boc-D-[3-Me]Cys(Trt)-OH to yield **63**. The trityl and Acm groups of the two residual cysteine residues of **63** were removed and oxidized with iodine to form the second cystine ring in the peptide **64**. This fully protected, bis(disulfide)-bridged hexapeptide was then desulfurized with $P(NEt_2)_3$ in a single-step reaction affording the bis(sulfide)-containing lanthionine peptide **59**.

Scheme 22 Synthesis of Rings D and E of Nisin by One-Pot Desulfurization of Multiple Cystines[73]



Nisin Segment IV Precursor (59):[71]

To a soln of **64** (350 mg, 0.376 mmol) in anhyd benzene (610 mL) was added $P(NEt_2)_3$ (3.70 g, 15.0 mmol). After stirring at rt for 3 d, the solvent was removed under reduced pressure and the residue dissolved in EtOAc and washed with 10% aq citric acid and brine. The organic layer was dried ($MgSO_4$) and then the solvent removed under reduced pressure. The residue was purified by column chromatography (silica gel, $CHCl_3$/MeOH 30:1) and the product **59** recrystallized (EtOAc/hexane); yield: 131 mg (40%); mp 154–158 °C; $[\alpha]_D^{23}$ –40.5 (*c* 0.306, MeOH).

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6.2.3 Selenocystine Peptides

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An increasing number of selenoproteins have been identified and isolated from prokaryotes and eukaryotes. Among these the oxidoreductases glycine-reductase, formate-dehydrogenase, glutathione peroxidase, tetraiodothyronine-5′-deiodinase, and mammalian thioredoxin reductase, contain in their sequence one selenocysteine (SeC) residue that confers unique biochemical properties to the enzymes because of its low pK_a value (5.73 for SeC vs 8.53 for Cys). Other proteins such as selenoprotein P contain larger numbers of SeC residues.[1,2] In terms of ribosome-mediated protein synthesis, selenocysteine can be viewed as the 21st amino acid because nature uses the UGA termination codon to specify the insertion of this selenoamino acid rather than a post-translational modification mechanism.[3] Bioincorporation of selenocysteine in the position of cysteine residues was successfully accomplished in the case of thioredoxin by using cysteine-auxotrophic *E. coli* strains leading to a *per*-SeC-protein variant.[4] Natural translation machinery has also been exploited in phage display experiments to express selenocysteine peptides.[46]

Redox Potential of Selenocysteine

From studies on aliphatic and aromatic diselenides it is known that quantitative reduction to selenols requires a large excess of a strongly reducing dithiol, such as dithiothreitol (DTT),[5] whereas reaction of diselenides with monothiols results in only partial reduction.[6] This was confirmed both in the case of the SeC^{32},SeC^{35}-analogue of thioredoxin[4] and of a selenocystine-containing pentapeptide dimer.[7] A comparative study of the redox potentials of the bis(cysteinyl) active-site fragment [10–17] of glutaredoxin with one or both cysteine residues replaced by selenocysteine[8–10] shows that the redox potential of the bis-SeC-peptide (−381 mV) is significantly more reducing than that of the mixed SeC/Cys-peptide (−326 mV), of DDT (−323 mV),[11] of glutathione (−205 mV),[11] and of the parent Cys,Cys-peptide (−180 mV).[12] These redox potential values clearly suggest that the formation of a diselenide bridge is more favored than that of a mixed selenide/sulfide bridge. This assumption was confirmed in the case of the synthetic [SeC^3,SeC^{11},Nle^7]-endothelin-1 analogue, as oxidative refolding of the fully reduced peptide leads to the quantitative production of the diselenide- and disulfide-bridged endothelin in the correct natural isomeric form.[13,14]

Selenocysteine as an Isomorphous Analogue of Cysteine

Between selenium and sulfur atoms there are slight differences in their electronegativities (2.55 vs 2.58) and covalent radii (1.17 vs 1.02 Å), there are also slight differences in the diselenide and disulfide bond lengths (2.28–2.33 vs 1.8–2.1 Å), dihedral angles (75–88° vs 90°), and energies (46 vs 64 kcal·mol^{-1}) as well as in the C–Se and C–S bond lengths (1.95–1.99 vs 1.82 Å) (for reviews see refs[42–44]). The largely isosteric character of selenocysteine has been exploited in the synthesis of a few bioactive cystine peptides, such as oxytocin,[15–18] somatostatin,[19] α-rat atrial natriuretic peptides,[20] and endothelin-1,[13,14] to obtain analogues that are possibly more stable under reducing conditions. As expected, the bioactivities of the SeC-containing analogues are fully retained. A comparison of the conformational preferences of [SeC^3,SeC^{11},Nle^7]-endothelin-1[13] and [SeC^1,SeC^{11}]-apamin[21] with those of the parent cystine peptides by NMR conformational analysis, fully confirms the isomorphous character of a Cys → SeC replacement even in the absence of the strong conformational restraints that are present in larger globular proteins.

Analytical Characterization of Selenocystine Peptides

Acid hydrolysis under standard conditions (6 M HCl, 110 °C, 24 h) leads to partial decomposition of selenocystine and selenocysteine derivatives, thus making quantification of this amino acid by amino acid analysis difficult. Similarly, acid hydrolysis of *Se*-[2-(4-pyridinyl)ethyl]selenocysteine peptides, obtained by reduction of the selenocystine peptides with $NaBH_4$ and reaction with 4-vinylpyridine, results in partial decomposition. This derivatization, however, is useful for the enantiomeric resolution of the acid hydrolysates by capillary zone electrophoresis by applying host–guest complexation with crown ethers.[10,22]

6.2.3.1 *per*-Selenocystine Peptides

So far, only peptides containing one disulfide bridge have been synthesized as *per*-selenocysteine analogues, because the selenol protecting groups that will allow regioselective crosslinking of selected selenocysteine pairs are missing and the behavior of multiple selenocysteine peptides in oxidative refolding is still unknown.

6.2.3.1.1 Selenocysteine Protection

For the synthesis of selenocysteine derivatives that are suitable for peptide synthesis essentially two approaches have been used to date: (1) conversion of β-chloroalanine[23] or serine-*O*-tosylate derivatives[24] into the desired selenocysteine derivatives by a nucleophilic displacement reaction with an areneselenol; and (2) full reduction of selenocystine and in situ reaction with aryl halides to produce the aryl selenides.[7,25] In this context, reduction of selenocystine in 2 M NaOH with 2-methyl-2-propanethiol for concomitant formation of the mixed selenide/sulfide derivative *Se*-(*tert*-butylsulfanyl)selenocysteine in analogy to the formation of *S*-(*tert*-butylsulfanyl)cysteine[26] fails as a consequence of the difficult reduction of the diselenide with monothiols.[27]

6.2.3.1.1.1 *Se*-Benzyl Protection

In the first syntheses of selenocysteine peptides the benzyl group was used exclusively for protection of the selenol group in analogy to the *S*-benzylcysteine derivative.[15–19,23,24,28] The SeC(Bzl) derivatives are very labile in aqueous alkaline media leading to decomposition and precipitation of metallic selenium and consistently low yields were reported in the deprotection step with sodium in liquid ammonia. Since the latter method is known to provoke significant desulfuration during deprotection of *S*-benzylcysteine derivatives,[29] β-elimination of phenylmethaneselenolate ($BzlSe^-$) most probably is the main cause of the low yields obtained with the benzyl-type protection. It is now well established that $BzlSe^-$ represents a much better leaving group than the corresponding thiolate.[30]

In these early syntheses, Z-SeC(Bzl)-OH[23,24,31] was predominantly used as an intermediate and for the peptide coupling steps the mixed anhydride or the active ester procedure was applied. Alternatively, smaller synthons containing the SeC(Bzl) are readily accessible from the related serine compounds via nucleophilic displacement of the *O*-tosyl moiety by $BzlSe^-$.[28]

H-SeC(Bzl)-OH:[24]

Z-Ser(Tos)-ODpm: To a cooled soln (−10 °C) of Z-Ser-ODpm (7 g, 18 mol) in dry pyridine was added TosCl (4.5 g, 24 mmol). After 3 h at 0 °C the mixture was poured onto crushed ice. The solid was collected by filtration, dried (P_2O_5), and recrystallized (EtOH); yield: 7.0 g (72%); mp 83–84 °C.

Z-SeC(Bzl)-ODpm: A soln of the Z-Ser(Tos)-ODpm (11.0 g, 20.4 mmol) in acetone (40 mL) was added to a soln of BzlSeH (4.0 g, 23.5 mmol) in DMF (10 mL) containing NaOH (0.8 g, 20 mmol) dissolved in

for references see p 222

H_2O (5 mL). After 2–3 min of vigorous stirring, the pH was adjusted to 5–6 with AcOH and the mixture was taken to dryness. Upon addition of ice/H_2O the product solidified and it was then recrystallized (EtOH); yield: 7.0 g (65%); mp 66–67 °C.

H-SeC(Bzl)-OH: To the soln of Z-SeC(Bzl)-ODpm (1 g, 1.86 mmol) in AcOH (3 mL) was added 3.3 M HBr/AcOH (5 mL). After 1 h the hydrobromide was precipitated with Et_2O. The hydrobromide was dissolved in 1 M NaOH and the soln was titrated to pH 5.5 with 2 M HCl. The precipitate was collected and dried; yield: 0.215 g (43%); mp 175–178 °C; $[\alpha]_D^{23}$ +36.4 (*c* 2, 1 M NaOH) [cf. lit.[23] mp 191–192 °C; $[\alpha]_D^{23}$ +35.6 (*c* 2, 1 M NaOH); lit.[31] mp 183–184 °C; $[\alpha]_D^{24}$ +35.2 (*c* 2, 1 M NaOH)].

Boc-SeC(Bzl)-Gly-OBzl:[28]
To a cooled soln (−10 °C) of Boc-Ser-Gly-OBzl (7 g, 20 mmol) in dry pyridine (20 mL) was added TosCl (7.6 g, 40 mmol). After 3 h at −10 °C the mixture was poured onto crushed ice, the precipitate was filtered off and recrystallized (EtOH); yield: 7.6 g (75%) of the tosylate derivative. To precooled DMF (10 mL at −10 °C) a mixture consisting of BzlSeH (6.8 g, 40 mmol) and NaOH (1.4 g, 35 mmol) dissolved in H_2O (3 mL), was added and immediately afterwards a soln of the tosylate (14.0 g, 27.6 mmol) in acetone (40 mL). After 3 min of vigorous stirring the pH was adjusted to 5–6 with dil AcOH. The solvent was removed and the product was solidified with ice-cold H_2O. It was filtered off and recrystallized (EtOAc/ligroin 2:1); yield: 11.5 g (82%); mp 90–92 °C; $[\alpha]_D^{24}$ −27.5 (*c* 2, MeOH).

6.2.3.1.1.2 *Se*-4-Methoxybenzyl Protection

More recently, 4-methylbenzyl[25] and 4-methoxybenzyl (Mob) protection[7] was proposed for the selenol group. The 4-methylbenzyl group is introduced by reacting selenocysteine with α-bromo-*p*-xylene following essentially the procedure for the preparation of the related cysteine derivative.[25] This type of protection is applied in combination with the Boc/Bzl strategy and its removal is achieved by HF treatment in the presence of 15% anisole and 2.5% ethyl methyl sulfide at 0 °C in one hour. Unfortunately, experimental details were not reported by Oikawa et al.[25]

The *Se*-(4-methoxybenzyl)selenocysteine is obtained by reduction of selenocystine with $NaBH_4$ and in situ reaction with 4-methoxybenzyl chloride.[7] The optimized procedure of Tanaka and Soda[32] is preferentially used for the synthesis of the starting selenocystine, which involves reaction of β-chloroalanine with a 2.3-fold excess of disodium diselenide in aqueous solution at pH 9. Alternatively, the significantly less selenium demanding synthesis of Stocking et al.[33] is used for the preparation of expensive ^{77}*Se*-selenocystine, this consists of the reaction of methyl (2*R*)-2-[(*tert*-butoxycarbonyl)amino]-3-iodopropanoate with equivalent amounts of dilithium diselenide. Subsequent conversion of SeC(Mob) into the N^{α}-Fmoc derivative[7] and finally into the pentafluorophenyl ester[10] is performed following standard procedures.

An alternative procedure has been proposed for the direct synthesis of Fmoc-SeC-(Mob)-OH that is based on activation of Fmoc-Ser-OAL as the *O*-tosylate followed by reaction with 4-methoxybenzyldiselenide reduced in situ with hypophosphoric acid to the selenol.[59] Selective reductive cleavage of the allyl ester generates Fmoc-SeC(Mob)-OH in good yields.

The 4-methoxybenzyl group is removed with 1 M TMSBr or 1 M TMSOTf in TFA containing thioanisole and *m*-cresol as scavengers, that is, it is performed under identical conditions to those reported for the related cysteine derivative.[7] The Mob group is removed more efficiently by the DMSO/TFA method, which leads to formation of the diselenide as discussed in Section 6.2.3.1.2.

H-SeC(Mob)-OH:[7]
To an ice-cold soln of selenocystine (1.9 g, 5.69 mmol) in 0.5 M NaOH (5 mL) was added portionwise $NaBH_4$ (1.8 g, 44.9 mmol) and the mixture was stirred at rt until the yellow color disappeared. The soln was then cooled in an ice bath and 2 M NaOH (15 mL) was added followed by dropwise addition of MobCl (4.06 mL, 29.9 mmol). After stirring for 4 h at 4 °C, the mixture was acidified with concd HCl. The

precipitate was collected and recrystallized (hot H_2O); yield: 2.44 g (74%); mp 173–175 °C; $[\alpha]_D^{26}$ +23.5 (*c* 0.22, 1 M HCl).
Fmoc-SeC(Mob)-OH was obtained by standard methods using Fmoc-OSu;[34] yield: 89%; mp 138–140 °C; $[\alpha]_D^{25}$ −33.1 (*c* 0.87, DMF).[7]
Fmoc-SeC(Mob)-OPfp was prepared according to known procedures[35] and precipitated from EtOAc with petroleum ether; yield: 95%; mp 155 °C.[10]

Fmoc-Ser(Tos)-OAl:[47]
A soln of Fmoc-Ser-OAl (2.29 g, 6.23 mmol; prepared by standard procedures from Fmoc-Ser-OH with allyl bromide) and TosCl (5.94 g, 31.16 mmol) in pyridine (11 mL) was stirred at 0 °C under argon for 11 h. The mixture was taken up in Et_2O (100 mL) and the organic phase was washed with H_2O (1 × 50 mL), 10% $KHSO_4$ (4 × 50 mL), sat. $NaHCO_3$ (1 × 50 mL), brine (1 × 50 mL), and H_2O (2 × 50 mL). The organic layer was dried ($MgSO_4$) and taken to dryness. Purification was accomplished by flash chromatography (hexane/EtOAc 2:1); yield: 3.01 g (94%); mp 65–67 °C.

Di-4-methoxybenzyl Diselenide:[47]
Se powder (5.0 g, 63.3 mmol) was suspended in THF (30 mL) and 1 M $LiBEt_3H$ in THF was added (63 mL, 63.0 mmol). The mixture was refluxed for 45 min under argon. In a separate 3-necked flask, Mob-Cl (9.02 mL, 66.5 mmol) was dissolved in THF (30 mL) and cooled to −78 °C. The refluxing mixture was cooled to rt and transferred to the cold flask via cannula. The mixture was stirred at −78 °C for 45 min, warmed to 0 °C over 30 min, and to rt over a further 30 min. The mixture was filtered through a silica plug (hexane/EtOAc 5:1), purified by flash chromatography (hexane/EtOAc 11:1), and finally recrystallized from EtOH to give a yellow solid; yield: 10.86 g (84%); mp 67–68 °C.

Fmoc-SeC(Mob)-OAl:[47]
Di-4-methoxybenzyl diselenide (6.71 g, 16.7 mmol) was placed in a round-bottomed flask fitted with a reflux condenser. The yellow solid was dissolved in THF (40 mL) and 50% H_3PO_2 (60 mL) and refluxed under argon for 1 h. The soln was cooled and extracted under argon with Et_2O (2 × 30 mL) and the organic layers were transferred via syringe to an argon-flushed round-bottomed flask containing degassed brine soln. The soln was shaken and the organic layer was transferred via syringe to an argon-flushed round-bottomed flask containing $MgSO_4$. After drying, the soln was rapidly filtered and the solvent removed. The resulting selenol was dissolved in degassed DMF (40 mL) under argon. NaOH (0.46 g, 11.5 mmol) dissolved in degassed H_2O (10 mL) was added immediately followed by Fmoc-Ser(Tos)-OAl (5.43 g, 10.4 mmol) in degassed acetone (40 mL). The mixture was stirred under argon at 0 °C for 4 h, taken up in EtOAc (100 mL), and washed with sat. NH_4Cl (3 × 40 mL) and brine (3 × 40 mL). The organic layer was dried ($MgSO_4$), filtered, and the solvent removed. The product was purified by flash chromatography (hexane/EtOAc 6:1) to give a white solid; yield: 5.29 g (92%); mp 92–93 °C.

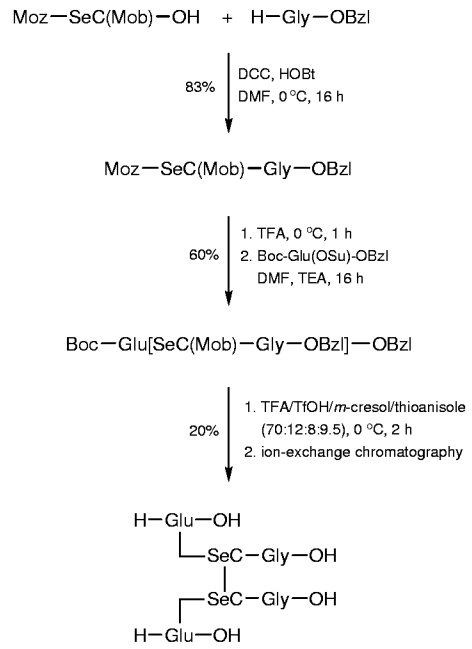
Fmoc-SeC(Mob)-OH:[47]
To a soln of Fmoc-SeC(Mob)-OAl (4.61 g, 8.4 mmol) and $Pd(PPh_3)_4$ (0.48 g, 0.42 mmol) in THF (50 mL) under argon was added morpholine (0.88 mL, 10.0 mmol) dropwise over 40 min. The soln was stirred for an additional 20 min, and then EtOAc (100 mL) was added. The soln was washed with 2 M HCl (3 × 30 mL), dried ($MgSO_4$), filtered, and the solvent was removed. The crude product was flushed through silica plugs, first with 30:1 and then 35:1 CH_2Cl_2/MeOH and finally with 40:1 CH_2Cl_2/MeOH containing 0.1% AcOH; yield: 4.23 g (99%); $[\alpha]^{21}_D = -31.6$ (*c* 5.7, DMF).

6.2.3.1.2 Synthesis of Selenocystine Peptides

A comparison of the early synthesis of the SeC-glutathione analogue with *Se*-benzyl protection[36] with a recent synthesis based on the use of SeC(Mob) derivatives[37] clearly reveals the advantage of the latter protecting group particularly in terms of overall yields. As shown in Scheme 1, with the acid-labile 4-methoxybenzyloxycarbonyl (Moz) and Boc groups an orthogonal protection of the α-amino group is assured, since the *Se*-4-methoxybenzyl derivative is sufficiently stable toward TFA treatments, while it is cleaved with TFA/TfOH (5:1) containing *m*-cresol and thioanisole as scavengers.

for references see p 222

Scheme 1 Synthesis of the Selenocystine Analogue of Glutathione[37]



Recent syntheses of selenocysteine peptides have been performed on solid supports mainly applying Fmoc/tBu chemistry[38] and correspondingly, Fmoc-SeC(Mob)-OH[7,20] or Fmoc-SeC(Mob)-OPfp[10,14,39] as building blocks. Using standard synthetic protocols, side products are formed which derive from the β-elimination of the 4-methoxyphenylmethaneselenolate ($MobSe^-$) with formation of dehydroalanine and piperidinoalanine in amounts strongly depending upon the reaction conditions used in the Fmoc cleavage and final deprotection steps.[10] Deselenation is largely suppressed by reducing the length of time the SeC(Mob) residues are exposed to piperidine, which is required for Fmoc cleavage, to a minimum and also by making use of pentafluorophenyl esters without the addition of base. Moreover, side reactions at the SeC(Mob) residue are minimized by using optimized acid cocktails (Table 1) in the resin cleavage and final deprotection step, although reattachment to the resin, particularly in the case of the Rink-type spacer, and thus significant losses of material still remain the main concern.[10]

Table 1 Optimized Protocol of the Synthesis of Selenocysteine Peptides on Rink-Amide-MBHA Resin According to the Fmoc Strategy[10]

Synthetic Step	Reagents and Solvents[a]	Reaction Time
Fmoc cleavage	20% piperidine/NMP	3 min
washings	1. 20% CH_2Cl_2/NMP	1. 2 min (2 ×)
	2. CH_2Cl_2	2. 2 min (2 ×)
	3. NMP	3. 2 min (2 ×)
coupling	pentafluorophenyl esters/HOBt (2:1, twofold excess), 20% CH_2Cl_2/NMP	90 min
washings	1. 20% CH_2Cl_2/NMP	1. 2 min (2 ×)
	2. CH_2Cl_2	2. 1 min (2 ×)
	3. NMP	3. 2 min (2 ×)
acetylation	AcOSu/HOBt(1:1, eightfold excess), 20% CH_2Cl_2/NMP	30 h
acidolytic cleavage	TFA/H_2O/CH_2Cl_2/TIS (89:5:5:1), 4 °C	1 h

[a] Unless otherwise stated v/v.

Among the methods established in cysteine chemistry for cleavage of the Mob group with concomitant formation of disulfide bridges, i.e. I_2, Tl(III)trifluoroacetate[40] and DMSO/TFA,[41] only the DMSO/TFA procedure allows for clean deprotection of SeC(Mob) residues and for their concomitant oxidation to selenocystine dimers[7] or intrachain cyclic structures.[10,21,39]

Ac-Gly-SeC(Mob)-Pro-Tyr-SeC(Mob)-Val-Lys-Ala-NH_2:[10]
The synthesis of the octapeptide was performed on Rink-amide-MBHA resin (2.0 g, 0.47 mmol·g^{-1}) following the general protocol described in Table 1 and using exclusively twofold excesses of the Pfp esters with HOBt (1 equiv) as catalyst in the coupling steps. Final acetylation was performed with an 8-fold excess of AcOSu/HOBt (1:1) for 30 h. Acidolytic cleavage was carried out with TFA/H_2O/CH_2Cl_2/TIS (85:5:5:1) at 4 °C for 1 h. The crude product (1.02 g) which contains about 90% according to HPLC (t_R 26.45 min) of the desired peptide, was purified by RP-chromatography on a Lichroprep RP18 column (2 × 230 cm) by elution at a flow rate of 4 mL·min^{-1} with 0.1 M NH_4OAc (pH 3.8)/iPrOH/BuOH (88:16:4) in isocratic manner for 2 h, followed by a linear gradient from 88:16:4 to 57:34:9 in 12 h. Fractions containing homogeneous product were pooled and lyophilized; yield: 0.7 g (61%); FAB-MS (*m/z*) Found: 1216.9 $[M+H]^+$. Calcd for $C_{54}H_{76}N_{10}O_{12}Se_2$: 1215.3.

Ac-Gly-SeC-Pro-Tyr-SeC-Val-Lys-Ala-NH_2:[10]
A soln of Ac-Gly-SeC(Mob)-Pro-Tyr-SeC(Mob)-Val-Lys-Ala-NH_2 (13 mg, 0.01 mmol) in ice-cold TFA (50 mL) was added dropwise to an ice-cold soln of DMSO (2 mL) in TFA (8 mL). The mixture was stirred for 30 min at 4 °C. The TFA was removed and the residue purified by preparative HPLC on a Nucleosil C18 column (2 × 25 cm) at a flow rate of 10 mL·min^{-1}; eluents: 0.1% aq TFA (A), 0.08% TFA in MeCN (B); isocratic elution for 15 min with A/B (95:5) followed by a linear gradient of A/B from 95:5 to 70:30 in 2 h. Fractions containing the desired product were pooled and lyophilized; yield: 6 mg (62%); FAB-MS (*m/z*) Found: 975.3 $[M+H]^+$. Calcd for $C_{38}H_{58}N_{10}O_{10}Se_2$: 974.2.

6.2.3.2 Mixed Selenocystine/Cystine Peptides

Replacement of one cysteine residue with selenocysteine in bis(cysteinyl) peptides raises the question of the optimal protection of both the SeC and Cys residue in terms of deprotection and subsequent oxidation to the mixed sulfide/selenide bridge. So far, the use of the 4-methoxybenzyl group for both the thiol and selenol group is the most advisable, since the DMSO/TFA deprotection/oxidation method provided satisfactory results in the case of model octapeptides.[10] This is further confirmed by the synthesis of the [SeC^7,Cys^{23}]- and [Cys^7,SeC^{23}]-α-atrial natriuretic peptide analogues.[20] Conversely, replacement of one disulfide bridge with a diselenide in multiple-cystine containing peptides requires a regioselective pairing of the selenocysteine and cysteine residues, respectively, and thus an orthogonal protection scheme. Two methods are presently available that both rely on the peculiar redox properties of selenocysteine. These differ strongly from those of cysteine (see Section 6.2.3) and thus allow for selective chemistry to be applied on these two residues.

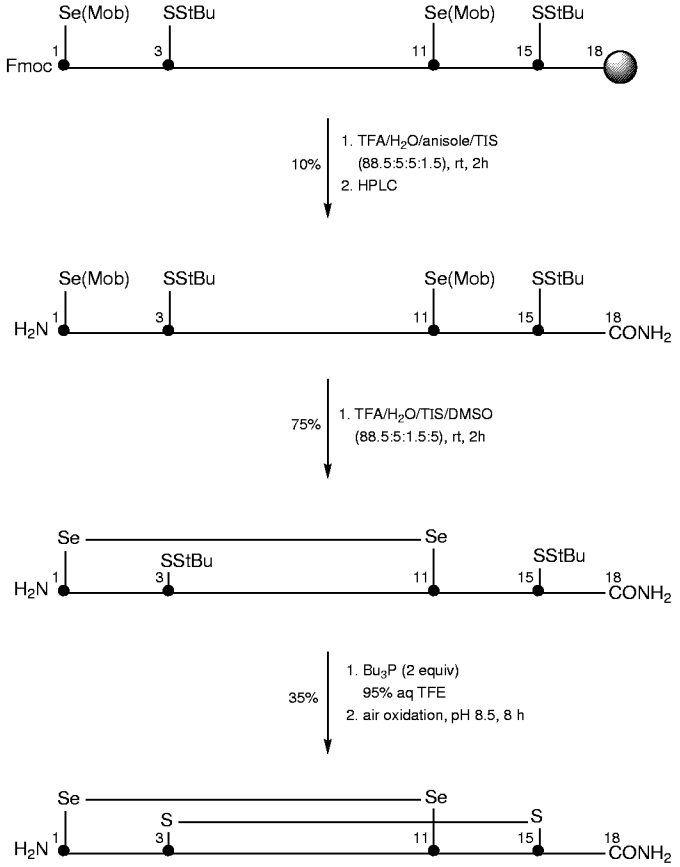
6.2.3.2.1 Selective Oxidation of Selenocysteine

As shown by the synthesis of [SeC^3,SeC^{11},Nle^7]-endothelin-1[13,14] and [SeC^1,SeC^{11}]-apamin[21,39,45] in which one of the two bridged cysteine pairs of the natural isomer is replaced by a selenocysteine pair, the desired diselenide and disulfide connectivities are obtained by using a combination of SeC(Mob) and Cys(StBu) derivatives. The linear peptides are synthesized using an automatic synthesizer, adapting the optimized synthetic protocol of Table 1 to the synthesizer protocols. The peptides are cleaved from the resin with concomitant deprotection of the acid-labile groups with the cocktail TFA/H_2O/CH_2Cl_2/TIS (89:5:5:1). By this procedure intermediate purification of the thiol- and selenol-protected peptides becomes possible. The full stability of the *S-tert*-butylsulfanyl group to acid treatment permits selective deprotection and oxidation of the two selenocysteine residues to form the diselenide-bridged intermediate, as shown in Scheme 2.[21,45] Due to the facile reduction of

for references see p 222

Cys(StBu) with phosphines only a slight excess of tributylphosphine is required for this purpose. Under these conditions reduction of the diselenide does not occur at all, and subsequent air oxidation of the two cysteine residues at high dilution leads in a highly selective manner to the diselenide- and disulfide-bridged peptides. The selectivity of the disulfide bridging is assured by the complete absence of thiol/diselenide exchange reactions even at alkaline pH values due to the very low reactivity of the diselenide toward monothiols[5] as a result of their highly differentiated redox potentials.[9]

Scheme 2 Regioselective Selenocysteine and Cysteine Pairings in [SeC[1],SeC[11]]-Apamin[21,45]



[SeC(Mob)[1],Cys(StBu)[3],SeC(Mob)[11],Cys(StBu)[15]]-Apamin:[21,45]
The peptide was synthesized on TentaGel S RAM resin (1 g, 0.25 mmol loading). Cleavage from the resin and deprotection was performed with TFA/H_2O/anisole/TIS (88.5:5:5:1.5, 20 mL) for 2 h at rt and the crude product was precipitated with tBuOMe/hexane (2:1); yield of crude product: 160 mg. The relatively low yield (28%) is due to reattachment as determined by amino acids analysis of the resin hydrolysate (40% reattachment). The title compound was obtained by preparative HPLC (linear gradient of 0.1% aq TFA/0.08% TFA in MeCN from 90:10 to 40:60 in 105 min): yield: 40 mg (10%); ESI-MS (*m/z*) Found: 2543.2. Calcd for $C_{103}H_{167}N_{31}O_{26}S_4Se_2$: 2542.2.

[SeC[1],SeC[11]]-Apamin:[21,45]
[SeC(Mob)[1],Cys(StBu)[3],SeC(Mob)[11],Cys(StBu)[15]]-apamin (30 mg, 0.012 mmol) was dissolved in TFA/H_2O/TIS (95:5:1.5) containing 5% DMSO and the reaction was monitored by HPLC. After 2 h at rt, the soln was concentrated and the product precipitated with tBuOMe/hexane (2:1). The Cys(StBu)-protected peptide was dissolved in 95% aq TFE (120 mL) and reacted with Bu_3P (6 µL, 0.025 mmol). The mixture was stirred at rt until complete reduction of the two Cys residues, then the solvent was removed and the residue lyophilized from H_2O. The product was dissolved in H_2O (120 mL), the pH adjusted to

8.5 with dilute NH_4OH, and air oxidation was allowed to occur at rt for 12 h. The soln was lyophilized and the crude product purified by HPLC (linear gradient of 0.1% aq TFA/0.08% TFA in MeCN from 95:5 to 45:55 in 95 min); yield: 10 mg (35%); ESI-MS (*m/z*) Found: 2122.2. Calcd for $C_{79}H_{131}N_{31}O_{24}S_2Se_2$: 2121.5

6.2.3.2.2 Oxidative Refolding of Mixed Selenocysteine/Cysteine Peptides

Reoxidation experiments on fully reduced [SeC^1,Cys^3,SeC^{11},Cys^{15}]-, [Cys^1,SeC^3,SeC^{11},Cys^{15}]-, and [SeC^1,SeC^3,Cys^{11},Cys^{15}]-apamin in which a cysteine pair is replaced by a selenocysteine pair leads to the selective diselenide bridging independently of the two cysteine residues and thus, to the three isomers without detectable formation of mixed sulfide/selenide bridges.[21,39,45] Although for apamin sufficient structural information is encoded in its sequence to drive the oxidative refolding pathway to the exclusive formation of the correct natural isomer, the high preference of selenocysteines for diselenide bridging completely predominates over these conformational preferences. Taking into account the isomorphous character of Cys → SeC mutations, these findings open a promising, new synthetic strategy for oxidative refolding of multiple-cystine peptides that relies on the replacement of one cysteine pair with a selenocysteine pair. In fact, the selective formation of one diselenide bridge can lead to a productive folding intermediate that consequently drives the folding pathway to the correct pairing of the residual cysteine residues.

6.2.3.3 Selenocysteine-Mediated Native Chemical Ligation

With the success of cysteine-mediated native chemical ligation,[48–50] and as a powerful extension of the expressed protein ligation[51–53] (see Vol. E 22a, Sections 1.2.4.2.2 and 4.1.5), it was clear that selenocysteine with the enhanced acidity of the selenol (pK_a ~ 5.5[54]) compared to a thiol should facilitate the transthioesterification between a peptide thioester and an N-terminal selenocysteine residue of the second peptide. Similarly, subsequent transfer of the acyl group from the Se to the N atom is also facilitated since aminolysis of selenoesters is known to be orders of magnitude faster than aminolysis of the corresponding thioesters.[55] Model studies and first syntheses of proteins by this new approach confirmed these expectations.[47,56–59] Although these ligations have to be performed under reducing conditions and an inert atmosphere because of the low redox potential of selenocysteine (see Section 6.2.3), addition of dithiothreitol, sulfanylethanesulfonic acid, PhSH, or TCEP is sufficient to generate the required selenol nucleophile.

The advantage of this synthetic approach over the total synthesis of selenocysteine peptides is the reduced number of synthetic steps to which the Se-protected selenocysteine is exposed, thus, largely preventing deselenation as an undesired side reaction. It also allows access to larger peptides and even proteins containing selenocysteine to exploit the Se atom for X-ray crystallography,[13] or as ^{77}Se nucleus for NMR spectroscopy.[60]

6.2.3.4 Selenocysteine Derivatives as Precursors of Dehydroalanine

The synthesis of dehydroamino acid containing peptides as versatile precursors for post-synthetic chemical manipulations generally involves incorporation of a masked residue into the peptide and its subsequent conversion into the dehydroamino acid (see Vol. E 22c, Section 11.1). Since oxidative elimination of *S*-alkyl or *S*-aryl cysteines represents one of the

for references see p 222

methods of choice[61–63] and *Se*-(phenyl)selenocysteine is known to undergo facile oxidative deselenation,[64–66] this derivative has recently been exploited, upon its conversion into dehydroalanine, for regioselective chemistry such as ligation of thioglycosides or lipidic thiols as well as the production of cyclic lanthionine peptides.[67–69]

Ring opening of *N*-Boc-protected serine lactone[70] with phenylselenide anion,[71] which is generated by the reaction of diphenyldiselenide with sodium trimethoxyborohydride[72] leads to Boc-SeC(Ph)-OH in good yields.[67] A procedure better suited for larger scale preparations was also reported where Fmoc-Ser-ODpm is converted into the tosyl derivative and reacted under basic conditions with phenylselenol.[47] Acidolysis of the ester with TFA generates the Fmoc-SeC(Ph)-OH derivative for use in Fmoc/tBu peptide synthesis. Studies on model peptides[67] showed that oxidative elimination is accomplished with sodium periodate (4 equiv) generally in less than two hours without affecting oxidation-sensitive residues such as Trp, Cys(Trt), or mixed and symmetrical Cys disulfides. In the presence of Met residues, the reaction is carried out with 1.1 equivalents of oxidant to avoid sulfoxide formation to larger extents.

Fmoc-Ser(Tos)-ODpm:[47]
A soln of Fmoc-Ser-ODpm[73] (11.21 g, 22.7 mmol) and TosCl (8.71 g, 45.7 mmol) in pyridine (30 mL) was immediately cooled to 0 °C and stirred under argon for 11 h. The mixture was taken up in Et_2O and the organic phase was washed with H_2O (1 ×), 10% $KHSO_4$ (4 ×), sat. $NaHCO_3$ (1 ×), brine (1 ×), and H_2O (2 ×). The organic layer was dried ($MgSO_4$) and the solvent removed. Purification was accomplished by flash chromatography (hexane/EtOAc 2:1); yield: 10.8 g (73%); mp 128.5–130 °C.

Fmoc-SeC(Ph)-OH:[59]
To a soln of PhSeH (1.88 mL, 17.7 mmol) in degassed DMF (26 mL) at 0 °C was added NaOH (0.54 g, 13.6 mmol) in degassed H_2O (8 mL) followed immediately by Fmoc-Ser(Tos)-ODpm (8.80 g, 13.6 mmol) in degassed acetone (26 mL). The reaction was stirred at 0 °C under argon for 3.5 h. The precipitated product was filtered off and recrystallized from EtOH; yield: 6.35 g (75%); mp 103–105 °C.
To Fmoc-SeC(Ph)-ODpm (6.36 g, 10.1 mmol) was added TFA (45 mL) and thioanisole (2.38 mL, 20.1 mmol) and the mixture was stirred under N_2 for 40 min. The solvent was removed and the oily residue was evaporated repeatedly from $CHCl_3$ and recrystallized (EtOAc/hexane); yield: 4.64 g (98%); mp 121–124 °C.

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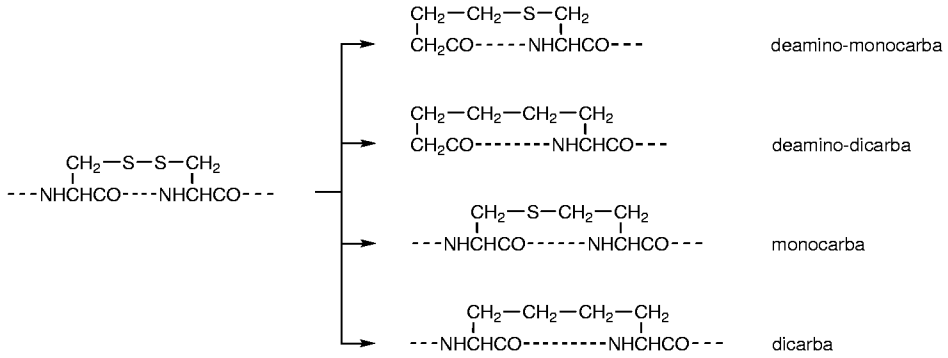
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6.2.4 Carba-analogues of Cystine Peptides

T. KIMURA

The concept of preparing synthetic carba-analogues of cystine peptides was originally developed for oxytocin (Scheme 1).[1–3] Knowing that deamidation of the N-terminal cysteine residue only marginally affects its hormonal activity, one or both sulfur atoms of the disulfide bridge in deamino-oxytocin were replaced by one or two methylene groups yielding the related mono- and dicarba-analogues, respectively. As a result of these first synthetic approaches, which also include the use of cystathionine,[4] several types of amino acids that can be used to replace cystine were prepared containing all the relevant functional groups and the length of the disulfide bridge was also varied together with related modifications.

Scheme 1 Carba-analogues of Cystine



6.2.4.1 Carba-analogues of α-Deaminocystine

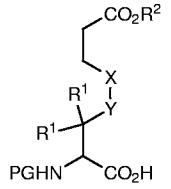
The simplest approach to isosteric replacement of one or both sulfur atoms of the cystine disulfide with a methylene or ethylene moiety is given for natural bioactive peptides when one cysteine residue is located in the N-terminal sequence position and the related amino group or peptide extension is not involved in the bioactivity. This allows for direct side chain to backbone (N-terminus) cyclization via amide bonds with suitable *S*-carboxyalkyl derivatives of the second cysteine residue, or with the ω-carboxy group of aminodicarboxylic acids containing an alkyl side chain that mimics the Cα to Cα′ spacer in cystine. Thereby, the length and degree of branching of the sulfide or alkyl spacer can additionally be varied.

6.2.4.1.1 Monocarba-analogues

This type of isosteric replacement of the cystine disulfide with the methylenethio group (-CH_2–S-) has been examined extensively in the deamino-oxytocin molecule.[2,3,5–9] Depending upon the sequence position of the cysteine residue in oxytocin where the replacement of the sulfur atom by the methylene group is to be performed, a 1- or 6-monocarba-deamino-oxytocin is obtained. The key cysteine derivatives for the syntheses of such analogues are *S*-(3-carboxypropyl)cysteine, Cys[$(CH_2)_3CO_2H$], for the 1-monocarba-analogue and *S*-(2-carboxyethyl)homocysteine, hCys[$(CH_2)_2CO_2H$], for the 6-monocarba-analogue. These α-aminodicarboxylic acids are readily prepared from L-cysteine and 3-bromobutanoic acid esters[5,6,10] and from L-homocysteine and acrylic acid esters,[6,10] respectively. This synthetic approach allows preselection, via the type of ester in the alkylating moiety, of the protection mode of the side-chain carboxy group with respect to the overall synthetic strategy (Scheme 2). Moreover, by replacing cysteine with penicillamine, *S*-(3-carboxy-

propyl)penicillamine, Pen[$(CH_2)_3CO_2H$] is obtained, which is possibly an interesting tool for constraining the cyclic structures by the side-chain branching of the penicillamine moiety. This type of cystine mimetic has been recently utilized in structure–function studies of oxytocin[11] and human atrial natriuretic factor (hANF).[12]

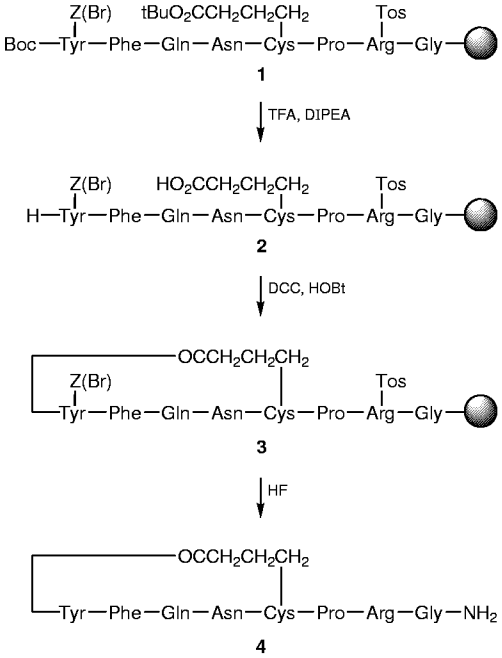
Scheme 2 Orthogonally Protected α-Deamino-Monocarba-analogues of Cystine



X = CH_2; Y = S; R^1 = H
X = CH_2; Y = S; R^1 = Me
X = S; Y = CH_2; R^1 = H
PG = Z, Boc, Fmoc
R^2 = Me, tBu, Fm

For peptide synthesis in solution or on solid support with this type of α-aminodicarboxylic acid standard procedures are applied using Z, Boc, or Fmoc derivatives for N^{α}-amino protection in suitable combination with the methyl, *tert*-butyl, or 9-fluorenylmethyl ester of the ω-carboxy group.[12–15] A higher degree of orthogonality in the protection scheme is required if side chain to backbone cyclization is to be performed in peptides containing additional carboxy and/or amino groups. In the simple cases, such as deamino-oxytocin or deamino-vasopressin, after final deprotection with TFA or HF, cyclization was performed with appropriate peptide coupling reagents.[5] This cyclization step can also be performed on a solid support using a selective deprotection step, as outlined in Scheme 3 for the synthesis of the 1-monocarba-deamino-vasopressin **4**.[16]

Scheme 3 Synthesis of a Monocarba-vasopressin Analogue by the Solid-Phase Cyclization Method[16]



for references see p 233

H-Cys[$(CH_2)_3CO_2$tBu]-OH:[10]
Cystine (10 g, 41.6 mmol) was dissolved in liq NH_3 (500 mL) and Na was added until the blue color persisted for 5 min. The soln was decolorized by addition of NH_4Cl, and *tert*-butyl 3-bromobutanoate (28 g, 0.125 mol) was added under stirring. The soln was concentrated under reduced pressure, the residue dissolved in H_2O (250 mL) and extracted with EtOAc. The aqueous layer was cooled to 0 °C and applied to a column of Dowex 50 (H^+ form, 300 mL). After washing the column with H_2O, the product was eluted with 5% aq NH_3. Removal of the eluent afforded the product; yield: 18.6 g (86%).

H-hCys($CH_2CH_2CO_2$tBu)-OH:[10]
The title compound was synthesized from homocystine (8.06 g, 30 mmol) and *tert*-butyl acrylate (17.4 mL, 120 mmol) as described for H-Cys[$(CH_2)_3CO_2$tBu]-OH; yield: 6.75 g (41%).

H-Pen[$(CH_2)_3CO_2$Fm]-OH:[12]
L-Penicillamine (1.2 g, 8.3 mmol) and 9-fluorenylmethyl 4-bromobutanoate[11] (3 g, 8.7 mmol) were reacted in EtOH/H_2O (1:1) containing DIPEA (4.2 mL, 23.6 mmol). The mixture was stirred for 18 h at rt and neutralized with 1 M HCl (pH 6). The precipitate was filtered and washed with H_2O and EtOH and recrystallized (aq EtOH); yield: 1.2 g (36%); mp 151–153 °C.

Boc-Cys[$(CH_2)_3CO_2$Fm)-OH:[12]
To a soln of H-Cys[$(CH_2)_3CO_2$Fm]-OH (1.9 g, 4.9 mmol) in 50% aq dioxane (100 mL) containing DIPEA (2.6 mL) was added Boc_2O (2.2 g, 10 mmol). The mixture was stirred at rt for 5 h and then diluted with H_2O (150 mL) and extracted with Et_2O. The aqueous phase was acidified with solid citric acid and extracted with EtOAc. The combined organic extracts were washed with 1 M HCl and H_2O, dried ($MgSO_4$), and the solvent was removed to give an oil; yield: 1.6 g.

Solid-Phase Cyclization to 4:[16]
The peptide resin **1** was treated for 1 h with 50% TFA in CH_2Cl_2 containing 2% anisole, washed with CH_2Cl_2, neutralized with 5% DIPEA in CH_2Cl_2, washed with CH_2Cl_2, and then DMF. The product **2** was treated with a 5-fold molar excess of HOBt in DMF, followed after 20 min with a 5-fold molar excess of DCC in CH_2Cl_2. The mixture was adjusted with DIPEA to pH 7.5. After 24 h, the resin was drained and after a neutralization step with DIPEA, the cyclization was repeated for another 24 h. The extent of the cyclization to **3** was estimated to be about 95% by the ninhydrin test.

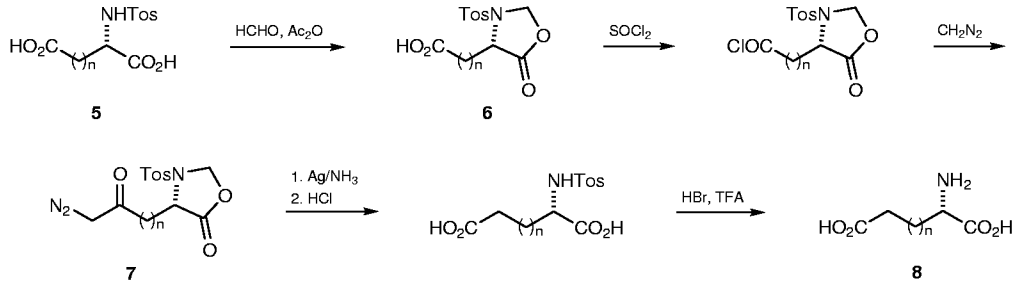
6.2.4.1.2 Dicarba-analogues

The α-aminodicarboxylic acid corresponding in side-chain length to α-deamino-cystine is α-aminosuberic acid (2-aminooctanedioic acid). A first synthesis of α-aminosuberic acid based on the hydrogenation of α-hydroxyiminosuberic acid is of no practical use because enantiomeric resolution of the resulting amino acid racemate cannot be achieved.[17] Therefore, alternative procedures have been developed for the preparation of optically pure L-α-aminosuberic acid as required for the synthesis of various neurohypophyseal hormones containing N-terminally a half-cystine residue.[3,18–21]

6.2.4.1.2.1 Synthesis of L-α-Aminosuberic Acid

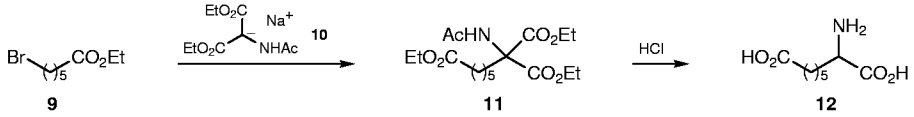
Following Scheme 4, L-α-aminosuberic acid can be obtained from *N*-tosyl-L-glutamic acid (**5**, n = 2), upon protection of the α-carboxy and tosylamino groups as an oxazolidone **6**, by extension of the side chain with the Arndt–Eistert method via the diazo ketone **7** by one methylene moiety in each cycle to produce in subsequent steps the L-α-aminoadipic acid [(*S*)-2-aminohexanedioic acid, **8**, n = 2], L-α-aminopimelic acid [(*S*)-2-aminoheptanedioic acid, **8**, n = 3] and finally L-α-aminosuberic acid [(*S*)-2-aminooctanedioic acid, **8**, n = 4].[22] Treatment with HBr in AcOH was the method of choice for the acidolytic hydrolysis of the tosyl group.[23]

Scheme 4 Synthesis of L-α-Aminosuberic Acid from *N*-Tosyl-L-glutamic Acid (n = 2) via L-α-Aminoadipic (n = 3) and L-α-Aminopimelic Acid (n = 4)[22]



Due to the laborious multiple steps of this synthesis and the correspondingly low yields, an alternative synthetic approach based on alkylation of the acetamidomalonate **10** with ethyl 6-bromohexanoate (**9**) giving the racemic α-aminosuberic acid **12** after deacetylation of the amino group in **11** and decarboxylation, as shown in Scheme 5, is significantly more efficient.[24]

Scheme 5 Synthesis of DL-α-Aminosuberic Acid[24]



For resolution of the racemate **12** two different procedures can be applied;[24] the enantioselective enzymatic deacylation of chloroacetyl-DL-α-aminosuberic acid at pH 7.2 with Taka-acylase or the enantioselective salt precipitation of Z-DL-Asu-OH with D-tyrosine hydrazide according to the method of Vogler et al.[25] Complete enzymatic digestion is achieved in about ten days at 37 °C, and the optically pure L-enantiomer is obtained in 80% yield; but the overall efficiency is lower than that of the chemical method. Fractional crystallization affords in good yields the Z-L-Asu-OH derivative **13** which is then used directly as a suitably protected intermediate in subsequent derivatization steps (see Scheme 6). Moreover, the recovery of the D-enantiomer from the mother liquors is also easy in this case.

H-DL-Asu-OH (12):[24]

Diethyl acetamidomalonate (**10**; 183 g, 0.82 mol) was added to a soln of NaOEt in abs EtOH [Na (18.4 g) in EtOH (800 mL)], and the mixture was refluxed gently. Then ethyl 6-bromohexanoate (**9**; 183 g, 0.82 mol; prepared from cyclohexanone according to Brown and Partridge;[26] bp 116–120 °C/14 Torr) was added under stirring over a period of 80 min. After 7.5 h refluxing, the mixture was kept overnight at rt. The precipitate was removed, the soln concentrated and the oily residue was dissolved in Et_2O. The soln was washed with H_2O and taken to dryness. The oily residue was refluxed with concd HCl (1.2 L) for 6 h and the solvent evaporated. The solid was dissolved in H_2O (350 mL), the pH of the soln adjusted to 3 with aq NH_3. Upon cooling compound **12** was collected, then washed with H_2O and MeOH; yield: 127 g (77%); mp 240–241 °C (dec).

Z-L-Asu-OH (13):[24]

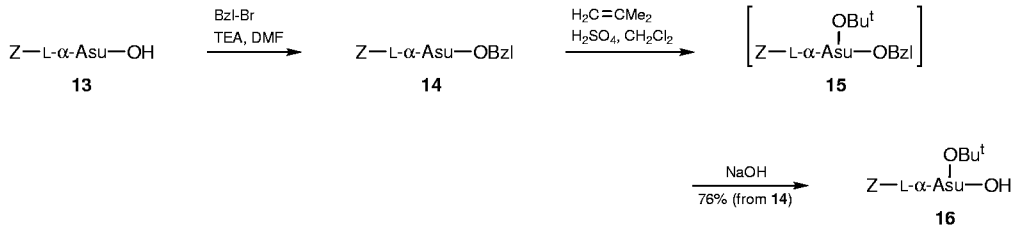
Racemic compound **12** was converted with ZCl under standard Schotten–Baumann conditions into the Z-DL-Asu-OH derivative [recrystallized (EtOAc) as needles; yield: 84%; mp 124–125 °C]. H-Tyr-N_2H_4 (29.3 g, 0.156 mol) and Z-DL-Asu-OH (32.3 g, 0.1 mol) were dissolved in boiling MeOH (600 mL). Upon filtration, the soln was concentrated to ca. 300 mL and EtOAc (~300 mL) was added carefully. The crystals were collected and recrystallized (EtOH) to obtain the H-D-Tyr-N_2H_2 · Z-L-Asu-OH salt; yield: 25.8 g (72%); mp 153–154 °C; $[\alpha]_D^{22}$ −56.1 (*c* 1.8, H_2O). The salt (25.6 g, 0.036 mol) was partitioned between EtOAc (120 mL) and 4 M HCl (30 mL). The EtOAc layer washed with H_2O, dried (Na_2SO_4) and upon removal of the solvent the residue was recrystallized (EtOAc/petroleum ether); yield: 10.8 g (93%); mp 119–121 °C; $[\alpha]_D^{22}$ −3.3 (*c* 7.7, AcOH).

for references see p 233

6.2.4.1.2.2 Synthesis of Selectively Protected α-Aminosuberic Acid

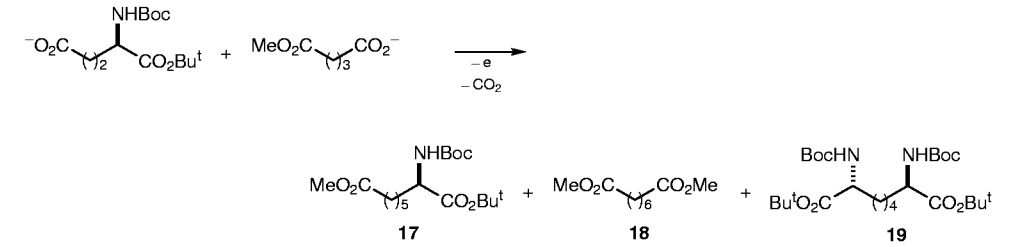
Derivatization of α-aminosuberic acid to produce selectively protected derivatives such as Z-L-Asu(OtBu)-OH **16** is performed as outlined in Scheme 6. Using Z-L-Asu-OH **13** as starting material, conversion into α-benzyl ester **14** by treatment with Bzl-Br, followed by reaction with isobutene/H_2SO_4 and saponification of the benzyl ester **15** leads to the desired product **16** as the piperazine salt.[24]

Scheme 6 Synthesis of N^{α}-Benzyloxycarbonyl-L-α-aminosuberic Acid ω-*tert*-Butyl Ester[24]



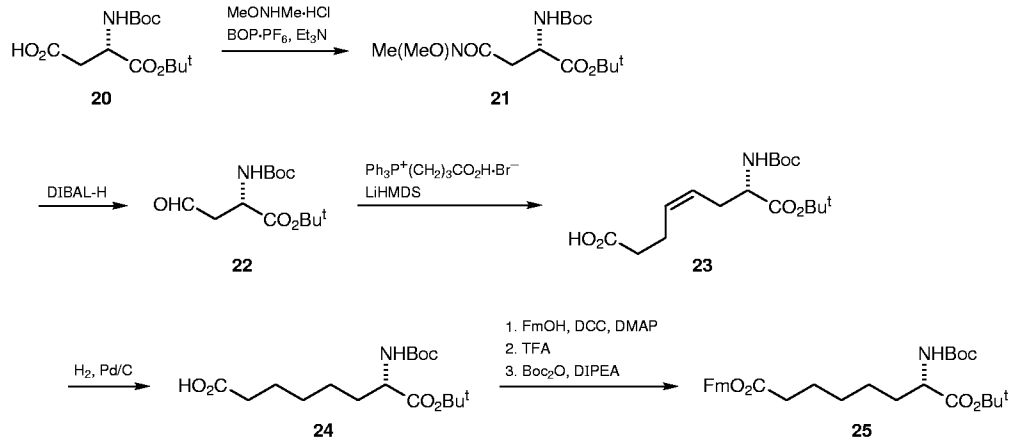
Two further stereospecific syntheses of selectively protected α-aminosuberic acid have been reported. One is based on the Kolbe electrolysis of an equimolar mixture of Boc-D-Glu-OtBu and monomethyl glutarate [4-(methoxycarbonyl)butanoate] to produce Boc-D-Asu(OMe)-OtBu **17** which, however, must be isolated from the statistical mixture containing besides the desired product also the two symmetrical dimers **18** and **19** as byproducts (Scheme 7).[27]

Scheme 7 Synthesis of N^{α}-*tert*-Butoxycarbonyl-D-α-aminosuberic Acid α-*tert*-Butyl Ester ω-Methyl Ester by the Kolbe Electrolytic Decarboxylation Dimerization Reaction[27]



The other stereoselective synthesis,[28] shown in Scheme 8, foresees conversion of Boc-L-Asp-OtBu **20** into the related β-aldehyde **22** via the Weinreb amide **21** and its reduction with diisobutylaluminum hydride (DIBAL-H). Wittig condensation of **22** with the ylide derived from (3-carboxypropyl)triphenylphosphonium bromide using lithium hexamethyldisilazanide at −78 to 0 °C, produces the unsaturated compound **23** which is catalytically hydrogenated to the protected L-α-aminosuberic acid derivative **24**. Conversion of the ω-carboxy group into the 9-fluorenylmethyl ester, followed by TFA treatment and reprotection of the N^{α}-amino group affords Boc-L-Asu(OFm)-OH (**25**).

Scheme 8 Enantioselective Synthesis of N^{α}-*tert*-Butyloxycarbonyl-L-α-aminosuberic Acid ω-9-Fluorenylmethyl Ester[28]



[Z-L-Asu(OtBu)-OH]$_2$ Piperazine Salt [(16)$_2$ · Piperazine]:[24]
A soln of Z-L-Asu-OBzl (**14**; 20.7 g, 0.5 mol) in CH_2Cl_2 (100 mL) was allowed to react with isobutene (50 mL) in the presence of H_2SO_4 (0.5 mL). After 2 d at rt, the mixture was poured into 5% Na_2CO_3, and then CH_2Cl_2 and the excess isobutene were removed under reduced pressure. The oily residue was taken up in Et_2O, and the Et_2O soln was washed successively with 5% $NaHCO_3$, H_2O, 0.2 M HCl, and H_2O. The organic layer was concentrated and the oily residue [Z-L-Asu(tBu)-OBzl, **15**] was saponified in acetone (200 mL) with 1 M NaOH (50 mL) for 3 h at −5 °C. The mixture was neutralized with 1 M HCl and the acetone removed. The residual soln was partitioned between Et_2O and 5% $NaHCO_3$. The aqueous layer was acidified with 1 M HCl and the oily precipitate was extracted with EtOAc. The soln was concentrated and upon addition of piperazine hexahydrate (4.4 g, 0.0225 mol) in acetone the soln was taken to dryness. The solid was triturated with EtOAc and recrystallized (EtOAc); yield: 16.0 g (76%); mp 130–131 °C; $[\alpha]_D^{22}$ −2.7 (*c* 2, AcOH).

Boc-(γ,δ)-(*Z*)-dehydro-L-Asu-OtBu (23):[28]
To a stirred mixture of $Ph_3P^+(CH_2)_3CO_2H \cdot Br^-$ (7.20 g, 16.75 mmol), anhyd THF (50 mL), and anhyd HMPA (20 mL) at 0 °C was added dropwise over 15 min 1 M LiHMDS in THF (28.5 mL, 28.5 mmol). Upon completion of the addition, the orange-red mixture was stirred at 0 °C for 1 h and then cooled to −78 °C. A soln of *tert*-butyl L-2-(*tert*-butoxycarbonylamino)-4-oxobutanoate (**22**) (2.7 g, 9.88 mmol) in anhyd THF (25 mL) was added dropwise (25 min). The resulting yellow mixture was stirred at −78 °C for 30 min and then at 0 °C for another 30 min. The mixture was poured into aq 10% citric acid and was extracted with Et_2O. The Et_2O soln was washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure to give a yellow, viscous oil. The oil was dissolved in EtOAc and the soln was washed with 10% LiCl, H_2O, and brine. The organic phase was dried and concentrated under reduced pressure to give a yellow oil that was purified by column chromatography (silica gel, EtOAc/hexane/AcOH 600:200:5) to give the title compound **23**. The product was crystallized (hexane); mp 80–83 °C; $[\alpha]_D^{22}$ −4.1 (*c* 1, EtOH).

The subsequent hydrogenation, deprotection and reprotection steps to produce the derivative **25** were performed according to standard procedures.

6.2.4.1.2.3 Synthesis of Peptides Containing α-Aminosuberic Acid

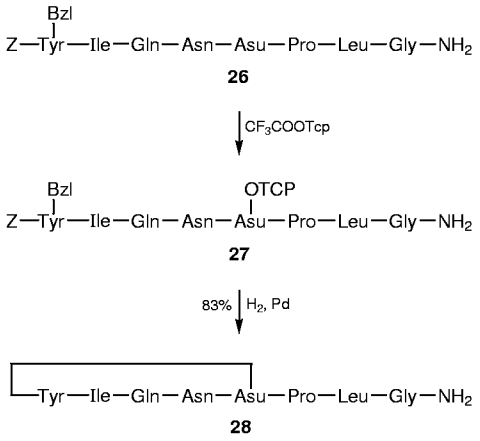
In the early synthesis of deamino-dicarba-oxytocin, the intermediate Z-Asu(OMe)-OH was used which requires a saponification step prior to cyclization.[1,2] Subsequently, a synthesis more consistent with the general protection strategies in peptide synthesis was developed with the intermediate Z-Asu(OtBu)-OH.[3,24] As outlined in Scheme 9, upon selective deprotection of the side-chain carboxy group of the Asu residue by exposure to TFA, the octapeptide derivative **26** is converted into the 2,4,5-trichlorophenyl ester **27** using the trifluoroacetate method.[29,30] Hydrogenolytic N^{α}-deprotection of **27** in dilute solution leads to

for references see p 233

in situ cyclization into **28** most probably on the catalyst surface and, thus, with good yields of cyclic monomer.[3]

The deamino-dicarba-analogues of various biologically active cystine-containing peptides, such as oxytocin,[1–3] calcitonin,[31,32] and somatostatin,[33–35] have shown high biological activity and enhanced metabolic and chemical stability owing to the absence of reducible disulfide linkages.

Scheme 9 Synthesis of [Asu^{1,6}]-Oxytocin by Cyclization via the Active Ester Method[3]



[Asu^{1,6}]-Oxytocin (28):[3]
The octapeptide **27** (419 mg, 0.3 mmol) was hydrogenated over Pd/C in DMF (300 mL). The catalyst was removed by filtration, the filtrate concentrated, diluted with H_2O (27 mL), and extracted with Et_2O. The aqueous soln was passed through an ion-exchange column and the oxytocin analogue was obtained upon chromatographic purification; yield: 62 mg (83%); $[\alpha]_D^{22}$ −98.4 (*c* 0.46, H_2O).

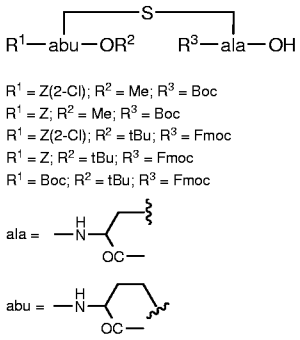
6.2.4.2 Carba-analogues of Cystine

More proper analogues of cystine than the α-aminodicarboxylic acids are α,α′-diaminodicarboxylic acids that contain all the functional groups of cystine as well as the correct length of the Cα to Cα′ spacer. For this purpose first synthetic approaches were based on the use of cystathionine as the monocarba-analogue[4,36] which were then followed by the use of dicarba-analogues.

6.2.4.2.1 Monocarba-analogues

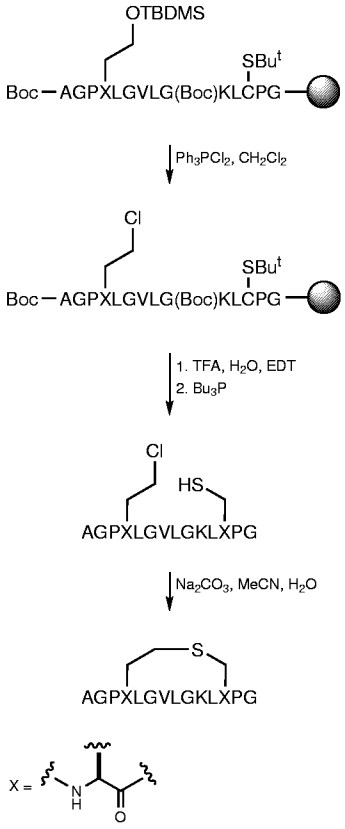
Selectively protected cystathionine derivatives (Scheme 10) are readily synthesized by S-alkylation of suitably protected cysteine or homocysteine with fully protected derivatives of L-2-amino-4-bromobutanoic acid.[4,36–41] The choice of the protecting groups is conditioned by the overall protection strategy applied, e.g. the cystathionine derivatives Boc-Cys-OH/H-abu-OMe (abu = -NH-CH(CH_2)$_2$CO-) and Z-hCys-OtBu/Boc-ala-OMe (ala = -NH-CH(CH_2)-CO-) were used for the synthesis of 7-monocarba-analogues of human calcitonin[39] and heat-stable enterotoxin (6- and 7-monocarba-analogues),[41] respectively. More recently, a combination of Boc-abu-OtBu/Fmoc-Cys-NH_2 was used for the synthesis of monocarba-analogues of disulfide bridged peptides.[42] Similarly, the combination Z(2-Cl)-Abu-OtBu/Fmoc-Cys-OH was applied in a successful synthesis of 1-carba-oxytocin on solid support.[40]

Scheme 10 Orthogonally Protected Cystathionine Derivatives as Intermediates for Peptide Synthesis



Postsynthetic intramolecular alkylation of a cysteine thiol group with a bromo amino acid has also been proposed,[43] but this approach is limited by the low coupling efficiency of the bromo amino acid in the peptide synthesis due to a competing intramolecular cyclization reaction. A promising new approach consists of using suitably protected cysteine and homoserine derivatives for the peptide synthesis, whereby the side-chain hydroxy group of homoserine is protected as the TBDMS derivative which is converted on resin into the chloro derivative with freshly prepared triphenylphosphine dichloride. Upon cleavage from the resin, cyclization is performed in solution as shown in Scheme 11.[44]

Scheme 11 Postsynthetic Sulfide Bridging of Peptides to Cystathionine Peptides[44]



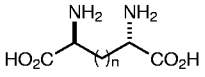
for references see p 233

Boc-Cys-OH/H-abu-OMe:[38]
N,N'-Bis(Boc)cystine (13.2 g, 30 mmol) was reduced by Na in liq NH_3 (300 mL). The soln was decolorized with NH_4Cl, and L-2-amino-4-bromobutanoic acid methyl ester hydrochloride (20.8 g, 90 mmol) was added and the soln lyophilized. The lyophilizate was dissolved in 0.5 M $NaHCO_3$ (250 mL) and extracted with Et_2O. Subsequently, the soln was acidified to pH 3.0 by 1 M HCl, evacuated, and filtered through a column of Dowex 50W-X4 (H^+ form, 1000 mL). The column was washed with H_2O and the product was displaced by 1% pyridine (cooled to 0 °C). The effluents were acidified with AcOH to pH 6.0 and taken to dryness. The residue was reprecipitated from MeOH with Et_2O and then separated from the contaminating *N,N'*-bis(Boc)cystine by countercurrent distribution; yield: 11.8 g (52%); mp 185–186 °C; $[\alpha]_D^{22}$ +37.2 (*c* 0.5, MeOH).

6.2.4.2.2 Dicarba-analogues

The constituent amino acid of dicarba-analogues of cystine is L,L-2,7-diaminosuberic acid [(2*S*,7*S*)-2,7-diaminooctanedioic acid, **29**, n = 4]. Despite the simplicity of this structure (Scheme 12), the two amino and two carboxy groups must be selectively protected for convenient application in solution and/or solid-phase peptide synthesis.

Scheme 12 α,α'-Diaminodicarboxylic Acids as Dicarba-analogues of Cystine

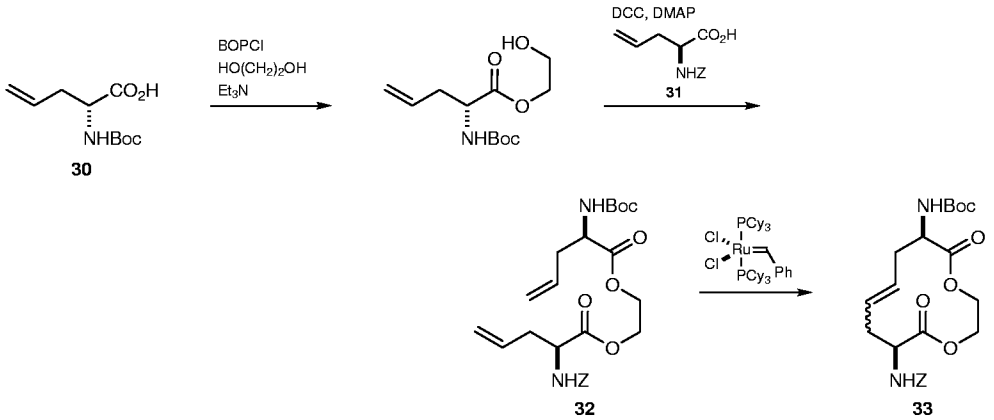


29

n = 0–6

N^{α}-Boc-$N^{\alpha'}$-Z-L,L-diaminosuberic acid α-*tert*-butyl ester [*tert*-butyl (2*S*,7*S*)-2-(benzyloxycarbonylamino)-7-(*tert*-butoxycarbonylamino)octanedioate] is synthesized by the mixed Kolbe electrolytic carboxylative dimerization of Boc-Glu-OtBu and Z-Glu-OMe as described in the synthesis of α-aminosuberic acid (Section 6.2.4.1.2.2), although this procedure yields a statistical mixture of the two undesired symmetrical dimers which must be separated from the desired unsymmetrical dimer.[27] An alternative asymmetric synthesis of L,L-diaminosuberic acid[45] or differentially protected derivatives such as N^{α}-Boc-$N^{\alpha'}$-Z-L,L-diaminosuberic acid[46] or N^{α}-Boc-L,L-diaminosuberic acid α-τ-phenyl ester[47] is carried out by a ring-closure metathesis approach using allylglycine-derived templates, as shown in Scheme 13 for N^{α}-Boc-$N^{\alpha'}$-Z-*meso*-diaminosuberic acid. Boc-D-allylglycine (**30**) and Z-L-allylglycine (**31**) are sequentially esterified on to ethylene glycol to form diester **32** (Scheme 13). Reaction with the Grubbs catalyst[48] produces the cyclic alkene **33**, which is hydrolyzed and hydrogenated under standard conditions to generate the protected 2,7-diaminosuberic acid derivative.

Scheme 13 Stereoselective Synthesis of Selectively Protected *meso*-2,7-Diaminosuberic Acid



N^{α}-Boc-$N^{\omega'}$-Z-L,L-diaminosuberic acid ω-*tert*-butyl ester or bis(N^{α}-Boc)-L,L-diaminosuberic acid synthesized by the mixed Kolbe electrolytic carboxylative dimerization method was successfully utilized in the synthesis of nonreducible bicyclic analogues of somatostatin[49] or hematoregulatory peptide (pGlu-Glu-Asp)$_2$-Dsu-(Lys)$_2$, respectively.[50]

The Grubbs method[48] is readily applicable to the synthesis of other stereoisomers and can be expanded to include homologues with additional methylene units spanning the glycyl substructures. This method can also be applied to the stereoselective synthesis of selectively protected *meso*- or L,L-2,6-diaminopimelic acid and related differentially protected derivatives.[46]

For the further shortened analogues reported in Scheme 12, such as L,L-2,5-diaminoadipic acid [(2*S*,5*S*)-2,5-diaminohexanedioic acid, **29**, n = 2],[50] L,L-2,4-diaminoglutaric acid [(2*S*,4*S*)-2,4-diaminopentanedioic acid, **29**, n = 1][51] and L,L-2,3-diaminosuccinic acid (**29**, n = 0)[52] stereoselective syntheses are known. The higher homologue of L,L-2,7-diaminosuberic acid is L,L-2,8-diaminoazelaic acid [(2*S*,8*S*)-2,8-diaminononanedioic acid, **29**, n = 5], which can be synthesized[50] using the Schöllkopf asymmetric alkylation method.[53] The same procedure is used for the synthesis of L,L-2,9-diaminosebacic acid [(2*S*,9*S*)-2,9-diaminodecanedioic acid, **29**, n = 7, a two-methylene extended analogue]. Their use for the interchain crosslinking of peptides is potentially promising, as recently observed for dimeric bradykinin antagonists.[54]

(6*R*,11*S*)-11-[(Benzyloxycarbonyl)amino]-6-[(*tert*-butoxycarbonyl)amino]-5,12-dioxo-1,4-dioxacyclododec-8-ene (33):[46]

All reagents and solvents were dried and degassed. A round-bottomed flask charged with catalyst bis(tricyclohexylphosphine)benzylidineruthenium dichloride (17 mg, 0.021 mmol) was evacuated and filled with argon three times before the addition of **32** (50.8 mg, 0.104 mmol) in degassed CH_2Cl_2 (15 mL) by an airtight syringe. The resulting orange soln was stirred under argon for 1.5 h and then stirred for 2 h exposed to the air. The soln color changed from orange to dark black. Removal of the solvent under reduced pressure gave an oily residue that was purified by flash chromatography (EtOAc/hexane 1:3) to afford **33** as an equal mixture of double-bond isomers; yield: 17 mg (35%).

[1] Rudinger, J.; Jošt, K., *Experientia*, (1964) **20**, 570.
[2] Jošt, K.; Rudinger, J., *Collect. Czech. Chem. Commun.*, (1967) **32**, 1229.
[3] Kobayashi, A.; Hase, S.; Kiyoi, R.; Sakakibara, S., *Bull. Chem. Soc. Jpn.*, (1969) **42**, 3491.
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6.3 Synthesis of Glycopeptides

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Carbohydrates and polysaccharides, on the one hand, and peptides and proteins, on the other, have been considered as separate classes of natural products for a long time. Fundamental chemical methodology for the synthesis of both saccharides and peptides was developed by Emil Fischer et al. at the beginning of the 20th century.[1,2] However, the harsh conditions employed in early solution and solid-phase peptide synthesis hindered the combination of peptide and carbohydrate chemistry, i.e. glycopeptide synthesis. Considerable efforts were made to combine the two branches of natural product chemistry, and the state of the art within glycopeptide synthesis has improved dramatically during the last decades, as described in a number of reviews.[3–23,512–514]

Since this is the first *Houben–Weyl* review of glycopeptides, we first present general considerations important for glycopeptide synthesis, then follow with selected examples of more specific targets.

6.3.1 Naturally Occurring Carbohydrate Peptide Linkages

Glycosylation is the most common posttranslational modification of proteins in eukaryotic cells. The carbohydrate portions of glycoproteins show a large diversity and play important roles in the organized distribution of these macromolecules within the cells and within multicellular organisms.[24–27] In particular, they are recognition signals involved in intracellular communication, e.g. in cell adhesion, regulation of cell growth, infectious processes, and immunological differentiation.[28–30] Glycopeptides can be divided into two principal classes: (1) the N-linked and (2) the more diverse O-linked glycopeptides.

Most frequently found are *N*-glycoproteins, with an amide bond between glycan and the side chain of asparagine, the so-called N-glycosidic bond. The sites of N-glycosylation in *N*-glycoproteins are characterized by sequons Asn-Xaa-Ser/Thr (where Xaa is any amino acid except proline). One reported exception to this consensus sequence is nephritogenoside, where the amino terminal sequence Asn-Pro-Leu is modified by a trisaccharide consisting of three glucose (Glc) residues, with α-Glc as the linkage unit.[26] The carbohydrate moiety is more variable, since several asparagine-linked monosaccharides have been reported in bacterial glycoproteins, including α- and β-Glc, β-*N*-acetylgalactosamine (GalNAc), and rhamnose (Rha).[28]

O-Linked glycoproteins are far more diversified than the *N*-glycoproteins. The most abundant is the mucin-type with an α-linkage between GalNAc and serine or threonine (Section 6.3.2.4). The first description of a β-D-GalNAc-Thr linkage in glycoproteins (S-layer) has been reported.[31] Other amino acids reported to be modified by O-linked glycosylation include hydroxyproline (Section 6.3.2.5), hydroxylysine (Section 6.3.2.5), and tyrosine (Section 6.3.2.6). Examples of other saccharides linked to serine or threonine include α-galactose (Gal) in potato lectin and some collagens,[28] Glc in bovine and human blood coagulation factors,[32] fucose (Fuc) (Section 6.3.2.1.1), mannose (Man) (Section 6.3.2.2.1), and xylose (Xyl) (Section 6.3.2.2.2). A separate class of O-linked glycosylation, which appears to be functionally distinct, is the *O*-GlcNAc modification found as monosaccharides attached to either serine or threonine (Section 6.3.2.4.3).[33,34] A series of O-linked *N*-GlcNAc bound oligosaccharides have also been isolated from *Trypanozoma cruzi*.[35] The C-glycosidic attachment of a Man residue to the side chain of tryptophan has been discovered (Section 6.3.4) and a few S-linked glycoproteins have been reported (Section 6.3.5). In the corn starch primer protein the β-Glc moiety of the starch molecule is

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linked to the protein via the guanidino group of an arginine residue.[36] Theonellamides are a family of bicyclic peptides with antifungal activity, which contain a glycopyranosyl unit attached to the π-nitrogen of a τ-histidinoalanine bridge.[37,38] There are also a number of different glycoproteins identified in slime molds and several unicellular parasites that contain oligosaccharides linked to serine or threonine in a peptide backbone via phosphodiesters (phosphoglycosylations).[39]

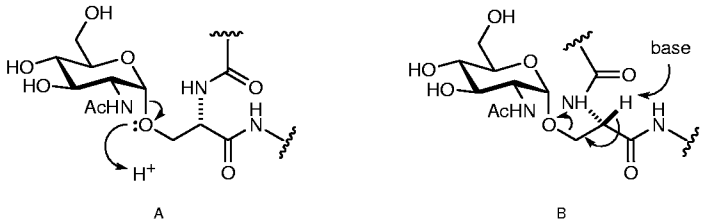
6.3.1.1 Strategies of Synthesis

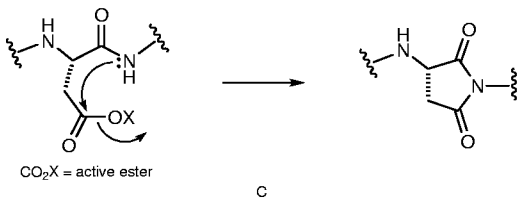
For glycopeptide synthesis, much attention has to be paid to the stereoselective formation of the natural-type glycosidic bond which must remain unchanged in the course of the synthesis. Therefore, in comparison to peptide synthesis, the compatibility and chemoselectivity of the applied reaction is fundamental in glycopeptide synthesis.

The assembly of glycopeptides can be performed either in solution or on solid phase. The carbohydrate moiety may be introduced in the final step of the synthesis by (1) selective N-acylation of a glycosylamine or O-glycosylation of the peptide (direct condensation approach) or (2) sequentially by the use of glycosylated amino acid building blocks. Although the direct condensation approach may be an appropriate choice with biologically isolated and very rare complex-type glycosylamines for optimal formation of N-linked glycopeptide, it is not well suited for the more difficult generation of O-glycosidic bonds. In the latter case, the nature of the peptide and the reaction conditions required for a successful synthesis are often incompatible with the postsynthetic glycosylation. Furthermore, problems arise with regioselectivity and anomeric selectivity. Conversely, the building block strategy, in which the selectivity problem is solved prior to the peptide assembly, is much more versatile and general and has been successfully employed for the synthesis of a large variety of both N- and O-linked glycopeptides. The building block approach has advanced to a state that allows glycosyl amino acids to be used in multiple[40–44] or library[45,46] syntheses to generate large ensembles of closely related glycopeptide compounds. Condensation of peptide and glycopeptide fragments in solution is only rarely employed, and is probably most suited for large-scale production of glycopeptides.

Additional problems must be taken into account when developing synthetic strategies for glycopeptides. Glycosidic bonds are labile toward strong acids, e.g. HF (Scheme 1, A),[47] and glycopeptides might undergo side reactions such as β-elimination of glycosylated serine or threonine residues (Scheme 1, B),[48] and/or epimerization of peptide stereocenters or aspartimide formation (Scheme 1, C; see also Section 6.3.3.2). Although less than initially anticipated,[49] this labile character limits the variety of protecting groups that can be applied for glycopeptide synthesis, and the judicious choice of compatible methods determines the success of the syntheses of these polyfunctional molecules.

Scheme 1 Potential Side Reactions of Glycopeptides: (A) with Acid, (B) with Base, and (C) by Aspartimide Formation[47–49]





C

6.3.1.1.1 Synthesis in Solution

The solution methods require orthogonal protection of the C- and N-termini of the *N*- or *O*-glycosyl amino acid or peptide as well as of the glycan throughout the synthesis. Furthermore, protection of the C- and N-termini should both be of the temporary type, which can be removed under mild selective conditions to have the peptide elongated in either direction.

Often, in situ coupling conditions have been applied in solution synthesis using EEDQ or the corresponding diisobutyl reagent IIDQ as the coupling reagents.[50–66] These reagents form very reactive mixed anhydride intermediates and it is important to be aware of the possible side reactions, in particular when they are used in difficult and slow couplings with glycosyl amino acid derivatives and in fragment condensations. The prevailing use of these reagents may be due to the ease of purification of the product. The EEDQ reaction has been compared with the BOP/HOBt procedure and the latter was found to be superior.[64]

Another popular procedure is the combined use of DCC or *N*-ethyl,*N*-(diisopropylaminoethyl)carbodiimide with HOBt.[66–78] In these reactions, very reactive *O*-acylisourea intermediates are formed and the reaction temperature and the mode of mixing of the reagents have a major influence on the outcome of the reaction.[79] *N*-Ethyl,*N*-(diisopropylaminoethyl)carbodiimide is used to avoid difficulties in the separation of reaction products. Both this reagent and EEDQ have frequently been employed in fragment coupling reactions.[53,57,64,80] When problems have been observed, other methods have been used. Thus, in the early and extremely elegant synthesis reported for bleomycin,[70] fragment coupling was achieved with diphenylphosphoric azide (DPPA); however, a similar reagent, *O*,*O*-diethyl cyanophosphate, has been less successfully used in the C-terminal elongation of glycopeptides.[81] The BOP/HOBt procedure was found to give high yields in fragment assembly of a glycohexapeptide from fibronectin.[64] Problems observed in obtaining pure products with the DCC procedure were avoided[82–84] with Woodward's Reagent K,[85] a reagent that was often employed in solution synthesis in the 1970s. The suppression of epimerization in fragment coupling may be affected by an order of magnitude[86] by replacing the HOBt with the catalyst HODhbt[50] in the DCC/HOBt procedure, but it is important to be aware of a side reaction which can cause the formation of 2-azidobenzoic acid. However, this side reaction can be avoided by the correct order of reagent mixing.[87] Another approach introduced by Pavia et al.[88–90] in 1976 for coupling in solution synthesis was the application of weakly activated active esters such as pentachlorophenyl, *N*-hydroxysuccinimide, or even 2- or 4-nitrophenyl esters that all are stable in solution and may be allowed to react with the nucleophile for a long period of time.

Synthesis of glycopeptides in solution has been successfully applied in the preparation of glycopeptides of moderate lengths (see Sections 6.3.2 and 6.3.3). However, yields in solution synthesis are often only modest and isolation of intermediates makes the approach inconvenient.

for references see p 322

6.3.1.1.2 Synthesis on Solid Support

The application of SPPS to the synthesis of glycopeptides began in the early 1980s on a Merrifield resin using a standard Boc-based strategy.[91] Polymer-supported synthesis of glycopeptides is now an established technique and has been reviewed.[92] Polystyrene polymers cross-linked with divinylbenzene (mostly used with DCC/HOBt coupling), the more polar kieselguhr-supported poly(dimethylacrylamide) resin[93] (mostly used with Pfp esters and in continuous flow synthesis), poly(ethylene glycol)–polyacrylamide copolymer[94] (PEGA resin, flexible and with high swelling capacity), and inert poly(ethylene glycol) crosslinked polymers[95] (polyethylene–polyoxypropylene, polyoxyethylene–polystyrene) have all been used successfully in glycopeptide synthesis. However, solid-phase synthesis involving cleavage of the product from the resin with HF is not compatible with the preparation of glycopeptides owing to the acid lability of the glycosidic linkages (Scheme 1, A).[47] Cleavage from the resin using strong base should also be avoided owing to the risk of β-elimination in O-linked glycopeptides (Scheme 1, B). Examples of linkers used in solid-phase glycopeptide synthesis are hydroxymethylphenoxyacetic acid, hydroxymethylbenzoic acid, Wang, Sasrin, Pal, Rink, Hycram, and a silyl ether type linker[515] that all allow cleavage from the resin under mild conditions.

Two methods of activation have prevailed in solid-phase synthesis of glycopeptides, i.e. the DCC/HOBt method[55,91,96–106] and the Pfp ester method.[107–117,119–120] The Pfp ester method has two obvious advantages: (1) the esters may be purified before use in the glycopeptide assembly and (2) they are easy to handle and therefore allow for automation and use in multiple methods, leading to many glycopeptides in a single operation.[109,119] The Pfp esters have often been used with addition of a catalyst, HOBt or HODhbt, the latter allowing the coupling reaction to be monitored.[121] When the Pfp ester is used without a catalyst, the excess of glycosylated amino acid Pfp ester building block may conveniently and quantitatively be recovered.[122] Excess of the activated glycosylated amino acid (1- to 1.5-fold) is sufficient to obtain complete and fast coupling.[119,123,124] Even an equivalent amount has been used without serious decrease of the overall yield.[125,126] The use of symmetrical anhydrides of glycosylated amino acids has also been reported,[114,116,127] as well as the use of HBTU with the addition of HOBt.[128] The almost exclusive use of the Fmoc group for N^{α}-protection in solid-phase synthesis of glycopeptides is mainly due to the mild conditions for its cleavage, e.g. with morpholine or piperidine.

The different techniques used in solid-phase glycopeptide synthesis, such as the low-cost plastic syringe technique or the multiple column technique, and also one type of fully automated synthesis, have been reviewed by Peters et al.[18]

6.3.1.1.3 Direct Condensation Strategy

The condensation of a synthetic peptide acceptor with an oligosaccharide donor to form an N- or an O-linked glycopeptide in solution or on solid phase is an appealing approach. There are, however, serious problems associated with this methodology. Most obviously, the nonpolar conditions required for the O-glycosylation are incompatible with the solubility properties of longer peptides. Therefore, direct condensations have been performed with small-size peptides, but almost always in low yields, e.g. lipophilic Z-Ala-Ser-Gly-Ile-OBzl has been glycosylated in nitromethane with the reactive donor Ac_4-β-D-Xyl and TMSOTf catalysis in 57% yield.[58] Similarly, the condensation of a tyrosine-containing pentapeptide with 2,3,4,6-tetra-*O*-benzyl-D-glucose promoted by DCC and CuCl afforded only 6% of the desired glycopeptide, whereas a stepwise assembly with the glycosylated tyrosine derivative via DCC/CuCl coupling was more efficient (26% yield).[129] The dipeptide Z-Ser-Ser-OBzl has been reacted with the T-antigen precursor Ac_4-β-D-Gal-(1→3)-Ac_2-α-D-$GalN_3$-Br in the presence of $AgClO_4/Ag_2CO_3$ to yield the diglycosylated peptide in 50% yield.[130] When

serine or threonine residues are present at the N-terminus of a peptide, they may be activated for glycosylation by hydrogen bonding to a Schiff base formed with the α-amino group.[76,131] Lipophilic hexapeptides have also been successfully O-glycosylated using triacetylgalactal or triacetylglucal and NIS, affording the 2-iodo-2-deoxy-α-glycoside.[132,133]

A comparison of direct O-mannosylation of serine or threonine on resin-bound peptides, with the incorporation of mannosylated serine or threonine building blocks, resulted in a higher overall yield for the building block approach (12.5 vs 5% for direct condensation).[134] Later it was demonstrated that resin-bound peptides are able to be glycosylated with different glycosyl trichloroacetimidates without protection of the hydroxy groups of the oligosaccharide moiety (Section 6.3.2.8).[135,136]

The succinimido ester of a tripeptide aspartic acid side chain was found to react with GlcNAc-NH_2 in 35% yield.[137,138] *O,O*-Diethylcyanophosphate has been used as an in situ activation reagent for the coupling of a trisaccharide glycosylamine with Boc-Asp-Pro-OMe, affording 64% of the glycopeptide.[139,140] The conditions for in situ activation with BOP and HBTU have been investigated, and both reagents were found to give good yields when a tri- or a pentapeptide was coupled with GlcNAc-NH_2. This method has successfully been used by Arsequell et al.[141] to produce a long peptide-containing chitobiose and a short peptide linked to a complex oligosaccharide.

Condensation between a glycosylamine and a peptide containing aspartic acid has been accomplished in the synthesis of N-linked glycopeptides. Using this approach, formation of aspartimide (succinimide) from the activated asparagine must be avoided and at the same time activation must be sufficient for the two large molecules to react. The mechanism of formation of aspartimide under base-catalyzed acetylation conditions (e.g., piperidine) is via abstraction of the aspartyl amide nitrogen proton with concurrent nucleophilic attack on the β-carboxy group (Scheme 1, C).

Several approaches to solve the problem of aspartimide formation during direct condensation glycopeptide synthesis have been reported. One strategy has been to substitute glutamyl for aspartyl as the residue for glycosylation, giving a nonnatural linkage which is useful for glycopeptide libraries.[142,143] Alternatively, the aspartyl amide bond contains a C-terminal proline residue.[142] Here the amide proton is absent and can no longer participate in formation of aspartimide. This technique, however, is not a general solution. The N-acylation of glycosylamines with C-terminal aspartic acid side chains is another strategy to avoid unwanted formation of aspartimide. The use of a large excess of glycosylamine to act both as base and substrate was recommended for aspartimide-prone amides.[144] However, this is prohibitive if limited quantities of glycan are available. Another approach is to replace the amide nitrogen proton of the aspartyl–Xaa bond with an acid-labile backbone-protecting group, e.g. Hmb, which has been applied to the solid-phase synthesis of N-linked glycopeptides through the on-resin activation (BOP/HOBt) of the aspartyl β-carboxy group (Section 6.3.3.2).[145] A bivalent N-linked glycopeptide was synthesized by HBTU-promoted coupling in solution of an unprotected sulfated lactosyl derivative, with a linker at the reducing end that terminates in an amino group, to two aspartic acid residues in an octapeptide, in excellent yields.[146] The success of this method is attributed to the use of the more reactive primary alkyl amine, which eliminates the aspartimide formation frequently observed when coupling glycosylamines. Coupling of an unprotected glycosylamine to a pentafluorophenyl carboxylate on the aspartic acid side chain of a resin-bound peptide was reported to be successful if the stoichiometry of glycosylamine, DIPEA, and HOBt were optimized.[143] Inversed convergent solid-phase synthesis of glycopeptides may circumvent some of the problems using the direct condensation approach[147] (see also Section 6.3.3.2).

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6.3.1.1.4 Building Block Strategy

The most versatile methodology in the chemical synthesis of glycopeptides is the use of glycosylated amino acids as building blocks. In this strategy, the difficulties of generating the crucial bond between the oligosaccharide and the amino acid are solved before the glycosyl amino acid is incorporated into the peptide, by stepwise or fragment assembly, in solution, or on solid phase. Most syntheses have been performed by solid-phase techniques where it is of importance that the building blocks can be easily and selectively converted into a C-terminal, highly activated derivative for coupling on the solid phase. Even amino acids that carry oligosaccharides as large as heptasaccharides have been found to perform well in the coupling steps.[123,148,149] The building block approach conceptually comprises eight crucial steps, which must all be successful to yield pure glycopeptides: (1) synthesis of an oligosaccharide which is readily converted into the appropriate glycosyl donor; (2) preparation of a suitably protected amino acid derivative; (3) condensation of the oligosaccharide and the amino acid; (4) deprotection of the C-terminus of the amino acid; (5) activation of the amino acid carboxy group; (6) assembly of the glycopeptide; (7) detachment and complete deprotection of the glycopeptide; and (8) purification of the product. Steps 4 and 5 may be omitted if preactivated amino acids are used. Many examples of the building block approach are given in Sections 6.3.2–6.3.6.

6.3.1.1.4.1 Protection of the Amino Acid

The application of Fmoc protection of the α-amino group serves most purposes in glycopeptide synthesis. The Fmoc group is stable to the acidic conditions normally used in O-glycosylations, and also survives the reaction conditions for the condensation of an aspartic acid side chain with glycosylamines. Furthermore, it can be removed under mild basic conditions, e.g. with 50% morpholine, 20% piperidine, or 2% DBU in DMF, all reaction conditions under which no β-elimination or epimerization of peptide stereocenters occurs.[49,120,150,151] Alternative base-labile protecting groups for the α-amino group are the 4,4′-dinitrophenylethyloxycarbonyl,[152] triethylphosphonium(methyloxycarbonyl),[153–155] 2-pyridylethyloxycarbonyl,[6] and 4-methylsulfonylphenylsulfonylethoxycarbonyl[156] groups. The Aloc[157] and Noc[158] groups can also be used for this purpose since they are removed under neutral conditions. The Noc, Aloc, and other allyl-derived protecting groups[159] are useful for the selective access to particular functionalities in side chains of assembled peptides.[160] An early approach to solid-phase synthesis of glycopeptides relied on N^{α}-Boc protection. However, repeated N^{α}-Boc deprotection with TFA and, in particular, the final cleavage from the solid phase under strong acid conditions, also hydrolyses glycosidic bonds. Synthesis of glycopeptides with Boc protection of the α-amino group, using the allylic-type of linker (Hycram), has been reported where the glycopeptide is released by allylic transfer with $Pd(PPh_3)_4$.[103]

In the synthesis of building blocks, the C-terminal carboxy group is used either protected or unprotected. For semipermanent C-terminal protection, where the N^{α}-protection is removed before the conversion into an activated amino acid, *tert*-butyl, benzyl, 1,3-dithian-2-ylmethyl,[161] 2-haloethyl,[162] and allyl esters have been employed.[6] Pac[65,101] and 9,10-anthraquino-2-yl methyl[69] esters, which can be removed by treatment with Zn in AcOH and sodium dithionite, respectively, have been used in combination with subsequent in situ activation with either EEDQ[65] or HOBt/DCC.[69,101] Alternatively, steps 3–5 mentioned earlier may be combined into one single step by use of an active ester, such as the Pfp ester, that is stable under the glycosylation reaction and therefore can subsequently be used directly in the glycopeptide synthesis. This method was introduced for solution synthesis[89] and later refined as a general method for solid-phase glycopeptide synthesis.[108,112,115] In some cases, it was more advantageous to glycosylate the more stable, yet more synthetically challenging,

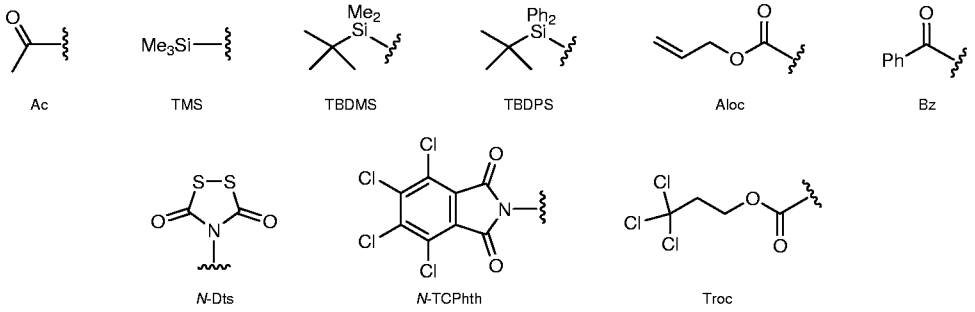
Fmoc-Ser/Thr-OtBu esters and cleave the ester to provide a building block suitable for synthesis.[43,163] The use of Fmoc amino acids, where the α-carboxy group of the amino acid is unprotected, has also been reported,[164–167] as well as use of the corresponding O'Donnell Schiff bases.[516]

6.3.1.1.4.2 Protection of the Glycosyl Moiety

For protection of the hydroxy groups of the glycosyl moiety, esters such as acetates and benzoates have been employed most frequently because of their easy removal under mild basic and homogeneous conditions.[101,168] It has also been found that cleavage of acetates with base does not epimerize the amino acid.[169] The acetyl groups appear to stabilize the glycosidic linkage, making it more resistant toward TFA hydrolysis.[170] More drastic conditions were required for the removal of benzoyl groups and side products have been observed.[169] Protecting groups that require heterogeneous catalysts for cleavage may work well with small molecules, but they are not as useful for the synthesis of larger glycopeptides. Furthermore, not all the functionalities of amino acids, e.g. those in Cys, Met, and Trp, are compatible with, for example, hydrogenolysis conditions. Therefore, protecting groups such as benzyl ethers have generally been exchanged with acetates before the building block was used in glycopeptide synthesis.

Ether-type acid-labile protecting groups have been introduced as an alternative to acyl protection for both O- and N-glycosylated amino acids. The TMS protecting group has been employed for 1-aminoalditols as well as oligosaccharides N-linked to asparagine.[123,171] The TBDMS and TBDPS groups are more stable than the TMS group and they are removed simultaneously with the acid cleavage of the peptide from the resin (Scheme 2).[125,172,517] Acid-labile protecting groups are incompatible with peptides carrying acid-labile deoxygenated sugars, e.g. Fuc.

Scheme 2 *O*- and *N*-Sugar Protecting Groups Used in Glycopeptide Synthesis



The use of glycosyl azides as precursors of glycosylamines has proven to be a very versatile approach[173,174] (Section 6.3.2.4.1.2.4). The introduction of the β-directing Dts as an oligosaccharide amino protecting group afforded a range of *N*-acetylglucosamines containing mucin glycosyl amino acids.[175–177] The Dts group is readily reduced to the amino functionality on solid phase under mild conditions and can also be directly converted into the *N*-acetate (Sections 6.3.2.4.1.2.4 and 6.3.2.4.3.1). Alternatively, the tetrachlorophthaloyl (TCPhth)[178] or the Troc group (Section 6.3.2.4.3.2) can be used quite efficiently for amino sugar protection and β-directing glycosylation. Also, the release of complex glycans by chemical means, followed by transformation into building blocks, has provided a versatile method of preparing large amounts of N-linked glycopeptides of a single glycoform.[149]

for references see p 322

6.3.1.1.5 Enzymatic Synthesis

There is substantial interest in developing methods that will permit modification of oligosaccharide structures on glycoproteins by removing and adding sugar units ("remodeling") and in making new types of protein–oligosaccharide conjugates. Modification of the sugar components of naturally occurring, or nonnatural, glycoproteins might increase solubility, increase stability in serum, decrease antigenicity, and promote uptake by target cells and tissues. In order to study the effects that carbohydrates have on glycoprotein structure and function, it is essential to be able to synthesize appropriate natural and nonnatural glycoprotein variants (glycopeptides) as single glycoforms. However, the microheterogeneity of the natural material makes the availability of pure glycoproteins difficult. Because available in vivo techniques, such as glycosyl engineering and cell-line mutations, provide only heterogeneous mixtures of different glycoforms, enzymatic in vitro methodologies have been developed. Enzyme-catalyzed reactions typically proceed in a mild and selective fashion, and therefore their application in the synthesis of glycopeptides and other glycoconjugates represents a promising approach.[518,519]

The use of glycosyltransferases to form glycosidic linkages promises a number of advantages. Many glycosyltransferases have been expressed and cloned by different laboratories; some are commercially available and are widely used in preparative work on oligosaccharides and glycopeptides.[179–181] These enzymes, which are often membrane proteins, are highly selective for the transfer of a few closely related analogues of the natural substrate to a specific position in the acceptor oligosaccharides.[71,182–184] They require uridine diphosphate sugars or other activated sugar phosphates, and the high selectivity and yield that can be obtained with the transferases make them attractive for the postsynthetic modification of glycopeptides[185] (Section 6.3.3.6). This is particularly true for the sialyltransferases,[174,186] which add the complex sialic acids to terminal 3- and 6-positions, since the sialic acids are very difficult to handle chemically.

The β-1,3- and β-1,4-galactosyltransferases are easily available and have been widely applied in oligosaccharide as well as glycopeptide synthesis.[187–190] Oligosaccharyltransferase catalyzes the transfer of an oligosaccharide, consisting of two GlcNAc, nine mannose, and three glucose units, from a dolichyl pyrophosphate intermediate to an asparagine residue of a nascent peptide or protein.[191] It also transfers the minimal unit β-D-GlcNAc-(1→4)-D-GlcNAc from the corresponding dolichyl pyrophosphate donor or from a derivative in which the lipid component is truncated or simplified.[192] Oligosaccharyltransferase has been utilized for the in vitro synthesis of several glycopeptides containing glycosylated asparagine residues.[71,193,194] The problem of the requirement of the C_{55} isoprenoid carrier has been addressed by substitution with a synthetic carrier,[193,195] allowing the transfer of chitobiose to asparagine-containing peptides.

The first successful solid-phase enzymatic glycosylation of a glycopeptide was performed on a nonswelling silica-based resin derivatized with a hexaglycyl aminopropyl spacer.[196] A small glycopeptide containing an Asn(*N*-β-D-GlcNAc) residue was enzymatically glycosylated, on the solid phase, first in the 4-position by galactosyltransferase (yield 55%) and then with 2,3-sialyltransferase (yield 65%). The obtained product could only be enzymatically fucosylated in a traditional solution-type reaction after enzymatic cleavage from the resin. To circumvent the problems due to inaccessibility of the interior of the solid matrix to enzymes, a new type of highly swelling and flexible PEG-based polymer in the form of beaded resin (PEGA resin) was designed. This resin allows quantitative solid-phase glycosylation of glycopeptides using glycosyltransferases.[197–199]

The improved availability of glycosyltransferases has led to increased use of chemoenzymatic synthesis, where chemical synthesis is combined with the use of glycosyltransferases to build up saccharides on a parent glycopeptide in solution or on solid phase (Section 6.3.3.6).[200–206]

As an alternative to the transferases, the glycohydrolases (glycosidases), which can be isolated in large amounts and are also commercially available, may serve as catalysts for glycosyl transfer from easily available starting materials such as lactose[207,208] or 2-nitrophenyl galactopyranoside.[208–210] Specificity for the anomeric configuration of the glycosidic bond formed and for the glycosyl donor is usually high, whereas the regioselectivity for the acceptor and the yields are very low. The method is therefore best applicable for small glycopeptides[207] and the glycosylation of serine derivatives.[208,211] In an alternative procedure, limiting amounts of the acceptor alcohol are used in combination with excess of the glycosyl donor.[212] Glycosylation of *N*-Boc-protected serine and threonine derivatives employing α-mannosidase and β-galactosidase as catalysts have been reported in poor yields (4–14%).[213] The first application of glycosynthases to solid-phase glycopeptide synthesis has been described.[520]

After significant effort, chemical syntheses of neoglycoconjugates have been developed and they usually involve multiple, laborious steps. Synthesis of the complex and high-mannose-type glycoconjugates proved to be especially difficult, even with enzymatic methods. *endo*-β-*N*-Acetylglucosaminidase from *Arthrobacter protophormiae* (Endo-A) cleaves the glycosidic bond in the core β-D-GlcNAc-(1→4)-D-GlcNAc residues of high-mannose-type and hybrid-type N-linked sugar chains in glycoproteins.[214] In addition, this enzyme is also reported to exhibit transglycosylation activity in the presence of GlcNAc and other monosaccharides.[215,216] Transglycosylation activity of *endo*-β-*N*-acetylglucosaminidase can be enhanced dramatically by inclusion of organic solvents, such as acetone, DMSO, or DMF, in the reaction medium.[217] Complex oligosaccharides have been successfully transferred from abundant and natural precursor glycoproteins or their pronase hydrolysates to synthetic GlcNAc-containing glycopeptides using endoglycanases in a reversed reaction in up to 8% yield.[201,218–222]

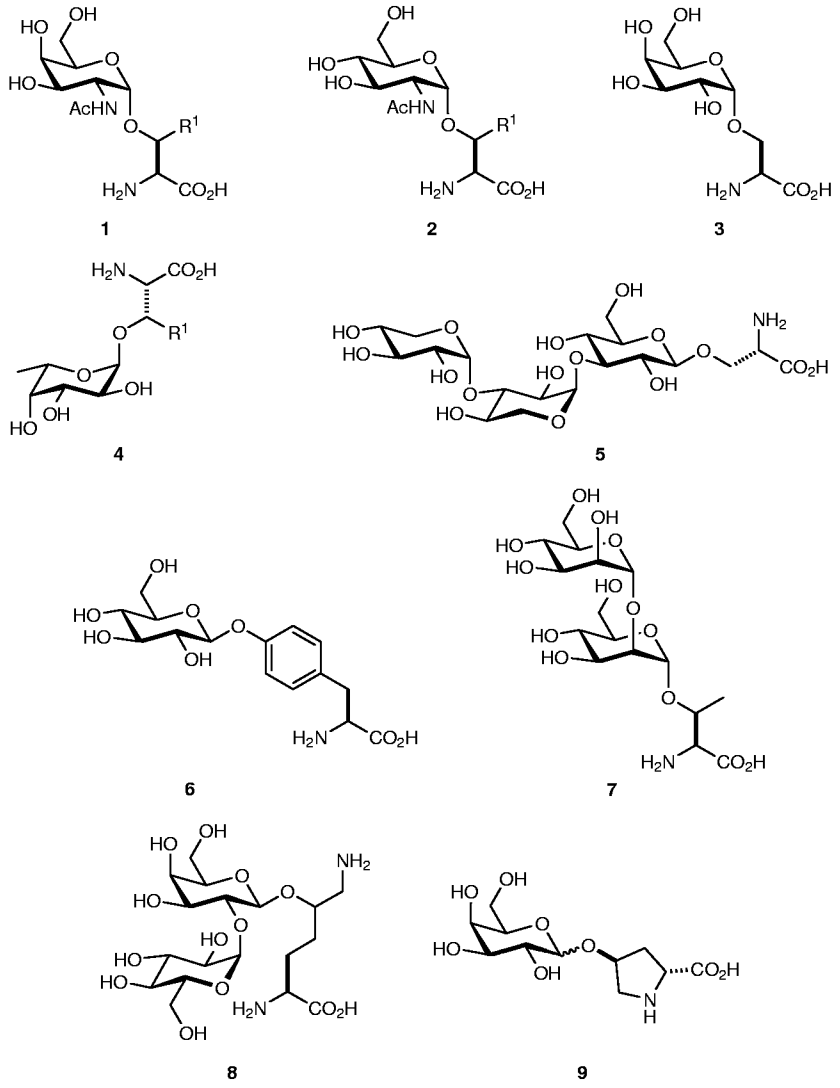
Another strategy for enzymatic glycopeptide synthesis is based on the use of engineered proteases coupled to glycosyltransferase reactions. Gene technologically transformed thiosubtilisin that favors aminolysis over hydrolysis has proven to efficiently catalyze the bond formation between a monoglycosylated peptide with another peptide. Both *N*- and *O*-glycopeptide fragments can be coupled in aqueous or organic solvents with nucleophilic amino acids, peptides, and their glycosylated derivatives based on subtilisin BPN′ and its variants designed to improve stability and selectivity. The glycopeptides obtained can be further elongated along the peptide backbone or glycosylated with glycosyltransferases.[9] Homogeneous RNase B glycoforms have been prepared via enzymatic glycopeptide condensation followed by glycosylations with glycosyltransferases,[203] and solution and solid-phase synthesis of N-protected glycopeptide esters of the benzyl type as substrates for subtilisin-catalyzed glycopeptide couplings has also been described.[223]

6.3.2 O-Linked Aliphatic Glycopeptides

Incidences of O-linked glycosylation occur primarily via the glycosylation of serine or threonine (**1**–**5**, **7**), although other amino acids such as tyrosine (**6**), hydroxylysine (**8**), and hydroxyproline (**9**) have been shown to be glycosylated in uncommon instances (Scheme 3). Usually these amino acids are glycosylated by any of the known monosaccharides in a 1,2-*cis* or 1,2-*trans* configuration. The glycan may be a simple monosaccharide or a complex glycan with five or more residues. The most abundant type of glycosylation is the mucin-type linkage between GalNAc and Ser or Thr, as shown in **1**.

for references see p 322

Scheme 3 Examples of Typical and Unusual O-Linked Glycosylations Found in Nature



Until recently, direct O-glycosylation of a peptide with an activated oligosaccharide had not been effective in solution or on solid phase. Partial success has been achieved in solution[58,224,225] and on solid phase[134,226,227] in low to fair yields, but only on very small peptides with very few glycosylation sites. Alternatively, the building block approach has proven to be the most versatile and efficient method for glycopeptide synthesis. Since one of the most challenging aspects of glycopeptide synthesis is the generation of glycosyl amino acid building blocks in a manner compatible with peptide chemistry, in this section, emphasis will be placed on the synthesis of suitable building blocks. The building blocks are classified according to the nature of the carbohydrate and the type of linkage to the amino acid. Once a suitable building block is synthesized, glycopeptide assembly is fairly routine. However, the cleavage and deprotection steps can be difficult and these will be addressed in Section 6.3.2.7.

6.3.2.1 1,2-*cis*-Glycosides

6.3.2.1.1 The Fucosyl Serine/Threonine Linkage

There have been increasing reports of a novel type of glycosylation consisting of fucose O-linked to Ser or Thr in the N-terminal epidermal growth factor (EGF) domains of coagulation and fibrinolytic proteins.[228–231] The same linkage was also found in a smaller peptide named PMP-C 1 isolated from the insect *Locusta migratoria*[232] and reported to be a serine protease inhibitor.[233] The biological function of this type of glycosylation is as yet unclear, thus making the chemical synthesis of fucosylated peptides of interest in order to elucidate their biochemical roles. Furthermore, L-fucose is known to be an important residue of branched oligosaccharides in glycoconjugates that bind selectins, and synthetic fucosylated peptides have proven to be mimics of sialyl Lewis X.[234]

6.3.2.1.1.1 Synthesis of O-Linked Fucosyl Amino Acids Using Benzyl Ether Type Protection

In nature, fucose is found α-linked to serine or threonine and consequently most syntheses focus on obtaining this stereochemistry. α-Linked fucosylated Thr and Ser derivatives have been prepared by the use of nonparticipating protecting groups such as benzyl ethers in the 2-position of the fucosyl donor. However, a major drawback of such a strategy is the acid lability of unprotected or benzyl-protected fucosides (5–6 times more labile than galactosides). Furthermore, benzyl protection is not fully compatible with peptide synthesis, particularly when sulfur-containing amino acids are present, which impair hydrogenolysis, and when susceptible amino acids such as tryptophans are hydrogenated under the hydrogenolysis conditions employed. Moreover, at the level of the building blocks, benzyl protection of fucose is not compatible with Fmoc-based solid-phase glycopeptide synthesis because the Fmoc group is not completely stable under the hydrogenolytic conditions required for the removal of benzyl ethers. Therefore, the amino group of the Ser or Thr building block is usually protected with a Z or Boc group and the carboxylic acid as the methyl ester. The resulting intermediates become useful for solid-phase procedures only after replacement of the benzyl groups with acetyl groups and conversion of the amino acid protecting groups into others that are more suitable for peptide synthesis.

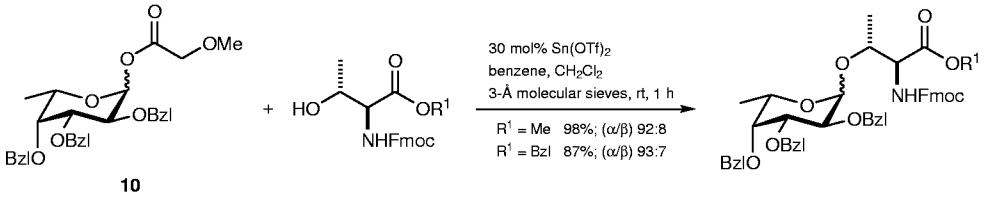
Despite these drawbacks, there are several reported procedures for the synthesis of fucosylated building blocks employing the established perbenzylated fucosyl phosphite[234] or bromide[235] as the donor. In the former case, treatment of Boc-Thr-OEt with the α-tribenzylfucosyl phosphite in the presence of catalytic TfOH yields the product as a mixture of anomers (α/β 71:29) in 72% yield. The Boc group was successfully removed without cleaving the fucosyl linkage by treatment with 30% TFA in CH_2Cl_2 for 30 min at 25 °C and the building block used in the solution-phase synthesis of sialyl Lewis X mimetics.[236] Attempts by the authors to synthesize a building block with acetyl groups on the fucose by reacting the triacetylfucosyl phosphite with the dipeptide Z-Ala-Ser-OMe resulted in a rather poor yield (18%), along with 35% of the orthoester and 68% of the phosphonate.[237] In the latter instance, glycosylation of Z-Ser-OBzl with α-tribenzylfucosyl bromide catalysed by Et_4NBr afforded the α-anomer in high yield (85%). For conversion into a building block suitable for Fmoc-based solid-phase synthesis, the benzyl groups were exchanged for acetyl groups and both the N- and C-terminal protecting groups were replaced. The overall yield of the building block suitable for SPPS was in fact 20%.

In another method, the need for exchanging the *N*-amino acid protecting group was precluded when Fmoc-Ser-OBzl and Fmoc-Ser-OMe were glycosylated stereoselectively and in high yield using the perbenzylated methoxyacetate **10** under $Sn(OTf)_2$ activation (Scheme 4).[238] Methyl esters of nonglycosylated amino acids can be selectively cleaved in the presence of the Fmoc group using $LiOH/CaCl_2$ solutions.[239] 2,3,4-Tri-*O*-benzylfucose has also

for references see p 322

been employed for the glycosylation of Z-Ser-OMe under TMSOTf activation at low temperatures, yielding the product in 74% yield (α/β 95:5).[240] The Z group could be selectively hydrogenolyzed in the presence of the benzyl protecting groups.

Scheme 4 Synthesis of Fucosyl Amino Acids Using the Perbenzylated Glycosyl Methoxyacetate[238]



10

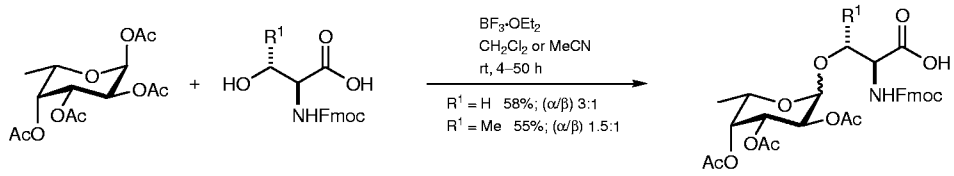
An alternative to the benzyl ether protection is the 4-methoxybenzyl group (Mob), which is acid labile and in principle can be simultaneously removed during acid cleavage of the glycopeptide from the solid support. The Mob-protected threonine building block was obtained in 57% yield by the reaction of Fmoc-Thr-OtBu with the thioethyl donor under dimethyl(methylsulfanyl)sulfonium triflate promotion.[241] However, even in this case, for use in SPPS the protecting groups had to be exchanged for acetyl protection for a more stable fucose derivative and the *tert*-butyl ester cleaved with formic acid. Obviously, benzyl or 4-methoxybenzyl protection adds several synthetic steps, reducing the overall yields.

Glycosylation with Fucosyl Methoxyacetate; General Procedure:[238]
To a stirred suspension of $Sn(OTf)_2$ (12.5 mg, 0.03 mmol), TMSOTf (37.2 mg. 0.3 mmol), and 3-Å molecular sieves (5 mg) in toluene/CH_2Cl_2 (3:1; 3 mL) was successively added 2,3,4-tri-*O*-benzyl-L-fucopyranosyl methoxyacetate (50.7 mg, 0.1 mmol) and the amino acid derivative (1.3 equiv) in CH_2Cl_2 (1.5 mL) at rt. The mixture was stirred for 20 h at rt, then quenched by adding sat. $NaHCO_3$. After workup and purification with preparative TLC (silica gel), the desired fucoside was isolated; yield: 98%. The ratio of anomers was determined by HPLC analysis.

6.3.2.1.1.2 Synthesis of O-Linked Fucosyl Amino Acids Using Other Carbohydrate Protection

Two methods that shorten the synthesis of fucosyl building blocks have also been described. In the first, persilylated fucose is converted into the glycosyl iodide and then coupled to Z-Ser-OBzl, giving exclusively the α-anomer in 68% yield.[242] Although this building block has not yet been used in solid-phase synthesis, protection of the glycan with acid-labile TMS groups provides the possibility of simultaneous deprotection along with acid cleavage of the peptide from the resin. In the second approach, peracetylated fucose is reacted with Fmoc-Ser/Thr-OH under Lewis acid catalysis, yielding an approximately 1:2 mixture of α/β anomers **11** in ca. 55% total yield (Scheme 5).[243] The building blocks can be directly used in SPPS. Although there are fewer synthetic steps and the overall yield is increased, the need for HPLC separation of the anomers makes bulk preparation of the building block difficult.

Scheme 5 Synthesis of Fucosyl Amino Acids Using Fucose Peracetate[243]



11

N^{α}-Fmoc-Thr(2,3,4-tri-*O*-acetyl-α-L-fucopyranosyl)-OH (11, R^1 = Me):[243]
To a soln of fucose peracetate (100 mg, 301 μmol) and Fmoc-Thr-OH (123 mg, 361 μmol) in dry CH_2Cl_2 (3 mL) at rt was added $BF_3 \cdot OEt_2$ (105 μL, 0.85 mmol). After 50 h, the mixture was diluted with CH_2Cl_2 (10 mL) and washed with H_2O (10 mL). The aqueous phase was extracted with CH_2Cl_2 (2×10 mL) and the combined organic phases were dried (Na_2SO_4), filtered, and concentrated. The crude product was purified by straight phase HPLC on a Kromasil silica column (100 Å, 5 μm, 20×250 mm) using a linear gradient of EtOH in hexane (0 to 60% over 250 min; both eluants contained 2% AcOH) to yield the β-product (~36 mg, ~20%) and the α-anomer **11** (R^1 = Me); yield: 64 mg (35%); ^{1}H NMR ($CDCl_3$, δ): 5.77 (d, *J* = 9.4 Hz, NH), 5.20 (d, *J* = 3.9 Hz, H1), 5.11 (dd, *J* = 10.0, 3.7 Hz, H2), 4.57 (dd, *J* = 9.6, 1.7 Hz, Hα), 4.27 (t, *J* = 7.0 Hz, Fmoc CH), 4.09 (q, *J* = 6.2 Hz, H5), 2.17, 2.07, and 2.02 (3 s, 3×Ac), 1.18 (d, *J* = 6.2 Hz, Hγ), 1.11 (d, *J* = 6.4 Hz, H6).

6.3.2.2 1,2-*trans*-Glycosides

6.3.2.2.1 The Mannose Serine/Threonine Linkage

Mannose is primarily found in mammalian N-linked glycoproteins as part of core structures containing the tri-Man core [α-D-Man-(1→3)-α-D-Man-(1→6)-Man]. However, in the glycoproteins of yeast and molds, mannose, primarily α-D-Man-(1→2)-α-D-Man and α-D-Man-(1→6)-α-D-Man, has been found α-linked to Ser or Thr.[244] Expression of glucoamylase G1 in *Aspergillus niger*[245] and human IGF-I (somatomedin C) in *Saccharomyces cerevisiae*[246] resulted in proteins glycosylated with mannose at Ser and Thr.

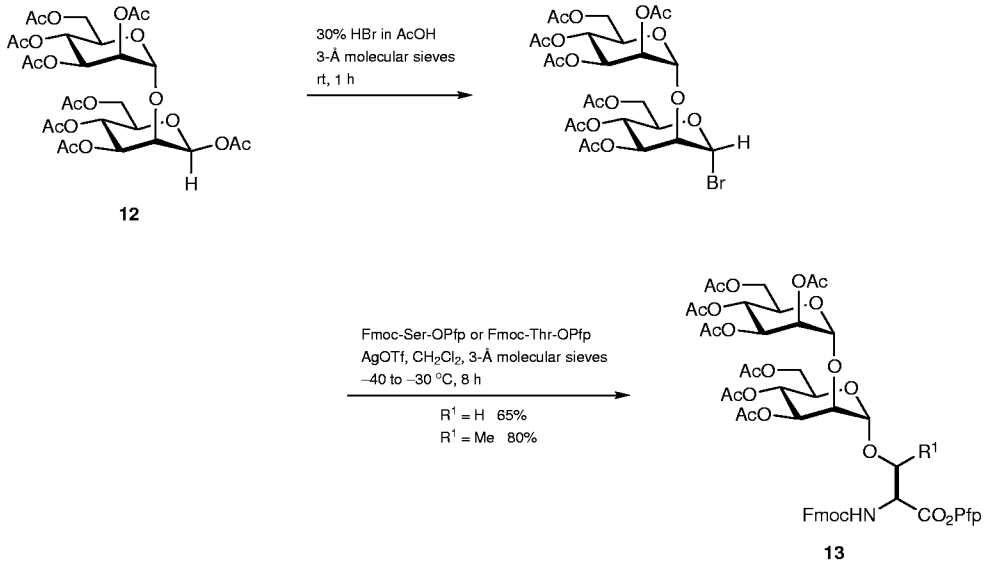
6.3.2.2.1.1 Synthesis of O-Linked Mannosyl Amino Acids Using Glycosyl Halides

Establishing an α-linkage of D-mannose is fairly straightforward. In a typical glycosylation, peracetylated mannosyl chlorides or bromides are coupled to the protected amino acid acceptor in the presence of a halophile. For example, glycosylation of Fmoc-Ser-OBzl and Fmoc-Ser-OPfp has been accomplished using D-mannosyl chlorides/bromides under silver triflate promotion in 89% and 60% yield, respectively.[111,134] Coupling of dimannosides has also been achieved by the glycosyl halide method.[111,117] In the synthesis of a dimannosylated heptadecaglycopeptide fragment from an IGF-I glycoprotein expressed in yeast, the 1→2-dimannose building block was synthesized by conversion of the peracetylated disaccharide **12** into its bromide with HBr (Scheme 6). Fmoc-Thr-OPfp was glycosylated in 80% yield. The Koenigs–Knorr methodology was also used for the synthesis of 6′-O-phosphorylated dimannosylated tripeptides as mimics of complex branched oligosaccharides that bind to the mannose-6-phosphate receptor.[42,122,247] Both the internal 1→2 and 1→6 Man linkages and the linkage to the amino acid were formed using glycosyl halides obtained from the oxidation of 1-phenylsulfanyls in high yield (70–82%). However, formation of the α-Man-(1→3)-Man internal linkage could only be accomplished using the trichloroacetamidate as the glycosyl donor. The efficiency of the trichloroacetimidate as a glycosyl donor for the formation of α-mannosides was recently validated during the coupling of an α-dystroglycan-derived tetrasaccharide to Fmoc-Ser/Thr-OX in high yield (84–91%).[248]

N^{α}-Fmoc-Thr{*O*-[4,6-di-*O*-acetyl-2-*O*-benzoyl-3-*O*-(2,3,4,6-tetra-*O*-acetyl-α-D-mannopyranosyl)-α-D-mannopyranosyl]}-OPfp (13, R^1 = Me):[117]
To a soln of **12** (0.30 g, 0.44 mmol) in dry CH_2Cl_2 (1.5 mL) with 3-Å molecular sieves was added 30% HBr in AcOH (1.1 mL, 4.4 mmol). The mixture was stirred at rt for 1 h, diluted with CH_2Cl_2, and then poured on to ice. The organic layer was washed successively with cold H_2O, sat. $NaHCO_3$, and H_2O. After drying and removal of the solvent, the resulting α-bromide (0.30 g, 0.43 mmol) was dissolved in dry CH_2Cl_2 (2.5 mL) and added to Fmoc-Thr-OPfp (0.22 g, 0.43 mmol), AgOTf (0.11 g, 0 43 mmol), and powdered 3-Å molecular sieves in dry CH_2Cl_2 (4.5 mL) under argon at −40 °C. The mixture was stirred at −30 to −40 °C for 8 h and the progress of the reaction was followed by TLC (EtOAc/light petroleum ether 3:2). 2,4,6-Collidine (56 μL, 0.43 mmol) was then added by syringe, the temperature was slowly raised to rt, and the mixture was stirred for 30 min. Dilution with dry CH_2Cl_2, filtration, and concentration of the filtrate gave a syrup, which was purified on dried silica gel with dry solvents (EtOAc/light petroleum ether 1:1) to give the product; yield: 0.39 g (80%).

for references see p 322

Scheme 6 Synthesis of Dimannosyl Amino Acids Using the Halide Method[117]

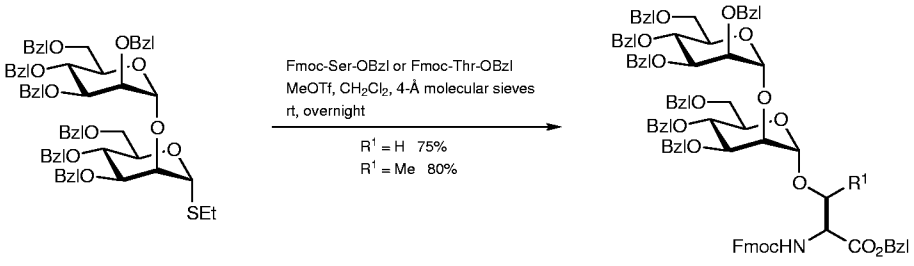


12 13

6.3.2.2.1.2 Synthesis of O-Linked Mannosyl Amino Acids Using Sulfanylglycosides

The α-linkage between a single mannose residue and Ser or Thr has also been accomplished by glycosylation of Fmoc-Ser-OAl using the peracetylated trichloroacetimidate with TMSOTf promotion in 70–75% yield.[249] Alternatively, the amino acid without carboxy group protection can be glycosylated by treatment with the peracetate under Lewis acid catalysis in 42% yield, providing exclusively the α-linked building block ready for peptide synthesis.[165] In another approach to the synthesis of the 1→2-dimannose, the internal linkage was achieved using the Koenigs–Knorr methodology, but the linkage to Fmoc-Ser-OBzl and Fmoc-Thr-OBzl was achieved by using the ethylsulfanyl disaccharide donor **14** under MeOTf catalysis in 75% and 80% yield, respectively (Scheme 7).[250] Sulfanylglycosides are easily synthesized and are generally more stable during storage than their corresponding acetimidate/halide counterparts, making them very attractive glycosyl donors.

Scheme 7 Synthesis of Dimannosyl Amino Acids Using the Sulfanylglycoside Method[250]



14 15

***N*$^{\alpha}$-Fmoc-Ser{*O*-[3,4,6-tri-*O*-benzyl-2-*O*-(2,3,4,6-tetra-*O*-benzyl-α-D-mannopyranosyl)-α-mannopyranosyl]}-OBzl** (15, **R^1 = H**):[250]

To a soln of **14** (0.21 g, 0.20 mmol) and Fmoc-Ser-OBzl (0.10 g, 0.25 mmol) in CH_2Cl_2 (5 mL) containing 4-Å molecular sieves was added MeOTf (75 μL, 0.69 mmol). The mixture was stirred overnight at rt and pyridine (100 μL) was added; the mixture was fractionated on a column (silica gel, toluene/EtOAc 19:1) to give the β-anomer; yield: 0.04 g (14%) and **15** (R^1 = H); yield: 0.21 g (75%); ^{13}C NMR ($CDCl_3$, δ): 169.8, 156.0, 143.8, 141.2, 138.6, 138.4, 138.3, 138.2, 135.4, 128.6–127.1, 125.2, 119.9, 100.0, 99.9, 79.7, 79.4, 75.1, 74.9, 74.8, 74.5, 73.3, 72.5, 72.3, 72.2, 69.7, 69.3, 69.1, 67.2, 54.6, 47.1.

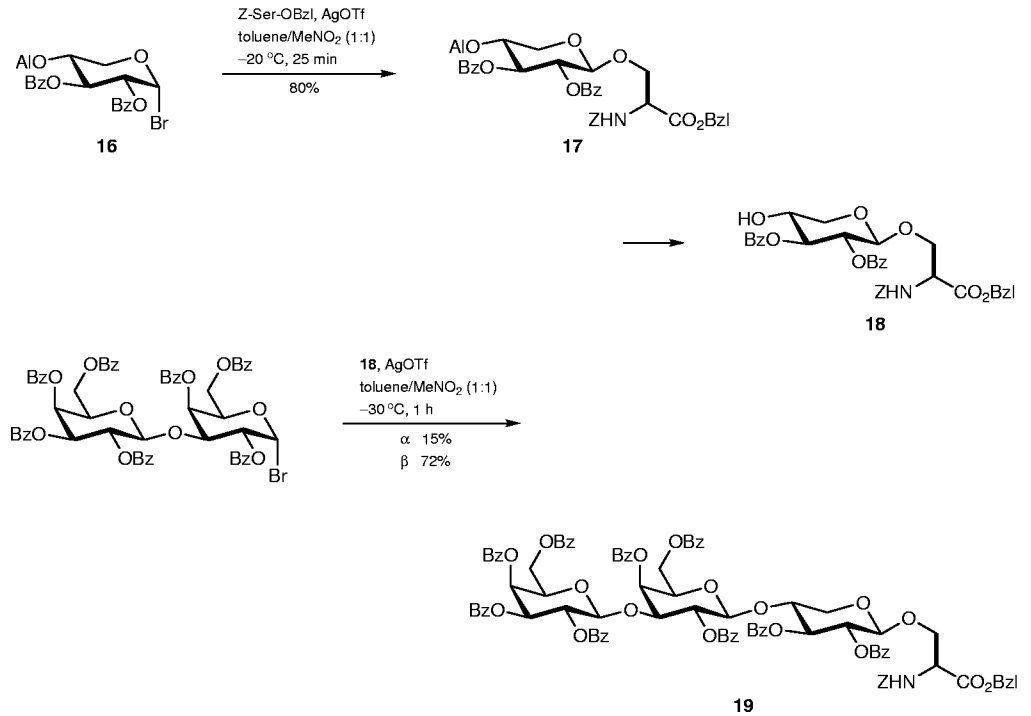
6.3.2.2.2 The Xylose Serine/Threonine Linkage

The second most abundant type of *O*-glycoproteins (proteoglycans) contains a β-linkage between D-xylose and serine. In mammalian proteoglycans, namely those carrying chondroitin sulfate, dermatan sulfate, heparin, and heparan sulfate, the glycosaminoglycan may be linked to a serine residue via D-xylose through a conserved tetrasaccharide sequence in which the most common structure is the trisaccharide β-D-Gal-(1→3)-β-D-Gal-(1→4)-β-D-Xyl-Ser.[251,252] This trisaccharide, as well as mono- and disaccharide fragments of it and other proteoglycan related structures, have been the target of several syntheses. The formation of the 1,2-*trans* linkage is easily achieved by the use of a participating group at C2, and although the β-selectivity is not as high as with the hexoses, glycosylation is not complicated and yields are often reasonable. The most common methodology is the use of the peracylated halide coupled to the suitably protected serine derivative.[253,254] The building blocks Z-Ser(Bzl_3-β-D-Xyl)-OH[255] and Fmoc-Ser(Ac_3-β-D-Xyl)-OH[253] are used in the solution-phase synthesis of tetrapeptides. A glycosylated tetrapeptide is also obtained by direct glycosylation with a xylose derivative.[58]

6.3.2.2.2.1 Synthesis of O-Linked Xylosyl Serine-Containing Trisaccharyl Amino Acids Using Glycosyl Halides

In the synthesis of the core structure β-D-Gal-(1→3)-β-D-Gal-(1→4)-β-D-Xyl-Ser as shown in Scheme 8, the trisaccharide glycosyl amino acid **19** was obtained by coupling of the perbenzoylated digalactosyl bromide to the C4 hydroxy of a xylosyl amino acid **18** in 72% yield along with 15% of the α-anomer.[256] The coupling of xylose derivative **16** to the Z-Ser-OBzl proceeded selectively, providing **17** in 80% yield. The Z and OBzl protecting groups were quantitatively removed, thus facilitating the application of a reprotection scheme to make the building block suitable for use in peptide synthesis.

Scheme 8 Synthesis of Xylose-Containing Trisaccharide Building Blocks Using the Halide Method[256]



for references see p 322

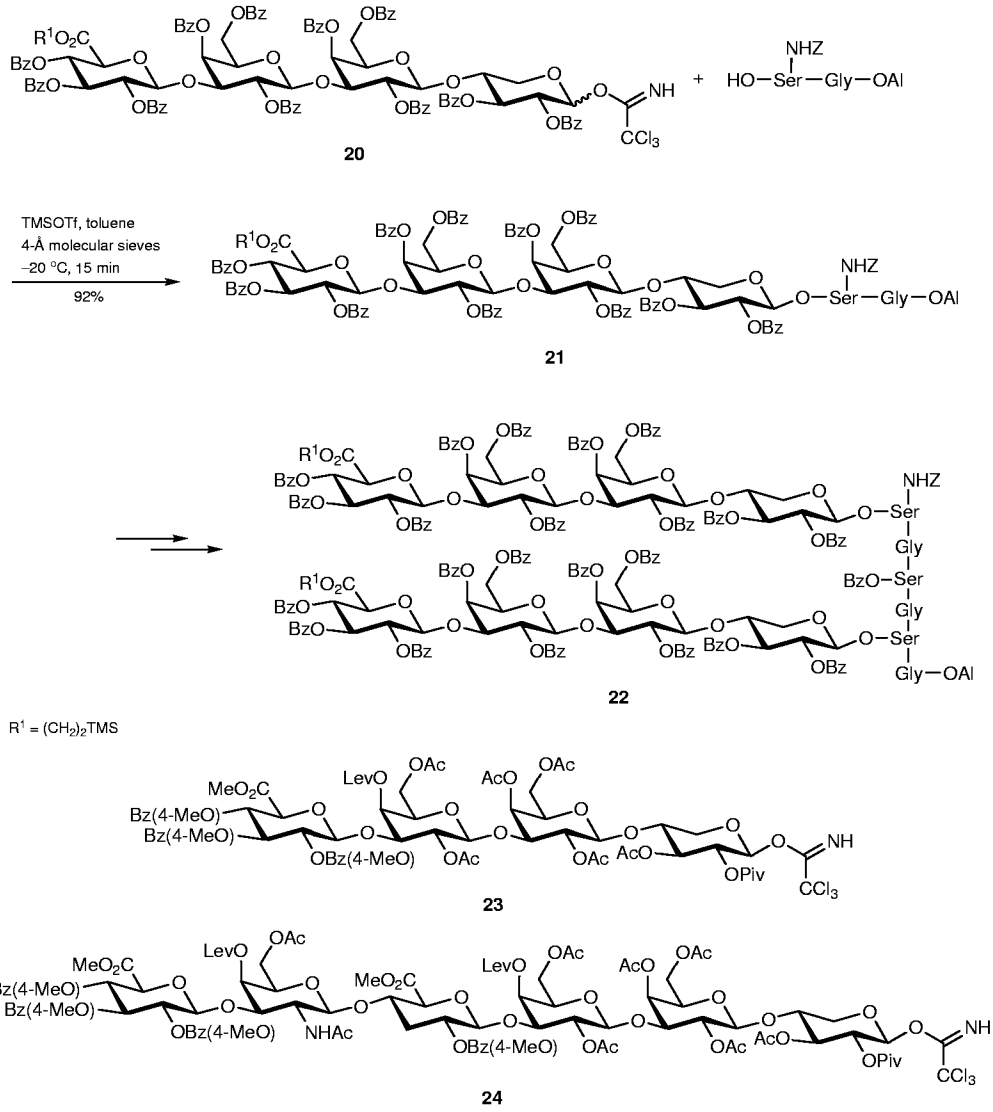
N^{α}-Z-Ser(4-*O*-allyl-2,3-di-*O*-benzoyl-β-xylopyranosyl)-OBzl (17):[256]
To a stirred soln of the bromide **16** (4.02 g, 8.7 mmol) and Z-Ser-OBzl (2.60 g, 7.92 mmol) in dry toluene/$MeNO_2$ (1:1, 30 mL) at −20 °C was added, over a 10-min period, a soln of AgOTf (2.90 g, 11.3 mmol) and 2,4,6-collidine (0.98 mL, 7.40 mmol) in dry toluene/$MeNO_2$ (1:1, 10 mL). After an additional 15 min at −20 °C the mixture was worked up and the crude product was chromatographed (silica gel, toluene/EtOAc 8:1); yield: 4.49 g (80%); ^{1}H NMR (400 MHz, $CDCl_3$ + 10% CD_3OD, δ): 5.77 (m, 1H, allyl H2), 5.50 (dd, 1H, $J_{2,3} = J_{3,4} = 7.7$ Hz, Xyl H3), 5.23–4.98 (m, 7H, Xyl H2, 4×benzyl H, allyl H1 and H1′), 4.65 (d, 1H, $J_{1,2} = 7.7$ Hz, Xyl H1), 4.53 (dd, 1H, $J_{\alpha,\beta} = 2.6$ Hz, $J_{\alpha,\beta} = 3.5$ Hz, Ser Hα), 4.32 (dd, 1H, $J_{\alpha,\beta} = 2.6$ Hz, $J_{\beta,\beta'} = -10.1$ Hz, Ser Hβ), 4.06 (m, 2H, allyl 2×H3), 3.96 (dd, 1H, $J_{4,5} = 4.5$ Hz, $J_{5e,5a} = -12.0$ Hz, Xyl H5e), 3.82 (dd, 1H, $J_{\alpha,\beta} = 3.5$ Hz, $J_{\beta,\beta'} = -10.1$ Hz, Ser Hβ), 3.64 (ddd, 1H, $J_{3,4} = 7.7$ Hz, $J_{4,5e} = 4.5$ Hz, $J_{4,5a} = 8.0$ Hz, Xyl H4), 3.37 (dd, 1H, $J_{4,5a} = 8.0$ Hz, $J_{5e,5a} = -12.0$ Hz, Xyl H5a).

N^{α}-Z-Ser[(*O*-2,3,4,6-tetra-*O*-benzoyl-β-D-galactopyranosyl)-(1→3)-*O*-(2,4,6-tri-*O*-benzoyl-β-D-galactopyranosyl)-(1→4)-*O*-(2,3-di-*O*-benzoyl-β-D-xylopyranosyl)]-OBzl (19):[256]
To a soln of the disaccharide bromide (4.99 g, 4.33 mmol) and **18** (2.61 g, 3.90 mmol) in dry toluene/$MeNO_2$ (1:1, 20 mL) at −30 to −20 °C was added a soln of AgOTf (1.44 g, 5.60 mmol) and 2,4,6-collidine (0.44 g, 3.32 mmol) in dry toluene/$MeNO_2$ (1:1, 10 mL) dropwise with stirring over a 5-min period. After stirring for an additional 1 h at −30 to −25 °C the mixture was worked up, and the crude product was chromatographed (silica gel, toluene/EtOAc 6:1). The α-trisaccharide (1.04 g, 15%) eluted first, followed by the title compound **19**; yield: 4.85 g (72%); ^{1}H NMR (400 MHz, $CDCl_3$ + 10% CD_3OD, δ): 5.86 (m, 2H, Gal-1 H4 and Gal-2 H4), 5.44 (dd, 1H, $J_{3,4} = 3.3$ Hz, $J_{2,3} = 10.5$ Hz, Gal-2 H3), 4.81 (d, 1H, $J_{1,2} = 8.0$ Hz, Gal-1 H1), 4.66 (dd, 1H, $J_{5,6} = 5.2$ Hz, $J_{6,6'} = -10.1$ Hz, Gal-2 H6), 4.62 (d, 1H, $J_{1,2} = 4.3$ Hz, Xyl H1), 4.50 (dd, 1H, $J_{\alpha,\beta} = 2.7$ Hz, $J_{\alpha,\beta'} = 5.5$ Hz, Ser Hα), 4.18 (dd, 1H, $J_{3,4} = 2.7$ Hz, $J_{2,3'} = 10.4$ Hz, Gal-1 H3), 4.09 (dd, 1H, $J_{5,6} = 4.3$ Hz, $J_{5,6'} = 7.3$ Hz, Gal-1 H5), 3.47 (dd, 1H, $J_{\alpha,\beta'} = 3.2$ Hz, $J_{\beta,\beta'} = -12.4$ Hz, Ser Hβ′), 3.02 (dd, 1H, $J_{4,5a} = 5.7$ Hz, $J_{5e,5a} = -12.4$ Hz, Xyl H5a); ^{13}C NMR (100 MHz, $CDCl_3$ + 10% CD_3OD, δ): 101.83 (Gal-1 C1), 101.33 (Gal-2 C1), 99.02 (Xyl C1), 77.74 (Gal-1 C3), 74.88 (Xyl C4), 71.78 (Gal-1 C5), 71.30 (Xyl C3 and Gal-2 C3), 71.12 (Gal-2 C5), 70.82 (Gal-1 C2), 70.15 (Xyl C2), 69.29 and 69.06 (Gal-1 C4 and Gal-2 C2), 68.32 (Ser C_β), 67.48 (Gal-2 C4), 66.93 and 66.81 (2×benzyl C), 62.68 (Xyl C5), 61.51 (Gal-1 C6), 60.22 (Gal-2 C6), 53.78 (Ser C_α).

6.3.2.2.2.2 Synthesis of O-Linked Xylosyl Serine-Containing Oligosaccharyl Amino Acids Using Glycosyl Imidates

For glycosylation with more complex, sulfated, or phosphorylated xylose tri-, tetra-, and hexasaccharide derivatives, as well as with the simpler disaccharides, the trichloroacetimidate method has proven quite effective.[257–260] In an illustrative example shown in Scheme 9, the complete tetrasaccharide is converted into the imidate **20** and coupled to Z-Ser-Gly-OAl to yield the desired product **21** in 92% yield.[257] Glycosylation was also carried out using Fmoc-Ser-OAl to furnish the analogous building block in 82% yield. The building block **21** was coupled in solution to give the bisglycosylated hexapeptide **22**. In another example, the tetrasaccharide imidate **23** was coupled to Z-Ser-OBzl in high yield.[258] The levunoyl (Lev) group was then chemoselectively removed by treatment with hydrazine acetate in toluene/EtOH and low temperature and the free hydroxy group was then sulfated to give the desired sulfated derivatives. In a related example, the same serine derivative was glycosylated with a hexasaccharide imidate **24** under $BF_3 \cdot OEt_2$ catalysis to yield the product in 64% yield.[259]

Scheme 9 Synthesis of Xylose-Containing Tetra- and Hexasaccharide Building Blocks Using the Imidate Method[257-259]



***N*$^{\alpha}$-(Benzyloxycarbonyl)-*O*-{*O*-[2-(trimethylsilylethyl)-2,3,4-tri-*O*-benzoyl-β-D-glucopyranosyluronate]-(1→3)-*O*-(2,4,6-tri-*O*-benzoyl-β-D-galactopyranosyl)-(1→3)-*O*-(2,4,6-tri-*O*-benzoyl-β-D-galactopyranosyl)-(1→4)-2,3-di-*O*-benzoyl-β-D-xylopyranosyl}serylglycine Allyl Ester (21):**[257]

A mixture of **20** (200 mg, 0.1 mmol) and Z-Ser-Gly-OAl (66 mg, 0.2 mmol) was stirred in the presence of activated, powdered 4-Å molecular sieves (100 mg), in CH_2Cl_2 (4 mL) at rt under dry argon. It was cooled to –20 °C and TMSOTf (9.2 mg, 0.041 mmol) in toluene (7.5 µL) was added. The mixture was stirred for 15 min at –20 °C. TEA (2 µL) was added, and the mixture was filtered and concentrated. The residue was chromatographed (silica gel, toluene/acetone 6:1), to give **21**; yield: 200 mg (92%); ^{1}H NMR ($CDCl_3$, δ): 7.58 (m, 60H, 12×Ph), 6.78 (t, 1H, J = 5.5 Hz, Gly NH), 5.82 (m, 1H, $OCH_2CH{=}CH_2$), 5.13 (dd, 1H, $J_{1,2}$ = 6.0 Hz, $J_{2,3}$ = 8.5 Hz, Xyl H2), 4.04 (d, 1H, $J_{4,5}$ = 9 5 Hz, GlcA H5), 0.75 (m, 2H, CH_2Si), –0.07 (s, 9H, 3×CH_3).

for references see p 322

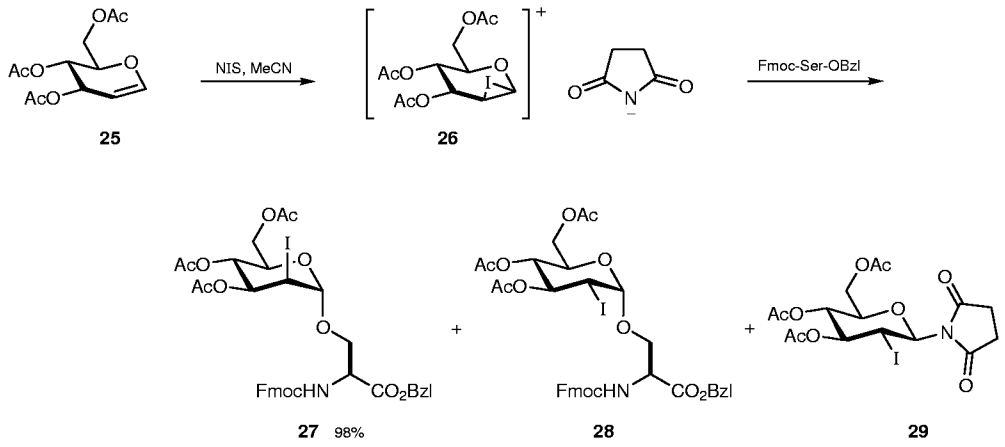
6.3.2.3 2-Deoxy O-Linked Glycosides

The synthesis of glycopeptides wherein the glycan portion is modified is a crucial tool for probing the structural function of glycoproteins in biological systems. Furthermore, such modifications are ways of modulating their pharmacokinetic behavior. There have been only a few examples where a 2-deoxy sugar has been used instead of the usual 2-deoxy-2-acetamido- or 2-hydroxyglycans.

6.3.2.3.1 Synthesis of O-Linked 2-Deoxyglycosyl Amino Acids via 2-Iodoglycosides

The synthesis of 2-deoxyglycosyl amino acids by direct coupling of a suitable 2-deoxyglycosyl donor is usually ineffective since the presence of two hydrogens at C2 makes stereocontrol challenging. There are few examples of the synthesis of 2-deoxyglycopeptides in the literature. The glycal **25**, protected as its benzyl or acyl derivative, is treated with NIS and the intermediate iodonium ion **26** is opened by nucleophilic attack of the amino acid hydroxy to give primarily the 2-iodo-α-glycoside **27** (Scheme 10), together with a little **28** and **29**.[133,225] Using this method, glucal and galactal were coupled to various amino acids and to di- to heptapeptides N-protected with the Aloc, Boc, Fmoc, or Z groups and C-protected as benzyl esters, in moderate to high yield. Cyclic hexa- and heptapeptides could also be glycosylated in 56–64% yield.[262] Glycosylation using the disaccharide glycals lactal, cellobial, and Gal(1→3)galactal yielded products in similar yields, except when the glycal was protected with benzyl groups. In this case, a greater portion of the β-anomer was formed because the C4 benzyl group forces the iodonium ion to acquire a more equatorial conformation. After formation of the glycoside, the 2-iodo group is removed by radical dehalogenation with Bu_3SnH. Since this method sometimes fails for larger glycopeptides, $NaBH_4/NiCl_2 \cdot 6H_2O$ in EtOH may be used as an alternative.

Scheme 10 Synthesis of α-2-Deoxyglycosyl Amino Acid Precursors via the 2-Iodo Intermediate[133]



***N*$^{\alpha}$-Fmoc-Ser-(3,4,6-tri-*O*-acetyl-2-deoxy-2-iodo-α-D-mannopyranosyl)-OBzl (27):**[133]

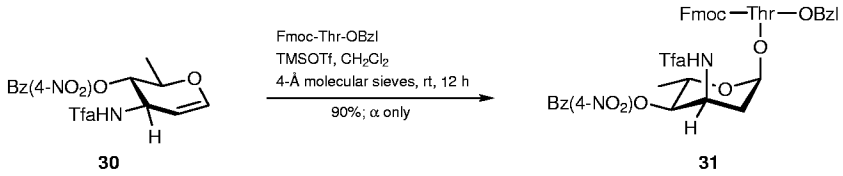
To a dry soln of Fmoc-Ser-OBzl (249.6 mg, 0.6 mmol) and **25** (326.4 mg, 1.2 mmol) in MeCN (5 mL) containing powdered 4-Å molecular sieves was added freshly recrystallized (CCl_4) NIS (270 mg, 1.2 mmol) at 0 °C; the mixture was stirred under inert gas for 1 h at rt. After 12–24 h, another portion of NIS was added; however, all together, not more than 1.1 equiv of the glycal. Then the solvents were removed and the residue resuspended in $CHCl_3$. After filtration over Celite, the filtrate was extracted with 10% NaS_2O_3 (2×). The aqueous phase was extracted with $CHCl_3$, the combined organic phases dried ($MgSO_4$), the solvent removed, and the residue purified by column chromatography (silica gel, $CHCl_3$/acetone 6.5:1); yield: 480 mg (98%); ^{1}H NMR (270 MHz, $CDCl_3$/TMS, δ): 5.87 (d, 1H, NH), 5.34, 5.18 (d, 1H, OCH_2Ph), 5.32 (dd, 1H, $J_{4,5}$ = 9.8 Hz, H4), 5.07 (d, 1H, $J_{1,2}$ = 1.1 Hz, H1), 4.62 (m, 1H, Ser Hα), 4.53 (dd, 1H, $J_{3,4}$ = 9.6 Hz, H3), 4.42 (m, 2H, Fmoc CH_2), 4.3 (dd, 1H, $J_{2,3}$ = 4.4 Hz, H2), 4.27 (dd, 1H,

Fmoc H9), 4.15 (m, 2H, H6), 3.99 (m, 2H, Ser Hβ), 3.95 (m, 1H, H5); ^{13}C NMR (75.47 MHz, $CDCl_3$, δ): 156 (urethane CO), 119.90–143.71 (aromatic Cs), 102.12 (C1), 69.78 (C5), 69.31 (Ser Cβ), 68.79 (C3), 67.59 (OCH_2Ph), 67.45 (C4), 67.3 (Fmoc CH_2), 62.05 (C6), 54.41 (Ser Cα), 47.1 (Fmoc C9), 28.65 (C2).

6.3.2.3.2 Synthesis of O-Linked 2-Deoxyglycosyl Amino Acids Using Trimethylsilyl Triflate

As an alternative method, with comparable yields and stereoselectivity but shorter reaction times than the iodo method, protected amino acids and cyclic and linear tri- to hexapeptides are glycosylated in solution using protected 3-amino glycals, L-acosaminal and L-ristosaminal **30**, under TMSOTf promotion (Scheme 11).[263] Yields range from 40 to 95% (95% for **31**) and the anomeric ratio depends on the nature of the acceptor, the α-anomer generally predominating. The 4-nitrobenzoyl group [Bz(4-NO_2)] is used for carbohydrate hydroxy protection while the trifluoroacetyl group served as the carbohydrate amine protection. Both groups are simultaneously cleaved by treatment with methanolic ammonia for 15 minutes without β-elimination of the carbohydrate.

Scheme 11 Synthesis of α-2-Deoxyglycosyl Amino Acids Using Trimethylsilyl Triflate[263]



***N*$^{\alpha}$-(9-Fluorenylmethoxycarbonyl)-*O*-[2,3,6-trideoxy-4-*O*-(4-nitrobenzoyl)-3-trifluoroacetamido-α-L-ribohexopyranosyl]-L-threonine Benzyl Ester (31):**[263]
A soln of Fmoc-Thr-OBzl (10.6 mg, 0.025 mmol) and glycal **30** (11.0 mg, 0.030 mmol, 1.2 equiv) in dry CH_2Cl_2 was stirred in the presence of 4-Å molecular sieves under argon for 1 h. TMSOTf (13.9 mg, 2.5 equiv) was then injected through a septum at rt, and stirring was continued for 12 h. The mixture was diluted with CH_2Cl_2, filtered, and washed with $NaHCO_3$. The organic layer was dried ($MgSO_4$) and the solvent removed. The crude product was purified by chromatography (silica gel, EtOAc/hexane 1:2) to give **31**; yield: 17.9 mg (89%); ^1H NMR (250 MHz, DMSO-d_6, δ): 8.38 (d, 1H, sugar NH), 7.89 (d, 1H, Thr NH), 5.19 (d, 2H, OBzl CH_2), 5.12 (s, 1H, H1), 4.88 (dd, 1H, $J_{3,4}$ = 3.9 Hz, $J_{4,5}$ = 9.0 Hz, H4), 4.60 (m, 1H, H3), 4.45 (dd, 1H, $J_{\alpha,\beta}$ = 3.6 Hz, $J_{\alpha,NH}$ = 8.6 Hz, Hα), 4.38–4.22 (m, 4H, CH, Fmoc CH_2, Hβ), 4.11 (m, 1H, H5), 2.13–2.27 (m, 1H, H2), 2.03–1.92 (m, 1H, H2′), 1.21 (d, 3H, 3×Hγ), 1.13 (d, 3H, 3×H6); ^{13}C NMR (62.9 MHz, $CDCl_3$, δ): 94.0 (Cl).

6.3.2.4 2-Amino-2-deoxy O-Linked Glycosides

6.3.2.4.1 1,2-*cis*-O-Linked Mucin Glycopeptides

The mucins are the predominant O-linked glycoproteins and may be divided into eight different core structures, depending on the relative occurrence of various glycosyltransferases at the location of their biosynthesis (Table 1).[23,182,183,264,265]

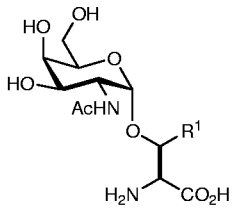
Table 1 The Eight Different Core Structures of Mucin Glycoproteins[23,182,183,264,265]

Core	Glycan Structure	Ref
core 1	β-D-Gal-(1→3)-α-D-GalNAc-Ser/Thr	[182,264,265]
core 2	β-D-GlcNAc-(1→6)-[β-D-Gal-(1→3)-]α-D-GalNAc-Ser/Thr	[183,265]
core 3	β-D-GlcNAc-(1→3)-α-D-GalNAc-Ser/Thr	[265]
core 4	β-D-GlcNAc-(1→6)-[β-D-GlcNAc-(1→3)-]α-D-GalNAc-Ser/Thr	[265]
core 5	α-D-Gal-(1→3)-α-D-GalNAc-Ser/Thr	[265]
core 6	β-D-GlcNAc-(1→6)-α-D-GalNAc-Ser/Thr	[265]
core 7	β-D-GalNAc-(1→6)-α-D-GalNAc-Ser/Thr	[23]
core 8	α-D-Gal-(1→3)-α-D-GalNAc-Ser/Thr	[23]

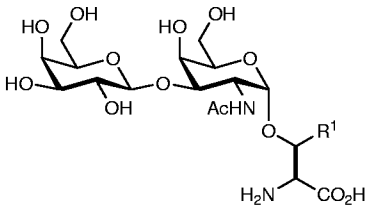
for references see p 322

Core structures 1, 2, and 3 are the most common in mucins and in other secreted cell-surface glycoproteins. Core structure 1, the so-called T-antigen, is widespread in nature and occurs in blood groups,[266] in the antifreeze glycoproteins of polar fish,[116,267] and in human oncofetal fibronectin.[268] Often these highly heterogeneous structures are further galactosylated at the 4-position,[269] monosialylated at the 3- and 6-positions,[182,264,270,271] or disialylated. Mucins are also highly expressed by epithelial cancer cells. A large body of synthetic effort has therefore been devoted to the preparation of fragments of the mucin glycoproteins. In particular, the T_N-antigen **32**, the T-antigen **33**, as well as their sialylated derivatives **34** and **35**, have gained much attention by virtue of their potential in cancer therapy, namely as vaccines or as immune response stimulants (Scheme 12).

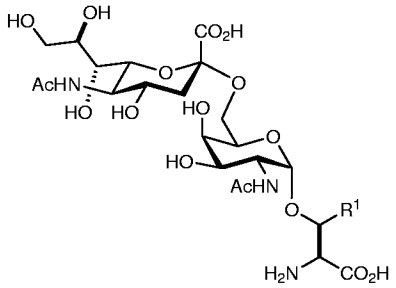
Scheme 12 Important Mucin Glycoprotein Glycan Structures



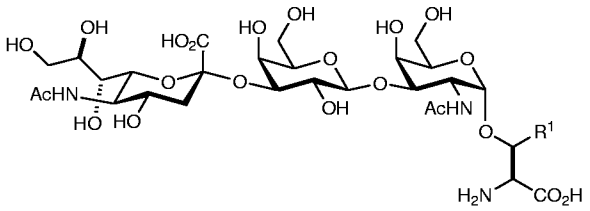
32 T_N-antigen



33 T-antigen



34 sialyl T_N-antigen



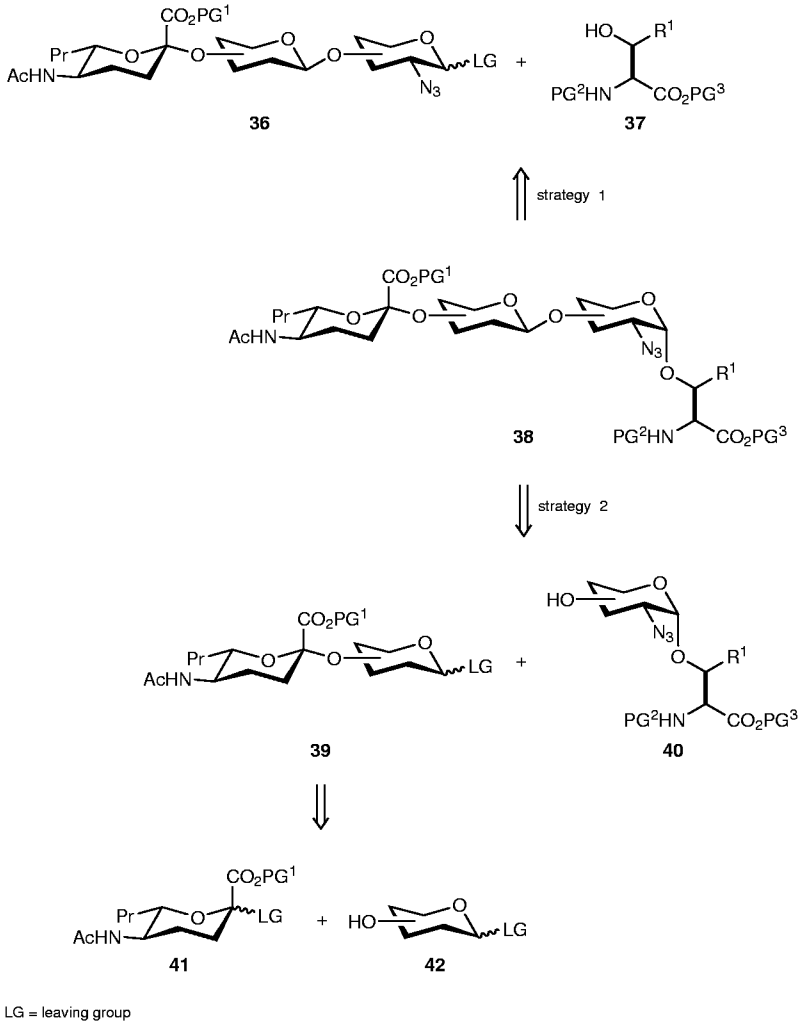
35 2,3-sialyl T-antigen

The assembly of the core structures can be accomplished in principally two different ways (Scheme 13). In strategy 1, the complete oligosaccharide is converted into a suitable donor **36** and then coupled to the suitably protected amino acid **37**, yielding the desired glycosyl amino acid **38**; in strategy 2, the components of the oligosaccharide (e.g., **39**, **41**, and **42**) are coupled in a stepwise manner to the preformed GalNAc precursor amino acid building block **40**. The couplings have been accomplished using various glycosyl donors and *N*- and *C*-amino acid protecting groups.

The synthesis of these structures is far from trivial and several problems need to be surmounted, the first being the stereoselective formation of the α-linkage between GalNAc and Ser or Thr. The use of GalNAc derivatives as the glycosyl donor leads to the formation of the oxazoline, which reacts only under conditions that are incompatible with the usual amino and acid protecting groups developed for peptide chemistry. Furthermore, formation of the desired α-anomer does not prevail. Alternatively, glycosylation can be carried out using glycosyl donors containing a nonparticipatory amine precursor group in the 2-position. For example, uncatalyzed condensations have been effected by reacting 3,4,6-tri-*O*-acetyl-2-deoxy-2-nitroso-α-D-galactopyranosyl chloride with glycosyl acceptors, followed by a completely stereoselective $NiCl_2/NaBH_4$ reduction and acetylation of the 2-hydroxyimino group to an equatorial acetamido group,[272] or base-catalyzed additions of Ser/Thr to a nitroglycal.[521] The use of 2-azido analogues has become the method of choice, leading to

problems related to the conversion of the azide precursor into the acetamido derivative. Another major consideration is the incorporation of the sialic acid unit by chemical or enzymatic means. Mucin-type glycosylations often occur in clusters, with many consecutive glycosylations, and efficient peptide synthesis with the synthesized building blocks is a prerequisite for the assembly of pure peptides.

Scheme 13 Different Strategies for the Synthesis of Mucin Glycan Core Structures

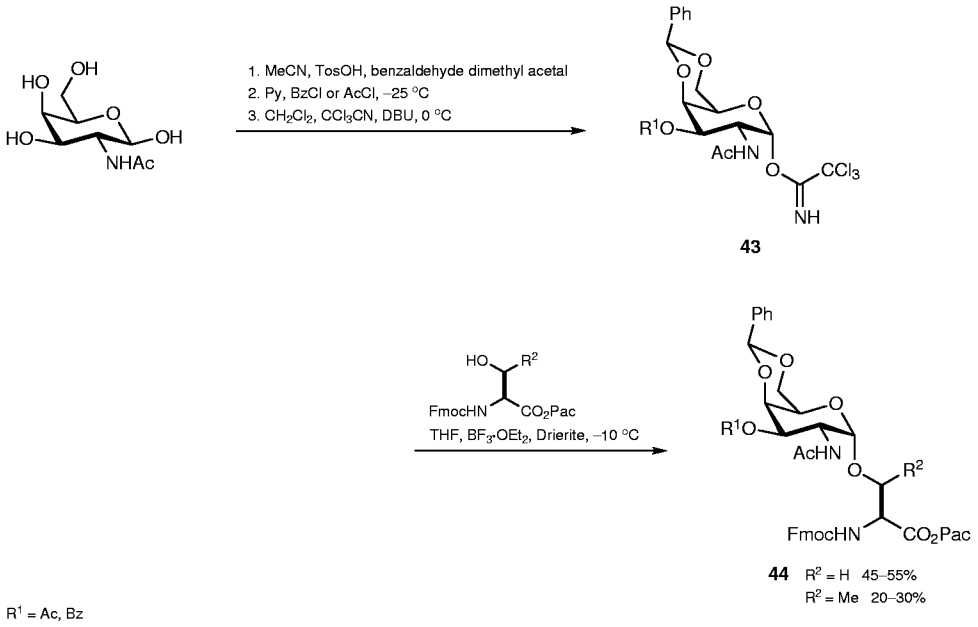


6.3.2.4.1.1 Synthesis of O-Linked 2-Acetamido-2-deoxyglycosyl Amino Acids Using 2-Acetamido Glycosyl Donors

In a short stereoselective synthesis of the T_N-antigen, glycosylation of Fmoc-Ser-OBzl or Fmoc-Thr-OBzl with 4,6-*O*-benzylidene-*N*-acetylgalactosamine trichloroacetimidate (**43**) under $BF_3 \cdot OEt_2$ catalysis gives the product **44** in 20–50% yield (Scheme 14).[273] Surprisingly, only traces of the β-glycoside and oxazoline were formed. The presence of the benzylidene clearly affected the stereochemical outcome of the reaction, although the mechanism is not known. Despite the low yield of the glycosylation step, synthesis of the glycosyl donor is short and the overall yield is higher than that sometimes obtained using glycosylamine protecting groups or azido precursors.

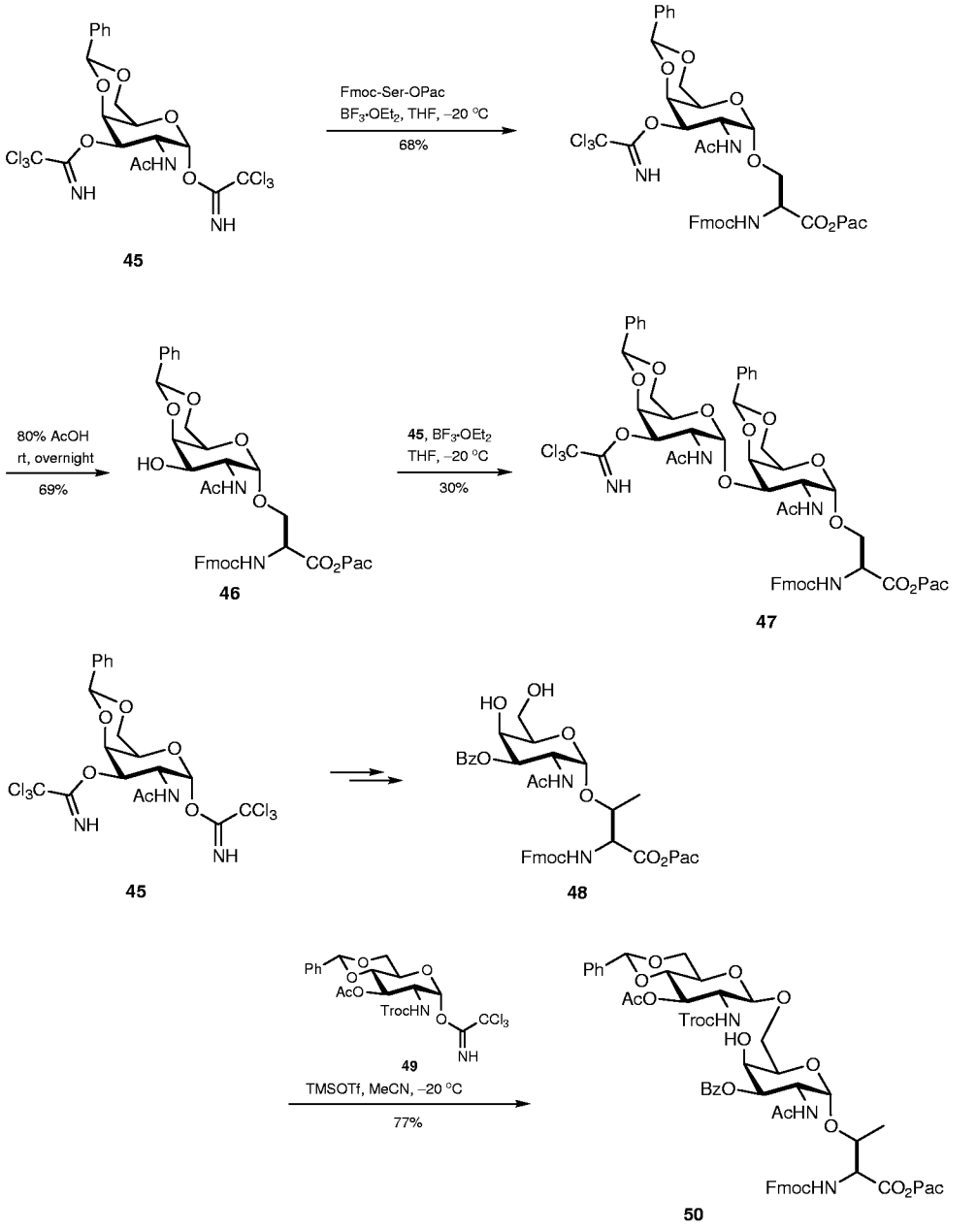
for references see p 322

Scheme 14 Synthesis of the T_N-Antigen Using the 2-Acetamido Glycosyl Donor[273]



2-Acetamido glycosyl donors and N-protected glycosyl donors have also been used in the synthesis of core 5 and core 6 structures using strategy 2. The syntheses are based on the stereoselective glycosylation of Ser or Thr using the 4,6-benzylidene-2-acetamido galactosyl tricloroacetimidate derivatives, as shown in Scheme 15.[274,275] In the synthesis of the core 5 structure, the trichloroacetimidate is used both as a leaving group and as an acid-labile protecting group for the C3 hydroxy in **45**.[274] The trichloroacetimidate was selectively cleaved in the presence of the 4,6-benzylidene group by treatment with 80% AcOH and its use precluded the need for additional protecting groups. Coupling of the deblocked glycosyl amino acid **46** to **45** gave the core 5 product **47** in 30% yield. Compound **45** was then converted in four steps into the glycosyl acceptor **48**, which was glycosylated with the GlcNTroc derivative **49** to yield the core 6 product **50** in 77% yield. The use of the Troc protecting group secures the formation of the β-linkage through the neighboring group effect. The Troc group can be converted into the *N*-acetamido group by reduction/acetylation using a mixture of Zn/AcOH/Ac_2O.

Scheme 15 Synthesis of the Core Structures 5 and 6 Using the 2-Acetamido Galactosyl Imidate as well as the *N*-Troc Intermediate[274,275]



T_N-Antigen 44; General Procedure:[273]

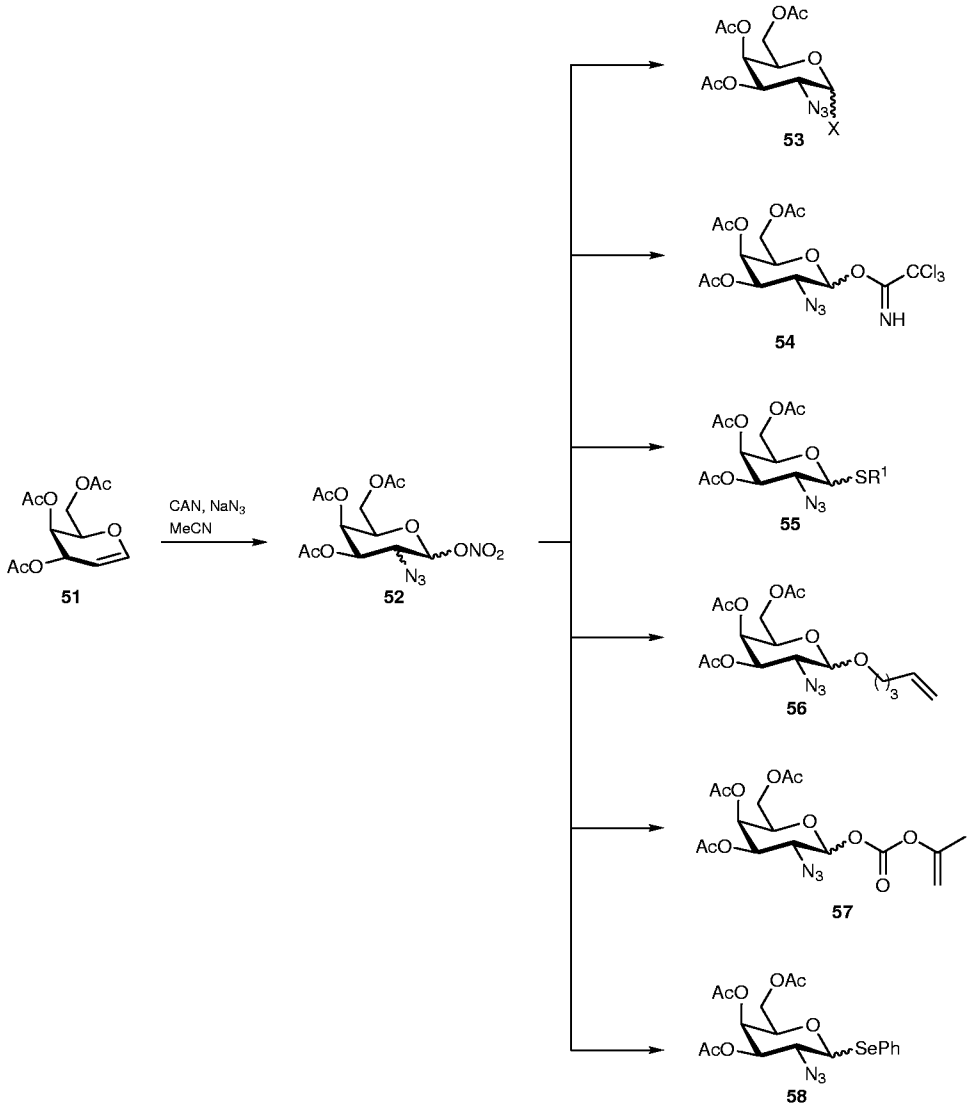
To donor **43** (1–2 g) and the acceptor (1.5 equiv) in dry THF was added $BF_3 \cdot OEt_2$ (0.1 equiv) in THF with stirring. The reaction was complete in about 1 h. The yield of the glycoside was lower with Thr (20–30%) compared to Ser (45–55%). ^1H NMR (for Ser) ($CDCl_3$, δ): 6.55 (d, 1H, J = 9.0 Hz, NH), 5.85 (d, 1H, J = 8.5 Hz, NH), 5.63 (d, 1H, J = 16.0 Hz, CH_2COPh), 5.51 (s, 1H, PhCH), 5.3–5.43 (m, 2H), 5.2 (d, 1H, $J_{1,2}$ = 3.5 Hz, H1), 5.15–5.07 (m, 2H), 4.79 (br s, 1H, Ser Hα), 4.2–4.58 (m, 7H), 3.84–3.99 (m, 2H), 1.8 (s, 3H, NAc); ^{13}C NMR ($CDCl_3$, δ): 191.57 (CH_2COPh), 169.75 (CO_2CH_2), 155.92 (NHCO), 100.76 (aryl CH), 99.52 (C1).

for references see p 322

6.3.2.4.1.2 Synthesis of O-Linked 2-Acetamido-2-deoxyglycosyl Amino Acids Using 2-Azido-2-deoxyglycosyl Donors

An important breakthrough in the use of 2-azido-2-deoxy precursors was made by Paulsen[276] and Lemieux,[277] when methods for generating the 2-azido glycosides via azido nitration of a glycal such as **51** were developed (Scheme 16). The resulting 2-azido-2-deoxy-1-nitropyranose **52** can be readily converted into several glycosyl donors such as glycosyl halides **53**,[277] glycosyl imidates **54**,[278] thioglycosides **55**,[279,280] pent-4-enyl glycosides **56**,[281] isopropenyl glycosyl carbonates **57**,[282] and selenoglycosides **58**.[283–286] The azide function plays the dual role of a nonparticipating group and an amine precursor. This methodology has played a pivotal role in the synthesis of mucin glycopeptide and is the most widely used today.

Scheme 16 Azidonitration of Glycals and Conversion of the Nitro Intermediate into Various Glycosyl Donors[276–286]

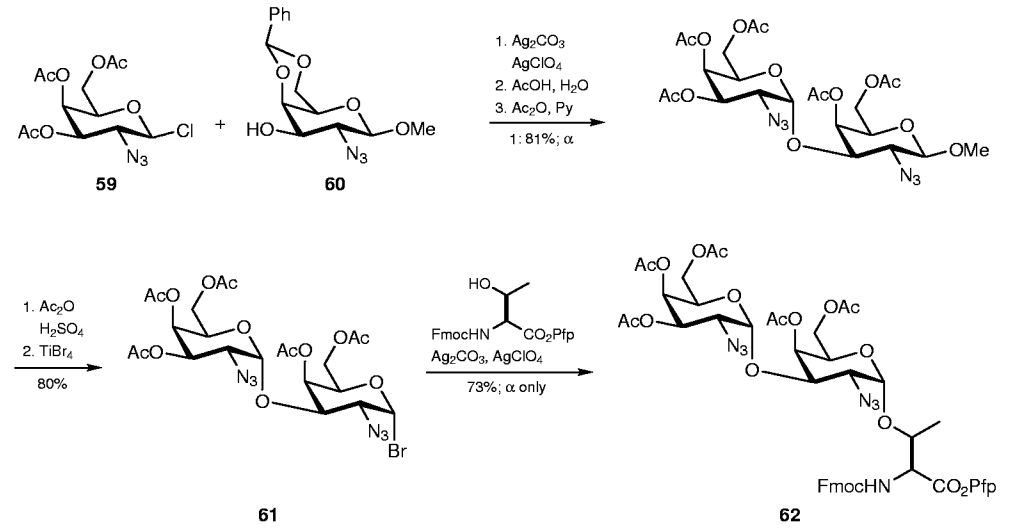


Many of the syntheses utilizing 2-azido precursors have been implemented utilizing strategy 1. Thus, the 2-azido-α-bromide or the 2-azido-β-chloride mono- or disaccharide has been used as donor in the synthesis of the core structures using several different catalysts, e.g. $AgClO_4/Ag_2CO_3$, TMSOTf, and AgOTf.[51,53,54,59,60,101,176,287–289] Typically, the galactal is first converted into the 2-azido derivative, judiciously protected, then coupled to a mono- or disaccharide. The entire oligosaccharide is then converted into a glycosyl donor and coupled to a protected amino acid. Attempts to vary that strategy by carrying out the azidonitration reaction on a completely formed trisaccharide glycal were not efficient.[290] The choice of *N*- or *C*-amino acid protecting groups using this approach has varied widely: Fmoc/Pac, Fmoc/tBu, Fmoc/Bzl, Fmoc/Al, Fmoc/Pfp, Z/Bzl, and Z/tBu have all been used. A good choice is the Fmoc/Pfp ester, which yields already activated glyco amino acid building blocks for stepwise solid-phase peptide synthesis in an average of 60% yield. Sulfanylglycosides[102,291,293] have also been used extensively in the synthesis of core structures, most often with dimethyl(methylsulfanyl)sulfonium triflate as the catalyst. MeOTf, dimethyl(methylsulfanyl) sulfonium tetrafluoroborate, alkenesulfenyl triflates, benzeneselenenyl triflate, or halonium-forming species have been applied as promotors in the glycosylation of amino acids bearing the following N/C protection pairs: Fmoc/Bzl, Fmoc/tBu, and Z/Bzl. The yields are generally comparable to those obtained using glycosyl halides.

6.3.2.4.1.2.1 Synthesis of O-Linked 2-Azido-2-deoxyglycosyl Amino Acids via the Glycosyl Halide

In contrast to the strategy outlined in Scheme 15, the core 5 structure has been successfully synthesized using the 2-azido Gal moiety, which serves both as the donor and acceptor (Scheme 17).[289] Using strategy 1, the α(1→3)-linked disaccharide is first formed in 81% yield by reacting the donor **59** and the acceptor **60** under $Ag_2CO_3/AgClO_4$ promotion. The resulting disaccharide is later converted into the bromide donor **61** by treatment with $TiBr_4$ after acetolysis. Coupling of the disaccharide bromide to Fmoc-Thr-OPfp is facilitated by the presence of the $Ag_2CO_3/AgClO_4$, yielding the desired building block **62** in 73% yield.

Scheme 17 Synthesis of the Core 5 Structure Using Glycosyl Halides and the 2-Azido Galactose Moiety as Both Glycosyl Donor and Acceptor[289]



for references see p 322

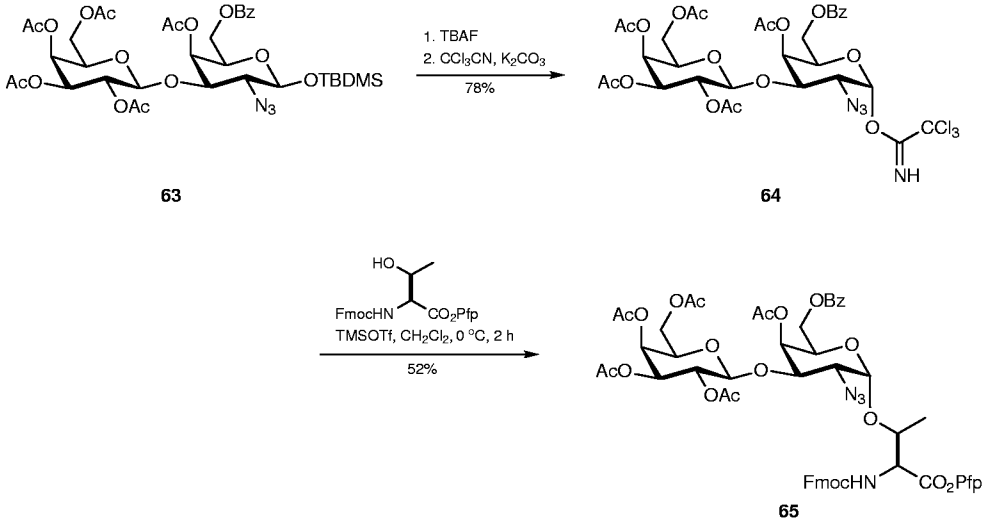
N^{α}-(Fluoren-9-ylmethoxycarbonyl)-*O*-[*O*-(3,4,6-tri-*O*-acetyl-2-azido-2-deoxy-α-D-galactopyranosyl)-(1→3)-(4,6-di-*O*-acetyl-2-azido-2-deoxy-α-D-galactopyranosyl)]-L-threonine Pentafluorophenyl Ester (62):[289]

A mixture of Fmoc-Thr-OPfp (168 mg, 0.33 mmol), activated powdered 4-Å molecular sieves, and Ag_2CO_3 (110 mg, 0.4 mmol) in dry CH_2Cl_2/toluene (1:1, 5 mL) was stirred at rt under N_2. After 1 h, $AgClO_4$ (11 mg, 0.05 mmol) and a soln of the bromide **61** (265 mg, 0.4 mmol) in dry CH_2Cl_2/toluene (1:1, 3 mL) were added. The mixture was stirred for 20 h at rt and then filtered. The organic layer was washed successively with $NaHCO_3$ and H_2O, dried (Na_2SO_4), and concentrated. The residue was chromatographed (silica gel, toluene/EtOAc 3:1) to give **62**; yield: 265 mg (73%); ^1H NMR (250 MHz, $CDCl_3$, δ): 1.42 (d, 3H, Thr CH_3), 3.68 (dd, 1H, $J_{1',2'}$ = 4.1 Hz, $J_{2',3'}$ = 11.2 Hz, H2′), 3.83 (dd, 1H, $J_{1,2}$ = 3.6 Hz, $J_{2,3}$ = 10.0 Hz, H2), 3.93 (dd, 1H, $J_{5,6a}$ = 6.6 Hz, $J_{6a,6b}$ = 10.2 Hz, H6a), 4.05 (dd, 1H, $J_{5,6a}$ = 7.1 Hz, $J_{6a,6b}$ = 11.2 Hz, H6a), 4.14 (dd, 1H, $J_{3,4}$ = 3.6 Hz, H3), 4.16 (dd, 1H, $J_{5,6b}$ = 7.1 Hz, $J_{6a,6b}$ = 11.2 Hz, H6b), 4.18–4.28 (m, 4H, Fmoc CH, H5, H5′, H6b), 4.47 (dd, 1H, $J_{CH,CH2b}$ = 7.1 Hz, $J_{CH2a,CH2b}$ = 10.7 Hz, Fmoc CH_2a), 4.52 (dd, 1H, $J_{CH,CH2b}$ = 7.2 Hz, Fmoc CH_2b), 4.58 (dd, 1H, $J_{CHβ,CH}$ = 6.1 Hz, $J_{CHα,CHβ}$ = 2.0 Hz, Thr Hβ), 4.79 (dd, 1H, Thr Hα), 5.19 (d, 1H, H1), 5.32 (d, 1H, H1′), 5.37 (dd, 1H, $J_{3',4'}$ = 3.05 Hz, H3′), 5.48 (dd, 1H, $J_{4',5'}$ = 1.0 Hz, H4′), 5.54 (dd, 1H, $J_{4,5}$ = 1.0 Hz, H4), 6.66 (d, 1H, $J_{NH,CHα}$ = 9.6 Hz, Thr NH).

6.3.2.4.1.2.2 Synthesis of O-Linked 2-Azido-2-deoxyglycosyl Amino Acids via the Glycosyl Imidate

Some synthetic procedures rely on 2-azido-2-deoxy imidates as glycosyl donors, in which case the stereochemical outcome of the glycosylation is also dependent on the nature of the promoter as well as the type and orientation of the leaving groups and the type of protection on the acceptor.[249,269,273,278,294,295] The Fmoc/Pfp ester also proved superior to other *N*- and *C*-amino acid protecting groups in the coupling of the imidates. In the synthesis of the T-antigen depicted in Scheme 18, the disaccharide anomeric hydroxy group in **63** is protected with a TBDMS group that can be selectively removed under conditions that leave both acid- and base-labile protecting groups untouched.[278] The disaccharide is converted into the imidate donor **64**, which is coupled to the Fmoc-Thr-OPfp with TMSOTf activation to give the α-product **65** in 52% yield.

Scheme 18 Synthesis of the Core 1 (T-Antigen) Structure Using the Glycosyl Imidate[278]



N^{α}-(Fluoren-9-ylmethoxycarbonyl)-*O*-[2,3,4,6-tetra-*O*-acetyl-β-D-galactopyranosyl-(1→3)-4-*O*-acetyl-2-azido-6-*O*-benzoyl-2-deoxy-α-D-galactopyranosyl]-L-threonine Pentafluorophenyl Ester (65):[278]

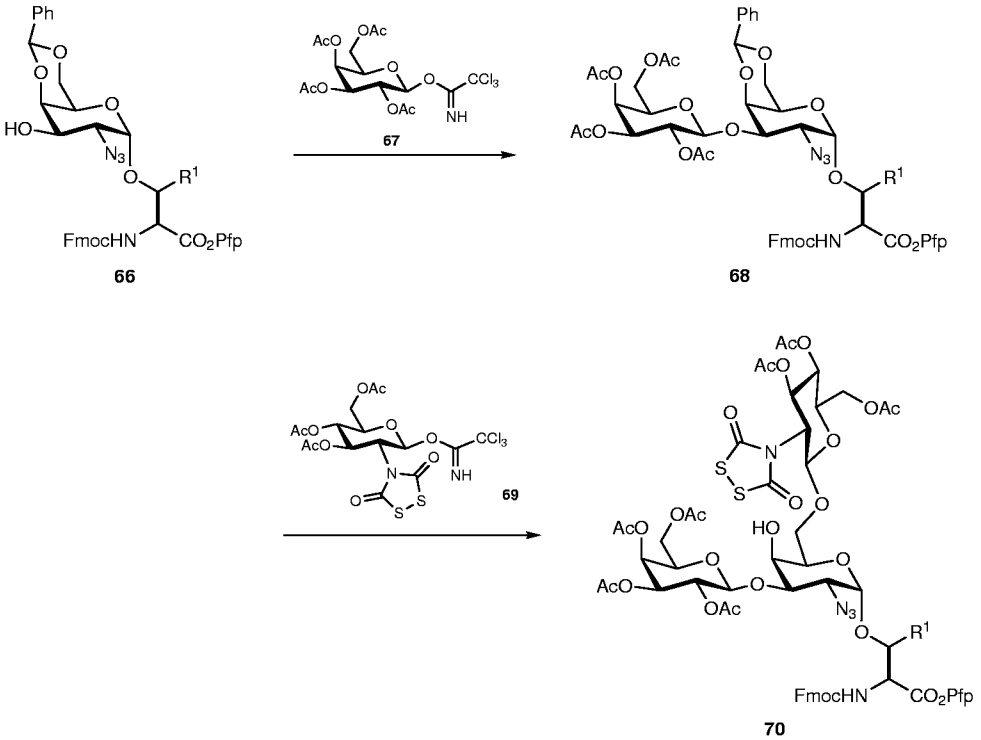
To a mixture of **64** (0.65 g, 0.78 mmol) and Fmoc-Thr-OPfp (0.5 g, 0.98 mol) in CH_2Cl_2 (3 mL) at 0 °C was slowly added TMSOTf (7 μL, 0.039 mmol, 0.05 equiv) in CH_2Cl_2 (1 mL). After 2 h, the mixture was neutralized by the addition of pyridine (3 μL, 0.039 mmol) and concentrated. The residue was purified by column chromatography (Florisil, toluene/acetone 12:1) and the anomers separated using MPLC (tol-

uene/acetone 9:1); yield: 483 mg (52%); ^{1}H NMR (250 MHz, $CDCl_3$, δ): 3.73 (dd, 1H, $J_{2,3}$ = 10.8 Hz, $J_{3,4}$ = 3.8 Hz, H3), 3.90–3.94 (m, 1H, H5′), 4.0–4.2 (m, 3H, Fmoc CH, 2×H6′), 4.2–4.5 (m, 5H, Fmoc CH_2, H5, 2×H6), 4.48–4.52 (m, 1H, Thr Hα), 4.53 (dd, 1H, $J_{1,2}$ = 3.9 Hz, H2), 4.71 (dd, 1H, $J_{CH\alpha,NH}$ = 9.2, $J_{\alpha,\beta}$ = 3.1 Hz, Thr Hα), 4.74 (d, 1H, $J_{1',2'}$ = 7.8 Hz, H1′), 5.03 (dd, 1H, $J_{2',3'}$ = 10.5 Hz, $J_{3',4'}$ = 3.1 Hz, H3′), 5.14 (d, 1H, H1), 5.19 (dd, 1H, H2′), 5.36 (d, 1H, H4′), 5.59 (d, 1H, H4).

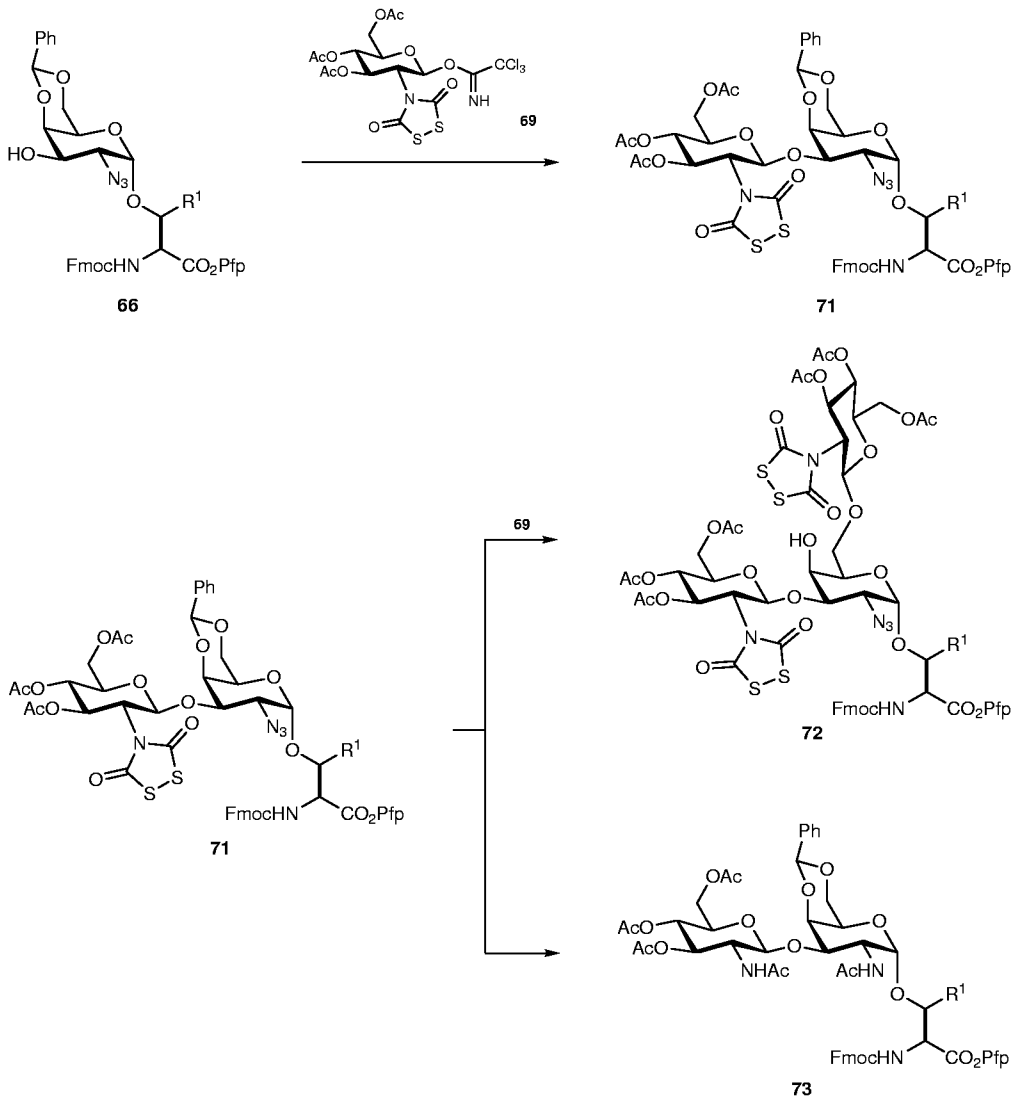
6.3.2.4.1.2.3 Versatile Syntheses of Cores 1, 2, 3, and 4 via 2-Azido-2-deoxy and N-Protected Precursors

While the preparation of core structures using strategy 1 has been fairly successful, in recent years, strategy 2, based on the glycosylation of preformed glycosylated amino acids, has proven to be a much more versatile method for the preparation of the other structurally more complex core structures. In an example of the versatility of strategy 2 for assembling core structures (Scheme 19),[296] glycosylation of various Fmoc-Thr(α-D-$GalN_3$)-OPfp derivatives with a peracetylated galactosyl imidate or *N*-Dts protected glycosyl donor afforded protected building blocks which were used directly in the solid-phase synthesis of core 1, core 2, core 3, and core 4 O-linked glycopeptides. First, the core 1 structure intermediate **68**, which is also the precursor to the core 2 building block, was obtained by stereoselective glycosylation of the *N*-Fmoc-Thr(4,6-benzylidene-α-D-$GalN_3$)-OPfp derivative **66** with the peracetylated galactosyl imidate **67**. The benzylidene group of the intermediate **68** was removed and the diol glycosylated with 2-*N*-Dts-2-deoxy-3,4,6-tri-*O*-acetylglucosyl trichloroimidate **69** under TMSOTf catalysis to yield the trisaccharide precursor **70** of core 2 in 67% yield. Similarly, the core 3 structure precursor **71** was obtained by glycosylation of **66** with **69** and the deblocked disaccharide intermediate coupled to the imidate **69** to yield the core 4 structure **72**. Simultaneous reduction and acetylation of both the azide and Dts groups could be effected using Zn/AcOH/Ac_2O (e.g., conversion of **71** into **73**). Simultaneous reduction can also be accomplished using DTT/DIPEA, while selective reduction of the Dts group in the presence of the azide was achieved by treatment with propanedithiol.

Scheme 19 Synthesis of the Core 1, 2, 3, and 4 Structures Using Both 2-Azido and *N*-Dts Building Blocks[296]



for references see p 322



6.3.2.4.1.2.4 Reduction of Dithiasuccinoyl and Azide Groups in Solution and on Solid Phase

The transformation of the azido into an acetamido group has been developed with the need for compatibility with other protection groups and for the efficiency required in solid-phase synthesis. The conversion can take place before or after the incorporation of the building block into the peptide on solid phase. The acetylated product can be obtained by first reduction to the amine followed by acetylation, or in a one-pot procedure that includes both reduction and acetylation.

A surprising selectivity, yielding up to 92% product, was observed in the reduction of the azido group by hydrogenolysis over Pd/C in the presence of a Z group for N^{α}-amino protection.[297–299] Subsequent acetylation of the amino group was most conveniently performed in a mixture of pyridine/Ac_2O. More homogeneous conditions were developed with a mixture of $NiCl_2/NaBH_4$, which did not lead to β-elimination of the saccharide;[51,269,300–302] the method also gives good yields when several azido groups are present in saccharide cluster

peptides.[266,303] With the use of Fmoc groups, less basic conditions were needed and it was found that H_2S in pyridine was compatible with the presence of the Fmoc group and gave a smooth conversion into the amino compound.[50,53,55,59,60]

Two newer types of reagents which both yield the acetamido group in a single reaction have been described. In the Staudinger method, phosphines in the presence of AcOH readily convert the azido group into the phosphinimine, which reacts with the dry AcOH to afford the target acetamido group.[56,288] Similarly, in the synthesis of the pentaglycosylated fragment from human intestinal mucin, thioacetic acid directly and quantitatively converts the azido group into the acetamido group.[98,105,107,120] However, reaction times are long on solid phase and *N*-thioacetates can be formed as impurities. To circumvent these problems, thiolytic reduction with dithiols such as DTT in the presence of DIPEA[43] or DBU[304] have been successfully employed both on solid phase and in solution. The progress of the reaction on solid phase can easily be monitored by FTIR. The azido group at the building block stage may also be converted directly by treatment with Zn/AcOH.[270] However, careful monitoring of the reaction is required to avoid the formation of the N-diacetylated derivatives.

One-Pot Azide Reduction and Acetylation Using Zn/Ac_2O; General Procedure:[296]
To a stirred soln of a compound containing the azide (42 μmol) in THF/Ac_2O/AcOH (3:2:1, 5 mL) was added Zn (75 mg, activated with 2% aq $CuSO_4$). After stirring for 4 h, the mixture was diluted with freshly distilled THF (20 mL), filtered through Celite, rinsed several times with THF, concentrated, and purified by chromatography on pre-dried silica gel.

Simultaneous Dts and Azide Reduction in Solution Using DTT; General Procedure:[43]
To a soln of the appropriate compound (1.1 mmol) in dry CH_2Cl_2 (20 mL) was added DTT (30.5 mg, 198 mmol) and DIPEA (20 μL) in dry CH_2Cl_2 (5 mL). The soln was then stirred at rt for 30 min. The reaction was followed by TLC and after complete N-deprotection and reduction, the solvent was removed and the residue dissolved in a mixture of Ac_2O (5 mL) and pyridine (10 mL). After 3 h at rt, the mixture was concentrated, any residual pyridine coevaporated with toluene, and the residue purified by chromatography on silica gel.

Simultaneous Dts and Azide Reduction on Solid Phase Using DTT; General Procedure:[43]
The resin (100 mg), with the Dts- and the azido group-containing glycopeptides, was washed thoroughly with CH_2Cl_2 (5×). The thiolysis and reduction were then performed by treatment of the resin with a soln of DTT (0.2 M) and DIPEA (0.1 mM) in CH_2Cl_2 (1×2 min+2×3 h). In the case of only the *N*-Dts-containing building blocks, the thiolysis procedure could be shortened to 2×10 min. The resin was then washed thoroughly with CH_2Cl_2 (5×) and DMF (10×).

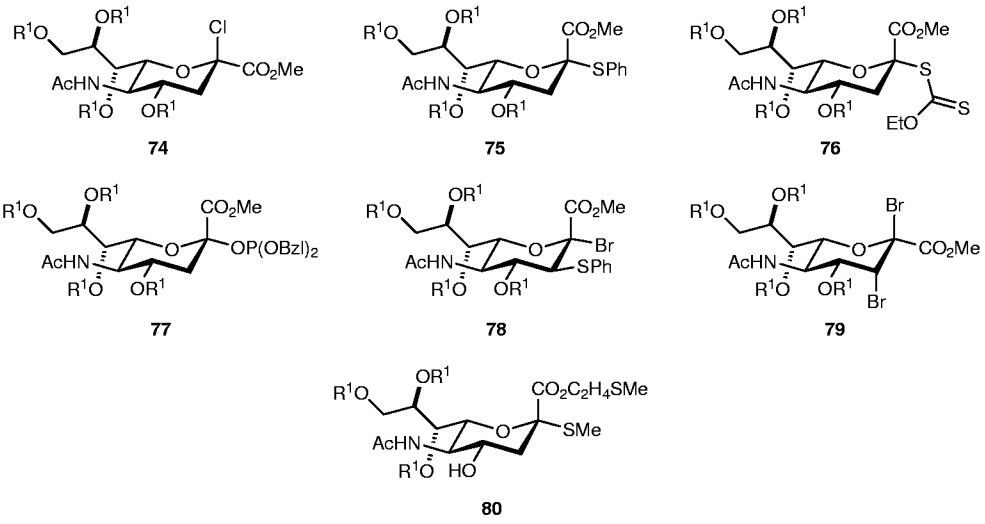
Selective Reduction of Dts in the Presence of an Azide on Solid Phase Using Propanedithiol; General Procedure:[43]
An *N*-Dts- and azido group-containing building block (1.6 mol equiv) was reacted with propanedithiol (0.2 M) and DIPEA (0.3 M) in CH_2Cl_2 (2×10 min). After a thorough wash with CH_2Cl_2 (5×) and DMF (10×), the amino group was acetylated by treatment with Ac_2O in DMF (20%) at rt for 30 min.

6.3.2.4.2 Sialylated Glycopeptides

Sialylated glycans play important roles in various biological phenomena. In the glycopeptide realm, the focus has mainly been on the synthesis of sialylated T_N- and T-antigens owing to their appearance on cancer cell surfaces and in the envelope glycoprotein gp120 of HIV. Sialylations are generally difficult to perform because of the deactivating effect of the carboxylic acid and the tendency to form the 2,3-glycal as a result of elimination. Early efforts involving the use of the glycosyl halide **74** (Scheme 20) as the donor resulted in low yields and stereoselectivity.[297,305,306] In recent years, two main approaches to obtaining the α-linked sialoside have emerged. The first approach advocates careful manipulation of the reaction conditions (donor, solvent, temperature, and catalyst). The second approach requires the use of a temporary participating group at C3 (see **78** and **79**)[307,308] or as part of the carboxylic acid ester (see **80**).[309] The latter approach has not yet been applied to the synthesis of glycosyl amino acids.

for references see p 322

Scheme 20 Various Sialic Acid Donors for the Synthesis of Sialylated Building Blocks



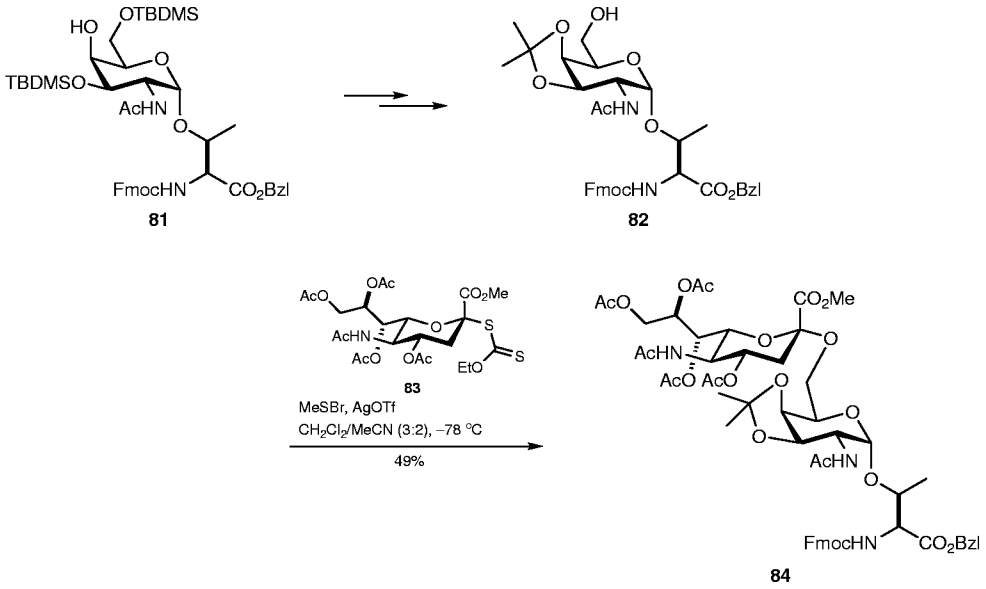
The first approach has led to the synthesis of several different efficient sialyl donors such as the thioglycosides **75**,[310] xanthates **76**,[311] and phosphites **77**,[312,313] which avoid the need for a C3 participating group. The xanthate has been found to be particularly effective, providing good yields and high stereoselectivity when employed for sialylations using methanesulfenyl or benzenesulfenyl bromide as the promoter in acetonitrile at low temperatures. Both the phenylsulfanyl **78** and bromide **79** have been used as C3 participating groups for sialylations; however, the phenylsulfanyl has gained wider usage. The SPh group is generally removed by treatment with AIBN and Ph_3SnH under reflux conditions. This method is not appropriate for glycosyl amino acids with *N*-Fmoc protection since the group is partially removed under these conditions. Removal of the SPh group should be carried out before coupling to the amino acid. Combinations of the above methodologies have been used for the synthesis of the 2,6-sialyl T_N-antigen[125,274,314,315] and the 2,6-[11] and 2,3-sialyl T-antigens.[11,274,316]

6.3.2.4.2.1 Synthesis of Sialylated 2-Azido-2-deoxyglycosyl Amino Acids Using Sialyl Donors without C3 Participating Groups

The synthesis of sialylated building blocks using strategy 2 requires that the glycosylated amino acid precursors are suitable for regioselective deprotection to furnish the 2,3- or the 2,6-sialyated derivatives. The prevalent strategy has been to use base-labile acyl groups for masking the 3- or 6-positions; their removal, however, may be incompatible with the base-labile *N*-Fmoc-protecting group typically utilized in solid-phase glycopeptide synthesis. Careful control of the pH is required for selective cleavage of acetates in the presence of the Fmoc group. Moreover, it must also be possible to deprotect the α-carboxy group of the amino acid selectively from the sialic acid ester, which in turn must be hydrolyzed from the glycopeptide without epimerization or β-elimination of the glycan side chain. Recent reports of selective cleavage of methyl esters in the presence of the Fmoc group may be helpful.[239] Attempts to circumvent this problem have led to the development of the acid-labile silyl protecting groups on the acceptor (Scheme 21)[125,291] or no protecting groups at all.[317] In the latter method, Fmoc-Thr(GalNAc)-OtBu was glycosylated using peracetylated sialic acid xanthate with MeSBr/AgOTf promotion at low temperature. The 2→6 linked product was obtained in 32% yield along with the bis- and tris-glycosylated products containing sialic acid in the 2- and 3-positions. The resulting disaccharide was completely acetylated to provide a

more stable building block for glycopeptide synthesis. The 2,3-sialyl T-antigen has also been synthesized using a sialyl phenylsulfanyl donor without a C3 participating group. Sialylation of the C3 hydroxy of Gal occurred in a high yield (65%) and the final trisaccharide trichloroacetimidate was coupled *en bloc* to Ser and Thr in 33 and 41% yield, respectively.

Scheme 21 Synthesis of the 2,6-Sialyl T_N-Antigen Building Block Using a Sialyl Donor without a C3 Participating Group[125]



***N*$^{\alpha}$-Fluoren-9-ylmethoxycarbonyl-3-*O*-{2-acetamido-2-deoxy-3,4-*O*-isopropylidene-6-*O*-[methyl(5-acetamido-4,7,8,9-tetra-O-acetyl-3,5,-dideoxy-D-glycero-α-galacto-2-nonulopyranosyl)onate]-α-D-galactopyranosyl}-L-threonine Benzyl Ester (84):**[125]

A mixture of **82** (124 mg, 184 μmol; from **81**), **83** (219 mg, 368 μmol), and 3-Å molecular sieves (180 mg) in MeCN/CH_2Cl_2 (3:2, 5 mL) was stirred under an inert atmosphere at rt for 30 min. AgOTf (118 mg, 459 μmol) was added and the mixture cooled to –78 °C. 4 M MeSBr in 1,2-dichloroethane (92 μL, 368 μmol) was added and after 80 min the reaction was quenched by addition of TEA (450 μL, 3.2 mmol). After an additional 70 min at –78 °C the mixture was allowed to attain rt. The crude mixture was first purified by chromatography over silica gel to give predominantly the α-product **84** contaminated with some of the β-anomer. Separation of the anomers was achieved by HPLC; yield: 103 mg (49%); ^{1}H NMR ($CDCl_3$, δ): 5.72 (d, 1H, J = 9.6 Hz, AcHN), 5.49 (d, 1H, J = 9.5 Hz, NHα), 5.36 (ddd, 1H, J = 8.1, 6.0, 2.7 Hz, H8′), 5.31 (dd, 1H, J = 7.5, 1.9 Hz, H7′), 5.20 (d, 1H, J = 9.3 Hz, AcHN′), 5.19 (Ab d, 1H, J = 12.0 Hz, $PhCH_2$), 5.08 (Ab d, 1H, J = 11.9 Hz, $PhCH_2$), 4.90 (ddd, 1H, J = 12.0, 9.6, 4.8 Hz, H4′), 4.67 (d, 1H, J = 3.2 Hz, H1), 4.48 (m, 2H, Fmoc CH_2), 4.38 (b d, 1H, J = 8.2 Hz, Hα), 4.34 (dd, 1H, J = 12.4, 2.7 Hz, H9′), 4.24 (t, 1H, J = 6.6 Hz, Fmoc CH), 4.20 (dd, 1H, J = 6. 1, 2.9 Hz, H2), 4.11 (dd, 1H, J = 12.4, 5.4 Hz, H9′), 3.95 (dd, 1H, J = 9.0, 4.8 Hz, H3), 3.92 (dd, 1H, J = 10.2, 6.9 Hz, H6), 3.78 (s, 1H, CO_2Me), 3.70 (dd, 1H, J = 10.2, 5.4 Hz, H6), 2.59 (dd, 1H, J = 12.9, 4.7 Hz, $H3'_{eq}$), 2.11 and 2.10 (2s, each 3H, 2×Ac), 2.0 (s, 9H, 3×Ac), 1.96 (t, 1H, J = 12.4 Hz, $J_{H3'ax,C\text{-}l}$ = 6.2 Hz, $H3'_{ax}$), 1.87 (s, 3H, Ac), 1.56 and 1.42 (2s, each 3H, isopropylidene CH_3), 1.29 (d, 3H, J = 6.2 Hz, Hγ).

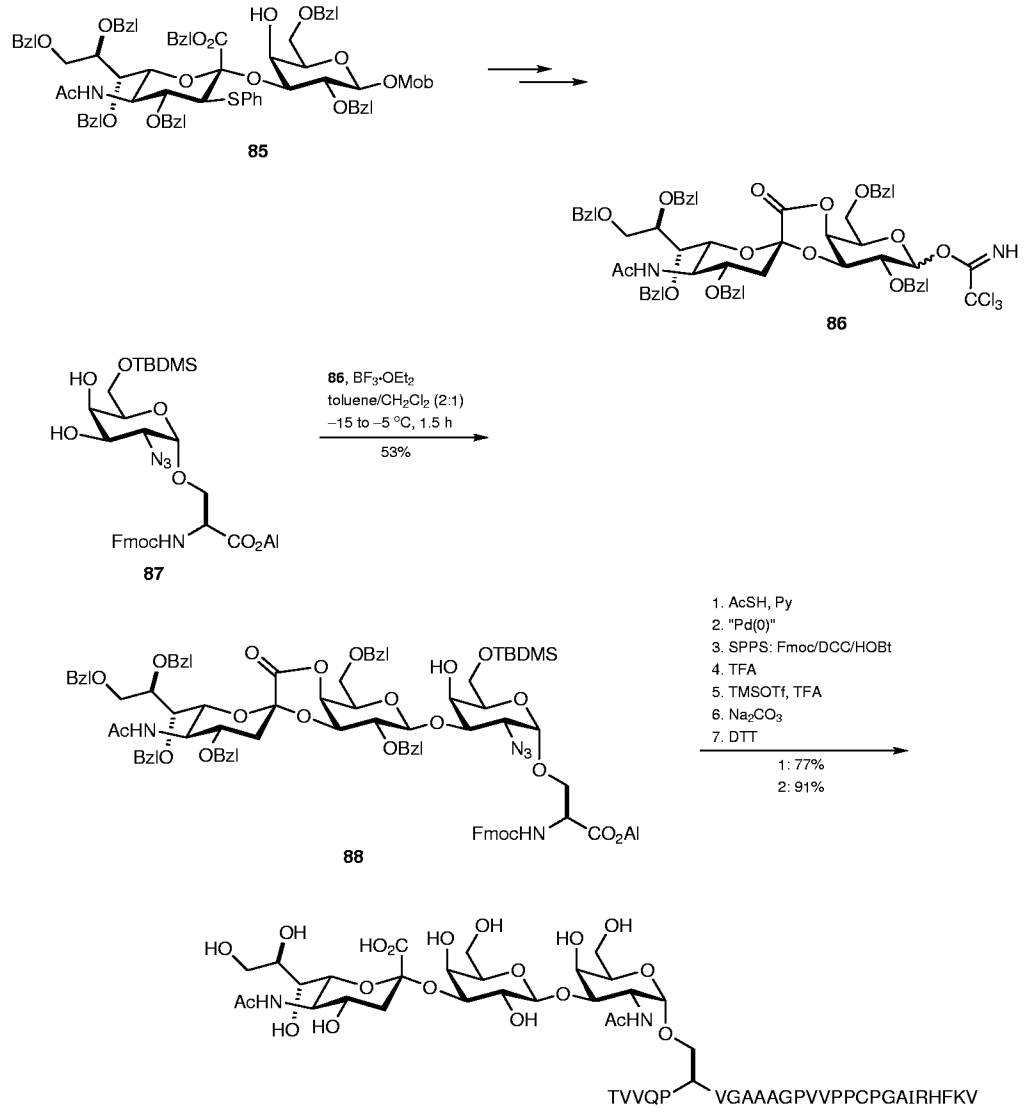
6.3.2.4.2.2 Synthesis of Sialylated 2-Azido-2-deoxyglycosyl Amino Acids Using Sialyl Donors with C3 Participating Groups

The synthesis of a 2,3-sialyl T-antigen cysteine-containing heptacosaglycopeptide has been described (Scheme 22).[318] The key features of this synthesis are the use of the SPh group (which is removed prior to coupling of the disaccharide to the amino acid) to direct the α-selectivity of the sialylation, and the use of benzyl protecting groups for the glycan portion. Thus the disaccharide **85** is converted into the imidate **86**, which is coupled to the T_N-antigen

for references see p 322

derivative **87** to yield the 2,3-sialyl T-antigen **88** in 53% yield. Generally, benzyl protecting groups are avoided in glycopeptide synthesis, particularly when Met, Cys, and Trp are present, since they are not compatible with hydrogenolytic removal of the benzyl groups. In this case, the benzyl groups were partially removed (30%) during TFA cleavage of the peptide from the resin and completely by treatment with TMSOTf/anisole in TFA.[318] Because of the presence of cysteine in the peptide sequence, dimerization was a problem. The correct glycopeptide was finally obtained after treatment with DTT. Another efficient synthesis of the 2,3-sialyl T-antigen has been recently described.[316] The carbohydrate protecting group in this case was the base-labile acetate.

Scheme 22 Synthesis of the 2,3-Sialyl T-Antigen Building Block and Glycopeptide Using a Donor with a C3 Participating Group[318]



6.3.2.4.2.3 Enzymatic Sialylation

As an alternative to chemical sialylations, glycosyl transferases have been used to accomplish sialylations, particularly of N-linked glycopeptide building blocks (see also Section 6.3.1.1.5). The 2,3- and 2,6-sialyltransferases have been easily expressed in preparative amounts. Their use in glycopeptide building block synthesis is described in Section 6.3.3.6.

6.3.2.4.3 1,2-*trans*-O-Linked *N*-Acetyl-D-glucosamine Glycosides

The compound β-D-GlcNAc-Ser has been found as an otherwise atypical glycosylation product in glycoproteins located in the nucleus and cytoplasm.[319,320] The function of these *O*-GlcNAc modified proteins is still unclear, although it has been suggested that they are involved in transcription control. This type of glycosylation product has been prepared for investigation of possible intermediates in *N*-glycoprotein biosynthesis[321] and as model compounds for mucin-type glycopeptides and β-elimination studies.[322,323]

Glycosylation with 2-amino-2-deoxy sugars to obtain the 1,2-*trans* product requires a participation group at C2. Initial efforts focused on the use of the 2-acetamido group itself as a participating group in glycosylations and use of the peracetate or the glycosyl halide as donor.[48,91,322,324–328] Thus, Lewis acid catalyzed glycosylation of Fmoc-Ser/Thr-OPfp and Fmoc-Ser/Thr-OH and with an excess of peracetylated GlcNAc furnish the building block in poor yield. Both reactions require refluxing in benzene for an extended period of time since they proceed through the oxazoline intermediate.

In an effort to improve yields and use milder conditions to generate the β-*N*-GlcNAc building block, recent efforts have concentrated on utilizing protecting groups at C2 that exert a neighboring group effect without the formation of the stable oxazoline intermediates. These protecting groups are often cyclic imides (e.g., phthalimides), carbamates, or protected amides deactivated with electron-withdrawing groups. As with the use of the azido group for the formation of the α-linked core structure, the protecting groups may be removed at the building block stage or on solid phase after peptide synthesis.

The phthalimido group has been used as an efficient β-directing amino protecting group in carbohydrate chemistry. However, the need for harsh, stringent conditions such as excess hydrazine, butylamine, hydroxylamine, ethylenediamine, or $NaBH_4$ at high temperatures and extended reaction times can compromise base-sensitive protecting groups elsewhere in the molecule, as well as epimerizable centers and delicate functionalities. To address this problem, derivatives such as the tetrachlorophthalimido (TCPhth),[329,330] dichlorophthalimido (DCPhth),[331] and nitrophthalimido (NO_2Phth)[332] analogues have been developed. Thus, Z-Ser-OBzl was glycosylated with the 2-deoxy-2-TCPhth-3,4,6-tri-*O*-benzoyl pentenyl glycoside in 76% yield.[329,330] Removal of the TCPhth group by treatment with 1.8 equivalents of ethylenediamine at 60 °C for 6 hours followed by acetylation yielded the desired product in only 42% yield. The other derivatives, NO_2Phth and DCPhth, have not yet been applied to the synthesis of glycosylated amino acids, but it is reported that they should be removed under much milder conditions than the phthalimido group.

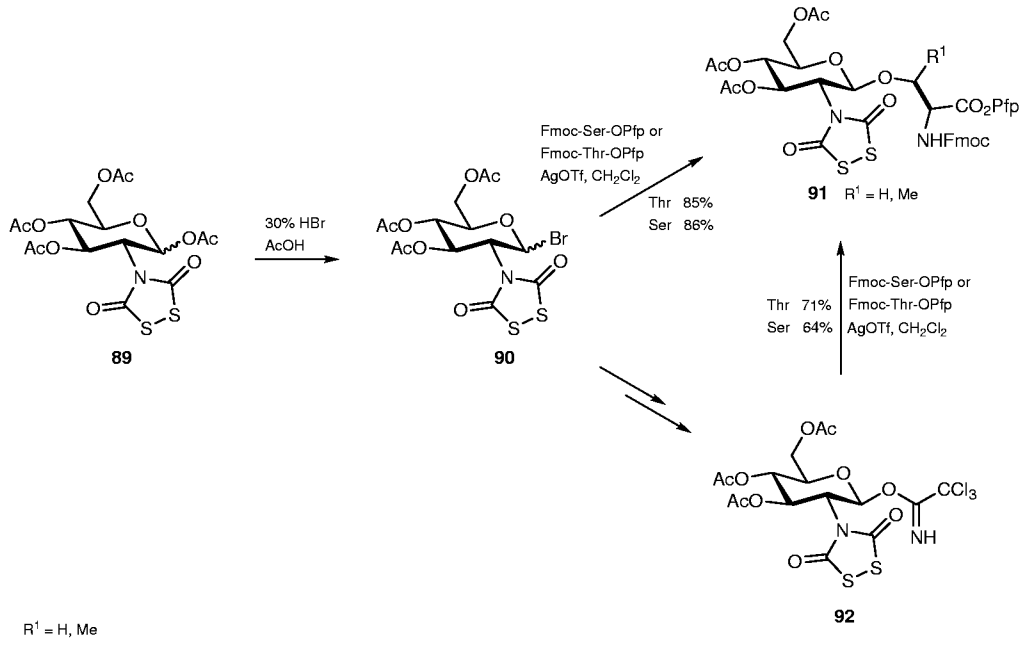
6.3.2.4.3.1 Synthesis of 1,2-*trans*-2-Acetamido-2-deoxyglycosyl Amino Acids Using Cyclic Imide (Dithiasuccinoyl) Protection of the Amine

The Dts group has been used to generate β-GlcNAc building blocks for SPPS.[43,177,333–335] For example (Scheme 23), the Dts glycosyl bromide **90**,[177] obtained from the peracetate **89**, was coupled to Fmoc-Ser/Thr-OPfp to yield exclusively the desired β-product **91** in high yield. In an alternative procedure, the bromide was first converted into the trichloroacetimidate **92** and coupled selectively to Fmoc-Ser/Thr-OPfp in 64 and 71% yield, respectively.[335] It must be noted, however, that while two additional steps are required to generate the glycosyl

for references see p 322

imidate, leading to slightly reduced overall yields, in contrast to the glycosyl bromide, the imidate is shelf stable. The Dts group can be removed under mild conditions that leave the OPfp ester untouched, such as with β-ME, *N*-methylsulfanylacetamide, or DTT. The DTT method seems to be the most trouble-free since the β-ME method has been reported to give thiourethane side products via thiolytic opening of the Dts ring carbonyl. Simultaneous reduction of both the Dts group and the azide has been achieved using DTT or Zn in Ac_2O/AcOH/THF, while selective reduction of the Dts group can be achieved by treatment with $NaBH_4$ in CH_2Cl_2/MeOH.[335]

Scheme 23 Synthesis of β-GlcNAc Gylcosyl Amino Acids via the Dts-Protected Glycosyl Halide or Imidate[335,509]



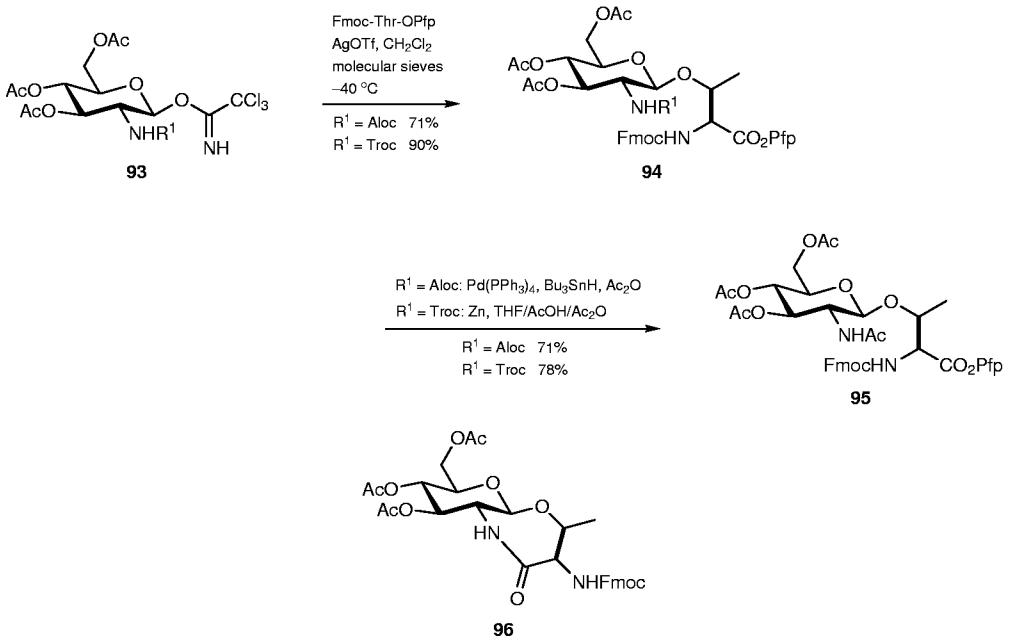
***N*$^{\alpha}$-(Fluoren-9-ylmethoxycarbonyl)-Thr[*O*-(3,4,6-tri-*O*-acetyl-2-deoxy-2-dithiasuccinimido-β-D-glucopyranosyl)]-OPfp (91):**[335]

Fmoc-Thr-OPfp (500 mg, 979 μmol), imidate **92** (618.3 mg, 1.08 mmol), and AgOTf (294 mg, 1.08 mmol) were placed in a pre-dried flask, which was evacuated for several hours at 0.1 Torr and filled with argon. Dry CH_2Cl_2 (30 mL) was injected and the mixture was stirred overnight at rt. The mixture was filtered through Celite and then directly purified by column chromatography (silica gel, EtOAc/light petroleum ether 2:3) to give exclusively the β-anomer; yield: 639 mg (71%).

6.3.2.4.3.2 Synthesis of 1,2-*trans*-2-Acetamido-2-deoxyglycosyl Amino Acids Using Carbamate (2,2,2-Trichloroethoxycarbonyl or Allyloxycarbonyl) Protection of the Amine

The two carbamates used as protecting groups, the Aloc and Troc groups (Scheme 24), both provide the β-glycoside through neighboring group participation. The Aloc group can be removed under practically neutral conditions using a palladium(0) catalyst in the presence of a nucleophile. However, sometimes the liberated amine group can be allylated. This side reaction can be prevented by using Bu_3SnH. Glycosylation of Fmoc-Thr-OPfp has been accomplished using the *N*-Aloc imidate donor **93** (R^1 =Aloc) and AgOTf catalysis to give the β-glycoside **94** (R^1 =Aloc) in 71% yield (Scheme 24).[343]

Scheme 24 Synthesis of β-GlcNAc Glycosyl Amino Acids via the Aloc- or Troc-Protected Glycosyl Donor[337,343]



The Troc group has the advantage of being stable under most of the conditions used for carbohydrate synthesis, including acidic conditions that allows the removal of *O*-acetyl groups by methanolic HCl. However, during selective O-deacetylation under basic conditions, the Troc group is affected. A mild selective O-deacetylation reagent has been described that leaves the *N*-Troc group intact.[336] *N*-Troc protection has been widely used with a variety of donors such as halides, 1-*O*-acetyl sugars, ethylsulfanyl glycosides, and imidates **93** (R^1 =Troc), yielding the desired β-GlcNAc building block **94** (R^1 =Troc) in varying yields (Scheme 24).[189,337–342] The Troc group can be removed and the resulting amine acetylated in the one-pot procedure utilizing Zn in Ac_2O/AcOH/THF, giving the product **95**. It must be noted here that, during the removal of the amine protecting groups and reduction of the azide groups of the β-glycoside containing an active ester (e.g., OPfp), a bicyclic lactam **96** can be formed in minor quantities as a result of the displacement of the ester by the amino group at C2, even though a large excess of Ac_2O is present.

Removal of Aloc Group: N^{α}-(Fluoren-9-ylmethoxycarbonyl)-Thr[*O*-(3,4,6-tri-*O*-acetyl-2-deoxy-2-acetamido-β-D-glucopyranosyl)]-OPfp (95):[343]

To a well-stirred soln of **94** (R^1 = Aloc; 302 mg, 0.344 mmol) in dry CH_2Cl_2 (15 mL) containing Ac_2O (1.9 mL) under argon was added $Pd(PPh_3)_4$ (14 mg, 0.012 mmol) followed by immediate addition of Bu_3SnH (0.181 mL). After 10 min at rt the solvent was removed and the residue was purified by column chromatography (silica gel, EtOAc/light petroleum 2:1) to give **95**; yield: 202 mg (71%).

Removal of Troc Group: N^{α}-(Fluoren-9-ylmethoxycarbonyl)-Thr[*O*-(3,4,6-tri-*O*-acetyl-2-deoxy-2-acetamido-β-D-glucopyranosyl)]-OPfp (95):[344]

To a stirred soln of compound **94** (R^1 =Troc; 2 g, 2.06 mmol) in THF/Ac_2O/AcOH (3:2:1, 60 mL) was added Zn dust (5 g, 325 mesh). After stirring for 2 h, the mixture was filtered through Celite, rinsed several times with freshly distilled THF, and concentrated. Purification by chromatography (silica gel, EtOAc/light petroleum ether 2:1) and subsequent crystallization afforded **95**; yield: 1.34 g (78%); ^{1}H NMR (250 MHz, $CDCl_3$, δ): 6.25 (1H, NH), 5.86 (1H, J = 5.2 Hz, Fmoc NH), 5.14 (1H, H4), 4.99 (1H, H3), 4.74 (1H, Thr Hα), 4.69 (1H, J = 7.73 Hz, H1), 4.43 and 4.37 (2H, Fmoc CH_2), 4.35 (1H, Thr Hβ), 4.21 and 4.11 (2H, H6a, H6b), 4.14 (1H, Fmoc CH), 3.67 (1H, H5), 3.62 (1H, H2), 1.31 (3H, J = 6.02 Hz, Thr Hγ); ^{13}C NMR (62.9 MHz, $CDCl_3$, δ): 96.05 (C1), 72.97 (C5), 72.89 (C3), 71.92 (Fmoc CH_2), 67.59 (Thr Cβ), 67.51 (C4), 62.05 (C6), 59.15 (C2), 58.41 (Thr Cα), 47.55 (Fmoc CH), 21.14, 20.94, 20.93 (3×OAc), 12.83 (Thr Cγ).

for references see p 322

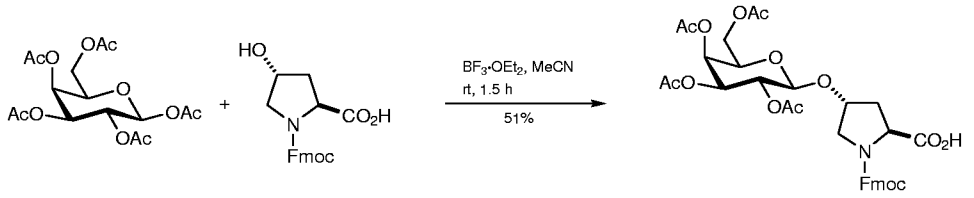
6.3.2.5 Unusual O-Linked Aliphatic Glycosides

In addition to the O-glycosylations of Ser or Thr discussed in Sections 6.3.2.1 – 6.3.2.4, in nature O-glycosylation of two additional amino acids, hydroxyproline (Hyp) and δ-hydroxylysine (Hyl), have also been observed. Glycosylation of Hyp by galactose in some collagens and by arabinose in an as-yet unknown stereochemistry in extensin, a hydroxyproline-rich plant glycoprotein, has been observed.[345] In addition, the mature protein of a class IV chitinase contains hydroxyproline residues gylcosylated with xylose or larger glycans.[346] In structure–function studies, proline is sometimes substituted by a glycosylated hydroxyproline in order to examine the effect of glycosylation on a particular system. To this end, a few glycosylated hydroxyproline-containing peptides have been synthesized.

6.3.2.5.1 Synthesis of 1,2-*trans*-O-Linked Glycosyl Hydroxyproline

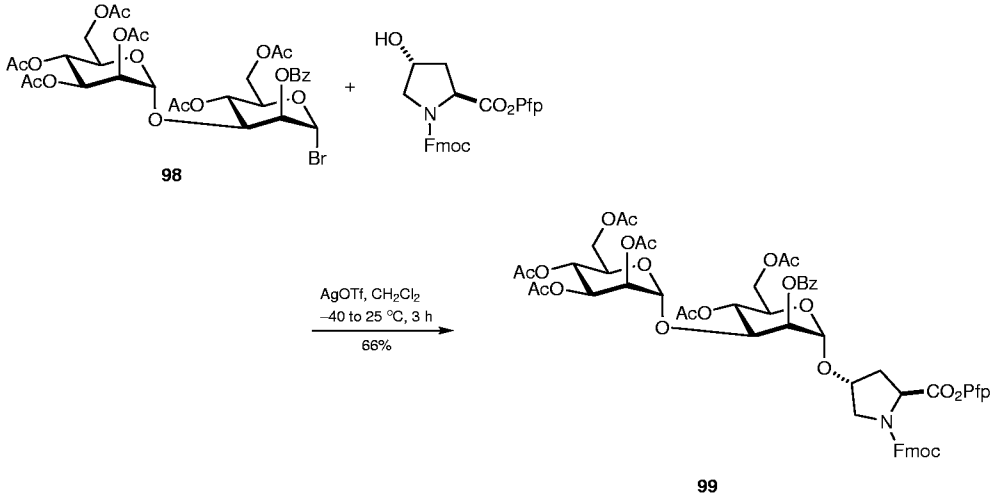
A common feature of the earlier procedures is Z protection of the amino group and the use of benzyl, methyl, or allyl esters for the α-carboxy protection of Hyp, providing building blocks that are not directly suited for peptide synthesis. Currently, there are two methods that generate glycosylated building blocks ready for incorporation into peptides. In the first and most straightforward method, the peracetylated sugar is coupled to Fmoc-Hyp-OH under Lewis acid catalysis at room temperature, yielding the β-product **97** in 51% yield for Gal (Scheme 25).[165] A byproduct formed in 15–20% derives from the acetylation of Hyp. This method has also been used to couple lactose and glucuronic acid (GlcA) in 45 and 2% yields, respectively.[166] The other method is the traditional Koenigs–Knorr procedure, which gives the product in higher yield (Scheme 26).[166,347] The suitably protected glycosyl bromide of dimannosides **98**,[347] Glc, Gal, lactose, and GlcA have been coupled to Fmoc-Hyp-OPfp in 56–66 (for **99**), 68, 73, 69, and 30% yields, respectively.

Scheme 25 Synthesis of a Glycosylated Hydroxyproline Building Block Using the Glycosyl Peracetate as a Donor[165]



97

Scheme 26 Synthesis of a Glycosylated Hydroxyproline Building Block Using the Glycosyl Halide as a Donor[347]



***N*$^{\alpha}$-(Fluoren-9-ylmethoxycarbonyl)-*O*-[4,6-di-*O*-acetyl-2-*O*-benzoyl-3-*O*-(2,3,4,6-tetra-*O*-acetyl-α-mannopyranosyl)-α-D-mannopyranosyl]-*trans*-L-hydroxyproline Pentafluorophenyl Ester (99):**[347]
The bromide **98** (1.55 g, 2.04 mmol), Fmoc-Hyp-OPfp (1.06 g, 2.04 mmol), and 3-Å molecular sieves (3.5 g) in dry CH_2Cl_2 (40 mL) were stirred under argon at −40 °C for 1.25 h. AgOTf (0.60 g, 2.34 mmol) was added quickly and the temperature was raised to −25 °C during 15 min. The mixture was then stirred at −25 °C for 3 h, then 2,6-di-*tert*-butyl-4-methylpyridine (0.42 g, 2.05 mmol) was added. The mixture was stirred without cooling for 15 min, filtered, concentrated to ~20 mL, and the residual soln was purified by column chromatography (silica gel, gradient of hexane to hexane/EtOAc 2:1) to afford **99**; yield: 1.61 g (66%).

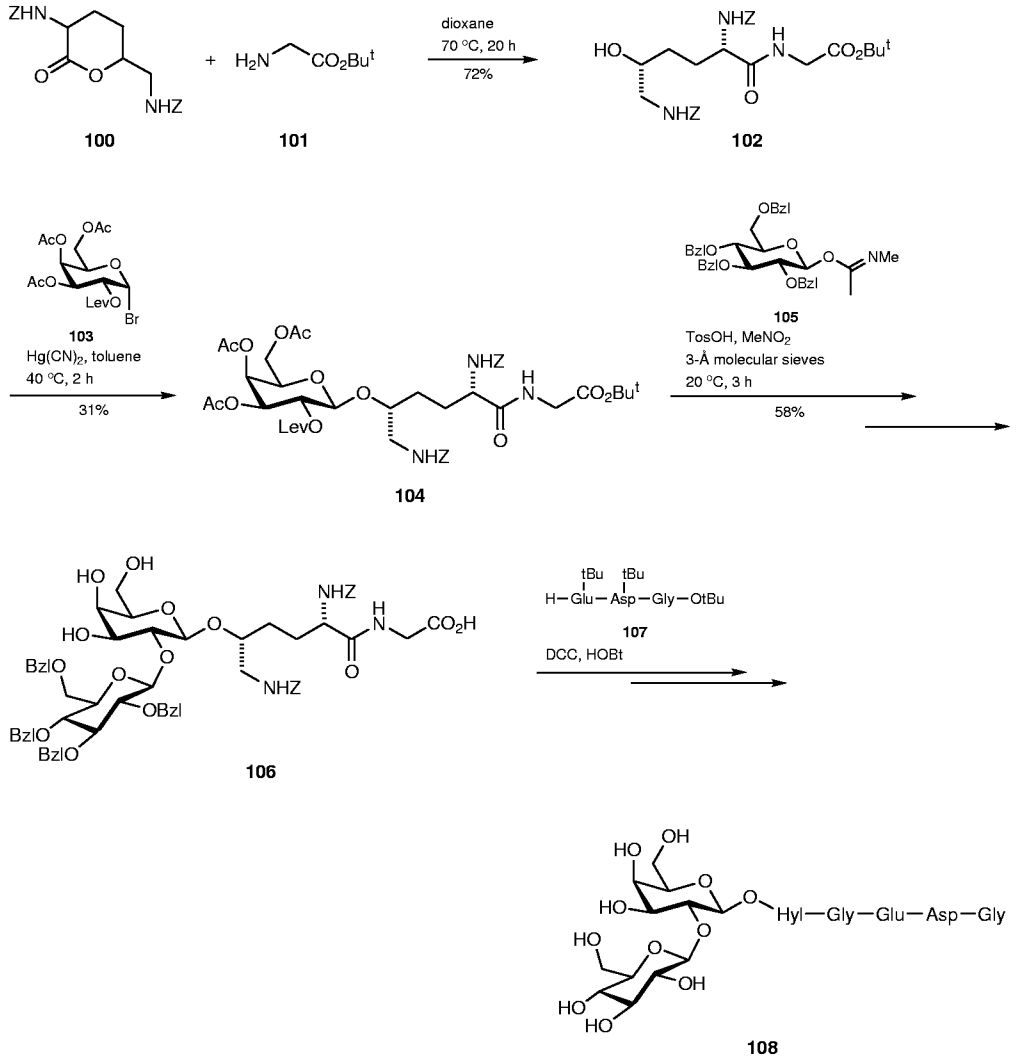
6.3.2.5.2 Synthesis of 1,2-*trans*-O-Linked Glycosyl Hydroxylysine via the Halide

The disaccharide α-Glc-(1→2)-β-Gal has been found linked to the side chain of Hyl in type II collagen in the glomerular basement membrane (GBM).[348] A small glycopeptide containing the disaccharide was isolated from the GBM and was reported to be immunogenic.[349] Efforts have thus focused on elucidating the role of glycosylation of lysine in type II collagen on the T-cell response.[350] Synthesis of the disaccharide amino acid building block is not trivial. Coupling of a preformed disaccharide donor to the amino acid leads to reduced selectivity because of the glycosidic linkage at C2. Alternatively, the galactose unit with a participating group at C2 is first coupled to the amino acid, ensuring the β-anomer. The C2 position must then be selectively deprotected for α-linkage of the glucose unit group.

The hydroxy group of Hyl is a secondary one and may be sterically affected by the protecting groups of the ε-amino group of Lys. Finally, the formation of the α-linkage to Gal requires a donor with nonparticipating groups at C2. There have been two approaches to solving this problem. In the first, the desired pentapeptide was elegantly synthesized by fragment condensation of a glycosylated dipeptide to a tripeptide in solution (Scheme 27).[74] The key step in the synthesis was the masking of the hydroxy group of Hyl as a lactone **100**. The hydroxy group is released for glycosylation when the N$^{\alpha}$-Z-protected lactone is coupled to H-Gly-OtBu (**101**). Glycosylation of the hydroxy group of **102** with bromide **103**, which contains a participating group at C2, yields the monosaccaride β-product **104** in only 31% yield. The C2 hydroxy is protected with a Lev group, which is removed for coupling to perbenzylated glucosyl imidate **105** under acid catalysis to form the 1→2 disaccharide **106** in 68% yield. Fragment condensation with the tripeptide **107** yields the glycopeptide **108** in high yield. A modification of this strategy using *N*-Boc protection has been carried out.[522]

for references see p 322

Scheme 27 Synthesis of Glycosylated Hydroxylysine Building Blocks and Glycopeptide in Solution Using the Glycosyl Halide as Donor[74]



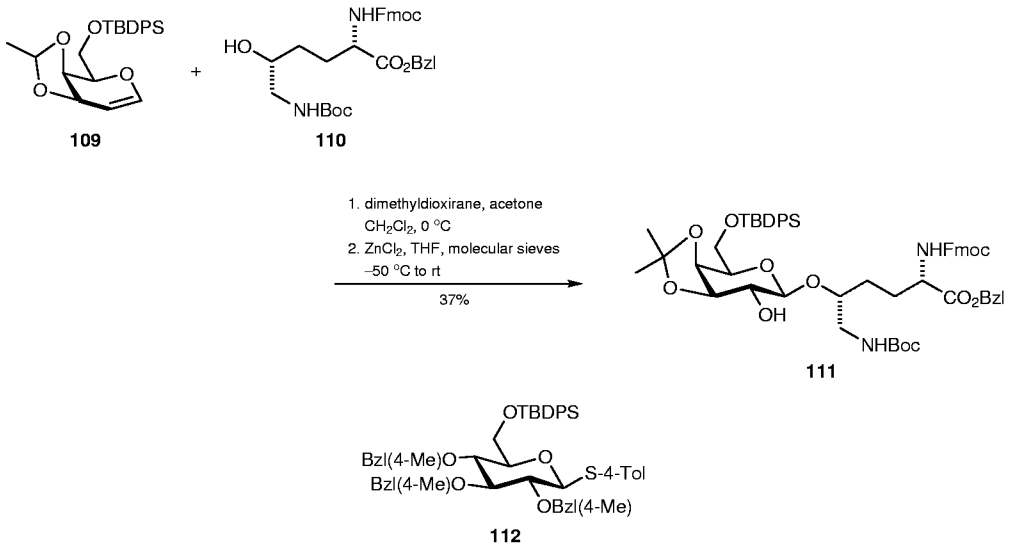
***N*$^{\alpha}$,*N*$^{\varepsilon}$-Bis(benzyloxycarbonyl)-δ-hydroxylysylglycine *tert*-Butyl Ester (102):**[74]
A soln of **100** (1.4 g, 3.4 mmol) and **101** (524 mg, 4.0 mmol) in dry dioxane (7.0 mL) was allowed to stand at 70 °C for 20 h. The solvent was removed and the residue dissolved in EtOAc (100 mL) and washed with $KHSO_4/K_2SO_4$ (pH 2, 25 mL), sat. NaCl (25 mL), 2.2 M K_2CO_3 (25 mL), and sat. NaCl. The organic layer was dried (Na_2SO_4), concentrated, and recrystallized (iPr_2O) to afford **102**; yield: 1.3 g (72%).

***N*$^{\alpha}$,*N*$^{\varepsilon}$-Bis(benzyloxycarbonyl)-δ-hydroxylysyl-*O*-(3,4,6-tri-*O*-acetyl-2-levunoyl-β-D-galactopyranosyl)glycyl *O*-*tert*-Butyl Ester (104):**[74]
To a stirred soln of **102** (1.3 g, 2.4 mmol) and **103** (2.2 g, 4.8 mmol) in dry toluene (30 mL) was added $Hg(CN)_2$ (1.36 g, 5.4 mmol). After 2 h at 40 °C the mixture was filtered and concentrated. The residue was dissolved in $CHCl_3$ (100 mL), washed with 1 M KBr (3×50 mL), and H_2O (50 mL), dried ($MgSO_4$), and concentrated. The product was purified by column chromatography (silica gel, $CHCl_3$/acetone 82:18) to yield **104** plus hydrolyzed donor. This mixture was treated with Ac_2O (3 mL) and pyridine (6 mL) for 2 h at 20 °C, then concentrated and coevaporated with toluene to remove the last traces of pyridine. The residue was dissolved in $CHCl_3$ (100 mL), washed with 10% $NaHCO_3$ (25 mL), H_2O (25 mL), dried ($MgSO_4$), and concentrated to a small volume which was purified by chromatography (silica gel, $CHCl_3$/acetone 84:16); yield: 670 mg (31%).

6.3.2.5.3 Synthesis of 1,2-*trans*-O-Linked Glycosyl Hydroxylysine via the Epoxide

In the second strategy, attempts were made to create a building block that could be immediately used in SPPS (Scheme 28).[350] Thus, the amino functionalities of Hyl were orthogonally protected with Fmoc and Boc, and the carboxy converted into the benzyl ester **110**. Stereoselective glycosylation resulting in a β-linkage to the amino acid and formation of the free hydroxy group at C2 of Gal was achieved in a one-pot procedure, wherein glycosylation with the protected glycal **109** via the epoxide intermediate yielded the monosaccharide product **111** in 37% yield. This yield is comparable to that obtained from glycosylation using the glycosyl bromide.[74] However, attempts to gylcosylate the free C2 hydroxy group with **112** were unsuccessful. An analogous compound, 5-hydroxynorvaline, could be successfully glycosylated, yielding a compound in which all the protecting groups were acid labile and could be simultaneously removed with cleavage of the peptide from the resin.[172]

Scheme 28 Synthesis of Glycosylated Hydroxylysine Building Blocks and Solid-Phase Glycopeptide Synthesis Using the Glycal[350]



***N*$^{\alpha}$-(Fluoren-9-ylmethoxycarbonyl)-*N*$^{\varepsilon}$-(*tert*-butoxycarbonyl)-5-*O*-[6-*O*-(*tert*-butyldiphenylsilyl)-3,4-*O*-isopropylidene-β-D-galactopyranosyl]-5-L-hydroxylysine Benzyl Ester (111):**[350]

A ca. 0.07 M soln of dimethyldioxirane in acetone (15.5 mL, 0.78 mmol) was added to **109** (274 mg, 0.646 mmol) in CH_2Cl_2 (7 mL) at 0 °C, and the soln was protected from light and stirred at 0 °C for 60 min. Concentration of the soln under reduced pressure yielded the corresponding 1,2-anhydro sugar. A soln of **110** (743 mg, 1.29 mmol) in THF (8 mL) containing crushed molecular sieves (AW-300, 500 mg) was added to the 1,2-anhydro sugar, and the mixture was stirred for 30 min at rt and then cooled to −50 °C. A 1.0 M soln of $ZnCl_2$ in Et_2O (970 μL, 0.970 mmol) was added, and the mixture was allowed to attain rt over 19 h. The mixture was then diluted with EtOAc, filtered (Hyflo-Supercel), and washed with H_2O. The aqueous phase was extracted with EtOAc (2×), and the combined organic phases were dried and concentrated. Flash column chromatography (toluene/MeCN 6:1→4:1) followed by purification by normal-phase HPLC gave the product; yield: 242 mg (37%); ^{1}H NMR (500 MHz, $CDCl_3$, δ): 5.59 (d, 1H, *J* = 7.4 Hz, NH), 5.36 (br s, 1H, NH), 5.20 (AB d, 1H, *J* = 12.1 Hz, PhC*H*$_2$O), 5.16 (AB d, 1H, *J* = 12.2 Hz, PhC*H*$_2$O), 4.44–4.38 (m, 1H), 4.42 (AB dd, 1H, *J* = 7.0, 10.6 Hz, OCOC*H*$_2$CH), 4.35 (AB dd, 1H, *J* = 7.2, 10.7 Hz, OCOC*H*$_2$CH), 4.23 (br d, 1H, *J* = 4.1 Hz, H4), 4.19 (t, 1H, *J* = 6.9 Hz, OCOCH$_2$C*H*), 4.15 (d, 1H, *J* = 8.1 Hz, H1), 4.02 (br t, 1H, *J* = 6.3 Hz, H3), 3.95–3.91 (m, 2H, H6), 3.90–3.86 (m, 1H, H5), 3.63 (br s, 1H), 3.51 (t, 1H, *J* = 7.7 Hz, H2), 3.29–3.23 (m, 1H, Hε), 3.18 (br s, 1H, OH), 3.10 (dt, 1H, *J* = 5.7, 14.4 Hz, Hε), 2.01–1.93 (m, 1H, Hβ), 1.92–1.84 (m, 1H, Hβ), 1.66–1.58 (m, 2H, Hγ), 1.49 and 1.33 (2 s, each 3H, C*H*$_3$), 1.28 and 1.04 (2 s, each 9H, tBu); ^{13}C NMR (126 MHz, $CDCl_3$, δ): 172.0, 156.2, 156.0, 143.8, 143.7, 141.2, 135.6, 135.5, 135.1, 133.2, 133.0, 129.8, 128.6, 128.5, 128.4, 127.8, 127.7, 127.1, 125.1, 119.9, 110.0, 102.8, 80.2, 79.0, 78.9, 73.6, 73.1, 67.4, 67.1, 62.5, 53.7, 47.2, 44.2, 28.4, 28.2, 28.1, 26.8, 26.3, 19.2.

for references see p 322

6.3.2.6 O-Linked Aromatic Glycosyl Amino Acids

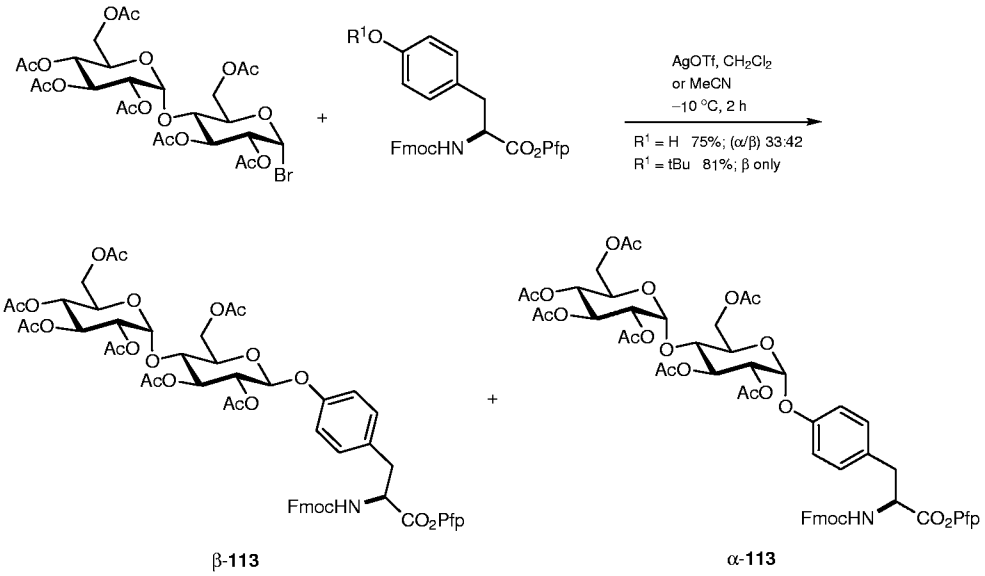
In addition to the many aliphatic O-glycosylation sites, a range of glycosylations involving tyrosines or 4-hydroxyphenylglycine has been discovered. Thus, Tyr(β-D-Glc) is found in insects as a way of storing tyrosine.[351,352] The antibiotics of the vancomycin and the related ristocetin group carry aromatic glycosylation with β-Glc and β-Man linked to phenolic oxygen atoms;[353] in the structure of ramnoplanose[354] a mannose trisaccharide is α-linked to 4-hydroxyphenylglycine. The protein linkage to sugar is Tyr(β-D-Glc) in S-layer glycoproteins of the eubacteria *Clostridium thermohydrosylfuricum*.[355,356] Finally, large glycogen particles are found to be synthesized on a self-glycosylating core protein and the synthesis is initiated by glucosylation of Tyr^{194} in the glycoprotein.[357–360]

6.3.2.6.1 Synthesis of 1,2-*trans*-O-Linked Glycosyl Tyrosine via the Halide

The glycosylation of aromatic hydroxy groups is thought to be inherently more difficult than aliphatic glycosylations owing to the low nucleophilicity of the phenols resulting from the donation of electrons into the neighboring π-electron system. Therefore, harsh conditions are often used for these glycosylations. Glycosylation of tyrosine is achieved under Koenigs–Knorr conditions in which Z-Tyr-OEt is reacted with acetobromoglucose in quinoline with Ag_2O as catalyst, yielding the recrystallized product in high yield (79–84%).[361] Similarly, glycosylation of Bz-Tyr-OEt with α-1-bromo-2,3,4-tri-*O*-acetylglucoronic acid methyl ester in quinoline under Ag_2O catalysis yields the recrystallized product in 32% yield.[362]

Enhancement of the acceptor reactivity by conversion into the phenolate as originally described[363] is also incompatible with labile protection groups; however, other means of enhancing the nucleophilicity, e.g. as acid-labile ethers, have been very successful.[364] The reactivity of the donor may be adjusted by protection of the carbohydrate as more reactive benzyl or less reactive acyl derivatives. It is important that the glycosyl donor is not too reactive, otherwise decomposition into glycals and other byproducts may occur. As illustrated in Scheme 29, it has been found that peracylated glycosyl bromides with AgOTf in CH_2Cl_2 at −10 °C or at ambient temperature could be used to furnish the β-product **113** in 42% yield from Fmoc-Tyr-OPfp (R^1 = H). The *tert*-butyl ether Fmoc-Tyr(tBu)-OPfp (R^1 = tBu), however, was found to be a superior glycosyl acceptor to give β-**113** in 81% yield, perhaps owing to enhanced nucleophilicity of the ether oxygen lone pairs. Access to both the α- and β-anomers of **113** was possible when the glycosylation was carried out in MeCN, despite acyl groups in the 2-position. By analytical HPLC the glycosylations were quantitative and the products, which were isolated in high yield, could be used directly in solid-phase glycopeptide synthesis of 25 amino acid long fragments from the glycogen core protein glycogenin.[126]

Scheme 29 Synthesis of α- or β-Glycosyl Tyrosine Building Blocks Using the Glycosyl Halide[364]



***N*$^{\alpha}$-(Fluoren-9-ylmethoxycarbonyl)-Tyr[2,3,4,6-tetra-*O*-acetyl-α-D-glucopyranosyl-(1→4)-2,3,6-tri-*O*-acetyl-β-D-glucopyranosyl]-OPfp (β-113):**[364]

To a mixture of Fmoc-Tyr(tBu)-OPfp (R^1 = tBu; 313 mg, 0.5 mmol), AgOTf (257 mg, 1.0 mmol), and 3-Å molecular sieves in dry CH_2Cl_2 (10 mL) under an inert atmosphere at −10 °C was added a soln of the glycosyl bromide (689 g, 1.0 mmol) in dry CH_2Cl_2 (5 mL). After 1 h, the suspension was neutralized with DIPEA (0.17 mL, 1.0 mmol) and centrifuged. The supernatant was concentrated, redissolved in MeCN (10 mL), and purified by preparative HPLC to give β-**113** (520 mg), which was recrystallized from THF/Et_2O/light petroleum ether; yield: 454 mg (81%).

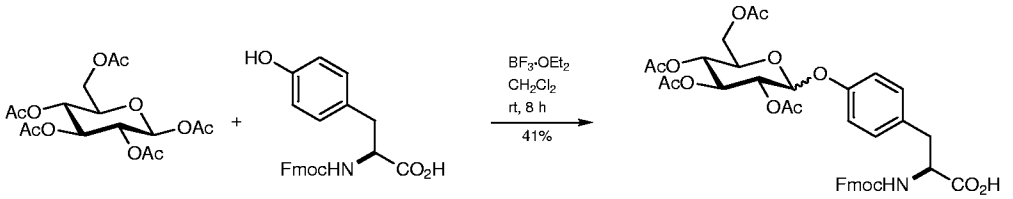
6.3.2.6.2 Synthesis of O-Linked Glycosyl Tyrosine Using Other Methods

Less reactive peracetyl glycosyl fluorides have been used with BF_3, affording Ac-Tyr(Ac_4-Glc)-OMe in high yield.[365] 3,4,6-Tri-*O*-acetyl-2-deoxy-2-nitroso-α-D-glucopyranosyl chloride reacts with Z- and Boc-Tyr-OMe without catalyst in DMF, as it does with Ser/Thr derivatives, yielding 25% of the glycosylation product.[272] A direct glycosylation of Z-Tyr-OtBu and Tyr-containing peptides in the presence of DCC and CuCl[129,366] has been reported. Only low yields of the glycosylated peptides were obtained, but the β-Glc-Tyr derivatives were formed in 18–68% yield and could be used for synthesis of a glyco-tetrapeptide after cleavage of the *tert*-butyl group.

The treatment of peracetylated sugars and phenols with Lewis acid catalysts at elevated temperatures requires very stable protecting groups and will not work with sensitive tyrosine derivatives and oligosaccharides. At room temperature, treatment of Fmoc-Tyr-OH with peracetylated Gal and Glc in CH_2Cl_2 under activation with $BF_3 \cdot OEt_2$ gave the product **114** in 34 and 41% yield, respectively after 4–8 hours (Scheme 30).[165] Whilst β-anomer formation was predominant (45:2), when the solvent was changed to MeCN the α-anomer predominated (1:2), as had been previously observed for glycosylation with glycosyl bromides.[364] Reaction was slower for glucose than for galactose. O-Acetylated tyrosine was obtained as a significant byproduct.

for references see p 322

Scheme 30 Synthesis of α- or β-Glycosyl Tyrosine Building Blocks Using the Peracetate[165]



114

***N*$^{\alpha}$-Fmoc-Tyr(2,3,4,6-tetra-*O*-acetyl-β-D-galactopyranosyl)-OH (114):**[165]
To glucose peracetate (53.5 mg, 137 μmol) and Fmoc-Tyr-OH (71.8 mg, 169 μmol, freeze-dried twice from 1.5 mL dioxane) in CH_2Cl_2 (2.6 mL) was added $BF_3 \cdot OEt_2$ (3 equiv) in CH_2Cl_2 and the reaction stirred for 4 h at rt. The reaction was monitored by TLC and analytical HPLC. When the reaction did not progress further, the mixture was diluted with CH_2Cl_2 (8 mL), washed with 1 M HCl (1 mL) and H_2O (1 mL), dried, and concentrated. The crude product was purified by HPLC; yield: 33.8 mg (41%); ^1H NMR (300 MHz, $CDCl_3$, δ): 2.02, 2.02, 2.03, and 2.17 (4s, 3H each, Ac), 3.06 (AB dd, 1H, J = 13.6, 6.5 Hz, Hβ), 3.17 (AB dd, 1H, J = 14.1, 4.9 Hz, Hβ), 3.92 (m, 1H, H5), 4.32 (m, 1H, Fmoc CHAr), 4.44 (m, 1H, Fmoc OCH_2), 4.66 (m, 1H, Hα), 4.93 (d, 1H, J = 8.0 Hz, H1), 5.07 (dd, 1H, J = 10.5, 3.4 Hz, H3), 5.29 (d, 1H, J = 8.2 Hz, NH), 5.42 (br s, 1H, H4), 5.45 (dd, 1H, J = 10.3, 7.9 Hz, H2), 6.91 and 7.08 (AB q, 4H, Hδ, Hε); ^{13}C NMR ($CDCl_3$, δ): 37.0 (Cβ), 47.1 (Fmoc CHAr), 54.6 (Cα), 61.3 (C6), 66.9, 67.2, 68.6, 70.8, and 70.9 (C2, C3, C4, C5, Fmoc OCH_2), 99.6 (C1), 117.3 (Cε), 130.5 (Cγ, Cδ).

6.3.2.7 O-Linked Glycopeptides

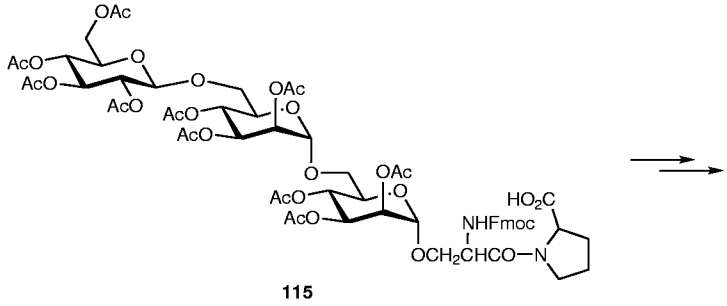
6.3.2.7.1 Assembly of O-Linked Glycopeptides in Solution Phase

Once the glycosylated building blocks are prepared, they are then assembled into glycopeptides of differing lengths and glycan clusters, either in solution or on solid phase. In an example of solution-phase synthesis, a trisglycosylated hexapeptide **116** with phytoalexin activity was synthesized by iterative coupling of a common building block **115** (Scheme 31).[367] Compound **115** was condensed with its amino deprotected analogue under EEDQ activation (in CH_2Cl_2) to yield the bisglycosylated tetrapeptide in 74% yield. The N-deprotected tetrapeptide was then acylated with **115** in 75% yield under the same conditions. The product was completely deprotected in one step by treatment with NaOMe in MeOH/H_2O (3:1). One of the main concerns that stem from the use of base-labile groups for carbohydrate protection is the possibility of racemization or glycan β-elimination under the deprotection conditions. In this case, there was no evidence of either occurring during the deprotection or coupling reactions.

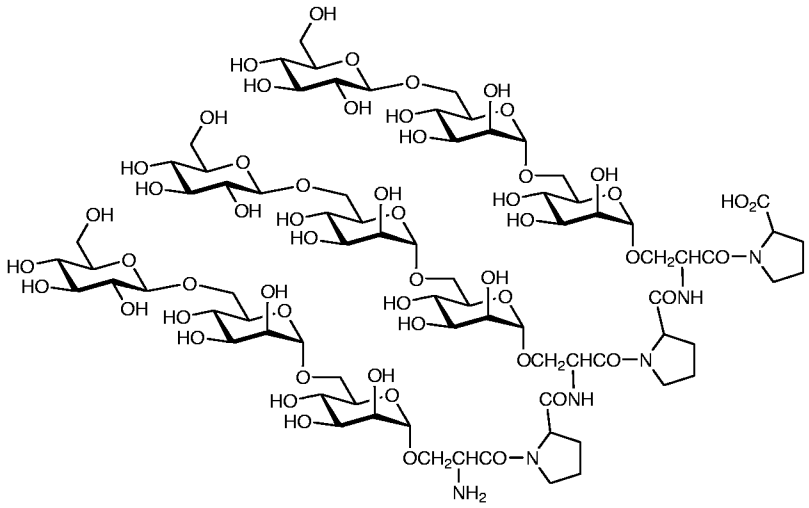
In another example, building block **117** was used for the synthesis of the sialylated cluster glycopeptide **118** (Scheme 31).[315] The building block contained a 2-acetamido group that was obtained after reduction and acetylation of the 2-azido precursor. It is often advantageous to convert the 2-azido moiety into the 2-acetamido group before incorporation into the peptide to avoid possible incomplete azide reduction or to enable incorporation into peptides containing groups that can be affected by the reduction conditions. Coupling of the building block and amino acids proceeded in high yield under IIDQ activation (85% average yield). In this synthesis, the Fmoc group was removed by treatment with KF instead of morpholine and comparable yields were obtained. Generally, during the synthesis of sialyl glycopeptides there is a concern that sialosides, known to hydrolyze faster than the common pyranosides, can be cleaved on treatment with strong acid. The evidence, however, is sparse and inconclusive. There have been anecdotal reports that glycopeptides containing acetylated sialosides are completely or partially desialylated by treatment with 95% TFA for 2.5 hours at room temperature. Alternatively, it has been reported that the sialic acid moiety, unprotected and acetylated, is not hydrolyzed even under the strongly acidic conditions

utilized for the cleavage of the peptide from the resin or during purification by RP-HPLC.[125] This stability is generally attributed to the stabilizing effect of the acyl protecting groups normally employed in the sialic acid building block, although a recent glycopeptide synthesis performed with benzylated building blocks suggests otherwise.[318] Possibly the formation of the intramolecular lactone between the sialic acid and the C4 Gal hydroxy group plays a role in the stabilization effect. In this example, no acid-labile groups were employed in the synthesis and the possibility of cleavage was avoided. As in the previous case, there was no evidence of β-elimination upon complete deprotection of the glycopeptide under basic conditions.

Scheme 31 Examples of Glycopeptide Synthesis by the Building Block Approach in Solution[315,510]

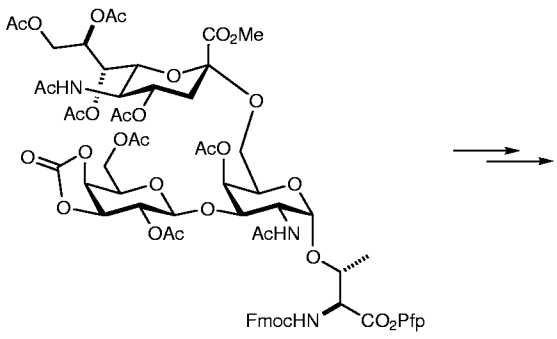


115

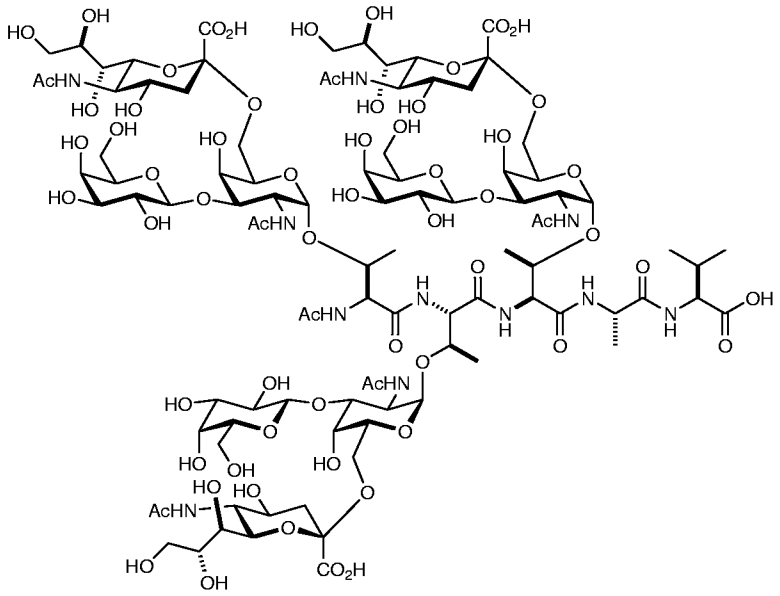


116

for references see p 322



117



118

6.3.2.7.2 Assembly of O-Linked Glycopeptides on Solid Phase

More frequently, the glycopeptide is assembled on solid phase, cleaved, and then completely deprotected in solution. If an acid-labile linker is not used, deprotection of the carbohydrates may take place on solid phase prior to cleavage. In a rather impressive example, a glycopeptide **121**, corresponding to the human glycophorin A^N and containing six adjacent glycosylation sites, was synthesized from two building blocks **119** and **120** (Scheme 32).[368] The synthesis was carried out on PEGA resin and proceeded smoothly. There was no difficulty stemming from steric hindrance in coupling six glycosylated building blocks sequentially. In fact, the acylation reaction was accelerated in the case of the glycosylated building block compared to the analogous non-glycosylated peptide. Thiolytic reduction and acylation of the 2-azido group was effected using thioacetic acid. A long reaction time (48 h) and repeated change of reagents to avoid thioacetamide byproduct formation was required. Cleavage from the resin under acidic conditions was followed by deacetylation using NaOMe (pH 8.5). Since the product was insoluble, water was added to the reaction mixture and some β-elimination of the glycan resulted. The glycopeptide could be effectively deacetylated without precipitation or β-elimination using hydrazine hydrate in MeOH.

6.3.2.8 Solid-Phase Glycosylations

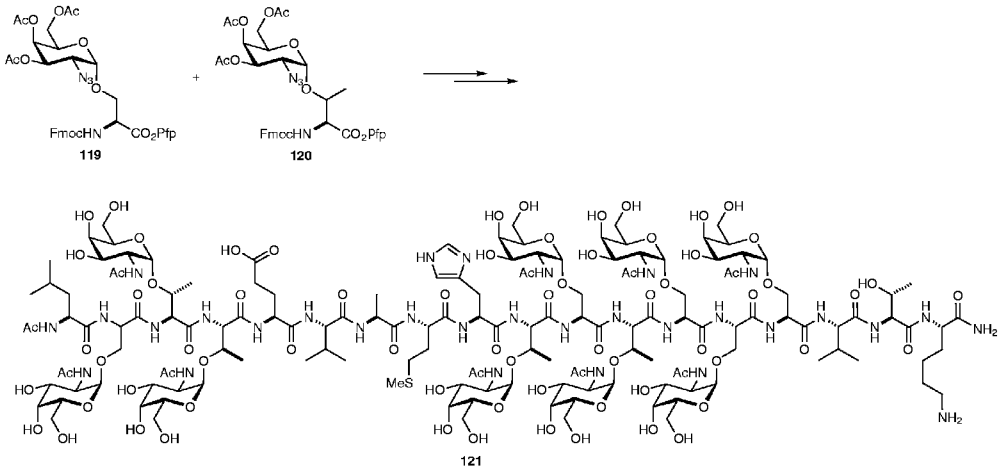
In an alternative to the building block approach, a glycopeptide can be synthesized by direct glycosylation of the resin-bound peptide or glycopeptide. Solid-phase glycosylation reactions have been highlighted in recent years, particularly by use of the highly reactive glycosyl sulfoxides or trichloroacetimidates, which can both be used under conditions of homogeneous catalysis. The success of this methodology for the synthesis of glycopeptides is strongly dependent on the nature of the solid support. In addition, chemical glycosylation usually requires conditions that are different from that of peptide coupling: anhydrous glycosylation in CH_2Cl_2.

6.3.2.8.1 Solid-Phase Glycosylation of Glycopeptides

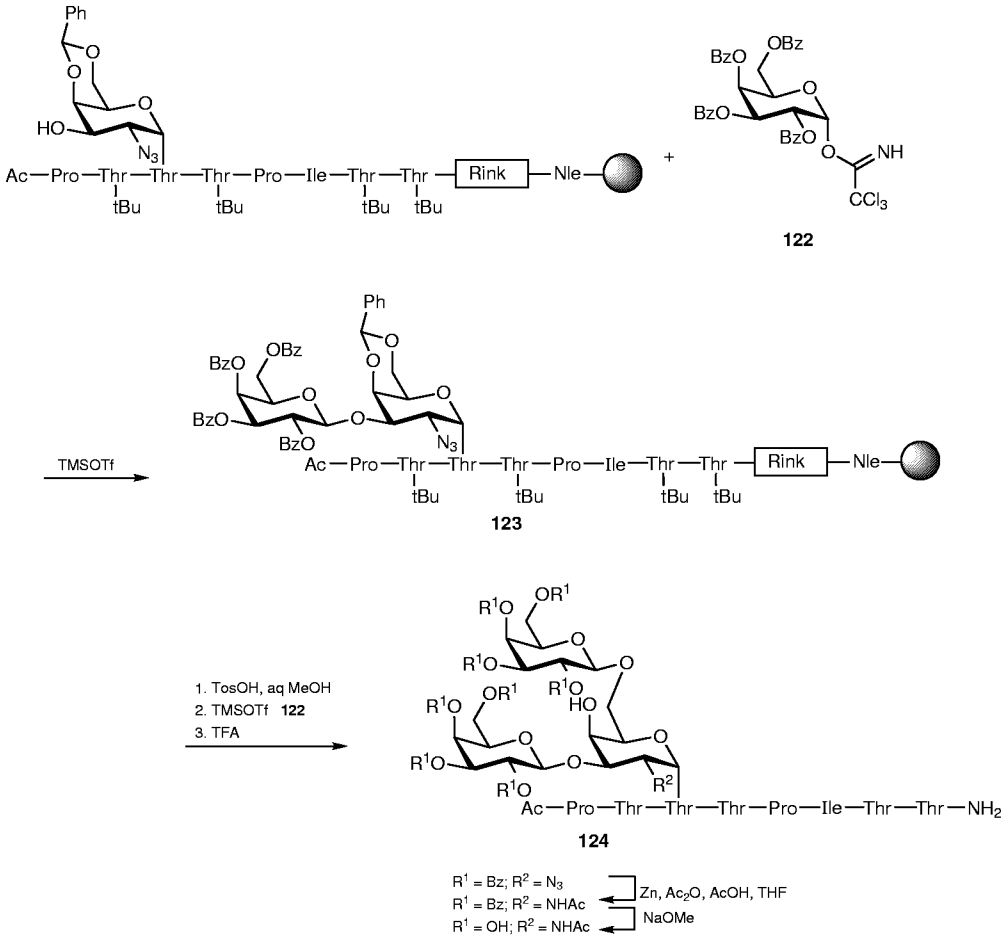
Paulsen et al.[136] have synthesized the partial sequence and variations thereof of the mucin MUC 2 glycoprotein by solid-phase glycosylation (Scheme 33). An initial glycopeptide synthesized on PEGA resin contained the 2-azidogalactosyl threonine incorporated by the building block approach. Good yields of glycopeptides containing disaccharides **123** and trisaccharides **124** were obtained using 5–8 equivalents of the perbenzoylated trichloroacetimidate **122** as the donor. Glycosylation was attempted on four resins: Polyhipe, TentaGel, $PEGA_{1900}$, and Macrosorb, but was successful only on PEGA and Polyhipe. Interestingly, the product anomeric ratio appears to depend on the resin.

for references see p 322

Scheme 32 Example of Glycopeptide Synthesis by the Building Block Approach on Solid Phase[511]



Scheme 33 Solid-Phase Glycosylation of a Glycopeptide[136]

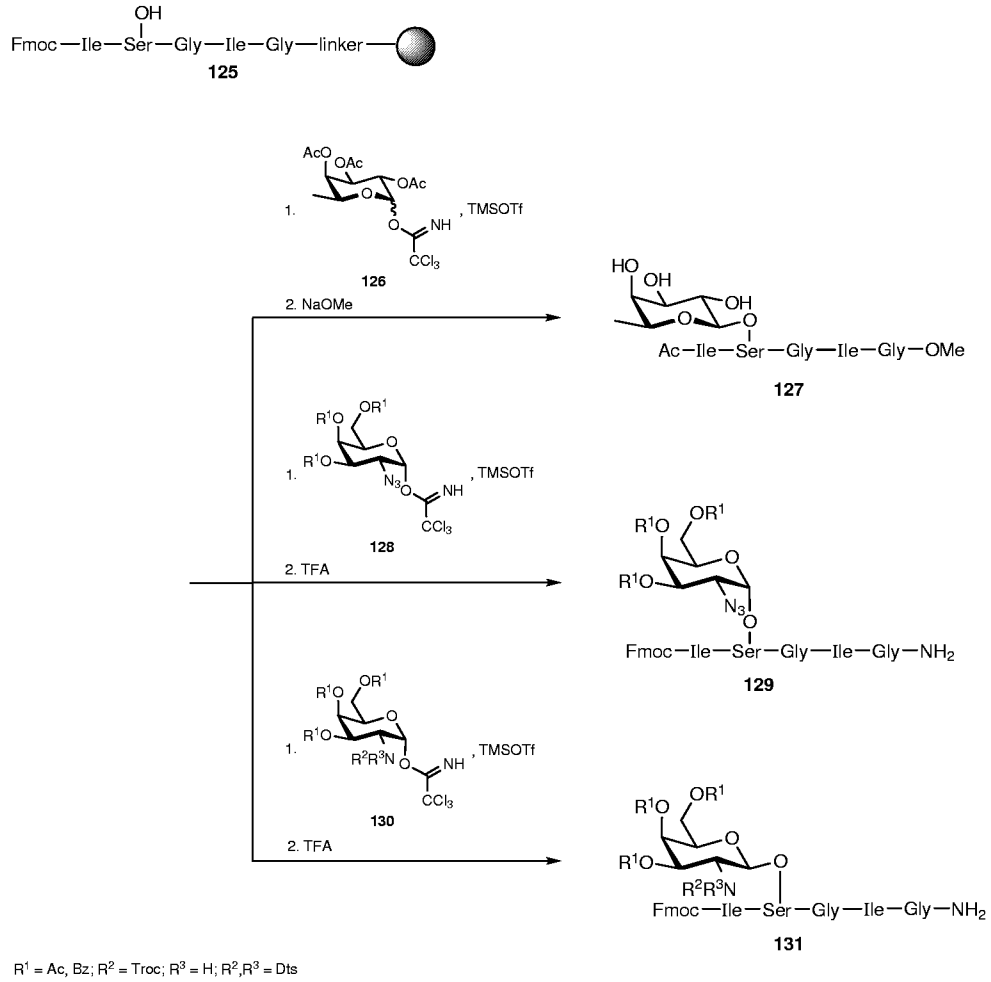


6.3.2.8.2 Solid-Phase Glycosylation of the Hydroxy Side Chain of Peptides

Attempts[134,136,226,227] to effect direct glycosylation of the amino acid side-chain hydroxy were partially successful, albeit low yielding (5–25%), particularly on inert resins such as Polyhipe. The failure of polar resins such as PEGA may be in part due to the amide backbone that interferes with the carbocation intermediate. However, since polar resins are required for solid-phase bioassays, new types of polar resin containing only ether bonds have been developed for the solid-phase glycosylation of peptides. Quantitative peptide glycosylation of peptide **125** to give the products **127**, **129**, and **131**[369,370] was achieved on a novel PEG-based resin (polyethylene–polyoxypropylene) containing only ether bonds[95] using 5–8 equivalents of the peracetylated or perbenzoylated glycosyl trichloroacetimidate (e.g., **126**, **128**, and **130**) (Scheme 34), and characterized by MAS-NMR (MAS = magic angle spinning).[523]

for references see p 322

Scheme 34 Solid-Phase Glycosylation of the Amino Acid Side-Chain Hydroxy Group on Resin-Bound Peptides[369,370]



R^1 = Ac, Bz; R^2 = Troc; R^3 = H; R^2,R^3 = Dts

Glycosylation using 2-amino sugars is possible. In these instances, the protection of the amino group of the glycosyl donors is important for stereoselectivity and for ensuring deprotection under mild conditions. The Troc and the Dts groups are best suited for this 2-amino protection. The perbenzoylated and peracetylated 2-azido-2-deoxy-D-galactopyranosyl trichloroacetimidates **128** were selected as glycosyl donors to form an α-linkage between the 2-acetamido-2-deoxy-D-galactopyranose and peptide acceptor peptide **125**. The 2-azido-2-deoxyglycosyl donors did not react with neighboring group participation in the 2-position and led to the corresponding α-glycosylated products **129**. It must be noted that the use of TMSOTf can sometimes lead to formation of the orthoester and cleavage of *tert*-butyl ether protecting groups present.

General Glycosylation Procedure:

The acceptor resin, in a syringe fitted with a Teflon filter, was washed with MeOH (5×) and CH_2Cl_2 (5×) and was dried under vacuum for 24 h. The donor imidate (5 equiv) and 4-Å molecular sieves (ca. 2-mm beads) were added and the mixture dried in vacuo for 6 h. Dry CH_2Cl_2 (2.5 mL) was added and the mixture was stirred at rt under N_2 for 1 h. Freshly distilled TMSOTf (0.15 equiv with respect to donor) was added and the reaction stirred for 2 h (reaction time depended on the donor but, on average, all reactions were complete after 2 h). The resin was filtered off, washed successively with MeOH (5×) and CH_2Cl_2 (5×), and dried.

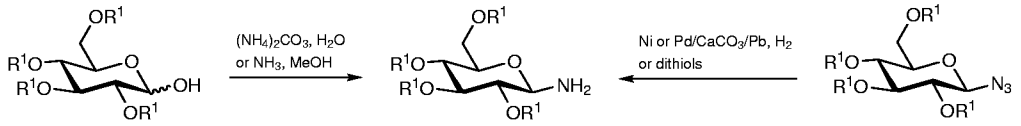
6.3.3 N-Linked Glycopeptides

Most work with N-linked glycopeptides has been carried out to improve the chemical methodology leading to short peptides carrying a single GlcNAc residue; the synthesis of biologically interesting[371] and fully deprotected glycopeptides has only been achieved during the last few years. Synthesis of such complex long-chain glycopeptides require SPPS techniques as well as synthesis[8,372] or isolation of larger oligosaccharides.[373–378] There are three specific and essential reactions during chemical assembly of complex N-linked glycopeptides: (1) the formation of the β-mannosidic linkage of the core oligosaccharide, (2) the generation of the glycosylamines, and (3) the acylation with aspartic acid or aspartic acid containing peptides. Methods described in this article are mainly those which are either simple or easy to perform or have proven suitable for synthesis of glycopeptides containing at least the pentasaccharide core of the complex N-linked oligosaccharides.

6.3.3.1 Synthesis of Glycosylamines

The introduction of the amino group at the reducing end of an oligosaccharide may be carried out by equilibration of the nonprotected carbohydrate with aqueous or methanolic solutions containing ammonia in one form or another (Scheme 35). The ammonium bicarbonate method, in which the free sugar is treated at 37 °C with a saturated solution of ammonium hydrogen carbonate in water adjusted to pH 8.5 with ammonia,[379,380] can afford 40–80% of the glycosylamine.[381] The reaction is, however, not clean, and often diglycosylamines are present in the crude product. Pure compounds may be obtained by ion-exchange chromatography.[379]

Scheme 35 Glycosylamine Syntheses[374,379,392,393]



Alternatively, the crude mixtures are reacted with a protected aspartic acid derivative, e.g. Fmoc-Asp-OtBu, after activating the carboxy group, and the product mixture is then separated by HPLC.[381] The crude amine can be protected with an Fmoc or a 2,2-bis(ethylcarboxy)vinyl[382] group and acetylated to yield crystalline derivatives.[112,383] For monosaccharides, however, Isbell's[118,384,385] procedure of dissolving the free sugar in methanol saturated with ammonia and isolating the pure crystalline glycosylamine is preferred. Apparently, no diglycosylamine is formed by this method[383] and it works well on a large scale.[118] The preparation of glycosylamines from a large range of complicated oligosaccharides and monosaccharides has been thoroughly investigated.[386]

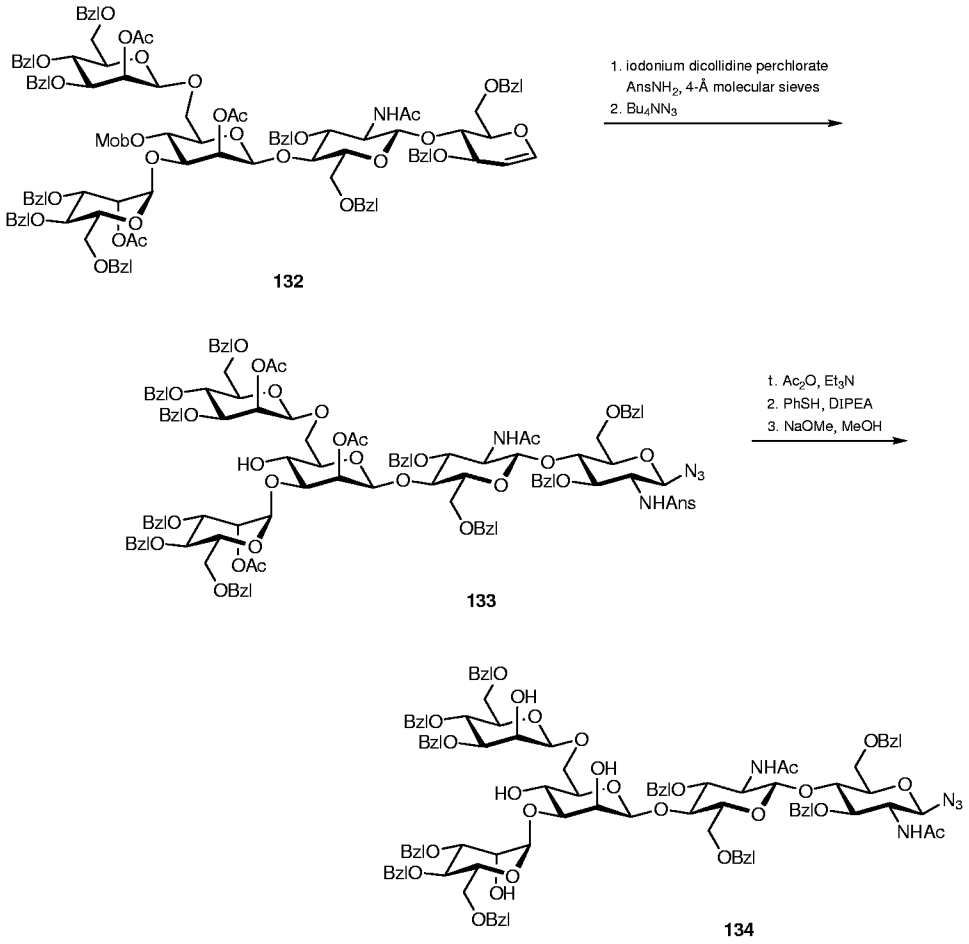
6.3.3.1.1 General Method for the Preparation of Glycosylamines

Previously, most protected glycosylamines were prepared by catalytic hydrogenation of a β-glycosyl azide[81,170,388–390] (Scheme 35) derived from a glycosyl halide or, preferably, from the oxazoline by treatment with, for example, trimethylsilyl azide.[374] A new method for the synthesis of 2-deoxy-2-tosylamidoglucosyl azides from D-glucal derivatives has been described.[391] Similarly, a number of syntheses of large oligosaccharyl amines have been based on the use of the azido group as a protected precursor for the glycosylamine (Scheme 36).[374,392,392,393] The conversion into the glycosylamine is best performed with $Pd/CaCO_3$

for references see p 322

doped with lead (Lindlar's catalyst),[374,394] by nickel-catalyzed hydrogenation, or, most advantageously, by reduction with dithiols, e.g. propanedithiol[392,395] or DTT.[175] The use of a nickel catalyst is known to give anomerization of the resulting glycosylamine[393] and Lindlar's catalyst seems to be preferred.[139,374] The time of addition of the acylating reagent and the conditions employed may be crucial for the success of the metal-catalyzed reactions.

Scheme 36 Synthesis of the Pentasaccharide Core as the Glycosyl Azide from Glycal Precursors[393]



However, the azide-based protocol often has problems with the formation of byproducts under both azide formation and subsequent reduction.[388] Therefore, mainly aminolysis is used. Glycosylamines are prepared from GlcNAc, $GlcNAc_2$, Glc, maltotriose, maltoheptaose, and di- and triantenneary complex glycans, isolated from asialofetuin by preparative hydrazinolysis. A modification of the method by Likhosherstov is also employed.[379]

Preparation of Glycosylamines; General Procedure:[149]

The parent saccharides (GlcNAc, $GlcNAc_2$, Glc, maltotriose, maltoheptaose, or di- or triantenneary complex glycans) were dissolved in sat. aq $(NH_4)_2CO_3$ (10 mg · 1 mL^{-1}) and stirred at 45 °C for 1–2 d. Solid $(NH_4)_2CO_3$ was added in fractions during the course of glycosylamine formation to ensure saturation. The amine formation was followed both by TLC (PrOH/EtOAc/H_2O 6:1:3, detection with ninhydrin reagent) and ^1H and ^{13}C NMR spectroscopy in D_2O. In all cases, conversions occurred in the range of 80–90% by comparison of the anomeric H1 signal of the exclusively β-glycosylamine with the anomeric H1 signals of the reducing hemiacetals. After the successful conversion into the glycosylamines, the mixture was diluted with H_2O and concentrated to half the volume. This procedure was

repeated (5×) until no further NH_3 could be detected. The glycosylamines were repetitively frozen, lyophilized, and dissolved. Gravimetrically determined yields were almost 100%. According to ^1H NMR spectroscopy (δ H1: 4.1–4.3 ppm for β-glycosylamines), yields were around 80–90%; however, yields determined by spectroscopy in D_2O may be artificially low owing to hydrolysis of the glycosylamines in aqueous soln. Formation of the diglycosylamine, a typical side reaction during the amination procedure, was not significant according to ^1H and ^{13}C NMR spectroscopy. The amination reaction yielded only the β-glycosylamines as white, amorphous solids, which could be stored at −20 °C under anhydrous conditions. Yields and analytical data were determined from ^1H NMR spectra in D_2O and are calculated on the integral ratio for the anomeric proton of the glycosylamines. Owing to the instability of the glycosylamine in aqueous solns, the amination reactions have alternatively been performed in DMSO. The saccharide was dissolved in DMSO (20 mg·5 mL^{-1}) and $(NH_4)_2CO_3$ was added until saturation. The mixture was kept for 24–48 h at 45 °C. Solid $(NH_4)_2CO_3$ was added frequently to secure saturation. The conversion was followed by ^1H NMR in DMSO-d_6. After complete conversion, the soln was lyophilized. In the case of unstable glycosylamines, the use of DMSO as a polar solvent system may prevent the hydrolysis of already formed glycosylamines. Overall higher yields and more reliable NMR spectra could be obtained when avoiding aqueous solns for the preparation of glycosylamines.

Opening of Anomeric Oligosaccharide Oxazolines with Trimethylsilyl Azide; Tetra-*O*-Ac-α-D-Man-(1→3)-[tetra-*O*-Ac-α-D-Man-(1→6)]-di-*O*-Ac-α-D-Man-(1→6)-[tetra-*O*-Ac-α-D-Man-(1→3)]-di-*O*-Ac-β-D-Man-(1→4)-di-*O*-Ac-β-D-GlcNAc-(1→4)-di-*O*-Ac-β-D-GlcNAc-(1→)-N_3:[374]

The oxazoline obtained from peracetylated $Man_5GlcNAc_2$ isolated from sheep urine (180 mg, 87.5 μmol) was treated at rt with a 0.2 M soln of $TMSN_3$ (1.5 mL) in CH_2Cl_2, followed by $SnCl_4$ (0.03 mL, 0.25 mmol) and DIPEA (0.03 mL, 0.17 mmol). The mixture was stirred for 86 h at 50 °C, when TLC ($CHCl_3$/MeOH 10:1) showed that the starting material (R_f 0.55) had been completely transformed into the azide (R_f 0.52). The mixture was directly applied to a column of silica gel (30 g), which was eluted with $CHCl_3$/MeOH (40:1 and then 10:1) at a flow rate of 1 mL/10 s. The latter eluent gave the glycosyl azide; yield: 78 mg (42%); $[\alpha]_D$ +3 (*c* 0.22, $CHCl_3$). The clean reaction according to TLC suggests that loss of material occurred only by adsorption to the tin compounds prior to chromatography. The ^1H NMR spectrum was recorded upon deacetylation and selected resonances were: ^1H NMR (500 MHz, D_2O, δ): 5.098 (2 d, 2H, $H1^4$,1^5), 4.910 (d, 1H, $H1^5$), 4.874 (d, 1H, $H1^4$), 4.810 (s, 1H, $H1^3$), 4.790 [d (10 °C), m (60 °C), 1H, $H1^1$], 4.598 (d, 1H, $H1^2$); MS *m/z*: 1283 [M+Na^+].

Preparation of Glycosyl Azides: Reaction of Iodonium Adduct with Anthracene-9-sulfonamide; 2-*O*-Ac-3,4,6-tri-*O*-Bzl-α-D-Man-(1→6)-[2-*O*-Ac-3,4,6-tri-*O*-Bzl-α-D-Man-(1→3)]-2-*O*-Ac-4-*O*-PMB-β-D-Man-(1→4)-3,6-di-*O*-Bzl-β-D-GlcNAc-(1→4)-di-*O*-Bzl-β-D-GlcNSO$_2$Anthracenyl-(1→)-N_3 (133):[393]

The pentasaccharide glycal **132** (112 mg, 0.0565 mmol) was dried by azeotropic distillation with benzene (3×6 mL). It was then mixed with anthracene-9-sulfonamide (dried in vacuo over P_2O_5 for 2 h) and activated 4-Å molecular sieves (500 mg) in dry THF (6.0 mL) and stirred at rt for 30 min in the dark under an inert atmosphere. The mixture was then cooled to 0 °C and freshly prepared iodonium di-collidine perchlorate (132 mg, 0.283 mmol) was added in one portion. The mixture was stirred in the dark at 0 °C for 3 h, then sat. $Na_2S_2O_3$ soln (10 mL) was added and the mixture shaken vigorously until the red color disappeared. The mixture was extracted with EtOAc (3×8 mL) and the combined organic extracts were washed with sat. $Na_2S_2O_3$ (5 mL), H_2O (5 mL), and finally brine (5 mL). The organic layer was then dried ($MgSO_4$), filtered, and the solvent removed to afford a yellow oil. The crude product was immediately dissolved in dry THF (3.0 mL) and stored in the dark under an inert atmosphere at rt, where dry tBu_4NN_3 (160 mg, 0.565 mmol) was added in one portion. The mixture was stirred for 30 min and then sat. $Na_2S_2O_3$ soln (10 mL) was added and the mixture extracted with EtOAc (2×10 mL). The organic extracts were washed with brine, dried ($MgSO_4$), and filtered. The solvent was removed and the residue chromatographed (silica gel, EtOAc/hexanes) to afford the anomeric azide of the pentasaccharide **133** as a yellow oil; yield: 83 mg (67%); R_f = 0.26 (EtOAc/hexane, 1:1); $[\alpha]_D$ = 4.0 (*c* 0.75, $CHCl_3$); ^1H NMR ($CDCl_3$, δ): 9.33 (d, 2H, *J* = 8.9 Hz), 8.61 (s, 1H), 7.97 (d, 2H, *J* = 8.3 Hz), 7.65–7.57 (m, 2H), 7.48–7.42 (m, 2H), 7.30–7.10 (m, 60H), 7.0–7.05 (m, 2H), 6.83 (d, 2H, *J* = 8.7 Hz), 6.29 (d, 1H, *J* = 9.1 Hz), 5 49 (br s, 1H), 5.44 (dd, 1H, *J* = 2.8, 2.0 Hz), 5.33 (d, 1H, *J* = 3.0 Hz), 5.11 (d, 1H, *J* = 1.4 Hz), 5.04 (d, 1H, *J* = 1.6 Hz), 4.88–4.77 (m, 5H), 4.69–4.35 (m, 22H), 4.28–4.22 (m, 3H), 4.12 (d, 1H, *J* = 10.6 Hz), 4.05–3.62 (m, 23H), 3.60–3.38 (m, 10H), 3.37–3.33 (m, 1H), 3.26–3.21 (m, 1H), 3.14 (br d, 1H, *J* = 9.5 Hz), 2.15 (s, 3H), 2.13 (s, 3H), 1.93 (s, 3H), 1.66 (s, 3H); ^{13}C NMR (anomer region, $CDCl_3$, δ): 113.83, 99.84, 99.36, 99.10, 89.00; HRMS *m/z*: 2302.8940.

Reduction of Glycosyl Azides using Lindlar's Catalyst; Tetra-*O*-Ac-α-D-Man-(1→3)-[tetra-*O*-Ac-α-D-Man-(1→6)]-di-*O*-Ac-α-D-Man-(1→6)-[tetra-*O*-Ac-α-D-Man-(1→3)]-di-*O*-Ac-β-D-Man-(1→4)-di-*O*-Ac-β-D-GlcNAc-(1→4)-di-*O*-Ac-β-D-GlcNAc-(1→)-NH_2:[374]

To a soln of *O*-Ac_{20}-[α-D-Man-(1→3)-{α-D-Man-(1→6)}-α-D-Man-(1→6)-{α-D-Man-(1→3)}-β-D-Man-(1→4)-β-D-GlcNAc-(1→4)-β-D-GlcNAc-(1→)]-N_3 (100 mg, 47.6 μmol) in MeOH (4 mL) was added

for references see p 322

Lindlar's catalyst (5% Pd on $CaCO_3$, poisoned with Pb, 250 mg) and the mixture was hydrogenated at 0.2 MPa for 35 min at rt. TLC showed the conversion of azide [R_f 0.48 ($CHCl_3$/MeOH 10:1)] into amine (R_f 0.21), and showed evidence of *O*- to *N*-acetyl migration, but no evidence of an anomeric mixture. The catalyst was filtered off and washed with MeOH, and removal of the solvent gave the amine as colorless oil. The crude oil was used immediately for amino acid coupling.

Reduction of a Glycosyl Azide with Raney Nickel and Peptide Coupling; N^{α}-Z-Asn[2-*O*-Ac-3,4,6-tri-*O*-Bzl-α-D-Man-(1→6)-{2-*O*-Ac-3,4,6-tri-*O*-Bzl-α-D-Man-(1→3)}-2-*O*-Ac-4-*O*-PMB-β-D-Man-(1→4)-3,6-di-*O*-Bzl-β-D-GlcNAc-(1→4)-di-*O*-Bzl-β-D-GlcNAc-(1→)-]-Val-Thr(Bzl)-OBzl:[393]

Raney Ni (50% slurry in H_2O, 75 mg) was washed intensively with deionized H_2O, followed by EtOH washes. It was then transferred with EtOH to a flask containing the azido pentasaccharide **134** (14.6 mg, 7.5 μmol). Reduction was carried out at rt for 30 min under H_2 (1 atm.). The crude amino sugar was then filtered through filter paper with EtOH, concentrated, and dried for 10 min under vacuum.

Tripeptide Z-Asp-Val-Thr(Bzl)-OBzl (19.4 mg, 30 μmol) was mixed with IIDQ (8.7 mg, 29 μmol) in CH_2Cl_2 (1 mL) at rt for 2 min. The soln was then added to the crude anomer mixture of sugar amines. After stirring for 2 h the crude protected glycopeptide was purified (silica gel, EtOAc/hexane/MeOH 60:40:3.5) to afford a 1:1 mixture of the anomers; yield: 12.0 mg (67%). The mixture was further separated on analytical TLC plates (EtOAc/hexanes/MeOH 60:40:3.5) to afford homogeneous β-linked glycopeptide; ^{1}H NMR ($CDCl_3$ δ): 7.42 (br d, 1H, J = 6.8 Hz), 7.32–7.10 (m), 6.78 (d, 2H, J = 8.8 Hz), 6.46 (br d, 1H, J = 8.8 Hz), 6.40 (d, 1H, J = 7.6 Hz), 5.13–5.07 (m, 5H), 4.95–4.90 (m, 3H), 4.85–4.10 (m), 4.05–3.95 (m, 4H), 3.90–3.40 (m), 3.72 (s, 3H), 3.40–3.28 (m, 2H), 3.20 (m, 1H), 2.70 (dd, 1H), 2.52 (dd, 1H), 2.05 (m, 1H), 1.73 (s, 3H) 1.63 (s, 5H), 1.15 (d, 3H, J = 6.0 Hz), 0.91–0.80 (dd, 6H, J = 6.8, 7.2 Hz); HRMS *m/z*: 2582.1401.

Reduction of an Oligosaccharyl Azide with Propanedithiol; N^{α}-Z-Asn[β-D-GlcNAc-(1→2)-α-D-Man-(1→3)-{β-D-GlcNAc-(1→2)-α-D-Man-(1→6)-}β-D-Man-(1→4)-*O*-Bzl$_2$-β-D-GlcNAc-(1→4)-*O*-Bzl$_2$-β-D-GlcNAc-(1-N^4)]-OBzl (160):[392]

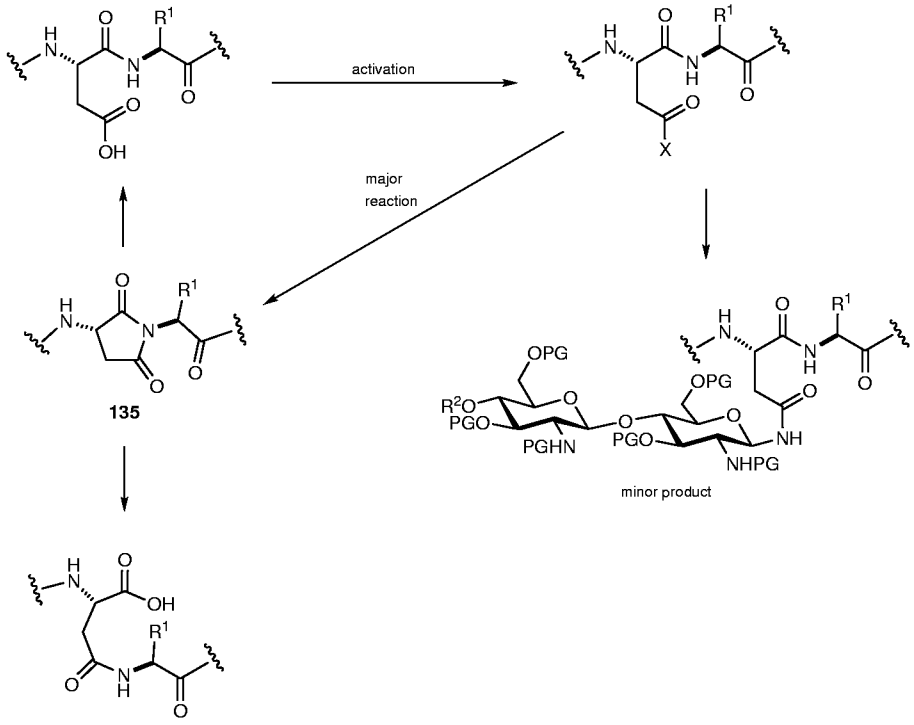
Heptasaccharide **158** (see Scheme 45) (20 mg, 11.7 μmol) was dissolved in a mixture of dry MeOH (600 μL) and DIPEA (20 μL). The flask was flushed with argon, followed by addition of propanedithiol (60 μL). Reduction to the amine was completed after 4 h (TLC; iPrOH/1 M NH_4OAc 4:1). Subsequently, the mixture was concentrated and dried in high vacuo. To the residue was added a soln of **159** (40 mg, 76 μmol), HOBt (10 mg, 65.3 μmol), and DIPEA (20 μL, 118 μmol) in NMP (450 μL). After consumption of the amine (TLC) the mixture was concentrated and the residue was purified by RP-HPLC (Nucleogel RP 100–8 column, gradient of 32–45% MeCN over 27 min). The pooled fractions were lyophilized to give **160**; yield: 15 mg (63.3%); R_f 0.58 (iPrOH/1 M NH_4OAc 4:1); $[\alpha]_D$ 0.3 (*c* 0.5, MeOH); FAB-MS: found 2017 ± 1 [M+1]; ^{1}H NMR (500 MHz, DMSO-d_6, δ): 8.44 (d, 1H, $J_{NH,1}$ = 8.8 Hz, Asn NHγ), 7.99 (d, 1H, $J_{NH,2}$ = 7.1 Hz, NH2^2), 7.83 (d, 1H, $J_{NH,2}$ = 8.9 Hz, NH2^1), 7.58 (m, 3H, NH2^5,2$^{5'}$, urethane NH), 7.37–7.18 (m, 20H, Ar), 5.11 and 5.06 (2 d, 2H, J_{gem} = 12.8 Hz, CH_2O), 5.05–4.94 (m, 11H, H1^1,1^4, OH3^5,3^5,4^3,4^5,4$^{5'}$, CH_2O), 4.75 (d, 1H, $J_{OH,4}$ = 4.8 Hz, OH4^4), 4.71 (d, 1H, $J_{OH,4}$ = 4.8 Hz, OH4), 4.66 (d, 1H, $J_{1,2}$ < 1.0 Hz, H1^4), 4.62–4.40 (m, 12H, Asn αCH, H1^2, H1^3, OH2^3, OH6^4, OH6^5, OH6$^{5'}$, CH_2O), 4.36 (d, 1H, $J_{1,2}$ = 8.1 Hz, H1^5), 4.31 (d, 1H, J_{gem} = 12.2 Hz, CH_2O), 4.23 (d, 1H, $J_{1,2}$ = 8.2 Hz, H1^5), 4.19 (m, 1H, OH6^4), 3.99–3.94 (m, 2H, H2^3, OH3^4), 3.90 (d, 1H, $J_{OH,3}$ = 8.0 Hz, OH3$^{4'}$), 3.88–3.83 (m, 2H, H2^4, H4^1), 3.14–3.02 (m, 5H, H4^5, H4^5, H5^3, H5^5, H5$^{5'}$), 2.67 (dd, 1H, J_{gem} = 16.1 Hz, J_{vic} = 5.0 Hz, Asn βCHa), 2.50 (dd, 1H, J_{vic} = 7.2 Hz, Asn βCHb), 1.80, 1.77, 1.70 (12H, 4×NAc); ^{13}C NMR (125 MHz, Me_2SO-d_6, δ): 171.2 (ester C=O), 169.7, 169.5, 169.4, 169.3 (NAc C=O), 155.8 (urethane C=O), 139.2, 138.5, 138.4, 136.7, 135.9 (Ar), 128.3–126.9 (Ar), 101.5 (C1$^{5'}$), 101.1 (C1^5), 100.2 (C1^3), 99.8 (C1^4), 99.6 (C1^2), 97.6 (C1$^{4'}$), 81.5 (C3^3), 81.4 (C3^1), 80.8 (C3^2), 79.2 (C2$^{4'}$), 78.8 (C2^4), 78.5 (C1^1), 77.0 (C5^5), 76.9 (C5$^{5'}$), 76.1 (C5^1), 75.9 (C4^2), 75.5 (C5^3), 74.6 (C4^1), 74.4 (C5^2), 74.2 (C5^4), 73.9 (C3^5, C3$^{5'}$), 73.8 (CH_2O), 73.6 (C5$^{4'}$), 73.2, 72.2, 71.5 (CH_2O), 70.5 (C4$^{5'}$), 70.3 (C4^5), 69.9 (C3^4, C3$^{4'}$), 69.1 (C2^3), 68.7 (C6^2), 68.3 (C6^1), 67.6 (C4^4), 67.1 (C4$^{4'}$), 66.0 (CH_2O, C6^3), 65.6 (CH_2O), 64.8 (C4^3), 61.5 (C6^4), 61.1 (C6$^{4'}$), 61.0 (C6^5), 60.8 (C6$^{5'}$), 55.6 (C2^5, C2$^{5'}$), 55.3 (C2^2), 53.2 (C2^1), 50.4 (Asn αC), 36.8 (Asn βC), 23.2, 22.9, 22.7 (NAc).

6.3.3.2 Convergent Condensation Strategy

In the convergent method of N-linked glycopeptide synthesis, an O-protected or unprotected oligosaccharyl amine is coupled to a protected peptide with a single free aspartic acid side-chain carboxylate. Either the peptide or the oligosaccharide may be attached to a solid support. Often the Asp residue is protected as the allyl ester, which is readily and selectively cleaved by palladium(0) catalysis. These methods have never been employed for the synthesis of larger oligosaccharides linked to large peptides. The main reason is the slow cou-

pling reaction between sterically hindered Asp-containing peptides and sterically hindered glycosylamines, which require strong activation of the carboxylate. This leads primarily to the peptide aspartimide byproduct **135**, as shown in Scheme 37 (see also Section 6.3.1.1.3).

Scheme 37 Aspartimide Formation during Convergent *N*-Glycopeptide Synthesis[395]



Therefore, while the method is useful for synthesis of glycopeptides containing N-linked GlcNAc or even chitobiose, it does not work for larger oligosaccharides.[395] Attempts have been carried out to protect the Asp-Xaa amide bond by reversible alkylation using the Hmb protecting group.[145,396] However, the additional steric hindrance introduced in this way prevents reaction with larger oligosaccharyl amines; this method has therefore little or no potential for application in complex glycopeptide synthesis.[396] The elegant solution synthesis of a core pentasaccharide linked to a pentapeptide by Danishefsky et al.[393] utilizes the glycal technology for assembly of the glycopeptide by a semiconvergent approach. First the O-protected pentasaccharide amine is coupled with a tripeptide containing an N-terminal Asp residue and subsequently the amino terminal of the peptide is elongated with a dipeptide.

Use of N^{α}-AcHmb Backbone Protection; Ac-Glu-Asn(β-D-GlcNAc-β-D-GlcNAc)-Ala-Ser-Lys-Ala-NH_2:[145]

Peptide-resin Ac-Glu(OtBu)-Asp-(AcHmb)Ala-Ser(tBu)-Lys(Boc)Ala-PepSyn-KR (50 mg) was treated with β-D-GlcNAc-β-D-GlcNAc-NH_2 (2.65 mg, 6.25 μmol) dissolved in DMSO (50 mL), followed by 1.25 mol equiv of BOP (2.8 mg, 6.25 μmol), HOBt (0.96 mg, 6.25 μmol), and DIPEA (0.8 mg, 6.25 μmol) added from stock solutions in DMSO. After 4 h, the reagents were washed out. The peptide-resin (40.0 mg) was cleaved with 95% aq TFA for 1 h. Analytical HPLC showed a minor and a major peak, the latter confirmed to be the expected compound by MS. The crude peptide was purified by preparative HPLC using a gradient. The soln of the peptide was lyophilized and the Hmb group deacetylated with a drop of DMF containing hydrazine hydrate (5% v/v) for 15 min. The glycopeptide was precipitated with Et_2O at −35 °C and analytical HPLC showed the presence of a single peak with the expected mass. The precipitate was treated with a TFA/scavenger mixture for 1 h. The product was isolated upon addition of

for references see p 322

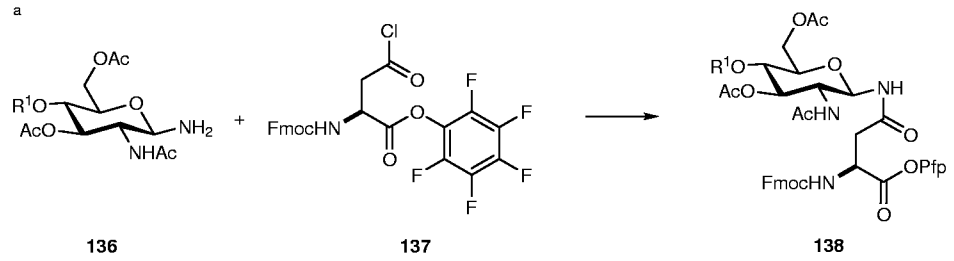
ice-cold Et_2O; yield: 0.7 mg (16.4%); ^1H NMR (300 MHz, D_2O, δ): 1.24–1.27 (m, 8H, 2×Ala βH_3, Lys γH_2), 1.53–1.8 (m, 6H, Lys βH_2, Lys δH_2, Glu βH_2), 1.87, 1.90, 1.92 (3×s, 9H, 3×$COCH_3$), 2.33 (t, 2H, Lys εH_2), 2.62 (dd, 1H, Asn CHα), 2.77 (dd, 1H, Asn CHβ), 2.87 (t, 2H, Glu γH_2), 3.35–3.38 (m, 14H, Ser βH_2+sugar H), 4.12–4.27 (m, 6H, Hα), 4.45 (d, 1H, J=9.0, H1'), 4.90 (d, 1H, J=9.5 Hz, H1).

6.3.3.3 The Building Block Strategy

The two approaches presented in Scheme 38 seem to have emerged as the preferred ones for the construction of asparagine glycosylated building blocks: (a) the protected mode[123,374,383,392–394] in which the building blocks are fully protected, preferably with silyl, acetyl, or benzoyl groups, allowing chemistry to be performed in organic solvents, or (b) the unprotected mode[114,149] in which the carbohydrate hydroxy groups are left unprotected as in structure **141**, requiring addition of water in the reaction mixtures in order to have sufficient solubility, at least during the coupling between the glycosylamine **139** and the activated aspartic acid **140**.

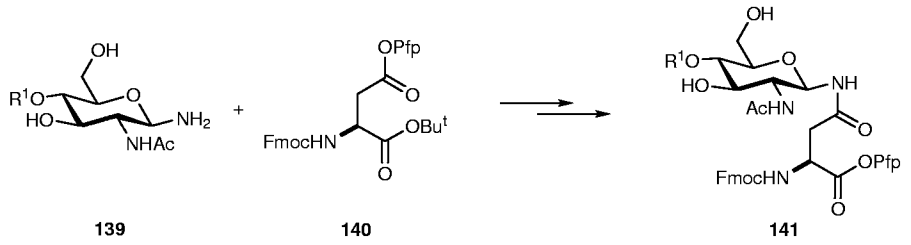
Scheme 38 The Two Prevailing Methods in the N-Linked Building Block Approach: (a) the Protected Mode and (b) the Unprotected Mode[114,123,149,374,383,392–394]

a



136 **137** **138**

b



139 **140** **141**

In the protected mode, O-protected glycosylamines have been synthesized and acylated by selective aminolysis of an activated aspartic acid side-chain. The C-terminal carboxy group is subsequently deprotected, activated (see below), and coupled to the peptide in a suitable organic solvent.[383,397] In a particularly efficient version of this strategy, the key compound Fmoc-Asp(Cl)-OPfp **137**[112,171] has been prepared quantitatively in a single-step procedure from commercially available Fmoc-Asp(OtBu)-OPfp, and when used in the coupling with the glycosylamine the activated building block has been obtained in high yield and then used directly for glycopeptide synthesis. At the end of the synthesis the hydroxy protecting groups are removed in a quantitative deacylation step.

6.3.3.3.1 Coupling of *O*-Acetate-Protected Mono- to Trisaccharyl Amines with Asparagine

Most frequently, sugar *O*-acetates have been used in N-linked glycopeptide synthesis. The protection of the glycosylamines as *O*-acetates requires two steps in the deprotection procedure after glycopeptide assembly.

N^{α}-Fmoc-Asp(Cl)-OPfp (137):[383]
N^{α}-Fmoc-Asp(OtBu)-OPfp (2.0 g, 3.47 mmol) was treated with a mixture of TFA (2.57 mL, 34.7 mmol) and $SOCl_2$ (25.0 mL, 347 mmol) at 40 °C for 24 h, then concentrated and coevaporated with THF. The resulting compound **137** was pure, according to ^1H and ^{13}C NMR, and was used without further purification; yield: 1.9 g (quant.); ^1H NMR (500 MHz, $CDCl_3$, δ): 5.758 (d, 1H, $J_{\alpha N,\alpha H}$ = 8.0 Hz, NαH), 5.024 (m, 1H, Hα), 3.792 (dd, 1H, $J_{\alpha,\beta}$ = 4.5 Hz, $J_{\beta,\beta'}$ = 19.0 Hz, Hβ), 3.670 (dd, 1H, $J_{\alpha,\beta'}$ = 3.5 Hz, Hβ′), 4.280 (t, 1H, Fmoc CH_2CH), 4.538 (br dd, 2H, Fmoc CH_2CH); ^{13}C NMR (125.77 MHz, $CDCl_3$, δ): 50.3 (Cα), 48.6 (Cβ), 47.0 (Fmoc CH_2CH), 67.7 (Fmoc CH_2CH); FAB-MS *m/z*: 539.87 [M + H^+].

1-*N*-Fmoc-3,4,6-tri-*O*-Ac-β-D-GlcNAc-NH:[383]
Crude 2-acetamido-2-deoxy-β-D-glucopyranosylamine (10.0 g, 45 mmol), prepared by the reaction of the reducing sugar with $(NH_4)_2CO_3$, was suspended in pyridine (125 mL), Fmoc-OSu (16.9 g, 50 mmol) was added and the mixture stirred overnight at rt. More pyridine (100 mL) and Ac_2O (40.0 mL, 424 mmol) were added. After 4 h, H_2O (25 mL) was added in small portions followed by additional H_2O (400 mL) and cooling to 0 °C. After stirring at 0 °C for 1 h, the crude product was isolated by filtration and recrystallization (THF/hexane); yield: 10.0 g (39%); $[\alpha]_D^{20}$ −15.2 (*c* 1.50, $CHCl_3$); mp 228–230 °C.

3,4,6-Tri-*O*-Ac-β-D-GlcNAc-NH_2 (136):[383]
1-*N*-Fmoc-2-acetamido-2-deoxy-3,4,6-tri-*O*-acetyl-β-D-glucopyranosylamine (1.5 g, 2.64 mmol) was suspended in THF (40 mL) and piperidine (10 mL) was added. After 10 min at rt the product was precipitated with hexane (110 mL) and filtered off; yield: 0.81 g (92%); $[\alpha]_D^{20}$ −19.3 (*c* 1.14, $CHCl_3$); mp 230 °C (dec).

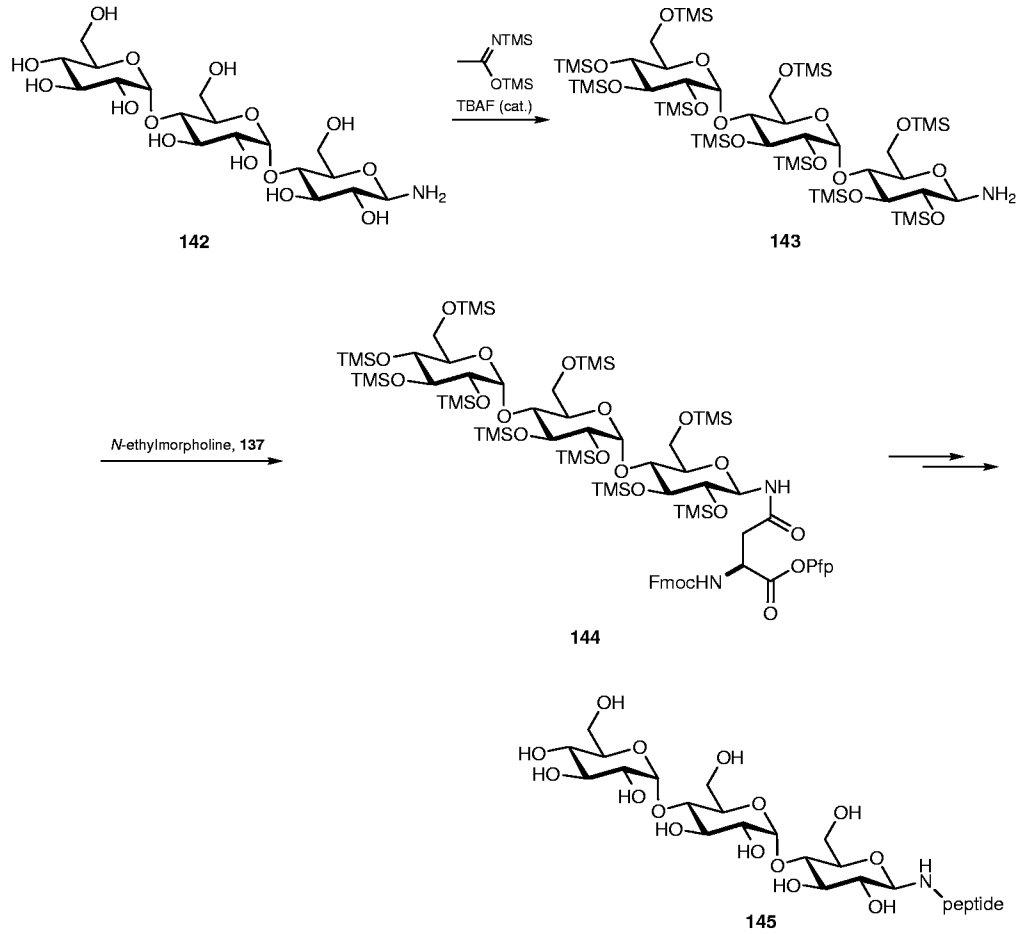
N^{α}-Fmoc-Asn(3,4,6-tri-*O*-Ac-β-D-GlcNAc)-OPfp (138):[383]
The O-acetylated glycosylamine **136** (R^1 = Ac; 0.58 g, 1.70 mmol) was dissolved in THF (17 mL) and *N*-ethylmorpholine (0.24 mL, 1.90 mmol) was added. The mixture was added dropwise to a soln of Fmoc-Asp(Cl)-OPfp (1.90 mmol) in THF (5 mL) at 0 °C. After filtration of the mixture the product was precipitated with hexane; yield: 1.20 g (84%); $[\alpha]_D^{20}$ +7.7 (*c* 1.03, $CHCl_3$); mp 190 °C (dec).

6.3.3.3.2 Silyl Protection of Glycosylamines

TFA-labile TMS protection as shown in Scheme 39 may be advantageously employed as an alternative to the acetates.[123] This has also the advantage that the crude glycosylamines can be protected directly, as the TMS reagent does not react with the anomeric amine.

for references see p 322

Scheme 39 Application of Trimethylsilyl Groups for O-Protection[123]



2,3,4,6-Tetra-*O*-TMS-β-D-Glc-NH_2:[123]
β-D-Glucopyranosylamine (1.0 g, 5.6 mmol) was suspended in pyridine (54 mL, 670 mmol) and BTMSA (8.2 mL, 33.5 mmol) was added. After 24 h at rt, MeOH (2 mL) was added and the mixture was extracted (hexane). The organic phase was washed with H_2O, dried ($MgSO_4$), and concentrated to give a syrup; yield: 1.84 g (84%); ES-MS *m/z*: 468.7 $[M+H]^+$.

Tetra-*O*-TMS-α-D-Glc-(1→4)-tri-*O*-TMS-α-D-Glc-(1→4)-tri-*O*-TMS-β-D-Glc-(1→)-NH_2 (143):[123]
Crude β-D-maltotriosylamine **142** was prepared from maltotriose (0.5 g) and converted into the TMS derivative as described above for the glucopyranosyl amine; yield: 1.16 g (95%); ES-MS *m/z*: 1224.8 $[M+H]^+$.

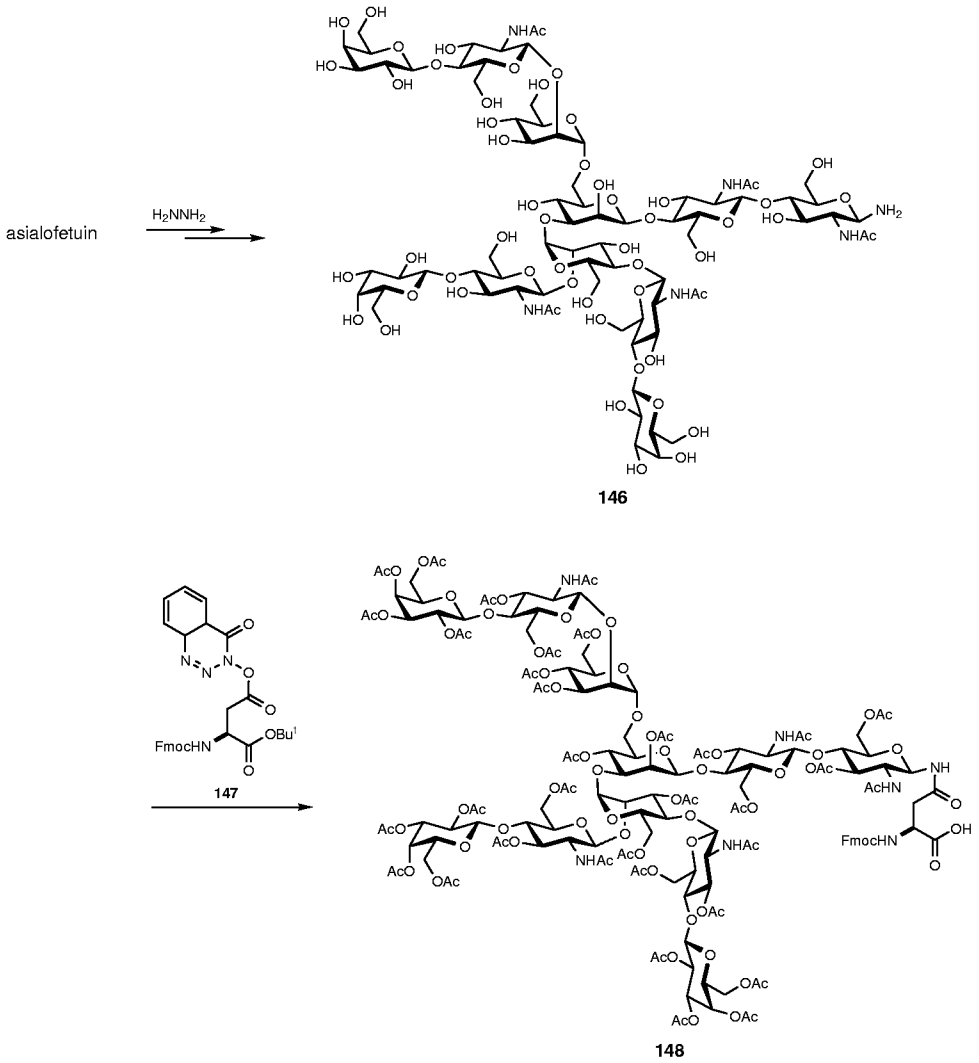
N^{α}-Fmoc-Asn(tetra-*O*-TMS-β-D-Glc)-OPfp:[123]
A mixture of tetra-*O*-TMS-β-D-Glc-NH_2 (1.84 g, 3.94 mmol) and *N*-ethylmorpholine (510 µL, 4.0 mmol) in THF (25 mL) was added dropwise to a soln of **137** (4.0 mmol) in THF (25 mL) at −40 °C. *N*-ethylmorpholine hydrochloride precipitated instantaneously and the mixture was allowed to warm to 20 °C. The hydrochloride was removed by filtration, the resulting filtrate concentrated, redissolved in dry CH_2Cl_2 (2 mL), and passed through a short column (silica gel, EtOAc/light petroleum ether 1:9) to give a syrup; yield: 2.1 g (56%); ^1H NMR (500 MHz, $CDCl_3$, δ): 5.13 ($J_{1,2}$ = 9 Hz, H1), 3.55, 3.46, 3.38 (H2, H3, H4), 3.27 (H5), 3.82 ($J_{5,6}$ = 1.5 Hz, $J_{6,6'}$ = 11.5 Hz, H6), 3.66 (H6′), 6.23, 5.80 (Asn NαH, NγH), 5.05 (Hα), 3.23 ($J_{\alpha,\beta}$ = 4 Hz, $J_{\beta,\beta'}$ = 17 Hz, Hβ), 2.93 ($J_{\alpha,\beta'}$ = 4 Hz, Hβ′), 4.28 (Fmoc H9), 4.51 ($J_{9,H}$ = 7.5 Hz, $J_{H,H'}$ = 10.5 Hz, CH_2), 4.38 ($J_{9,H'}$ = 7.5 Hz, CH_2′); ^{13}C NMR (125.77 MHz, $CDCl_3$, δ): 79.8, 78.7, 78.3, 74.6, 71.3 (C1, C2, C3, C4, C5), 62.1 (C6), 50.2 (Asn Cα), 37.9 (Cβ), 47.0 (Fmoc C9), 67.6 (CH_2), 136.8–142 (Pfp); ES-MS *m/z*: 971.6 $[M+H]^+$.

N^{α}-Fmoc-Asn(O-TMS$_3$-α-D-Glc-O-TMS$_3$-α-D-Glc-O-TMS$_3$-β-D-Glc)-OPfp (144):
Compound **144** was prepared from silylated maltotriosylamine **143** and Fmoc-Asp(Cl)-OPfp (**137**) as described above, except for the fact that the crude product was not filtered through silica gel but used directly in the synthesis of glycopeptides **145**; ES-MS *m/z*: 1772.4 $[M+HCO_2H]^+$.

6.3.3.3.3 Glycosylated Asparagine Building Blocks from Large Oligosaccharides

In the second approach (Scheme 40), crude glycosylamines **146**[379] were coupled in water/DMF with Fmoc-Asp-OtBu in situ activated as, for example, the Pfp ester or the Dhbt ester **147**.[149] The product may be acetylated (to give **148**) to increase the solubility for the peptide synthesis. The condensation product may be purified by HPLC before the C-terminal protection is removed and the glycosyl amino acid subsequently activated in situ for glycopeptide synthesis. The final fully deprotected peptide is released in a single step and subjected to HPLC purification.[113] This approach is very appealing for the conversion of small amounts of large isolated oligosaccharides into glycopeptides.[149]

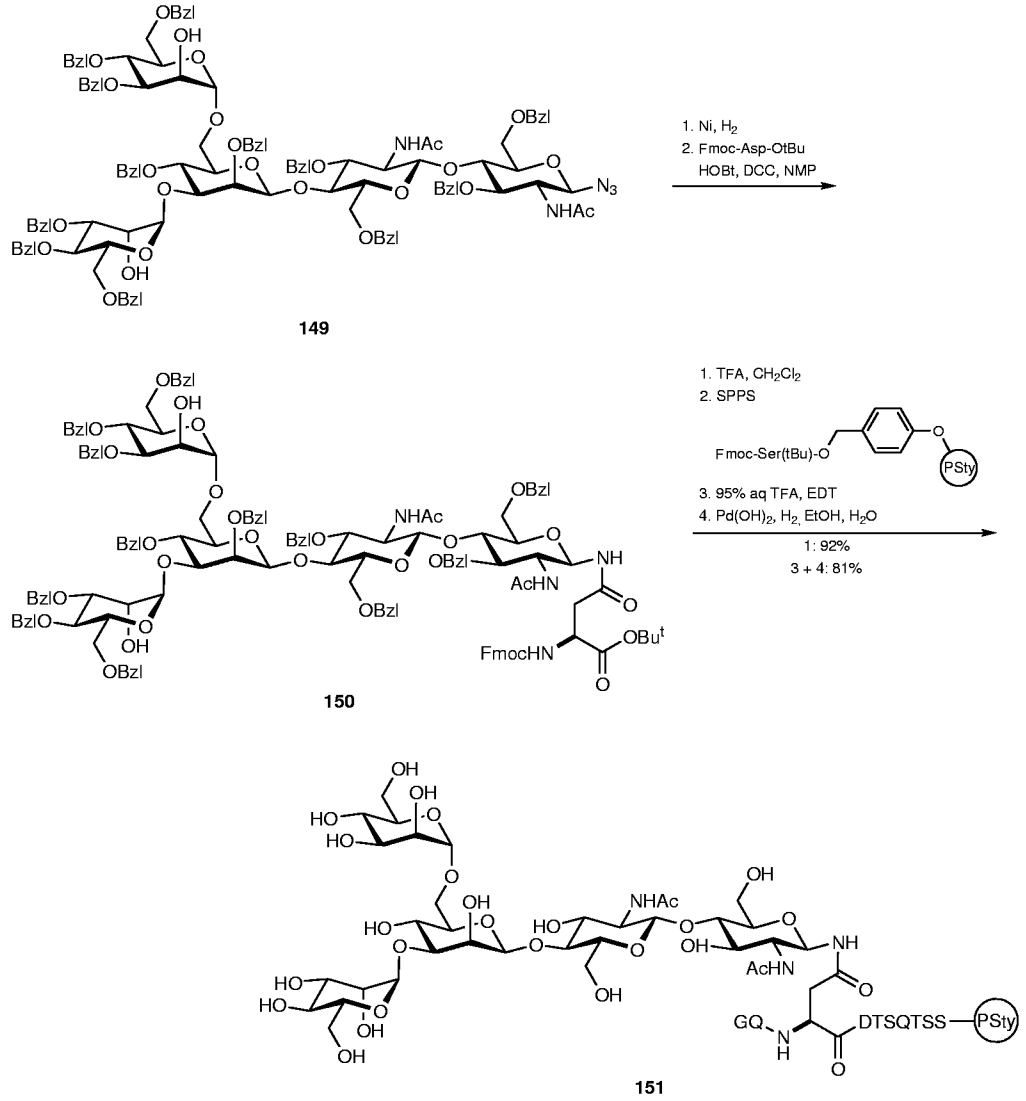
Scheme 40 Large Oligosaccharides Isolated from Asialofetuin (Complex-Type) and Ribonuclease B (High-Mannose-Type) by Hydrazinolysis[149]



for references see p 322

It has been advocated that protected building blocks are less well incorporated into the peptide than unprotected ones.[114] This is in sharp contrast to the results of others[96,97,103,383,394] and may instead derive from the higher reactivity of Pfp esters versus symmetrical anhydrides. As an example, compound **149** (Scheme 41), prepared via a silver silicate promoted glycosylation (see Section 6.3.3.5), was converted into building block **150**, which was incorporated in almost quantitative yield during glycopeptide assembly of compound **151** on solid phase. The previous failure with protection may be due to an unsuccessful attempt to prepare the symmetrical anhydride of the large glycosylated amino acid.

Scheme 41 Conversion of a Glycosyl Azide into a Glycodecapeptide[394]



A third interesting method is the coupling of glycosyl isothiocyanates,[382,398–400] generated in high yield from the fully protected oligosaccharide oxazolines by reaction with potassium isothiocyanate in the presence of a crown ether,[372] with C- and N-terminally protected aspartic acid derivatives.[372,401,402] Here the glycosyl isothiocyanate is functioning as the activating agent and only the addition of the free aspartic acid is required. Under strictly dry

conditions it has been thus possible to obtain a 75% yield of peracetylated Aloc-Asn[β-D-Man-(1→4)-β-GlcNAc-(1→4)-β-D-GlcNAc]-OtBu, corresponding to the core trisaccharide of N-linked glycoproteins.

N^{α}-Fmoc-Asn{tetra-*O*-Ac-α-D-Man-(1→6)-[tetra-*O*-Ac-α-D-Man-(1→3)]di-*O*-Ac-α-D-Man-(1→6)-[tetra-*O*-Ac-α-D-Man-(1→3)-]di-*O*-Ac-β-D-Man-(1→4)-di-*O*-Ac-β-D-GlcNAc-(1→4)-di-*O*-Ac-β-D-GlcNAc-(1→*N*)}-OBzl:[374]

A soln of tetra-*O*-Ac-α-D-Man-(1→6)-[tetra-*O*-Ac-α-D-Man-(1→3)-]di-*O*-Ac-α-D-Man-(1→6)-[tetra-*O*-Ac-α-D-Man-(1→3)-]di-*O*-Ac-β-D-Man-(1→4)-di-*O*-Ac-β-D-GlcNAc-(1→4)-di-*O*-Ac-β-D-GlcNAc-(1→)-NH_2 (100 mg, 47.6 μmol), obtained from reduction of the glycosyl azide with Lindlar's catalyst, Fmoc-Asp-OBzl (42 mg, 95 μmol), and DIPEA (125 μmol) in DMF (0.6 mL) was cooled to −10 °C and treated with a 0.22 M soln of diethyl cyanophosphonate (110 μmol) in DMF (0.5 mL). The red-colored mixture was stirred for 2.5 h at 5 °C, when TLC ($CHCl_3$/MeOH 10:1) showed the conversion of amine (R_f 0.21) into the product (R_f 0.51). After the addition of AcOH (0.2 mL) and EtOAc (30 mL), the organic phase was separated and washed with H_2O (5×10 mL). Removal of the solvent gave a residue that was applied to five preparative TLC plates, which were developed with $CHCl_3$/MeOH (10:1). The band containing the product was located by UV irradiation and by cutting a 0.5-cm strip from each plate and spraying with the anisaldehyde reagent. Extraction of the silica gel by stirring for 10 min with $CHCl_3$/MeOH (2:1), then washing on a filter with $CHCl_3$/MeOH (1:1), was followed by concentration of the combined filtrates; yield: 60 mg (50%); $[\alpha]_D$ +12.4 (*c* 3.0, CH_3Cl).

Fmoc-Asp(ODhbt)-OtBu (147):[149]

To a soln of Fmoc-Asp-OtBu (1.5 g, 3.65 mmol) in freshly distilled THF (20 mL), cooled to −35 °C, DCC (0.752 mg, 3.65 mmol) was added with stirring. After 10 min, HODhbt (595 mg, 3.65 mmol) was added and the mixture was stirred for 2 h at −35 °C. It was allowed to warm to rt, filtered through Celite, and concentrated. The residue was redissolved in THF (15 mL), filtered through Celite, and concentrated. Compound **147** was obtained by chromatography of the residue (silica gel, EtOAc/light petroleum ether 1:4); yield: 1.78 g (88%); ES-MS *m/z*: 579.22 $[M+Na]^+$.

Derivatization of the Mixture of the Tri- and the Diantennary Asialofetuin Oligosaccharides with Fmoc-Asp(ODhbt)-OtBu (150):[149]

The purified mixture of two asialofetuin oligosaccharides (20 mg, 10 μmol), obtained by hydrazinolysis, was converted into oligosacharide glycosylamines, redissolved in H_2O (2 mL), and lyophilized several times to yield quantitatively the glycosylamines (20 mg). The derivatization of the dibranched and the tribranched (see **146**) glycosylamines was accomplished by dissolving the oligosaccharyl amine mixture (20 mg) in DMSO (2 mL), adding DIPEA (1.2 equiv) and a 2-fold excess of Fmoc-Asp(ODhbt)-OtBu. The reaction was allowed to proceed at rt for 3 h. The formation of the oligosaccharide–amino acid building blocks was monitored by analytical HPLC and TLC (EtOAc/MeOH/H_2O 2:3:3). When the reaction was complete, the mixture was lyophilized exhaustively. Separation of the derivatized oligosaccharides from the asparagine amino acid derivatives was achieved by preparative RP-HPLC. The two N-linked oligosaccharides eluted separately as single peaks and purity was confirmed by analytical HPLC. They were pooled separately and lyophilized. Each of the major peaks was chromatographed again on analytical RP-HPLC and analyzed by ^1H NMR. Analytical RP-HPLC [gradient 100:0 (A:B) for 5 min, 50:50 in 35 min, and then 0:100 in 10 min; A 0.1% aq TFA, B 0.1% aq TFA in 90% MeCN] gave for Fmoc-Asn($Gal_2GlcNAc_2Man_3GlcNAc_2$)-OtBu a t_R of 16.06 min and for Fmoc-Asn($Gal_3GlcNAc_3Man_3GlcNAc_2$)-OtBu a t_R of 14.38 min. For the synthesis of the N-linked glycopeptides containing the dibranched ($Gal_2GlcNAc_2Man_3GlcNAc_2$) and the tribranched ($Gal_3GlcNAc_3Man_3GlcNAc_2$) complex oligosaccharides, two different synthesis strategies were utilized. The first strategy used the peracetylated building blocks. Attempts were made to peracetylate the separated oligosaccharide–asparagine derivatives (2 mg and 8 mg, respectively) with pyridine/Ac_2O (1:1, 2 mL) at 50 °C for 12 h. The mixtures were concentrated and coevaporated with toluene and analyzed by analytical HPLC and ^1H NMR. Both techniques showed that complete O-acetylation of the building blocks could not be achieved. However, synthesis was continued by cleaving the tBu ester of the oligosaccharide–asparagine derivatives by treatment with TFA for 1 h. The TFA was removed and the residue was coevaporated with toluene and purified by preparative HPLC. The resulting building blocks, diantennary (2.6 mg) and **148** (12 mg), were used in the synthesis of the hemoglobin-derived glycopeptides VITAFN(Ac_n-$Gal_2GlcNAc_2Man_3GlcNAc_2$)EGLK and VITAFN($Ac_nGal_3GlcNAc_3Man_3GlcNAc_2$)EGLK.

Due to the difficulties in peracetylation of the oligosaccharide–asparagine building blocks, another strategy was also followed utilizing the building blocks with completely unprotected carbohydrate moieties. In this second approach the unprotected oligosaccharide-asparagine derivatives were purified and subsequently directly subjected to TFA treatment. The building blocks (2 and 8 mg, respectively) were kept in neat TFA for 1 h in order to cleave the tBu ester. Purification by preparative HPLC

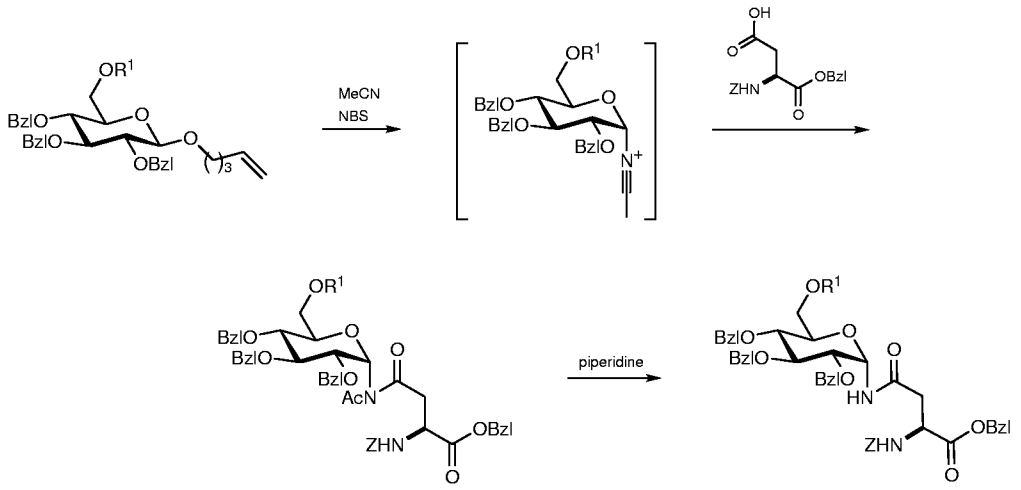
for references see p 322

afforded the diantennary (1.2 mg) and triantennary (5.2 mg) building blocks, respectively. These were characterized by ES-MS and ^{1}H NMR spectroscopy. The unprotected building blocks have also been used in the synthesis of the hemoglobin-derived glycopeptides VITAFN(Gal$_2$GlcNAc$_2$Man$_3$GlcNAc$_2$)-EGLK and VITAFN(Gal$_3$GlcNAc$_3$Man$_3$GlcNAc$_2$)EGLK (see also Section 6.3.3.5).

6.3.3.4 α-N-Linked Glycosylamides

Glycosylamines are characterized by a property named the reversed anomeric effect[403] that leads to destabilization of axial glycosylamines compared to equatorial ones. This in turn leads to rapid anomerization of an axial amine to the equatorial isomer and has caused difficulties in preparing asparagine carrying an α-linked glucose present in the rat basement membrane.[139,404–407] In these studies the α-glycosyl azide was, therefore, reduced by $NaBH_4$, H_2S, or by catalytic hydrogenation over PtO_2 or Lindlar's catalyst; the latter in the presence of Et_3N gave the best yield of the α-glycosylamine.[140,408] The mixture of anomers was directly acylated with Boc-Asp-Pro-OMe by activation with *O,O*-diethylcyanophosphate. Ratcliffe et al.[409,410] reported a new reaction (Scheme 42) which elegantly solved the problem. During activation of pentenyl glycosides with NBS in MeCN, the kinetically formed α-acetonitrilium ion may be trapped by the presence of a carboxylic acid. The intermediate rearranges and loses an acetyl group originating from the MeCN upon treatment with piperidine. The method has been improved and perbenzylated Z-Asn[α-D-(1→6)-Glc-β-D-Glc-(1→6)-α-D-Glc]-OBzl could be obtained (in 74% yield) by NBS/TfOH activation of the ethyl thioglycoside in propionitrile at low temperature.[140]

Scheme 42 N-Linked α-Glycosides Obtained through Bromonium Activation of Pentenyl Glycosides in Acetonitrile[409,410]



R^1 = Bzl$_4$-α-D-Glc-(1→6)-Bzl$_3$-α-D-Glc

Reduction of α-Glycosyl Azide and Peptide Coupling; *N*^{α}-Fmoc-Asn[tetra-*O*-Bzl-α-D-Glc-(1→6)-tri-*O*-Piv-β-D-Glc-(1→6)-tri-*O*-Bzl-α-D-Glc-(1→)-*N*]-Pro-OtBu:[408]

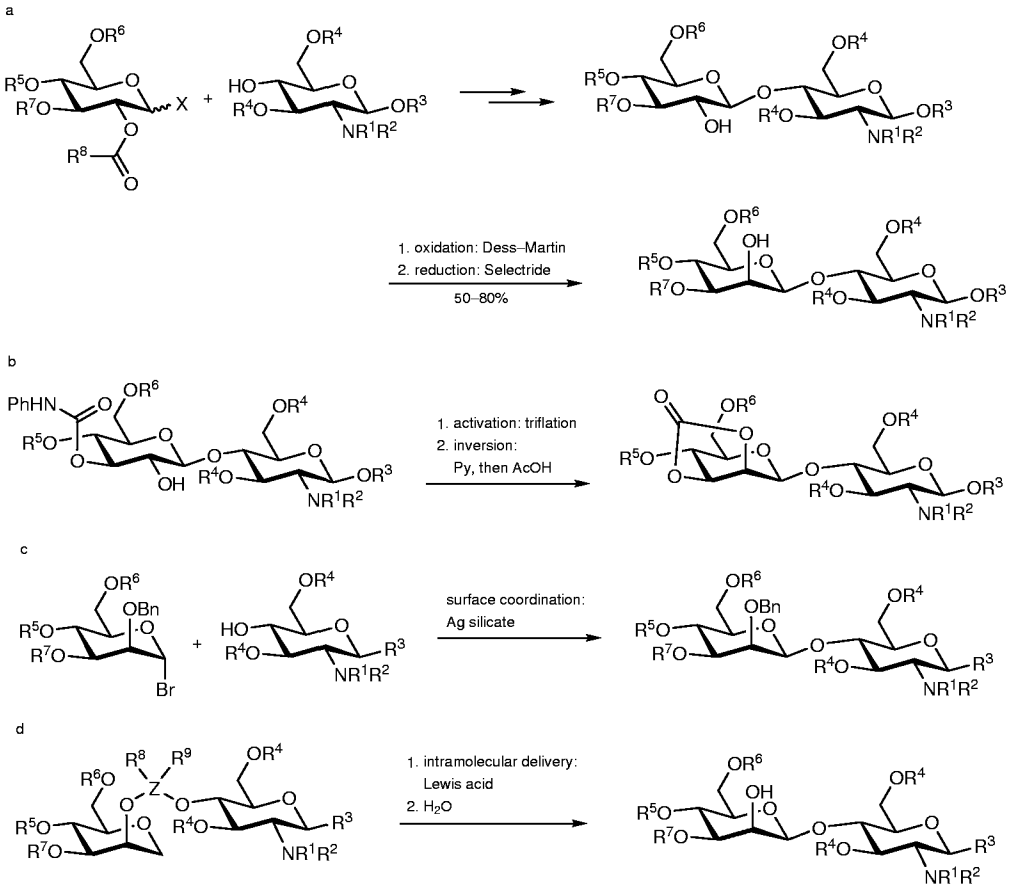
A soln of tetra-*O*-Bzl-α-D-Glc-(1→6)-tri-*O*-Piv-β-D-Glc-(1→6)-tri-*O*-Bzl-α-D-Glc-(1→)-N_3 (141.3 mg, 0.1 mmol) in THF/MeOH (1:1, 10 mL), was hydrogenated under H_2 pressure (1520 Torr) in the presence of Lindlar's catalyst (70 mg) and TEA (130 mg, 1.25 mmol) for 16 h at rt. The catalyst was removed and the filtrate was concentrated to dryness to give an α/β mixture of the O-protected trisaccharyl amine. This mixture was dissolved in CH_2Cl_2/DMF (9:1, 2 mL) and then added to a soln of *N*^{α}-Fmoc-Asp-Pro-OtBu (61 mg, 0.12 mmol), TBTU (38.5 mg, 0.12 mmol), HOBt (18.4 mg, 0.12 mmol), and DIPEA (15.5 mg, 0.12 mmol) in CH_2Cl_2/DMF (9:1, 5 mL). The resulting mixture was stirred at 0 °C for 1 h, then left at rt for 12 h, diluted with CH_2Cl_2, washed with H_2O, 10% citric acid, H_2O, dried, and concentrated. The product was isolated by chromatography of the residue (silica gel, CH_2Cl_2/EtOAc 10:1); yield:

142 mg (76%) of a mixture (α/β 5:1). The α-linked compound could be isolated by TLC (CH_2Cl_2/EtOAc 6:1, R_f = 0.47); $[\alpha]_D$ = +22.8 (*c* 0.5, $CHCl_3$); MS *m/z*: 1876.8 [M^+]; ^{1}H NMR (400 MHz, CD_2Cl_2, δ): 5.96 (d, 1H, *J* = 6.2 Hz), 5.70 (t, 1H, *J* = 6.2 Hz), 5.18 (t, 1H, *J* = 9.4 Hz), 5.08 (t, 1H, *J* = 9.4 Hz), 4.90 (t, 1H, *J* = 8.7 Hz), 2.73–2.50 (dd, 2H, *J* = 15.4, 6.0 Hz, $COCH_2CH$), 1.32 (s, 9H, Me_3CO), 1.05, 1.03, 1.01 (3 s, 27H, 3×tBuCO); ^{13}C NMR (100.6 MHz, CD_2Cl_2, δ): 176.83, 176.17, 175.98 (3C, 3×tBuCO), 170.54, 170.08, 168.69, 156.02 (4C, 4×C=O), 100.36 (1C, C1b), 97.00 (C1c), 81.18 (C1a), 81.01–65.72 (24C), 59.90 (1C, Pro C), 49.80 (1C, $COCH_2CH$), 46.94, 46.80 (2C, Pro C, Fmoc CH), 40.20 (1C, $COCH_2CH$), 38.42 (3C, 3×tBuCO), 28.84 (1C, Pro C), 27.55 (3C, tBuCO), 26.82 (9C, 3×tBuCO), 2467 (1C, Pro C).

6.3.3.5 Preparation of Larger Building Blocks by Synthesis

Many studies have been carried out to investigate the synthesis of complex[8,411] and high-mannose oligosaccharide fragments.[3,412,413] Although elegant designs of protection schemes are required, different condensation strategies, e.g. thioglycosides, glycosyl halides, or pentenyl glycosides, work equally well for the synthesis of the outer antennary structures, whereas the most difficult problem is the assembly of the trisaccharide core linking these structures to the proteins. The core of the N-linked glycosides contain a β-mannosidic linkage, which is extremely difficult to obtain by conventional glycosylation procedures. Four conceptually different methods have been used for the formation of the β-mannosidic linkage (Scheme 43).

Scheme 43 Four Methods for β-Glycosylation[8,393,414–416,418,419,422–425]



for references see p 322

The most practical one uses a 2-*O*-acyl-glucosyl donor with the *gluco* configuration directing β-glycosylation of a chitobiose aglycon with anchimeric assistance from the 2-position by intermediate acyloxonium ion formation as the β-mannosyl precursor (Scheme 43, a). The product is selectively deprotected in the 2-position of the glucosyl unit and the hydroxy group is inverted. This may be effected either by oxidation (Dess–Martin), reduction (Selectride),[393,414] or by intramolecular inversion (Scheme 43, b) of a triflate in the 2-position by a carbamoyl group in the 3-position of a glucopyranose.[415,416] The latter gives the 2,3-carbonate as an intermediate product. Alternatively, intermolecular nucleophilic reactions may be used for the inversion.[417]

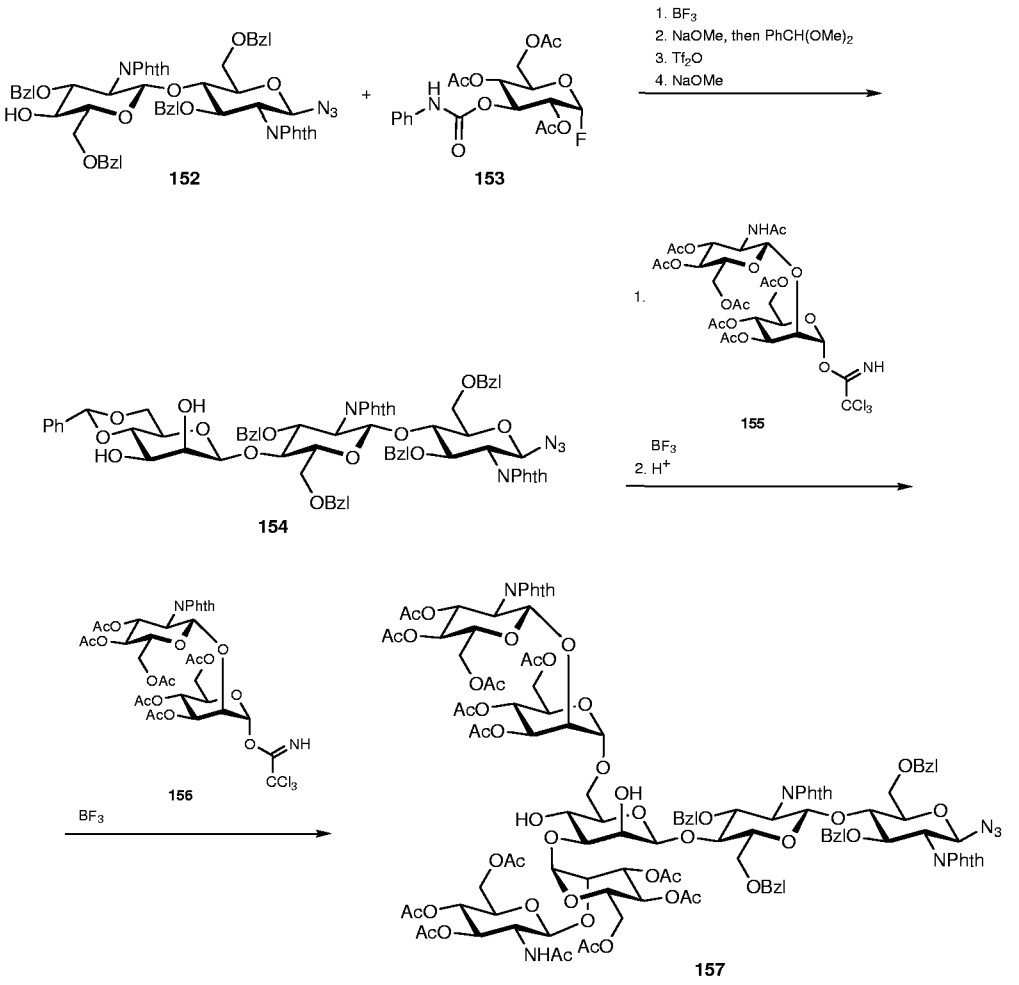
The β-mannosidic linkage of complex oligosaccharides has also been obtained[8,418] by the method of Paulsen and Lockhoff[419] using silver alumina silicate as a heterogeneous catalyst coordinating to the α-mannosyl bromide and thus affording the β-glycoside by a S_N2-type reaction (Scheme 43, c). Paulsen et al.[8] elegantly demonstrated the synthesis of very large complex oligosaccharides containing the β-mannosidic linkage by silver silicate promoted glycosylation of 1,6-anhydro-2-azido-2-deoxy-3-*O*-benzyl-β-D-glucopyranose as the key reaction. The method has also been used by Ogawa et al. to construct complete complex-type penta- (see **149**) to heptasaccharides.[420,421] However, only a few of these procedures have ever appeared as detailed experimental descriptions.

The last method is by intramolecular delivery (Scheme 43, d);[422–425] although elegant, this method has not yet found practical application in the synthesis of complex oligosaccharides. It remains to be seen if the method will work with the rather unreactive 4′-hydroxy group of chitobiose derivatives. This reactivity has been increased both by application of phthaloyl protection of the 2-amino groups and by application of the inverted conformation of 1,6-anhydro-2-azido-2-deoxyglucopyranose.[8]

A more efficient scheme for the preparation of the complex and high-mannose N-linked glycosyl amino acids has also been described.[174,204,392,411,426,427] Thus di-, tri-, and tetra-antennary glycosyl asparagine building blocks are synthesized by chemoenzymatic synthesis. The procedure minimizes the number of operations and combines efficiently the chemical assembly of core penta- to nonaglycosyl azides with the use of enzymes to elongate the structures. This method readily competes in terms of yield and purity with the mainly enzymatic procedures described in Section 6.3.3.6, which utilize reverse endoglycosylase reactions combined with the natural pools of carbohydrates isolated from pronase digests of glycoprotein.[222,428–430] The use of chemical synthesis to build the central core penta- to nonasaccharides followed by enzymatic glycosylation with galactosyl- and sialyltransferases to attach the terminal sugars has been carefully optimized to yield most of the commonly observed glycoform structures.

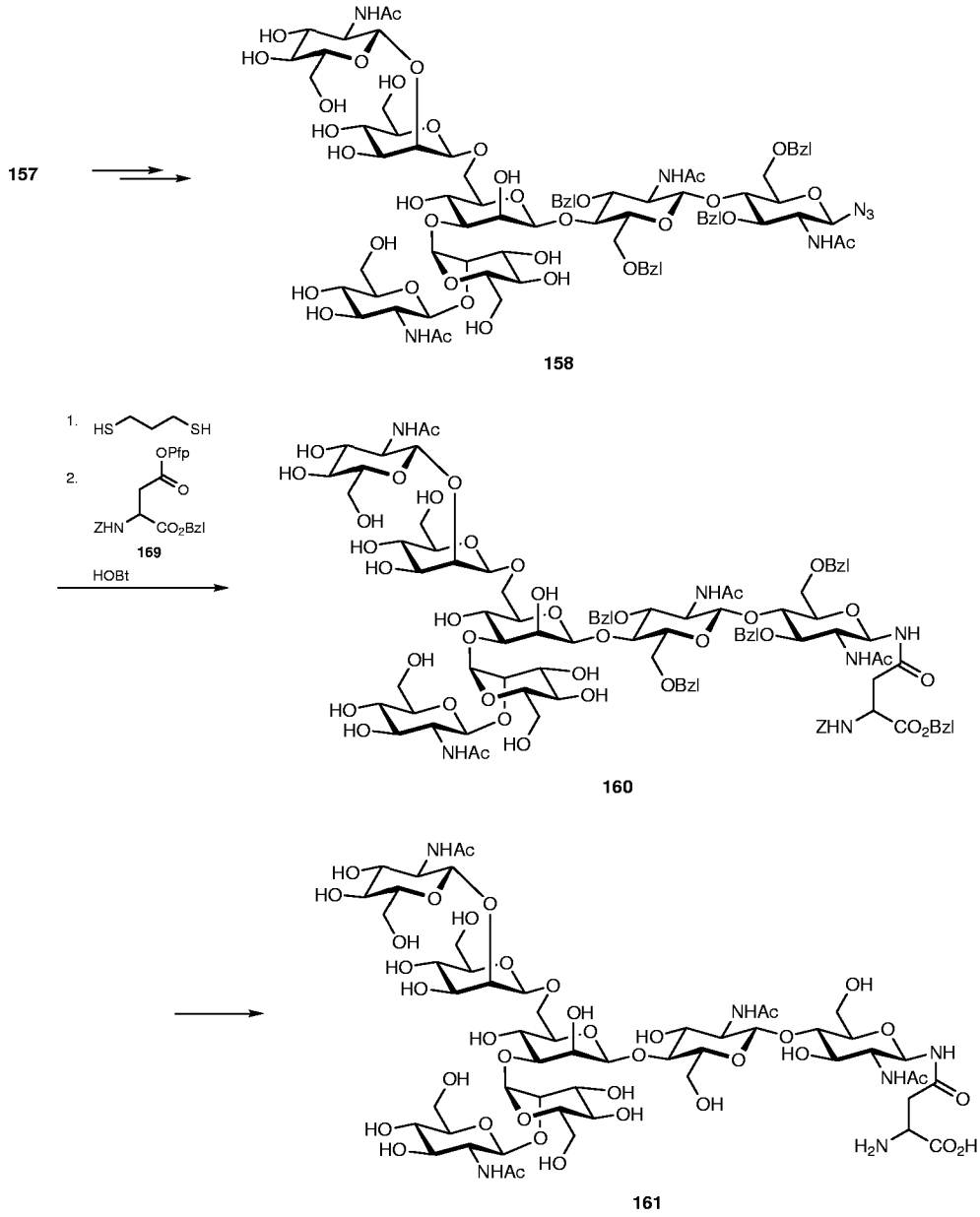
Thus reaction of 2-*N*-phthaloyl-3,6-di-*O*-benzyl-β-D-glucosaminosyl azide with 2-*N*-phthaloyl-3,6-di-*O*-benzyl-4-chloroacetyl-β-D-glucosaminosyl fluoride affords exclusively the protected chitobiosyl azide derivative in high yield. The chloroacetyl group is removed to give compound **152** and β-glycosylation is achieved with 2,4,6-tri-*O*-acetyl-3-*O*-(*N*-phenylcarbamoyl)-α-D-glucopyranosyl fluoride **153** in the presence of BF_3 (Scheme 44). After exchange of the acetates for a 4,6-benzylidene group, the free hydroxy group of the trisaccharide is triflated and the intramolecular inversion catalyzed with NaOMe to afford **154**. Disaccharide trichloroacetimidates **155** and **156** were used as building blocks catalyzed by BF_3 for the succesive elongation in the 3- and 6-positions with GlcNAc-(1→2)-Man units. All these reactions were performed in high yield, affording **157**. The heptasaccharide was deprotected, the anomeric azide **158** reduced with propanedithiol, and the amine coupled with Z-Asp(OPfp)-OBzl (**159**) to give the glycosylated asparagine building block **160**, which was fully deprotected to afford **161** ready for enzymatic elongation (Scheme 45).[411]

Scheme 44 Optimized Assembly of the Core of Complex Carbohydrates[411]



for references see p 322

Scheme 45 Synthesis and Deprotection of a Glycosylated Asparagine Building Block[411]



β-Mannosides by Inversion via Oxidation–Reduction; 2-*O*-TBDMS-3,4,6-tri-*O*-Bzl-α-D-Man-(1→6)-[2-*O*-TBDMS-3,4,6-tri-*O*-Bzl-α-D-Man-(1→3)]-4-*O*-PMB-β-D-Man-(1→4)-3,6-di-*O*-Bzl-β-D-GlcNAc-(1→4)-3,6-di-*O*-Bzl-D-Glucal:[393]

The free equatorial 2-hydroxy group of the Glc unit of 2-*O*-TBDMS-3,4,6-tri-*O*-Bzl-α-D-Man-(1→6)-[2-*O*-TBDMS-3,4,6-tri-*O*-Bzl-α-D-Man-(1→3)]-4-*O*-PMB-β-D-Glc-(1→4)-3,6-di-*O*-Bzl-β-D-GlcNAc-(1→4)-3,6-di-*O*-Bzl-D-Glucal, the partially protected pentasaccharide precursor to **132**, was oxidized according to the following procedure. The precursor (992 mg, 0.454 mmol) was azeotropically distilled with benzene (3×50 mL) and dried under vacuum. It was then dissolved in CH_2Cl_2 (31 mL) and freshly redistilled pyridine was added (431 mg, 5.45 mmol). Dess–Martin reagent (770 mg, 1.81 mmol) was added in one portion in a glove bag. The reaction was carefully monitored by TLC and quenched with sat. $Na_2S_2O_3$ (30 mL) as soon as the reaction went to completion (4 h). The mixture was extracted with EtOAc (3×100 mL) and the combined extract was thoroughly washed with sat. $Na_2S_2O_3$ (3×50 mL),

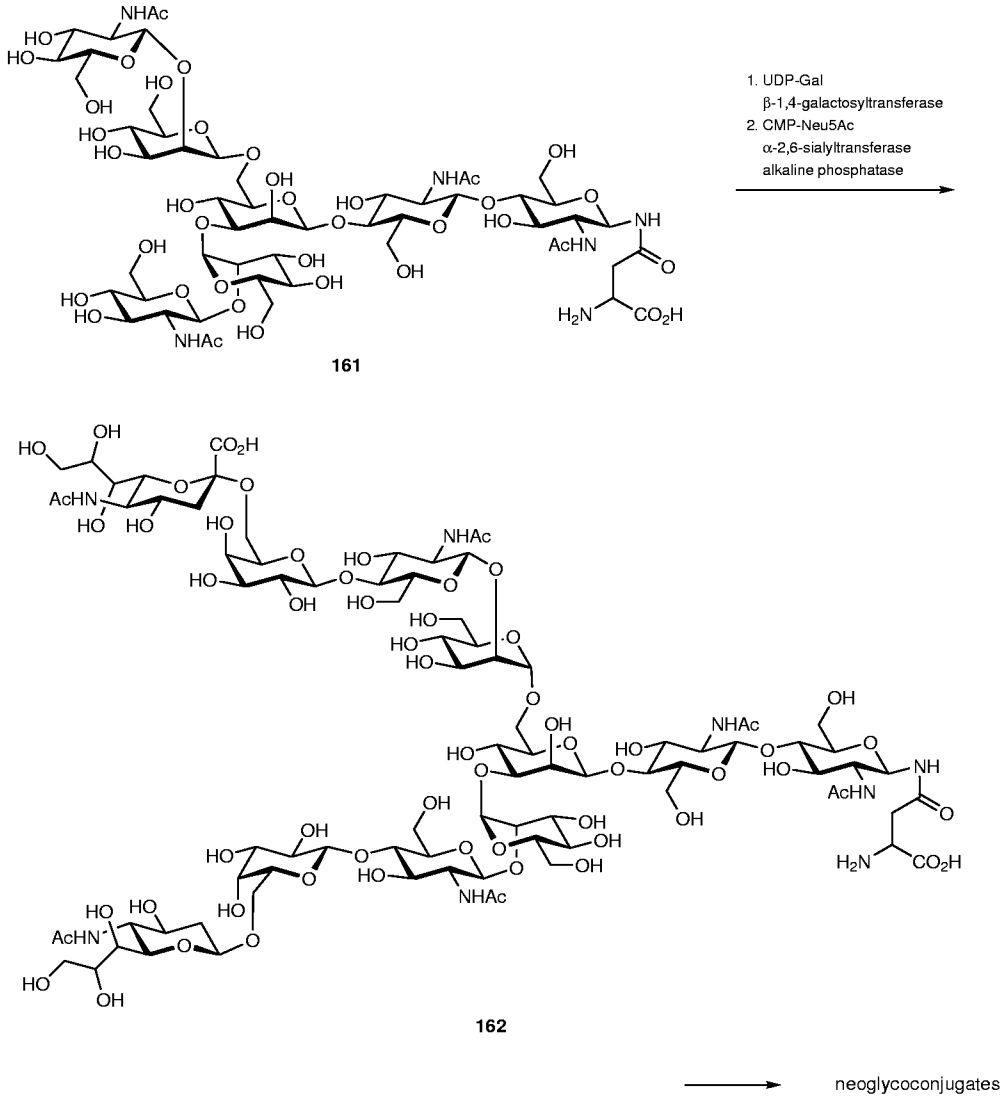
followed by sat. $NaHCO_3$ (50 mL), and sat. NaCl (100 mL). After drying (Na_2SO_4) and concentration, the crude ketone was azeotropically distilled with benzene (3×20 mL) and further dried under vacuum for 20 min. It was then dissolved in THF, and cooled to −42 °C. 1 M L-Selectride (1.36 mmol) in THF was added slowly. The mixture was then allowed to warm to rt and further stirred for 2 h. It was quenched with sat. $NaHCO_3$ (10 mL) and extracted with EtOAc (3×50 mL). The combined extracts were washed with sat. NaCl and dried ($MgSO_4$). After concentration, purification was by chromatography (silica gel, EtOAc/hexane 2:9); yield: 817 mg (83%); $[\alpha]_D$ 2.6 (*c* 2.05, CH_2Cl_2); ^1H NMR ($CDCl_3$, δ): 7.79 (d, 2H, *J* = 7.5 Hz), 7.38–7.18 (m, 54H), 6.84 (d, 2H, *J* = 8.5 Hz), 6.36 (d, 1H, *J* = 6.1 Hz), 4.96 (br s, 1H), 4.93 (br s, 1H), 4.88 (br s, 1H), 4.81 (br s, 1H), 4.90–4.39 (m, 24H), 4.33–4.30 (m, 2H), 4.28 (br s, 1H), 4.19 (br s, 1H), 4.10–3.50 (m, 30H), 3.80 (s, 3H), 3.49–3.45 (m, 3H), 3.80–3.30 (m, 3H), 0.99 (s, 18H), 0.16 (s, 3H), 0.12 (s, 6H), 0.11 (s, 3H); ^{13}C NMR ($CDCl_3$, δ): 159.05, 144.31, 141.74, 138.80, 138.72, 138.61, 138.54, 138.19, 138.14, 138.05, 138.00, 137.74, 137.57, 132.10, 130.05, 128.67, 128.53, 128.39, 128.28, 128.21, 128.16, 128.13, 128.07, 128.04, 128.02, 127.97, 127.91, 127.55, 127.41, 127.37, 127.32, 127.26, 127.17, 127.07, 113.70, 102.20, 101.14, 100.57, 100.47, 100.13, 84.20, 80.12, 79.74, 75.88, 75.72, 75.30, 74.85, 74.52, 74.50, 73.40, 73.36, 73.14, 73.07, 72.99, 72.88, 72.76, 72.26, 70.33, 70.19, 69.78, 69.15, 69.00, 67.62, 66.48, 58.40, 55.08, 25.79, 18.05, 4.57, 4.69, 4.79, −4.96; HRMS *m/z*: 2204.9700.

6.3.3.6 Enzymatic Elongation

Large building blocks as well as complete glycopeptides may conveniently be elongated by enzymatic transfer of a monosaccharide from the uridine diphosphate or guanosine monophosphate derivative. After removal of the amino acid protecting groups in **160** the heptasaccharyl amino acid **161** was consecutively used as an acceptor for GlcNAc-β-1,4-galactosyltransferase, using galactosyl-1-*O*-uridine-5′-diphosphate (UDP-Gal) as the donor, and for β-galactoside-α-2,6-sialyltransferase, using N-acetylneuraminyl-2-*O*-cytidine-5′-monophosphate acid (CMP-Neu5Ac) as the donor, affording undecaglycosyl amino acid **162** (Scheme 46). The method has been shown to work also for tri- and tetraantennary structures.[392]

for references see p 322

Scheme 46 Synthesis of an Undecasaccharyl Asparagine[392]



Enzymatic Elongation of Building Blocks; H-Asn[α-Neu5Ac-(2→6)-β-D-Gal-(1→4)-β-D-GlcNAc-(1→2)-α-D-Man-(1→3)-[α-Neu5Ac-(2→6)-β-D-Gal-(1→4)-β-D-GlcNAc-(1→2)-α-D-Man-(1→6)-]β-D-Man-(1→4)-β-D-GlcNAc-(1→4)-β-D-GlcNAc-(1-*N*⁴)]-OH (162):[392]

H-Asn[β-D-Gal-(1→4)-β-D-GlcNAc-(1→2)-α-D-Man-(1→3)-[β-D-Gal-(1→4)-β-D-GlcNAc-(1→2)-α-D-Man-(1→6)-]β-D-Man-(1→4)-β-D-GlcNAc-(1→4)-β-D-GlcNAc-(1-*N*⁴)]-OH (4 mg, 2.79 μmol) was dissolved in 20 mM sodium cacodylate buffer (pH 7.4, 1400 μL) containing BSA (1.0 mg), NaN_3 (162 μg, 2.5 μmol), $MnCl_2$ (176 μg, 1.4 μmol), UDP-Gal (5.6 mg, 8.4 μmol), alkaline phosphatase (EC 3.1.3.1, 6 U), and GlcNAc-β-1,4-galactosyltransferase (EC 2.4.1.22, 120 mU). The mixture was incubated for 24 h at 37 °C, maintaining the pH at 7.0 by addition of 1 M NaOH. After complete reaction (TLC: iPrOH/1 M NH_4OAc 2:1), CMP-Neu5Ac (4.8 mg, 6.2 μmol) and β-galactoside-α-2,6-sialyltransferase (EC 2.4.99.1, 25 mU) were added. Incubation was continued for 24 h at 37 °C (pH 7.0), followed by another addition of CMP-Neu5Ac (4.8 mg, 6.2 μmol) and β-galactoside-α-2,6-sialyltransferase (25 mU). After 24 h, the precipitate was removed by centrifugation. The supernatant was concentrated to 400 μL and purified by gel filtration (Hi Load Superdex 30, 600×16 mm; mobile phase: 100 mM NH_4HCO_3; flow rate: 750 μL/min; detection: 220 and 260 nm). The positive fractions were collected and lyophilized; yield: 5.6 mg (86%); R_f nonasaccharide (digalactoside) 0.13 (iPrOH/1 M NH_4OAc 2:1); R_f undecasaccharide 0.08 (iPrOH /1 M NH_4OAc 2:1); $[\alpha]_D$ −9.1 (*c* 0.5, H_2O); ^1H NMR (500 MHz, D_2O, δ): 5.19 (d, 1H, $J_{1,2}$ <

1.0 Hz, H1^4), 5.12 (d, 1H, $J_{1,2}$ = 9.7 Hz, H1^1), 5.01 (d, 1H, $J_{1,2}$ < 1.0 Hz, H1$^{4'}$), 4.83 (d, 1H, $J_{1,2}$ < 1.0 Hz, H1^3), 4.69–4.65 (m, 3H, H1^2, H1^5, H1$^{5'}$), 4.505, 4.502 (2 d, 2H, $J_{1,2}$ = 7.8 Hz, H1^6, H1$^{6'}$), 4.31 (dd, 1H, $J_{2,3}$ = 1.9 Hz, H2^3), 4.25 (dd, 1H, $J_{2,3}$ = 1.9 Hz, H2^4), 4.17 (dd, 1H, $J_{2,3}$ = 1.9 Hz, H2^4), 3.00 (dd, 1H, J_{gem} = 17.2 Hz, J_{vic} = 4.2 Hz, Asn βCHa), 2.92 (dd, 1H, J_{vic} = 7.0 Hz, Asn βCHb), 2.76–2.71 (m, 2H, H3eq^N, H3eq$^{N'}$), 2.14, 2.13, 2.09, 2.07, 2.06 (18H, 6×NAc), 1.78 (dd, 2H, J_{gem}, J_{vic} = 12.1 Hz, H3ax^N, H3ax$^{N'}$); ^{13}C NMR (DMSO-d_6 as internal standard, chemical shifts were determined by a HMQC spectrum, 125 MHz, D_2O, δ): 104.8 (C1^6, C1$^{6'}$), 102.6 (C1^2), 101.8 (C1^3), 100.7 (C1^4), 100.6 (C1^5, C1$^{5'}$), 98.1 (C1$^{4'}$), 81.9 (C4^5, C4$^{5'}$), 81.8 (C3^3), 80.9 (C4^2), 80.2 (C4^1), 79.4 (C1^1), 77.8 (C2^4), 77.6 (C2$^{4'}$), 77.5 (C5^1), 75.7 (C5^2, C5^3, C5^5, C5$^{5'}$) 75.0 (C5^6, C5$^{6'}$), 74.9 (C5^4), 74.1 (C5$^{4'}$), 74.0 (C3^1), 73.8 (C3^6, C3$^{6'}$, C3^N, C3$^{N'}$) 73 3 (C3^2, C3^5, C3$^{5'}$), 73.0 (C8^N, C8$^{N'}$), 72.0 (C2^6, C2$^{6'}$), 71.4 (C2^3) 70 7 (C3^4, C3$^{4'}$), 69.7 (C7^N, C7$^{N'}$), 69.6 (C4^6, C4$^{6'}$), 69.4 (C4^N, C4$^{N'}$), 68.6 (C4^4, C4$^{4'}$), 67.1 (C6^3), 67.0 (C4^3), 64.6 C6^6, C6$^{6'}$), 64.0 (C9^N, C9$^{N'}$), 62 9 (C6^4, C6$^{4'}$), 61.5 (C6^5, C6$^{5'}$) 61.2 (C6^2), 61.1 (C6^1), 56.2 (C2^2), 55.9 (C2^5, C2$^{5'}$) 54.9 (C2^1), 53.2 (C5^N, C5$^{N'}$), 52.1 (Asn αC), 39.0 (C3^N, C3$^{N'}$), 36.3 (Asn βC), 23.9, 23.8, 23.7 (NAc); ESI-MS *m/z*: 2337.8 [M+1].

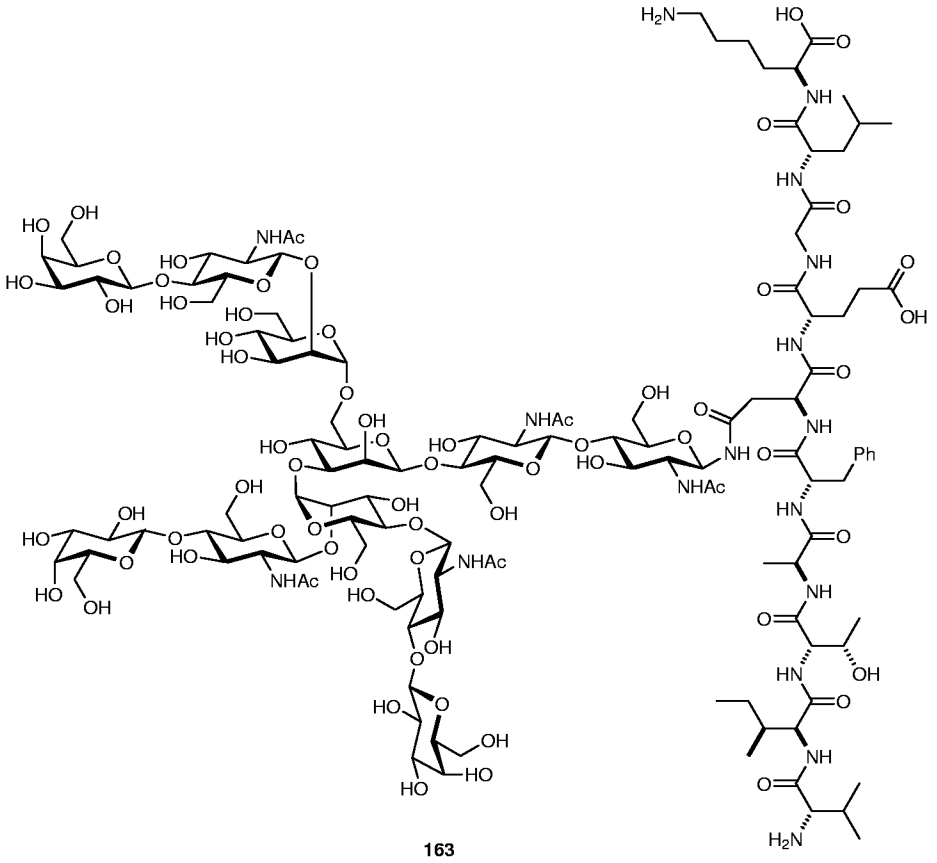
6.3.3.7 Assembly of Complex N-Linked Glycopeptides

The assembly of complex glycopeptides can readily be performed on a solid support. There are two reports[149,418] of solid-phase synthesis of glycopeptide-carrying complex glycans. Guo et al.[418] synthesized Fmoc-Asp-OH glycosylated with the pentasaccharide core structure. This building block (**150**, Scheme 41) was used in solid-phase assembly of the C-terminal sequence of CD52 containing the only glycosylation site of CD52. The peptide synthesis was performed with HOBt/DCC activation and deprotection involved TFA treatment and $Pd(OH)_2$-catalyzed hydrogenation. High yield and purity of the glycopeptide **151** were obtained. In a second report,[149] the synthesis is described of a protected glycosyl amino acid building block using a triantennary undecasaccharide obtained from asialofetuin by hydrazinolysis (Scheme 47). The resulting glycosyl hydrazide was hydrolyzed and the oligosaccharide converted into the glycosylamine. It was coupled with Fmoc-Asp(ODhbt)-OtBu and, after cleavage of the *tert*-butyl group, the purified building block was used in the assembly of a hemoglobin-derived MHC class II E^k binding glycosylated decapeptide, obtained in good yield as a homogeneous material. A PEGA resin with a HMPAA linker and Fmoc-(amino acid)-OPfp esters were used except for the glycosyl amino acid, which was coupled in a mixture of DMSO and DMF using TBTU/DIPEA for the activation.

In conclusion, the synthesis of single glycoform glycopeptides has been achieved by several groups and these elegant techniques may be utilized by the biochemist for the elucidation of the significance of the occurrence of many different glycoforms on the mature glycoproteins. The combined use of chemical and enzymatic synthesis has been the key to much of the success and has allowed synthesis of complex oligosaccharyl asparagine derivatives with the size of nona- to decasaccharides. Solid-phase glycopeptide assembly has been achieved with very good results and little synthetic challenge is left in N-linked glycopeptide synthesis.

for references see p 322

Scheme 47 Glycopeptide Assembly with Triantennary Building Block[149]



163

Solid Phase Synthesis of Di- and Triantennary N-Linked Glycopeptides; H-Val-Ile-Thr-Ala-Phe-Asn{β-D-Gal-(1→4)-β-D-GlcNAc-(1→2)-[β-D-Gal-(1→4)-β-D-GlcNAc-(1→4)]α-D-Man-(1→3)-[β-D-Gal-(1→4)-β-D-GlcNAc-(1→2)-α-D-Man-(1→6)-]β-D-Man-(1→4)-β-D-GlcNAc-(1→4)-β-D-GlcNAc-(1-N^4)}-Glu-Gly-Leu-Lys-OH (163); General Procedure:[149]

Compounds **163** and the diantennary analogue were synthesized on a highly porous and swelling PEGA resin with the acid-labile HMPAA linker. The PEGA resin (500 mg, 0.25 mmol·g^{-1}) was placed in a syringe and allowed to swell in DMF (5mL). After treatment of the resin with piperidine in DMF (20%) for 20 min, the resin was rinsed thoroughly with DMF (10 vol), dried under reduced pressure, and derivatized with the HMPAA linker. HMPAA (45.5 mg, 250 μmol), TBTU (75 mg, 237 μmol), and *N*-ethylmorpholine (63 μL, 500 μmol) were dissolved in DMF (3 mL) and added to the resin after 5 min. After 2 h, the resin was washed with DMF (20 vol) and CH_2Cl_2 (10 vol) and dried under reduced pressure for 24 h. Fmoc-Lys(Boc)-OH (175 mg, 375 μmol), NMI (22.3 μL, 281 μmol), and MSNT (111 mg, 371 μmol) were dissolved in CH_2Cl_2 (6 mL) under an argon atmosphere. After 5 min the soln was added to the PEGA resin and kept for 1 h. The resin was washed with CH_2Cl_2 (10 vol) and DMF (10 vol) and unreacted amino groups of the resin were acetylated with 20% Ac_2O in DMF. The resin was washed with DMF (10 vol) and CH_2Cl_2 (10 vol) and dried under reduced pressure. The level of incorporation of the first amino acid, Lys, was determined by measuring the absorbance of the column eluate after Fmoc cleavage at 290 nm. The subsequent amino acids were coupled as their Pfp esters (3 mol equiv) with HODhbt (1 mol equiv) as auxiliary nucleophile. The resin was thoroughly dried under reduced pressure prior to the coupling of the di- and triantennary asparagine building blocks. The acetylated building blocks Fmoc-Asn($Ac_nGal_2GlcNAc_2Man_3GlcNAc_2$)-OH and Fmoc-Asn($Ac_nGal_3GlcNAc_3Man_3GlcNAc_2$)-OH (**148**) were dissolved in DMF (0.5 and 1 mL, respectively). TBTU (1.5 mol equiv) and DIPEA (1 mol equiv) were added. The mixture was added to the resin and kept for 12 h. After rinsing with DMF (10 vol) the peptide synthesis was continued as described above. Cleavage of the glycopeptide from the resin was achieved with 95% aq TFA for 2 h. It was filtered from the resin, concentrated, coevaporated with toluene, and the residue was triturated with Et_2O. The acetylated glycopeptides were dissolved in dry MeOH (1 mg/mL) and O-deacetylated by addition of catalytic amounts of NaOMe

(1%) in MeOH to pH 9.0. The mixtures were stirred for 12 h at rt, neutralized by addition of solid CO_2, and concentrated. The residue was dissolved in H_2O and purified by semipreparative HPLC to yield VITAFN($Gal_2GlcNAc_2Man_3GlcNAc_2$)EGLK and VITAFN($Gal_3GlcNAc_3Man_3GlcNAc_2$)EGLK (**163**), which were characterized by ES-MS (found 2714.76 and 3080.56, respectively); diantennary structure: ^{1}H NMR (600 MHz, acetic acid-d_4, D_2O, δ): 5.02, 4.64, 4.77; 5.12, 4.94, 4.58, 4.48, 4.47 (H-1 of the sugars), 3.93 V, 4.32 I, 4.39 T, 4.38 A, 4.59 F, 4.72 N, 4.38 E, 3.98 G, 4.41 L, 4.46 K (α-protons); triantennery structure **163**: ^{1}H NMR (600 MHz, acetic acid-d_4, D_2O, δ): 5.04, 4.64, 4.79; 5.12, 4.91, 4.58, 4.47, 4.47, 4.55, 4.47 (H-1 of the sugars), 3.92 V, 4.33 I, 4.36 T, 4.35 A, 4.60 F, 4.68 N, 4.39 E, 3.99 G, 4.42 L, 4.40 K (α-protons). Alternatively, the unprotected oligosaccharide–asparagine building blocks Fmoc-Asn($Gal_2GlcNAc_2$-$Man_3GlcNAc_2$)-OH and Fmoc-Asn($Gal_3GlcNAc_3Man_3GlcNAc_2$)-OH were directly utilized in the multicolumn peptide synthesis. The building blocks were dissolved in DMF/DMSO (1:1) soln (0.5 mL and 1 mL, respectively). DIPEA and TBTU were added and the solns were added to the resin. It was kept for 16 h at rt and rinsed with DMF (10 vol). The peptide synthesis was continued as described above. Cleavage and deprotection of the glycopeptides were achieved with 95% TFA. The crude glycopeptides VITAFN($Gal_2GlcNAc_2Man_3GlcNAc_2$)EGLK and VITAFN($Gal_3GlcNAc_3Man_3GlcNAc_2$)EGLK were dissolved in H_2O and directly purified by preparative HPLC to give the triantennery glycopeptide VITAFN($Gal_3GlcNAc_3Man_3GlcNAc_2$)EGLK (**163**); yield: 35%. The purified glycopeptides were then characterized by MALDI-MS and ^{1}H NMR spectroscopy.

6.3.4 *C*-Glycopeptides

Carbon-linked glycopeptides may be employed as mimics of naturally occurring O- or N-linked glycopeptides.[524] They are attractive synthetic targets owing to their increased stability to chemical and enzymatic cleavage.[431] Replacing the more labile glycosidic C–O bond or C–N bond with the more stable C–C bond results in a *C*-glycopeptide which is not subject to deglycosylation in vivo, and which is perfectly stable to reagents used in solution and solid-phase peptide chemistry. Additionally, *C*-glycosides may populate conformations similar to those of *O*-glycopeptides in solution[432,433] and may show biological activities similar to naturally occurring glycopeptides.[434] However, such compounds are not easily obtained. Direct formation of the C–C bond, through reactions between a sugar and an amino acid, usually gives diastereomeric mixtures.

Naturally occurring *C*-glycopeptides have recently been discovered. A C-linked α-mannose, where the mannopyranosyl C1 atom is directly linked to the C2 atom in a Trp indole ring, has been identified in humans[435–438] and in nonhuman organisms.[525] Stereocontrolled synthesis of α-*C*-mannosyltryptophan has been reported.[439,440,526,527]

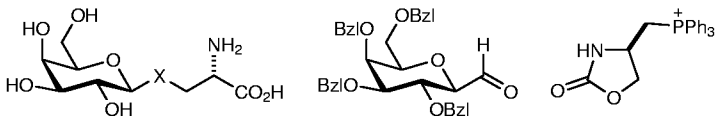
6.3.4.1 C-Analogues of Serine

6.3.4.1.1 Stereoselective Synthesis of the C-Linked Analogue of β-D-Galactosyl Serine

O-Linked galactosyl serine **164** is present in collagens and other associated structural glycopeptides (Scheme 48).[441,442] The stereoselective synthesis of the C-analogue of β-D-galactopyranosyl-L-serine (**165**), wherein the glycosidic oxygen atom has been replaced by a methylene group, has been carried out in two different ways by Wittig reactions, starting from the *C*-glycosyl aldehyde **166**.[128,443,444] In the first case, tetra-*O*-benzyl *N*-α-Fmoc and *N*-α-Boc derivatives of the serine analogue **165** were prepared by Wittig condensation of the α-D-galactosyl aldehyde **166**, with a suitable phosphorane **167** serving as a β-alaninol anion equivalent. Therefore, the β-D-linkage at the anomeric center of the sugar and the *S*-configuration at the carbon bearing the amino group were already in place in the reagents employed. Forming the C-linkage by reaction of the aldehyde **166** with the cyclic carbamate of a (2-amino-3-hydroxy-D-propyl)triphenylphosphonium salt **167** afforded a product (in 34% yield) which was then saturated, converted into the Fmoc amino alcohol, and the alcohol oxidized.

for references see p 322

Scheme 48 Structures Used in the Synthesis of C-Analogues of Serine



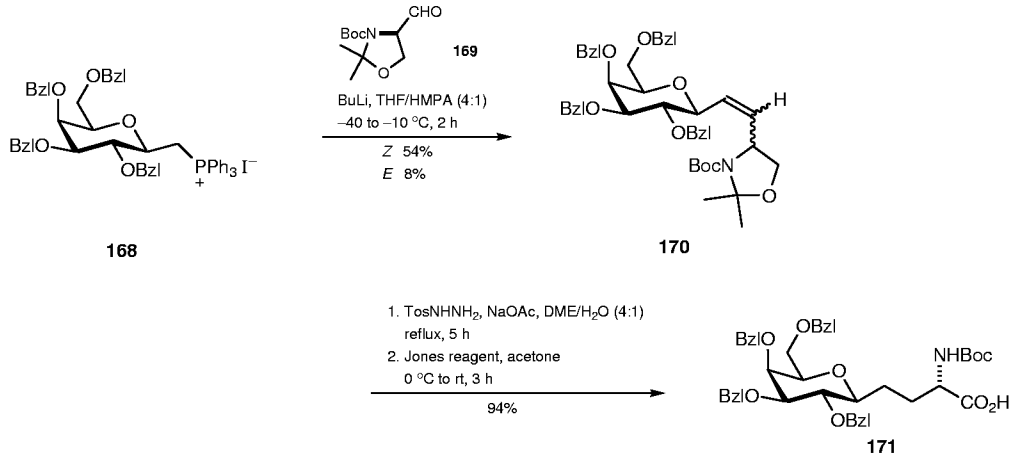
164 X = O
165 X = CH_2

166 **167**

The tetra-*O*-benzylated *N*-Fmoc derivative of **165** was then incorporated by solid-phase synthesis into a helix-forming glycoheptadecapeptide, Ac-Tyr-Lys-Ala-Ala-Ala-Ala-Lys-Ala-Ala-Cgaa(β-D-Gal-*C*-)-Ala-Lys-Ala-Ala-Ala-Ala-Lys-NH_2 (Cgaa = C-glycosylated amino acid).[128]

The same concept as above, used in a reverse manner, improved the synthesis of the serine analogue **171** by condensation of a β-linked D-galactose phosphorane ylide, derived from the phosphonium iodide **168**, with a D-serine-derived aldehyde **169** as the key coupling step as shown in Scheme 49.[444] The sugar phosphonium salt **168** was prepared in gram quantities from the aldehyde **166** and proved to be a nonhygroscopic material, stable on storage for long periods of time. The phosphorane of **168** was generated and reacted with the readily available *N*-Boc-*N*,*O*-isopropylidene-D-serinal **169**, followed by reduction of the *Z*- and *E*-alkenes **170** with diimide. To complete the synthesis, deacetonation and oxidation with Jones reagent afforded, in a single step, the C-analogue of β-D-galactopyranosyl-L-serine (**171**) in an overall yield of 44% from the sugar phosphoniun salt **168**. This procedure led to higher yield in the Wittig condensation and the alkene reduction was simple. It may be extended to other C-linked glycosyl serines starting from suitable sugar phosphoranes.

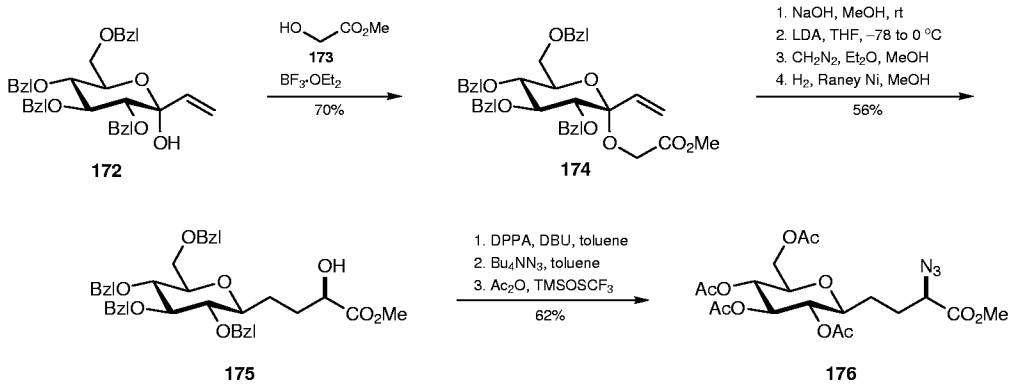
Scheme 49 Synthesis of a C-Linked Analogue of β-D-Galactopyranosyl Serine[444]



168 **169** **170** **171**

A stereoselective synthesis of the C-analogue of β-D-glucopyranosyl serine has been described using a [2,3]-Wittig sigmatropic rearrangement of an appropriate glucose derivative in which the chirality of the sugar induces the stereoselective formation of the amino acid moiety, as shown in Scheme 50.[445]

Scheme 50 Synthesis of a C-Linked Analogue of β-D-Glucopyranosyl Serine[445]

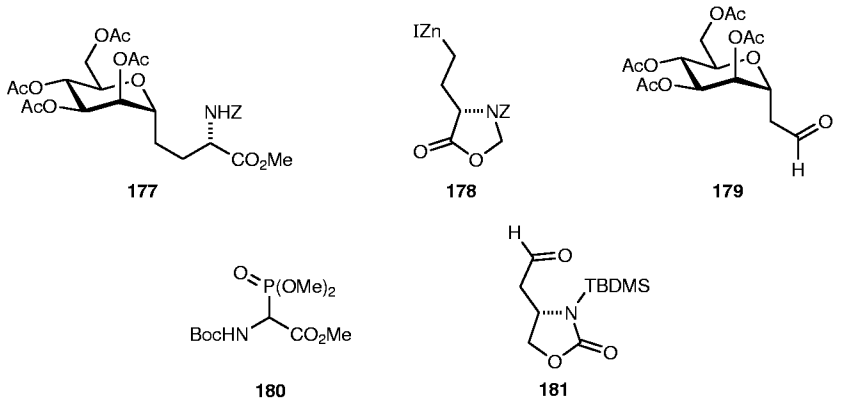


Synthesis of the peracetylated C-linked serine analogue **176** started by oxidation–vinylation of tetrabenzylglucose to compound **172**, followed by glycosylation with methyl glycolate (**173**) to yield the 1-vinyl-α-glucoside **174** (Scheme 50). Stereoselective [2,3]-Wittig sigmatropic rearrangement using lithium diisopropylamide and subsequent reduction of the double bond yielded the 2-hydroxy-β-C-linked serine precursor **175**. The α-hydroxy group was substituted, with inversion of configuration, to yield the azide using DPPA and Bu_4NN_3. Finally, acetolysis of the benzyl groups gave compound **176**. The optical purity of **176** was confirmed by NMR spectroscopy. The unprotected *C*-glycosyl azido acid derivative of **176** was then used as a building block in the solid-phase synthesis of a C-linked glycosyl decapeptide analogue of CBA/J mouse hemoglobin Hb(67–76), VITAFNEGLK, where it was introduced at the position of Asn^{72}.[446] It was incorporated by TBTU coupling and subsequently reduced using DTT in DMF[447] to allow further chain elongation with Fmoc-(amino acid)-OPfp esters. Since NMR experiments revealed that racemization had occurred at the α-carbon of compound **176** during ester hydrolysis, the two stereoisomers of the *C*-glycodecapeptide were separated by RP-HPLC.

The α-mannosyl C-linked α-amino acid derivative **177** has been synthesized from tri-*O*-acetylglucal using the organozinc reagent **178** (Scheme 51).[448] Another route to the corresponding N^{α}-Boc derivative of **177** relies on Horner–Emmons alkenation of the aldehyde **179** with phosphonate **180** followed by a rhodium-catalyzed asymmetric hydrogenation of the resulting C-linked enamide ester.[449] The corresponding α-glucosyl and α-galactosyl derivatives were prepared by the same procedure, and the synthesis proceeds in greater than 45% overall yield from the corresponding peracetylated sugar. Both D and L stereochemistry at the α-amino center can be generated in high diastereomeric excess from the same enamide precursor, depending on the choice of rhodium ligands.[528] A *C*-glycosidic analogue of the T_N-antigen (α-D-GalNAc→*O*-Ser/Thr) has been efficiently synthesized employing a SmI_2-promoted coupling of the anomeric pyridyl sulfone of tri-*O*-benzyl-protected *N*-acetylgalactosamine with the aldehydoamino acid derivative **181**.[450]

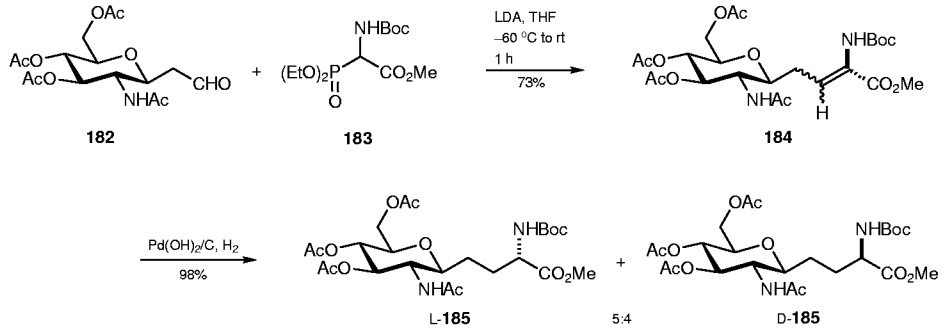
for references see p 322

Scheme 51 Structures Used in the Synthesis of C-Analogues of Serine



In order to investigate the specific functions of *O*-GlcNAc attachment to proteins, the hydrolytically stable C-analogue was synthesized according to Scheme 52.[451] Glucosamine was first transformed into the tetrachlorophthaloyl-protected glycosyl fluoride, which with allyltrimethylsilane and $BF_3 \cdot OEt_2$ gave the β-*C*-allyl derivative. Removal of tetrachlorophthaloyl group, introduction of the *N*-acetyl group, and then ozonolysis afforded the *C*-glycosyl acetaldehyde derivative **182**. A Wittig–Horner reaction of compound **182** with the glycine-derived phosphonate **183** gave a mixture of the α-acrylaminoacrylates **184**. Hydrogenolysis of (*E*/*Z*)-**184** was performed with Pearlman's catalyst [$Pd(OH)_2$/C] in order to obtain both potential diastereomers of **185**. The yield was practically quantitative and after separation of the mixture, performed by MPLC eluting with toluene/acetone (3:2), the L/D ratio was 5:4. None of the common chiral hydrogenation catalysts for the selective synthesis of either L- or D-**185** worked well in this case, presumably due to the acetylamino group present in the sugar residue.

Scheme 52 Synthesis of a C-Linked Analogue of β-D-GalNAc-Serine[451]



(*Z*/*E*)-5,9-Anhydro-6,7,8,10-tetra-*O*-benzyl-2-(*tert*-butoxycarbonylamino)-2,3,4-trideoxy-1,2-*N*,*O*-isopropylidene-D-threo-L-galactodec-3-enitol (170):[444]

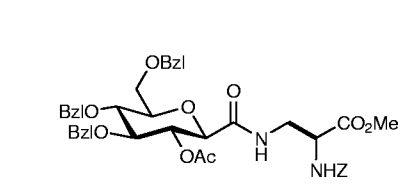
To a stirred, cooled (–40 °C) mixture of phosphonium salt **168** (923 mg, 1.00 mmol), powdered 4-Å molecular sieves (1.00 g), anhyd HMPA (2 mL), and anhyd THF (6 mL), a 2.5 M soln of BuLi in hexane (400 μL, 1.00 mmol) was slowly added. After 5 min, to the resulting red-colored suspension was slowly added a soln of the aldehyde **169** (228 mg, 1.00 mmol) in anhyd THF (2 mL). The mixture was allowed to warm to –10 °C over 2 h, then diluted with Et_2O (100 mL) and filtered through a pad of Celite. The soln was washed with 1 M phosphate buffer at pH 7 (30 mL), dried ($MgSO_4$), and concentrated. The residue was eluted from a column (silica gel, cyclohexane/EtOAc 6:1) to afford first (*Z*)-**170** as a syrup; yield: 405 mg (54%); $[\alpha]_D$ –22.3 (*c* 1, $CHCl_3$); MALDI-TOF MS *m*/*z*: 773.4 $[M+Na]^+$, 789.4 $[M+K]^+$; ^1H NMR ($C_2D_2Cl_4$, 120 °C, δ) selected data: 5.73 (dd, 1H, *J* = 8.2, 11.3 Hz, CH=), 5.63 (dd, 1H, *J* = 5.2, 11.3 Hz, CH=); elemental analysis calcd for $C_{46}H_{55}NO_8$: C, 73.67; H, 7.39; N, 1.87; found: C, 74.01; H, 7.48; N, 1.68. Eluted second was (*E*)-**170**, also a syrup and contaminated by small amounts of uncharacterized by-products; yield: 60 mg (8%); ^1H NMR (DMSO-d_6, 160 °C, δ) selected data: 5.77 (dd, 1H, *J* = 6.0, 16.0 Hz, CH=), 5.70 (dd, 1H, *J* = 5.1, 16.0 Hz, CH=); MALDI-TOF MS *m*/*z*: 773.2 $[M+Na]^+$, 789.2 $[M+K]^+$.

6.3.4.2 C-Analogues of Asparagine

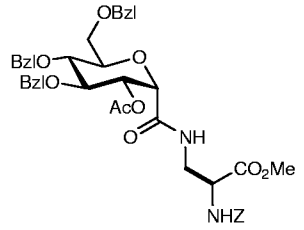
6.3.4.2.1 Anomeric Glycosyl Dianions as Intermediates in the Synthesis of *C*-Glycopeptides

Several *C*-glycosidic analogues of N^{β}-glycosylated asparagine have been synthesized to improve the metabolic stability (Scheme 53). Kessler et al. have introduced a method where an anomeric glycosyl dianion (e.g., **195**, **198**, or **201**, derived from the sugar derivatives **194**, **197**, or **200**, respectively, Scheme 54) is used as a key intermediate in the synthesis of several C-analogues of asparagine, e.g. the retroamides **186**, **187**, and **188** and the C-analogue **190**.[452–454] The two retroamides of *N*-glucoasparagine, **186** and **187**, were prepared by stereoselective synthesis from the corresponding sugar acids **196** and **199**, derived from D-glucose, and an amino acid derivative **191** containing an amino group in the β-position.[452]

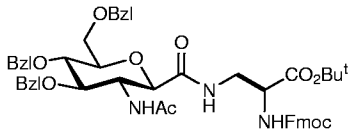
Scheme 53 Structures Used in the Synthesis of C-Analogues of Asparagine



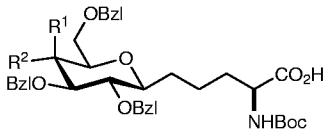
186



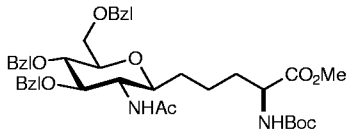
187



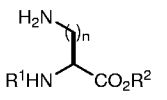
188



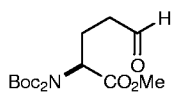
189a R^1 = OBzl; R^2 = H
b R^1 = H; R^2 = OBzl



190

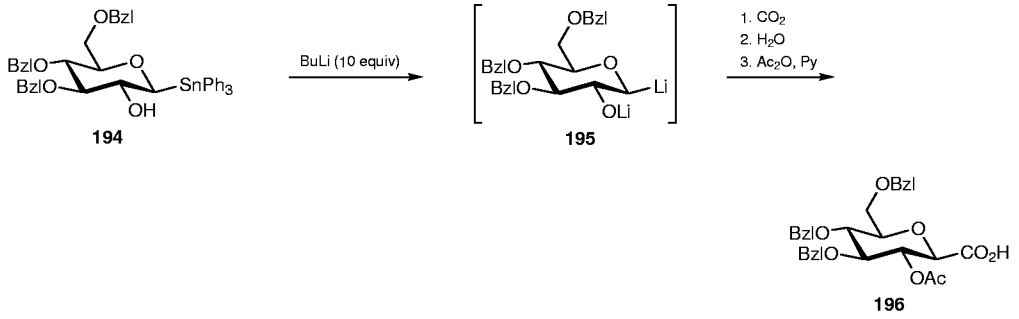


191 R^1 = Z; R^2 = Me; n = 1
192 R^1 = Fmoc; R^2 = tBu; n = 1



193

Scheme 54 Formation of Glycosyl Dianions Used in C–C Bond Formation

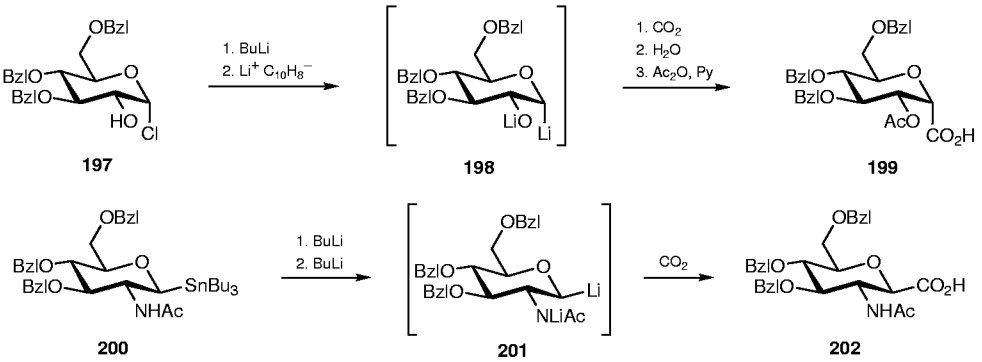


194

195

196

for references see p 322



197 198 199

200 201 202

The couplings were performed with TBTU and HOBt in the presence of 2 equiv DIPEA in NMP as solvent, in high yields. The α- and β-configurations at the anomeric centers were deduced from the NMR spectra, which were found to be 3.2 Hz for compound **187** and 9.2 Hz for compound **186**. The *N*-acetylglucosmine retroamide **188** was synthesized according to the same procedure from tin derivative **200**, which was deprotonated at −78 °C and transmetallated at −65 °C by butyllithium to give the intermediate **201**. Experiments at lower temperatures than −65 °C for the second step showed slow transmetalation. Subsequent addition of carbon dioxide provided compound **202** and coupling with the amine **192** using TBTU/HOBt as coupling reagents afforded **188** in high yield.[453] To demonstrate that the glycosylated amino acid can be used in standard solid-phase peptide synthesis (after deesterification with TFA), the peptide *c*[-D-Pro-Phe-Ala-Cgaa-Phe-Phe-] was synthesized using TBTU/HOBt for activation. Amines with different chain lengths (**191**, n = 2, 3, 4) were also reacted with compound **202**, giving retroamides with different linker lengths.

Mukaiyama-type coupling of benzylated galactosyl and glucosyl aldehydes with a four-carbon silyl enol ether carrying the oxazolidine ring affords the corresponding sugar-containing aldols.[455] Deoxygenation and oxidative cleavage of the heterocyclic ring leads to the two β-D-Gal- (R^1 = OBzl; R^2 = H) and β-D-Glc- (R^1 = H; R^2 = OBzl) asaparagine ethylene isosters **189a** and **189b**.

A set of 12 *C*-glycosyl asparagines has been synthesized from lithium *C*-glycoside acetylides and the Garner aldehyde **169**.[529]

Formation of a C–C bond from a glycosyl dianion with an aldehyde has also been demonstrated in the synthesis of the *C*-glycosidic N^{β}-glucosylasparagine analogue **190**.[454] The aldehyde derivative **193** was added to the dianion **201**. After one Boc-protecting group had been removed with $Mg(ClO_4)_2$, the hydroxy group was transformed with 1,1′-thiocarbonyldiimidazole and DMAP to give the corresponding thiocarbonyldiimidazole. Subsequent radical reduction with Bu_3SnH and AIBN gave the desired C-glycosylated amino acid **190** in 49% overall yield (from tin derivative **200**), with pure β-configuration at C1.

A chemoenzymatic approach to the synthesis of a α-GlcNAc-CH_2-Asn containing glycopeptide of the high-mannose-type has been described.[201] Chemical synthesis of a *C*-glycopentapeptide, H-Tyr-Ile-Asn(N^4-β-GlcNAc-CH_2)-Ala-Ser-NH_2, was followed by enzymatic transfer of a $Man_9GlcNAc$ residue to the 4-hydroxy of the terminal GlcNAc residue by *endo*-β-*N*-acetyl-glucosaminidase (Endo-A). The chemical synthesis of the glycosylated building block started by hydrogenation of 3,4,6-tri-*O*-acetyl-2-deoxy-2-phthalimido-β-D-glucopyranosyl cyanide and subsequent *N*-Boc protection of the resulting glycosylmethylamine. Treatment with hydrazine hydrate and subsequent N-acetylation was followed by a condensation reaction with Boc-Asp-OBzl to obtain fully protected *N*-(2-acetamido-2-deoxy-β-D-glucopyranosylmethyl)asparagine. De-O-benzylation followed by coupling with H-Ala-Ser-OMe in solution gave the tripeptide. This peptide was first elongated from the N-terminal to the pentapeptide derivative and then deprotected before the enzymatic transfer.

3-Acetamido-2,6-anhydro-4,5,7-tri-*O*-benzyl-3-deoxy-β-D-glycero-D-guloheptonic Acid (202); General Procedure:[453]

Under argon at −78 °C, a 1.6 M soln of BuLi in hexanes (1 equiv) was added within 10 min to a soln of the tin derivative **200** in THF (~0.1 mol/L). The mixture was warmed to −65 °C and more BuLi (1.3 equiv) was added within 5 min (→ deep red). The subsequent addition of CO_2 (or other electrophiles) changed the color of the soln to pale yellow. After 1 h, the reaction was stopped with sat. NH_4Cl or $KHSO_4$ soln and the mixture was warmed in an ultrasonic bath to 20 °C. The mixture was diluted with H_2O, extracted twice with EtOAc, dried ($MgSO_4$), and concentrated. Purification was achieved by flash chromatography (hexane/EtOAc 1:0, 1:1, 1:4; EtOAc/MeOH/AcOH 45:5:3, 2:2:1) to furnish compound **202** as a pale yellow solid; yield: 4.60 g (83%); TLC ($CHCl_3$/MeOH 3:1 + 0.1% TFA): R_f 0.34; $[\alpha]_D^{20}$ +29.4 (*c* 1.1, THF); ^1H NMR (500 MHz, DMSO-d_6, δ): 8.05 (d, 1H, *J* = 9.2 Hz, NH) 7.35–7.13 (m, 15 arom H), 4.70 (m, 1H, PhCH_2), 4.67 (s, 2H, PhCH_2), 4.54 (d, 1H, *J* = 12.1 Hz, PhCH_2), 4.50, (d, 1H, *J* = 10.5, PhCH_2), 4.47 (d, 1H, *J* = 12.1 Hz, PHCH_2), 3.94 (m, 1H, H4), 3.79 (d, 1H, *J* = 10.4 Hz, H3), 3.70–3.56 (m, 3H), 3.53–3.37 (m, 2H), 1.76 (s, Me); FAB-MS *m/z*: 520 $[M+H]^+$.

***tert*-Butyl 3-[(3-Acetamido-2,6-anhydro-4,5,7-tri-*O*-benzyl-3-deoxy-β-D-glycero-D-guloheptanoyl)amino]-2-[(9*H*-fluorenyl-9-yloxycarbonyl)amino]propanoate (188); General Procedure:**[453]

TBTU (1.2 equiv) and HOBt (1.5 equiv) were added to a soln of compound **202** (1 equiv) in NMP at rt. The pH of the soln was adjusted to 6–7 by DIPEA. The diamino acid **191** or **192** (n = 1–4, ca. 1.5 equiv) was added, and the pH was maintained at 6–7 by addition of DIPEA. After 1 h, the mixture was quenched with H_2O, concentrated, and partitioned between H_2O and EtOAc. The organic phase was washed ($NaHCO_3$, $KHSO_4$, and brine), dried (Na_2SO_4), and concentrated. The residue was purified by flash chromatography (acetone/hexane). According to this general procedure, using the diamino acid **192** (n = 1) the desired product **188** was obtained as a white solid; yield: 795 mg (90%); TLC (hexane/acetone 1:1): R_f 0.64; $[\alpha]_D^{20}$ −16.3 (*c* 0.8, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$, δ): 7.73 (d, 2H, *J* = 7.8 Hz, 2 arom H), 7.69 (d, 1H, *J* = 7.8 Hz, 1 arom H), 7.67 (d, 1H, *J* = 7.3 Hz, 1 arom H), 7.36–7.15 (m, 19 arom H), 6.74 (d, 1H, *J* = 8.4 Hz, NH Cα), 6.70 (dd, 1H, *J* = 5.5, 7.1 Hz, NH Cβ), 5.40 (d, 1H, *J* = 8.8 Hz, NH C3), 4.83 (d, 1H, *J* = 11.7 Hz, PhCH_2), 4.80 (d, 1H, *J* = 10.7 Hz, PhCH_2), 4.66 (d, 1H, *J* = 11.7 Hz, PhCH_2), 4.54 (d, 1H, *J* = 10.8 Hz, PhCH_2), 4.52 (d, 1H, *J* = 12.0 Hz, PhCH_2), 4.47 (d, 1H, *J* = 12.0 Hz, PhCH_2), 4.41 (m, 1H, Hα), 4.33 (ddd, 1H, *J* = 7.9, 7.8, 10.4 Hz, Fmoc CH_2), 4.28 (ddd, 1H, *J* = 7.4, 7.5, 10.4 Hz, Fmoc CH_2), 4.21 (t, 1H, *J* = 7.6 Hz, Fmoc CH), 3.91 (m, H3), 3.88 (m, Hβ), 3.66 (m, H2), 3.65 (m, H4, H6), 3.63 (m, H5), 3.43 (m, Hβ), 1.81 (s, Me), 1.46 (s, tBu); FAB-MS *m/z*: 884 $[M+H]^+$.

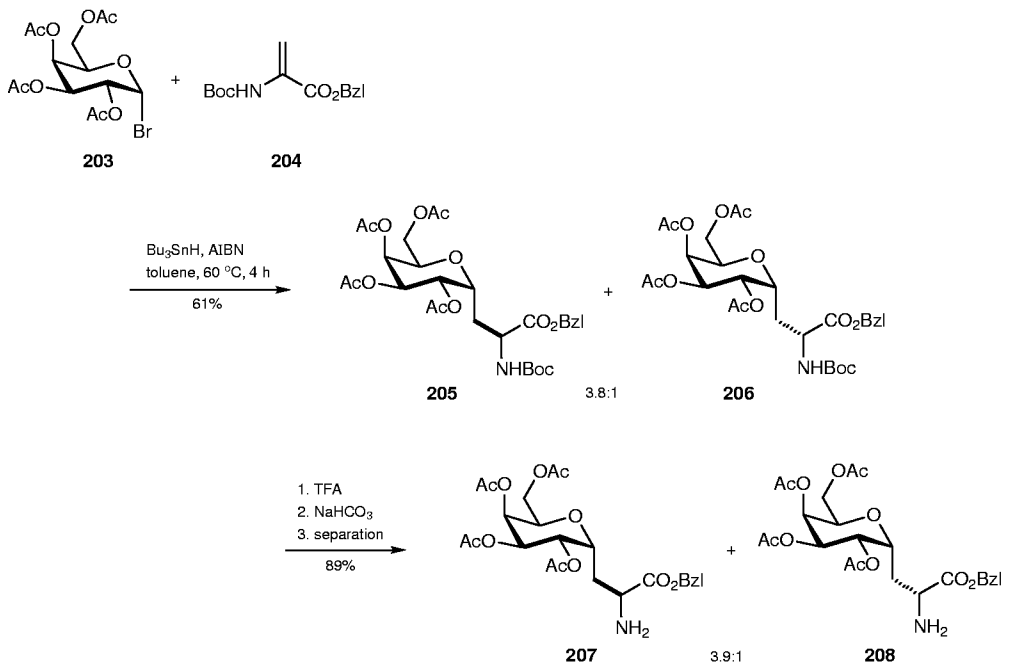
6.3.4.3 Other Analogues of Glycosyl Amino Acids

6.3.4.3.1 Synthesis of *C*-Glycopeptides by Radical Addition of Glycosyl Bromide to a Dehydroalanine Derivative

Glycosyl moieties linked to the side chain of the amino acid through one carbon bond have been generated by reaction of a glycosyl radical with dehydroalanine, leading to diastereoisomers which are then separated.[456,457] D-Alanine carrying a galactosyl moiety bound as *C*-glycoside has been successfully synthesized by this method and used in the synthesis of a C-glycosylated nonapeptide analogue of the LHRH agonist busserelin.[457] A mixture of the epimeric *C*-galactosyl alanine derivatives **205** and **206** was obtained by adding peracetylated galactopyranosyl bromide to Boc-ΔAla-OBzl in the presence of AIBN (Scheme 55). Gram-scale separation was easily carried out by silica gel column chromatography after removal of the Boc group with TFA and neutralization with $NaHCO_3$ (Scheme 55).

for references see p 322

Scheme 55 Synthesis of an Epimeric Mixture of *C*-Galactosyl Alanine Derivatives[457]

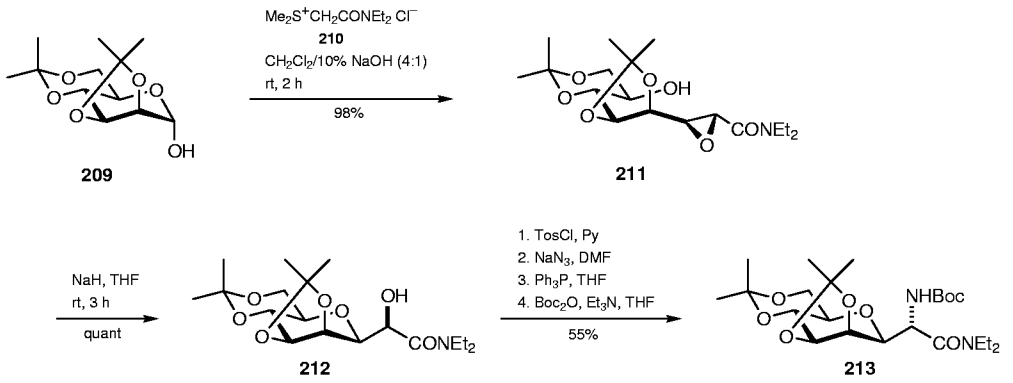


The major epimer **207** was converted into a suitable building block and applied to Fmoc solid-phase synthesis of *C*-glycopeptides.[458] The minor isomer **208** was coupled with the tripeptide Fmoc-Trp-Ser-Tyr-OH under activation with EDC and HODhbt, and further used in the synthesis of the nonapeptide pGlu-His-Trp-Ser-Tyr-D-Ala[C^{β}-(α-D-GalAc$_4$)]-Leu-Arg-Pro-NHEt, by segment coupling in solution.[457] Diastereoselective synthesis of β-(3,4,6-tri-*O*-benzyl-2-deoxy-β-D-galactopyranosyl)-*N*-*tert*-butyloxycarbonyl-D-alanine has been performed, starting from methyl 3,4,6-tri-*O*-benzyl-2-deoxy-β-D-galactopyranoside and triphenylphosphonium tetrafluoroborate. Wittig reaction of the α/β mixture of the phosphonium salt and the Garner aldehyde **169** (see Scheme 49) to the two exocyclic enol ethers was followed by diastereoselective hydrogenation and oxidation; the overall yield was 46%.[459]

α-Carbon-linked glycosylglycins are also obtained by an efficient stereoselective two-chiral carbon integration of pyranoses to acyclic epoxy amides, as shown in Scheme 56.[460] The mannopyranose derivative **209** was reacted with the sulfonium salt **210** in CH_2Cl_2 and 10% sodium hydroxide solution to obtain almost quantitatively, and with complete stereoselectivity, the epoxy amide **211**. Treatment of this epoxy amide with NaH in anhydrous THF at room temperature led to *C*-mannopyranose **212** in virtually quantitative yield and with complete stereoselectivity. Compound **212** was successfully converted by conventional reactions into the Boc derivative **213**, which was readily purified.

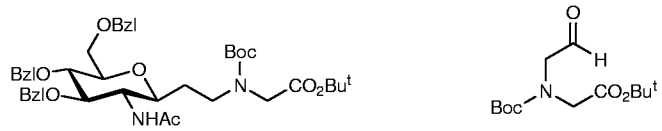
Synthesis of a range of glycosylacetylene–phenylalanine amino acid building blocks has been reported[530] as well as Fmoc-protected *C*-glycosyltyrosines.[531]

Scheme 56 Synthesis of a C-Linked Glycosylglycin[460]



A C-glycosylated peptoid analogue **214** of the naturally occurring Ser(*O*-GlcNAc) has been prepared by coupling of the peptoid aldehyde **215** (see Scheme 57) with the glycosyl dianion of glucosamine **201** (see Scheme 54). The dianion was generated from the tin compound **200** which was first deprotonated with MeLi and then transmetalated with BuLi.[461]

Scheme 57 Structures Used in the Synthesis of a C-Glycosylated Peptoid Analogue[461]



Boc-L-Ala[C$^{\beta}$-(α-D-GalAc$_4$)]-OBzl (205) and Boc-D-Ala[C$^{\beta}$-(α-D-GalAc$_4$)]-OBzl (206):[457]
Boc-ΔAla-OBzl (**204**; 10.00 g, 36.06 mmol) and 2,3,4,6-tetra-*O*-acetyl-α-D-galactopyranosyl bromide (**203**; 29.66 g, 72.12 mmol) were twice coevaporated with dry toluene, dissolved in dry toluene (360 mL), and heated to 60 °C. Under argon, Bu_3SnH (29.1 mL, 108.2 mmol) and AIBN (3.00 g, 18.03 mmol) were added. After stirring the mixture for 4 h at 60 °C, the solvent was evaporated, the residue dissolved in MeCN and extracted (5×) with hexane to remove the tin compounds. Excess of glycosyl bromide was hydrolyzed by adding ca. 25% (v/v) of H_2O and heating the homogeneous soln to 60 °C for 30 min. The solvent was removed and the remaining oil purified by flash chromatography (toluene/EtOAc 2.2:1) to give a mixture of the epimers **205** and **206** in a ratio of 3.8:1; yield: 13.40 g (61%).

H-L-Ala[C$^{\beta}$-(α-D-GalAc$_4$)]-OBzl (207) and H-D-Ala[C$^{\beta}$-(α-D-GalAc$_4$)]-OBzl (208):[457]
The mixture of the epimers **205** and **206** (5.11 g, 8.38 mmol) was dissolved in dry $CHCl_3$ (13 mL) and treated with TFA (12.6 mL). After stirring for 30 min, the mixture was diluted with $CHCl_3$ (150 mL), neutralized by washing with sat. $NaHCO_3$ soln, and dried (Na_2SO_4). Removal of the solvent gave 4.26 g (quant.) of a mixture of the amines **207** and **208**. Separation by flash chromatography with $CHCl_3$/iPrOH (95:5) gave 0.79 g (18%, syrup) of the minor isomer **208** (D, less polar) and 3.01 g (71%, syrup) of the major isomer **207** (L, more polar). Compound **207**: R_f 0.39 ($CHCl_3$/iPrOH 9:1); ^1H NMR (250 MHz, $CDCl_3$, 300 K, δ): 7.46–7.26 (m, 5H, aromatic), 5.39 (dd, 1H, Gal H4), 5.21 (m, 1H, Gal H2), 5.18 (m, 1H, Gal H3), 5.15 (s, 2H, benzyl CH_2), 4.42 (ddd, 1H, Gal H1), 4.32 (dd, 1H, Gal H6), 4.12 (ddd, 1H, Gal H5), 4.07 (dd, 1H, Gal H6′), 3.64 (dd, 1H, Hα), 2.06, 1.93 (2 ddd, 2H, Hβ, Hβ′), 2.09–2.02 (4s, 12H, 4×OAc). Compound **208**: R_f 0.46 ($CHCl_3$/iPrOH 9:1); ^1H NMR (500 MHz, $CDCl_3$, 300 K, δ): 7.40–7.30 (m, 5H, aromatic), 5.41 (dd, 1H, Gal H4), 5.27 (m, 1H, Gal H2), 5.22–5.12 (m, 2H, benzyl CH_2) 5.16 (m, 1H, Gal H3), 4.50 (ddd, 1H, Gal H1), 4.23 (dd, 1H, Gal H6), 4.12 (ddd, 1H, Gal H5), 4.10 (dd, 1H, Gal H6′), 3.60 (dd, 1H, Hα), 2.18, 1.58 (2 ddd, 2H, Hβ, Hβ′), 2.11–2.00 (4s, 12H, 4×OAc).

for references see p 322

6.3.5 *S*-Glycopeptides

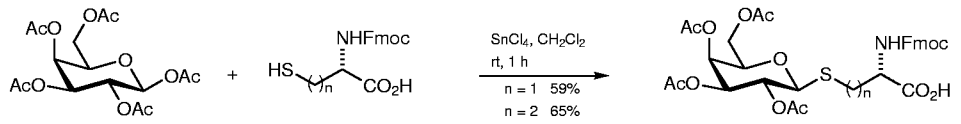
Replacement of the anomeric oxygen of *O*-glycopeptides by sulfur, thus leading to *S*-glycopeptides, increases the stability of the peptide–carbohydrate linkage against chemical degradation as well as against enzymatic cleavage.[462] Although *O*- and *S*-glycosides are isosteric, differences in bond lengths as well as in bond polarity have to be considered when structures of corresponding compounds are compared. The high stability of thioglycosides toward glycohydrolases may have a great pharmacological potential for biologically active *S*-glycopeptides. They may also play an important role in the synthesis of peptidomimetics with modified activity, stability, and bioavailability. S-Linked glycoproteins appear to be very rare in nature but a few examples have been reported.[292,387]

Peptides containing cysteine glycosylated with β-D-galactose have been synthesized by solution methodology.[463,464] For the preparation of Boc-protected glycosylated cysteine with a free carboxy group, *N,N'*-bis-Boc-cystine was first reduced with sodium in liquid ammonia, giving the S,O-dianion. Removal of the ammonia, dissolution in dry THF, and treatment with acetobromogalactose afforded the β-thiogalactoside Boc-Cys(tetra-*O*-acetyl-β-D-galactopyranosyl)-OH in 75% yield.[463] Synthesis of the S-glycosylated cyclic hexapeptide *c*[-D-Pro-Phe-Cys(tetra-*O*-acetyl-β-D-galactopyranosyl)-Trp-Lys(Z)-Phe-] was performed in solution following the strategy of fragment coupling using the galactosylated cysteine building block described above. EDCI in combination with HOBt was employed as coupling reagent.[464] A nonapeptide containing D-cysteine glycosylated with α-L-rhamnose, pGlu-His-Trp-Ser-Tyr-D-Cys[*S*-(α-L-Rha)]-Leu-Arg-Pro-NHEt·TsOH, has also been synthesized via segment condensation in solution, starting with the coupling reaction of the glycosylated cysteine and a tripeptide.[457] The thioglycosylation was achieved in a yield of 50% by reaction of 2,3,4-tri-*O*-acetyl-α-L-rhamnosyl trichloroacetimidate with Z-D-Cys-OAl in the presence of $HgBr_2$ as catalyst. Synthesis of the S-β-D-glucosylated cysteine derivative has been performed by alkylation of 2,3,4,6-tetra-*O*-acetyl-1-thio-β-D-glucopyranose with a serine-derived β-iodoamino acid.[465] Serine- and threonine-derived cyclic sulfamidates have been used in the synthesis of S-linked glycosyl amino acids.[261]

6.3.5.1 S-Linked Glycosylated Fmoc Amino Acids with the 1,2-*trans* Anomeric Configuration

The S-β-D-galactosylated cysteine and homocysteine building blocks **218** (n = 1 and 2, respectively) were prepared by a simple Lewis acid promoted glycosylation of Fmoc-cysteine **217** (n = 1) and Fmoc-homocysteine **217** (n = 2) with β-D-galctose pentaacetate (**216**), as shown in Scheme 58.[466] Use of $SnCl_4$ as promotor in CH_2Cl_2 was found to give significantly better yields (20–25% improved) than when $SnCl_4$ in MeCN or $BF_3 \cdot OEt_2$ in CH_2Cl_2 were employed.[165]

Scheme 58 Synthesis of S-β-D-Galactosylated Cysteine and Homocysteine Building Blocks[165]



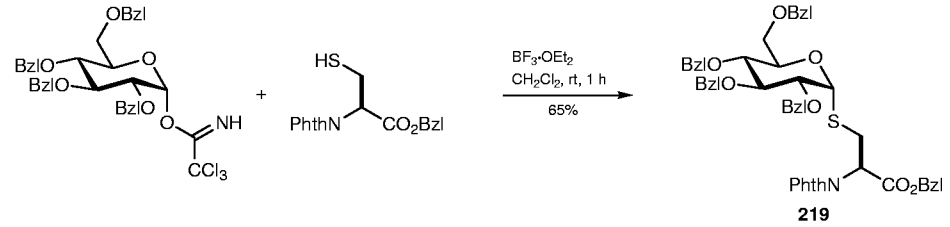
216 **217** **218**

N^{α}-(9-Fluorenylmethoxycarbonyl)-3-*S*-(2,3,4,6-tetra-*O*-acetyl-β-D-galactopyranosyl)cysteine (218, **n = 1**):[165] $SnCl_4$ (20 μL, 170 μmol) was added to β-D-galactose pentaacetate (**216**; 50 mg, 128 μmol) and Fmoc-Cys-OH (52.8 mg, 154 μmol) in dry CH_2Cl_2 (2.0 mL) under N_2 at rt. After 1 h, the mixture was diluted with CH_2Cl_2/H_2O (1 mL), dried (Na_2SO_4), and concentrated. Purification of the crude product by HPLC (0.1% aq TFA/MeCN 1:1) gave the product; yield: 48 mg (59%); $[\alpha]_D^{20}$ +1.7 (*c* 4.0, $CHCl_3$); ^1H NMR (500 MHz, acetone-d_6, δ): 6.79 (d, 1H, *J* = 8.0 Hz, NH), 5.43 (d, 1H, *J* = 2.3 Hz, H4), 5.23 (t, 1H, *J* = 9.1 Hz, H2), 5.19 (dd, 1H, *J* = 3.2 Hz, H3), 4.90 (d, 1H, *J* = 9.3 Hz, H1), 4.54 (m, 1H, Hα), 4.40 (dd, 1H, *J* = 7.2, 10.3 Hz, Fmoc OCH_2), 4.34 (dd, 1H, *J* = 7.2 Hz, Fmoc OCH_2), 4.29 (t, 1H, *J* = 7.0 Hz, Fmoc CHAr), 4.21 (t, 1H, *J* = 6.4 Hz, H5), 4.13 (d, 1H, *J* = 8.0 Hz, H6, H6'), 3.43 (dd, 1H, *J* = 3.7, 14.2 Hz, Hβ), 3.01 (dd, 1H, *J* = 8.8, 14.1 Hz, Hβ'), 1.93, 1.95, 2.00, 2.10 (4s, 4×Ac); HR-MS *m/z*: 674.1907 $[M+H]^+$.

6.3.5.2 Synthesis of *S*-α-D-Glucosyl Cysteine (1,2-*cis*)

While S-β-D-glucosylated cysteine is easily obtained by using neighboring group participation,[164,465] the isomeric *S*-α-D-glucosyl amino acid **219** requires another procedure.[467] The glycosylation of highly nucleophilic *N*-phthaloylcysteine benzyl ester (or allyl ester) with 2,3,4,6-tetra-*O*-benzyl-α-D-glucopyranosyl trichloroacetimidate in the presence of 1 equivalent of $BF_3 \cdot OEt_2$ provided the protected cysteinyl-*S*-α-D-glycosides in high yield, as shown in Scheme 59.

Scheme 59 Synthesis of a S-α-D-Glucosylated Cysteine Building Block[467]



***N*$^{\alpha}$-(Phthaloyl)-3-*S*-(2,3,4,6-tetra-*O*-benzyl-α-D-glucopyranosyl)-L-cysteine Benzyl Ester (219):**[467]
A soln of Phth-Cys-OBzl (3.3 g, 9.7 mmol) and 2,3,4,6-tetra-*O*-benzyl-α-D-glucopyranosyl trichloroacetimidate (6.7 g, 9.8 mmol) in dry CH_2Cl_2 was stirred for 30 min in the presence of 4-Å molecular sieves under argon. The glycosylation was then catalyzed by injection of $BF_3 \cdot OEt_2$ (1.2 mL, 9.7 mmol) at rt. After stirring for 1 h, the mixture was neutralized with TEA (1 equiv), filtered, and washed (H_2O). The organic layer was dried ($MgSO_4$) and the solvent was removed. Purification by chromatography (silica gel, hexane/EtOAc 2:1) gave compound **219** as a colorless oil; yield: 7.2 g (86%). A sample of the product (13.4 mg) was purified by HPLC (hexane/EtOAc 7:1), which afforded pure **219**; yield: 8.7 mg (65%); TLC R_f 0.3 (hexane/EtOAc 3:1); $[\alpha]_D^{20}$ +25.5 (*c* 0.8, $CHCl_3$); ^{1}H NMR (500 MHz, DMSO-d_6, δ): 7.97–7.88 (m, 4H, phthaloyl), 7.31–7.13 (m, 25H, 5×Ph), 5.66 (d, 1H, $J_{1,2}$ = 5.4 Hz, H1), 5.18 (dd, 1H, $J_{\alpha,\beta}$ = 4.7 Hz, $J_{\alpha,\beta'}$ = 10.9 Hz, Hα), 4.80–4.40 (m, 10H, 5×OCH_2Ph), 4.03 (dt, 1H, *J* = 2.7, 10 Hz, H5), 3.70 (dd, 1H, $J_{2,3}$ = 9.4 Hz, H2), 3.65 (br s, 2H, H6, H6′), 3.52 (t, 1H, *J* = 9.4 Hz, H3), 3.47–3.36 (m, 3H, H4, Hβ, Hβ′); FAB-MS *m/z*: 886 $[M+Na]^+$.

6.3.6 Neoglycopeptides

A neoglycopeptide is a peptide that has been modified with carbohydrate derivatives in a nonnatural manner. A number of glycosylated peptides have been synthesized as models of natural glycopeptides or to modify their properties or activity; also, a large variety of multivalent neoglycopeptides have been designed, spanning from low-valency clusters to multiantennary glycopeptides.[468–471] Access to "glycoclusters" of various shapes, conformations, and valencies is important for understanding the interactions of multivalent carbohydrate binding proteins. This section will illustrate a few of these neoglycopeptide applications.

O-Glycosylation of peptides with monosaccharides (e.g., Glc, Gal, Rha, or Xyl) by the methods described previously (see Section 6.3.2) has been applied to modify the activity of hormones.[67,73,472] Selective acylation of the primary hydroxy group with a peptide pentachlorophenyl ester has also been used to modify hormonal activity.[80,473] The effect of glycosylation on the interactions between peptides and T-cell receptors as well as MHC molecules has been investigated with synthetic glycopeptides. An otherwise nonimmunogenic branching point pentasaccharide from amylopectin and starch, α-D-Glc-(1→4)-α-D-Glc-(1→6)-[α-D-Glc-(1→4)-]α-D-Glc-(1→4)-D-Glc, was transformed into its peracetylated glycosyl bromide, used for glycosylation of Fmoc-Ser-OPfp, and the product was incorporated by solid-phase glycopeptide synthesis into the middle and N-terminus of the hen egg lysozyme (HEL) fragment 81–96, known to be a T-cell determinant.[474] The influence on the MHC class II binding, as well as the T-cell stimulation, was highly dependent on the position of the glycosylation.

for references see p 322

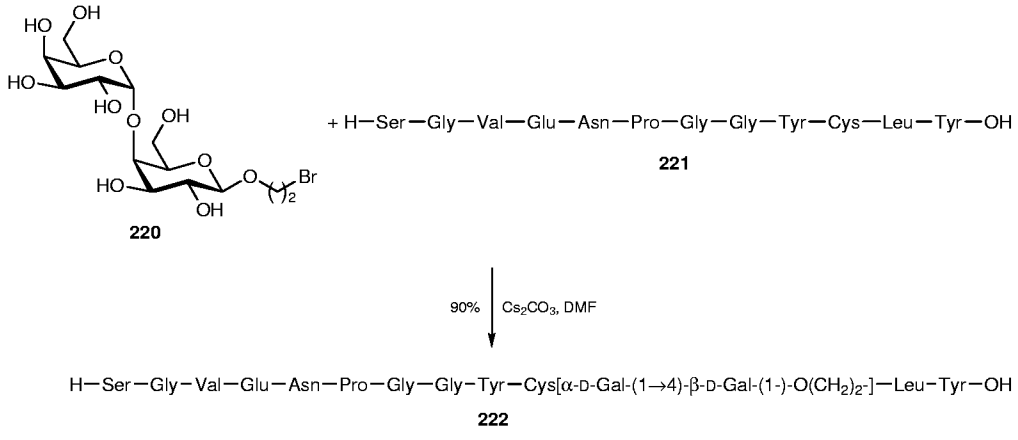
In N-linked neoglycopeptides the carbohydrate may be linked to the peptide through an amide bond or through an amine linkage formed by reductive amination or by Amadori rearrangement (see also Section 6.3.3). The amide bond can be formed by condensation of the C-terminal end of a biologically active peptide with an amino group in the sugar. Thus glucose substituted with an amino group in the 1-, 2-, or 6-position[475] and 1-amino-1-deoxy-β-D-galactose have been conjugated with opioid peptides.[472,476] Complex-type oligosaccharides were similarly linked to small labeled peptides, e.g. with biotin.[477,478] The α-carboxy group of aspartic acid has been linked to Ac_3-GlcNAc-NH_2 and the β-carboxy group was then used for peptide coupling to obtain modified substrates for oligosaccharyl transferases isolated from yeast.[71] The glycosylation of asparagine in OVA(323–339)[389] as well as in HEL(81–96)[474] influenced the binding to MHC class II molecules and resulted in loss of activity. Glycosylation of asparagine was found to influence the secondary structure of α-helical T-cell determinants,[479] as determined by CD spectroscopy in TFE/water mixtures. The N-glycosylation of an HIV protective peptide T analogue did not change the conformational properties significantly under these conditions,[383] although some stabilization of the dominant reverse turn structure has been observed in neat TFE.[480]

O-Glycopeptides and glycoconjugates, in which the saccharide is linked to a peptide or a protein through a linear or a branched aliphatic spacer, have been used to enhance biological activity.[298,301,373,472,481–483] Peracetylated galabiose and lactose react with 3-sulfanylpropanoic acid in the presence of BF_3 and the products have been used in the synthesis of HEL(46–61) and shorter fragments thereof.[164]

6.3.6.1 Convergent Synthesis of Neoglycopeptides by Coupling of 2-Bromoethyl Glycosides to Cysteine (or Homocysteine)

A straightforward convergent strategy to neoglycopeptides that allows different carbohydrates to be linked to specific positions in preformed, unprotected peptides has been reported (Scheme 60).[484] The nucleophilic properties of the thiol group in cysteine or homocysteine were utilized to create S-linked neoglycopeptides of T-cell stimulating peptides by substitution of the bromine atom in 2-bromoethyl β-glycosides of the disaccharide galabiose [α-D-Gal-(1→4)-D-Gal], the trisaccharide globotriose [α-D-Gal-(1→4)-β-D-Gal-(1→4)-D-Glc], and 3′-sialyllactose [α-D-Neu-5-Ac-(2→3)-β-D-Gal-(1→4)-D-Glc].

Scheme 60 Convergent Synthesis of a Neoglycopeptide by Coupling of a 2-Bromoethyl Glycoside to Cysteine[484]



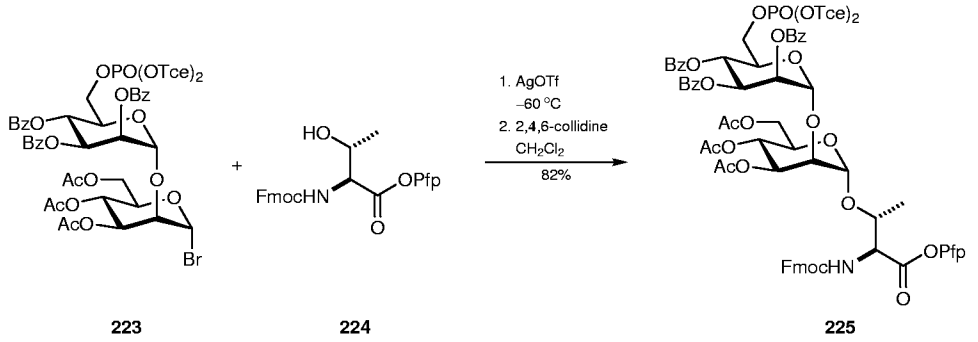
H-Ser-Gly-Val-Glu-Asn-Pro-Gly-Gly-Tyr-Cys[α-D-Gal-(1→4)-β-D-Gal-(1-)-O(CH_2)$_2$-]-Leu-Tyr-OH (222):[484]

To a mixture of 2-bromoethyl glycoside **220** (17 mg, 40 μmol) and Cs_2CO_3 (32 mg, 100 μmol) in dry DMF (1.0 mL), peptide **221** (13 mg, 10 μmol) was added under argon. The mixture was sonicated for 5 min and then stirred at rt until analytical RP-HPLC indicted that all the peptide was consumed. Then 0.1% aq TFA (20 mL) was added and the mixture was freeze-dried. Purification of the residue by preparative RP-HPLC gave the target neoglycopeptide **222**; yield: 14 mg (90%); FAB-MS *m/z*: 1564 $[M+H]^+$; selected ^{1}H NMR (500 MHz, H_2O with 10% D_2O, δ): 8.44 (Cys NH), 4.83 (Gal H1′), 4.40 (Cys Hα), 4.39 (Gal H1), 4.28 (Gal H5′), 3.97 (SCH_2CH_2O), 3.92 (Gal H4), 3.90 (Gal H4′), 3.80 (Gal H3′, Gal H6), 3.74 (SCH_2CH_2O), 3.70 (Gal H2′), 3.66 (Gal H5), 3.64 (Gal H3), 3.59 (Gal H6′), 3.47 (Gal H2), 2.90 (Cys Hβ), 2.76 (Cys Hβ′), 2.71 (SCH_2CH_2O).

6.3.6.2 Synthesis of Mannose 6-Phosphate Containing Disaccharide Threonine Building Blocks and Their Use in Peptide Template Synthesis

Neoglycopeptides composed of peptide templates as a molecular framework for the spatial positioning of the carbohydrate ligands have been constructed in order to mimic the activity of natural ligands to receptors.[63,122,481] Peptide templates carrying 6-O-phosphorylated mannose have been synthesized with great difficulty, owing to protection–deprotection of the phosphate, and the best result was obtained with protection of the phosphate as the bis-*O*-trichloroethyl (Tce) ester.[122,481] This approach has been used in a synthesis of fully protected 6′-O-phosphorylated Fmoc-Thr[α-D-Man-(1→2)-α-D-Man]-OPfp (**225**) and Fmoc-Thr[α-D-Man-(1→6)-α-D-Man]-OPfp building blocks by AgOTf-promoted glycosylation of Fmoc-Thr-OPfp with the respective glycosyl bromides, as shown in Scheme 61. The products were used in the synthesis of putative tripeptide ligands for the mannose 6-phosphate receptor (MPR).[122] Cyclic templates were also used to present 6-O-phosphorylated α-D-Man-(1→2)-α-D-Man ligands to MPR, but with less affinity than the linear analogues.[42] Multiantennary high-mannose oligosaccharide mimics were prepared after computational design and found to be unable to inhibit the binding of mannan-binding protein to polymannan.[41]

Scheme 61 Synthesis of a Fully Protected 6′-O-Phosphorylated Fmoc-Thr[α-D-Man-(1→2)-α-D-Man]-OPfp Building Block[122]



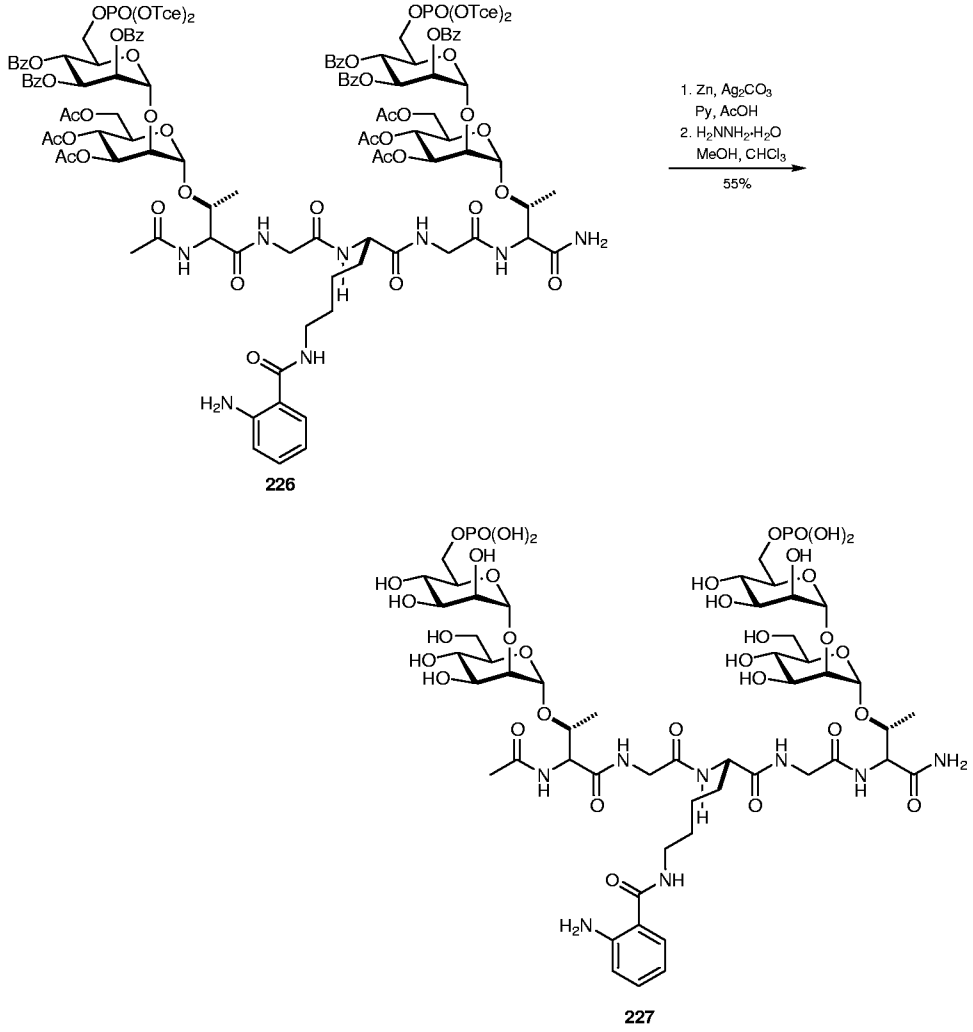
223 224 225

The glycosylated amino acid derivative **225** was used in solid-phase glycopeptide synthesis of the glycosylated tripeptide **227** (Scheme 62).[247] The synthesis was carried out with N^{α}-Fmoc-protected amino acid Pfp esters with the addition of HODhbt as an auxiliary nucleophile. In order to incorporate a fluorescent moiety into the glycopeptide, the free N^{ε}-amino group in N^{α}-Fmoc-Lys-OH was reacted with 3,4-dihydro-4-oxo-1,2,3-benzotriazin-3-yl 2-(*tert*-butoxycarbonylamino)benzoate (Boc-Abz-ODhbt), followed by conversion into the corresponding Pfp ester. Since removal of the Tce groups from the phosphate moiety by catalytic hydrogenation was not quantitative,[122] the conditions for this step were optimized.[247] Treatment with Zn and Ag_2CO_3 in pyridine containing 10% AcOH resulted in a quantitative

for references see p 322

reaction. When using Na in MeOH for the deacetylation, the 2-aminobenzoyl group was partially cleaved. Therefore, hydrazine hydrate in MeOH/$CHCl_3$ was used and no elimination of either the 2-aminobenzoyl group or the carbohydrate moieties was observed.

Scheme 62 Deprotection of a Tripeptide Template for the Mannose 6-Phosphate Receptor[247]



Fmoc-Thr[Bz_3-α-D-Man-6-$(Tce)_2PO_3$-(1→2)-Ac_3-α-D-Man]-OPfp (225):[122]

To a soln of Bz_3-α-D-Man-6-$(Tce)_2PO_3$-(1→2)-Ac_4-β-D-Man (2.67 g, 2.29 mmol) in dry CH_2Cl_2 (12 mL) was added 4 M HBr in AcOH (5.7 mL, 22.9 mmol). The soln was stirred at rt for 30 min, diluted with CH_2Cl_2, and then poured onto ice. The organic layer was washed successively with cold H_2O, sat. $NaHCO_3$, and H_2O. Drying ($MgSO_4$) and concentration gave the corresponding glycosyl bromide **223**; yield: 2.40 g, (88%). This bromide (2.33 g, 1.97 mmol) and Fmoc-Thr-OPfp (**224**; 1.0 g, 1.97 mmol) were dissolved in dry CH_2Cl_2 (13 mL) and stirred for 0.5 h under argon over 3-Å molecular sieves (0.5 g) at −40 °C. AgOTf (0.62 g, 2.41 mmol) was added quickly and the mixture was stirred at −40 °C for 1 h. 2,4,6-Collidine (0.54 mL, 4.04 mmol) was then added, and the temperature was slowly raised to 20 °C. After dilution with dry CH_2Cl_2 and filtration through Celite, the filtrate was extracted (3×) with a mixture of 10% aq TFA and ice, and then with H_2O (1×). Drying ($MgSO_4$) and concentration were followed by vacuum liquid chromatography (silica gel, light petroleum ether/EtOAc 3:1), which gave the title compound; yield: 2.59 g (82%); $[\alpha]_D^{25}$ −34.3 (*c* 1.4, CH_2Cl_2); selected ^1H NMR (500 MHz, $CDCl_3$, δ): 6.20 (NH) 5.32, 5.24 (anomeric protons), 4.84 (Hα), 4.58 (Hβ), 4.50–4.62 (CH_2CCl_3), 1.49 (Hγ).

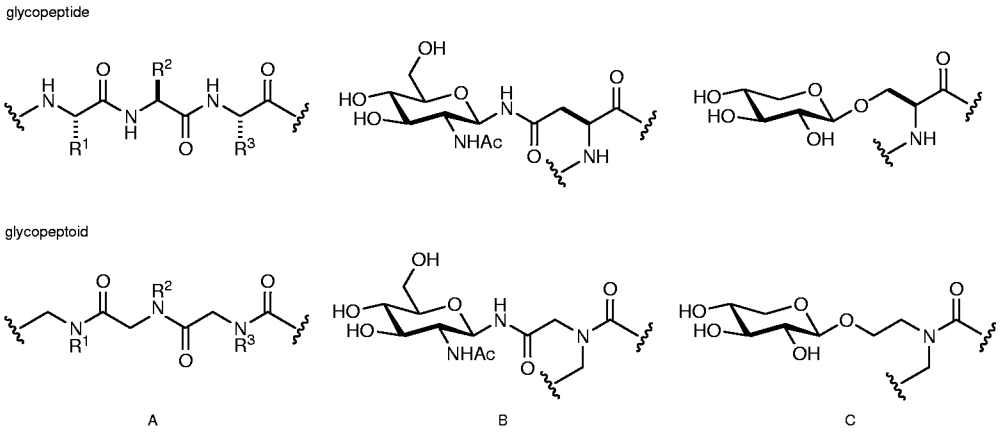
Ac-Thr[α-D-Man-6-PO_3H_2-(1→2)-α-D-Man]-Gly-Lys(2-aminobenzoyl)-Gly-Thr[α-D-Man-6-PO_3H_2-(1→2)-α-D-Man]-NH_2 (227):[247]

After solid-phase synthesis and purification, the fully protected glycopeptide **226** (11.5 mg) was dissolved in pyridine containing 10% AcOH (3 mL). Zn (17 times the weight of glycopeptide) and Ag_2CO_3 (6 times the weight of the glycopeptide) were added, and the suspension was stirred at 50–60 °C for 18 h and, after filtration, directly purified by HPLC using first 0.1% aq TFA for 10 min, then a linear gradient of 0–100% of 0.1% TFA in 90% MeCN/10% H_2O over 100 min, which gave the partially deprotected glycopeptide; yield: 5.1 mg (55%). This compound (5.1 mg) was then dissolved in $CHCl_3$ (200 μL), and MeOH (800 μL) and hydrazine hydrate (200 μL) were added. After 2 h the mixture was directly purified by filtration followed by HPLC, using the same gradient; yield: 2.9 mg (93%); ES-MS *m/z*: 1429.6 [M – H]$^-$; selected ^{1}H NMR (500 MHz, D_2O/acetic acid-d_4 1:1, δ): 7.89, 7.70, 7.61, 7.57 (2-aminobenzoyl), 5.16, 5.13, 5.02, 4.97 (anomeric protons), 4.11–4.19, 4.22–4.28 (phosphorylated H6), 4.43, 4.53 (Thr Hα), 4.41 (Lys Hα).

6.3.6.3 Synthesis of an O-Linked Glycopeptidomimetic (Glycopeptoid) Corresponding to the Linkage Region of Proteoglycans

A new class of neoglycopeptides has been described in which the glycan is attached to the peptide scaffold via the backbone amide nitrogen, giving rise to a novel type of conformationally flexible mimetics, the glycopeptoids. A schematic representation showing the structural analogies between glycopeptides and glycopeptoids is illustrated in Scheme 63. Conformationally flexible glycopeptoids have been synthesized as both N-[485–487] and O-linked[488,489] glycopeptidomimetics. α-O-Linked *N*-acetylgalactosamine clusters with two, three, four, six, and eight repeating units were also synthesized using a reiterative blockwise approach and N-substituted oligoglycine peptoid backbones.[490] Attachment of peptoids (*N*-alkenyl-containing oligoglycins) to the solid phase and cross-metathesis with several *C*-allyl glycosides produced, by using Grubbs' ruthenium benzylidene catalyst, $(Cy_3P)_2Cl_2Ru{=}CHPh$, cross-metathesis products in more than 70% yields.[491] This strategy can be used to generate combinatorial libraries by utilizing different peptoids, glycosides, and repeating units.

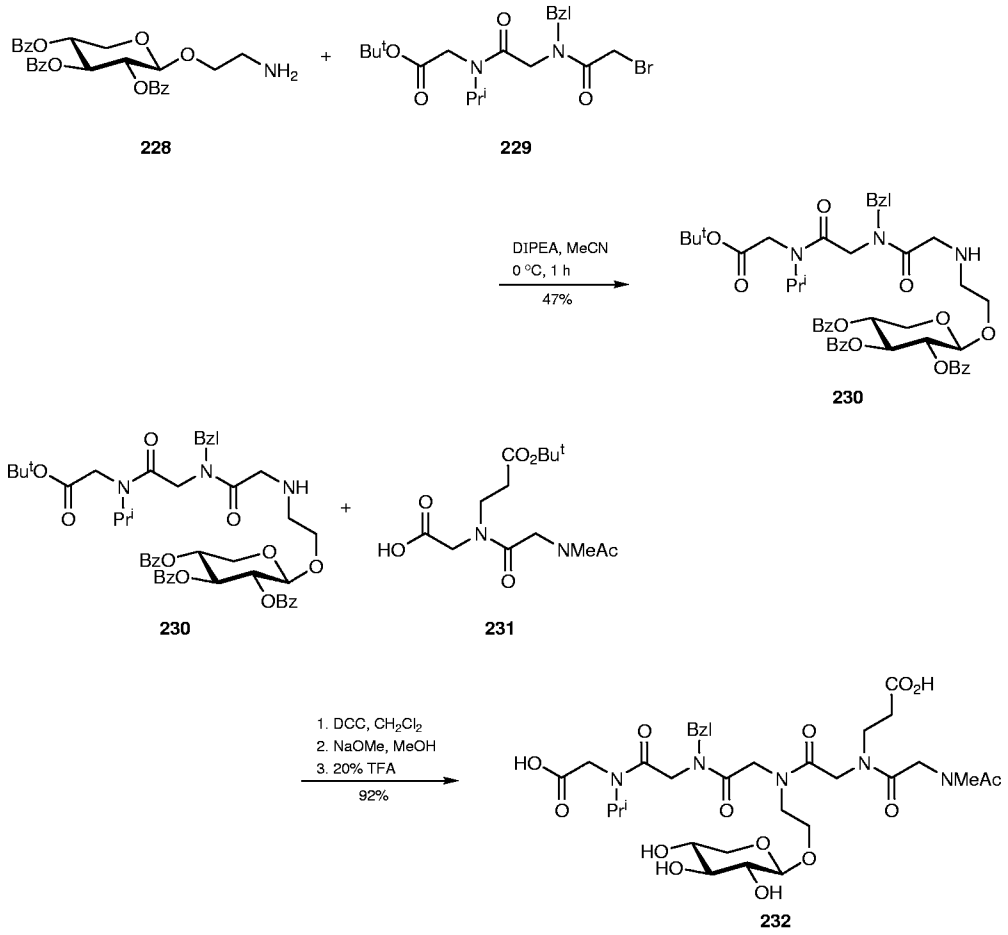
Scheme 63 Comparison of (A) Glycopeptide and Glycopeptoid, (B) a Typical Asparagine-Linked *N*-Acetylglucosaminide and its N-Linked Peptoid Mimetic, and (C) a Typical β-D-Xylose-(1→3)-Ser Linkage and its Homoserine Glycopeptidomimetic[485,489]



As illustrated in Scheme 64, pentaglycopeptoid **232**, composed of an N-substituted oligoglycine (peptoid) mimicking the β-D-Xyl-(1→3)-*O*-Ser linkage region of human bone and cartilage proteoglycans, was synthesized using a convergent approach.[489]

for references see p 322

Scheme 64 Synthesis of a Xyloyl Pentapeptoid (O-Linked Glycopeptidomimetic)[489]



Xylodipeptoid 230:[489]

To a soln of dipeptoid **229** (154 mg, 350 μmol) and DIPEA (54 mg, 420 μmol) in MeCN (15 mL) was added 2-aminoethyl 2,3,4-tri-*O*-benzoyl-β-D-xylopyranoside (**228**; 212 mg, 420 μmol). The soln was stirred for 1 h at 0 °C. The mixture was then concentrated and the residue purified by chromatography (silica gel, $CHCl_3$/MeOH/MeCN 18:1:1, R_f 0.40) to give a white foam; yield: 152 mg (47%); FAB-MS *m/z*: 866 $[M+H]^+$; ^1H NMR ($CDCl_3$, δ): 7.93–8.00 (m, 6H, 2-Ar), 7.44–7.53 (m, 3H, 4-Ar), 7.17–7.40 (m, 11H, 3-Ar, Ph), 5.72 (t, 1H, $J_{3,4}$ = 7.3 Hz, H3), 5.32 (t, 1H, $J_{2,3}$ = 7.1 Hz, H2), 5.26 (m, 1H, H4), 4.85 (d, 1H, $J_{1,2}$ = 5.3 Hz, H1), 4.58–4.61 (m, 2H), 4.41 (dd, 1H, $J_{5a,5e}$ = 12.1 Hz, $J_{5e,4}$ = 4.2 Hz, H5e), 3.38–4.21 (m, 10H), 2.83, 2.91, (2dd, 2H, *J* = 5.5, 2.0 Hz, NCH_2CH_2Oxyl), 2.08 (br s, 1H, NH), 1.34, 1.39, 1.44 (3s, 9H, CMe_3), 1.04, 1.12, 1.13 (3d, 6H, *J* = 6.9 Hz, $CHMe_2$).

Xylopentapeptoid 232:[489]

To a mixture of **230** (119 mg, 137 μmol) and **231** (43.4, 137 μmol) in CH_2Cl_2 (10 mL) was added DCC (28.4 mg, 137 μmol). The mixture was stirred for 2 h at rt. The white precipitate was filtered and the soln was concentrated. The residue was purified by column chromatography using the same eluent as above (R_f 0.31) to give a white solid; yield 147 mg (92%). A soln of this compound (40 mg, 34.4 μmol) in MeOH (5 mL) was treated with a 1 M NaOMe soln in MeOH until pH 9. The soln was then stirred for 2 h at rt. The mixture was treated with H^+ resin for 30 min, filtered, and then concentrated. The residue was dissolved in CH_2Cl_2 (2 mL) containing 20% TFA, and the resulting soln was stirred for 1 h at rt to give the fully deprotected pentapeptoid **232** in essentially quantitative yield.

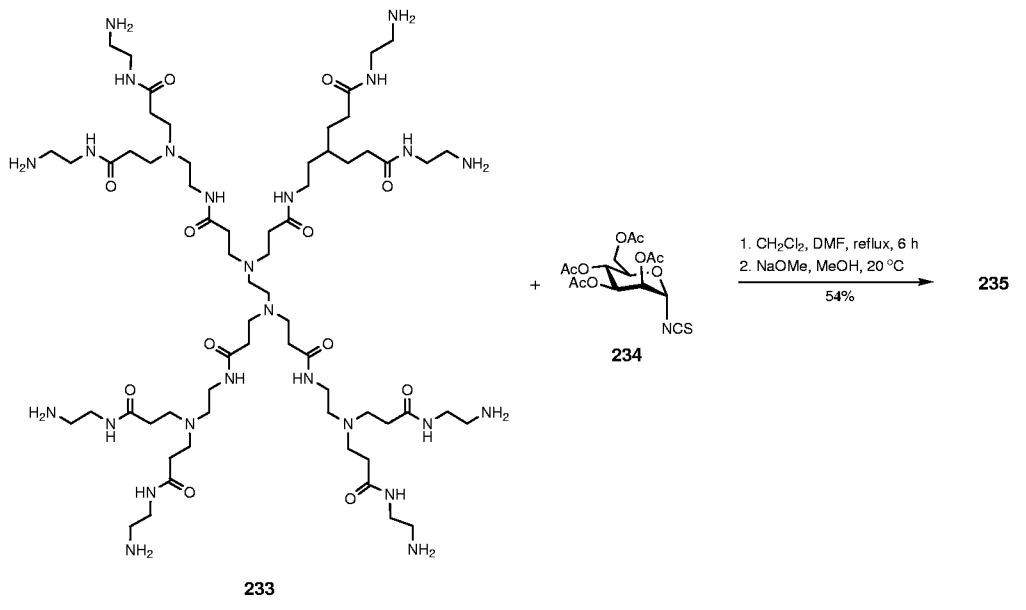
6.3.6.4 Synthesis of Thiourea-Bridged Glycosides from Glycosyl Isothiocyanates with Dendritic Polyamines

Carbohydrate-containing dendrimers, glycodendrimers, are highly symmetric branched macromolecules with a branch point at each repeating unit. They represent ideal clusters mimicking multiantennary glycoproteins and can be synthesized with the desired carbohydrate density.

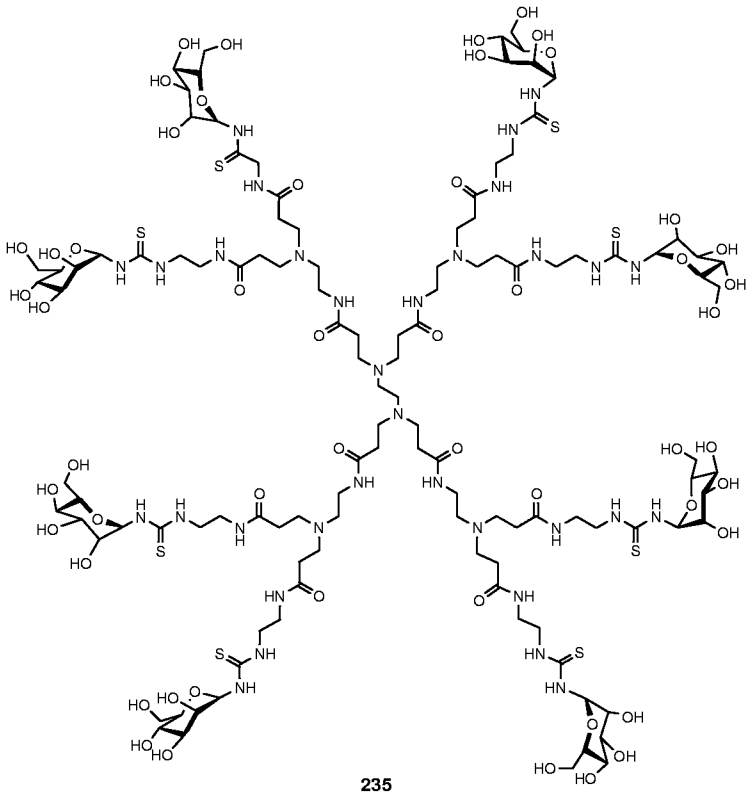
Synthesis of a number of glycodendrimers, based on a lysine core, has been reported.[492–494] The hyperbranched dendritic cores have been conjugated to α-thiosialosides,[492] β-D-lactosides,[493] *N*-acetyllactosaminides,[493] *N*-acetylglucosaminides,[493] α-D-mannosides,[494] and a 3′-sulfo-Lewis^x-(Glc) analogue.[495] Spherical Starburst poly(amidoamine) dendrimers,[496,497] are commercially available and have been used to attach various carbohydrate derivatives. Disaccharide lactones of lactose and maltose have been directly conjugated to the amine-terminated dendrimers,[498] and syntheses of poly(amidoamine) α-thiosialo dendrimers have been described.[499] An isothiocyanate coupling strategy has also been employed to conjugate poly(amidoamine) dendrimers to α- and β-D-mannose, β-D-glucose, β-D-galactose, β-cellobiose, and β-lactose isothiocyanate derivatives.[500,501]

Glycosyl isothiocyanates are ideally suited for the preparation of cluster glycosides. The ease of the transformation of isothiocyanates and amines to thiourea derivatives can generally be applied to the reaction of glycosyl isothiocyanates with oligovalent, multibranched amines, to yield oligoantennary, thiourea-bridged glycoconjugates directly.[500] The reaction with trivalent tris(2-aminoethyl)amine and mono- as well as disaccharide glycosyl isothiocyanates of different configurations affords the triantennary products in good yields (Scheme 65). The sugar units can be deprotected easily in a Zémplen reaction with NaOMe in MeOH without attack on the thiourea bridges. The acetylated gluco-, manno-, galacto-, cellobio-, and lacto-configurated glycosyl isothiocayanates were converted in two steps into the tris(2-glycosylthiourea-ethyl)amines in overall yields of 70–80%. This method can also be applied to the synthesis of higher branched clusters, in which dendritic polyamines serve as suitable core molecules.

Scheme 65 Synthesis of an Octaantennary Mannoside Cluster by Reaction of a Glycosyl Isothiocyanate with a Dendritic Polyamine[500]



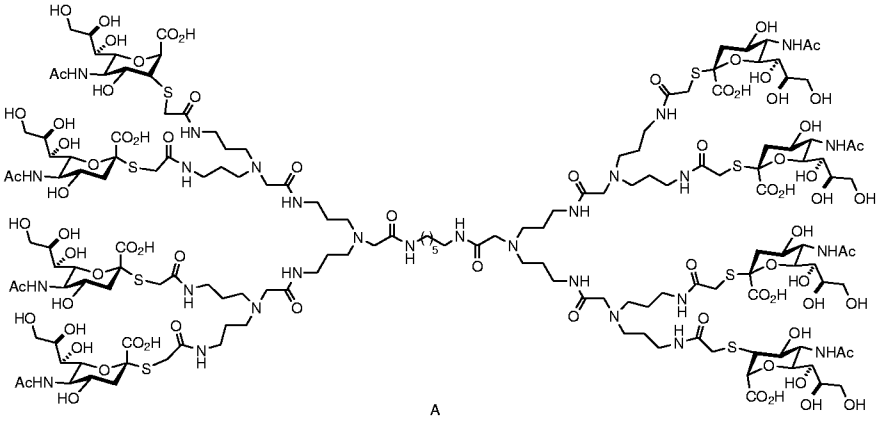
for references see p 322



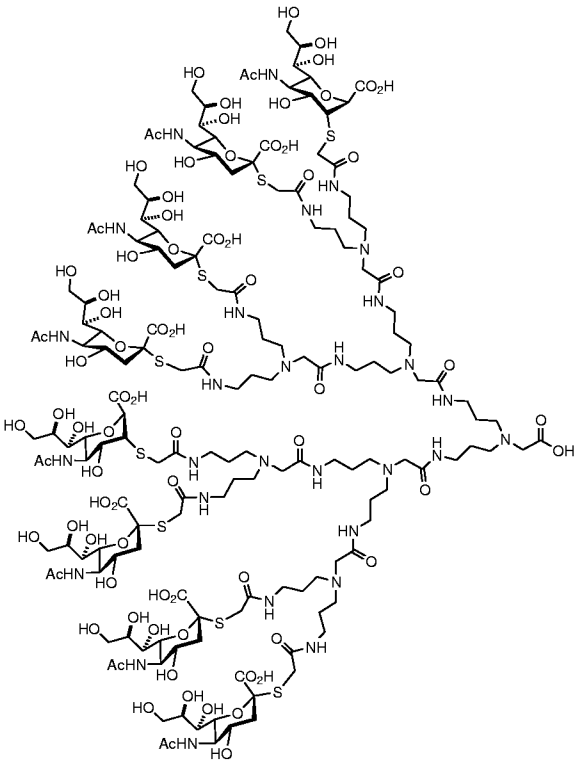
235

Two types of thiosialo glycopeptide dendrimers have been reported, and their ability to inhibit or crosslink and precipitate *Limax flavus* lectin was compared.[502,503] Both types of glycodendrimers were scaffolded onto a 3,3′-iminobis(propylamine) core: one type was synthesized as the divergent cascade dendrimer (B, Scheme 66), the other multidirectionally (A, Scheme 66). Although the structural difference between the dendrimers is not obvious, the results indicated that the multidirectional dendrimers were more effective inhibitors of the binding of *Limax flavus* to human α1-acid glycoprotein. A series of N-linked α-sialo dendrimers with valencies of 2, 4, and 8 has been scaffolded on an *N,N*-bis(3-aminopropyl)succinamic acid core using Wang resin and Fmoc/HBTU solid-phase chemistry.[504]

Scheme 66 Two Types of Thiosialo Glycopeptide Dendrimers: (A) Multidirectional and (B) Divergent[502,503]



A



B

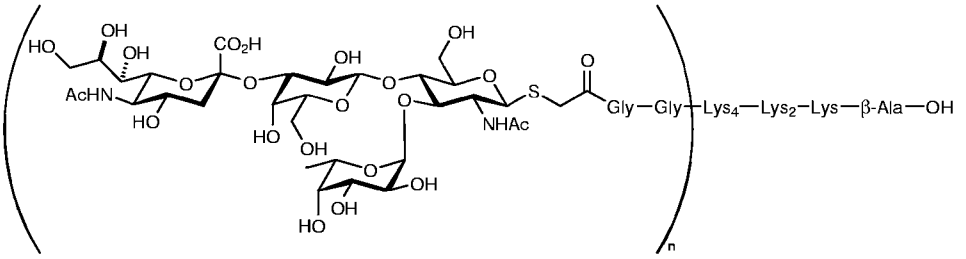
In a convergent approach to glycodendrimers, building blocks called dendrons are first constructed followed by attachment of these dendrons to a multipodent core unit in the final steps of the dendrimer construction.[505,506] The synthetic strategy involves: (1) the synthesis of the triglucosylated derivative of tris(hydroxymethyl)aminomethane (Tris), (2) the introduction of a glycine-derived spacer and 3,3′-iminodipropanoic acid derived branching units on the Tris derivative by amide bond formation, (3) condensation of the above saccharide-containing dendrons with a trifunctional 1,3,5-tricarbonylbenzene derivative used as the core, by formation of amide bonds, and (4) deprotection of the saccharide units. A 9-mer and an 18-mer, carrying nine and 18 saccharide units at the periphery, respectively, have been synthesized, in high yields at each step, by this synthetic strategy.[506] The preparation of even larger dendritic carbohydrate derivatives possessing 24 and 36 peripheral D-glucopyranosidic units has been reported, using a similar synthetic strategy.[505] The 36-mer glucodendrimer was convergently synthesized using a 12×3 reaction sequence for the final coupling of the glycodendrimer wedges to the 1,3,5-tricarbonylbenzene central core.[505] A conclusion from this work is that steric hindrance and filling of the interstitial space around the core component of these dendrimers are at least two factors that are believed to limit completion of the reactions at the core by the convergent approach. The same approach to the synthesis of α-D-mannopyranoside-containing dendrimers of various generations has been used successfully to obtain high molecular weight and monodisperse dendrimers in which 3–36 identical copies of the mannopyranoside residues are located at their peripheries.[507]

A chemoenzymatic synthesis of di-, tetra-, and octavalent sialyl Lewis X ligands scaffolded on dendrimers has been reported (Scheme 67).[508] Hypervalent L-lysine cores with covalently attached 2-acetamido-2-deoxy-D-glucose residues were chemically prepared and enzymatically transformed into sialyl Lewis X-containing dendrimers using β-(1→4) ga-

for references see p 322

lactosyltransferase, α-(2→3) sialyltransferase, and α-(1→3/4) fucosyltransferase. All chemical and enzymatic yields were good to excellent, indicating that this chemoenzymatic approach is an effective and viable route to well-defined, structurally similar series of dendritic sialyl Lewis X, differing only in valency.

Scheme 67 Chemoenzymatically Prepared Dendritic Multivalent Sialyl Lewis X Ligands[508]



Octaantennary Mannoside Cluster 235:[500]
A soln of generation 1 poly(amidoamine) dendrimer **233** (40 mg, 0.028 mmol) in DMF (10 mL) was added dropwise to 2,3,4,6-tetra-*O*-acetyl-α-mannopyranosyl isothiocyanate **234** (122 mg, 0.31 mmol) in CH_2Cl_2 (50 mL) at reflux temperature. It was stirred at reflux for 6 h, then the solvent was removed by distillation, and the residual syrup was purified by flash column chromatography (silica gel, CH_2Cl_2/MeOH 9:1). The acetylated product was dissolved in dry MeOH (80 mL) and the pH was adjusted to 8–9 with a freshly prepared 1 M NaOMe solution. The mixture was stirred at 40 °C until the reaction was complete, then it was neutralized with ion-exchange resin (Dowex, H^+ W50X8); the ion exchanger was filtered off and washed with H_2O. The resulting soln was concentrated to remove the MeOH followed by lyophilization. The octaantennary cluster mannoside **235** was obtained as a white lyophilizate; yield: 49 mg (54%); MALDI-TOF MS: found $[M+H]^+$ 3201; selected ^1H NMR (400 MHz, D_2O, δ); 5.37 (8H, glycosyl H1), 3.65 (16H), 3.31 (24H), 2.99 (12H), 2.90 (16H), 2.73 (8H), 2.51 (8H), 2.41 (16H) (CH_2 groups of the core region); selected ^{13}C NMR (69.89 MHz, D_2O, δ): 182.60 (C=S), 175.01 (C=O), 82.72 (glycosyl C1), 51.89, 49.97, 49.60, 49.19, 44.43, 39.26, 37.04, 32.69, 31.97 (CH_2 groups of the core region).

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6.4 Synthesis of Lipopeptides

S. Pegoraro and L. Moroder

Co- and posttranslational lipidation by acylation with fatty acids and prenylation as well as by covalent conjugation of a structurally more complex lipid component, for instance, the glycosylphosphatidylinositol (GPI) moiety occurs on a wide variety of cellular signaling proteins. The enzymes that catalyze specific attachment of these lipid groups to proteins at defined sequence positions have been identified and characterized (for extensive reviews see ref[1]). Each lipid moiety confers unique properties to the modified protein, resulting in alteration of protein–protein interactions, membrane binding and targeting, and intracellular signaling. Lipidation is thus a key event that determines subcellular or subcompartmental localization of proteins and their biological functions in signal transduction. The biochemistry and cell biology of protein lipidation has been extensively reviewed.[2–5] For a detailed study of the molecular mechanisms of signaling pathways, the need for tailored synthetic molecules structurally similar to the lipoproteins has fostered the development of synthetic procedures that allow the preparation of such natural lipo-derivatives as well as of related mimetic structures.

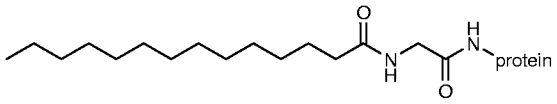
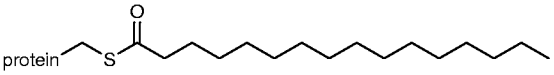
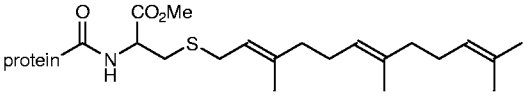
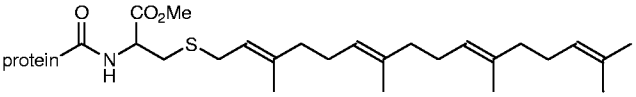
Additionally, a vast array of natural lipopeptides with remarkable structural diversity is produced by microorganisms living in different habitats, from aquatic to terrestrial environments. These products are not gene encoded, but are synthesized nonribosomally by large multifunctional enzymes, the peptide synthetases.[6–8]

6.4.1 Natural Co- and Posttranslational Lipidation

Although the first S-prenylated peptide, the mating factor from the fungus *Rhodospiridium toruloides*,[9] was discovered some time ago, the notion that entire proteins could be post-translationally lipidated has only recently been realized. In fact, an ever-increasing number of intra- and extracellular proteins and peptides are known to be modified with one or more covalently bound lipid groups. While in cell membranes a large variety of lipid molecules is found, only relatively few lipids are covalently attached to proteins by the co- or post-translational modification mechanisms. So far, the covalent protein modifications are known to involve the N-terminal myristoylation of a glycine,[10,11] S-prenylation[12,13] of cysteine residues at or close to the C-terminus, and S-palmitoylation[11,14] of cysteines throughout proteins (Table 1). In addition, a few O-acylated peptides and proteins have been identified.[15,16] With the fast progress in the field, it may not be surprising to discover that nature uses alternative functionalities for lipidation.

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Table 1 Natural Monolipidation Motifs of Peptides and Proteins

Lipidation	Lipid–Protein Structure	Consensus Sequence	Bond Type
myristoylation		Gly-(Xaa)$_3$-Ser/Thr	amide
palmitoylation		–	thioester
farnesylation		Cys-Xaa^1-Xaa^2-Xaa^3 [a]	thioether
geranylgeranylation		Cys-Xaa^1-Xaa^2-Xaa^3 [a] Cys-Cys Cys-Xaa^1-Cys	thioether

[a] The sequence motif for C-terminal cysteine prenylation, Cys- Xaa^1-Xaa^2-Xaa^3 (also known as the CaaX motif), contains two aliphatic amino acids (Xaa^1 and Xaa^2) followed by Xaa^3 which dictates S-farnesylation (Xaa^3 = Met, Ser) or S-geranylgeranylation (Xaa^3 = Leu), respectively.

In most tissues, protein N-acylation with fatty acids is highly specific for a myristoyl chain, although in visual cells lauroyl and C_{14}-unsaturated fatty acid chains are also coupled to protein N-termini.[10] In microorganisms, N-acylation with a large variety of fatty acids is mediated by *N*-acyl tranferases.[17] Protein S-acylation is often referred to as palmitoylation, although a spectrum of *S*-acyl chains, with palmitoyl and stearoyl generally predominating, has been found coupled to various cellular proteins. S-Acylation leads to a thioester bond with the thiol group of a cysteine residue which, in contrast to the amide bond formed by N-myristoylation, is much more labile and thus is readily reversed by enzymes allowing the cell to control the lipidation state of *S*-acyl proteins. S-Prenylated proteins contain two types of unsaturated lipidic chains: either the C_{15}-farnesyl or the C_{20}-geranylgeranyl residue.[12,13] The lipid chains are attached to cysteine thiol groups by a sulfide bond. In prenylated proteins, the lipidated cysteine residues are often located in the C-terminal portion of a protein; moreover, in some proteins with two cysteines positioned in the C-terminus as vicinal residues (Cys-Cys or Cys-Xaa-Cys) two isoprenoid lipids, predominantly geranylgeranyl, are linked via sulfide bonds. Dual lipidation with the simultaneous presence of two distinct lipid chains is also very common (see Section 6.4.1.4). The final class of lipidated proteins is those containing glycosylphosphatidylinositol, a complex lipid moiety that contains a phospholipid associated with a carbohydrate chain and ethanolamine. The anchor is linked to the proteins via an amide bond with the C-terminal carboxy group of the polypeptide chain (see Section 6.4.4.5).[18–20]

To better understand the structural properties of the lipoproteins involved in intracellular signaling pathways, synthetic lipopeptides are required as model compounds. In this context, to bypass the enzyme-catalyzed lipidation occurring in nature, optimized synthetic procedures have been elaborated for the more simple lipo-derivatives (for comprehensive reviews see ref[1]). However, similar efficient synthetic procedures for the glycosylphosphatidylinositol anchor are still lacked.

To make these substrates suitable for biological assays, the introduction of functional groups that can be traced with the proper analytical techniques is essential. The use of radio-, fluorescent-, and biotin-labeled lipidated peptides has been reported. The synthesis of fluorescent substrates is chemically straightforward and allows for production of larger quantities than the enzymatic synthesis used for radiolabeled peptides and is thus preferred over the use of radioactive compounds.[1,21] Common fluorescent probes can be introduced by conjugation to a free functional group present in the peptide. The fluorescent moiety is

usually linked to the peptide prior to lipidation. In the case of prenylated peptides, the fluorescent group can be introduced at the N-terminal amino group. For N-myristoylated peptides, the C-terminal carboxylate is used to anchor the fluorescent dye. Even lipids containing fluorescent[22] or photoactivatable groups[23] have been incorporated into peptides to probe lipid–lipid or lipid–protein interactions.

6.4.1.1 N-Myristoylation

N-Myristoylation is achieved by the covalent attachment of the 14-carbon saturated myristic acid ($C_{14:0}$) to the N-terminal glycine residue of various proteins with formation of an irreversible amide bond (Table 1).[10] This process is cotranslational and is catalyzed by a monomeric enzyme called *N*-myristoyltransferase.[24] Several proteins of diverse families, including tyrosine kinases of the Src family, the alanine-rich C kinase substrate (MARKS), the HIV Nef phosphoprotein, and the α-subunit of heterotrimeric G protein, carry a myristoylated N-terminal glycine residue which in some cases is in close proximity to a site that can be S-acylated with a fatty acid. Functional studies of these proteins have shown an important structural role for the myristoyl chain not only in terms of enhanced membrane affinity of the proteins, but also of stabilization of their three-dimensional structure in the cytosolic form. Once exposed, the myristoyl chain promotes membrane association of the protein.[5] The myristoyl moiety however, is not sufficiently hydrophobic to anchor the protein to the membrane permanently,[25,26] and in vivo this interaction is further modulated by a variety of "switches" that operate through covalent or noncovalent modifications of the protein.[4,5,27] In MARKS, for example, multiple phosphorylation of a positively charged domain moves the protein back to the cytosolic compartment due to the mutated electrostatic properties of the protein, a so-called myristoyl-electrostatic switch.[28]

6.4.1.1.1 Synthesis of *N*-Myristoyl Peptides

N-Myristoylated peptides represent a very useful tool for studying the mechanism of interaction of myristoylated proteins with the cellular membrane, and for the development of substrates and inhibitors of molecules involved in cellular signaling. Due to the chemical nature of the *N*-myristoyl group as a saturated fatty acid derived acyl moiety and the nature of the linkage with the peptide chain, i.e. amide bond, with the N-terminal amino acid, the synthesis of *N*-myristoyl peptides is relatively straightforward. It can be performed by solution methods as well as on solid supports by building up the peptide chain with standard protocols and coupling the fatty acid to the suitably protected peptide derivatives.[29,30] The myristoyl group is stable toward the more or less strong acidic conditions used for the peptide deprotection and resin-cleavage steps in both the Boc- and Fmoc-based strategies. Difficulties may be encountered with the resulting peptide amphiphiles in the chromatographic purification steps due to their tendency to aggregate.

In particular cases where small amounts of myristoylated peptides are required, enzymatic N-myristoylation of unprotected and purified peptides can be achieved using the yeast *N*-myristoyltransferase. This procedure has been applied for N-myristoylation of the N-terminal deca-, dodeca-, and tetradecamer fragment of $pp60^{v\text{-}src}$, the transforming protein of Rous sarcoma virus.[31] This method is particularly useful when [^{3}H]-labeled myristoylated compounds are required.

Myr-Asp-His-D-Phe-Arg-Trp-Lys-NH_2 by On-Resin Acylation with Myr-OH and DIC/HOBt; Typical Procedure:[29]

To H-Asp-His(Bom)-D-Phe-Arg(Tos)-Trp(For)-Lys-MBHA-resin (0.25 g; 0.31 mmol · g^{-1} substitution) in CH_2Cl_2/NMP (1:1) was added Myr-OH (590 mg, 2.58 mmol), 1 M HOBt in DMF (3 mL), and 1 M DIC in CH_2Cl_2 (3 mL). The slurry was shaken for 45 min and completion of the reaction was monitored by the

for references see p 370

ninhydrin test. The peptide-resin was then washed with CH_2Cl_2 and DMF, and dried. After HF cleavage and extraction of the peptide with glacial and 30% aq AcOH, the crude peptide was purified by RP-HPLC (conditions not given); yield: 13.2 mg (36%); FAB-MS *m/z*: 1080 $[M+H]^+$.

Myr-Gly-Cys(Plm)-Thr-Leu-Ser-Ala-OH by Acylation with Myr-OH and EDC/HOBt in Solution; Typical Procedure:[32]
To a soln of Myr-OH (8.5 mg, 0.037 mmol), HOBt (5 mg, 0.037 mmol), and EDC (7 mg, 0.037 mmol) in CH_2Cl_2 (2 mL) was added after 15 min H-Gly-Cys(Plm)-Thr-Leu-Ser-Ala-OH·TFA [43 mg, 0.037 mmol; obtained by TFA treatment of Boc-Gly-Cys(Plm)-Thr-Leu-Ser-Ala-OH; yield: 93%] in DMF (2 mL) containing TEA (5 μL, 0.037 mmol). The mixture was stirred at 20 °C for 2 h, the solvents were removed, and the desired product was isolated by flash chromatography (silica gel, gradient CH_2Cl_2/MeOH 95:5 to CH_2Cl_2/MeOH 80:20); yield: 27 mg (73%); MALDI-TOF *m/z*: 1002.6 $[M+H]^+$.

Myr-Gly-Cys($C_{16}H_{33}$)-OAl by Acylation with Myr-Cl in Solution; Typical Procedure:[24]
To a soln of H-Gly-Cys($C_{16}H_{33}$)-OAl·TFA (90 mg, 0.16 mmol) [obtained by TFA treatment of Boc-Gly-Cys($C_{16}H_{33}$)-OAl; yield: 100%] in CH_2Cl_2 (20 mL) was added TEA (44 μL, 0.32 mmol) and Myr-Cl (50 μL, 0.17 mmol). The mixture was stirred at 20 °C for 16 h and the soln was washed with brine. The organic layer was dried ($MgSO_4$) and concentrated. The product was recrystallized from CH_2Cl_2/Et_2O; yield: 78 mg (74%); mp 68 °C.

[^{3}H]Myr-Gly-Ser-Ser-Lys-Ser-Lys-Pro-Lys-Asp-Pro-Ser-Tyr-OH by Enzymatic N-Myristoylation:[31]
The mixture contained peptide (10 μL; 5 mg·mL^{-1}), buffer [40 μL; 10 mM Tris·HCl(pH 7.4)/1 mM DTT/0.1 mM ethylenebis(oxyethylenenitrilo)tetraacetic acid/0.03% Triton X-100/10 μg of aprotinin per mL], [^{3}H]-myristoyl-CoA (50 μL; 1 μCi), and a cellular extract containing 50 to 150 μg of *N*-myristoyltransferase. The final reaction volume was adjusted to 120 μL. Following incubation for 20 min at 30 °C, the reaction was quenched by the addition of MeOH (120 μL) and TFA (10 μL). Most of the relatively large proteins were precipitated by cooling the mixture on ice for 10 min, whereas more than 90% of free and acylated peptides remained in soln. The mixture was centrifuged for 4 min at 7000*g*, and the supernatant filtered through a Spin-X filter (Costar, Cambridge, MA) by spinning for 5 min at 7000*g*. Samples (100 μL) of the mixture were analyzed by RP-HPLC on a C-18 column (eluent A: 0.1% aq TFA; eluent B: 0.1% TFA in H_2O/MeCN 4:6; flow rate: 1 mL·min^{-1}). The gradient run was initiated with 35% buffer B and proceeded with a 1%·min^{-1} linear gradient increase of buffer B. Fractions (1 mL) were collected, mixed with scintillation fluid, and analyzed in a scintillation counter. Under these conditions the free peptide eluted with the void volume, myristoylated peptide at 20–21 min, and myristic acid and myristoyl-CoA eluted with $t_R > 25$ min.

6.4.1.2 S-Prenylation

S-Prenylation of proteins, i.e. addition of a lipid chain formed by three (farnesyl) or four (geranylgeranyl) isoprene units, to a free thiol group represents a key initial step in a series of posttranslational events occuring during cellular signal transduction (Table 1).[2,12] Different classes of proteins, among them the family of Ras proteins, are known to be farnesylated at specific consensus sites corresponding to the C-terminal sequence Cys-Xaa^1-Xaa^2-Xaa^3 reported in Table 1.[11,12] Upon enzymatic prenylation, proteolytic removal of the C-terminal tripeptide followed by enzyme-mediated methylation of the S-prenylated cysteine leads to the mature lipidated protein or peptide. The best known S-prenylated peptides are the pheromones.[33,34]

Other C-terminal sequences, i.e. Cys-Xaa^1-Cys and Cys-Cys in Rab proteins, are consensus sequences for dual geranylgeranylation.[12] Two heterodimeric enzymes, farnesyltransferase and geranylgeranyltransferase I, are responsible for the lipidation of most monoprenylated proteins. Geranylgeranyltransferase II was more recently characterized.[12] The obvious function of S-prenylation is to increase the hydrophobicity of a protein, thereby promoting association of the protein with cellular membranes. However, there is an important distinction between farnesylation and geranylgeranylation based on the relative lipophilicity of the two groups. It has been demonstrated that the introduction of a single farnesyl chain provides barely enough hydrophobicity to result in stable membrane association.[35,36] In contrast, the longer geranylgeranyl chain provides sufficent hydrophobicity

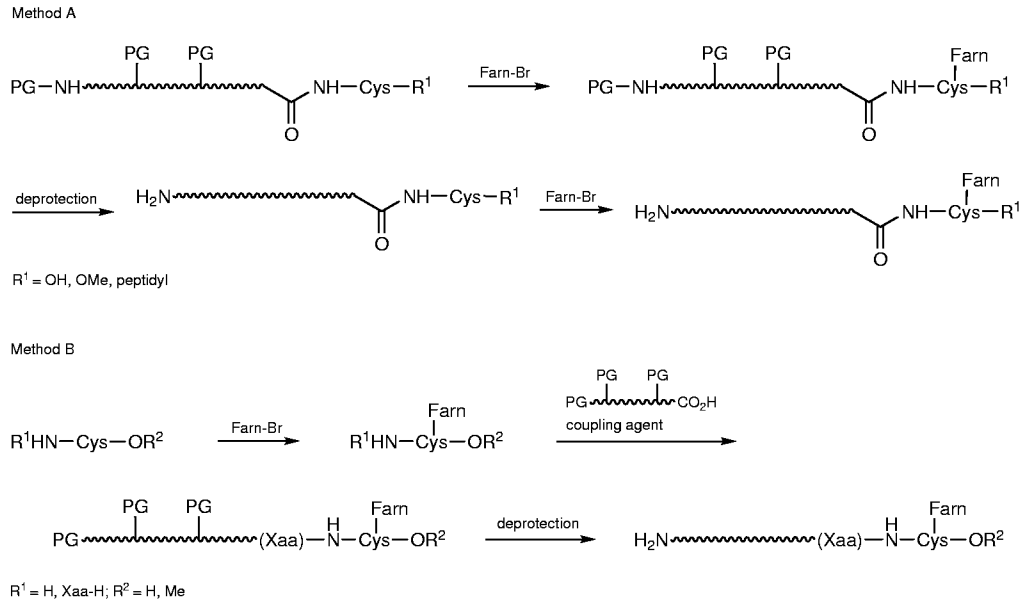
for a stable membrane anchorage. Using model peptides and proteins, it has been possible to understand the fundamental role of the C-terminal methylation that increases the hydrophobicity by eliminating the charged carboxylate group and consequently increases membrane affinity of the farnesylated and, to a lesser extent, of the geranylgeranylated peptides.[37,38]

6.4.1.2.1 Synthesis of S-Prenylated Peptides

The synthetic difficulties encountered in the S-prenylation of peptides has been reviewed.[39] In fact, the synthesis of peptides containing farnesyl or geranylgeranyl groups attached as sulfides to a cysteine residue is severely hampered by the pronounced acid-lability of the prenyl groups, in particular of the stereochemistry of the double bonds which are all *trans*-configured in the farnesyl and geranylgeranyl chains. These double bonds are sensitive to the acid treatments which are normally applied in standard Boc- and Fmoc-based peptide synthesis. If the C-terminal carboxy group has to be methylated to mimic the natural lipidation state, the use of solid-phase techniques is even more questionable. A pioneering work for the synthesis of S-prenylated peptides has been performed by Naider and Becker on the a-factor,[39] a farnesylated dodecapeptide pheromone from *S. cerevisiae* which is essential for mating of a- and α-haploids in yeast.

Two major synthetic strategies have been applied for the synthesis of S-prenylated peptides as shown in Scheme 1. In the first approach (method A), S-alkylation is performed on the final peptide in a protected or unprotected form.[36,40–45] In method B, which has produced better results in terms of overall yield, thiol alkylation is carried out on the cysteine residue or on a small peptide fragment and then the S-prenylated synthons are coupled to the remaining peptide fragment to produce the final products.[45–50] Most of the procedures reported for the synthesis of farnesylated peptide derivatives, can also be applied to the synthesis of geranylgeranylated peptides.

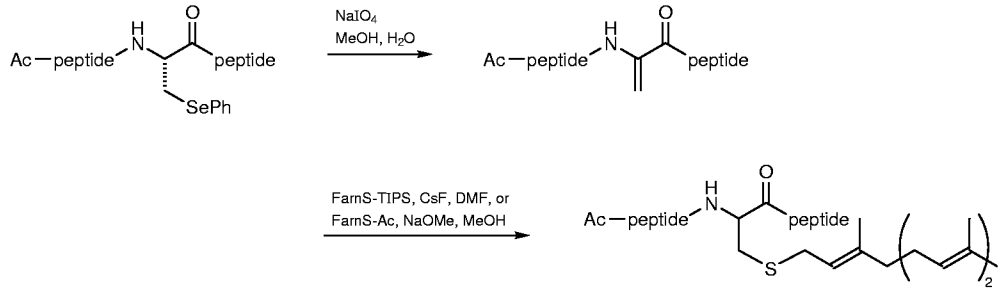
Scheme 1 Synthetic Routes for the Preparation of S-Prenylated Peptides[39]



for references see p 370

An elegant new procedure has been proposed for the synthesis of S-prenylated peptides[51] which is based on chemoselective transformations of peptides containing *Se*-phenylselenocysteine at preselected positions.[52] After deprotection–resin cleavage and chromatographic purification of the *Se*-phenylselenocysteine peptide, mild oxidation of the selenide and elimination of the selenoxide provides the dehydroalanine peptide which allows for chemoselective ligation of isoprenoid lipids via Michael addition of triisopropylsilyl-protected derivatives of farnesylthiol or geranylgeranylthiol in the presence of cesium fluoride, or of farnesylthiol or geranylthiol acetate in combination with sodium methoxide (Scheme 2). Despite the lack of diastereoselectivity in the Michael addition, which requires an additional chromatographic separation step, the ease and versatility of this approach provides an attractive alternative to other established routes.

Scheme 2 Synthesis of S-Prenylated Peptides via *Se*-Phenylselenocysteine Precursor Peptides[52]



6.4.1.2.1.1 Peptide Synthesis with S-Farnesylated Cysteine Derivatives and Cysteine Fragments

S-Alkylation of the cysteine thiol group is usually performed with appropriate alkyl bromides in the presence of a base. This type of reaction has been carried out under a wide variety of conditions with organic solvents as well as aqueous organic media, and as base, potassium fluoride, potassium hydrogen carbonate, guanidinium carbonate, or organic bases have been used in various amounts, and employing various excesses of the alkyl halide (1.5–18 equiv). Furthermore, a wide range of pH values has been analyzed from partially acidic to neutral to alkaline conditions (Table 2). In optimized procedures, organic bases and anhydrous polar aprotic solvents were found to generally improve the final yields.

S-Farnesylated cysteine methyl ester represents in most of the cases the C-terminal residue of the synthetic model peptides. The choice whether to use this amino acid derivative or related larger fragments for the final coupling step, strongly depends upon the peptide sequence.

The S-prenylated cysteine fragment, normally protected at the carboxy group as the methyl ester, can be used as the amino component in fragment couplings, where the choice of the protecting groups of the two fragments is essential as the S-prenyl group is not stable under acidic conditions (*vide supra*). Therefore, the base-labile Fmoc group is used for protection of N^{α}- and N^{ε}-amino groups in combination with the 9-fluorenylmethyl ester for side-chain carboxy groups.[46–49,53] Two alternative N^{α}-protections have been proposed by Hinterding et al. for the synthesis of S-farnesylated peptides,[21] these are the enzyme-labile 4-acetoxybenzyloxycarbonyl group [Z(OAc)] which is enzymatically removed in aqueous buffer under mild conditions (see Vol. E 22a, Section 2.5.1.2)[50] and the allyloxycarbonyl group (Aloc) which is removed by triphenylphosphine/palladium(0) in the presence of morpholine.[54,55] The fragment coupling steps are performed with common reagents.

Table 2 Summary of Various S-Farnesylation Methods[36,40–43,45,47–49,54,56–59]

Peptide	Reagent (Equiv)	Conditions	pH	Yield[a] (%)	Ref
H-Cys-OH	Farn-Br (0.5)	BuOH/MeOH/H_2O (1:1:1), rt, 18 h	basic guanidinium carbonate	44	[56]
YPEISWTRNGC	Farn-Br (1.5)	DMF/H_2O (1:1), rt, 16 h	basic MgO (5 equiv)	n.r.	[40]
Dns-GCVLS	Farn-Br (1)	DMF, rt	basic DIPEA (3 equiv)	n.r.	[58]
Dns-GCVLS	Farn-Br (1.1)	DMF, 1 h	basic DIPEA (1.1 equiv)	34	[59]
LNSSDDGTQG-C(StBu)MGLPC-OMe	Farn-Br (2)	DMF/H_2O (8:2), followed by Ellman test	basic (8.5) $KHCO_3$	40	[57]
H-Cys-OMe	Farn-Br (0.7)	DMF/EtOAc (2:1), rt, overnight	basic DIPEA (1.1 equiv)	46	[47]
H-Cys-OMe	Farn-Br (1)	DMF/EtOAc (5:3), rt, 1 h	basic DIPEA (2.2 equiv)	95	[49]
Fmoc-GR(Pmc)D(OtBu)-N(Trt)GS(tBu)PI-GY(tBu)S(tBu)-S(tBu)C-OMe	Farn-Br (2)	DMF, rt, 1–2 h	basic DIPEA (8 equiv)	18	[49]
H-Cys-OMe	Farn-Br (1)	DMF, 0 °C, 30 min	basic TEA (2.5 equiv)	75	[54]
KCVLS	Farn-Cl (1.8)	liq NH_3, reflux, 2 h	basic	> 95	[41]
H-Cys-OH	Farn-Cl (1)	4 M NH_3 in MeOH, 0 °C for 3 h then 20 °C for 1 h	basic	80	[41,45]
Dns-WDPACVIA	Farn-Br (3)	DMSO/DMF/MeCN (3:3:1), 0 °C, 3h	basic DIPEA (8 equiv)	86	[42]
H-Ala-Cys-OMe	Farn-Br (4)	DMF/MeCN 1:2.5, rt, 1 h	neutral KF·$2H_2O$ (cat.)	82	[48]
Fmoc-RERC-OMe	Farn-Br (2)	DMF/MeCN, rt, 1 h	neutral KF·$2H_2O$ (cat.)	n.r.	[36]
H-Ala-Cys-OMe	Farn-Br (1)	DMF/MeCN/0.025% aq TFA (2:1:1), 0 °C, 1 h	acidic TFA $Zn(OAc)_2·2H_2O$ (cat.)	88	[43]
YIIKGVFWDPACV-IA	Farn-Br (4)	DMF/BuOH/0.025% aq TFA (2:1:1), rt, 1 h	acidic TFA $Zn(OAc)_2·2H_2O$ (cat.)	74	[43]

[a] n.r. = not reported.

H-Cys(Farn)-OH by Farnesylation in Aqueous Media; Typical Procedure:[56]

To H-Cys-OH (150 mg, 1.24 mmol) and freshly prepared Farn-Br[60] (200 mg, 0.70 mmol) in BuOH/MeOH/H_2O (1:1:1; 20 mL) guanidinium carbonate (150 mg) was added. After 18 h at rt, H_2O (20 mL) was added and the mixture extracted with BuOH (3×10 mL). Removal of the solvent gave the crude product which was recrystallized from aq MeOH; yield: 100 mg (44%); mp 148–150 °C (dec).

H-Cys(Farn)-OMe by Farnesylation in Organic Media with DIPEA as Base; Typical Procedure:[49]

H-Cys-OMe·HCl (1 g, 5.9 mmol) was dissolved in degassed, argon-sat. DMF (5 mL) and treated with DIPEA (0.973 mL, 7 mmol). Farn-Br (1.7 g, 5.9 mmol) in argon-sat. EtOAc (3 mL) was added in one portion, and the brown soln was stirred for 1 h. After addition of another equivalent of DIPEA (0.820 mL, 5.9 mmol) and 1 h stirring, the solvent was removed and the residue was taken up in EtOAc and washed several times with H_2O. Drying ($MgSO_4$) and removal of the solvent gave the pure product as a brown oil; yield: 1.89 g (95%); ES-MS *m/z*: 339 [M+H$^+$].

H-Ala-Cys(Farn)-OMe by Farnesylation in Organic Media with Potassium Fluoride as Base; Typical Procedure:[48]

(Boc-Ala-Cys-OMe)$_2$ (338 mg, 0.554 mmol) was treated with 50% TFA in CH_2Cl_2 (10 mL) for 30 min and the soln was concentrated. The residue was taken up in 90% AcOH (10 mL) and Zn powder (1 g) was added with stirring. After 1 h, the Zn was filtered off and the soln lyophilized. The product was dissolved in DMF (2 mL) and MeCN (5 mL) and reacted with Farn-Br (316 mg, 1.11 mmol) and

for references see p 370

$KF \cdot 2H_2O$ (1 g). The soln was stirred for 1 h and diluted with EtOAc and H_2O. The organic phase was washed with 1% aq TFA and H_2O, dried ($MgSO_4$), and concentrated. The residue was chromatographed (silica gel, $CHCl_3$/MeOH/AcOH 85:10:5); yield: 0.37 g (82%).
Using the same method, the geranylgeranyl and isoprenyl derivatives were produced in 84 and 88% yield, respectively.[48]

H-Ala-Cys(Farn)-OMe by Farnesylation in Aqueous Organic Media with Zinc Acetate at pH 4; Typical Procedure:[43]
To an ice-cold soln of H-Ala-Cys-OMe (10 mg, 50 μmol) and $Zn(OAc)_2 \cdot H_2O$ (33 mg, 150 μmol) in DMF/MeCN/0.025% aq TFA (2:1:1; pH 4; 4 mL) was added dropwise over a few min Farn-Br (14.3 mg, 50 μmol) and the mixture was stirred for 1 h. The mixture was diluted with EtOAc, washed with H_2O, and dried ($MgSO_4$). The crude product was purified by flash chromatography (silica gel); yield: 20 mg (88%).

Fragment Coupling To Give H-Tyr-Ile-Ile-Lys-Gly-Val-Phe-Trp-Asp-Pro-Ala-Cys(Farn)-OMe; Typical Procedure:[47]
The dipeptide derivative Fmoc-Ala-Cys(Farn)-OMe[47] [17 mg, 27 μmol; obtained after coupling of Fmoc-Ala-OH with H-Cys(Farn)-OMe using DCC/HOBt] was reacted with 5% piperidine in DMF (5 mL) at rt for 30 min then the soln was concentrated to dryness. The residue was dissolved in DMF (2 mL) and added to a soln of Fmoc-Tyr-Ile-Ile-Lys(Fmoc)-Gly-Val-Phe-Trp-Asp(OFm)-Pro-OH (50 mg, 27 μmol) in DMF (3 mL) at 0 °C followed by BOP (12 mg, 27 μmol) and DIPEA (100 μl). After stirring at 0 °C for 1 h, the reaction was terminated and the Fmoc/OFm groups were removed by adding piperidine (0.5 mL). After stirring for 1 h, the soln was added to Et_2O (100 mL) and the precipitate was filtered off and washed with Et_2O. The solid was dissolved in MeOH (5 mL), acidified with 1% aq TFA (2 mL) and purified by RP-HPLC with a linear gradient of 0 to 70% MeCN in 0.025% aq TFA over 60 min. The product was pooled and lyophilized; yield: 14 mg (28%).

H-Leu-Pro-Cys(Farn)-OMe by Enzymatic Cleavage of the Protecting Group; Typical Procedure:[50]
To a soln of Z(OAc)-Leu-Pro-OH (0.63 g, 1.5 mmol) and of H-Cys(Farn)-OMe (0.5 g, 1.5 mmol) in CH_2Cl_2 (10 mL) under argon was added a soln of HOBt (0.4 g, 3 mmol) and EDC (0.34 g, 1.8 mmol) in CH_2Cl_2 (10 mL). The soln was stirred for 24 h at rt, then it was extracted with 1 M HCl, 1 M $NaHCO_3$, and H_2O, and then dried ($MgSO_4$). The solvent was removed and Z(OAc)-Leu-Pro-Cys(Farn)-OMe was isolated from the residue by flash chromatography (silica gel, hexane/EtOAc 1:1); yield: (94%). Z(OAc)-Leu-Pro-Cys(Farn)-OMe (20 mg, 27 μmol) was dissolved with ultrasonication in a mixture of 0.2 M KI soln (pH 5, 160 mL) and MeOH (40 mL). Next, 25 units of lipase from *M. miehei* was added, and the mixture was stirred at 37 °C and at pH 5 for 12 h. The soln was extracted with CH_2Cl_2 and the organic layer was dried ($MgSO_4$). The solvent was removed, and the product was isolated from the residue by flash chromatography (silica gel, EtOAc/MeOH 4:1); yield: 9.6 mg (65%); MS *m/z*: 549.357 $[M+H]^+$.

H-Leu-Pro-Cys(Farn)-OMe by Palladium(0)-Mediated Cleavage of the Protecting Group; Typical Procedure:[55]
At 0 °C, a soln of EEDQ (793 mg, 3.03 mmol) in CH_2Cl_2 (10 mL) was added to a soln of H-Cys(Farn)-OMe (640 mg, 1.89 mmol) and the dipeptide Aloc-Leu-Pro-OH (500 mg, 1.60 mmol) in CH_2Cl_2 (25 mL). The resulting mixture was stirred for 12 h at rt. The soln was extracted with 1 M HCl (10 mL), $NaHCO_3$, and H_2O. The organic layer was dried ($MgSO_4$) and concentrated. The N-protected S-farnesylated tripeptide was purified by flash chromatography (silica gel, hexane/EtOAc 2:1); yield: 98%. Morpholine (10 mg, 0.115 mmol) and $Pd(PPh_3)_4$ (25 mg, 0.02 mmol) were added to a soln of the tripeptide (45 mg, 0.071 mmol) in THF (10 mL), and the mixture was stirred under argon in the dark for 1 h at rt. The solvent was removed and the product was purified by flash chromatography (silica gel, hexane/EtOAc 1:4); yield: 34.3 mg (88%); MS *m/z*: 549.358 $[M+H]^+$.

6.4.1.2.1.2 Postsynthetic S-Prenylation of Peptides

An alternative and possibly more simple approach to S-prenylation is based on S-alkylation of suitably protected or unprotected peptides. This method is widely applied particularly when small amounts of S-prenylated peptides are required, whereby preparative RP-HPLC is generally applied for the isolation of the lipopeptides. The S-alkylation methods used are generally those described above. Whilst with suitably protected peptides solubility represents a major problem, which can be bypassed by S-alkylation of the peptide on resin,[61] with unprotected peptides a very efficient tuning of the reaction conditions is required

to allow for regiospecific thiol alkylation because alkyl halides can easily react with other functional groups present in the peptide chain. When an excess of alkyl halide is used, the formation of sulfonium ions is possible, which can also be responsible for the alkylation of other functional groups. Many side products are normally obtained when using bases, particularly in the case of unprotected peptides.[40–42,49,57,62] Therefore, S-prenylation at neutral or acidic pH is more regioselective and unprotected peptides can be used.[36,37,43,44] In all cases the reaction has to be performed in the absence of air/oxygen to prevent concomitant oxidation to disulfides, and generally the use of anhydrous solvents contributes to higher yields.

Dns-Trp-Asp-Pro-Ala-Cys(Farn)-Val-Ile-Ala-OH by Farnesylation in Anhydrous Media; Typical Procedure:[42]
The peptide derivative Dns-Trp-Asp-Pro-Ala-Cys-Val-Ile-Ala-OH (0.01 M) was reacted in DMSO/DMF/MeCN (3:3:1) with Farn-Br (3 equiv) in the presence of DIPEA (8 eq) at 0 °C under an argon atmosphere for 3 h. The mixture was cooled to 0 °C, was acidified with AcOH to pH 4, and lyophilized. The product was isolated by RP-HPLC on a C-18 column; yields: >80%; FAB-MS *m/z*: 1311.8 $[M+H]^+$.

H-Tyr-Ile-Ile-Lys-Gly-Val-Phe-Trp-Asp-Pro-Ala-Cys(Farn)-Val-Ile-Ala-OH by Farnesylation in Acidic Aqueous Organic Media; Typical Procedure:[43]
To a soln of H-Tyr-Ile-Ile-Lys-Gly-Val-Phe-Trp-Asp-Pro-Ala-Cys-Val-Ile-Ala-OH (10 μmol) in DMF/BuOH/0.025% aq TFA (2:1:1, 2 mL) was added Farn-Br (2.85 mg, 40 μmol) and dropwise over 2 min a 1 M soln of $Zn(OAc)_2 \cdot 2H_2O$ (50 μL, 50 μmol). The turbid mixture was stirred vigorously for 1 h and then the reaction was quenched by addition of β-ME (200 μL). After 3 h, the product was purified by RP-HPLC on a C-18 column using a linear gradient of 0.025% aq TFA and 0.025% aq TFA in MeCN from 0 to 30% MeCN in 60 min; yield: 74%.

6.4.1.2.1.3 Enzymatic S-Prenylation of Peptides

Small amounts of farnesylated or geranylgeranylated peptides can be obtained by enzymatic methods, whereby the C-terminal sequence (Table 1) for recognition of the peptides as substrates by the farnesyltransferase and geranygeranyltransferase I is required. This enzymatic approach is particularly useful for the synthesis of [^{3}H]-labeled derivatives.

H-Lys-Lys-Ser-Lys-Lys-Thr-Lys-Cys(Farn)-Val-Ile-Met-OH:[63]
The peptide substrate (400 pmol dissolved in 10 mM DTT immediately prior to use), [1-^{3}H]farnesyl diphosphate (200 pmol; 15 Ci·mmol^{-1}) and farnesyltransferase (~200 units enzyme, specific activity 100–300 nmol·h^{-1}·mg^{-1} protein) were mixed in buffer [30 μL; 50 mM Tris·HCl (pH 7.7)/20 mM KCl/5 mM $MgCl_2$/2 mM DTT]. The mixture was incubated for 20 min at 37 °C, then 0.1% octylglucoside was added and the reaction was allowed to proceed for 60 min. The product was purified by RP-HPLC on a C-18 column using a linear gradient of 10 to 100% MeCN in 0.1% aq TFA over 30 min. Typical yield of the reaction is about 90%.

6.4.1.3 O- and S-Palmitoylation

As discussed for N-myristoylation and S-prenylation, even S-acylation of proteins with a fatty acid which in the vast majority of cases is the $C_{16:0}$ palmitic acid, plays a fundamental role in the cellular signal-transduction process (Table 1).[2–5,14] While N-myristoylation and S-prenylation are permanent protein modifications due to the amide- and sulfide-type linkage, the thioester bond between palmitic acid and the peptide chain is rather labile and palmitoylation is referred to as a dynamic modification.[64] This reversibility plays a crucial role in the modulation of protein functions since the presence or absence of a palmitoyl chain can determine the membrane localization of the protein and can also be used to regulate the interactions of these proteins with other proteins. Furthermore, a unique consensus sequence for protein palmitoylation has not been found, in contrast to the strict consensus sequences required for N-myristoylation and S-prenylation. Palmitoylation can occur at N- or C-terminal parts of the polypeptide chain depending on the protein family and often coexists with other types of lipidation (see Section 6.4.1.4). Given the diversity of protein sequences

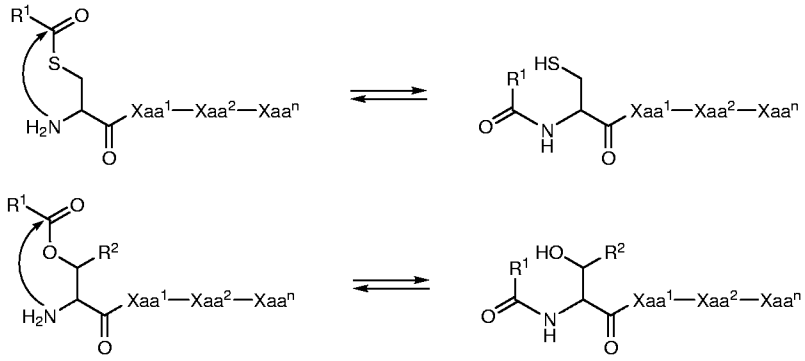
for references see p 370

that are palmitoylated, it is likely that more than one enzyme is involved with varying substrate specificity. As well as the S-palmitoylated proteins involved in signaling pathways,[14,64–66] there are other examples of natural S-palmitoylation such as the surfactant protein C (SP-C),[67] a 35-residue peptide which contains two thioester-linked palmitoyl groups. It has also been demonstrated that palmitoylation can also occur through an ester linkage with the hydroxy group of a threonine residue, two such cases are the myelin lipophilin[15] and a toxin isolated from *Plectreurys* spider venom (PLTX-II).[16]

6.4.1.3.1 Synthesis of S-Palmitoylated Peptides

S-Palmitoylation of a cysteine residue is generally performed via the acid chloride. Due to the base-lability of the thioester bond, particular precautions have to be taken in subsequent synthetic steps. The thioester bond is rapidly hydrolyzed by bases at pH $\leqslant 7$ in aqueous media,[55] but it is stable under the acidic conditions normally used in deprotection and resin-cleavage steps. An additional difficulty arises from the high tendency of such thioesters to undergo an S→N acyl shift when the *S*-palmitoylcysteinyl-peptide is deprotected at its amino group, as shown in Scheme 3.[68–70] To bypass this side reaction, palmitoylation is carried out post-synthetically; alternatively, suitably protected aminoacylcysteinyl peptides are used as synthons.[32,50,54]

Scheme 3 Mechanisms of S→N and O→N Acyl Shift[68–72]



6.4.1.3.1.1 Synthesis in Solution

To prevent the S→N acyl shift, S-palmitoylation with palmitoyl chloride is performed in anhydrous solvents on N^{α}-aminoacylated cysteines, e.g. dipeptide building blocks, whereby the residual functionalities of the peptide have to be protected taking into account the base-lability of the thioester bond.

Boc-Gly-Cys(Plm)-OAl:[32]
To a soln of (Boc-Gly-Cys-OAl)$_2$ (1.00 g, 1.57 mmol) in CH_2Cl_2 (50 mL) was added DTT (1.21 g, 7.85 mmol) and TEA (0.44 mL, 3.14 mmol). The mixture was stirred at 20 °C for 1 h, was washed twice with H_2O (25 mL), and dried ($MgSO_4$). To the crude Boc-Gly-Cys-OAl in CH_2Cl_2 was added TEA (0.44 mL, 3.14 mmol) and DMAP (cat.) and Plm-Cl (2.15 g, 7.85 mmol). After stirring the mixture at 20 °C for 1 h, the soln was concentrated and the product was isolated by flash chromatography (silica gel, hexane/EtOAc 70:30); yield: 1.22 g (70%); mp 39–40 °C; EI-MS *m/z*: 556.4 $[M+H]^+$.

6.4.1.3.1.2 Synthesis on Solid Supports

Due to the S→N acyl shift, the stepwise synthesis using a S-palmitoylated cysteine derivative is not advisable. A more convenient approach is to assemble the peptide sequence and to acylate the cysteine thiol group upon its selective deprotection, retaining, however, the protection of other reactive groups. For this purpose the Cys(StBu) and Cys(Acm) deriv-

atives are used, but other protections can be envisaged depending upon the overall protection strategy applied.[73,74] Using the Cys(StBu) derivative, the double palmitoylated surfactant protein C (SP-C)[16] was synthesized on solid support and, upon reductive cleavage of the StBu groups with β-ME, selective S-palmitoylation of the resin-bound peptide was performed with a large excess of palmitoyl chloride in the presence of pyridine and DMAP as catalyst.[75]

Ac-Trp-Lys-Thr-Pro-Gly-Cys(Plm)-Val-Lys-Ile-Lys-Lys-Ala-OH:[76]
The peptide was assembled using the Fmoc/tBu strategy on UltraSyn-KA resin. The side-chain protections were: Lys(Boc), Thr(tBu), and Cys(Acm). After acetylation of the N-terminal amino group, the peptide-resin (100 mg) was suspended in 30% aq AcOH (2 mL) and $Hg(OAc)_2$ (2 equiv) was added. The mixture was shaken gently for 1 h at rt, then β-ME (20 equiv) was added to precipitate the excess of $Hg(OAc)_2$ as sulfide. Shaking was continued for 1 h and after decantation of the solvent and of suspended HgS, the resin was washed with *tert*-amyl alcohol, Et_2O, and DMF. The resin was then suspended in DMF and Plm-Cl (2 equiv) was added followed by DIPEA (2 equiv). After 12 h, the resin was washed with $CHCl_3$ and Et_2O and dried. The fattyacylated peptide was cleaved from the resin using TFA/thioanisole/*m*-cresol/EDT (10:1:1:1). Removal of the TFA followed by trituration with Et_2O, gave the peptide; yield: 8 mg (70%). The peptide was purified by FPLC using a Pharmacia PepRC 5/5 RP column and a shallow gradient of 0.1% aq TFA (A) and 0.1% TFA in MeCN (B) starting from 100% A to 85% A in 8 min followed by isocratic (85% A) elution for 22 min with a flow rate of 0.5 mL·min^{-1}. The yield of the purified peptide was 65% from the crude.

6.4.1.3.2 Synthesis of O-Palmitoylated Peptides

Similarly to the S→N shift occuring in S-acylated cysteine residues, the O→N shift of O-acylated threonine and serine residues (Scheme 3) is a well-established side reaction in peptide chemistry[71,72] that occurs preferentially in alkaline media, but was also observed under acidic conditions.[77] In the synthesis of the toxin PLTX-II, from the *Plectreurys* spider venom, and related C-terminal fragments containing an O-palmitoylated threonine residue, this was fully confirmed.[16,78,79] Tertiary bases such as DIPEA were not found to induce significant O→N acyl shift over a wide pH range (5 to 9), whilst piperidine leads to formation of over 40% N^{α}-Plm-Thr-NH_2 from H-Thr(Plm)-NH_2 in 1 h[79] as reported also by Fields et al.[78] Thus, the Boc strategy is more suitable, although problems were encountered in the HF-cleavage step.[79] A comparative study between incorporation of the palmitoylated residue during stepwise solid-phase synthesis and on-resin palmitoylation clearly revealed the advantage of the latter method[78] as already observed for S-palmitoylated peptides.

Fmoc-Thr(Plm)-OH:[78]
Fmoc-Thr-OH (5.0 g, 14.7 mmol) was reacted with Plm-Cl (10.0 g, 36.4 mmol) and pyridine (3 mL, 38.7 mmol) in CH_2Cl_2 (100 mL) for 5 h. Upon concentration and cooling, a byproduct was filtered off (PA), the filtrate concentrated to dryness and the residue recrystallized (MeOH/hexane 1:9); yield: 2.9 g (34%).

H-Cys(β-4-pyridylethyl)-Asp-Thr(Plm)-NH_2:[78]
Boc-Cys(β-4-pyridylethyl)-Asp(OtBu)-Thr-Rink-resin (~0.1 mmol) was treated with PA (0.256 g, 1.0 mmol), DIC (0.157 mL, 1.0 mmol), and DMAP (0.122 g, 1.0 mmol) in CH_2Cl_2/DMF (1:1) for 2 h. The peptide was cleaved from the resin with TFA/H_2O (95:5) and HPLC analysis of the crude product showed 95% purity. ES-MS *m/z*: 680.4 $[M+H]^+$.

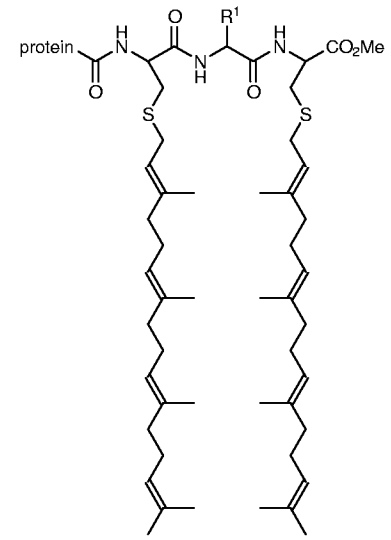
6.4.1.4 Dual Lipidation

It is well-established that in the case of short lipid chains such as farnesyl and myristoyl, the binding affinity of monolipidated proteins to membranes, is not sufficient to promote stable association.[25,26,36,80] Among the various intracellular proteins that are covalently modified by an acyl group from a fatty acid or a prenyl group, several carry more than one such hydrophobic substituent per protein molecule. The addition of a second hydrophobic chain to a

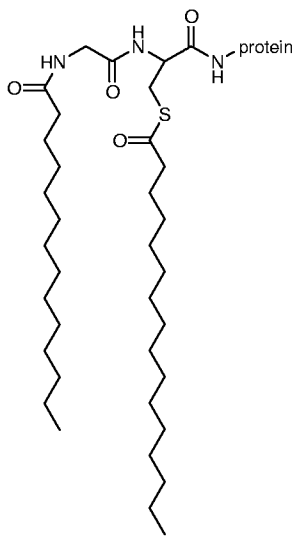
for references see p 370

protein shifts the membrane–cytoplasma equilibrium of the modified protein in a decisive manner in favor of the membrane phase.[37,38] Various double lipidations have been characterized, and several double-lipid combinations have been found, e.g. N- and S-acylation with a fatty acid, S-farnesylation and S-palmitoylation, S- and S-geranylgeranylation,[2–5] or S- and S-palmitoylation (Scheme 4).[67]

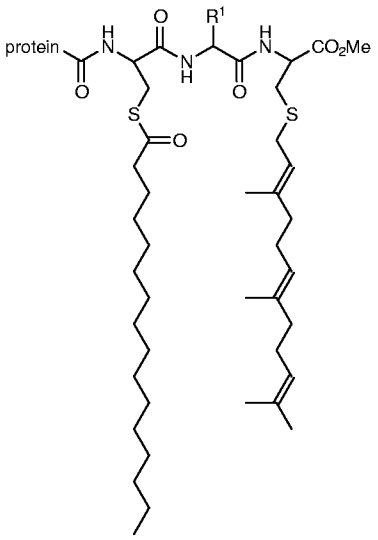
Scheme 4 Natural Double-Lipidation Motifs



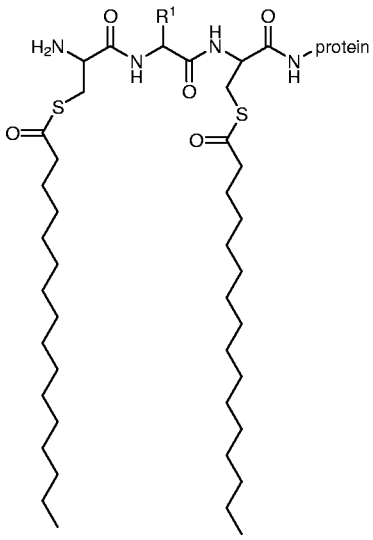
di-*S*-geranylgeranyl protein



N-myristoyl/*S*-palmitoyl protein



S-farnesyl/*S*-palmitoyl protein



di-*S*-palmitoyl protein

6.4.1.4.1 Synthesis of Doubly Lipidated Peptides

For the synthesis of doubly farnesylated and geranylgeranylated peptides the methods used for monoprenylation can be applied as for the postsynthetic selective alkylation of the two cysteine residues (see Section 6.4.1.2.1.2).[37] When different types of lipidation are present in

the same polypeptide chain, e.g. *S*-farnesyl and *S*-palmitoyl derivatives, the chemistry to be applied has to take into account the characteristics of the sulfide and thioester bonds.[21,50,54,55]

The difficulties resulting from the limited number of suitable protecting groups and synthetic strategies proposed have been extensively rewiewed by Waldmann et al.[81–83] The synthetic route is usually based on the condensation in solution of small monolipidated, suitably protected fragments using EDC/HOBt as the condensing agent in the absence of base. Protecting groups that are removed under mild chemoenzymatic conditions (pH = 6–8, rt) have to be used in order to preserve the integrity of the acid-/base-labile *S*-farnesyl and *S*-palmitoyl derivatives, respectively, or by applying noble-metal-mediated transformations which also proceed under virtually neutral conditions.[50,84–87]

For carboxy protection the allyl ester is used which is removed by tetrakis(triphenylphosphine)palladium(0) and morpholine (see Vol. E 22a, Section 2.2.2.7); these are conditions where the *S*-palmitoyl group is stable.[88] Alternatively, or in combination, the enzyme-cleavable choline ester can be applied, if the double-lipidated peptide is sufficiently soluble in aqueous media as required for the enzymatic-cleavage step (see Vol. E 22a, Section 2.5.3.1.1).[24,32,54,55] Again, for amino protection enzyme-removeable protecting groups such as the 4-acetoxybenzyloxycarbonyl [Z(OAc)] or 4-phenylacetoxybenzyloxycarbonyl [Z(OAcPh)] groups are used (for their cleavage see Vol. E 22a, Sections 2.5.1.2 and 2.5.1.3), as well-demonstrated in the synthesis of C-terminal *N*-Ras peptides[50] and N-myristoylated and doubly S-palmitoylated peptides of nitric oxide synthase.[86] To increase the solubility of lipidated intermediates for enzyme-mediated cleavage, addition of organic solvents as well as of dimethyl-β-cyclodextrin has been recommended.[54,87] Again, as an alternative, the Aloc group has found interesting applications in this context.[54,84]

Z(OAc)-Gly-Cys(Plm)-Met-Gly-Leu-Pro-Cys(Farn)-OMe:[50]
To a soln of H-Met-Gly-Leu-Pro-Cys(Farn)-OMe (50 mg, 67 μmol) in DMF (5 mL) at 0 °C under argon, was added Z(OAc)-Gly-Cys(Plm)-OH (41 mg, 67 μmol), EDC (19 mg, 100 μmol), and HOBt (18 mg, 130 μmol). The mixture was stirred for 12 h at rt, the solvent was removed and the residue dissolved in CH_2Cl_2. The soln was extracted with 0.5 M HCl, the organic layer was dried ($MgSO_4$), and the solvent removed under reduced pressure. The product was isolated by flash chromatography (silica gel, hexane/EtOAc/MeOH 5:5:1); yield: 27 mg (61%).

H-Gly-Cys(Plm)-Met-Gly-Leu-Pro-Cys(Farn)-OMe:[50]
A soln of Z(OAc)-Gly-Cys-(Plm)-Met-Gly-Leu-Pro-Cys(Farn)-OMe (8 mg, 6 μmol) and dimethyl-β-cyclodextrin (720 mg, 600 μmol) in MeOH (5 mL) was diluted with 20 mM phosphate buffer, pH 6 (100 mL). Then acetylesterase (8 units) was added, and the mixture was stirred at 37 °C and at pH 6 for 24 h. The soln was extracted with $CHCl_3$ (4 × 50 mL) and the organic layer was dried ($MgSO_4$). The solvent was removed and the product was isolated by flash chromatography (silica gel, EtOAc/MeOH 9:1); yield: 2.5 mg (35%); FAB-MS *m/z*: 1135 $[M+H]^+$.

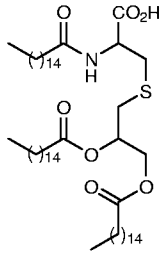
6.4.1.5 Palmitoylated *S*-Glycerylcysteine Derivatives

The lipoproteins from the outer envelope of Gram positive or Gram negative bacteria both have as their N-terminal amino acid the *S*-glycerylcysteine residue which is palmitoylated at the α-amino group via an amide linkage and at the two hydroxy groups as esters (Scheme 5).[89] Since the discovery that the small lipopentapeptides derived from the N-termini of these lipoproteins act as immune modulators, and thus are able to replace the immunogenic function of Freunds adjuvant or the carrier proteins normally used to increase the immunogenicity of antigens,[90,91] lipopeptides containing such lipidated *S*-glycerylcysteine residues [Plm-Dhc$(Plm)_2$-OH, **1**, also known as Plm_3Cys] have encountered large interest in the field of immunology and vaccine production.[90–94] Essential for the immunogenic activity is the N-terminal lipo-amino acid and the minimal structure for its bioactivity is the Plm-Dhc$(Plm)_2$-Ser dipeptide portion; however, variations in the residual pentapeptide sequence are without effect.[93,94] Furthermore, the activity of these lipopeptides depends only mar-

for references see p 370

ginally on the length of the acyl chains attached to the *S*-glycerylcysteinyl moiety, but a reduction or increase in the total number of fatty acid acyl chains causes a decrease or total suppression of the activity.[95] Finally, the chirality of C2 of the glyceryl moiety plays an important role with the *R*-configured isomer being more active than the *S*-isomer.

Scheme 5 Structure of *N*-Palmitoyl-*S*-[2,3-bis(palmitoyloxy)propyl]cysteine[89]

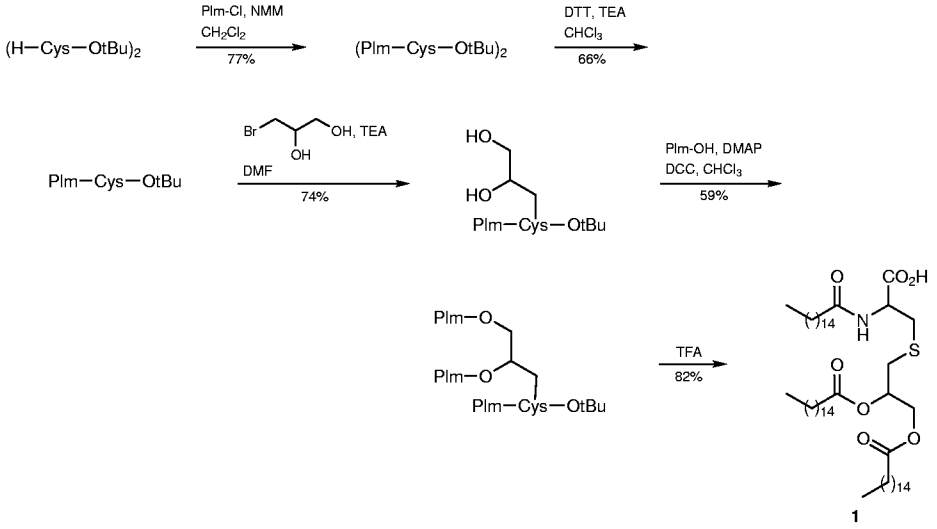


1

6.4.1.5.1 Synthesis of *N*-Palmitoyl-*S*-[2,3-bis(palmitoyloxy)propyl]cysteine

An early synthesis of *N*-palmitoyl-*S*-[2,3-bis(palmitoyloxy)propyl]cysteine employed cysteine methyl ester, however, this leads to difficulties in the saponification step of the tripalmitoylated residue.[96] The optimized procedure, in which the cystine di-*tert*-butyl ester is used,[90] is outlined in Scheme 6; after N-acylation with palmitoyl chloride, the ester is reduced to the cysteine derivative for S-alkylation with 1-bromopropane-2,3-diol to yield chirally defined isomers if optically pure bromo derivatives are used. Esterification of the hydroxy groups is best carried out with a 1.25-fold excess of palmitic acid, DCC, and DMAP. The use of a larger excess of palmitoyl chloride is not recommended due to purification problems. The diastereomeric mixture can be separated by silica gel chromatography using CH_2Cl_2/EtOAc (20:1) as eluent and the configuration was assigned by comparison with an optically pure sample obtained with (2*R*)-1-bromopropane-2,3-diol.

Scheme 6 Solution Synthesis of *N*-Palmitoyl-*S*-[2,3-bis(palmitoyloxy)propyl]cysteine[90]



1

(Plm-Cys-OtBu)$_2$:[90]
Plm-Cl (17.6 g, 64 mmol) and NMM (14.3 mL, 130 mmol) in CH_2Cl_2 (100 mL) were added dropwise under stirring to (H-Cys-OtBu)$_2$[90] (10.9 g, 31.0 mmol) in CH_2Cl_2 (50 mL). After 3 h at rt, H_2O (3 mL) was

added and the mixture concentrated to dryness. The residue was dissolved in CH_2Cl_2 (150 mL) and was washed with 5% citric acid, 5% $NaHCO_3$, and H_2O. After drying (Na_2SO_4), the solvent was removed and the residue crystallized (MeOH); yield: 19.8 g (77%); mp 76 °C.

Plm-Cys-OtBu:[90]
$(Plm\text{-}Cys\text{-}OtBu)_2$ (3.8 g, 4.58 mmol) was reduced in $CHCl_3$ (60 mL) with DTT (3 g, 19.45 mmol) in the presence of TEA (1.45 g, 14.35 mmol). After 2 h under N_2, the soln was washed with oxygen-free 5% citric acid and H_2O. After drying (Na_2SO_4) and removal of the solvent, the oily residue was crystallized (MeOH) (30 mL) at −20 °C; yield: 2.5 g (66%); mp 46–48 °C.

***N*-Palmitoyl-*S*-[2,3-dihydroxy-(2*R*,*S*)-propyl]-(*R*)-cysteine *tert*-Butyl Ester, Plm-Dhc-OtBu:**[90]
1-Bromo-2,3-propanediol (3.7 mL, 36.1 mmol) freshly distilled on a 20-cm Vigreux column (bp 86 °C, 50 Pa) and TEA (5 mL, 36.1 mmol) were added to Plm-Cys-OtBu (1.5 g, 3.61 mmol) in DMF (30 mL). After stirring for 15 min at 80 °C and 1 h at rt, the mixture was concentrated under reduced pressure. The oily residue was dissolved in $CHCl_3$ (100 mL), saturated with N_2, and washed with 5% citric acid and N_2-saturated H_2O. After drying (Na_2SO_4), the solvent was removed and the oily residue solidified at −20 °C with light petroleum ether; yield: 1.3 g (74%); mp 43 °C.

***N*-Palmitoyl-*S*-[2,3-bis(palmitoyloxy)-(2*R*,*S*)-propyl]-(*R*)-cysteine *tert*-Butyl Ester, Plm-Dhc$(Plm)_2$-OtBu:**[90]
PA (1.6 g, 6.24 mmol), DMAP (80 mg, 0.65 mmol), and DCC (1.3 g, 6.3 mmol) were added to a stirred soln of the diol (1.3 g, 2.65 mmol) in $CHCl_3$ (25 mL). After 2 h at rt, the mixture was diluted with $CHCl_3$ (50 mL) and the precipitated DCU was filtered off. The organic soln was washed with 5% citric acid, H_2O, 5% $NaHCO_3$, and H_2O. After drying (Na_2SO_4) and concentrating to dryness, the residue was crystallized ($MeOH/CHCl_3$ 5:1); yield: 1.5 g (59%); mp 60 °C.

***N*-Palmitoyl-*S*-[2,3-bis(palmitoyloxy)-(2*R*,*S*)-propyl]-(*R*)-cysteine, Plm-Dhc$(Plm)_2$-OH (1):**[80]
The *tert*-butyl ester (440 mg, 0.48 mmol) was treated with TFA (5 mL) at rt. After 1 h, the TFA was removed, the residue was treated with H_2O (10 mL), and the precipitate washed with cold H_2O and crystallized from $CHCl_3$ (3 mL) at −20 °C by addition of light petroleum ether; yield: 360 mg (82%); mp 63 °C.

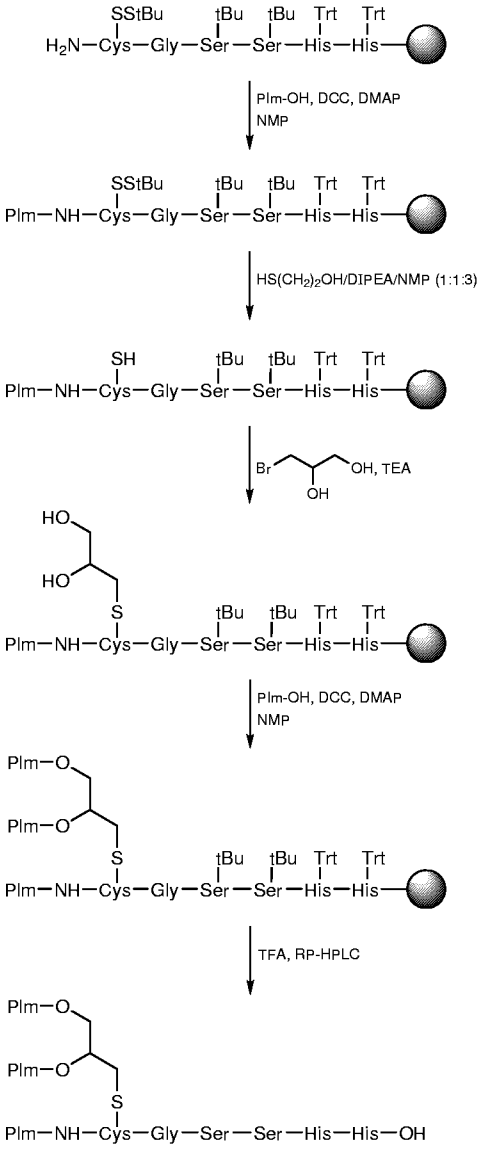
6.4.1.5.2 Synthesis of *N*-Fluorenylmethoxycarbonyl-*S*-[2,3-bis(palmitoyloxy)propyl]cysteine

For the synthesis of analogues of Plm-Dhc$(Plm)_2$-OH, N^{α}-Fmoc-protected *S*-glycerylcysteine was prepared,[97] this can be acylated with fatty acids directly on the resin, upon incorporation into the peptide, or in solution prior to its use.[97,98] Due to the presence of the base-labile Fmoc-group, the reduction of $(Fmoc\text{-}Cys\text{-}OtBu)_2$ is performed under acidic conditions with zinc. By using optically pure 2,3-epoxypropan-1-ol [(−)-(*S*)-glycidol; (+)-(*R*)-glycidol] the stereospecific S-alkylation leads to diastereomerically pure *N*-Fmoc-*S*-glycerylcysteine derivatives.[97] Alternatively, in solid-support peptide synthesis using Fmoc-Cys(StBu)-OH, upon cleavage of the thiol protecting group, S-alkylation is performed on the resin followed by acylation with a fatty acid (Scheme 7).[99]

The carba-analogue of Plm-Dhc$(Plm)_2$-OH was also synthesized.[95,100] The four possible diastereoisomers were all obtained by stereospecific synthesis via the construction of dide-hydro-amino acids using the phosphonoglycinate method and subsequent enantio- or diastereoselective hydrogenation.

for references see p 370

Scheme 7 Solid-Phase Synthesis of Tri(palmitoyl)-*S*-(2,3-dihydroxypropyl)cysteine-Containing Peptides by On-Resin S-Alkylation[99]



***N*-Fluorenylmethoxycarbonyl-*S*-[2,3-dihydroxy-(2*S*)-propyl]-(*R*)-cysteine *tert*-Butyl Ester, Fmoc-Dhc-OtBu:**[97]

To a soln of (Fmoc-Cys-OtBu)$_2$[97] (1.92 g, 2.4 mmol) in CH_2Cl_2 (15 mL) and Zn (1.1 g, 16.8 mmol) was added a freshly prepared soln of MeOH, 32% HCl (d 1.16), and concd H_2SO_4 (d 1.84) (100:7:1; 8 mL) under vigorous stirring. After 15 min (−)-(*S*)-glycidol (1.6 mL, 2.4 mmol) was added. The mixture was stirred for 5 h at 40 °C. The soln was concentrated and diluted with 5% $KHSO_4$ (2 mL). This mixture was kept at −4 °C for 16 h and then extracted with CH_2Cl_2. The organic phase was dried ($MgSO_4$) and concentrated to dryness; yield: 2.06 g (91%).

Using (+)-(*R*)-glycidol, the 2*R*-isomer is obtained; yield: 85%, and with racemic glycidol the (2*R*,*S*)-diastereomeric mixture; yield: 82%.

N-Fluorenylmethoxycarbonyl-S-[2,3-bis(palmitoyloxy)-(2S)-propyl]-(R)-cysteine *tert*-Butyl Ester, Fmoc-Dhc(Plm)$_2$-OtBu:[97]
To the above intermediate (2*S*-isomer; 1.21 g, 2.55 mmol), PA (2.1 g, 8.18 mmol), and DIC (1.53 mL, 9.86 mmol) in THF (25 mL) was added DMAP (128 mg, 1.045 mmol), and the mixture was stirred for 2 h. After addition of AcOH (1 mL), the mixture was concentrated to dryness. The residue was recrystallized (CH_2Cl_2/MeOH 1:3) (40 mL) at −20 °C; yield: 2.07 g (85%).

Analogously, the 2*R*-isomer; yield: 85%, and diastereomeric mixture; yield: 82%, were obtained by acylation of the corresponding diols.

N-Fluorenylmethoxycarbonyl-S-[2,3-bis(palmitoyloxy)-(2S)-propyl]-(R)-cysteine, Fmoc-Dhc(Plm)$_2$-OH:[97]
Cleavage of the *tert*-butyl ester group was performed with TFA at rt for 1h in quantitative yield.

6.4.1.5.3 Synthesis of Tri(palmitoyl)-*S*-(2,3-dihydroxypropyl)cysteine Peptides

As mentioned in Section 6.4.1.5.2, the N-terminal pentapeptide containing the Plm-Dhc(Plm)$_2$- derivative (the natural lipoprotein contains a mixture of fatty acids with palmitic acid present in larger amounts) coupled to Ser-Ser-Asn-Ala (the natural C-terminal sequence of murine lipoprotein from *E. Coli*) possesses properties similar to those of the intact protein and the activity is retained, though reduced, by reducing the size of the lipopeptide to Plm-Dhc(Plm)$_2$-Ser-OH. Among the many analogues synthesized, a water-soluble analogue, Plm-Dhc(Plm)$_2$-Ser-(Lys)$_4$-OH, a cationic lipopeptide, represents one of the most promising lipopeptides used in synthetic vaccines.[94]

Peptides containing N-terminal Plm-Dhc(Plm)$_2$- are synthesized in solution or on solid support using Plm-Dhc(Plm)$_2$-OH in the last coupling step. Some precautions are required in this coupling step due to the fact that Plm-Dhc(Plm)$_2$-OH is an N-acylated amino acid and thus may racemize in the presence of bases. The coupling is normally carried out in the absence of base, using a preactivation with DCC/HOBt or DCC/HOSu.[94] The use of cysteine derivatives with urethane-type protection, i.e. Fmoc-Dhc(Plm)$_2$-OH, lowers the tendency to racemize in the acylation step. As an alternative to the Fmoc- or Boc-protected derivatives, Troc-Dhc(Plm)$_2$-OH was proposed.[100,101] The Troc group is selectively removed with Zn in AcOH to yield the free amino group which is finally acylated. The Plm-Dhc(Plm)$_2$-OH residue is stable toward the common reagents used for SPPS such as piperidine, TFA, and HF.[94,97–99] Saponification of the fatty acid esters and partial removal of the *S*-glyceryl group were observed under strong basic conditions (1 M NaOH).[94] Plm-Dhc(Plm)$_2$-OH shows good solubility in typical coupling solvents, thus allowing for relatively high concentrations of this component. The solubility of the lipopeptides strongly depends on the amino acid sequence flanking the Plm-Dhc(Plm)$_2$- residue. Whilst the natural pentapeptide Plm-Dhc(Plm)$_2$-Ser-Ser-Asn-Ala-OH is almost insoluble in most solvents including DMSO, DMF, alcohols, and chlorinated solvents, Plm-Dhc(Plm)$_2$-Ser-(Lys)$_4$-OH shows a strong amphiphilic character. It is soluble in organic solvents such as $CHCl_3$ and DMF, but also in water.[94,97]

Solution Synthesis:

Plm-Dhc(Plm)$_2$-Ser(tBu)-Ser(tBu)-Asn-Ala-OtBu:[90]
To Plm-Dhc(Plm)$_2$-OH (337 mg, 0.37 mmol) in CH_2Cl_2 (10 mL), HOBt (50 mg, 0.37 mmol) in DMF (1 mL) and DCC (76 mg, 0.37 mmol) were added and the mixture was stirred for 30 min at 0 °C. Concomitantly, Fmoc-Ser(tBu)-Ser(tBu)-Asn-Ala-OtBu (284 mg, 0.37 mmol) was treated with piperidine/DMF (1:10; 10 mL) for 1 h at rt. After removal of the solvents, the residue was dissolved in CH_2Cl_2 (10 mL) and added to the activated Plm-Dhc(Plm)$_2$-OH. After 12 h at rt, the product was precipitated with MeOH (30 mL), filtered off, and washed with MeOH and light petroleum ether; yield: 430 mg (81%); mp 209 °C (dec).

Plm-Dhc(Plm)$_2$-Ser-Ser-Asn-Ala-OH:[90]
The protected pentapeptide (300 mg, 0.21 mmol) was treated with TFA (3 mL) for 1 h at rt. After removal of the acid, H_2O (20 mL) was added and removed again. The residue was dissolved in warm

for references see p 370

$CHCl_3$ (30 mL) and the product was precipitated with MeOH (30 mL); yield: 206 mg (77%); mp 214 °C (dec).

Solid-Phase Synthesis:

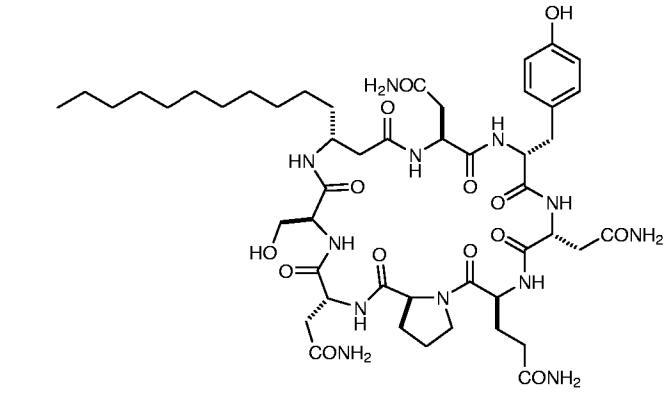
Fmoc-Dhc(Plm)$_2$-Ser-Lys-Lys-Lys-Lys-OH:[97]
Fmoc-Dhc(Plm)$_2$-OH (200 mg, 0.48 mmol) was activated in CH_2Cl_2/DMF (1:1; 5 mL) with *N*-hydroxynorborn-5-ene-2,3-dicarboxy imide (86 mg, 0.48 mmol) and DIPEA (74 µL, 0.48 mmol) at 0 °C. After 5 min, the mixture was added to H-Ser(tBu)-[Lys(Boc)]$_4$- linked to 4-benzyloxybenzyl alcohol-resin (0.5 g; 0.48 mmol · g^{-1}). After shaking for 2 h, the solvent was removed by filtration, and the resin washed with CH_2Cl_2 and DMF. The Fmoc-protected peptide-resin was treated with a mixture of TFA (5 mL) and thioanisole (0.5 mL) for 1 h. The resin was filtered off and the filtrate was concentrated to dryness. The Fmoc-protected hexapeptide was recrystallized (acetone) at −20 °C. By lyophilization from tBuOH, a powder was obtained; yield: 80 mg (87%).

Plm-Dhc(Plm)$_2$-Gly-Ser-Ser-His-His-OH:[99]
DCC (2.06 g, 10 mmol) and DMAP (0.123 g, 1 mmol) were added to PA (2.57 g, 10 mmol) in NMP (20 mL). The soln was added to H-Cys(StBu)-Gly-Ser(tBu)-Ser(tBu)-His(Trt)-His(Trt)- linked to 4-benzyloxybenzyl alcohol-resin (0.5 g; 0.2 mmol · g^{-1}) in NMP and the slurry was stirred at rt for 90 min. The precipitated DCU was dissolved with MeOH/CH_2Cl_2 and filtered off, and the peptide-resin was washed extensively with MeOH and CH_2Cl_2. The Cys(StBu) was deprotected using a mixture of β-ME/DIPEA/NMP (1:1:3, 20 mL) for 2 h at rt; then the peptide-resin was washed extensively with NMP, CH_2Cl_2, and MeOH, and dried. The peptide-resin was swollen in NMP (15 mL), then 1-bromopropane-2,3-diol (1.85 mL, 18.1 mmol) and TEA (2.5 mL, 18.1 mmol) were added. After stirring for 15 min at 80 °C and for 1 h at rt, the peptide-resin was washed with DMF, CH_2Cl_2, and MeOH. PA (0.8 g, 3.12 mmol), DMAP (40 mg, 0.325 mmol), and DCC (0.65 g, 3.15 mmol) were added to the resin in $CHCl_3$ (12.5 mL). After 2 h stirring at rt, the peptide-resin was washed with $CHCl_3$, CH_2Cl_2, and MeOH, and dried. The lipopeptide was cleaved from the resin with TFA/H_2O (95:5) for 90 min at rt. The crude product (0.113 g) was purified by HPLC on a PR-300 column (Applied Biosystems) using a gradient of 0.1% aq TFA (A) and 0.1% TFA in THF (B) starting from 50% B to 80% B in 40 min at flow rate 1.2 mL · min^{-1}; yield: 35 mg (25%); MALDI-MS *m/z*: 1413.9 $[M+H]^+$.

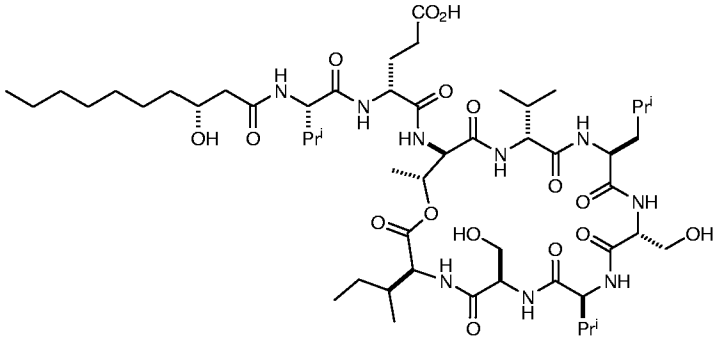
6.4.2 Lipopeptides from Nonribosomal Biosynthesis

An exceptionally rich source of structurally unique natural lipopeptides produced by microorganisms living in different habitats, is synthesized nonribosomally by peptide synthethases.[6–8,102] Generally, they show a large diversity of structures and properties due to the variable nature of both the fatty acid, that differs in length, double bond number, and possible ramification of the fatty acid chain, and of the peptide moiety.[103] A general feature of these peptides is the abundant presence of D-amino acids, N-methylated, and other noncoded amino acids. Another frequent feature is the absence of free, charged N- and C-termini. The N-terminal amino group may be acylated, while the C-terminal carboxy group is often involved in the formation of macrocyclic ring structures of the lactam (amide) or lactone (ester) type of different sizes with large lipophilic amino or hydroxy acids (α- or β-),[103] as shown with a few selected examples in Scheme 8. The most prominent representatives of these cyclic lipopeptides are cyclosporin with (4*R*)-4-[(*E*)-butenyl]-4,*N*-dimethyl-L-threonine as the lipo-amino acid[104,105] and the microcystins and nodularins with the β-amino acid (2*S*,3*S*,8*S*,9*S*)-3-amino-9-methoxy-2,6,8-trimethyl-10-phenyldeca-4,6-dienoic acid.[106] In the linear peptides, which are less frequent, the N-termini are acylated, sometimes with fatty acids, and the C-termini are esterified or reduced to α-amino alcohols.[103] Due to their amphiphilic character and their peculiar structures most of these peptides possess important pharmacological activities, e.g. as immunosuppressants (cyclosporin),[104] hepatotoxins (microcystins and nodularins),[106] biosurfactants (surfactin),[107] and antibiotics (iturin,[108] gramicidin K[109]).

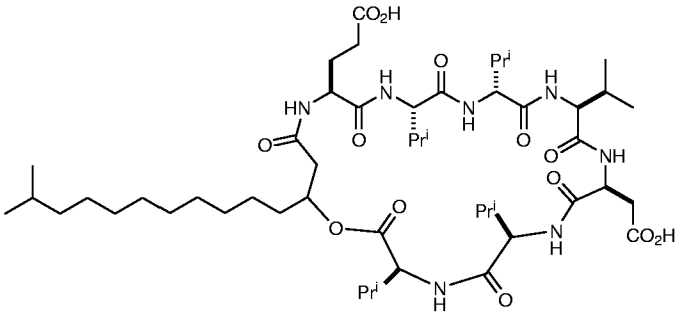
Scheme 8 Examples of Nonribosomal Lipopeptides[104,106–109]



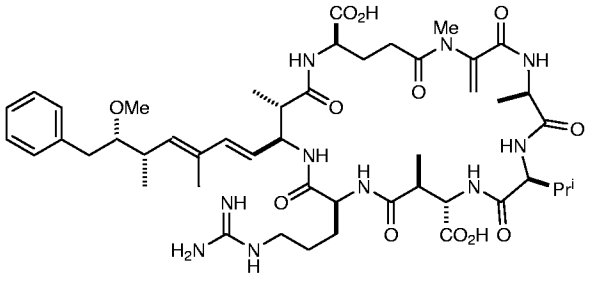
iturin-A2



viscosin

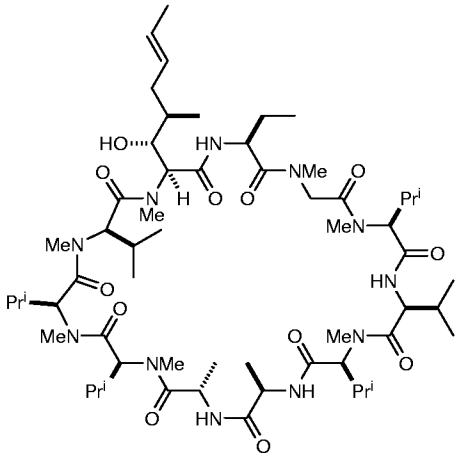


surfactin

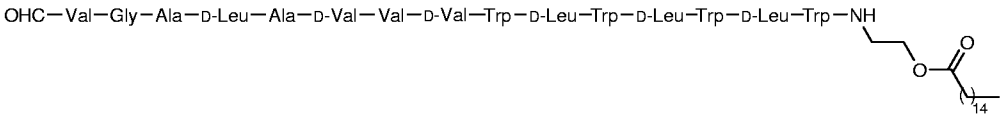


microcystin

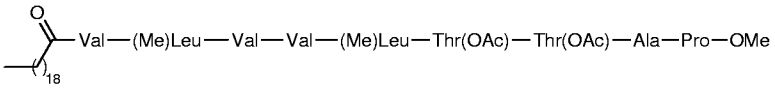
for references see p 370



cyclosporin A



gramicidin K



fortuitin

6.4.2.1 Synthesis of Nonribosomal Lipopeptides

The synthesis of these complex microbial molecules represents a challenging synthetic task and for many of these natural products it has not yet been achieved, although often it constitutes the only way to confirm or verify the proposed structure. The syntheses are normally hampered by the nature of the single building blocks such as the lipo-amino acids and some noncoded α- or β-amino and hydroxy acids. An accurate retrosynthetic analysis of the target molecule, with selection of the appropriate building blocks, represents the first required step. Once these target molecules are determined, they are prepared separately, in solution or on solid phase, for their final assembly into the microbial peptides. Two selected examples are described below; for more information regarding the synthetic strategies in the cyclization steps, see Section 6.8, and for additional examples of the synthesis of such natural products, see Vol. E 22d, Section 16.

6.4.2.1.1 Synthesis of Iturin-A2

Iturins are a group of fungicidal cyclic lipopeptides produced by *Bacillus subtilis*.[103,108,110] All members of this group are cyclic octapeptides with seven α-amino acids and a unique β-amino fatty acid homologue of iturinic acid. The predominant iturin-A isomer, A2, contains the n-C_{14}-isomer of iturinic acid (Itu) (Scheme 8). The synthesis of the molecule can be divided into two steps with the first stage consisting of the assembly of the linear octapeptide on solid phase followed by the backbone cyclization in solution.

H-Itu-Asn-D-Tyr-D-Asn-Gln-Pro-D-Asn-Ser-OH:[111]
The synthesis of the linear octapeptide was performed applying Fmoc/tBu chemistry and starting from 0.82 g (0.39 mmol·g^{-1}, 0.32 mmol) of Fmoc-Ser(tBu)-HMPAA-resin (Millipore). Iturinic acid was coupled as the *N*-Boc derivative. After deprotection and cleavage with TFA/H_2O (95:5), the product was purified by preparative RP-HPLC; yield: 37 mg (74%).

Iturin-A2:[111]
To a soln of the linear peptide (245 mg, 209 μmol) in DMF (150 mL) under an N_2 atmosphere and cooled to 0 °C, were added dropwise DPPA (49 μl, 229 μmol) and $NaHCO_3$ (87 mg, 1.04 mmol). After 8 days at 0 °C, H_2O (40 mL) and Bio-Rad AG501-X8 (D) mixed-bed resin (5 g) were added to the mixture. The mixture was stirred for 3 h, the resin was removed by filtration and washed with DMF/H_2O (4:1; 2×20 mL). The combined filtrates were concentrated; yield: 197 mg (91%) of crude product that was purified by preparative RP-HPLC.

6.4.2.1.2 Synthesis of Viscosin

The antibiotic viscosin from *Pseudomonas viscosa* (Scheme 8) is a cyclo-depsipeptide acylated at the N-terminus with D-3-hydroxydecanoic acid.[103,112] The C-terminal carboxy group is esterified with the hydroxy group of the Thr^3 residue. The synthesis of this natural product presents the problem of an ester ring closure and was performed in two steps with initial solid-phase synthesis of the Thr^3-O-branched linear intermediate followed by ring closure with formation of the amide bond.[113]

D-3-Hydroxydecanoyl-Leu-D-Glu(OBzl)-D-*allo*-Thr(H-Ile)-D-Val-Leu-D-Ser(Bzl)-Leu-D-Ser(Bzl)-OH:[114]
The synthesis was performed on Wang resin by standard Fmoc/tBu protocols up to the D-Val residue. Fmoc-D-*allo*-Thr-OH was coupled with unprotected hydroxy group to allow for its esterification with Boc-Ile-OH. To avoid the well-known O→N acyl shift under the conditions of Fmoc deprotection, the unprotected Thr residue was first N-acylated with Fmoc-D-Glu(OBzl)-OPfp as Pfp esters are known to react rapidly with amines but sluggishly with alcohols. Then the Thr residue was O-acylated with Boc-Ile-OH using DIC (1 equiv) in the presence of DMAP (0.1 equiv). The synthesis was completed by coupling the remaining residues, Fmoc-Leu-OH with DCC/HOBt and the fatty acid with unprotected hydroxy group as a Pfp ester. Upon deprotection and cleavage from the resin with TFA/CH_2Cl_2 (1:1) for 30 min, the crude product was purified by preparative RP-HPLC using a C-18 column and a H_2O/MeCN gradient system; yield: 25%.

Cyclization to Viscosin:[114]
Cyclization was carried out at 1 mM peptide concentration in DMF with bis(2-oxo-3-oxazolidinyl)phosphinic chloride in the presence of TEA. The cyclic product was isolated by RP-HPLC; yield: 24%. Deprotection was carried out by hydrogenolysis in MeOH over 10% Pd/C with ammonium formate.[115] The product was isolated by RP-HPLC; yield: 78%.

6.4.3 Nonnatural Mono- and Multiple-Lipidation of Peptides

To mimic nature in its irreversible monolipidation of proteins by N-myristoylation and S-prenylation, peptides are N-acylated with fatty acids or N-alkylated with linear or branched, saturated or unsaturated aliphatic chains as well as by incorporation of newly synthesized lipo-amino acids.

6.4.3.1 Acylation of Peptides with Fatty Acids

The acylation of peptides with fatty acids is easily achieved by acylation of a free amino group that can be N-terminal or side-chain located. After the discovery that short tripalmitoylated peptides can elicit strong immunological responses against the peptide,[90,91]

for references see p 370

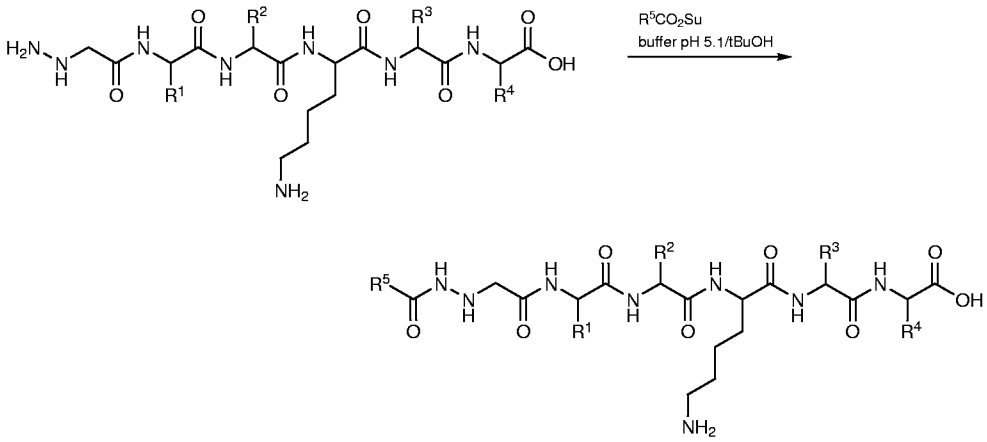
further studies demonstrated that even simple palmitoylation of peptide immune epitopes results in increased immune responses comparable to those induced by the addition of adjuvants.[88,114,116,117] Although the mechanism of the enhanced immunogenicity is still poorly understood, the tendency of the peptide amphiphiles to aggregate may play a key role.[118,119]

In the absence of a strict functional role, the fatty acid acyl group is generally introduced at the N- or C-terminal part of the peptide sometimes using spacer groups. This acylation step is performed in solution or on resin as a final synthetic step, either using the acyl chlorides of fatty acids or fatty acids and coupling reagents.[116,117] Double acylation of N-terminal lysine residues with fatty acids has also been reported.[118–120] For selective side-chain lipidation of Lys residues, the Fmoc derivative is used for orthogonal ε-amino protection via the Boc strategy, whereas the Dde or Aloc derivatives are used in the Fmoc/tBu strategy.

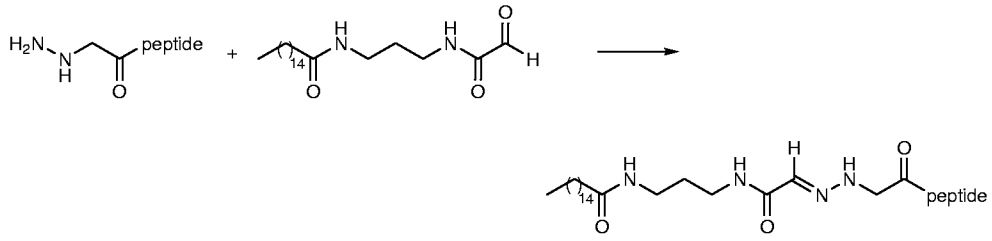
An interesting new approach, which relies on the chemoselective acylation of α-hydrazinoacetyl derivatives of unprotected peptides, has been proposed.[121] A pK_a of 6.45 was determined for the hydrazino group which is about two pK_a units below that of an α-amino group and thus if a nucleophile is introduced that allows for discrimination among the different nucleophiles present in the unprotected peptide (Scheme 9). For N-terminal incorporation of the α-hydrazino acetyl moiety (Boc)NHN(Boc)-CH_2CO_2H or $(Boc)_2$NN-(Boc)-CH_2CO_2H is efficiently coupled with HBTU/HOBt or the BOP reagents. While the N–N bond is partially unstable to HF, it is stable to TFA as used in the Fmoc/tBu strategy in the final cleavage and deprotection steps.[122] Upon chromatographic purification, selective N^{β}-acylation via fatty acid active esters is performed at pH 5.1 in buffer/*tert*-butanol mixtures. The α-hydrazino acids are known to be regioselectively acylated at the N^{β}-position, so that only a single isomer is formed.[123]

Since this postsynthetic, chemoselective lipidation procedure bypasses all the difficulties encountered during preparative HPLC purification of lipopeptides, the α-hydrazinoacetyl peptides were further developed into a strategy of lipidation by chemoselective hydrazone formation between lipophilic glyoxaldehyde vectors and prepurified hydrazino peptides (Scheme 10).[124] The hydrazino handle can be located at the N-termini or even at preselected lysine side-chain positions. Various lipophilic vectors were proposed containing one or more fatty acids coupled via amides to spacers which in turn are linked to glyoxylic acid via an amide bond.[124–127]

Scheme 9 Chemoselective N^{β}-Acylation of α-Hydrazinoacetyl Peptides[121]



Scheme 10 Lipidation by Chemoselective Hydrazone Formation[124]



Selective Acylation of α-Hydrazino Acetyl Peptides with Fatty Acids:[121]
A soln of NH_2-NH-CH_2-CO-Arg-Gly-Arg-Ile-Leu-Lys-Glu-Pro-Val-His-Gly-Val-OH (6.0 mg, 3.0 μmol) in citrate–phosphate buffer pH 5.2 (900 μL) was adjusted to pH 5.1 with 0.2 M aq Na_2HPO_4. To the soln Plm-OSu (1.16 mg, 3.3 μmol) in tBuOH (900 μL) was added and the reaction was monitored by HPLC. After 72 h at rt the mixture was diluted with H_2O/AcOH (4:1, 5 mL) and purified by HPLC on a C3 Zorbax column using buffer A: 0.05% aq TFA and buffer B: MeCN/H_2O, 4:1, containing 0.05% TFA in a linear gradient from 0 to 60% B in 70 min, 3 mL·min^{-1}, 50 °C, detection at 215 nm; yield: 4.4 mg (69%); MALDI-TOF *m/z*: 1672.2 $[M+H]^+$; Calcd 1672.2.

6.4.3.2 N-Alkylated Lipopeptides

Lipo-amino acid derivatives are readily obtained in good yields by direct alkylation of amino acids esters with the related alkyl halides, e.g. farnesyl bromide, under careful control of the reaction conditions to avoid exhaustive alkylation of the amino group.[128] Alternatively, peptoid chemistry is applied for N-alkylation of glycine ester via reaction of alkyl amines, e.g. hexadecylamine, with ethyl bromoacetate.[36,98]

The difficulties encountered in aminoacylation of such lipo-amino acid derivatives are similar to those of the N-alkylated amino acids (see Vol. E 22c, Sections 10.1.1 and 10.1.2), but may be even worse due to steric hindrance of the large alkyl chains. Similarly, acylation with such amino acids leads to poor yields by the use of even stronger acylating reagents, e.g. HBTU, BOP, and PyBroP, with a marked improvement by the use of the mixed anhydride method although long reaction times were required.[128]

H-(Farn)Met-OMe:[128]
To an ice-cold soln of H-Met-OMe·HCl (20 mg, 10 mmol) in DMF (20 mL), TEA (2.78 mL, 20 mmol) was added and the mixture was immediately filtered and cooled again. Farn-Br (2.7 mL, 10 mmol) was added dropwise. The mixture was allowed to reach rt and, after 4–6 h, 5% $NaHCO_3$ (300 mL) was added and the mixture was extracted with EtOAc (200 mL). The organic phase was washed with 5% $NaHCO_3$, brine, and H_2O, dried ($MgSO_4$), and concentrated. The crude product was purified (silica gel, hexane/EtOAc 6:4): yield: 90%; FAB-MS *m/z*: 368 $[M+H]^+$.

H-($C_{18}H_{33}$)Gly-OEt:[14]
Within 1.5 h, a soln of ethyl bromoacetate (290 mmol) in toluene (100 mL) was added dropwise under cooling with ice to a soln of hexadecylamine (300 mmol) and TEA (300 mmol) in toluene (200 mL). After additional stirring for 48 h at rt, the precipitated ammonium salt was filtered off and the solvent removed. Purification of the residue by fractionated destillation in vacuo gave the final product; yield: 41%; mp 93 °C; EI-MS *m/z*: 323 $[M+H]^+$.

Saponification with NaOH gave the corresponding *N*-hexadecylglycine in quantitative yield.

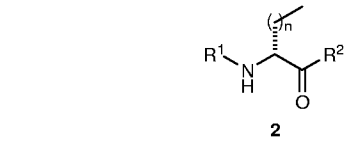
6.4.3.3 Lipo-Amino Acid Peptides

The lipidated α-amino acids (e.g., **2**) are nonnatural α-amino acids with long aliphatic side chains covalently attached to the Cα-carbon.[130] The side chains are saturated or unsaturated, and vary in the total number of carbon atoms (8 to 24), as shown with a few examples in Scheme 11. These amino acids possess the peculiar dual nature of amphiphiles,

for references see p 370

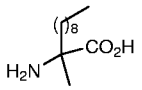
which combine the structural properties of long alkyl chains with those of amino acids and peptides and it makes them possible candidates for many interesting applications. Lipidated α-amino acids or related derivatives have been used in drug delivery systems,[131] as an adjuvant/carrier system,[132,133] and as building blocks in biologically active lipid–peptide mimetics.[134–136]

Scheme 11 Examples of Synthetic Lipidated α-Amino Acids

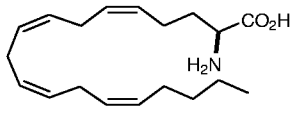


2

n = 7, 9, 11, 13, 17
R^1 = H, Boc, Z
R^2 = OH, OMe



3



4

6.4.3.3.1 Synthesis of Lipo-Amino Acids and Enantiomeric Resolution

The lipo-amino acids are generally synthesized as racemic mixtures which are resolved into optically pure α-amino acids by chemical or enzymatic methods.[130,137,138] The synthesis is based on the alkylation of diethyl acetamidomalonate followed by hydrolysis and decarboxylation.[129,138–140] Also, 20% DMF has been used in the hydrolysis step, as it is suitable for industrial scale preparation.[138] Alternatively, lipidated α-amino acids are synthesized by reacting α-bromoalkanoic acid with ammonium hydroxide.[141]

For the enzymatic resolution of the racemic mixtures, lipidated α-amino acids are converted into the *N*-chloroacetyl derivatives that are stereospecifically hydrolyzed by acylase I from porcine kidney or from *A. mellius*,[129,137,140–142] to yield the (2*S*)-enantiomers. The (2*R*)-enantiomer is then obtained by acid hydrolysis of the related *N*-chloroacetyl derivatives. This method, however, can generally not be applied for lipidated α-amino acids with more than 12 carbon atoms,[137,141] although it was succesfully used for α-aminohexadecanoic acid using a longer incubation time and a large excess of the enzyme.[129] Alternatively, diastereomeric Schiff bases are formed with the chiral derivative (−)-(1*S*,2*S*,5*S*)-2-hydroxypinan-3-one that can be separated by preparative TLC.[138,143,144] A new C^{α}-methylated lipidated α-amino acid, called 2-amino-2-methyldecanoic acid (**3**), was recently used for the synthesis of model peptides (Scheme 11).[145–147] The enantiomerically pure amino acid is obtained on a large scale by applying a chemoenzymatic synthesis which combines a partial Strecker synthesis for the preparation of the racemic amino acid followed by the use of a broadly specific amino acid amidase from *Mycobacterium neoaurum* to achieve optical resolution.[148,149]

Stereoselective alkylation with aliphatic bromides and iodides of the Schiff bases of *tert*-butyl glycinate with (−)-(1*S*,2*S*,5*S*)-2-hydroxypinan-3-one or (+)-(1*R*,2*R*,5*R*)-2-hydroxypinan-3-one[150] was reported to produce lipidated amino acids as D- and L-enantiomers in 80 to over 90% ee.[151] Similarly, the asymmetric synthesis of a derivative of arachidonic acid (**4**) has also been reported. The pure enantiomer was obtained via regioselective functionalization of a chirally pure glutamic acid.[152]

(2*S*,2*R*)-2-Aminodecanoic Acid, H-DL-Ade-OH, from Diethyl Acetamidomalonate; Typical Procedure:[138]

Diethyl acetamidomalonate (24.3 g, 0.11 mol) and 1-bromooctane (29 g, 0.15 mol) were heated under reflux in a soln of Na (2.5 g, 0.11 mol) in EtOH (85 mL) for 24 h. Upon cooling, the mixture was poured into ice water (160 mL), and the precipitate was filtered and washed with H_2O. The crude solid was heated at reflux for 24 h with 6 M HCl (180 mL) and DMF (20 mL). After cooling, the mixture was poured into EtOH/H_2O (3:1) and neutralized with 25% NH_4OH. The product was precipitated and washed with EtOH/H_2O; yield: 17.5 g (85%).

Enantiomeric Resolution by Enzymatic Methods:

$ClCH_2CO$-DL-Ade-OH:[129]

To a suspension of racemic α-aminodecanoic acid (H-DL-Ade-OH; 200 mmol) in MeCN (200 mL), chloroacetyl chloride was added and the mixture was refluxed for 2 h. The yellow-brown soln was concentrated, the residue dissolved in EtOAc and the soln filtered and concentrated. The pure product was obtained by recrystallization (EtOAc/hexane); yield: 91%; mp 88 °C.

H-L-Ade-OH:[129]

A suspension of $ClCH_2CO$-DL-Ade-OH (10 g, 38 mmol) in H_2O (1 L) was heated to 80 °C and adjusted to pH 7 by addition of 2 M LiOH. After cooling to rt, acylase I (1 g) from *A. mellius* was added and the mixture was left to react until precipitation of the product was completed. The precipitate was centrifuged, washed with hot H_2O and EtOAc, and dried; yield: 2.3 g (63%); mp 278 °C; $[\alpha]_D^{24}$ +26.4 (*c* 0.5, AcOH); FAB-MS *m/z*: 188 $[M+H]^+$.

H-D-Ade-OH:[129]

The filtrate from the enzymatic reaction was acidified to pH 2 with 6 M HCl. The precipitate was filtered, washed with H_2O, and dried. The chloroacetyl compound was dissolved in EtOAc and the soln was filtered and concentrated. The residue was suspended in 3 M HCl (160 mL) and refluxed for 8 h. After neutralization with solid KOH, the precipitate was filtered, washed with H_2O and EtOAc, and dried; yield: 2.5 g (70%); mp 282 °C; $[\alpha]_D^{24}$ −26.4 (*c* 0.5, AcOH); FAB-MS *m/z*: 188 $[M+H]^+$.

Enantiomeric Resolution by Chemical Methods:

H-L-Ade-OMe:[138]

To a suspension of H-DL-Ade-OMe·HCl (freshly prepared; 0.99 g, 5.29 mmol) in toluene (40 mL) were added NMM (0.54 mL, 5.29 mmol), (−)-(1*S*,2*S*,5*S*)-2-hydroxypinan-3-one (0.8 g, 5.29 mmol), and $BF_3 \cdot OEt_2$. The mixture was refluxed for 6 h under N_2 and the reaction progress followed by TLC (hexane/EtOAc 3:7). The solvent was removed at 30 °C and the residue taken up in EtOAc (40 mL), filtered, washed briefly with brine and H_2O, and dried. The diastereomers were separated by TLC (silica gel, hexane/EtOAc 3:7); yield: 0.65 g (44%) [(2*S*)-Schiff base]; 0.72 g (49%) [(2*R*)-Schiff base].
The (2*S*)-Schiff base (0.75 g, 1.52 mmol) was dissolved in MeOH (2 mL) and a 0.5 M soln of hydroxylamine hydrochloride in MeOH/H_2O (7:3; 5mL) was added. The mixture was stirred at rt and concentrated. The residue was taken up in $CHCl_3$ (25 mL) and washed with 0.1 M HCl, 0.5 M $NaHCO_3$, brine, and H_2O, and dried. The solvent was concentrated and the residue purified by chromatography (silica gel, CH_2Cl_2/MeOH 20:1); yield: 0.23 g (80%); mp 132–133 °C; CD (Δε): +1.60 (λ 210 nm, *c* 1.0, MeOH).

6.4.3.3.2 Asymmetric Synthesis of Unsaturated Lipo-Amino Acids

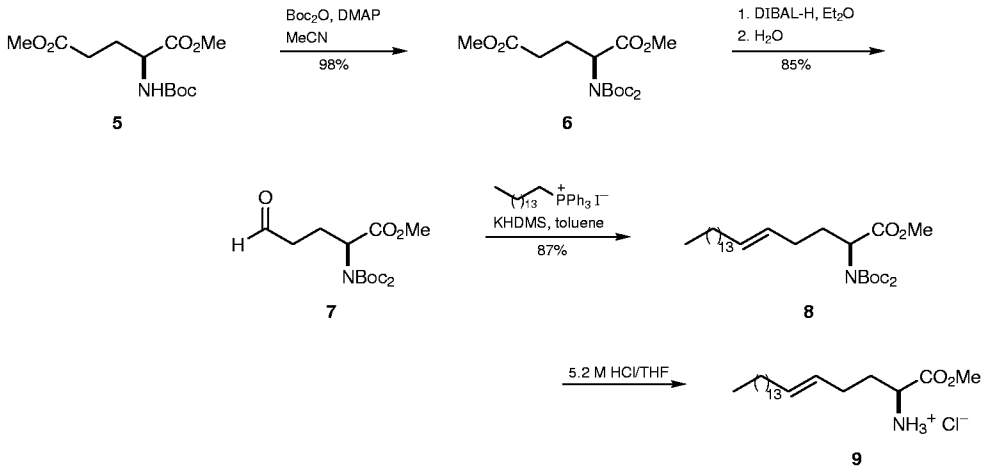
A simple and general method for producing chirally pure α-amino acids with an unsaturated fatty chain takes advantage of the chiral glutamic acid **5** which is reduced at the side-chain carboxy group to the aldehyde (Scheme 12).[152] The aldehyde **7** is obtained in good yield from dimethyl *N*,*N*-di-Boc-glutamate (**6**) using DIBAL-H under controlled conditions. With the mono-Boc derivative, low yields and a very heterogeneous product mixture is obtained. The alkyl chain is introduced using a Wittig-type reaction with formation of a double bond (e.g., **8**). Following the alkylation, the methyl ester can be hydrolyzed effectively from **9** only upon removal of the two Boc protecting groups.

The identical chemistry can be applied to the preparation of enantiopure ω-hydroxy amino acids by reaction of the (2*S*)-2-[bis(*tert*-butoxycarbonyl)amino]-5-oxopentanoate with ω-trityloxyalkylidene triphenylphosphoranes. After hydrogenation of the double bond, the

for references see p 370

ω-hydroxy α-amino acids may serve for conversion of the hydroxy group into carboxy or amino groups.[153] (The side-chain lengths of the acids depend upon the alkanediol or dibromide used to prepare the phosphoranes.) Alternatively, suitably protected Glu and Asp ω-Weinreb amides are used to assemble the peptides which are then subjected on resin to the Wittig reaction to produce peptides containg the desired unsaturated lipo-amino acids.[154,155]

Scheme 12 Synthesis of Methyl 2-Aminoicosa-5-enoate[152]



(Boc)$_2$-Glu(OMe)-OMe (6):[152]
To a stirred soln of Boc-Glu(OMe)-OMe (**5**; 2.5 g, 9 mmol) and DMAP (220 mg, 1.8 mmol) in dry MeCN (30 mL) was added Boc_2O (2.2 g, 9.9 mmol) at rt. The mixture was stirred for 2 h, and then more Boc_2O (1.1 g, 4.9 mmol) was added. After 12 h, the solvent was removed and the crude product purified by chromatography (silica gel) to give **6** as an oil; yield: 3.32 g (98%); MS *m/z*: 376 $[M+H]^+$.

Methyl (2*S*)-2-[Bis(*tert*-butoxycarbonyl)amino]-5-oxopentanoate (7):[152]
To a stirred soln of (Boc)$_2$-Glu(OMe)-OMe (**6**; 1 g, 2.7 mmol) in dry Et_2O (27 mL) was added dropwise 1.0 M DIBAL-H in hexane (3 mL, 3 mmol) at −78 °C. After 5 min, the reaction was quenched with H_2O (0.4 mL) and allowed to warm to rt. The mixture was stirred for 30 min, dried ($MgSO_4$), and filtered through Celite. The solvent was removed and the residue was purified by column chromatography (silica gel) to give **7** as an oil; yield: 790 mg (85%); MS *m/z*: 302 $[M-43]^+$.

Methyl (5*Z*,2*S*)-2-[Bis(*tert*-butoxycarbonyl)amino]icos-5-enoate (8):[152]
To a stirred suspension of pentadecyltriphenylphosphonium bromide (3.85 g, 7 mmol) in dry toluene (40 mL) under argon was added dropwise 0.5 M KHMDS in toluene (12.8 mL, 6.4 mmol) at 0 °C. After 15 min, the mixture was cooled to −78 °C and **7** (2 g, 5.8 mmol) in toluene (5 mL) was added dropwise. The mixture was stirred for 2 h, and then it was quenched with a sat. soln of NH_4Cl (50 mL) and extracted with Et_2O (3 × 10 mL). The combined organic phases were washed with brine, dried ($MgSO_4$), and concentrated. The crude product was purified by chromatography (silica gel) to give **8** as an oil; yield: 2.7 g (87%); MS *m/z*: 382 $[M-157]^+$.

The *N*,*N*-di-Boc protection can be removed with HCl (5.2 M soln in THF) at rt to give **9**. The mono N^{α}-Boc protected amino acid derivative is then obtained with Boc_2O under standard procedures.

Boc-Glu[N(Me)OMe]-OH:[155]
To a soln of Boc-Glu-OBzl (3.37 g, 10 mmol) in CH_2Cl_2 (100 mL), BOP (4.42 g, 10 mmol), and NH(Me)OMe·HCl (1.07 g, 11 mmol) were added followed by DIPEA to reach pH 9–10 (moistened pH paper). After 2 h, the soln was washed with 5% aq $KHSO_4$ (3×), sat. $NaHCO_3$ (3×), and brine, then dried (Na_2SO_4). The solvent was removed to give an oil that was purified by flash chromatography; yield: 3.4 g (89%).

This intermediate (1.9 g, 5 mmol) was hydrogenated over 10% Pd/C in EtOH for 6 h. The catalyst was filtered off and the soln concentrated under reduced pressure to give an oil; yield: 1.4 g (96%).

6.4.3.3.3 Synthesis of Lipo-Amino Acid Peptides

The N^{α}-Z- or N^{α}-Boc-protected lipo-amino acids and their esters are obtained by standard protocols.[138] In contrast, for the synthesis of the N^{α}-Fmoc derivatives intermediate bis-silylation and the use of Fmoc-Cl as acylating agent are recommended.[129] The standard procedure with Fmoc-OSu leads to low yields (25%) mainly due to low solubility of the amino acids.[133] Alternatively, the improved solubility of the 4-toluenesulfonate salt of the benzyl esters is exploited for reaction with Fmoc-OSu. The benzyl ester is then removed by catalytic hydrogenation, although the Fmoc group is known to be reduced under these conditions unless particular precautions are taken.[133,156–159]

Small C- and/or N-protected hetero-oligomers containing both lipidated α-amino acids and coded amino acids have been prepared stepwise using the procedure described for the preparation of the lipidated α-amino acids homo-oligomers in solution.[138] Boc- and Fmoc-protected lipo-amino acids are used without particular problems in stepwise solid-phase peptide synthesis, e.g. Fmoc-Ade-OH (Ade = 2-aminodecanoic acid) was incorporated into peptides by the Fmoc/tBu strategy using TBTU/HOBt/DIPEA[129] or HATU/NMM.[133]

Boc-L-Ade-OH:[138]
To a suspension of H-L-Ade-OH (8.6 g, 46 mmol) in tBuOH/H_2O (2:3; 180 mL) was added 8 M NaOH until the soln was pH 13. Boc_2O (15 g, 69 mmol) in tBuOH (30 mL) was added at rt and the pH adjusted to 11–12. After 2 h, the mixture was diluted with H_2O (50 mL) and acidified with solid citric acid to pH 3. The oil was extracted with EtOAc, the organic phase was dried ($MgSO_4$), and the solvent removed. The residue was triturated with MeCN; yield: 12.2 g (93%); mp 64–66 °C.

Z-L-Ade-OH:[138]
To a soln of H-L-Ade-OH (7.0 g, 37.4 mmol) in 1 M NaOH (37.4 mL) and tBuOH (10 mL), benzyl chloroformate (6.39 g, 37.7 mmol) and 1 M NaOH (37.4 mL) were added simultaneously over a period of 45 min at 0 °C. The mixture was stirred for 3 h at 0 °C and 1 h at rt. The mixture was extracted with Et_2O, the organic solvent was removed, and the residue stirred with 3% HCl (100 mL). The acidic soln was extracted with Et_2O, the combined organic phases dried (Na_2SO_4), and the solvent removed to give an oil; yield: 8.0 g (67%).

Fmoc-L-Ade-OH:[129]
A soln of H-L-Ade-OH (1.86 g, 10 mmol) and TMSCl (2.54 mL, 20 mmol) were refluxed in CH_2Cl_2 (100 mL) for 15 min, then TEA (2.77 mL, 20 mmol) was added and the mixture refluxed for an additional 60 min. After cooling to 0 °C, Fmoc-Cl (2.57 g, 10 mmol) was added in one portion and the mixture was stirred overnight. After concentration, the residue was acidified with 0.01 M HCl (100 mL) and extracted with CH_2Cl_2. The organic phases were washed with H_2O, dried ($MgSO_4$), and concentrated. The crude compound was recrystallized (CH_2Cl_2/hexane); yield: 2.93 g (72%); mp 129 °C; CI-MS *m/z*: 409 $[M+H]^+$.

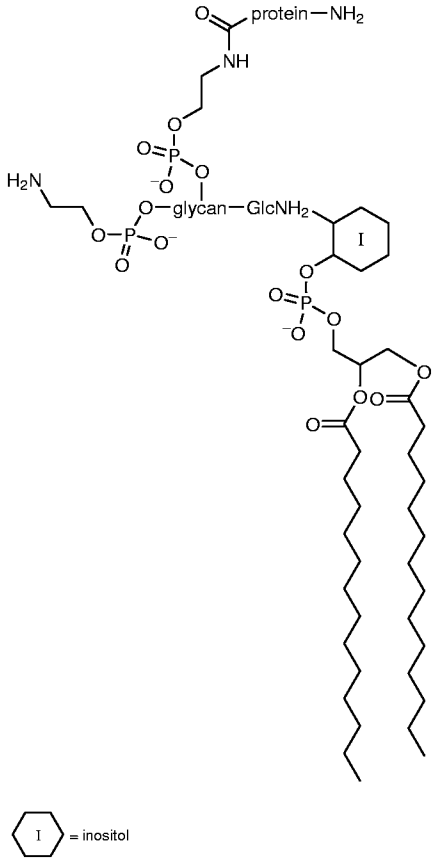
6.4.4 Building Blocks for Lipidation of Peptides

It is well-established that two vicinal lipid chains are required for an almost irreversible capture of peptides and proteins by lipid bilayers via interdigitation of the lipid chains with artificial vesicles or cell membranes.[37,38] Nature uses the glycosylphosphatidylinositol (GPI) anchor, shown in Scheme 13, for this purpose. To mimic nature, various strategies have been developed for the preparation of doubly lipidated building blocks that can be conjugated to peptides at preselected positions with or without spacers to minimize direct effects on their bioactivities (Scheme 10). The conjugation of such lipo-moieties to peptides results in peptide amphiphiles that exhibit a high tendency to self-associate into larger aggregates of micellar or bilayer type which rapidly fuse with artificial and natural membranes.[160–165] The capture of the lipopeptides by the membrane can facilitate their interaction with membrane-bound receptors.[37,38,166–168] Moreover, the pharmacokinetic properties of bioactive peptides

for references see p 370

can be improved by facilitating their passage through hydrophobic membranes and finally, the high tendency to form aggregates such as micelles and liposomes can be exploited for the design of peptide-based drug delivery systems to well-defined targets. One of the major advantages of lipidation with building blocks in comparison to the other lipidation methods is the relatively free choice of the site and chemistry of conjugation. By this approach the lipid moieties are incorporated in a straightforward manner into preselected positions of the preassembled and prepurified peptide.

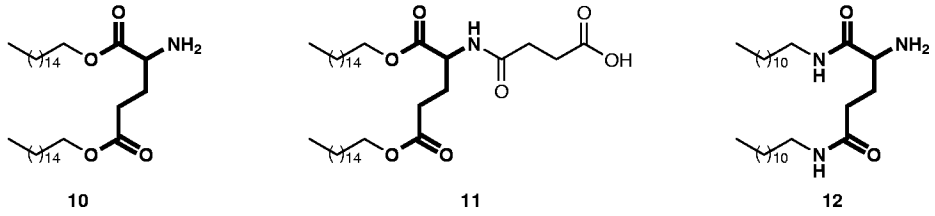
Scheme 13 Structure of the Natural Protein Membrane Anchor Glycosylphosphatidylinositol



Mainly two types of building blocks containing vicinal lipid chains have been developed these are based on diesters (**10** and **11**) or di-*N*-(alkyl)amides (**12**) of the dicarboxylic amino acids, *N*-dialkylamides of dicarboxylic acids (**13** and **14**), as well as on diesters of glycerol and related derivatives (**15–21**) (Scheme 14).

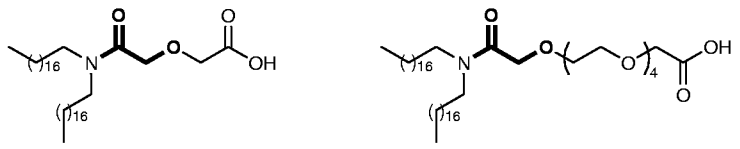
Scheme 14 Structures of Several Doubly Lipidated Building Blocks

Asp- and Glu-based scaffolds



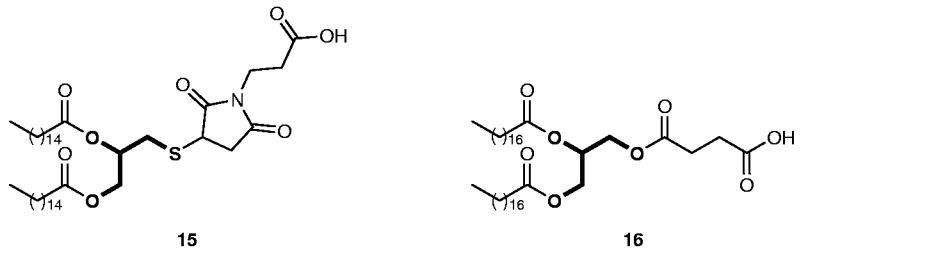
10 11 12

bis(alkyl)amide-based scaffolds

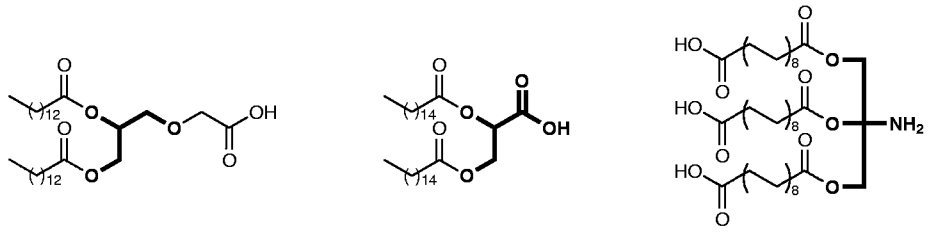


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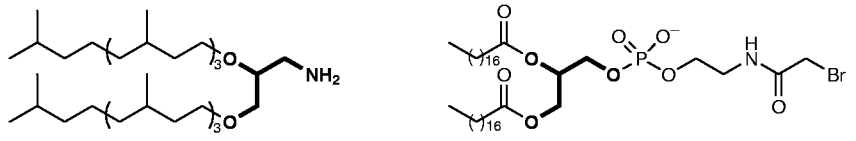
glycerol-based scaffolds



15 16



17 18 19



20 21

6.4.4.1 Lipophilic Building Blocks Based on Dicarboxylic Amino Acid Esters and Amides

Among the large variety of synthetic amphiphiles with one, two, three, or four lipid chains that have been synthesized,[160,161,169] glutamic acid and aspartic acid are natural scaffolds to mimic natural phospholipid molecules and to display two vicinal lipid chains for interdigitation with synthetic and natural bilayers.[170–175] The lipophilic chains are attached via ester or amide bonds to the two carboxy groups of Glu or Asp residues which can then be coupled via the α-amino group to other amino acids, peptides, or different polar groups (e.g., alkyl ammonium moieties) directly (e.g., **10**, **12**) or via suitably functionalized spacers (e.g.,

for references see p 370

11). The first peptide amphiphiles were obtained by simple esterification of glutamic acid with lipid alcohols.[176] To increase the acid stability the di-*N*-(alkyl)amide derivatives were developed.[172,174,175] The synthesis of Glu-based amphiphiles has to be performed in solution when the lipid moiety is positioned at the C-terminus and can be performed on solid support when it is introduced into the N-terminal position. For the latter purpose the glutamic acid diester or diamide has to be functionalized, e.g. by reaction with succinic anhydride to introduce a free carboxy group that can be activated as an ester for acylation of the peptide chain.[174,175] Synthesis on solid support using the Boc strategy is not advisable due to the lability of the lipid esters to HF treatment. In contrast, Fmoc/TFA based SPPS has been extensively used for the preparation of this type of lipopeptide conjugate.[171,172,177] While synthesis in solution raises solubility problems, such syntheses are relatively straightforward by the solid-phase strategy.[162,174,175]

As an alternative to the Glu and Asp scaffolds, the *N*-(dialkyl)amide of diglycolic acid has been proposed (**13**). This is readily prepared by heating equivalent amounts of diglycolic anhydride and, for example, bis(octadecyl)amine in refluxing toluene for 48 hours followed by recrystallization from chloroform (mp 81–82 °C).[178] In a similar mode, polyoxyethylene of selected chain length has been used to prepare dicarboxylic acids which, upon mono-amidation with bis(alkyl)amines, yield lipo moieties (e.g., **14**) that are coupled to selected amine groups of peptides to produce the peptide amphiphiles.[179,180]

TosOH·H-Glu($OC_{16}H_{33}$)-$OC_{16}H_{33}$ (10):[171]
A soln of H-Glu-OH (10 g, 0.068 mol), TosOH (14 g, 0.074 mol), and hexadecanol (38 g, 0.16 mol) in toluene was refluxed with azeotropic separation of the H_2O by a Dean–Stark trap. The reaction was continued until the stoichiometric amount of H_2O was recovered. Toluene was removed and the solid was recrystallized (acetone); yield: 43 g (81%).

***N*-Succinyl-Glu($OC_{16}H_{33}$)-$OC_{16}H_{33}$ (11):**[174]
To a soln of **10** (20 g, 26 mmol) and TEA (5.5 mL, 39 mmol) in THF/$CHCl_3$ (1:1) was added succinic acid (3.9 g, 39 mmol) under stirring. The mixture was kept for 4 h at 30 °C. The product obtained after removal of the solvent was recrystallized (acetone and EtOH); yield: 18 g (94%); mp 74.5 °C; FAB-MS *m/z*: 696.6 $[M+H]^+$, 694.6 $[M^-]$.

***N*-Succinyl-Glu($OC_{16}H_{33}$)-$OC_{16}H_{33}$ 4-Nitrophenyl Ester:**[174]
To a soln of **11** (6.9 g, 9.9 mmol) and 4-nitrophenol (1.65 g, 11.9 mmol) in CH_2Cl_2, was added DCC (2.05 g, 10 mmol) at 0 °C as well as DMAP (80 mg, 0.65 mmol, cat.). After 2 h in the cold and 24 h at rt, the urea was filtered off and the product precipitated with EtOH; yield: 6.9 g (85%); mp 89 °C.

HBr·H-Glu($NHC_{12}H_{25}$)-$NHC_{12}H_{25}$ (12):[172]
A soln of Z-Glu(OSu)-OSu (1.0 g, 2.1 mmol) and dodecylamine (0.78 g, 4.21 mmol) in $CHCl_3$ (20 mL) was allowed to stand at rt for 2 days. The $CHCl_3$ soln was washed with 4% $NaHCO_3$ and H_2O, dried ($NaSO_4$), and concentrated; yield: 1.17 g (91%); mp 138–139 °C.

A soln of this compound (0.83 g, 1.35 mmol) in TFA (3 mL) was treated with 25% HBr/HOAc (4.2 mL) at rt for 2 h. The precipitate was filtered and taken up in Et_2O. The oily residue obtained by concentration of the Et_2O was solidified with a H_2O/MeOH/$CHCl_3$ mixture; yield: 0.74 g (98%); mp 118–122 °C.

***N*,*N*-1-Bis(octadecyl)amide-3,6,9,12-tetraoxatetradecanoic Acid Potassium Salt (14):**[180]
1-Benzyl-15-ethyl-3,6,9,12-tetraoxatetradecanoate ester[180] (0.96 g, 2.24 mmol) was hydrogenated for 12 h over 5% Pd/C in CH_2Cl_2 (40 mL). The catalyst was filtered off, and to the filtrate at 0 °C was added DMAP (395 mg, 3.24 mmol), HOSu (372 mg, 3.25 mmol), and DCC (668 mg, 3.24 mmol). After 1 h, bis(octadecyl)amine (1.69 g, 3.24 mmol) was added and the mixture was heated at 30 °C for 12 h. DCU was filtered off and the soln was diluted with CH_2Cl_2 (50 mL) and washed with 0.5 M HCl (10 mL), sat. $KHCO_3$ (10 mL), and brine (10 mL). The organic phase was dried (Na_2SO_4), concentrated, and the residue purified by chromatography (silica gel, EtOAc); yield: 1.2 g (49%).

The ester (1.2 g, 1.44 mmol) and KOH (720 mg, 12.8 mmol) were dissolved in EtOH (6 mL) and the soln was refluxed for 6 h. Upon cooling, the product was isolated; yield: 89 mg (76%). The salt was used directly for coupling to an amino group via TBTU/HOBt.

Synthesis of Lipopeptides in Solution:

HCl·H-Pro-Pro-Pro-Pro-Glu($NHC_{12}H_{25}$)-$NHC_{12}H_{25}$:[172]
To Boc-$(Pro)_4$-OH (0.53g, 1.04mmol) and H-Glu($NHC_{12}H_{25}$)-$NHC_{12}H_{25}$·HBr (0.70g, 1.24mmol) in DMF (50mL) was added at 0°C with stirring a soln of DPPA (0.27mL, 1.24mmol) in DMF (5mL), followed by TEA (0.31mL, 2.29mmol) in DMF. Stirring was continued at 0°C for 6h and at rt overnight. The mixture was diluted with $CHCl_3$ (100mL) and was washed with 10% citric acid, 4% $NaHCO_3$, brine, and H_2O. The organic layer was dried (Na_2SO_4) and concentrated. The residue was purified by column chromatography (silica gel, $CHCl_3$/MeOH 5:1). The resulting pale yellow oil (0.65g) was dissolved in EtOAc (10mL) and treated with 3.5M HCl in EtOAc (10mL) at rt for 1h. Upon removal of the solvent, the oily residue was recrystallized (H_2O/MeOH/$CHCl_3$); yield: 400mg (77%).

Synthesis of Lipopeptides on Solid Support:

***N*-Succinyl-[Gly-Val-Lys-Gly-Asp-Lys-Gly-Asn-Pro-Gly-Trp-Pro-Gly-Ala-Pro-Tyr-OH]-Glu($OC_{16}H_{33}$)-$OC_{16}H_{33}$:**[174]
The lipopetide was synthesized by reacting the N^{α}-deprotected-peptide-resin with a fourfold molar excess (over the substitution level of the resin) of *N*-succinyl-Glu($OC_{16}H_{33}$)-$OC_{16}H_{33}$ 4-nitrophenyl ester and HOBt in DMF/CH_2Cl_2 (1:1). After 4h, the amphiphilic peptide was deprotected and cleaved from the resin with reagent K.[181] The crude product was washed with cold Et_2O and purified by RP-HPLC on a Vydac C-4 column using a gradient of 0.1% aq TFA and 0.1% TFA in MeCN, from 55% to 90% MeCN in 40min or 80% MeCN isocratic elution. LD-MS *m/z*: 2277.2 $[M+H]^+$.

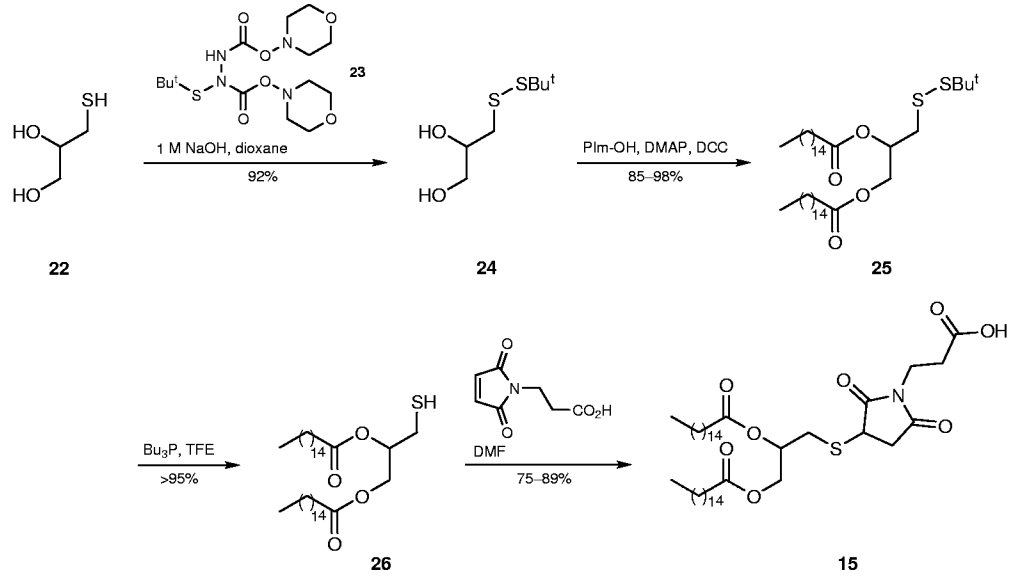
6.4.4.2 Lipophilic Building Blocks Based on Di-Fatty Acid Substituted Glycerol

Glycerol (propane-1,2,3-triol) is a very important structural part of the phospholipids, the major components of natural membrane bilayers. Based on this natural scaffold, diesters of glycerol and of related derivatives have been extensively used for the construction of synthetic building blocks carrying vicinal lipid chains (for examples, see Scheme 14).

6.4.4.2.1 Di-Fatty Acid Substituted Thioglycerol

Racemic thioglycerol (3-sulfanylpropane-1,2-diol) was used for the attachment of two lipid chains via ester bond formation with the hydroxy groups[182] while the free thiol group serves for selective cross-linking to other molecules via disulfide or sulfide bonds utilizing mild thiol–disulfide interchange or thiol addition reactions (Scheme 15).[163,164,167]

Scheme 15 Synthesis of the Di-Fatty Acid Substituted Thioglycerol Building Block[163,164,167,182]



for references see p 370

6.4.4.2.1.1 Synthesis of Di-Fatty Acid Substituted Thioglycerol

As shown in Scheme 15, for the synthesis of the di-fatty acid substituted thioglycerol building blocks the thiol group was protected as the *S-tert*-butylsulfanyl derivative, which is selectively cleaved with phosphines to give a diacylated thioglycerol derivative.[183] Besides acylation of both hydroxy groups with the identical fatty acid, mixed diacyl derivatives can also be synthesized taking advantage of the different reactivities of the primary and secondary alcohol groups of the glycerol. The primary hydroxy group is selectively protected in high yields using trityl chloride allowing the secondary hydroxy to undergo esterification with a fatty acid. Detritylation cannot be carried out using the standard acid-catalyzed cleavage procedures due to substantial C2→C1 acyl migration, and the best results were obtained with zinc bromide followed by silica gel chromatography. The resulting mono-acyl glycerol derivative is then esterified with a different fatty acid at the secondary hydroxy group to yield the mixed diacyl derivatives.[182] Since interdigitation of di-fatty acid substituted glycerol groups with phospholipid bilayers is affected only negligibly by the chirality at the C2 of the glycerol[184] only racemic derivatives were synthesized.

3-*tert*-Butyldithiopropane-1,2-diol (24):[182]
To a stirred ice-cold soln of thioglycerol (**22**; 10.8 g, 0.1 mmol) and hydrazine **23** (52.0 g, 0.15 mmol) in argon-sat. dioxane[185] was added dropwise 1 M NaOH (100 mL). After 2 h at rt, the solvent was removed and the residue was partitioned between EtOAc and 2% $KHSO_4$. The organic phase was washed with H_2O, dried (Na_2SO_4), and concentrated to small volume. The product crystallized upon addition of petroleum ether; yield: 18 g (92%); mp 48.5–49.5 °C.

3-*tert*-Butyldithio-1,2-bis(palmitoyloxy)propane (25):[182]
3-*tert*-Butyldithiopropane-1,2-diol (**24**; 0.2 g, 1 mmol) and PA (0.77 g, 3 mmol) in dry THF (10mL) were reacted with DCC (0.62 g, 3 mmol) and DMAP (37 mg, 0.3 mmol) at rt for 12 h. The DCU was removed by filtration, the solvent removed, and the residue purified by column chromatography (silica gel, hexane/tBuOMe/AcOH 88:10:2). The pure fractions were combined and the solvent removed; yield: 0.64 g (95%); mp 39–41 °C.

2,3-Bis(palmitoyloxy)propanethiol (26):[182]
To an argon-sat. soln of **25** (0.18 g, 0.27 mmol) in TFE (5 mL), MTBE (5 mL), and H_2O (0.1 mL), was added Bu_3P (92 μL, 0.37 mmol). The mixture was stirred at rt for 12 h, then the solvent was concentrated and the residue taken up in cyclohexane. The soln was washed with TFE and concentrated to afford the product as a white wax-like solid; yield: 0.16 g (quant).

2,3-Bis(myristoyloxy)propanethiol:[182]
The title compound was prepared following the procedures outlined above.

6.4.4.2.1.2 Synthesis of 3-[2-(1,2-Diacyl-3-thioglyceryl)succinimidyl]propanoic Acid

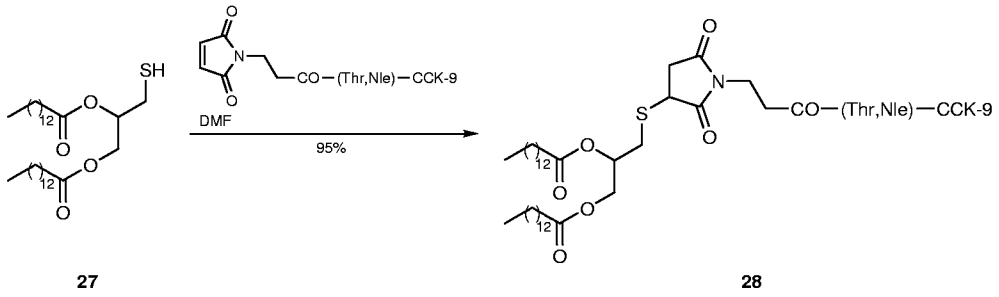
Thiol addition to maleinimide groups is a well-established, mild and efficient conjugation method,[186] that was exploited for functionalizing the di-fatty acid substituted thioglycerol moiety via a spacer containing a flexible ethene moiety with a carboxy group (Scheme 15). The carboxy group can be used for acylating peptides at defined amino groups.[182]

3-{[1-(2-Carboxyethyl)-2,5-dioxo-3-pyrrolidin-3-yl]sulfanyl}-1,2-bis(palmitoyloxy)propane (15):[182]
To an argon-sat. soln of **26** (165 mg, 0.28 mmol) in DMF (2 mL) was added 3-maleinimido-propanoic acid (47 mg, 0.28 mmol) in DMF (1mL). After stirring at rt for 30 min, the solvent was concentrated and the residue was partitioned between MTBE and 5% AcOH. The organic layer was washed with H_2O, dried (Na_2SO_4), concentrated to dryness, and the solid recrystallized (hexane); yield: 0.18 g (85%); mp 55–57 °C.

6.4.4.2.1.3 Lipo-Derivatization of Peptides

Maleimide-functionalized peptides react quantitatively with thiols and thus with di-fatty acid substituted thioglycerol, a fact which facilitates purification of the amphiphilic lipopeptide derivatives by simple precipitation as required for separation of the excess lipid building block (Scheme 16).[163,164,167]

Scheme 16 Conjugation of a Di-Fatty Acid Substituted Thioglycerol Building Block to a Peptide[164]



27 28

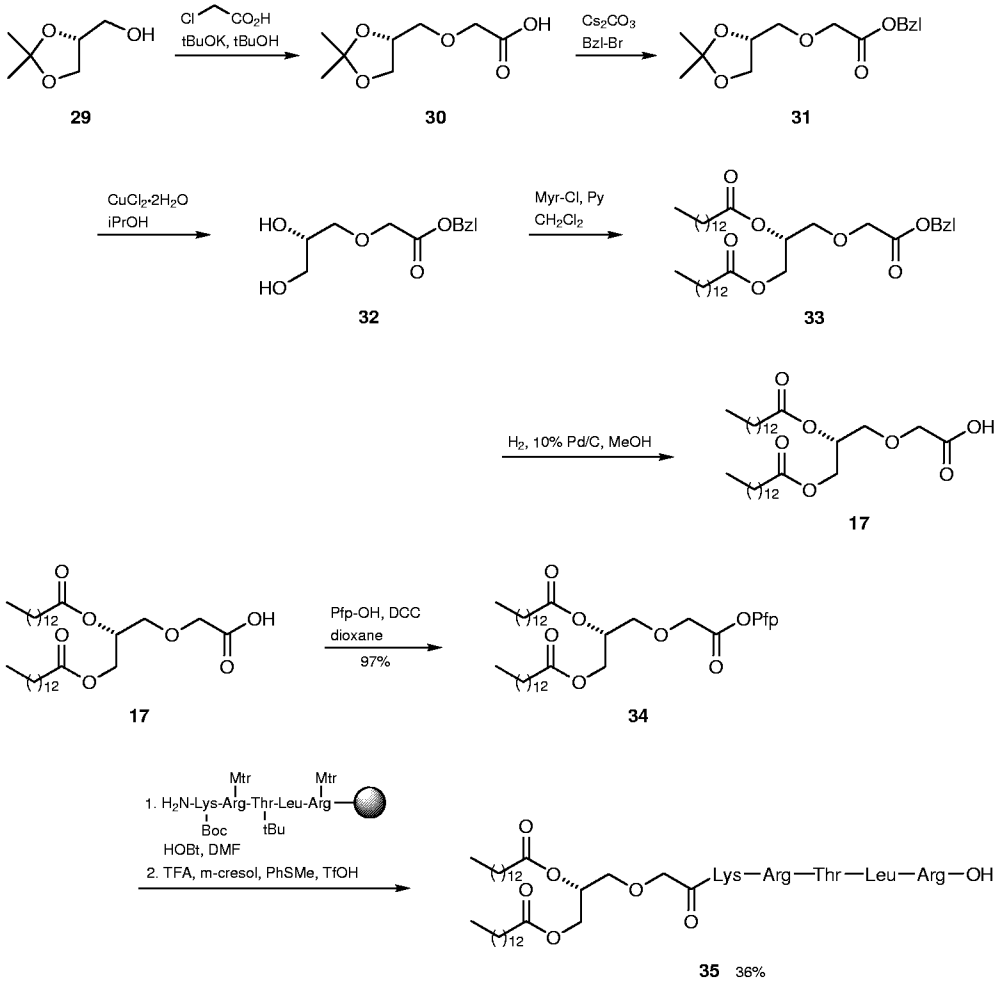
1,2-Bis(myristoyloxy)propanethiol/*N*-Maleoyl-β-alanyl-(Thr,Nle)-CCK-9 Adduct (28):[164]
1,2-Bis(myristoyloxy)propanethiol [**27**; obtained by reduction of the StBu-protected precursor (74 mg, 0.12 mmol)] was dissolved in degassed, argon-sat. DMF (5 mL) and added to a soln of *N*-maleoyl-β-alanyl-(Thr,Nle)-CCK-9[163] (35 mg, 0.021 mmol) in argon-sat. DMF (5 mL). The mixture was allowed to react for 15 h at rt. The soln was then concentrated and the product precipitated with EtOAc/hexane (1:1). The precipitate was collected by centrifugation, and washed extensively with EtOAc to remove the excess of the thioglycerol derivative. The solid residue was redissolved in tBuOH/H_2O and freeze-dried; yield: 42 mg (95%); FAB-MS *m/z*: 1852.4 $[M - SO_3 + H]^+$.

6.4.4.2.2 Functionalized 1,2-Di-Fatty Acid Substituted Glycerol

1,2-Di-fatty acid substituted glycerol can be esterified with succinic anhydride in the presence of DMAP to produce the succinate ester (**16**).[187] A second approach relies on etherification of 1,2-isopropylidene glycerol with chloroacetic acid and subsequent fattyacylation of the diol to give **17** as shown in Scheme 17.[188] Both reagents contain a carboxy group which can be activated for the acylation of peptides directly or via suitable spacers.

for references see p 370

Scheme 17 Synthesis of (1,2-Dimyristoyl-*sn*-3-glyceroxy)acetate and Conjugation to a Peptide[188]



1,2-Distearoyl-3-succinyl-*sn*-glycerol (16) by Reaction of Distearoylglycerol with Succinic Anhydride and DMAP:[187]

Distearoyl-*sn*-glycerol (1 g, 1.6 mmol), succinic anhydride (0.2 g, 2 mmol), and DMAP (0.24 g, 2 mmol) were dissolved in CH_2Cl_2 (10 mL). The mixture was stirred at rt for 1 h, the solvent was removed and the product was isolated (silica gel, EtOAc/hexane 9:1).

Benzyl (1,2-Isopropylidene-*sn*-3-glyceroxy)acetate (31):[188]

Commercial 1,2-isopropylidene-*sn*-glycerol (**29**; 620 µL, 5 mmol) was added to a soln of tBuOK (1.4 g, 12.5 mmol) in tBuOH (10 mL). After 30 min at rt, a soln of chloroacetic acid (0.52 g, 5.5 mmol) in tBuOH (5 mL) was added dropwise and the mixture was refluxed for 4 h. The mixture was diluted with EtOAc and washed with 5% citric acid. The aqueous layer was extracted with EtOAc (2 ×) and the combined organic layers were washed with H_2O and brine, dried (Na_2SO_4), and concentrated. The residue **30** was dissolved in a mixture of MeOH (20 mL) and H_2O (2 mL), neutralized with 20% aq Cs_2CO_3 (6 mL), and the solvent was then removed. The resultant Cs salt was dissolved in dry DMF and concentrated (2 ×). Subsequently, dry DMF (12.5 mL) and Bzl-Br (890 µL, 7.5 mmol) were added. The soln was stirred for 2 h at rt. After completion of the reaction as indicated by TLC, the solvent was removed. The residue was dissolved in Et_2O and washed with H_2O. The organic layer was dried ($MgSO_4$) and concentrated. The residue was purified by chromatography (silica gel, Et_2O/petroleum ether, gradient 0 to 25% Et_2O). The product **31** was obtained as a colorless oil; yield: 980 mg (70%).

Benzyl (1,2-Dimyristoyl-*sn*-3-glyceroxy)acetate (33):[188]
To compound **31** (280 mg, 1 mmol) in iPrOH (10 mL) was added $CuCl_2 \cdot 2H_2O$ (900 mg, 5 mmol). After 1 h at rt, 1 M Na_2CO_3 (5.5 mL, 5.5 mmol) was added and the mixture was filtered and concentrated. The residue was dissolved in EtOAc, washed with H_2O and brine, dried (Na_2SO_4), and the solvent concentrated to give **32**; yield: 216 mg (90%). Then a soln of Myr-Cl (665 mg, 2.61 mmol) and pyridine (200 µL, 3 mmol) in dry CH_2Cl_2 (6 mL) was added to a soln of the diol **32** (209 mg, 0.87 mmol) in dry CH_2Cl_2 (4 mL). After 3 h the mixture was concentrated, the residue was dissolved in Et_2O, washed with 1 M $KHSO_4$, 5% Na_2CO_3, and H_2O, and dried ($MgSO_4$). Purification by chromatography (silica gel, Et_2O/petroleum ether, gradient 0 to 15% Et_2O) gave the product **33**; yield: 400 mg (70%).

(1,2-Dimyristoyl-*sn*-3-glyceroxy)acetate (17):[188]
The myristoylated derivative **33** (300 mg, 0.45 mmol) was hydrogenated in MeOH (20 mL) for 12 h at rt in the presence of 10% Pd/C (10 mg). After usual workup, **17** was obtained; yield: 252 mg (97%).

Pentafluorophenyl (1,2-Dimyristoyl-*sn*-3-glyceroxy)acetic Acid (34):[188]
Glycerol **17** was converted into the Pfp ester **34** in quantitative yield following the procedure of Kisfaludy and Schön.[189]

N^{α}-(1,2-Dimyristoyl-*sn*-3-glyceroxy)acetyl-Lys-Arg-Thr-Leu-Arg-OH (35):[188]
Compound **34** (738 mg, 0.24 mmol, 4 equiv) and HOBt (33 mg, 0.24 mmol, 4 equiv) dissolved in DMF (5 mL) were allowed to react with H-Lys(Boc)-Arg(Mtr)-Thr(tBu)-Leu-Arg(Mtr)-Pepsyn-K-resin (functionalized with the Wang linker) for 1 h. After usual workup, the peptide was deprotected and cleaved from the resin with TFA (3.8 mL), *m*-cresol (550 µL), thioanisole (600 µL), and TfOH (300 µL) for 1 h at 0 °C. The final lipopeptide was purified by Sephadex LH-20 gel filtration (MeOH/CH_2Cl_2 1:1), followed by RP-HPLC using a CN (nitrile) column and an appropriate gradient from 0.1% aq TFA to 0.1% TFA in MeCN; yield: 27 mg (36%); FAB-MS *m/z*: 1226.5 $[M+H]^+$.

6.4.4.3 Lipophilic Building Blocks Based on 2,3-Di-Fatty Acid Substituted Glyceric Acid

A simple approach for lipidation of peptides with di-fatty acid substituted glycerol moieties is based on the use of glyceric acid.[190] For this purpose (2*R*)-glyceric acid is esterified at the two hydroxy groups with fatty acid acyl chlorides and the resulting lipophilic synthon (**18**) is used directly as an active ester, e.g. Pfp ester, to acylate selected amino groups of peptides, or is used to acylate suitably functionalized spacers.

(2*R*)-2,3-Bis(palmitoyloxy)propanoic Acid (18):[190]
(2*R*)-glyceric acid (1.64 g, 15.4 mmol) obtained from the Ca salt with a Dowex H^+ (AG 50W-X8) cation-exchange resin, was dissolved in dioxane (20 mL). To the cooled soln was added TEA (12.95 mL, 92.6 mmol) and dropwise a soln of Plm-Cl (14.05 mL, 46.3 mmol) in dioxane (10 mL), maintaining the temperature below 35 °C. The soln was stirred for 2 h at rt and after removal of the solvent, the residue was dissolved in MeOH and acidified to pH 4 with 2 M HCl. After removal of the solvent, the residue was dissolved in CH_2Cl_2 and was washed with sat. $NaHCO_3$ and H_2O. The organic layer was dried (Na_2SO_4) and concentrated. The product was purified by chromatography (silica gel, CH_2Cl_2 then EtOAc/MeOH 9:1); yield: 0.81 g (9%); CI-MS *m/z*: 600 $[M+18]^+$.

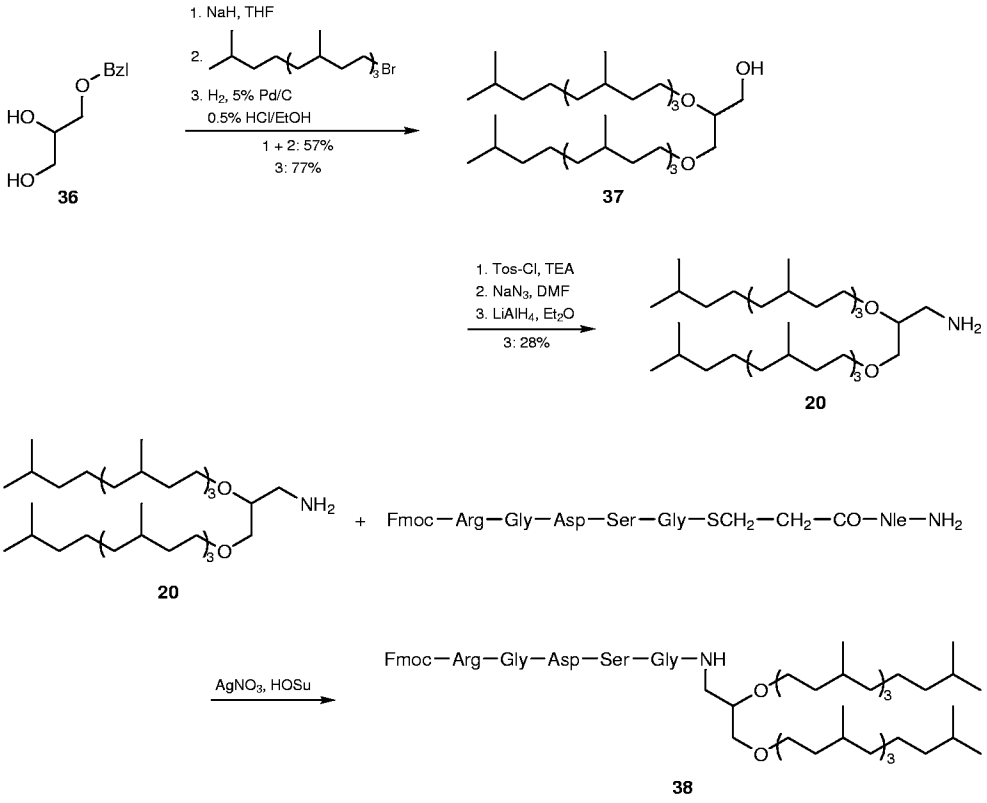
N^{α}-Acetyl-N^{ε}-(2*R*)-2,3-bis(palmitoyloxy)propanoyl-Lys-Gly-Arg-Gly-Asp-Gly-NH_2:[190]
Compound **18** (100 mg, 0.172 mmol), Pfp-OH (38 mg, 0.206 mmol), and DCC (42.5 mg, 0.206 mmol) were dissolved in CH_2Cl_2 (5 mL). The soln was stirred for 1 h at rt, filtered, and concentrated to yield the active ester.[189] Ac-Lys-Gly-Arg-Gly-Asp-Gly-NH_2 (55 mg, 0.087 mmol) was dissolved in H_2O (2.5 mL). To this soln were added DIPEA (30 µL, 0.173 mmol) and active ester (1.5 equiv) dissolved in dioxane (2.5 mL). The soln was stirred for 20 h at rt, and the pH was mantained at 9.5 with DIPEA. The solvent was removed and the residue dissolved in H_2O and lyophilized. The product was obtained after RP-HPLC on a C-8 column, using MeCN and 0.1% aq TFA; yield: 15.5%.

for references see p 370

6.4.4.4 Lipophilic Building Blocks Based on Aminoglycerol

For incorporation of lipoglycerol derivatives into the C-termini of peptides the use of aminoglycerol as a scaffold has been proposed.[191,192] For this purpose 1-*O*-benzylglycerol is dialkylated with aliphatic chains at the remaining two hydroxy groups as shown in Scheme 18. Upon removal of the benzyl group, the primary hydroxy group is converted into the amino group via the azide procedure to give **20**. The 2-amino-triglycerides (e.g., **19**)[193] have also been synthesized as lipophilic groups that can be utilized for the multiple attachment of peptides.[193]

Scheme 18 Synthesis of 2,3-Bis(phytanoxy)propanamine and Conjugation to a Peptide Thioester Synthesized on Solid Support[191,192]



1,2-Di(3*RS*,7*R*,11*R*-phytanyl)glycerol (37):[191]
A mixture of NaH (60% in oil; 4.4 g, 0.11 mol) and 1-*O*-benzylglycerol (**36**; 3.63 g, 20.1 mmol) in dry THF (85 mL) was stirred at rt for 1 h. (3*RS*,7*R*,11*R*)-phytanyl bromide (17.7, 49.0 mmol) was added to the resulting alkoxide soln and the mixture was heated in an oil bath at 145 °C for 4 h; the solvent was removed by distillation during an initial stage of the heating. To the residue was added additional phytanyl bromide (3.2 g, 8.8 mmol), and heating was continued for another 2.5 h. The mixture was partitioned between $CHCl_3$ and H_2O. The organic layer was dried (Na_2SO_4), concentrated, and chromatographed (silica gel, hexane/EtOAc 20:1); yield: 8.36 g (57%).

The Bzl ester was hydrogenated over 5% Pd/C (1.49 g) in EtOH (60 mL) containing HCl (0.3 mL) at rt for 4 h. After usual workup, the residue was purified (silica gel, hexane/EtOAc from 5:1 to 2:1); yield: 4.91 g (77%).

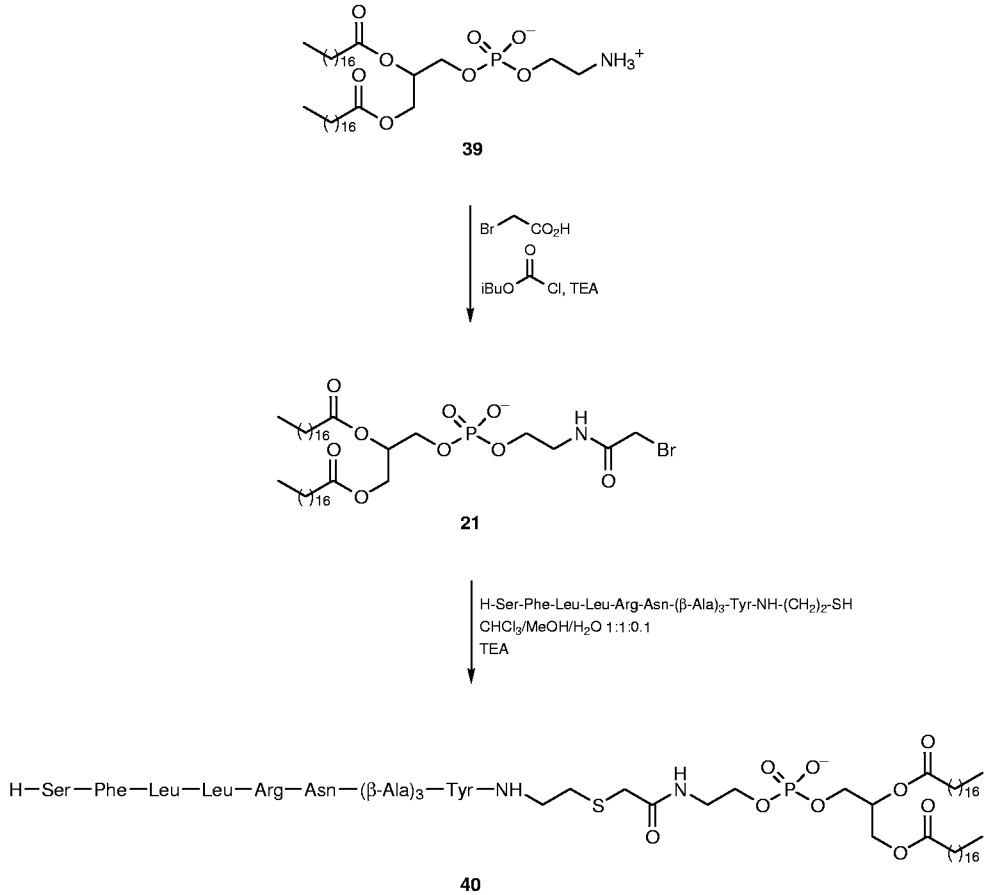
1,2-Di(3*RS*,7*R*,11*R*-phytanyl)glyceryl-1-amine (20):[192]
1,2-Di(3*RS*,7*R*,11*R*-phytanyl)glycerol (**37**) was tosylated with Tos-Cl in the presence of TEA. The product was treated with NaN_3 in DMF, followed by reduction with $LiAlH_4$ in Et_2O to obtain the product; yield: 28% based on the starting 1,2-diphytanylglycerol.

6.4.4.5 Lipophilic Building Blocks Based on Phosphatidylethanolamine

Many eukaryotic cell-surface proteins are anchored to the cellular membrane by a C-terminal amide linkage to glycosylphosphatidylinositol.[19,20] This lipophilic moiety is much more complex than the lipid groups already described and it includes, in addition to the phospholipid, a branched glycan core consisting of mannose and several phosphoethanolamine units (Scheme 13). The glycan portion can vary depending on the identity of the sugar units and on the number of phosphoethanolamine molecules attached to it. Glycosylphosphatidylinositol is nomally linked to a protein after the biosynthesis is completed in a one-step transfer reaction directed by a C-terminal signal sequence on the protein.[194]

The synthesis of peptides containing this natural lipid anchor has never been performed. However, phosphatidyl ethanolamine, the membrane-anchoring portion of glycosylphosphatidylinositol, has been used as a building block to construct synthetic lipopeptides. Functionalization of phospholipids with dicarboxylic acid active esters yields carboxy-activated phospholipid derivatives suitable for acylation of amino groups.[195] Due to the instability of such compounds to the acid treatments required in peptide chemistry, selective aminoacylation with these lipid moieties is only possible when amino protecting groups are used that can be cleaved under mild conditions. Alternatively, phospholipids are acylated with bromoacetic acid via the mixed anhydride and the resulting bromo derivatives such as compound **21** are reacted with free thiol groups on the synthetic peptide for selective conjugation, e.g. to give **40** (Scheme 19).[196]

Scheme 19 Synthesis of Phospholipid–Peptide Conjugates[196]



for references see p 370

Bromoacylation of Distearoylphosphatidylethanolamine (39) To Give 21:[196]
To a commercial suspension of 1,2-distearoyl-*sn*-phosphatidylethanolamine (**39**) in $CHCl_3$ (3 mL of a 20 mg·mL^{-1} soln; 0.058 mmol) was added TEA (10 µL, 0.058 mmol). A mixture of bromoacetic acid (11.4 mg, 0.081 mmol), isobutyl chloroformate (12 µL, 0.081 mmol), and TEA (15 µL, 0.081 mmol) in dry $CHCl_3$ (1 mL) was allowed to react at 0 °C for 5 min separately and then combined with the lipid mixture. After 15 h at rt, the mixture was concentrated to dryness at 45 °C under a strong N_2 stream and the residue was chromatographed (silica gel, $CHCl_3$/MeOH/AcOH, linear gradient 99.5:0:0.5 to 85.5:14:0.5). The fractions of interest were pooled, concentrated under a strong stream of N_2, vortexed in $CHCl_3$/MeOH/H_2O (10:10:80, 5×) and freeze-dried overnight; yield 40%; the compound was characterized by FAB-MS.

6.4.4.6 Lipophilic Building Blocks Based on *N*-Alkylamides

Taking advantage of the side-chain linkage of C-terminal trifunctional amino acids to solid supports, related *N*-mono(alkyl)amides or *N*-bis(alkyl)amides serve for the production of C-terminally lipidated peptides. So far, this approach has been reported only for the synthesis of a C-terminally mono-alkylated peptide.[30]

Fmoc-Asp(OH)-$NHC_{13}H_{27}$:[30]
Fmoc-Asp(OtBu)-OH (1.1 g, 2.5 mmol), HOBt (0.337 g, 2.5 mmol), and BOP (1.2 g, 2.5 mmol) were dissolved in a mixture of CH_2Cl_2 and DMF (1:1, 25 mL). TEA (0.52 mL, 3.75 mmol) and tridecylamine (0.6 g, 3 mmol) were then added. After 1 h at rt, the solvent was removed and the oily residue dissolved in EtOAc (100 mL), washed with 5% $NaHCO_3$, brine, 2.5% $KHSO_4$, and brine. The residue was treated with 3 M HCl in EtOAc for 1 h. The solvent was removed and the residue triturated with Et_2O. The resulting solid was filtered and dried; yield 1.1 g (96.6%).

H-His-Arg-Phe-Ala-Arg-Lys-Gly-Ala-Leu-Arg-Gly-Lys-Asn-$NHC_{13}H_{27}$:[30]
Fmoc-Asp(OH)-$NHC_{13}H_{27}$ (0.2 g, 0.45 mmol) was coupled to polyamide resin Pepsyn KR (1 g) with HOBt (0.061 g, 0.45 mmol), BOP (0.2 g, 0.45 mmol), and TEA (0.094 mL, 0.67 mmol) in DMF (3 mL) for 1 h at rt. The solvent was removed and acylation repeated with a fresh soln of the above components. After washing, the resin was transferred to an automatic synthesizer. The synthesis of the peptide was performed using standard Fmoc/tBu protocols. The final peptide was cleaved with TFA/anisole (9:1) for 5 min at rt and 15 min at 80 °C. The acidic treatment was repeated (4×). The product was isolated by RP-HPLC; yield: 34.6 mg; FAB-MS *m/z*: 1628 $[M+2H]^+$.

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6.5 Synthesis of Phosphopeptides

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The posttranslational modification of proteins via phosphorylation, sulfation, or glycosylation provides an added dimensional facility to a protein's biochemical and physiological activity via chemical changes at a particular site or changes in tertiary structure. Protein phosphorylation plays a major role in many physiological processes and occurs most frequently on the hydroxy side chain of serine and threonine residues and, to a lesser extent, on the phenolic side chain of tyrosine residues. The landmark discovery of phosphorylase kinase in the 1950s and its subsequent relationship with cAMP established that the process of glycogen metabolism involves a complex cascade mechanism which is initiated by activation of a cAMP-dependent protein kinase.[1–3] These pioneering studies demonstrate that protein serine/threonine phosphorylation plays an important role in metabolic regulation and have heralded a new era of protein regulation. Since 1960, protein kinases have been discovered at an exponential rate such that, at the turn of the millennium, the intracellular protein phosphorylation of a multitude of serine and threonine residues has been extensively documented. These processes provide an essential and critical mechanism for the regulation of cell physiology.[4]

In comparison to the level of cellular serine or threonine phosphorylation, protein tyrosine phosphorylation occurs at quite low levels in normal cells but dramatically increases upon oncogenic transformation or stimulation. Since the first discovery in 1978 that the transforming protein from Rous sarcoma virus ($pp60^{v\text{-}src}$) exhibited intrinsic kinase activity,[5] protein kinase activity has also been shown to be inherent to other growth factor receptors such as epidermal growth factor receptor and the insulin receptor,[6–9] and to involve autophosphorylation processes. The diverse biochemical activity exhibited by protein tyrosine phosphorylation has stimulated the development of chemical methods for the preparation of phosphorylated peptides for use as substrates in elucidating the biochemical and physiological activity of phosphorylated site(s).

Since the first chemical studies into the preparation of simple phosphoserine-peptides in 1959, several synthetic approaches have been investigated for the solution or solid-phase synthesis of phosphorylated peptides containing phosphoserine Ser(*P*), phosphothreonine Thr(*P*), and phosphotyrosine Tyr(*P*) residues. This section outlines two main methods for the synthesis of phosphorylated peptides which involve (1) the specific incorporation of protected phosphoamino acids in Boc or Fmoc peptide synthesis or (2) the "post-assembly" phosphorylation of Ser-, Thr- or Tyr-containing peptide-resins. For comprehensive recent reviews on the synthesis of phosphorylated peptides, the reader is directed to refs[141,142].

6.5.1 Synthesis of Phosphotyrosine-Containing Peptides Using Protected Derivatives

Although protein tyrosine phosphorylation accounts for only a small percentage of cellular phosphate, these kinases are implicated as playing critical roles in growth regulation and mitogenesis via the control of key signal pathways through changes in the phosphorylation state of intracellular protein tyrosine residues. This process is accomplished by the phosphorylation of specific tyrosine sites via a variety of protein kinases and/or dephosphorylation of the phosphorylated site via protein phosphatase action. This mechanism is responsible for the activation, inactivation, or localization of these proteins and causes a signal to be propagated, amplified, interrupted, or localized to a specific region of a cell. The key step in this pathway involves the sequence-specific recognition of a phosphorylated tyrosine residue by

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specific regions called Src Homology (SH2) domains, which occur in numerous intracellular proteins associated with cell signaling and generally possess a kinase or phosphatase domain. In association with a phosphorylated tyrosine, these SH2 domains cause phosphorylation or dephosphorylation of other tyrosine residues and play a critical role in the assemblage of numerous multiprotein complexes and the final propagation of intracellular signals.

The increasing importance of protein tyrosine phosphorylation processes in cellular signal transduction provides a continuous impetus for the ongoing development of synthetic methods for the synthesis of larger and more complex Tyr(*P*)-containing peptides and peptidomimetics for use in the study of cell regulation processes. One particular approach to the preparation of Tyr(*P*)-peptides involves the incorporation of phosphorylated tyrosine derivatives into peptides and this approach was first demonstrated in 1968 by the active ester coupling of Boc-Tyr[PO(OH)$_2$]-ONp to a tetrapeptide.[10] The viability of this approach was subsequently advanced in the late 1980s through the use of phosphate protection in which the requisite Boc-Tyr[PO(OR^1)$_2$]-OH (R^1 = alkyl) derivatives were prepared using a range of novel protected phosphorylation reagents. A major synthetic advantage of employing phosphate protection is that the incorporation of protected Boc-Tyr[PO(OR^1)$_2$]-OH derivatives facilitates peptide assembly and minimizes chemical side reactions. However, the selection of the phosphate ester is particularly critical since the efficiency of a successful peptide synthesis is dependent on the clean and quantitative removal of the phosphate protecting groups. The use of simple alkyl groups for protection of the phosphate group has been the subject of particular interest with methyl, benzyl, *tert*-butyl, 2-(trimethylsilyl)ethyl, and 2-(methyldiphenylsilyl)ethyl groups exhibiting a range of acidolytic susceptibilities as well as being prone to various silylitic-based cleavage procedures.

In addition to acid-labile alkyl phosphates, other alkyl esters such as 2,2,2-trichloroethyl and allyl have been investigated, which are acid stabile but prone to reductive cleavage. Both Fmoc-Tyr[PO(OTce)$_2$]-OH[11] and Fmoc-Tyr[PO(OAl)$_2$]-OH[12] have been employed in peptide synthesis with the final cleavage of the 2,2,2-trichloroethyl phosphate esters being effected by palladium-catalyzed hydrogenolysis in aqueous ethanol and the cleavage of the allyl esters being effected by tetrakis(triphenylphoshine)palladium(0) treatment. In a departure from phosphate protection, the phosphoramidate derivative Fmoc-Tyr[PO(NHiPr)$_2$]-OH has been successfully employed in peptide synthesis with the isopropylamine groups being hydrolyzed from the phosphoramidate by 5% aqueous TFA (4 h) treatment.[13] While these examples provide interesting alternatives and may be of use in particular cases, current focus remains on the use of acid-labile protecting groups that are quantitatively cleaved using conventional deprotection procedures.

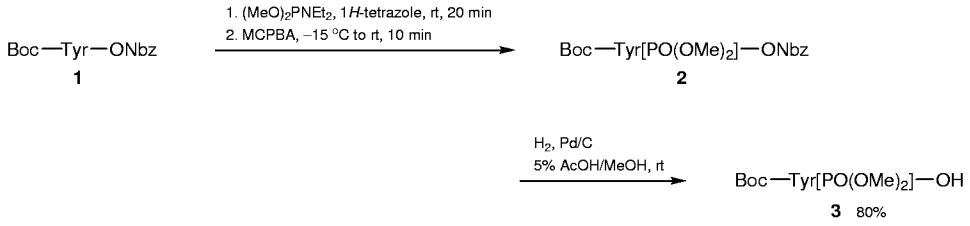
6.5.1.1 Synthesis of Phosphotyrosine-Containing Peptides via Phosphate Dimethyl Ester Protection

Initial peptide synthesis studies using protected Boc-Tyr[PO(OR^1)$_2$]-OH derivatives (R^1 = Et, Me, Bzl) showed that the methyl group was the preferred phosphate protecting group for Boc-based peptide synthesis as it is stabile under peptide assembly conditions and because it is readily removed by silylitic or acidolytic deprotection.[14] The benzyl and ethyl phosphate esters were both found to be unsuitable for Boc-based peptide assembly since the benzyl group displayed marked lability[15] to Boc-cleavage deprotection conditions (i.e., 40% TFA) whilst the ethyl group exhibited sufficient stability such that its slow silylitic or acidolytic cleavage required prolonged treatment at elevated temperatures. Despite the acid lability of the benzyl group, the Boc-Tyr[PO(OBzl)$_2$]-OH derivative has found application in the synthesis of particular Tyr(*P*)-containing peptides in which the phosphorylated residue is located at the N-terminus with the benzyl phosphate esters being cleaved during the final TFA treatment.[16,17]

6.5.1.1.1 Synthesis of Phosphotyrosine Derivatives

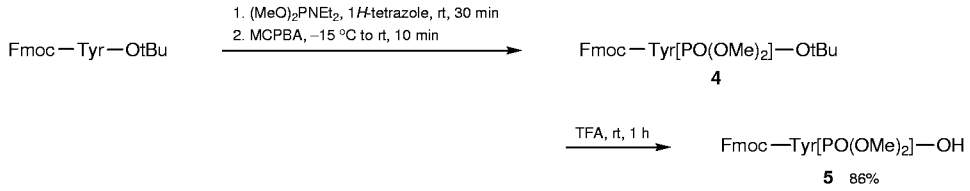
The synthesis of Boc-Tyr[PO(OMe)$_2$]-OH (**3**) is readily accomplished by phosphorylation of Boc-Tyr-ONbz (**1**) to give Boc-Tyr[PO(OMe)$_2$]-ONbz (**2**) followed by hydrogenolytic cleavage of the 4-nitrobenzyl ester. While the phosphorylation of Boc-Tyr-ONbz (**1**) is possible by treatment of its sodium phenolate (generated using NaH) with dimethyl phosphorochloridate,[14] a simpler and more convenient procedure involves the use of dimethyl dialkylphosphoramidite for the "phosphite-triester" of Boc-Tyr-ONbz (**1**) or Boc-Tyr-OMaq followed by MCPBA or tBuOOH oxidation of the intermediate phosphite triester (Scheme 1).[18] In addition to hydrogenolytic cleavage of the benzyl group, the 2-methylanthraquinone (Maq) group is also suitable for carboxy protection and offers additional removal by sodium dithionite reduction.

Scheme 1 Synthesis of Boc-Tyr[PO(OMe)$_2$]-OH[18]



The corresponding Fmoc derivative, Fmoc-Tyr[PO(OMe)$_2$]-OH (**5**), is prepared in 86% yield from Boc-Tyr[PO(OMe)$_2$]-OH by acidolytic cleavage of the Boc group using 40% TFA/CH_2Cl_2 followed by treatment of the resultant Tyr[PO(OMe)$_2$]-OH·TFA with Fmoc-OSu.[19] Alternatively, this derivative is prepared by a simple two-step procedure which employs Maq or tBu carboxy protection with the "phosphite-triester" phosphorylation of Fmoc-Tyr-OMaq or Fmoc-Tyr-OtBu being effected with dimethyl dialkylphosphoramidite[18] followed by sodium dithionite reduction of the Maq group[18,20] or acidolytic cleavage (TFA) of the *tert*-butyl ester **4** (Scheme 2).[21] While the efficiency of dithionite reductions is dependent on the variable quality of the commercially available reagent,[22] the straightforward acidolytic cleavage of the *tert*-butyl group provides the simpler synthetic procedure for the preparation of this derivative.

Scheme 2 Synthesis of Fmoc-Tyr[PO(OMe)$_2$]-OH[20,21]



Boc-Tyr[PO(OMe)$_2$]-OH (3):[18]

A soln of dimethyl diethylphosphoramidite or dimethyl diisopropylphosphoramidite (12.0 mmol) in dry THF (10 mL) was added to a soln of Boc-Tyr-ONbz (**1**) or Boc-Tyr-OMaq (10.0 mmol) and 1*H*-tetrazole (1.54 g, 22.0 mmol) in dry THF (40 mL) and the soln stirred at rt for 1 h. The soln was cooled to −15 °C (ice-salt bath) and a soln of 85% MCPBA (2.4 g, 12.0 mmol) in CH_2Cl_2 (20 mL) was added. After 10 min, EtOAc (100 mL) and 10% aq $Na_2S_2O_5$ (50 mL) were added and the aqueous layer discarded. The organic soln was washed with 10% aq $Na_2S_2O_5$ (2 × 50 mL), 5% aq $NaHCO_3$ (2 × 50 mL), and 1 M HCl (50 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure to give crude Boc-Tyr[PO(OMe)$_2$]-ONbz (**2**) as a light yellow oil.

A soln of the oil in 5% AcOH/MeOH (50 mL) containing 10% Pd/C (1.0 g) was charged with H_2 at rt and stirred until the cessation of H_2 uptake (generally 1 to 5 h). The catalyst was removed by gravity

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filtration and the solvent removed under reduced pressure. The residue was dissolved in Et_2O (100 mL), washed with 1 M HCl (2 × 50 mL) and extracted with 5% aq $NaHCO_3$ (3 × 30 mL). The combined aqueous extracts were washed with Et_2O (50 mL), acidified to pH 1 with 3 M HCl, and extracted with CH_2Cl_2 (3 × 50 mL). The soln was dried (Na_2SO_4), filtered, and the solvent removed under reduced pressure to give Boc-Tyr[PO(OMe)$_2$]-OH (**3**) as a colorless oil; yield: 3.11 g (80%); $[\alpha]_D^{24}$ 36.9 (*c* 1, $CHCl_3$); lit.[14] $[\alpha]_D^{20}$ −0.1 (*c* 1, MeOH); ^{31}P NMR ($CDCl_3$, δ): −3.7; FAB-MS *m/z*: 390 [M+H], 412 [M+Na].

Fmoc-Tyr[PO(OMe)$_2$]-OH (5):[20,21]

Fmoc-Tyr[PO(OMe)$_2$]-OtBu (**4**) was prepared as described for compound **2** by phosphorylation of Fmoc-Tyr-OtBu (10.0 mmol) and was obtained as a light yellow oil.

The oil was dissolved in TFA (3 mL), allowed to stand for 1 h, and the solvent removed under reduced pressure. The oil residue was dissolved in Et_2O (10 mL), washed with 1 M HCl (2 × 10 mL), and extracted with 5% aq $NaHCO_3$ (3 × 8 mL). The combined base extracts were washed with Et_2O (20 mL) and acidified with 3 M HCl. The aqueous soln was extracted with CH_2Cl_2 (3 × 20 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure and the residue dried under high vacuum to give Fmoc-Tyr[PO(OMe)$_2$]-OH (**5**) as a crisp white foam; yield: 0.440 g (86%); lit.[19] $[\alpha]_D^{18}$ +61.4 (*c* 1, $CHCl_3$); lit.[12] $[\alpha]_D^{20}$ +42.3 (*c* 0.7, $CHCl_3$); ^{13}C NMR ($CDCl_3$, δ): 37.1, 47.2, 54.4, 55.0 (d, $J_{P,C}$ = 5.5 Hz), 67.0, 119.9 (s, $J_{P,C}$ = 0.0 Hz), 125.0, 127.0, 127.7, 130.9, 133.2, 141.3, 143.7, 149.9 (d, $J_{P,C}$ = 6.6 Hz), 155.7, 172.9; ^{31}P NMR ($CDCl_3$, δ): −4.6; FAB-MS *m/z*: 512 [M+H].[12]

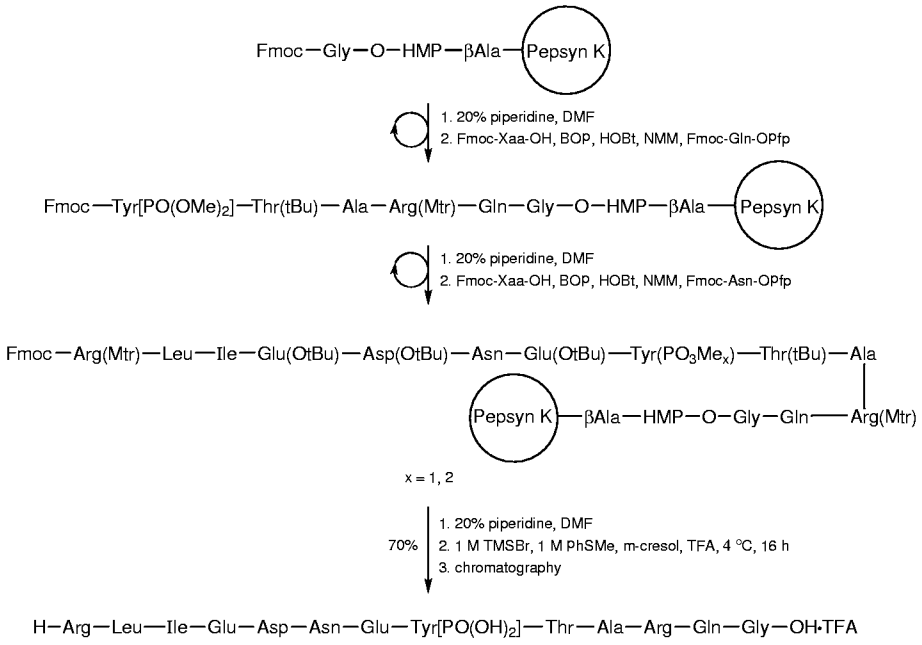
6.5.1.1.2 Solution and Solid-Phase Synthesis of Phosphotyrosine Peptides

The Boc-based solution-phase synthesis of Tyr[PO(OMe)$_2$]-peptides using Boc-Tyr-[PO(OMe)$_2$]-OH has been performed using the mixed anhydride coupling and the use of 40% TFA/CH_2Cl_2 for removal of the Boc group during peptide assembly. This approach was demonstrated by the synthesis of both Boc-Asn-Glu(OBzl)-Tyr[PO(OMe)$_2$]-Thr(Bzl)-Ala-OBzl[23] and Boc-Pro-Tyr[PO(OMe)$_2$]-Val-OMaq[18] with the coupling of the phosphorylated derivative proceeding in high yield and without byproduct formation. Despite the labor-intensive nature of solution-phase synthesis, an advantage of this approach is that it permits the efficiency of each coupling step to be monitored by ^{13}C NMR spectroscopy.

The generation of larger Tyr[PO(OMe)$_2$]-peptides is best accomplished using Boc-based solid-phase peptide synthesis with peptide assembly performed using Merrifield or PAM-polystyrene resins as the solid support and DCC/HOBt, PyBOP, or HBTU as the coupling procedure. The use of conventional HF procedures for the deprotection of Tyr[PO(OMe)$_2$]-peptides is not possible as this procedure causes extensive dephosphorylation of the Tyr-[PO(OMe)$_2$]-residue.[24,25] This problem is circumvented through the use of acid- (TfOH) or silyl-based (TMSBr, TMSOTf) procedures[26] for the cleavage of the Tyr[PO(OMe)$_2$]-peptide from its resin support, which also effects simultaneous cleavage of the methyl phosphate groups (see Section 6.5.1.1.3). A particular application of Boc-Tyr[PO(OMe)$_2$]-OH is that this derivative can be used in conjunction with both Boc-Ser[PO(OMe)$_2$]-OH and Boc-Thr[PO(OMe)$_2$]-OH for the Boc-based solid-phase synthesis of multi-phosphorylated peptides.[27]

The synthesis of Tyr(*P*)-peptides is also readily accomplished by Fmoc-based solid-phase synthesis and was first demonstrated by the synthesis of RLIEDNEY(*P*)TARQE (**6**) using Fmoc-Tyr[PO(OMe)$_2$]-OH (Scheme 3). However, partial loss of the methyl phosphate group during repeated piperidine-mediated cleavage of the Fmoc group requires that subsequent couplings are best performed using phosphonium- or uronium-mediated couplings as the activated form of the phosphorodiester[28] generated by these reagents does not undergo amidation.[29] In contrast, DCC/HOBt-mediated coupling is not recommended since activation of the phosphorodiester by this coupling procedure promotes dimer formation along with aberrant byproducts.

Scheme 3 Fmoc-Based Solid-Phase Synthesis of RLIEDNEY(*P*)TARQG Using Fmoc-Tyr[PO(OMe)$_2$]-OH[20]



6

6.5.1.1.3 Acidolytic and Silylitic Deprotection of Phosphotyrosine Dimethyl Ester Containing Peptides

The deprotection of Tyr[PO(OMe)$_2$] residues can be effected using a variety of TfOH, TMSBr, or TMSOTf based deprotection procedures and requires the addition of thioanisole or dimethyl sulfide as a nucleophilic cleavage agent.[18] The cleavage of methyl groups from H-Pro-Tyr[PO(OMe)$_2$]-Val-OH was effected using TfOH/TFA/PhSMe (or DMS)/*m*-cresol, TMSBr/TFA/PhSMe/*m*-cresol, and TMSOTf/TFA/PhSMe (or DMS)/*m*-cresol with thioanisole giving slightly higher rates of cleavage than dimethyl sulfide.[18] Methyl cleavage proceeds via a "push-pull" mechanism in which the phosphonyl oxygen of the dimethyl phosphorotriester is protonated or silylated and results in the methyl group of the dimethyl phosphonium intermediate being activated to thioanisole- or dimethyl sulfide mediated cleavage. Deprotection of Tyr[PO(OMe)$_2$]-peptides are best monitored by RP-HPLC or ^{31}P NMR spectroscopy since the rate of methyl cleavage is sequence dependent and can require up to 24 hours. While these deprotection procedures are suitable for the cleavage of peptides from Merrifield, PAM, MBHA, or Wang polystyrene supports, the two silyl-based procedures are preferred over TfOH deprotection since this acidolytic procedure fails to effect complete methyl cleavage[18,25] and generally gives products of lower purity.

The TMSBr/TFA/PhSMe/*m*-cresol procedure provides a simple one-step procedure for the deprotection of Tyr[PO(OMe)$_2$]-peptide-resins and has been routinely used for the preparation of numerous 8 to 26-mer Tyr(*P*)-peptides with methyl cleavage being complete after 16 to 24 hours (see Tables 1 and 2). Rapid and efficient demethylation of Tyr[PO(OMe)$_2$]-peptides is also possible using TMSOTf deprotection in which dimethyl sulfide was found to be superior to thioanisole for effecting demethylation with the rate of cleavage being dependent on the ratio of DMS/TMSOTf.[27] This procedure involves initial TMSOTf/TFA/PhSMe/*m*-cresol/EDT (1.5 h) cleavage of the Tyr[PO(OMe)$_2$]-peptide from the resin support (and partial demethylation) followed by the in situ addition of DMS and additional TMSOTf to effect complete methyl cleavage. A list of selected Tyr(*P*)-peptides prepared using Boc-Tyr[PO(OMe)$_2$]-OH and Fmoc-Tyr[PO(OMe)$_2$]-OH is displayed in Tables 1 and 2.

for references see p 421

Table 1 A Selection of Tyr(*P*)-Peptides Prepared Using Boc-Tyr[PO(OMe)$_2$]-OH[14,18,23,25,27,56]

Peptide	Protein	Phase	Resin Cleavage	Methyl Cleavage	Ref
NEY(*P*)TA	pp60$^{v\text{-}src}$	solution (MA)		TMSBr/PhSMe/TFA TfOH/TFA/DMS/*m*-cresol	[23]
PY(*P*)V		solution (MA)		TMSBr/PhSMe/TFA TfOH/TFA/DMS/*m*-cresol TMSOTf/TFA/DMS/*m*-cresol	[18]
LRRAY(*P*)VLG	[Y(*P*)]^5-kemptide	Merrifield (DCC/HOBt)	HBr/TFA/PhOMe	45% HBr/AcOH	[14]
FTSTEPQY(*P*)QPGENL	pp60^{src}	Merrifield (DCC/HOBt)	HF/DMS/PhOMe	TfOH/TFA/DMS/*m*-cresol	[25]
TEPQY(*P*)QPGE	pp60^{src}	Merrifield (DIC/HOBt)	TMSOTf/TFA/PhSMe/EDT/*m*-cresol+DMS/TMSOTf		[27]
FLT(*P*)EY(*P*)VATRW-TRAPEIMLN-NH$_2$	MAP kinase	MBHA (DIC/HOBt)	TMSOTf/TFA/PhSMe/EDT/*m*-cresol+DMS/TMSOTf		[27]
c[-GY(*P*)VPML-]	PDGF receptor	Wang (DIC/HOBt)	TMSBr/PhSMe/TFA/*m*-cresol/EDT		[56]

Table 2 A Selection of Tyr(*P*)-Peptides Prepared Using Fmoc-Tyr[PO(OMe)$_2$]-OH[12,19,20,30–32]

Peptide	Protein	Coupling	Resin	Resin Cleavage +Methyl Cleavage	Ref
RLIEDNEY(*P*)TARQG	pp60^{src}	BOP/HOBt	Pepsyn K	TMSBr/TFA/PhSMe	[20]
DRVY(*P*)IHPF	angiotensin II	BOP/HOBt	Pepsyn K	TMSBr/TFA/PhSMe	[19]
AEY(*P*)LRV	EGFR	BOP/HOBt	Wang	TMSBr/TFA/PhSMe	[30]
TAENAEY(*P*)LRVA	EGFR	BOP/HOBt	Wang	TMSBr/TFA/PhSMe	[30]
TRDIY(*P*)ETDY(*P*)YKK	IR	BOP/HOBt	Wang	TMSBr/TFA/PhSMe	[30]
TRDIYETDY(*P*)YKK	IR	BOP/HOBt	Wang	TMSBr/TFA/PhSMe	[30]
TRDIYETDYY(*P*)KK	IR	BOP/HOBt	Wang	TMSBr/TFA/PhSMe	[30]
EDNEY(*P*)TARE	p56^{lck}	TPTU	Wang	TMSBr/TFA/PhSMe	[30]
TEGQY(*P*)QPQP	p56^{lck}	TPTU	Wang	TMSBr/TFA/PhSMe	[30]
TEPQY(*P*)QPGE	p60^{src}	TPTU	Wang	TMSBr/TFA/PhSMe	[12, 30]
EAENTITY(*P*)SLLKH	FcγRIIB1	BOP/HOBt	Wang	TMSBr/TFA/PhSMe	[31]
DMPDDYEDENLY(*P*)EGLNLDDCSMYEDI	Ig-α	BOP/HOBt	unknown	TMSBr/TFA/PhSMe	[32]
DMPDDYEDENLYEGLNLDDCSMY(*P*)EDI	Ig-α	BOP/HOBt	unknown	TMSBr/TFA/PhSMe	[32]
DMPDDYEDENLY(*P*)EGLNLDDCSMY(*P*)EDI	Ig-α	BOP/HOBt	unknown	TMSBr/TFA/PhSMe	[32]

H-Pro-Tyr(*P*)-Val-OH·TFA; TfOH Deprotection:[18]
Pro-Tyr[PO(OMe)$_2$]-Val-OH·TFA (50.0 mg, 0.084 mmol) was dissolved in TfOH/TFA/PhSMe/*m*-cresol (1:5:3:1, 3.0 mL) and stirred under N_2 at 0 °C for 4 h. The soln was removed under reduced pressure and chilled Et_2O (50 mL) added. The precipitated oil was triturated with Et_2O (3 × 30 mL), dried under high vacuum and purified by semipreparative RP-HPLC (0.1% TFA, 0–60% MeCN over 30 min) to give H-Pro-Tyr(*P*)-Val-OH·TFA as a fluffy white powder; yield: 25.0 mg (52%); ^{31}P NMR ($CDCl_3$, δ): −3.6; FAB-MS *m/z*: 458 [M+H].

H-Pro-Tyr(*P*)-Val-OH·TFA; TMSBr Deprotection:[18]
TMSBr (0.40 mL) was added by syringe to a soln of H-Pro-Tyr[PO(OMe)$_2$]-Val-OH·TFA (38.0 mg, 0.06 mmol) in PhSMe (0.35 mL), *m*-cresol (0.30 mL), and TFA (1.95 mL) and the soln stirred under N_2 at 0 °C for 6 h. MeOH (2 mL) was added and the solvent removed under reduced pressure. The residue was dissolved in H_2O (20 mL), washed with Et_2O (2 × 20 mL) and CH_2Cl_2 (2 × 20 mL), lyophilized, and purified by semipreparative RP-HPLC to give H-Pro-Tyr(*P*)-Val-OH·TFA as a fluffy white powder; yield: 27.0 mg (75%).

H-Pro-Tyr(*P*)-Val-OH·TFA; TMSOTf Deprotection:[18]
TMSOTf (0.50 mL) was added by syringe to a soln of H-Pro-Tyr[PO(OMe)$_2$]-Val-OH·TFA (47.0 mg, 0.08 mmol) in PhSMe (0.35 mL), *m*-cresol (0.30 mL), and TFA (1.85 mL) and the soln stirred under N_2 at 4 °C for 22 h. MeOH (2 mL) was added, the soln removed under reduced pressure, and chilled Et_2O (50 mL) added. The precipitated oil was triturated with Et_2O (2 × 10 mL), dried under high vacuum and purified by semipreparative RP-HPLC to give H-Pro-Tyr(*P*)-Val-OH·TFA as a fluffy white powder; yield: 20.0 mg (53%).

H-Thr-Glu-Pro-Gln-Tyr(*P*)-Gln-Pro-Gly-Glu-OH; TMSOTf Deprotection:[27]
A soln of 1 M TMSOTf/PhSMe in TFA (2.0 mL), *m*-cresol (100 µL) and EDT (100 µL) was added to the Tyr[PO(OMe)$_2$]-Merrifield-resin (29.0 mg, 11.0 µmol) and the soln stirred under N_2 at 4 °C. After 90 min, TMSOTf (0.4 mL) and DMS (0.6 mL) were added and the soln stirred at 4 °C for a further 3 h. The resin was filtered and the peptide was precipitated by the addition of Et_2O (12 mL). The peptide was collected by centrifugation, washed with Et_2O (3 × 12 mL), and purified by RP-HPLC to give the Tyr(*P*)-nonapeptide as a white powder; yield: 10.9 mg (81%); IS-MS/MS *m/z*: 1128.2 [M+H].

6.5.1.2 Fmoc-Based Solid-Phase Synthesis of Phosphotyrosine-Containing Peptides via Acid-Labile Phosphate Alkyl Ester Protection

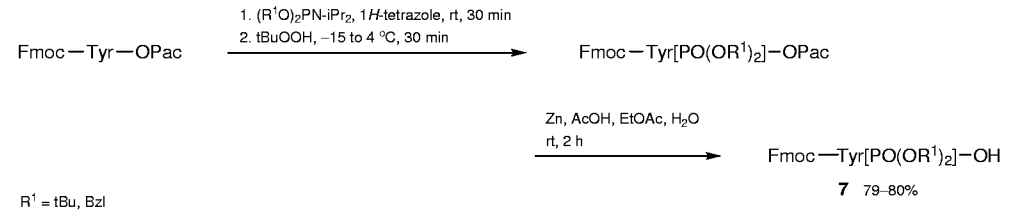
A major advancement in the Fmoc-based solid-phase synthesis of Tyr(*P*)-peptides was realized with the Fmoc-Tyr[PO(OR^1)$_2$]-OH derivatives **7** (R^1 = Bzl, tBu, Mdpse) [Mdpse = 2-(methyldiphenylsilyl)ethyl] in which the acid-labile phosphate esters offered simple cleavage by TFA treatment. While the initial difficult synthesis of derivatives **7** restricted general synthetic usage, the subsequent commercial availability of these derivatives and their compatibility with solid-phase synthesis has provided a simple and efficient procedure for the routine synthesis of large complex Tyr(*P*)-peptides.

6.5.1.2.1 Synthesis of Acid-Labile Phosphate Alkyl Ester Derivatives of Fmoc-Protected Phosphotyrosine

The chemical preparation of the requisite Fmoc-Tyr[PO(OR^1)$_2$]-OH derivatives **7** is now particularly straightforward with the advent of commercially available (BzlO)$_2$PNR$^2_2$ and (tBuO)$_2$PNR$^2_2$ (R^2 = Et, iPr) reagents for the "phosphite-triester" phosphorylation of protected Tyr derivatives. Both Fmoc-Tyr[PO(OBzl)$_2$]-OH and Fmoc-Tyr[PO(OtBu)$_2$]-OH are conveniently prepared at the 20-mmol scale by a simple three-step procedure which involves: (1) phenacyl protection of Fmoc-Tyr-OH, (2) (R^{1}O)$_2$PNR$^2_2$ (R^1 = Bzl, tBu; R^2 = Et, iPr) mediated phosphitylation of Fmoc-Tyr-OPac followed by *tert*-butyl hydroperoxide oxidation to give Fmoc-Tyr[PO(OR^1)$_2$]-OPac (R^1 = Bzl or tBu), followed by (3) Zn/AcOH reduction of the phenacyl group (Scheme 4).[33] Since the *tert*-butyl derivative is prone to slow autocatalytic decomposition, this derivative needs to be isolated free of acid contamination and is best stored at low temperature under an inert atmosphere.

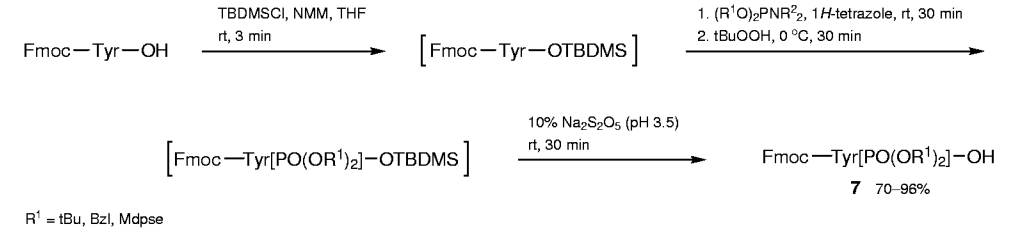
for references see p 421

Scheme 4 Synthesis of Fmoc-Tyr[PO(OBzl)$_2$]-OH and Fmoc-Tyr[PO(OtBu)$_2$]-OH[33]



The synthesis of the three derivatives of Fmoc-Tyr[PO(OR^1)$_2$]-OH **7** (R^1 = tBu, Bzl, Mdpse) is also readily accomplished at the 5-mmol scale via a one-pot procedure and utilizes temporary *tert*-butyldimethylsilyl protection of the carboxy group (Scheme 5).[34,35] This procedure involves initial protection of the carboxy group of Fmoc-Tyr-OH with *tert*-butyldimethylsilyl chloride, "phosphite-triester" phosphitylation of the tyrosine hydroxy group with (R^{1}O)$_2$PNR$^2_2$/1*H*-tetrazole (tBuOOH oxidation) and final cleavage of the *tert*-butyldimethylsilyl group by mild acidolysis at pH 3.5. The 2-(methyldiphenylsilyl)ethyl group (Mdpse) is extremely sensitive to mild acidic conditions and is an extension of the 2-(trimethylsilyl)ethyl (TMSE) group, which lacks sufficient stability to survive prolonged storage.[36]

Scheme 5 One-Pot Synthesis of Fmoc-Tyr[PO(OR^1)$_2$]-OH Derivatives[34–36]



Fmoc-Tyr[PO(OtBu)$_2$]-OH (7, R^1 = tBu):[33]
Di-*tert*-butyl diethylphosphoramidite (7.47 g, 30.0 mmol) in dry THF (10 mL) was added to a stirred soln of Fmoc-Tyr-OPac (9.90 g, 19.0 mmol) and 1*H*-tetrazole (4.97 g, 71.0 mmol) in dry THF (85 mL) and the soln stirred at rt for 30 min. The soln was cooled to −15 °C (ice-salt bath) and 70% tBuOOH (8.4 mL) in H_2O (30 mL) was added dropwise such that the temperature was kept below 4 °C. After 30 min at 4 °C, Et_2O (250 mL) and H_2O (500 mL) were added and the aqueous layer discarded. The organic soln was washed with 10% aq $Na_2S_2O_5$ (2 × 50 mL), 5% aq $NaHCO_3$ (3 × 50 mL), 10% aq citric acid (2 × 50 mL), and sat. NaCl (50 mL). The soln was dried (Na_2SO_4), filtered, and the solvent removed under reduced pressure to give crude Fmoc-Tyr[PO(OtBu)$_2$]-OPac as a thick yellow oil.

Fresh acid-washed Zn dust (8.54 g) was added to a soln of the crude Fmoc-Tyr[PO(OtBu)$_2$]-OPac in EtOAc (60 mL), AcOH (150 mL), and H_2O (30 mL) and the soln vigorously stirred at rt for 2 h. The unreacted Zn was removed by suction filtration, EtOAc (200 mL) and H_2O (1000 mL) were added to the filtrate, and the organic soln washed with H_2O (3 × 400 mL). The solvent was removed under reduced pressure, the oily residue dissolved in Et_2O (100 mL) and extracted with 5% aq $NaHCO_3$ (6 × 40 mL). The combined base extracts were washed with Et_2O (2 × 60 mL), acidified to pH 3.5 with solid citric acid, and extracted with Et_2O (2 × 100 mL). The organic phase was washed with H_2O (50 mL) and sat. NaCl (50 mL), dried (Na_2SO_4), filtered, and the solvent removed under reduced pressure to give compound **7** (R^1 = tBu) as a crisp white foam; yield: 8.94 g (79%); R_f ($CHCl_3$/MeOH/AcOH 90:8:2) 0.3; ^{13}C NMR ($CDCl_3$, δ): 29.7 (d, $J_{P,C}$ = 4.4 Hz, PtBu CH_3), 37.0, 47.0, 54.7, 66.9, 82.9, and 84.3 (each d, $J_{P,C}$ = 7.7 Hz, tBu Cq), 119.8 ($J_{P,C}$ = 0.0 Hz, Tyr Ar C2), 120.0, 125.0, 126.9, 127.6, 130.4, 132.4, 141.1, 143.6, 150.0 (d, $J_{P,C}$ = 6.6 Hz, Tyr Ar C1), 155.8, 173.8; ^{31}P NMR ($CDCl_3$, δ): 16.0; FAB-MS *m/z*: 596 [M+H]; IS-MS *m/z*: 596.0 [M+H].

Fmoc-Tyr[PO(OBzl)$_2$]-OH (7, R^1 = Bzl):[33]
Fmoc-Tyr[PO(OBzl)$_2$]-OH was prepared as described for derivative **7** (R^1 = tBu) using dibenzyl diethylphosphoramidite (9.51 g, 30.0 mmol) (note, the Na salt of the benzyl derivative partially separates

as a thick brown oil from the aqueous soln); yield: 10.08 g (80%); ^{13}C NMR ($CDCl_3$, δ): 37.0, 47.0, 54.5, 66.9, 70.1 (d, $J_{P,C}$ = 5.5 Hz, Bzl Cb), 119.8 ($J_{P,C}$ = 0.0 Hz, Tyr Ar C2), 120.0, 124.9, 127.0, 127.6, 128.0, 128.5, 128.6, 130.7, 133.1, 135.0 (d, $J_{P,C}$ = 6.6 Hz, Bzl Ar C1), 141.2, 143.6, 149.2 (d, $J_{P,C}$ = 7.7 Hz, Tyr Ar C1), 155.7, 173.2; ^{31}P NMR ($CDCl_3$, δ): 7.0; FAB-MS *m/z*: 664 [M+H].

One-Pot Synthesis of Fmoc-Tyr[PO(OtBu)$_2$]-OH (7, R^1 = tBu):[34,35]
NMM (0.152 g, 1.5 mmol) in dry THF (1 mL) and TBDMSCl (0.214 g, 1.43 mmol) in THF (1 mL) were successively added to a soln of Fmoc-Tyr-OH (0.605 g, 1.5 mmol) in THF (3 mL) at rt. After 3 min, 1*H*-tetrazole (0.49 g, 7.0 mmol) was added in one portion followed by the addition of di-*tert*-butyl diethylphosphoramidite (0.747 g, 3.0 mmol) in THF (1 mL) at rt. After 30 min, the soln was cooled to −5 °C and 14% aq tBuOOH (2.6 mL, 4.0 mmol) was added. After 30 min at +5 °C, 10% aq $Na_2S_2O_5$ (10 mL) was added at 0 °C and the mixture rapidly stirred at rt for 30 min. The soln was transferred to a separatory funnel using Et_2O (40 mL) and the aqueous phase discarded. The organic phase was washed with 10% aq $Na_2S_2O_5$ (2 × 15 mL) and extracted with 5% $NaHCO_3$ (3 × 12 mL). The combined aqueous extracts were washed with Et_2O (30 mL), acidified to pH 3.5 (pH electrode) with 30% citric acid, and extracted with CH_2Cl_2 (3 × 30 mL). The solvent was removed under reduced pressure, the residue dissolved in Et_2O (50 mL), washed with H_2O (5 × 30 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure and dried under high vacuum to give Fmoc-Tyr[PO(OtBu)$_2$]-OH (**7**, R^1 = tBu) as a crisp, white foam; yield: 0.73 g (85%).

One-Pot Synthesis of Fmoc-Tyr[PO(OBzl)$_2$]-OH (7, R^1 = Bzl):[34,35]
The one-pot synthesis of Fmoc-Tyr[PO(OBzl)$_2$]-OH was prepared as described for derivative **7** (R^1 = tBu) using dibenzyl diethylphosphoramidite (0.951 g, 3.0 mmol) in the above procedure and gave the title compound as a white, crisp foam; yield: 0.96 g (96%) (note, the Na salt of the benzyl derivative partially separates as a thick brown oil from the aq $NaHCO_3$).

One-Pot Synthesis of Fmoc-Tyr[PO(OMdpse)$_2$]-OH (7, R^1 = Mdpse):[36]
NMM (0.505 g, 5.0 mmol) and TBDMSCl (0.748 g, 5.0 mmol) were successively added to a soln of Fmoc-Tyr-OH (2.02 g, 5.0 mmol) in THF (15 mL). After 15 min at rt, 1*H*-tetrazole (1.05 g, 15.0 mmol) was added followed by the addition of (MdpseO)$_2$PN-iPr$_2$ (6.13 g, 10.0 mmol) in THF (5 mL). After 2 h at rt, the soln was cooled to 0 °C and 90% aq tBuOOH (15.0 mmol) was added. After 3 h at 0 °C, the soln was removed under reduced pressure, the oily residue dissolved in EtOAc (50 mL), washed with 5% aq $KHSO_4$ (30 mL) and sat. NaCl (30 mL), dried (Na_2SO_4), and filtered. The soln was removed under reduced pressure, purified by column chromatography (silica gel, $CHCl_3$/MeOH/AcOH 9:1:0.1) and lyophilized from dioxane to give derivative **7** (R^1 = Mdpse) as a white powder; yield: 3.26 g (70%); ^{31}P NMR (DMSO-d_6, δ): −5.84.

6.5.1.2.2 Solid-Phase Synthesis of Phosphotyrosine Peptides with Acid-Labile Phosphate Alkyl Ester Protection

The three protected Fmoc-Tyr[PO(OR^1)$_2$]-OH derivatives **7** (R^1 = Bzl, tBu, Mdpse) are generally incorporated into Fmoc-based SPPS in a threefold excess (30–60 min) using either BOP, PyBOP, HBTU, or HATU as the coupling reagent.[12,33,35–40] In the case of large-scale peptide syntheses, the consumption of derivatives **7** is minimized by performing couplings in a 1.5- to 2-fold excess for extended reaction times (1–2 h). While both the Bzl and Mdpse phosphate esters are susceptible to piperidine-mediated cleavage, the generated phosphorodiester functionality does not compromise subsequent phosphonium/uronium coupling cycles.[12,33,36,37,39] A feature of these acid-labile phosphate groups is that complete cleavage is effected during the routine acidolytic cleavage (1–4 h) of the peptide/polymer-support linkage.[26]

The ready incorporation of derivatives **7** in Fmoc-based solid-phase synthesis has permitted the simple and straightforward synthesis of numerous Tyr(*P*)-peptides ranging from 5- to 17-mer in length (see Table 3). A particular advantage of the Fmoc-Tyr[PO(OR^1)$_2$]-OH derivative is that the *tert*-butyl phosphate ester is stable to piperidine-mediated ester hydrolysis and thereby avoids the introduction of multiple negatively charged phosphorodiester groups into a peptide chain. This is particularly crucial in the case of contiguous incorporation of phosphotyrosine residues where the repulsive effect of negative charges is known to retard the efficiency of residue couplings[41] and may promote peptide aggregation in specific peptide sequences.

for references see p 421

Table 3 Tyr(*P*)-Peptides Prepared Using Fmoc-Tyr[PO(OR^1)$_2$]-OH Derivatives[12,21,33,36–40]

Peptide	Protein	Resin	Resin Cleavage	Ref
R^1 = Bzl				
GVY(*P*)AASG	$p85^{gag\text{-}fes}$	Pepsyn K	TFA/PhSMe (95:5)	[37]
GVY(*P*)AASG	$p85^{gag\text{-}fes}$	Wang	TFA/H_2O (95:5)	[33]
GVY(*P*)AASG	$p85^{gag\text{-}fes}$	Wang	TFA/PhSMe/H_2O/PhOMe/ EDT (82.5:5:5:5:2.5)	[33]
TFLPVPEY(*P*)INQSV	EGFR	Pepsyn K	TFA/PhOH (95:5)	[37]
TFLPVPEY(*P*)INQSV		Wang	TFA/H_2O (95:5)	[33]
TFLPVPEY(*P*)INQSV		Wang	TFA/PhSMe/H_2O/PhOMe/ EDT (82.5:5:5:2.5)	[33]
ETGTKAELLA-KY(*P*)EATHK	helix	PAL-PEG-PS	TFA/PhSMe/H_2O/Et_3SiH (95:2:2:1)	[39]
TEPQY(*P*)QPGE	$p60^{src}$	Pepsyn K	TFA/H_2O/PhSH (95:2.5:2.5)	[12]
EDNEY(*P*)TARE	$p56^{lck}$	Pepsyn K	TFA/H_2O/PhSH (95:2.5:2.5)	[12]
TEGQY(*P*)QPQP	$p56^{lck}$	Pepsyn K	TFA/H_2O/PhSH (95:2.5:2.5)	[12]
TAENAEY(*P*)KRVA	EGFR	Pepsyn K	TFA/H_2O/PhSH (95:2.5:2.5)	[12]
TRDIY(*P*)ETDYYRK	IR	Pepsyn K	TFA/H_2O/PhSH (95:2.5:2.5)	[12]
TRDIYETDY(*P*)YRK	IR	Pepsyn K	TFA/H_2O/PhSH (95:2.5:2.5)	[12]
TRDIYETDYY(*P*)RK	IR	Pepsyn K	TFA/H_2O/PhSH (95:2.5:2.5)	[12]
R^1 = tBu				
KVEKI-GEGTY(*P*)GVVYK	cdc2	Wang	TFA/H_2O (95:5)	[40]
GVY(*P*)AASG	$p85^{gag\text{-}fes}$	Wang	TFA/H_2O (95:5)	[33]
GVY(*P*)AASG		Wang	TFA/PhSMe/H_2O/PhOMe/ EDT (82.5:5:5:5:2.5)	[33]
TFLPVPEY(*P*)INQSV	EGFR	Wang	TFA/H_2O (95:5)	[33]
TFLPVPEY(*P*)INQSV	EGFR	Wang	TFA/PhSMe/H_2O/PhOMe/ EDT (82.5:5:5:5:2.5)	[33]
TEPQY(*P*)QPGE	$p60^{src}$	Wang	TFA/H_2O/PhSH (95:2.5:2.5)	[12]
DIY(*P*)ETDY(*P*)Y(*P*)RK	IR	Wang	TFA/H_2O/Et_3SiH (95:3:2)	[21]
SSSY(*P*)Y(*P*)		Wang	TFA/PhOMe (95:5)	[38]
DRIY(*P*)VHPF	[Y(*P*)]^4-Ang II	Wang	TFA/PhSMe (95:5)	[38]
RRPY(*P*)IL	[Y(*P*)]^4-NT (8–13)	Wang	TFA/PhSMe (95:5)	[38]
R^1 = Mdpse				
PDDY(*P*)EDEN-NH_2	MB-1	LinkerAM	TFA/PhOH/Et_3SiH (95:3:2)	[36]
ENLY(*P*)EGLN-NH_2	MB-1	LinkerAM	TFA/PhOH/Et_3SiH (95:3:2)	[36]
CSMY(*P*)EDIS-NH_2	MB-1	LinkerAM	TFA/PhOH/Et_3SiH (95:3:2)	[36]

6.5.1.3 Synthesis of Phosphotyrosine Peptides Using Fmoc-Protected Phosphotyrosine or the Related Monobenzyl Ester

In contrast to full phosphate protection, a directional change in the synthesis of Tyr(*P*)-peptides was demonstrated in 1993[42] through the successful use of Boc-Tyr[PO(OH)$_2$]-OH or Fmoc-Tyr[PO(OH)$_2$]-OH by the simple change of DCC-mediated coupling[43] to BOP coupling. ^{31}P NMR studies have shown that both DCC/HOBt and HBTU cause activation of the phosphate group of Fmoc-Tyr[PO(OH)$_2$]-OH but that only the DCC-activated phosphate undergoes further reactions during the acylation step.[21] While Fmoc-Tyr[PO(OH)$_2$]-OH is readily incorporated using various phosphonium/uronium reagents (BOP, PyBOP, HBTU, HATU, etc.) (Table 4),[40,42,44–46] both HBTU and HATU provide improved coupling efficiency over both BOP and PyBOP.[46] Despite the improved performance of uronium reagents, HBTU-mediated couplings of Fmoc-Tyr[PO(OH)$_2$]-OH are not as problem-free as promoted in one report[26] since variable quantities of diphosphotyrosine [Tyr(*PP*)] peptides, diphosphate bridged Tyr(*P*)-peptide dimers $[2M - 16]^+$, and other high-molecular-weight phosphorylated byproducts have been isolated from various peptide syntheses.[40,44,46]

Table 4 Tyr(*P*)-Peptides Prepared Using Fmoc-Tyr[PO(OH)$_2$]-OH[40,42,44,45]

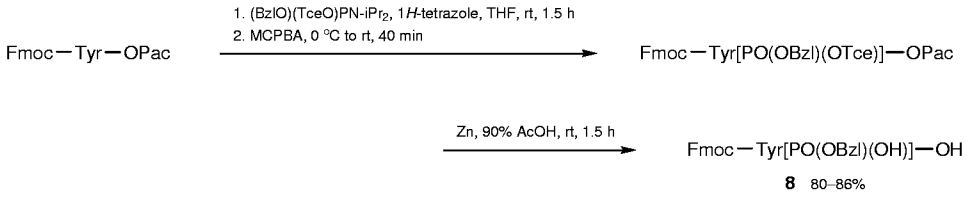
Peptide	Coupling Reagent	Cleavage[a]	Ref
DY(*P*)MK-NH$_2$	BOP/HOBt/NMM	reagent K	[42]
SSENFDDY(*P*)MK-NH$_2$	BOP/HOBt/NMM	reagent K	[42]
AEEEIY(*P*)GEFEAKKK	PyBOP/HOBt/NMM	TFA/H$_2$O (95:5)	[40]
ADFGLARIEDNEY(*P*)TAREG	PyBOP/HOBt/NMM	reagent K	[40]
KKTHQEEEEPQY(*P*)EEIPIYL	PyBOP/HOBt/NMM	TFA/H$_2$O/TES (95:2.5:2.5)	[40]
KVEKIGEGTY(*P*)GVVYK	PyBOP/HOBt/NMM	TFA/H$_2$O (95:5)	[40]
Y(*P*)LRVAP-NH$_2$	BOP/HOBt/NMM	TFA/H$_2$O (95:5)	[44]
NAEY(*P*)LRVAP-NH$_2$	BOP/HOBt/NMM	TFA/H$_2$O (95:5)	[44]
VIEDNEY(*P*)TAR	BOP/HOBt/NMM	TFA/H$_2$O/PhOMe	[45]

[a] Reagent K: TFA/PhOH/H$_2$O/PhSMe/EDT (82.5:5:5:5:2.5).

In some instances, the coupling of Fmoc-Tyr[PO(OH)$_2$]-OH is sluggish and requires extended coupling times and/or double couplings and has been attributed to loss of available coupling reagent for carboxy activation through competing phosphate activation. The poor acylation with Fmoc-Tyr[PO(OH)$_2$]-OH/HBTU is particularly pronounced in the coupling of contiguous Tyr[PO(OH)$_2$]-residues as evident by the synthesis of EIY(*P*)ETDY(*P*)Y-(*P*)A being contaminated with 10% EIY(*P*)ETDY(*P*)A [the Tyr(*P*)-deleted peptide] in spite of double coupling of the phosphotyrosine residues.[46] Despite the inherent synthetic problems involved with the use of Fmoc-Tyr[PO(OH)$_2$]-OH, this derivative nevertheless provides a facile means for the synthesis of small- to medium-length simple Tyr(*P*)-peptides.

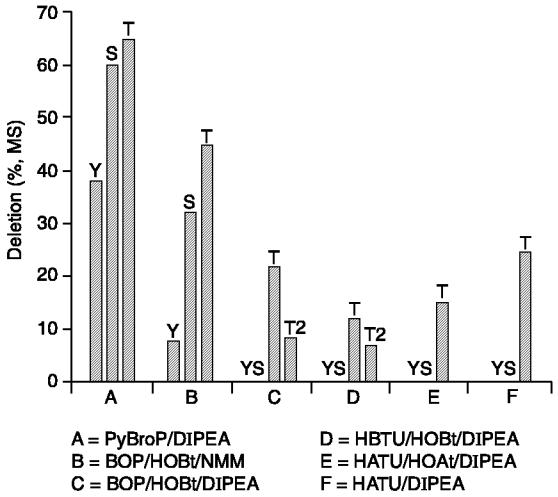
A major improvement in the synthesis of Tyr(*P*)-peptides was realized through the use of Fmoc-Tyr[PO(OBzl)(OH)]-OH (**8**)[26] in which the side-chain phosphate functionality is protected as its benzyl phosphorodiester. The synthesis of **8** is accomplished by phosphorylation of Fmoc-Tyr-OPac using the difunctional phosphoramidite, (TceO)(BzlO)PN-iPr$_2$, followed by simultaneous reductive removal of the Tce and Pac groups from Fmoc-Tyr-[PO(OBzl)(OTce)]-OPac by Zn/AcOH treatment (Scheme 6).[47] Despite partial phosphate protection, the coupling efficiency of **8** is also effected by the phosphorodiester functionality as comparative studies showed that both HBTU/HOBt/DIPEA and HATU/HOAt/DIPEA provided complete incorporation whereas BOP failed to provide efficient coupling and PyBroP performed poorly (Figure 1).[48] The effectiveness of Fmoc-Tyr[PO(OBzl)(OH)]-OH/HBTU coupling was demonstrated by the synthesis of the problematic α-helical ETGTKAELLAKY(*P*)EATHK-NH$_2$ peptide that gave comparable results to those obtained by a comparative synthesis using BOP coupling of Fmoc-Tyr[PO(OBzl)$_2$]-OH.[39]

Scheme 6 Synthesis of Fmoc-Tyr[PO(OBzl)(OH)]-OH[47]



for references see p 421

Figure 1 Proportion of the Xaa(*P*)-Deleted Peptide, ASQGLEDPA-NH_2, Determined in Crude Products Obtained from the PyBroP/DIPEA, BOP/HOBt/NMM, BOP/HOBt/DIPEA, HBTU/HOBt/DIPEA, HATU/HOAt/DIPEA, and HATU/DIPEA Coupling of Fmoc-Tyr[PO(OBzl)(OH)]-OH (Y), Fmoc-Ser[PO-(OBzl)-(OH)]-OH (S), and Fmoc-Thr[PO(OBzl)(OH)]-OH (T) (3 equiv) for 1 h.[48][a,b]



[a] The T2 bars represent 2 h coupling times using Fmoc-Thr[PO(OBzl)OH].

[b] Reprinted from [Perich; Ede; Eagle; Bray, *Letters in Peptide Science*, (1999) Vol. 6], Copyright (1999), p 91 with permission from Kluwer Academic Publishers.

Fmoc-Tyr[PO(OH)$_2$]-OH:[42]

A suspension of phosphotyrosine[49] (0.50 g, 1.9 mmol), Fmoc-OSu (0.78 g, 2.3 mmol), and TEA (0.27 mL, 1.9 mmol) in H_2O (5 mL) and MeCN (5 mL) was rapidly stirred at rt for 30 min with the soln maintained at pH 9 by addition of TEA. The solvent was removed under reduced pressure, EtOAc (50 mL) and H_2O (50 mL) added and the soln acidified to pH 2 with 12 M HCl. The organic phase was collected and the aqueous phase extracted with EtOAc (2 × 50 mL). The combined organic phases were combined, filtered, and washed with 1 M HCl (2 × 40 mL), H_2O (2 × 40 mL), and sat. NaCl (2 × 40 mL). The soln was dried ($MgSO_4$), filtered, and the solvent removed under reduced pressure to give Fmoc-Tyr[PO(OH)$_2$]-OH as a white crystalline solid; yield: 0.76 g (81%); mp 132–135 °C; ^{31}P NMR ($CDCl_3$, δ): −4.2; FAB-MS *m/z*: 484 [M+H], 504 [M+Na].

Fmoc-Tyr[PO(OBzl)(OH)]-OH (8):[47]

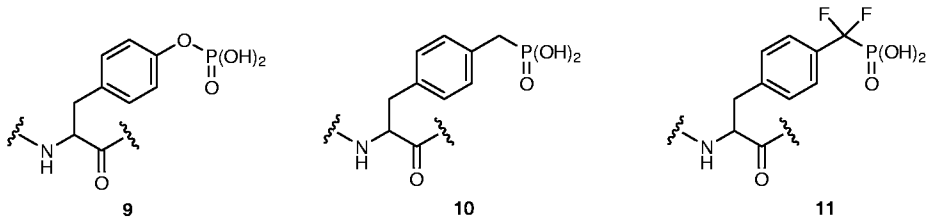
1*H*-Tetrazole (2.36 g, 33.6 mmol) and (TceO)(BzlO)PN-iPr_2[47] (6.08 g, 15.7 mmol) were added to a soln of Fmoc-Tyr-OPac (5.84 g, 11.2 mmol) in THF (25 mL) under N_2 and the soln stirred at rt for 1.5 h. The soln was cooled to 0 °C, 80% MCPBA (6.05 g, 28.0 mmol) added and the soln stirred at 0 °C for 10 min and then at rt for 30 min. A soln of $NaHSO_3$ (2.93 g, 28.0 mmol) in H_2O (10 mL) was added, the mixture stirred for 5 min at 0 °C and the soln removed under reduced pressure. The residue was dissolved in EtOAc (50 mL), washed successively with 10% aq citric acid (3 × 20 mL), NaCl soln (20 mL) sat. $NaHCO_3$ (6 × 20 mL), and NaCl soln (3 × 20 mL), dried ($MgSO_4$), and the solvent removed under reduced pressure to give crude Fmoc-Tyr[PO(OBzl)(OTce)]-OPac as a thick oil.

Zn dust (11.0 g, 168.0 mmol) was added in several portions to a soln of the crude oil in 90% AcOH (110 mL) and stirred at rt for 1.5 h. The insoluble material was filtered and the solvent removed under reduced pressure. The oil was dissolved in Et_2O (50 mL), filtered, and the solvent removed under reduced pressure [the two filtered solids were combined, stirred in EtOAc (100 mL) and 1 M HCl (30 mL) containing sat. NaCl, the aqueous phase was extracted with EtOAc (2 × 20 mL), the organic extractions were combined, washed with NaCl soln (4 × 30 mL), dried ($MgSO_4$), and the solvent removed under reduced pressure]. The two residues were combined and crystallized (EtOAc/Et_2O/hexane) to give **8** as a white solid; yield: 80–86%.

6.5.2 Synthesis of Phosphotyrosine Peptide Mimetics

The regulation of cellular tyrosine phosphorylation levels by protein tyrosine kinases and phosphatases has focused attention on the use of synthetic nonhydrolyzable phosphonate mimetics of the Tyr(*P*)-residue as competitive substrates for PTPase inhibition. In the search for suitable peptidomimetics of Tyr(*P*)-peptides **9** (Scheme 7), the synthesis of peptides containing Phe[$PO(OH)_2$]-, Phe[$CH_2PO(OH)_2$]-, Phe[$CHFPO(OH)_2$]-, and Phe[CF_2-$PO(OH)_2$]-mimetics has been particularly pursued in which the phenolic oxygen atom of the Tyr(*P*)-residue is either removed[50] or replaced by the CH_2, CHF, or CF_2 group. While early studies focused on the synthesis of Phe[$CH_2PO(OH)_2$]-peptides **10** using various Fmoc-Phe[$CH_2PO(OR^1)_2$]-OH derivatives (R^1 = tBu, Me, Et),[51–56] this mimetic was subsequently surpassed by the Phe[$CF_2PO(OH)_2$] group on the basis that Phe[$CF_2PO(OH)_2$] peptides **11** exhibited increased inhibitory potency in comparative inhibition studies.[57–59] This improved inhibitory performance is attributed to the pK_a of the Phe[$CF_2PO(OH)_2$] group (approx. 5.71) being closer to that of the Tyr(*P*)-residue (approx. 6.22) than either the Phe[CH_2-$PO(OH)_2$] (pK_a 7.72) or Phe[$CHFPO(OH)_2$] groups (pK_a 6.60).[60] The SPPS synthesis of various Phe[$CF_2PO(OH)_2$] peptides has been accomplished using both Boc-Phe[CF_2-$PO(OEt)_2$]-OH or Fmoc-Phe[$CF_2PO(OEt)_2$]-OH (**12**) or Fmoc-Phe[$CF_2PO(OH)_2$]-OH (**13**) with phosphonate protection being limited to the stable ethyl group for derivatives Boc-Phe[$CF_2PO(OEt)_2$]-OH or Fmoc-Phe[$CF_2PO(OEt)_2$]-OH (**12**) due to the unavailability of suitable CF_2-based phosphonylating reagents bearing other labile alkyl groups.

Scheme 7 Tyr(*P*)-Peptides and Peptidomimetics Thereof



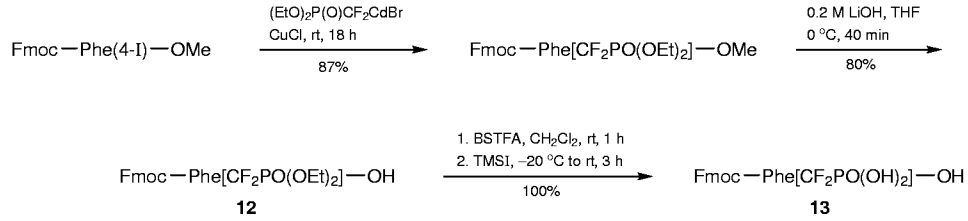
9 10 11

6.5.2.1 Synthesis of N-Protected Phenylalanine (Difluoromethyl)phosphonic Acids and Their Alkyl Esters

The synthesis of the target Boc-Phe[$CF_2PO(OEt)_2$]-OH and Fmoc-Phe[$CF_2PO(OEt)_2$]-OH derivatives has been achieved using five different procedures[61–65] with the key step based on the generation of the α,α-difluorobenzylic phosphonate functionality. The simplest and most efficient of these methods involves the copper(I) chloride mediated coupling of the cadmium or zinc derivative of commercially available diethyl bromodifluoromethylphosphonate[64] with Boc-Phe(I)-OMe or Fmoc-Phe(I)-OMe. Subsequent mild saponification of the methyl ester is effected using lithium hydroxide and gives both the target Boc-Phe[$CF_2PO(OEt)_2$]-OH and Fmoc-Phe[$CF_2PO(OEt)_2$]-OH (**12**) in good yield (Scheme 8). In addition, the unprotected phosphonate derivative, Fmoc-Phe[$CF_2PO(OH)_2$]-OH (**13**), is prepared by treatment of Fmoc-Phe[$CF_2PO(OEt)_2$]-OH (**12**) with *N*,*O*-bis(trimethylsilyl)trifluoroacetamide (BSTFA) followed by in situ treatment with TMSI.[65,66] An advantage in the use of this derivative in peptide synthesis is that it obviates problems in effecting the complete cleavage of ethyl phosphonate esters from large sterically hindered Phe[$CF_2PO(OEt)_2$]-peptides.

for references see p 421

Scheme 8 Synthesis of Fmoc-Phe[$CF_2PO(OEt)_2$]-OH and Fmoc-Phe[$CF_2PO(OH)_2$]-OH[64–66]



Fmoc-Phe[$CF_2PO(OEt)_2$]-OH (12); Typical Procedure:[64]

A soln of diethyl bromodifluoromethylphosphonate (0.89 mL, 5.0 mmol) in dry DMF (5 mL) was added dropwise over 5 min to acid-washed Cd (or Zn) (100 mesh) (5.5 mmol) under N_2 and stirred at rt for 3 h. The unreacted Cd powder was allowed to settle over 30 min and an aliquot (2.5 mL, 2.5 mmol) of the Cd reagent was added to a flame-dried round-bottomed flask containing Fmoc-Phe(I)-OMe (0.5 g, 0.95 mmol) and CuCl (0.38 g, 3.8 mmol). After stirring at rt for 18 h, a second soln of the Cd reagent (1.5 mmol) was added and the soln was stirred at rt for a further 18 h. Et_2O (100 mL) was added, the soln was filtered through Celite and then washed with 1 M HCl (50 mL) and 5% $NaHCO_3$ (2 × 50 mL), dried (Na_2SO_4), and filtered. The soln was removed under vacuum and purified by flash column chromatography (EtOAc/hexane 45:55). The solvent was removed under vacuum to give Fmoc-Phe[$CF_2PO(OEt)_2$]-OMe as a thick oil; yield: 0.49 g (87%); $[\alpha]_D^{24}$ +28.0 (*c* 1.08, $CHCl_3$); lit.[63] $[\alpha]_D^{24}$ +21.4 (*c* 1.05, $CHCl_3$); ^{13}C NMR ($CDCl_3$, δ): 16.5, 38.1, 47.3, 52.6, 54.8, 65.0, 67.1, 118.1 (td, J = 265.0, 219.0 Hz), 120.1, 125.1, 125.2, 126.6, 127.2, 127.9, 129.5, 131.5 (td, J = 23.0, 14.0 Hz), 140.0, 141.4, 143.8, 143.9, 155.6, 171.7; ^{19}F NMR ($CDCl_3$, δ): −109.6 (d, $J_{P,F}$ = 116.0 Hz); ES-MS *m/z*: 588.4 [M+H].

0.2 M LiOH (0.24 mmol) was added to a soln of Fmoc-Phe[$CF_2PO(OEt)_2$]-OMe (0.070 g, 0.12 mmol) in THF (1.5 mL) at 0 °C and stirred for 40 min.[63,64] EtOAc (20 mL) and 0.2 M HCl (20 mL) were added to the soln, the organic phase collected, the aqueous layer extracted with EtOAc (30 mL), the organic solns combined, dried (Na_2SO_4), and filtered. The soln was removed under reduced pressure to give Fmoc-Phe[$CF_2PO(OEt)_2$]-OH (**12**) as a hygroscopic light yellow foam;[63] yield: 0.055 g (80%); lit.[66] $[\alpha]_D$ 42.0 (*c* 1.0, $CHCl_3$); lit.[63] $[\alpha]_D^{24}$ 41.6 (*c* 1.10, $CHCl_3$); lit.[61] FAB-MS *m/z*: 574 [M+H]; ES-MS (+ve) *m/z*: 574.2 (M+H).

Boc-Phe[$CF_2PO(OEt)_2$]-OH:[64]

This derivative was prepared from Boc-Phe(I)-OMe (0.95 mmol) according to the procedure for derivative **12** and gave Boc-Phe[$CF_2PO(OEt)_2$]-OMe as a thick oil; yield: 0.385 g (87%); ^{13}C NMR ($CDCl_3$, δ): 16.6, 28.5, 38.4, 52.6, 54.4, 65.0, 80.3, 118.2 (td, J = 263.0, 220.0 Hz), 126.6, 129.6, 131.5 (td, J = 23.0, 14.0 Hz), 139.3, 155.2, 172.2; ^{19}F NMR ($CDCl_3$, δ): −109.6 (d, $J_{P,F}$ 117.0 Hz); ES-MS *m/z*: 466.4 [M+H].

The hydrolysis[63,64] of Boc-Phe[$CF_2PO(OEt)_2$]-OMe was performed according to the procedure for derivative **12** to give Boc-Phe[$CF_2PO(OEt)_2$]-OH as a gummy solid; yield: 0.246 g (66%);[63] lit.[63] $[\alpha]_D$ +8.06 (*c* 1.08, MeOH); lit.[62] $[\alpha]_D$ +7.96 (*c* 1.08, MeOH); lit.[61] FAB-MS *m/z*: 452.1 [M+H]; ES-MS *m/z*: 450.2 [M−H].

Fmoc-Phe[$CF_2PO(OH)_2$]-OH (13):[65,66]

N,O-Bis(trimethylsilyl)trifluoroacetamide (BSTFA, 6.4 mL, 24.2 mmol) was added to a soln of Fmoc-Phe[$CF_2PO(OEt)_2$]-OH (2.1 mmol) in dry CH_2Cl_2 (20 mL) and stirred under N_2 at rt for 1 h. TMSI (2.5 mL, 17.6 mmol) was added by syringe at −20 °C and the soln stirred at −20 °C for 1 h. The soln was stirred at rt for a further 1 h, the solvent removed under reduced pressure, and the residue dissolved in MeCN/TFA/H_2O (4.0:0.5:1.0, 6.5 mL) and stirred at rt for 2.5 h. The soln was removed under reduced pressure, the brown oil dissolved in EtOAc (50 mL), washed with 5% aq $Na_2S_2O_5$ (20 mL) and sat. NaCl (20 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure to give Fmoc-Phe[$CF_2PO(OH)_2$]-OH (**13**) as a yellow glass; yield: 1.19 g (>100%); lit.[66] $[\alpha]_D$ 7.0 (*c* 0.18, DMSO); lit.[65] ^{13}C NMR (CD_3OD, δ): 38.1, 57.6, 67.9, 120.9, 126.9, 126.2, 127.5, 128.2, 128.7, 130.0, 140.6, 142.5, 145.1, 145.2, 158.4, 174.9; lit.[65] ESI-MS *m/z*: 518 [M+H].

6.5.2.2 Synthesis of Peptides Using Fmoc-Protected Phenylalanine (Difluoromethyl)phosphonate Dialkyl Ester Derivatives

The incorporation of Fmoc-Phe[$CF_2PO(OEt)_2$]-OH in Fmoc-based solid-phase synthesis is readily performed using DIC/HOBt, PyBOP, and other uronium-based reagents (HBTU, HATU). The deprotection of the peptide-resin is effected by the one-step treatment of the

Phe[$CF_2PO(OEt)_2$]-peptide-resin with TMSOTf/DMS/TFA/EDT/*m*-cresol or via initial cleavage of the peptide from the resin and TMSOTf treatment of the purified Phe[$CF_2PO(OEt)_2$]-peptide (Table 5).[61,67] The cleavage of ethyl phosphonate esters is generally incomplete and requires the minor proportion of the Phe[$CF_2PO(OEt)(OH)$]-peptide to be separated from the target Phe[$CF_2PO(OH)_2$]-peptide by RP-HPLC. While the routine use of Fmoc-Phe[$CF_2PO(OEt)_2$]-OH is limited by the slow acidolytic cleavage of ethyl phosphonate esters, the availability of alternative derivatives such as Fmoc-Phe[$CF_2PO(OMe)_2$]-OH or Fmoc-Phe[$CF_2PO(OBzl)_2$]-OH is expected to facilitate peptide deprotection on the basis of the greater lability of these esters to acidolytic cleavage.

In the case of Fmoc-Phe[$CF_2PO(OH)_2$]-OH,[65,66] the incorporation of this derivative in Fmoc-based solid-phase synthesis is best performed using uronium-based reagents (HBTU, HATU) in order to minimize side reaction of the free phosphonate group (Table 5). While the incorporation of derivative **13** simplifies the final peptide deprotection step,[66] the coupling efficiency of Fmoc-Phe[$CF_2PO(OH)_2$]-OH and following Fmoc amino acids should be monitored (e.g., Kaiser test for batch synthesis) since the incorporation of the side-chain phosphonate group to the peptide chain has been found to compromise the coupling efficiency of subsequent residues.

Table 5 Phe[$CF_2PO(OH)_2$]-Peptides Prepared Using Boc/Fmoc-Phe[$CF_2PO(OEt)_2$]-OH and Fmoc-Phe[$CF_2PO(OH)_2$]-OH Derivatives[59,61,63–67]

Peptide[a]	Amino Acid	Cleavage	Ref
G-DL-ZVPML-NH_2	Boc-DL-Phe[$CF_2PO(OEt)_2$]-OH[61]	TMSOTf/DMS/TFA/EDT/*m*-cresol	[61]
G-DL-ZVPML	Fmoc-DL-Phe[$CF_2PO(OEt)_2$]-OH[61]	TMSOTf/DMS/TFA/EDT/*m*-cresol	[61,67]
Ac-N-DL-ZIDLD-NH_2	Fmoc-DL-Phe[$CF_2PO(OEt)_2$]-OH[61]	TMSOTf/DMS/TFA/EDT/*m*-cresol	[67]
Ac-N-DL-ZVNIE-OH	Fmoc-DL-Phe[$CF_2PO(OEt)_2$]-OH[61]	TMSOTf/DMS/TFA/EDT/*m*-cresol	[67]
Ac-DADEZL-NH_2	Fmoc-L-Phe[$CF_2PO(OEt)_2$]-OH[63–66]	TMSOTf/DMS/TFA/EDT/*m*-cresol	[59]
Ac-NZVNIE-NH_2	Fmoc-L-Phe[$CF_2PO(OEt)_2$]-OH[63–66]	TMSOTf/DMS/TFA/EDT/*m*-cresol	[59]
Ac-QZEEIP-NH_2	Fmoc-L-Phe[$CF_2PO(OEt)_2$]-OH[63–66]	TMSOTf/DMS/TFA/EDT/*m*-cresol	[59]
Ac-DZVPML-NH_2	Fmoc-L-Phe[$CF_2PO(OEt)_2$]-OH[63–66]	TMSOTf/DMS/TFA/EDT/*m*-cresol	[59]
Ac-ZINQ-NH_2	Fmoc-L-Phe[$CF_2PO(OH)_2$]-OH[65,66]	not required	[66]
Ac-EZINQ-NH_2	Fmoc-L-Phe[$CF_2PO(OH)_2$]-OH[65,66]	not required	[66]

[a] Z = Phe[$CF_2PO(OH)_2$].

6.5.3 Synthesis of Phosphoserine and Phosphothreonine Peptides

In addition to enzymatic protein regulation, phosphorylation of serine plays an important role in high-molecular-weight phosphoproteins, which have the ability to bind extraordinary amounts of calcium phosphate or serve as nucleators and/or regulators of biomineralization. Since the first studies in the early 1960s that examined the ability of food phosphoproteins from milk (caseins), egg white (riboflavin-binding protein) and egg yolk (phosvitin) to bind calcium phosphate,[68,69] other phosphoproteins such as dentine phosphophoryn, dentine matrix phosphoprotein, osteopontin, matrix-Gla protein, and statherin have been identified and have the common feature of possessing a clustered region of phosphorylated serine residues.[70–75]

The first general and practicable method for the chemical synthesis of Ser(*P*)-containing peptides was developed during the early 1960s in which diphenyl and dibenzyl phosphorochloridate were employed for the phosphorylation of various protected Ser-containing tripeptides followed by hydrogenolytic deprotection of the protected Ser[$PO(OR^1)_2$]-containing peptides.[68] Despite the successful preparation of various simple Ser(*P*)-tripeptides, the extension of this method to the synthesis of larger Ser(*P*)-peptides method was com-

for references see p 421

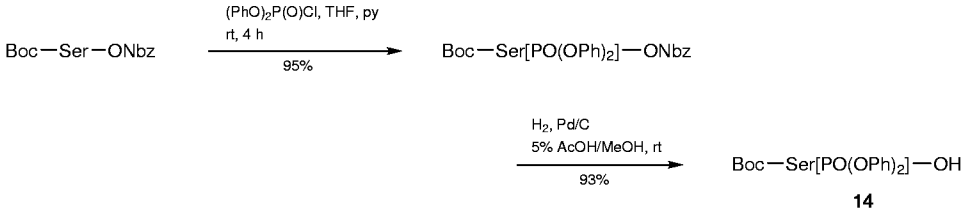
plicated due to inherent limitations of the phosphorylation/hydrogenation combination and that the phosphorylation of specific peptide sequences was followed by a dephosphorylative rearrangement. A more efficient and reliable procedure for the synthesis of Ser(*P*)-peptides was possible through the use of protected Boc-Ser[PO(OR^1)$_2$]-OH derivatives in which the phosphorylated amino acid is directly incorporated into solution or solid-phase peptide synthesis. A key requirement of the phosphate protecting group is that it has sufficient stability to survive the course of peptide assembly and yet, provide sufficient lability to permit efficient cleavage without concomitant decomposition of the sensitive Ser(*P*)-residue. To this purpose, the synthesis of Ser(*P*)- and Thr(*P*)-containing peptides has been accomplished by the use of protected Boc-Ser[PO(OR^1)$_2$]-OH and Boc-Thr[PO(OR^1)$_2$]-OH derivatives (R^1 = Ph, Me) in Boc-based solid-phase synthesis or through the use of Fmoc-Ser[PO(OBzl)(OH)]-OH in Fmoc-based solid-phase synthesis. However, despite the successful development of these two particular methods, the development of efficient solid-phase approaches is of further interest, its aim is to generate large phosphorylated peptides containing multiple phosphorylated residues.

6.5.3.1 Boc-Based Solution-Phase Synthesis of Phosphoserine and Phosphothreonine Peptides

The first studies into the synthesis of Ser(*P*)-containing peptides used protected Boc-Ser-[PO(OR^1)$_2$]-OH derivatives with phenyl and alkyl (R^1 = Ph, Et, Me, Bzl) phosphate protection.[76] Of these groups, phenyl phosphate protection has proved most suitable for peptide synthesis since the alkyl phosphate groups are either too acid sensitive to survive peptide assembly conditions (Bzl) or required harsh acidolytic conditions for their final removal (Me, Et). While early studies reported problems in effecting the complete platinum-catalyzed hydrogenolytic cleavage of phenyl groups from Ser[PO(OPh)$_2$]-peptides in AcOH (presumably through "clumping" of the catalyst), this situation was remedied by using a platinum-mediated hydrogenation procedure with TFA as the hydrogenation solvent.[77]

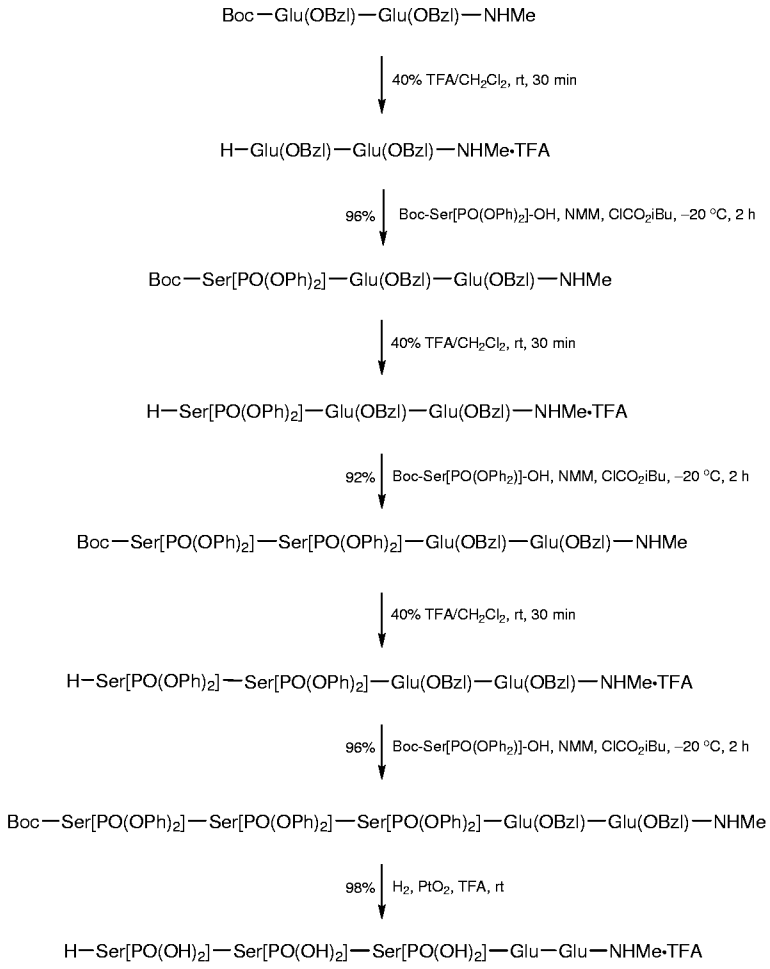
The synthesis of Boc-Ser[PO(OPh)$_2$]-OH (**14**) and Boc-Thr[PO(OPh)$_2$]-OH is accomplished by a simple three-step procedure which employs diphenyl phosphorochloridate for the phosphorylation of the hydroxy group of serine and threonine (Scheme 9). This procedure involves: (1) 4-nitrobenzyl protection of the carboxy group of Boc-Ser-OH or Boc-Thr-OH; (2) phosphorylation of Boc-Ser-ONbz or Boc-Thr-ONbz with diphenyl phosphorochloridate in pyridine followed by; (3) palladium-catalyzed hydrogenolytic cleavage of the 4-nitrobenzyl protecting group.[76] While phosphorylation of Boc-Ser-ONbz proceeds quickly and is complete within four hours, the phosphorylation of Boc-Thr-ONbz is much slower due to increased steric hindrance about the secondary hydroxy group. In this case, complete phosphorylation is effected by the supplementary addition of phosphorochloridate reagent during the phosphorylation and extending the reaction time to 16 hours.

Scheme 9 Synthesis of Boc-Ser[PO(OPh)$_2$]-OH[76]



Both Boc-Ser[PO(OPh)$_2$]-OH (**14**) and Boc-Thr[PO(OPh)$_2$]-OH can be readily coupled in high yields[78,79] using the mixed anhydride coupling procedure and no major steric problems have been encountered for the coupling of the sterically hindered Thr derivative.[79] On completion of peptide assembly, the protected Ser[PO(OPh)$_2$]- and Thr[PO(OPh)$_2$]-containing peptide is deprotected by platinum-mediated hydrogenolysis in TFA solution with phenyl cleavage generally being complete within 60 minutes. The hydrogenation procedure requires the use of 1.1 equivalents of platinum oxide per phenyl group so as to ensure rapid and complete reductive cleavage of the phenyl group without premature "clumping" of the platinum. The target Ser(*P*)- or Thr(*P*)-peptide is simply isolated by removal of platinum by gravity filtration, removal of the solvent, and ethereal precipitation of the phosphopeptide. (Scheme 10).

Scheme 10 Boc-Based Solution-Phase Synthesis of H-Ser(*P*)-Ser(*P*)-Ser(*P*)-Glu-Glu-NHMe[78]



for references see p 421

Although Boc-based solution-phase synthesis is laborious and time-consuming, this approach has generally been found to generate protected Ser[PO(OPh)$_2$]- and Thr[PO(OPh)$_2$]-peptides in such high purity that little, or no, chromatographic purification of the Ser(*P*)- or Thr(*P*)-peptide is necessary after hydrogenolytic deprotection. This has been particularly critical in the preparation of multiply phosphorylated peptide sequences such as H-Ser(*P*)-Ser(*P*)-Ser(*P*)-Glu-Glu-NHMe (Scheme 10) and Ac-Glu-Ser(*P*)-Leu-Ser(*P*)-Ser(*P*)-Ser(*P*)-Glu-Glu-NHMe which were unsuitable for purification by reversed-phase chromatography due to their high hydrophilicity.[80] However, despite the many successful syntheses using this approach, a limiting factor of this method is that it precludes the synthesis of phosphopeptides containing Phe, Tyr, Trp, His, Cys, and Met residues due to the incompatibility of these residues with the hydrogenolytic deprotection conditions.

The Boc-based solution-phase synthesis of Ser(*P*)- and Thr(*P*)-containing peptides has also been performed through the use of the protected Boc-Ser[PO(OTce)$_2$]-OH and Boc-Thr[PO(OTce)$_2$]-OH derivatives.[81,82] In this procedure, the cleavage of the 2,2,2-trichloroethyl groups from the protected Ser[PO(OTce)$_2$]-peptide is effected by palladium-catalyzed hydrogenolysis and is performed in aqueous ethanol solution since reductive cleavage does not proceed in TFA solution. An economical advantage of this procedure is that the less expensive Pd/C is used for the reduction of the Tce groups. Although the solution-phase synthesis of Ser[PO(OTce)$_2$]-containing peptides may be limited due to the lower solubilities of these peptides in comparison to Ser[PO(OPh)$_2$]-containing peptides, both Boc-Ser[PO(OTce)$_2$]-OH and Boc-Thr[PO(OTce)$_2$]-OH may be more amenable for use in Boc-based solid-phase synthesis.

Boc-Ser[PO(OPh)$_2$]-ONbz:[76]
A soln of diphenyl phosphorochloridate (4.0 g, 15.0 mmol) in THF (5 mL) was added to a soln of Boc-Ser-ONbz (3.40 g, 10.0 mmol) in pyridine (5 mL) at 4 °C and the soln stirred at rt for 4 h. 1 M HCl (10 mL) was added, the soln stirred for a further 30 min and Et_2O (50 mL) added. The aqueous phase was discarded and the ethereal phase was washed with 1 M HCl (3 × 30 mL) and 5% $NaHCO_3$ (2 × 30 mL), dried (Na_2SO_4), filtered, and the soln removed under reduced pressure. The white solid was crystallized (EtOAc/petroleum ether) to give Boc-Ser[PO(OPh)$_2$]-ONbz as fine off-white needles; yield: 5.43 g (95%); mp 113–114.5 °C; $[\alpha]_D^{14}$ +6.71 (*c* 1, $CHCl_3$); ^{13}C NMR ($CDCl_3$, δ): 28.2, 54.0 (d, $J_{P,C}$ = 7.3 Hz), 66.0, 68.6 (d, $J_{P,C}$ = 5.9 Hz), 80.6, 119.9 (d, $J_{P,C}$ = 4.4 Hz), 123.7, 125.6, 128.4, 129.8, 142.0, 147.7, 150.2 (d, $J_{P,C}$ = 7.3 Hz), 155.0, 168.7; ^{31}P NMR ($CDCl_3$, δ): −12.2.

Boc-Thr[PO(OPh)$_2$]-ONbz:[79]
A soln of diphenyl phosphorochloridate (4.0 g, 15.0 mmol) in THF (5 mL) was added to a soln of Boc-Thr-ONbz (3.54 g, 10.0 mmol) in pyridine (5 mL) at 4 °C. After 5 h at 4 °C, a second soln of diphenyl phosphorochloridate (2.0 g, 7.5 mmol) in THF (2.5 mL) was added and the soln stirred at 4 °C for 16 h. 1 M HCl (10 mL) was added, the soln was stirred for a further 30 min, and Et_2O (50 mL) added. The organic soln was washed with 1 M HCl (3 × 30 mL) and 5% aq $NaHCO_3$ (2 × 30 mL), dried (Na_2SO_4), and filtered. The soln was removed under reduced pressure to give Boc-Thr[PO(OPh)$_2$]-ONbz as a light yellow oil; yield: 5.74 g (98%); ^{13}C NMR ($CDCl_3$, δ): 18.0 (d, $J_{P,C}$ = 2.2 Hz), 27.8, 57.7 (d, $J_{P,C}$ = 6.6 Hz), 65.6, 76.8 (d, $J_{P,C}$ = 5.5 Hz), 80.0, 119.6 (d, $J_{P,C}$ = 4.4 Hz), 123.2, 125.1, 128.1, 129.4, 141.8, 147.3, 149.9 (d, $J_{P,C}$ = 6.6 Hz), 155.4, 168.8; ^{31}P NMR ($CDCl_3$, δ): −12.9.

Boc-Ser[PO(OPh)$_2$]-OH (14):[76]
A soln of Boc-Ser[PO(OPh)$_2$]-ONbz (4.86 g, 8.50 mmol) in 5% AcOH/MeOH (50 mL) containing 10% Pd/C (0.85 g) was charged with H_2 at rt and rapidly stirred until the cessation of H_2 uptake. The catalyst was removed by gravity filtration and the soln removed under reduced pressure. The residue was dissolved in Et_2O (50 mL) and washed with 1 M HCl (2 × 20 mL). The organic phase was extracted with 5% $NaHCO_3$ (3 × 8 mL) and the combined aqueous extracts washed with Et_2O (20 mL). The soln was acidified to pH 1 with 3 M HCl, extracted with CH_2Cl_2 (3 × 30 mL), dried (Na_2SO_4), filtered, and the soln removed under reduced pressure. The white solid was crystallized (Et_2O) to give Boc-Ser[PO(OPh)$_2$]-OH as white needles; yield: 3.47 g (93%); mp 126–128 °C; $[\alpha]_D^{14}$ +40.0 (*c* 1, $CHCl_3$); ^{13}C NMR ($CDCl_3$, δ): 28.2, 53.7 (d, $J_{P,C}$ = 5.9 Hz), 68.9 (d, $J_{P,C}$ = 5.9 Hz), 80.4, 120.0 (d, $J_{P,C}$ = 4.4 Hz), 125.6, 129.8, 150.1 (d, $J_{P,C}$ = 7.3 Hz), 155.4, 170.7; ^{31}P NMR ($CDCl_3$, δ): −12.5; FAB-MS *m/z*: 438 [M+H].

Boc-Thr[PO(OPh)$_2$]-OH:[79]
Boc-Thr[PO(OPh)$_2$]-ONbz (2.64 g, 4.50 mmol) was hydrogenated as described for the hydrogenation of Boc-Ser[PO(OPh)$_2$]-ONbz and gave the title compound as a clear oil; yield: 1.79 g (88%); ^{13}C NMR ($CDCl_3$, δ): 18.3 (d, $J_{P,C}$ = 2.2 Hz), 28.1, 57.6 (d, $J_{P,C}$ = 7.7 Hz), 77.7 (d, $J_{P,C}$ = 6.6 Hz), 80.4, 119.9 (d, $J_{P,C}$ = 5.5 Hz), 125.4, 129.6, 150.1 (d, $J_{P,C}$ = 6.6 Hz), 155.8, 171.6; ^{31}P NMR ($CDCl_3$, δ): −13.4; FAB-MS *m/z*: 452 [M+H].

Ac-Glu-Ser(*P*)-Leu-Ser(*P*)-Ser(*P*)-Glu-Glu-NHMe:[80]
Peptide assembly commenced with Boc-Glu(OBzl)-NHMe, Boc-Glu(OBzl)-OH, Boc-Ser-[PO(OPh)$_2$]-OH, Boc-Leu-OH, and AcOH being coupled as their isobutoxycarbonyl mixed anhydrides [repetitive excess mixed anhydride (REMA) method] and the Boc group cleaved by 40% TFA/CH_2Cl_2 (30 min).

A rapidly stirred soln of the protected octapeptide Ac-Glu(OBzl)-Ser[PO(OPh)$_2$]-Leu-Ser[PO(OPh)$_2$]-Ser[PO(OPh)$_2$]-Ser[PO(OPh)$_2$]-Glu(OBzl)-Glu(OBzl)-NHMe (0.212 g, 0.1 mmol) in TFA/AcOH (1:1; 4 mL) containing 80% PtO_2 (0.25 g, 0.88 mmol) was hydrogenated at atmospheric pressure until H_2 uptake ceased (60 min). The Pt was removed by gravity filtration, washed with TFA (4 mL) and the solvent removed under reduced pressure. The residue was triturated with Et_2O (3 × 20 mL) and the white solid lyophilized from H_2O to give the title octapeptide as a fluffy white solid; yield: 0.123 g (99%); ^{13}C NMR (400 MHz) (D_2O, δ): 20.50 and 22.25 (Leu Cδ), 21.69 (acetyl C), 24.15 (Leu Cγ), 25.74, 25.84, 25.98, and 26.13 (NHMe and Glu^{1,7,8} Cβ), 29.94 (Glu^{1,7,8} Cγ; 39.54, Leu Cβ), 52.36 (Leu Cα), 53.13, 53.18, and 53.29 (Glu^{1,7,8} Cα), 53.90, 54.08, 54.82, and 54.95 (d, $J_{P,C}$ = 7.3, 8.8, 7.3, 7.3 Hz, Ser^{2,4,5,6} Cα), 64.05–64.20 (m, Ser^{4,5,6} Cβ), 64.27 (d, $J_{P,C}$ = 8.8 Hz, Ser^2 Cβ), 170.65, 170.94, 171.31, and 171.37 (Ser^{2,4,5,6} CO), 172.97 (acetyl CO), 173.35 and 173.56 (Glu^{7,8} CO), 174.45 (Glu^1 CO), 174.59 (Leu CO), 176.80, 176.86, and 176.88 (Glu^{1,7,8} δ-CO); ^{31}P NMR (D_2O, δ): −0.1; FAB-MS *m/z*: 1242 [M+H]; amino acid analysis: Glu 2.99 (3), Ser 3.99 (4), Leu 1.03 (1).

H-Thr(*P*)-Thr(*P*)-Thr(*P*)-Glu-Glu-NHMe:[79]
Peptide assembly was performed using Boc-Thr[PO(OPh)$_2$]-OH and hydrogenation of the protected Thr[PO(OPh)$_2$]-peptide was performed as above and gave the title pentapeptide as a fluffy white solid; yield: 0.094 g (99%); ^{13}C NMR (400 MHz) (D_2O, δ): 17.7 (Thr^{1,2,3} Cγ), 25.8 (Glu^{4,5} Cβ), 25.9 (NHMe), 29.8 and 29.9 (Glu^{4,5} Cγ), 53.2 (Glu^{4,5} Cα), 57.8, 58.4, and 58.9 (d, $J_{P,C}$ = 6.6, 6.6, 5.5 Hz, Thr^{1,2,3} Cα), 70.8, 72.0, and 72.4 (d, each $J_{P,C}$ = 4.4 Hz, Thr^{1,2,3} Cβ), 167.3, 170.4, and 170.8 (Thr^{1,2,3} CO), 172.9 and 173.3 (Glu^{4,5} CO), 176.9 (Glu^{4,5} CO-δ); ^{31}P NMR (D_2O, δ): −0.9; FAB-MS *m/z*: 833 [M+H]; amino acid analysis: Glu 2.07 (2), Thr/Thr(*P*) 2.93 (3).

6.5.3.2 Boc-Based Solid-Phase Synthesis of Phosphoserine and Phosphothreonine Peptides

In contrast to the inherent limitations of synthesis in solution, solid-phase peptide synthesis provides a key method for the generation of many large and complex peptides. The application of phosphorylated amino acids to solid-phase methodology has been the subject of particular interest in consideration of the synthetic potential of this approach for the rapid and routine preparation of complex phosphopeptides. Unlike other amino acids, the generation of Ser(*P*)- and Thr(*P*)-peptides is complicated due to the sensitivity of these residues to harsh acid or base conditions and the selection of suitable phosphate derivatives that are compatible with solid-phase peptide synthesis.

for references see p 421

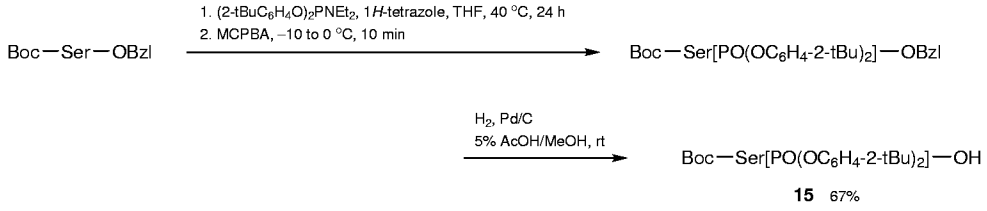
6.5.3.2.1 Boc-Based Solid-Phase Synthesis Using Aryl Phosphate Ester Protection

The Boc-based solid-phase assembly of Ser[PO(OPh)$_2$]- and Thr[PO(OPh)$_2$]-peptides has generally been performed using DCC/HOBt for the coupling of both Boc-Ser[PO(OPh)$_2$]-OH and Boc-Thr[PO(OPh)$_2$]-OH. While phosphonium and uronium reagents are also suitable coupling reagents, as demonstrated by the problem-free assembly of LRRAS-[PO(OPh)$_2$]LG using PyBOP/NMM,[82] further study of coupling methods is warranted in consideration that the BOP/DIPEA coupling of Boc-Ser[PO(OcPe)$_2$]-OH was reported to result in significant byproduct formation.[83]

In the development of a solid-phase approach, the cleavage of the Ser[PO(OPh)$_2$]- and Thr[PO(OPh)$_2$]-peptides from Merrifield or PAM supports has presented the major synthetic hurdle. While high-pressure Parr hydrogenolysis[85] ($PdCl_2$, DMF, 2.72 atm, 50 °C) was initially employed for the hydrogenolytic deprotection of Boc-Glu(OBzl)-Ser[PO(OPh)$_2$]-Leu-Merrifield-resin, this procedure has not been adopted for routine use due to the need for specialized chemical equipment and the incompatibility of Cys and Met residues to hydrogenolytic deprotection. A major drawback in the use of HF deprotection of Ser-[PO(OPh)$_2$]- and Thr[PO(OPh)$_2$]-peptides from Merrifield or PAM supports[83] is that this procedure causes extensive dephosphorylation of the phosphorylated residues. Although dephosphorylation is minimized using 30 minute HF contact times and the Thr[PO(OPh)$_2$] residue exhibits greater stability than the Ser[PO(OPh)$_2$] residue, the isolated yields of Ser/Thr[PO(OPh)$_2$]-peptides using HF deprotection are generally low to moderate.[83,86,87] A major development in HF-mediated deprotection was the finding that dephosphorylation of Ser[PO(OR^1)$_2$] residues decreased with increased steric bulk in the 2-position of the phenyl group[88] such that the Ser[PO(OC$_6$H$_4$-2-tBu)$_2$]-residue is inert to HF-mediated dephosphorylation.

The synthesis of Boc-Ser[PO(OC$_6$H$_4$-2-tBu)$_2$]-OH (**15**) is achieved by phosphorylation of Boc-Ser-OBzl using bis(2-*tert*-butylphenyl) diethylphosphoramidite followed by hydrogenolytic cleavage of the benzyl group from Boc-Ser[PO(OC$_6$H$_4$-2-tBu)$_2$]-OBzl (Scheme 11).[89] The phosphitylation of Boc-Ser-OBzl using the sterically hindered (2-tBuC$_6$H$_4$O)$_2$P-NEt$_2$ proceeds slowly and requires overnight treatment at 40 °C to effect completion prior to MCPBA oxidation of the diaryl phosphite-triester.

Scheme 11 Synthesis of Boc-Ser[PO(OC$_6$H$_4$-2-tBu)$_2$]-OH[89]



The suitability of this derivative for Boc-based solid-phase synthesis has been demonstrated by the syntheses of KSSPGS[PO(OC_6H_4-2-tBu)$_2$]PGTPGSR-NH_2 (tau 139–151) and LRRAS[PO(OC_6H_4-2-tBu)$_2$]LG (kemptide). Peptide assembly was performed using DCC/HOBt as the coupling procedure and the Ser[PO(OC_6H_4-2-tBu)$_2$]-peptides were cleaved from their respective MBHA or Merrifield resin supports by HF treatment (4 °C, 30 min). The hydrogenolytic cleavage of the 2-*tert*-butylphenyl group is slower than the phenyl group[88] and therefore requires the use of specific hydrogenation conditions to effect rapid aryl reduction. The role of hydrogenation solvent is particularly critical since the platinum-mediated hydrogenation (2.72 atm) of the tau-tridecapeptide using AcOH as solvent was complicated due to slow and incomplete aryl cleavage,[88] whereas hydrogenolytic deprotection (1 atm) of LRRAS[PO(OC_6H_4-2-tBu)$_2$]LG in TFA solvent proceeded smoothly and gave LRRAS(*P*)LG in quantitative yield.[21]

Bis(2-*tert*-butylphenyl) Diethylphosphoramidite:[88]
A soln of 2-*tert*-butylphenol (6.01 g, 40.0 mmol) and TEA (4.85 g, 48.0 mmol) in dry Et_2O (20 mL) was added to a stirred soln of Cl_2PNEt_2 (3.48 g, 20.0 mmol) in dry Et_2O (10 mL) at 0 °C and the soln stirred at rt for 18 h under N_2. A soln of 5% Na_2CO_3 (30 mL) was added, the soln transferred to a separating funnel using Et_2O (30 mL), and the aqueous phase discarded. The organic phase was washed with 5% Na_2CO_3 (3 × 30 mL) and sat. NaCl (30 mL), dried (Na_2SO_4), filtered, and the solvent removed under reduced pressure to give a yellow oil which was crystallized (Et_2O) to give colorless cubic crystals; yield: 1.81 g (24%); mp 74–78 °C; ^{13}C NMR ($CDCl_3$, δ): 14.3 (d, $J_{P,C}$ = 3.0 Hz, NCH_2CH_3), 29.9 (tBu CH_3), 34.8 (tBu Cq), 38.5 (d, $J_{P,C}$ = 22.1 Hz, NCH_2CH_3), 117.7 (d, $J_{P,C}$ = 23.7 Hz, tBuPh C6), 121.7 (Phe C4), 126.7 (Ph C5), 127.0 (Ph C3), 139.3 (d, $J_{P,C}$ = 2.5 Hz, Ph C2), 153.3 (d, $J_{P,C}$ = 8.1 Hz, Ph C1); ^{31}P NMR ($CDCl_3$, δ): 134.6.

Boc-Ser[PO(OC_6H_4-2-tBu)$_2$]-OH (15):[88]
Bis(2-*tert*-butylphenyl) diethylphosphoramidite (0.402 g, 1.00 mmol) was added to a soln of Boc-Ser-OBzl (0.443 g, 1.50 mmol) and 1*H*-tetrazole (0.105 g, 1.5 mmol) in dry THF (1.0 mL) under N_2 and the soln stirred at 40 °C for 24–48 h. The soln was cooled to −10 °C and a soln of 85% MCPBA (0.236 g, 1.20 mmol) in CH_2Cl_2 (1 mL) added at −10 °C and stirred at rt for 10 min. Et_2O (20 mL) was added and the soln washed with 10% $Na_2S_2O_5$ (2 × 10 mL), 5% $NaHCO_3$ (2 × 10 mL), and 1 M HCl (2 × 10 mL). The organic phase was dried (Na_2SO_4), filtered, and the solvent removed under reduced pressure to give the crude Boc-Ser[PO(OC_6H_4-2-tBu)$_2$]-OBzl/Boc-Ser-OBzl mixture as a white foam; ^{31}P NMR ($CDCl_3$, δ): −14.8.

A soln of the crude Boc-Ser[PO(OC_6H_4-2-tBu)$_2$]-OBzl/Boc-Ser-OBzl mixture (0.744 g) and 10% Pd/C (130 mg) in 5% AcOH/MeOH (5 mL) was charged with H_2 and stirred until the cessation of H_2 uptake. The soln was filtered, the solvent removed under reduced pressure and the oil dissolved in Et_2O (10 mL). The excess Boc-Ser-OH was removed by extraction with 5% aq $NaHCO_3$ (4 × 10 mL), the organic phase washed with 1 M HCl (2 × 10 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure and dried under high vacuum to give Boc-Ser[PO(OC_6H_4-2-tBu)$_2$]-OH (**15**) as a white foam; yield: 0.368 g (67%); ^{13}C NMR ($CDCl_3$, δ): 28.2 (Boc Me), 30.0 (tBuPh Me), 34.6 (tBuPh Cq), 53.6 (d, $J_{P,C}$ = 7.3 Hz, Ser Cα), 68.7 (d, $J_{P,C}$ = 7.7 Hz, Ser Cβ), 80.4 (Boc Cq), 119.3 (d, Ar C6), 125.0 (d, Ar C4), 127.4 (Ar C4), 127.6 (d, Ar C3), 139.3 (d, Ar C2) 149.4 (d, Ar C1), 155.3 (Boc CO), 171.1 (Ser CO); ^{31}P NMR ($CDCl_3$, δ): −15.2.

H-Leu-Arg-Arg-Ala-Ser(*P*)-Leu-Gly-OH · 3 TFA:[89]
Peptide assembly was performed using Boc-Gly-Merrifield-resin (0.76 mmol · g^{-1}) (0.066 g, 0.05 mmol) and Boc-Arg(Tos)-OH. Coupling reactions were performed using Boc amino acid/DCC/HOBt (3 equiv each) in CH_2Cl_2 (3 mL) for 1 h and the Boc group removed using 50% TFA/CH_2Cl_2. A Kel-F vessel containing the Ser[PO(OC_6H_4-2-tBu)$_2$]-peptide-resin (15.0 mg), *m*-cresol (0.25 mL), and thiocresol (0.25 mL) was cooled in an dry ice/acetone bath, charged with HF (4.5 mL) and stirred for 30 min in an ice–water bath. The HF was removed under reduced pressure and Et_2O (20 mL) was added. The precipitated peptide was collected by filtration, triturated with Et_2O (2 × 10 mL), dissolved in 10% AcOH, and purified by RP-HPLC (0.1% TFA, 0–80% MeCN over 60 min).

A soln of the Ser[PO(OC_6H_4-2-tBu)$_2$]-peptide H-Leu-Arg-Arg-Ala-Ser[PO(OC_6H_4-2-tBu)$_2$]-Leu-Gly-OH · 3 TFA (5.8 mg, 0.004 mmol) and PtO_2 (30.0 mg) in TFA (4 mL) was charged with H_2 and stirred rapidly until the cessation of H_2 uptake. The platinum was removed by gravity filtration and the solvent removed under reduced pressure. The peptide was precipitated by the addition of Et_2O (10 mL) and the solid triturated with Et_2O (2 × 10 mL). The crude peptide was dissolved in H_2O, filtered (20-μm pore)

for references see p 421

and lyophilized to give the title peptide as a fluffy white powder; yield: 4.4 mg (93%); ^{31}P NMR (D_2O, δ): −0.2; MALDI-MS *m/z*: 852.5 [M+H].

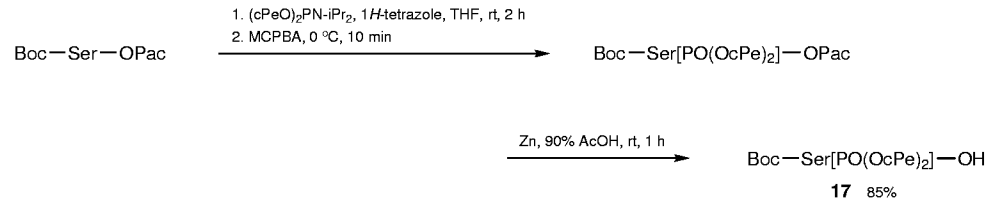
6.5.3.2.2 Boc-Based Solid-Phase Synthesis Using Alkyl Phosphate Ester Protection

The Boc-based solid-phase synthesis of Ser(*P*)- and Thr(*P*)-peptides is also possible via methyl or cyclopentyl phosphate protection with their removal effected by silylitic or acidolytic treatment. The synthesis of Boc-Ser[PO(OMe)$_2$]-OH (**16**) and Boc-Thr[PO(OMe)$_2$]-OH is accomplished in two steps by phosphorylation of Boc-Ser-ONbz or Boc-Thr-ONbz with dimethyl phosphorochloridate/pyridine [or (MeO)$_2$PNEt$_2$/1*H*-tetrazole, MCPBA] followed by hydrogenolytic cleavage of the benzyl group from Boc-Ser/Thr[PO(OMe)$_2$]-OBzl (Scheme 12).[76] In a similar manner, both Boc-Ser[PO(OcPe)$_2$]-OH (**17**) and Boc-Thr[PO(OcPe)$_2$]-OH are prepared by phosphorylation of Boc-Ser/Thr-OPac with (cPeO)$_2$PN-iPr$_2$ followed by zinc reduction of the phenacyl group (Scheme 13).[84]

Scheme 12 Synthesis of Boc-Ser[PO(OMe)$_2$]-OH[76]



Scheme 13 Synthesis of Boc-Ser[PO(OcPe)$_2$]-OH[84]



The above derivatives are incorporated into peptides using DCC/HOBt or DIC/HOBt procedures with both the methyl and cyclopentyl phosphate groups being stable to repetitive 40% TFA/CH_2Cl_2 treatments. Initial deprotections of Ser[PO(OMe)$_2$]-peptides using 45% HBr/AcOH or TMSBr were problematic as these reagents caused degradation of the phosphorylated residue.[76] However, the successful deprotection of Ser[PO(OMe)$_2$]- and Ser[PO(OcPe)$_2$]-peptides was later accomplished using TfOH- and TMSOTf-based deprotection systems. The efficiency of TMSOTf deprotection is illustrated through the syntheses of RRVS(*P*)VAAE and KRT(*P*)LRRLL in which the two-stage procedure involves initial liberation of the Ser/Thr[PO(OMe)$_2$]-peptide from their respective Merrifield and PAM supports by treatment with 1 M TMSOTf/PhSMe in TFA (1.5 h) followed by in situ cleavage of the methyl phosphate groups through the further addition of TMSOTf and DMS.[27] Alternatively, the synthesis of RALNRQLS(*P*)SGVC-NH_2 was accomplished by TfOH/PhSMe/*m*-cresol/EDT/TFA (9:12:10:4:65) treatment (0 °C, 2 h) of the protected Ser-[PO(OcPe)$_2$]-peptide-MBHA-resin.[84]

Boc-Ser[PO(OMe)$_2$]-OH (16)**:**[76]
A soln of dimethyl phosphorochloridate (2.17 g, 15.0 mmol) in THF (5 mL) was slowly added to a stirred soln of Boc-Ser-ONbz (10.0 mmol) in pyridine/THF (1:1, 10 mL) so that the reaction temperature was maintained at −40 °C. After 3 h at −40 °C, a second soln of dimethyl phosphorochloridate (1.08 g, 7.5 mmol) in THF (2.5 mL) was added and the soln stirred for a further 3 h. A soln of 1 M HCl (5 mL)

was added, the soln stirred at rt for 30 min and Et_2O (50 mL) was added. The soln was transferred to a separating funnel, the aqueous phase discarded and the organic phase washed with 1 M HCl (5 × 30 mL) and 5% $NaHCO_3$ (3 × 30 mL), dried (Na_2SO_4), and filtered. The soln was removed under reduced pressure to give Boc-Ser[PO(OMe)$_2$]-ONbz as a light yellow oil; yield: 4.30 g (96%); $[\alpha]_D^{14}$ +12.88 (*c* 1, $CHCl_3$); ^{13}C NMR ($CDCl_3$, δ): 28.0, 53.9 (d, $J_{P,C}$ = 7.3 Hz, Ser Cα), 53.9 (d, $J_{P,C}$ = 7.3 Hz, POMe), 65.7, 67.0 (d, $J_{P,C}$ = 5.9 Hz), 80.1, 123.5, 128.2, 142.2, 147.5, 155.0, 168.8; ^{31}P NMR ($CDCl_3$, δ): +1.3.

A soln of Boc-Ser[PO(OMe)$_2$]-ONbz (3.81 g, 8.50 mmol) in 2% AcOH/EtOAc (50 mL) containing 10% Pd/C (0.85 g) was charged with H_2 at rt and stirred until the cessation of H_2 uptake. The catalyst was removed by gravity filtration and the soln removed under reduced pressure. The residue was dissolved in Et_2O (50 mL) and washed with 1 M HCl (2 × 20 mL). The organic phase was extracted with 5% aq $NaHCO_3$ (3 × 8 mL) and the combined aqueous extracts washed with Et_2O (10 mL). The soln was acidified to pH 1 with 3 M HCl and extracted with CH_2Cl_2 (3 × 30 mL). The soln was dried (Na_2SO_4), filtered, and the soln removed under reduced pressure to give a light yellow oil that changed to a white solid; yield: 2.53 g (95%); mp 68–70 °C; $[\alpha]_D^{14}$ +43.1 (*c* 1, $CHCl_3$); ^{13}C NMR ($CDCl_3$, δ): 27.9, 53.5 (d, $J_{P,C}$ = 7.3 Hz, Ser Cα), 54.4 (d, $J_{P,C}$ = 5.9 Hz, POMe), 67.5 (d, $J_{P,C}$ = 5.9 Hz), 79.9, 155.1, 170.5; ^{31}P NMR ($CDCl_3$, δ): +0.5; FAB-MS *m/z*: 314 [M+H].

Boc-Thr[PO(OMe)$_2$]-OH:[76]

Boc-Thr[PO(OMe)$_2$]-OH was prepared as described for derivative **16** except that the reaction was stirred overnight and the hydrogenation of Boc-Thr[PO(OMe)$_2$]-ONbz gave the title compound as a clear oil, yield: 3.04 g (93%); ^{13}C NMR ($CDCl_3$, δ): 18.4, 28.2, 54.4 (d, $J_{P,C}$ = 5.9 Hz, POMe), 57.4 (d, $J_{P,C}$ = 5.9 Hz, Thr Cα), 75.9 (d, $J_{P,C}$ = 5.9 Hz, Thr Cβ), 80.0, 155.9, 171.1; ^{31}P NMR ($CDCl_3$, δ): −1.0; FAB-MS *m/z*: 328 [M+H].

RRVS(*P*)VAAE; TMSOTf Cleavage of Methyl Groups:[27]

Solid-phase synthesis was performed using Boc-Glu(OBzl)-Merrifield-resin (0.2 mmol) (0.67 mmol·g^{-1}) with the coupling of Boc amino acids and Boc-Ser[PO(OMe)$_2$]-OH performed using DIC/HOBt (2.5 mol equiv) in DMF. A soln of 1 M TMSOTf/PhSMe in TFA (1.0 mL) was added under N_2 to a mixture of the peptide-resin (20.0 mg, 7.4 μmol), *m*-cresol (50 μL), and EDT (50 μL) at 4 °C. After stirring for 1.5 h, DMS (0.7 mL) and TMSOTf (0.3 mL) were successively added to the soln and the soln was stirred at 4 °C for 2 h. The soln was filtered and the peptide precipitated by the addition of Et_2O (12 mL). The white solid was collected by centrifugation, washed with Et_2O (2 × 10 mL), and purified by RP-HPLC to give a white powder; yield: 7.8 mg (81%); IS-MS/MS *m/z*: 966.8 [M+H].

KRT(*P*)LRRLL; TMSOTf Cleavage of Methyl Groups:[27]

This peptide was prepared as above using Boc-Leu-PAM-resin (0.2 mmol) (0.5 mmol·g^{-1}) and Boc-Thr[PO(OMe)$_2$]-OH. The deprotection of the peptide-resin (31.0 mg, 8.3 μmol) was performed as above with the second deprotection step extended to 3 h at 4 °C followed by 1 h at rt. The crude solid was purified by RP-HPLC to give a white powder; yield: 9.7 mg (69%); IS-MS/MS *m/z*: 1136.0 [M+H].

Boc-Ser[PO(OcPe)$_2$]-OH (17):[84]

1*H*-Tetrazole (2.94 g, 42.0 mmol) and (cPeO)$_2$PN-iPr$_2$ (5.90 g, 19.6 mmol) were added to a stirred soln of Boc-Ser-OPac (4.52 g, 14.0 mmol) in THF (30 mL) and stirred at rt for 2 h. 80% MCPBA (7.51 g, 35.0 mmol) was added, the soln stirred at 0 °C for 10 min, a soln of $NaHSO_3$ (3.64 g, 35.00 mmol) in H_2O (3 mL) added, and the THF removed under reduced pressure. The oil was dissolved in EtOAc (60 mL) and washed successively with 10% aq citric acid (3 × 10 mL), NaCl soln (10 mL), and sat. $NaHCO_3$ (3 × 10 mL). The organic soln was dried ($MgSO_4$), filtered, and the solvent removed under reduced pressure to give crude Boc-Ser[PO(OcPe)$_2$]-OPac as a thick oil.

Zn dust (13.7 g, 210.0 mmol) was added in several portions to a soln of the oil in 90% AcOH (80 mL) and the soln stirred at rt for 1 h. The insoluble material was filtered, the solvent removed under reduced pressure, the residue dissolved in Et_2O (130 mL) and the insoluble material filtered [the insoluble solid fractions were combined, treated with 0.5 M HCl (40 mL), extracted with Et_2O (3 × 10 mL) and added to the main ethereal fraction]. The ethereal soln was washed with 10% aq citric acid (3 × 20 mL) and NaCl soln (3 × 20 mL), dried ($MgSO_4$), and filtered. The soln was removed under reduced pressure, the oil triturated with hexane to give a solid, which was recrystallized (EtOAc/hexane) to give Boc-Ser[PO(OcPe)$_2$]-OH as a crystalline solid; yield: 4.99 g (85%); mp 110–113 °C; $[\alpha]_D^{28}$ +15.0 (*c* 1.01, EtOH); ^1H NMR (DMSO-d_6, δ): 1.39 (s, 9H, Boc Me), 1.51–1.76 (m, 16H, cPe CH_2), 4.05–4.19 (m, 2H, Ser β-CH_2), 4.23 (br m, 1H, Ser α-CH), 4.75 (m, 2H, cPe CH), 6.77 (br s, 1H, NH); ^{31}P NMR (DMSO-d_6, δ): −2.7.

Boc-Thr[PO(OcPe)$_2$]-OH:[84]

Prepared as above to give Boc-Thr[PO(OcPe)$_2$]-OH as a crystalline solid; yield: 5.50 g (90%); mp 120–123 °C; $[\alpha]_D^{25}$ +18.2 (*c* 1.00, EtOH); ^1H NMR (DMSO-d_6, δ): 1.30 (d, 3H, $J_{P,H}$ = 6.5 Hz, Thr CH_3),

for references see p 421

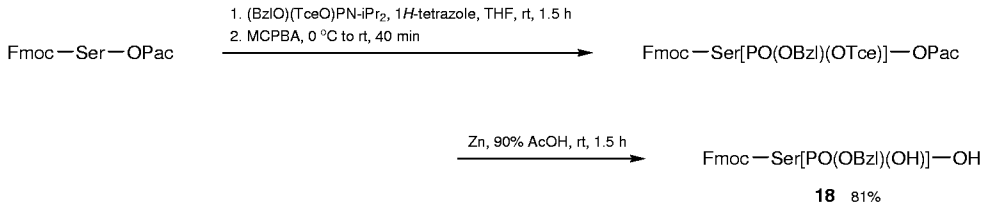
1.41 (s, 9H, Boc Me), 1.50–1.79 (m, 16H, cPe CH_2), 4.16 (m, 1H, Thr α-CH), 4.71–4.77 (m, 3H, Thr β-CH and cPe CH), 6.46 (br s, 1H, NH); ^{31}P NMR (DMSO-d_6, δ): −3.7.

RALNRQLS(*P*)SGVC-NH_2 · 3 TFA; TfOH Cleavage of cPe Groups:[84]
Solid-phase synthesis was performed using MBHA resin (1.22 g, 0.5 mmol) with the coupling of Boc amino acids and Boc-Ser[PO(OcPe)$_2$]-OH performed using DCC/HOBt (4 mol equiv) in NMP. The peptide-resin (0.101 g) was treated with TfOH (0.408 mL), PhSMe (0.544 mL), *m*-cresol (0.453 mL), and EDT (0.181 mL) in TFA (2.95 mL) (9:12:10:4:65) and stirred at 0 °C for 2 h. The resin was removed by filtration under N_2 and the resin washed with TFA (4 mL). The combined TFA fractions were poured into chilled Et_2O (200 mL) and allowed to stand at 4 °C for 1 h. The precipitated peptide was collected by filtration (Fluoropore FP-022 0.22-μm filter), dried, dissolved in 0.1% aq TFA, filtered (Millipore HA 0.45-μm filter), and lyophilized. DTT (0.179 g, 1.16 mmol) was added to the peptide (35.1 mg) in 0.1% aq TFA, the pH of the soln adjusted to pH 7.5 (NH_4OAc) and the soln stirred at rt for 1 h. The soln was purified by RP-HPLC to give the product Ser(*P*)-peptide as a white powder; yield: 11.4 mg (29%); FAB-MS *m/z*: 1380 [M − H].

6.5.3.3 Fmoc-Based Solid-Phase Synthesis of Phosphoserine and Phosphothreonine Peptides Using Phosphate Monobenzyl Ester Protection

The Fmoc-based solid-phase synthesis of Ser(*P*)- and Thr(*P*)-peptides using protected Fmoc-Ser[PO(OR^1)$_2$]-OH and Fmoc-Thr[PO(OR^1)$_2$]-OH derivatives was initially precluded on the basis that these phosphorylated residues underwent piperidine-mediated β-elimination during Fmoc cleavage steps.[90,91] However, this situation changed in 1994 when both Fmoc-Ser[PO(OBzl)(OH)]-OH (**18**) and Fmoc-Thr[PO(OBzl)(OH)]-OH were demonstrated to be suitable for peptide assembly since ionization of the phosphorodiester functionality suppressed base-mediated β-elimination.[92] Both these derivatives are readily prepared by phosphorylation of Fmoc-Ser-OPac or Fmoc-Thr-OPac using (BzlO)(TceO)PN-iPr_2[47] followed by simultaneous reductive removal of the Tce and Pac groups by Zn/AcOH treatment (Scheme 14). The synthesis of derivatives Fmoc-Ser[PO(OBzl)(OH)]-OH (**18**) and Fmoc-Thr[PO(OBzl)(OH)]-OH is also possible through the phosphorylation of Fmoc-Ser-OAl or Fmoc-Thr-OAl with dibenzyl dialkylphosphoramidite followed by Pd(0)-mediated cleavage of the allyl group and final sodium iodide mediated debenzylation of the dibenzyl phosphorotriester to give Fmoc-Ser[PO(OBzl)(ONa)]-OH and Fmoc-Thr[PO(OBzl)(ONa)]-OH in 37 and 38% yields, respectively.[93]

Scheme 14 Synthesis of Fmoc-Ser[PO(OBzl)(OH)]-OH[92]



The incorporation of derivatives Fmoc-Ser[PO(OBzl)(OH)]-OH (**18**) and Fmoc-Thr[PO(OBzl)(OH)]-OH is best effected using uronium reagents such as HBTU/HOBt/DIPEA and HATU/HOAt/DIPEA since both BOP and PyBroP failed to effect complete incorporation in comparative test couplings.[48] In a separate work, the BOP/HOBt/DIPEA coupling of Fmoc-Ser[PO(OBzl)(OH)]-OH was found to be only 85% complete after double coupling and the incorporation of the anionic phosphorodiester caused a decrease in the coupling efficiency of following residues.[94] Despite these potential problems, both Fmoc-Ser[PO(OBzl)(ONa)]-OH and Fmoc-Thr[PO(OBzl)(ONa)]-OH have been successfully employed for the Fmoc-based solid-phase synthesis of various Ser(*P*)- and Thr(*P*)-

containing peptides[26,48,84] using HBTU, HATU, TBTU, or TPTU coupling reagents (Table 6).

Table 6 Ser(*P*)- and Thr(*P*)-Peptides Prepared Using Fmoc-Ser[PO(OBzl)(OH)]-OH and Fmoc-Thr[PO(OBzl)(OH)]-OH[26,48,84,92,93]

Peptide	Coupling	Resin	Cleavage	Ref
SGVS(*P*)EI	BOP	Alko-resin[a]	TFA/PhSMe/*m*-cresol (78:12:10)	[92]
LLRSPS(*P*)WEPFRC-NH_2	HATU	Rink amide	TFA/PhOH/H_2O/PhSMe/EDT (82.5:5:5:5:2.5)	[84]
ASQGS(*P*)LEDPA-NH_2	HBTU	Rink amide pins	TFA/PhOMe/EDT (95:2.5:2.5)	[48]
ASQGT(*P*)LEDPA-NH_2	HBTU	Rink amide pins	TFA/PhOMe/EDT (95:2.5:2.5)	[48]
FFKNIVT(*P*)PPPS(*P*)EGK	TPTU	NovaSyn TGR	TFA/TIS/H_2O (95:2.5:2.5)	[26]
TGFLT(*P*)EYVAT-NH_2	TPTU	TMBHA-TentaGel S	TFA/DTT/H_2O (95:2.5:2.5)	[93]
CNNTSS(*P*)PQPKK-NH_2	TPTU	TMBHA-TentaGel S	TFA/DTT/H_2O (95:2.5:2.5)	[93]
TEPQS(*P*)QPGE	TPTU	TMBHA-TentaGel S	TFA/DTT/H_2O (95:2.5:2.5)	[93]
TEPQT(*P*)QPGE	TPTU	TMBHA-TentaGel S	TFA/DTT/H_2O (95:2.5:2.5)	[93]

[a] Alko-resin = $-OCH_2C_6H_4-OCH_2C_6H_4$.

Fmoc-Ser[PO(OBzl)(OH)]-OH (18):[47]
1*H*-Tetrazole (2.36 g, 33.6 mmol) and (TceO)(BzlO)PN-iPr_2[47] (6.08 g, 15.7 mmol) were added to a soln of Fmoc-Ser-OPac (5.00 g, 11.2 mmol) in THF (25 mL) under N_2 and the soln stirred for 1.5 h at rt. The soln was cooled to 0 °C, 80% MCPBA (6.08 g, 15.7 mmol) added and the soln stirred at 0 °C for 10 min and then at rt for 30 min. A soln of $NaHSO_3$ (2.93 g, 28.0 mmol) in H_2O (10 mL) was added, the mixture stirred at 0 °C for 5 min and the soln removed under reduced pressure. The residue was dissolved in EtOAc (50 mL), washed successively with 10% aq citric acid (3 × 20 mL), NaCl soln (20 mL), sat. $NaHCO_3$ (6 × 20 mL), and NaCl soln (3 × 20 mL), dried ($MgSO_4$), and the solvent removed under reduced pressure.

Zn dust (11.0 g, 168.0 mmol) was added in several portions to a soln of the crude oil in 90% AcOH (110 mL) and stirred at rt for 1.5 h. The insoluble material was filtered and the solvent removed under reduced pressure. The oil was dissolved in Et_2O (50 mL), filtered, and the solvent removed under reduced pressure [the two filtered solids were combined, stirred in EtOAc (100 mL) and 1 M HCl (30 mL) containing sat. NaCl, the aqueous phase extracted with EtOAc (2 × 20 mL), the organic extractions combined, washed with NaCl soln (4 × 30 mL), dried ($MgSO_4$), and removed under reduced pressure]. The two oil residues were combined and crystallized (EtOAc/Et_2O/hexane) to give Fmoc-Ser[PO(OBzl)(OH)]-OH as a white solid; yield: 4.50 g (81%); mp 110–111 °C; $[\alpha]_D^{24}$ +6.3 (*c* 1.1, EtOH); ^{1}H NMR (DMSO-d_6, δ): (partial) 4.92 (2H, d, $J_{P,C}$ = 6.9 Hz, PO_3CH_2Ph); ^{31}P NMR (DMSO-d_6, δ): −0.89.

Fmoc-Thr[PO(OBzl)(OH)-OH:[47]
This derivative was prepared as described for Fmoc-Ser[PO(OBzl)(OH)]-OH (**18**) and gave Fmoc-Thr[PO(OBzl)(OH)]-OH as a white solid; yield: 4.92 g (86%); mp 154.5–156.5 °C; $[\alpha]_D^{26}$ +5.5 (*c* 1, EtOH); ^{1}H NMR (DMSO-d_6, δ): (partial) 4.92 (2H, d, $J_{P,C}$ = 6.9 Hz, PO_3CH_2Ph); ^{31}P NMR (DMSO-d_6, δ): −1.48.

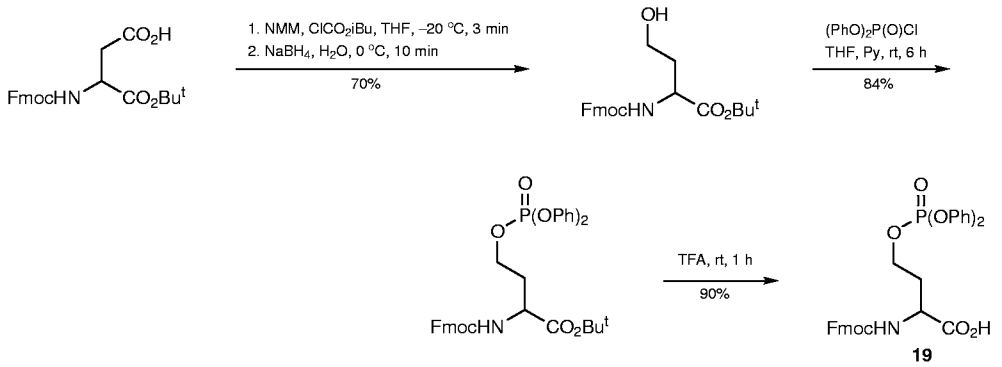
6.5.4 Synthesis of Phosphohomoserine Peptides

The rationale behind the synthesis of phosphohomoserine-containing peptides [Hse(*P*)] is that this phosphorylated residue is inert to base-mediated β-elimination and thereby may provide an interesting surrogate for the study of Ser(*P*)-based processes. In initial work, Boc-Hse[PO$(OPh)_2$]-OH[21] was prepared by a simple four-step procedure but was found to be unsuitable for use in Boc-based solution synthesis due to extensive amine-mediated intramolecular dephosphorylation of the Hse[PO$(OPh)_2$] residue during mixed anhydride coupling of its N-terminus. This side reaction was subsequently minimized through the use of Fmoc-Hse[PO$(OPh)_2$]-OH in a Fmoc-based solid-phase approach in which the use of a large

for references see p 421

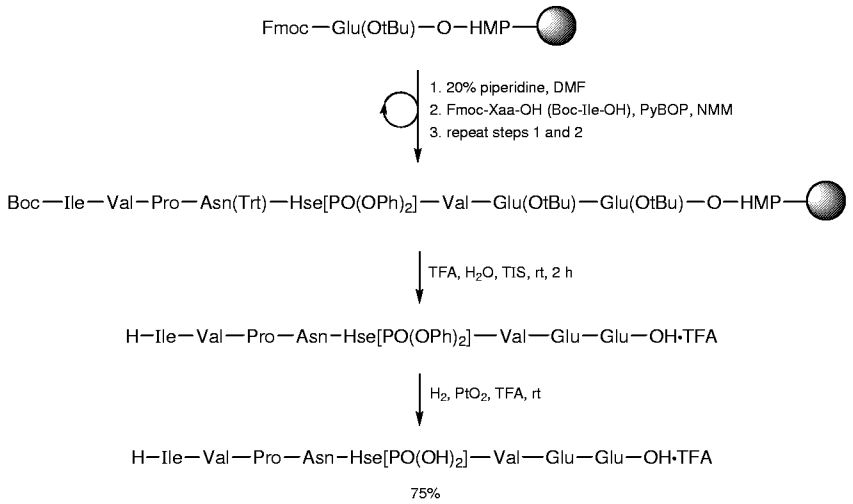
excess of acylating reagent (5–10 equiv) competed with the dephosphorylation step. The synthesis of Fmoc-Hse[PO(OPh)$_2$]-OH (**19**) is prepared by a three-step procedure which involves initial $NaBH_4$ reduction of the isobutoxycarbonyl mixed anhydride of Fmoc-Asp-OtBu, phosphorylation of Fmoc-Hse-OtBu with diphenyl phosphorochloridate in pyridine/THF, and final acidolytic cleavage (TFA) of the *tert*-butyl group from Fmoc-Hse[PO(OPh)$_2$]-OtBu (yields of 70, 84, and 90% respectively) (Scheme 15).[21] The selection of the *tert*-butyl group (R^1 = tBu) for carboxy protection is necessary as extensive lactonization of Fmoc-Hse-OR^1 derivatives occurs with the use of other alkyl esters (R^1 = Me, Bzl). Derivative **19** is unsuitable for long-term storage since the acid form of this derivative undergoes intramolecular acid-catalyzed displacement of the diphenyl phosphate to give Fmoc-homoserine lactone and diphenyl phosphate.

Scheme 15 Synthesis of Fmoc-Hse[PO(OPh)$_2$]-OH[21]



The Fmoc SPPS of H-Ile-Val-Pro-Asn-Hse(*P*)-Val-Glu-Glu-OH (Scheme 16) using Fmoc-Hse[PO(OPh)$_2$]-OH was accomplished with Fmoc-Glu(OtBu)-Wang-resin as the polymer support with the PyBOP-coupling of Fmoc-Asn(Trt)-OH to the Hse[PO(OPh)$_2$]-peptide-resin being performed with a fivefold excess so as to maximize acylation of the Hse-[PO(OPh)$_2$] residue over its dephosphorylative rearrangement. The deprotection of the Hse[PO(OPh)$_2$]-peptide-resin was effected by TFA treatment followed by platinum-mediated hydrogenolytic cleavage of the phenyl phosphate groups. Despite the successful use of excess Fmoc-Hse[PO(OPh)$_2$]-OH, the formation of numerous byproducts confirmed that acylation of the Hse[PO(OPh)$_2$] residue is complicated and further work is necessary to overcome the dephosphorylative rearrangement.

Scheme 16 Fmoc-Based Solid-Phase Synthesis of H-Ile-Val-Pro-Asn-Hse(*P*)-Val-Glu-Glu-OH·TFA[21]



Fmoc-Hse-OtBu:[21]
NMM (0.505 g, 5.0 mmol) in THF (1 mL) and isobutyl chloroformate (0.685 g, 5.0 mmol) in THF (1 mL) were successively added to Fmoc-Asp-OtBu (2.055 g, 5.00 mmol) in THF (10 mL) at −15 °C. After 3 min, the precipitated salt was removed by suction filtration and washed with THF (2 × 5 mL). The soln was cooled to −30 °C and a soln of $NaBH_4$ (0.285 g, 7.5 mmol) in H_2O (5 mL) was added in 1-mL aliquots (**CAUTION:** *an exothermic reaction occurs with vigorous evolution of gas*). After 5 min, Et_2O (30 mL) was added and the soln washed with 5% $NaHCO_3$ (2 × 20 mL) and 1 M HCl (20 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure to give Fmoc-Hse-OtBu as a clear oil: yield: 1.39 g (70%); ^{13}C NMR ($CDCl_3$, δ): 27.6, 35.0, 46.8, 51.6, 58.1, 66.7, 81.8, 119,6, 124.7, 126.7, 127.3, 140.9, 147.4, 143.4, 143.5, 156.4, 171.3.

Fmoc-Hse[PO(OPh)$_2$]-OtBu:[21]
A soln of diphenyl phosphorochloridate (0.501 g, 1.865 mmol) in THF (1 mL) was added to a soln of Fmoc-Hse-OtBu (0.617 g, 1.55 mmol) in pyridine/THF (1:1, 2 mL) at 0 °C and then stirred at rt for 16 h. H_2O (1 mL) was added and the soln was partitioned with Et_2O (30 mL). The organic phase was washed with 5% $NaHCO_3$ (4 × 20 mL) and 1 M HCl (20 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure to give Fmoc-Hse[PO(OPh)$_2$]-OtBu as a yellow oil; yield: 0.823 g (84%); ^{13}C NMR ($CDCl_3$, δ): 27.7, 32.5 (d, $J_{P,C}$ = 6.59 Hz), 47.0, 51.3, 65.3 (d, $J_{P,C}$ = 6.59 Hz), 66.8, 82.4, 119.8 (d, $J_{P,C}$ = 4.39 Hz), 119.9, 124.8, 125.1, 126.8, 127.5,129.6, 141.0, 143.5, 143.6, 150.2 (d, $J_{P,C}$ = 6.59 Hz), 155.6, 170.3; ^{31}P NMR ($CDCl_3$, δ): −12.4.

Fmoc-Hse[PO(OPh)$_2$]-OH (19):[21]
Fmoc-Hse[PO(OPh)$_2$]-OtBu (1.145 g, 1.82 mmol) was dissolved in TFA (3 mL) and allowed to stand for 1 h. The soln was removed under reduced pressure, dissolved in Et_2O (10 mL) and washed with Et_2O (3 × 10 mL). The organic layer was extracted with 5% $NaHCO_3$ (4 × 8 mL) (note: the Na salt precipitates as a brown oil) and the combined base extracts washed with Et_2O (10 mL). The soln was acidified to pH 3 using 30% citric acid, extracted with CH_2Cl_2 (3 × 30 mL), dried (Na_2SO_4), and filtered. The soln was removed under residue pressure to give Fmoc-Hse[PO(OPh)$_2$]-OH as a yellow oil; yield: 0.937 g, (90%); ^{13}C NMR ($CDCl_3$, δ): 32.1 (d, $J_{P,C}$ = 6.59 Hz), 47.0, 50.8, 65.8 (d, $J_{P,C}$ = 6.59 Hz), 67.4, 119.9 (d, $J_{P,C}$ = 5.49 Hz), 119.9, 124.9, 125.7, 127.0, 127.7, 129.9, 141.2, 143.4, 150.0 (d, $J_{P,C}$ = 6.59 Hz), 156.4, 174.3; ^{31}P NMR ($CDCl_3$, δ): −12.7; FAB-MS *m/z*: 574 [M+H].

H-Ile-Val-Pro-Asn-Hse(*P*)-Val-Glu-Glu-OH·TFA:[21]
Solid-phase synthesis was performed using Wang resin (0.57 mmol·g^{-1}; 0.35 g, 0.2 mmol) with the coupling of Fmoc amino acids and Fmoc-Hse[PO(OPh)$_2$]-OH performed using PyBOP (3 mol equiv) and NMM (5 mol equiv) in DMF. The peptide-resin was treated with TFA/H_2O/TIS (95:2.5:2.5; 5 mL) at rt for 2 h, the resin filtered under N_2, and washed with TFA (4 mL). The solvent was removed under reduced pressure, the peptide precipitated by the addition of Et_2O (20 mL), triturated with Et_2O (2 × 10 mL), and dried under high vacuum.

for references see p 421

A soln of the crude Hse[PO(OPh)$_2$]-peptide H-Ile-Val-Pro-Asn-Hse[PO(OPh)$_2$]-Val-Glu-Glu-OH·TFA (62.0 mg, 0.05 mmol) and PtO_2 (30.0 mg) in TFA (4 mL) was charged with H_2 and stirred rapidly until the cessation of H_2 uptake. The Pt was removed by gravity filtration, the solvent removed under reduced pressure and the peptide precipitated by the addition of Et_2O (10 mL). The solid was triturated with Et_2O (2 × 10 mL), dissolved in H_2O, filtered (20 μm pore), and purified by RP-HPLC (0.1% TFA, 0–30% MeCN over 30 min) to give the title Hse(*P*)-peptide as a white powder; yield: 40.1 mg (75%); ^{31}P NMR (D_2O, δ): −0.5; FAB-MS *m/z*: 980 [M+H].

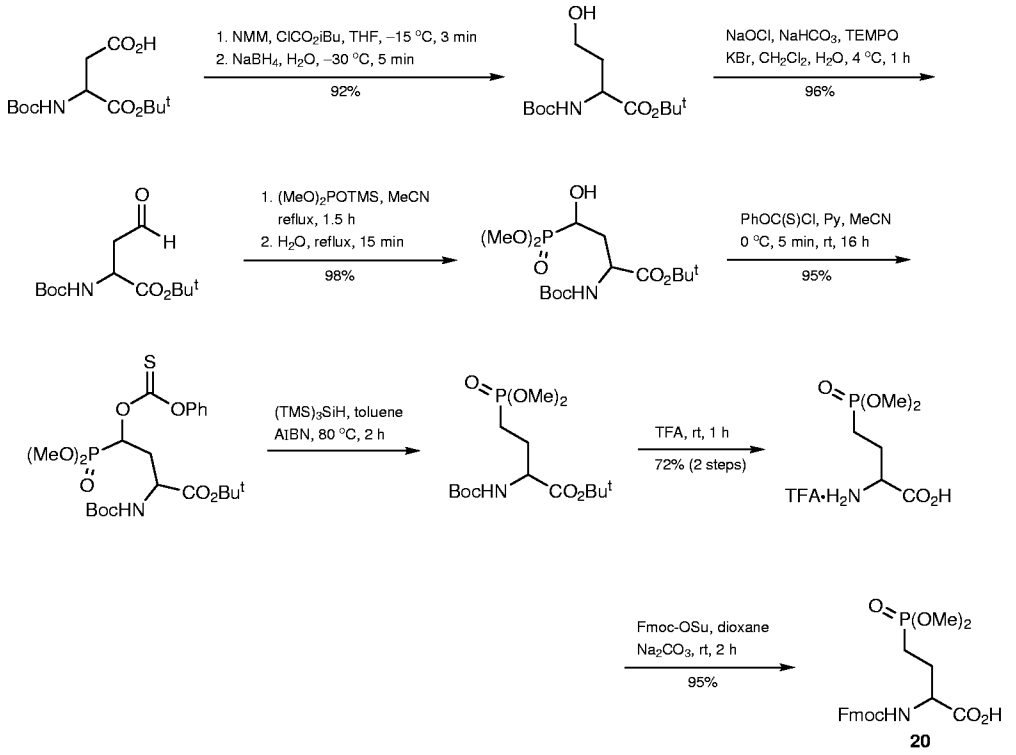
6.5.5 Synthesis of Phosphoserine Peptide Mimetics

The application of phosphatase-resistant Ser(*P*)-peptides in biochemical studies has focused research on the synthesis of peptides containing the 2-amino-4-phosphonobutanoic acid Abu[PO(OR^1)$_2$] residue in which the O–P linkage of the serine phosphate functionality is replaced by the stable CH_2–P linkage.[95] The chemical synthesis of Abu[PO(OR^1)$_2$] containing peptides is performed by Boc or Fmoc solid-phase synthesis using protected Boc-Abu[PO(OR^1)$_2$]-OH and Fmoc-Abu[PO(OR^1)$_2$]-OH (R^1 = Et, Me, Al) derivatives, respectively, which are all readily incorporated into peptides using carbodiimide, phosphonium, or uronium reagents. The Fmoc solid-phase synthesis of Abu[PO(OR^1)$_2$]-peptides is particularly straightforward since the Abu[PO(OR^1)$_2$]-residue is not prone to piperidine-mediated β-elimination as occurs with Ser[PO(OR^1)$_2$]-residues during repetitive Fmoc-cleavage cycles.

6.5.5.1 Synthesis of 2-Amino-4-phosphonobutanoic Acid Containing Peptides via Methyl Phosphonate Protection

The synthesis of both Boc-Abu[PO(OMe)$_2$]-OH[96] and Fmoc-Abu[PO(OMe)$_2$]-OH (**20**) starts from commercially available Boc-Asp-OtBu by an efficient seven-step procedure[97,98] which involves initial sodium borohydride reduction of the isobutoxycarbonyl mixed anhydride of Boc-Asp-OtBu to Boc-Hse-OtBu followed by TEMPO-catalyzed sodium hypochlorite oxidation to give the related aldehyde derivative (Scheme 17). The aldehyde is converted into the γ-hydroxy phosphonate by reflux with dimethyl trimethylsilyl phosphite followed by in situ aqueous hydrolysis of the silyl ether. The hydroxy group is readily removed by homolytic deoxygenation, which involves conversion into the thiocarbonate using *O*-phenyl chloridothiocarbonate followed by treatment with tris(trimethylsilyl)silane/AIBN to give Boc-Abu[PO(OMe)$_2$]-OtBu. The Boc and tBu groups are cleaved by treatment with TFA and the target derivatives are obtained by treatment of Abu[PO(OMe)$_2$]-OH with Boc_2O or Fmoc-OSu. The selection of the tBu ester protecting group is a key feature of this synthetic approach as it prevents lactonization of both the homoserine and γ-hydroxy phosphonate derivatives.

Scheme 17 Synthesis of Fmoc-Abu[PO(OMe)$_2$]-OH[98]

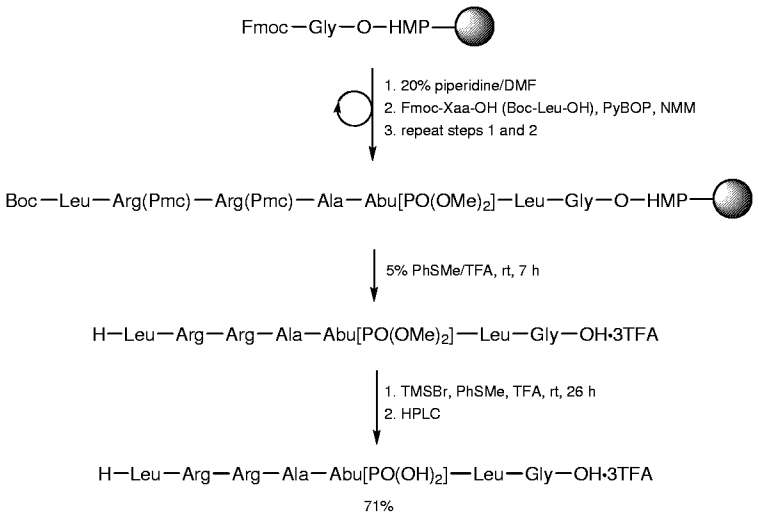


The incorporation of Boc-Abu[PO(OMe)$_2$]-OH and Fmoc-Abu[PO(OMe)$_2$]-OH (**20**) into peptides using solid-phase synthesis is performed with conventional coupling reagents, which include DCC, PyBOP, and HBTU. In the Boc mode of solid-phase synthesis, the assembly of H-Glu-Abu[PO(OMe)$_2$]-Leu-OH was accomplished via PyBOP couplings and the use of HBr/TFA or PhOMe/HF for cleavage of the Abu[PO(OMe)$_2$]-peptide from its Merrifield support.[99] However, in the Fmoc mode of solid-phase synthesis, couplings are best performed using phosphonium (PyBOP) and uronium (HBTU) reagents on account of the slow piperidine-mediated cleavage of the methyl phosphonate group during Fmoc-cleavage cycles. This side reaction, therefore, precludes the use of DCC for subsequent residue couplings since the interaction of this reagent with the phosphorodiester functionality generates aberrant products.

The final hydrolysis of the phosphonate dimethylester of Abu[PO(OMe)$_2$]-peptides is readily effected using TMSBr, TMSOTf, or TfOH based procedures as described for the deprotection of Ser[PO(OMe)$_2$]-peptides and is facilitated by the greater lability of methyl phosphonate groups over methyl phosphate groups. In the deprotection of small peptides such as H-Glu-Abu[PO(OMe)$_2$]-Leu-OH where product isolation is often complicated, both 30% TMSBr/MeCN and 33% HBr/AcOH effected the rapid cleavage of phosphonate methyl esters and provided the straightforward isolation of H-Glu-Abu[PO(OH)$_2$]-Leu-OH through simple removal of solvent.[100] In the case of larger peptides, the effectiveness of the TMSBr/PhSMe/TFA procedure for the deprotection of Abu[PO(OMe)$_2$]-peptide-resins was demonstrated through the high yielding syntheses of H-Leu-Arg-Arg-Ala-Abu[PO(OH)$_2$]-Leu-Gly-OH [Ser(*P*)-kemptide-related analogue] (Scheme 18) and H-Ile-Val-Pro-Asn-Abu-[PO(OH)$_2$]-Val-Glu-Glu-OH [α_{s1}-casein-related analogue].[98]

for references see p 421

Scheme 18 Synthesis of {Abu[PO(OH)$_2$]}^{5}-Kemptide[98]



Boc-Hse-OtBu:[98]
NMM (1.01 g, 10.0 mmol) in THF (1 mL) and isobutyl chloroformate (1.37 g, 10.0 mmol) in THF (1 mL) were successively added to Boc-Asp-OtBu (2.89 g, 10.0 mmol) in THF (10 mL) at −15 °C. After 3 min, the precipitated salt was removed by suction filtration and washed with THF (2 × 5 mL). The soln was then cooled to −30 °C and a soln of $NaBH_4$ (0.57 g, 15.0 mmol) in H_2O (7.5 mL) was added in 1-mL aliquots (**CAUTION:** *an exothermic reaction occurs with vigorous evolution of gas*). After 5 min, Et_2O (30 mL) was added and the organic phase washed with 5% $NaHCO_3$ (2 × 20 mL) and 1 M HCl (20 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure to give a clear oil; yield: 2.53 g (92%); ^{13}C NMR ($CDCl_3$, δ): 27.7 and 28.0 (Boc and tBu Me), 35.6 (Hse Cβ), 51.2 (Hse Cα), 58.1 (Hse Cγ), 79.7 (Boc Cq), 81.7 (tBu Cq), 156.0 (Boc CO), 171.7 (Hse CO); FAB-MS *m/z*: 276 [M+H].

Boc-Aspartaldehyde-OtBu:[98]
A soln of 0.35 M NaOCl (23 mL) containing $NaHCO_3$ (1.15 g, 13.7 mmol) was added dropwise to a vigorously stirred soln (>1200 rpm) of Boc-Hse-OtBu (2.72 g, 9.89 mmol), TEMPO (0.016 g, 0.1 mmol), and KBr (0.119 g, 1.00 mmol) in CH_2Cl_2 (12 mL) and H_2O (2 mL) such that the temperature was maintained at 4 °C. The soln was vigorously stirred at 4 °C for 1 h and transferred to a separating funnel using Et_2O (50 mL). The organic soln was washed with 5% $NaHCO_3$ (2 × 20 mL) and 1 M HCl (2 × 20 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure to give a colorless oil, which slowly changed into a white solid; yield: 2.59 g (96%); mp 65–66 °C; ^{13}C NMR ($CDCl_3$, δ): 27.7 and 28.1 (Boc and tBu Me), 46.1 (Asa Cβ), 49.3 (Asa Cα), 79.8 (Boc Cq), 82.4 (tBu Cq), 155.2 (Boc CO), 179.8 (Asa CO), 197.7 (Asa CHO); FAB-MS *m/z*: 274 [M+H]. (Note: in the NMR section Asa is aspartaldehyde.)

Boc-Abu{4-[PO(OMe)$_2$],4-OH}-OtBu:[98]
A soln of Boc-aspartaldehyde-OtBu (3.82 g, 14.0 mmol) and dimethyl trimethylsilyl phosphite (3.87 g, 20.7 mmol) in MeCN (15 mL) was refluxed for 1.5 h. H_2O (15 mL) was added and the soln refluxed for a further 15 min. The soln was cooled, EtOAc (70 mL) added and the organic phase washed with NaCl (2 × 20 mL) and H_2O (2 × 20 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure to give a colorless oil which slowly changed to a white solid; yield: 5.25 g (98%); mp 107–108 °C; ^{13}C NMR ($CDCl_3$, δ): 27.7 and 28.0 (Boc and tBu Me), 33.6 and 35.0 (each d, $J_{P,C}$ = 3.3 Hz, Abu C3), 50.1 and 52.3 (each br s, Abu C2), 53.0 and 53.3 (each d, $J_{P,C}$ = 2.44 Hz, POMe), 63.7 and 64.3 (each d, $J_{P,C}$ = 168.5 Hz, Abu C4), 79.7 and 80.2 (Boc Cq), 81.7 and 82.2 (tBu Cq), 155.2 and 156.6 (Boc CO), 170.9 (Abu CO); ^{31}P NMR ($CDCl_3$, δ): 26.4 and 26.6; FAB-MS *m/z*: 384 [M+H].

H-Abu[PO(OMe)$_2$]-OH·TFA:[98]
A soln of *O*-phenyl chloridothiocarbonate (0.65 g, 3.75 mmol) in MeCN (1 mL) was added to a soln of Boc-Abu{4-[PO(OMe)$_2$],4-OH}-OtBu (1.046 g, 2.72 mmol) and pyridine (0.79 g, 10.0 mmol) in dry MeCN (4 mL) at 0 °C. After stirring at rt for 16 h, H_2O (1 mL) was added, the mixture stirred for a further 30 min and the solvent removed under reduced pressure. Et_2O (30 mL) was added and the organic phase washed with 5% $NaHCO_3$ (2 × 10 mL) and 1 M HCl (10 mL), dried (Na_2SO_4), and filtered.

The solvent was removed under reduced pressure to give crude (4*RS*)-thionocarbonate as a yellow oil; yield: 1.36 g (95%); ^{13}C NMR ($CDCl_3$, δ): 28.0 (Boc Me), 28.3 (tBu Me), 31.9 and 32.0 (br s, Abu C3), 50.7 (br d, Abu C2), 53.5 and 53.9 (J_{PC} = 6.6, 5.5 Hz, Me), 74.0 and 74.3 (d, J_{PC} = 167.0 Hz, Abu C4), 80.0 (Boc Cq), 82.5 (tBu Cq), 115.4 (Ph C1), 121.9 (Ph C3), 126.6 (Ph C4), 129.4 and 129.6 (Phe C2), 153.7 (CS), 155.2 and 155.4 (Boc CO), 170.8 (Abu CO); ^{31}P NMR ($CDCl_3$, δ): 21.3 and 20.9 (2:1 ratio).

AIBN (0.197 g, 1.20 mmol) was added to a soln of the crude thionocarbonate (1.08 g, 2.15 mmol) and tris(trimethylsilyl)silane (1.19 g, 4.80 mmol) in dry toluene (10 mL) and stirred at 80 °C for 2 h. The solvent was removed under reduced pressure, Et_2O (30 mL) added and the organic phase washed with 5% $NaHCO_3$ (2 × 10 mL) and 1 M HCl (10 mL), dried (Na_2SO_4), and filtered. The solvent was removed under reduced pressure and dried under high vacuum to give a yellow oil, which was dissolved in TFA (5 mL) and allowed to stand for 1 h. The solvent was removed under reduced pressure, the residue partitioned between H_2O (5 mL) and Et_2O (5 mL), the aqueous phase collected and the ether soln extracted with H_2O (5 mL). The combined aqueous solns were combined, washed with Et_2O (5 mL), and lyophilized to give H-Abu[PO(OMe)$_2$]-OH·TFA as a hygroscopic light yellow foam; yield: 0.503 g (72% based on the thionocarbonate).

Boc-Abu[PO(OMe)$_2$]-OH:[98]

A soln of Boc_2O (2.06 g, 9.4 mmol) in dioxane (5 mL) was added to a soln of H-Abu[PO(OMe)$_2$]-OH·TFA (1.20 g, 3.69 mmol) and Na_2CO_3 (0.391 g, 3.69 mmol) in H_2O (5 mL) and the soln vigorously stirred for 2 h. The dioxane was removed under reduced pressure, the aqueous phase washed with Et_2O (2 × 5 mL), acidified to pH 1 (2 M HCl) and extracted with CH_2Cl_2 (3 × 20 mL). The solvent was removed under reduced pressure to give a colorless oil; yield: 0.758 g (66%); ^{13}C NMR ($CDCl_3$, δ): 20.4 (d, J_{PC} = 142.8 Hz, Abu C4), 25.4 (d, J_{PC} = 4.5 Hz, Abu C3), 28.3 (Boc Me), 52.8 (d, J_{PC} = 6.1 Hz, Me), 53.1 (d, J_{PC} = 19.5 Hz, Abu C2), 79.2 (Boc Cq), 155.7 (Boc CO), 173.6 (Abu CO); ^{31}P NMR ($CDCl_3$, δ): 34.6; FAB-MS *m/z*: 312 [M+H].

Fmoc-Abu[PO(OMe)$_2$]-OH (20):[98]

A soln of Fmoc-OSu (1.042 g, 3.09 mmol) in dioxane (3 mL) was added to a soln of H-Abu[PO(OMe)$_2$]-OH·TFA (0.503 g, 1.548 mmol) in 10% aq Na_2CO_3 (3 mL) and vigorously stirred for 2 h. The dioxane was removed under reduced pressure and Et_2O (10 mL) added. The aqueous layer was collected and the ethereal layer extracted with 5% aq $NaHCO_3$ (2 × 8 mL). The aqueous solns were combined, washed with Et_2O (5 mL), acidified to pH 1 (3 M HCl) and extracted with CH_2Cl_2 (3 × 10 mL). The soln was removed under reduced pressure to give a light yellow foam; yield: 0.637 g (95%); ^{13}C NMR ($CDCl_3$, δ): 20.3 (d, J_{PC} = 140.6 Hz, Abu C4), 25.2 (d, J_{PC} = 1.1 Hz, Abu C3), 47.0 (Fmoc CH), 52.8 (d, J_{PC} = 6.6 Hz, Me), 53.7 (d, J_{PC} = 17.6 Hz, Abu C2), 67.1 (Fmoc CH_2), 119.8, 124.9, 126.9, 127.6, 141.1, 143.6, and 143.7 (Fmoc Ar C), 156.1 (Fmoc CO), 173.3 (Abu CO); ^{31}P NMR ($CDCl_3$, δ): 34.9; FAB-MS *m/z*: 434 [M+H].

H-Leu-Arg-Arg-Ala-Abu(*P*)-Leu-Gly-OH·3 TFA:[98]

Fmoc-based solid-phase peptide assembly was performed manually using Fmoc-Gly-Wang-resin (0.52 mmol·g^{-1}, 0.21 g, 0.11 mmol), Fmoc-Arg(Pmc)-OH, and Boc-Leu-OH. The Fmoc group was cleaved by 20% piperidine/DMF (7 mL) treatment (10 min) and amino acid couplings were performed using PyBOP (5 equiv), the N-protected amino acid (5 equiv), NMM (7.5 equiv) in DMF/CH_2Cl_2 (1:1; 7 mL) for 20 min. On completion of peptide assembly, the peptide-resin was treated with 5% PhSMe/TFA (5 mL) for 7 h and the soln removed under reduced pressure and Et_2O (20 mL) added to give the crude Abu[PO(OMe)$_2$]-peptide as a white powder (0.115 g, theoretical 0.134 g). The crude peptide was treated with 1 M TMSBr–PhSMe/TFA (5 mL) at rt for 26 h, the solvent was removed under reduced pressure and Et_2O (20 mL) added. The precipitated residue was triturated with Et_2O (2 × 10 mL) and a portion of the brown residue (23.0 mg) was purified by semipreparative RP-HPLC (0.1%, 0–48% MeCN over 30 min) to give a fluffy white powder; yield: 17.7 mg (71% based on resin); ^{13}C NMR (D_2O, δ): 16.31, 20.57, 21.11, 21.56, 22.07, 23.74 {d, J_{PC} = 135.0 Hz, Abu[PO(OH)$_2$] Cγ}, 23.74. 24.20, 24.31, 24.33, 25.34 {br d, Abu[PO(OH)$_2$] Cβ}, 28.10, 39.66, 39.82, 40.48, 41.00, 49.69, 51.57, 52.28, 53.23, 53.34, 53.91 {d, J_{PC} = 17.7 Hz, Abu[PO(OH)$_2$] Cα}, 156.66 (× 2), 170.24, 172.84, 172.97, 173.03 (× 2), 174.72, 174.88; ^{31}P NMR (D_2O, δ): 25.1; ES-MS *m/z*: 849.6 [M+H]; FAB-MS *m/z*: 850 [M+H].

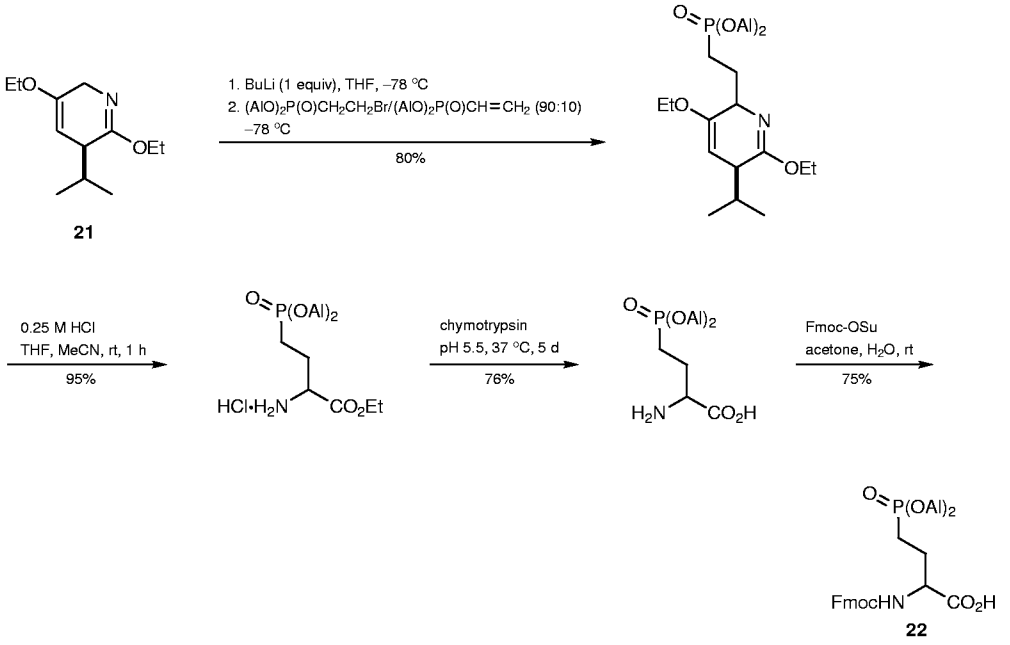
6.5.5.2 Synthesis of 2-Amino-4-phosphonobutanoic Acid Containing Peptides via Allyl Phosphonate Protection

The synthesis of Abu[PO(OH)$_2$]-peptides is also accomplished via the use of Fmoc-Abu-[PO(OAl)$_2$]-OH (**22**) with the allyl phosphate ester being cleaved by Pd(0)-mediated treatment. Fmoc-Abu[PO(OAl)$_2$]-OH (**22**)[101] is prepared from Schöllkopf's (−)-bislactim ethyl ether **21**[102] by a four-step procedure that involves initial treatment of the lithium salt of

for references see p 421

the bislactim ether **21** with an equimolar proportion of diallyl (2-bromoethyl)phosphonate containing 10% diallyl vinylphosphonate at −78 °C (Scheme 19). The target derivative is obtained by mild acid hydrolysis (0.25 M HCl) of the bislactim ether, enzymatic cleavage of the ethyl ester from H-Abu[PO(OAl)$_2$]-OEt·HCl (chymotrypsin, pH 5.5, 5 d), and final introduction of the Fmoc group using Fmoc-OSu. The optical purity of Fmoc-Abu-[PO(OAl)$_2$]-OH was determined to be >98% L-(+) through HPLC estimation of the (−)-(*S*)-phenylethylamine amides of both the (+)- and (−)-derivatives.

Scheme 19 Synthesis of Fmoc-Abu[PO(OAl)$_2$]-OH[101]



Cleavage of the phosphonate allyl esters from Abu[PO(OAl)$_2$]-peptide-resins is accomplished by treatment with Pd(PPh$_3$)$_4$ (50 °C, 5 h)[103] followed by acidolytic deprotection of the Abu[PO(OH)$_2$]-peptide from the resin support. The viability of employing allyl phosphonate protection for phosphopeptide synthesis has been demonstrated by the Fmoc solid-phase synthesis (DIC/HOBt couplings) of several 6–9-mer Abu[PO(OH)$_2$]-containing peptides and has been extended to the synthesis of a 19-mer calcineurin-related Abu[PO(OH)$_2$]-peptide.[104]

Fmoc-Abu[PO(OAl)$_2$]-OH (22):[101,102]

A soln of 1.60 M BuLi in hexane (3.50 mL, 5.6 mmol) was injected by syringe into a soln of (−)-bislactim ethyl ether **21**[102] (1.15 g, 5.4 mmol) in THF (10 mL) at −78 °C and stirred for 10 min. A soln of diallyl (2-bromoethyl)phosphonate (1.45 g, 5.40 mmol) and diallyl vinylphosphonate (0.102 g, 0.54 mmol) in THF (4 mL) was added at −78 °C and the soln stirred at −78 °C for 2 h. AcOH (0.33 g, 5.5 mmol) in THF (1 mL) was added, the soln allowed to warm to rt and the solvent removed under reduced pressure. The residue was dissolved in Et$_2$O (20 mL), washed with H$_2$O (2 × 10 mL), dried (Na$_2$SO$_4$), and filtered. The solvent was removed under reduced pressure and purified by flash column chromatography (silica gel) to give the phosphorylated bislactim ethyl ether; yield: 80%; $[\alpha]_D^{20}$ −1.7 (*c* 1.2, MeOH).

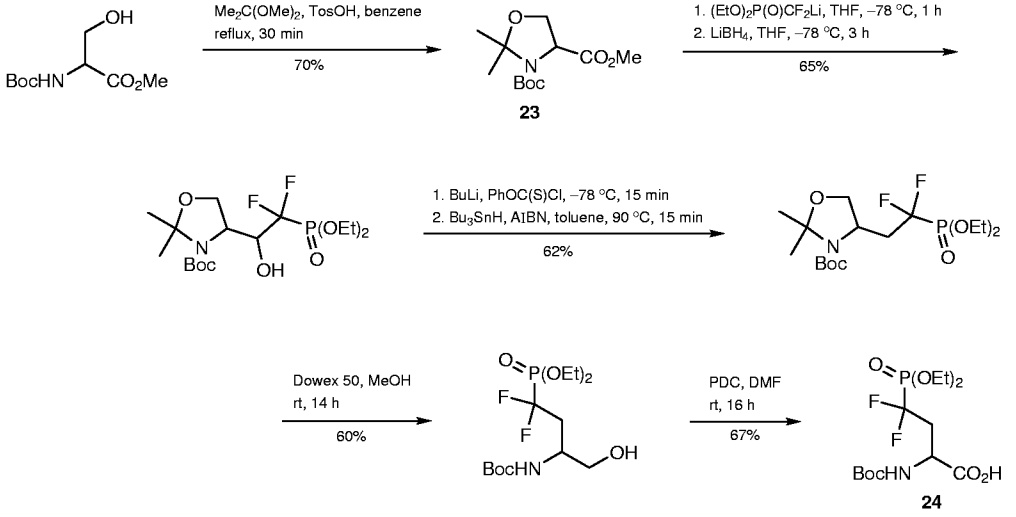
The phosphorylated bislactim ethyl ether was dissolved in 0.25 M HCl and THF/MeCN (3:1) and the soln vigorously stirred at rt for 1 h to give H-Abu[PO(OAl)$_2$]-OEt·HCl; yield: 95%. Hydrolysis of the ethyl ester was effected by treatment with chymotrypsin in KH$_2$PO$_4$/Na$_2$PO$_4$ (pH 5.5) at 37 °C for several days (no details given) to give H-Abu[PO(OAl)$_2$]-OH; yield: 76%; $[\alpha]_D^{20}$ +9.7 (*c* 0.65, H$_2$O). The Fmoc group was introduced by treatment with Fmoc-OSu in acetone/H$_2$O at rt (no details given) to give a solid which was crystallized (hexane/CH$_2$Cl$_2$) to give Fmoc-Abu[PO(OAl)$_2$]-OH as a white solid; yield: 75%, 43% overall; mp 136–138 °C; $[\alpha]_D^{20}$ −20.1 (*c* 1.1, CH$_2$Cl$_2$).

H-Asp-Leu-Asp-Val-Pro-Ile-Pro-Gly-Arg-Phe-Asp-Arg-Arg-Val-Abu[PO(OH)$_2$]-Val-Ala-Ala-Glu-OH:[104] Fmoc-based solid-phase peptide assembly was performed on an ABI 430A Peptide Synthesizer (standard DIC/HOBt coupling cycle programs) using Wang resin (0.50 mmol) and Pmc protection of Arg. On completion of peptide assembly, the peptide-resin was treated with a soln of $Pd(PPh_3)_4$ (20 mol%), $BuNH_2$ (4 equiv), HCO_2H (16 equiv) in THF at 50 °C for 5 h[101] and the soln removed by suction filtration. The Abu[PO(OH)$_2$]-peptide-resin was treated with a soln of TFA/EDT/PhSMe/H_2O (10:0.25:0.5:0.5; 11.25 mL) containing PhOH (0.75 g) for 2 h, the soln removed under reduced pressure, and Et_2O (20 mL) added. The precipitated peptide was purified by semipreparative RP-HPLC and desalted (Biorad AG 4X-4, 100–200 mesh, aq 2 M AcOH eluant) to give the title peptide as a white powder; yield: 11% (based on resin); FAB-MS *m/z*: 2000 [M+H].

6.5.6 Synthesis of (Difluoromethyl)phosphonate Mimetics of Phosphoserine and Phosphothreonine Peptides

The generation of CF_2-peptide mimetics of Ser(*P*)- and Thr(*P*)-peptides for the study of phosphorylation systems is of particular interest in consideration of the inhibitory action of various phosphorylated biomolecule CF_2-mimetics.[105–107] The improved inhibitory performance of CF_2-peptide substrates over CH_2-substrates is attributed to the similar pK_a of the CF_2–$PO(OH)_2$ group with that of the native O–$PO(OH)_2$ group. The synthesis of Boc-Ala[$CF_2PO(OEt)_2$]-OH (**24**)[108,109] is carried out in six steps from the Garner aldehyde or methyl ester[110] using commercially available diethyl (difluoromethyl)phosphonate (Scheme 20). The key phosphorylation step is accomplished by treatment of the Garner methyl ester **23** (derived from Boc-D-Ser-OMe) with diethyl (difluorolithiomethyl)phosphonate at −78 °C to give the β-oxo α,α-difluorophosphonate, which is immediately reduced to the β-hydroxy α,α-difluorophosphonate by lithium borohydride (65% overall yield).[109] Homolytic deoxygenation is effected by conversion into its phenyl thionocarbonate using phenyl chlorothionoformate and treatment with Bu_3SnH/AIBN. Subsequent ring opening of the oxazolidinone is achieved by Dowex-50 treatment and final oxidative conversion of the hydroxy group into a carboxylic acid is effected using PDC to generate the target Boc-Ala[$CF_2PO(OEt)_2$]-OH derivative **24** as a crystalline solid.

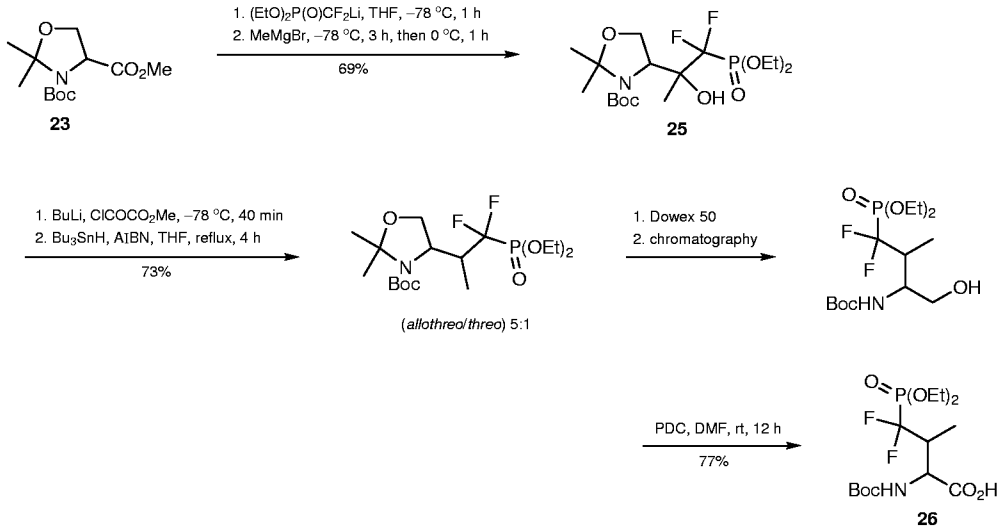
Scheme 20 Synthesis of Boc-Ala[$CF_2PO(OEt)_2$]-OH[109]



for references see p 421

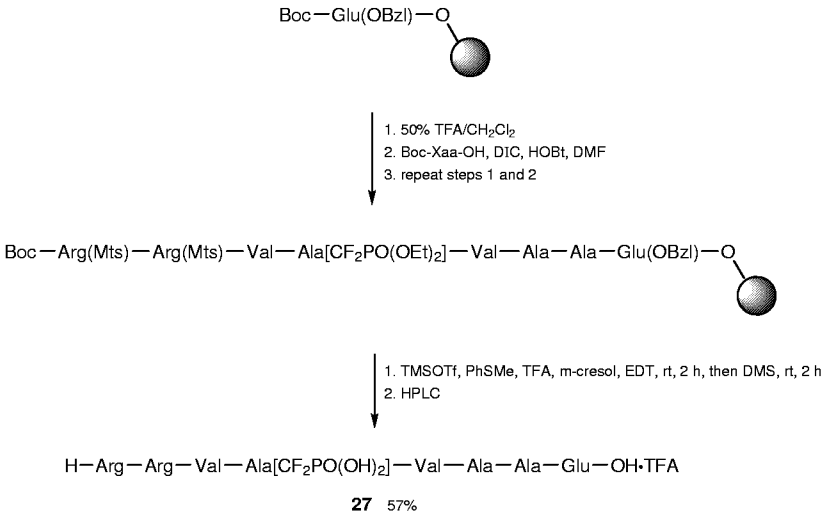
A feature of the above approach is that the generated β-oxo α,α-difluorophosphonate also permits entry to the synthesis of the CF_2-mimetic of Thr[PO(OEt)$_2$]-peptide through a Grignard reaction with methylmagnesium bromide which generates the β-methyl-β-hydroxy α,α-difluorophosphonate **25**.[109] Subsequent homolytic reduction of the secondary alcohol via its methyl oxalate ester and ring opening of the oxazolidinone generates a diastereomeric mixture of the δ-hydroxy difluorophosphonate in a 5:1 *allothreo/threo* ratio. The isomers are separated by silica gel chromatography and final PDC oxidation of the minor *threo* isomer generates the target Boc-Gly[CH(Me)CF_2PO(OEt)$_2$]-OH (**26**) (Scheme 21).

Scheme 21 Synthesis of Boc-Gly[CH(Me)CF_2PO(OEt)$_2$]-OH[109]



The successful application of Boc-Ala[CF_2PO(OEt)$_2$]-OH in Boc-based solid-phase synthesis (Merrifield resin) has been demonstrated by the preparation of H-Arg-Arg-Val-Ala[CF_2PO(OH)$_2$]-Val-Ala-Ala-Glu-OH (DIC/HOBt couplings) using the TMSOTf/DMS/TFA procedure for the one-step deprotection of the Ala[CF_2PO(OEt)$_2$]-peptide-resin (Scheme 22).[108] Although the silylitic cleavage of the phosphonate ethyl esters by TMSOTf/DMS is incomplete and gives a mixture of the Ala[CF_2PO(OEt)(OH)]- and Ala[CF_2-PO(OH)$_2$]-peptides, these peptides are generally separable by RP-HPLC. In consideration of the successful use of methyl phosphonate protection for the synthesis of Abu[PO(OH)$_2$]-peptides, it is expected that improved chemical efficiency will be attained through the use of Boc-Ala[CF_2PO(OMe)$_2$]-OH and thereby provide a reliable synthetic route for the preparation of various CF_2-mimetics of biochemically relevant Ser(*P*)- and Thr(*P*)-containing peptides.

Scheme 22 Synthesis of an Ala[$CF_2PO(OH)_2$]-Containing Peptide[108]



Diethyl (*RS*)-2-[(4*R*)-3-(*tert*-Butoxycarbonyl)-2,2-dimethyloxazolidin-4-yl]-1,1-difluoro-2-hydroxyethylphosphonate:[109]

A soln of 1.3 M BuLi in hexane (14.5 mL, 18.9 mmol) was added dropwise by syringe to a stirred soln of iPr_2NH (2.66 mL, 18.9 mmol) in dry THF (4 mL) at −78 °C and the temperature of the soln allowed to increase to 0 °C and stirred for a further 25 min. The soln was then cooled to −78 °C and a soln of diethyl (difluoromethyl)phosphonate (2.96 mL, 18.9 mmol) in THF (4 mL) was added. The soln was stirred at −78 °C for 30 min and a soln of the oxazolidinecarboxylic acid methyl ester **23** (3.79 g, 14.6 mmol) in THF (4 mL) added dropwise via cannula. The soln was stirred at −78 °C for 60 min and AcOH (1.925 mL, 33.75 mmol) was added followed by sat. NH_4Cl (40 mL). The aqueous phase was extracted with EtOAc (3 × 40 mL), the combined organic extracts dried ($MgSO_4$), and filtered. The solvent was removed under reduced pressure and the residue dried under high vacuum.

The crude ketone was dissolved in Et_2O (75 mL), cooled to −78 °C and a soln of 2 M $LiBH_4$ in THF (1.75 mL, 3.51 mmol) added dropwise by syringe. The soln was stirred at −78 °C for 3 h, a soln of sat. NH_4Cl (50 mL) added, the organic phase collected and the aqueous phase further extracted with Et_2O (4 × 50 mL). The organic extracts were combined, dried ($MgSO_4$), filtered, and the solvent removed under reduced pressure. The residue was purified by chromatography (silica gel, EtOAc/hexane 30:70) to give the pure alcohol as a colorless oil; yield: 3.97 g (65% overall yield); ^{13}C NMR ($CDCl_3$, δ): 16.30, 16.34, 24.1, 27.1, 28.3, 56.8, 64.5, 64.6, 65.1 (br), 73.9 (br), 82.0. 93.7, 156.6.

Diethyl (*RS*)-2-[(4*R*)-3-(*tert*-Butoxycarbonyl)-2,2-dimethyloxazolidin-4-yl]-1,1-difluoro-2-[(phenoxy)thiocarbonyloxy]ethylphosphonate:[109]

A soln of 1.4 M BuLi in hexane (6.79 mL, 9.51 mmol) was added dropwise by syringe to a stirred soln of the diethyl (*RS*)-2-[(4*R*)-3-(*tert*-butoxycarbonyl)-2,2-dimethyloxazolidin-4-yl]-1,1-difluoro-2-hydroxyethylphosphonate (3.96 g, 9.51 mmol) in dry THF (50 mL) at −78 °C followed immediately by the addition of phenyl chlorothionoformate (1.97 mL, 14.3 mmol). After 15 min at −78 °C, a soln of $NaHCO_3$ (25 mL) was added and extracted with Et_2O (4 × 5 mL). The combined organic extracts were dried ($MgSO_4$), filtered, and the solvent removed under reduced pressure. The residue was purified by flash chromatography (silica gel, EtOAc/hexane 10:90 to 30:70) to give the pure product; yield: 4.05 g (79%).

Diethyl (*RS*)-2-[(4*S*)-3-(*tert*-Butoxycarbonyl)-2,2-dimethyloxazolidin-4-yl]-1,1-difluoroethylphosphonate:[109]

A soln of the above thionocarbonate (0.836 g, 1.52 mmol), Bu_3SnH (0.610 mL, 2.27 mmol), and AIBN (0.05 g, 0.30 mmol) in toluene (7.6 mL) under N_2 was heated at 90 °C for 15 min. The soln was removed under reduced pressure and the residue was purified by flash column chromatography (silica gel, Et_2O/toluene 10:90 to 30:70) to give the pure (*S*)-acetal as a colorless oil; yield: 0.479 g (79%); ^{13}C NMR ($CDCl_3$, δ) major [minor] rotamer: 16.1 [16.1], 16.2 [16.2], 23.0 [24.3], 26.6 [27.4], 28.2 [28.2], 36.9–37.3 [35.9–36.1], 51.3 [51.7], 64.42 [64.30], 64.38, [64.30], 67.6 [67.2], 79.9 [80.4], 93.2 [92.8], 120.0 (m) [120.0 (m)], 151.1 [151.1].

for references see p 421

***N*-(*tert*-Butoxycarbonyl)-3-[(diethoxyphosphoryl)difluoromethyl]-L-alaninol:**[109]
A soln of the (*S*)-*O*,*N*-acetal (0.40 g, 1.00 mmol) and Dowex-50×8 (2.00 g) in MeOH (4 mL) was stirred at rt for 14 h. The soln was filtered, washed with MeOH (40 mL) and the combined filtrates removed under reduced pressure and purified by column flash chromatography (silica gel, EtOAc/hexanes 50:50 to 100:0) to give pure *N*-(*tert*-butoxycarbonyl)-3-[(diethoxyphosphoryl)difluoromethyl]alaninol as a clear oil; yield: 0.215 g (60%); FAB-MS (3-NOBA/LiI) *m/z*: 368.1625 [M+Li].

Boc-L-Ala[$CF_2PO(OEt)_2$]-OH (24):[109]
A soln of *N*-(*tert*-butoxycarbonyl)-3-[(diethoxyphosphoryl)difluoromethyl]alaninol (0.521 g, 1.44 mmol) and PDC (3.23 g, 8.65 mmol) in DMF (4 mL) was stirred at rt overnight. EtOAc (60 mL) and H_2O (60 mL) were added, the organic phase collected and the aqueous phase extracted with EtOAc (4 × 50 mL). The combined organic extracts were dried ($MgSO_4$), filtered, and the solvent removed under reduced pressure. The residue was purified by chromatography (silica gel, MeOH/EtOAc 0:100 to 5:95) to give the product as a clear oil; yield: 0.362 g (67%); $[\alpha]_D$ −10.8 (*c* 2.2, MeOH); ^{13}C NMR ($CDCl_3$, δ): 16.27, 16.29, 28.3, 35.3–35.4 (m), 65.0, 65.1, 80.5, 155.5, 173.0; FAB-MS (3-NOBA/Na_2CO_3) *m/z*: 398.1 [M+Na].[109] The oil changed to a solid on standing and was crystallized from Et_2O to give a white crystalline solid; mp 112–113 °C; $[\alpha]_D^{27}$ +9.7 (*c* 1.5, $CHCl_3$).[108]

H-Arg-Arg-Val-Ala[$CF_2PO(OH)_2$]-Val-Ala-Ala-Glu-OH·TFA (27):[108]
Boc-based peptide assembly of Boc-Arg(Mts)-Arg(Mts)-Val-Ala[$CF_2PO(OEt)_2$]-Val-Ala-Ala-Glu-Merrifield-resin was performed using Boc-Glu(OBzl)-Merrifield-resin (0.67 mmol·g^{-1}) and DIC/HOBt (2.5 equiv each) couplings of the Boc amino acids (2.5 equiv) in DMF. A soln of 1 M TMSOTf/1 M PhSMe in TFA (1.0 mL) was added by syringe under N_2 to a mixture of the peptide-resin (0.01 mmol), *m*-cresol (50 μL), and EDT (50 μL) and the mixture stirred at rt for 2 h. DMS (200 μL) was added to the soln and the soln stirred at rt for a further 2 h. The soln was filtered, chilled Et_2O (10 mL) added and the precipitated peptide collected by centrifugation. The peptide was washed with Et_2O (3 × 10 mL) and purified by semipreparative RP-HPLC (0.1% TFA, 0–30% MeCN over 30 min) to give the product as a white powder; yield: 57% (based on resin); IS-MS *m/z*: 1000.24 [M+H]; amino acid analysis: Glu (1) 0.97, Ala (2) 2.00; Val (2), 2.00, Arg (2) 1.84.

6.5.7 Synthesis of Phosphopeptides Using the "Post-Assembly" Phosphorylation Approach

The general and simple synthesis of phosphorylated peptides is also possible by an alternative approach which is based on the "post-assembly" (global) phosphorylation of protected peptides containing unprotected Tyr, Ser, and Thr residues in solution[111,112] or on the solid-phase.[113] The first "post-assembly" phosphorylations of nonprotected Tyr-containing peptides had limited success due to substantial peptide degradation caused by the harsh phosphoryl chloride/pyridine procedure and tedious phosphopeptide isolation.[114] The subsequent development of the milder diphenyl or dibenzyl phosphorochloridate/pyridine procedure during the 1950s provided improved access to small phosphorylated peptides but was complicated due to inefficient phosphorylation (dibenzyl phosphorochloridate) or phosphate deprotection (diphenyl phosphorochloridate) procedures.[68] Despite early successes, the extension of this approach to larger peptides proved unsatisfactory since diphenyl or dibenzyl phosphorochloridate phosphorylations of Ser- and Thr-containing peptides were compromised by a base-mediated dephosphorylative rearrangement[115] and through poor yields for phosphorylations performed on the solid phase.[90]

6.5.7.1 "Post-Assembly" Phosphorylation of Peptide-Resins

A major development in "global" phosphorylation was realized through the application of highly reactive diorganyl dialkylphosphoramidites for the simple "phosphite-triester" phosphorylation of target sites.[76] While this phosphorylation procedure has been success-

fully employed for the phosphorylation of protected peptides in solution,[111,112,116] the principal application of this procedure has been for the phosphorylation of Tyr-, Ser-, and Thr-containing peptide-resins. The assembly of the requisite Tyr-, Ser-, and Thr-containing peptide-resin is accomplished via the incorporation of side-chain unprotected Fmoc-Xaa-OH (Xaa = Tyr, Ser, and Thr) derivatives and performed using phosphonium (BOP, PyBOP) or uronium reagents (HBTU, TBTU) in order to prevent acylation of the side-chain hydroxy group[117] as occurs with the use of either DCC- or DIC-mediated couplings. In the case of solid-phase peptide synthesizers using DCC or DIC coupling chemistry, the assembly of Ser-containing peptide-resins is possible through the incorporation of Fmoc-Ser(TBDMS)-OH or Fmoc-Ser/Thr(Trt)-OH derivatives with the side-chain TBDMS group being cleaved from the assembled peptide-resin by TBAF treatment[118] and the Trt group cleaved by 1% TFA/CH_2Cl_2 (10×2 min).[119]

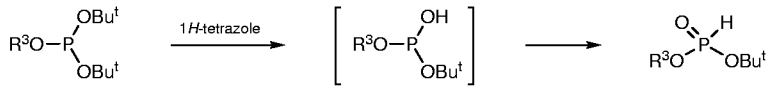
The general synthesis of phosphopeptides is best accomplished using the acid-labile Wang and MBHA resins with $(tBuO)_2PNR^1_2$, $(BzlO)_2PNR^1_2$ and $(TMSEO)_2PNR^1_2$ (R^1 = Et, iPr) being commonly used for the phosphitylation step.[120–133] A common feature of these phosphoramidite reagents is that the alkyl group of the generated dialkyl phosphorotriester is particularly sensitive to acid conditions and is readily cleaved during the final TFA deprotection step. The various diorganyl dialkylphosphoramidites (alkyl = Et, iPr) are readily prepared by the treatment of $Cl_2PNR^1_2$ with 2 equivalents of alcohol in the presence of excess triethylamine.[76,134] While $(tBuO)_2PNR^1_2$ is isolated by high vacuum distillation, both $(BzlO)_2PNR^1_2$ and $(TMSEO)_2PNR^1_2$ (R^1 = Et, iPr) are best purified by silica gel chromatography since both these reagents undergo disproportionation at high distillation temperatures. The susceptibility of phosphoramidite reagents to oxidation and hydrolysis requires their purity to be checked by ^{31}P NMR spectroscopy on a regular basis, especially in the case of commercially available $(tBuO)_2PNR^1_2$ and $(BzlO)_2PNR^1_2$ (R^1 = Et, iPr) where significant deterioration occurs during long-term storage.

The phosphitylation of a target hydroxy site is generally performed by activation of the $(R^2O)_2PNR^1_2$ reagent by a weak acid such as 1*H*-tetrazole in DMF. This solvent is preferred over THF or MeCN since its better resin swelling properties permits reactions to be performed in lesser volumes and at a higher phosphitylation rate. The reaction is particularly sensitive to moisture and requires thorough high vacuum drying of both the peptide-resin and 1*H*-tetrazole. While best results are obtained using freshly distilled anhydrous DMF, the use of commercially available anhydrous DMF (99.8%, <0.005% water) has proved to be a satisfactory alternative.

The general "phosphite-triester" phosphorylation of peptide-resins is best performed under "concentrated" reaction conditions using 10–20 equivalents of $(R^2O)_2PNR^1_2$ per hydroxyamino acid residue in a minimum volume of DMF (for this purpose, high-loading resins are preferred over lower-loading "continuous flow" resins). However, since high molar concentrations of 1*H*-tetrazole causes acid-mediated conversion of $(tBuO)_2POR^3$ and $(TMSEO)_2POR^3$ phosphite triesters into their alkyl H-phosphonate forms, aqueous iodine must be used for the following oxidation so as to effect conversion of the H-phosphonate byproduct into its phosphorodiester (Scheme 23).[135] Notably, the benzyl phosphite group is not subject to acid-mediated cleavage and thereby permits the use of tBuOOH, MCPBA, or aqueous iodine for phosphite-triester oxidation. Since MCPBA and I_2/H_2O oxidation are both incompatible with Trp, Met, and Cys residues, the oxidation of these peptide-resins are best effected using tBuOOH since this procedure does not cause oxidation of the Met residue.[119] In the event of Met oxidation, the Met(O) residue can be reduced by overnight treatment with *N*-methyl(sulfanyl)acetamide.[124] The acidolytic deprotection of Xaa-$[PO(OR^1)_2]$-containing peptide-resins (R^1 = tBu, TMSE) is readily accomplished by a 1 hour treatment with an appropriate TFA/scavenger solution[26] with the slower cleavage of benzyl phosphate groups from Ser$[PO(OR^1)_2]$ and Thr$[PO(OR^1)_2]$ residues requiring between 2 to 8 hours.

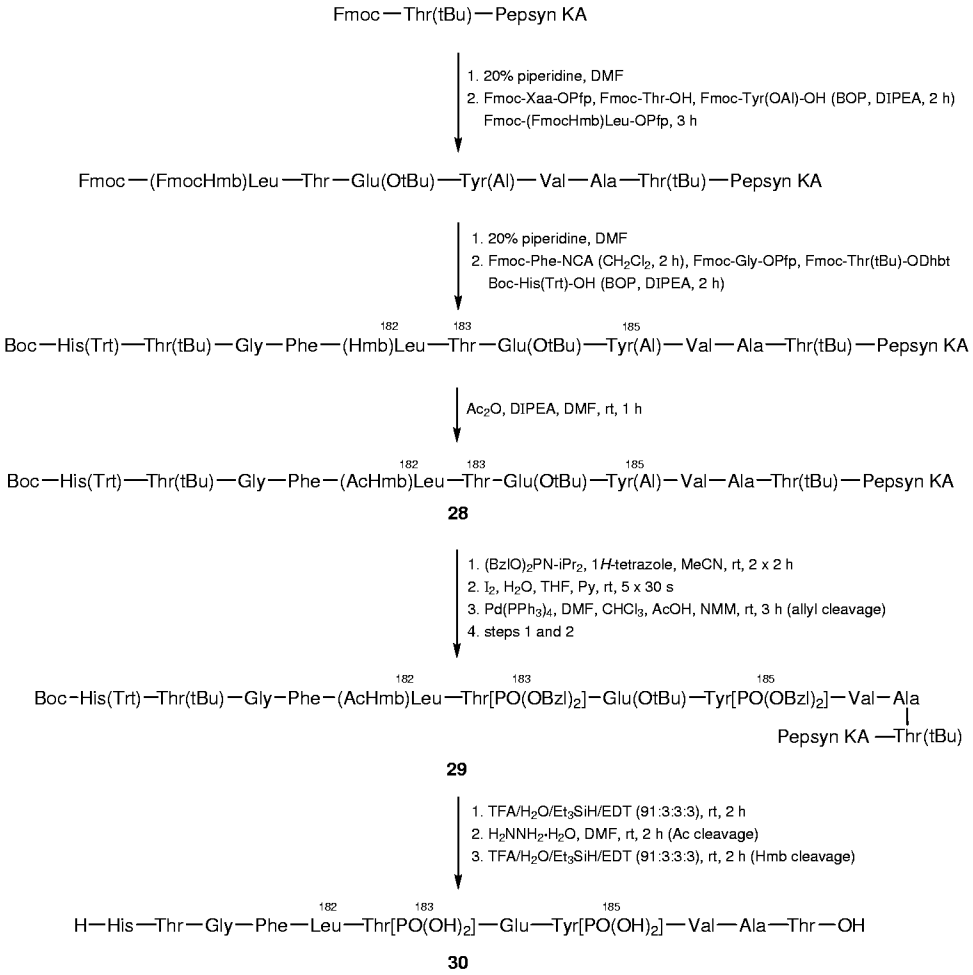
for references see p 421

Scheme 23 Acid-Mediated Conversion of Di-*tert*-butyl Phosphite Triester into the H-Phosphonate[135]



While post-assembly phosphorylation has been used successfully in many instances, the slower phosphorylation of Thr-peptides and other sterically hindered Tyr- and Ser-peptide sequences generally requires repeated phosphitylations to ensure efficient phosphorylation levels. However, the incomplete phosphorylation of some particular peptide-resins after repeated phosphitylation treatments has generally been attributed to inaccessibility of the target site through peptide-chain aggregation.[94,116] In such instances, improved phosphorylative conversions have been obtained through the use of Hmb-amide backbone protection (see Vol. E 22a, Section 2.3.2) or through the adoption of a segment condensation approach. In the first instance, improved phosphitylation through disruption of peptide-chain aggregation was demonstrated by the synthesis of the bis-phosphorylated MAP kinase ERK 2(178–188) (**30**) in which Hmb-amide protection was introduced at Leu^{182} during peptide assembly (Scheme 24).[94] Successive phosphorylation of Thr^{183} and Tyr^{185} of the Hmb-protected peptide-resin **28** was effected by repeated phosphitylations using the $(BzlO)_2PN\text{-}iPr_2$ and I_2/H_2O/THF combination with peptide deprotection of **29** effected by TFA treatment, hydrazine cleavage of the acetyl group, and final TFA cleavage of the Hmb group. The only drawback of this approach is that additional chemical steps are required for introduction of the acetyl group of the Hmb group and its subsequent cleavage.

Scheme 24 Synthesis of MAP Kinase ERK 2(178–188) via "Post-Assembly" Phosphorylation and Hmb-Amide Protection[94]

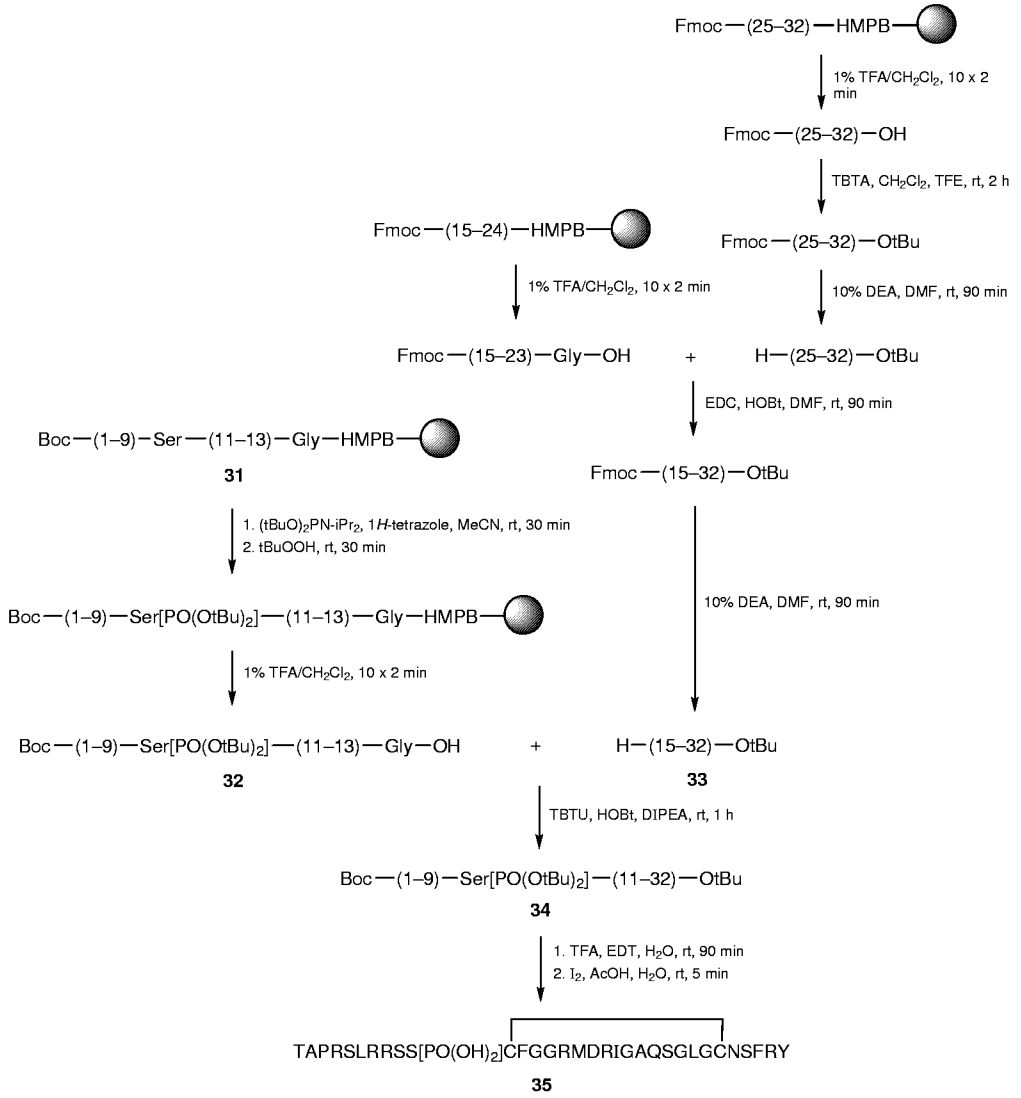


In addition to the above approach, the synthesis of large phosphopeptides containing suitably placed Gly residues offers synthesis by a segment condensation strategy in which the "global" phosphorylation procedure is applied to the phosphorylation of smaller, less sterically hindered segments. In contrast to the steric problems associated with the final "global" phosphorylation of Ser-urodilatin(95–126),[116] Ser(*P*)-urodilatin(95–126) **35** was readily prepared by the TBTU-mediated segment condensation of the protected 14-mer Ser[PO(OtBu)$_2$]-peptide segment **32** with the 18-mer C-terminal segment **33** followed by TFA/EDT/H_2O deprotection of the Ser[PO(OtBu)$_2$]-peptide(1–32) **34** and final generation of the disulfide linkage by I_2/AcOH/H_2O treatment. The protected phosphopeptide segment was obtained by "global" phosphorylation of the Ser-peptide-HMPB-resin **31** using the $(tBuO)_2PNEt_2$/tBuOOH combination followed by 1% TFA cleavage of the acid-labile HMPB-linkage (Scheme 25). In this instance, the acid-mediated decomposition of the di-*tert*-butyl phosphite triester to its H-phosphonate was fortuitously minimized by a short 30 minute phosphitylation treatment and the use of "dilute" reaction conditions [e.g., 0.469 M of 1*H*-tetrazole (17.5 mg in a total solvent volume of 533 mL)]. However, it should be noted that formation of the H-phosphonate byproduct is inconsequential in this particular synthesis since the final I_2/AcOH treatment also effects oxidative conversion of the H-phos-

for references see p 421

phonate group to its phosphate form. Notwithstanding the obvious advantages of the segment approach over the steric problems associated with a stepwise approach, this strategy also simplifies the final separation of the target phosphopeptide from the parent segment peptides and other non-phosphorylated or deletion peptides.

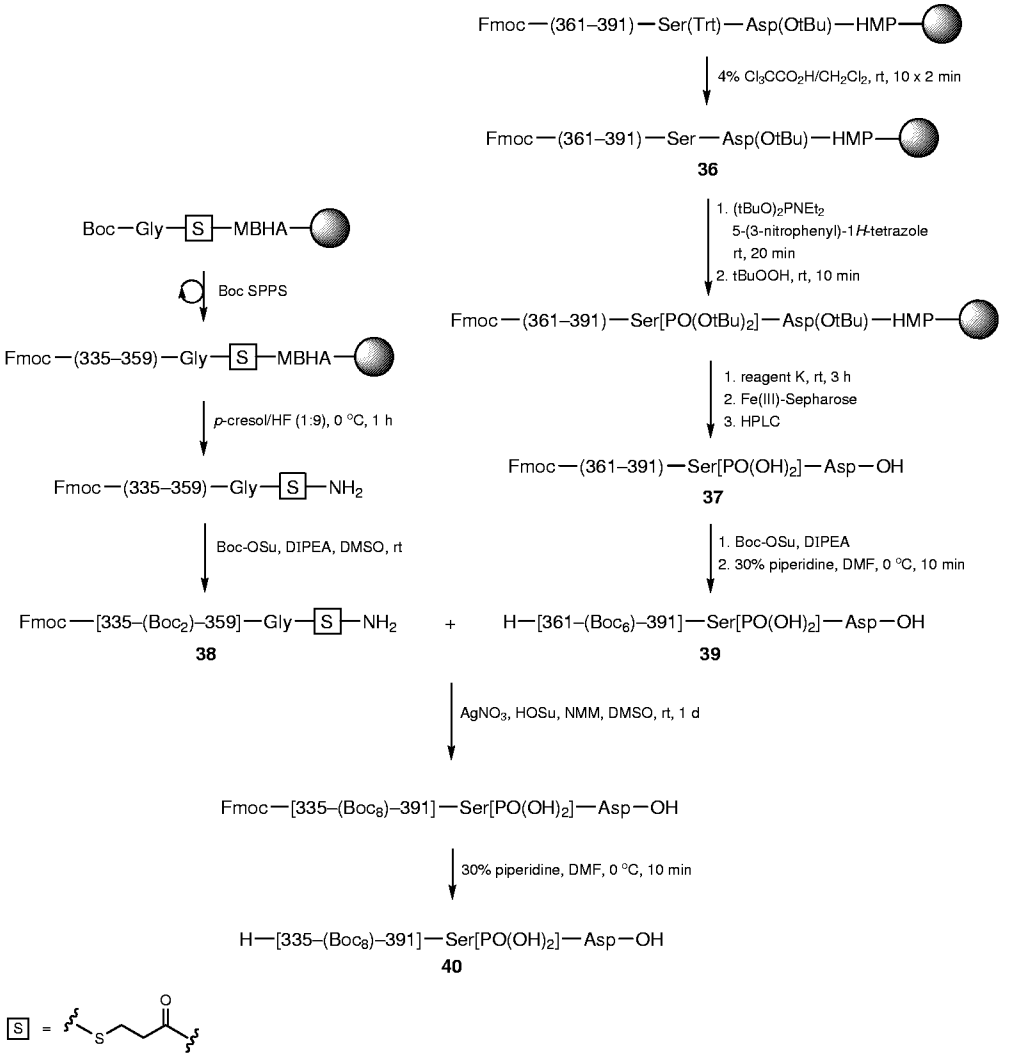
Scheme 25 Synthesis of Ser(*P*)-Urodilatin via "Post-Assembly" Phosphorylation[116]



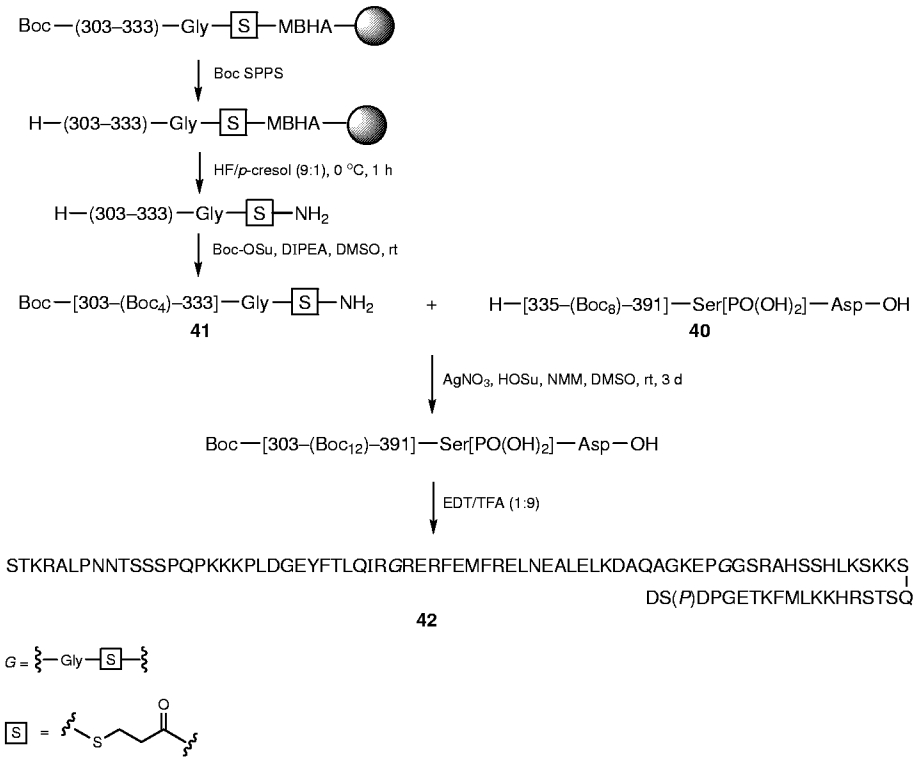
With respect to the future synthesis of biochemically relevant phosphopeptides, the "global" phosphorylation procedure provides an important synthetic tool for the synthesis of phosphorylated peptides for use in peptide assembly via the thioester ligation of peptide segments. The viability of this strategy was elegantly demonstrated by the synthesis of a 91-mer Ser(*P*)-peptide **42** corresponding to human p53 tumor suppressing protein(303–393) using a Ser(*P*)-containing C-terminal segment.[119] The synthesis of the phosphorylated segment was accomplished by "global" phosphorylation of Ser-peptide-resin **36** which was assembled using Fmoc-Asp(OtBu)-HMP-resin with the target Ser-residue incorporated as Fmoc-Ser(Trt)-OH and the Trt group selectively cleaved by repeated 4% trichloroacetic acid/

CH_2Cl_2 treatments. The Ser-peptide-resin was phosphorylated using the $(tBuO)_2PNEt_2$/tBuOOH combination and employed 5-(3-nitrophenyl)-1*H*-tetrazole for activation of the phosphoramidite. The phosphorylated peptide-resin was deprotected by treatment with reagent K and the Fmoc-protected Ser(*P*)-peptide **37** converted into a form suitable for segment coupling by reprotection of the side-chain lysine amino groups with Boc-OSu and final piperidine cleavage of the N-terminal Fmoc group. The Ser(*P*)-peptide was assembled by the segment condensation (HOSu/$AgNO_3$/NMM) of peptide thioester (335–360) **38** with Ser(*P*)-peptide segment (361–393) **39** and subsequent condensation of peptide thioester (303–334) **41** with Ser(*P*)-segment (335–393) **40** (Scheme 26). The only drawback of this synthesis was that the target Ser(*P*)-peptide was contaminated with 10% of the H-phosphonate-peptide, which originated from the "global" phosphorylation of Ser-peptide-resin **36**.

Scheme 26 Synthesis of a Ser(*P*)-Peptide via Fragment Condensation[119]



for references see p 421



The inherent acid sensitivity of the *tert*-butyl phosphite group necessitates that the "global" phosphorylation of peptide-resins is performed using a suitable phosphitylation/oxidation combination so as to remedy H-phosphonate formation. In the case where protected Tyr-[$PO(OtBu)_2$]- or Ser[$PO(OtBu)_2$]-peptide fragments are required for a segment condensation strategy, the "global" phosphorylation is best performed under "dilute" $(tBuO)_2PNEt_2$/1*H*-tetrazole phosphitylation conditions followed by tBuOOH oxidation. The use of 5-(3-nitrophenyl)-1*H*-tetrazole or *N*-methylanilinium trifluoroacetate is not recommended in these cases as the higher acidity of these acids promotes *tert*-butyl phosphite cleavage during the phosphitylation step.

In general, the "global" phosphorylation of peptide-resins using $(tBuO)_2PNEt_2$/1*H*-tetrazole [or 5-(3-nitrophenyl)-1*H*-tetrazole or *N*-methylanilinium trifluoroacetate] requires the use of aqueous iodine for the subsequent oxidation step so as to effect conversion of the generated H-phosphonate into the desired phosphate form. Alternatively, the greater acid stability of the benzyl phosphite group permits "global" phosphorylations using $(BzlO)_2P$-NEt_2/1*H*-tetrazole to be performed using either tBuOOH, H_2O/I_2 [Xaa $\neq$ Trp] or MCPBA [Xaa $\neq$ Cys, Met, Trp] oxidation procedures with the oxidation of Cys-, Met- and Trp-containing peptide sequences being kept to a minimum in order to avoid undesired oxidation of these residues. In consideration of the humble origins of the "global" phosphorylation procedure for the synthesis of H-Glu-Ser(*P*)-Leu-OH in 1986, the successful syntheses of both Ser(*P*)-urodilatin and the Ser(*P*)-human p53 peptide in 1996 demonstrates that this procedure will play a key role within a segment condensation strategy for the efficient preparation of numerous biochemically relevant phosphorylated peptides.

Di-*tert*-butyl Diethylphosphoramidite; Typical Procedure:[76,134]

CAUTION: *Diorganyl dialkylphosphoramidites should not be distilled at high temperature due to their propensity to decomposition. Skin contact should also be avoided due to the unknown pharmacology of these reagents.*

A soln of anhyd tBuOH (0.74 g, 0.10 mmol) and TEA (12.12 g, 0.12 mmol) in dry Et_2O (35 mL) was added over 10 min to a stirred soln of Cl_2PNEt_2 (Aldrich Chem Co.) (0.05 mmol) in dry Et_2O (15 mL) at 0 °C (ice/salt bath). On completion of addition, the soln temperature was allowed to rise to rt and stirred for 3 h. A soln of 5% $NaHCO_3$ (25 mL) was added and the aqueous phase was discarded. The organic layer was washed with 5% $NaHCO_3$ (2 × 60 mL) and sat. NaCl (60 mL), dried (Na_2SO_4), filtered, and the solvent removed under reduced pressure. Distillation of the crude light yellow residue gave $(tBuO)_2PNEt_2$ as a clear liquid; yield: 7.84 g (63%); bp 54–58 °C/ 1.0 Torr; ^{31}P NMR ($CDCl_3$, δ): 134.3.

Dibenzyl Diethylphosphoramidite:[76]
This reagent was prepared as described above using BzlOH; yield: 11.57 g (73%); bp 131–132 °C/ 0.0001 Torr; bp 150–152 °C/0.01 Torr (**CAUTION:** *extensive disproportionation of this reagent occurs with the use of higher distillation temperatures*); ^{31}P NMR ($CDCl_3$, δ): 148.6. Alternatively, this reagent is also purified by chromatography [silica gel, CH_2Cl_2/petroleum ether (bp 40–60 °C)/TEA 80:15:5].[124]

Bis[2-(trimethylsiloxy)ethyl] Diethylphosphoramidite:[36]
This reagent was prepared as described above using TMSE-OH and purified by chromatography (silica gel, hexane/EtOAc/TEA 90:10:4); yield: 81%; ^{31}P NMR ($CDCl_3$, δ): 145.1 (**CAUTION:** *extensive disproportionation of this reagent occurs during distillation*[21]).

Phosphorylation of Peptide-Resins (Xaa ≠ Trp, Cys, Met) Using $(tBuO)_2PNEt_2$, $(BzlO)_2PNEt_2$, or $(TMSEO)_2PNEt_2$; General Procedure:[94,124,135]

CAUTION: *Diorganyl dialkylphosphoramidites should not be distilled at high temperature due to their propensity to decomposition. Skin contact should also be avoided due to the unknown pharmacology of these reagents.*

Dry 1*H*-tetrazole (0.210 g, 3.0 mmol) was added to a suspension of the vacuum-dried peptide-resin (0.075 mmol) and $(tBuO)_2PNEt_2$, $(BzlO)_2PNEt_2$, or $(TMSEO)_2PNEt_2$ (0.75 mmol, 10 equiv) in a minimum volume of anhyd DMF (0.3 to 1.0 mL depending on peptide-resin loading and assembled peptide length) and the vessel rotated for 3–24 h under N_2. The soln was removed by vacuum suction, the resin washed with CH_2Cl_2 (3 × 3 mL) and treated with a soln of I_2 (0.095 g, 0.375 mmol) in THF/pyridine/H_2O (7:2:1, 3 mL) for 30 min. The solvent was removed by vacuum suction, the peptide-resin washed with DMF (2 × 3 mL) and CH_2Cl_2 (5 × 3 mL), and dried under high vacuum. The peptide-resin was treated with the appropriate TFA/ scavenger soln for the recommended period[26] and the solvent removed under reduced pressure. The phosphopeptide was precipitated by the addition of Et_2O (10 mL) and purified by semipreparative RP-HPLC.

Phosphorylation of Peptide-Resins Containing Trp, Cys, or Met Residues Using $(BzlO)_2PNEt_2$; General Procedure:[94,124,135]

CAUTION: *Diorganyl dialkylphosphoramidites should not be distilled at high temperature due to their propensity to decomposition. Skin contact should also be avoided due to the unknown pharmacology of these reagents.*

Dry 1*H*-tetrazole (0.210 g, 3.0 mmol) was added to a suspension of the vacuum-dried peptide-resin (0.075 mmol) and $(BzlO)_2PNEt_2$ (0.236 g, 0.75 mmol, 10 equiv) in anhyd DMF (0.3 to 1.0 mL depending on peptide-resin loading and assembled peptide length) and the vessel rotated for 3–24 h. The soln was removed by vacuum suction, the resin washed with CH_2Cl_2 (3 × 3 mL) and treated with a soln of 14% tBuOOH in DMF (3 mL) for 1 h. The solvent was removed by vacuum suction, the peptide-resin washed with DMF (2 × 3 mL) and CH_2Cl_2 (5 × 3 mL), and dried under high vacuum. The peptide-resin was treated with the appropriate TFA/scavenger soln for the recommended period[26] and the solvent removed under reduced pressure. The phosphopeptide was precipitated by the addition of Et_2O (10 mL) and purified by semipreparative RP-HPLC.

6.5.7.2 Synthesis of Phosphotyrosine-Containing Peptides via "On-Line" Phosphorylation

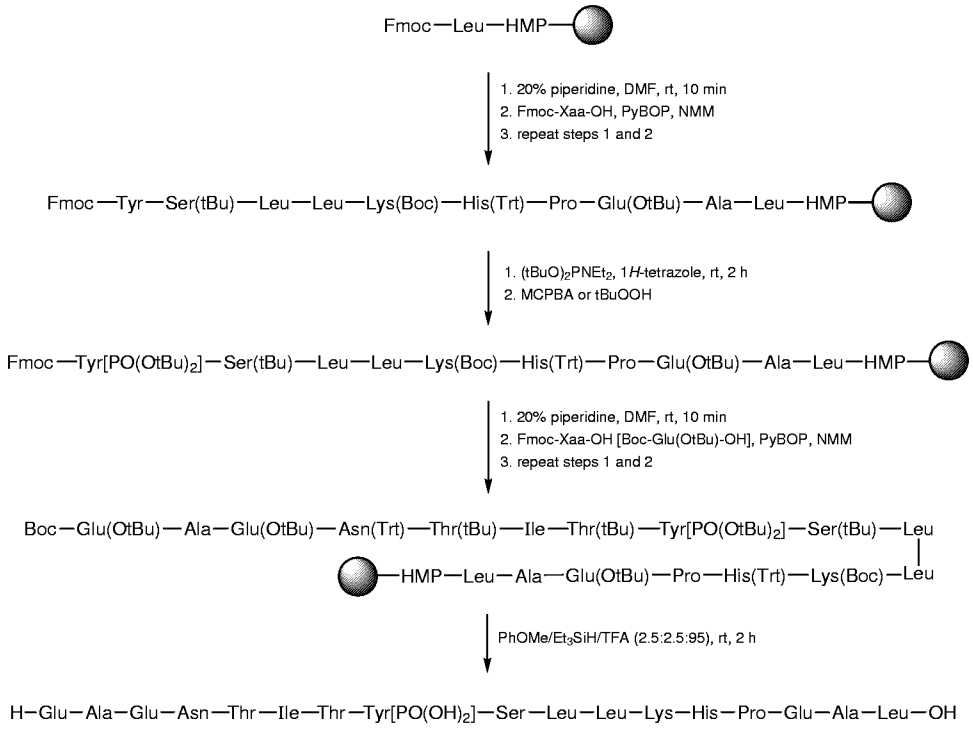
The synthesis of Tyr(*P*)-containing peptides via "on-line" phosphorylation is effected by immediate phosphitylation of the Tyr-residue using $(tBuO)_2PNEt_2$ or $(BzlO)_2PNEt_2$ after incorporation of Fmoc-Tyr-OH to the peptide-resin. This approach is particularly suited for

for references see p 421

the preparation of sterically hindered peptide sequences in which the target Tyr residue is situated near the N-terminus of the peptide sequence such that the phosphorylation of the Fmoc-Tyr-peptide-resin avoids the potential steric constraints imposed by a bulky N-terminal segment. This method also finds particular application in the synthesis of Trp-containing peptides in which the Trp residue is located upstream from the target Tyr site such that phosphitylation/oxidation of the Tyr residue is performed prior to the introduction of the oxidation sensitive Trp residue.

The "on-line" phosphorylation of a Fmoc-Tyr-peptide-resin using $(tBuO)_2PNEt_2$ is performed under "dilute" phosphitylation conditions and in conjunction with MCPBA or tBuOOH oxidation. A feature of this approach is that repeated phosphorylations can be performed to ensure complete formation of the Fmoc-Tyr[PO(OtBu)$_2$]-peptide-resin prior to continuation of peptide assembly. Unlike the benzyl or 2-(trimethylsilyl)ethyl phosphorotriester groups, the *tert*-butyl phosphorotriester group is inert to piperidine-mediated ester hydrolysis during repetitive cleavage of the Fmoc group. The utility of this approach has been demonstrated[136,137] by the syntheses of the Fcγ-receptor peptide EAENTITY(*P*)SLLKHPEAL (**43**) (Scheme 27) and the Stat 91 peptide GPKGTGY(*P*)IKTELIS. After acidolytic deprotection, the level of H-phosphonate contamination, if any, is best determined by MALDI or ES mass spectrometry {the Tyr[PO(OH)H]-peptide generally co-elutes with the target Tyr(*P*)-peptide on RP-HPLC}. In the case of peptides not containing Trp, Cys, or Met residues, remedial oxidation of the H-phosphonate byproduct is simply effected by aqueous iodine treatment of the fully assembled peptide-resin prior to acidolytic deprotection.

Scheme 27 Synthesis of a Tyr(P)-Peptide via "On-Line" Peptide Phosphorylation[136]



The "on-line" phosphorylation of Fmoc-Tyr-peptide-resins can also be performed using $(BzlO)_2PNEt_2$ in conjunction with MCPBA or tBuOOH oxidation. Although the dibenzyl

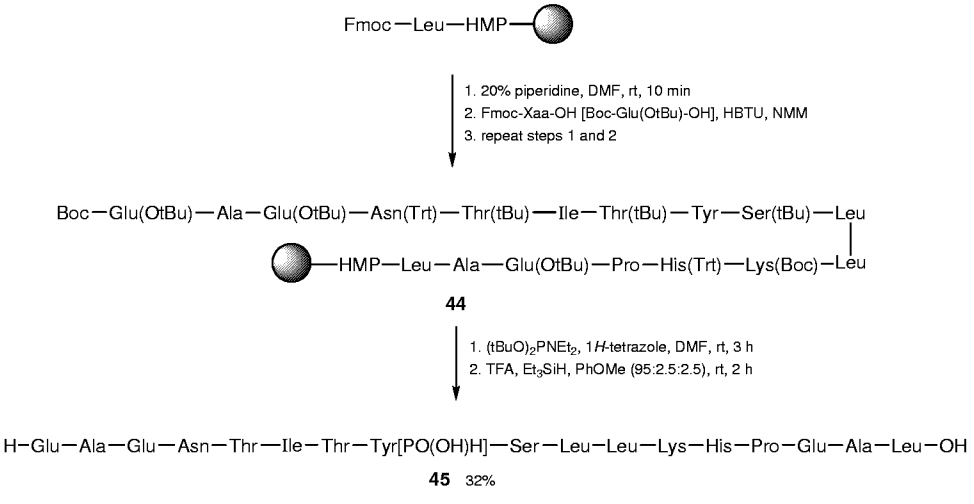
phosphorotriester functionality undergoes slow piperidine-mediated debenzylation during peptide assembly, the generated phosphorodiester does not compromise peptide assembly providing that phosphonium or uronium coupling reagents are used. While 1 M DBU/DMF[19] has been employed as an alternative Fmoc-cleavage procedure to minimize debenzylation, the use of this nonnucleophilic base can be problematic in that its use has been reported to promote aspartimide rearrangement.

H-Glu-Ala-Glu-Asn-Thr-Ile-Thr-Tyr(*P*)-Ser-Leu-Leu-Lys-His-Pro-Glu-Ala-Leu-OH (43):[136]
Dry 1*H*-tetrazole (0.316 g, 4.5 mmol) was added to a soln of the protected Fmoc-Tyr-peptide-Wang-resin (0.075 mmol) and di-*tert*-butyl diethylphosphoramidite (0.375 g, 1.50 mmol) in anhyd DMF (1.0 mL) under N_2 and the vessel rotated for 3 h. The soln was removed by vacuum suction, the resin washed with CH_2Cl_2 (3 × 3 mL) under N_2 and treated with a soln of 80% MCPBA (0.152 g, 0.75 mmol) in CH_2Cl_2 (3 mL) for 60 min. The solvent was removed by vacuum suction, the peptide-resin washed extensively with CH_2Cl_2 (5 × 3 mL), and dried. The Fmoc group was removed by treatment with 20% piperidine/DMF and peptide assembly continued. On completion of peptide assembly, the peptide-resin was treated with I_2 (0.095 g, 0.375 mmol) in H_2O/pyridine/THF (1:2:7, 3 mL) for 60 min (not recommended for peptide sequences containing Trp, Cys, or Met). The soln was removed by suction filtration and the resin washed with 5% aq $Na_2S_2O_5$/DMF (2:1, 2 × 2 mL), H_2O (2 × 2 mL), DMF (2 × 2 mL), and CH_2Cl_2 (2 × 2 mL). The peptide-resin was treated with PhOMe/Et_3SiH/TFA (2.5:2.5:95, 2 mL), the solvent removed under reduced pressure and the peptide precipitated by the addition of Et_2O (10 mL) to give a white solid (48 mg, expect 66 mg). The peptide (10 mg) was purified by RP-HPLC (0.1% TFA, 0–40% MeCN over 40 min, flow rate 3.0 mL·min^{-1}) and gave the title peptide as a white fluffy powder; yield: 6.1 mg (44% based on resin).

6.5.7.3 Synthesis of H-Phosphonate Peptides

The post-assembly (global) phosphorylation approach also provides a general and simple method for the synthesis of Tyr[PO(OH)H]-, Ser[PO(OH)H]-, and Thr[PO(OH)H]-peptide isosteres. This procedure was employed for the synthesis of the 17-mer EAENTITY-[PO(OH)H]SLLKHPEAL (**45**) in which the phosphitylation reaction was performed as per the "global" phosphitylation of a Tyr-peptide-resin **44** using $(tBuO)_2PNEt_2$/1*H*-tetrazole and was followed by immediate TFA treatment of the phosphitylated peptide-resin (Scheme 28).[21] Since the H-phosphonate group of Tyr[PO(OH)H]-peptides is labile and undergoes complete hydrolysis under acidic aqueous conditions, the HPLC purification of Tyr-[PO(OH)H]-peptides is best performed using the basic triethylammonium acetate solvent system.

Scheme 28 Synthesis of a Tyr[PO(OH)H]-Peptide[21]



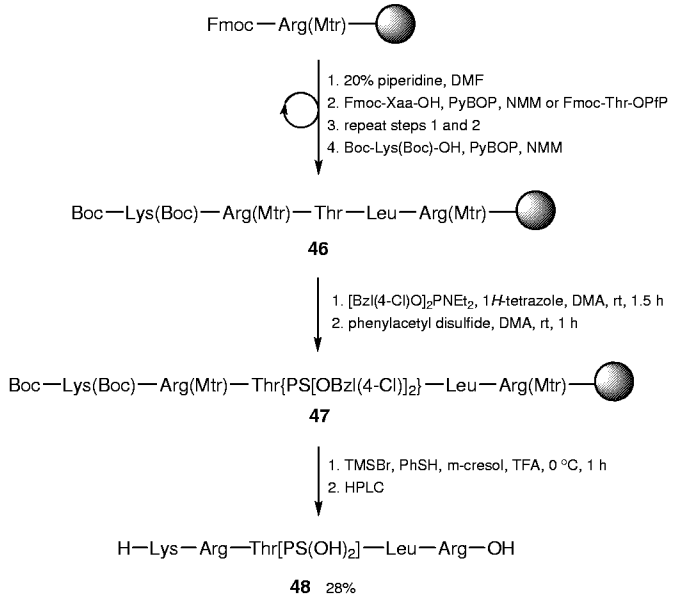
for references see p 421

H-Glu-Ala-Glu-Asn-Thr-Ile-Thr-Tyr[PO(OH)H]-Ser-Leu-Leu-Lys-His-Pro-Glu-Ala-Leu-OH (45):[21] Dry 1*H*-tetrazole (0.131 g, 1.875 mmol) was added to a suspension of Tyr-peptide-Wang-resin **44** (0.075 mmol) and $(tBuO)_2PNEt_2$ (0.187 g, 0.750 mmol) in anhyd DMF (0.5 mL) and the sintered reaction vessel rotated for 3 h under N_2. The soln was removed under positive N_2 pressure, the resin washed with CH_2Cl_2 (3 × 3 mL) and treated with TFA/Et_3SiH/PhOMe (95:2.5:2.5, 2 mL) for 2 h. The soln was collected, the solvent removed under reduced pressure and the peptide precipitated by the addition of Et_2O (10 mL). The peptide was purified by RP-HPLC [0.1 M TEA (adjusted to pH 7.5 with AcOH), 0–40% MeCN over 40 min, flow rate 2.0 mL·min^{-1}] and gave the product as a white powder; yield: 41%; MALDI-MS *m/z*: 1992.5 [M+H].

6.5.7.4 Synthesis of Phosphorothioate Peptides

The "post-assembly" (global) phosphorylation approach also provides a general and simple route to the preparation of thiophosphorylated peptides. The synthesis of KRT[PS(OH)$_2$]LR (**48**) was demonstrated by the phosphitylation of Thr-peptide-resin **46** using [Bzl(4-Cl)O]$_2$PNEt$_2$ followed by phenylacetyl disulfide thionation (or tetraethylthiuram disulfide thionation) to generate the phosphothioate functionality. The deprotection of the bis(4-chlorobenzyl) phosphorothionate peptide-resin **47** was effected by TMSBr/TFA/PhSH (10:80:10) treatment followed by HPLC separation of the target Thr[PS(OH)$_2$]-peptide (**48**) from the *S*-benzyl phosphorothiolate byproduct (Scheme 29).[138] In the case of Tyr[PS(OH)$_2$]-peptides, HPLC purification is performed using triethylammonium acetate buffer since the Tyr[PS(OH)$_2$] residue is unstable in aqueous TFA solution and decomposes to the Tyr-peptide.[139] While TFA/PhSH has been reported to successfully effect the acidolytic deprotection of dibenzyl phosphorothionates,[139] other studies[21] have shown that the TFA/PhSH deprotection generates a high proportion of the *S*-benzyl phosphorothiolate peptide along with the formation of substantial decomposition products.

Scheme 29 Synthesis of a Thr[PS(OH)$_2$]-Peptide via "Post-Assembly" Phosphitylation and Thionation[138]



The synthesis of thionophosphorylated peptides has also been reported using $(tBuO)_2PNEt_2$ followed by dibenzoyl tetrasulfide[140] thionation. The use of elemental sulfur is not practical due to its low solubility and the use of tetraethylthiuram disulfide is precluded since the use of this reagent does not generate the di-*tert*-butyl phosphorothionate.[21] The viability of the

above approach was demonstrated by the synthesis of the EGFR-peptide AE-NAEY[PS(OH)$_2$]LRVAPQS-NH$_2$ with peptide assembly being effected by Fmoc continuous SPPS using TentaGel S RAM Fmoc resin and final deprotection of the thiophosphorylated peptide-resin using anhydrous TFA. However, the procedure is not straightforward since the *S-tert*-butyl phosphorothiolate byproduct is formed from *tert*-butylation of the phosphorothionate.

Lys-Arg-Thr[PS(OH)$_2$]-Leu-Arg-OH (48):[138]
Dry 1*H*-tetrazole (0.133 g, 1.90 mmol) was added to a suspension of Boc-Lys(Boc)-Arg(Mtr)-Thr-Leu-Arg(Mtr)-HMP-kieselguhr-polydimethylacrylamide-resin (0.52 g, 38 μmol) and [Bzl(4-Cl)O]$_2$PN-iPr$_2$ (0.16 g, 0.38 mmol) in anhyd DMA (3.5 mL) and the vessel slowly agitated for 1.5 h under argon. The solvent was filtered, the resin washed with DMA (2 × 2.5 mL) and treated with phenylacetyl disulfide (0.230 g, 0.76 mmol) in DMA (2 mL) for 1 h. The soln was removed, the peptide-resin washed with DMA (3 × 2.5 mL), 2-methylbutan-2-ol (3 × 2.5 mL), and Et$_2$O (3 × 2.5 mL) and dried (P_2O_5 in vacuo). A cooled 0 °C soln of TMSBr (0.62 mL, 4.79 mmol), PhSH (0.56 mL, 5.3 mmol), *m*-cresol (0.56 mL, 5.3 mmol), in TFA (3.6 mL) was added to the peptide-resin and the soln slowly agitated at 0 °C for 1 h. The solvent was filtered into a rapidly stirred soln of hexane (160 mL) and Et$_2$O (40 mL), stirred for 1 h, and the soln centrifuged at 3000 rpm for 20 min. The pellet was collected, dissolved in H_2O and purified using Sephadex LH-20 (MeOH/H_2O 85:15) to give the product as a white powder; yield: 30.5 mg (100%); ^{31}P NMR (D_2O, δ): 43.7; MS *m/z*: 769.3875 [M+H].

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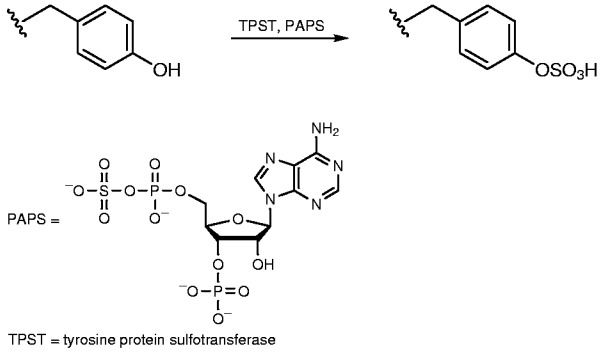
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6.6 Synthesis of Sulfated Peptides

H.-J. Musiol, A. Escherich, and L. Moroder

Tyrosine O-sulfation is an ubiquitous posttranslational modification of secretory, plasma membrane, and lysosomal proteins that occurs in all multicellular organisms.[1–3,139] Its functional importance is in part reflected by the fact that as much as 1% of the total protein of an organism may be tyrosine O-sulfated.[4] Tyrosine phosphorylation in proteins is quite uncommon in normal cells compared to phosphorylation of serine and threonine residues. This phosphorylation of tyrosine increases dramatically upon oncogenic transformation or stimulation by various growth factors.[5] Tyrosine sulfation in proteins, on the other hand, appears to be part of the normal cellular physiology. Apparently sulfation occurs exclusively on tyrosine residues,[6,7] although there are indications of some serine residue sulfation.[6] Sulfation is catalyzed by the tyrosine protein sulfotransferase, a *trans*-membrane protein residing in the *trans*-Golgi network, that uses 3′-phosphoadenosine-5′-phosphosulfate as the sulfate donor (Scheme 1).[8] A major breakthrough in the study of tyrosine sulfation has occurred with the cloning of tyrosine protein sulfotransferase-1 and -2 from both mouse and human[2,140,141] and its discovery even in plants.[142]

Scheme 1 Enzymatic Sulfation of Tyrosine Residues in Proteins[8]



The general biological effect of this modification appears to be enhancement of protein–protein interactions and thus modulation of biological activities, but also reduced protein degradation rates as well as intracellular protein targeting and transport possibly mediated by a receptor, the tyrosine *O*-sulfate [Tyr(SO_3H)] binding protein.[9] Reports describing tyrosine O-sulfation in P-selectin glycoprotein ligand (PSGL)-1[143–145] and the chemokinin receptors CCR2, CCR4, and CCR5[146,147] have dramatically increased interest in this modification particularly because CCR5 acts as a coreceptor for HIV infection.[148]

Consensus features were identified from comparison of sequence homologies of known sulfation sites[1,10,139] and from in vitro sulfation of synthetic peptides using TPST-enriched membrane preparations[11–13] as well as from analysis of mutations.[14] These structural features mainly consist of a preferential type of charge distribution around the sulfation site, i.e. acidic residues on the amino-terminal side of the tyrosine, particularly in position 1. Full accessibility of the site to the TPST in surface-exposed loops and the absence of disulfide bridges in the proximity are also important for sulfation.

for references see p 450

6.6.1 Sulfated Tyrosine Peptides

After the early discovery of a tyrosine *O*-sulfate residue in bovine fibrinopeptide B,[15] this posttranslational modification which occurs ubiquitously in proteins was also detected in a series of biologically active peptides such as the neurohormones of the gastrin/cholecystokinin (CCK) family of peptides, phyllokinin, Leu-enkephalin, and the thrombin inhibitor hirudin listed in Table 1.

Table 1 Amino Acid Sequences of Known Naturally Occurring Bioactive Peptides Sulfated at Tyrosine Residues[16–30]

Peptide	Amino Acid Sequence	Ref
big gastrin II, HG-34-II	<Glu-Leu-Gly-Pro-Gln-Gly-Pro-Pro-His-Leu-Val-Ala-Asp-Pro-Ser-Lys-Lys-Gln-Gly-Pro-Trp-Leu-Glu-Glu-Glu-Glu-Glu-Ala-Tyr(SO_3H)-Gly-Trp-Met-Asp-Phe-NH_2	[16]
gastrin II, HG-17-II	<Glu-Gly-Pro-Trp-Leu-Leu-Glu-Glu-Glu-Glu-Glu-Ala-Tyr(SO_3H)-Gly-Trp-Met-Asp-Phe-NH_2	[17]
minigastrin II	H-Trp-Leu-Leu-Glu-Glu-Glu-Glu-Glu-Ala-Tyr(SO_3H)-Gly-Trp-Met-Asp-Phe-NH_2	[18]
porcine cholecystokinin, CCK-33	H-Lys-Ala-Pro-Ser-Gly-Arg-Val-Ser-Met-Ile-Lys-Asn-Leu-Gln-Ser-Leu-Asp-Pro-Ser-His-Arg-Ile-Ser-Asp-Arg-Asp-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp-Phe-NH_2	[19]
human cholecystokinin, CCK-33	H-Lys-Ala-Pro-Ser-Gly-Arg-Met-Ser-Val-Lys-Asn-Leu-Gln-Asn-Leu-Asp-Pro-Ser-His-Arg-Ile-Ser-Asp-Arg-Asp-Tyr-(SO_3H)-Met-Gly-Trp-Met-Asp-Phe-NH_2	[20]
CCK-8	H-Asp-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp-Phe-NH_2	[21]
phyllocaerulein	<Glu-Glu-Asp-Tyr(SO_3H)-Thr-Gly-Trp-Met-Asp-Phe-NH_2	[22]
caerulein	<Glu-Gln-Asp-Tyr(SO_3H)-Thr-Gly-Trp-Met-Asp-Phe-NH_2	[23]
leucosulfakinin I, LSK-I	H-Glu-Gln-Phe-Glu-Asp-Tyr(SO_3H)-Gly-His-Met-Arg-Phe-NH_2	[24]
leucosulfakinin II, LSK-II	<Glu-Ser-Asp-Asp-Tyr(SO_3H)-Gly-His-Met-Arg-Phe-NH_2	[25]
drosulfakinin, DSK-I	H-Phe-Asp-Asp-Tyr(SO_3H)-Gly-His-Met-Arg-Phe-NH_2	[26]
cionin	H-Asn-Tyr(SO_3H)-Tyr(SO_3H)-Gly-Trp-Met-Asp-Phe-NH_2	[27]
sulfated Leu-enkephalin	H-Tyr(SO_3H)-Gly-Gly-Phe-Leu-OH	[28]
phyllokinin	H-Arg-Pro-Pro-Gly-Phe-Ser-Pro-Arg-Phe-Ile-Tyr(SO_3H)-OH	[29]
hirudin	H-Val-Val-Tyr-Thr-Asp-Cys-Thr-Glu-Ser-Gly-Gln-Asn-Leu-Cys-Leu-Cys-Glu-Gly-Ser-Asn-Val-Cys-Gly-Gln-Gly-Asn-Lys-Cys-Ile-Leu-Gly-Ser-Asp-Gly-Glu-Lys-Asn-Gln-Cys-Val-Thr-Gly-Glu-Gly-Thr-Pro-Lys-Pro-Gly-Ser-His-Asn-Asp-Gly-Asp-Phe-Glu-Glu-Ile-Pro-Glu-Glu-Tyr(SO_3H)-Leu-Gln-OH	[30]

It was observed that for CCK and related peptides sulfation of the tyrosine residue is essential for triggering the full hormonal response.[31,32] More recently, additional sulfated peptides have been identified as CCK-58,[135] phytosulfokinine-α, as peptide growth factor in plants[136] as well as various toxins from *Conus* venom.[137,138] This observation has fostered intensive research to develop synthetic methods for efficient production of tyrosine O-sulfated peptides by chemical and enzymatic approaches.

Stability of Tyrosine *O*-Sulfate

Tyrosine *O*-sulfate is stable under alkaline conditions, thus allowing for its quantification by amino acid analysis upon alkaline hydrolysis [0.2 M $Ba(OH)_2$, 110 °C, 24 h] of sulfated tyrosine peptides and proteins.[6,33] Conversely, more than 95% of the ester is hydrolyzed after five minutes in 1 M hydrogen chloride at 100 °C. Despite this pronounced acid lability, sulfated tyrosine peptides are sufficiently stable to short exposures of TFA[34,35] or aqueous TFA[36] as required in peptide synthesis for removal of acid-labile protecting groups.

Conversely, exposure to hydrogen fluoride at 0 °C over short periods of time leads to almost quantitative hydrolysis.[37–39] In aprotic organic solvents, decomposition takes place unless the sulfate hemi-ester is neutralized with strong counterions.[37,38] The results of a detailed study are summarized in Figures 1 and 2[40] which show the effect of various acids used in peptide chemistry in deprotection and in cleavage from the resin as well as the effect of temperature on the hydrolysis of tyrosine *O*-sulfate as a sodium salt.

Figure 1 Hydrolysis of Tyrosine *O*-Sulfate as a Sodium Salt[40]

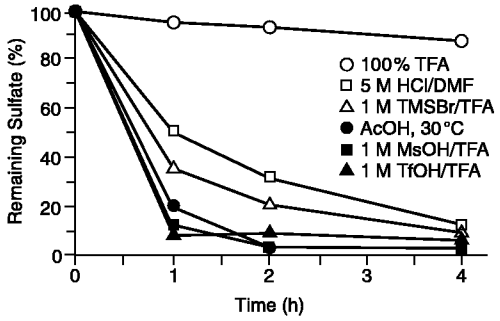
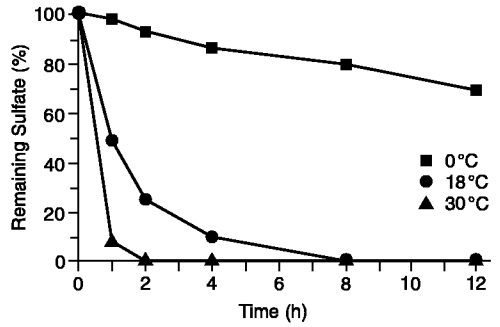


Figure 2 Effect of the Temperature on the Stability of Tyrosine *O*-Sulfate as a Sodium Salt in Trifluoroacetic Acid[40]

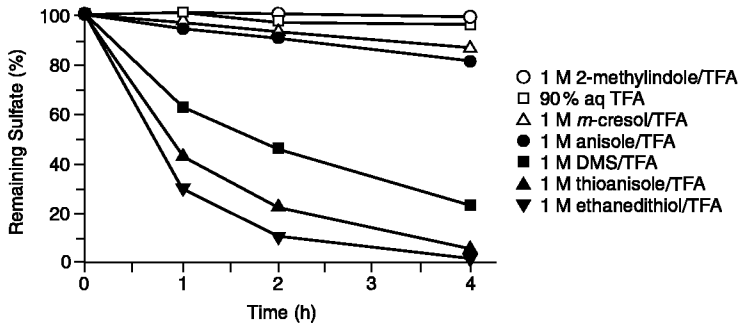


With TFA at a lower temperature, the results clearly indicate that the hydrolysis of the tyrosine *O*-sulfate ester is sufficiently low to be acceptable for preparative purposes.

Since scavengers are usually required in deprotection steps to prevent alkylations as serious side reactions, the effect of various additives is shown in Figure 3.[40]

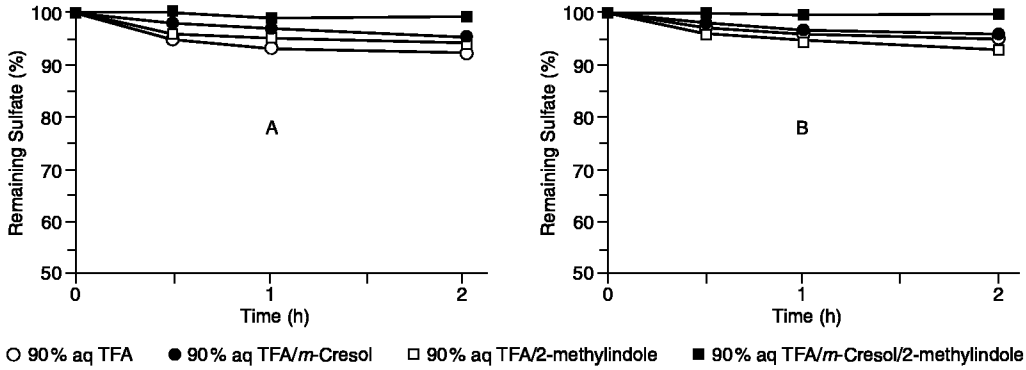
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Figure 3 Hydrolysis of Tyrosine *O*-Sulfate as a Sodium Salt at 4 °C by Exposure to Various Scavengers[40]



Among the scavengers generally applied in peptide chemistry, highly efficient thiols must be avoided since rates of hydrolysis of tyrosine *O*-sulfate residues are significantly enhanced (Figure 3) as previously reported in the case of partially protected CCK-peptides.[41] Conversely, by using 2-methylindole in 90% aqueous TFA to protect tryptophan residues from alkylation, hydrolysis is significantly reduced, as shown in Figure 4 for leucosulfakinin-II and CCK-12[40] and as reported for sulfated CCK-peptides[36,41,42] and gastrins.[43]

Figure 4 Desulfation Rates of Leucosulfakinin-II (A) and CCK-12 (B) in Acidic Media at 4 °C[40a]



[a] *m*-Cresol (50 equiv) and 2-methylindole (25 equiv) are added as scavengers.

The sulfate hemi-ester must be converted into a salt prior to the deprotection steps with TFA in order to avoid a partial O→C shift with formation of the tyrosine-3-sulfonic acid derivatives.[38] This side reaction must have occurred in the first syntheses of CCK-8.[34,35] In this context it was found that salts of the sulfate ester with barium, sodium, potassium, or ammonium as counterions are less efficient in preventing hydrolysis than intramolecular salt-bridging to proximal arginine guanidino groups. In fact, hydrolysis of the barium salt of tyrosine *O*-sulfate as a free amino acid[41] or incorporated into gastrin,[43] by 90% aqueous TFA/2-methylindole, occurs at rates similar to those reported in Figure 3 for the sodium salt. Conversely, no desulfated product was detected in the crude deprotection mixture[41] upon treatment under identical acid conditions of partially protected sulfated CCK-9 analogues containing an N-terminal arginine residue. The beneficial effect of the intramolecular salt bridge is further confirmed by the observation that in the case of CCK-10, where the molecule is N-terminally extended with an additional aspartic acid residue, partial hydrolysis of the sulfate ester takes place.[42] Even addition of excesses of guanidine proved to be useful in this context.[44] The intramolecular salt-bridge formation in CCK-peptides has been confirmed by the slow rate of enzymatic hydrolysis of the Arg^{25}-Asp^{26} bond in sulfated porcine

CCK-33 with trypsin[45] as well as by the negative-ion MALDI-TOF-MS spectra of CCK-12 that show intense $[M-H]^-$ peaks indicating a stabilized sulfate ester moiety even in the gas phase (see Analytical Characterization).[44,149]

As a consequence of the high stability of tyrosine *O*-sulfate under alkaline conditions, no side effects are noted in treatments of sulfated tyrosine peptides with bases in water or in organic solvents as required for removal of base-labile protecting groups, e.g. Ac,[47,48] Msc,[49] or Fmoc.[42,50,149]

Analytical Characterization

For the analytical characterization of sulfated tyrosine peptides, spectroscopic methods as well as amino acid analysis and, more recently, mass spectrometry are employed. In Table 2 the spectroscopic data of tyrosine *O*-sulfate are compared to those of the related sulfonic acid derivatives as possible byproducts in the chemical sulfation of the tyrosine or tyrosine peptides.[36] In the course of the synthesis of tyrosine *O*-sulfate peptides and, particularly in the final deprotection step, desulfation may occur which limits the characterization of the final compounds in terms of quantitative identification of the tyrosine *O*-sulfate. This is achieved by amino acid analyses of basic hydrolysates of the sulfated tyrosine peptides or preferably by analyses of the enzymatic hydrolysates with aminopeptidase M or leucine-aminopeptidase.

Tyrosine *O*-sulfate elutes on the amino acid analyzer (a citrate buffer of pH 3.25 is changed to a citrate buffer of pH 4.30 after 85 minutes) before aspartic acid, at a t_R value similar to cysteic acid.[51] Acidic treatment (6 M HCl, 110 °C, 24 h or 3 M 2-mercapto-ethanesulfonic acid, 110 °C, 22 h) leads to complete hydrolysis of the sulfate hemi-ester, whereas upon alkaline hydrolysis [0.2 M $Ba(OH)_2$, 110 °C, 24 h] Tyr(SO_3H) is recovered almost quantitatively[33] as assessed for synthetic peptides such as leucosulfakinin II[52] and CCK-peptides.[53] Nevertheless, the most efficient procedure is the enzymatic hydrolysis by aminopeptidase M or leucine aminopeptidase since it allows for detection of even traces of desulfated tyrosine in the crude deprotection products.[54] This enzymatic procedure establishes the integrity of the O-sulfated amino acid residues in the final products as demonstrated in the case of synthetic gastrin-II,[43] human[55] or porcine CCK-33,[56] and related fragments.[41,42,50,57]

Table 2 Spectroscopic Data of Tyrosine *O*-Sulfate, Tyrosine-3-sulfonic Acid, and Tyrosine-3,5-disulfonic Acid[36]

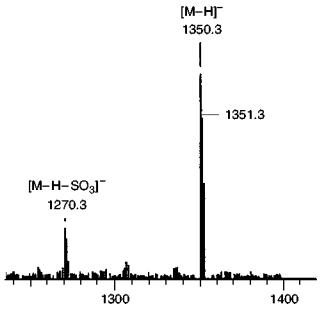
Compound	UV (0.01 M HCl)		UV (0.1 M NaOH)		IR (KBr): $\tilde{\nu}(SO_3)$ (cm^{-1})	^1H NMR (90 MHz): 2% ND_3/D_2O δ (ppm), *J* (Hz)
	λ_{max}(nm)	ε(10^3)	λ_{max}(nm)	ε(10^3)		
H-Tyr-OH	274	1.4	293	2.4	–	6.74 (d, 2H, 2,6H), 7.08 (d, 2H, 3,5H), $J_{2,3}=J_{5,6}=8.7$, $J_{2,6}=J_{3,5}=2$, $J_{2,5}=J_{3,6}=0.5$
H-Tyr(SO_3H)-OH	260.5	0.28	262	0.32	1050	7.27 (d, 2H, 3,5H), 7.31 (t, 2H, 2,6H), $J_{2,3}=J_{5,6}=8.5$, $J_{2,6}=J_{3,5}=1$, $J_{2,5}=J_{3,6}=0.5$
H-Tyr(3-SO_3H)-OH	283.5	2.5	310	4.2	1025	6.98 (d, 1H, 5H), 7.32 (dd, 1H, 6H), 7.59 (d, 1H, 2H), $J_{5,6}=8.5$, $J_{2,6}=2.3$
H-Tyr-[3,5-(SO_2H)$_2$]-OH	291	3.9	328.5	7.2	1035	7.69 (s, 2H, 2,6H)

More recently, mass spectrometry studies on the tyrosine *O*-sulfate containing caerulein, and CCK peptides[58,59,149,158] and on multiple tyrosine *O*-sulfate peptides,[60,61] clearly revealed

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that in the positive mode of FAB-MS the ester moiety is unstable yielding small but detectable $[M+H]^+$ ions with the $[M+H-SO_3]^+$ peaks approximately five to ten times higher in intensity. In the negative mode the FAB-MS spectra show $[M-H]^-$ ions approximately ten times higher in intensity than the $[M-SO_3-H]^-$ ions. A similar FAB-MS spectrum, shown in Figure 5, was obtained for synthetic caerulein.[62]

Figure 5 Negative Mode FAB-MS Spectrum of Caerulein Showing the Relative Intensities of the $[M-H]^-$ and $[M-H-SO_3]^-$ Ions[62]



Presently, FAB-MS spectra are routinely used to characterize synthetic tyrosine *O*-sulfate peptides.[52,57,63–67] Since partial hydrolysis of the sulfate ester occurs in the gas phase, quantification of the tyrosine *O*-sulfate residue by mass spectrometry is not possible, but combined with one-peak assignment in HPLC, FAB-MS represents a powerful analytical tool. On the other hand, partial hydrolysis in the gas phase excludes the presence of sulfonated species which should be perfectly stable. In early studies the presence of such species were excluded by quantitative recovery of tyrosine upon acid hydrolysis or upon hydrolysis with arylsulfatase.[36] Recently, even MALDI-TOF-MS spectra of CCK-peptides[44] and of conotoxins α-PnIA and α-PnIB[138] were reported which show that in the positive-ion mode the $[M+H-SO_3]^+$ ions represent the base peaks, while in the negative-ion mode, $[M-H]^-$ ions consistently correspond to the base peaks. In the CCK peptides intramolecular salt bridging of the sulfate hemi-ester with proximal positive charges of arginine or lysine side chains was found to reduce the extent of hydrolysis in the gas phase significantly.[44,149]

6.6.1.1 Chemical Sulfation of Tyrosine and Tyrosine Peptides

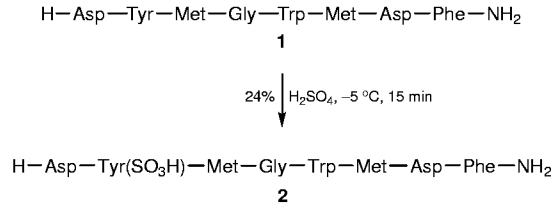
The first studies on the sulfation of organic compounds, amino acids, and proteins have shown that pyridine/sulfur trioxide complex (pyridine/SO_3 or pyridine/$ClSO_3H$),[68–72] concentrated sulfuric acid,[73,74] sulfuric acid/*N*,*N*′-dicyclohexylcarbodiimide,[75,76] and chlorosulfonic acid[77] are the most efficient reagents for the sulfation of tyrosine. More recently, alternative methods based on dimethylformamide/sulfur trioxide complex (DMF/SO_3),[52,78] trimethylamine/sulfur trioxide (Me_3N/SO_3),[152,153] pyridinium acetylsulfate,[37,79] and pyridinium trifluoroacetylsulfate[80] have been proposed to minimize side reactions which are difficult to control for the chemical sulfation of tyrosine peptides.

6.6.1.1.1 Sulfation with Concentrated Sulfuric Acid

In addition to quantitative sulfation of hydroxy and thiol groups, with concentrated sulfuric acid, extensive sulfonation of tyrosine occurs if the reaction is carried out at room temperature.[73] However at a lower temperature (–5 °C) and with shortened reaction times tyrosine *O*-sulfate (K^+ salt) is obtained, although in low yield.[74]

In the first syntheses of CCK-related peptides that lack serine, threonine, or hydroxyproline residues, sulfation of tyrosine residues was accomplished by treatment of the fully deprotected peptides with sulfuric acid. Under optimized conditions good separation between sulfated, sulfonated, and unreacted species (Scheme 2) was achieved using ion-exchange chromatography.[34,35] Sulfonation of the indole group, as observed in model studies,[73] is apparently not a dominant side reaction.

Scheme 2 Sulfation of the CCK-Octapeptide with Concentrated Sulfuric Acid after Acidolytic Removal of the Acid-Labile Protecting Groups to Yield CCK-8[34]



H-Tyr(SO_3K)-OH · $2H_2O$:[74]
The amino acid H-Tyr-OH (2 g, 11 mmol) was added in small portions over a period of 20 min with stirring to concd H_2SO_4 (11 mL) at −5 °C. Stirring was continued at −5 °C for an additional 30 min. The mixture was then decanted into crushed ice and H_2O (300 mL) containing $Ba(OH)_2$ (70 g). Precipitated $BaSO_4$ was removed by filtration, and excess Ba^{2+} ions were precipitated by addition of solid CO_2. After removal of $BaCO_3$ the soln was taken to dryness, the residue dissolved in H_2O (60 mL) and exchanged on a Dowex 50 (H^+ form) column. The eluate was adjusted to pH 5 with 5% aq KOH and then taken to dryness. The product crystallized as the dihydrate from H_2O (2 mL) by addition of EtOH (20 mL) and Et_2O (18 mL); yield: 0.4 g (11%).

H-Asp-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp-Phe-NH_2 (2):[34]
H-Asp-Tyr-Met-Gly-Trp-Met-Asp-Phe-NH_2 (**1**) as its trifluoroacetate (984 mg, 0.84 mmol) was added to precooled (−5 °C) concd H_2SO_4 (60 mL) with stirring. After 15 min the mixture was poured into precooled (−70 °C) Et_2O (850 mL) the precipitate was collected by centrifugation and washed several times with Et_2O. The residue was then dissolved in 0.1 M $(NH_4)_2CO_3$ (20 mL) and chromatographed on a DEAE-Sephadex A-25 column by elution with a linear gradient from 0.1 to 1.5 M $(NH_4)_2CO_3$. Three major fractions were collected corresponding to the unreacted peptide, the sulfonated derivative, and compound **2** which was isolated by repeated freeze-drying from H_2O; yield: 231 mg (24%); $[\alpha]_D^{23}$ −18.4 (*c* 0.7, 1 M NH_3); IR (KBr) $\tilde{\nu}_{max}$: 1050, 1250 cm^{-1}; UV (0.1 M NaOH) λ_{max} (ε): 280 (4850), 288 (4230).

6.6.1.1.2 Sulfation with Chlorosulfonic Acid

A detailed comparative study on sulfation of hydroxyamino acids, tyrosine and tryptophan by sulfuric acid and chlorosulfonic acid in TFA clearly revealed that the latter reagent had a significantly higher reactivity.[77] While the hydroxyamino acids are converted at room temperature into the corresponding *O*-sulfates (see Section 6.6.3), tyrosine and tryptophan are sulfonated yielding tyrosine-3-sulfonic acid and a bis-sulfonated tryptophan derivative of unknown structure. However, at low temperature (−20 to 0 °C), tyrosine *O*-sulfate is formed in high yields.[38,63] By increasing the temperature to above 0 °C a fast O→C shift occurs under anhydrous acidic conditions to produce exclusively the tyrosine-3-sulfonic acid derivative. Since tryptophan sulfonation can be prevented by its N^{in}-formylation[38] (see also Vol. E 22a, Section 2.6.3.1) the chlorosulfonic acid/TFA method has been applied for sulfation of the CCK-8 derivative H-Asp-Tyr-Met-Gly-Trp(For)-Met-Asp-Phe-NH_2 (5 h, −20 °C), and the related CCK-8 was obtained in 18% yield upon HPLC chromatography.[38]

H-Tyr(SO_3Na)-OH · $2H_2O$:[63]
To a precooled soln (−20 °C) of H-Tyr-OH (9.01 g, 0.05 mol) in TFA (50 mL), was added dropwise over 2 min $ClSO_3H$ (5 mL, 0.075 mol) whilst the temperature was maintained at −20 °C. After 5 min stirring,

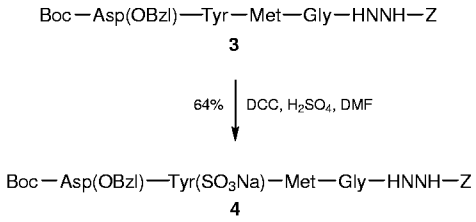
for references see p 450

the excess $ClSO_3H$ was reacted with EtOH (1.5 mL) and the product was precipitated with Et_2O, filtered and washed extensively with Et_2O. It was then dissolved in 1 M NaOH adjusting the pH of the soln to 5.5. The soln was lyophilized, the crude product was dissolved in H_2O (50 mL) and then EtOH (450 mL) and Et_2O (400 mL) were added. The product crystallized upon cooling as the dihydrate [H-Tyr(SO_3Na)-OH·$2H_2O$]; yield: 13.4 g (80%) as colorless crystals; mp 228–230 °C; $[\alpha]_D^{20}$ −19.1 (*c* 1, H_2O); IR (KBr) $\tilde{\nu}_{max}$: 1050 cm^{-1}.

6.6.1.1.3 Sulfation with Sulfuric Acid/Dicyclohexylcarbodiimide

After the sulfation of alcohols and phenols was reported to proceed efficiently with DCC/H_2SO_4,[75,76] Wieland et al.[81] applied this procedure for the sulfation of the tyrosine residue in D-Tyr^6-*all*-D-retro-antamanide. This contains no reactive group in addition to the phenolic group, and the desired product was obtained in satisfactory yields. Despite these promising initial results, this type of sulfation reaction has been adopted only sporadically, e.g. for sulfation of the CCK-8 derivative Boc-Asp(OtBu)-Tyr-Met-Gly-Trp(For)-Met-Asp-(OtBu)-Phe-NH_2 with 70% yield[82] or for the preparation of an intermediate fragment in the synthesis of CCK-related peptides (Scheme 3).[83]

Scheme 3 Sulfation of a Suitable Protected Tetrapeptide with Dicyclohexylcarbodiimide/Sulfuric Acid as an Intermediate Fragment in the Synthesis of CCK-8[83]



More recently sulfation with H_2SO_4/DCC has been proposed for the preparation of tyrosine *O*-[^{35}S]sulfated peptides, e.g. of CCK-8 in nanomolar amounts, since analytical HPLC or two-dimensional TLC can be applied for the separation of the sulfated peptide from the unsulfated starting material as well as from side products.[84] The low yields of sulfation reported (max 40% even upon optimization with excess DCC and H_2SO_4) most probably derive from the difficulties encountered by working under strictly anhydrous conditions with lyophilized peptide samples.

Boc-Asp(OBzl)-Tyr(SO_3Na)-Met-Gly-N_2H_2-Z (4):[83]
Boc-Asp(OBzl)-Tyr-Met-Gly-N_2H_2-Z (**3**; 1 g, 1.2 mmol) was dissolved in DMF (4 mL). The soln was cooled to 0 °C and DCC (2.5 g, 12 mmol) and H_2SO_4 (0.3 mL, 5.4 mmol) in DMF (3 mL) were added. After 4 min the soln was diluted with DMF (20 mL) and the pH adjusted to 7 with 1 M NaOH. The DCU was removed by filtration and the soln taken to dryness under reduced pressure and the residue was triturated with EtOAc. The solid was collected by filtration and chromatographed (silica gel); yield: 0.7 g (64%); mp 124–126 °C; $[\alpha]_D^{23}$ −9.3 (*c* 1, DMF).

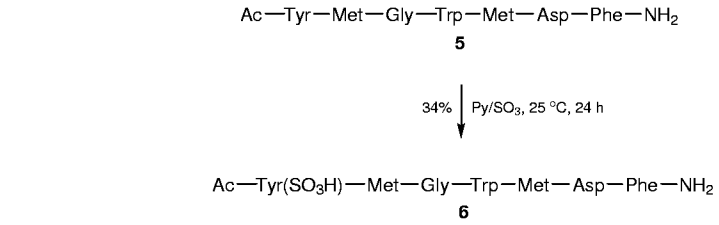
6.6.1.1.4 Sulfation with Pyridine/Sulfur Trioxide Complex

With pyridine/SO_3 even at higher temperatures, sulfation of the phenolic group is reportedly the predominant reaction rather than sulfonation.[70] This finding was confirmed by the high yields of sulfation of Z-Tyr-OH and by the analytical characterization of the tyrosine *O*-sulfate (Table 2) isolated upon its hydrogenolytic deprotection.[36] See also H-Tyr-($SO_3Ba_{1/2}$)-$OBa_{1/2}$ (*vide infra*).

With pyridine/SO_3 complex at higher temperatures (70–80 °C), in addition to sulfation of phenolic, hydroxy, and thiol groups, sulfation of amino, guanidino, and primary amido

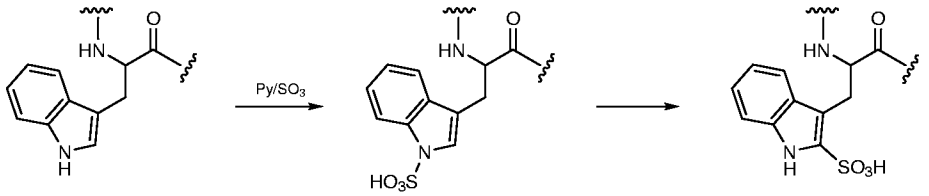
groups also occurs as well as partial modification of the indole and imidazole groups.[70] Correspondingly, for an application of this procedure to the sulfation of tyrosine peptides, suitably protected peptide derivatives must be used. Sulfation of partially protected phyllocaerulein,[47] caerulein and related peptides,[48,85,86] CCK peptides,[34,35,49,82,87,88] as well as GPR1 related peptides[150] leads to the desired compounds generally in low yields, even if the reaction is performed under mild conditions, e.g. at room temperature, in anhydrous solvents, and under a nitrogen atmosphere, as representatively shown in Scheme 4 for the N^{α}-acetyl-CCK-heptapeptide.[88] Significant improvement of these yields has been reported by use of a workup protocol involving a methanol quench and immediate silica gel chromatography.[151]

Scheme 4 Sulfation of the N^{α}-Protected CCK-Heptapeptide with Pyridine/Sulfur Trioxide Complex[88]



The low yield of this rather simple N^{α}-acetyl-CCK-7 peptide and the need for tedious purification steps can only be explained by serious side reactions which most probably involve the tryptophan and methionine residues. Treatment of gastrin in aqueous solution at pH 10.5 with pyridine/SO_3 leads only to partial sulfation of tyrosine and to the destruction of the methionine residue in ca. 30–50%.[89] Since homoserine is formed, a type of reaction similar to that of cyanogen bromide with methionine has been proposed, although cleavage of the peptide chain was not observed. The fact that methionine residues may be critically involved in side reactions even under anhydrous conditions is well supported by the significantly improved yield (74%) obtained in the sulfation of desamino-Tyr-Nle-Gly-Trp-Nle-Asp-Phe-NH_2 (a CCK analogue in which both methionine residues are replaced by norleucine) using reaction conditions practically identical to those reported for the production of Ac-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp-Phe-NH_2.[90] Sulfonation of the tryptophan indole group has also been observed on exposure to pyridine/SO_3 or tertiary amine/SO_3 complexes.[82] As the N^{in}-formyl derivatives proved to be a convenient protection in this context, sulfonation is thought to proceed via an N→C shift as outlined in Scheme 5.[82]

Scheme 5 Reaction Mechanism of Sulfonation of the Indole Group[82]



The observation that sulfation of N-terminally extended CCK peptides containing the aspartyl residue adjacent to tyrosine, e.g. of CCK-8 or CCK-10, with pyridine/SO_3 leads to partial sulfonation of the aromatic ring is surprising. It suggests involvement of the proximal carboxylate in the side reaction.[91]

Additional interesting side reactions can be observed in the sulfation of C-terminal decapeptide analogues of hirudine containing a free α-carboxy group (Scheme 6) when

for references see p 450

pyridine/SO_3 is used.[92,93] With the analogue containing a glutamine residue at the C-terminus, in addition to sulfation of the two tyrosine residues, partial sulfation of the carboxamide group has been detected, a side reaction that was predicted by model studies.[70] The extensive epimerization (>20%) at the C-terminal amino acid residue was unpredicted for sulfation with ten equivalents of pyridine/SO_3 in DMF/pyridine (1:1) mixtures. The rate of epimerization is found to be strongly dependent on the pyridine content in the reaction mixture, as shown in Figure 6.

Scheme 6 Sulfation of Peptides Related to the C-Terminal Sequence 56–65 of Hirudin[92,93]

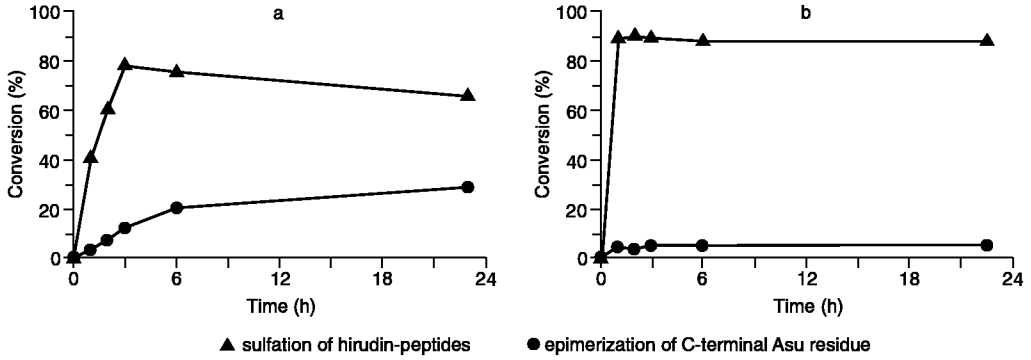
Suc—Phe—Glu—Pro—Ile—Pro—Glu—Tyr—Tyr—Leu—Xaa—OH

↓ Py/SO_3 (10 equiv), DMF/Py (1:1), 25 °C, 24 h

Suc—Phe—Glu—Pro—Ile—Pro—Glu—Tyr(SO_3H)—Tyr(SO_3H)—Leu—Xaa—OH

Xaa = Gln, Asu

Figure 6 Rates of Sulfation of the Hirudin-Peptides with Pyridine/Sulfur Trioxide and of Epimerization of the C-Terminal Asu Residue[93][a]



[a] a: in Pyridine/DMF (1:1); b: in DMF.

Kenner and Stedman[94] proposed the use of DMF/SO_3 for peptide bond formation via the mixed acylsulfuric acid anhydride. In these studies rapid racemization of the activated amino acid residue of N^{α}-protected dipeptides was detected under basic conditions whereas at pH 6.8 racemization was suppressed.[95] Formation of the mixed acylsulfuric acid anhydride at the C-terminal carboxy group of the hirudine peptides could explain the observed racemization in the presence of pyridine via the oxazol-5(4*H*)-one mechanism. This phenomenon has not been previously observed, since most of the sulfation studies were performed on peptide amides related to gastrin and CCK sequences. The observation of Bodanszky regarding sulfonation of tyrosine facilitated by a proximal aspartic acid residue[91] could possibly derive from the intermediate formation of the mixed β-aspartylsulfuric acid anhydride which itself represents an efficient sulfating agent (see Section 6.6.1.1.6). The ability of base/SO_3 complexes to promote peptide bond formation has been exploited with trimethylamine/SO_3 for oligomerization of tyrosine with concomitant O-sulfation.[152]

Ac-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp-Phe-NH_2 (6):[88]

Ac-Tyr-Met-Gly-Trp-Met-Asp-Phe-NH_2 (**5**; 0.55 g, 0.56 mmol) was dissolved in a mixture of anhyd DMF (10 mL) and anhyd pyridine (5 mL) and was then treated with pyridine/SO_3 (2 g, 13 mmol) at rt for 24 h. The solvents were removed and the residue was dissolved in ice-cold H_2O (20 mL) while keeping the pH at 7.0 with 10% aq Na_2CO_3. The aqueous soln was then concentrated and after addition of DMF the inorganic salts were removed by filtration. The filtrate was taken to dryness and the resulting crude product purified by countercurrent distribution in the solvent system BuOH/EtOH/H_2O (5:1:8). Frac-

tions containing homogeneous product were pooled and taken to dryness. The residue was re-chromatographed on Sephadex G-25 with H_2O as eluent, this gave the disodium salt pentahydrate of the title compound; yield: 227 mg (34%); $[\alpha]_D^{20}$ −18 (*c* 0.5, H_2O).

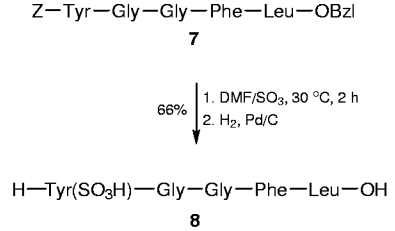
H-Tyr($SO_3Ba_{1/2}$)-O$Ba_{1/2}$·H_2O:[36]
A suspension of Z-Tyr-OH (12.7 g, 40.3 mmol) and pyridine/SO_3 (25.8 g, 161.1 mmol) in pyridine was warmed to 60 °C and stirred at this temperature until dissolution of pyridine/SO_3 (30 min). The mixture was cooled in an ice bath and filtered; the filtrate was concentrated and the insoluble excess pyridine/SO_3 was removed by filtration. The soln was diluted with H_2O, extracted with EtOAc and then treated with an excess of $Ba(OH)_2$. The $BaSO_4$ was removed by filtration and the excess $Ba(OH)_2$ was precipitated by passing CO_2 through the soln while keeping the pH at 7 through addition of pyridine. The $BaCO_3$ was removed by filtration and the filtrate was concentrated to about 100 mL. Then the product was precipitated with EtOH to give Z-Tyr($SO_3Ba_{1/2}$)-O$Ba_{1/2}$·$3H_2O$; yield: 21.4 g (91%); $[\alpha]_D^{20}$ +18.9 (*c* 1, DMF) and $[\alpha]_{546}^{20}$ +22.7 (*c* 1, DMF).

Z-Tyr($SO_3Ba_{1/2}$)-O$Ba_{1/2}$·$3H_2O$ (5 g, 8.55 mmol) was hydrogenated in DMF over Pd black. The catalyst was removed by filtration and the filtrate was taken to dryness. The residue was dissolved in H_2O, filtered again, and then the product was precipitated with EtOH. It was dissolved again in H_2O and the pH was adjusted to 6.5 with CO_2. The precipitated $BaCO_3$ was removed by filtration, the filtrate was concentrated and the product precipitated with EtOH to give H-Tyr($SO_3Ba_{1/2}$)-O$Ba_{1/2}$·H_2O; yield: 2.6 g (84%); mp 232 °C; $[\alpha]_D^{20}$ −24.8 (*c* 1, DMF/H_2O, 95:5) and $[\alpha]_{546}^{20}$ −29.6 (*c* 1, DMF/H_2O, 95:5); IR (KBr) $\tilde{\nu}_{max}$: 1050 cm^{-1}; for additional analytical characterization see Table 2.

6.6.1.1.5 Sulfation with Dimethylformamide/Sulfur Trioxide Complex

The reagent DMF/SO_3 which has been previously used by Kenner and Stedman for peptide bond formation[94] contains dimethylformamide as a base which is weaker than pyridine, a property that is expected to enhance its sulfating properties. The effect of pyridine as a stronger base than DMF is well documented in Figure 6. The exchange of the pyridine/DMF mixture as a solvent for pure DMF leads to a significant increase in the sulfation rates of tyrosine residues with pyridine/SO_3. The use of DMF/SO_3 in DMF/pyridine (4:1) as proposed by Futaki et al.[52] is, therefore, contradictory since a fast base exchange is expected to occur under such conditions (Scheme 7). Nevertheless, sulfation of tyrosine with DMF/SO_3 in the DMF/pyridine mixture was found to be significantly enhanced (ca. 20%). This sulfation procedure combined with hydrogenolytically cleavable protecting groups has also been applied for the synthesis of sulfated dermorphin and deltorphin peptides.[96] In more recent years trimethylamine/SO_3 in pyridine was proposed as an efficient O-sulfation agent.[152,153]

Scheme 7 Sulfation of Protected Leu-enkephalin with Dimethylformamide/Sulfur Trioxide Complex[52]



H-Tyr(SO_3H)-Gly-Gly-Phe-Leu-OH (8):[52]
Z-Tyr-Gly-Gly-Phe-Leu-OBzl (**7**; 50 mg, 0.064 mmol) in DMF/pyridine (4:1, 1 mL) was allowed to react with DMF/SO_3 (200 mg, 20 equiv) at 30 °C for 2 h. The solvent was removed and the residue treated with 10% aq NH_3 in an ice bath for 30 min while maintaining the pH at 8.0. The product was extracted with BuOH, the organic layer was washed with H_2O and was then concentrated to a small volume. The product was precipitated with Et_2O and subsequently hydrogenated over Pd/C in DMF/MeOH (1:2; 10 mL) containing a few drops of AcOH. The catalyst was removed by filtration, the filtrate was taken to dryness and the residue partitioned between H_2O and EtOAc. The aqueous soln was lyophilized and the crude product purified by HPLC (isocratic elution, 28% MeCN in 0.1 M NH_4OAc). Fractions with t_R = 8 min were collected and lyophilized to give **8**; yield: 27 mg (66%); FTIR $\tilde{\nu}_{max}$: 1047, 1253 cm^{-1}.

for references see p 450

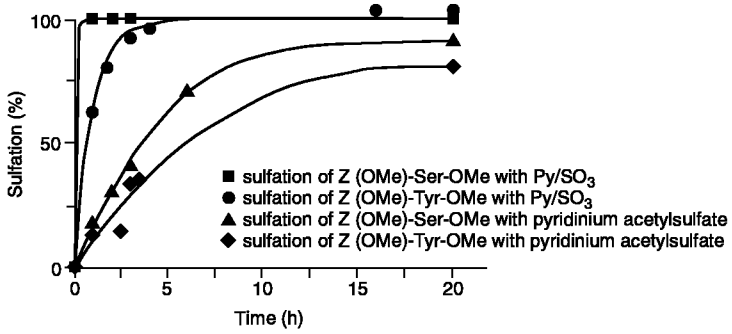
Similarly, sulfation of **7** with pyridine/SO_3 (30 °C, 4 h) followed by an identical workup procedure gave **8**; yield: 22.5 mg (55%).

6.6.1.1.6 Sulfation with Pyridinium Acetylsulfate

The search for milder sulfating agents led Penke et al.[37,38,79] to propose acetylsulfuric acid which itself is not stable,[97] but can be isolated as a pyridinium salt in a solid form which can be stored in the cold. This pyridinium salt pyridinium acetylsulfate leads to quantitative sulfation of hydroxy groups while the tyrosine phenolic group is sulfated at a slower rate. As expected from the studies of Kenner and Stedman,[94] amino groups are acetylated using this reagent. Model experiments on amino acid derivatives indicated that pyridinium acetylsulfate is unreactive toward imidazole, indole, guanidino, and sulfide groups and that it is particularly suited for the sulfation of peptides such as gastrin and CCK.[38] The use of acidic media, e.g. acetic acid or TFA, efficiently suppresses acetylation of amino groups, but above 0 °C leads to sulfonation of tyrosine and sulfonation of tryptophan in low quantities,[38] as previously observed with chlorosulfonic acid.[77] Note that N^{in}-protection prevents the latter side reaction.

The main drawback of the pyridinium acetylsulfate reagent arises from its reduced reactivity that is well documented by the rates of sulfation of tyrosine and serine derivatives in comparison to pyridine/SO_3, as shown in Figure 7.[55]

Figure 7 Sulfation of Z(OMe)-Tyr-OMe and Z(OMe)-Ser-OMe[55][a]



[a] Sulfation of peptide (0.05 mmol) in DMF/pyridine (4:1, 1 mL) with pyridine/SO_3 complex (5 equiv) or pyridinium acetylsulfate (10 equiv) at 25 °C.

Sulfation of tyrosine peptides with pyridinium acetylsulfate has been mainly carried out in DMF/pyridine mixtures at room temperature (Scheme 8), where various excess reagents and reaction times are required for quantitative sulfation depending on the peptide and reaction conditions.

Scheme 8 Sulfation of an N^{α}-Acetylated CCK-Heptapeptide Analogue[53]

Ac—Tyr—Met—D-Ala—Trp—Met—Asp—Phe—NH_2
9

↓ pyridinium acetylsulfate (5 equiv), 25 °C, 24 h

Ac—Tyr(SO_3Na)—Met—D-Ala—Trp—Met—Asp—Phe—NH_2
10

The relatively low reactivity of pyridinium acetylsulfate led to the proposal that pyridinium trifluoroacetylsulfate can be employed as an alternative reagent.[80] The preparation of this

reagent as a homogeneous material proved to be even more difficult than pyridinium acetylsulfate, but sulfation of the tyrosine derivatives is found to proceed at significantly higher rates (DMF/pyridine 3:1, 3–5 h, 25 °C) as compared to pyridinium acetylsulfate (8–12 h).

Pyridinium Acetylsulfate:[53]
To a mixture of Ac_2O (28.4 mL, 0.3 mol) and pyridine (8.05 mL, 0.1 mol) was added concd H_2SO_4 (5.4 mL, 0.1 mol), dropwise at −15 °C. After 5 min Et_2O (300 mL) was added to the suspension formed. The solid was removed by filtration and was washed three times with Et_2O, and dried in vacuo over P_2O_5 and KOH; yield: 18.4 g (83%); mp 75 °C; IR (KBr) $\tilde{\nu}_{max}$: 4050 cm^{-1}. The reagent is highly hygroscopic and usually contaminated by varying amounts of a second component, most probably acetylpyridinium sulfate. Even changing the order of the addition of the reactants, i.e. pyridine to the mixture of Ac_2O/H_2SO_4, does not prevent formation of the second component that is easily identified by ^1H NMR in D_2O.[80]

Ac-Tyr(SO_3Na)-Met-D-Ala-Trp-Met-Asp-Phe-NH_2 (10):[53]
The unsulfated peptide **9** (401 mg, 0.4 mmol) was allowed to react in DMF/pyridine (1:1,4 mL) with pyridinium acetylsulfate (440 mg, 2 mmol) at rt for 24 h. The mixture was diluted with H_2O, and the pH adjusted to 7.0 with 4 M NaOH. The mixture was filtered through Celite, and the filtrate was purified by RP-HPLC [linear gradient elution, 12 to 14% MeCN/0.1 M NH_4OAc buffer (pH 6.5)] to give **10**; yield: 210 mg (52%); $[\alpha]_D^{20}$ −21.2 (*c* 1, DMF); IR (KBr) $\tilde{\nu}_{max}$: 1050 cm^{-1}.

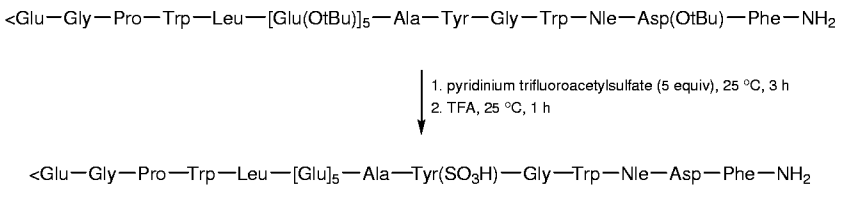
6.6.1.1.7 Protection Schemes for Postsynthetic Sulfation of Tyrosine Peptides

All sulfating agents are known to react faster with alcoholic rather than with phenolic groups. Additionally, α- and ε-amino groups are sulfated with base/SO_3 complexes to form sulfamic acids and are acetylated with pyridinium acetylsulfate. With concentrated sulfuric acid or pyridinium acetylsulfate in TFA sulfamic acids are not formed. Correspondingly, a selective postsynthetic chemical sulfation of tyrosine residues in peptides can be achieved upon suitable protection of the remaining reactive groups. It was for this reason that the threonine hydroxy group was protected as an acetyl derivative. The acetyl group was cleaved after sulfation of the tyrosine residue with pyridine/SO_3 by saponification with sodium hydroxide at pH 11 over 3 hours in the first syntheses of phyllocaerulein[47] and caerulein.[48] These compounds do not contain amino groups. Conversely, for the syntheses of CCK-related peptides with an N-terminal amino group the alkali-labile Msc group[49] or N^{α}-Boc derivatives of the peptides were used.

For the sulfation of peptides that do not contain methionine residues, the use of protecting groups that are cleaved by hydrogenolysis represents, independent of the sulfation procedure followed, the safest approach (see Scheme 7).[52,96]

Chemical sulfation of unprotected gastrin with pyridine/SO_3 in aqueous media can lead to serious side reactions and correspondingly to very low yields.[89] In order to solubilize gastrin in DMF/pyridine for sulfation with pyridinium acetylsulfate and trifluoroacetylsulfate the protected derivative shown in Scheme 9 was used.[80]

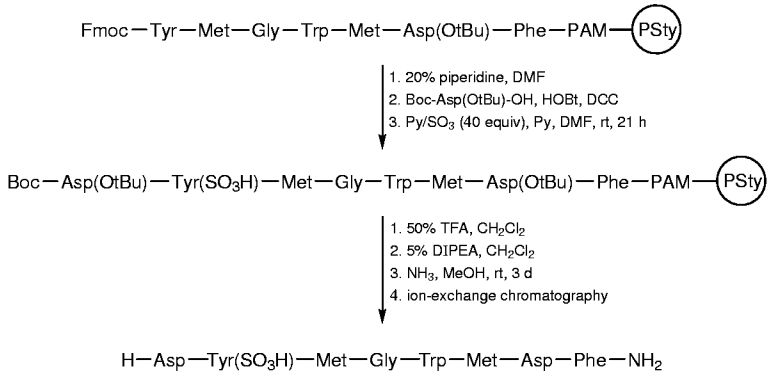
Scheme 9 Sulfation of Gastrin Protected at the Side-Chain Carboxy Groups as *tert*-Butyl Esters by Treatment with Pyridinium Trifluoroacetylsulfate/Dimethylformamide[80]



for references see p 450

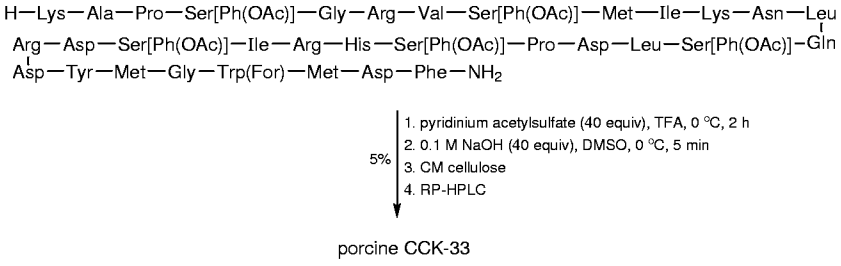
Quantitative sulfation is not achieved even with a larger excess of sulfating reagent (max sulfation 80%), a fact which is not understood. On the other hand, sulfation of a suitably protected CCK-8 linked to PAM resin with pyridine/SO_3 proceeds smoothly (Scheme 10) and upon ammonolysis, the CCK-octapeptide is isolated in 23% overall yield.[98]

Scheme 10 Synthesis of CCK-8 Including Sulfation on Solid Support[98]



The major improvement in protection strategies over the past years has permitted scientists to undertake the syntheses of demanding targets such as CCK-33 with all its numerous reactive groups in postsynthetic sulfation steps. The approach proposed by Sakakibara et al.[45,56] is based on sulfation with pyridinium acetylsulfate in TFA with unprotected amino groups. As shown in Scheme 11, the five serine residues are protected as phenoxyacetyl derivatives since this group is stable to hydrogen fluoride. The hydrogen fluoride medium is required for removal of all the residual protecting groups used in the synthesis in solution of unsulfated porcine CCK-33. Moreover, tryptophan is protected as a formyl derivative to prevent sulfation of the indole group. Upon sulfation of the free phenolic group of this partially protected and purified 33-membered synthetic peptide, the phenoxyacetyl and formyl groups are removed by very short treatment with sodium hydroxide, but the yield of purified porcine CCK-33 is disappointingly low (5%). This has been attributed mainly to oxidation of the methionine residues; it could, however, also result from partial sulfonation of the tyrosine residue with pyridinium acetylsulfate in acidic medium.[38]

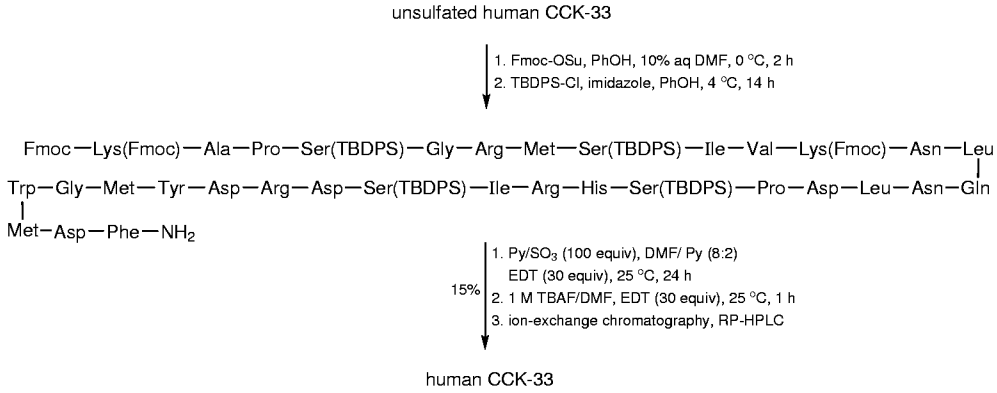
Scheme 11 Protection Strategy for Postsynthetic Sulfation of CCK-33 with Pyridinium Acetylsulfate in TFA[56]



To allow for the use of the stronger sulfating reagent pyridine/SO_3, an alternative protection strategy has been proposed by Fujii et al.[55,99] that is based on postsynthetic protection of the amino groups as Fmoc derivatives and of the serine hydroxy groups as *tert*-butyldiphenylsilyl derivatives as shown in Scheme 12. The histidine side chain is not protected since model studies indicate that sulfated histidine is readily hydrolyzed by water within 60 minutes.

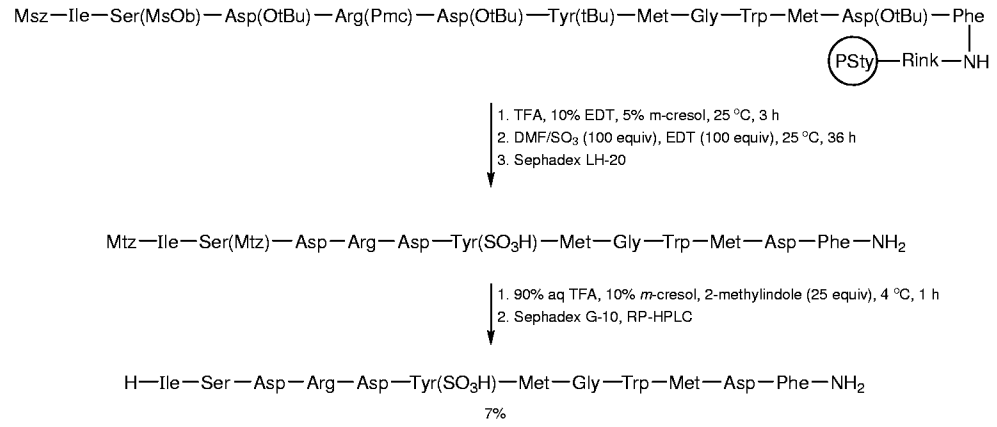
Ethanedithiol is added in the sulfating step to prevent oxidation of the methionine residues and in the deprotection step with TBAF to quench the dibenzofulvene, although thiol groups are expected to be sulfated at higher rates than phenolic groups.

Scheme 12 Postsynthetic Protection of Unsulfated Human CCK-33 at Amino and Hydroxy Groups prior to Sulfation with Pyridine/Sulfur Trioxide Complex Followed by Suitable Deprotection Steps[55]



The solid-phase synthesis of peptides suitably protected for chemical sulfation can be achieved using a combination of the 4-(methylsulfinyl)benzyloxycarbonyl group for protection of the amino groups and 4-(methylsulfinyl)benzyl group for protection of the hydroxy groups. Both reagents are stable to TFA treatment and can be used in combination with acid-labile groups for those groups that do not react with sulfating agents.[57,100] This approach is illustrated in Scheme 13 for the synthesis of CCK-12.[57] It has also been successfully applied for the synthesis of leucosulfakinin-II[57] and cionin.[64] This protection strategy foresees, upon cleavage of the protected peptide from the resin, reduction of the sulfinyl groups to sulfide groups with SO_3/thiols[101,102] with concomitant sulfation of the tyrosine residue. Since a large excess of ethanedithiol is used, sulfation of the thiol groups must occur preferentially; therefore, the resulting sulfate thioester acts as the sulfating agent in a *trans*-esterification reaction under anhydrous conditions. This hypothetical mechanism has as yet not been experimentally investigated.

Scheme 13 Synthesis of Sulfated CCK-12[57]



for references see p 450

6.6.1.2 Synthesis of Sulfated Tyrosine Peptides with Tyrosine *O*-Sulfate Synthons

Difficulties encountered in the postsynthetic chemical sulfation of peptides and the correspondingly low yields have led to the proposal of an alternative approach. This approach makes use of appropriate tyrosine *O*-sulfate derivatives for the chain elongation steps in solution and on solid supports by applying protection strategies compatible with the acid sensitivity of the sulfate ester. Moreover, the analytical characterization of the peptides synthesized with tyrosine *O*-sulfate derivatives is greatly facilitated since contaminations deriving from the preparation of the intermediates are easily detected by chromatographic (HPLC and CE) and spectroscopic methods (see Table 2).

6.6.1.2.1 Synthesis of Tyrosine *O*-Sulfate Derivatives

For the direct use of tyrosine *O*-sulfate in peptide synthesis a set of N^{α}-protected Tyr(SO_3H) derivatives were synthesized that can be combined with the most common protection schemes. In addition to Z-Tyr($SO_3Ba_{1/2}$)-$OBa_{1/2}$ (**11**)[36] (see Section 6.6.1.1.4), the related Boc and Fmoc derivatives have been synthesised as barium, sodium, potassium salts, and even as tetrabutylammonium salts due to their better solubility in solvents generally used for automated synthesis on solid supports.[152,154,155]

Boc-Tyr(SO_3Na)-ONa:[83]
$ClSO_3H$ (17 mL, 0.254 mmol) was dissolved in $CHCl_3$ (17 mL) and pyridine (60 mL) at −10 °C and was then added to a soln of Boc-Tyr-OH (19.3 g, 68 mmol) in pyridine (70 mL). The mixture was warmed up to 50 °C. After 2 d the solvents were removed under reduced pressure and the residue neutralized with aq NaOH (50 mL). Insoluble material was removed by filtration and the filtrate concentrated to about 15 mL. Upon addition of EtOH (500 mL), the precipitate was collected; yield: 19 g (77%); mp 285–290 C (dec); $[\alpha]_D^{20}$ –52.4 (*c* 1, DMF).

Boc-Tyr(SO_3K)-OK:[103]
Boc_2O (0.4 g, 1.83 mmol) was added to a soln of H-Tyr(SO_3K)-OH (0.4 g, 1.34 mmol) in a mixture of dioxane (1.6 mL) and 1 M KOH (1.6 mL). After stirring at rt for 4 h, the dioxane was removed under reduced pressure and the aqueous soln washed with Et_2O and filtered through a column of Dowex 50 (12 mL). The eluate was neutralized with 0.5 M KOH to pH 5.0 and taken to dryness. The residue was triturated with Et_2O and crystallized from EtOH/Et_2O to give Boc-Tyr(SO_3K)-OK·$3H_2O$; yield: 0.465 g (79%); mp 285–288 °C; $[\alpha]_D^{20}$ +51.7 (*c* 0.3, DMF). A part of the product was converted into its corresponding sodium salt by means of an ion exchanger; $[\alpha]_D^{20}$ +51.9 (*c* 0.3, DMF) {therefore, the value of $[\alpha]_D^{20}$ −52.4 (*c* 1, DMF) reported by Tóth et al.[83] is probably wrong}.

Fmoc-Tyr(SO_3Na)-OH:[40]
Fmoc-Tyr-OH (1.30 g, 3.22 mmol) was dissolved in a mixture of DMF/pyridine/dioxane (11 mL, 8:2:1) and the DMF/SO_3 complex (1.48 g, 9.66 mmol) was added in one portion. The mixture was stirred overnight at rt, the solvent was removed, and the residue dissolved in ice-cold H_2O (20 mL). The pH of the soln was adjusted to 6.5 with 5% aq Na_2CO_3, within 1 h and the soln was lyophilized. The residue was taken up in DMF (20 mL), insoluble material was removed by filtration and the solvent was removed. The residue was triturated with EtOH and the solid was precipitated twice from MeOH with Et_2O. For further purification, the product was chromatographed on silica gel ($CHCl_3$/ MeOH/AcOH 9:3:0.5). Fractions containing the desired material were pooled and taken to dryness to give the product as its sesquihydrate; yield: 1.41 g (87%); mp 225 °C (dec); $[\alpha]_D^{20}$ +27.9 (*c* 1, MeOH); FAB-MS (*m/z*): 506 $[M+H]^+$, 528 $[M+Na]^+$; FTIR (KBr) $\tilde{\nu}_{max}$: 1055, 1250 cm^{-1}.

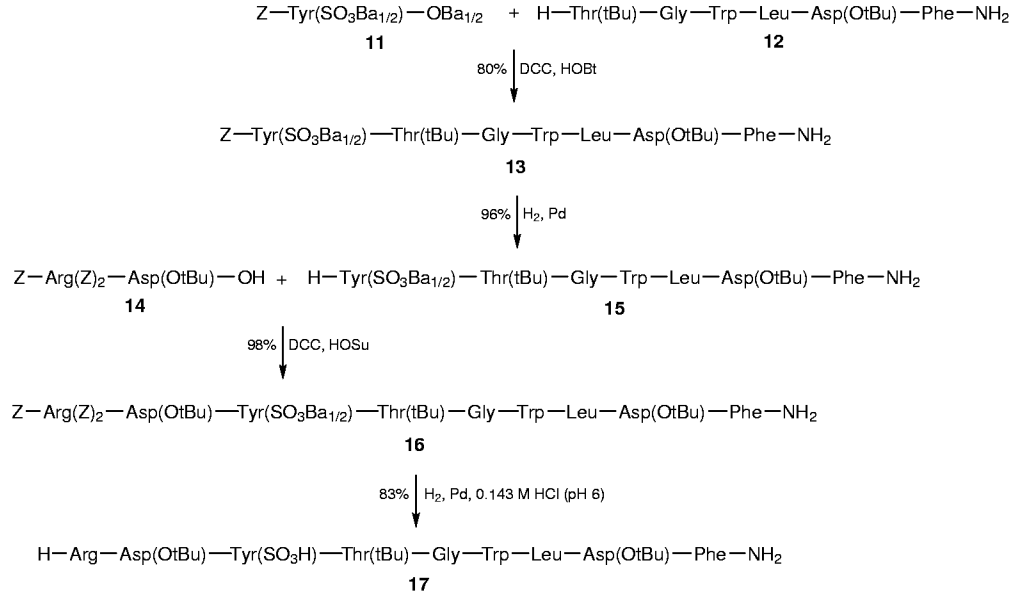
Fmoc-Tyr($SO_3Ba_{1/2}$)-$OBa_{1/2}$ (18):[42]
To the ice-cold aq soln of H-Tyr($SO_3Ba_{1/2}$)-OH·H_2O (2.08 g, 60 mmol), Fmoc-Cl (1.7 g, 66 mmol) in dioxane was added together with a sat. soln of $Ba(OH)_2$ to maintain the pH at 8. After 10 h the dioxane was removed and the soln was diluted with H_2O and filtered. The filtrate was poured into EtOH and the precipitate was removed by filtration and redissolved in H_2O. Upon acidification to pH 6.5 with CO_2, $BaCO_3$ was removed by filtration, the filtrate was concentrated and chromatographed on Sephadex LH-20 with H_2O as the eluent. Fractions containing the product were collected and lyophilized to give **18** as its dihydrate; yield: 1.9 g (49%); $[\alpha]_D^{20}$ +18.3 and $[\alpha]_{546}^{20}$ +21.4 (*c* 1, DMF).

Fmoc-Tyr(SO_3NBu_4)-OH:[152]
A soln of Fmoc-Tyr-OH (4.02 g, 10.0 mmol) and pyridine/SO_3 (4.77 g, 30.0 mmol) in DMF (10 mL) was stirred under argon at rt for 1 h, and then slowly poured into ice-cooled 5% aq $NaHCO_3$ (300 mL). After gas evolution ceased, a small quantity of $CHCl_3$ and a soln of Bu_4NHSO_4 (5.09 g, 15.0 mmol) in aq $NaHCO_3$ (pH > 7) were added and the pH of the soln was adjusted to about 4 by addition of 5% citric acid. The organic layer was separated and the aqueous layer extracted with $CHCl_3$. The combined extracts were washed with 5% citric acid (3×), H_2O (2×), and brine, then dried (Na_2SO_4). The solvent was removed again and the residue solidified with petroleum ether. The solid was lyophilized from dioxane/H_2O (10:1, 33 mL); yield: 6.88 g (95%) as hemihydrate; $[\alpha]_D^{26} = +8.0$ (*c* 1.0, MeOH).

6.6.1.2.2 Synthesis of Sulfated Tyrosine Peptides in Solution

In the absence of methionine residues Z-Tyr($SO_3Ba_{1/2}$)-$OBa_{1/2}$ (**11**) proved to be well suited for the incorporation of tyrosine *O*-sulfate residue into peptides. With DCC/HOBt as a coupling reagent, the HOBt is sufficiently acidic to displace the hemi-barium salt from the carboxy group. Subsequent hydrogenolysis allows for removal of the N^{α}-protection and further chain elongation steps as shown in Scheme 14.[41]

Scheme 14 Synthesis of Tyrosine *O*-Sulfate Peptides with the Benzyloxycarbonyl Group for Temporary N^{α}-Protection[41]

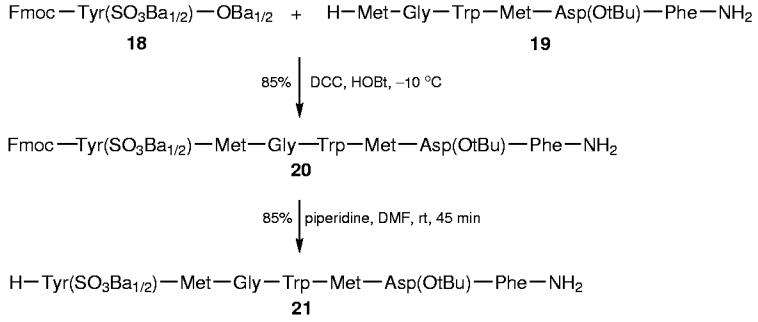


With the use of the benzyloxycarbonyl group for temporary N^{α}-protection the [Thr^{28},Leu^{31}]-CCK-[25–33] and [Thr^{28},Nle^{31}]-CCK-[25–33] peptides[41] as well as [Nle^{15}]-gastrin-II and des-Trp^1-[Nle^{12}]-minigastrin-II[43] were obtained in good overall yields as protected intermediates.

For the synthesis of methionine-containing peptides the Fmoc group is recommendable for temporary protection in the chain elongation steps as shown in Scheme 15.

for references see p 450

Scheme 15 Synthesis of Tyrosine *O*-Sulfate Peptides with the 9-Fluorenylmethoxycarbonyl Group for Temporary N^{α}-Protection[42]



The derivative Fmoc-Tyr(SO_3Na)-OH is coupled in high yield to peptides via the DCC/HOBt procedure.[62] Moreover, cleavage of the Fmoc group can be achieved under milder conditions than those reported for compound **21**, e.g. 10% piperidine in DMF. An N^{α}-Boc-protection has been used as an alternative to the Fmoc protecting group for the synthesis of CCK-related peptides containing methionine residues.[83] In these syntheses the pentachlorophenyl ester and pentafluorophenyl ester were used for chain elongation with tyrosine *O*-sulfate. Larger forms of CCK peptides have been assembled by acylation of the tyrosine-*O*-sulfate containing fragments via the thioester segment condensation approach.[156,157]

Z-Tyr($SO_3Ba_{1/2}$)-Thr(tBu)-Gly-Trp-Leu-Asp(OtBu)-Phe-NH_2 (13):[41]
To a soln of H-Thr(tBu)-Gly-Trp-Leu-Asp(OtBu)-Phe-NH_2 (**12**; 0.5 g, 0.58 mmol) and Z-Tyr($SO_3Ba_{1/2}$)-$OBa_{1/2}$ (**11**; 0.4 g, 0.69 mmol) in DMF, HOBt (0.1 g, 0.75 mmol) and DCC (0.143 g, 0.69 mmol) were added at –20 °C. The mixture was stirred for 12 h at 5 °C and for an additional 48 h at rt. The DCU was removed by filtration, and the product precipitated by addition of Et_2O, which was then reprecipitated from DMF with H_2O; yield: 0.6 g (80%); mp 168 °C (dec); $[\alpha]_D^{20}$ –18.8 (*c* 1, DMF) and $[\alpha]_{546}^{20}$ –22.7 (*c* 1, DMF).

H-Tyr($SO_3Ba_{1/2}$)-Thr(tBu)-Gly-Trp-Leu-Asp(OtBu)-Phe-NH_2 (15):[41]
Z-Tyr($SO_3Ba_{1/2}$)-Thr(tBu)-Gly-Trp-Leu-Asp(OtBu)-Phe-NH_2 (**13**; 4.07 g, 3.14 mmol) was hydrogenated in DMF over Pd black. The catalyst was removed by filtration, the filtrate taken to dryness and the residue triturated with Et_2O; yield: 3.7 g (96%); mp 178 °C (dec); $[\alpha]_D^{20}$ –29.85 (*c* 1, DMF); $[\alpha]_{546}^{20}$ –36.0 (*c* 1, DMF).

Z-Arg(Z)$_2$-Asp(OtBu)-Tyr($SO_3Ba_{1/2}$)-Thr(tBu)-Gly-Trp-Leu-Asp(OtBu)-Phe-NH_2 (16):[41]
To a soln of H-Tyr($SO_3Ba_{1/2}$)-Thr(tBu)-Gly-Trp-Leu-Asp(OtBu)-Phe-NH_2 (**15**; 3.0 g, 2.43 mmol) and Z-Arg(Z)$_2$-Asp(OtBu)-OH (**14**; 2.18 g, 2.92 mmol) in DMF at –20 °C, were added HOSu (0.36 g, 3.16 mmol) and DCC (0.63 g, 3.04 mmol). The mixture was stirred for 24 h at 5 °C and for an additional 24 h at rt. The DCU was filtered off, the solvent removed and the product precipitated twice from DMF with Et_2O to give **16** as its dihydrate; yield: 4.6 g (98%); mp 195–198 °C; $[\alpha]_D^{20}$ –14.3 (*c* 1, DMF) and $[\alpha]_{546}^{20}$ –17.3 (*c* 1, DMF).

H-Arg-Asp(OtBu)-Tyr(SO_3H)-Thr(tBu)-Gly-Trp-Leu-Asp(OtBu)-Phe-NH_2 (17):[41]
Z-Arg(Z)$_2$-Asp(OtBu)-Tyr($SO_3Ba_{1/2}$)-Thr(tBu)-Gly-Trp-Leu-Asp(OtBu)-Phe-NH_2 (**16**; 3.7 g, 1.9 mmol) was hydrogenated in DMF over Pd black at pH 6 (titration with 0.14 M HCl in MeOH). The catalyst was removed by filtration and the product precipitated with H_2O containing TEA. The precipitate was collected, washed with H_2O and reprecipitated from DMF with H_2O; yield: 2.24 g (83%); mp 208 °C (dec); $[\alpha]_D^{20}$ –21.2 (*c* 1, DMF) and $[\alpha]_{546}^{20}$ – 25.5 (*c* 1, DMF).

Fmoc-Tyr($SO_3Ba_{1/2}$)-Met-Gly-Trp-Met-Asp(OtBu)-Phe-NH_2 (20):[42]
To a soln of H-Met-Gly-Trp-Met-Asp(OtBu)-Phe-NH_2 (**19**; 0.37 g, 0.43 mmol) and Fmoc-Tyr($SO_3Ba_{1/2}$)-$OBa_{1/2}$ (**18**; 0.34 g, 0.52 mmol) in DMF at – 10 °C were added HOBt (0.076 g, 0.56 mmol) and DCC (0.107 g, 0.52 mmol). After 2 h stirring at 0 °C and 12 h at rt, additional **18** (0.17 g, 0.258 mmol), HOBt (0.038 g, 0.280 mmol), and DCC (0.053 g, 0.258 mmol) were added at – 20 °C. The mixture was kept at rt for 12 h. The DCU was then removed by filtration and the filtrate taken to a small volume. The product

was precipitated with EtOAc and reprecipitated from DMF with H_2O to give **20** as its dihydrate; yield: 0.52 g (85%); mp 197–203 °C (dec); $[\alpha]_D^{20}$ – 18.6 (*c* 1, DMF) and $[\alpha]_{546}^{20}$ – 22.0 (*c* 1, DMF).

H-Tyr($SO_3Ba_{1/2}$)-Met-Gly-Trp-Met-Asp(OtBu)-Phe-NH_2 (21):[42]
To compound **20** (0.78 g, 0.55 mmol) in DMF (3 mL) was added piperidine (10 mL) and the mixture was stirred for 45 min at rt. The soln was then concentrated and the product precipitated with EtOAc which was then reprecipitated from DMF with iPrOH to give **21** as its sesquihydrate; yield: 0.55 g (85%); mp 205–207 °C; $[\alpha]_D^{20}$ –30.4 (*c* 1, DMF) and $[\alpha]_{546}^{20}$ – 36.1 (*c* 1, DMF).

Boc-Tyr(SO_3Na)-OPcp:[83]
To an ice-cold soln of Boc-Tyr(SO_3Na)-ONa (6.7 g, 17 mmol) and pentachlorophenol (5.3 g, 20 mmol) in DMF (50 mL) was added DCC (3.9 g, 19 mmol) and the mixture was kept cold for 12 h. The DCU was removed by filtration and the filtrate was taken to dryness, the residue was crystallized from iPrOH/Et_2O; yield: 7.1 g (66%).

Boc-Tyr(SO_3Na)-OPfp:[83]
The title compound was prepared as described for Boc-Tyr(SO_3Na)-OPcp; yield: 84%.

6.6.1.2.3 Synthesis of Sulfated Tyrosine Peptides on Solid Supports

The acid-lability of tyrosine *O*-sulfate residues limited the synthetic strategies on solid supports with tyrosine *O*-sulfate as a building block to the Fmoc route with acid-labile side-chain protecting groups and resin anchors that can be cleaved under mild acidic conditions. Therefore, Fmoc-Tyr(SO_3Na)-OH, its hemi-barium salt, and the tetrabutylammonium salt have been exclusively used to date. Chain elongation steps with these derivatives were carried out with the related pentafluorophenyl ester[63] or by applying the BOP,[104] PyBOP,[105] HBTU/HOBt,[152,154] or the DIC/HOBt method as shown in Table 3.

The main drawback for synthesis on solid support results from the cleavage of the peptide from the resin which usually requires strong acid conditions. In this context the recently developed xanthenylamide handles represent a promising approach.[106] However, the best results are apparently obtained with the 2-chlorotrityl resin[107] with the two step cleavage/deprotection reaction proposed by Kitagawa et al.[65,66,158] This procedure is illustrated in Scheme 16 and foresees cleavage of the fully protected peptides from the resin in almost quantitative yields followed by deprotection under optimized conditions as used for the peptides synthesized in solution (see Section 6.6.1.2.4).

Table 3 Synthesis of Sulfated Tyrosine Peptides on Solid Support and Cleavage/Deprotection Conditions Used[40,63–65,106,108,149,152]

Amino Acid	Coupling Method	Resin	Cleavage Step	Peptide	Yield[a] (%)	Ref
Fmoc-Tyr(SO_3Na)-OPfp (3 equiv)	active ester, CH_2Cl_2/DMF (1:1)	2,4-DMBHA-resin	TFA/thioanisole, 10 min, rt	CCK-8	6	[108]
Fmoc-Tyr(SO_3Na)-OPfp (2×3 equiv)	active ester, CH_2Cl_2/DMF (1:1)	SAMB-HA-resin	50% TFA/CH_2Cl_2, 5% ethanedithiol, 2% DMS	CCK-33; RP-HPLC	6	[63]
Fmoc-Tyr(SO_3Na)-OH (3 equiv)	PyBOP (3 equiv), NMM (9 equiv), DMF, 90 min	PAL-resin (0.33 mmol/g)	90% aq TFA/2-methylindole, *m*-cresol, 4 °C, 15 h	CCK-12; Sephadex G-10; RP-HPLC	9	[40]
Fmoc-Tyr(SO_3Na)-OH (3 equiv)	PyBOP (3 equiv), NMM (9 equiv), DMF, 90 min	PAL-resin (0.33 mmol/g)	90% aq TFA/2-methylindole, *m*-cresol, 4 °C, 5 h	cionin; RP-HPLC	6	[64]

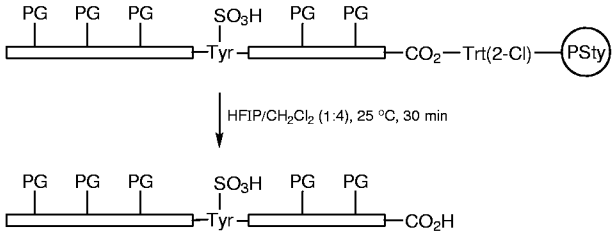
for references see p 450

Table 3 (cont.)

Amino Acid	Coupling Method	Resin	Cleavage Step	Peptide	Yield[a] (%)	Ref
Fmoc-Tyr($SO_3Ba_{1/2}$)-OH (5 equiv)	BOP (5 equiv), HOBt (10 equiv)/ NMP (15 equiv), DMF	XAL_1-PEG-resin (0.21 mmol/ g)	1) allyl esters, $Pd(PPh_3)_4$, $CHCl_3$/ AcOH/NMM; 2) TFA/CH_2Cl_2/H_2O (1:18:1), 25 °C, 15 min	CCK-8	71	[106]
Fmoc-Tyr(SO_3Na)-OH (3 equiv)	PyBOP (3 equiv), NMM (9 equiv), DMF, 90 min	2-chloro-trityl-resin	AcOH/TFE/CH_2Cl_2 (1:1:3), 25 °C, 45 min	protected gastrin II, big gastrin II, and CCK-33	n.r.	[65]
Fmoc-Tyr(SO_3Na)-OH (3 equiv)	DIC (3 equiv)/ HOBt (3 equiv)	2-chloro-trityl-resin	AcOH/TFE/CH_2Cl_2 (1:1:3), 20 °C, 20 min	*N*-acetyl-sulfated Leu-en-kephalin	n.r.	[149]
Fmoc-Tyr-(SO_3NBu_4)-OH (3 equiv)	HBTU (3 equiv)/HOBt (3 equiv)	trityl-resin	HFIP/CH_2Cl_2 (1:4)	oligo-tyrosine-*O*-sulfate peptides	18	[152]

[a] n.r. = not reported.

Scheme 16 Synthesis of Sulfated Peptides with Tyrosine *O*-Sulfate Synthons on 2-Chlorotrityl Resin as Proposed by Kitagawa[158]



6.6.1.2.4 Acidolytic Deprotection of Sulfated Tyrosine Peptides

Cleavage of acid-labile side-chain protecting groups from the fully protected tyrosine *O*-sulfate hemi-barium, monosodium, or arginine salt-bridged peptides is best achieved with 90% aqueous TFA containing 2-methylindole as a scavenger.[36,41] The 10% water represents a compromise between lower cleavage rates of *tert*-butyl based protecting groups, particularly of *tert*-butyl ethers, and the reduced rates of hydrolysis of the sulfate ester. Moreover, the above noted conditions prevent tryptophan *tert*-butylation[18] which is well documented in the final deprotection step of the CCK-peptide **17**.[41]

The mild acidolytic cleavage procedure noted above is used to deprotect various side chain protected peptides synthesized in solution or on chlorotrityl-resin with tyrosine *O*-sulfate synthons as listed in Table 4. The overall yields are significantly superior to those obtained by postsynthetic sulfation of the purified peptides, since they are typical for synthetic peptides after the final deprotection and purification step. The additional main advantage of this approach derives from a facile analytical characterization, since sulfonated byproducts at the tyrosine and tryptophan level, as well as oxidation of the methionine residues resulting from postsynthetic sulfation of tyrosine peptides are avoided.

Table 4 Deprotection of Synthetic Sulfated Tyrosine Peptides[41–43,50,62,65,66,158]

Protected Peptides with Built-In Tyr(SO_3H)	Deprotection Step	Chromatographic Purification	Peptide	Yield (%)	Ref
H-Arg-Asp(OtBu)-Tyr(SO_3H)-Thr(tBu)-Gly-Trp-Nle-Asp(OtBu)-Phe-NH_2	90% TFA/2-methylindole, 25 °C, 90 min	Sephadex LH-20, DEAE-cellulose	[Thr^{28},Nle^{31}]-CCK-[25–33]	57	[41]
H-Asp(OtBu)-Arg-Asp(OtBu)-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp(OtBu)-Phe-NH_2	90% TFA, 25 °C, 90 min	Biogel P6, DEAE-cellulose	CCK-[24–33]	58	[42]
H-Ile-Ser(tBu)-Asp(OtBu)-Arg-Asp(OtBu)-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp(OtBu)-Phe-NH_2	90% TFA/2-methylindole, 25 °C, 90 min	DEAE-Sephadex, RP-chromatography (Lichroprep-RP18)	CCK-[22–33]	40	[50]
<Glu-Gly-Pro-Trp-Leu-[Glu(OtBu)]$_5$-Ala-Tyr($SO_3Ba_{1/2}$)-Gly-Trp-Nle-Asp(OtBu)-Phe-NH_2	90% TFA/2-methylindole, 25 °C, 60 min	RP-HPLC (C18)	[Nle^{15}]-gastrin-II	38	[43]
Boc-Leu-[Glu(OtBu)]$_5$-Ala-Tyr($SO_3Ba_{1/2}$)-Gly-Trp-Nle-Asp(OtBu)-Phe-NH_2	90% TFA/2-methylindole, 25 °C, 60 min	RP-HPLC (C18)	des-Trp^1-[Nle^{15}]-minigastrin-II	39	[43]
<Glu-Gln-Asp(OtBu)-Tyr(SO_3Na)-Thr-(tBu)-Gly-Trp-Met-Asp(OtBu)-Phe-NH_2	90% TFA, 5 °C, 3 h	RP-chromatography (Lichroprep-RPl8)	caerulein	44	[62]
<Glu-Leu-Gly-Pro-Gln-Gly-Pro-Pro-His-(Boc)-Leu-Val-Ala-Asp-(OtBu)-Pro-Ser-(tBu)-Lys(Boc)-Lys(Boc)-Gln-Gly-Pro-Trp-Leu-[Glu(OtBu)]$_5$-Ala-Tyr(SO_3Na)-Gly-Trp-Nle-Asp(OtBu)-Phe-NH_2	90% TFA, 4 °C, 8 h	RP-HPLC	human big-gastrin-II	18	[65,66]
H-Lys(Boc)-Ala-Pro-Ser(tBu)-Gly-Arg(Pbf)-Met-Ser(tBu)-Ile-Val-Lys-(Boc)-Asn-Leu-Gln-Asn-Leu-Asp-(OtBu)-Pro-Ser(tBu)-His(Boc)-Arg(Pbf)-Ile-Ser(tBu)-Asp(OtBu)-Arg(Pbf)-Asp(OtBu)-Tyr(SO_3Na)-Met-Gly-Trp-Met-Asp(OtBu)-Phe-NH_2	90% TFA, 4 °C, 7 h and 25 °C, 1 h	RP-HPLC	human CCK-33	10	[65,158]

H-Arg-Asp-Tyr(SO_3H)-Thr-Gly-Trp-Leu-Asp-Phe-NH_2:[41]
The nonapeptide **17** (0.5 g, 0.35 mmol) was dissolved in ice-cold argon-saturated 90% aq TFA (50 mL) containing 2-methylindole (0.46 g, 3.5 mmol). After 90 min at rt the crude product was precipitated with Et_2O. It was then collected and dried (KOH pellets). The crude product was chromatographed on Biogel-P6 with 0.05 M NH_4OAc (pH 8.1) followed by ion-exchange chromatography on DEAE-cellulose with 0.025 M NH_4OAc (pH 6.3) as eluent; yield: 0.28 g (65%); IR(KBr) $\tilde{\nu}_{max}$: 1050 cm^{-1}.

6.6.1.3 Enzymatic Sulfation of Tyrosine Peptides

The tyrosine protein sulfotransferase preparations from Golgi-enriched membranes have been used for sulfation of synthetic mono- and multiple-tyrosine peptides related to known sulfation sites in proteins and peptides at analytical levels to establish the enzyme specificities.[11–13] Preparative sulfations have not been carried out to date. A novel type of arylsulfotransferase produced by *Eubacterium A-44* which is part of the human intestinal flora, has recently been discovered.[109,110] This enzyme catalyzes the transfer of a sulfate group from phenolic sulfate, but not from 3′-phosphadenosine-5′-phosphasulfate, to other phenolic compounds. Using 4-nitrophenylsulfate as a donor substrate and tyrosine-containing peptides and proteins as acceptor substrates it catalyzes the specific sulfation of the tyrosine residues.[111,112] While this enzyme very efficiently sulfates tyrosine derivatives, the

for references see p 450

yields of sulfation of peptides are relatively low (<10%).[112,113] Nevertheless, under optimized conditions CCK-related peptides are sulfated in semi-preparative experiments in acceptable yields.[111]

The enzyme is unstable on incubation and storage. It can be stabilized by immobilization on different supports such as covalent attachment on AH-Sepharose-4B by the use of water soluble EDC. With this immobilized enzyme which retains a high sulfation efficiency for tyrosine derivatives, CCK-peptides have been sulfated to a low extent. Human CCK-33 was obtained in 15% yield with concomitant recovery of the non-sulfated peptide.[115] Using similar conditions the [Tyr^{61},Tyr^{62}]-hirudin analogue as well as the related C-terminal fragment [Tyr^{61},Tyr^{62}]-hirudin-[54–65] have been sulfated with arylsulfotransferase.[114,116] The main products obtained for hirudin were the sulfated-Tyr^{63} species (natural sulfation site by tyrosine protein sulfotransferase[117]) contaminated by sulfated Tyr^{62}, Tyr^{63}, or Tyr^{62} analogues.

Although enzymatic sulfation on a larger scale may represent a challenge, it certainly offers an interesting bypass to postsynthetic chemical sulfation.

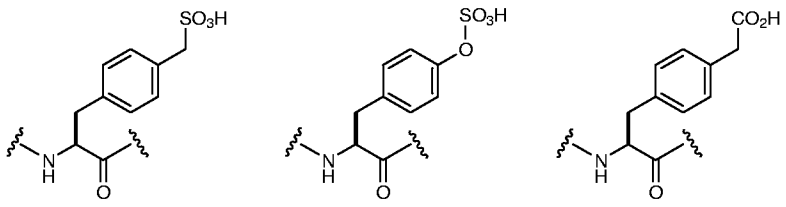
More recently a glycosulfopeptide was prepared by enzymatic sulfation of three tyrosine residues in a 23-mer glycopeptide.[46]

Sulfation of Peptides with Arylsulfotransferase from *Eubacterium A-44*:[114]
A 500 μL soln was prepared with the following concentrations: peptide (0.2 mM), 4-nitrophenylsulfate (1 mM) in glycine/0.1 M NaOH (pH 8.6), $MgCl_2$ (25 mM), and 1 M AST. The reaction was started by the addition of 4-nitrophenylsulfate, and incubation was carried out at 37 °C for 24 h. The progress of the reaction was monitored by HPLC and the sulfated products were isolated by RP-HPLC (linear gradient from 15 to 60% MeOH in H_2O). Fractions containing the sulfated peptides were neutralized with NH_3 to pH 7 and lyophilized; yields: CCK-8 (74%), CCK-12 (47%), phyllocaerulein (68%), caerulein was sulfated only to 25%.

6.6.2 Synthesis of Analogues of Sulfated Tyrosine Peptides

Since the discovery of bioactive tyrosine *O*-sulfate containing peptides, attempts have been made to replace the relatively labile sulfate hemiester with groups capable of mimicking its function in the receptor recognition processes. The first replacements of the tyrosine *O*-sulfate residue in caerulein with tyrosine-4-sulfonic acid, tyrosine-4-sulfonamide, tyrosine-4-sulfamic acid, or with tyrosine *O*-phosphate led to negative results.[118,119] Although the sulfate ester is somewhat larger in size than the phosphate ester, the major difference between them is that the former has a single negative charge at all physiological pH values. Moreover, the sulfate and phosphate esters participate in electrostatic interactions. The sulfate ester can act only as a hydrogen-bond acceptor whereas the phosphate ester acts as both an acceptor and donor for hydrogen bonds.[120] A sulfate ester may therefore interact differently from the phosphate ester at the molecular recognition level.

Scheme 17 Structures of 4-(Sulfomethyl)- and 4-(Carboxymethyl)phenylalanine as Mimetics of Tyrosine *O*-Sulfate



From all the peptidomimetic studies carried out so far, particularly on CCK-peptides, two stable analogues of tyrosine *O*-sulfate have emerged that apparently retain the properties for expression of the full bioactivity of the peptide analogues, i.e. 4-(sulfomethyl)phenylalanine[121] and 4-(carboxymethyl)phenylalanine[122] shown in Scheme 17.

6.6.2.1 Synthesis of 4-(Sulfomethyl)phenylalanine Peptides

Among the methods described for the synthesis of the key intermediate 4-(chloromethyl)-L-phenylalanine,[123–125] direct chloromethylation of a protected L-phenylalanine followed by deprotection represents the simplest procedure.[126] This intermediate is converted into the 4-sulfomethyl derivative which is then used to prepare the N^{α}-*tert*-butyloxycarbonyl-4-(sulfomethyl)phenylalanine monosodium salt[125] and the related Fmoc derivative.[126]

The N^{α}-*tert*-butyloxycarbonyl-4-(sulfomethyl)phenylalanine monosodium salt is used for the synthesis of CCK-related peptides in solution and on solid support using the BOP/DIPEA coupling reagent.[121,125] The full stability of the sulfomethyl group even to hydrogen fluoride treatment (0 °C, 1 h) indicates that there are no limitations in the protection strategies and in the resin cleavage/deprotection step, as demonstrated by the synthesis of human CCK-33 and CCK-58 related analogues with both Boc and Fmoc protection.[126,127]

H-Phe(4-CH_2Cl)-OH · HCl:[126]
A mixture of Ac-Phe-OEt (20 g, 86 mmol), chloromethyl methyl ether (100 mL, 1.3 mol) (**CAUTION:** *highly carcinogenic*) and $ZnCl_2$ (44 g, 320 mmol) was stirred for 50 h at 0 °C. After removal of the chloromethyl methyl ether, the oily residue was dried and distributed between EtOAc and H_2O. The aqueous soln was extracted with EtOAc and the combined extracts were washed with sat. NaCl, sat. $NaHCO_3$, and sat. NaCl, dried ($MgSO_4$) and concentrated. Ac-Phe-4-(CH_2Cl)-OEt crystallized on cooling to give colorless needles; yield: 18.5 g (76%); mp 93–95 °C; ^{1}H NMR (DMSO-d_6, δ): 4.80 (s, 2H, CH_2Cl). Ac-Phe(4-CH_2Cl)-OEt (6.0 g, 21 mmol) was refluxed in 12 M HCl (60 mL) for 4 h. The mixture was concentrated, and the resulting colorless precipitate was filtered, washed with cold 6 M HCl, and dried; yield: 3.8 g (72%); mp 180 °C (dec); ^{1}H NMR (DMSO-d_6, δ): 3.07 (d, 2H, β-CH_2), 4.12 (m, 1H, α-CH), 4.71 (s, 2H, CH_2Cl), 7.22 and 7.36 (d, 4H, aromatic protons), 8.37 (bs, 3H, NH_3^+).

H-Phe(4-CH_2SO_3Na)-ONa:[125,126]
H-Phe(4-CH_2Cl)-OH · HCl (0.78 g, 3.1 mmol) was refluxed for 2 h in H_2O (20 mL) in the presence of Na_2SO_3 (3 g, 24 mmol). After removal of the H_2O, the resulting solid was washed with EtOH, collected and dried giving a mixture of the product and salts; yield: 3.56 g. This mixture was used in the following step without further purification.

Boc-Phe(4-CH_2SO_3Na)-OH:[125]
To an ice-cold soln of H-Phe(4-CH_2SO_3Na)-ONa (3.56 g) in H_2O/dioxane (1:2, 60 mL), 1 M NaOH (15 mL) and Boc_2O (0.84 g, 1.8 equiv) were added. The soln was stirred at 0 °C for 1 h and overnight at rt before addition of a second portion of Boc_2O (1 equiv). After 1 h the solvents were removed and the residue was dissolved in H_2O. The soln was extracted with Et_2O, acidified with 1 M HCl to pH 2, and then saturated with NaCl. The product was extracted with THF and the soln was dried (Na_2SO_4). The solvent was removed and the residue chromatographed on silica gel ($CHCl_3$/MeOH 7:3); yield: 0.5 g {42% calcd from H-Phe-4-(CH_2Cl)-OH · HCl}; $[\alpha]_D^{20}$ +13 (*c* 1, H_2O). ^{1}H NMR (D_2O, δ): 1.32 (s, 9H, Boc), 2.88 and 3.15 (dd, 2H, β-CH_2), 4.16 (s, 2H, CH_2SO_3Na), 4.18 (m, 1H, α-CH), 7.27 and 7.37 (d, 4H, aromatic protons).

Fmoc-Phe(4-CH_2SO_3Na)-OH:[126]
To an ice-cold soln of H-Phe(4-CH_2SO_3Na)-ONa (6.0 g) in H_2O (120 mL) was added Fmoc-OSu (4.2 g, 12.5 mmol) in dioxane (120 mL) at pH 9. The mixture was stirred at 0 °C for 30 min and at pH 9 overnight at rt. The dioxane was removed, the excess Fmoc-OSu extracted with Et_2O, and the aqueous soln acidified with 1 M HCl to pH 5 and lyophilized. The residue was dissolved in MeOH, insoluble material was removed by filtration, and the filtrate taken to dryness. The residue was dissolved in H_2O, the soln was extracted with EtOAc and lyophilized; yield: 5.3 g {88% calcd. from H-Phe(4-CH_2Cl)-OH · HCl}. An analytical sample was chromatographed on silica gel ($CHCl_3$/MeOH/AcOH 65:5:5); mp 184 °C (dec); EI-MS *m/z*: 480.1 [M+H].

for references see p 450

6.6.2.2 Synthesis of 4-(Carboxymethyl)phenylalanine Peptides

The 4-(carbomethoxymethyl)-D,L-phenylalanine benzyl ester is obtained by catalytic phase-transfer alkylation of the dibenzophenone imine of glycine benzyl ester with methyl 4-(bromomethyl)phenyl acetate according to the well established method described by O'Donnell and Bennett,[128] followed by mild acid hydrolysis of the Schiff base.[122] Its N^{α}-*tert*-butyloxycarbonylation and subsequent hydrogenolysis generates the intermediate *rac*-Boc-Phe(4-CH_2CO_2Me)-OH, selectively protected at the N^{α}-amino and ω-carboxy group for the synthesis of peptides. It is used for the synthesis of a CCK-7 analogue on an MBHA-resin.[122] Upon cleavage from the resin with hydrogen fluoride/*p*-cresol (9:1) (0 °C, 1 h), the ω-methyl ester is saponified (0.5 M NaOH), a reaction that can be carried out on simple peptides, but is not advisable for more complex synthetic peptides.

Enantiomeric resolution via the acetyl derivative with acylase or chymotryptic hydrolysis of an alkyl ester was not attempted. Conversely, a stereoselective synthesis of the Boc-Phe(4-CH_2CO_2tBu)-OH has recently been reported[129] which, however, lacks selective protection for incorporation to any position of a synthetic peptide.

***rac*-Boc-Phe(4-CH_2CO_2Me)-OH:**[132]
Crude *rac*-Phe(4-CH_2CO_2Me)-OBzl obtained by the O'Donnell and Bennetts'[122] method was converted in the standard manner into the N^{α}-Boc derivative with Boc_2O; yield: (77%) and then hydrogenated in MeOH over Pd/C. The product was chromatographed on silica gel (CH_2CH_2/MeOH 9:1); yield: 90%; mp 127 °C; ^{1}H NMR (DMSO-d_6, δ): 1.27 [s, 9H, $(CH_3)_3C$], 2.77 and 2.97 (dd, 2H, β-CH_2), 3.52 (s, 3H, CH_3), 3.55 (s, 2H, 4-CH_2), 3.94 (m, 1H, α-CH), 6.67 (brs, 1H, NH), 7.11 (s, 4H, aromatic protons).

6.6.3 Sulfated Hydroxyamino Acid Peptides

To date, the O-sulfation of hydroxyamino acids as residues in proteins or as free amino acids has not been detected. Similarly, desulfation by sulfatases has been observed only for sulfated serine, but it could not be attributed unambiguously to a sulfatase.[130] This lack of natural occurrence explains why only marginal attention has been paid to the synthesis of sulfated hydroxyamino acid peptides.

The O-sulfated hydroxyamino acids show in IR (KBr) strong absorption bands in the ranges of $\tilde{\nu}_{max}$ 1210–1250 and 770–810 cm^{-1}.[51,131] Both acid (6 M HCl, 110 °C, 24 h or 3 M 2-sulfanylethanesulfonic acid, 110 °C, 22 h) and alkaline hydrolysis [0.2 M $Ba(OH)_2$, 110 °C, 24 h] lead to hydrolysis of the O-sulfated hydroxyamino acids.[130] Correspondingly, for quantitative identification of O-sulfated hydroxyamino acid residues in synthetic peptides, amino acid analysis of the enzymatic hydrolysates obtained with aminopeptidase M or leucine aminopeptidase has been used, e.g. to assess the integrity of serine *O*-sulfate[132] and ε-hydroxynorleucine *O*-sulfate.[132] The O-sulfated amino acids elute using an amino acid analyzer under conditions where a citrate buffer of pH 3.25 is changed to a citrate buffer of pH 4.30 before aspartic acid at t_R values similar to cysteic acid in the following order: Hyp(SO_3H)<Ser(SO_3H)<Thr(SO_3H)<Tyr(SO_3H).[51]

6.6.3.1 Synthesis of Hydroxyamino Acid *O*-Sulfate Derivatives

Sulfation of hydroxyamino acids can possibly be carried out with all the sulfating agents described in Section 6.6.1.1,[38] but experimental details have only been reported for the method of sulfation with concentrated sulfuric acid[130] and with chlorosulfonic acid/TFA.[77]

H-Ser(SO_3K)-OH:[130]
The amino acid H-Ser-OH (1 g, 9.5 mmol) was added at rt to concd H_2SO_4 (5 mL) and the mixture was stirred until the material dissolved (10 min). The mixture was kept in vacuo for 1 h and then poured into anhyd Et_2O (400 mL) at −17 °C. After 15 min the supernatant was decanted and the residue was washed with Et_2O (200 mL portions) at −17 °C. The sticky white residue was dissolved in ice-cold H_2O (100 mL) and inorganic SO_4^{2-} was precipitated with ice-cold sat. aq $Ba(OH)_2$ until the solution was slightly alkaline. Precipitated $BaSO_4$ was removed by centrifuging and the excess of Ba^{2+} ions were precipitated as $BaCO_3$ by passing CO_2 through the cold soln. The suspension was centrifuged and taken to dryness. The residue was extracted with ice-cold H_2O (20 mL) and, after centrifuging, the clear extract was passed through a Dowex 50 ion-exchange resin (H^+ form). The column was washed with H_2O and the combined eluates were adjusted to pH 7.8 with 5% aq KOH before being taken to dryness. The residue was dissolved in the minimum amount of cold H_2O and precipitated by the dropwise addition of ice-cold EtOH to give H-Ser(SO_3K)-OH as its hemihydrate as colorless crystals; yield: 0.81 g (37%); $[\alpha]_D^{16}$ −9.5 (*c* 5.0, H_2O); $[\alpha]_D^{16}$ +8.7 (*c* 3.4, 1 M HCl).

H-Ser(SO_3H)-OH·HCl:[77]
To a soln of H-Ser-OH (1.05 g, 0.01 mol) in TFA (10 mL) was added $ClSO_3H$ (1.75 g, 0.015 mol) in small portions under vigorous stirring. A colorless crystalline product separated within a few min. After standing for 20 min at rt the excess reagent was extracted with EtOH (1 mL) and the product was precipitated by adding Et_2O. The precipitate was dissolved in a minimum amount of H_2O and crystallized by addition of acetone giving H-Ser(SO_3H)-OH·HCl; yield: 2.04 g (92%); mp 230 °C; $[\alpha]_D^{20}$ +9.8 (*c* 3.4, 1 M HCl).

H-Thr(SO_3K)-OH:[130]
Sulfation of H-Thr-OH was carried out as described for H-Ser(SO_3K)-OH; after 1 h the mixture was poured into butan-2-one (400 mL). The final product was precipitated from the aqueous soln with Et_2O/butan-2-one (1:1) to give the product as its monohydrate; yield: (40%); $[\alpha]_D^{18}$ –20.6 (*c* 9.7, H_2O).

H-Thr(SO_3H)-OH·HCl:[77]
The title compound was prepared as described for H-Ser(SO_3H)-OH·HCl; yield: 89%; mp 240 °C; $[\alpha]_D^{20}$ −3.7 (*c* 2, H_2O).

H-Hyp(SO_3K)-OH:[130]
This preparation was identical to that of H-Ser(SO_3K)-OH except that the mixture was poured into crushed ice and H_2O (300 mL) containing $Ba(OH)_2$ (50 g) rather than into Et_2O. The final product was crystallized by dropwise addition of EtOH/Et_2O (1:1) to give the product; yield: 0.6 g (32%); $[\alpha]_D^{18}$ −26.1 (*c* 10.5, H_2O); $[\alpha]_D^{18}$ −12 (*c* 9.8, 1 M HCl).

6.6.3.2 Synthesis of O-Sulfated Hydroxyamino Acid Peptides

Taking into account the relatively high stability of O-sulfated hydroxyamino acids under acidic conditions (see Table 5), postsynthetic sulfation of suitably protected peptides with pyridine/SO_3 followed by TFA cleavage of the acid-labile protecting groups allows for the preparation of CCK analogues where tyrosine *O*-sulfate is replaced by sulfated ε-hydroxynorleucine[133] or serine,[132] respectively. Similarly sulfation of LHRH with pyridinium acetylsulfate (3 equiv) in pyridine produces within minutes the [Ser(SO_3H)]^4-LHRH analogue.[38]

Table 5 Stability of O-Sulfated Hydroxyamino Acids under Various Conditions[38]

Amino Acid *O*-Sulfate	0.2 M $Ba(OH)_2$, 110 °C, 24 h	110% TFA, 20 °C, 2 d	HF, 0 °C, 45 min	Organic Solvents, 20 °C, 8 h
H-Ser(SO_3Na)-OH	100% cleavage	stable	5% cleavage	unstable
H-Thr(SO_3Na)-OH	100% cleavage	stable	5% cleavage	unstable
H-Hyp(SO_3Na)-OH	95% cleavage	stable	5% cleavage	unstable

An interesting method for the selective persulfation of hydroxy groups in the presence of tyrosine has been proposed by Berg et al.[134] with the use of 100% H_2SO_4/DMS/TFA (10:30:60) in the "low acidity" deprotection cocktail for resin-bound or free peptides with

for references see p 450

benzyl-type protection. Since the hydroxy groups react with the sulfating reagents, e.g. with pyridinium acetylsulfate, much faster than the phenolic group, essentially selective sulfation of hydroxyamino acid residues in the presence of unprotected tyrosine is achieved. This is exploited in the synthesis of Ac-[Tyr^{27},Thr^{32}]-CCK-[27–33] analogues monosulfated at the threonine or bis-sulfated at the tyrosine and threonine residues.[67] Sulfation is carried out with pyridinium acetylsulfate (38 equivalents) or pyridine/SO_3 (38 equivalents) in pyridine at 20 °C for 16 hours on the PAM-resin-linked peptide for monosulfation and at 45 °C for 4 hours, followed by resin cleavage with dry ammonia. Similarly, bis-sulfation in solution of the Ac-[Tyr^{27},Xaa^{32}]-CCK-[27–33] analogues with Xaa = Ser, Thr, Hyp can be achieved in DMF/pyridine (1:1) with pyridinium acetylsulfate (5 equiv, 24 h).[53]

For the synthesis of peptides in solution with sulfated hydroxyamino acids as synthons N^{α}-Boc protection has been used. Coupling steps are accomplished with DCC using HOBt or pentafluorophenol as additives, and identical coupling procedures are applied for the synthesis on solid supports.[38] Experimental details of these syntheses have not been reported.

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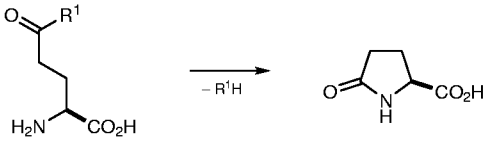
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6.7 Pyroglutamic Acid Peptides

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Direct dehydration of glutamic acid and desamination of glutamine leads to 5-oxo-2-pyrrolidinecarboxylic acid also named pyroglutamic acid (pGlu) or 5-oxo-L-proline, the latter name being misleading as chemically the five-membered lactam ring is no longer an amino acid.[1] The commercially available pyroglutamic acid, first discovered by Haitinger,[2] is generally synthesized by acid-catalyzed cyclization of glutamic acid or related derivatives as outlined in Scheme 1, and the effect of the catalyst on the kinetics of this reaction has been studied in detail.[3] The biological significance of pyroglutamic acid, which is able to permeate the blood brain barrier and acts as a glutamic acid precursor and partial agonist at the glutamate receptor,[4] as well as its use as a versatile building block in asymmetric synthesis[5] have been reviewed.

Scheme 1 Formation of Pyroglutamic Acid from Glutamine, Glutamic Acid, and Related Derivatives



R^1 = OH, NH_2, alkoxy, aryloxy

Natural Occurrence of Pyroglutamic Acid and Pyroglutamyl Peptides

In nature, pyroglutamic acid occurs as the free amino acid in a number of tissues, especially in the skin, in the plasma, and brain[4] or it occupies the N-terminal position of a large variety of peptides and proteins,[6] e.g. of thyrotropin releasing hormone (TRH), of peptides of the gastrin/CCK family, luteinizing hormone releasing hormone (LHRH) and neurotensin, and of depsipeptides, e.g. didemnins.[7] The mechanism by which the pyroglutamyl residue is formed, i.e. by spontaneous or enzyme-assisted cyclization of glutamine, is still not fully understood. The identification of a mammalian glutaminyl cyclase[8] would support a post-translational cyclization of N-terminal glutaminyl residues. Similarly, the enzyme 5-oxoprolinase[9] catalyzes the conversion of pyroglutamic acid into glutamic acid. Since the N-terminal pyroglutamyl residue prevents degradation of peptides by aminopeptidases, there are several bacterial and mammalian pyroglutamyl peptidases known, that specifically remove the pyroglutamic acid residue from the related peptides.[6,10]

Analytical Characterization

During acid hydrolysis under standard conditions (see Section 7.2), pyroglutamic acid is quantitatively converted into glutamic acid. Thus, for analytical characterization of synthetic pyroglutamyl peptides by amino acid analysis of the enzymatic digests, preincubation with pyroglutamyl peptidases is essential.[11] Also, sequence analysis of pyroglutamyl peptides or proteins is only possible upon enzymatic cleavage of the N-terminal pGlu residue.[12] In synthetic steps the integrity of the pGlu residue can be visualized on TLC by a color reaction if it is protected as the urethane derivative. Selective reaction with 4-(4-nitrobenzyl)pyridine followed by exposure to TEA vapor leads to a blue color.[13] The reaction is not observed with other derivatives such as Ac-pGlu-OH, For-pGlu-OH, Tos-pGlu-OH, and pGlu-OH itself.

6.7.1 Synthesis of Pyroglutamic Acid Peptides

Whilst cyclization of α-aminoadipic acid to 6-oxopiperidine-2-carboxylic acid occurs at rates even higher than formation of pyroglutamic acid, cyclization of aspartic acid and α-aminopimelic acid to the corresponding four- and seven-membered lactam rings occurs with some difficulty. Correspondingly, peptides containing such differently sized lactams *per se* are accessible, but have not yet been studied because all naturally occurring bioactive peptides contain exclusively the pyroglutamic acid residue in the N-terminal position.

6.7.1.1 Synthesis of N-Terminal Pyroglutamyl Peptides

Pyroglutamyl peptides are synthetically accessible either by side-chain cyclization of related glutaminyl and glutamyl peptides or by direct acylation of suitably protected peptides with N^{α}-protected or unprotected pyroglutamic acid.

6.7.1.1.1 Cyclization of N-Terminal Glutaminyl Peptides

Although N-terminal glutamyl(γ-alkyl ester) residues, except the *tert*-butyl ester, can be converted into the pyroglutamyl residue,[14,15] N-terminal glutaminyl peptides are generally preferred as precursors of the pyroglutamyl derivatives because cyclization of the glutaminyl residue can be performed under very mild acidic conditions that are usually harmless to the rest of the peptide unless acid-sensitive sequences are present, e.g. Asp-Xaa which under identical conditions leads to the succinimide derivatives (see Vol. E 22a, Section 2.2.2).[14,16–18] Thereby, the rate of cyclization was found to be sequence-dependent at least to some extent.[14,19]

This simple synthetic approach has been extensively investigated in the synthesis of TRH (pGlu-His-Pro-NH_2) and it may be useful for the synthesis of relatively short pyroglutamyl peptides using the optimized conditions reported in Table 1. Furthermore, cyclization to pyroglutamyl peptides and cleavage of acid-labile protecting groups can be performed in a one-pot reaction with TFA under reflux conditions as reported for the synthesis of the N-terminal fragment of human big-gastrin I (HG-34 I).[22]

Table 1 Conditions for Cyclization of H-Gln-His-Pro-NH_2 to Thyrotropin Releasing Hormone[20,21]

Conditions	Reaction Time	Ref
1 M AcOH in H_2O, 40 °C	$t_{1/2}$ 7.4 h	[20]
1 M AcOH in dioxane, 40 °C	$t_{1/2}$ 0.8 h	[20]
8 M AcOH in H_2O, 40 °C	$t_{1/2}$ 2.8 h	[20]
8 M AcOH in dioxane, 40 °C	$t_{1/2}$ 1.5 h	[20]
2 M AcOH in H_2O, rt	100% in 4 d	[21]
glacial AcOH, reflux	100% in 2 min	[21]
30% aq NH_3 in MeOH, reflux	high yield in 2 h	[21]

pGlu-His-Pro-NH_2:[20]
Cyclization of H-Gln-His-Pro-NH_2 (100 mg, 0.264 mmol) was performed in 4 M AcOH in dioxane (20–30 mL) at 40 °C for 8–12 h until the starting material was no longer detectable by TLC. The mixture was diluted with dioxane and lyophilized. The product was chromatographed (Sephadex LH-20, BuOH/MeOH/AcOH/H_2O 6:6:18:70). Fractions were pooled, concentrated, and lyophilized from 10% AcOH; yield: 77 mg (81%); $[\alpha]_D^{23}$ −55.9 (*c* 1, MeOH).

for references see p 459

pGlu-Leu-Gly-Pro-Gln-Gly-OH:[22]
The hexapeptide derivative H-Gln(4,4′-Mbh)-Leu-Gly-Pro-Gln-Gly-OtBu (2.3 g, 2.6 mmol) was refluxed in TFA/anisole (10:1, 22 mL) for 30 min. The solvent was removed and the residue solidified with Et_2O. The solid was partitioned between EtOAc and H_2O, the aqueous phase concentrated, and the residue crystallized (MeOH); yield: 1.13 g (70%); mp 165–166 °C; $[\alpha]_D^{23}$ −55.5 (*c* 0.7, DMF).

6.7.1.1.2 Peptide Synthesis with Pyroglutamic Acid

Since the nitrogen in pyroglutamic acid is part of an amido group and thus difficult to acylate, there is no compelling reason for its protection, besides the observed amelioration of the solubility properties. In fact, acylation of peptides with active esters of unprotected pyroglutamic acid has been reported[23–26] as well as direct coupling of the free acid using coupling reagents such as CDI or DCC.[24,27,28] A convenient synthesis of pGlu-Cl has been described by Rigo et al.;[29] however, in basic media this compound tends to self-condense which limits its use in coupling reactions.

Although acetylation of pyroglutamic acid with acetyl chloride produces the related Ac-pGlu-OH in 95% yield, acylation with benzoyl chloride or with benzyl chloroformate under standard conditions leads to only 20% of the N-acylated products.[30] Correspondingly, the synthesis of N-protected pyroglutamic acid derivatives is generally based on the cyclization of N-protected glutamine or glutamic acid. For the synthesis of Z-pGlu-OH[31] and Boc-pGlu-OH[32] the related Z- and Boc-protected glutamic acid anhydrides are exposed to DCHA to catalyze the cyclization. Conversely, Fmoc-pGlu-OH is synthesized from Fmoc-Glu-OH with thionyl chloride in two steps.[33] The first reaction step results in Fmoc-pGlu-Cl that can be used directly in peptide synthesis or can be converted into active esters suitable for synthesis in solution and on solid supports. Despite protection of the amide nitrogen of the pGlu residue, side reactions may occur as exemplified by the observation of Mazurov et al.[34] that in the synthesis of pGlu-Gln-Pro-NH_2 with Z-pGlu-OPfp about 60% pyroglutamylglutamine piperazine-2,5-dione (diketopiperazine, DKP) is formed as side product. A similar amount of piperazine-2,5-dione was obtained even when unprotected pGlu-OPfp was used.

Z-Glu-Anhydride:[31]
To Z-Glu-OH (5.6 g, 0.02 mol) in THF (50 mL) was added DCC (4.12 g, 0.02 mol) and the reaction allowed to proceed overnight. The DCU was filtered off, the solvent concentrated and the resulting anhydride recrystallized ($CHCl_3/Et_2O$); yield: 97%; mp 86–88 °C.

Z-pGlu-OH · DCHA:[31]
A soln of DCHA (98 mL, 0.5 mol) in Et_2O (150 mL) was added dropwise to Z-Glu-anhydride (131.5 g, 0.5 mol) in THF (200 mL) and Et_2O (400 mL) under stirring. After 12 h, Z-pGlu-OH · DCHA was filtered off, washed carefully with Et_2O, and recrystallized ($CHCl_3/Et_2O$ or MeOH); yield: 221.6 g (quant); mp 199–200 °C.

Z-pGlu-OH:[31]
The Z-pGlu-OH · DCHA salt (44.4 g, 100 mmol) was stirred with 1 M HCl (300 mL) and EtOAc (350 mL) for 1 h. After filtration, the organic phase was washed with H_2O, dried (Na_2SO_4), concentrated, and the residue recrystallized (EtOAc/petroleum ether); yield: 23.2 g (88%); mp 134–135 °C.

Boc-pGlu-ONp:[32]
To Boc-pGlu-OH (1.15 g, 5 mmol; prepared as described for Z-pGlu-OH[31]) and 4-nitrophenol (0.84 g, 6 mmol) in THF (5 mL) was added DCC (1.23 g, 5.3 mmol) at 0 °C. After 6 h, the DCU was filtered off, the solvent removed, and the residue washed in EtOAc with 5% aq $NaHCO_3$, dil citric acid, and H_2O. The organic layer was concentrated and the crude product recrystallized twice (EtOAc/petroleum ether); yield: 0.6 g (34%); mp 146 °C.

Fmoc-pGlu-Cl:[33]
A soln of Fmoc-Glu-OH (369.4 mg, 1 mmol), $SOCl_2$ (2 mL, 27.5 mmol), and a drop of DMF in CH_2Cl_2 (20 mL) were stirred for 3 days. The solvent was removed, and CH_2Cl_2 was added and then removed.

The residue was taken up in CH_2Cl_2 and washed with 1 M HCl, 5% aq $NaHCO_3$, and H_2O. After drying ($MgSO_4$), the solvent was removed and the residue recrystallized (CH_2Cl_2/EtOH) by slowly removing the CH_2Cl_2; yield: 314 mg (85%); mp 40–50 °C; $[\alpha]_D$ −3.9 (*c* 1, $CHCl_3$).

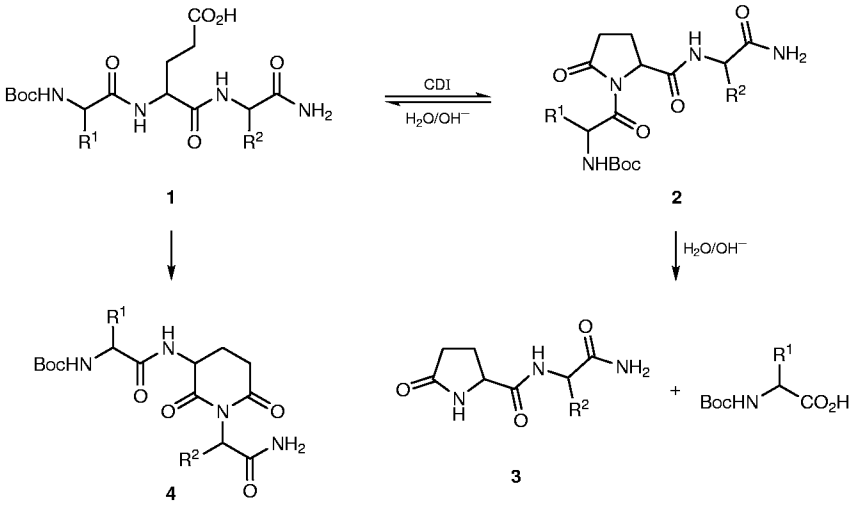
Fmoc-pGlu-OH:[33]
To freshly prepared Fmoc-pGlu-Cl (from 1 mmol of Fmoc-Glu-OH) in THF (10 mL) was added 1 M HCl (10 mL). The soln was stirred for 24 h and then made alkaline by the addition of 5% aq $NaHCO_3$. The aqueous layer was washed with CH_2Cl_2, acidified with 1 M HCl, and extracted with CH_2Cl_2. The soln was dried ($MgSO_4$) and concentrated. The residue was crystallized (CH_2Cl_2/petroleum ether); yield: 282 mg (80%); mp 157–158 °C; $[\alpha]_D$ −21.6 (*c* 1, $CHCl_3$).

Fmoc-pGlu-OSu:[33]
To freshly prepared Fmoc-pGlu-Cl (from 1 mmol of Fmoc-Glu-OH) in CH_2Cl_2 (10 mL) was added a soln of HOSu (115 mg, 1 mmol) and NMM (110 mL, 1 mmol) in CH_2Cl_2 (10 mL). After 24 h, the soln was washed with 1 M HCl, 5% aq $NaHCO_3$, and H_2O. After drying ($MgSO_4$), the solvent was removed and the residue recrystallized (CH_2Cl_2/EtOH) by slowly removing the CH_2Cl_2; yield: 377.5 mg (84%); mp 105–108 °C; $[\alpha]_D$ −43.5 (*c* 1, $CHCl_3$).

6.7.1.2 Synthesis of Peptides with *endo*-Pyroglutamyl Residues

Cyclization of glutamyl residues located in *endo*-positions of the peptide chain to give pyroglutamyl residues **2** is performed by activation of the free γ-carboxy group with CDI[35] as shown in Scheme 2. Alternatively, the activated γ-carboxy group can react with the next amido group to form the six-membered cyclic imide **4**. The extent of formation of this byproduct is dependent on the relative bulkiness of the R^1 and R^2 side chains, but usually is less than 1%. However, similar reaction of Boc-Ile-Glu-Gly-NH_2 provides only 2% of the five-membered lactam and 95% of the six-membered imide. Peptides with internal pyroglutamyl residues are sensitive to hydrolysis. In fact, at pH 8 such peptides are readily cleaved to form the more stable peptide fragments with N-terminal pGlu residues **3**. To a lesser extent, pGlu is hydrolyzed to the free glutamate **1**.[35]

Scheme 2 Synthesis of Peptides with Internal Pyroglutamyl Residues[35]



for references see p 459

Due to the low reactivity of the nitrogen, incorporation of the pyroglutamic acid into *endo*-positions by stepwise chain elongation is difficult (see ref[49] and refs cited therein). It may be achieved using suitably protected aminoacyl-pyroglutamic acid derivatives as dipeptide synthons. These are accessible in satisfactory yields, e.g. by acylation of pGlu-OBzl with N-protected amino acid pentafluorophenyl esters in the presence of NaH or LiHMDS.[49]

Boc-Gly-pGlu-Val-NH_2:[35]
To a soln of Boc-Gly-Glu-Val-NH_2 (0.40 g, 1.0 mmol) in DMF (2.0 mL) was added CDI (0.20 g, 1.2 mmol) at −20 °C. The soln was stirred at −20 °C for 0.5 h and at 20 °C for 1 h and then concentrated. Upon addition of Et_2O, the crude product was filtered off and reprecipitated from DMF/Et_2O; yield: 304.5 mg (79%).

6.7.2 Pyroglutamyl Peptides as Side Products in Peptide Synthesis

The high tendency of glutamine and to a lesser extent of glutamic acid and related derivatives to form pyroglutamic acid which is exploited for the synthesis of pyroglutamyl peptides (Section 6.7.1.1.1), at the same time represents a source of undesirable side reactions in peptide chemistry (for reviews, see refs[16–18]).

6.7.2.1 Peptides with N-Terminal Glutamic Acid

Since cyclization of N-terminal side-chain protected glutamyl residues to pyroglutamyl derivatives is a slow reaction that does not occur in the case of γ-*tert*-butyl ester protection, this side reaction usually raises no major problem during chain elongation steps. However, it might affect the stability of already assembled, fully deprotected peptides with glutamic acid at their N-termini. Cyclization is catalyzed both by bases and carboxylic acids,[4] but also near neutral conditions favor this reaction.[36] HF-mediated peptide deprotection has also been reported to cause significant 5-oxo-2-pyrrolidinecarbonyl formation even in *endo*-peptide positions,[37] that leads to autohydrolysis of the glutamyl peptides. Autohydrolysis of glutamyl peptides on storage has similarly been attributed to intermediate 5-oxo-2-pyrrolidinecarbonyl formation.[38]

Base-catalyzed pyroglutamic acid formation has been reported for Fmoc-Glu(OBzl)-peptide in piperidine deprotection steps. By the use of short deprotection times (1 min) with 50% piperidine/DMF this side reaction was no longer detectable.[39]

6.7.2.2 Peptides with N-Terminal Glutamine

Base-catalyzed pyroglutamyl formation with N-terminal glutaminyl peptides is an extremely slow reaction (0.1% per hour) as determined by exposure of H-Gln-Ala-Val-PAM-resin to 50% piperidine/DMF or 5% DIPEA/CH_2Cl_2.[40] Correspondingly, formation of this type of side product using the Fmoc/tBu chemistry is negligible. Cyclization may, however, occur during the coupling reactions since the amino acid carboxy group and HOBt promote this side reaction (see Table 2).[40] Therefore, acylation of glutaminyl peptides with N^{α}-protected amino acids as active esters is recommended. In solution synthesis this side reaction was found to strongly depend upon the rates of peptide-bond formation[41] and generally it is observed when N^{α}-protecting groups of glutaminyl peptides are cleaved by acidolysis[42,43] or by catalytic hydrogenolysis in acetic acid as solvent.[44] In the latter case it can be largely prevented by titration of the amino group with hydrogen chloride[45] and generally the rates of this side reaction are markedly reduced by side-chain protection of the glutaminyl resi-

dues (see Vol. E 22a, Section 2.3.2).[46] Several attempts to examine in more detail the influence of the peptide sequence as well as of different solvents and reagents on pGlu formation have led to inconsistent results.[19] In general, however, it was found that weak acids and bases favor this side reaction. In mixtures of TFA in dichloromethane, as generally used in SPPS for the Boc/Bzl strategy, it occurs to a lesser extent.[47,48]

Table 2 pGlu Formation in H-Gln-Ala-Val-OCH_2-PAM-Resin by Treatment with Various Reagents[40]

Reagent	Solvent	Concentration	Rate (% · h^{-1})	Ref
HCl	dioxane	4 M	0.0	[40]
TFA	CH_2Cl_2	50%	1.0	[40]
TFA	CH_2Cl_2	0.5%	4.7	[40]
DIPEA	CH_2Cl_2	5%	0.1	[40]
piperidine	DMF	50%	0.1	[40]
Boc-Ile-OH	CH_2Cl_2	0.01 M	9.2	[40]
Boc-Ile-OH/0.02 M DIPEA	CH_2Cl_2	0.01 M	3.3	[40]
Boc-Ile-OH	DMF	0.01 M	3.6	[40]
HOBt	CH_2Cl_2	0.1 M	10.2	[40]
HOBt	DMF	0.1 M	5.2	[40]

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6.8 Synthesis of Cyclic Peptides

C. Gilon, C. Mang, E. Lohof, A. Friedler, and H. Kessler

In an era when combinatorial peptide libraries are being actively used for the discovery of drug lead compounds, this section covers natures multi-millennial lead discovery process in providing a rich harvest of bioactive molecules, which have become the focus of pharmacophoric exploration over many years, i.e. the cyclic peptides. Peptides containing cyclic, bicyclic, and polycyclic structural elements occur naturally as products of posttranslational modifications or of multienzyme synthesis[1,2] and are classified into two categories, i.e. homodetic and heterodetic cyclic peptides. In homodetic cyclic peptides the constituent amino acids are exclusively connected through peptide bonds, while in heterodetic cyclic peptides other functions such as esters (resulting in peptolids, also termed depsipeptides), ethers, thioethers (sulfides), and most commonly, disulfide bridges are part of the ring structure, but C–C bonds are also encountered. With the exception of the disulfide- and thioether-bridged cyclic structures, as well as some heterocycles such as oxazoles and thiazoles found in peptides derived from posttranslational processing of protein precursors, the largest diversity of homodetic and heterodetic cyclic peptides is produced in bacteria, fungi, and in lower animals and plants, and many of these exhibit unusual biological properties. Due to the exceedingly diverse structures of these natural compounds, the emphasis of this section will be on the synthesis of homodetic cyclic peptides that exploit the conformational and functional requirements of constrained cyclic peptides (ring size of cyclic di- to heptapeptides) for pharmacological purposes.

The interest in homodetic cyclic peptides began in the 1940s when the antibiotic gramicidin S was found to be a cyclic decapeptide.[3–5] Gramicidin S was also the first naturally occurring cyclic peptide synthesized in the laboratory.[6,7] Commercially available drugs such as the immunosuppressant cyclosporin indicate that research in this field still continues, despite the pharmaceutical thrive for low-molecular-weight compounds.[8,9] Many other antibiotics, antitumor, immunosuppressing agents, and toxins from different sources have been shown to contain cyclic structures that result in significantly higher metabolic stability, increased potency, receptor selectivity, and improved bioavailability, and above all, they are synthetically challenging.

In addition to the synthetic challenge of natural cyclic peptides, these compounds are often isolated in very low yields and thus their structure cannot always be determined without ambiguity. Consequently, total synthesis represents the safest way to prove their constitution, configuration, and even biological activity, since frequently this results from highly active minor contaminants.[10–12] Moreover, synthesis of the natural compounds allows for structural variations as required for structure–activity relationship (SAR) studies to identify the important pharmacophores and their orientation in space. Herein the reader is not provided with a comprehensive overview of the vast list of different natural cyclic peptides, but rather directed to comprehensive reviews.[1,13] Nonetheless, some important and characteristic representative members will be reported in the context of the synthetic methods.

In addition to the synthesis of natural compounds, the constrained geometry of cyclic peptides has been extensively exploited to investigate correlations between primary and secondary structures of protein sequences and thus gaining insight into protein folding events, as well as analyzing the function of selective metal ion chelating agents (ionophores). However, artificial cyclization of peptides is also an important, effective, and attractive approach to restrict the conformational space of peptidic structures,[14–17] thus, exploiting the reduced entropic penalty for improving binding and often selectivity toward a given biological receptor.[18,19] Consequently, much higher bioactivities can be achieved, when the rigidified structure is closer to the biologically active conformation (often identical to the receptor-bound conformation).[20] However, it should be kept in mind that this only applies when the conformational space of the cyclic peptide overlaps with the bioactive conformation,[14,21–23] and that often cyclization may lead to inactive peptides because of con-

for references see p 529

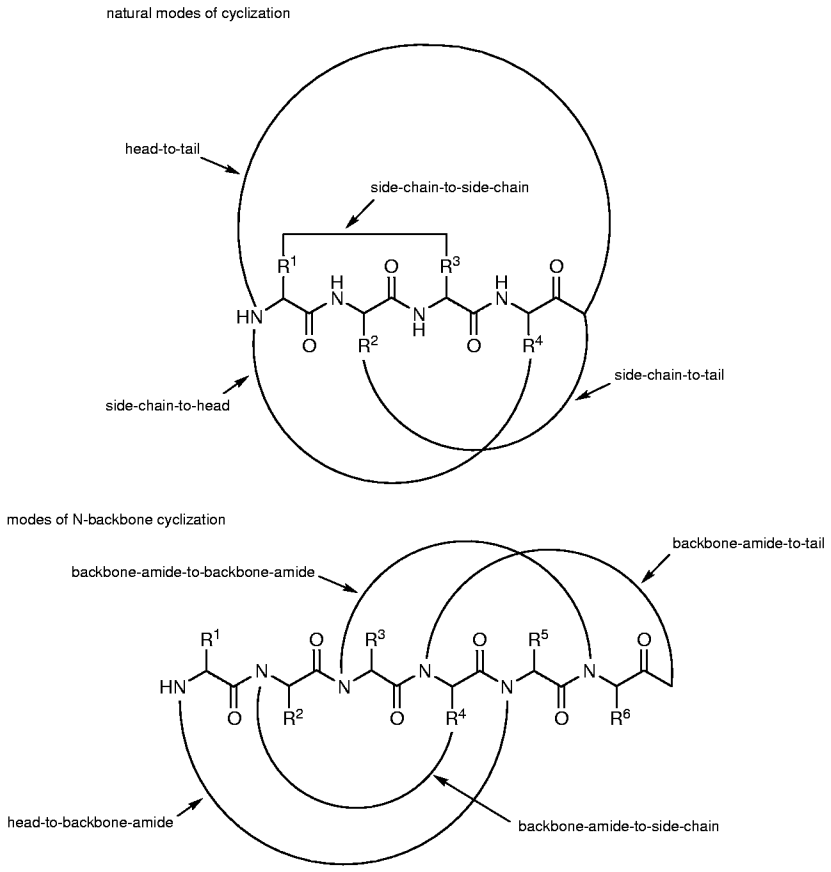
formational mismatch. Since different receptor subtypes are frequently known to bind the identical bioactive sequence in different conformations, different cyclic structures of the identical peptide sequence may serve to address selectively the receptor subtypes as well documented for cyclic RGD peptides.[24,25] Cyclization of peptides may also serve to probe the optimal stereochemical orientation of side chains for bioactivity.[26] The information derived may then be applied in the design of nonpeptidic drugs, i.e. of peptidomimetics, which contain the pharmacophoric groups as in the conformational space of the rigidified cyclic peptides, thus mimicking the "active loop" of a macromolecular ligand with its spatial and electronic as well as binding properties.[27,28]

6.8.1 General Principles

6.8.1.1 Modes of Cyclization to Homodetic Cyclic Peptides

Homodetic cyclic peptides are classified according to the location of the ring juncture as shown in Scheme 1. The most common mode of cyclization is head-to-tail where ring closure occurs by amide bond between the N-terminal amino group and the C-terminal carboxy group. Additional ring closures involve side-chain functionalities such as head-to-side-chain, side-chain-to-side-chain or side-chain-to-tail cyclizations. Moreover, in backbone cyclizations[29] (see Section 6.8.3.2.4) the backbone amides may be linked to a second backbone amide or to side chains as well as to the termini of the peptide.

Scheme 1 Natural Modes of Cyclization and Backbone Cyclization

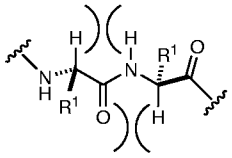


6.8.1.2 Conformational Properties of Cyclic Peptides

The important difference between linear and cyclic peptides is the reduced conformational flexibility imposed by the cyclization.[14,15] Cyclic peptides have, therefore, been under intense investigation with respect to their conformational preferences[30,31] and their different turn motifs, e.g. reverse turns.[32–35] To differentiate between the structural influences of the numerous sequence elements the following special concepts have been described:[16] (1) Global constraints are structural characteristics that restrict the whole molecule, e.g. the macrocyclic structure of the peptide framework and (2) local constraints comprise a large number of structure-inducing building blocks, such as proline, D-amino acids, and turn mimetics.

Although cyclization significantly reduces the accessible conformational space, cyclic peptides often maintain considerable flexibility.[36,37] The more the backbone prefers a single conformation, the more the side chains do the same, at least for the χ_1-angle. The preferred backbone conformation is primarily determined by the tendency to minimize the allylic strain,[24,38] as shown in Scheme 2. If possible, the C^{α}-proton is *syn* to the carbonyl oxygen of the amino acid preceding it in the sequence ($\varphi \sim -120°$) and to the NH of the following amino acid ($\psi \sim +120$–$180°$). For this reason glycine (two α-hydrogen atoms) introduces flexibility into the peptide chain/ring. Frequently a flip of the peptide bond is observed which is associated with conformational changes of both adjacent φ- and ψ-bonds, namely the interconversion of βI and βII turns.[39,40]

Scheme 2 Minimized Allylic Strain in a Peptide Chain[38]

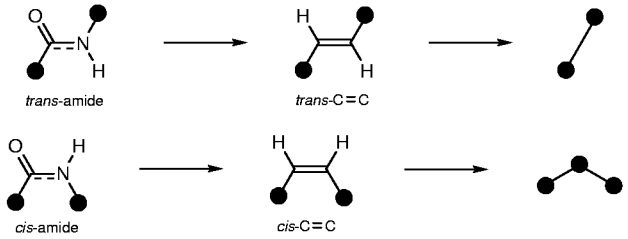


Another important characteristic of cyclic peptides is the hydrogen bonding network. The resulting turn topology is essentially determined by the chirality of the amino acid residues in the sequence.[41] Generally only a limited number of intramolecular hydrogen bonds is possible, which contributes to the stability of a given spatial structure. As a consequence, in nonpolar environments, such as membranes, constraints are imposed on the peptide conformation, while in polar environments the intramolecular hydrogen bonds provide only a minor energy contribution, which should not be overestimated in its structural effect.[42]

The Dunitz–Waser concept,[43] based on systems with isolated double bonds, has been used to give a better understanding of the conformational behavior of cyclic peptides.[26,44–46] This concept utilizes the similarity of a *cis*-C=C bond and a methylene group with respect to conformational flexibility. In peptides the *cis*-amide bond is compared and considered as equivalent to a *cis*-C=C bond because of its partial double bond character and thus it is replaced by a methylene group. Conversely, a *trans*-peptide bond is considered as a *trans*-C=C bond and thus is replaced according to Dunitz and Waser by an elongated single bond (see Scheme 3). Correspondingly, cyclic pentapeptides with all peptide bonds in *trans*-configuration correlate with the conformations of the highly flexible cyclopentane,[26,47] whereas cyclic hexapeptides are correlated to cyclohexane.[26]

for references see p 529

Scheme 3 Application of the Dunitz–Waser Concept[43] to Peptide Bonds[26,44–46]



Conformational analysis can be performed using a variety of methods.[16] Most important are X-ray analysis and NMR spectroscopic techniques, which allow detailed insights into the topology of the peptides. NMR combined with molecular dynamic calculations provides the spatial structure of the peptide, but also its dynamics in solution.[36,48,49] Additional information can be obtained from FTIR,[50–52] CD,[51,53–55] or Raman spectroscopy.[56,57] The results derived from such conformational analysis of cyclic peptides have a considerable impact on the study of the bioactive conformations of peptides and on the design of cyclic peptides as proteinomimetics.[24,58,59,630]

6.8.1.3 Cyclization of Peptides

In principle cyclization can be performed in solution and on solid support (on-resin cyclization). Usually peptides prepared in solution are also cyclized in solution, whereas peptides prepared by SPPS may be cyclized either on the solid support or detached from the resin as suitably protected linear precursors for subsequent cyclization in solution. In both methods the principle of high dilution is applied to avoid oligomerization reactions,[60] since at higher concentrations of the activated peptide bimolecular reactions such as dimerization, trimerization,[61,62] or polymerization may occur.[63,64] Thereby, regioselective cyclization is essential to avoid the formation of unwanted side products. This may be achieved in two ways: (1) The cyclization step is performed on the fully protected peptide with only the sites of amide bond formation in deprotected form. In this case a second order of orthogonality in the protection of the sites of ligation is needed (for potential protection schemes see Vol. E 22a, Section 2). Even when peptidomimetic structures or other bonds (e.g., ester bonds) are incorporated into cyclic peptides, cyclization via an amide bond is generally preferred as the last step prior to the final deprotection (e.g., synthesis of virginiamycines[65,66]) and (2) the cyclization step is performed by a selective chemical reaction such as disulfide bond formation or chemical ligation, where other functionalities present in the peptide do not interfere. In such cases, the cyclization step may be performed even on the deprotected peptide (see Section 6.8.2.1.6).

For the synthesis of homodetic cyclic peptides, many methods of peptide bond formation can be applied in the ring-closure reaction, although cyclization steps require special consideration.[67–69] In fact, cyclization yields depend upon many factors such as the ring size, the site of cyclization, the number and position of structure-inducing amino acid residues such as D-configured or N-alkylated amino acids, glycine, or proline, the sequence characteristics of the linear precursor, and particularly upon the nature of the side chains of residues distal to the site of cyclization, and finally the reaction conditions (coupling reagents, solvent, temperature, concentration, etc.). When cyclization is performed with protected peptides, the number and size of the protecting groups of the amino acid residues between the sites of cyclization may also dramatically affect the rate of cyclization. Although some rules have been derived for specific cases it is difficult to predict the correct cyclization strategy for maximum yields, and thus optimal conditions have to be elaborated in each case.

Several reviews are available that contain additional information on the synthesis of cyclic peptides.[70–72]

6.8.1.3.1 Head-to-Tail Cyclization

The factor that mostly affects head-to-tail cyclization is the ring size. For that reason cyclic tripeptides are only accessible when both peptide bonds in the linear precursor are in the *cis* configuration.[31] This restriction limits, among other problems, the sequences to those containing N-alkylated amino acids or proline (see Section 6.8.2.2.2). In cyclic tetrapeptides only one amide bond has to be *cis*-configured,[31] and cyclization of all-L-sequences is difficult if not impossible.[73] These limitations are not as severe in cyclic pentapeptides, although the cyclization reaction remains difficult and epimerization occurs frequently.[58] The synthesis, however, is facilitated by the use of new more reactive and racemization-suppressing coupling reagents.[74,75] Cyclic hexapeptides share the same problems as cyclic pentapeptides, although to a significantly lesser extent,[76] and cyclization of heptapeptides or of even larger precursors generally is straightforward; with larger ring sizes cyclization rates can decrease significantly.

A second crucial factor that affects the cyclization yield is the conformation of the linear precursor peptide.[19,58,62,63,77,78] Obviously, cyclization occurs without major difficulties if the bridgeheads of the linear precursor are located in spatial proximity; this was well-evidenced by measuring the dielectric increments and their correlation with the cyclization tendency as early as 1963,[79] and by conformational analysis of the linear precursors by CD spectroscopy.[62] This conformational aspect is also responsible for the difficult cyclization of all-L- and all-D-sequences. Such peptides are mostly in the preferred *trans*-configured secondary amide form and thus in an extended conformation in solution[69,80] that causes the sites of cyclization to be in distal position.[58,73,80] These aspects have also been investigated theoretically in detail.[78] Experimental approaches that facilitate such difficult cyclizations make use of temporary protection of the backbone amide with Boc or Hmb[81–83] (see Vol. E 22a, Section 2.3.2) to generate tertiary amides that can assume both the *cis* and *trans* configuration, and thus force homochiral peptides (i.e., all-L- and all-D-sequences) into conformations more favorable for cyclization. However, routine application of this approach has not yet been reported.[69,75] A *cis*-peptide bond can also be forced by an introduction of "pseudoprolines" [N, X (X = O, S) acetals from side chain to the backbone nitrogen, preferably in the dimethylated acetal structure] in a Xaa-Ser, -Thr, or -Cys sequence.[84–86] This modification is readily removed after successful cyclization. Introduction of single L- or D-proline residues into homochiral sequences appears to play a key role in facilitating cyclization.[62,87] Earlier reports on the favorable influence of proline on cyclizations[58] have been disproved in other cases and it has even been found to exert the opposite effect.[62,87] Conversely, a D-Pro-L-Pro unit apparently affects the cyclization tendency in positive manner,[62] probably because of its high tendency to induce β-turns.[88–90] Similarly, other structure-inducing elements[19,58,90] such as D-amino acids,[69,78,80] which also exert turn-inducing effects,[33] or achiral amino acids that can assume the conformational space of both an L- or a D-amino acid[31] favor cyclizations. In addition, sequences containing N-alkylated amino acids, tend to cyclize more readily, since the tertiary amide bonds, in contrast to secondary amides, can be configured in the energetically equivalent *cis* and *trans* form (see Vol. E 22c, Section 10.1.3).[69,91] Consequently, the kink formed by the *cis*-amide bond shortens the distance between the bridgeheads and thus facilitates cyclization. Thereby, the location of the different structural elements within the sequence seems to play a minor role.[80]

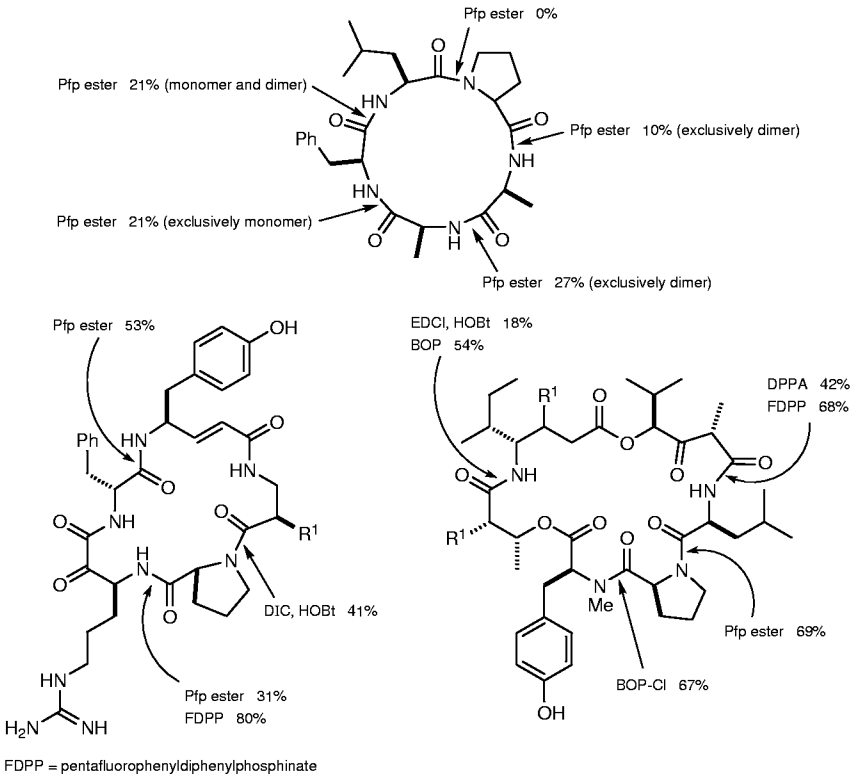
An additional factor that governs cyclization yields is the correct choice of the cyclization site.[67] In this context from the experience gained in the field a few simple rules for head-to-tail cyclizations have been derived: (1) Cyclization is usually a relatively slow process, and thus long reaction times are required. If the C-terminal amino acid is chiral, the risk of epimerization is increased, although this side reaction strongly depends upon the coupling reagent used.[58,92] Therefore, the choice of glycine or of amino acids of low racemization tendency such as proline as C-termini are recommended,[67,93] (2) to avoid steric hindrance

for references see p 529

that can slow down the rate of cyclization, amino acids with bulky side chains such as valine, leucine, or isoleucine should be avoided in the linear precursor at both sites of cyclization,[67,76,93] (3) D-amino acids should be placed at the N- and/or C-termini, although this rule is tentative and has only been useful in selected cases;[62,77,94] therefore, it remains the least imperative rule,[73,80] and (4) the difficulty of cyclization of sequences containing only homochiral amino acids (e.g., all-L) can be overcome, in sequences containing Ser, Thr, or Cys, by introduction of "pseudoprolines" (especially the dimethyl derivative, *vide supra*).[86,95]

The importance of the right choice of the cyclization site is outlined in Scheme 4 with selected examples.

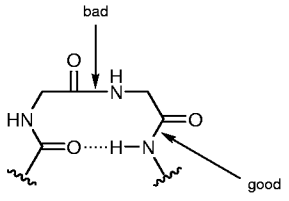
Scheme 4 Effects of the Ring-Closure Site and Coupling Reagent in Head-to-Tail Cyclizations[73,80]



FDPP = pentafluorophenyldiphenylphosphinate

If the dominant conformation of the cyclic peptide is already known, the tendency for β-turns can, at least to a certain extent, be exploited to bring the termini of the linear precursor into closer proximity. Correspondingly, the peptide bond in $i+1/i+2$-position usually is a bad choice, whereas the peptide bond $i+2/i+3$ often represents one of the best cyclization sites (Scheme 5).

Scheme 5 Good and Bad Cyclization Sites in Peptides of Known β-Turn Structure



For successful cyclizations the choice of the coupling method is similarly crucial,[62,94] since the yields may vary significantly with the procedure employed for a given precursor. Several comparative studies have been reported in this context, but the results are difficult to rationalize.[19,63,93,96–99] In fact, yields were found to strongly depend upon the individual peptides,[100] and optimization in terms of cyclization site and coupling reagents is required to maximize yields.[19,101] Nonetheless, the majority of cyclic peptides have been synthesized in acceptable yields without excessive optimization efforts. Hence, it is difficult to derive general rules from literature data that are valid for a broader range of compounds.

6.8.1.3.2 Cyclizations Involving Side Chains

The difficulties associated with head-to-tail cyclization are relatively minor in the case of cyclic peptides involving side chains. For the synthesis of side-chain-to-side-chain, head-to-side-chain, or side-chain-to-tail homodetic cyclic peptides, routinely the amino groups of diamino acids (Dap, A_2bu, Orn, or Lys) and the ω-carboxy groups of Asp or Glu are exploited.[102–105] Due to the usually increased ring size and thus reduced ring strain of this type of cyclic peptide compared to head-to-tail cyclic peptides, amide bond formation is considerably facilitated. This is particularly important in the case of small cyclic tri- or tetrapeptides.[106,107] The high conformational flexibility of the side chains compared to the "normal" peptide backbone leads to favored cyclization reactions, a fact that is important in all-L-peptides with their unfavorable extended conformation that results from the minimization of allyl strain (*vide supra*). This also means that the introduced conformational restrictions by cyclization via side chains are smaller than by head-to-tail cyclization. Even the choice of the cyclization site becomes less problematic when side-chain functionalities are involved. Moreover, the carboxy group of the side chains is usually not linked to a chiral carbon (e.g., in Asp and Glu), and thus problems associated with epimerization are not encountered, a fact that allows the use of strongly activating coupling reagents.[105] Similarly, because of the absence of α-carbon substituents the steric requirements for proximity of the functions that have to be linked are minimized.

6.8.1.3.3 Cyclization in Solution versus On-Resin

Of the two possible alternatives, cyclization in solution or on-resin, the reaction in solution is the most commonly used and generally the preferred method for head-to-tail cyclizations, whereas ring closure on-resin is preferably applied when side-chain cyclizations are performed. Very few comparative studies of these two methods have been reported.[108,109] The classical advantages of solid-phase synthesis (e.g., the easy removal of excess coupling reagents through washing steps) are in many cases offset by the lower cyclization yields compared to those in solution. Conversely, cyclization in solution suffers from the problems of tedious workup procedures, so that the advantage of the efficiency of the cyclization step is offset by the material losses in the purification processes. Therefore, the difference in the overall yield of the two methods generally is marginal. In summary, both methods offer reliable access to cyclic peptides, and the method chosen remains a matter of opinion. In cases where large quantities of cyclic peptides are required, the cyclization reaction is preferably performed in solution. Several different protocols for large-scale preparation of cyclic peptides (up to 50 g) have been reported.[77,110]

for references see p 529

6.8.1.3.4 Side Reactions in the Cyclization of Peptides

Great care has to be taken in the analytical characterization of synthetic cyclic peptides.[73] The major side reactions during cyclization are epimerization of the C-terminal amino acid residue and cyclodimerization. Cyclodimers can be detected by mass spectrometry, although the analysis is not trivial, because artifacts do occur in some ionization techniques such as ES-MS as a result of aggregation.[111,112] Real dimers can be detected as double-charged particles with *m/z* values identical to the cyclic monomers, but with a mass difference of 0.5 amu in the resolved isotope signals. The mass difference of the corresponding monomer is 1 amu. The cyclodimerization has received some attention as a direct method for the synthesis of C_2-symmetrical cyclic peptides.[62,67,94,113–115]

C-terminal epimerization of the linear precursor peptides during cyclization is a frequently observed side reaction that is strongly dependent upon the rate of cyclization. Since in many cases introduction of a D-amino acid facilitates cyclization, the rate of cyclization of the C-terminal D-diastereomer is much faster than that of the C-terminal L-diastereomer, a fact that may lead to 100% chiral inversion of this residue in the cyclic peptide.[58,92] To avoid such pitfalls chiral amino acid analysis (e.g. by chiral gas chromatography; see Vol. E 22b, Section 7.4) or synthesis of both epimers for comparative analysis, e.g. by HPLC, is strongly recommended. Synthesis of head-to-tail cyclic peptides in which the C-terminal amino acid has the opposite chirality to the sequence (e.g., C-terminal D-amino acid in an L-sequence) or cyclization at a different position along the sequence can also be used. The first case is usually confined to difficult sequences, such as homochiral peptides. "Normal" sequences (containing at least one structure-inducing element such as an N-alkylated, achiral, or D-amino acid) usually cyclize without particular difficulty and with a low degree of epimerization that generally ranges between 1 to 5% (depending upon the coupling reagent used).

6.8.2 Synthesis of Homodetic Cyclic Peptides in Solution

The linear peptide precursor can obviously be prepared either in solution or by SPPS. In both cases the linear peptide should have all functionalities protected except those involved in the cyclization reaction (either on the side chain or as N- and C-terminus). Regarding the mode of cyclization, the head-to-tail cyclization via amide bond is by far the most commonly used for peptides (*vide supra*) and it is generally performed in solution using the principle of high dilution. Thereby, in analogy to the segment condensation strategies, both maximum and minimum protection schemes are possible depending upon the coupling procedure selected for the ring closure. Conversely, for the various types of cyclization involving side chains, the on-resin procedures are preferred (see Section 6.8.3).

6.8.2.1 General Cyclization Procedures

Historically, the methods used for ring closure of linear precursor peptides via amide bond formation evolved in parallel to the methods applied in segment condensations from the azide and active ester procedures to the use of coupling reagents such as DCC in the presence of additives, or of the more recently developed phosphonium and uronium/guanidinium reagents. In all cases the choice of method is mainly dictated by the epimerization problem when chiral amino acids act as the carboxy component in the cyclization reaction, and by other side reactions.

6.8.2.1.1 Azide Procedure

For more details on the acid azide procedure including the preparation of the hydrazide precursors, the direct conversion of carboxy groups into azides, as well as possible side reactions of this amide forming reaction see Vol. E 22a, Section 3.1. The precursor linear peptide hydrazides are prepared from related peptide alkyl or aryl esters by hydrazine treatment. Since hydrazine can be used to cleave directly the ester bond of resin-bound peptides in the classical Boc/Bzl strategy (Vol. E 22a, Section 3.1.1.5),[116–118] the hydrazide precursors for cyclization can also be prepared by SPPS using this type of chemistry. Additionally, suitable hydrazide linkers have been developed that allow application of Fmoc/tBu chemistry for the synthesis of peptides hydrazides.[119,120] Alternatively, the required hydrazide precursors may be obtained by deprotection of suitably N′-protected peptide hydrazides, a procedure that is generally used to perform the cyclization of peptides with minimum side-chain protection (Vol. E 22a, Section 3.1.1.4). With peptide hydrazides as precursors, transformation into the reactive azides occurs under acidic conditions preferentially with sodium nitrite or alkyl nitrites. The fact that the conversion of hydrazides to azides can be performed in the presence of the unprotected N-terminal amino group (because of the very slow potential deamidation reaction), has made peptide cyclization through azides via hydrazide precursors fairly popular.[121,122] As the acid conditions required in this transformation reaction protect the amino group from reaction with the azide, conversion into the reactive azide can be performed at higher concentration or the azide be isolated prior to dilution and neutralization for the ring-closure reaction.

A favorable alternative is the in situ conversion of the C-terminal carboxy group into the acid azide with phosphorus azide reagents (see Vol. E 22a, Section 3.1.2.3). For this purpose the most commonly used reagent is diphenyl phosphorazidate[123] with TEA or DIPEA as the base, however solid inorganic salts, such as $NaHCO_3$, Na_2CO_3, or K_2HPO_4, are preferred as the base because of their insolubility in organic solvents such as DMF.[77,97,98,124,125] With these salts the excess of base does not augment the risk of epimerization and higher yields are generally observed. This may derive from the fact that the intramolecular cyclization possibly occurs at the solid particle surface.

As the main drawback of the azide method, the relatively long reaction times may be mentioned and as a possible side reaction in the case of slow and difficult cyclizations the occurrence of Curtius rearrangement with formation of ureido compounds.[126]

c[-Pro-Phe-Gly-Phe-Gly-]; Typical Procedure for the Cyclization of Peptide Hydrazides by the Azide Procedure:[127]

To a chilled (−15 °C) soln of H-Pro-Phe-Gly-Phe-Gly-$NHNH_2 \cdot HCl$ (1.1 g, 1.91 mmol) in DMF (10 mL) were added concd HCl (0.76 mL, 9.5 mmol) and 14% aq $NaNO_2$ (1.87 mL, 3.8 mmol). The mixture was kept at −15 °C for 30 min and then diluted (to 3×10^{-3} mol·L^{-1}) with cold DMF (640 mL) and neutralized by the addition of NMM (1.48 mL, 13.3 mmol). The mixture was kept at rt for 4 d and then taken to dryness. The residue was dissolved in MeOH/H_2O (2:1, 100 mL), and treated for 2 h with a mixed-bed ion exchanger (10 g). The solvent was removed under reduced pressure from the filtrate and the residue purified by gel chromatography (Sephadex LH 20, DMF). The first fraction that showed optical activity contained 10 mg of the dimer. The next fraction was chromatographed again (Sephadex LH60); yield: 378 mg (39%); mp 266–267 °C; $[\alpha]_D^{25}$ −149.5 (*c* 0.5, DMF).

Cyclization with DPPA; General Procedure:[77,97,98,123]

To a soln of the linear peptide in DMF (1.5×10^{-3} mol·L^{-1}), DPPA (3 equiv), and $NaHCO_3$ (5 equiv) were added and the soln was stirred for 1 h at rt. After completion of the reaction (monitored by HPLC), $NaHCO_3$ was filtered off. A few drops of H_2O were added to the soln to hydrolyze excess DPPA and the mixture was concentrated, the residue dissolved in tBuOH/H_2O (or dioxane/H_2O) and lyophilized. The protecting groups were removed after optional HPLC purification and the cyclic peptide purified and characterized.

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c[-D-Trp-Lys(Z)-Val-Phe-(Me)Ala-Tyr-]; Typical Procedure for Cyclization by DPPA:[124]
A soln of H-D-Trp-Lys(Z)-Val-Phe-(Me)Ala-Tyr-OH·HCl (24.0 g, 24.1 mmol) in DMF (degassed, 3 L) was cooled to 0 °C under a stream of N_2. DPPA (5.71 mL, 26.5 mmol) was slowly added and followed by the addition of solid $NaHCO_3$ (10.1 g, 120 mmol). The mixture was stirred briskly at 0–5 °C for 3 d, at which time TLC showed <2% of starting peptide left. Then H_2O (750 mL) was added and the mixture was combined with four other similarly proceeded 24 g preparations for treatment with a Bio-Rad AG501-X8D mixed-bed ion-exchange resin (2.27 kg) under stirring (3 h). The resin was separated by filtration, washed with DMF/H_2O (4:1), and the combined filtrate was concentrated to give a viscous oil. Addition of H_2O (500 mL) and trituration afforded the crude product as a white solid; yield: 111.0 g (98%).

6.8.2.1.2 Active Ester Procedure

The first successful synthesis of a biologically active cyclic peptide, gramicidin S, was accomplished by Schwyzer and Sieber[6,7] via the 4-nitrophenyl ester. The fact that N^{α}-protected peptide esters can be deprotected to give the peptide active ester salts has made this approach popular not only in the synthesis of sequential polypeptides but also of cyclic peptides. Among the various active esters examined for this purpose, the pentafluorophenyl esters have emerged as the most reactive ones, although a high risk of epimerization is encountered when C-terminal chiral amino acids are involved.

In the original variation a combination of N^{α}-Z protection and C-terminal pentafluorophenyl ester activation is used.[73,128–131] The N^{α}-protection is removed by hydrogenolysis over Pd/C catalyst where concomitantly intramolecular cyclization takes place. Yields are significantly increased by addition of alcohols such as EtOH, MeOH, or tBuOH and 4-pyrrolidinopyridine can be replaced by DMAP. As an alternative to the Z/tBu strategy, the Boc/Bzl strategy has been applied where accordingly N^{α}-deprotection is achieved by use of acid and in a second step cyclization is performed upon neutralization and dilution. Despite the high yields and relatively short reaction times, the active ester procedure has found only limited application because of the elaborate experimental handling required and the difficult control of the reaction conditions in terms of minimization of C-terminal epimerization.

Cyclization via the Pentafluorophenyl Ester; General Procedure:[73,128–131]
To a soln of N^{α}-Z protected linear peptide and HOPfp (1.2 equiv) in anhyd CH_2Cl_2 (0.3 M) was added EDC (1.2 equiv) with stirring at −15 °C. The soln was allowed to warm to rt over a period of 20 h, and then the mixture was taken to dryness. After extraction with EtOAc the pentafluorophenol ester was isolated as a solid in quantitative yield.

To a rapidly stirred mixture of abs dioxane and abs EtOH (40:1) at 95 °C, containing 4-pyrrolidinopyridine (0.1–1 equiv of the OPfp ester) and Pd/C, a soln of the peptide pentafluorophenyl ester in abs dioxane (one-tenth of the total volume used above, to yield a 1–2 mM soln) was added dropwise at 95 °C over 5 h, while simultaneously passing H_2 through the soln. After completion of the reaction (monitored by HPLC), the mixture was cooled to rt, filtered, and the solvent was removed under reduced pressure. The crude cyclization product was purified by standard procedures.

6.8.2.1.3 Carbodiimide Procedure

In analogy to the experience gained in the coupling of peptide segments with carbodiimides (for details see Vol. E 22a, Section 3.5), independently of the carbodiimide used (DCC, DIC, or EDC) the use of additives is recommended to increase cyclization yields as well as to minimize racemization when chiral amino acids are involved in the amide bond formation; generally, HOBt is used as additive.[106,132] To further improve the reaction yields addition of DMAP has been recommended, although comparative analysis clearly revealed that by enhancing the acylation rates concomitant epimerization may occur to a greater extent in particular cases.[58] Thus the use of HOAt is recommended.[133–135]

6.8.2.1.4 Onium Salts Procedure

As expected from the mechanism of carboxy activation, which requires the carboxy group in the anionic form and thus the use of a base, cyclization via phosphonium or uronium/guanidinium salts, significantly increases the risk of epimerization compared to the carbodiimide/additive procedure. Correspondingly, these coupling reagents, which would be potentially very useful particularly for sterically hindered ring-closure reactions, must be used with due care. Cyclization with TBTU/HOBt[97,98,136] or with HBTU/HOBt proceeds rapidly and is generally complete within one hour. Similarly efficient cyclization is observed when HBTU is replaced by HATU.[74,75,133,135] Moreover, the use of guanidinium/uronium salts of HATU, namely *O*-(7-azabenzotriazol-1-yl)-*N*,*N*,*N*′,*N*′-pentamethyluronium tetrafluoroborate (TAPipU) and HAPyU,[74,75] as cyclization reagents results in higher cyclization yields with reduced epimerization. Of the phosphonium salts, BOP is predominately used[19,137–139] either by itself or with the addition of DMAP.[132] Generally DMF is used as solvent, which can be replaced by DMSO, NMP, or DMA. Cyclization yields depend markedly on the sequence and ring size and are usually below 50%; higher yields are exceptional.

Cyclization with TBTU/HOBt; General Procedure:[97,98,136]
A 1.5 mM soln of the linear peptide in DMF was added to a soln of HOBt (3 equiv) and TBTU (3 equiv). The soln was adjusted to pH 8 with 1% DIPEA and was stirred at rt. Upon completion of the reaction (monitored by HPLC), the mixture was concentrated and the residue dissolved in tBuOH and lyophilized. The product was worked up as by standard procedures.

Cyclization with HATU; General Procedure:[74,75,133,135]
To a 1 mM soln of the linear peptide in DMF were added HATU (1 equiv) and 2,4,6-trimethylpyridine (2 equiv). The mixture was stirred at rt and the reaction monitored by HPLC. Upon completion, the solvent was removed and the resulting residue redissolved in MeOH and precipitated with EtOAc/Et_2O. The product was worked up by standard procedures.

***c*[-Nal-D-Cpa-D-Pal-Gln-Tyr-D-Arg-Leu-Arg-Pro-Lys(Ac)-]; Typical Procedure, Cyclization with HAPyU:**[74]
To a 0.1 M soln of H-Nal-D-Cpa-D-Pal-Gln-Tyr-D-Arg-Leu-Arg-Pro-Lys(Ac)-OH (0.1 mmol) in DMF was added under stirring HAPyU (0.11 mmol) followed by DIPEA (0.052 mL, 0.3 mmol). The mixture was stirred at rt and the reaction monitored by RP-HPLC. Upon completion (10 min), the solvent was removed and the resulting residue redissolved in MeOH and precipitated with EtOAc/Et_2O; yield: 98% (of crude product). The crude product was worked up by standard procedures.

A similar procedure was applied to cyclization of H-Val-Arg-Lys(Ac)-Ala-Val-Tyr-OH except that the peptide concentration was 0.01 M and the reaction time 30 min; yield: 55%; D-Tyr: 0.5%.

***c*[-D-αAbu-(Me)Arg(Tos)-Gly-Asp(OtBu)-Mamb-]; Typical Procedure, Cyclization with HBTU:**[125]
A soln of H-D-Abu-(Me)Arg(Tos)-Gly-Asp(OtBu)-Mamb-OH [Mamb = 3-(aminomethyl)benzoic acid] (393 g, ~0.35 mol) (note: this quantity was quoted in the original literature but is clearly incorrect) and DIPEA (100 mL, 0.574 mol) in MeCN (3.4 L) was added dropwise to a stirred soln of HBTU (136 g, 0.359 mol) in MeCN (3 L) over 5.5 h. The mixture was stirred overnight, cooled to 0 °C for 2 h, and the solids were collected by filtration and washed with MeCN/DIPEA (400 mL/4 mL). The crude product was dissolved in a soln of 18% MeOH in $CHCl_3$ (1215 mL) at 34 °C. Silica gel (527 g) was added to the soln, the slurry was filtered, and the filter cake was further extracted with 12% MeOH in $CHCl_3$. The filtrate was concentrated and the solvent was replaced with MeCN until GC analysis of the pot contents indicated the $CHCl_3$ content was <0.5%. The slurry was stirred at 34 °C overnight, and the purified cyclic peptide was collected by filtration. The solids were washed with MeCN (200 mL) and dried; yield: 158.6 g (58%).

6.8.2.1.5 Other Cyclization Reagents

In addition to the most common methods for cyclization of linear peptides (*vide supra*), alternative coupling reagents have been used with varying success such as BOP-Cl,[140–142] norborn-5-ene-2,3-dicarboximido diphenyl phosphate (NDPP),[19,143,144] pentafluorophenyl diphenylphosphinate (FDPP),[145–148] diethyl phosphorocyanidate (DEPC),[138,149] or T3P.[97,150]

for references see p 529

A comparative analysis of the latter reagent with HOAt-derived coupling reagents showed this phosphonic anhydride to be superior when sterically hindered amino acid residues are involved in the cyclization reaction (see Vol. E 22a, Section 3.9.1.2).[99]

6.8.2.1.6 Cyclization by Enzymatic and Native Chemical Ligation Procedures

In addition to the chemical methods of amide bond formation used for cyclization of linear peptide precursors, in the case of larger peptides (>12 amino acid residues) enzyme-mediated ring closure such as with subtiligase has been proposed,[151] although the general applicability of this procedure has not yet been proven. More promising in this context is the application of the native chemical ligation procedure[152] (for detailed information see Vol. E 22a, Section 4.1.4) for the preparation of cyclic peptides via regioselective intramolecular transthioesterification and ring contraction.[153] This reaction exploits the ring–chain tautomeric equilibrium between the linear precursor containing an N-terminal cysteine and a C-terminal thioester and the cyclic thioester. The thioester subsequently undergoes an irreversible proximity-driven ring contraction by S-to-N acyl migration. Due to the chemistry involved, this procedure allows cyclizations to be performed with unprotected peptides. Moreover with this newly developed method larger ring sizes are accessible, which would otherwise be difficult to synthesize by the standard cyclization methods.[154] When in addition to the required N-terminal cysteine residue more cysteines are present in internal positions the reaction proceeds via successive ring expansion by transthioesterification reactions ("thiazip" cyclization) to produce the final head-to-tail cyclic peptide at significantly enhanced rates.[155] The limitation is the required presence of an N-terminal cysteine residue. This can be overcome by the use of a removable auxiliary group such as oxyethanethiol[156] or hydroxynitrobenzyl.[157] An alternative approach utilizes the silver ion assisted cyclization.[153]

Due to the chemistry involved in this method, the risk of epimerization of the C-terminal amino acid residue is low (generally <1%).[156] No side-chain protection is required, since only the α-amino group of the N-terminal amino acid residue can react by entropical activation,[158] and the reaction proceeds in neutral to weakly acidic buffered aqueous solution under denaturing conditions. Another characteristic feature of native chemical ligation is the rare formation of higher oligomers. This has been attributed to the ring–chain tautomerism that results from the weak C-terminal activation required for the regioselective coupling;[153,158] it also explains why considerably higher concentrations can be used compared to the usual cyclization methods, i.e. up to 2 mM. Similarly, the size of the peptide formed has no apparent effect on the yield of the cyclization, although the formation of larger rings proceeds at increasingly slower rates.

Although this methodology has been successfully used for a number of peptides, its efficacy for the cyclization of peptides with sterically hindered cyclization sites or difficult sequences of preferred extended conformation has not been investigated.

In addition to homodetic cyclic peptides, chemical ligation methods have been developed that lead to heterodetic cyclic peptides containing thiazolidine.[158–161]

Cyclization by Native Chemical Ligation; General Procedure:[153]
The linear peptide thioester was dissolved in 0.2 M phosphate buffer (pH 7.2, 0.1–0.6 mM soln) containing 50% DMF and TCEP (6 equiv). After completion of the reaction (3–24 h, monitored by HPLC) the cyclic peptide was purified by preparative HPLC (small amounts of byproducts result from hydrolysis of the peptide thioester).

Synthesis of Cyclopsychotride; Typical Procedure:[155]
The crude peptide thioester H-Cys-Ser-Cys-Lys-Ser-Lys-Val-Cys(Acm)-Tyr-Lys-Gln-Ser-Ile-Pro-Cys-Gly-Glu-Ser-Cys-Val-Phe-Ile-Pro-Cys(Acm)-Thr-Val-Thr-Ala-Leu-Leu-Gly-$COS(CH_2)_2CONH_2$
(50–80 mg), obtained after cleavage from the resin support by HF (HF/PhOMe 9:1), was extracted into a 8 M urea soln (150 mL) at pH 7.6 in a highly reductive environment containing 5- to 10-fold excess of TCEP and then dialyzed against 8 M urea (2000 mL) to allow cyclization in a descending concentration

of 8 to 2 M urea. The whole dialysis process with concomitant thiazip cyclization was completed within 20 h. A single major peak of cyclic peptide with four reduced thiols and $Cys^{3,6}$(Acm) was observed in the RP-HPLC profile (30–85% MeCN, 30 min) (M_r calcd: 3379.1, found: 3380.0; t_R 23.8 min). Disulfide formation was achieved by the addition of 15% DMSO to the peptide soln and was complete in 10 h, yielding the three disulfide isomers (M_r calcd: 3376.0, found: 3376.0; t_R 19.1, 21.3, 28.3 min) with the correctly folded isomer obtained; yield: 32%. The third disulfide bond was formed by adjusting the pH to 4 and treatment with I_2/MeOH under N_2 to remove the Acm protecting groups of $Cys^{3,6}$. The peptide was purified by preparative RP-HPLC and characterized by MS giving the expected molecular weight (M_r calcd: 3230.9, found: 3232.2); overall yield: 5% (based on HPLC of the crude peptide).

6.8.2.2 Synthesis of Head-to-Tail Cyclic Peptides of Varying Ring Sizes

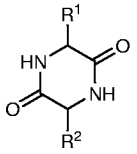
Different ring sizes exhibit distinct characteristics, especially in the case of smaller cyclic peptides (di- to hexapeptides), including ring-closing reactions, which will be treated briefly, and related conformational characteristics.

6.8.2.2.1 Piperazine-2,5-diones

Piperazine-2,5-diones (2,5-dioxopiperazines, diketopiperazines, DKP, Scheme 6) are head-to-tail cyclic dipeptides with both amide bonds in the *cis* configuration. DKPs are often encountered as unwanted side products in peptide synthesis, particularly when peptides contain proline, proline analogues, or *N*-alkylamino acid residues, which favor a *cis*-peptide bond (see also Vol. E 22a, Section 2.1.1.1.1.3.3). DKPs are also among the most common peptide derivatives found in nature and are characterized by a large spectrum of bioactivities.[162] Most of these cyclic dipeptides appear to emerge as byproducts of fermentation and food processing and many are endogenous in the animal and plant kingdoms.[163]

Due to the rigidity of this cyclic structure, there is great interest in the synthesis of DKPs as inhibitors of cell cycles and enzymes[164–166] and as receptor agonists and antagonists.[167,168] Moreover this core molecule with its well-defined stereochemistry and stereoorientation is readily polyfunctionalized for use in the design of peptidomimetics[169–171] or for the generation of compound libraries for drug discovery.[172–176] However, the normally low solubility is a disadvantage for such applications. In addition, piperazine-2,5-diones are used for the enantioselective synthesis of amino acids according to Schöllkopf.[177,178]

Scheme 6 Piperazine-2,5-diones with $R^1 = R^2$ (Symmetric) and $R^1 \neq R^2$ (Asymmetric)



Piperazine-2,5-diones can be symmetric or asymmetric. Symmetric DKPs are readily obtained by heating amino acid esters,[179–181] whereas asymmetric DKPs are obtained directly from the related dipeptides under basic or, more properly, acid catalysis, or by cyclocondensation of dipeptide esters.[182–185] As an alternative procedure hexafluoroacetone can be used to protect/activate the amino acid for the synthesis of symmetric DKPs or of the second amino acid residue for synthesis of the dipeptide ester and subsequent direct cyclocondensation to DKPs.[186] The use of active esters for the cyclocondensation is less appropriate since it may lead to epimerization when a chiral amino acid is involved as the carboxy component in the cyclization reaction. Resin-bound DKPs as scaffolds for further on-resin transformations are readily prepared using the backbone amide linker (BAL) approach, where the amino acid ester is attached to the BAL resin by its α-amino group and then acylated with a Fmoc-protected amino acid by the HATU procedure, N^{α}-deprotection leads to on-resin DKP formation[172] (see Section 6.8.3.2.2.3).

for references see p 529

N^{α}-(2-Aminobenzoyl)-peptides undergo acid-catalyzed cyclization to 1,4-benzodiazepine-2,5-diones.[187] Thus, Boc-Abz-Xaa-OH, where Xaa corresponds to various *N*-alkyl-amino acids such as Pro, Hyp, Thz, Pip, and azetidine-2-carboxylic acid and their methyl esters are readily converted into the corresponding 1,4-benzodiazepine-2,5-diones in the acidolytic Boc-cleavage step by treatment with 95% aq TFA or 100% TFA (rt, 1 h).

Symmetric Piperazine-2,5-diones; General Procedure:[186]
The hexafluoroacetone-protected amino acid[188,189] (10 mmol) was dissolved in anhyd MeOH (15 mL) and stirred at rt for 24 h. After additional refluxing for 1 h, the product was precipitated with Et_2O and the solid recrystallized (H_2O, hot H_2O, MeOH, or EtOH); average yields: 50%.

***c*[-Asp-Phe-]; Typical Procedure for Asymmetric Piperazine-2,5-diones:**[186]
The dipeptide H-Asp-Phe-OMe (1.0 g, 3.4 mmol) was heated at 200 °C for 20 min. The solid was washed with acetone; yield: 92%.

***c*[-D-Asp(OtBu)-L-Asp(OtBu)-]; Typical Procedure:**[190]
The dipeptide derivative Z-D-Asp(OtBu)-L-Asp(OtBu)-OMe (6.0 g, 11.69 mmol) was hydrogenated in MeOH (500 mL) over Pd catalyst. After 7 h, the catalyst was filtered off and the soln was refluxed for 3 d. The solvent was removed under reduced pressure and the residue recrystallized (MeOH); yield: 3.2 g (80%).

6.8.2.2.2 Cyclic Tripeptides

Cyclic tripeptides constitute a small, but well-defined class of cyclic structures that have been studied intensively in terms of synthetic accessibility and conformational properties,[1] although naturally occurring representatives have not, so far, been identified.

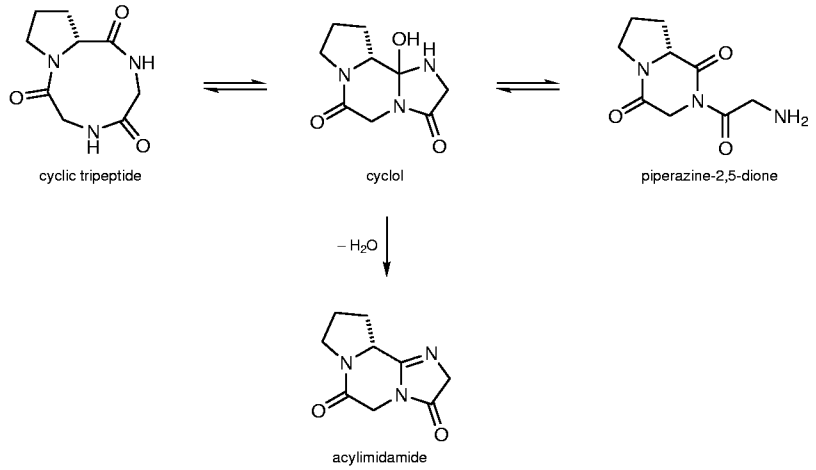
The structural properties of cyclic tripeptides have been determined by X-ray crystallography[191–196] and NMR spectroscopy.[46,197,198] As the nine-membered ring is strongly constrained and allows (with few exceptions[199]) only all-*cis* peptide bonds,[46,192] a clear understanding of their conformational behavior can be achieved using the Dunitz–Waser concept (see Section 6.8.1.2).[200] The all-*cis* cyclic tripeptides correspond to all-*cis*-cyclonona-1,4,7-triene and consequently to cyclohexane. As in cyclohexane, for cyclotripeptides both the rigid (chair in cyclohexane, crown in cyclic tripeptides) and flexible (boat) conformations are observed. Their relative populations are determined by the substitution pattern and are interchangeable with activation energies of ca. 20 kcal·mol^{-1} (see Table 1).[201]

Table 1 The Possible Conformations of Cyclic Tripeptides Depend upon the Chirality of the Amino Acid Residues[45,46]

SSS/RRR	SS0/RR0	S00/R00	000	SSR/RSS	SR0/RS0
1 crown *c*[-Pro_3-] *c*[-Pro_2-Hyp-]	1 crown + 1 boat *c*[-Pro_2-(Bzl)Gly-] *c*[-Pro_2-[Bzl(2-NO_2)]Gly-] *c*[-Pro_2-Sar-]	1 crown + 3 boats *c*[-Pro-(Bzl)Gly_2-] *c*[-Pro-Sar_2-]	2 crowns + 6 boats *c*[-(Bzl)Gly_3-] *c*[-Sar_3-]	1 boat *c*[-L-Pro_2-D-Pro-]	2 boats *c*[-L-Pro-(Bzl)Gly-D-Pro-]

The stability of cyclic tripeptides containing nontertiary amide bonds in their sequences is considerably reduced by cyclol formation.[199,202,203] This degradation reaction probably proceeds by *cis*/*trans* isomerization of an amide bond near a neighboring carbonyl carbon atom (see Scheme 7), but in some cases a stable cyclol may also be generated by nucleophilic attack of the secondary amide nitrogen.[202] The formation of the cyclol structure is reversible, but may proceed further to generate a stable piperazine-2,5-dione by cleavage of the second C–N bond. In fact, cleavage of a protecting group from the secondary amide of a cyclic tripeptide induces transformation to the piperazine-2,5-dione.[201] Alternatively, a bicyclic acylimidamide is formed by dehydration; this reaction can be only reversed in selected cases. This facile degradation occurs whenever cyclic tripeptides contain secondary amide bonds, and it is responsible for the lack of stability of such compounds.

Scheme 7 Interconversion of Cyclic Tripeptides into Cyclols and Piperazine-2,5-diones; Dehydration to Acylimidamide is an Almost Irreversible Transformation[199,202,203]



Due to the constrained nature of the nine-membered ring of cyclic tripeptides that often results in cyclol formation and the need to have both amide bonds in the *cis* configuration for successful cyclization,[204] cyclization is not possible for many tripeptide sequences and in these cases only formation of cyclodimers and cyclooligomers is observed.[64,199,205,206] The limitation of suitable building blocks to those forming tertiary amides such as proline, hydroxyproline, sarcosine, *N*-benzylglycine, and other N-alkylated amino acid residues, which should have minimal or no racemization tendency, enables the use of harsh reaction conditions for cyclization.[207] Proline provides a geometry favorable for ring closure, while because of the absence of a side chain at the α-carbon N-alkylated glycines do not hamper cyclization. Cyclization yields strongly depend upon the sequence of the linear tripeptide precursor, but generally range between 10 and 50%, except *c*[-Pro_3-], which can be obtained in nearly 90% yield.[205]

***c*[-(Bzl)Gly-(Bzl)Gly-(Bzl)Gly-]; Typical Procedure:**[46]

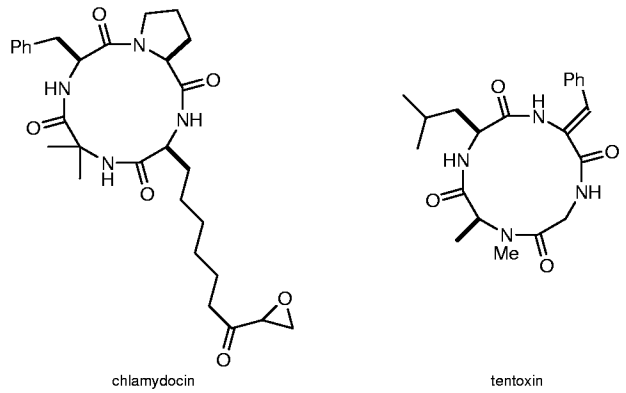
A soln of HCl·H-(Bzl)Gly-(Bzl)Gly-(Bzl)Gly-OTcp (2.15 g, 3 mmol) in pyridine (1.5 L) was stirred at 90 °C for 3 h. After removal of the solvent, the residue was treated with toluene several times to remove traces of pyridine. The residue was dissolved in MeOH and H_2O and stirred overnight with mixed-bed ion-exchange resin (5 g). The resin and the precipitate {primarily *c*[-(Bzl)Gly-(Bzl)Gly-]} were filtered off; the filtrate was taken to dryness and column chromatographed (Sephadex, DMF). The peptide containing eluate was concentrated to an oil that crystallized on addition of Et_2O; it was recrystallized (MeOH): yield: 0.3 g (19.5%).

6.8.2.2.3 Cyclic Tetrapeptides

Contrary to the cyclic tripeptides, cyclic tetrapeptides are found as natural products such as chlamydocin,[208–211] HC-toxin,[212,213] WF 3161,[214,215] or tentoxin,[73,216–219] which all exhibit interesting pharmacological properties, e.g. phytotoxic (HC-toxin and tentoxin) or cancerostatic activities (chlamydocin) (Scheme 8). Correspondingly, this class of compounds has been studied extensively in terms of synthesis and conformational preferences.

for references see p 529

Scheme 8 Structures of the Cyclic Tetrapeptides Chlamydocin[208–211] and Tentotoxin[73,216–219]



The conformational properties of cyclic tetrapeptides are far more complex than those of the smaller cyclic peptides; details are given in specialized reviews.[1,31] The most striking difference compared to piperazine-2,5-diones and cyclic tripeptides is that in addition to a *cis*-amide bond, for stable ring structures *trans*-amide bonds may also be present as established by NMR spectroscopy and X-ray crystallographic analysis. In most cases a *cis*/*trans*/*cis*/*trans* arrangement of the amide bonds is found[220–230] along with a *cis*/*trans*/*trans*/*trans* arrangement,[212] while an all-*trans* arrangement is rather unusual.[231,232] The preferred conformation is considerably dependent upon the polarity of the solvent,[215] and the turn topology is influenced by the side chains.[103] The phenomenon of cycloenantiomerism has also been observed and investigated.[227,230,233] Rules based on the chirality of the constituent amino acid residues and their N-alkylation were proposed by Kato et al. for tentative predictions of preferred backbone conformations of cyclic tetrapeptides,[234,235] and these were further refined by Rich et al.[236] Again the conformational properties of cyclic tetrapeptides are better understood with the Dunitz–Waser concept (see Section 6.8.1.2).[26] It is noteworthy that axial substituents across the peptide bond are avoided in order to minimize "1,3-diaxial interaction". The reason for this is that the preferred φ,ψ-angles that prevent allylic strain lead to conformational preferences similar to those of cyclohexanes. The synthetic accessibility of cyclic tetrapeptides has been studied both in theory[236] and practice,[19,69,73,229,237] and the cyclization tendency of the linear precursors critically depends on their conformation. In fact, structure-inducing elements such as cyclic or N-alkylated amino acids enhance the yields of cyclic tetrapeptides,[69,230,231] as do sequences with mixed L- and D-amino acids or achiral amino acids that can be cyclized by standard synthetic protocols.[19,211,215,225,238,239] The success of the synthesis, however, largely depends on the cyclization site,[73,237] which can be predicted by means of molecular modeling.[69,240] Yields generally range between 10 to 25%.[69,216,237]

Problems do, however, occur during cyclization with all-L- or all-D-residues (see Section 6.8.1.3.1). Cyclodimerization can be triggered by an extended backbone conformation of the linear precursor, and C-terminal epimerization derived from slow amide bond formation. These lead to dimers or epimerized side products.[73,241] Cyclization of such tetrapeptides are successful only in a few exceptional cases.[69,73] To overcome this problem, protection of the backbone amide with Boc has been proposed since this approach favors the *cis*-amide bond configuration which induces one or more suitable conformations for cyclization.[69] As a consequence, the cyclization yield of *c*[-Ala-]$_4$[242] improved from 1 to 27%[69] (see also the use of *N*-Hmb protection in Section 6.8.1.3.1).

As an alternative, a rather unusual strategy for the synthesis of cyclic tetrapeptides utilizes the deprotonation of the backbone amides of cyclic tetrapeptides with LDA and subsequent alkylation to produce N-alkylated peptides, which can be achieved regio- and

stereospecifically with the appropriate precursors. This method, however, is limited because of the sequence-specific requirements.[243,244]

Table 2 Examples for the Synthesis of Cyclic Tetrapeptides[19,73,77,141,210,233,243–247]

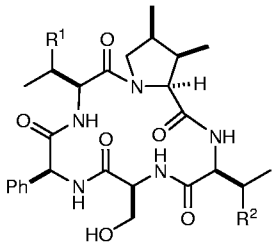
Peptide[a]	Cyclization Method[b]	Yield (%)	Ref
c[-(Me)Ala-Leu-(Me)(Z)ΔPhe-Gly-]	Tcp ester/pyridine	18	[233]
c[-Val-D-Val-Val-D-Val-]	mixed anhydride	3[c]	[245]
c[-Ala-Sar-Sar-Sar-]	Tcp ester/pyridine	25[d]	[224]
c[-DL-Ade-Aib-Phe-D-Pro-]	OSu ester	12	[210]
c[-Phe(4-I)-D-Ala-Asu(OtBu)-D-Pro-]	BOP-Cl/TEA	28	[141]
c[-Asu(OtBu)-D-Pro-Tyr-D-Ala-]	BOP-Cl/TEA	20	[141]
c[-D-Orn(Boc)-Phe-D-Pro-Gly-]	NDPP	64	[19]
c[-D-Orn(Boc)-Phe-D-Pro-Gly-]	DPPA	50	[19]
c[-Leu-Sar-Sar-Gly-]	Z/Pfp ester	42	[243]
c[-Val-Sar-Sar-Gly-]	Z/Pfp ester	13	[243]
c[-(Me)Leu-Gly-D-Ala-Sar-]	Boc/Pfp ester	30	[243]
c[-Leu-Sar-(Me)Ala-Gly-]	Boc/Pfp ester	35	[244]
c[-Aclpr-Aib-Phe-D-Pro-]	Z/Pfp ester	94	[246]
c[-D-Phe-Leu-Pip-Aclpr-]	Z/Pfp ester	74	[247]
c[-Pro-Val-Pro-Tyr(OTBDMS)-]	Z/Pfp ester	31	[73]
c[-Pro-D-Phe-Pro-D-Phe-]	DPPA/TEA	18[e]	[77]
c[-Orn(Boc)-Phe-D-Pro-Gly-]	DPPA/TEA	50	[19]
c[-Orn(Boc)-Phe-D-Pro-Gly-]	EDC/DMAP	26	[19]
c[-Orn(Boc)-Phe-D-Pro-Gly-]	BOP/DMAP	15	[19]
c[-Orn(Boc)-Phe-D-Pro-Gly-]	HBTU/DMAP	13	[19]

[a] Ade = 2-aminodec-9-enoic acid; Aclpr = 8,9-*O*-isopropylidene-protected (*S,S,R*)-amino-10-chloro-8,9-dihydroxydecanoic acid; Phe(4-I) = 4-iodophenylalanine.
[b] NDPP = norborn-5-ene-2,3-dicarboximido diphenyl ester.
[c] Strong tendency for cyclodimerization.
[d] In addition 13% *c*[-Sar-Sar-Sar-], 16% *c*[-Ala-Sar-], and traces of *c*[-Ala-Sar-Sar-Sar-] were formed.
[e] In addition 6% *c*[-Pro-D-Phe-Pro-Phe-] was formed.

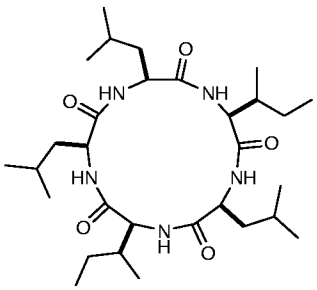
6.8.2.2.4 Cyclic Pentapeptides

The class of cyclic pentapeptides is inherently more numerous than those of the smaller ring sizes and many representatives are found among natural products from marine organisms such as pseudostellarin A,[248,249] astins A–C,[250] cyclotheonamides A and B,[251–253] or viscumamide,[73,254] the latter being the only example of a naturally occurring all-L-cyclic pentapeptide (Scheme 9).

Scheme 9 Structures of the Cyclic Pentapeptides Astin A–C[250] and Viscumamide[73,254]



astin A $R^1 = H$; $R^2 = OH$
astin B $R^1 = OH$; $R^2 = H$
astin C $R^1 = R^2 = H$

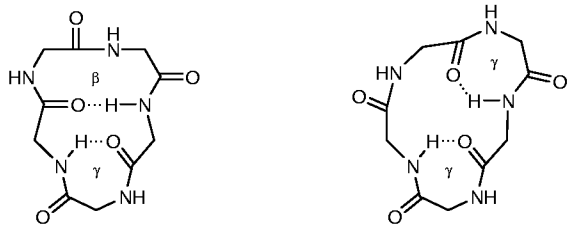


viscumamide

for references see p 529

Cyclic pentapeptides are important model systems for studying conformational preferences of peptides,[1,31,90,255–257] particularly, with regard to reverse turns.[34,35,258–261] They also offer interesting and wide-ranging aspects for structure–activity relationships as a result of their limited conformational flexibility (*vide infra*). This property was successfully utilized in the spatial-screening approach[80,260] to backtrack the receptor-bound active conformation of RGD peptides.[24,262] Cyclic pentapeptides exhibit a far more complex picture in their conformational preferences than the cyclic tetrapeptides.[31] Generally, all-*trans*-configured peptide bonds are allowed, although incorporation of N-alkylated amino acids such as Pro, Sar, or (Bzl)Gly often results in a high or exclusive occurrence of the related peptide bond in the *cis* configuration. Sporadically, *cis*-peptide bonds are also found in non-alkylated amides, when the specific arrangement of D- and L-amino acid residues provokes steric constraints in the all-*trans* conformation.[263] According to the Dunitz–Waser concept[26] (see Section 6.8.1.2) a cyclic pentapeptide with one *cis*-peptide bond should correspond conformationally to a cyclohexane, whereas the all-*trans*-cyclopeptide is a highly dynamic system similar to cyclopentane, i.e. the five α-carbons can adopt envelope or twist arrangements. The conformational preference is governed by the number and position of D- and L-residues.[90,264,265] Within these conformations there is again the tendency to minimize the allylic strain. For example, one D-amino acid residue in a pentapeptide with all other residues being L-configured prefers the $i+1$ position of a βII′-turn, whereas the rest of the cycle often includes a γ-turn.[1,257,266–268] Conformations with two γ-turns as shown in Scheme 10 have also been found.[256]

Scheme 10 Two Frequently Observed Turn Topologies in Cyclic Pentapeptides



Since there is only a small energy difference between the different conformational states depending upon the presence or absence of a Pro, Gly or N-alkylated amino acid residue, and upon the chirality of the constituent amino acid residues and also to a lesser extent upon the side-chain functionalities, it is not possible to unambiguously predict the conformation of a cyclic pentapeptide. These molecules have often been studied in different solvents and solvent effects were neglected, and/or the methodology to handle such conformational equilibrium was not available. It is only recently that modern NMR techniques and computational procedures have become available to treat this complex problem of fast exchanging conformational equilibria.[36,269,270]

As seen from Table 3, the synthesis of cyclic pentapeptides with mixed L- and D-amino acid residues or containing N-alkylated or achiral building blocks generally results in good yields;[1,19,73] although the choice of the cyclization site is crucial to the overall yields.[73,75,87] While a single Pro residue may hamper cyclization, structural elements such as D-Pro-Pro, Sar-Pro, or Gly-Pro facilitate ring closure.[87] Problems are encountered with all-L- or all-D-pentapeptides because of their unfavorable extended conformation in solution.[11,73–75] The tendency of these linear precursors to cyclize is so low that applying standard coupling reagents often only results in cyclic peptides epimerized at the juncture being isolated.[59,74] In most cases, this problem can be solved by the use of HOAt-derived cyclization reagents, e.g. HAPyU or HATU.[74,75] With such highly activating reagents even all-L- or all-D-cyclic peptides are obtained with a relatively low extent of epimerization (<10%) and in acceptable yields. Formation of cyclic dimers almost always occurs with yields ranging from 10 to

30%. Alternatively, one or more backbone amides may be protected with the Hmb group to simulate transiently the presence of N-alkylated amino acid residues.[75] However, the use of *N*-Hmb protected amino acid residues is limited, since most of these derivatives lead to rather poor coupling yields making the accessibility of the linear precursors problematic.[271] Despite these intrinsic difficulties, this approach seems promising, especially in combination with the HOAt-derived coupling reagents.

Table 3 Examples for the Synthesis of Cyclic Pentapeptides[58,63,87,110,114,272–277]

Peptide	Cyclization Method[a]	Yield (%)	Ref
c[-Gly-Pro-Ser-D-Ala-Pro-]	ONp ester/pyridine	38	[272]
c[-Gly-Pro-D-Phe-Ala-Pro-]	ONp ester/pyridine	85	[273]
c[-D-Trp-D-Cys(O_3Na)-Pro-D-aIle-Leu-]	azide	12	[274]
c[-D-Trp-D-Cys(O_3Na)-Glu-D-Val-Leu-]	azide	50	[274]
c[-Pro-Pro-Phe-Phe-Gly-]	azide	12	[275]
c[-Gly-Pro-Pro-Phe-Phe-]	azide	31	[275]
c[-D-Trp-D-Cys(O_3Na)-Pro-D-Val-Nle-]	EDC/HOBt	15	[274]
c[-Ala-Arg(NO_2)-Lys(Z)-Glu(OBzl)-D-Val-]	EDC/DMAP	38	[276]
c[-Tyr-Arg(NO_2)-Ala-Glu(OBzl)-D-Val-]	EDC/DMAP	46	[276]
c[-D-Trp-D-Asp(OtBu)-Pro-D-Val-Leu-]	HBTU	67	[110]
c[-Gly-Ala-D-Val-Leu-Ile-]	azide	40	[144]
c[-Gly-Ala-D-Val-Leu-Ile-]	ONp ester/pyridine	57	[144]
c[-Gly-Ala-D-Val-Leu-Ile-]	NDPP	18	[144]
c[-Gly-Ala-D-Val-Leu-Ile-]	DCC/HOBt	traces	[144]
c[-Pro-Val-Orn(Z)-Leu-D-Phe-]	azide (DMF)	58[b]	[63]
c[-Pro-Val-Orn(Z)-Leu-D-Phe-]	azide (pyridine)	65[c]	[63]
c[-Pro-Val-Orn(Z)-Leu-D-Phe-]	DCC/HOSu (30:2)	18[d]	[63]
c[-Pro-Val-Orn(Z)-Leu-D-Phe-]	OSu ester (DMF)	36[e]	[63]
c[-Tyr-Arg(NO_2)-Lys(Z)-Glu(OBzl)-D-Val-]	azide	29	[275]
c[-Tyr-Arg(NO_2)-Lys(Z)-Glu(OBzl)-D-Val-]	EDC/DMAP	46	[58]
c[-Ala_5-]	HAPyU	44[f]	[87]
c[-Ala-Ala-Pro-Ala-Ala-]	HAPyU	20[g]	[87]
c[-Ala-Pro-Pro-Ala-Ala-]	HAPyU	0[h]	[87]
c[-Ala-D-Pro-Pro-Ala-Ala-]	HAPyU	96[i]	[87]
c[-Ala-D-Ala-Pro-Ala-Ala-]	HAPyU	82[i]	[87]
c[-Arg(Mtr)-Gly-Asp(OtBu)-Lys(Z)-D-Val-]	DPPA/$NaHCO_3$	24	[277]
c[-Arg(Mtr)-Gly-Asp(OtBu)-D-Phe-Ala]	DPPA/$NaHCO_3$	64	[277]
c[-Arg(Mtr)-Gly-Asp(OtBu)-D-Phe-Lys(Z)-]	DPPA/$NaHCO_3$	44	[277]
c[-Arg(Mtr)-Gly-Asp(OtBu)-Trp-D-Val-]	DPPA/$NaHCO_3$	3	[277]

[a] NDPP = norborn-5-ene-2,3-dicarboximido diphenyl phosphate.
[b] 0.5% epimerization.
[c] 0.6% epimerization.
[d] 1.2% epimerization.
[e] 10.9% epimerization.
[f] 46% cyclodimer.
[g] 53% cyclodimer.
[h] Cyclization to higher homologues exclusively.
[i] No cyclodimer.

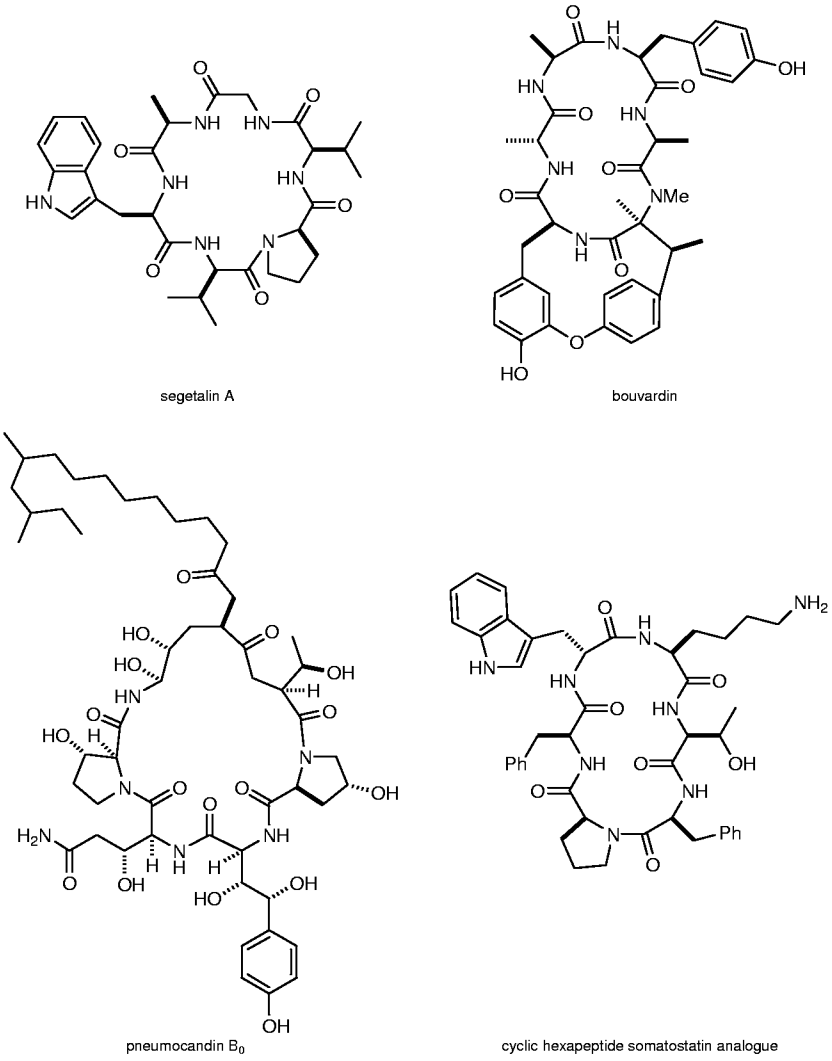
A typical method for the large-scale synthesis of a cyclic pentapeptide is reported in Section 6.8.2.1.1.

6.8.2.2.5 Cyclic Hexapeptides

Cyclic hexapeptides can be considered as the classical and the most prominent representatives among cyclic peptides. Natural products are numerous and widespread (see Scheme 11), e.g. segetalin A,[278] bouvardin,[279,280] RA-III,[281–283] or pneumocandin B_0,[284–286] and this type of ring structure has frequently been used to stabilize conformational preferences of bioactive portions of larger peptides or even proteins.

for references see p 529

Scheme 11 Structures of the Naturally Occurring Cyclic Hexapeptides Segetalin A,[278] Bouvardin,[279,280] Pneumocandin B_0,[284–286] and of a Synthetic Cyclic Somatostatin Analogue[287]



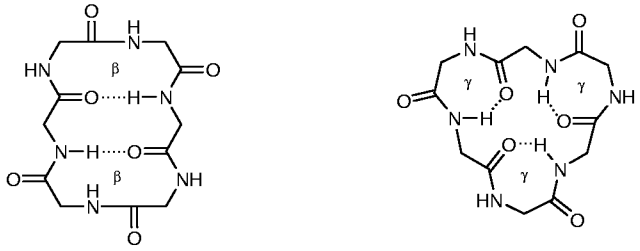
The conformational properties of cyclic hexapeptides have been extensively investigated over decades by the use of suitable model systems.[37,288] With appropriate peptide sequences the phenomena of cycloenantiomerism and cyclodiastereomerism have been defined[289,290] and reverse turns have been classified for model ring structures.[206] Similarly, the retro-inverso concept was established for the first time with cyclic hexapeptides;[30] in this context it is noteworthy that the total retro-inverso structure of a cyclic peptide is especially easy to synthesize, as it does not require specific building blocks but only inverted chirality and full-sequence reversal.[265]

Due to the increased ring size of cyclic hexapeptides, the peptide backbone adopts conformations similar to those of linear peptides. In fact, *cis* configurations of non-alkylated amides are observed only in rare cases,[291] but as usual, *cis*-peptide bonds are encountered in proline-containing cyclic hexapeptides. Cyclic hexapeptides allow the formation of two energetically favored β-turns,[206] which generate a rectangular mini-sheet structure as shown in Scheme 12. In the absence of a strong preference of an amino acid residue for a distinct position, three degenerated conformations are observed in rapid equilibrium.[1] Conversely,

incorporation of a single D-amino acid residue may result in a preferred conformation of the ring caused by preference of such a residue for the $i+1$ position in a βII′-turn.[24,292] A similar effect is found with proline and sometimes with glycine as well.

Cyclic hexapeptides with alternating chirality (LDLDLD) prefer a C_3 symmetric structure (three γ-turns).[31] For symmetry reasons six γ-turns (three above and three below the ring plane) can also be adopted, although this structure is difficult to prove and to distinguish because of the rapid flip-flop between two structures each with three γ-turns.[31] These types of structures lead to self-organization by strong intermolecular interactions into nanotubes (see Section 6.8.6.1).

Scheme 12 Two Frequently Observed Turn Topologies in Cyclic Hexapeptides



The use of cyclic hexapeptides to mimic and stabilize β-turn conformations of bioactive sequences from larger peptides and proteins was pioneered by Veber et al. with the synthesis of a highly active somatostatin analogue.[124,287,293] This approach has been subsequently applied to many different target molecules.[294–299] The synthesis of cyclic hexapeptides is achieved without major difficulties by the use of standard coupling procedures when achiral, N-alkylated, or D-amino acids are present in the sequence.[80,97,300] Thereby the position of the D-amino acid within the sequences is of minor importance. The cyclization is rendered more difficult and increased cyclodimerization is observed if a proline occurs in the linear precursor along with five L-(or D-)amino acids.[87] Conversely, a D-Pro-Pro unit with its strong tendency for a β-turn formation[88] considerably enhances the cyclization yield. This is also observed when Gly-Pro, Sar-Pro, or D-Ala-Pro units are present.[68,87] Although not as severe as with cyclic pentapeptides, difficulties are encountered with the cyclization of homochiral sequences,[68,74,75,80] which are prone to cyclodimerization and C-terminal epimerization.[75] These complications can generally be solved by the use of HATU or HAPyU as cyclization reagent.[74,75] The tendency for cyclization can be predicted qualitatively with computational methods.[68,301]

A typical procedure for the large-scale synthesis of a cyclic hexapeptide is reported in Section 6.8.2.1.1

for references see p 529

Table 4 Some Examples for the Synthesis of Cyclic Hexapeptides[74,87,97,113,275,302–304]

Peptide	Cyclization Method	Yield (%)	Ref
c[-βAla-Pro-Phe-βAla-Pro-Phe-]	DCC	21	[300]
c[-Arg(NO_2)-Lys(Z)-Asp(OBzl)-Val-Tyr-Gly-]	azide	15	[275]
c[-Arg(NO_2)-Lys(Z)-Asp(OBzl)-Val-Tyr-Gly-]	EDC/DMAP	38	[275]
c[-Arg(NO_2)-Lys(Z)-Glu(OBzl)-Val-Tyr-Gly-]	EDC/DMAP	51	[275]
c[-Gly-Pro-Phe-Val-Phe-Phe-]	azide	79	[302]
c[-Gly-Pro-Phe-Val-D-Phe-Phe-]	azide	84	[302]
c[-Gly-Pro-Phe-Val-D-Pro-Phe-]	azide	67	[302]
c[-Tyr-Asp(OtBu)-Phe-Phe-Ser(tBu)-D-Ala-]	TBTU/HOBt	61	[97]
c[-Tyr-Asp(OBzl)-Phe-Phe-Ser(Bzl)-D-Ala-]	TBTU/HOBt	18	[97]
c[-Tyr(tBu)-Asp(OBzl)-Phe-Phe-Ser(Bzl)-D-Ala-]	TBTU/HOBt	31	[97]
c[-Lys(Z)-Lys(Z)-Gly-Lys(Z)-Lys(Z)-Gly-]	DPPA/$NaHCO_3$[a]	20	[113]
c[-Phe-Gly-Arg-Gly-Asp(OFm)-Phe-]	DPPA/$NaHCO_3$	12	[303]
c[-Asp(OFm)-D-Ser-Phe-D-Phe-Arg-Gly-]	DPPA/$NaHCO_3$	3	[303]
c[-Asp(OFm)-D-Ser-Phe-D-Phe-Arg-Gly-]	DPPA/$NaHCO_3$, 2,4-dinitrophenol	39	[303]
c[-Asp(OFm)-D-Ser-Phe-Gly-Arg-Gly-]	DPPA/$NaHCO_3$	35	[303]
c[-D-isoAsp(OBzl)-Thr-Thr-Asn-Tyr-Thr-]	DPPA/$NaHCO_3$	45	[304]
c[-Val-Arg-Lys(Ac)-Ala-Val-Tyr-]	HAPyU	25[b]	[87]
c[-Val-Arg-Lys(Ac)-Ala-Val-Tyr-]	HAPyU/DIPEA[c]	55	[74]

[a] No product with Azide or Pfp ester.
[b] 42% of cyclodimer.
[c] No product with BOP or TBPipU.

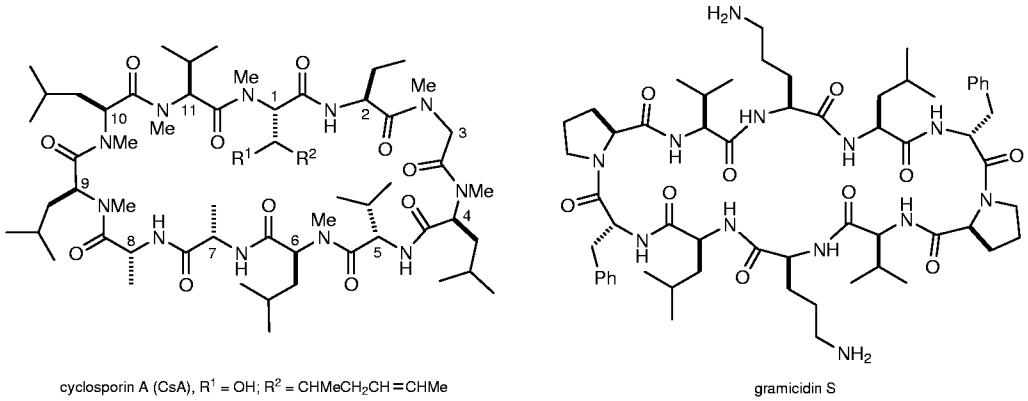
A typical procedure for the large-scale synthesis of a cyclic hexapeptide is reported in Section 6.8.2.1.1.

6.8.2.2.6 Cyclic Heptapeptides and Larger Ring Structures

Structurally highly diverse cyclic peptides comprising of sequences with seven or more amino acid residues are produced in huge numbers by bacteria, yeast, fungi, plants, and other microorganisms and many of them have unusual biological activities. Among these larger cyclic peptides cyclosporin is the most important (Scheme 13);[305–308] it is a highly N-methylated cyclic undecapeptide that is used as an immunosuppressive agent in human medicine.[309] In spite of its relatively high mass and the large number of hydrogen donors and acceptors, which violates "Lipinski's Rules of Five",[8,9] (but see also ref[633]), it is orally available and also one of the most widely marketed drugs. Its conformation has been determined in the crystal state,[310] in different solvents[310–312] and when bound to its natural receptor cyclophyllin.[22,23] Like cyclosporin, which inhibits phosphatase 2B, microcystins, hepatotoxic cyclic heptapeptides from cyanobacteria, are potent inhibitors of phosphatases 1 and 2A (for a review see ref[313]). Their conformation in solution and related dynamics has been determined by NMR[314] and when bound to the enzyme by X-ray crystallography.[315] Gramicidin S[114] and antamanide,[316–318] were also investigated in detail both in crystal and solution, the latter as a model peptide to study internal dynamics on a very different timescale.[318–321,628] These cyclic decapeptides are of minor medicinal importance. More recently, many biologically active cyclic peptides from marine sources have been isolated, synthesized, and conformationally investigated. In several cases, however, the synthetic cyclic peptides were found to show lower or no activity at all, which indicates that small, but highly active contaminants in the natural material may lead to wrong conclusions.[10]

Most, if not all of the larger size natural cyclic peptides contain one or more N-alkylated amino acid residue. Their ring conformation depends on the substitution patterns as well as on environmental conditions (crystal, solvent). As a result of the large number of energetically similar and thus allowed conformational states, their 3D structure can not be predicted with great certainty.

Scheme 13 Structure of Cyclosporin A[305-308] and Gramicidin S[114]



Due to the relatively high flexibility of cyclic peptides of larger ring size an additional cyclization is sometimes used to constrain their conformation, even by nature, e.g. amanitins and phalloidins.[316] For this purpose primarily side-chain-to-side-chain cyclization is adopted, e.g. Glu/Asp versus Lys or disulfide bridges;[322] alternatively, even the principle of backbone cyclization is applied (see Section 6.8.4).[29,80,323,324]

As summarized in Table 5, generally, the synthesis of homodetic monocyclic peptides with rings containing more than six amino acid residues is straightforward. However, the cyclization yield may strongly depend upon the cyclization site. When the conformation of the desired ring is not known, the best juncture cannot be predicted and therefore it is advisable to attempt several possible sites if optimized yields are the goal. In this context, cyclosporin may serve as an example. In fact, cyclization yields could only be optimized by exploring different juncture sites.[309,325,326] It has also been shown that the triphosgene method[424] (see Section 6.8.3.2.4) allows very efficient coupling and cyclization of N-alkylated cyclic peptides such as cyclosporin A and omphalotin A.[629]

Table 5 Examples for the Synthesis of Cyclic Heptapeptides and Larger Systems[11,136,309,327-329]

Peptide[a]	Cyclization Method[b]	Yield (%)	Ref
c[-Asn(Trt)-Pro-Phe-Val-Leu-Pro-Val-] (axinastatin 2)	DPPA/$NaHCO_3$	24	[11]
c[-Asn(Trt)-Pro-Phe-Ile-Leu-Pro-Val-] (axinastatin 3)	DPPA/$NaHCO_3$	23	[11]
c[-Thr(tBu)-Pro-Leu-Trp-Val-Pro-Leu-] (axinastatin 4)	DPPA/$NaHCO_3$	19	[11]
c[-Leu-Ile-Phe-Ser-Pro-Ile-Pro-] (stylopeptide 1)	TBTU/DIPEA	67	[327]
c[-Gly_2-Lys(Z)-Gly_3-Lys(Z)-Gly-]	DPPA/DIPEA	70	[138]
c[-Gly_2-Lys(Z)-Gly_3-Lys(Z)-Gly-]	DEPC/DIPEA	65	[138]
c[-Gly_2-Lys(Z)-Gly_3-Lys(Z)-Gly-]	BOP/DIPEA	68	[138]
c[-Gly_2-Lys(Z)-Gly_3-Lys(Z)-Gly-]	DCC/HOBt	20	[138]
c[-Gly_2-Lys(Z)-Gly_3-Lys(Z)-Gly-]	DCC/HOSu	25	[138]
c[-Phe-Phe-Pro-Pro-Phe-Phe-Val-Phe-Phe-Ala-]	DCC/HOSu	31	[328]
c[-Phe-Phe-Pro-Pro-Phe-Phe-Ala-Phe-Phe-Gly-]	DCC/HOSu	49	[328]
c[-Dah(Boc)-Pro-Pro-Phe-Phe-Val-Pro-Pro-Ala-Phe-]	DCC/HOSu	36	[329]
c[-D-Ala-(Me)Leu-((Me)Leu-(Me)Val-(Me)Bmt-L-αAbu-Sar(Me)Leu-Val-(Me)Leu-Ala-] (cyclosporin A)	DPPA	65	[309]

[a] (Me)Bmt = (4*R*)-4-[(*E*)-but-2-enyl]-*N*,4-dimethyl-L-threonine; Dah = diaminohexenoic acid.
[b] DEPC: diethyl phosphorocyanidate.

Cyclic peptides with an even number of amino acids and alternating D- and L-configuration have been shown to form nanotubes by a continuous array of hydrogen bonds above and below the ring planes (see Section 6.8.6.1).

Selected examples of the synthesis of complex natural cyclic peptides of larger ring sizes are described in Vol. E 22d, Section 16.

for references see p 529

6.8.3 Synthesis of Homodetic Cyclic Peptides on Solid Support

With the increasingly optimized solid-phase methodology for synthesis of peptides in terms of resins, linker groups, and, particularly, selectivity of protection schemes, it was appealing to develop methods for efficient cyclizations of peptides, in the various modes described in Section 6.8.1.1, directly in the resin-bound state prior to the resin-detachment/deprotection step. At least on a laboratory scale this approach reduces the number of synthetic steps and once the synthesis of a desired cyclic peptide is optimized, it allows for manual or automated preparation of related analogues and of libraries as discussed in Section 6.8.4. Although the solid-phase method may not be superior to the cyclization methods in solution in terms of final yields, it is time saving and even cost effective due to the generally facilitated purification processes.

First reports of successful cyclization on resin appeared already in the 1960s;[330,331] e.g. for the successful synthesis of gramicidin S the chloroformate derivative of the classical Merrifield resin[332] was used for urethane-based side-chain linkage of the ornithine residue to allow for a final head-to-tail cyclization of the suitably protected linear precursor prior to resin cleavage/deprotection. Since then, great progress has been made in the field, which has been comprehensively reviewed.[70–72,333–335]

In the following discussion a selection of optimized procedures is presented for the synthesis on solid support of homodetic monocyclic and bicyclic as well as mixed homodetic/heterodetic bicyclic peptides that contain at least one lactam ring, whereas heterodetic cyclic peptides containing other than amide bonds or peptidomimetics are not discussed here because of their unlimited diversity.

6.8.3.1 General Factors Affecting On-Resin Cyclization

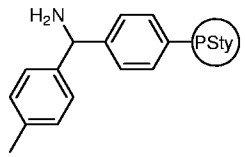
In addition to the general factors affecting yields of cyclization independent of whether it is performed in solution or on resin (see Sections 6.8.1.3.1 and 6.8.1.3.2), for cyclization of peptides in the resin-bound state particular features are known to play a crucial role. The success of the on-resin approach strongly depends upon the physicochemical characteristics of the solid support, the compatibility of the overall protection scheme including the resin linker moiety, i.e. the handle, and the chemistry employed for the cyclization reaction; but it also strongly depends upon the intrinsic properties of the target molecule in terms of amino acid composition and conformational preferences. Correspondingly, an a priori prediction of the optimal conditions for a particular synthesis is difficult, as yields are frequently affected by factors of opposing effects.

6.8.3.1.1 Solid Supports and Handles for On-Resin Cyclization

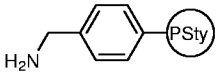
A large variety of solid supports such as polystyrene (PSty), poly(ethylene glycol)-polystyrene graft polymers (PEG-PS), polyamide-Kieselguhr supports (PepsynK), and cellulose-based polymers and membranes have been successfully used for the synthesis and on-resin cyclization of peptides. The most common resins and handles used for on-resin cyclization are shown in Scheme 14. All the resins and many others are commercially available. The handles are sorted according to their functional groups (amine resins, hydroxy resins, and electrophilic resins) and according to the basic protection strategy applied in the synthesis, i.e. Fmoc/tBu (resins type A) or Boc/Bzl chemistry (resins type B). For additional information on handles see Vol. E22a, Section 4.3.1.2, manufacturers catalogues, and specialized reviews.[336]

Scheme 14 Resins and Handles Most Commonly Used in the Synthesis and Cyclization of Peptides on Resin; (A) Compatible with Fmoc/tBu and (B) with Boc/Bzl Chemistry[a]

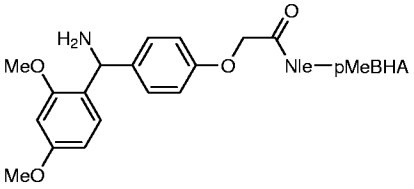
amine resins



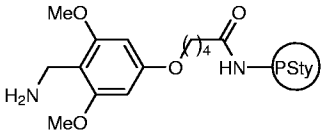
pMeBHA-PSty (B)



benzylamine-PSty (B)

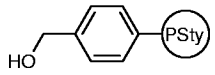


Rink-Nle-pMeBHA (A)

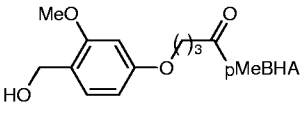


PAL-amino-PSty (A)

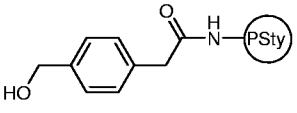
hydroxy resins



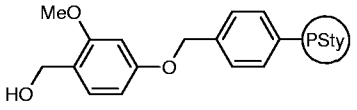
hydroxymethyl-PSty (B)



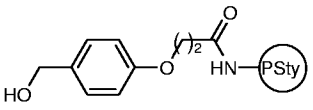
HMPB-pMeBHA (A)



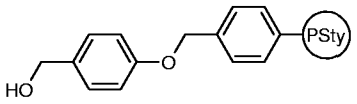
hydroxymethyl-PAM-PSty (B)



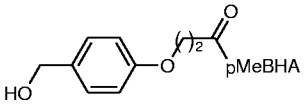
SASRIN-PSTy (A)



HMPPA-amino-PSty (A)

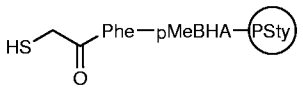


4-alkoxybenzylalcohol-PSty (Wang resin) (A)

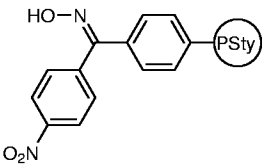


HMPPA-pMeBHA (A)

electrophilic resins

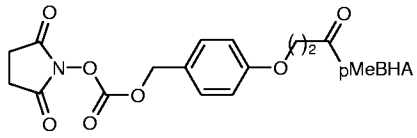


thioester-pMeBHA (B)

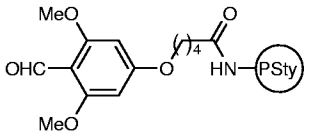


oxime-based-PSty (B)
(Kaiser resin)

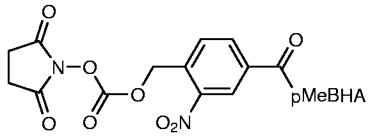
miscellaneous resins



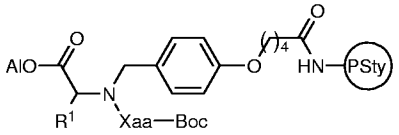
succinimide-carbonate-HMPPA-pMeBHA (A)



BAL-PSty (A)

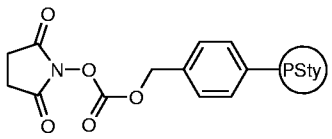


succinimide-carbonate-Bzl(2-NO_2)-pMeBHA (B)



allyl *N*-{4-[(5-amino-5-oxopentyl)oxy]benzyl}alkylglycinate-PSty (BOV-PSty) (B)

for references see p 529



succinimide-carbonate-hydroxymethyl-PSty (B)

[a] See also Vol. E22a, Section 4.3.1.2, Table 1.

Intramolecular reactions of resin-bound substrates supposedly take advantage of a pseudodilution phenomenon[337–340] that arises from the fact that the precursor molecules are attached to the solid support at distinct locations. This should favor intramolecular reactions that lead to cyclomonomers over intermolecular reactions that generate cyclodimers, cyclooligomers, and polymers. However, based on comparative studies between cyclization in solution and on resin it was argued that the on-resin method does not confer any significant benefit by the pseudodilution effect,[341] since this is offset by higher "local" concentration of the precursor molecules that may arise from their nonhomogenous distribution on the resin beads. Correspondingly, the physicochemical properties and particularly, the degree of cross-linking and functionalization of the polymeric support is expected to affect the yield and quality of the cyclic peptides in terms of cyclomonomer/cyclooligomers ratio significantly.

The role of the degree of substitution of resins on the ratio cyclomonomer/cyclooligomers is controversial. In fact, various comparative studies have confirmed that the amount of unwanted cyclodimers, cyclooligomers, and polymers is usually higher with resins of high loading capacities.[342–344] Correspondingly, partial capping of the resin prior to the incorporation of the handle in order to minimize formation of cyclodimers and polymers has been recommended.[344] However, studies on oxime resin have shown that in head-to-tail cyclizations, the extent of cyclooligomerization does not depend on the degree of resin substitution, but rather on the choice of the C-terminal residue; yields of cyclomonomer in the examined case were significantly superior, e.g. with C-terminal Phg rather than with Arg(Tos).[341,345] Moreover, it has been shown that when the repeating unit of gramicidin S [D-Phe-Pro-Val-Orn(Z)-Leu] is constructed on an oxime resin (loading: 0.54 and 0.17 mmol oxime $\cdot g^{-1}$ resin which yielded in both cases 0.163 mmol Leu $\cdot g^{-1}$) the resin of lower oxime loading led to larger amounts of cyclodimer due to intermolecular reactions.[346] The monocyclic peptide is formed at a faster rate and in higher yields with the higher oxime-loaded resin. However, it should be kept in mind that these studies were performed on oxime resins where the site of cyclization is attached to the polymeric support and thus may exert a marked steric effect on the rate of cyclization arising from the C-terminal amino acid residue, and this could well become the dominant factor.

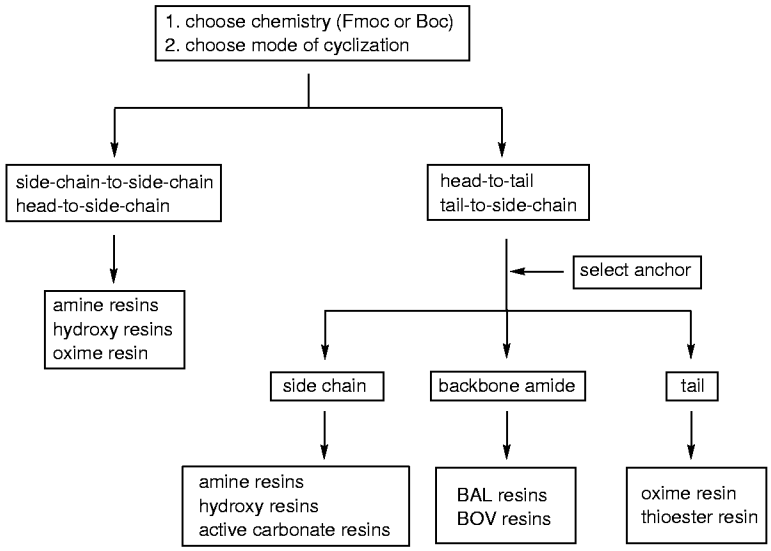
A variety of polymeric supports are presently commercially available; however, the "same" resin from different manufacturers has been found to produce markedly different results in terms of yields and purity of the identical peptide under identical experimental conditions.[340] In many cases the PEG-PSty resins have proved to be superior compared to PSty resins, a fact which is attributed to the better physicochemical properties of PEG-PSty, e.g. narrow size distribution, high diffusion rate, and excellent swelling with a wide range of solvents.

A detailed comparative study regarding the effect of the degree of cross-linking of PSty on the yield and purity of on-resin head-to-tail cyclization using the antibacterial cyclic undecapeptide kawaguchipeptin B as model system has been reported.[347] For this purpose Fmoc-Asp-Gly-OAl was anchored by the aspartic acid side-chain carboxy group to Rink-Nle-PSty resins cross-linked with 11 different ratios of DVB (cross-linking ranged from 0.3–6.0%). All synthetic steps were performed under identical conditions and yields and purity of the precyclic as well as of the cyclic peptides were determined by HPLC/MS and NMR. The highest yield and best quality of both precyclic and cyclic peptides were obtained

on resins with 3–6% DVB cross-linking. Yields of cyclization improved with increasing cross-linking from 28% purity for the 0.3% resin to >50% for the higher cross-linked polymer. These results contrast previous ones where resins with high swelling properties were recommended. On the other hand, such results could also reflect a greater reactivity and accessibility of sites on the lower cross-linked PSty resins that permit more side reactions to occur or increased intermolecular interactions and reactions. Thus, for small cyclopeptide synthesis where access may not be such an issue, higher cross-linking of the resins could well offer some advantages.

All these comparative studies have been performed on one mode of cyclization and with specific peptides. For more general conclusions and thus, for a more precise selection of the most suitable resin for cyclization of specific peptides in the desired mode, more experimental data is required. Nonetheless, a guideline for the selection of at least the most appropriate handles on the basis of the protection scheme applied for the synthesis and cyclization in the various modes is summarized in Scheme 15. Generally, side-chain-to-side-chain and head-to-side-chain cyclizations are less troublesome than those involving the C-terminal carboxy group. The choice of chemistry whether Fmoc/tBu or Boc/Bzl depends on the laboratory setup and on personal preferences. For cyclizations involving the C-terminus three different strategies for anchoring the resin are available and scopes and limitations of each of these strategies are discussed in Section 6.8.3.2.2.

Scheme 15 Guideline for the Selection of Handles for On-Resin Cyclization



6.8.3.1.2 Protection Schemes for the On-Resin Synthesis of Cyclic Peptides

A variety of orthogonal combinations of protecting groups is available for the synthesis of monocyclic peptides on resin.[339,348] They must satisfy an additional level of selectivity that is required upon assembly of the fully protected peptide on resin, which enables the subsequent regioselective chemistry to be performed at the functionalities involved in the cyclization reaction while the substrate is still bound to the resin and protected at all residual functions.

The most common protection schemes applied in the synthesis of cyclic peptides on solid supports are listed in Table 6. To facilitate understanding of these combinations of protecting groups, the single types are classified according to the order of removal along the synthetic route. Thus, the 1st order protecting groups (also called temporary or transient

for references see p 529

protecting groups) are those used for N^{α}-protection. These are removable while retaining the 2nd and 3rd order protecting groups. The 2nd order protecting groups (also called semi-permanent protecting groups) are used to protect the sites of cyclization and are removable while retaining the 1st order and the 3rd order protection. The 3rd order protecting groups (also called permanent protecting groups) are those used to protect all other functional groups in the peptide that are not involved in either assembly of the peptide chain or cyclization. These 3rd order protecting groups including the resin linker are those that are removed in the final resin-cleavage/deprotection step.

In the case of bi- or tricyclic peptides more than three levels of orthogonality in the protecting schemes is required (see Section 6.8.3.2.5).

Table 6 Common Protection Schemes Used for SPPS and On Resin

Entry	Mode[a]	1st Order	Removal	2nd Order[b]	Removal	3rd Order[c]	Deprotection/ Cleavage[d]
1	all	Boc	TFA	Fmoc/OFm	piperidine	Bzl	HF
2	all	Boc	TFA	Aloc/OAl	Pd(0)	Bzl	HF
3	all	Fmoc	piperidine	Boc/OtBu	TFA	Bzl	HF
4	all	Boc	TFA	oxime resin	free NH_2	Bzl	HF
5	all	Fmoc	piperidine	Aloc/OAl	Pd(0)	OtBu	TFA
6	all	Fmoc	piperidine	Dde/ODmab	hydrazine	OtBu	TFA
7	all	Fmoc	piperidine	Ddiv/ODmab	hydrazine	OtBu	TFA
8	C	Fmoc	piperidine	Cam	NaOH	OtBu	TFA
9	C	Fmoc	piperidine	OTMSE	TBAF	OtBu	TFA
10	C	Fmoc	piperidine	2,4-Dmb	1% TFA	OtBu	TFA
11	C	Fmoc	piperidine	OPp	1% TFA	OtBu	TFA
12	B, C	Fmoc	piperidine	Mtt	1% TFA	OtBu	TFA
13	B, C	Fmoc	piperidine	Adpoc/OPp	1% TFA	OtBu	TFA

[a] B = side-chain-to-side-chain; C = head-to-side-chain.
[b] Ddiv = 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl (also called ivDde); Adpoc = 2-(1-adamantyl)-2-propoxycarbonyl; OPp = 2-phenyl-2-propyl ester.
[c] Only the type of protection is given. Thus Bzl type protecting groups represents those used for 3rd order protection with 1st order Boc chemistry and are removed by HF or equivalent and the OtBu protecting group represents those used for 3rd order protection with 1st order Fmoc chemistry and are removed by TFA.
[d] HF or equivalent acid such as TfOH.

The 2nd order protecting groups listed in Table 6 are divided into two types: (1) 2nd order protecting groups that are removed by a different mechanism than the other protecting groups (e.g., Aloc/OAl for both Fmoc and Boc chemistries and Fmoc/OFm for Boc chemistry). In these cases, side reactions due to either partial removal of the 2nd order protecting groups during cleavage of the 1st order protecting groups (e.g., peptide elongation) or partial removal of the 3rd order protecting groups during cleavage of the 2nd order protecting groups (e.g., prior to cyclization), do not occur, and (2) 2nd order protecting groups that are removed by the same, or similar mechanism as the other protecting groups but at different rates (e.g., Dde/ODmab in Fmoc chemistry and 2,4-Dmb in Boc chemistry). In such cases some side reactions may occur. In fact, the *N*-Dde protecting group (Table 6, entry 6) is susceptible to cleavage on prolonged or repeated treatment with 20% piperidine in DMF.[349,350] Moreover, this group is also susceptible to migration, in the presence of free amines, by a transamination pathway. In this context the *N*-Ddiv protecting group (Table 6, entry 7) is less prone to side reactions. In addition, removal of the ODmab protecting group (Table 6, entries 6 and 7) with 2% hydrazine monohydrate in DMF may also cause partial cleavage of the N^{α}-Fmoc protecting group. Nevertheless, the use of both Dde/ODmab and Ddiv/ODmab as 2nd order protecting groups has distinct advantage over the Aloc/OAl or the OTMSE groups, since its cleavage is fast and facile and avoids extensive post-column

manipulations.[351,352] Another example is the removal of the carboxy protecting group, 2,4-Dmb (Table 6, entry 10) with 1% TFA. These conditions lead to partial cleavage of the third order Trt protecting group from Cys or His. Minor loss of the third-order tBu protecting group during this procedure has also been reported.[353]

6.8.3.1.3 Reagents for On-Resin Cyclization

In principle all methods used for peptide bond formation can be applied in all the various modes of cyclization of peptides as discussed in Section 6.8.2.1, although for cyclization on resin some limitations are given depending upon whether the peptide is anchored C-terminally or by side-chain functions to the solid support. While in the former case the active ester procedure is used, for all other modes of cyclization, in addition to the active esters, coupling reagents are predominantly employed for the generation of the lactam bridge. Conversion of the side-chain carboxy groups to active esters occurs either by standard carbodiimide-mediated esterification (see Vol. E 22a, Section 3.2.1.1) or by transesterification, e.g. with pentafluorophenyl trifluoroacetate[354] or 4-nitrophenyl trifluoroacetate.[355] Carbodiimides in the presence of 1,2-dinucleophiles as additives (see Vol. E 22a, Section 3.5.1.2) are generally used as coupling reagents; mixed anhydrides mediated by ethyl 2-ethoxy-1,2-dihydroquinoline-1-carboxylate are also occasionally used.[356,357] The carbodiimide reaction proceeds rapidly and C-terminal epimerization as well as other side reactions are minimized. Among the carbodiimides, *N,N'*-diisopropylcarbodiimide is the preferred reagent, since the resulting urea is soluble in the solvents generally used. More recently HOAt has been shown to be a more efficient additive than HOBt, even in amide bond formation on resin.[134] With the development of the phosphonium (see Vol. E 22a, Section 3.7) and uronium/guanidinium reagents (see Vol. E 22a, Section 3.8) lactam rings are increasingly generated with these classes of coupling agents. Since the latter reagents can react with the amino group to produce linear tetramethylguanidino derivatives,[358–361] phosphonium reagents such as BOP or PyAOP are preferred.

The alternative is to exploit the attachment of the carboxy group to the solid support in the form of an active ester that undergoes aminolysis by the amino group in the cyclization reaction with concomitant release from the resin. For this purpose various groups have been proposed such as resin-bound 2-nitrophenol,[330] 4-sulfonylphenol,[362] or sulfanylacetyl resin[363] and 4-nitrobenzophenone oxime resin (see Section 6.8.3.2.2).[364,365] A selection of the optimal amide forming reaction for lactamization on resin is difficult, since various factors are known to affect yields and correspondingly, amounts of side products such as unreacted precyclic peptide, epimerization in cases where the cyclization reaction involves a chiral center, or formation of cyclomonomers, cyclodimers, cyclooligomers, and polymers. As listed in Table 7, numerous studies have been performed to analyze this aspect by comparing various coupling reagents in their efficiency of cyclization of selected model peptides under identical experimental conditions. The results clearly underline the strong impact of the intrinsic propensity of the resin-bound linear precursor to cyclize. To analyze this aspect, preliminary experiments on the synthetic targets are recommended in which efficacious coupling reagents such as BOP, PyBOP, or BOP/HOBt and less efficient ones such as DIC in the presence or absence of HOBt should be used with DMSO/NMP (1:4) as solvent,[366] since this solvent mixture is the most efficient for favoring cyclomonomer over cyclodimer formation.[125] Side products are formed in addition to the cyclic peptide when using phosphonium reagents, even in the case of peptides that have a low cyclization tendency, however the use of DIC in such cases results mainly in the recovery of the linear precursor but extending the reaction time gives enhanced yields of cyclomonomer.

for references see p 529

Table 7 Reagents Used for On-Resin Cyclization[340,366–374]

Mode of Cyclization[a]	Resin[b] (Final Deprotection)	Peptide	Cyclization Reagent(s)	Detection Method	Preferred Reagent	Ref
A (SCA)	HMPPA-PEG-PSty (TFA)	*c*[-Tyr-D-Trp-Val-D-Trp-D-Trp-Arg-Asp-]	PyAOP/HOAt, PyBOP/HOBt, HATU/HOAt, HBTU/HOBt	Kaiser test and HPLC	PyAOP/HOAt> then rest	[367]
B	pMeBHA (HF)	dynorphin analogue	HBTU, HATU, HAPyU, BOP, PyBOP, PyAOP	Kaiser test	phosphonium >>>uronium salts PyBOP>BOP	[368]
A (SCA)	benzylamine-PSty (HF)	*c*[-Ala-Ala-Arg-D-Phe-Pro-Glu-Asp-Gln-Tyr-Glu-]	BOP/HOBt, DIC/HOBt in DMF, CH_2Cl_2, and THF	Kaiser test and HPLC	BOP/HOBt ~ DIC/HOBt in DMF and THF DIC/HOBt> BOP/HOBt in CH_2Cl_2	[340]
A (SCA)	HMP (TFA)	*c*[-Trp-Ile-Val-D-Pro-Asp-] and other peptides containing Asp at the C-terminus	BOP/HOBt	FAB-MS, Marfey's test for racemization	racemization of Asp is minimized by using only equiv DIPEA	[369]
B	pMeBHA (HF)	analogues of influenza virus hemagglutinin	HATU/HOAt BOP/HOBt, DIC/HOBt	Kaiser test	BOP/HOBt (3 equiv) in DMF, DIPEA (7.5 equiv), 2 h	[370]
B	pMeBHA (HF)	$\alpha MSH_{4\text{-}10}\text{-}NH_2$ analogue containing D-Xaa	BOP, DIC/HOBt, DPPA	quant ninhydrin test	BOP (6 equiv) in DMF, DIPEA (excess), 4 h	[371]
B	pMeBHA (HF)	hGRF 1–29 analogues	BOP, DCC/HOBt, BOP-Cl, DPPA	quant ninhydrin test	BOP (6 equiv) in DMF, DIPEA (excess), 4 h	[372]
A (SCA)	PAL-PEG-PSy (TFA)	*c*[-D-Val-Arg-Gly-Asp-Glu(εAhx-Cys-NH_2)-]	TBTU, BOP, HATU	Kaiser test	HATU >>TBTU >>BOP	[373]
A (SCA)	1. HMPPA-pMeBHA (TFA) 2. Nbz-pMeBHA (HF)	somatostatin and gramicidin S analogues	BOP/HOBt, PyAOP/HOAt	Kaiser test	yield and purity depends on anchoring Xaa and handle	[374]
B	pMeBHA (HF)	cyclic analogues of hGH (6–13)	BOP, BOP/HOBt, HBTU/HOBt, DIC/HOBt	HPLC/MS	BOP, BOP/HOBt, HBTU/HOBt (3 equiv) in DMF, 7 h	[366]

[a] A = head-to-tail; B = side-chain-to-side-chain; (SCA) = side-chain anchor.
[b] For structures and abbreviation of resins, see Scheme 14.

In addition to the mentioned side products, formation of the piperidyl amide of the linear precursors has been observed as deriving from the piperidine-mediated cleavage of the Fmoc/OFm groups used for intermediate protection of the functionalities involved in the cyclization. In these cases, a brief washing with 0.4% (v/v) concentrated aqueous HCl in DMF[375] or with tertiary amines, e.g. DIPEA,[369] prior to cyclization is recommended in order to displace the piperidine. Moreover, both these extra washings were found to shorten the cyclization time to 2 hours with BOP and 0.5 hours with HATU.

Generally, monitoring of the cyclization reaction on resin is recommended; it is readily performed by the qualitative Kaiser[376] test and/or more correctly by HPLC/MS upon cleavage/deprotection of aliquots from the resin. In this context, attention must be paid to the "false" positive Kaiser test as resulting from the piperidyl amide formation.[366,369,375]

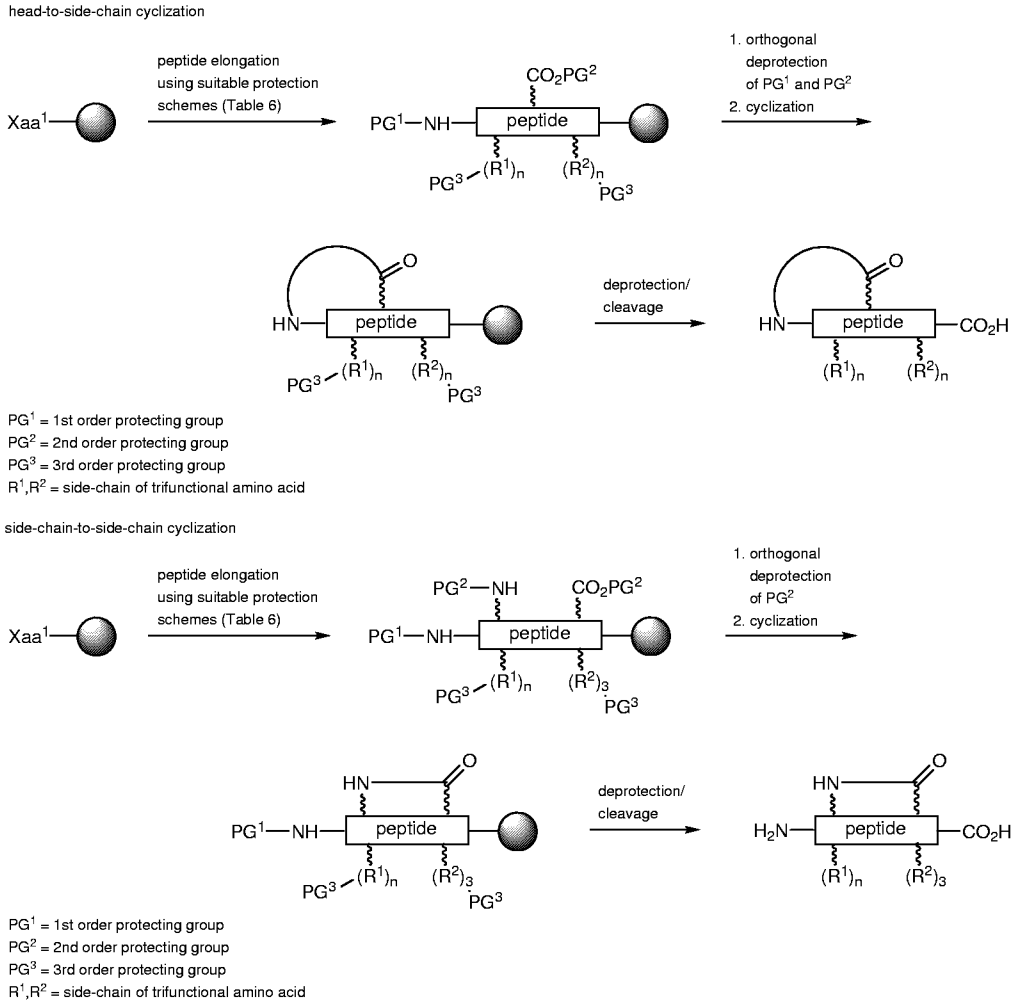
6.8.3.2 Strategies for Different Modes of On-Resin Cyclization

The choice of protecting groups and of the type of resin-anchor required for synthesis of cyclic peptides on solid supports not only depends upon the particular amino acid sequence of the target molecule, but decisively upon the mode of cyclization, particularly whether the C-terminal carboxy group should act as bridgehead or not. Correspondingly, the synthetic routes can be subdivided into two main classes either based on the attachment of the C-terminus to the solid support by suitable anchors or where side-chain functionalities are exploited for this purpose.

6.8.3.2.1 Synthesis of Side-Chain-to-Side-Chain and Head-to-Side-Chain Cyclic Peptides

The majority of cyclic peptides synthesized on solid support are cyclized in the head-to-side-chain or side-chain-to-side-chain mode. For this purpose the amino acids involved in cyclization must be side-chain protected in a manner that allows for an additional level of orthogonal deprotection. Thus, upon assembly of the fully protected linear precursor on-resin, deprotection of the functionalities involved in the lactam ring formation is performed, followed by regioselective cyclization by amide bond formation, and finally by the resin-cleavage/deprotection step as outlined in Scheme 16. In Table 8, examples of syntheses of such cyclic peptides are listed with the relevant information regarding protection scheme, resin anchor, and mode of cyclization.

Scheme 16 General Synthetic Routes for Synthesis and On-Resin Cyclization



for references see p 529

Table 8 Examples of Head-to-Side-Chain and Side-Chain-to-Side-Chain[206,334,372,377–394]

Mode of Cyclization[a]	Resin[b] (Final Deprotection)	Peptide[c]	Protecting Groups (Order of Orthogonality)[d]			Ref
			1st	2nd	3rd	
C	pMeBHA (HF)	carba vasopressin	Boc	OtBu	Bzl	[377]
C	Rink amide (TFA)	*c*[-ETRPMD-]	Fmoc	OTMSE	tBu	[378]
B	pMeBHA (HF)	enkephalin analogues dynorphin analogues	Fmoc	Boc/OtBu	Bzl	[106,361,379,380]
B	pMeBHA (HF)	hGRF 1–29; analogues of F&M disease virus	Boc	Fmoc/OFm	Bzl	[372,381–388]
B	pMeBHA (HF)	α-MSH (4–11)-NH_2	Fmoc	Boc/OtBu	Bzl	[334,389]
B	pMeBHA (HF)	α-MSH (4–10)-NH_2	Boc	Fmoc/OFm	Bzl	[334,390,391]
B	pMeBHA (HF)	glucagon-NH_2	Boc	Fmoc/OFm	Bzl	[334,391]
B, C	PAL, HMPPA (TFA)	model peptides	Fmoc	Dde/ODmab	tBu	[392]
C	Pepsyn K (TFA)	model peptides	Fmoc	Cam	z/tBu	[393]
B	HMPPA-pMeBHA PAL (TFA)	astressin; endothelin B antagonist; neurotensin and somatastatin analogues	Fmoc	Adpoc/OPp	tBu	[394]

[a] B = side-chain-to-side-chain; C = head-to-side-chain.
[b] For structures and abbreviation of resins see Scheme 14.
[c] F&M = foot and mouth.
[d] Adpoc = 2-(1-adamantyl)-2-propoxycarbonyl; OPp = 2-phenyl-2-propyl ester.

Head-to-Side-Chain and Side-Chain-to-Side-Chain Cyclized Peptides with Boc/Bzl Chemistry; General Procedure:

The peptide was synthesized with amino acid derivatives compatible with the Boc/Bzl chemistry and similarly, using anchor groups suitable for this protection scheme. Conversely, amino acids involved in the cyclization reaction were side-chain protected in a selective manner (see Table 6), e.g. as Boc-Asp(OFm)-OH or Boc-Asp-(OAl)-OH and/or Boc-Lys(Fmoc)-OH or Boc-Lys(Aloc)-OH. For side-chain-to-side-chain cyclization the N^{ω}-Fmoc and ω-Fm ester were removed with piperidine and N^{ω}-Aloc and ω-OAl ester with Pd(0), while for head-to-side chain cyclization the N^{α}-Boc group was additionally cleaved with TFA. After deprotection with piperidine the resin was briefly washed with 0.4% (v/v) concd aq HCl in NMP (1 × 1 min) to remove quantitatively piperidine and then with DIPEA/NMP (1:9, 3 × 2 min). After deprotection with TFA the resin was washed with CH_2Cl_2 (3 × 2 min), neutralized with DIPEA/CH_2Cl_2 (1:9, 3 × 2 min), and washed with NMP (3 × 2 min) followed by the cyclization solvent (3 × 2 min), preferentially DMSO/NMP (1:4). When phosphonium reagents were used for cyclization, BOP or PyBOP (5 equiv), HOBt (5 equiv), and DIPEA (10 equiv) were mixed for 1 min in solvent (1 mL) and the mixture was added to the resin (100 mg; loading: 0.6 mmol·g^{-1}) suspended in solvent (1 mL). The resin was shaken at rt for 2 h. When carbodiimides were used DIC (5 equiv), HOBt (5 equiv), and DIPEA (5 equiv) were mixed for 1 min in solvent (1 mL) and the mixture was added to the resin suspended in solvent (1 mL). The resin was shaken at rt for 7 h. To monitor the progress of the reaction a small sample of the resin was removed from the mixture and washed with NMP (3 × 2 min), CH_2Cl_2 (3 × 2 min), and Et_2O (3 × 2 min); after drying the resin was subjected to the Kaiser test. The cyclic peptide-resin was washed (3 × 2 min) with NMP, EtOH, and Et_2O and after drying, the cyclic peptide was cleaved and fully deprotected by treatment with HF/scavengers (10:1, 10 mL for 1 g peptide-resin) at 0 °C for 2 h. The product was isolated by removal of HF, washing with EtOAc, extraction with TFA, concentration, and trituration with Et_2O. After drying, the peptide was dissolved in dil aq AcOH or H_2O/dioxane (1:9) and lyophilized.

Ac-Nle-Asp-His-D-Phe-Arg-Trp-Lys-NH_2; Typical Procedure for Side-Chain-to-Side-Chain Cyclized Peptides with Boc Chemistry:[390]

The protected peptide-resin was prepared on pMeBHA-resin (1.5 g, 0.6 mmol of NH_2/g of resin) by first coupling of Boc-Lys(Fmoc)-OH to the resin (substitution level: 0.3 mmol·g^{-1} of resin). The residual amino groups were capped by acetylation with acetic anhydride. Stepwise chain elongation with Boc-Trp(For)-OH, Boc-Arg(Tos)-OH, Boc-D-Phe-OH, Boc-His(Bom)-OH, Boc-Asp(OFm)-OH, and Boc-Nle-OH was performed by standard methods. Each coupling reaction was carried out with a threefold excess of DIC and a 2.4-fold excess of HOBt. After coupling the last amino acid, the Fmoc and OFm side-chain protecting groups were removed with 50% piperidine in DMF for 1 h. The peptide-resin was washed with DMF (3 × 40 mL), CH_2Cl_2 (3 × 40 mL), 10% DIPEA in CH_2Cl_2 (3 × 40 mL), and CH_2Cl_2 (3 × 40 mL). The peptide-resin was suspended in DMF (15 mL) and mixed with a sixfold excess of BOP in the presence of an eightfold excess of DIPEA for 6 h. The cyclization was repeated twice until the

resin resulted negative to the ninhydrin test. After washings as above, the N-terminal Boc group was removed and the amino group was neutralized and acetylated with 25% Ac_2O in CH_2Cl_2 for 0.5 h. The resin was washed as above and dried (1.86 g). The peptide-resin (0.5 g) was treated with anhyd HF in the presence of anisole (~10%) and EDT (~0.8%) at 0 °C for 45 min in the usual manner. The crude peptide powder was purified by cation-exchange chromatography on a CMC column with discontinuous NH_4OAc buffer soln followed by gel filtration (Sephadex G-15, 15% aq AcOH). The final purification was effected by preparative RP-HPLC (C18 column, Vydac 218TP1010, 1 × 2.5 cm, linear 20–40% MeCN gradient with a constant concentration of 0.1% TFA); yield: 87 mg (58%).

H-Thr-Thr-Glu-Thr-Ala-Ser-Ala-Arg-Gly-Asp-Leu-Ala-His-Leu-Thr-Thr-Thr-His-Ala-Lys-His-Leu-OH; Typical Procedure for Side-Chain-to-Side-Chain Cyclized Peptides with Boc/Bzl Chemistry:[395]
The protected peptide-resin Boc-Thr(Bzl)-Thr(Bzl)-Glu(OFm)-Thr(Bzl)-Ala-Ser(Bzl)-Ala-Arg(Tos)-Gly-Asp(OCy)-Leu-Ala-His(Dnp)-Leu-Thr(Bzl)-Thr(Bzl)-Thr(Bzl)-His(Dnp)-Ala-Lys(Fmoc)-His(Dnp)-Leu-pMeBHA-resin (300–400 mg) was treated with PhSH/DIPEA/DMF (3:3:4, 6 × 5 mL, 60 min) in DMF, then with 20% piperidine and 2% DBU in DMF (1 × 20 min). The cyclization on resin was carried out with BOP/DIPEA (20 and 60 equiv; every 6 h, respectively) in DMF at rt for 18 h and followed by Boc removal with neat TFA (1 × 10 min). Cleavage and total deprotection of the cyclic peptide-resin was accomplished with anhyd HF/*p*-cresol (9:1, 5 mL, 0 °C, 1 h). After HF removal, the residue was triturated with dry Et_2O and CH_2Cl_2 and dissolved in 0.1 M AcOH (14 mL). After lyophilization, the product was purified by preparative HPLC [Vydac C18 column (5 µm, 1 × 25 cm), linear 14–18% gradient CH_2Cl_2/H_2O (both containing 0.05% TFA), 45 min, flow rate 3 mL·min^{-1}]. Fractions corresponding to cyclic peptide were pooled and lyophilized; yield: 3 mg (5%).

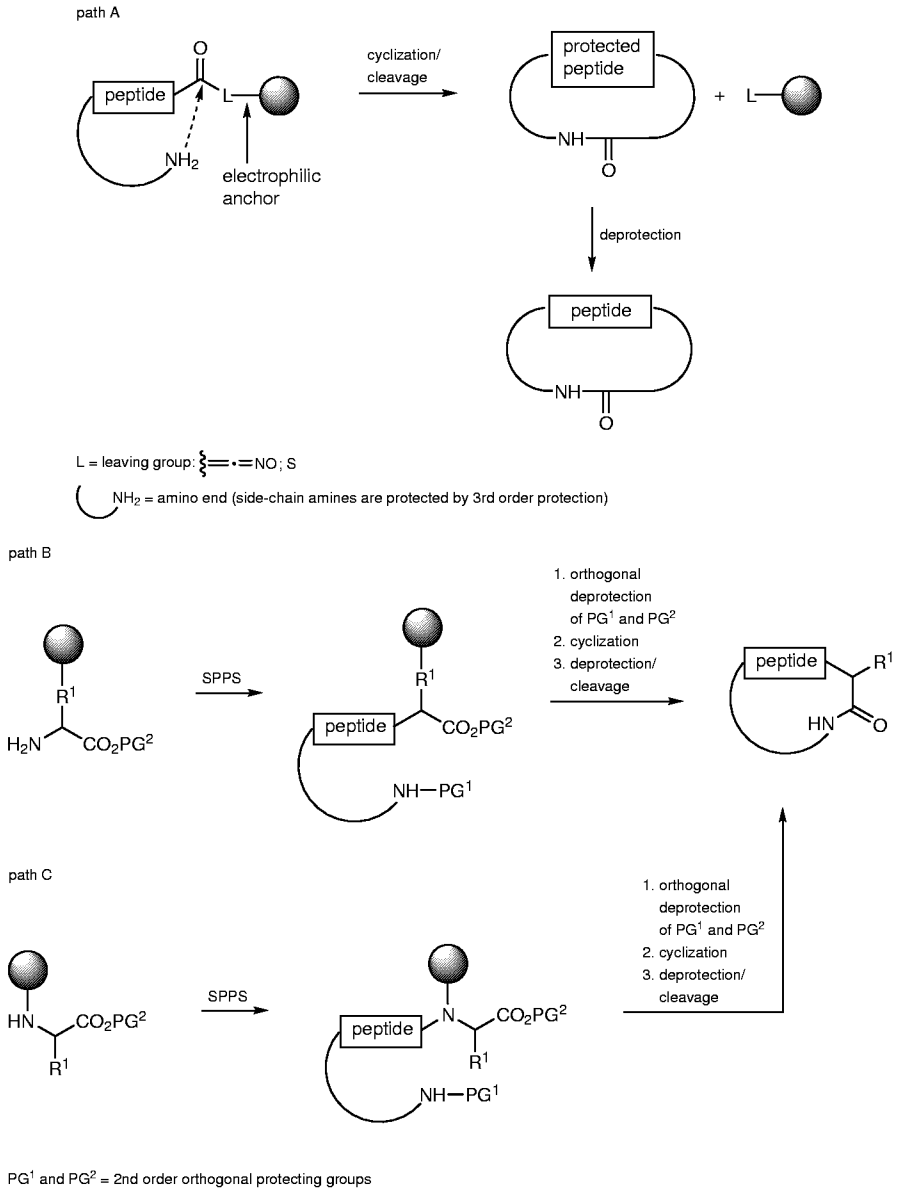
Head-to-Side-Chain and Side-Chain-to-Side-Chain Cyclized Peptides with Fmoc/tBu Chemistry; General Procedure:
The peptide was synthesized with amino acid derivatives compatible with the Fmoc/tBu chemistry and similarly, using anchor groups suitable for this protection scheme. Conversely, amino acids involved in the cyclization reaction were side-chain protected in a selective manner (see Table 6), e.g. as Fmoc-Asp-(OAl)-OH and/or Fmoc-Lys(Aloc)-OH. For side-chain-to-side-chain cyclization the N^{ω}-Aloc and ω-OAl ester were removed with Pd(0), while for head-to-side-chain cyclization the N^{α}-Fmoc group was cleaved with piperidine. The resin was washed (3 × 2 min) with NMP, CH_2Cl_2, and NMP, and then with the cyclization solvent (3 × 2 min), preferentially DMSO/NMP (1:4). When phosphonium reagents were used, BOP or PyBOP and HOBt (5 equiv) and DIPEA (10 equiv) were mixed for 1 min in solvent (1 mL) and the mixture was added to the resin (100 mg; loading: 0.6 mmol·g^{-1}) suspended in solvent (1 mL). The resin was shaken at rt for 2 h. When carbodiimides were used DIC (5 equiv), HOBt (5 equiv), and DIPEA (5 equiv) were mixed for 1 min in solvent (1 mL) and the mixture was added to the resin suspended in solvent (1 mL). The resin was shaken at rt for 7 h. To monitor the progress of the reaction a small sample of the resin was removed from the mixture and washed with NMP (3 × 2 min), CH_2Cl_2 (3 × 2 min), and Et_2O (3 × 2 min); after drying the resin was subjected to the Kaiser test. The cyclic peptide-resin was washed (3 × 2 min) with NMP, EtOH, and Et_2O and after drying, the cyclic peptide was cleaved and fully deprotected by treatment with TFA/scavengers. The resin was washed with TFA and the solvent was removed from combined filtrates by a stream of N_2 and the residue triturated with Et_2O. After drying, the peptide was dissolved in dil aq AcOH or H_2O/dioxane (1:9) and lyophilized.

6.8.3.2.2 Synthesis of Head-to-Tail and Side-Chain-to-Tail Cyclic Peptides

For cyclizations on resin involving the C-terminal carboxy group such as in the head-to-tail and side-chain-to-tail mode, three different strategies have been developed as shown in Scheme 17. These are based on three different types of linkage of the peptide to the resin: (1) the C-terminal amino acid is attached to the resin by its α-carboxy group by an electrophilic linker that permits nucleophilic attack of an N-terminal or side-chain amino group on the C-terminal carbonyl to generate the desired amide bond with concomitant release of the cyclic product from the resin (Scheme 17, path A); (2) the C-terminal amino acid is attached to the resin by its side-chain functionality, while the α-carboxy group is judiciously protected according to the overall protection strategy to allow for its selective removal prior to cyclization (Scheme 17, path B), and (3) the C-terminal amino acid is attached to the resin by a suitable N^{α}-aryl derivative that acts as reversible amide protecting group; again the α-carboxy group must have appropriate protection to allow for subsequent regioselective cyclization (Scheme 17, path C).

for references see p 529

Scheme 17 General Strategies for Peptide Cyclizations On Resin Involving the C-Terminal Carboxy Group



6.8.3.2.2.1 Active Ester Type Resin Handle

Among the electrophilic handles proposed for head-to-tail and side-chain-to-tail cyclization of peptides on solid support by intrachain aminolysis with concurrent detachment of the product from the resin in the protected form (see Section 6.8.3.1.3), generally the oxime resin (also called Kaiser resin)[364,365] and a thioester resin[363] are recommended (see Scheme 14). In addition to the classical head-to-tail cyclization,[343] the oxime resin is used for side-chain cyclizations as well as for the synthesis of multicyclic peptides (*vide infra*). Due to its dual functions, the oxime resin can be employed only with Boc/Bzl chemistry; it is not compatible with Fmoc/tBu chemistry where the basic N^{α}-deprotection leads to free amino groups and thus to premature cyclization reactions. To avoid this premature cleavage of the

peptide from the resin during peptide chain assembly according to the Boc/Bzl strategy, after TFA-mediated N^{α}-Boc cleavage, the neutralization step is omitted and coupling is performed on the TFA salt using phosphonium or uronium/guanidinium reagents with in situ neutralization. Intramolecular aminolysis is only induced upon neutralization of the amino group in the absence of strong acylating agents. The cyclization reaction is relatively slow in CH_2Cl_2 (24 h),[343] but can be catalyzed with acetic acid (10 equiv) in dioxane (3–4 h),[396] or acetic acid in DMF (50 °C, 72 h).[397]

In a similar manner, the thioester resin[363] is used only with Boc/Bzl chemistry, and the intramolecular aminolysis of the thioester by the amino group occurs upon deprotection and neutralization with DIPEA (3.0 equiv) and is catalyzed by DMAP (0.1 equiv) in DMA (cyclization time: 2–7 d).

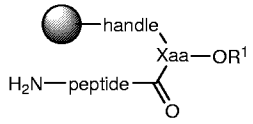
***c*[-D-Val-(Me)Arg-Gly-Asp-Mamb-]; Typical Procedure for Head-to-Tail Cyclization on Oxime Resin:**[397] To a soln of Boc-Mamb-OH [Mamb = 3-(aminomethyl)benzoic acid] (0.126 g, 0.5 mmol) in DMF (6 mL) were added HBTU (0.194 g, 0.5 mmol), oxime resin (0.52 g, substitution level: 0.96 mmol·g^{-1}), and NMM (0.165 mL, 1.50 mmol). The suspension was mixed at rt for 24 h and then the resin was washed thoroughly with DMF (3 ×), MeOH(1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and CH_2Cl_2 (3 ×) each 10–12 mL; substitution level: 0.389 mmol·g^{-1} (by quantitative ninhydrin assay). Unreacted oxime groups were blocked by treatment with 0.5 M pivaloyl chloride/0.5 M DIPEA in DMF for 2 h. The resin was washed with DMF (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and CH_2Cl_2 (3 ×) and the Boc group cleaved with 25% TFA in CH_2Cl_2 for 30 min. The resin was washed as before with CH_2Cl_2 (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and DMF (3 ×), and Boc-Asp(OCy)-OH (0.613 g, 1.94 mmol) was coupled with HBTU (0.753 g, 1.99 mmol) in DMF (8 mL) and NMM (0.642 mL, 5.84 mmol) for 2.5 h. Quantitative ninhydrin assay was used to monitor the reaction. The identical deprotection and coupling steps were repeated with Boc-Gly-OH, Boc-(Me)Arg(Tos)-OH, and Boc-D-Val-OH. Couplings of Boc-(Me)Arg(Tos)-OH and Boc-D-Val-OH were monitored by the picric acid test. After the linear peptide was assembled, the N-terminal Boc group was removed with 25% TFA in CH_2Cl_2 (30 min) and the resin was washed thoroughly with CH_2Cl_2 (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), and neutralized by washing with 10% DIPEA in CH_2Cl_2 (2 × 1 min). The resin was washed again with CH_2Cl_2 (3 ×) and MeOH (1 ×) and then dried. Half of the resin (0.101 mmol) was used for cyclization by treatment with DMF (6 mL) containing AcOH (5.8 mL, 0.101 mol) at 50 °C for 72 h. The resin was then removed by filtration and washed with DMF. The combined filtrates were concentrated to an oil, and the residue was dissolved in MeCN/H_2O (1:1), and lyophilized (49 mg of protected crude product). The peptide (42 mg) was treated with anhyd HF at 0 °C, in the presence of *m*-cresol for 30 min. The crude material was precipitated with Et_2O, redissolved in 10% AcOH, and lyophilized. The crude product was purified by RP-HPLC [Vydac C18 column, 2.5 cm, 0.23%/min gradient 7–18% MeCN containing 0.1% TFA]; overall yield: 9.4%.

6.8.3.2.2.2 Resin Attachment by Side-Chain Functionalities

Another convenient approach for the synthesis of head-to-tail or side-chain-to-tail cyclic peptides on resin is attachment of the peptide by the side-chain function of the C-terminal or penultimate amino acid to a suitable anchor as firstly proposed for a C-terminal aspartyl residue.[359] This strategy, outlined in Scheme 17 (path B), has been successfully utilized for various peptides containing Asp, Glu, Asn, and Gln by using resin handles of the hydroxy (ester linker) or amine type (amide linker), respectively, or Ser, Thr, and Tyr with resin handles of the halobenzyl (benzyl ether linker) or mixed carbonate type (ester linker), as well as Lys, Orn, and A_2bu, with resin handles of the mixed carbonate or chloroformate type (both leading to urethane-based linkers). Depending upon the compatibility of the linker group both the Boc/Bzl and Fmoc/tBu strategies can be used with appropriate selection of an additional level of orthogonality for protection of the N^{α}- or N^{ω}-amino and the C^{ω}- and C-terminal carboxy groups involved in the cyclization. An overall view of the possible combinations of anchors and suitable protection schemes is summarized in Table 9.

for references see p 529

Table 9 Examples for the Synthesis of Head-to-Tail and Side-Chain-to-Tail Cyclic Peptides Using the Side-Chain Anchor Method[109,340,347,352,373–375,398–407]



Xaa	Chemistry	R^1	Handle/Resin[a,c,d,e]	Ref
Asp	Fmoc (TFA)	allyl	HMPPA	[398]
Asp	Fmoc (TFA)	allyl	HMPPA	[352,375]
Asp	Fmoc (TFA)	Dmb	HMPPA-L-amino-PSty	[340,399]
Asp	Fmoc (TFA)	allyl	HMB-V-pMeBHA	[109]
Asp	Boc (HF)	Fm	hydroxymethyl-PAM-PSty	[400]
Asp	Boc	Fm	MHMB-pMeBHA	[401]
Asp	Boc	Fm	pMeBHA	[402]
Asp	Boc	Fm	HMB-V-pMeBHA	[109]
Asp	Boc	Fm	HMPPA	[403]
Glu	Fmoc (TFA)	Dmb	HMPPA-L-amino-PSty	[340,399]
Glu	Fmoc (TFA)	allyl	HMPPA	[352,375]
Glu	Boc	Fm	hydroxymethyl-PSty	[403]
Asn	Fmoc (TFA)	allyl	Rink Nle-pMeBHA	[398]
Asn	Fmoc (TFA)	allyl	PAL	[352,375]
Asn	Fmoc (TFA)	Gly-OAl	Rink Nle-pMeBHA	[347]
Asn	Fmoc (TFA)	allyl	AH-C-PAL	[404]
Asn	Boc	Fm	pMeBHA	[403]
Gln	Fmoc (TFA)	allyl	AH-C-PAL	[373]
Gln	Boc	Fm	pMeBHA	[403]
Ser	Boc (UV)	allyl	Nbz-A-pMeBHA	[374,405]
Tyr	Boc (HF)		HMPPA	[374]
Ser	Boc (NH_3)	Fm	amino-PSty	[403]
		Bzl(4-NO_2)[b]		
Tyr	Fmoc (TFA)	Me	HMPPA, sasrinHMPB-MBHA	[406]
Tyr	Boc (HF)	Fm	amino-PSty	[403]
		Bzl(4-NO_2)[b]		
Lys	Fmoc (TFA)	allyl	carbonate	[374,405,407]
Orn	Fmoc (TFA)	Bzl(4-NO_2)[b]	HMPPA-F-pMeBHA	[374]
Lys	Boc (HF)	Fm	amino-PSty + dipentafluorophenyl carbonate	[403]
Orn	Boc (HF)	Fm		
A_2bu	Boc (HF)	Fm		

[a] For the structures and abbreviation of the handles see Scheme 14.
[b] Removed prior to cyclization by (1) $SnCl_2$/DMF, (2) benzenesulfinic acid.
[c] MHMB-pMeBHA = 4-hydroxymethylbenzoyl-MBHA.
[d] HMB-V-MBHA = 4-hydroxymethylbenzoyl-valyl-MBHA.
[e] AH-C-PAL = 6-aminohexanoyl-cysteinyl-PAL.
[f] Succ = $COCH_2CH_2CO$.

The use of the C-terminal esters imposes two synthetic hurdles: (1) efficient and racemization-free attachment of the C-terminal amino acid ester to the resin by its side-chain functionality and (2) prevention of C-terminal epimerization both during chain assembly and cyclization.

During attachment of a chiral amino acid ester to the resin, the base required in the reaction, particularly when used in excess, leads to facile racemization.[369,401] This occurs by C^α-proton abstraction and enolization of the amino acid ester (see Vol. E 22a, Section 1.2.1.2 and Section 7.4), which may also occur during chain assembly in the repetitive base treatments, especially with Fmoc/tBu chemistry and in case of razemization-prone amino acid residues. Thus careful control of the base is required or the use of dipeptide building blocks such as Fmoc-Asp-Gly-OFm,[369] as C^α-proton abstraction is not stabilized by enolization.

Of the various esters used in this synthetic strategy (see Table 9), the fluorenylmethyl ester may additionally undergo partial hydrolysis in the neutralization cycles of the Boc/Bzl chemistry.[408,409] A possibly interesting alternative are Dmab esters[351] combined with *N*-Dde protection (see Vol. E 22a, Section 2.1.2.2.4); both protecting groups are compatible with Fmoc/tBu chemistry and are cleaved concurrently with 2% hydrazine in DMF. Even silyl-based amino and carboxy groups lend themselves for selective chemistry in this context.

(Bromomethyl)phenyl-pMeBHA-Resin:[401]
[(Bromomethyl)phenyl]acetic acid (4 equiv) and DCC (4 equiv) were added to pMeBHA-resin (1 equiv) in DMF. The suspension was stirred overnight at rt. Washing with DMF (6 × 2 min) yielded the (bromomethyl)phenyl-pMeBHA-resin; IR (KBr) $\tilde{\nu}$: 3350 (strong), 1620 cm^{-1} (sharp); gel-phase ^{13}C NMR (300 MHz, CD_2Cl_2, δ): 43.6 (C_6H_4-CH_2-CO), 44.4 (Br-CH_2); ES-MS and CID-MS of the residue resulting from acidolysis (HF/PhOMe, 9:1, 0 °C, 1 h) of the polymer, confirmed the integrity of the product.

Attachment of Boc-Asp-OFm to (Bromomethyl)phenyl-pMeBHA-Resin via the Cesium Salt:[401,410]
A soln of Boc-Asp-OFm (2 g, 4.9 mmol) in EtOH (10–15 mL) and H_2O (3–5 mL) was neutralized to pH 7.0 by adding aq $CsHCO_3$, and the solvent repeatedly azeotroped with benzene and removed under reduced pressure to eliminate traces of H_2O. After drying over P_2O_5 for 5 h, Boc-Asp-OFm Cs salt (0.54 g, 1 mmol) and (bromomethyl)phenyl-pMeBHA-resin (1 equiv) were stirred in DMF (6–8 mL/g resin) overnight at 50 °C. The resin was collected by filtration, washed with DMF (3 × 10 mL, 2 min), DMF/H_2O (9:1, 3 × 10 mL, 2 min), DMF (3 × 10 mL, 2 min), and EtOH (3 × 10 mL, 2 min) and dried; substitution: 0.17 mmol·g^{-1}; D-Asp: 2%.

Attachment of Boc-Asp-OFm to (Bromomethyl)phenyl-pMeBHA-Resin via the Zinc Salt:[401,411]
To a soln of Boc-Asp-OFm (0.3 g, 0.75 mmol) in EtOH (2 mL) and H_2O (1 mL) solid $ZnCO_3$ (45 mg, 0.36 mmol) was added and the mixture was sonicated for 10 min until neutral pH. The translucent soln was repeatedly concentrated with benzene and the solid dried over P_2O_5 for 5 h. A soln of Boc-Asp-OFm Zn salt (0.4 mmol) in DMF (5 mL) was added to (bromomethyl)phenyl-pMeBHA-resin (1 equiv) and the mixture was sonicated in a bath at rt for 1 h. The resin was collected by filtration, washed with DMF (3 × 10 mL, 2 min), DMF/H_2O (9:1, 3 × 10 mL, 2 min), DMF (3 × 10 mL, 2 min), and EtOH (3 × 10 mL, 2 min) and dried; substitution: 0.17 mmol·g^{-1}; D-Asp: 2%.

Attachment of Boc-Asp-OFm to HMP Resin:[401]
A soln of Boc-Asp-OFm (3.06 g, 7.49 mmol) in CH_2Cl_2 (40 mL) was added to HMP resin (5.0 g, 3.745 mmol). Pyridine (0.60 mL, 7.49 mmol) was added to this suspension under stirring, followed by a soln of $(Boc)_2O$ (1.88 g, 8.61 mmol) in CH_2Cl_2 (20 mL). The mixture was stirred at rt for 72 h, the resin was collected by filtration, washed with CH_2Cl_2 (6 × 30 mL), EtOH (3 × 30 mL), Et_2O (3 × 20 mL), and hexane (2 × 20 mL), and dried; yield: 6.70 g; theory 6.47 g (0.579 mmol·g^{-1}). Spectrophotometric determination of OFm ester on the resin provided 0.57 mmol·g^{-1} (98.5% substitution) and quantitative amino acid analysis 0.47 mmol Asp·g^{-1} (87.7% substitution).

6.8.3.2.2.3 Resin Attachment via the Backbone Amide

Based on the successful use of the TFA labile N^{α}-2-hydroxybenzyl (Hbz) and N^{α}-2-hydroxy-4-methoxybenzyl (Hmb) backbone amide protecting groups (see Vol. E 22a, Sections 2.3.2.1 and 2.3.2.2), an extension of this principle to resin-bound amide protecting groups[81] is well-suited for attachment of peptides to the solid support. Such a strategy allows exploitation of, in addition to the amino and carboxy terminus, all sequence-intrinsic side-chain functionalities for all types of monocyclizations and multiple cyclizations via lactam as well as other bridges by the correct choice of various levels of orthogonal protecting groups (see also Section 6.8.3.2.5). This strategy is still in its infancy and it has some disadvantages arising mainly from the temporary introduction of an N^{α}-benzyl group on the C-terminal amino acid. In the chain-elongation step such backbone linkage to the resin converts the peptide bond into a tertiary amide that favors a *cis* configuration and thus markedly facilitates piperazine-2,5-dione formation with consequent chain termination.[412] Moreover, as observed for the N^{α}-Hbz- or N^{α}-Hmb-amino acid derivatives the benzyl substituents increase steric hindrance in subsequent acylation reactions, which is difficult to overcome (see Vol.

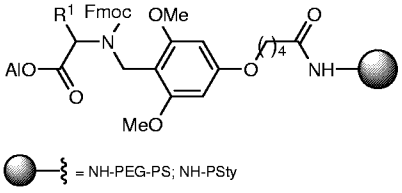
for references see p 529

E 22a, Sections 2.3.2.1 and 2.3.2.2, and Vol. E22c, Section 10.1.2.1). Additionally, N-alkylation of C-terminal amino acid residues significantly increases the risk of epimerization by intermediate 4-alkyloxazol-5(4*H*)-one formation (see Vol. E 22a, Section 1.2.1.2 and Section 7.4). Improved handles have been proposed that provide a wider array of amide derivatives.[413,414]

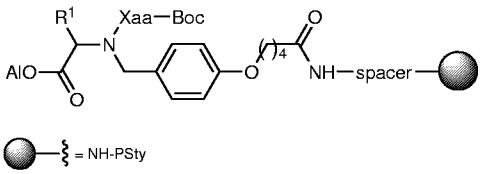
The two main resin linkers developed so far are shown in Scheme 18, i.e. tris(alkoxy)benzylamide-[412] and 4-alkoxybenzylamide-type linkers,[341] the former being TFA labile and thus fully compatible with Fmoc/tBu and the latter strongly acid labile and correspondingly compatible with Boc/Bzl chemistry. As shown in the case of the tris(alkoxy)benzaldehyde handle such handles may be introduced into the C-terminal amino acid ester by reductive amination, and after suitable N^{α}-protection coupled to amino-functionalized resins (see Scheme 18). Alternatively, the tris(alkoxy)benzaldehyde-functionalized resin, BAL resin, (see Scheme 14) is used to link the C-terminal amino acid ester by reductive amination. To overcome the difficult acylation of the N^{α}-arylamino acid ester derivative on resin (best results with 10 equivalent symmetrical anhydrides), synthesis in solution of the C-terminal dipeptide building block containing the amide handle followed by its attachment to the resin has been proposed.[341]

Scheme 18 Linkers Employed for Attachment of Peptides by a Backbone Amide to Solid Supports[341,412]

tris(alkoxybenzyl) linker used for Fmoc chemistry



4-alkoxybenzyl linker used for Boc chemistry



To avoid formation of piperazine-2,5-diones the use of mild acid labile N^{α}-protecting groups such as Trt or Ddz for the penultimate amino acid residue have been proposed in the case of the tris(alkoxy)benzyl linker, and subsequent acylation of the dipeptide as TFA salt.

Fmoc-N^{α}-(BAL-OH)Xaa-OAl:[412]

5-(4-Formyl-3,5-dimethoxyphenoxy)pentanoic acid (0.14 g, 0.5 mmol), H-Gly-OAl·TFA(140 mg, 0.6 mmol) and $NaBH_3CN$ (31 mg, 0.5 mmol) were suspended in MeOH (5 mL) and stirred at 25 °C for 60 min. The suspension was taken to dryness, and the oily residue was suspended in dioxane/H_2O (1:1, 4 mL). Solid $NaHCO_3$ (0.13 g, 1.5 mmol) was added, the suspension was cooled in an ice bath, and Fmoc-OSu (0.22 g, 0.66 mmol) dissolved in dioxane (1 mL) was added. Stirring was continued for 90 min while cooling in an ice bath and at 25 °C for another 90 min. The pH was then adjusted to approximately 9 by the addition of solid $NaHCO_3$. The suspension was diluted with H_2O (20 mL) and washed with Et_2O (2 × 25 mL). The aqueous layer was acidified to pH 3 with 4 M HCl and extracted with EtOAc (2 × 25 mL). The combined EtOAc phases were taken to dryness; yield: 0.29 g (95%).

In the same manner the BAL handle derivatives of Fmoc-Gly-OMe, Fmoc-glycinal dimethylacetal, Fmoc-Ala-OtBu, Fmoc-Leu-OtBu, Fmoc-Phe-OMe, and Fmoc-Phe-OtBu were prepared.

Fmoc-N^{α}-(BAL-PEG-PSty)Phe-OtBu:[412]
Amino-PEG-PSty-resin (1.0 g, 0.18 mmol·g^{-1}) was washed with CH_2Cl_2 (5 × 1 min), neutralized with DIPEA/CH_2Cl_2 (1:19, 5 × 2 min), and washed further with CH_2Cl_2 (5 × 1 min) and DMF (5 × 1 min). Then a soln of Fmoc-(BAL-OH)Phe-OtBu (1 g, 0.38 mmol), BOP (0.24 g, 3 equiv), and HOBt (84 mg, 3 equiv) in DMF (5 mL) and NMM (0.12 mL, 6 equiv) was preactivated for 5 min and added to the resin. After 4 h, the resin was negative by the Kaiser test.[376] The resultant Fmoc-(BAL-PEG-PSty)Phe-OtBu-resin was washed with DMF (2 × 2 min) and CH_2Cl_2 (2 × 2 min), and dried. A sample (11 mg) of the dried resin was swollen in DMF, treated with piperidine/DMF (1:4, 1 × 2 min, 2 × 10 min, and 2 × 2 min) and the supernatant was analyzed spectroscopically (301 nm, $\varepsilon = 7800\,cm^{-1}M^{-1}$) for the resultant piperidine/dibenzofulvene adduct. A second sample (6 mg) was similarly deprotected and hydrolyzed with aq 12 M HCl/propionic acid (1:1) at 160 °C for 1 h and evaluated by amino acid analysis. These complementary methods showed a final loading of Phe = 0.43 mmol·g^{-1}. Aliquots of the resin (50 mg) were taken, treated to remove Fmoc, washed with CH_2Cl_2 (5 × 2 min) and coupled with the second Fmoc amino acid (e.g., Gly, Ala, and Leu; final reaction volume: 1 mL). Upon completion of the reaction (or douple coupling), resins were washed with DMF (2 × 2 min) and CH_2Cl_2 (2 × 2 min), and dried for 15 h. A sample (5–10 mg) of each resin was treated to remove the Fmoc group, and analyzed by UV or upon hydrolysis by amino acid analysis, as described above. In the same manner the (BAL-PEG-PSty)-resin of Fmoc-Sty-Fmoc-glycinal dimethyl acetal, Fmoc-Ala-OtBu, Fmoc-Leu-OtBu-Fmoc-Ser-OAl were prepared.

***c*[-Arg-D-Phe-Pro-Glu-Asp-Asn-Tyr-Glu-Ala-Ala-]:**[412]
The fully protected BAL-anchored peptide-resin (20 mg, 0.13 mmol·g^{-1}) was washed with DMF (5 × 0.5 min), and the C-terminal Ala-OAl ester cleaved by treatment with $Pd(PPh_3)_4$ (15 mg, 5 equiv) in DMSO/THF/aq 0.5 M HCl/morpholine (2:2:1:0.1, 1.5 mL) under argon at 25 °C for 3 h. The peptide-resin was then washed with THF (3 × 2 min), DMF (3 × 2 min), CH_2Cl_2 (3 × 2 min), DIPEA/CH_2Cl_2 (1:19, 3 × 2 min), CH_2Cl_2 (3 × 2 min), 0.03 M sodium *N*,*N*-dimethyldithiocarbamate in DMF (3 × 15 min), DMF (5 × 2 min), CH_2Cl_2 (3 × 2 min), and DMF (3 × 1 min). The Fmoc group was removed with piperidine/DMF (1:4, 3 × 1 min, 2 × 5 min, and 2 × 1 min) followed by washings with DMF (5 × 0.5 min) and CH_2Cl_2 (5 × 0.5 min). Next, a soln of PyAOP (6.8 mg, 5 equiv) and HOAt (1.8 mg, 5 equiv) in CH_2Cl_2 (0.3 mL) was added and cyclization was initiated by addition of DIPEA (4.4 µL, 10 equiv). After 2 h at rt, the peptide-resin was negative by the Kaiser test.[376] Treatment with TFA/Et_3SiH/H_2O (92:5:3) at rt for 3 h gave the cyclic peptide; cleavage yield: 85%, which was characterized by analytical HPLC (L-Ala-peptide: t_R 22.5 min; D-Ala-peptide: t_R 23.4 min; ratio: 88:12; overall purity for both peaks >95%; linear precursor: t_R 20.5 min, negligible amounts). The expected [M + H$^+$] ion peak was obtained by MS.

Other cyclization conditions: (a) in DMF; complete cyclization; L/D-ratio: 75:25; (b) 2,4,6-trimethylpyridine as base; 26% linear precursor; L/D-ratio: 76:24; (c) in DMF and 2,4,6-trimethylpyridine as base: complete cyclization; L/D-ratio: 50:50; (d) 2,4,6-trimethylpyridine as base in $CHCl_3$/TFE (7:3); 75% linear precursor (after 3 × 2 h coupling); L/D-ratio: 78:22.

Boc-Ala-{*N*-[4-(4-carboxybutoxy)benzyl]}Ile-OAl:[341]
To a soln of Boc-Ala-OH (2.89 g, 15 mmol) in CH_2Cl_2 (30 mL) was added DAST (4.12 g, 25.5 mmol), and after 10–15 min, the mixture was washed with cold H_2O, and dried ($MgSO_4$). The solvent was removed and the acid fluoride immediately added to a soln of *N*-(4-{4-[2-(trimethylsilyl)ethoxycarbonyl]butoxy}benzyl)Ile-OAl (4.78 g, 10.0 mmol) and DIPEA (2.67 mL, 15 mmol) in THF (20 mL). After stirring at rt for 16 h, EtOAc (100 mL) was added, and the mixture was washed with 10% K_2CO_3, brine, and H_2O, and dried ($MgSO_4$). The solvent was removed and the residue purified by flash chromatography (hexane/Et_2O 1:5); yield: 2.86 g (44%).

To Boc-Ala-[*N*-(4-{4-[2-(trimethylsilyl)ethoxycarbonyl]butoxy}benzyl)]Ile-OAl (2.0 g, 2.82 mmol) in THF (20 mL) was added under stirring at rt 1 M TBAF in THF (3 mL) dropwise, and saponification of the ester proceeded for 3 h. The H_2O (100 mL) and AcOH (3 mL) were added to the mixture. The acid was extracted into EtOAc and the combined EtOAc extracts were washed with brine and H_2O, and dried ($MgSO_4$). The solvent was removed, and the resulting oil was purified by preparative HPLC (C18, 2.2 × 25 cm; 0–60% MeCN over 60 min) to give a colorless oil; yield: 2.54 g (44%) (note: this is the yield given in the original paper, clearly it is incorrect).

***c*[-Pro-Phe-Asn-Ser-Leu-Ala-Ile-], Stylostatin I:**[341]
Boc-Ala-{*N*-[4-(4-carboxybutoxy)benzyl]}Ile-OAl was coupled to H-Gly-Leu-Leu-PSty by established methods using in situ neutralization/HBTU activation protocols for Boc chemistry. The peptide was further elongated after removal of the Boc protecting group with neat TFA (2 min) using Boc-Ser(Bzl)-OH and Boc-Asn(Xan)-OH as side-chain-protected derivatives. Coupling reactions were monitored by quantitative ninhydrin assay and were typically >99.9% complete. After chain assembly was complete, the removal of the allyl protecting group from the peptide-resin (250 mg, 0.167 mmol·g^{-1}) was achieved by the addition of $Pd(PPh_3)_4$ (580 mg, 0.5 mmol, 3 equiv) to the resin in a soln of $CHCl_3$/AcOH/NMM

for references see p 529

(37:2:1), and the soln was stirred at rt for 14 h. The resin was washed with a 10% sodium *N,N*-diethyldithiocarbamate trihydrate in DMF, DMF, MeOH/CH_2Cl_2 (1:1), and finally CH_2Cl_2. The Boc group was removed with neat TFA (2 × 1 min), and the peptide-resin in DMF (20 mL) was cooled to −10 °C, and BOP (221 mg, 0.5 mmol) was added. Next, 2,6-lutidine (194 μL, 1.66 μmol) was added dropwise, and the reaction was monitored by ninhydrin test. After completion of the reaction (amine <1% by ninhydrin), the resin was filtered, and the cyclic peptide was cleaved from the resin with HF/*p*-cresol (11 mL, 10:1) at 0 °C for 1 h. After removal of the HF under reduced pressure, the crude peptide was precipitated in anhyd Et_2O before being dissolved in the HPLC buffer and lyophilized. The crude peptide was purified to yield: 3.1 mg (10%); as side products the cyclodimer (24.5%) and cyclotrimer (1%) were also isolated.

6.8.3.2.3 Synthesis of Extended and Reverse-Extended Cyclic Peptides

Both homodetic and heterodetic cyclizations via the peptide termini and/or the sequence-encoded side-chain functionalities limits the variability in ring size and chemistry. In order to enhance the diversity in these terms, the concept of extended and reverse-extended lactam ring (also called branched peptides) was introduced.[415] As shown in Scheme 19, this necessitates an additional spacer consisting of an amino acid residue or of a peptide chain inserted between the two side-chain functionalities involved in cyclization. Depending upon the parallel or antiparallel orientation of the inserted spacer, the resulting constructs are termed extended (branched) and reverse-extended peptides.[415]

Scheme 19 Synthesis of Extended and Reverse-Extended Cyclic Peptides

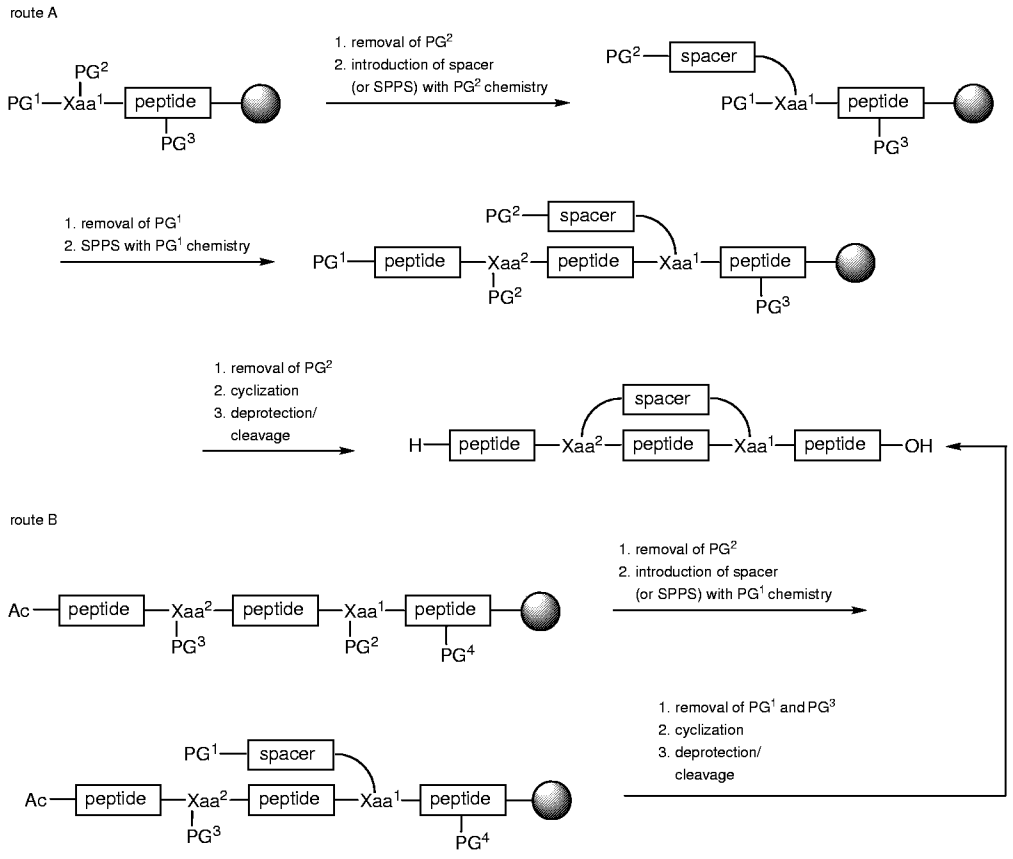


Table 10 Extended and Reverse-Extended Peptides Synthesized According to Routes A and B[344,415]

Type	Route[a]	Protecting Groups (PG_{order})				Final Deprotection/ Cleavage	Ref
		PG_1	PG_2	PG_3	PG_4		
extended	A	Boc	Fmoc	Bzl	–	HF	[415]
reverse-extended	A	Boc	OFm	Bzl	–	HF	[415]
extended	B	Fmoc	Dde	OAl	Boc/Trt	TFA	[344]
reverse-extended	B	Fmoc	OAl	Dde	Boc/Trt	TFA	[344]

[a] See Scheme 19.

Route A utilizes Boc/Bzl chemistry for the synthesis of the main peptide with Fmoc/OFm at the side-chain functions where additional chemistry is performed. For synthesis of extended cyclic peptides, the N^{ω}-Fmoc group is removed and the extension is synthesized with Fmoc/Bzl chemistry, whereas for reverse-extended cyclic peptides, the ω-Fm ester is hydrolyzed and the peptide spacer is built up in the N-to-C direction with fluorenylmethyl esters with all the risk of epimerization or more correctly, it is incorporated by fragment condensation.

Route B utilizes a combination of four levels of orthogonal protecting groups and one blocking group. The main chain is synthesized by the Fmoc/tBu chemistry and the assembled peptide is N^{α}-acetylated. For the synthesis of reverse-extended cyclic peptide, the positions of the amino acid residues N^{ω}-protected as Dde derivatives and ω-allyl esters are reversed. The Dde group is removed and elongation is accomplished by Fmoc/tBu chemistry in the standard C-to-N direction. Cleavage of the Fmoc/OAl groups, followed by cyclization and resin-detachment/deprotection yields the desired peptide.

H-Tyr-Ala-Asp-Ala-Ile-Phe-Thr-Asp(Gly)-Ser-Tyr-Arg-Orn-Val-Leu-Ala-Gln-Leu-Ser-Ala-Arg-Orn(Gly)-Leu-Leu-Gln-Asp-Ile-Met-Ser-Arg-OH, Bicyclic GRF Analogue Containing an Extended and Reverse-Extended Lactam Ring; Typical Procedure:[415]

pMeBHA resin (605 mg, 0.25 mmol·g^{-1}, 0.15 mmol) was swollen in DMF and then reacted with Boc-Arg(Tos)-OH (3 equiv) and BOP (3 equiv)/DIPEA (6 equiv) for 2.5 h. The following steps were then performed: (1) The resin was washed with DMF (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and CH_2Cl_2 (3 ×). (2) The Boc group was deprotected with 25% TFA in CH_2Cl_2 for 30 min. (3) The resin was washed with CH_2Cl_2 (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and DMF (3 ×). (4) Boc-Ser(Bzl)-OH (3 equiv) was coupled to the resin using BOP (3 equiv)/DIPEA (6 equiv) and the reaction was allowed to proceed for 2.5 h. (5) The coupling reaction was found to be complete as assessed by the quantitative ninhydrin assay. Steps 1–5 were repeated with Boc-Met-OH, Boc-Ile-OH, and Boc-Asp(OFm)-OH. The Fm ester was saponified with 20% piperidine in DMF (1 × 5 min and 1 × 20 min) and the resin washed with DMF (10 mL, 5 × 2 min). Attachment of HCl·H-Gly-OFm on the side chain of Asp^{25} was accomplished by double coupling using BOP (3 equiv)/DIPEA (6 equiv) for 2.5 h. After Boc deprotection/washings, steps 1–5 were repeated with Boc-Gln-OH, Boc-Leu-OH, Boc-Leu-OH, and Boc-Orn(Fmoc)-OH. The Fmoc and OFm groups were simultaneously removed by 20% piperidine in DMF (1 × 5 min and 1 × 20 min) and the resin washed with DMF (10 mL, 5 × 2 min). Cyclization was carried out in DMF using BOP (3 equiv)/DIPEA (6 equiv) for 48 h and was followed by quantitative ninhydrin test. After formation of the reverse-extended lactam ring, steps 1–5 were repeated with Boc-Arg(Tos)-OH, Boc-Ala-OH, Boc-Ser(Bzl)-OH, Boc-Leu-OH, Boc-Gln-OH, Boc-Ala-OH, Boc-Leu-OH, Boc-Val-OH, and Boc-Orn(Fmoc)-OH. The Fmoc protection was removed and after standard washing steps, Fmoc-Gly-OH was coupled to the side chain of Orn^{12}. Steps 1–5 were repeated with Boc-Arg(Tos)-OH, Boc-Tyr[Bzl(2,6-Cl_2)]-OH, Boc-Ser(Bzl)-OH, and Boc-Asp(OFm)-OH. The Fmoc and OFm groups were simultaneously removed by 20% piperidine in DMF (1 × 5 min and 1 × 20 min) and the resin washed with DMF (10 mL, 5 × 2 min). Cyclization was carried out in DMF using BOP (3 equiv)/DIPEA (6 equiv) for 24 h and was followed by quantitative ninhydrin test. After formation of the extended lactam ring, steps 1–5 were repeated with Boc-Thr(Bzl)-OH, Boc-Phe-OH, Boc-Ile-OH, Boc-Ala-OH, Boc-Asp(OCy)-OH, Boc-Ala-OH and Boc-Tyr[Bzl(2,6-Cl_2)]-OH. The resin was washed with DMF (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and CH_2Cl_2 (3 ×), and dried (1.18 g of peptide-resin). A portion of the protected peptide-resin (760 mg) was treated with HF using the low/high procedure.[416] The solvents were removed at 0 °C (high vacuum; CaO trap), triturated with EtOAc, and extracted with TFA. The TFA was removed under reduced pressure and the residue triturated with anhyd Et_2O and dried; yield: 192 mg of crude peptide that contained one major product in analytical

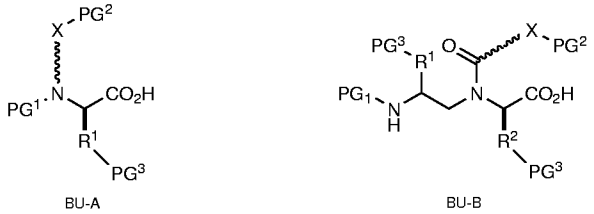
for references see p 529

HPLC. The crude product was dissolved in 0.1% aq TFA (10 mL), filtered, and purified by preparative HPLC [Preppak, Cartridge YMC ODS column (4.8 × 30 cm), gradient elution MeCN/0.1% TFA/H_2O from 20 to 50% MeCN in 90 min (flow rate of 50 mL·min^{-1})]. The fractions containing pure product were combined, concentrated, and lyophilized; yield: 22.4 mg (44%); the peptide was characterized by HPLC (purity 98%), amino acid analysis, and FAB-MS.

6.8.3.2.4 Synthesis of Backbone-Cyclized Peptides

In natural bioactive peptides the modes of cyclization described previously may be prevented either by the lack of suitable side-chain functionalities for lactamization or because these as well as the amino and carboxy termini are crucially involved in the bioactivity itself, and thus cannot be modified. In order to overcome these potential limitations, the concept of backbone cyclization has been proposed.[29] According to this, the cyclization is performed by a covalent interconnection of two backbone amides by artificial spacers or of one backbone amide by a correctly functionalized spacer with side-chain functions or with the N- and C-terminus of the peptide (Scheme 21). This type of strategy significantly increases the diversity of possible ring structures (see Scheme 22) and of their related libraries (see Section 6.8.4). Its potential for enhancing the stability of the related peptide derivatives toward proteolytic digestion,[417–419] potency,[419,420] and selectivity,[417–419] is well-established.

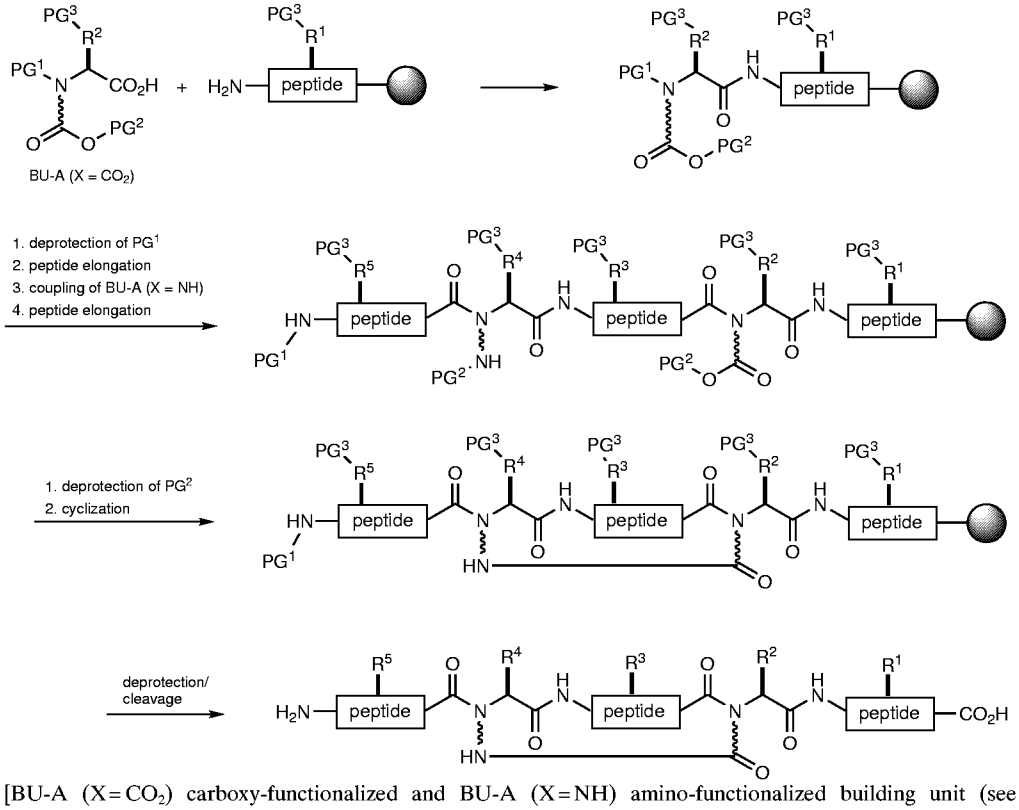
Scheme 20 Building Blocks for Backbone Cyclization; BU-A: Protected *N*-(ω-Functionalized Alkyl)amino Acids; BU-B: *N*-(ω-Carboxy, ω-Amino, or ω-Sulfanyl)acyl Derivatives of a Reduced Dipeptide



X = NH, CO_2, S
PG^1 = 1st order protecting group
PG^2 = 2nd order protecting group
PG^3 = 3rd order protecting group
~~~ = alkyl chain of various length
R^1,R^2 = side chain with protection

Two different approaches are proposed for this cyclization strategy, which are based on suitable building units that are to be incorporated into the peptide chain (Scheme 20). The first approach A, corresponds to an amino acid which is N^{α}-substituted with a suitably functionalized alkyl moiety.[421–423] For details on the synthesis of these building blocks and the inherent difficulties encountered in peptide-chain extension with these sterically hindered N^{α}-alkylamino acid derivatives and their subsequent acylations on-resin, see Section 10.1.1.1.4; excellent results are obtained with bis(trichloromethyl) carbonate (triphosgene).[424] The second type of building block is derived from dipeptides reduced at the amide group to a secondary amine; this amino group is acylated with ω-functionalized carboxylic acids.[425–427] With these building blocks the difficult coupling and acylation reactions are prevented. Examples of syntheses of backbone-cyclized peptides are listed in Table 11.
~~~

Scheme 21 Synthesis of Backbone-to-Backbone Cyclic Peptides[a]



[a] [BU-A (X = CO_2) carboxy-functionalized and BU-A (X = NH) amino-functionalized building unit (see Scheme 20).

Table 11 Examples of Backbone Cyclic Peptides[29,417,418,420,426,428–433]

Mode/BU[a]	Resin/Final Deprotection	Peptide[b]	Protecting Groups (Order of Orthogonality)			Ref
			1st	2nd	3rd	
E/BU-A	pMeBHA (HF)	substance P, C-terminal hexapeptide	Fmoc	Boc	Bzl	[29,417,428]
G/BU-A	Rink-pMeBHA (TFA)	somatostatin	Fmoc	Aloc/OAl	tBu	[418]
E/BU-A	MBHA (HF)	NLS of the HIV-1 MA	Fmoc	Boc	Bzl	[429,430]
E/BU-A	MBHA (HF)	NLS of the HIV-1 Tat/Rev	Fmoc	Boc	Bzl	[429,430]
E/BU-A	Rink-MBHA (TFA)	PBAN C-terminal heptapeptide	Fmoc	Boc[c]	tBu	[420,431]
F, G/BU-A	HMPPA (TFA)	BK	Fmoc	Aloc/OAl	tBu	[432,433]
F, G/BU-B	HMPPA (TFA)	BK	Fmoc	Aloc/OAl	tBu	[426]

[a] Modes (see Scheme 1) E = head-to-backbone-amide; F = backbone-amide-to-side-chain; G = backbone-amide-to-backbone-amide; for structure of BU-A see Scheme 20.
[b] NLS = nuclear localization signal; MA = matrix protein; PBAN = pheromone biosynthesis activating neuropeptide.
[c] Removed by trichloro(iodo)silane.

Experimental procedures for the synthesis of the N-alkylated amino acid building blocks are described in Section 10.1.1.1.4.

Synthesis of Backbone-Cyclized Peptides; General Procedure:[424]
Fmoc-Rink-Nle-pMeBHA-resin (1 g, 0.55 mmol · g^{-1}) was swollen for 1.5 h in NMP in a reaction vessel equipped with a sintered glass bottom, and placed on a shaker. The Fmoc group was removed with 20% piperidine in NMP (8 mL, 2 × 15 min) and after washing with NMP (8 mL, 5 × 2 min), the Fmoc removal was monitored by the ninhydrin Kaiser test. Coupling of building blocks such as N^{α}-Fmoc-N^{ω}-carboxyalkyl(OAl)Xaa-OH or N^{α}-Fmoc-N^{ω}-aminoalkyl(Aloc)Xaa-OH, or N^{α}-alkylated amino acids such as

for references see p 529

Fmoc-(Me)Xaa-OH (3 equiv) to amino acyl- or peptide-resin was performed with PyBroP (3 equiv)/DIPEA (6 equiv) in NMP (8 mL) at rt for 1 h by monitoring with the Kaiser test. The resin was washed with NMP (8 mL, 5 × 2 min) and coupling of Fmoc-Xaa-OH or N^{α}-alkylated amino acids such as Fmoc-(Me)Xaa-OH to *N*-alkylamino acyl-resin, *N*-methylamino acyl-peptide-resin, building-blocks-resin or building-blocks-peptide-resin was performed as follows: In a well-ventilated fume hood, Fmoc-Xaa-OH, Fmoc-(Me)Xaa-OH or building blocks such as N^{α}-Fmoc-N^{ω}-carboxyalkyl(OAl)Xaa-OH or N^{α}-Fmoc-N^{ω}-aminoalkyl(Aloc)Xaa-OH (5 equiv) and bis(trichloromethyl) carbonate (1.65 equiv) were dissolved in dry THF, dioxane, diglyme, or 1,3-dichloropropane (to give a 0.15 M soln) and cooled in an ice-bath. Then 2,4,6-trimethylpyridine (25 equiv) was added to give a white suspension. After 1 min, the suspension was poured into the *N*-methylamino acyl-, *N*-methylamino acyl-peptide-resin, building-block-resin, or building-block-peptide-resin prewashed with the appropriate solvent and preheated to 50 °C. The mixture was shaken at 50 °C for 1 h, filtered, and washed with CH_2Cl_2 (10 mL, 5 × 2 min) and DMF (10 mL, 5 × 2 min). Peptide elongation was performed by repeating the removal of the Fmoc protecting group and the coupling cycle described above. Then the resin was washed with a mixture of AcOH/NMM/CH_2Cl_2 (5:2.5:92.5, 10 mL, 3 × 2 min) and the OAl/Aloc protecting groups were removed by the addition of a premade soln of $Pd(PPh_3)_4$ (1.5 g/1 g resin) in a mixture of AcOH/NMM/CH_2Cl_2 (5:2.5:92.5, 10 mL, under argon) The reaction was kept under shaking at rt for 2 h in the dark. The peptide-resin was washed with $CHCl_3$ (10 mL, 5 × 2 min) and NMP (10 mL, 5 × 2 min). Cyclization was carried out in NMP with PyBOP (3 equiv)/DIPEA (6 equiv) at rt for 1 h and monitored as follows: a sample of resin (10 mg) was washed with CH_2Cl_2 (5 × 2 min), dried by a stream of N_2, and treated with a few drops of TFA/TIS/H_2O (95:2.5:2.5). After removal of the resin, the filtrate was taken to dryness, the residue dissolved in H_2O/MeCN/0.1% TFA, and analyzed by MS. After washing with NMP the cyclization was repeated if necessary. The resin was washed with NMP (5 × 2 min) and CH_2Cl_2 (5 × 2 min) and dried. The backbone cyclized peptide was deprotected and cleaved from the resin by treatment with of TFA/TIS/H_2O (95:2.5:2.5, 10 mL) at 0 °C for 30 min, and at rt for 1–3 h. The resin was removed by filtration and washed with an additional quantity of TFA, the combined TFA was removed by a N_2 stream to give oils that solidified upon treatment with cold Et_2O (40 mL). The solvent was removed by centrifugation and the solids dried. The crude peptides were purified by preparative RP-HPLC [column C18, 2.5 × 30 cm, flow rate of 10 mL · min^{-1}, linear gradient from 95% H_2O/0.1% TFA/MeCN to 100% MeCN, 35 min]. Fractions containing pure product were combined, concentrated, and lyophilized.

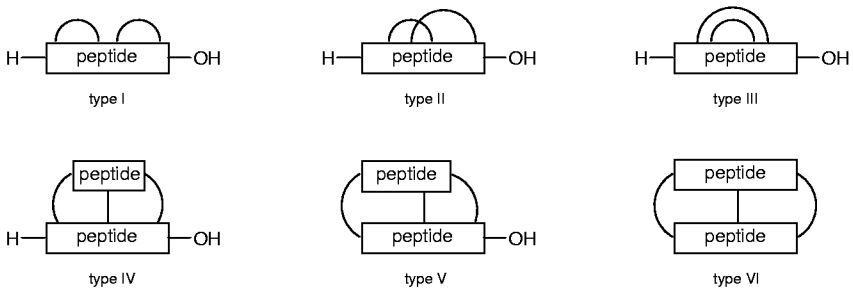
6.8.3.2.5 Synthesis of Bi- and Polycyclic Peptides

Nature uses multiple disulfide bridges even in relatively small peptides such as protease inhibitors or toxins to induce and stabilize compact globular structures. To mimic nature in these conformational restrictions the synthesis of bi- and polycyclic peptides containing lactam bridges or combinations of lactam/disulfide bridges has become a challenging task. Protection schemes have evolved to a degree of selectivity that permits such complex syntheses. In fact, taking into account the selectivity of the required pairs of amino and carboxy protecting groups, e.g. Aloc/OAl versus Dde/ODmab versus Fmoc/OFm versus Adpoc (or Mtt)/2-phenyl-2-propyl ester and the additional possible combinations with Npys, Troc/OPac, or the silyl-based protecting groups that are all used in the context of the basic Fmoc/tBu or Boc/Bzl chemistries, a priori, a large diversity of polycyclic structures is possible. This is even exponentially enhanced when mixed lactam/disulfide bridges are employed. The main question, however, remains the sequence-dependent side-chain functionalities available for such cyclizations and more importantly, the sequence-encoded conformational preferences of bioactive peptides or protein fragments that may prevent such multiple structural restrictions. Nature has mastered this crucial aspect in the eons of evolutionary history. Nevertheless bicyclic peptides have been synthesized and the imparted conformational constraints allowed to further improve or modulate bioactivities of peptides (see, for example, ref[434]) or of protein fragments (see, for example, ref[435]).

The bicyclic peptides that have been synthesized with at least one lactam ring correspond to the structures shown in Scheme 22, and examples of such syntheses of homodetic or mixed homodetic/heterodetic structures are listed in Table 12. For bicyclic or polycyclic cystine peptides see Sections 6.1 and 6.2. In the ring structures of types I – III both rings derive from side-chain-to-side-chain cyclizations. In the structures of types IV – VI an ω-functionalized amino acid is employed as bridgehead. This can be linked directly to three

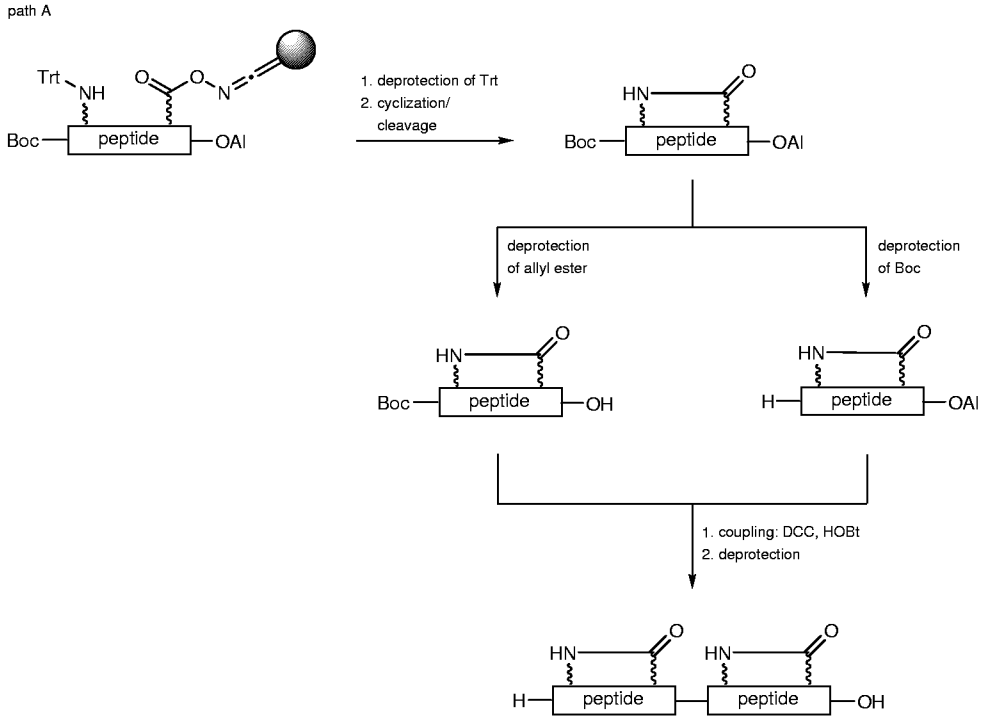
side chains of the peptide (type IV) or the bridgehead as part of the main peptide chain. In the type V structure one ring is formed by a bridgehead connected through its side chain to another side chain and through its carboxy group to a backbone amide. In type VI one ring is formed by head-to-tail cyclization of the peptide and the second ring by cross-linking of two side chains within the first ring.

Scheme 22 Various Types of Bicyclic Peptides Synthesized on Solid Supports



The synthesis of type I bicyclic peptides is the most simple as the two side-chain-to-side-chain cyclizations can be performed successively on the assembled peptide or more appropriately by condensation of the two monocyclic segments preferably in solution. As shown in Scheme 23 (path A), for the synthesis of suitably protected monocyclic peptides, side-chain attachment of Asp or Glu residues to the oxime resin is proposed,[436,437] which leads to the desired cyclization and release of the protected segments as Pac or preferably as Al esters.[396] These are C-terminally deprotected, when required, and then assembled in solution into bi- or polycyclic peptides of type I.

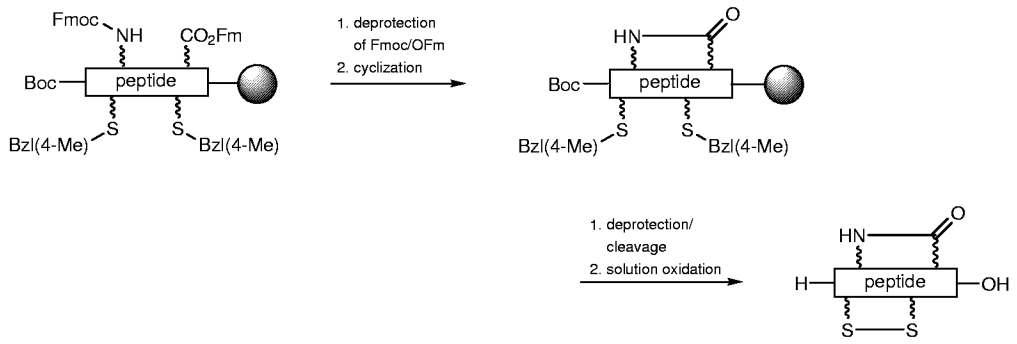
Scheme 23 Synthesis of Bicyclic Peptides of Types I – VI[a]



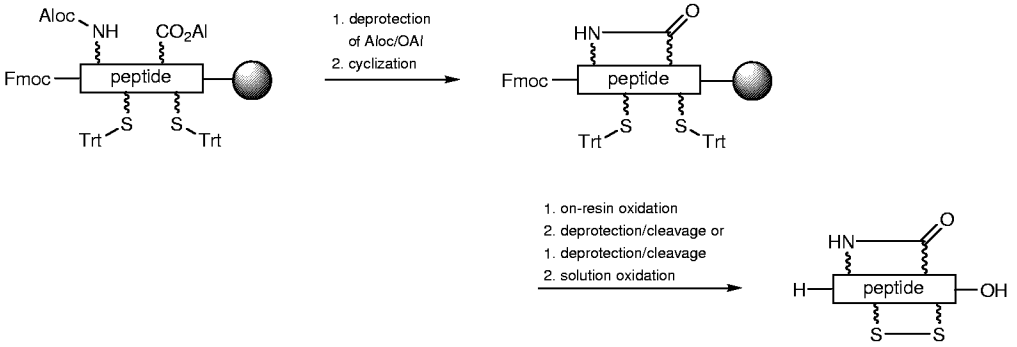
[a] See Scheme 22.

for references see p 529

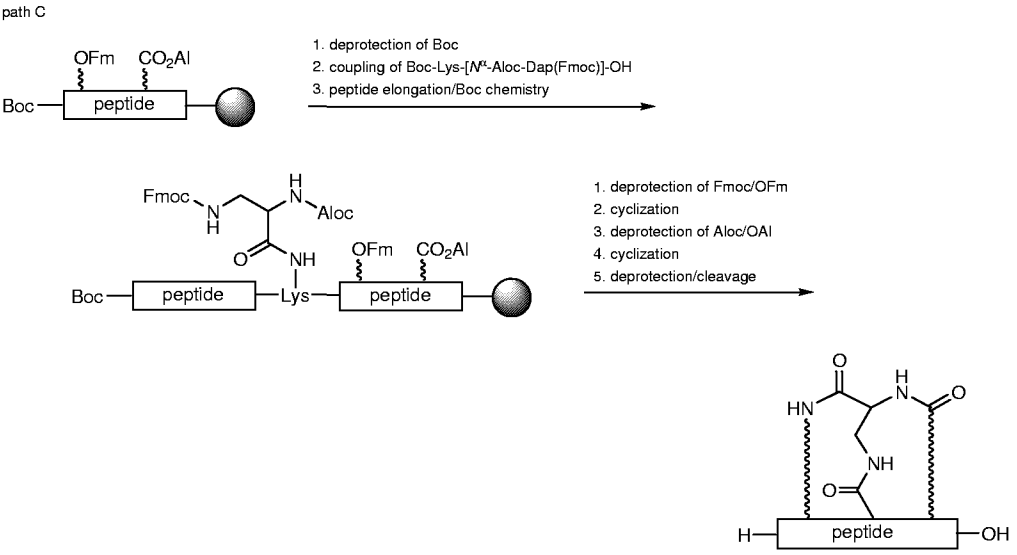
path B1

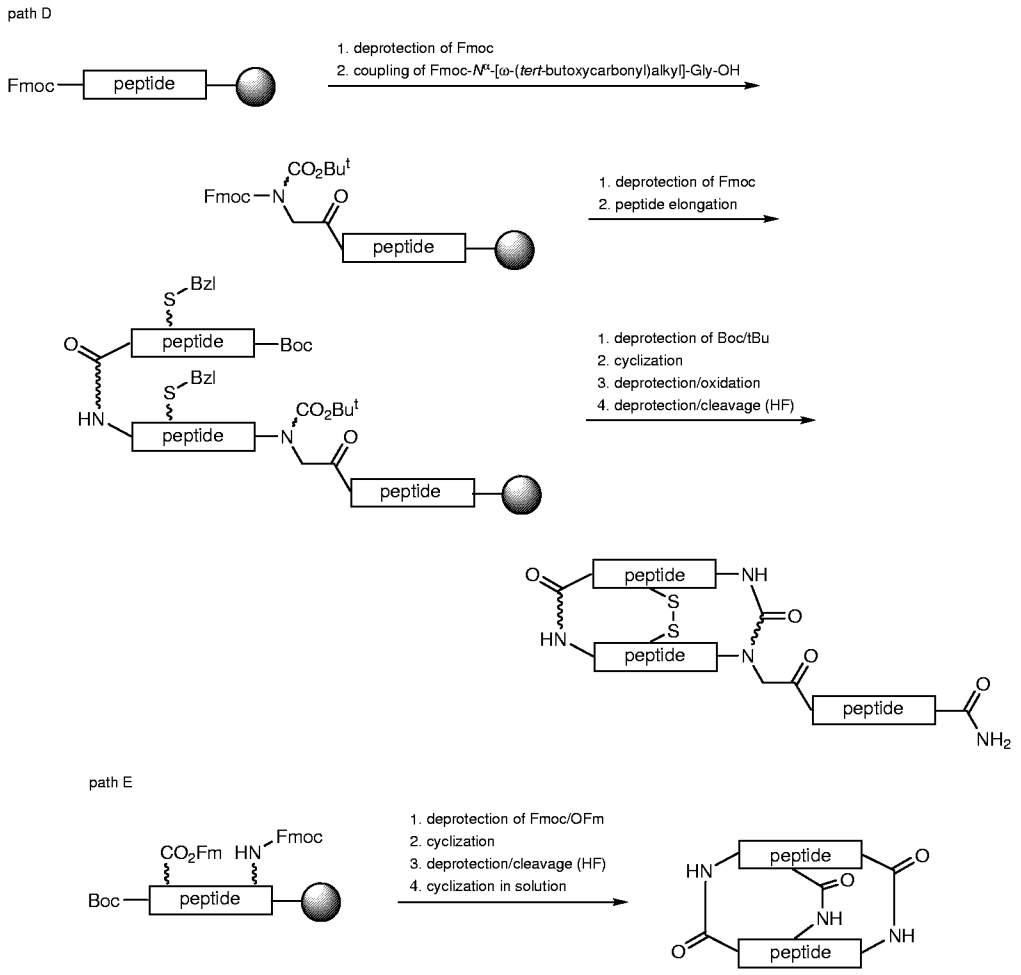


path B2



path C





Similarly, the mixed homodetic/heterodetic bicyclic structures containing a disulfide bridge are preferably produced as monocyclic compounds that are then typically oxidized in solution adopting standard procedures as described in Section 6.1.1 (Scheme 23, path B1). If even disulfide formation is performed on resin by the procedures reported in Section 6.1.2, an additional level of selective protection is required for the cysteine thiol functions. The synthetic paths B1 and B2 are also applicable for the synthesis of type II and III bicyclic peptides, independent of whether only lactam bridges are produced or mixtures of lactam/disulfide bridges.

More complex are the synthesis of bicyclic peptides containing a bridgehead, as shown in Scheme 23 (path C), or a backbone cyclization unit[422,435,438] (path D). In fact, in addition to the required higher levels of orthogonal protecting groups, the key steps in the former case are the introduction of the bridgehead as a suitably protected iso-dipeptide building block prepared separately in solution, and in the latter case of the suitably N^{α}-(ω-functionalized alkyl)amino acid derivative (see Section 6.8.3.2.4).

for references see p 529

Table 12 Selected Bicyclic Peptides That Represent the Six Types of Bicyclization Shown in Scheme 22[125,391,434–452]

Type[a]	Mode of Cyclization[b]		Ring Chemistry[c]		Peptide	Ref
	Ring 1	Ring 2	Ring 1	Ring 2		
I (A)	B	B	$(Lys/Glu)_3$	Lys/Asp	amphiphilic helix	[436,437,439]
	B		$(Lys/Asp)_3$	Lys/Asp	PTH/PTHrP	
II (B1)	B	B	Mpa/Cys	Glu/Lys	oxytocin	[391,434,440]
			Cys/Cys	Ala/Lys	α-MSH	
II (B2)	B	B	Glu/Lys	Cys/Cys	α-conotoxin SI mimetics	[441]
			Cys/Cys	Glu/Lys		
			Lys/Glu	Cys/Cys		
			Cys/Cys	Lys/Glu		
III	B	B	Asp/Dap	Glu/Lys	GnRH	[442,443]
III (B1)	B	B	Dap/Asp	Cys/Cys	endothelin 1	[444]
VI (D)	A	B	*c*[peptide]	Asp/Dap	tachykinin NK2 antagonists, type I/II β-turn scaffolds	[445–448]
II	B	B	Glu/A_2bu	Asp/Dap	GnRH	[449,450]
V (E)	head-to-BH	SC-BH/BH-BN	lactam	thioproline/Cys	BPTI	[438]
V (E)	head-to-BH	SC-BH	lactam	BU/Cys	BPTI	[435,451]
		BH-BN	lactam	BU/Cys	substance P	
IV (C)	SC-BH	SC-BH	Lys/Dap	Dap/Asp	α-helix model	[125,452]
		SC-BH		Dap/Aad		

[a] Shown in parentheses is the synthetic path in Scheme 23.
[b] B = side-chain-to-side-chain; A = head-to-tail; BH = bridgehead; SC= side chain; BN = backbone nitrogen. In bicyclic peptides of type IV and V the two modes of cyclization of ring 2 are described.
[c] BU = N^{α}(thioalkyl)Gly.

Synthesis of Type I Polycyclic Peptides:
The synthesis and on-resin cyclization of polycyclic peptides of type I is described in the experimental part of Section 6.8.3.2.3. If side-chain-to-side-chain cyclization is needed the couplings of H-Gly-OFm and Fmoc-Gly-OH should be omitted from that procedure.

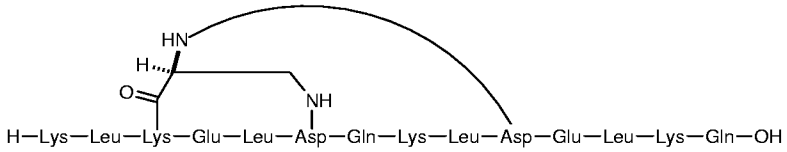
Boc-{Lys[Z(2-Cl)]-Leu-Lys-Glu(OBzl)-Leu-Lys[Z(2-Cl)]-Glu}$_2$-OH; Typical Procedure with Oxime Resin:[436,437]
The DCHA salt of Boc-Lys[Z(2-Cl)]-Glu-OPac (1.55 g, 2.34 mmol) was coupled to *p*-nitrobenzophenone oxime resin (25 g) using ethyl 2-cyano-2-(hydroxyamino)acetate[453] (0.67 g, 4.68 mmol) and DCC (0.53 g, 2.57 mmol) at −10 °C for 30 min and at 20 °C overnight. After the resin was drained and washed with CH_2Cl_2 (2 ×), EtOH/CH_2Cl_2 (1:1, 2 ×), and CH_2Cl_2 (2 ×), it was acetylated with Ac_2O (37.5 mL)/DIPEA (12.5 mL) in CH_2Cl_2 (250 mL) for 2.5 h. The resin was washed with CH_2Cl_2 (3 ×), EtOH/CH_2Cl_2 (1:1, 2 ×), and EtOH (2 ×), and dried; substitution: 0.066 mmol·g^{-1} based on picric acid titration.[454] The peptide-resin was then subjected to two cycles of the standard oxime resin protocol (see experimental procedure in Section 6.8.3.2.2.1). In the first cycle Boc-Leu-OH·H_2O (1.42 g, 5.7 mmol) was mixed with BOP (2.52 g, 5.7 mmol) in CH_2Cl_2 (70 mL) followed by DIPEA (0.99 mL, 5.7 mmol). The mixture was added to the Boc-deprotected and washed peptide-resin, and shaken for 2 h. A second coupling (1 equiv reagents) was performed and the resin was washed with DMF (2 ×), EtOH/CH_2Cl_2 (1:1, 1 ×), DMF (2 ×), and CH_2Cl_2 (2 ×) then acetylated with Ac_2O (7 mmol)/DIPEA (7 mmol) in CH_2Cl_2 for 1 h. After deprotection and washings as before the resin was neutralized with DIPEA (5% in CH_2Cl_2, 2 × 2 min) and washed with CH_2Cl_2 (2 ×). Then, freshly prepared symmetric anhydride of Boc-Glu(OBzl)-OH in CH_2Cl_2 (110 mL) was used for the next coupling in the presence of DIPEA (0.35 mL, 2 mmol). After 2 h, the resin was washed with CH_2Cl_2 (3 ×), EtOH/CH_2Cl_2 (1:1, 2 ×), and EtOH (2 ×) and dried; yield of peptide-resin: 1.72 mmol of peptide (99%); substitution level: 0.065 mmol·g^{-1} based on Leu as determined by amino acid analysis. The peptide-resin (25 g, 1.72 mmol of peptide) was deprotected washed, neutralized, and washed as above and coupled with the tripeptide Boc-Lys[Z(2-Cl)]-Leu-Lys(Trt)-OH using HOBt (0.63 g, 4.7 mmol), DIC (0.27 mL, 2 mmol), and DIPEA (0.35 mL, 2 mmol) in DMF (125 mL) at rt for 3 d. On the second day, further quantities of DIC (2 mmol) and DIPEA (2 mmol) were added to the mixture. The resin was washed and dried; yield of peptide-resin: 64%, 0.041 mmol·g^{-1} based on Leu as determined by amino acid analysis. Peptide-resin (900 mg, 36.9 μmol) was swollen in CH_2Cl_2 (10 mL) and the Trt group was removed with 5% TFA in CH_2Cl_2/TFE (1:1, 4 × 15 min, 15 mL).

The peptide-resin was drained and washed with CH_2Cl_2 (2 × 10 mL), iPrOH (2 × 10 mL), CH_2Cl_2 (2 × 10 mL), iPrOH (2 × 10 mL), and CH_2Cl_2 (2 × 10 mL). The resin was neutralized with 5% DIPEA in CH_2Cl_2 (2 × 1.5 min) and then washed with CH_2Cl_2 (4 ×). The cyclization reaction was carried out by shaking the resin with AcOH (10 equiv) in CH_2Cl_2 (14 mL) at rt for 72 h. The cyclic product was collected by draining and washing the resin with CH_2Cl_2 (3 × 10 mL). The combined filtrates were concentrated and washed at 0 °C with H_2O, 1 M HCl, 5% $NaHCO_3$, and brine. The soln was taken to dryness and the crude product was purified by flash chromatography (silica gel, 2 × 20 cm, $CHCl_3$/MeOH/AcOH, 18:1:1). The appropriate fractions were pooled and the solvent was removed. The product was recrystallized (MeOH/Et_2O); yield: 33.9 mg (60%); mp 90–94 °C; $[\alpha]_D$ −15.3 ± 0.3 (*c* 1.7, $CHCl_3$). The protected cyclic peptide was divided into two parts for saponification of the Pac ester and N^{α}-deprotection respectively.

The Boc-protected peptide Pac ester (150 mg, 0.1 mmol) was dissolved in 90% AcOH (2.5 mL), and Zn dust (2 × 120 mg) was added to the vigorously stirred soln. The reaction was followed by TLC (silica gel, $CHCl_3$/MeOH/AcOH 18:1:1). After 1 h at rt, the Zn was removed by filtration and the soln concentrated. The product was dissolved in CH_2Cl_2, the soln was filtered and taken to dryness. The residue was flash chromatographed (silica gel, 2 × 20 cm, $CHCl_3$/MeOH/AcOH 90:7:3) and recrystallized (MeOH/Et_2O); yield: 117 mg (84%); mp 112–116 °C; $[\alpha]_D$ −27.9 ± 0.8 (*c* 0.5, AcOH).

The Boc-protected peptide Pac ester (90.8 mg, 60 μmol) was treated with TFA (4 mL) at 20 °C for 15 min. The soln was concentrated and the product was precipitated with Et_2O. The precipitate was collected, washed with Et_2O, and dried (NaOH). This material was dissolved in DMF (2 mL) together with the Boc-protected peptide (83.4 mg, 60 μmol) and HOBt (18.4 mg, 136 μmol). The soln was cooled to −5 °C and DCC (13.6 mg, 66 μmol) and DIPEA (11.5 μL, 66 μmol) were added. After stirring at 0 °C for 1 h and at 20 °C overnight, the mixture was concentrated, dissolved in CH_2Cl_2, and washed with H_2O, 5% citric acid, 5% $NaHCO_3$, and brine (each 10 mL at 0 °C). The CH_2Cl_2 soln was concentrated and the residue flash chromatographed (silica gel, 2 × 20 cm, $CHCl_3$/MeOH/AcOH 18:1:1). The appropriate fractions were pooled and the solvent was removed; yield: 110 mg (66%); mp 68–72 °C; $[\alpha]_D$ −8.5 ± 0.3 (*c* 2, MeOH); the peptide was characterized by amino acid analysis and MS.

Synthesis of the Bicyclic Peptide Type IV:[455]



Boc-Gln-OH was attached to Cl-Bzl-PSty (Merrifield resin; 1 g, 0.2 mmol · g^{-1}) using the Cs salt method described in the experimental part of Section 6.8.3.2.2.2. The following steps were then performed: (1) The resin was washed with DMF (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and CH_2Cl_2 (3 ×); (2) the Boc group was removed using 25% TFA in CH_2Cl_2 for 30 min; (3) the resin was washed with CH_2Cl_2 (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and DMF (3 ×); (4) Boc-Lys[Z(2-Cl)]-OH (3 equiv) was coupled to the resin using BOP (3 equiv) and DIPEA (6 equiv) and the reaction was allowed to proceed for 2.5 h; (5) the completeness of the coupling reaction was assessed by the quantitative ninhydrin assay. Steps 1–5 were repeated with Boc-Leu-OH, Boc-Glu(OBzl)-OH, Boc-Asp(OAl)-OH, Boc-Leu-OH, Boc-Lys[Z(2-Cl)]-OH, Boc-Gln-OH, Boc-Asp(OFm)-OH, Boc-Leu-OH, Boc-Glu(OBzl)-OH, Boc-Lys[Aloc-Dap(Fmoc)]-OH, Boc-Leu-OH, and Boc-Lys[Z(2-Cl)]-OH. The Fmoc and OFm protecting groups were simultaneously removed by 20% piperidine in DMF (1 × 5 min and 1 × 20 min) and the resin washed with DMF (10 mL, 5 × 2 min). The resin was washed with DMSO/NMP (1:4, 10 mL, 2 × 2 min) and cyclization was carried out in DMSO/NMP (1:4, 7.56 mL) using BOP (1.5 equiv) and DIPEA (1.5 equiv) at rt for 2 h, this was followed by quantitative ninhydrin assay. Then the Aloc/OAl groups were deprotected by $Pd(PPh_3)_4$ with *N*-methylaniline in THF/DMSO/1 M HCl (4:4:1) after washing and prewashing with THF/DMSO/1 M HCl (4:4:1) for 4 h. The resin was washed with DMSO/NMP (1:4, 10 mL, 5 × 2 min). The second cyclization was carried out in DMSO/NMP (1:4, 7.56 mL) using BOP (1.5 equiv) and DIPEA (1.5 equiv) at rt for 2 h, this was followed by quantitative ninhydrin test. The Boc group was removed by using steps 1 and 2, and the resin was washed with DMF (3 ×), MeOH (1 ×), CH_2Cl_2 (3 ×), MeOH (2 ×), and CH_2Cl_2 (3 ×), and dried. A portion of the protected peptide-resin was cleaved with HF/PhOMe/DMS (9:1:1) at 0 °C for 1 h. The solvents were removed at 0 °C and the crude product triturated with Et_2O, extracted into 10% AcOH, and lyophilized. The crude peptide contained one major product by analytical HPLC. The crude bicyclic peptide was purified by preparative HPLC and characterized by amino acid analysis and MS.

***c*[-Phe-Asp-Trp-Phe-Dap-Trp-]; Synthesis of Bicyclic Peptide Type VI:**[448]

The peptide was synthesized on Boc-Trp(For)-PAM-PSty-resin (0.67 mmol · g^{-1}, 0.5 mmol scale). The peptide was assembled by consecutive addition of the Boc-protected amino acids Boc-Dap(Fmoc)-OH,

for references see p 529

Boc-Phe-OH, Boc-Trp(For)-OH, Boc-Asp(OFm)-OH, and Boc-Phe-OH. All amino acids (2 mmol) [except Boc-Asp(OFm)-OH and Boc-Dap(Fmoc)-OH] were activated as symmetric anhydrides with 0.5 M DCC in CH_2Cl_2 (2 mL, 1 mmol) over 10 min. Boc-Asp(OFm)-OH (2 mmol) was activated with 0.5 M HOBt in DMF (2 mmol) and 0.5 M DCC in CH_2Cl_2 (4 mL, 2 mmol) over 45 min. Each cycle consisted of a 45 min coupling step. For Boc-Trp(For)-OH and Boc-Asp(OFm)-OH the coupling was repeated twice. Boc-Dap(Fmoc)-OH (1.5 mmol) was coupled by preactivation with PyBOP (3 equiv) and DIPEA (6 equiv) for 10 min in DMF and then coupling for 2 h. Completeness of the reactions, after each coupling, was monitored by the Kaiser test. Boc protecting groups were removed by prewash with 33% TFA in CH_2Cl_2, followed by deprotection with 50% TFA in CH_2Cl_2 for 18 min, washing, and 2 min neutralization with 10% DIPEA in DMF. Before coupling of Boc-Phe-OH the Fmoc/OFm side-chain protection was removed by 20% piperidine in DMF (for 3 min and then for 6 min) and the linear precursor was cyclized by treatment with PyBOP (3 equiv) and DIPEA (6 equiv) in DMF overnight. Coupling of Boc-Phe-OH was followed by deprotection of the N-terminal protecting group. The monocyclic peptide was cleaved from the resin with concomitant removal of the formyl group by the low/high HF procedure. The two deprotection procedures were: HF/DMS/*p*-cresol/*p*-thiocresol (5:13:1:1) at 0 °C for 4 h and HF/DMS/*p*-cresol (8.5:1:0.5) at 0 °C for 0.5 h. After removal of HF and scavengers, the mixture was washed with Et_2O, and the crude product was extracted into 50% AcOH and lyophilized; yield: 260 mg of crude product with a major peak according to HPLC. The crude material was purified by preparative RP-HPLC (Vydac C18, 5 × 25 cm, 10 µm, linear gradient 20% MeCN in H_2O (0.1% TFA) to 80% MeCN in 40 min, flow rate of 114 mL · min^{-1}). The pooled fractions containing the desired product were lyophilized; yield: 120 mg (28%); the peptide was characterized by FAB-MS.

To the monocyclic peptide (120 mg, 0.138 mmol) in dry DMF (138 mL), PyBOP (1 equiv) and DIPEA (3 equiv) were added and the soln stirred at rt overnight. The solvent was removed and the residue lyophilized from MeCN/H_2O. The crude peptide was purified by preparative HPLC as described above; yield: 59 mg (50%); the peptide was characterized by FAB-MS.

6.8.4 Cyclic Peptide Libraries

Combinatorial linear peptide libraries are constituted from large numbers of peptides where each individual peptide has a unique structure, but allows diverse elements such as size and connectivity and also sequential, chiral, and chemical diversity. The advantages of such libraries in terms of discovery of new pharmacological lead compounds and the various strategies applied in their design and synthesis are discussed in details in Vol. E 22a, Section 4.3.7. Due to the reduced conformational space of cyclic peptides compared to the flexible linear peptides, libraries based on this class of peptidic structures offer an additional dimension in the degree of diversity, which can be collectively defined as spatial diversity (*vide infra*). This spatial diversity allows very large numbers of cyclic peptides to be generated based on the sequence of single linear or cyclic peptides that may be natural peptides, active sites in proteins, or peptidic compounds discovered by combinatorial peptide libraries (either synthetic or phage display libraries), and by peptide mapping. Taken together with the distinct pharmacological advantages of cyclic peptides such as selectivity, metabolic stability, and improved bioavailability, cyclic peptide libraries represent a powerful tool for lead discovery and optimization.

Early work in the field has established the synthetic strategies and analytical tools for such class of libraries (for reviews see refs[111,112,456]). As listed in Table 13, the first generation of cyclic peptide libraries focused on biologically active sequences such as the cell adhesion RGD motif, the antileukemic heptapeptide stylostatin, or endothelin antagonists, but also on metal-binding sequence motifs and on the de novo discovery of bioactive cyclic peptides without sequence-biased motifs. Moreover, synthetic questions were addressed such as the sequence dependency of peptide cyclization reactions (see Table 13).

Table 13 Examples of Cyclic Peptide Libraries[111,402,403,457–462]

Mode of Cyclization[a]	Peptide[b,e]	Library Size	Diversity[c,e]	Chemistry[d]	Target[e]	Ref
A (SCA)	*c*[-XXXXN-]	130321	S	tea bags: mixtures	α-glycosidase inhibitor *c*[-iWyR-yN-] Ki = 1.7μM	[457]
	c[-XXXXXN-]	2476099				
	c[-XXXXXXN-]	47045881				
	c[-xxxxN-]	130321		pMeBHA resin (HF)		
	c[-xxxxxN-]	2476099		SPPS: HATU		
	c[-xxxxxxN-]	47045881		ORC: PyBOP/HOBt		
	c[-XxXxN-]	130321				
	c[-xXxXN-]	130321				
	c[-XxXxXN-]	2476099				
	c[-xXxXxN-]	2476099				
	c[-XxXxXxN-]	47045881				
	c[-xXxXxXN-]	47045881				
A (SCA)	*c*[-XXXPD-], *c*[-XXPXD-]	216	S	SPPS: PM mixtures	NIH screen, antibiotics	[111]
	c[-XPXXD-], *c*[-PXXXD-]	216				
	c[-XXXXPD-], *c*[-XXXPXD-]	1296		pMeBHA resin (HF)		
	c[-XXPXXD-], *c*[-XPXXXD-]	1296		BOP/HOBt		
	c[-PXXXXD-]	1296		ORC: HATU/HOAt		
	c[-XXXXXPD-], *c*[-XXXXPXD-]	7776				
	c[-XXXPXXD-], *c*[-XXPXXXD-]	7776				
	c[-XPXXXXD-], *c*[-PXXXXXD-]	7776				
A	*c*[-XKYGGFL-]	19	S	mixtures	model for Ag^+ assisted orthogonal cyclization	[458]
D	X[-KGGFL-]	19		SPPS: pMeBHA resin (HF) HATU/BOP		
D	XK[-YGGFL-]	19		SC: Ag^+		
A	*c*[-DFXXXa-]	5832	S	mixtures SPPS:	chiral selectors in CE	[459]
	c[-RKXXXa-]	5832				
	c[-RMXXXa-]	5832		Trt-resin (TFA)		
A (SCA)	*c*[-fψ(CH_2NH) O^1RGD-][d]	4	S	SC: HATU	RGD and enkephalin analogues; methodology of ψ scan	[460]
	c[-Nalψ(CH_2NH) O^1RGD-]	4	ψ scan	SPPS: PM		
	c[-aψ(CH_2NH) O^1RGD-]	4				
	c[-ANalψ(CH_2NH) O^1RGD-]	4				
	c[-Aaψ(CH_2NH) O^1RGD-]	4		mixtures		
	c[-O^4 O^3 O^2 O^1Yψ(CH_2NH) D-]	4		pMeBHA resin (HF)		
	c[-O^4 O^3 O^2 Yψ(CH_2NH) O^1D-]	4		BOP/HOBt		
	c[-O^4 O^3 Yψ(CH_2NH) O^2 O^1 D-]	4		ORC: DCC/HOBt		
	c[-O^4 O^1Yψ(CH_2NH) O^3 O^2 D-]	4				
A (SCA)	*c*[-$O^5O^5O^5$Pψ(CH_2S)GE-]	343	S	SPPS: PM mixtures	60 biological assays, NIH screen for anti HIV-1	[461]
	c[-O^5O^5Pψ(CH_2S)G O^5E-]	343	ψ scan			
	c[-O^5Pψ(CH_2S)G O^5O^5E-]	343				
	c[-Pψ(CH_2S)G $O^5O^5O^5$E-]	343		pMeBHA resin (HF),		
	c[-Pψ(CH_2SO)G$O^5O^5O^5$E-]	686		BOP/HOBt		
	c[-$O^6O^6O^6$Pψ(CH_2S)GB-]	216		ORC: HATU/HOAt		
	c[-O^6O^6Pψ(CH_2S)GO^6B-]	216				
	c[-O^6Pψ(CH_2S)GO^6O^6B-]	216				
	c[-Pψ(CH_2S)G$O^6O^6O^6$B-]	216				
	c[-$O^6O^6O^6O^6$Pψ(CH_2S)GB-]	1296				
	c[-$O^6O^6O^6$Pψ(CH_2S)GO^6B-]	1296				
	c[-O^6O^6Pψ(CH_2S)GO^6O^6B-]	1296				
	c[-O^6Pψ(CH_2S)G$O^6O^6O^6$B-]	1296				
	c[-Pψ(CH_2S)G$O^6O^6O^6O^6$B-]	1296				
	c[-$O^6O^6O^6O^6O^6$Pψ(CH_2S)GB-]	7776				
	c[-$O^6O^6O^6O^6$Pψ(CH_2S)GO^6B-]	7776				
	c[-$O^6O^6O^6$Pψ(CH_2S)GO^6O^6B-]	7776				
	c[-O^6O^6Pψ(CH_2S)G$O^6O^6O^6$B-]	7776				
	c[-O^6Pψ(CH_2S)G$O^6O^6O^6O^6$B-]	7776				
	c[-Pψ(CH_2S)G$O^6O^6O^6O^6O^6$B-]	7776				

for references see p 529

Table 13 (cont.)

Mode of Cyclization[a]	Peptide[b,e]	Library Size	Diversity[c,e]	Chemistry[d]	Target[e]	Ref
A (SCA)	*c*[-WIVpO^7-]	5	S	SPPS (Boc) on various resins (HF) ORC: BOP/HOBt	model for the synthesis of SCA library	[403]
A (SCA)	*c*[-RGDSPX-]	19	S	mixtures	evaluation of SC reagents: HATU, TBTU, PPA.	[462]
	c[-RGDSPx-]	19		SPPS: Trt resin (TFA) ORC: HATU		
A (SCA)	*c*[-PO^{8}KRGD-]	4	S	SPPS (Boc) HMP-resin and CO-HMP-resin (HF)	RGD model for the synthesis of SCA library	[403]
	c[-RGDPO^{8}K-]	4		ORC: BOP/HOBt		
A (SCA)	*c*[-PO^{9}PO^{9}N-]	25	S	SPPS (Boc), PM	metal ion host–guest interaction	[402]
	c[-PO^{9}NPO^{9}N-]	15		pMeBHA resin (HF)		
	c[-GdpO^{10}DpD(Aca-F-NH$_2$)-]	3		ORC: BOP/DIPEA		

[a] A = head-to-tail; D = side-chain-to-tail; SCA = side-chain anchor.
[b] One letter symbols are used. Small letters represent D-amino acids; X = proteinogenic amino acids; x = D-amino acids; in cyclization other than head-to-tail amino acids participating in the cyclization are in bold. In each one equimolar ratio of the amino acids were used: O^1 = A, S, V, D; O^2 = F, G, E, T; O^3 = R, K, I, H; O^4 = L, Y, G, A; O^5 = K, L, R, E, S, Y; O^6 = K, L, R, E, S, Y, A; O^8 = A, S, L, Y; O^9 = A, E, L, K, F; O^{10} = G, Sar, (Me)Ala; O^7 = the library was synthesized with an equimolar mixture of the SCA resin bound Boc derivatives of D, E, N, Q, and K-OFm esters. B = Asp or Asn.
[c] S = sequential diversity.
[d] ORC = on-resin cyclization; PM = portioning mixing; SC = solution cyclization.
[e] SCA = side-chain anchor; SC = solution cyclization; NIH = National Institute of Health; ψ = amide bond surrogate.

6.8.4.1 Cyclic Peptide Libraries with Sequential and Chemical Diversity

There are two general types of cyclic peptide libraries with sequential and chemical diversity. The first type is similar to that of liner peptide libraries except that the peptides are cyclic.[456] In the second type the ring structure is used as a scaffold and the sequential and chemical diversity is attained by attaching various groups to the functional side chains of the cyclic peptide template. Basically all the strategies used for the generation of linear peptide libraries are applicable to the synthesis of cyclic peptides libraries (for detailed information see Vol. E 22a, Section 4.3.7). In most cases the synthesis is completed on resin, and the libraries are deprotected/detached from the resin for biological evaluation and/or analyses. Resins and synthetic conditions are described in Section 6.8.3. The main problem in the preparation of such homodetic cyclic peptide libraries is the cyclization reaction itself, since the extent of ring closure is strongly dependent on the cyclization propensity of each linear precursor (see Section 6.8.1.3). If this propensity is low, side reactions such as C-terminal epimerization when the carboxy terminus is involved, as well as cyclodimerization may become predominant. As a consequence complex mixtures with undefined product distributions are obtained. Similarly, great attention has to be paid to optimize synthetic procedures to prevent suppression of single components. Moreover, during isolation of such libraries, e.g. by precipitation with diethyl ether/hexane, partial loss of unpolar components may occur, whilst the polar components are recovered quantitatively. For general experimental procedures see Vol. E 22a, Section 4.3.7.2.1.

An additional problem arises with the difficult analytical characterization of the cyclic peptide libraries. This is preferably performed by mass spectrometry, although there are limitations with the ionization method. In fact, FAB-MS leads to overexpression of hydrophobic components by ion suppression, whereas in MALDI-MS preferentially hydrophilic

components, e.g. arginine-containing compounds, are ionized. In ESI-MS in addition to ionization of aggregates, multiple ionization occurs which utterly complicates analysis of oligomers in the mixtures.

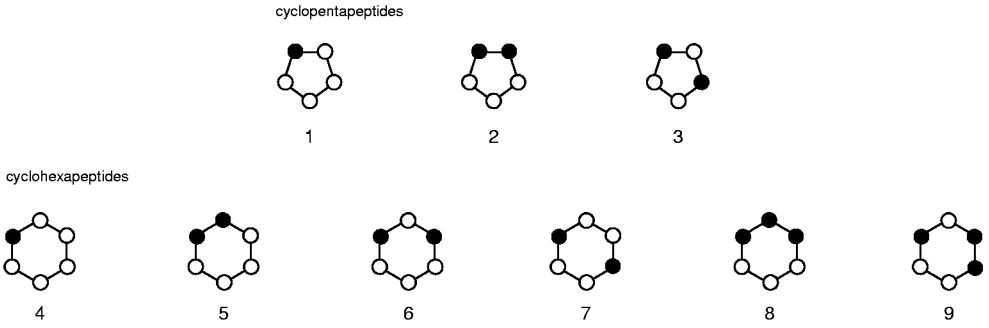
6.8.4.2 Cyclic Peptide Libraries with Conformational Diversity

In addition to sequential and chemical diversity, cyclic peptide libraries offer the possibility of conformational diversity where all components of the library differ from each other in their conformation, despite the identical connectivity. In this context two complementary approaches have been proposed, i.e. "spatial screening" based on head-to-tail cyclic peptides and "cycloscan" where the rest of the modes of cyclization are exploited.

6.8.4.2.1 Stereoisomeric Cyclic Peptide Libraries

For "spatial screening", stereoisomeric peptide libraries are constructed from conformationally constrained head-to-tail cyclic penta- and hexapeptide scaffolds that exhibit distinct backbone conformations,[24,26,260,265,463] often inappropriately defined as rigid instead of preferred conformations. These scaffolds were derived from NMR/MD studies[464] of a variety of cyclic head-to-tail penta- and hexaalanines that contain one to three D-Ala residues. Of these cyclic peptides only those that exhibit one distinct backbone conformation serve as scaffolds for the synthesis of stereoisomeric libraries. Similarly, NMR/MD studies of scaffolds where D-Ala is replaced with other structure-inducing amino acid residues such as D-amino acids, glycine (that can serve both as L- and D-amino acid), or D- or L-proline are characterized by distinct conformations that directly depend upon the position of these residues in the ring structure. Thus, each position in a given scaffold is conformationally defined. Correspondingly, each of the cyclopentapeptide and cyclohexapeptide scaffolds shown in Scheme 24 is characterized by a distinct conformation.

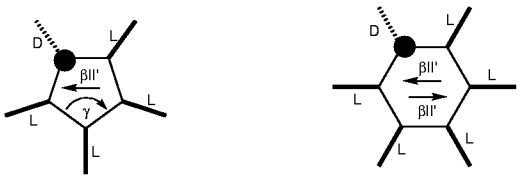
Scheme 24 Cyclopentapeptide and Cyclohexapeptide Scaffolds for Stereoisomeric Libraries



In Scheme 25, the conformations of scaffolds (1) and (4) from Scheme 24 are depicted for ring structures consisting of L-Ala and one D-Ala as models. In both cases the D-Ala residue (and in the same manner other D-amino acid or structure-inducing residues) occupy the $i+1$ position in the β-turn thus forcing each position in the scaffolds to be conformationally unique and defined.

for references see p 529

Scheme 25 The Conformations of Scaffolds (1) and (4)[a]

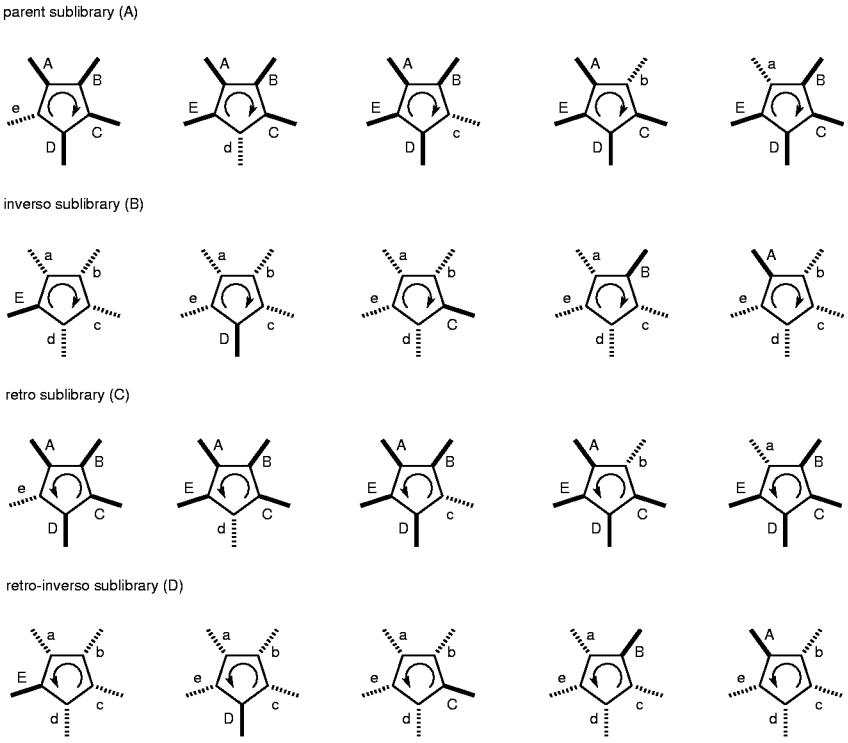


1 4

[a] Cyclopenta- and cyclohexaalanine are used as model compounds.

Each of the scaffolds reported in Scheme 24 can be used for the production of a stereoisomeric sublibrary based on the appropriate peptide sequence. For example, with the sequence A-B-C-D-E and scaffold (1) two types of stereoisomeric sublibraries can be prepared. One type includes the sublibraries A and B of Scheme 26 in which within a given sequence the configuration of each residue is successively inverted; thus, retaining the identical connectivity as in the parent linear peptide. In the second type[265] of sublibraries C and D (Scheme 26) the direction of the amide bond is inverted and hence the connectivity is not maintained. In most members of these sublibraries the overall conformation of the scaffold is maintained and therefore these components constitute stereoisomeric sublibraries of the parent library. Conversely, by introducing amide bond surrogates such as reduced amide bonds[465] or thioamide bonds[260,466] the conformation of the scaffolds are changed and their conformational flexibility enhanced.

Scheme 26 Stereoisomeric Cyclic Pentapeptide Sublibraries Having the Sequence A-B-C-D-E Based on Scaffold (1)[a]



[a] The arrow indicates the direction of the amide bond from N-to-C; dashed line and lower case letters indicate D- or other structure-inducing residues.

The concept of spatial screening with stereoisomeric libraries has been successfully applied to the RGD sequence motif and resulted in the discovery of superactive, highly selective $\alpha_v\beta_{3,5}$ integrin antagonists.[25,265] Similar work has been done on $\alpha_4\beta_7$ integrin antagonists which recognize the sequence LDT.[630]

Guidelines for the Synthesis of Stereoisomeric Libraries:
Bioactive sequences of up to six amino acid residues known to assume β- or γ-turns in the bioactive conformation are suitable for such libraries. If the sequence is short, residues have to be added in a manner to retain the desired physicochemical properties of the peptide (e.g., to short polar active sequences hydrophobic residues are preferentially added and vice versa). The choice of the scaffold depends on the number of structure-inducing amino acids such as Gly or Pro present in the native sequence. In absence of such residues scaffolds (1) or (4) (Scheme 24) are selected, whereas if Gly or Pro is present alternative scaffolds can be considered. Then the components of the four stereoisomeric sublibraries of Scheme 26 (or their equivalents if other scaffolds are chosen) are synthesized according to procedures described in the preceding sections.

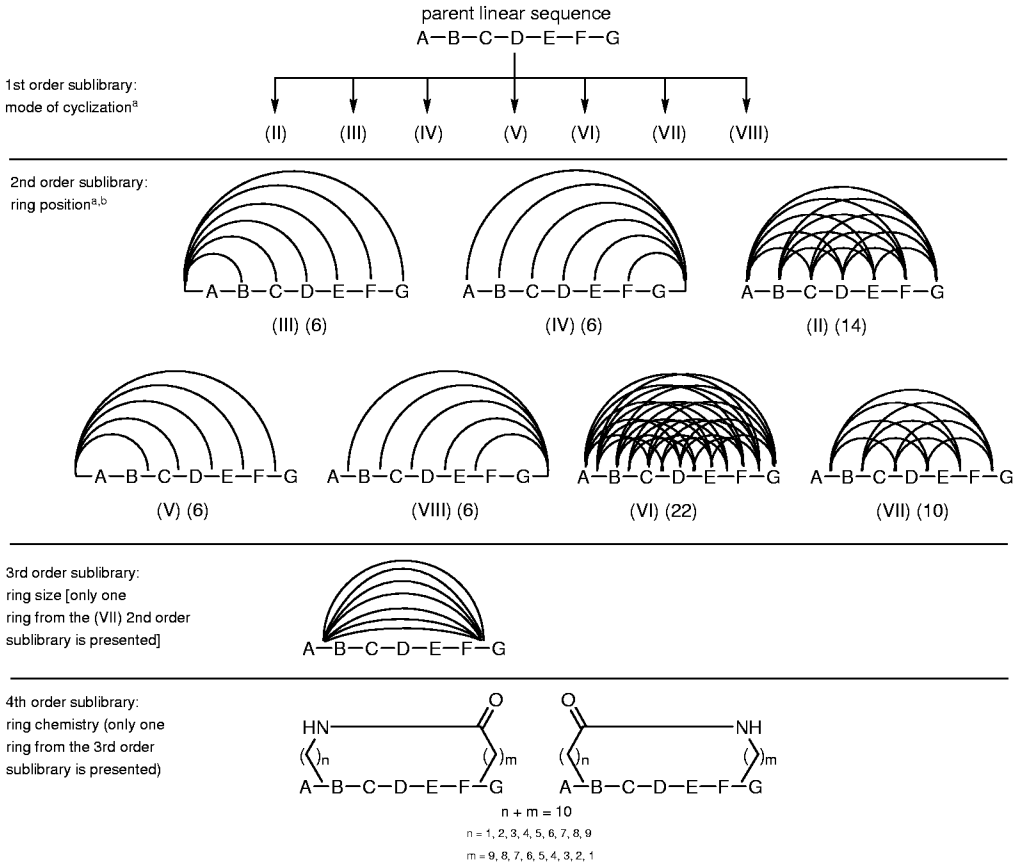
6.8.4.2.2 Cycloscan

For the cycloscan, conformational libraries are synthesized by cyclization of continuous or noncontinuous bioactive epitopes and not by their insertion into a scaffold. Originally, the concept of cycloscan was introduced for the generation of backbone-cyclized peptide libraries;[467] however, cycloscan can also be applied to other modes of cyclization. In this approach all components of each sublibrary bear the identical sequence, and differ from each other in distinct parameters that affect their conformation, but do not alter their connectivity, and hence their potential bioactivity. This is achieved by gradually introducing discrete conformational perturbations, which allow an efficient screening of the conformational space of the parent peptide. The majority of the components of such libraries should be inactive, because they do not overlap the bioactive conformation. However, the peptide that does fit the bioactive conformation should be very potent and have all the pharmacological advantages of cyclic peptides.

While in normal combinatorial peptide libraries (either chemical or phage display) each component has a unique sequence that is different from all others, in the cycloscan libraries all components have the same sequence, but differ in their conformation. This conformational diversity is generated in a dendrimeric hierarchy as shown exemplarily in Scheme 27 for the parent linear heptapeptide A-B-C-D-E-F-G. The diversity of the 1st order sublibrary (this nomenclature was adopted from Furka[468]) is based on the mode of cyclization. Excluding the head-to-tail cyclization there are seven different modes of cyclization that can be used for cycloscan; three natural modes of cyclization and four modes of N-backbone cyclization. In addition there are five theoretical modes of C-backbone cyclization (see Scheme 1) which are not included in Scheme 27.

for references see p 529

Scheme 27 Diversity of Cycloscans



[a] Modes of cyclization: (II) side-chain-to-side-chain; (III) side-chain-to-head; (IV) side-chain-to-tail; (V) head-to-backbone-amide; (VI) backbone-amide-to-side-chain; (VII) backbone-amide-to-backbone-amide; (VIII) backbone-amide-to-tail.
[b] Number of possibilities in brackets.

The diversity of the 2nd order sublibrary results from the ring position. For each one of the above modes of cyclization, the ring can be constructed at various positions along the parent sequence. Each mode of cyclization has a limited number of possible ring positions that depend on the size of the parent peptide, but crucially also on the amino acid composition and thus on the available side-chain functionalities. The diversity of the 3rd order sublibrary is the ring size. This sublibrary contains cyclopeptides that have again the same sequence, the same mode of cyclization, and the same position of the ring but differ in their ring size.

Finally, the diversity of the 4th order sublibrary is the chemistry of the ring. Retaining again the same sequence, the same mode of cyclization, the same ring position, and the same ring size, the components of this sublibrary differ from each other in the chemistry of the ring. As exemplarily shown in Scheme 27 for the 4th order sublibrary based on a backbone cyclic peptide with a lactam bridge of 12 atoms, the components (18 cyclic peptides) differ from each other in the direction of the lactam bond and its location along the ring. The chemical diversity of such 4th order sublibraries is actually limited only by the synthetic accessibility. It should be mentioned that some of the different cyclizations shown in Scheme 27 are theoretical and have not yet been synthesized, e.g. all the sublibraries derived from mode VIII. The synthesis of cyloscan libraries is generally performed manually on resin using methods described in preceding sections,[469] since the number of peptides in these libraries is limited and rationally designed.

Using the C-terminal hexapeptide of substance P as a model compound, each one of cycloscan diversity parameters have been shown to affect the conformation and hence the biological activity of the peptide.[417,470,471] This approach has been successfully applied to various peptides such as somatostatin,[419] the insect neuropeptide pheromone biosynthesis activating neuropeptide (PBAN),[420,431] BPTI,[435] and on the nuclear localization signal (NLS) of the HIV-1 matrix protein (MA),[429] and HIV-1 Tat/Rev.[430]

The diversity of cycloscan libraries may be extended by introducing diversity elements typical of linear combinatorial libraries such as sequential diversity, unnatural amino acids, and modifications or replacements of the peptide bond. For example DNA-encoded libraries containing 10^9 backbone-cyclized peptides on 7 mm glass beads have been developed[472] using the "portioning-mixing" technique[468] and the "one-bead-one-compound" concept.[473] For typical examples of design and synthesis of cycloscan libraries, see refs[419,420,430,435].

6.8.5 Cyclic Peptides Containing Hybrid Structural Elements

Cyclic peptides have been synthesized not only for the purpose of improving biological activities and selectivity, but also to explore basic features of secondary structures in peptides and to investigate with such mimetic compounds the conformational behavior of proteins. For this purpose artificial building blocks have been frequently used or amide bonds have been modified isosterically. Nature also offers a variety of modifications in cyclic peptides that are critically involved in their bioactivity. Some of the most common natural and synthetic modifications including unusual structural elements such as thiazoles (and dihydrothiazoles) and oxazoles (and dihydrooxazoles) with broad synthetic applications will be presented in the following section.

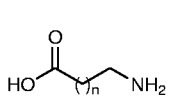
6.8.5.1 Cyclization with Nonnatural Amino Acids

Nonnatural amino acids are incorporated into cyclic peptides for predominately the following reasons. They may act as a clamp between the N- and C-terminus, side chains, and/or the backbone artificially forcing the bioactive peptide sequence into the desired conformation (see also Section 6.8.3.2.3), but may also serve as mimics of natural cyclic structures such as the carbaanalogues of disulfide bridges (see Section 6.2.4) and/or simply aid in the ring closure.[28,474,475] In matched cases superactivity and selectivity may be achieved, especially when the ring size is sufficiently small. Alternatively, some nonnatural amino acids may inherently carry functional groups involved in molecular recognition in the required spatial array, although only in rare cases both the desired structural and functional mimetics can be obtained. In this context, great care must be taken in the replacement of supposedly unnecessary amide bonds, which might indeed be important in the binding process.

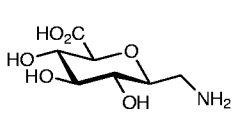
In principle every compound with an amino and a carboxy group can be used for such purpose ranging from simple ω-amino acids [e.g., 5-aminopentanoic acid (δ-aminovaleric acid) (**1**, n = 3)][54] or 6-aminohexanoic acid (ε-aminocaproic acid) (**1**, n = 4)][35,57,476–479] and related derivatives of 3-aminobenzoic acid;[480] or more sophisticated structures. A few examples (**1** – **6**) are shown in Scheme 28. Numerous cyclic turn mimetics have been developed in the past years and for details on this subject the reader is directed to Vol. E 22c, Section 12. To explore the rigidification introduced by nonnatural amino acids or equivalent structures into cyclic peptides, a careful NMR conformational analysis is required, since frequently the so-called β-turn mimetics do not enable such turns to be established, conversely other secondary structure elements may be induced.[481]

for references see p 529

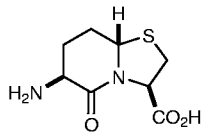
Scheme 28 Examples of Different Turn Structures Incorporated into Peptides[28,475]



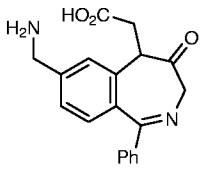
1 5-aminopentanoic acid (n = 3)
6-aminohexanoic acid (n = 4)



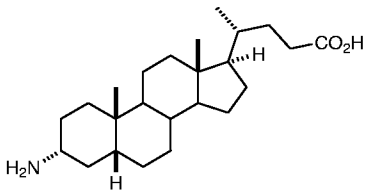
2 glucosyluronic acid methylamine



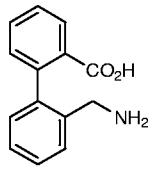
3 bicyclic β-turn dipeptide



4 benzodiazepine



5 steroid amino acid



6 2'-(aminomethyl)biphenyl-2-carboxylic acid

The incorporation of these artificial structural elements as well as the subsequent cyclizations are generally performed according to standard procedures; e.g. in the case of 5-aminopentanoic acid (**1**, n = 3) and 6-aminohexanoic acid (**1**, n = 4) the cyclization proceeds in good yields without particular difficulties.[35] Conversely, real turn mimetics may increase cyclization yields as a result of the anticipated rearrangement of the linear precursors.[28,87,481,482] In most cases, the nonnatural amino acids or equivalent building blocks are used in the synthetic steps in complete analogy to natural amino acids.

6.8.5.2 Backbone-Modified Cyclic Peptides

6.8.5.2.1 Thioamide Bonds

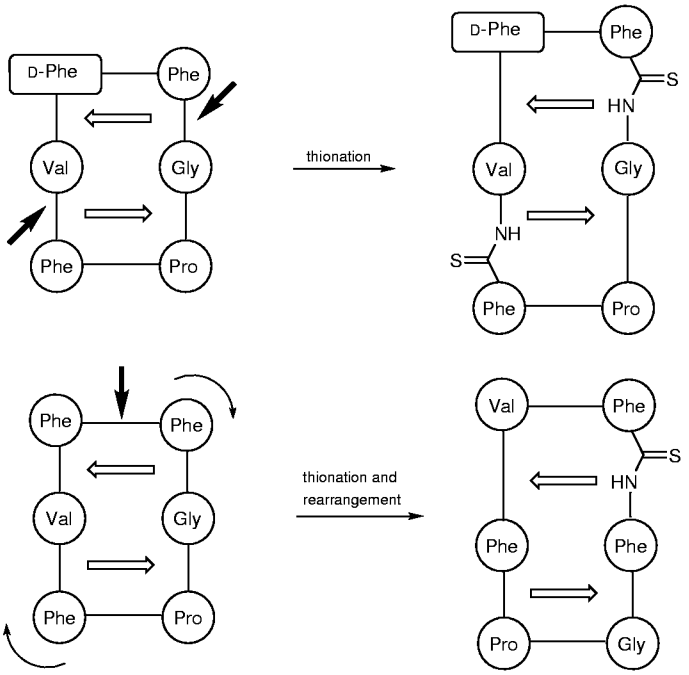
Replacement of the amide by a thioamide bond in peptides represents a minimal modification that may be of interest as it increases resistance against enzymatic degradation.[483,484] Such modification has been reported to both increase[485,486] and decrease biological activities, and the effect is not readily predictable.[487–489]

Scheme 29 Conversion of an Amide into a Thioamide Group



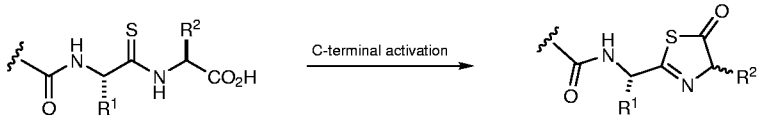
Structurally there are many similarities between the amide and thioamide bond (see Scheme 29).[490] The differences originate mainly from the different atomic radii and polarizability of oxygen and sulfur.[491] Thus different spatial arrangements,[492] higher rotation barriers,[493,494] and an altered donor–acceptor behavior in hydrogen bonding are observed.[491,492] The substitution of oxygen by sulfur results in enhanced donor, but decreased acceptor capability.[495] As a consequence of these different properties, the overall conformation of a cyclic peptide may change dramatically after thionation.[490,495] In cases where the thioamide NH is involved in the secondary structure as a hydrogen bond donor, the effect is generally small.[495] Conversely, the conformation favored in the oxo-peptide is destabilized if the newly introduced thioamide is not a hydrogen bond donor.[488,496] In fact, a complete shift of the conformational equilibrium may occur,[497] with the thioamide NH acting as the donor in the so-called mixed intramolecular hydrogen bond as outlined in Scheme 30.[491,492]

Scheme 30 Different Effects of a New Thioamide Bond on the Secondary Structure of Cyclic Peptides[495,497]



Several methods have been developed for the incorporation of thioamide bonds at defined positions into linear peptides (for details see Vol. E 22c, Section 10.9.1).[498,499] The resulting linear precursors can be cyclized by standard procedures, although the thioamide bond should not be located in penultimate position, since activation of the carboxy group leads to thiazolone formation as outlined in Scheme 31.[500–502] The cyclization yield of thioamide-containing peptides is generally rather low. This effect results from the additional local steric restrictions exerted by the thioamide bond, which leads to decreased conformational flexibility.[488]

Scheme 31 C-Terminal Activation of Peptides Containing a Terminal Thioamide Group[500–502]



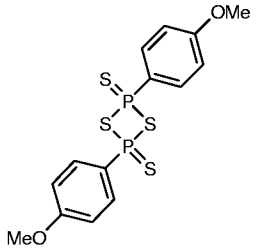
C-Terminal activation of peptides containing a terminal thioamide group results in thiazolone formation; this leads to epimerization at the marked chiral centers (see Scheme 35); the thiazolone is only a weak acylating agent.[500] The coupling yield may still be satisfactory, if achiral C-terminal amino acids are employed.[488,503]

Alternatively, the thioamide bonds can be inserted directly into the backbone of a cyclic peptide by the use of the Lawesson's reagent[504] or related variations (shown in Scheme 32).[505] In contrast to the linear peptides where normally only a very low degree of selectivity can be obtained, cyclic peptides often yield mono- or dithionylated products that are generally contaminated only by relatively minor amounts of side products. However, a mixture of thionylated cyclic peptides is frequently formed with thioamide bonds in different positions of the backbone.[486,506] In small-size, constrained cyclic peptides with a high degree of rigidity single reaction products may be obtained, although prediction of the preferred thionylation site is difficult. Hence direct thionylation of cyclic peptides as a general pro-

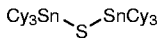
for references see p 529

cedure for systematic variations (e.g., a "thioamide scan" of biologically interesting peptides) is of limited value.[466] With increasing ring size selectivity decreases in a mode similar to linear peptides, and several positions are thionylated including multiple thionylations. Only in exceptional cases it is possible to favor formation of a single product by varying the reaction conditions such as temperature and solvent.[507–509]

Scheme 32 Examples of Thionylating Reagents[510,511]



Lawesson's reagent



bis(tricyclohexylstannyl) sulfide

As discussed prediction of the thionylation site with thionylating reagents is difficult. It can solely be based on (1) the electronic situation of single amide bonds which affects thionylation (the most positively charged carbonyl carbon reacts preferentially with thionylating reagents as confirmed by the enhanced reactivity of secondary compared to tertiary amide bonds[512]) and (2) the steric effects of the spatially demanding thionylation reagents (the accessibility of individual carbonyl oxygens dictates the thionylation sites[506]). Furthermore, hydrophobic interactions of the aromatic rings of the thionylating reagents with the peptide backbone may play a considerable role.[504]

Another possible method for the selective introduction of thioamide bonds into cyclic peptides is the use of oxazoles.[513–515] The oxazoles, which can be generated in serine- or threonine-containing peptides by the use of the Burgess reagent [(methoxycarbonylsulfamoyl)triethylammonium hydroxide inner salt) or under Mitsunobu conditions (see Section 6.8.5.2.2.2), are thionylated with hydrogen sulfide and triethylamine. A disadvantage of this method is that it is limited to serine- and/or threonine-containing peptides.

Cyclic peptides containing thioamide bonds can also be used for interesting follow-up chemistries. The thioamide bond can be regarded as a mandatory cleavage point, since a macrocycle can be selectively cleaved at this position after S-alkylation and subsequent acid hydrolysis. This approach was utilized in the semisynthesis of novel cyclosporin analogues for SAR studies on cyclosporin A. This cyclic natural product to some extent can be selectively thionylated with Lawesson's reagent and via the seco derivative, Edman degradation and subsequent recyclization gives the desired analogues.[516] Another application of thionylated cyclic peptides is the selective reduction of the thioamide bond to the reduced peptide bond analogues.[465,485]

Thionylation with Lawessons Reagent; General Procedure:[517]
Lawesson's reagent (3.56 g, 0.0088 mol) and peptide (0.005 mol) in anhyd benzene (10 mL) were stirred at 80 °C until the starting material was consumed (monitored by TLC or HPLC). The solvent was removed and the residue purified by chromatography (silica gel). Excessive Lawesson's reagent was eluted with CH_2Cl_2 and the peptide eluted with EtOAc/CH_2Cl_2 (9:1). The product was recrystallized (MeOH/Et_2O).

***c*[-Gly-Pro-Phe-Val-Pheψ{CS-NH}-Phe-], Thionylation with Yokoyamas Reagent; Typical Procedure:**[497]
To a soln of Yokoyama's reagent, 2,4-bis(phenylsulfanyl)-1,2,3,4-dithiadiphosphetane 2,4-disulfide[505] (60 mg, 0.15 mmol) in dry THF (15 mL) was added *c*[-Gly-Pro-Phe-Val-Phe-Phe-] (100 mg, 0.144 mmol) and the mixture was stirred at rt for 16 h. The solvent was removed under reduced pressure and the

peptidic fraction separated on a small column (silica gel, $CHCl_3$/MeOH in different ratios), the resulting material was purified by flash chromatography; yield: 27 mg (27%).

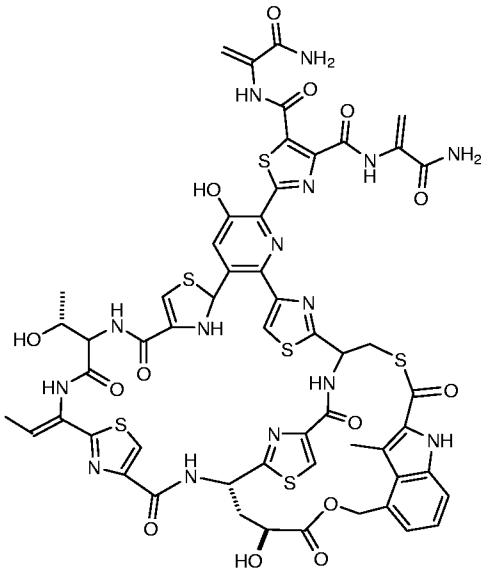
c[-D-Valψ{(C=S)NH}-Ser-D-Pheψ{(C=S)NH}-Ser-Phe-Pro-*allo*-Thr-], Thionylation via Oxazole Derivatives; Typical Procedure:[513]

A soln of c[-D-Val-Ser-D-Phe-Ser-Phe-Pro-*allo*-Thr(TIPS)-] (40.5 mg, 0.0439 mmol) was dissolved in THF (4 mL) and treated at rt with Burgess reagent (32.0 mg, 0.134 mmol). The mixture was stirred at 65 °C for 2.5 h, cooled to rt, and concentrated. The residue was dissolved in MeOH/TEA (1:1) and saturated with H_2S gas. The soln was stirred at rt for 4 h and then concentrated. Chromatography (silica gel, $CHCl_3$ to 4% MeOH/$CHCl_3$) afforded the thiolysis product (40.6 mg). The crude material was dissolved in THF (1 mL), treated with 1 M TBAF in THF (49 μL), and heated at 37 °C for 3 h. The soln was cooled to rt and directly chromatographed (silica gel, 6% MeOH/$CHCl_3$) and also purified by HPLC (silica gel, 2% MeOH/EtOAc); yield: 14.4 mg (41%); mp 195–197 °C (dec); $[\alpha]_D$ −118 (*c* 0.9, MeOH, 23 °C); IR (neat) $\tilde{\nu}$: 3292, 1657, 1649 cm^{-1}; ^1H NMR and FAB-MS were consistent with the assigned structure.

6.8.5.2.2 Thiazole- and Dihydrothiazole-Containing Cyclic Peptides

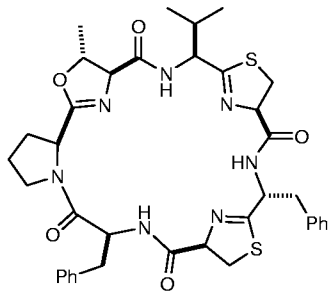
The five-membered heterocycles thiazoles and dihydrothiazoles are structural elements that often occur in natural cyclic peptides isolated from marine and microorganisms (Scheme 33).[514,518–520] Some members of this group of secondary metabolites are the lissolinamides (1–8),[521] tawicyclamides (A and B),[522] ulicyclamide,[523] ulithiacyclamide,[523,524] bistatramide C,[525] several keramamides (F–J),[526] and as the most known representatives cyclic dolastatins.[527,528] Most of these natural products exhibit interesting pharmacological properties such as antineoplastic,[529] antitumor,[529] or cytotoxic[515,522,530] properties. Another group comprises of peptide antibiotics[531–533] and among these the most representative natural products are nosiheptide[534–536] and thiostreptone.[537]

Scheme 33 Examples of Naturally Occurring Thiazole- and Dihydrothiazole-Containing Cyclic Peptides[521,522,534–536]

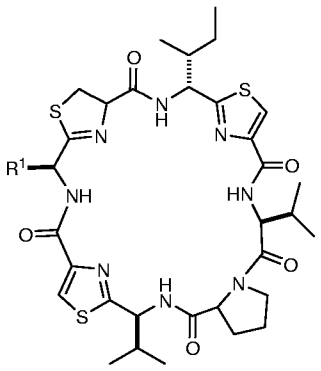


nosiheptide

for references see p 529



lissoclinamide 7



tawicyclamide A (R^1 = Phe)
tawicyclamide B (R^1 = Leu)

In biosynthesis the heterocyclic units are generated from dipeptide precursors via a multienzyme complex mediated condensation of the β-thiol group of a cysteine residue with the carbonyl group of the preceding amide bond without or with subsequent dehydration, thus leading to the amino- and carboxy-functionalized thiazolyl and dihydrothiazolyl derivatives shown in Scheme 34.[538]

Scheme 34 Common Structural Elements of Dihydrothiazole- and Thiazole-Containing Cyclic Peptides



The mechanisms of action of this class of natural products and in this context the function of the heterocyclic components are not fully understood,[539] although several SAR studies with thiazole- and dihydrothiazole-containing cyclic peptides have been performed.[515,540] Incorporation of these dipeptide mimetics into cyclic peptides has been shown to exert severe effects on the overall conformation[519,530,541,542] and may, thus, serve to additionally constrain larger-ring structures.[543] Furthermore, their potential role in metal ion transport as ionophores has been discussed.[539,544] In this context, several metal ion containing complexes were structurally investigated,[545–547] and the X-ray and solution structures of patellin 2,[548] patellamide D,[549] tawicyclamide A and B[522] have been resolved. Numerous other cyclic peptides of this class of compounds have been intensively investigated to derive possible structure–activity relationships.[530,550–553]

The most interesting pharmacological properties of these thiazole- and dihydrothiazole-containing cyclic peptides have spurred considerable synthetic efforts.[513,514,554–557]

6.8.5.2.2.1 Synthesis of Thiazole-Containing Cyclic Peptides

Various more or less efficient methods have been reported for the synthesis of 2-(1-aminoalkyl)thiazole-4-carboxylic acids and their suitably protected derivatives.[237,539,541,558–568] Optimal conditions must be selected in these syntheses to prevent racemization at the chiral aminoalkyl moiety, e.g. when applying a modified Hantzsch synthesis[559] racemization has been observed to occur at the level of the starting N^{α}-protected amino acid thioamide as well as in the base-mediated dehydration step of the intermediate hydroxydihydrothiazoles.[558] The 2-(aminoalkyl)thiazole-4-carboxylic acids are incorporated into the linear precursors by standard procedures of peptide synthesis,[237,514,529,539,552,555,558,564,569] and cyclization is pref-

erably not performed via the thiazole carboxy group as it undergoes epimerization during the ring-closure reaction.[570] The macrolactamization is performed at another site for which, again, standard coupling procedures are applied.[529,541,554,557,558,564]

N-(*tert*-Butoxycarbonyl)glycinethioamide:[568]
To a soln of Boc-Gly-NH_2 (17.4 g, 100 mmol) in DME (400 mL) was added Lawesson's reagent (30 g, 75 mmol) and the reaction was allowed to proceed at rt until consumption of Boc-Gly-NH_2 as monitored by TLC ($CHCl_3$/MeOH 9:1). The solvent was removed and the residue was washed in EtOAc with 10% aq $NaHCO_3$. The aqueous phase was reextracted with H_2O (2 ×), the combined organic phases were washed with brine, and dried. The solvent was removed and the residue recrystallized (EtOAc/hexane); yield: 17 g (89%).

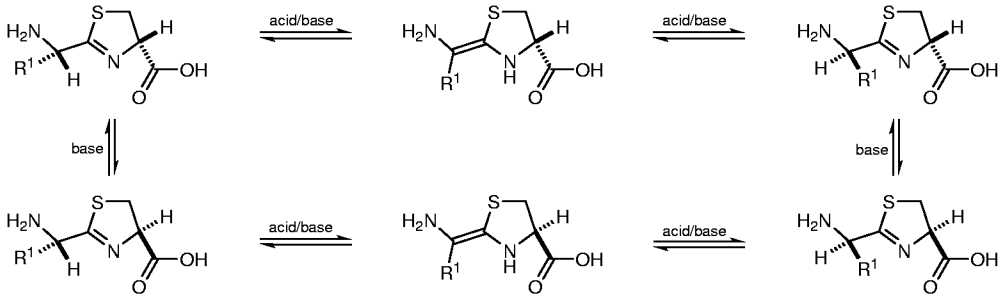
2-[(*tert*-Butoxycarbonylamino)methyl]thiazole-4-carboxylic Acid:[568]
3-Bromo-2-oxopropanoic acid (1.93 g, 11.6 mmol), *N*-(*tert*-butoxycarbonyl)glycinethioamide (1.52 g, 8 mmol), and $CaCO_3$ (2.3 g, 23 mmol) were dissolved in dry EtOH (60 mL) and stirred under argon at rt for 5 h. The mixture was filtered and the residue washed with EtOH (2 × 20 mL). The filtrates were concentrated and the residue was washed in EtOAc (150 mL) with 5% aq $KHSO_4$ and brine. The soln was dried, the solvent removed, and the residue crystallized (EtOAc/hexane); yield: 1.48 g (74%).

The related Fmoc derivative was obtained by cleavage of the Boc group with 50% TFA in CH_2Cl_2 and reaction with Fmoc-OSu in aq Na_2CO_3/THF under standard conditions; yield: 1.9 g (100%).

6.8.5.2.2.2 Synthesis of Dihydrothiazole-Containing Cyclic Peptides

For the synthesis of dihydrothiazole-containing cyclic peptides alternative synthetic routes are presently available.[514,570] In one of these approaches suitably protected 2-(aminoalkyl)dihydrothiazole-4-carboxylic acids are prepared and then used as building units for the assembly of the target peptides by standard synthesis on solid supports[519,539,561,571,572] or more frequently in solution,[539,571] followed by cyclization again by standard protocols.[552,564] However, great care has to be taken in all these synthetic steps because of the facile racemization at both chiral centers of the dihydrothiazole building blocks (see Scheme 35).[513,519,573,574]

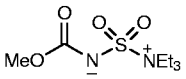
Scheme 35 Acid/Base-Catalyzed Racemization of 2-(Aminoalkyl)dihydrothiazole-4-carboxylic Acids[513]



Alternatively, the dihydrothiazole group is produced in the cyclic peptide at the desired positions with the Burgess reagent [(methoxycarbonylsulfamoyl)triethylammonium hydroxide inner salt, Scheme 36][575] by dehydration of a thioamide bond next to a side-chain unprotected serine or threonine residue, i.e. of Xaaψ[C(=S)NH]-Ser/Thr units.[570] This type of synthesis is presently the most simple and efficient, since the single steps rely upon well-established techniques, and the final step, i.e. thiazolidine formation, proceeds in good yields with a low degree of epimerization. By using a similar reaction dihydrooxazoles are produced, thus, allowing for simultaneous generation of both structural elements as often encountered in natural cyclic peptides.[576]

for references see p 529

Scheme 36 Structure of the Burgess Reagent[575]



This conversion of serine/threonine-containing peptides into oxazolidines may also be exploited for a subsequent dihydrooxazole → dihydrothiazole heterocycle interconversion in order to bypass both the problem of selective thioamide formation at the desired position and the facile racemization of the dihydrothiazole unit during incorporation into cyclic peptides (*vide supra*).[513–515] For this purpose, in the first instance the dihydrooxazole ring is formed which is then converted into the thioamide by degradation with hydrogen sulfide in TEA/MeOH. The final dehydration with the Burgess reagent yields the dihydrothiazole group at the desired position. The selective thiolysis of serine-derived dihydrooxazoles in the presence of threonine-derived oxazolidines was obtained in acyclic test systems, but could not be achieved in cyclic peptides.[513]

Another similar, but less frequently used method for incorporation of dihydrothiazole units into cyclic peptide precursors is the Mitsunobu reaction.[513,564] This reaction, however, is not suitable for the simultaneous production of dihydrooxazole rings, because of predominant aziridine formation.[571]

Synthesis of Dihydrothiazole-Containing Cyclic Peptides with the Burgess Reagent; Typical Procedure:[513]

A soln of *c*[-Valψ{C(=S)NH}Ser-D-Pheψ{C(=S)NH}Ser-Phe-Pro-*allo*-Thr-] (19.8 mg, 0.0248 mmol) in THF (3 mL) was treated at rt with Burgess reagent (29.6 mg, 0.124 mmol). The mixture was stirred at rt for 10 min and then heated to 40 °C for 20 min and further to 70 °C for 1.5 h. The soln was cooled, filtered, concentrated, and the residue was chromatographed (silica gel) and further purified by HPLC (silica gel); yield: 13.1 mg (71%) of the cyclic peptide with the sequence portions Val-Ser and D-Phe-Ser converted into the related thiazolidine derivatives and the Pro-*allo*-Thr into the oxazolidine derivative.

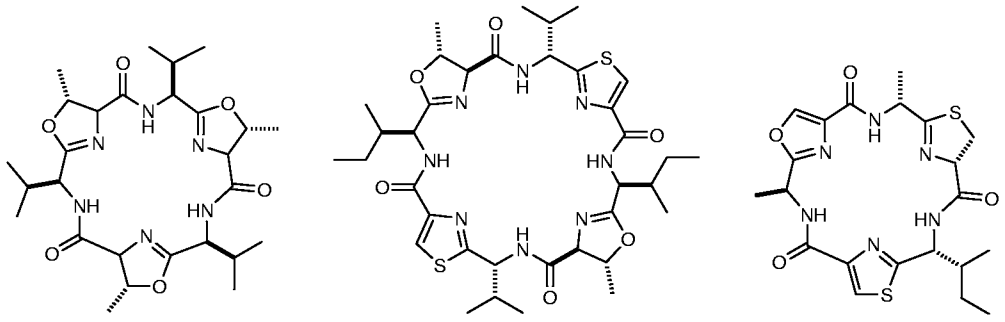
Dihydrooxazole → Dihydrothiazole Interconversion in Cyclic Peptides; Typical Procedure:[513]

The *c*[-Val-Ser-D-Phe-Ser-Phe-Pro-*allo*-Thr(TIPS)-] with Val-Ser and D-Phe-Ser cyclized to the related dihydrooxazoles (65.9 mg, 0.0743 mmol) was dissolved in MeOH/TEA (1:1, 5 mL) and saturated with H_2S gas. The soln was stirred at rt for 4 d and then concentrated. Chromatography on silica gel (MeOH/$CHCl_3$) afforded *c*[-Valψ{C(=S)NH}Ser-D-Pheψ{C(=S)NH}Ser-Phe-Pro-*allo*-Thr(TIPS)-] (51 mg). This thiolysis product was deprotected at the Thr residue with TBAF and purified by chromatography (silica gel) and HPLC (silica gel); yield: 22.0 mg (35%) of the desired intermediate as a white solid. This intermediate (19.8 mg, 0.0248 mmol) was dissolved in THF (3 mL) and treated at rt with Burgess reagent (29.6 mg, 0.124 mmol). The mixture was stirred at rt for 10 min and heated to 40 °C for 20 min and to 70 °C for 1.5 h. The soln was cooled, filtered, concentrated, and purified by chromatography (silica gel, MeOH/$CHCl_3$) and HPLC (silica gel, MeOH/$CHCl_3$); yield: 13.1 mg (90%) of cyclic peptide containing the Val-Ser and D-Phe-Ser derived dihydrothiazole units as well as the Pro-*allo*-Thr related dihydrooxazole unit as a colorless solid.

6.8.5.2.3 Oxazole- and Dihydrooxazole-Containing Cyclic Peptides

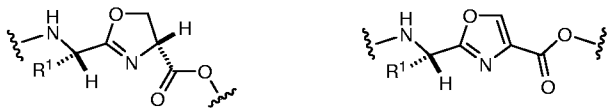
Similarly to thiazole- and dihydrothiazole, oxazoles and dihydrooxazoles also occur as backbone constituents in natural cyclic peptides isolated from marine and other microorganisms,[2,518,577,578] often together with the related sulfur-containing heterocycles. Representatives of this group of secondary metabolites are the keramamides B–E,[526,579] the lissoclinamides (1–8),[521,540] cyclooxazoline (westiellamide),[571,580] dolastatine E,[564,581] or ascidiacylamide,[551,569,582] sulfomycin,[531] or berninamycin[531] (see Scheme 37).

Scheme 37 Structures of Cyclooxazoline (Westiellamide),[571,580] Ascidialcyclamide,[551,569,582] and Dolastatine E[564,581]



The biosynthesis[237,538] involves enzymatic dehydration of serine and threonine residues in a manner similar to the formation of thiazoles and dihydrothiazoles (*vide supra*) with or without subsequent oxidation to yield the 2-(1-aminoalkyl)oxazole-4-carboxylic acid and 2-(1-aminoalkyl)dihydrooxazole-4-carboxylic acid shown in Scheme 38. These cyclic peptides display interesting physiological properties such as cytotoxicity,[541,569,583] antitumor activities, or antineoplastic effects;[523,529] but as for the sulfur-containing compounds the mechanism of action is not yet understood despite extensive SAR studies.[515,521,540,543]

Scheme 38 Structures of Dihydrooxazole- and Oxazole-Containing Peptides



Even for this type of natural product the function of the heterocycle within the cyclic peptide is yet unknown.[549,551] One function might certainly be the conformational restriction imposed by these heterocycles.[541,584] In this context it has been postulated that the easy formation and subsequent hydrolysis of the dihydrooxazole rings[554,569] may possibly serve for a conformational switch of the macrocyclic peptides.[541] Several X-ray and solution structures of different dihydrooxazole-containing cyclic peptides have been reported[530,543,549–551,553] and a possible role of the dihydrooxazole unit in inducing β-turns has been proposed.[543] Furthermore, as for the sulfur-containing heterocycles the ionophoric properties could suggest a role in metal transport or concentration.[544,584] Metal complexes of dihydrooxazole-containing cyclic peptides have been synthesized and investigated by X-ray analysis.[546,547,585] Based on these properties even a potential function of the cyclic peptides in carbon dioxide activation and mobilization has been proposed.[546]

The synthesis of oxazole- and dihydrooxazole-containing cyclic peptides has gained considerable interest in view of the structural characterization of the natural products. In fact, in various cases a final structural elucidation was provided only by total synthesis as originally proposed structures had to be corrected because of epimerization, difficult stereochemical assignment, and/or impurities.[523,524,554,555]

6.8.5.2.3.1 Synthesis of Oxazole-Containing Cyclic Peptides

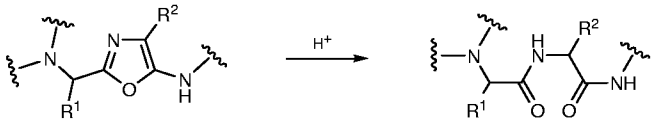
The oxazole-containing cyclic peptides are synthesized predominately by the direct use of suitably protected 2-(1-aminoalkyl)oxazole-4-carboxylic acids since oxidation of the oxazolidine precursors to the oxazole ring with MnO_2[558] or NiO_2[586] (or even with H_2O_2[564] or

for references see p 529

CuBr[587] and DBU in CCl_4/MeCN/Py[588]) is not compatible with the nature of preassembled peptides. For cyclization standard protocols are applied as in the case of the thiazole-peptides.[558,564]

It may be worthy to note that 2-(aminoalkyl)oxazole-4-carboxylic elements are masked Xaa-Ser/Thr dipeptide units and, thus, may also serve in particular cases to enhance the synthetic potential, since after acid hydrolysis both amide bonds are released restoring Xaa-Ser/Thr (Scheme 39).

Scheme 39 Oxazoles as Masked Dipeptide Equivalents



2-[(Benzyloxycarbonylamino)methyl]oxazole-4-carboxylic Acid:[588]
DBU (3 mL, 20 mmol) was added to methyl 2-[(benzyloxycarbonylamino)oxazolidine-4-carboxylate (2.8 g, 9.6 mmol) in CCl_4/MeCN/Py (2:3:3). After 3 h at rt, the solvent was extracted with 0.5 M HCl and the aqueous phase reextracted with EtOAc (2 ×). The EtOAc phase was washed with brine, dried ($MgSO_4$), and the solvent removed. Chromatography (silica gel, EtOAc/hexane 1:1) afforded methyl 2-[(benzyloxycarbonylamino)methyl]oxazole-4-carboxylate; yield: 0.94 g (33%). The oxazole ester (0.29 g, 1 mmol) was dissolved in dioxane (15 mL) and NaOH (0.12 g, in 5 mL H_2O) was added. The mixture was stirred at rt for 1 hr. The soln was neutralized with 10% aq $KHSO_4$ to pH 6, and the dioxane was removed. The soln was acidified to pH 3 and taken to dryness. The residue was crystallized (EtOAc/hexane); yield: 0.24 g (90%).

6.8.5.2.3.2 Synthesis of Dihydrooxazole-Containing Cyclic Peptides

As already shown for dihydrothiazole-containing cyclic peptides (Section 6.8.5.2.2.2), basically two different synthetic routes are used for the introduction of dihydrooxazoles into cyclic peptides. In the first one, 2-(aminoalkyl)dihydrooxazole-4-carboxylic acids and related derivatives are synthesized,[554,571,572,589,590] then incorporated directly into the linear precursors, which are finally cyclized by standard protocols.[541,554,567,569,571] Again the main disadvantage to this synthetic approach is the facile racemization of the dihydrooxazole synthon. Therefore, the preferred method is the production of the oxazolidine ring in the preassembled cyclic peptide. For this purpose various methods have been proposed.

The cyclic serine- or threonine-containing peptide is reacted with thionyl chloride (sometimes with additional silver triflate) to produce the related β-chloroalkyl intermediate which is then cyclized to the dihydrooxazole.[529,554,555,591] Alternatively, dihydrooxazoles can be produced from serine and allothreonine residues in cyclic peptides under Mitsunobu reaction conditions,[513,572] although in the case of threonine mainly aziridines are formed.[571] A simple and effective procedure is the reaction of serine and threonine residues in the preformed peptides with Burgess reagent.[513,570,575] As mentioned in Section 6.8.5.2.2.2, this procedure permits simultaneous generation of the dihydrooxazole and dihydrothiazole heterocycles,[576] but also generates the dihydrooxazole rings as precursors of the dihydrothiazole rings.[513]

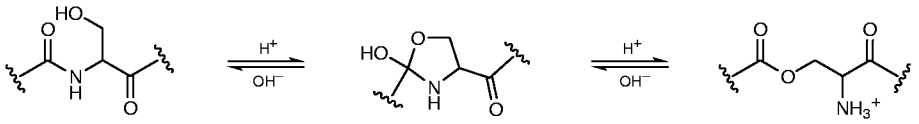
Synthesis of Dihydrooxazole Rings by Mitsunobu Reaction:[513]
A soln of *c*[-D-Val-Ser-D-Phe-Ser-Pro-Phe-*allo*-Thr-] containing the Phe-*allo*-Thr derived dihydrooxazole ring (42.7 mg, 0.057 mmol) in THF (10 mL) was treated at −60 °C with a soln containing Ph_3P (149.5 mg, 0.57 mmol), THF (2 mL), and DIAD (118 μL, 0.57 mmol). The mixture was warmed to 0 °C over a period of 45 min, concentrated, and chromatographed (silica gel); yield: 34.6 mg (85%) of *c*[-D-Val-Ser-D-Phe-Ser-Pro-Phe-*allo*-Thr-] with D-Val-Ser, D-Phe-Ser, and Phe-*allo*-Thr converted into the dihydrooxazole units.

Synthesis of Dihydrooxazole Rings with Burgess Reagent:[513]
A soln of *c*[-D-Val-Ser-D-Phe-Ser-Pro-Phe-*allo*-Thr-] (40.8 mg, 0.0532 mmol) in THF (7 mL) was treated at rt with Burgess reagent (63.4 mg, 0.266 mmol). The mixture was kept at 55 °C for 1 h and at 72 °C for 4 h, cooled to rt, and concentrated. The residue was chromatographed (silica gel, EtOAc followed by 5–10% MeOH/EtOAc) and again on silica gel (3–4% MeOH/$CHCl_3$); yield: 27.9 mg (74%) of the cyclic peptide with D-Val-Ser, D-Phe-Ser, and Phe-*allo*-Thr converted into dihydrooxazoles.

6.8.5.2.4 Conversion of Cyclic Peptides into Cyclic Depsipeptides by N→O Acyl Shift

Peptides containing serine or threonine may undergo an N→O acyl shift upon exposure to strong acids (Scheme 40).[592,594] This reaction has been exploited in the structure elucidation of cyclosporin A, since the acid-catalyzed acyl shift with formation of an ester allowed its selective hydrolysis to the linear peptide for further stepwise degradation.[593]

Scheme 40 Mechanism of the N→O Acyl Shift[592,594]



This is a reversible reaction that probably proceeds through a hydroxyoxazolidine derivative (Scheme 40)[592,594] and leads to cyclic depsipeptides, which can be trapped by acylation of the intermediately liberated amino group. Serine residues are known to react considerably faster and more exhaustively than threonine. The reaction rates in larger systems seems to be independent of the substrate (linear or cyclic peptide), as demonstrated on compounds analogous to cyclosporin A;[595] however, N-alkylation was found to considerably increase the rates.[595] A variety of acids can be used for such purpose such as H_2SO_4, TosOH, HCl, $BF_3 \cdot OEt_2$, and TFA. The yields are often low, especially in the case of threonine residues,[283] because of the prolonged reaction times and rather harsh reaction conditions required such as heating, which lead to further degradation.[596] Nevertheless this reaction has been exploited, e.g. in the synthesis of the RA-III-lactone where a dihydrooxazole was formed within the cyclic peptide under Mitsunobu conditions (see Section 6.8.5.2.3.2), which was then hydrolyzed under mild conditions with TFA in good yields to the desired cyclic depsipeptide.[283] The preparative usefulness of the N→O acyl shift in the synthesis of peptide lactones has been well-demonstrated with the preparation of an actinomycin-related depsipeptide.[596]

6.8.6 Miscellaneous Applications of Cyclic Peptides

Synthetic cyclic peptides are of eminent interest not only as replicates or analogues of natural products or as conformationally restricted bioactive peptides and peptide or protein fragments, but may also serve for specific purposes as listed in the following sections.

6.8.6.1 Synthesis of Nanotubes

The ability of cyclic peptides consisting of amino acids with alternating chirality to undergo self-assembly into synthetic nanotubes, this was already predicted theoretically,[597] has been realized through a pH-triggered reaction.[598–600] The single layers are associated by hydrogen bonds between the carboxamide groups that are perpendicular to the plane of the macro-

for references see p 529

cycles.[601,602] This leads to the formation of an antiparallel β-sheet structure between the cyclic peptide planes where the side chains are pointing to the outside and the inside becomes a tube-like cavity with diameters depending on the number of the amino acid residues in the ring.[603] By a suitable choice of side chains the lipophilic profile of the outer sphere can be controlled to enable embedment of the nanotubes into lipid bilayers[604,605] and thus formation of transmembrane channels. With appropriate ring sizes channels for both ions (8-membered cyclic peptides form channels of approximately 7.5-Å diameter that are permeable for sodium and potassium ions[606]) and neutral molecules (10-membered cyclic peptides form nanotubes of approximately 10-Å diameter that can be permeable to glucose[607]) are formed.

6.8.6.2 Synthesis of Dendrimers

Alternatively, cyclic peptides may be used as end groups in dendrimers[153,161,608,609] for the multiple attachment of bioactive peptides to a single template for their polypresentation to molecular recognition processes. The core structure may be formed by a lysine tree or pentaerythrol to build up several symmetric branched structures to which the peptides are attached mainly by Michael addition of a cysteine thiol group to a terminal acrylamide or acryl ester unit on the outer level of the dendrimers. Such constructs are used as immunogens, diagnostic agents, or artificial enzymes.

6.8.6.3 Synthesis of Cyclic Peptides as Structural Templates

Cyclic peptides have also been used in the concept of template-assembled synthetic proteins (TASP) (see Vol. E 22d, Section 13.1). The cyclic peptide templates are used as a structural motif to assemble secondary structural elements thereby enforcing defined folding patterns.[610] Such cyclic peptides generally consist of 10 to 16 amino acid residues, which are constructed with the β-turn inducing Gly-Pro sequences or β-turn mimetics at opposite sides.[482,611,612] As the functionalized peptide segments generally are attached to the template by lysine side chains, the selectivity has to be ensured by a multidimensional protecting group strategy.[613,614] For this purpose, new orthogonal ligation techniques have been developed.

6.8.6.4 Synthesis of Cyclic Peptides as Ionophores

Great attention has been paid to the excellent metal-complexation properties of cyclic peptides.[1,615,631] Naturally occurring examples include enniatin, valinomycin,[616] or antamanide,[316] and cyclosporin A.[632] Several libraries of cyclic peptides have been synthesized to address these properties,[402] but also bicyclic peptides have been synthesized in this context.[617–620]

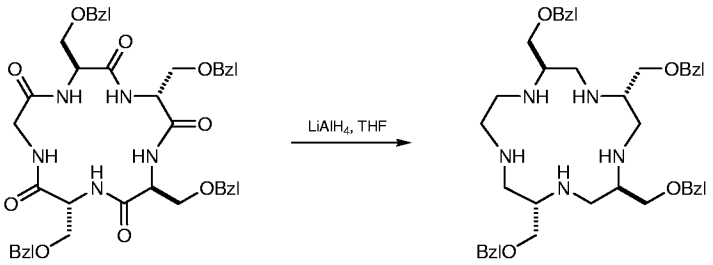
6.8.6.5 Synthesis of Glycosylated Cyclic Peptides

Glycoproteins are known to be involved in many biological processes such as cell recognition, differentiation, growth control, and cell regulation processes. The use of conformationally restricted glycosylated cyclic peptides offers the possibility of addressing such recognition processes with a defined display of the glycosyl moieties.[621] Despite the interesting perspectives of this class of compounds, only few homodetic glycosylated cyclic peptides have been synthesized.[39,621–625] For this purpose generally glycosylated amino acid derivatives (see Section 6.3) are used as building blocks,[621] since direct glycosylation of assembled peptides is difficult to achieve.[622]

6.8.6.6 Synthesis of Polyazamacrocycles

Another application of cyclic peptides is their conversion to polyazamacrocycles via reduction of the amide bonds with $LiAlH_4$ (Scheme 41).[626,627] This reaction proceeds without racemization and yields compounds that are otherwise hard to access.

Scheme 41 Reduction of a Cyclic Peptide to the Corresponding Polyazamacrocycle[626]



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7 Analytics of Synthetic Peptides

M. GOODMAN

This chapter commences with Section 7.1 which deals with reversed-phase chromatography (RPC) and related techniques as applied to synthetic peptides.[1–3] A detailed discussion on RPC is presented. Basic considerations are covered as are issues of fundamental physical chemistry. Many examples of chromatography and quantitative relationships are described for peptides and peptide derivatives. There is also an extensive table of naturally occurring peptides that have been isolated and purified by RPC techniques. The section includes many examples of RPC and HPLC profiles of peptidic systems.

Following the above section, is an overview of chromatographic techniques (Section 7.2). Various chromatographic approaches are described including thin-layer chromatography, high-pressure liquid chromatography (HPLC), normal and reversed-phase, size-exclusion, ion-exchange chromatographies, HPLC, and capillary electrophoresis.[1,4–6] Detection, solid supports, and elution systems are also discussed.

In Section 7.3, the subject of amino acid analysis is covered including a brief description of hydrolytic techniques, column preparations, and data analysis.[7,8] This is followed by a discussion of racemization assays (Section 7.4). These areas of analysis are critically important for researchers in the field of peptide synthesis.[9–11] In Section 7.4, a systematic approach is outlined for the study of racemization and mechanisms are discussed for epimerizations at asymmetric sites. A comparative study is presented for determining the intrinsic rates of racemization, based upon urethane-protected *N*-carboxyanhydrides (UNCAs). These approaches are extremely important for understanding the tendencies of novel amino acids and other building blocks to racemize.

Section 7.5 covers NMR spectroscopy[12,13] and deals with high-resolution NMR spectroscopy, primarily focusing on the constitution and identification of peptides. Peptide sequences are also noted, as are peptides with side-chain modifications. The section also contains a description of the use of NMR spectroscopy together with mass spectrometry (Section 7.6) to identify and characterize intermediate and target structures. Section 7.6 deals with techniques of mass spectrometry as applied to peptide research.[14] The basis for ionization and mass platform configurations are covered, as are issues of peptide sequence. This primer will be most useful to those concerned with the power of chromatography (Sections 7.1 and 7.2), and with mass spectrometry (Section 7.6). Together, they form a most useful source that encapsulates these techniques which modern peptide scientists employ to isolate, purify, and begin the identification of target molecules.

The next three sections (Sections 7.7.1, 7.7.2, and 7.7.3) cover fluorescence spectroscopy,[15–18] infrared, and circular dichroism, three powerful approaches to characterize the structure and conformational considerations of synthetic peptides. Section 7.7.1 deals with the use of fluorophores and broad aspects of fluorescence spectroscopy to characterize conformational aspects of peptide structure. In a similar manner, Section 7.7.2 covers a broad aspect of the uses of infrared (IR) techniques to study peptide conformations.[19–22] Many IR techniques are discussed, as are approaches for the study of specific peptidic structures including amyloid, β-turn, and membrane peptides. Finally, there is a section on circular dichroism (Section 7.7.3) that covers the major issues of concern for peptide synthetic chemists such as the assignments of α-helix, 3_{10}-helix, β-sheets and β-turns, and polyproline helices.[23–25] There is also a brief description of cyclic peptides.

A section on quality standards and comparative syntheses concludes the entire chapter. This section (Section 7.8) sets the scope and approach necessary to compare and contrast closely related syntheses and target structures.[26] In this section, a broad utilization of many of the techniques described in the chapter are applied to peptide target structures.

for references see p 544

Although the focus of these Houben–Weyl volumes is primarily on synthesis, it is important to recognize the context and criteria of characterizations of purity and structure that are necessary in our field of peptide science.

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7.1 Purification of Synthetic Peptides

M. T. W. Hearn

Since the 1980s advances in peptide chemistry have underpinned many facets of the biological and biomedical sciences. The ability to rapidly purify and characterize synthetic or naturally occurring peptides based on high-resolution procedures has become integral and essential components of these developments. Both chromatographic and electrophoretic procedures dominate these experimental approaches. Of the various options available to peptide chemists or, in general, to all biological scientists, reversed-phase chromatography (RPC) is currently the most important and widely used separation technology for the analysis and purification of synthetic and naturally occurring peptides.

A central feature of all applications with RPC methods is the use of hydrophobic sorbents, involving, for example, straight-chain alkyl (*n*-alkyl) or diphenyl groups chemically attached to the surface of a hydrophilic support material such as porous silica, zirconia, or other type of ceramic-based medium. Alternatively, the RPC sorbent can be formed from macroreticular polymeric resins, such as the polystyrene–divinylbenzene microparticulate or monolithic materials. A large range of hydrophobic sorbents, suitable for use in high-performance RPC separations of peptides at analytical or laboratory-scale purification levels are now commercially available from over 500 different vendors. Additional sorbent materials more suited to large-scale preparative applications have also been developed. In all cases, retention and, thus, resolution of the various peptide components within a crude mixture are achieved through discrete interactions involving hydrophobic regions, primarily associated with the nonpolar amino acid side chains of the peptide(s), and chemically immobilized nonpolar ligands on the RPC sorbent. This feature sets RPC apart from the other chromatographic or electrophoretic procedures that exploit different physicochemical properties, such as the charge characteristics or Stokes radii of peptides, in order to achieve resolution of a complex mixture.

In this section the focus is on how the practical opportunities of RPC methods can be exploited by peptide chemists from a basic understanding of the principles, concepts, and procedures implicit to this technique. Drawing upon this information, and experience gained from the wide usage of RPC procedures in peptide analysis and purification, the opportunities offered by other chromatographic approaches such as high-performance size-exclusion (HP-SEC), ion-exchange (HP-IEX), hydrophilic interaction (HP-HILIC), and metal ion affinity (HP-IMAC) chromatographic procedures will then be briefly discussed and contrasted. In addition, the application of high-performance capillary zone electrophoresis (HP-CZE) and capillary electrochromatography (HP-CEC) in peptide analysis and nano-isolation tasks will be briefly described. A perspective will thus be developed on the experimental opportunities provided by RPC and how this fundamental separation technique can be integrated with these other methodologies to achieve the productive isolations and skilful analyses of peptides. Importantly, the comprehensive knowledge available on these different experimental procedures can now be collectively integrated into multidimensional capabilities, facilitating the rapid and selective isolation and rigorous physicochemical characterization of synthetic as well as naturally occurring bioactive peptides, and peptides derived from the proteolytic fragmentation of proteins.

Following an introductory section on the historical background that has led to the development of RPC techniques as practiced today, the principles and theory of RPC will be discussed, incorporating the role of amino acid sequence and hierarchical structural effects that determine the outcome of the interaction of a peptide(s) with various types of nonpolar chromatographic surfaces. Subsequently, the influence of operating parameters such as the effects of different (1) surface morphologies or chemistries of nonpolar sorbents; (2) concentrations of organic solvents, salts, or other mobile-phase additives; (3) pH conditions; or

for references see p 622

(4) temperatures will be examined. The impact on resolution of conformational changes and other secondary equilibrium phenomena that larger polypeptides can undergo in these nonpolar environments will then be explored from the perspective of their thermodynamic implications and practical consequences during their isolation. Finally, a survey of selected applications is presented to highlight some of the useful attributes and versatility of RPC methods in modern peptide chemistry. This background will then be used to introduce other high-performance separation technologies, which have found specific niches in peptide isolation and characterization, but have not yet gained the same level of prominence or encompassed the same diverse range of applications as RPC procedures in peptide analysis or preparative isolation at different scales. Because continuous growth in knowledge in this field have occurred since the 1980s, RPC procedures can be expected to maintain their dominance as the exquisite laboratory and process separation technologies par excellence for the analysis, quantitation, purification, and manufacture of peptides.

Some of the symbols used in this section are defined in Table 1.

Table 1 Symbols Used in this Section

Symbol	Definition
α	selectivity
χ	molar fraction of charged species of a peptide with different ionization states
δ_i, δ_m, δ_s	solubility parameters of the peptide P_i, the mobile phase m, and the stationary phase s
$\Delta A_{contact}$	molecular contact area
ΔA_{hydr}	hydrophobic contact area
$\Delta G_{assoc,i}^{\ominus}$	change in Gibbs free energy
$\Delta H_{assoc,i}^{\ominus}$	change in enthalpy of interaction
$\Delta S_{assoc,i}^{\ominus}$	change in entropy of interaction
ΔC_p	change in heat capacity of interaction
$\Delta G_{fold,i}^{\ominus}$	change in Gibbs free energy associated with folding
ΔP	pressure drop across the column
ε	molar absorptivity
ε_o	permittivity in vacuo
ε_r	relative permittivity (dielectric constant) of mobile phase
Φ	phase ratio of the chromatographic system
φ	column resistance factor
γ_τ	packing tortuosity parameter
γ	surface tension
η	mobile-phase viscosity
κ	structural packing parameter
κ^e	microscopic cavity factor
λ	column packing term
μ_a	apparent electrophoretic mobility of analyte (= $\mu_{eo}+\mu_e$)
μ_{eof}	electrophoretic mobility of EOF
μ_e	electrophoretic mobility of analyte
ν	linear flow velocity
σ	charge density of the surface of shear
ψ	volume fraction of organic solvent
ξ	zeta potential of the liquid–solid interface
A	eddy diffusion term (= $2\lambda d_p$)
B	longitudinal diffusion term (= $2\gamma D_m$)
C	mass transfer term {= $[k'/(1+k')]^2 d_p^2/D_5$}
C	the electrolyte concentration
d_c	column diameter
D_m	bulk diffusion coefficient in mobile phase
d_p	average particle diameter of sorbent material
D_s	overall diffusion coefficient of analyte in sorbent microenvironment
E	electric field strength
F	Faraday constant
F	flow rate
$h_{e,i}$	height equivalent to a theoretical plate

Table 1 (cont.)

Symbol	Definition
α	selectivity
K_a	equilibrium distribution constant
k'	unitless capacity factor in the μ-HPLC mode
κ_{cec}	unitless capacity factor in CEC mode
K_p	permeation coefficient
L	column length
M	molal concentration
$n_{i,j}$	slope dependency for selectivity in HPSEC
P_c	partition coefficient
P_d	average pore diameter
Q	ion charge
R	ion radius
R	gas constant
R_g	radius of gyration
R_S	Stokes radius
S, S',	slope dependencies of the ln k' versus ψ plots
T	temperature
t_e	elution time
V_e	elution volume
V_m	mobile-phase volume
V_s	stationary phase volume
w	peak width

Historical Origins of Reversed-Phase Chromatography and the Concept of the Hydrophobic Effect

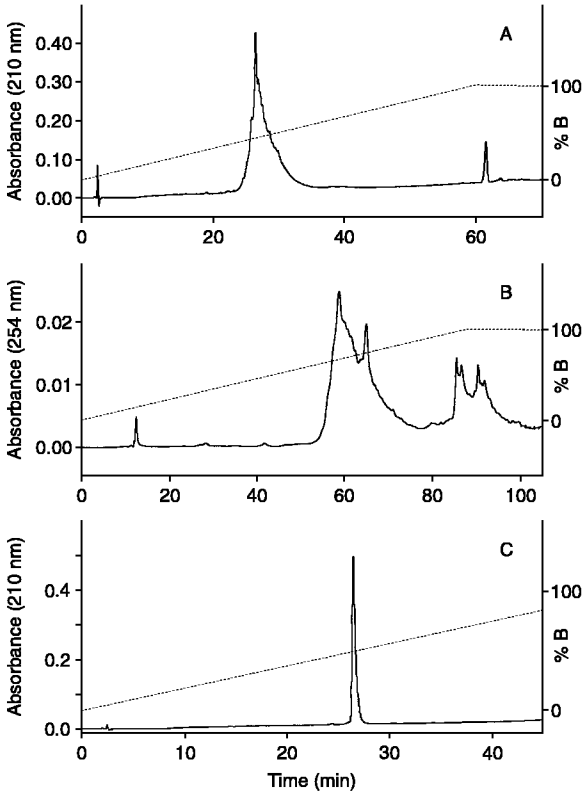
In science, important concepts often coalesce historically near to the same dates. Such was the case leading to the initial observations on the use of the RPC concept in peptide and protein separation and the role of the hydrophobic effect in protein folding. In 1950, Howard and Martin[1] introduced a new variant of partition chromatography with elution of polar compounds achieved by an increasing concentration of an organic solvent. This development was called "reversed-phase chromatography" to distinguish it from the other chromatographic methods in vogue at that time, such as paper partition chromatography or soft gel ion-exchange chromatography, represented a novel concept because the polarities of the chromatographic phases were "reversed", i.e. encompassed a nonpolar stationary phase and a polar mobile phase. Near to this time, Kauzmann[2] was exploring the physicochemical basis of protein folding; this work led to his seminal observations published in 1959 on the origin and role of the hydrophobic effect. Previously, in 1949, Tiselius and Sheppard[3] had described a technique, now known as hydrophobic interaction chromatography (HIC) but then called by the acronym "salting-out chromatography", whereby a more polar eluent was used in conjunction with a less polar paper chromatographic sorbent for the separation of dyes in the presence of relatively high concentrations of salts, e.g. sulfate and phosphate salts. In subsequent years, soft gel systems attracted some interest for the open column, low-pressure partition chromatographic separation of peptides, such as the purification of oxytocins,[4] β-endorphins,[5] somatostatin analogues,[6] or insulin-related peptides[7] on modified Sephadex G-25 or LH-60 support material. However, it was not until 1976 when two reports[8,9] introduced high-performance RPC procedures based on the mechanically more robust *n*-alkylsilicas that these techniques started to gain ascendancy as the rapid and highly selective separation method that we now take for granted for the isolation of synthetic and naturally occurring peptides. Subsequent to these initial reports, a large number of investigations have greatly expanded upon these observations, directly shedding light on the physicochemical basis not only of the retention mechanisms per se that operate during the RPC separation of peptides, but also how different elution strategies can be optimally implemented.

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In RPC, elution of a peptide(s) is achieved with a mobile phase containing a defined (isocratic) or an increasing (gradient) concentration of an organic solvent that forms part of a water/organic solvent mobile phase, often of low pH. The fundamental physicochemical principle that is exploited in RPC separation of peptide mixtures involves the manipulation of the surface tension, γ, of the mobile phase. Retention of specific peptides can then be increased or decreased with a defined RPC sorbent, buffer composition, pH, or flow rate through the simple expediency of changing the concentration of an organic solvent used as a surface tension modifier as part of the mobile phase. Illustrative of the use of an ascending organic solvent gradient in RPC to achieve this surface tension change is the preparative and analytical separations, shown in Figure 1, of a crude mixture and a purified sample obtained following solid phase peptide synthesis (SPPS) of the 26-mer zinc finger CCCH motif peptide related to the ribosomal L36 protein isolated from *Thermus thermophilus*.[10,11] Clearly evident in the case of the crude sample (Figure 1A) is the presence of deletion products of similar sequence that arise as a consequence of the repetitive nature of the synthetic steps of the SPPS procedures. In order to achieve by preparative RPC the resolution of the desired synthetic polypeptide (Figure 1C) from the various deletion peptides and other chemical components present in this sample, optimized low pH, linear 0–100% gradient elution conditions of 90 minutes duration from H_2O/MeCN/TFA (98:2:0.1), pH 2.1, to H_2O/MeCN/TFA (39.1:60:0.09) were used with a TSK-ODS-120 T column (300 × 21.5 mm i.d.) packed with 10 µm octadecylsilica and operated at a flow rate of 7.5 mL · min^{-1} with UV detection at 254 nm. The first stage of the preparative fractionation of the crude polypeptide mixture is shown in Figure 1B. How, and more importantly why, were these specific conditions chosen? In this section, the basis behind this selection will be elaborated.

Results comparable to those shown in Figure 1 can readily be achieved with most crude mixtures containing synthetic or naturally occurring peptides, using similar RPC strategies. The resolution of synthetic peptides prepared by solid- or solution-phase chemical methods is now so routine by analytical and preparative RPC methods that an April 2000 CD-ROM database search of relevant scientific journals revealed that over 2500 publications arise each year on the use of RPC for the purification or analysis of peptides as part of the scientific literature related to peptide chemistry and its application in various fields of the biomedical or biological science and in biotechnological applications related to the food, environmental, and pharmaceutical industries.

Figure 1 Analytical RP-HPLC Profiles for the 26-mer Zinc Finger CCCH Motif Polypeptide Derived from the *Thermus thermophilus* TthL36 Ribosomal Protein[a]



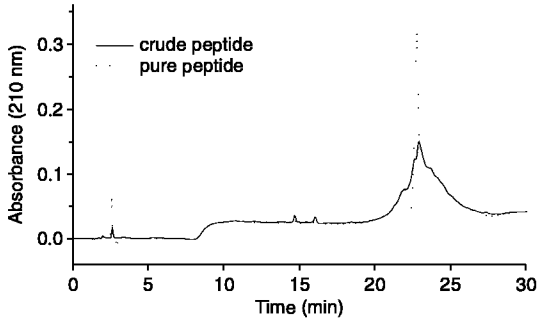
[a]Analytical RP-HPLC profiles for the crude (A) and purified (C) synthetic 26-mer zinc finger CCCH motif polypeptide derived from the *Thermus thermophilus* TthL36 ribosomal protein. The Acm-protected crude 26-mer polypeptide was purified by gradient elution RP-HPLC (B) [Waters 600/486 HPLC system; TSK-ODS-120 T column (300 × 21.5 mm i.d., Tosoh Corp. Yamaguchi, Japan); 10 µm octadecylsilica; eluent: A: 0.1% TFA/H_2O and B: 0.09% TFA in 60% MeCN/H_2O, linear gradient 0–100% B over 90 min; UV detection (254 nm); flow rate: 7.5 mL·min^{-1}]. The crude synthetic 26-mer polypeptide and the purified polypeptide obtained from the recovered fractions were analyzed by analytical RP-HPLC [column as above (150 × 4.6 mm i.d.); 5 µm octadecylsilica; eluent: linear gradient of 0–100% B over 60 min; UV detection (214 nm); flow rate: 1 mL·min^{-1}]. The gradient slopes are illustrated as dotted lines.

From this literature, it is apparent that the analytical and preparative applications of RPC methods have been more heavily focused on the separation of peptides up to about 40 amino acid residues in size, although analogous procedures are now increasingly being employed with larger polypeptides and small proteins prepared by total or semisynthetic ligation/condensation approaches. A concern expressed by some investigators about the use of RPC is that the technique may be too harsh with large synthetic polypeptides, resulting in impaired recovery from the RPC column in large-scale preparative applications (and similar concerns have often been raised for the isolation of proteins in bioactive state by RPC methods). Yet, numerous examples can be found in the literature[12,13] whereby bioactive, large polypeptides, i.e., with 50–120 amino acid residues, and globular proteins can be successfully purified on laboratory scale as well as on industrial scale by RPC procedures. Illustrative of the use of such approaches is the purification of human activin β_A subunit analogue, a 110-residue polypeptide prepared by total synthesis in this laboratory by SPPS procedures[14] (Figure 2). As evident from Figure 2, a synthetic product of very high purity can be obtained, with the correct molecular mass as determined by electrospray ionization

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mass spectrometry (ESI-MS). Biological and biophysical investigations can then be pursued with such large synthetic polypeptides with confidence. Again, sequential use of preparative RPC techniques based on low pH H_2O/MeCN mobile phases proved essential for the separation of the desired synthetic 110-mer from other deletion, α-, β-, γ-,.... side-chain isomerization or L-α- → D-α-racemization products. By exploiting algorithmic procedures such as DryLab/G, Hephaestus–Pek-N-Eze, or the G2-based ExpertSystem, which permit the optimization of the selectivities (i.e., the α values) and the peak shapes (i.e., the theoretical plate numbers, *N*), high resolution separations and improved recoveries can be achieved even with such large, fully synthetic polypeptides. Achieving optimal recoveries, whilst at the same time maximizing the resolution between large polypeptides from closely related deletion peptides still, however, represents a major challenge. To achieve these outcomes, investigators require an understanding of the origin of the selectivity (i.e., the thermodynamic) and zone broadening (i.e., the dynamic) behavior of peptides in RPC environments.

Figure 2 The Analytical RP-HPLC Profiles for the Crude and Purified Synthetic 110-mer Activin β_A Subunit Polypeptide[14][a,b]



[a] This polypeptide was purified by gradient elution RP-HPLC [Waters 600/486 HPLC system; TSK-ODS-120 T column (300 × 21.5 mm i.d., Tosoh Corp. Yamaguchi, Japan); 10 μm octadecylsilica; eluent: A: 0.1% TFA/H_2O and B: 0.09% TFA in 60% MeCN/H_2O, linear gradient of 0–100% B over 90 min; UV detection (254 nm); flow rate: 6.0 mL·min^{-1}]. The crude synthetic and purified 110-mer polypeptide, which was obtained from the recovered fractions, was analyzed by analytical RP-HPLC [column as above (150 × 4.6 mm i.d.; 5 μm octadecylsilica; eluent: linear gradient 0–100% B over 60 min; UV detection (214 nm); flow rate: 1 mL·min^{-1}].
[b] Adapted from Keah and Hearn,[14] with permission.

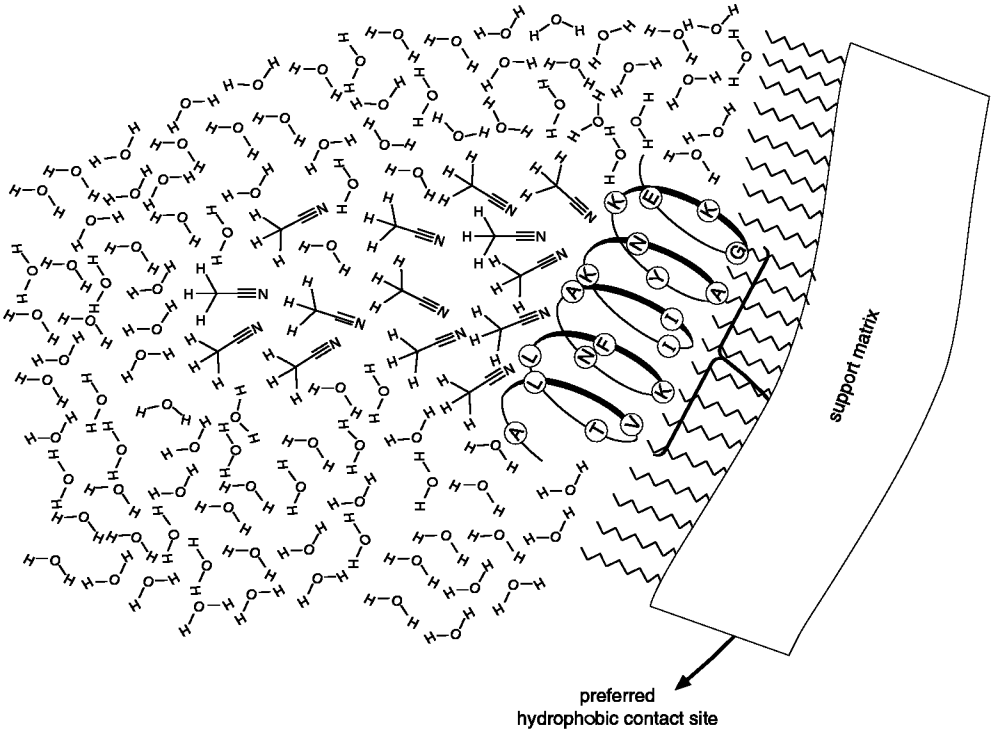
Various studies have also noted that RPC techniques represent very powerful avenues to explore the molecular basis of the intrinsic hydrophobicity and conformational status of synthetic peptides. Since the initial attempts by Martin and Synge, over 50 years ago, to exploit the hydrophobic effect as part of partition chromatographic procedures[15] based on impregnated paper for the separation of amino acid derivatives, RPC techniques in association with sensitive spectroscopic methods have emerged as powerful alternatives to the traditional, and more time consuming, liquid–liquid partition methods to explore and quantify (1) the role of the hydrophobic effect in the stabilization and folding of polypeptides and proteins; (2) the nature of the molecular forces that are involved in peptide–nonpolar ligand interactions; (3) the thermodynamics and biophysics of peptide interactions with relatively well-defined nonpolar chemical and lipid-like surfaces, as models of peptide–membrane interactions; and (4) the effects, more generally, of amino acid sequence and chirality on biorecognition phenomena. As a consequence, along the way, RPC procedures have emerged as powerful techniques to study the role of the hydrophobic effect per se in the generation of supramolecular structures with peptides such as α_2-, α_3-, or α_4-coiled coils, βαβ motifs or parallel and antiparallel β-sheet fibril structures.

Representative of the types of interactions that a peptide can undergo in a full aqueous RPC environment, such as a silica-based sorbent that has been chemically modified with *n*-alkyl chains, is the schematic illustration shown in Figure 3. The bracket/arrow in Figure 3 identifies the preferred (hypothetical) hydrophobic contact region of the polypeptide, P_i, illustrated in α-helical amphipathic conformation. When binding occurs, a reduction in the cavity volume, $V_{cav,i}$, of the interacting polypeptide–nonpolar ligand system occurs. Under such conditions, a change in the overall free energy, $\Delta G_{assoc,i}^{\ominus}$, for the RPC chromatographic process arises, with the magnitude of these changes represented by the summation of the various free energy contributions, i.e. by $\Delta G_{assoc,i}^{\ominus} = \Delta G_{g/p,i}^{\ominus} + \Delta G_{cav,i}^{\ominus} + \Delta G_{int,i}^{\ominus} - RT\log(RT/PV)$, where $\Delta G_{g/p,i}^{\ominus}$, $\Delta G_{cav,i}^{\ominus}$, and $\Delta G_{int,i}^{\ominus}$ are the free energy changes for the association process in the hypothetical gas phase, the formation of the appropriate cavity, and the interaction between the protein species and the surrounding molecules, respectively. R is the gas constant, and T, P, and V are the standard temperature, pressure, and volume terms. When acetonitrile is employed as the organic solvent, clustering of the solvent molecules can occur as schematically represented, with the potential for the polypeptide to adopt a preferred secondary structure reinforced under the specific RPC conditions by the peptide-backbone–solvent dipole interactions.

Typically, in a water/organic solvent mobile phase, the peptide will exist in a highly solvated cavity surrounded by water and solvent molecules that interact by hydrogen bonding and/or other dipole–dipole effects. On approaching the nonpolar surface, incremental desolvation of the peptide will occur from regions of highest hydrophobicity and lowest polar solvent accessibility, permitting the generation of a preferred contact region for intercalation of the peptide with the nonpolar ligands on the sorbent through partition and/or adsorption effects. As shown in Figure 3, with H_2O/MeCN eluents, the acetonitrile molecules prefer to exist as clusters due to their dipolar characteristics. This property results in a different interaction mechanism, when compared to a hydrogen-bonding solvent such as methanol, although overall with either dipolar or hydrogen-bonding solvents in RPC, elution of a specific peptide can be achieved by a progressive decrease in the surface tension of the eluent through an increase in the concentration of the organic solvent. Because acetonitrile-based solvents tend to generate greater entropic changes, this property also tends to result in partial stabilization of secondary structures of polypeptides, i.e. induce α-helical or β-sheet features either in the bulk solution or at the nonpolar sorbent–liquid interface. For this reason the CD spectra of polypeptides in the presence of acetonitrile at a concentration approximating the value at which the polypeptide desorbed from the RPC sorbent frequently will show some secondary structure, whilst the corresponding methanol-based system, which tends to solvate by hydrogen-bonding effects, may not. Irrespective of whether enthalpic or entropic effects dominate the solvation, the free energy change for a favorable binding of the peptide to the immobilized nonpolar ligands, given by $\Delta G_{assoc,i}^{\ominus}$, will be negative. If the peptide adopts a preferred secondary structure in the bulk solvent, the magnitude of the molecular forces associated with this peptide–ligand interaction may exceed the corresponding $\Delta G_{fold,i}^{\ominus}$ value associated with the folding of the peptide into this preferred secondary structure. In this case, it is likely that the peptide will assume a more extended (random coil) conformation on the surface of the nonpolar sorbent. When mixtures containing individual peptide components that can manifest either stabilized versus destabilized secondary structures are separated, the consequences of this behavior are evident as eluted peaks of different peak widths, and changing selectivities in response to small changes in mobile-phase composition, temperature, or even RPC sorbent usage.

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Figure 3 Schematic Representation of the Hydrophobic Interaction of Polypeptide Solvated in a H_2O/MeCN Mobile Phase with a RPC Sorbent Containing Immobilized *n*-Alkyl Chains



Depending on its sequence and structure (i.e., whether the peptide is linear, cyclic, has disulfide bonds, etc.) the peptide may thus partially or completely unfold on contact with the nonpolar sorbent surface. Alternatively, the interaction with the nonpolar ligands may be associated with favorable $\Delta G_{assoc,i}^{\ominus}$ and $\Delta G_{fold,i}^{\ominus}$ terms, with the peptide then adopting a more highly stabilized conformation(s) on the surface of the RPC sorbent than occurs in the bulk mobile phase. The participation of conformational interconversions that are relatively long-lived compared to the chromatographic timescale will directly impact on the selectivity and resolution of complex mixtures of synthetic peptides. When long-lived conformational processes occur with polypeptides, different selectivities will be observed for the "native" compared to the "unfolded" state of the polypeptide. As a result of these selectivity differences, the elution profiles of peptides in RPC can be fundamentally different from those obtained by other chromatographic or electrophoretic procedures. Appropriate precautions must be taken to ensure that effectively only a single (time-averaged) conformational species of each peptide is present during the RPC separation. Attention has to be given to the role of different adsorption phenomena for synthetic polypeptides within a complex mixture when they interact with hydrophobic surfaces. The issue whether the peptide–nonpolar ligand interaction involves a "dwell"- or "residency"-induced conformational change,[16–19] whereby these molecules change their conformation at the solid–liquid interface, is thus highly relevant to the selection of RPC conditions that achieve the best resolution, and ultimately the highest purity of the recovered fractions of the desired peptide.

Since ca. 1985 in particular, a great deal of practical experience has been accumulated related to the characteristics of different RPC systems, their use in computer-optimized method development, and the physicochemical origin of the retention and kinetic phenomena that peptides manifest in these systems. Peptide retention in RPC is largely due to the docking of certain amino acid side-chain and main-chain functional groups of the peptide

with the immobilized ligands on the stationary phase surface. Typically, the dominant interaction between a peptide and a RPC sorbent involves the immobilized nonpolar ligands, such as an *n*-alkyl group chemically attached to porous silica or other types of inorganic oxide materials. However, the backbone functionalities of the support material itself also influence the selectivity and peak width for a peptide(s), either directly by providing competitive binding sites or through their effect on the zeta potential of the surface. Incomplete surface derivatization of the support material will thus result in electrostatic, polar phase, and secondary "nonspecific" contributions to the retention. The participation of these latter effects becomes evident in peptide separations as badly tailing peaks or reduced recoveries. Once a particular RPC sorbent has been selected, the choice of the mobile phase dictates the relative affinity and selectivity changes that can be achieved. The presence of different types of organic solvents; kosmotropic, neutral, or chaotropic salts; or other types of additives in the mobile phase such as ion-pairing agents or detergents has a dual effect on the retention and conformational equilibrium of the system. Compared to many other techniques, the RPC separation of synthetic and naturally occurring peptides can be achieved from the microgram through to the kilogram level utilizing the same sorbent and mobile-phase conditions. This outcome is a consequence of the very reproducible procedures that are now available for the manufacture of high-performance, macroporous, microparticulate RPC sorbents of excellent stability in terms of their resolving power, selectivity, and peak efficiencies. For this reason, some key features of the use of macroporous *n*-alkylsilica and other types of hydrophobic sorbent materials will be summarized later.

Future Developments

In this section, some of the practical opportunities of RPC and other high-resolution separation procedures for peptide analysis and isolation are examined. Several common themes inherent to these separation technologies are highlighted, including the nature of the retention processes, the fundamental basis, the rich practical opportunities to explore the physicochemical properties of peptides in situ, and the powerful capabilities to isolate and analyze peptides to a very high purity and at a very high detection sensitivity. From the perspective of advances in the separation sciences, further refinement of RPC and these other techniques can be expected to continue unabated, due to their importance at the research as well as industrial levels. The development of new classes of RPC and other classes of interactive HPLC sorbents with tailored selectivities and greater stability over wider pH ranges can be readily envisioned. Collectively, these techniques provide new opportunities to assess the purity, relative structural stability, and associated information on the secondary structure of peptides in solution and at nonpolar interfaces. Because of the ability to assess the intrinsic hydrophobicities of peptides in subtly different solvent environments, RPC methods may become even more useful through correlative principal component analysis and expert-system chemometric methods as procedures par excellence for the determination of the structure–function relationships of peptides, leading to a considerable expansion of knowledge on peptide design, biorecognition, and molecular folding in the post-genomic era of biotechnology.

From a regulatory perspective, the development of new quality control procedures will be increasingly dependent on faster, more sensitive separation procedures. The procedures in this section will facilitate these objectives through acceptance by the peptide community of "tailored" microparticulate sorbents, with greater pH operation ranges and improved mass transfer kinetics, and miniaturized instrumentation, including novel micromachined systems for very high speed, nano-analytical separations. The opportunities provided by sequential multidimensional separation principles, including advanced forms of HP-CEC in multichannel/capillary micromachined systems will also become apparent. Greater utiliza-

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tion of capillary column systems in novel multichannel formats with gradient elution times of 30 seconds to 1 minute can be anticipated for such ultra-analytical separations. Such developments will become the engine for the development of new detection systems, including micromachined devices. Integration of several chromatographic and detection systems into a single instrument is now feasible at the research laboratory level. Already, various computer-assisted procedures can be employed to run simultaneously multiple equipment systems, change columns, or adjust the composition of mobile phases, whilst achieving concurrently the optimization of the resolution, peak spectral scanning, or delivering the peak effluent to a second and third detector as a multidimensional hyphenated separation. Full implementation of such control and monitoring systems will change how the analysis and quality control of peptide synthesis is carried out in the research laboratory as well as in larger-scale production procedures with peptides. Possibly, the greatest excitement, however, will come from experimental work on the structure–retention-function correlations with peptides that these new instrumental systems and procedures will offer. Realization of this potential, in their various manifestations, will ensure that high resolution separation procedures will expand their fundamental importance as innovative tools in the peptide sciences revolution arising from genome/proteome mining over the next several decades.

7.1.1 Retention of Peptides in Reversed-Phase Chromatography

7.1.1.1 General Characteristics of Reversed-Phase Chromatography Separations of Peptides

In RPC separation of peptides, the fundamental structural properties of the amino acids within the sequence and the relative accessibility of the nonpolar amino acid residues to a large measure determine the overall selectivity that can be achieved with a defined RPC system.[20–23] As a consequence, peptides typically elute from RPC sorbents in the order of their relative hydrophobicities, for a pre-selected mobile-phase composition, pH, and temperature. However, the relative hydrophobicities of different peptides are also conditional on the solvation environment in which they are placed. The exposure or greater accessibility of previously sequestered polar or hydrophobic amino acid side chains in polypeptides with well-developed secondary structures will thus significantly affect the relative binding affinities of these peptides to hydrocarbonaceous-bonded phase surfaces.

What sets peptides apart from low-molecular-weight organic compounds in their RPC behavior is their structural hierarchy. Thus, the chemical organization of the amino acid sequence and various levels of assembled structure (i.e., whether any secondary, tertiary, or supramolecular characteristics exist) that dictates their overall topographical features also determines how they will interact with the solvent or nonpolar ligands. Superimposed upon the molecular features of the peptide are the chemical and physical attributes of the chromatographic system per se. Various studies[23–30] have shown that a major contributor underpinning the differences in retention and selectivity of larger polypeptides separated on various RPC sorbents are associated with the extent of changes in conformation of the polypeptides. These effects are mediated case by case by differences in the sequence of the polypeptide, the composition and/or density of the immobilized ligands at the surface of the sorbent, and/or the mobile-phase composition. These latter characteristics of RPC systems directly affect the interactive behavior of synthetic peptides at the molecular level, irrespective of whether or not the side chains of the peptide are fully or partially protected/deprotected. Fundamental properties of the amino acid residues in terms of their molecular mass, volume, accessible surface area, partial specific volume, pK_a, and effective hydro-

phobicity have relevance in this context, and can be used as part of the experimental strategy to predict the relative retention and selectivity of homologous peptides in RPC.

Typically, small peptides in bulk solution have no or very poorly developed secondary structure, i.e. largely exist in random coil conformations. If a preferred secondary structure does exist, then the favorable free energy for folding, $\Delta G_{fold,i}^{\ominus}$, will usually be very small and negative. For monomeric, highly α-helical polypeptides of about 25–30 residues in size, the magnitude of the $\Delta G_{fold,i}^{\ominus}$ in bulk solution is typically about -5 kcal·mol^{-1}. Interestingly, the change in free energy, $\Delta G_{assoc,i}^{\ominus}$, associated with the interaction of the same peptide with immobilized nonpolar ligands of a RPC sorbent is frequently of similar magnitude. Although thermodynamic studies have elucidated some aspects of the interaction behavior observed when peptides are adsorbed from water/organic solvent mixtures to solvated nonpolar RPC sorbents,[18,31] most interpretations of the RPC separation process with peptides have relied more on empirical assumptions about the properties of the nonpolar stationary phase or the mobile-phase composition, and disregarded the linkages between the structural, thermodynamic, and extra-thermodynamic properties of peptides. Along the pathway to an energetically favored bound state, many polypeptides can sample a relatively large proportion(s) of their conformational space in the RPC environment. It is thus essential when undertaking a RPC separation of larger peptides to assess constantly whether any changes in conformational status have arisen from experimental adjustments of the chromatographic conditions per se, such as the choice of the RPC sorbent, composition of the mobile phase, or from changes in flow rate or temperature. When they occur in an uncontrolled manner, then complex peak shapes and selectivity variations can arise, and substantial difficulties will be experienced in the scale-up of an analytical separation to the preparative level.

It is well known that conformational stability, and frequently the solubility, of peptides in solution or in liquid–liquid two-phase or multiphase systems can be manipulated by exposure to different salts, organic solvents, organic polymers, or pH changes, particularly if they are near to the isoelectric point. In all of these cases, advantage is taken of the changes in the bulk hydration and molecular cavity properties of the peptides when they are placed in solution. Depending on the solution conditions, all peptides can, in principle, undergo self-associated (isodesmic) processes leading to finite and/or infinite networks of solvated monomeric species, which finally lead to aggregate formation. Analogous effects also occur within the chromatographic environment. Whether the peptide exists under RPC conditions as a monomer or as an aggregate will thus depend on its concentration and its solubility parameter under a particular set of solvent, pH, and ionic strength conditions. Participation of peptide–peptide interaction processes involves secondary solution chemical equilibrium. Hence, it is potentially possible to rationally modulate separation selectivities and zone broadening processes of peptides in RPC by taking advantage of the molecular propensities of peptides to aggregate, while the influence of the experimental conditions on their conformational status is examined at the same time. Without this attention, suboptimal chromatographic performances will arise, leading to lower levels of resolution in the separation of the various peptides in a mixture.

7.1.1.2 Hydrophobic Effects and Solvophobic Considerations for the Isolation of Peptides by Reversed-Phase Chromatography Methods

The interaction between a solvated peptide and an RPC sorbent in a fully or partially aqueous solvent environment can be discussed in terms of the interplay of weak physical forces. Based on linear free energy considerations, the effects of these forces can, to a first approximation, be linearly summated. Consequently, knowledge of the amino acid sequence of a peptide permits, to a first approximation, the effective hydrophobicity of the peptide to be derived by correlation analysis methods using data derived from other techniques, or

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from the databases available on peptide retention times measured in various RPC systems.[32–35] The dominant types of physical interactions that control the selectivity of peptides in RPC systems are in order of relevance (1) hydrophobic interactions and related phenomena mediated by nonbonding, polarized electron-donor or electron-acceptor processes; (2) Lifshitz–London forces and van der Waals and associated weak dipolar interactions; (3) π → π and π → dipole interactions; (4) hydrogen bond interactions; (5) electrostatic interactions; (6) metal ion coordination interactions; and (7) secondary molecular interactions involving force field effects. Close packing interactions mediated by van der Waals effects between the nonpolar amino acid side chains of a peptide have been shown[36] to play an important role in the establishment of stable secondary structures. However, these weak van der Waals interactions make only a small contribution to the negative change in enthalpy that arises when a peptide is dissolved in an aqueous mobile phase. When a peptide is bound to the immobilized nonpolar ligands of an RPC sorbent, van der Waals interactions probably make a greater contribution to the overall free energy change. The London–van der Waals forces are influenced by the size and geometry of the molecules involved in the interaction(s) and by the ease of polarization of their electron clouds. Momentary shifts in the symmetry of the electron clouds of a peptide molecule arise when it approaches the nonpolar surface of an RPC sorbent. These effects will be conditional on the chemical nature (i.e., the zeta potential) of the support material that forms the RPC sorbent, the type of nonpolar ligand (i.e., *n*-alkyl versus phenyl ligands), the temperature, and the solvent composition. Of all of the forces, the secondary contributions from hydrogen-bonding effects probably have, after the hydrophobic effect has been taken into account, the greatest impact on the selectivity characteristics of polypeptides with well-developed secondary structures. For this reason, significant differences in resolution and retention can arise when a poor hydrogen-bonding but good dipole-intercalating solvent such as acetonitrile is changed to a good hydrogen-bonding solvent, such as methanol, and vice versa for the separation of synthetic or naturally occurring peptides.

7.1.1.3 Basic Considerations Linking Retention, Resolution, and Peak Efficiency with the Other Chromatographic Parameters

In RPC, as in all modes of chromatography, when a peptide is eluted under isocratic conditions, the retention can be expressed in either time, t_e, or volume, V_e, units. Expressions of the elution time or elution volume of a peptide, eluting with a peak width w ($= 4\sigma_t = 4\sigma_V$), from a column packed with RPC particles of mean particle diameter d_p, incorporate the physical aspects of the column (diameter d_c, length L), the flow rate F (or linear flow velocity, $\nu = LF/V_m$) and the phase ratio, $\Phi = V_s/V_m$, where V_s is the volume of sorbent and V_m is the volume of the mobile phase in the chromatographic column. Usually, retention dependencies for a peptide P_i are represented in terms of a capacity factor k'_i as follows:

$$t_e = \frac{V_e}{F} = \frac{L}{\nu}(1 + k'_i) \quad \text{eq 1}$$

In the absence of secondary effects as part of the RPC distribution process(es), i.e., when no temperature- or ligand-mediated conformational effects, slow chemical equilibrium, or pH-dependent ionization events occur, then the resolution, $R_{s,i}$, between two adjacent peptides separated under equilibrium or near-equilibrium conditions can be expressed as

$$R_{s,i} = \frac{1}{4}\left(\frac{L}{h_{e,i}d_p}\right)^{1/2}(\alpha - 1)\left(\frac{1}{1 + k'_i}\right) \quad \text{eq 2}$$

or

$$R_{s,i} = \frac{1}{4}(N)^{1/2}(\alpha - 1)\left(\frac{1}{1 + k'_i}\right) \quad \text{eq 3}$$

where α is the selectivity, N is the number of theoretical plates, d_p is the particle diameter, L is the column length, and $h_{e,i}$ is the reduced plate height. Moreover, the column efficiency for a specific peptide can be defined in terms of the retention time and the peak width for that peptide. Hence, N and $h_{e,i}$ can be expressed as follows:

$$N = 16\left(\frac{t_e}{w}\right)^2 \quad \text{and} \quad h_{e,i} = \frac{L}{Nd_p} \quad \text{eq 4}$$

Equations 2–4 provide an important linkage between the three essential parameters of all RPC separations of peptides; they dictate the overall quality of resolution, namely the relative retention, expressed in terms of the unitless capacity factor, k'_i; the relative selectivity, α, expressed as the ratio of the capacity factors of two adjacently eluting peptides, k'_j/k'_i; and the extent of peak dispersion, expressed in terms of N_i or $h_{e,i}$. If larger values of $R_{s,i}$ per unit time (or alternatively greater peak capacities, *PC*s) are achieved, then higher system performances can be realized in analytical or preparative separations. When such criteria are met at the preparative/process level, the resultant high peak capacity performance in the RPC separation of a mixture of synthetic peptides equates with lower costs for each kilogram of product isolated at a predefined purity level per unit time, i.e. in higher overall throughput per system cost. Similarly, at the analytical level, this outcome results in lower costs of analysis per unit time. As evident from eqs 2–4, resolution in RPC separations of complex mixtures of peptides can thus be enhanced by optimizing the incremental $\Delta k'_i$ values, by increasing the α values, by decreasing the mean particle diameter, or by increasing N_i (and thus decreasing $h_{e,i}$). For a particular analytical separation of a complex mixture of synthetic peptides, each peptide in the mixture will have an optimal k'_i value for maximum resolution with a designated column, flow rate, and mobile-phase composition. Similar considerations apply in process scale, preparative RPC operated in the overload mode with multicomponent peptide mixtures, when again the resolution can be enhanced following optimization of the chromatographic selectivity and the zone bandwidth.

7.1.1.4 Relationships between Retention, the Equilibrium Association Constant, and Mobile-Phase Composition

When a peptide is dissolved in a solvent, according to solvophobic considerations a cavity is formed in the solvent with the molecular dimensions of the peptide. If water is the solvent, then a negative change in enthalpy will arise and there will be overall a net reduction in the number of hydrogen bonds in the system. Concomitantly, there will be an increase in the order of the water molecules near to the cavity containing the peptide, resulting in a negative change in entropy. When interaction of the peptide occurs with the immobilized hydrophobic ligands of the RPC sorbent, an increase in entropy will arise as the solvent molecules leave the more structured environment of the partially solvated peptide for the unstructured environment of the bulk mobile phase. Although the outcome is a positive change in enthalpy, this enthalpic change does not normally control the spontaneous association of the peptide with the nonpolar ligands, at least within the usual range of temperatures employed in RPC separations of peptides, since this energy term contribution is typically smaller than the change in the entropy of association, i.e. the process of peptide retention in RPC is entropically driven. Consequently, temperature provides an important opportunity to enhance resolution of peptides separated under RPC conditions.

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The various enthalpic and entropic contributions associated with solvation, hydrogen-bonding formation, and other nonbonding phenomena associated with changes in the organization and degrees of freedom of the peptide that occur in solution or following interaction at the liquid–solid interface of a chromatographic sorbent can, however, have large positive or negative values depending on the process. Overall, however, the result is a compensatory effect with the summated values leading to the observed values of $\Delta G_{\text{fold,i}}^{\ominus}$ and $\Delta G_{\text{assoc,i}}^{\ominus}$. The propensity of the peptide to adopt preferred secondary structures such as an α-helix or β-sheet, either in solution or at the nonpolar-solid–polar-liquid interface, is dictated by structural features of unique regions within the amino acid sequence of the peptide, the chemical nature of the surface of the RPC sorbent and by the characteristic features of the surrounding solvent. Thus, for a polypeptide initially with random coil structure, the participation of thermodynamic processes associated with the $\Delta G_{\text{fold,i}}^{\ominus}$ term sets the stage in the bulk solution for different sequence regions within the unfolded polypeptide to progressively nucleate, and, if the conditions are favorable, to adopt a preferred secondary structure on the surface of the RPC sorbent.

Moreover, the influences of these enthalpic and entropic processes associated with the interaction of peptides with immobilized nonpolar ligands vary considerably with temperature. It has been confirmed by available data that the hydrophobic effect for the binding of peptides to RPC sorbents changes from an entirely entropically driven process at temperatures below 25 °C to an entirely enthalpically driven process at ca. 110 °C, assuming that the change in the heat capacity term ($\Delta C_{p,\text{i}}^{\ominus}$) for the different states of the peptide remains constant at higher temperatures.[37,38] Retention of peptides in RPC can thus be considered to be primarily due to a solvent-mediated hydrophobic effect. From a conceptual perspective, the influence of mobile-phase composition can thus be accommodated in terms of the noncovalent association of nonpolar moieties present within the peptide with the nonpolar ligands, both of which undergo solvational changes as a consequence of the retention process.

Peptide retention in RPC can formally be described from a number of theoretical perspectives, either in terms of the more rigorous treatments based on the thermodynamics of interaction, in terms of preferential solvation models or from solely empirical considerations. Of the approaches that have gained currency, the solvophobic model,[39,40] the "close-contact" model[41–43] with the interactions mediated via van der Waals and other short range forces, and the stoichiometric solvent-displacement model[44] have attracted the most interest. In all of these models, the "near-equilibrium" binding assumption is applied to the interactions of the peptide with the nonpolar ligands. The peptide is also assumed to bind to the RPC sorbents in such a way that close contact is achieved between the hydrophobic moieties accessible at the surface of the peptide and the hydrophobic ligands of the sorbent. No distinction is made in these models between an adsorption event, with the retention dependent on surface area relationships, or a partition event, whereby the peptide is thought to penetrate into the immobilized ligands with the retention dependent on a partition coefficient relationship. Existing theories, of which the solvophobic approach is the most comprehensive, attempt to explain the changes in retention in terms of mobile-phase-mediated changes in the affinity of the binding, as measured by the logarithmic capacity factor, ln k'. In the adsorption model, ln k' is proportional to the molecular contact area, $\Delta A_{\text{contact}}$, whilst in the partition model, ln k' has a dependency on ln P_c, where P_c is the partition coefficient of the peptide.

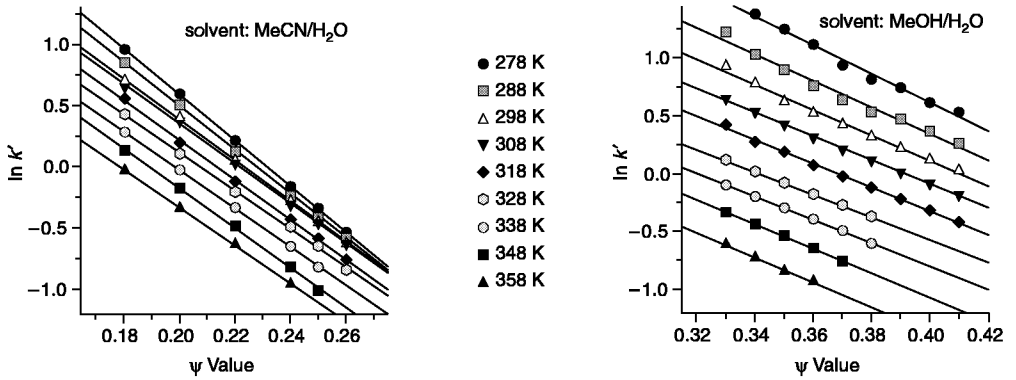
All of these models predict that the hydrophobic effect provides a significant driving force for the exclusion of even highly polar, charged peptides from an aqueous environment to the nonpolar environment of the RPC sorbent. According to the solvophobic model, in order to place a peptide into a mobile phase, a cavity of the same molecular dimensions must first be created. The energy required to create this cavity is related to the cohesive energy density or the surface tension of the mobile phase. Conceptually, each solvent-accessible unit

of the peptide, e.g. each amino acid side chain, requires the formation of an individual microcavity. Fusion of these microcavities into a global cavity corresponding in molecular dimensions to that of the peptide will effectively reduce the total surface area of the cavity, and hence the free energy of the system. In RPC of peptides, a driving force exists to sequester these nonpolar moieties away from a highly polar (water) solvent environment, resulting in an energetically favorable interaction between the peptide and the nonpolar ligands. The surface area dependency for the free energy of transfer of these amino acid side chains for different peptides can then be employed to describe the magnitude of the hydrophobic effect of these peptides at the microscopic level. Similarly, at the macroscopic level, the differences in effective hydrophobicity of several peptides can be assessed from the magnitude of the changes in surface tension of the solvent that is required to effect desorption of each peptide from the nonpolar surface.

The duality of these microscopic and macroscopic interpretations permits a rigorous framework to be elaborated for the role of hydrophobic processes, in terms of linear free energy concepts and the molecular parameters of peptides in association with nonpolar surfaces. Moreover, this duality enables selection criteria for the choice of the mobile-phase composition(s) to be established and conditions to be optimized to achieve the highest resolution of peptide mixtures with a defined type of RPC sorbent/column. Thus, to a first approximation, the retention of a peptide in RPC is expected to decrease exponentially with increasing organic solvent concentration. Plots of ln k' of a peptide against the volume fraction, ψ [or volume percentage (v/v)], of the organic solvent modifier in RPC are thus anticipated to be quasi-linear.[45–48] Illustrative of such dependencies are the plots[49] of ln k' versus the volume fraction, ψ, for the polypeptide, bombesin, separated on a butylsilica sorbent with low pH H_2O/MeCN and H_2O/MeOH mobile phases shown in Figure 4. In common with many other small peptides, the slopes of these ln k' versus ψ plots for the 14-mer bombesin can be approximated to linear dependencies. For this reason, use of the DryLab, Hephaestus–Pek-N-Eze, etc., software can be productively employed to interchange gradient and isocratic retention data, and vice versa, permitting optimization of the elution conditions for mixtures of peptides of related size, hydrophobicity, and/or sequence. What is also clearly apparent from the results shown in Figure 4 is that the solvent choice has a major influence on the slope dependencies of the ln k' versus ψ plots, whilst the relative retention and selectivity relationships are strongly influenced by temperature. By employing the corresponding correlation tables for the ψ and γ values of different organic solvents available from the scientific literature, water/organic solvent mobile phases of similar relative elutropicities and surface tension relationships can be generated. In this manner, comparable ln k' versus ψ or ln k' versus γ dependencies can be maintained for one or more peptide in a crude mixture, permitting similar selectivities and resolution to be achieved for the separation of peptide mixtures by RPC or the influences of different water/organic solvent mobile-phase compositions to be rationalized.

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Figure 4 Plots of the Logarithmic Capacity Factor, ln k', versus the Volume Fraction of the Organic Solvent, ψ, for the 14-mer Peptide, Bombesin[46][a,b]



[a] $MeCN/H_2O$ or $MeOH/H_2O$ mixtures containing 0.1% TFA and butylsilica at temperatures $T = 278$–358 K were used. The plots show the experimental data and the lines of best fit according to the quadratic form of the relationship given in eq 7.

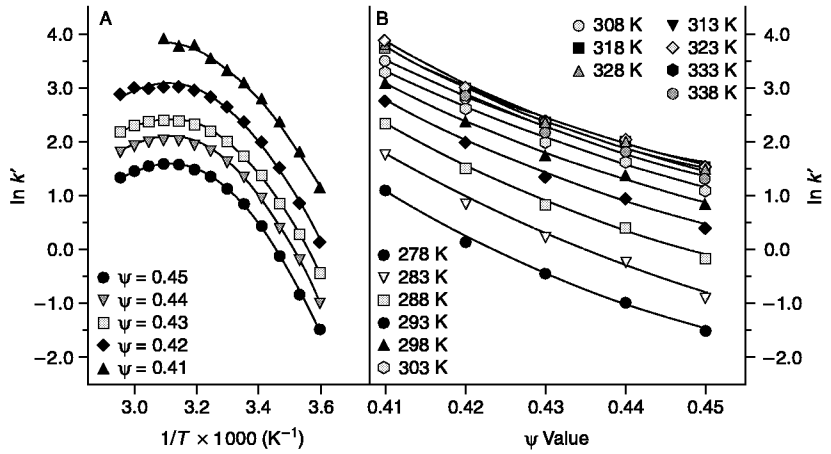
[b] Adapted from Hearn and Zhao,[46] with permission.

The two most frequently used binary solvent mobile-phase systems in the RP-HPLC of peptides are $MeCN/H_2O$ and $MeOH/H_2O$, although many other solvent combinations have been employed over the years. When mobile phases of comparable relative elutropicities are employed, i.e. under conditions of equivalent surface tension, advantage can be taken of the well-known differences[46,49–55] in solution chemistries and molecular organization of $MeCN/H_2O$ and $MeOH/H_2O$ mixtures to influence the selectivity. Differences in the molecular organization of $MeCN/H_2O$ and $MeOH/H_2O$ mobile phase of the same ψ value give rise to distinctive selectivity changes for peptides in RPC, which can be anticipated from an evaluation of the molar surface tension increment of the solvent. Related selectivity effects can similarly be achieved with peptides when other organic solvents, e.g. ethanol, propanol, propan-2-ol, tetrahydrofuran, or dioxane, are employed in combination with water in RPC.

Illustrative of these differences in elution behavior of peptides in RPC when solvents of different elutropicities are employed are the plots of ln k' versus the volume fraction, ψ, of the "displacing" solvent with $MeCN/H_2O$ mixtures containing 0.1% TFA for the HIV-1 gp160 leucine zipper [N^{558}-Q^{595}] polypeptide separated[24] on butylsilica, shown in Figure 5. As evident from Figure 5, the slopes of the ln k' versus ψ plots for this polypeptide in this RPC system are again negative, but curvilinear, whilst the van't Hoff plots of ln k' versus $1/T$ are also curvilinear, as the volume fraction, ψ, of the displacing solvent or the temperature, T, are increased. Moreover, analogous studies with other dipolar and hydrogen-bonding solvent systems have revealed that fundamental differences also exist for the corresponding plots, namely the slopes and intercepts of the ln k' versus ψ plots and the ln k' versus $1/T$ plots change according to the solvent characteristics. When the results shown in Figure 5 for this 36-mer polypeptide are compared, for example, to the data for bombesin (Figure 4), overall similarities in the retention behavior are evident, i.e. ln k' values decrease as ψ values increase, consistent with a dominant role of the hydrophobic effect in the retention process. What discriminates these two sets of results (and numerous similar comparisons between small peptides and polypeptides larger that 20 amino acid residues) is the shapes of the ln k' versus ψ dependencies, that is, larger polypeptides tend to exhibit more curvilinear dependencies reflecting the involvement of a number of secondary equilibrium processes, such as conformational, silanolphilic, and isodemic effects. When conformational effects contribute to the retention process, the corresponding plots of ln k' versus $1/T$ often pass through a maximum value, at which $\Delta G_{assoc,i}^{\ominus} = 0$, at a specific inflection temperature, T_H, with mobile phases of different organic solvent content; this is evident for the HIV-1 gp160 leucine zipper

[N^{558}-Q^{595}] polypeptide shown in Figure 5A. At one level, the availability of the ln k' versus ψ plots and the ln k' versus $1/T$ plots permit pragmatic considerations to be implemented in achieving enhanced resolution. On the other hand, the same information provides important biophysical insight into the fundamental molecular properties of the peptides in different solvent and interactive environments. These distinctive features of RPC provide enormous scope to manipulate the separation selectivity during a particular peptide isolation study. In all cases, the slope is roughly proportional to the molecular size/contact area established between the peptide and the immobilized ligands at the surface of the RPC sorbent. Due to the profound differences in the chromatographic behavior of small and large polypeptides in RPC, the ln k' values become very sensitive to small changes in the organic solvent concentration with a particular RPC sorbent. As a consequence with many polypeptides, the "window" of ψ values over which isocratic desorption can be achieved is very small, i.e. typically $\Delta\psi \leq 0.05$.

Figure 5 Plots of the Logarithmic Capacity Factor, ln k', versus the Reciprocal of the Absolute Temperature ($1/T$) (A) and ln k' versus the Volume Fraction of the Organic Solvent, ψ, (B) for the 37-mer Peptide, HIV-1 gp160 Leucine Zipper [N^{558}-Q^{595}] Polypeptide[24]a,b



[a] MeCN/H_2O mixtures of volume fractions from $\psi = 0.41$ to $\psi = 0.45$ containing 0.1% TFA and butylsilica at temperatures $T = 278$–338 K. The plots show the experimental data and the lines of best fit according to the quadratic form of the relationship given in eq 7 and the polynomial form of the van't Hoff relationship.
[b] Adapted from Jong, Boysen, and Hearn,[24] with permission.

Since other secondary interactive effects, e.g. electrostatic effects, can also contribute with a particular type of RPC sorbent to the retention of peptides as the mobile-phase composition is changed, the shape of these ln k' versus ψ plots then over the range $0.0 \leq \psi \leq 1.0$ takes on the characteristic U-shaped dependencies.[12,21] As a consequence of these dependencies of ln k' on ψ, for mixtures of peptides elution usually requires the use of gradient elution conditions, even when closely related or homologous peptides are being separated.

7.1.1.5 The Nature of the ln k' versus $[\Im]_m$ Dependencies and Their Impact on Peptide Separation in Reversed-Phase Chromatography

In common with all other interactive modes of chromatography, the relative retention of a peptide in RPC can be expressed in terms of the equilibrium association constant, K_{assoc}, the concentrations (in mol·L^{-1}) of the peptide present in the stationary and mobile phases, i.e. $[X]_s$ and $[X]_m$, and the volumes of the stationary and mobile phases, V_s and V_m respectively, within a column of length L and internal diameter d_c. Since K_{assoc} is the ratio $[X]_s/[X]_m$, whilst

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the phase ratio Φ of the chromatographic system is defined by the ratio V_s/V_m, the capacity factor, k', for a peptide P_i can be expressed in the following familiar form:

$$k'_i = \Phi[X_i]_s/[X_i]_m = \Phi K_{assoc,i} \quad \text{eq 5}$$

Extensive literature has developed related to the preferential interaction of different solvents with proteins or peptides in bulk solution.[56–58] Similar concepts can be incorporated into descriptions of the RPC behavior of peptides and employed as part of the selection criteria for optimizing the separation of a particular peptide mixture. As noted previously, the dependency of the equilibrium association constant, $K_{assoc,i}$, of a peptide and the concentration of the solvent required for desorption in RPC can be empirically described[44] in terms of nonmechanistic, stoichiometric solvent displacement or preferential hydration models, whereby the mass distribution of a peptide P_i with n nonpolar ligands, each of which is solvated with solvent molecules D_o is given by the following:

$$[P_{(+aD_o),i}]_m + n[L_{(+bD_o)}]_s \rightleftharpoons [(P_{(+cD_o),i}]_s \Longleftrightarrow [nL_{(+dD_o)}]_s + [(a-c)+(nb-nd)]D_o \quad \text{eq 6}$$

where $[P_{(+aD_o),i}]_m$ is the peptide in bulk mobile phase solvated with aD_o solvent molecules; $[P_{(+cD_o),i}]_s$ is the peptide adsorbed to the RPC sorbent surface with cD_o solvent molecules; $[L_{(+bD_o)}]_s$ is the nonpolar ligand in the free state solvated with bD_o solvent molecules; $[L_{(+dD_o)}]_s$ is the nonpolar ligand in the peptide-bound state solvated with dD_o solvent molecules; $[(a-c)+n(b-d)]$ represents the number of solvent molecules that are released from the peptide and the nonpolar ligands on adsorption of the peptide; $[(a-c)]$ represents the number of solvent molecules required to solvate the contact area of the peptide docked to the nonpolar ligand surface; and the subscripts m and s refer to the mobile and the stationary phase, respectively. When "near-equilibrium" chromatographic conditions prevail for the distribution process, i.e. when T, P, V, Φ, and the flow rate, F, are constant, and the adsorption process approximates a linear (Langmuirean) isotherm, then the dependency of $K_{assoc,i}$, and hence the logarithm of the corresponding capacity factor $\ln k'_i$, on the solvent concentration, $[\Im]_m$, in the mobile phase for RPC separations can be empirically written as follows:

$$\ln k'_i = \ln k'_{i,o} - S\psi_{solvent} + S'(\psi_{solvent})^2 - S''(\psi_{solvent})^3 + \cdots \quad \text{eq 7}$$

$$\ln k'_i = \ln k'_{i,o} + S_c \ln(1/[\Im]_m) + S'_c \ln(1/[\Im]_m)^2 + S''_c \ln(1/[\Im]_m)^3 + \cdots \quad \text{eq 8}$$

These representations also form the basis of the empirical stoichiometric model discussed below. For the RPC of peptides, the change in $K_{assoc,i}$ can be derived from knowledge of the change in the retention behavior (usually expressed as $\Delta \ln k'_i$ or τ^*) following an increase in the concentration, $[\Im]_m$, or volume fraction, $\psi_{solvent}$, of an organic solvent in the mobile phase. The coefficients S, S', S'', and S_c, S'_c, S''_c correspond to peptide-specific parameters associated with the reversed-phase interaction processes. In fact, $k'_{i,o}$ is the capacity factor of the peptide when $\psi_{solvent} \rightarrow 0$, whilst the coefficients S, S', S'' are the tangents of the plot of $\ln k'_i$ versus $\psi_{solvent}$, and the coefficients S_c, S'_c, S''_c are the tangents of the plot of $\ln k'_i$ versus $\ln(1/[\Im]_m)$ at defined values of $\psi_{solvent}$ or $[\Im]_m$ respectively. For $k'_{i,o}$ to have physical meaning, i.e. to become equivalent to the partition coefficient, P_c, of the peptide, the value of $[\Im]_m$ for an organic solvent will need to approach a value near to 1 M for RPC separations. This condition will be achieved for acetonitrile and methanol when $\psi_{MeCN} \approx 0.04$ and $\psi_{MeOH} \approx 0.03$, respectively. Such conditions agree with good experimental practice, as the minimum concentration of an organic solvent is selected for an RPC separation of peptides undertaken with a water/organic solvent mobile-phase system, where $[H_2O]$ is typically varied between 55 and 30 M.

When "ideal" retention dependencies prevail in the RPC separations of peptides, i.e. no secondary equilibrium effects occur, these empirical dependencies reduce to the familiar form given by equations 9 and 10.

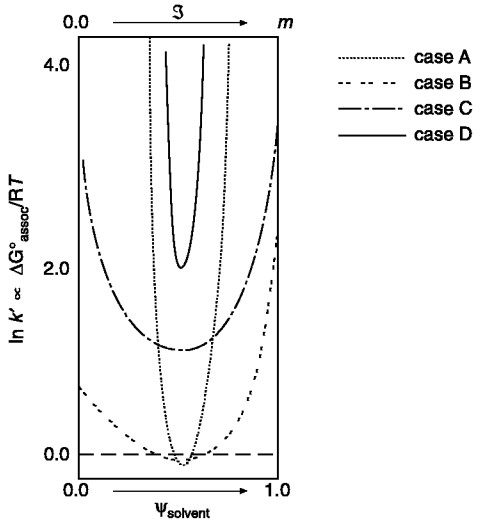
$$\ln k'_i = \ln k'_{i,o} - S\psi_{solvent} \quad \text{eq 9}$$

$$\ln k'_i = \ln k'_{i,o} + S_c \ln(1/[\Im]_m) \quad \text{eq 10}$$

Depending on the magnitude of the S and S_c parameters and the corresponding $k'_{i,o}$ values, a variety of retention time versus mobile-phase elutropic strength scenarios can thus be anticipated for the interaction of any peptide with immobilized nonpolar ligands of an RPC sorbent. Knowledge of these parameters greatly assists in the establishment of a suitable elution protocol for the isolation of a particular peptide from a crude synthetic mixture or biological extract. Figure 6 represents typical examples of four limiting cases where such ln k'_i versus $\psi_{solvent}$ or ln k'_i versus $[\Im]_m$ retention dependencies for peptides separated on *n*-alkylsilicas have been found (cf. Figures 4 and 5 and eqs 8–10). Over the complete operation range, the $\psi_{solvent}$ values will typically vary between 0.05 and 0.95 whilst $[\Im]_m$ will typically vary between 0.01 and 19.9 M. Case A represents a typical scenario for the behavior of larger, polar polypeptides in the presence of RPC sorbents.[17,24,31,46,59–63] For polypeptides that exhibit case A behavior, a steep dependence of ln k'_i occurs on the change in concentration of the displacing solvent $[\Im]_m$, leading to a minimal ln k'_i value at a specific value of $[\Im]_m$ or $\psi_{solvent}$, but is followed by an increase in ln k'_i as $[\Im]_m$ or $\psi_{solvent}$ is increased further. As the size of the polar polypeptide increases, and secondary equilibrium effects such as conformational transitions become more significant, the window over which the polypeptide can be eluted decreases, whilst the shape of the dependencies becomes more curvilinear (cf. Figure 5). The commonly observed situation with small peptides separated under high-performance RPC conditions, where relatively small free energy changes arise over the range of elution strength employed,[24,28,59,64,65] is typified in case B, where shallow ln k'_i versus $[\Im]_m$ or ln k'_i versus $\psi_{solvent}$ dependencies arise with small values of ln k'_i as $[\Im]_m$ and $\psi_{solvent} \rightarrow 0$. Over a limited range of ln k'_i values these dependencies can be approximated to linear relationships (cf. Figure 4). Shallow ln k'_i versus $[\Im]_m$ dependencies can also be observed under some conditions with very hydrophobic peptides, but in this case much larger values of ln k'_i are usually obtained (case C). Since the ln $k'_{i,o}$ values are large with case C peptides, small changes in a secondary mobile-phase component or condition, e.g. pH or buffer salt type, can lead to significant secondary retention effects and potentially to significant reduction in peptide yield. When such behavior is evident, a limiting chromatographic condition must be chosen based on the practical expediency of ensuring that the minimum in the plot of ln k'_i versus $[\Im]_m$ or ln k'_i versus $\psi_{solvent}$ corresponds to a k'_i value of ≤ 20. In situations associated with the purification of large nonpolar polypeptides of limited solubility with a propensity to aggregate, retention dependencies approaching case D are often observed, with a characteristic narrow desorption window that is often associated with significant secondary high-affinity sorption effects and low mass balance recoveries. Such polypeptides typically elute well after the completion of the gradient, i.e. in mobile phases of high organic solvent content, yet are frequently poorly soluble at these organic solvent concentrations.

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Figure 6 Four Limiting Cases of the Retention Dependencies between ln k'_i and the Concentration of the Organic Solvent in the Mobile Phase[a]



[a]Schematic illustration of four limiting cases of the retention dependencies between ln k'_i (proportional to the change in free energy $\Delta G_{assoc,i}{}^{\ominus}$) and the concentration of the organic solvent in the mobile phase, $[\Im]_m$, often expressed in terms of the mole fraction of the displacing species, ψ, over the concentration range $0 < [\Im]_m < m$, where m is typically less than 20 M, or the mole fraction range $0 < \psi < 1.0$ (cf. eqs 7–10). As the contact area and the retention factor, k'_i, of the polypeptide with the RPC sorbent increase, the slope of the plot of ln k'_i versus $[\Im]_m$ or ln k'_i versus ψ increases, resulting in a narrowing of the elution window over which the polypeptide can be desorbed. Cases A and B are typically observed in the RP-HPLC with polar peptides and polypeptides, whilst cases C and D are more representative of the behavior evident with more hydrophobic polypeptides. As evident from these plots, the shape of these log k' versus $[\Im]_m$ or ln k'_i versus ψ plots can take on the characteristic U-shaped dependencies typical of multimodal interactions, such as those found with many silica-based or polymer-based sorbents.

Changes in the magnitude of ln k'_i with solvent concentration in peptide–ligand interactions in RPC involving hydrophobic effects can also be related in solvophobic terms to the chemical potential, $\Delta\mu^\circ$, for the process, or to the surface tension, γ, of the mobile phase through the following expressions:

$$\ln k'_i = \ln \Phi + \frac{\Delta\mu^o}{RT} = \frac{Z_{mol}}{RT}\left[W_{sc} - W_{sa} + \frac{W_{aa}}{2} - \frac{W_{cc}}{2}\right] \quad \text{eq 11}$$

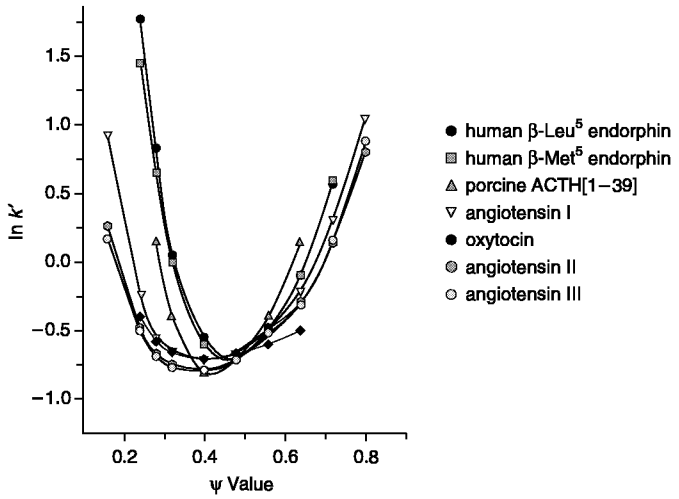
or

$$\ln k'_i = \ln k'_o - \gamma\frac{[N\Delta A_{hydr} + 4.836\,N^{1/3}(\kappa^e - 1)V^{2/3}]}{2.3033RT} \quad \text{eq 12}$$

where Z_{mol} is the number of nearest neighboring (solvent) molecules that surround the peptide in solution, R is the gas constant, W_{sc}, W_{sa}, W_{aa}, and W_{cc} are the reversible work required to achieve adsorption/partition of the peptide into a liquid film associated with the RPC sorbent, N is Avogadro's number, ΔA_{hydr} is the hydrophobic contact surface area, V is the partial molar volume, and κ^e is defined as the ratio of the energy required to form a cavity with surface area equal to the surface area of the peptide and the energy required to extend the planar surface of the liquid by the same area amount. Equation 11 assumes that the interaction can be described in terms of a partition model, whilst eq 12 accounts only for the solvent cavity formation in the mobile phase and does not include any stationary phase effects; that is, for a specific sorbent, the retention is assumed to be independent of the structural organization of the nonpolar ligands on the surface of the RPC sorbent. As

documented elsewhere,[25,29,66,67] with nonpolar *n*-alkylsilica sorbents of different ligand densities, this assumption is sustainable only when the ligand density is $\geq 2.7\,\mu mol \cdot m^{-2}$. When well-bonded, high coverage *n*-alkylsilicas are employed, solvophobic predictions have proved to be remarkably precise in anticipating the elution behavior of peptides that lack secondary structure, i.e. linear relationships have been observed between ln k'_i and γ. Illustrative examples of these dependencies are the plots of the experimental ln k'_i (proportional to $\Delta G_{assoc,i}^{\ominus}$) versus the volume fraction of the organic solvent, ψ, or alternatively ln k'_i versus the mobile-phase surface tension, γ, for several small peptide homologues analyzed by RPC procedures[68] using butylsilica sorbents (Figures 7 and 8). With larger polypeptides, such as the hormonal polypeptides illustrated in Figure 7, the curvilinear nature of the ln k'_i versus $\psi_{solvent}$ dependencies results in more complex ln k'_i versus γ behavior over a wide range of solvent conditions. For this reason, linearity in the ln k'_i versus γ plots for larger polypeptides is evident over only a narrow range of $[\Im]_m$ or $\psi_{solvent}$ values. However, with small peptides, particularly homologues and analogues, linearity in the ln k'_i versus γ plots is often observed (Figure 8) over a wider range of solvent compositions, consistent with the more shallow dependencies of ln k'_i on $[\Im]_m$ or $\psi_{solvent}$ in these cases. As discussed elsewhere in this section, variations in the ln k'_i dependencies on $\psi_{solvent}$, $\ln(1/[\Im]_m)$ or γ of the mobile phase, as the temperature, *T*, is systematically changed, provide fundamental information on the thermodynamic properties and physicochemical characteristics of the interaction of peptides with the nonpolar surfaces of RPC sorbents. Equally important is that the same data provide the basis to appropriately optimize a particular RPC separation system to achieve the best resolution of a particular peptide mixture.

Figure 7 Plots of ln k'_i versus ψ for Several Polypeptide Hormones[343,344][a,b]

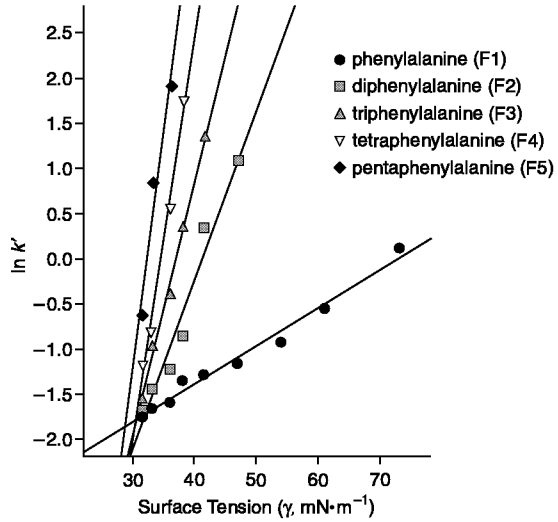


[a] Experimental conditions: column: μBondapak C_{18} (4.0 × 300 mm, d_p 10 μm); flow rate: 2.0 mL·min^{-1}; mobile phase: 15 mM TEA/4 mM H_2SO_4/15 mM H_3PO_4, volume fraction, ψ, of MeCN: 0.0–0.8. pH 2.2 was used for the 100% H_2O experiment.

[b] Adapted from Hearn and Grego,[343,344] with permission.

for references see p 622

Figure 8 Illustrative Examples of the Dependency of the Experimental ln k'_i on the Surface Tension for Several Polypeptides[68]a,b



[a] Plots of ln k'_i versus γ for several peptide homologues analyzed by RP-HPLC procedures using MeOH as the organic solvent modifier. These examples illustrate the dependency of the experimental ln k'_i (proportional to $\Delta G_{assoc,i}^{\ominus}$) on the surface tension, γ, of the polypeptides analyzed by RPC procedures using *n*-alkylsilica sorbents. Experimental conditions: column: μBondapak C_{18} (4.0×300 mm, $d_p = 10$ μm); flow rate: 2.0 mL·min^{-1}; mobile phase: H_2O/20 mM H_3PO_4, with the surface tension adjusted from 78 to 32 mN·m^{-1} by varying the volume fraction, ψ, of MeOH from 0.0 to 0.56. The pH value for the 100% H_2O experiment was pH 2.2. Also shown are the linear regression lines for each plot, where the corresponding correlation coefficients were obtained: $r_{F5}^2 = 0.982$; $r_{F4}^2 = 0.998$; $r_{F3}^2 = 0.991$; $r_{F2}^2 = 0.988$; $r_{F1}^2 = 0.967$.

[b] Adapted from Hearn and Grego,[68] with permission.

7.1.1.6 Basic Thermodynamic Considerations Relevant to the Optimization of Peptide Separation by Reversed-Phase Chromatography

In the RPC of peptides, the "near-equilibrium" condition has generally been assumed to prevail for binding and desorption with nonpolar ligands. As indicated above, the equilibrium distribution coefficient, $K_{assoc,i}$, for the interaction of a peptide, P_i, with the nonpolar ligands of a RPC sorbent can be related to the overall energy change in Gibbs free energy ($\Delta G_{assoc,i}^{\ominus}$) through the following relationship:

$$\Delta G_{assoc,i}^{\ominus} = -RT \log K_{assoc,i} \quad \text{eq 13}$$

As the Gibbs standard free energy for the peptide–nonpolar ligand interaction decreases, and $\Delta G_{assoc,i}^{\ominus}$ becomes increasingly negative, enhanced retention will occur. The relationship between the relative retention of a specific peptide in a RPC separation process carried out at constant pressure, *P*, and constant molar volume, *V*, can thus be expressed in terms of the following well-known, fundamental thermodynamic relationship:

$$\ln k'_i = \ln \Phi - \frac{\Delta G_{assoc,i}^{\ominus}}{RT} \quad \text{eq 14}$$

where *R* is the gas constant and *T* is the absolute temperature (in kelvin). The use of the capacity factor, k'_i, concept thus permits the separation of peptides in a particular RPC process, operated at a constant flow rate and temperature, to be defined both in empirical as well as thermodynamic terms. Temperature effects can thus be exploited for the optimization of a particular RPC separation of a peptide mixture. Changes in the thermodynamic para-

meters for the peptide–nonpolar ligand interaction can then be depicted in terms of criteria specified by the Gibbs–Helmholtz relationship:

$$\Delta G^{\ominus}_{\text{assoc,i}} = \Delta H^{\ominus}_{\text{assoc,i}} - T\Delta S^{\ominus}_{\text{assoc,i}} \quad \text{eq 15}$$

Hence, the logarithm of the capacity factor, $\ln k'_i$, can be related in terms of fundamental thermodynamic considerations to the temperature, T, through the expression

$$\ln k'_i = -\Delta H^{\ominus}_{\text{assoc,i}}/RT + \Delta S^{\ominus}_{\text{assoc,i}}/R + \ln \Phi \quad \text{eq 16}$$

where $\Delta H_{\text{assoc,i}}{}^{\ominus}$ and $\Delta S_{\text{assoc,i}}{}^{\ominus}$ are the changes in enthalpy and entropy associated with the interaction, and Φ is the phase ratio of the column packed with a defined sorbent and operated with a defined mobile-phase composition.

Understanding the consequences of the thermodynamic behavior manifested by peptides in their interactions with the different classes of RPC ligands is clearly important if high throughput, high-resolution separations are to be achieved. For example, this knowledge can be used to more precisely interpret the molecular processes associated with unexpectedly poor selectivity of peptide mixtures due, for example, to surface-induced conformational changes that one or more peptides in the mixture may undergo in the RPC environment.

Aberrant retention and peak broadening behavior of peptides in different nonpolar ligand systems will affect the isolation performance in RPC, and thus needs to be rigorously examined from theoretical as well as experimental perspectives. Moreover, the derived information can be employed as useful guidelines for the selection of alternative chromatographic conditions to remedy unsatisfactory resolution or recovery. Access to this information simultaneously yields insight into the extra-thermodynamic characteristics of the peptide–nonpolar ligand interaction. Such information links the retention behavior to the molecular properties, including the electrostatic, hydrogen bonding, and other characteristics of the peptides, when peptides are placed in the hydrophobic milieu of a RPC system. In addition, the retention behavior expressed as the total free energy change incorporates the contributions from all of the free energy terms, i.e.

$$\Delta G^{\ominus}_{\text{assoc,i}} = \Delta G^{\ominus}_{\text{es,i}} + \Delta G^{\ominus}_{\text{hydr,i}} + \Delta G^{\ominus}_{\text{vdw,i}} + \Delta G^{\ominus}_{\text{sol,i}} + \Delta G^{\ominus}_{\text{aggr,i}} + \Delta G^{\ominus}_{\text{red,i}} \quad \text{eq 17}$$

where $\Delta G_{\text{es,i}}{}^{\ominus}$, $\Delta G_{\text{hydr,i}}{}^{\ominus}$ (= $\Delta G_{\text{cav,i}}{}^{\ominus}$), $\Delta G_{\text{vdw,i}}{}^{\ominus}$, $\Delta G_{\text{sol,i}}{}^{\ominus}$, $\Delta G_{\text{aggr,i}}{}^{\ominus}$, and $\Delta G_{\text{red,i}}{}^{\ominus}$ are the free energy differences associated with electrostatic effects (subscript es,i), hydrophobic processes including the formation of a solvent cavity of dimensions of the peptide (subscript hydr,i or cav,i), van der Waals interactions (subscript vdw,i), solvational effects (subscript sol,i), self-association or heterogeneous association of the peptide in the presence and absence of solvent (subscript aggr,i), and the reduction in free energy due to non-ideal effects (subscript red,i), respectively. As a consequence, the fundamental relationship between $\Delta G_{\text{assoc,i}}{}^{\ominus}$, $\ln k'_i$, and the surface tension, γ, of the mobile phase thus provides the basis to understand the composite interplay between the dominant solvophobic processes and secondary size exclusion, coulombic, or other interactive processes that are manifested by all current HPLC packing materials, including *n*-alkyl silica RPC sorbents. Depending on the magnitudes of the individual Gibbs free energy contributions to the capacity factor, k'_i, the retention behavior for any peptide in the presence of nonpolar ligand systems can in this manner be formalized and the temperature or mobile-phase conditions selected for a predetermined set of RPC sorbent and sample loading requirements to achieve an appropriate isocratic or gradient retention time, based on the k'_i relationships given as eqs 13–17.

As is evident from the above discussion, the composition and primary structure of the peptide as well as the characteristics of the mobile and stationary phases all play critical roles in determining the magnitude of retention in RPC. Solvophobic theory[17,39,40,47,56,69–71] can be

for references see p 622

employed as a theoretical framework to describe quantitatively the effect of organic solvents on peptide retention in RPC. According to eqs 11–17, the change in the interfacial free energy upon adsorption of a peptide to solvated *n*-alkyl ligands immobilized onto the surface of a support material can be related to the change in the solvational state of the peptide. This process can be treated in terms of two distinct processes, the first of which involves the formation of a cavity with a size and shape that can accommodate the peptide, and the second involves the introduction of the peptide into this cavity and its subsequent interaction with the surrounding solvent environment. According to eq 11, the retention of a peptide, P_i, in a RPC system can be related to the surface tension of the mobile phase, γ, the magnitude of the cavity factor, κ^e_i, and to the hydrophobic contact area, $\Delta A_{hydr,i}$, which is established between the peptide and the stationary phase surface. To a first approximation, $\Delta A_{hydr,i}$ is proportional to the total surface area of the peptide, $A_{total,i}$, and thus can be related to the total molecular volume, $V_{total,i}$, as calculated from the individual amino acid residues according to the procedures of Privalov et al.[71] In turn, the total surface area, $A_{total,i}$, can be calculated using the individual amino acid surface area values as determined by Rose et al.[72] Similarly, the total van der Waals surface area, $A_{vdw,i}$, and the solvent accessible surface area, $A_{asa,i}$, can be derived using the Connolly algorithm based on the partitioning of the total surface area into the area accessible to a 1.4 Å solvent probe,[73] whilst the hydrophobic surface area, $A_{hydr,i}$, can be calculated again using the Connolly algorithm, including only those amino acid residues with a hydrophobicity coefficient value >2 as determined, for example, from the Hearn hydrophobic coefficient sets.[32,34,35] Finally, the solvent-accessible hydrophobic surface area, $A_{solacc,i}$, can be derived, again with the Connolly algorithm, from the total hydrophobic surface area accessible to solvent probes of different size, i.e. from 1.4 Å (H_2O) to say 3.6 Å (MeCN). When hydrophobic interactions prevail in the RPC of peptides, the change in free energy for cavity formation for a peptide in the mobile phase is related to the surface tension, γ, and surface area of the molecule, $A_{total,i}$, according to the following:

$$\Delta G^{\ominus}_{cav,mp} = [NA_{total,i} + 4.8N^{1/3}(\kappa^e_i - 1)V^{2/3}]\gamma \quad \text{eq 18}$$

The surface tension of an aqueous solution containing an organic solvent (or salt) is a function of the molal solvent (or salt) concentration, m, and can be given by

$$\gamma = \gamma^o + \sigma m \quad \text{eq 19}$$

where γ° is the surface tension of pure water (72 dyn·cm^{-1}) and σ is molal surface tension increment, a characteristic property of each solvent (or salt), and m is the molality of the solvent (or salt). Information on σ values for various water-soluble organic solvents can be extracted or derived from the scientific literature as tables of surface tension values for solvents (or salts) at different mole fraction values. Assuming that the magnitude of the solvent concentration has no effect on κ^e_i, $A_{total,i}$, or V, then the free energy change associated with cavity formation ($\Delta G^{\ominus}_{cav,mp,i}$) in a mobile phase of defined composition can be evaluated as follows:

$$\Delta G^{\ominus}_{cav,mp,i} = [NA_{total,i} + 4.8N^{1/3}(\kappa^e_i - 1)V^{2/3}]\gamma^o + [NA_{total,i} + 4.8N^{1/3}(\kappa^e_i - 1)V^{2/3}]\sigma m \quad \text{eq 20}$$

which can be expressed in a simplified form as

$$\Delta G^{\ominus}_{cav,mp,i} = \Delta A_{contact}\sigma m + \text{constant} \quad \text{eq 21}$$

where $\Delta A_{contact}$ (proportional to $A_{hydr,i}$) is the difference in surface area of the peptide exposed to the mobile phase in the bound and unbound states, i.e. equivalent to the molecular contact area upon binding. Thus, the greater the value of γ that is required to affect elution

of a peptide, or the greater the value of $\Delta A_{contact}$, the greater will be the retention. When the respective free energy change favors peptide–peptide rather than peptide–ligand interaction, then aggregates will form in solution due to isodesmic self-association. Under these conditions, the solvophobic model predicts that a "precipitation" phenomenon analogous to the "salting-out" effect with proteins[47,56–58] will occur with peptides, whereby the distribution occurs between a solvated state of the peptide and a solid-phase state, rather than between a solvated monomer and a chromatographically bound state. Obviously, isodesmic self-association can also occur on the stationary phase surface, and these effects[74] will lead to bizarre retention processes with synthetic peptides. In such cases, the elution behavior will be highly condition dependent, as found[24] for the 26-mer peptide derived from the N-terminal leader sequence of the sperm tail protein, tpx-1.

In RPC, immediately prior to the interaction of a peptide(s) with the nonpolar ligands, both the peptide and the nonpolar ligands will have structured water molecules associated with their nonpolar surfaces. When these two surfaces are brought into close contact, solvent exclusion events occur, releasing structured water molecules; this, in turn, increases the entropy of the bulk mobile phase. This process of solvent exclusion can be viewed as the driving force for adsorption. Since an increase in van der Waals forces is expected, whilst the area of the peptide in contact with the nonpolar ligands will be relatively small, $\Delta H_{assoc,i}^{\ominus}$ may also be small (and either negative or positive), whilst $\Delta S_{assoc,i}^{\ominus}$ may be large. In this case, $\Delta G_{assoc,i}^{\ominus}$ is controlled by a large positive entropy change. Such entropy-driven processes are characterized by greater retention at higher temperatures, as expected from eq 16. Indeed, in contrast to affinity or ion-exchange chromatography, higher column temperatures in well-controlled RPC separations of peptides generally lead to longer retention. One of the advantages of RPC, then, is that column operation at subambient temperatures will tend to result in a weaker interaction between the peptide and the stationary phase, relative to similar experiments carried out at ambient or more elevated temperature. Use of lower temperatures will often translate into a stabilizing influence for peptides with a propensity to form preferred secondary structures. The ability to operate at subambient temperature provides an opportunity to separate and isolate conformational forms of some classes of peptides, including the *cis/trans*-isomers of proline-containing peptides.[75,76]

When the structural and conformational properties of the peptide, the surrounding bulk solvent, and the interactive surface of the RPC sorbent are invariant with regard to the temperature, T, then the contributions from the corresponding changes in enthalpy, $\Delta H_{assoc,i}^{\ominus}$, and entropy, $\Delta S_{assoc,i}^{\ominus}$, of the system to the overall Gibbs free energy change, $\Delta G_{assoc,i}^{\ominus}$, associated with the peptide–sorbent interaction will also be independent of temperature. The dependency of log k'_i on $1/T$ is then anticipated to take the form of the well-known linear van't Hoff plot when such conditions prevail, as observed experimentally with small peptides.[12,24,31,35,37,46,49,77–79] For many larger polypeptides separated under RPC conditions, significant divergencies from this ideal behavior have been found.[17,24,31,37,46,80]

For RPC separation systems where the phase ratio, Φ, and the properties of the peptide and sorbent are invariant with temperature, the corresponding values of the enthalpy change, $\Delta H_{assoc,i}^{\ominus}$, and the entropy change, $\Delta S_{assoc,i}^{\ominus}$, associated with the interaction of P_i with the nonpolar ligands can be derived by linear regression analysis of the ln k'_i versus $1/T$ plot from the slope and Y-axis intercept values, respectively. In fact, for synthetic peptides separated by RPC, three situations have been documented[17,24,31] for their interaction with immobilized hydrocarbonaceous ligands: the change in heat capacity, $\Delta C_{p,i}^{\ominus}$, of the system is (1) zero and remains invariant with regard to T (the so-called "isothermic" scenario); (2) not zero but is linearly dependent on T (the so-called "homothermic" scenario); or (3) not zero and shows a nonlinear dependency on T (the so-called "heterothermic" scenario). In the bulk state, the heat capacity, $\Delta C_{p,i}^{\ominus}$, of a peptide is usually defined as the quantity of heat necessary to raise the temperature of a unit mass of the peptide by 1 K. In the case of RPC, the change in heat capacity, $\Delta C_{p,i}^{\ominus}$, for the interaction of a peptide with a nonpolar ligands

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system can be expressed as a function of T in terms of the enthalpy and entropy changes according to the Kirchoff relationships:[81,82]

$$\Delta H^{\ominus}_{\text{assoc,i}} = \int \Delta C^{\ominus}_{pi} \mathrm{d}T \quad \text{eq 22}$$

$$\Delta S^{\ominus}_{\text{assoc,i}} = \int \frac{\Delta C_{pi}}{T} \mathrm{d}T \quad \text{eq 23}$$

Evaluations of the interaction of a peptide P_i with immobilized hydrocarbonaceous ligands of a RPC sorbent in terms of eqs 13–21, however, do not provide complete descriptions of the binding process, because the condition(s) under which the thermal energy of the system is increased has neither been specified, nor will all peptides within the same mixture necessarily exhibit exactly the same relative changes in the magnitudes in their heat capacity dependencies, although the trends for analogous and homologous peptides may be similar. However, such approaches provide a rigorous method to select optimal temperature conditions with a particular isocratic or gradient mobile-phase composition for the resolution of mixtures of synthetic peptides.

A further effect arises in the RPC of peptides when the temperature of the system is changed, namely the viscosity, η, of the mobile phase will also change. As a result, the pressure drop along the column (ΔP) will change according to the following relationship:

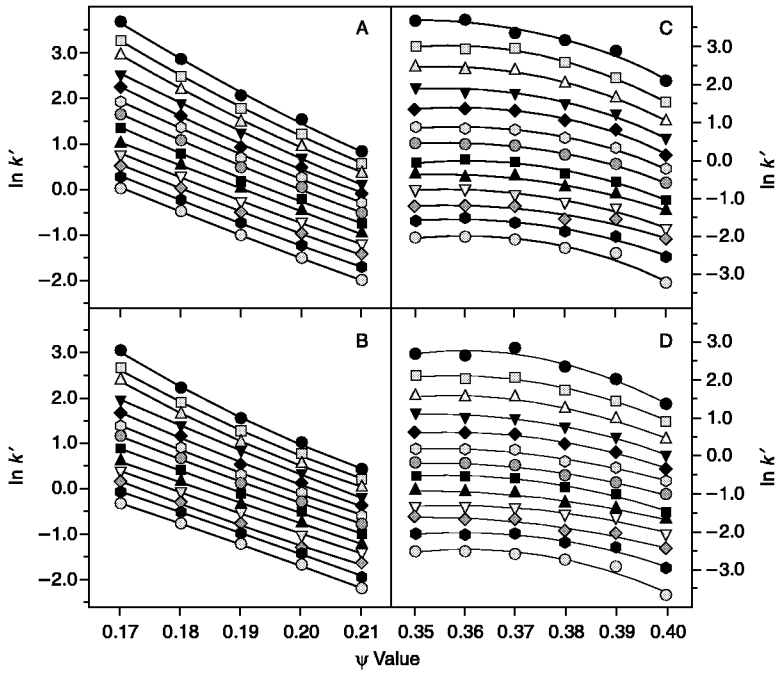
$$\Delta P = \frac{\nu \eta L}{\varepsilon_p d_p^2} \quad \text{eq 24}$$

where ν is the linear flow velocity of the eluent, L is the length of the column, ε_p is the specific permeability of the sorbent, and d_p is the average particle diameter. Over the temperature range employed in most laboratories for the RPC separation of peptides, i.e. from 5–65 °C, the ΔP change for well-packed HPLC columns should typically not exceed more than 15–20% of the original value. Consequently, higher ν values can be achieved without sacrificing peak efficiency of the separated peptides or alternatively smaller $h_{e,i}$ values (larger N values, cf. eq 4) for a selected ν at higher temperatures. Various studies have shown that isocratic conditions at constant flow rate are the preferred experimental option[24,31,80,83] for the determination of either the relevant thermodynamic parameters or optimization of the temperature conditions to achieve enhanced resolution for peptides in RPC systems. Application of eqs 13–21 thus provides a useful approximation of the physicochemical features for the nonpolar ligand interaction with peptides in RPC, enabling evaluation on the one hand of the relationship between k'_i, T, and the associated thermodynamic parameters $\Delta H_{\text{assoc,i}}{}^{\ominus}$, $\Delta S_{\text{assoc,i}}{}^{\ominus}$, and $\Delta C_{p,i}{}^{\ominus}$ for a specific peptide to be determined, and on the other hand permitting experimental conditions to be selected that permit optimal resolution of closely eluting peptides for a defined flow rate and mobile-phase composition.

With peptides that exhibit in RPC separations nonlinear plots of ln k'_i versus $1/T$ (cf. Figure 5 or 10) the value of $\Delta H_{\text{assoc,i}}{}^{\ominus}$ may approach zero, i.e. $\Delta H_{\text{assoc,i}}{}^{\ominus} \rightarrow 0$. In these circumstances, ln k'_i may reach a maximal value at a temperature value (corresponding to an inflection temperature, T_H, where $\Delta C_{p,i}{}^{\ominus}$ becomes constant for the interactive process). When such processes prevail, the RPC retention of closely eluting peptides may show selectivity reversals, or substantial changes in peak shape over relatively narrow ranges of mobile-phase composition or temperature. Exploitation of such effects can be very beneficial with difficult separation tasks, such as the separation of diastereomeric peptides that are present in mixtures of all-L-α-peptides and their *retro-inverso* all-D-α-analogues. Illustrative of this approach are the results[31] for the all-L-α-polypeptide H_2N-DDALYDDKNW-DRAPQRCYYQ-CO_2H (**1**), *retro-inverso* D-α-isomer H_2N-QYYCRQPARDWNKDDY-

LADD-CO_2H (**2**) and various synthetic analogues with an octylsilica sorbent as plots of ln k'_i versus the volume fraction, ψ, of the organic solvent over the range of $\psi = 0.17$ to 0.21 with H_2O/MeCN mixtures and from $\psi = 0.35$ to 0.40 with H_2O/MeOH mixtures, for the temperature range of 278 to 338 K in 5 K increments shown in Figure 9. For these polypeptides, the trends expected for the hydrophobic effect dominating the interaction mechanism is clearly evident, with the ln k'_i values of polypeptide **1** and **2** decreasing as the ψ value increased, but increasing as T was increased. Again, solvent-dependent differences in the ln k'_i versus ψ plots are evident, whilst the curvilinear nature of the dependencies is indicative of heterothermic conformational processes (see: ref[31]).

Figure 9 Plot of ln k'_i versus the Volume Fraction, ψ, for the Polypeptide Pair, the all-L-α-H_2N-DDALYDDKN-WDRAPQRCYYQ-CO_2H (**1**) and the all-D-α-H_2N-QYYCRQPARDWNKDDYLADD-CO_2H (**2**)[31]a,b



[a] Conditions: MeCN/H_2O (A and B) and MeOH/H_2O (C and D) mixtures at different temperatures. In each panel the data shown as the top ln k'_i versus ψ plot (shown as the -λ-λ- values) corresponds to $T = 278$ K and the bottom ln k'_i versus ψ plot (shown as -λ-λ- values) corresponds to $T = 338$ K, respectively.
[b] Adapted from Boysen, Wang, Keah, and Hearn,[31] with permission.

According to eqs 7–10, the S values of peptides separated with a particular RPC system can be derived from the experimental plots of ln k'_i versus ψ at specified T values by regression analysis. Moreover, the S value of a peptide in the presence of a RPC sorbent can be related[17,31,32,34] via implicit linear free energy dependencies on extra-thermodynamic parameters, such as the accessible molecular surface area, ΔA_{mol}, through the expression

$$S = a'\Delta A_{hyd} + b'\kappa^e + c' \quad \text{eq 25}$$

where ΔA_{hyd} is the hydrophobic contact area (proportional to the accessible molecular surface area, ΔA_{asa}), κ^e is a solvophobic cavity factor, and a', b', and c' are constants related to the molecular properties of the polypeptide. Since ΔA_{asa} can be predicted from knowledge of the peptide sequence and molecular mass, and κ^e can be correlated with the surface area/volume relationships of a peptide, or when available with the partition coefficient, P_c, values

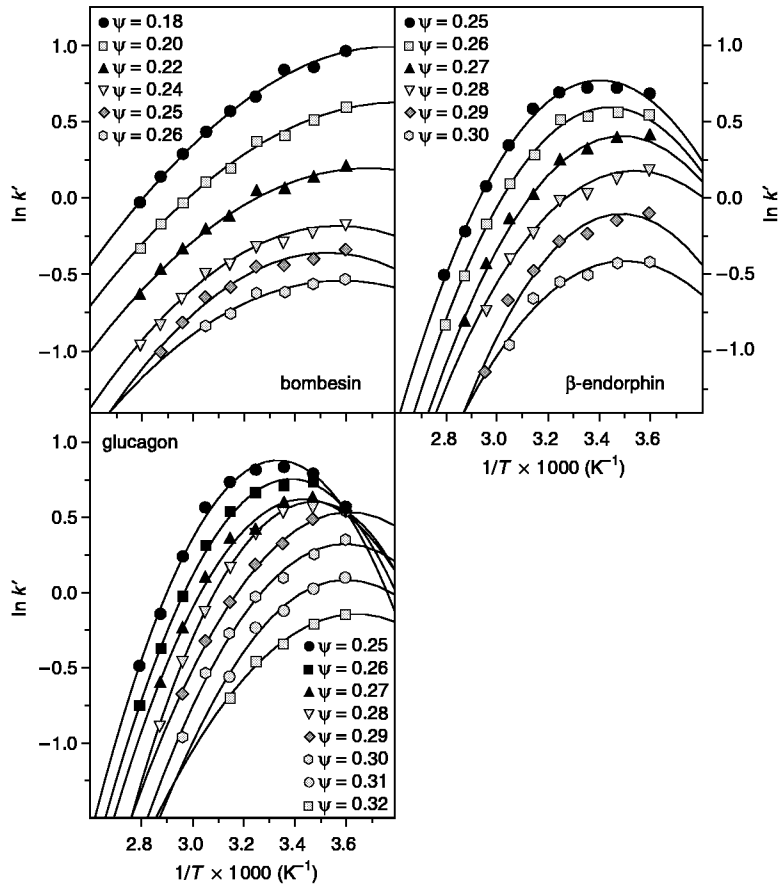
for references see p 622

of the peptide, the magnitude of the S term can be calculated for any peptide. The S value so derived can then be imported into various optimization algorithms such as DryLab or Hephaestus–Pek-N-Eze to enable isocratic or gradient elution conditions to be simulated and sets of alternative elution conditions selected and compared, initially computationally, and then by experimentation. Arising from such approaches, several features relevant to the final selection of elution conditions for the preparative isolation of peptides by RPC techniques becomes evident. Firstly, these methods can be used to rapidly reveal resolution differences (or difficulties) that may arise from a change in organic solvent, e.g. when an acetonitrile solvent system is changed for a MeOH solvent system. Secondly, these methods can be used to anticipate or rapidly reveal resolution difficulties that may arise when one type of RPC sorbent is changed for another. Thirdly, these methods can be used to rapidly evaluate or anticipate resolution differences (or difficulties) that may arise when the temperature is raised or, for that matter, reduced to subambient values.

When nonlinear (parabolic) plots of ln k'_i versus $1/T$ are obtained in the RPC of polypeptides, as found, for example, with the hormonal polypeptides, bombesin, β-endorphin, and glucagon shown in Figure 10, it can be concluded that heterothermic processes dominate the RPC retention process, in accordance with Kirchoffs relationships, i.e. the change in the heat capacity, $\Delta C_{p,i}^{\ominus}$, for the interaction $\neq 0$, and is dependent on the temperature. Exploitation of such heterothermic processes has proved useful in a variety of investigations with peptides that are "difficult" to resolve, e.g. very hydrophobic peptides or peptides with significant β-sheet content. For example, in Figure 11A and 11B for an MeCN/H_2O system and Figure 11C and 11D for a MeOH/H_2O system the corresponding ln k'_i versus $1/T$ experimental data are illustrated. The all-L-α-polypeptide H_2N-DDALYDDKNWDRAPQRCYYQ-CO_2H in Figure 11A and 11C and its *retro-inverso* D-α-isomer H_2N-QYYCRQPARDWNKDDYLADD-CO_2H in Figure 11B and 11D are used as examples. Although the ln k'_i versus $1/T$ and the corresponding ln k'_i versus ψ dependencies for these polypeptides are very similar, as would be expected for polypeptides that have the same composition and essentially the same topology, subtle but more significant differences in retention behavior are, however, evident with the acetonitrile-based mobile-phase systems rather than with the methanol-based mobile-phase systems, and these differences are sufficient to permit the resolution of these polypeptides under optimized gradient RPC conditions with low pH, H_2O/MeCN mobile phases.

In RPC separations of peptides, electrostatic effects can be used to modulate selectivity at several levels, since most peptides will assume a "fully protonated" state under the low pH conditions. The inclusion of an ion-pairing reagent as an additive to the mobile phase is then required to achieve a "near-neutral" state for the peptide. If electrostatic/polar interactions can be excluded from the retention process with peptides, the free energy changes associated with the cavity formation and the Lifshitz–London–van der Waals interaction represents the dominant basis of the RPC separations. Solvents that have large negative σ values will give rise to significant decreases in the surface tension of the mobile phase and hence cause more significant reduction in retention. Knowledge of the effect of the molal surface tension increment of the organic solvent, thus, provides a quantitative measure of the effect on retention, and thus can be used to optimize the separation of a particular peptide from a complex synthetic mixture.

Figure 10 van't Hoff Plots for Bombesin, β-Endorphin, and Glucagon at Different ψ Values[46,345]a,b



[a] van't Hoff plots for bombesin over the range $\psi = 0.18$ to 0.26; β-endorphin over the range $\psi = 0.25$ to 0.30; glucagon over the range $\psi = 0.25$ to $\psi = 0.32$, using MeCN/H_2O mixtures containing 0.1% TFA, with a butylsilica sorbent at temperatures $T = 278$–358 K. The plots, characteristic of heterothermic binding to RPC sorbents, show the experimental data and the lines of best fit according to the polynomial form of the van't Hoff relationship.

[b] Adapted from Hearn, Zhao, Purcell, Zhao, and Aguilar,[46,345] with permission.

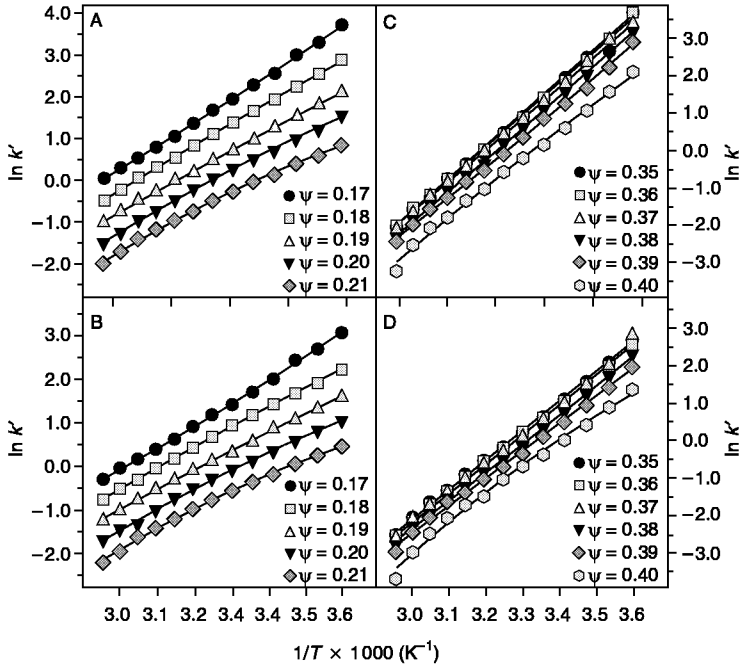
As apparent from eqs 11, 12, and 18, the free energy of cavity formation is proportional to the surface area of the peptide. Therefore, retention differences, i.e. selectivities, between peptides will reflect the incremental changes in their surface areas as a consequence of adsorption to the stationary phase ligands. This relationship permits hydrophobicity coefficients or side-chain fragmental coefficients to be used with good effect to predict the retention behavior of synthetic peptides with a defined RPC system. Illustrative of this approach has been, for example, the studies with the 115 synthetic peptides related to myohemerythrin[35] and peptide epitopes.[84] In addition, in the presence of an organic solvent in RPC, the retention relationship for a specific peptide takes the simplified linear form:[37,47,68]

$$\ln k'_i = \ln k'_o - \lambda m \quad \text{eq 26}$$

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where λ is the slope of the plot of ln k'_i versus solvent molality, m, and ln k'_o corresponds to the peptide specific chromatographic parameter when $m \rightarrow 0$. Moreover, experience has shown that λ strongly correlates to σ with the slopes of the plots for a specific peptide tending to closely follow the relative elutropicity ranking of solvents of the same family type (i.e., alcohols, etc.). For many larger polypeptides in RPC, the slope/shape of these plots are not coincidental for different classes of solvents (i.e., whether they are hydrogen-bond donors, dipolar, etc.) when the retention data are normalized (cf. comments above on the influence of surface tension differences between MeCN and MeOH). Moreover, for the RPC of larger polypeptides, it can be argued that α-helix-inducing or secondary-structure-stabilizing solvents should be used in preference to other solvent combinations. These solvent conditions will tend to favor induction of a more stable secondary structure for these molecules under a specific RPC condition.

Figure 11 van't Hoff Plots for the Synthetic L-α-Polypeptide H_2N-DDALYDDKNWDRAPQRCYYQ-CO_2H (**1**) (A and C) and Its *retro-inverso* D-α-Isomer H_2N-QYYCRQPARDWNKDDYLADD-CO_2H (**2**) (B and D) for MeCN/H_2O (A and B) and the MeOH/H_2O (B and D) Mobile-Phase Systems Containing 0.09% TFA at Different ψ Values[31][a,b]



[a] Solvents were filtered and degassed by sparging with helium. Measurements of the ln k'_i versus $1/T$ dependencies were performed with Zorbax 300SB-C8 (Rockland Technologies, Inc., USA); 150 × 4.6 mm i.d.; MeCN or MeOH contents in H_2O within the range 17–21% or 35–40%, respectively, increased in increments of 1%; flow rate: 1.0 mL·min^{-1}; temperatures: 278, 283, 288, 293, 298, 303, 308, 313, 318, 323, 328, 333, and 338 K. Solutions of the 20-mer polypeptides **1** and **2** were prepared by dissolving the polypeptides at a concentration of 0.4 mg·mL^{-1} in 0.09% TFA. Injection sizes varied between 2–8 μg. Under these mass loading conditions, the polypeptides **1** and **2** exist in solution as monomeric species with partial α-helical structure in the C-terminal region as assessed by circular dichroism and 2D ^{1}H NMR measurements.[346] All data points were derived from at least duplicate measurements. The column dead volume, V_o, and phase ratio, Φ, of the system were determined with the non-interactive solute $NaNO_3$.

[b] Adapted from Boysen, Wang, Keah, and Hearn,[31] with permission.

Just as different organic modifiers can be employed to generate different selectivities in the RPC separation of peptides, similar effects can be independently achieved through the use of different RPC sorbents. As the hydrophobicity of the stationary phase increases, the λ value

for a particular peptide often also increases for a defined mobile-phase composition. Such behavior would be in accord with the concept that a larger nonpolar contact area will be generated when a peptide interacts with a more nonpolar ligand system. Several studies have also shown that for topologically similar peptides, with (nearly) identical types of hydrophobic patches accessible on their surfaces, the λ value or S_c value dependencies remain essentially constant for RPC sorbents of similar ligand densities and *n*-alkyl chain length.[24,31]

7.1.1.7 Consequences of the Preferential Interaction Model for Peptide Isolation by Reversed-Phase Chromatography

In the preferential interaction model, the equilibrium association constant, K_{assoc}, for a peptide docked to a RPC sorbent can be related via eq 6 to the changes in the preferential interaction of the solvent and water with the peptide P_i, the ligands L, and the peptide–ligand complex P_iL as a function of solvent concentration, $[\Im]_m$. The effect of preferential hydration/solvation of a peptide on k' can be represented in terms of the following expression:[85]

$$\frac{d(\ln k'_i)}{d(\ln m_s)} = 3\left[\Delta\nu_s - \frac{m_s}{m_w}\right] = Z^* \quad \text{eq 27}$$

where m_s is the molal concentration of solvent, i.e. $[\Im]_m$, and m_w is the molal concentration of water (with a maximum value of 55.5 mol · L^{-1}), $\Delta\nu_w$ and $\Delta\nu_s$ refer to the number of moles of water and solvent, respectively, displaced upon adsorption per mole of peptide whilst Z^* corresponds to the slope of the plot of ln k' versus ln m_s. The ratio m_s/m_w will thus vary with the solvent concentration and is a weighting factor for the water concentration term. Since Z^* provides a measure of the changes in both solvent and water bound to a peptide as a consequence of the chromatographic distribution process, it can be related through solvophobic considerations to λ through the relationships

$$\lambda = \frac{d(\ln k'_i)}{dm_s} = \frac{Z^*}{m_s} \quad \text{eq 28}$$

The preferential interaction theory and the solvophobic theory thus have a common linkage through the slopes of these ln k'_i versus $\ln(1/[\Im]_m)$ plots. From these dependencies, and the relationship of ln k' on the free energy change, $\Delta G_{assoc,i}^{\ominus}$, a unified approach can be employed, linking the thermodynamic and extra-thermodynamic relationships for the interaction of peptides with any type of hydrocarbonaceous ligand in RPC. As a consequence, knowledge of the molecular properties of the different synthetic peptides in a mixture, such as molecular weight, size, predicted surface area, etc., can be employed as described in Section 7.1.1.1 to aid the selection of the surface tension or mole fraction characteristics of the mobile phase or the temperature at which the RPC system can be operated for optimal resolution. In many cases, with synthetic peptides prepared by solid-phase Fmoc or tBoc chemistries with well-characterized side-chain deprotection protocols, information on these molecular parameters of the peptides is readily available from the knowledge of the sequence and from molecular mass analysis by ESI-MS. Consequently, the selection of the mobile-phase characteristics based *inter alia* on the preferential solvation model can be achieved using various algorithmic approaches, i.e. DryLab, Hephaestus–Pek-N-Eze, etc., with minimal experimental effort.

for references see p 622

7.1.1.8 Use of the Stoichiometric Solvent Displacement Model in Peptide Isolation by Reversed-Phase Chromatography

According to this theoretical treatment, the slope of the plots of ln k'_i versus the solvent concentration, $[\Im]_m$, can be employed to derive the contact area associated with the peptide–nonpolar ligand interaction. The retention and elution of a peptide in RPC can then be treated as a series of microequilibriums between the different components of the system, as represented by eq 6. The stoichiometric solvent displacement model addresses a set of considerations analogous to that of the preferential interaction model, but from a different empirical perspective. Thus, the affinity of the organic solvent for the free peptide P_i in the mobile phase can be represented as follows:

$$P_{i,m} + aD_o \rightleftharpoons P_{(+aD_o)_{i,m}} \quad \text{eq 29}$$

whilst the affinity of the organic solvent for the free nonpolar ligand L in the RPC system can be represented by:

$$L + bD_o \rightleftharpoons L_{(+bD_o)} \quad \text{eq 30}$$

If n solvated nonpolar ligands simultaneously interact with the solvated peptide, then the binary complex that forms can be represented by the equilibrium shown in eq 6. When the concentration and thus the surface tension of the organic solvent reaches a critical value, then the partially solvated peptide $P_{(+cDo)i,s}$ bound to the RPC stationary phase will desorb and become re-solvated to generate the species $P_{(+aDo)i,m}$. Thus, overall desorption can be represented as:

$$[P_{(+cD_o)i,s} \Leftrightarrow nL_{(+dD_o)}] + [(a-c) + n(b-d)]D_o \rightleftharpoons P_{(+aD_o)i,m} + nL_{(+bD_o)} \quad \text{eq 31}$$

According to the stoichiometric displacement model, the equilibrium constant for peptide adsorption with the solvated nonpolar ligands can be expressed as follows:

$$K_{adsorb} = \frac{[P_{(+cD_o)i,s}][D_o]^{[(a-c)+n(b-d)]}}{[P_{(+aD_o)i,m}][L_{(+bD_o)}]^n} \quad \text{eq 32}$$

and, thus, the capacity factor for the peptide can be derived from the expression

$$k' = \frac{([L_{(+bD_o)}]^n \Phi K_{adsorb})}{[D_o]^{[(a-c)+(b-d)]}} \quad \text{eq 33}$$

By convention, the number of solvent molecules shed by the peptide on adsorption to the nonpolar RPC ligands has been represented by either Z, where

$$Z = (a-c) + n(b-d) \quad \text{eq 34}$$

and hence

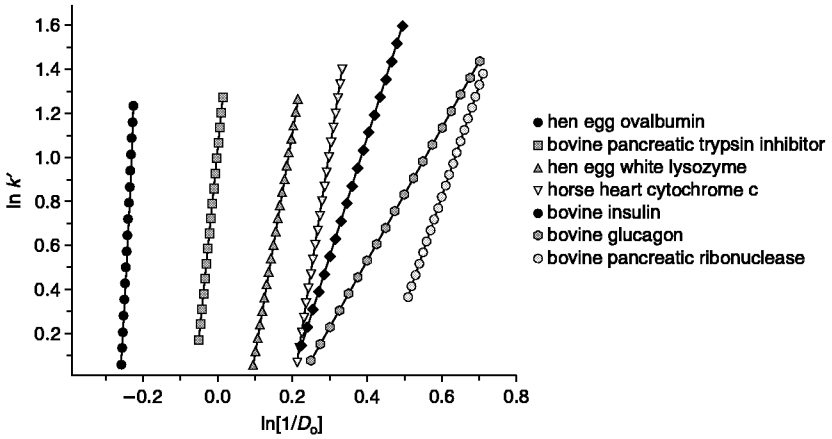
$$\ln k' = Z \ln(1/[D_o]) + \ln C \quad \text{eq 35}$$

where

$$C = ([L_{(+bD_o)}]^n \Phi K_{adsorb}) \quad \text{eq 36}$$

A formal relationship thus exists between the Z and S_c values for all peptide–nonpolar ligand interactions since Z is the slope of the plot of ln k' versus the logarithm of $1/[D_o]$ assuming that a first order dependency prevails. Linearity can be observed for the dependencies of ln k' on $\ln(1/[D_o]$ such as the example[44] shown in Figure 12 for several polypeptides and small proteins eluted with a 50% formic acid mobile phase containing different concentrations of propan-2-ol from *n*-alkylsilica under RPC conditions.

Figure 12 Representative Dependencies of ln k' on the Reciprocal of the Logarithm of the Organic Solvent Concentration, $\ln(1/[D_o])$, for Several Large Polypeptides and Small Globular Proteins[44]a,b



[a] In this figure, several small proteins were eluted at different organic solvent percentages under RPC conditions [octylsilica; 50% HCO_2H mobile phase, iPrOH as the organic solvent modifier; flow rate 1.0 mL·min^{-1}].
[b] Adapted from Geng and Regnier,[44] with permission.

As noted previously,[17,19,31,46] the Z (or S_c) and ln C terms represent average values characteristic of the overall adsorption–desorption phenomena that a peptide undergoes at the nonpolar surface of a RPC sorbent. Clearly, the derived values of Z (or S_c) incorporate the influences of all additives in the mobile phase that interact with the peptide. Thus, the effects of different ion-pairing reagents, acids, or salts, the variations in the molal surface tension increment of the mobile phases due to the use of different solvents or detergent additives, or the type of buffer species present can be accommodated, and the mobile phase most suited to the resolution task at hand can be determined. In common with the other models used to interpret the retention behavior of peptides in RPC, with the stoichiometric displacement model the assumption is again made that the structure of the peptide remains constant throughout the chromatographic migration, and that there is a constant number of ligands available for interaction with the peptide(s) over the complete range of mobile-phase compositions. Consistent with the underpinning physicochemical relationships linking molecular structure and retention behavior, various studies have shown that the Z (or S_c) values are proportional to the molecular weights of peptides with a similar ratio of hydrophilic to hydrophobic amino acid residues. This empirical correlation permits a more fundamental question related to the interaction of peptides with nonpolar ligands to also be explored, namely the nature of the relationships between the hydrophobic surface area, the total surface area, and the contact surface area of a peptide when in free solution or when bound to a nonpolar surface, and the effect of changes in these molecular parameters on retention behavior. Again, coefficient scales of amino acid side-chain hydrophobicities[32,35] in different solvent environments can be productively used to anticipate with reasonable precision the relative elution order in RPC of many small peptides, with <20 amino acid residues, of known sequence, thus greatly facilitating the isolation of synthetic peptides.

for references see p 622

7.1.1.9 Influence of van der Waals Effects and Other Weak Forces on Peptide Interactions in Reversed-Phase Chromatography

An alternative view can be advanced to facilitate the interpretation and optimization of the selectivity of peptides in RPC, namely that the interaction of a peptide with immobilized hydrophobic ligands is mediated via van der Waals and other close range forces.[41,42] In this scenario, the driving force for retention of a peptide in RPC is dependent on the interfacial characteristics of the interaction, which show different distance dependencies. Thus, in this model, the force of the interaction decays according to a distance dependency of $1/r^4$ for uncharged atoms and $1/r^3$ for carboxylate oxygens, where r is the distance between the two interacting surfaces. When electrostatic interactions occur, e.g. between the charge groups within the peptide and silanol groups of an *n*-alkysilica RPC sorbent, the forces mediated by formation of electrical double layers between the interacting surfaces has a dependency of $1/r^2$ and operates over considerably larger atomic distances. Similarly, the distance dependency of polar hydrogen-bonding effects, mediated primarily by hydrogen-bonding interactions between the peptide and type I or type II silanol groups on the RPC sorbent surface, takes the form of a $1/r^3$ relationship, and is also believed to be operative over relatively small atomic distances, i.e. 1.5–5 Å. These short-range RPC interactions, and their relevance to the interdigitation of the contact region of a peptide with the hydrocarbonaceous surface leads to a number of outcomes that have bearing on the choices available to the peptide chemist towards enhancing the separation selectivity in a particular separation of a peptide mixture. Incomplete control of the secondary effects associated with these short-range force effects leads to impaired resolution. Remedies to such situations are examined in greater detail in Section 7.1.2.

7.1.2 Sorbent and Mobile-Phase Considerations for Reversed-Phase Chromatography and Hydrophobic Interaction Chromatography Separations

7.1.2.1 General Characteristics of Reversed-Phase Chromatography Sorbents

The popularity of RPC methods for the separation of peptides on both analytical and preparative scale owes its origins to a number of key operational features. One of these features is the chemical and physical nature of the sorbents used. Most RPC sorbents in use today for peptide separations consist of a microparticulate rigid porous support, usually silica, with hydrocarbonaceous ligands covalently bound to the surface. The surface chemistry and physical properties of the stationary phase define the overall performance of the chromatographic system, and hence can be considered as one of the essential choices underpinning the development of a successful separation method. Numerous illustrations of the very high resolution that can be achieved by analytical RP-HPLC procedures with *n*-alkylsilicas encompassing the particle size range of 3–10 μm can be found in the scientific literature for the separations of synthetic peptides. Excellent resolution can be achieved for peptides with nonporous *n*-alkylsilicas, with average d_p of 1.5 μm, at low pH, with linear gradient MeCN/H_2O elution conditions, and with separation times of less than 10 minutes feasible in such applications with these high-performance microparticulate sorbents packed into columns of 50 × 4.5 mm i.d. or smaller capillary systems.

7.1.2.2 Sorbent Characteristics

Substantial effort has been expended on the development of more stable silica-based sorbents specifically suited to peptide separation as well as sorbents derived from polymerization of organic monomeric materials for use in both analytical and preparative RPC applications. Most silica-based RPC materials now available in the particle size range of 3–25 μm have the desirable features of excellent mechanical stability and good chemical stability up to pH 7.5. High-performance macroporous, polymeric sorbents (with d_ps in the range 5–50 μm), large pore diameters (p_d >500 Å), with limited deformation up to pressure drops of 200 bar or reduced swelling in different mobile-phase compositions are now also available. For example, various types of styrene–divinylbenzene, vinyl alcohol–polymethacrylate copolymers, perfluoro polymers, and other polymeric matrices have found their way into the application arsenal of RPC media for the separation of synthetic peptides. However, chemically modified porous silicas still attract the greatest amount of attention for analytical, laboratory-scale preparative and large-scale process separations of synthetic peptides by RPC,[86–88] e.g. that of recombinant human insulin.[89]

Different types of naked silicas are now readily available in narrow ranges of average pore and particle diameters. Particles can be produced in spherical and irregular geometries, resistant to significant changes in their morphologies by static or dynamic pressure variations, and with excellent mechanical/handling characteristics. Moreover, with the current generations of silica-based packing materials, changes in packing morphologies are infrequent, resulting in many consecutive (i.e., >500) uses of the same column under non-hostile elution conditions at low pH values. With current manufacturing technologies, spherical silicas of defined particle sizes can be produced with excellent control over the distribution of pore diameter and tortuosity, although micropores may still contribute to loss of efficiency with some types of porous silica. With polymer-based RPC sorbents, micropore effects can make a significant contribution to loss of efficiency and peak dispersion with peptides.

Because silica is rapidly dissolved in high pH solutions,[86–89] this property represents a major disadvantage of silica-based RPC sorbents when buffers of higher pH values are required to achieve satisfactory resolution of peptide mixtures. In these cases, other types of surface-modified inorganic oxides, including zirconias,[90–94] aluminas,[95] titanias,[96,97] yttrium-doped silicas,[86–88,98,99] polymer-coating silicas,[86–88,100–102] as well as graphite or glassy carbon materials[103,104] can be employed as alternatives. Due to their more limited availability, many of these alternatives have yet to gain significant popularity for the isolation of synthetic peptides. Nevertheless, with alkaline mobile phases, i.e. up to pH 11, these new types of surface-modified, doped, or clad-silica or ceramic oxide RPC sorbents show improved chemical stability for considerable periods of time at elevated temperatures, conditions that would degrade conventional silica-based sorbents very rapidly. The hydrolytic instability of silica-based RPC sorbents has occasionally been claimed as the major impediment to their use in high-performance process-scale purification of synthetic peptides by the pharmaceutical, biotechnology, and food industries. Experience has shown that *n*-alkylsilicas can still be efficiently employed in such applications, provided suitable cleaning-in-place and column-regeneration protocols are employed. Such cleaning-in-place protocols can be based on the knowledge gained on the elutropic characteristics of different organic solvents from analytical or small-scale laboratory separations of the same mixture of synthetic peptides.

Although macroreticular polymeric resins have shown promise[105–110] for the laboratory-scale purification or analysis of synthetic peptides by RPC procedures, generally they have yet to assume a significant place in larger-scale purification tasks. The major limitation of some of these polymeric sorbents destined for RPC applications is that they tend to swell when in contact with the organic solvents typically employed in RPC separations (the exceptions being the highly cross-linked materials produced from specialized polymer sys-

for references see p 622

tems). This swelling process can cause peak tailing due to a loss in permeability of the sorbent. Since styrene–divinylbenzene copolymer supports are already hydrophobic, no further derivatization may be required for these sorbents when used in the RPC mode of separation. Moreover, resins made from styrene–divinylbenzene copolymers and other polymer combinations, e.g. methacrylates or polyvinyl alcohol copolymers, are commercially available both with and without surface-immobilized *n*-alkyl ligands. The presence of unreacted vinyl or acidic groups in some commercial polymeric phases, however, means that these polymer-based sorbents can show "mixed mode" effects with synthetic peptides when low ionic strength mobile phases are used.

7.1.2.3 Particle Size, Pore Diameter, and Ligand Density Effects in the Reversed-Phase Chromatography Separation of Peptides

In order to achieve good performance with RPC sorbent, attention must be given to the choice of the particle shape and size. These physical attributes are key characteristics, after the issue of chemical stability of the support, if efficient, high-resolution separations of synthetic peptides are to be achieved by RPC procedures. In contrast to the soft, deformable gel materials used in classical chromatography, where particle sizes are typically in the range of 15–200 μm, for analytical and process-scale RPC separations of peptides, rigid particles in the range of average diameters (d_ps) of 1.5–25 μm are employed as high-efficiency packings, coated with relatively stable hydrophobic phases. As evident from eq 2, the selection of the average particle diameter (d_p) of a RPC sorbent will directly control the resolution that can be achieved. For most analytical peptide separations by RP-HPLC, *n*-alkylsilicas in the range of d_p = 1.5–5 μm are, for example, employed, with significant improvements in efficiencies, separation speed, and resolution evident when smaller particles are used. The penalty here is reduced sample capacity and/or increased column backpressure.

Currently, a diverse range of irregular and spherically shaped silica-based particles suitable for peptide separation are available from a variety of manufacturing processes, typically as extensions of the sol-gel and spray-drying procedures, often followed by hydrothermal treatments. In excess of 500 different vendors[111] of RPC sorbents can currently be identified, and the obvious question is which sorbent and which vendor should be used. Although it may sound a little trite, the answer lies in experience. Firstly, the RPC sorbent(s) should have a good reputation for minimal interbatch variations, an ability to be scaled from the ultra-micropreparative through to the preparative/industrial scale, robustness in terms of the number of repetitive usages, and the vendor should be willing to market different columns of similar performance characteristics over an extended period of time at (relatively) low cost. There is nothing worse that a vendor who fails to be able to provide reproducible packing materials or one whose marketing practices gouge the end-user!

The workhorses for most RPC applications with synthetic peptides have been the C_3-, C_4-, C_8-, and C_{18}-*n*-alkylsilicas available as d_p = 3–10 μm particles, with higher column efficiencies typically achieved with spherical particles rather than irregular particles. Because spherical sorbents can now be manufactured to high specifications, columns packed with spherical sorbents tend to have greater durability, since close-packed morphology can be achieved with the "balanced density" packing procedure. The use of surface-modified spherical and irregular silica and zirconia particles with mean d_ps in the range of 5–20 μm has had great ascendancy during the 1990s for preparative- and process-scale applications, partly because of the high productivities and lower unit operation costs that can be achieved at the process level in the purification of synthetic peptides.[112]

Synthetic peptides have relatively large Stokes radii and hydrodynamic volumes, and smaller diffusion coefficients, compared to low-molecular-weight organic pharmaceutical molecules. As a consequence, the mean pore size and the pore size distribution of the

support can have a significant influence on the separation efficiency, and thus the resolution (through the dependency $R_s \propto \sqrt{N}$). To achieve optimal resolution, RPC sorbents need to have pore diameters significantly larger than the mean molecular diameter of the largest peptide in the mixture,[113,114] ideally by a factor of about 10. In order to achieve a reasonable level of capacity through adequate utilization of the surface area of the RPC sorbent, the peptide has to gain access to the internal regions of the porous particle. Particles with pore diameters of 100 Å are suitable for many small peptides (with <20 amino acid residues), although the trend for both analytical and preparative applications has been to use the so-called wide-pore RPC sorbents with pore diameters of 300 Å or larger. However, it should always be remembered that this gain in accessibility with larger-pore RPC sorbents is offset by the loss in particle surface area, and therefore loss in column loading capacity, as well as the potential for decreased mechanical stability and increased pressure drop fragility, resulting in reduced column life. Gigaporous materials, i.e. sorbents with very large pores (up to 10,000 Å), based on silica- or polymer-based RPC materials have also gained some popularity. With these macroporous materials, as well as with the monolithic types of RPC sorbents, very high flow rates can be achieved without generating high backpressures. The use of sorbents with gigapores or monolithic channels effectively acting as interconnecting tunnels result in convective or perfusive flow[105–110,115] of the mobile-phase fluid along the main branches, and diffusive flow in the side branches where pores of smaller mean diameter may exist. Although RPC sorbents based on macroporous organic polymers, such as polystyrene, have been available since 1990 and have been widely employed at high superficial fluid velocity for rapid analytical and semipreparative applications with synthetic peptides, their application in large-scale purification of synthetic peptides has been much more limited. Similar applications with the monolithic types of RPC sorbents at high flow rates have yet to gain popularity due to their lack of commercial availability.

Micropellicular RPC sorbent materials[116–118] suitable for peptide separations provide an alternative approach to exploit the particle size, pore size, and surface area accessibility effects to achieve acceptable mass transfer kinetics. These RPC sorbents are based on microspherical, fluid-impervious particles with appropriate types of nonpolar coatings immobilized to their surfaces. Since intra-particle diffusion resistances are absent, micropellicular RPC sorbents show improved peak efficiencies at high linear flow velocities, at the penalty of their greatly reduced mass loadability. To circumvent this constraint, particles of smaller average diameter, i.e. $d_p = 1.5–3\ \mu m$, can be employed,[59,119–121] but clearly this application area targets only the ultra-fast, high-flow-rate analytical separations of peptides, rather than preparative isolation requirements.

Reducing the average d_p of an RPC sorbent is a simple way to increase column efficiency. In this context the nonporous silicas, such as the Micra NPS-1, or the polymeric TSKgel NPR reversed-phase sorbents have gained some popularity with synthetic peptides. When RPC sorbents of small particle diameters are used, an elevated temperature is often required to decrease the viscosity of the mobile phase and permit low operating backpressure to be generated, otherwise the consequences of the pressure dependency on particle diameter will become evident, e.g. $\Delta P \propto 1/(d_p)^2$ at very high backpressures. When such RPC sorbents are used in well-packed columns, peak efficiencies of >300,000 plates $\cdot\ m^{-1}$ can be achieved. For many analytical RPC separations of peptides, such minicolumns of 25–33 mm length and 4.5–5.0 mm i.d. have been the norm for many years.

The need for enhanced detection sensitivity and automation has steadily increased for the separation and analysis of peptides from natural sources or proteolytic digestion of low abundance proteins; this is also partly a consequence of the greater usage of combinatorial solid-phase synthetic approaches. Narrow bore (1–2 mm i.d.), microbore (0.5–1.0 mm i.d.), and capillary (100–500 µm i.d.) columns have provided attractive solutions to these problems.[122] An important attribute of very small particle diameter micropellicular, porous, or nonporous RPC sorbents is that they are ideally suited to such microbore or capillary

for references see p 622

columns. With appropriate engineering of the endfittings, frits, and detector cells, considerable improvement in detector response can be achieved, permitting ultra-micropreparative separations to be integrated on-line with increased detection sensitivity using UV/DAD or ESI-MS techniques.[123–128] With smaller diameter columns, the advantage of decreased solvent consumption due to the lower flow rates also becomes a consideration with these "miniaturized" systems. Peptides will typically elute from a 1-mm i.d. microbore column packed with the appropriate 3-μm RPC sorbent in smaller peak volumes and with a 5–10-fold increase in detection sensitivity compared to a separation carried out on a conventional 4.6-mm i.d. column, when the same sample amount of a peptide mixture is injected.

The advantages of employing capillary columns packed with suitable RPC sorbents for the resolution of peptide samples at the analytical and ultra-micropreparative scale are obvious not only for synthetic peptide mixtures of limited abundance, but also for peptides derived from enzymatic or chemical cleavage of proteins. For example, as little as 2 pmol of a tryptic digest of horse skeletal muscle myoglobin can be readily separated[129] on a 180-μm i.d. capillary column, packed with a $d_p = 5$ μm, 300-Å pore diameter C_{18}-silica using a standard gradient system at a flow rate of 100 μL · min^{-1}, whilst with a 50-μm i.d. capillary packed with $d_p = 1.5$ μm, 300-Å pore diameter C_{18}-silica using the same gradient system at a flow rate of 25 μL · min^{-1}, as little as 150 fmol of the same peptide mixture can be separated and individual components analyzed by ESI-MS. Although the gain in sensitivity [expressed as detection sensitivity gain = $(d_{c1})^2/(d_{c2})^2$ where $d_{c,1}$ is the column diameter] with microbore or capillary columns does not always correspond to the theoretical value predicted by comparison with conventional columns, up to 80% of the theoretical advantage in sensitivity can typically be achieved with conventional HPLC equipment. Importantly, such systems readily permit the effluent stream to be interfaced with an ESI-MS spectrometer or gas-phase Edman automated microsequencer without additional handling steps, thus greatly facilitating the determination of the structure or associated characterization of the isolated low-abundance peptides.

A variety of commercial instruments are available to reliably deliver flow rates in the range of 5–100 μL · min^{-1}, thus enabling microbore and capillary columns packed with suitable RPC sorbents to be generally employed. These developments have also permitted exciting new modes of RPC to emerge, including high-performance capillary electrochromatography (HP-CEC) of peptides.[130,131] With these new CEC systems, as well as with capillary μRPC, peak efficiencies of more than 500,000 theoretical plates per meter can be obtained, permitting very high sensitivity detection of the separated components, i.e. at the level of ca. 20–200 pg of peptide samples. As discussed later in this section, the selectivity of peptides in CEC with nonpolar sorbents is fundamentally different from that manifested in the conventional RPC mode. As a consequence, peptide mixtures that have proved recalcitrant to standard RPC techniques have proved responsive to HP-CEC procedures.[132,133] Additional gains in detection sensitivity can be made if the separation of peptides on capillary columns are followed where the separated peptide can be detected by laser-induced fluorescence techniques. Finally, as noted above, by interfacing the effluent directly with a continuous-flow fast atom bombardment (FAB), matrix-assisted laser desorption ionization time-of-flight (MALDI-TOF), or electrospray ionization (ESI) mass spectrometer, peptides derived from solid-phase peptide synthesis or from the enzymatic digestion of proteins and separated by RPC under isocratic or gradient elution conditions with such microbore or capillary columns can be directly analyzed on-line, and their mass, composition, and sequence can be determined within tens of seconds.

7.1.2.4 Surface Modification Considerations for the Selection of Reversed-Phase Chromatography Sorbents for Peptide Isolation

Naked silicas contain surface-reactive silanol groups, thus permitting the covalent attachment of a large variety of different nonpolar ligands by conventional siloxane[86–89] and disilazane chemistries.[67] The acidic nature of residual silanol groups accessible on the sorbent surface, even when extensively endcapped, gives rise to the electrostatic characteristics of the retention process due to this silanophilic effect. Depending on the composition and sequence of the peptide(s), these silanophilic contributions may represent a constraint or an advantage, i.e. highly basic peptides require special attention[10,17,23,78,80] with regard to mobile-phase selection, to ensure that adequate resolution and elution is achieved without undue peak tailing. When optimized mobile-phase conditions are employed, with higher ionic strength buffers, even highly basic peptides can be separated successfully and recovered in good to high yield from non-endcapped *n*-alkyl silicas of high zeta potential.

With silica-based sorbents, in particular, a variety of procedures have been used to immobilize *n*-alkyl groups of different chain length and structure, permitting the impact of *n*-alkyl chain length on RPC behavior of peptides to be studied. Arising from these investigations,[120,134–136] a consensus view has emerged that, with silica-based RPC sorbents, *n*-alkyl chain lengths between C_3 and C_8 are preferred for middle to higher molecular weight polypeptides, whilst the C_8 to C_{18} chain length RPC sorbents offer more versatile selectivity options with smaller peptides.[20,137] In the case of macroreticular polystyrene–divinylbenzene, alkylated polyhydroxy–methylacrylate, or other types of derivatized polymeric sorbents, the presence of nonpolar clusters, which form the interactive sites for peptide binding, can be achieved via a variety of different polymer and surface-derivatization chemistries. Depending on the nature of the immobilization chemistry, as well as the type of hydrocarbonaceous ligand and support matrix, a great many types of nonpolar stationary phases with different surface characteristics can be prepared, all of which can loosely be encompassed within the category of RPC sorbents.

In the case of silica- or ceramic-based sorbents, the selectivity of RPC sorbents is clearly dependent on the quality of the bonding chemistry, whilst with the polymeric RPC sorbents, the type of the monomer units used in the production of these materials has a significant bearing on the resolution. Treatment of silica or other inorganic oxide particles is usually carried out with mono-, di-, or trifunctional organosilane reagents[68,86–89,119–121,138–142] under anhydrous conditions in the presence of a small amount of catalyst. Alternatively, triethoxy-*n*-alkylsilanes can be used under either anhydrous or aqueous modification conditions. Monofunctional alkyl(chloro)(dimethyl)silanes, including the sterically shielded alkylsilanes,[98,135] generally yield more efficient and stable RPC sorbents. Higher surface coverages may be attained with di- and trifunctional silanes, although this carries the penalty of reduced mass transport kinetics with peptides, and thus reduced peak efficiencies. When trifunctional silanes, such as triethoxy-*n*-alkylsilane, are employed for the surface modification, greater sorbent stability of the RPC sorbent at low pH can be obtained compared to the more traditional monofunctionalized *n*-alkylsilane, although, typically, hydrolytic degradation occurs to a greater extent with buffers containing inorganic anions or cations at higher pH values such as pH 9–10.

As an alternative, stable high-coverage nonpolar RPC sorbents phases have been prepared by cross-linking hydrophobic polymers at the silica surface, either via free radical[143] or condensation[101] polymerization chemistry. In this case, the underlying silica becomes partly protected from hydrolytic degradation due to the presence of the hydrophobic polymer film coating that effectively shields the support material. Similar procedures have been employed to chemically modify the surface of other support materials, such as porous zirconia, titania, or alumina, to further impart resistance to degradation when alkaline mobile-phase conditions are employed. Porous polystyrene–divinylbenzene sorbents, be-

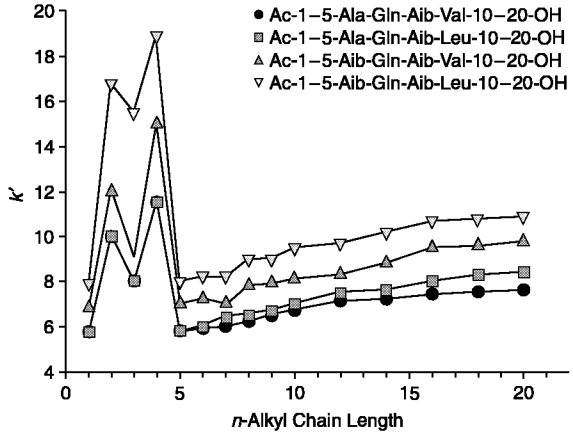
for references see p 622

cause of their intrinsic hydrophobicities, also fulfill useful roles in the analytical RPC of peptides. Although these types of particles are resistant to alkaline pHs, most applications still favor low pH elution conditions to achieve resolution of peptides. Moreover, with macroporous polystyrene–divinylbenzene sorbents, such as the perfusive Poros RPC materials,[105,107] very high superficial velocities can be achieved without sacrificing resolution. The advent of monolithic hydrophobic sorbents has introduced[106–110] a further dimension for the modulation of the selectivity and resolution of peptides under conditions of high superficial velocities and mobile-phase compositions that encompass high pH values.

The shape and pore morphology of the support material, the chemical nature of the immobilized ligand, and the chemistry employed in the bonding reaction have all been shown to affect the performance of RPC sorbents in peptide separations.[86–89] Both linear and branched *n*-alkyl chains up to C_{30} have been immobilized to silica surfaces for use as RPC sorbents in the separations of peptides. Figure 13 provides a representative illustration of the influence of *n*-alkyl chain length from C_1 to C_{20} on the retention of several paracelsin-related peptides separated[120] under the same elution conditions. The progressive increase in the k' values of the different paracelsin peptides as the *n*-alkyl chain length is increased is consistent with the expected behavior for hydrophobic effects dominating the interaction, whilst the significant changes in the k' values from C_1 to C_5 are consistent with the transition from the rigid picket-fence-type nonpolar sorbent surface typical of C_4 phases to a more relaxed, collapsed phase frequently found for immobilized C_8 to C_{18} chains. Paracelsin A-D are 20-meric, amphipathic α-helical peptides that involve α-aminoisobutyric acid (Aib) substitution for alanine (Ala), and/or leucine (Leu) substitutions for valine (Val) residues at positions 6 and 9 in their amino acid sequences, respectively. The observed changes in retention, expressed in terms of the k' values, as a function of the length of the *n*-alkyl chain immobilized to the same densities by the same chemistries onto the same porous silica support material clearly reflect differences in the hydrophobic contact areas that these closely related peptides generate on binding to the nonpolar ligands. Edmundsen wheel projections of paracelsin A-D reveal that the Ala → Aib and Val → Leu replacement residues are in close proximity on the same face when these peptides fold into their α-helical conformations. The elution order reflects these differences in amino acid sequence and secondary structure assembly, with the peptides with more hydrophobic amphipathic faces eluting later in the chromatogram. Moreover, the changes in k' values as the *n*-alkyl length is varied reflects the intrinsic differences in the surface characteristics of the RPC sorbent in terms of the partitional/adsorptive behavior of the *n*-alkyl groups. The *n*-alkyl length of an RPC sorbent is well-known to influence the retention and kinetic behavior of polypeptides, and this consideration must be factored into the choice of a particular sorbent, if optimal resolution is to be achieved.

Although higher selectivities with peptides can be achieved with *n*-alkylsilicas of longer chain length, better mass balances have typically been found when RPC sorbents containing shorter *n*-alkyl chains are employed with either nonporous or porous silicas. Consequently, the propyl-, butyl-, or octylsilica phases are the more widely used ones in polypeptide RPC. However, when very short linear *n*-alkyl chains (i.e., $<C_4$) are used without endcapping, the RPC sorbents tend to be less stable than those obtained with longer nonpolar ligands, whilst the C_8–C_{18} chains at an equivalent ligand density tend to shield the underlying silica surface more effectively from hydrolysis by the aqueous mobile phase. For this reason, shielded phases, such as the *n*-alkyldiisobutylsilicas or the bridged *n*-alkylsilicas, have emerged[98,135] as one preferred option for the RPC of synthetic peptides.

Figure 13 Plots of the Capacity Factors, k', of the Paracelsin A-D as a Function of the *n*-Alkyl Length of Different RPC Silica Sorbents of Constant Ligand Density in H_2O/MeOH/MeCN (22:39:39)[120]a,b



[a] The amino acid sequences of the paracelsin A-D peptides are as follows: Ac-Aib-Ala-Aib-Ala-Aib-6-Gln-Aib-9-Aib-Gly-Aib-Aib-Pro-Val-Aib-Aib-Gln-Gln-PheOH.

[b] Adapted from Lork, Unger, Bruchner, and Hearn,[120] with permission.

Cyanopropyl- and diphenyl-substituted and acylated aminopropyl ligands have also been employed as alternatives to *n*-alkyl groups in peptide RPC.[86–89] The cyanopropyl groups are less hydrophobic than the *n*-alkyl functionalities, and hence tend to be less retentive with peptides. Due to their aromatic $\pi \rightarrow \pi$ bonding characteristics, diphenyl groups provide different selectivities to the more conventional *n*-alkyl systems but also tend to be less retentive. As a consequence, synthetic peptides can be eluted from diphenyl- and cyanopropyl-bonded silicas under milder RPC elution conditions, resulting in some cases in better recoveries than can be achieved with the more nonpolar *n*-alkyl hydrocarbonaceous phases. However, the selectivity differences that can be achieved may not be as significant as found with *n*-alkylsilica sorbents, whilst the long-term column stability can be lower with cyanopropyl- and diphenyl-substituted and acylated aminopropyl phases immobilized onto silica. The so-called ether-type phases and other mildly hydrophobic stationary phases developed for use in high-performance hydrophobic interaction chromatography[18,134,144–147] can also be employed in the RPC mode for the separation of very hydrophobic polypeptides with water/organic solvent eluents.

Since the 1980s, several different bonding strategies have been developed for the preparation of RPC sorbents based on the chemical modification with an *n*-alkylsilane of the silica support material. One approach involves the reaction of the silica, frequently preactivated by hydrothermal treatment with an *n*-alkylsilane of the type $RSiX_3$ or R_2SiX_2 (X = Cl or OEt). Silica derivatized with R_2SiX_2 silanes often exhibit multifunctional surface characteristics, since this strategy usually leads to the generation of a polymer layer deposited at the silica surface, partially stabilizing the silica to hydrolysis. The thickness of the polymeric layer can vary, leading to reduced mass transport and impaired column efficiencies compared to the corresponding monolayer-bonded phase materials. In terms of reproducibility of the manufacturing process, RPC sorbents made with $RSiX_3$ or R_2SiX_2 silanes tend, moreover, to be harder to reproduce, with variation in the total organic coverage evident from batch to batch. For peptide isolation applications, this type of RPC sorbent has consequently been largely displaced by sorbents prepared with monochlorosilanes.

The reaction of an *n*-alkyl(chloro)(dimethyl)silane or *n*-trialkylchlorosilane of the type RMe_2SiX or R_3SiX (X = Cl) with a hydrothermally activated silica represents the second and more popular approach. Excellent mass transfer characteristics with peptides can be

for references see p 622

achieved with this monolayer coverage approach, with the resulting sorbent exhibiting high column efficiencies for most peptides. The silanization reaction can be carried out under anhydrous or aqueous conditions, using a small amount of a basic catalyst such as lutidine. Moreover, when shielded *n*-alkyl(chloro)diisopropylsilanes or *n*-alkyldi-*tert*-butylchlorosilanes are employed, RPC sorbents with excellent pH and temperature profiles can be prepared. In addition to the reactive *n*-alkylchlorosilanes, di-*n*-alkyltetramethyldisilazane, such as dibutyltetramethyldisilazane, organoaminosilanes, *n*-alkylenolate, and hydrosilanization with Speiers catalyst can also be employed, permitting both proximal and geminal silanol groups to be more effectively modified.

Typical *n*-alkyl ligand densities that can be achieved with *n*-alkylchlorosilanes are within the range of 2.5–3.2 $\mu mol \cdot m^{-2}$, whilst with disilazanes, ligand densities approaching the limited values can be reached under optimized conditions, i.e. between 3.50 and 4.20 $\mu mol \cdot m^{-2}$. Surface-modified zirconia or other metal oxide based RPC sorbents can be similarly prepared by either of the above two strategies. Compared to *n*-alkylsilicas, these ceramic RPC sorbents show different selectivities with synthetic peptides, as well as different chemical stability profiles. Consequently, RPC sorbents based on these types on surface-modified, porous metal oxide materials fulfill useful and complementary roles, but at this point in time, have achieved a more limited range of applications for the resolution of synthetic peptides due to their limited availability.

As an alternative strategy, weakly hydrophobic polymers, such as polyvinyl alcohols, oligoamino alcohols or polyoxyethylenes, of various molecular weights can be attached via multistep reactions to propoxy-modified silicas to generate RPC sorbents of different hydrophobic characteristics. Similar multistep procedures have been employed to introduce[154] *n*-alkylamines of different chain lengths into previously prepared polyaspartic acid modified silicas, whilst polyamide-bonded silica support materials have been employed as a scaffold for the subsequent immobilization of various nonpolar ligands. With these more complex types of RPC sorbents, additional care is required in terms of the control over the derivatization chemistries. As a consequence, immobilization of the preformation hydrophobic ligand represents a more elegant approach. Based on this strategy, different types of RPC sorbents custom-made for peptide separations have been described, including those derived from polyether silanes containing *n*-alkyl groups of different ligand chain length and hydrophobicity or from acetamido-*n*-alkylsilanes. Due to issues of commercial sensitivity about the actual characteristics of the polymer backbone, and the proprietary nature of the chemical formulations used in emulsion or free radical polymerization procedures to generate the porous network or to control particle size distribution with these cross-linked organic polymers/copolymers, considerably less information is in the public domain on methods for the commercial production of polymeric RPC sorbents.

In RPC, the length of the *n*-alkyl chain determines the sorbent hydrophobicity, which must be balanced against the properties of the peptides to be separated. Because the retention process is dictated by the affinity of the interaction between hydrophobic "patches" on the peptide surface and the nonpolar ligands, RPC sorbents of inappropriate hydrophobicity can lead to "irreversible" binding, particularly if the mobile-phase composition has not been optimized. Theory predicts that retention should be greatest in RPC when the mobile-phase pH is near the isoelectric point of the peptide. Moreover, under these conditions, peptide solubility may be reduced, particularly for more hydrophobic peptides. Ionization changes of specific amino acids located within the chromatographic contact region can particularly affect the retention. Consequently, pH changes can be indirectly used to significantly affect the magnitude of hydrophobic interactions of peptides in RPC, with the opportunity, in principle, to carry out separations over a wide range of pH values. Mobile-phase pH conditions can thus often be chosen for RPC separations of synthetic peptides based on the peptide titration data derived from HPCZE to obtain appropriate selectivities without necessarily compromising the solubility of the peptide to be purified.

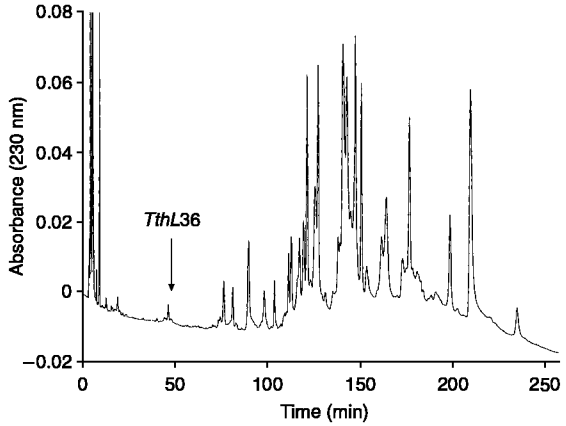
7.1.2.5 Mixed Modality Sorbents in Peptide Isolations

Besides the role of hydrophobic interactions, peptides can also manifest a combination of electrostatic, size-exclusion, and hydrogen-bonding effects with all current RPC sorbents. As discussed earlier (cf. Section 7.1.1.4), selectivity reversals and changes in resolution can thus be achieved during the separation of peptide mixtures on a single column by enhancing or suppressing a particular type of secondary interaction by exploiting the effects of changes in pH, solvent composition, ionic strength, or the presence of different classes of ionic additives. Alternatively, different retention modes and selectivities can be achieved for peptides by purposefully incorporating or "tailoring" into the surface of an RPC sorbent a "mixed-mode" property as part of the chemical synthesis strategy. Mixed modality can be achieved in one of two ways with RPC sorbents, with the diluted nonpolar phase approach being more popular. Although various manufacturers have introduced different RPC sorbents that take advantage of these mixed-mode effects,[148–150] their popularity has not been high, partly because of reasons of chemical instability or lack of availability of "user-friendly" guidelines for their use for a particular type of polypeptide separation. The so-called "membrane-mimic" sorbents containing chemically immobilized phospholipid moieties fall into this category of limited usefulness for polypeptide isolation.

Additional modulation of the selectivity of synthetic peptides can be achieved through changes in the chemistry used at the endcapping stage of the synthesis of *n*-alkylsilica RPC sorbents. In these circumstances, advantage is taken of the silanophilic characteristics evident with all silica-based RPC sorbents or the chemical characteristics of the endcapping reagent. Thus, in this approach, RPC sorbents of lower *n*-alkyl chain density or alternatively those not endcapped are employed. Such RPC sorbents tend to exhibit peak tailing with polypeptides when low ionic strength elution buffers are employed. To circumvent this limitation and to adjust peptide selectivity, water/organic solvent eluents containing relatively high concentrations of a compatible salt, such as 100–500 mM NaH_2PO_4 are used. Although a subsequent RPC purification step is required to remove the salt, one advantage of using this type of mixed-mode approach lies in the potential to generate selectivity features that are different from those obtained with standard RPC surfaces. Exemplifying this approach is the example shown in Figure 14 for the resolution and purification of the relatively hydrophobic ribosomal polypeptides and proteins from *Thermus thermophilus*, including the *TthL*36 polypeptide, which contains a zinc finger CCCH motif. In this case a non-endcapped 10 µm LiChrospher C-18 sorbent of 30-nm average pore size was employed. With this strategy, two important properties of RPC system are simultaneously adjusted. Firstly, the surface tension of the mobile phase is changed, leading to a variation in selectivity in its own right. Secondly, electrostatic and hydrogen-bonding effects mediated by the geminal/vicinal type I or type II silanol groups are masked, again leading to a change in selectivity and retention. The literature contains a large number of examples of the use of such mixed/multimodal RPC adaptations for the separation and/or purification of peptides, particularly at the analytical or small-scale preparative levels, with relatively high ionic strength buffers, such as 300 mM KH_2PO_4 added to the mobile phase with a weakly endcapped RPC sorbent (e.g., LiChrospher 1000).[10,151]

for references see p 622

Figure 14 Separation of the Ribosomal Proteins of the 50S Subunit of *Thermus thermophilus* by RP-HPLC[a]



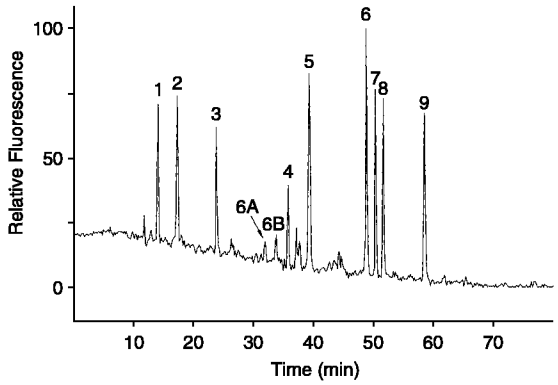
[a]The 50S ribosomal proteins were separated using a Beckman System Gold HPLC; thermostatically (20 ± 0.1 °C) controlled LiChrospher C-18 column (10 μm, 300 Å, 250 × 4 mm i.d.); optimized MeCN/H_2O linear gradient of 257.6-min duration (derived using the DryLab G/plus software); eluent: A: 500 mM NaH_2PO_4/500 mM H_3PO_4, pH 2.1, and B: 200 mM H_3PO_4 pH 2.1/MeCN (25:75); flow rate: 1.0 mL·min^{-1}; detection: 215 nm. The collected fractions were desalted by a second RP-HPLC procedure [Nucleosil C-4 column (10 μm, 300 Å, 30 × 4 mm i.d.); linear gradient: 0.1% TFA/H_2O to 0.09% TFA/60% aq MeCN over 60 min; flow rate: 1 mL·min^{-1}; detection: 215 nm]. The elution position of the CCCH motif zinc finger polypeptide, *TthL*36, is illustrated by the arrow.

7.1.2.6 Mobile-Phase Characteristics in the Reversed-Phase Chromatography of Peptides

One of the major attributes of RPC for peptide analysis and purification is the ease with which relatively inexperienced investigators can achieve within a short period of time the desired level of selectivity and resolution for a peptide mixture by manipulating the composition of the mobile phase. As noted above, in RPC the mobile phase is more polar than the sorbent, and is typically composed of a low pH, buffered water/organic solvent mixture. For most peptide mixtures, the profound dependency of ln k' on the volume fraction of the organic solvent modifier, ψ, necessitates that gradient elution protocols are usually applied in RPC separations with the ψ value progressively increased over a preselected time or solvent-composition interval. When gradient elution is carried out, the corresponding median capacity factor, ln k, and median volume fraction, ψ, are used to derive the optimal flow rate, gradient steepness, and gradient duration for a specific range of mobile-phase compositions. With current HPLC instrumentation, many types of gradient profiles can be generated. However, linear gradient profiles are to be preferred. Besides the issue of reproducibility under many RPC conditions, it is important that a linear change in ln k' per unit time is achieved for most if not all of the peptides within a mixture if optimization routines are to be efficiently applied as part of the isolation of peptides. Such conditions satisfy the criteria of the linear solvent strength (LSS) gradient elution.[12,17,20,152,153] As a consequence, the effect of gradient time and slope on the retention times and peak widths of the RPC separation of a peptide mixture can be predicted *de novo* utilizing these linear solvent strength criteria and with the effects of column length and particle diameter rationalized. Computational algorithmic approaches such as DryLab, Pek-N-Eze, or Hephaestus, can be employed to optimize the resolution of peptides under these gradient elution RPC conditions, and the choice of mobile-phase compositions to achieve a specific retention behavior for peptide mixtures can be based on these or related simulation models. The example shown in Figure 15 utilizes such an optimization approach. In this example, advantage was taken[264] of a high-coverage butylsilica RPC sorbent together with an optimized concentration of H_3PO_4 as a "hydro-

philic" ion-paring reagent to achieve the resolution of each of the tryptic peptides, fluorescently labeled at cysteine residues with the reagent 5-[(iodoacetamidoethyl)amino]-naphthalene-1-sulfonic acid (5-I-AEDANS), of the bovine thyrotropin β-subunit, thus permitting the characterization via molecular footprinting procedures of the receptor binding site of this glycoprotein. Good correlation has generally been obtained for other peptides generated by enzymatic digestion of proteins[17,27,48,49] in terms of the linearity of the ln k versus ψ plots over limited ranges of ψ values, provided secondary effects, such as conformational changes, self-aggregation, specific metal ion binding, or temperature-induced effects, do not significantly impact on the chromatographic performance. Similar procedures can be employed to determine the choice and concentration of a particular organic solvent, the concentration of different ion-pairing reagents or other additives such as nonionic detergents, as well as the buffer species and operational pH, in order to ensure that a successful RPC separation of the proteolytically derived peptide mixture is achieved. Analogous considerations apply to the separation by RPC procedures of peptide mixtures available from solution or SPPS methods.

Figure 15 Representative RP-HPLC Separation of the Tryptic/Chymotryptic Peptides of Bovine Thyrotropin β-Subunit bTSHβ Labeled In Situ with the Fluorescent Reagent 5-[(Iodoacetamidoethyl)amino]naphthalene-1-sulfonic acid (5-I-AEDANS)[360][a,b]



[a] The separation of these peptides was achieved under low pH, linear gradient MeCN/H_2O elution conditions. The bovine TSHβ was reduced with 1000-fold molar excess of dithiothreitol (DTT) at pH 8.2 for 3 h on ice, then alkylated by the addition of 5-I-AEDANS at a 5-fold molar excess over the DTT present. The remaining disulfide bonds were then fully reduced and the fluorescently labeled bTSH β-subunit preparation was digested firstly with trypsin and then by chymotrypsin. The fluorescently labeled peptides (1–9) were then separated by RP-HPLC [Bakerbond C_{18} silica sorbent, average $d_p = 5$ μm; 4.6 × 25 cm column; pre-equilibrated in 0.1% H_3PO_4; elution with linear gradient 0–50% MeCN in 0.1% H_3PO_4 over 120 min]. All fluorescently labeled peptides, except for the tryptic/chymotryptic glycopeptide, bTSHβ [CY^{19}***C***LTI***N***TTV***C***AGY^{30}], where the symbol ***C*** corresponds to AEDANS-labeled Cys and the symbol ***N*** to the N-linked glycosylation site, were eluted during the first 80 min and could be subsequently characterized. Fluorescent peptides, 1–9, were monitored using a fluorescence detector fitted with a 340-nm excitation filter and 500-nm emission filter, at gain 128. The major peptide peaks (1 to 9) were subsequently identified by automated gas-phase Edman amino acid sequence analysis, permitting the disulfide connectivities and the relative stabilities of the bTSHβ subunit to be ascertained. The peaks 6A and 6B corresponded to further enzymatic digestion of peak 6, namely fragmentation of the peptide ***C***^{88}NTDYSD***C***IHEAIK.[101]

[b] Adapted from Burke, Mant, Black, and Hodges,[347] with permission.

For most peptide separations by RPC, acidic pH conditions are used with either ca. 15 mM TFA, 15 mM H_3PO_4, 25–50 mM AcOH,[154] up to 60% HCO_2H,[155,156] or 10 mM HCl[157,158] added to the water/organic solvent mobile phase. Volatile organic acids (heptafluorobutanoic acid, pentafluoropropanoic acid, TFA, AcOH, or HCO_2H) allow direct recovery of the separated peptide samples via lyophilization. Low pH conditions have a desirable outcome with silica-based sorbents, since under these conditions the silanol groups are not ionized and peak tailing

for references see p 622

with basic peptides can be partially suppressed. Moreover, when volatile organic acids are used as ionic modifiers, direct injection into an ESI-MS can be carried out for subsequent analysis of the separated peptides. Experience has shown that the resolution by RPC methods of very closely related peptides, such as deamido forms or methionine *S*-oxide variants, are best achieved if mobile-phase conditions of slightly higher pH values, i.e. at pH 4.5–7.5, are chosen.

Besides the issue of pH control, when the perfluoroalkanoic acids are present in low concentrations, they also act as ion-pairing reagents with peptides, thus permitting selectivities to be modulated, peak efficiencies to be improved and recoveries to be enhanced. When higher pH values are required for the RPC separations of peptide mixtures, 100 mM ammonium hydrogen carbonate, 25–200 mM ammonium acetate, 50 mM triethylammonium phosphate, or 500 mM pyridine acetate have been employed, although in the latter case post-column derivatization with fluorescence detection is required. Higher pH ranges, i.e. pH > 8.5, are incompatible with the use of many silica-based RPC sorbents; polymeric RPC sorbents can be used instead, although most investigators still favor the use of elution conditions of lower pH values, i.e. pH $\leq$ 4.0, even with these polymer-based RPC sorbents.

In order to take advantage of the ionization status of the amino acid side chains at these and lower pH values, a very large number of different ion-pairing reagents have been developed[20,56,159,160] for use in the resolution of peptides by RPC procedures. Selected examples of these ion-pairing reagents are listed in Tables 2 and 3. Several of these reagents, e.g. cetyltrimethylammonium salts, permit RPC separations to be carried out at neutral and even alkaline pH values rather than acidic pH values, providing an opportunity to resolve peptides that may not be separable under low pH conditions. Because of the structural properties of many of these "ion-pairing" reagents, changes in both the bulk mobile-phase secondary chemical equilibriums, and also equilibriums associated with dynamic coating of the nonpolar sorbent, can be achieved. Concentration-dependent modulation of the selectivity of peptides results with these ion-pairing reagents with RPC sorbents.

Table 2 Selection of Commonly Used Ion-Pairing/Dynamic Liquid–Liquid Ion Exchange/Mobile-Phase Additive Species, Which at pH <7.0 Modify the Retention Characteristics of Unprotected Peptides on Chemically Bonded Hydrocarbonaceous Stationary Phases[a,b]

Anionic Pairing Ion		Cationic Pairing Ion	
$H_2PO_4^-/HPO_4^{2-}$	Cl^-	NH_4^+	$MeNH_3^+$
ClO_4^-	SO_4^-	$C_3H_7NH_3^+$	$Me(CH_2)_{11}NH_3^+$
CO_3^{2-}	BO_3^{3-}	$HOCH_2CH_2NH_3^+$	$Me(CH_2)_5NH_3^+$
HCO_2^-	$MeCO_2^-$	$(HOCH_2CH_2)_3NH^+$	Et_3NH^+
$CF_3CO_2^-$	$CF_3CF_2CO_2^-$	*N*-methylmorpholinium	Me_4N^+
$CF_3(CF_2)_3CO_2^-$	$CF_3(CF_2)_5CO_2^-$	*N*-methylpiperidinium	Et_4N^+
$BuSO_3^-$	$C_5H_{11}SO_3^-$	triammonium propane	Pr_4N^+
$Me(CH_2)_5SO_3^-$	$Me(CH_2)_6SO_3^-$	Bu_4N^+	
$Me(CH_2)_7SO_3^-$	$Me(CH_2)_{11}SO_3^-$		
$Me(CH_2)_{11}SO_4^-$	camphor-10-sulfonate		

[a] Many of the ions identified in this selection of ion-pairing species are generally compatible with UV detection < 235 nm, or in the case of those reagents with an aromatic nucleus, with fluorometric or UV (> 254 nm) detection. (Typically, the concentration of these components in the mobile phase is <5 mM for peptide separation and analysis, although concentrations > 5 mM can be employed to optimize resolution.)

[b] For a more complete compendium of applications, conditions, etc., see refs[17,20,160,342].

Table 3 Selection of Commonly Used Neutral Polar-Phase Modifying Reagents, Which at pH <7.0 Modify the Retention Characteristics of Unprotected Peptides on Chemically Bonded Hydrocarbonaceous Stationary Phases[a,b]

Neutral Polar-Phase Modifying Reagents
$Me(CH_2)_{11}OH$
$Me(CH_2)_7OH$
$Me(CH_2)_{10}CN$
$PhCH_2OH$
$tBuCH_2OH$
Pr_3COH

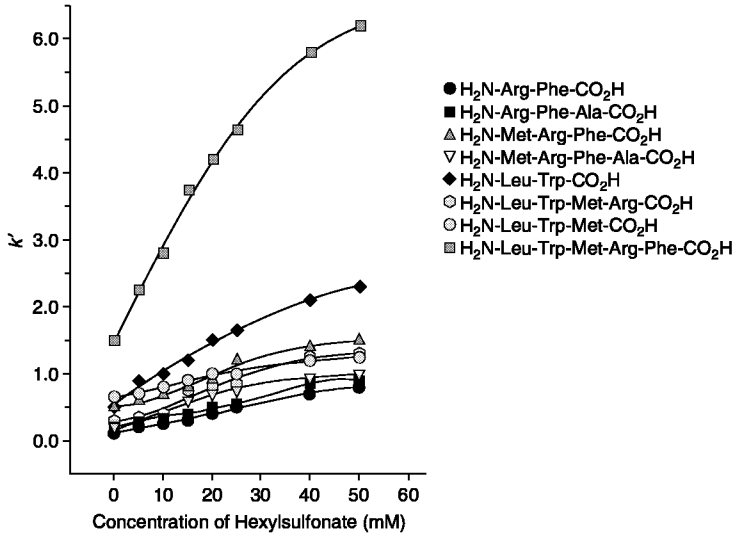
[a] Many of the ions identified in this selection of neutral species are generally compatible with UV detection < 235 nm, or in the case of those reagents with an aromatic nucleus, with fluorometric or UV (> 254 nm) detection. (Typically, the concentration of these components in the mobile phase is < 5 mM for peptide separation and analysis, although concentrations > 5 mM can be employed to optimize resolution.)

[b] For a more complete compendium of applications, conditions, etc., see refs[17,20,160,342].

Illustrative of the effects of these reagents on the retention and selectivity of peptides with high-performance *n*-alkylsilica sorbents are the plots of k' versus the concentration of the ion-pairing reagent, hexylsulfonic acid, shown in Figure 16. This example highlights several characteristics implicit in all nonpolar in hydrophobic ion-pairing reagents, irrespective of whether they are anionic or cationic. Firstly, as the concentration $[X]$ of the ion-pairing reagent increases, the k' value of the peptide increases. This behavior is in accord with the concept that the ion-pairing reagent selectively interacts with the peptide, changes its effective hydrophobicity in a concentration-dependent manner. Such interactions can occur either in the bulk mobile phase or as a dynamic liquid–solid binding event at the surface of the RPC sorbent. Secondly, the increase in the k' value follows an asymptotic dependency on $[X]$, with a maximum value determined by the nature of the ion-pairing reagent. With most anionic ion-pairing reagents, maximal retention occurs at $[X]$ values typically in the range of 15–50 mM. For most applications with synthetic peptides, the chosen concentrations of these ion-pairing reagents usually falls within the range of 10–15 mM for optimal control over selectivity, although in some cases with more hydrophobic peptides, higher concentrations, i.e. up to 50 mM, have been employed. Thirdly, the shape of the k' versus $[X]$ dependencies for polypeptides will be dependent on the composition and sequence of the polypeptides in the mixture. Consequently, selectivity reversals can be achieved through use of different concentrations of the ion-pairing reagent. Finally, the shape of the k' versus $[X]$ dependencies for polypeptides depends on the mobile-phase pH. Thus, different approaches can be used with ion-pairing reagents with mobile phases of different pH values for the separation of complex, heterogeneous mixtures of synthetic or naturally occurring peptides. In the first approach, the heterogeneity of the peptide mixture is associated with substitution, deletion, or incomplete deprotection of specific amino acid residues related to the peptide of interest. In this case, it is prudent to use pH conditions that maximize the differences in charge state between the different peptide species. The correct choice of these mobile-phase conditions in terms of the selection of organic solvent, pH, or ion-pairing reagent thus becomes important to achieve appropriate selectivity differences. This selection can often be guided using high-performance capillary electrophoresis (HPCZE) methods as a prelude, whereby the charge dependencies of the peptides can be elucidated. In the second approach, the heterogeneity of the peptide mixture is due to racemization effects that lead to significantly different conformational species and thus different effective hydrophobicities. In this case, elution conditions that favor solvent-mediated peptide dipole–dipole interactions, i.e. the use of mixed ternary organic solvent systems, are often more appropriate. Such approaches have been generally applied to many diastereomeric or deletion peptides prepared by solid-phase peptide synthesis.

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Figure 16 Representative Asymptotic Dependency of the Relative Retention of Peptides on the Concentration of an Ion-Pairing Reagent in RPC[23][a,b]



[a] In this figure, plots of k' versus the concentration of the ion-pairing reagent, hexylsulfonic acid, in the mobile phase for several peptides, are shown [μ-Bondapak C18 sorbent; flow rate: 2.0 mL·min^{-1}; $T = 20\,^{\circ}C$; eluent: 50% MeOH/50% H_2O/50 mM NaH_2PO_4 with H_3PO_4 added to pH 3.0, containing various concentrations of the ion-pairing reagent].

[b] Adapted from Hearn and Grego,[23] with permission.

7.1.2.7 Impact of the Organic Solvent and Other Mobile-Phase Properties on Peptide Separation in Reversed-Phase Chromatography

A large variety of organic solvents can be employed for the RPC separation of peptides. As discussed above, acetonitrile, methanol, ethanol, propanol, and propan-2-ol have been the most commonly used organic solvents with synthetic or naturally occurring peptides, although other organic solvents such as butanol, acetone, tetrahydrofuran, and dioxane, typically between $0 \leq \psi \leq 0.2$ have also found application with more hydrophobic molecules. Although ternary organic solvent systems have been used in some applications, binary water/organic solvent mobile phases are favored, in part because they can be prepared with greater reproducibility.

Several criteria can be applied to guide the selection of the organic solvent employed for a particular RPC separation of a peptide mixture. Firstly, the issues of purity and UV optical transparency of the solvent must be taken into account. As detection of most synthetic or naturally occurring peptides is commonly carried out by UV spectroscopic methods, very high purity solvents with good optical transparency at 215 nm are essential in order to ensure small baseline changes and good detectability of the separated peptides. Secondly, attention must be given to the relative elutropicity of the solvent. As discussed above, the molecular architecture of MeCN/H_2O mobile phases is fundamentally different to that found, for example, with mobile phases prepared with methanol or the other alkyl alcohols. These differences in hydrogen bonding and dipole characteristics will be reflected in the selectivity of an RPC separation through the dependencies of ln k' on ψ. Moreover, higher alcohols, e.g. the propanols, are "stronger" solvents in RPC than the shorter-chain alcohols or acetonitrile, in accord with their relative elutropicity values. Thirdly, solvents with larger molal surface tension increments will show steeper dependencies of ln k' on γ (and hence on ψ) and thus will tend to generate higher resolution when shallow gradients are used with peptides. Fourthly, solvents that tend to stabilize the conformation of larger polypeptides as a "single"

species are more likely to result in higher peak efficiencies and thus higher purification factors for recovered fractions. Finally, organic solvents that generate mobile phases of low viscosity are to be favored simply because they do not generate high column backpressures under most commonly employed flow rate regimes.

Since small increases in eluent strength can dramatically change the binding properties of peptides in RPC, the mobile-phase composition, in terms of the organic modifier content, must be prepared with very high precision for both isocratic and gradient elution separations. Attention has to be given to the order of adding the different components of the mobile phase, since volume expansion or contraction can arise due to exothermic or endothermic heats of mixing, thus generating an error in the value of the concentration or volume fraction of the organic solvent modifier. Care in the preparation of mobile phases cannot be overemphasized. Attention must also be taken when degassing or storing these mobile phases to avoid compositional changes caused by evaporation of the organic solvent. Filtration of all mobile phases through 0.25-μm pore size membranes prior to use is essential, whilst appropriate degassing and purging of the mobile phase with a neutral gas is important prior to or during use with all high-pressure systems.

Most HPLC equipment currently available has a high tolerance to most mobile-phase conditions that can be contemplated for use in RPC applications with peptides. If it is intended to use mobile phases containing halide salts in RPC separations of peptides with standard HPLC equipment made from type 316 stainless steel, it is essential that the equipment is properly flushed with neat water when not in operation to avoid corrosion by the residual halide ions, especially at low pH. Otherwise, the use of the less popular "biocompatible" metal-free HPLC equipment, marketed by several manufacturers, avoids potential problems of equipment malfunction due to corrosion of the stainless steel or the contamination of peptide samples by low levels of leached metal ions. With such metal-free HPLC equipment, titanium, glass, or perfluoro-polymeric components have been used to replace any wettable stainless steel components.

7.1.2.8 Surface Tension Effects Mediated by Solvents, Salts, and Detergents and Their Consequences on the Resolution of Peptides by Reversed-Phase Chromatography

In the RPC of peptides, irrespective of what type of organic solvent is employed, selective desorption of individual peptides is achieved through lowering the surface tension of the mobile phase. As discussed above, advantage can be taken of the molal surface tension increment, σ, of a specific solvent (or the presence of other additives such as a nonionic detergent) to affect the desorption, either via isocratic, step, or gradient elution methods. Retention of a particular peptide will thus be proportional to the relative change in the σ value of a solvent. This physical relationship can be used to provide the basis for all preparative applications with synthetic peptides when multicomponent mobile phases are required for a specified surface tension range.

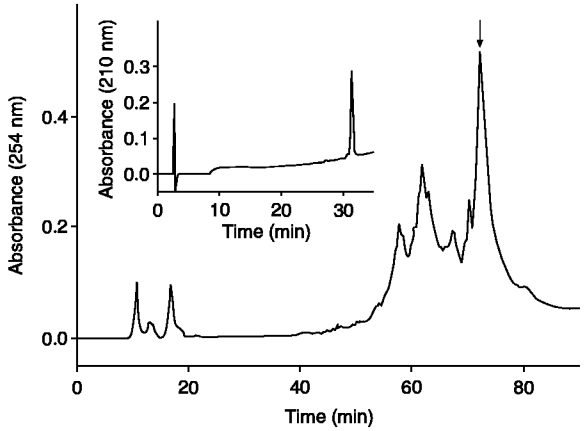
Since retention of peptides in RPC is largely controlled by the surface tension of the mobile phase, pH changes per se are not expected to be a dominant contributor to the global selectivity exhibited by homologous peptides, in accordance with the predictions of solvophobic theory. On the other hand, pH changes affect the net charge of the peptides and thus ionization differences will be manifested as selectivity changes within specific regions of a RPC profile. Perturbation of the structure of the hydrophobic contact sites through titration of ionizable groups or "salt" bridges such as Glu → Lys will thus affect the overall resolution of the RPC separation. When the pH of the mobile phase approximates the isoelectric point of the peptide, such changes in retention will become particularly evident. When such cir-

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cumstances prevail, a second phenomenon due to an intermolecular hydrophobic effect can arise, namely the preferential salting-out of the peptide, leading to precipitation.

Obviously, with very hydrophobic peptides, specially tailored elution conditions must be designed to achieve successful RPC separations. Often peptides such as the N-terminal secretory leader sequences of proteins, e.g. the very hydrophobic 22-mer peptide[161] from the sperm tail protein, tpx-1, illustrated in Figure 17, or alternatively peptide fragments of integral membrane proteins have low water solubility and are frequently strongly adsorbed to RPC sorbents even with mobile phases of high organic solvent content. In these circumstances, desorption of such hydrophobic peptides from the RPC sorbent can often be improved by use of a higher concentration of formic acid (or acetic or another alkanoic acid), e.g. 40–60% in the mobile phase, or by the presence of a low concentration of nonionic (e.g., Triton X-100), ionic (e.g., sodium dodecyl sulfate, cetyltrimethylammonium bromide), or zwitterionic (e.g., CHAPS) detergent,[162,163] together with an organic solvent of high elutropicity (e.g., iPrOH or BuOH). From solvophobic considerations, detergents/surfactants can be anticipated to alter the surface tension of mobile phase, and thus affect the retention behavior of synthetic peptides in RPC systems. The magnitude of the retention, the selectivity, and the adsorption kinetics will all be altered in a concentration-dependent manner by including these additives in the mobile phase. When the tpx-1 peptide described above is rechromatographed on the same types of RPC sorbents but with more hydrophilic ion-pairing regents in the mobile phase, multiple peak zones are all generated with the same composition. The participation of conformational and/or isodesmic self-association effects are the hallmark of such behavior.

Figure 17 Elution Profiles for the Analytical RP-HPLC Analysis for the Purified (Insert Panel) and the Preparative RPC of the Crude Synthetic 22-mer Polypeptide Derived from the N-Terminal Secretory Leader Sequence of the Sperm Tail Protein, tpx-1, a Member of the Cystine-Rich Secretory Protein Superfamily[161][a,b]



[a] The crude synthetic 22-mer polypeptide was purified by gradient elution RP-HPLC [Waters 600/486 HPLC system; TSK-ODS-120 T column (300 × 21.5 mm i.d.), 10 µm octadecylsilica; eluents: A: 0.1% TFA/H_2O, B: 0.09% TFA in 90% MeCN/H_2O, linear gradient of 0–100% B over 90 min; UV detection (254 nm); flow rate: 6.0 mL · min^{-1}]. The purified 22-mer polypeptide, which eluted after completion of the gradient, was analyzed by analytical RP-HPLC [TSK-ODS-120 T column (150 × 4.6 mm i.d.); 5 µm octadecylsilica; eluent: linear gradient of 0–100% B over 25 min; UV detection (214 nm); flow rate: 1 mL · min^{-1}].

[b] Adapted from Jong, Keah, Tawados, and Hearn,[161] with permission.

A better alternative is to use nonpolar additives or ion-pairing reagents that have the ability to modify the RPC sorbent surface into dynamic liquid–liquid phases. Because of their properties, such dynamic coatings frequently provide improved solubility for highly hydrophobic peptides in the mobile phase as well as providing enhanced resolution. One downside of employing such dynamic liquid–liquid phases is that long re-equilibration times are re-

quired due to the slow mass-transfer kinetics. Illustrative of the use of nonionic detergents in the RPC resolution of nonpolar peptides is the analysis and purification of polypeptide fragments of human erythrocyte acetylcholine esterase and other membrane proteins[164,165] with wide-pore RPC sorbents, e.g. TSK-phenyl-RP or Vydac C_4, and water/organic solvent mobile phases containing 1% (w/v) octylglucoside or Brij 35. Adaptation of this dynamic coating concept with RPC systems provides a very interesting approach to study the behavior of amphipathic peptides in artificial biomembrane-like environments utilizing phospholipids. Under such conditions, RPC sorbents can be dynamically coated with various phospholipids. Densities approaching 2 $\mu mol \cdot m^{-2}$ can be generated,[166,167] which are close in value to those found for lipid vesicles. The alternative approach involving the chemical immobilization of phospholipids, such as phosphatidylglycerol, has proved to be less reliable due to the heterogeneity of the sorbent surface and constraints in their synthesis. Various experimental results confirm that the retention behavior of peptides with noncovalent, chemisorptively immobilized artificial biomembrane-like RPC sorbents show excellent correlations[166] with the lipid-binding affinities of phospholipid vesicles. In these cases, correlations based on indices of accessible hydrophobic surface areas of these peptides can be developed and related to the relative elution order. This knowledge can then be used to facilitate the purification of amphipathic nonpolar peptides and their analogues from complex mixtures.

7.1.3 Operational Considerations with the Reversed-Phase Chromatography of Peptides

7.1.3.1 Basic Strategy Considerations

Important progress, based on the application of various predictive algorithms,[152,153,168–172] has been made over the past 10 years in the selection of isocratic and gradient elution procedures as part of the optimization of the resolution of a particular peptide mixture. Furthermore, it is feasible in circumstances of regular retention behavior and high mass recovery to apply data derived from analytical or ultra-micropreparative experiments as normalized integrals of the elution volume, column diameter, etc., to scale up the performance of the chromatographic bed in terms of the configuration and length of the preparative column or the choice of the physical characteristics of the sorbent to achieve large-scale RPC peptide separations.[117,173–175] Such approaches underpin the current popularity of RPC procedures for the process purification of many synthetic or recombinant polypeptides.

In preparative applications, irrespective of whether linear, overload, or displacement elution chromatographic conditions are employed, it is mandatory in the design of preparative RPC separation systems that attention be specifically given to the issues of mass recovery and selectivity. In this regard, some key parameters can often be readily controlled, for example, by operating the preparative RPC separation at lower or higher temperatures, by making a better pH choice, or by minimizing column residency times, etc. Also, inherent to all RPC separations of peptides is the question of to what end use the separated products be applied. If the objective is only the subsequent primary structure determination of the peptide, then the purification task can usually be undertaken at the ultra-micro- to semipreparative scales with standard laboratory columns (e.g., 250 mm × 4.6 mm i.d.) and traditional HPLC instrumentation. Obviously, with a new or partially characterized naturally occurring peptide, high mass and bioactivity recoveries are essential, even at these ultra-micro- to semipreparative levels. When the task involves large-scale preparative procedures, it is similarly important that the process RPC separation system also addresses these recovery and selectivity issues, particularly if a pharmaceutical end-point is contemplated for the use of a very high purity peptide. The issues of achieving high recovery, high resolution,

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high capacity, and high speed within the same separation system have challenged peptide chemists for many years. Current practice usually sacrifices separation capacity in order to optimize the other three parameters in process applications in accordance with the heuristic concepts developed in the late 1980s for peptide and protein purification by HPLC techniques.

At the microscale level on the other hand, the application of multidimensional and multimodal RPC separation techniques with peptides in conjunction with mass spectrometric procedures has been in vogue[176–178] for a number of years. Computer-assisted routines form an integral part of these approaches with RPC systems in the selection and optimization of the multidimensional chromatographic conditions, including the choice of mobile-phase composition with different ion-pairing systems, pH conditions, or organic solvents.[10,17,24,37,179] Moreover, when an ion-exchange step has been employed immediately prior to a RPC procedure for peptide fractionation, advantage can be taken of the ability of the RPC conditions to de-salt the peptide sample with minimal handling losses. Clearly, care must be taken in the choice of multidimensional and multimodal RPC protocols to ensure that the chromatographic conditions are fully compatible with the task in hand, i.e. complex elution conditions always tend to be less robust than simple isocratic elution procedures.

7.1.3.2 Monitoring the Structure, Conformation, and Interactions of Peptides in Reversed-Phase Chromatography Systems by Different Spectroscopic Procedures

In order to monitor the purity of a synthetic or naturally occurring peptide following separation by RPC procedures (or for that matter by other chromatographic or electrophoretic methods) it is essential to use several orthogonal detection methods, preferably on-line as the peak zones elute, such as electrospray ionization mass spectroscopy (ESI-MS)[128,180,181] and diode array (DAD) or derivative ultraviolet–visible spectroscopy (UV–vis). The opportunity also exists to carry out on-line circular dichroism–optical rotary dispersion (CD-ORD) spectroscopic assessments[182,183] to assess the α-helical, β-sheet, and random coil content of a polypeptide as it elutes from an RPC column. In order to evaluate fully the effects of the molecular forces involved in peptide retention with hydrophobic chromatographic surfaces, a comprehensive biophysical experimental strategy is similarly required for a particular separation. Ideally, these measurements should be carried out as part of the purification tasks as "on-line" and in situ activities, but experimental constraints can preclude these possibilities. Time-resolved polarization fluorescence,[184,185] diffuse reflectance Fourier transform infrared spectroscopy (DRIFT),[186,187] laser activated light scattering (LALLS), Raman, and other spectroscopic techniques[188] can also be employed to probe different physicochemical aspects of peptide interaction with chromatographic ligands at the molecular level. On-line, stop-flow ^{1}H NMR techniques do not currently have the spectral sensitivity to permit rapid characterization of peptides as they elute from an RPC column, although progress with miniaturized probes for coupling microscale HPLC systems with on-line NMR is encouraging.[189] With the imminent availability of very high field ^{1}H NMR or solid-state heteronuclear NMR spectrometers with miniaturized probes,[190,191] such measurements may become more accessible in the near future. Similarly, the current generation of matrix-assisted laser-desorption ionization time-of-flight (MALDI-TOF) spectrometers[126,192–197] or scanning differential microcalorimetery[197,198] can be employed as off-line techniques to examine other molecular features of the interaction.

For rapid confirmation of the molecular mass of a recovered peptide following RPC purification, either ESI-MS or MALDI-MS procedures are now indispensable. Studies on the interaction of peptides with solvent molecules, ions, and free or immobilized ligands in chromatographic or electrophoretic environments have, however, been largely confined to methods that examine global properties of different peptides in the bulk mobile phase. In

this case, the assumption has frequently been made that interconversion between different conformational or association states following desorption from an RPC sorbent are associated with fast kinetics. Precise information on the microenvironmental changes that occur in the physical state or chemical properties of a peptide in contact with a particular type of chromatographic surface and shortly after desorption has usually not been obtained to substantiate claims made, for example, about peptide orientation or conformational relaxation. As a consequence, caution must be observed to avoid over-simplified interpretations of experimental results, as has arisen in some studies[199] related to the elution profiles of antimicrobial peptides with nonpolar chromatographic sorbents. Availability of powerful computational modeling methods, coupled with experimental data derived with orthogonal spectroscopic procedures, now provides a more rigorous basis to explore in greater detail the precise mechanisms of interaction and the molecular forces involved in the recognition events when peptides dock with solvated immobilized nonpolar ligands. For example, application of $^{1}H \rightarrow {}^{2}H$ amide exchange ESI-MS techniques,[180,200] surface-activated ionization mass spectrometry (SAI-MS), or in situ heteronuclear 2D and 3D NMR[201,202] permits the solvational features and molecular orientation of peptides in association with chromatographic surfaces to be more accurately delineated.

Significant peak shape distortions of very hydrophobic peptides can result from self-association in RPC, a feature commonly seen with these peptides in most other modes of interactive chromatography. Such behavior is a consequence of a specific molecular property of the peptide, the propensity to form amphipathic helices or fibril-like structures in solution, leading to isodesmic self-associate and multiple peak zones for an otherwise compositionally homogeneous sample. Under well-characterized RPC conditions of defined pH and temperature, the participation of such self-association processes can be assessed[203,204] by on-line light scattering (LALLS), thermometric HPLC procedures, and, in some cases, by CD-ORD spectroscopic procedures through an examination of the peak shape response as a function of sample concentration as part of the purification activities. Again, the use of low concentrations of a detergent or high concentrations of formic acid or other alkanoic acids in the mobile phase may remedy this behavior in the RPC separation and permit adequate resolution and recovery.

The elution profiles of peptides achieved in size-exclusion chromatography (SEC) or other non-interactive separation techniques such as HP-CZE largely reflect a bulk property of a peptide in terms of its average molecular weight or average Stokes hydrodynamic radius, which is often taken to imply a fixed physical state for rapidly interconverting conformers. When used in conjunction with RPC methods, such procedures also enable the participation of slowly interconverting conformers to be monitored, and their effect on the chromatographic resolution to be assessed. In reality, all peptides undergo continuous, dynamic changes in their shape, self-association and mobility in response to variations in the properties of the surrounding liquid environment. These environmental changes affect the chemical potential of the peptide and result in reversible solution equilibrium processes. When such phenomena are rapid (i.e., with half-lives in the range of nanosecond to millisecond) they have minimal influence on RPC selectivities or peak shapes; to all intents and purposes, the immobilized nonpolar ligands are sampling the same set of rapidly averaging molecules. However, when these phenomena have long relaxation times (i.e., of the order of seconds or even minutes), major variations in selectivity can arise with the RPC separation. As noted above, these effects are particularly evident with proline-rich peptides, where the long-lived proline *cis/trans*-isomerization effects can give rise to multiple peak zones for the same peptide. When such effects arise, cognizance has to be given at the preparative level to the resulting very complex adsorption isotherms, since these directly affect the column loadability and resolution. Again, knowledge of effects associated with the interaction of a particular peptide with components in the mobile phase or those present on the stationary phase surface, and optimization of their consequences using, for example, the G2-based

for references see p 622

Expert artificial intelligence system,[205] are particularly pertinent to the development of robust separations strategies for the RPC isolation of larger synthetic peptides.

Overall, a panel of different experimental procedures is needed to substantiate the purity of peptides eluted from RPC systems. Spectroscopic techniques readily provide much of this information. Wavelengths that are typically monitored during peptide isolation by RPC methods include the regions of isobestic absorbance of the amide backbone (210–220 nm), or the phenylalanine (254 nm), tyrosine and tryptophan (280 nm), cystine (240 nm), or histidine (228 nm) side chains. On-line photodiode array detection (DAD), utilizing difference spectroscopy, absorbency ratio spectroscopy, or second derivative spectroscopy at different wavelengths can be employed to assess the purity of the individual eluted polypeptide peaks from the extent of exposure of aromatic amino acid residues, and tyrosine in particular, to solvent. Moreover, peak identification of peptides via peak tracking algorithms[206,207] enabling conclusive determinations of peptide purity can also be achieved with UV–vis DAD methods. Similarly, fluorescence detectors provide higher sensitivity, as well as greater selectivity, in the detection of peptides containing tryptophan residues through their intrinsic fluorescence. The extent of solvent exposure of tryptophan residues can be assessed from the time-resolved fluorescence polarization spectral measurements, providing an opportunity to assess the conformational status of the purified peptide(s) that contain these residues.

Mass spectrometry in its different modes is now the ultimate method for the characterization of the mass, composition, and sequence of peptides in a mixture, and is particularly well-suited to the mobile phases employed in RPC. Electrospray ionization (ESI) and matrix-assisted laser desorption ionization time-of-flight (MALDI-TOF) mass spectroscopic methods, and to a lesser extent thermospray and fast atom bombardment (FAB) mass spectrometry, can all be very successfully employed for these purposes. The routine application of tandem RP-HPLC–ESI-MS of peptides, with suitable stream-splitting devices and electrospray interfaces has greatly expanded, permitting mass spectra to be acquired on-line for most classes of peptides. In particular, ESI-MS or, alternatively, continuous-flow FAB-MS[208,209] can be operated at flow rates of $\leq 50\,\mu L \cdot min^{-1}$ or ca. $1–10\,\mu L \cdot min^{-1}$, respectively, with in-line microbore and capillary columns with/without postcolumn stream splitting. These methods represent two very powerful approaches to characterize peptides derived from enzymatic digestions of proteins or from combinatorial solid-phase synthesis.

Finally, when RPC methods are used in preparative studies with peptides, the opportunity routinely exists for subsequent analysis of the recovered fractions by a variety of analytical methods including high-speed RP-HPLC, HP-IEX, HP-HILIC, or HP-IMAC, zonal or micellar electrokinetic high-performance capillary electrophoresis (HP-CZE and MECK-CZE), capillary electrochromatography (CEC), or capillary isotachophoresis. The combination of the RPC information, drawn from the ln k' versus ψ plots, with the data derived from on-line spectroscopic detection thus readily provides a comprehensive opportunity to assess the purity of an isolated peptide, many of the physicochemical features of the interaction, as well as a means to optimize the resolution in the RPC separation.

7.1.3.3 Resolution Effects with Peptides Undergoing Interconversion in Reversed-Phase Chromatography Systems

As evident from the discussion above, during the isolation of a peptide, RPC procedures also offer considerable scope as biophysical procedures to reveal important information on the nature and magnitude of the changes in Gibbs free energy for the interaction of the peptide with nonpolar ligands and any associated conformational changes, and the enthalpy or entropy changes associated with these processes. Thus, analytical and micropreparative RPC techniques have also been employed to resolve conformational intermediates of peptides with

relaxation half-lives of similar magnitude to the mass transport time (i.e., $\tau_{conf} \approx \tau_{mass\ transfer}$, where $\tau_{mass\ transfer} > 10$ s) of the peptide in the RPC system. These attributes have important ramifications for the selection of RPC conditions in a purification procedure. For example, if a proline-rich peptide were to undergo, in a highly simplified case, only a two-stage interconversion between the *cis* and *trans* backbone isomer forms in the mobile phase or at the surface of an RPC sorbent, then peak broadening or even individually resolved peaks could become evident as the temperature was lowered. At higher temperatures, on the other hand, only a single peak zone might be evident. Various investigations have confirmed that a variety of peptides exhibit such conformational interconversions, particularly if the peptide contains structural constraints due to an L-α-/D-α-proline, β-amino acid residue or α-/γ-lactam residue replacements. The example of the tpx-1 leader peptide[161] described above is illustrative of this behavior. In this case, the relative abundance of the two species is dependent on the temperature, residency time, and mobile-phase composition, despite the fact that both species have the same sequence and molecular mass. In such cases, distorted and/or multiple peak zones for a specific peptide, homogeneous in terms of its composition and mass spectral properties, can arise as a consequence of the chromatographic residency and temperature conditions when a peptide mixture is subjected to RPC separation. When such interconversion processes, involving transitions of the peptide between $P_N \rightarrow P^*_U$ states, occur in the mobile or stationary phase, the binding process can still be represented as a reversible event involving pseudo first-order interconversion kinetics, but with the column now also effectively behaving as a "chemical reactor" whose characteristics can be specified in terms of unique Damkohler numbers, D_a, characteristic of the peptides. The Damkohler number represents the ratio of the time taken by a peptide to passage along the RPC column to the overall relaxation time for all conformation interconversions of the peptide in the chromatographic system. When the D_a is either very small or very large, kinetic effects associated with conformational interconversion essentially vanish. However, at intermediate values, whereby the peptides behave within the RPC column such that $1 < D_a < 10$, complex elution profiles will be obtained, which are dependent on the intrinsic column efficiency, the equilibrium distribution constants for the various species P_N, P^*_U .., etc, the rate of interconversion of $P_N \rightarrow P^*_U$ .., etc, and the type of RPC conditions used. The *cis*/*trans* interconversion of proline-rich peptides in RPC systems noted above represents a classic example of this type of interconversion behavior. As a consequence, alertness is needed with peptides that can undergo this type of conformational reordering in either the mobile phase or stationary phase, since peptide elution in multiple zones can be anticipated, with the relative mole fraction of each species dependent on the nature of the RPC sorbent, the mobile-phase composition, time, peptide concentration, and temperature.

A further type of multizoning phenomenon can occur in the RPC separation of peptides due to nonlinear isotherm behavior arising from the heterogeneity of the support matrix and non-uniformity in the distribution of the nonpolar ligands over the stationary-phase surface. This effect is most noticeable when results for "virgin" columns are compared with those obtained for "conditioned" columns. For this reason, the use of a peptide "standard" often fulfills a useful role in calibrating and comparing the performance of different columns and different batches of the (notionally) same RPC sorbent. With the improved manufacturing technologies now available, batch-to-batch variability does not represent the same level of problem, although with process-scale RPC sorbents, a concern of some end-users still remains the issue of batch equivalence.

for references see p 622

7.1.4 Practical Aspects of the Reversed-Phase Chromatography Separation of Peptides

7.1.4.1 General Considerations

Due to the high resolving power of these techniques, sample preparation is usually straightforward. The sample should always be appropriately filtered prior to loading onto a RPC column. Selection of the operating variables for the RPC separation of peptides follows the common set of criteria outlined above, with the flow rate, mobile-phase composition, and temperature the key parameters to optimize once a particular RPC sorbent has been selected. With high-performance RPC sorbents, rapid separations of peptides can be achieved with crude feedstocks at low to high temperatures under conditions that also favor the desalting of the sample. Because of their flexibility, RPC procedures can be applied either early or late in a purification protocol, depending on the task. This versatility readily permits purified peptide(s) to be obtained in the mass range from micrograms to grams. Since RPC utilizes an organic solvent as the stronger eluent, recovery of peptide fractions from RPC separations typically requires only a lyophilization step. In order to obtain the eluted peptide in fractions of higher concentration, various approaches have been examined, including the use of displacement modes of RPC,[210,211] techniques that have been found with low-molecular-weight organic compounds to result in an increase in the concentration of the eluted components by about an order of magnitude compared to what can be achieved by a zonal separation in a single run. Despite extensive explorative work, application of self-displacement, tandem-displacement, or multicomponent-displacement RPC for the separation of peptides has not gained much popularity, due to the limited availability of suitable displacement compounds, although some initial success with some low-molecular-weight displacers with small peptides have been reported.[212,213]

Following the selection of the mobile and stationary phase conditions, the other operating parameters in an RPC separation of a peptide mixture must be chosen with care so as to achieve optimal resolution and column efficiency without sacrificing adequate control over the binding strength or conformational state. The efficiency of the chromatographic separation is determined in part by the mobile-phase flow rate. Peptides typically have considerably smaller molecular diffusivities than small organic molecules, and thus column efficiencies in the RPC separations of peptides are much more flow-rate sensitive. With micropellicular or perfusive RPC sorbents, high linear flow velocities can be employed due to the reduced diffusional resistances, resulting in higher efficiencies as well as reduced distortion of peak shapes compared to the many more conventional porous sorbents.

As noted above, the effects of temperature on the retention of peptides in RPC is more complex than found for small molecules. For example, for small molecules in RPC, column efficiencies typically increase whilst retention decreases as the temperature is increased. With peptides, on the other hand, retention often increases with temperature due to conformational re-ordering of the peptide structure in the RPC mode, although decreased retention times can arise at some specific temperature values due to the "non-ideality" effects described earlier. The choice of mobile-phase conditions in isocratic or gradient elution also more significantly influences the retention and peak-broadening behavior of peptides than found for small organic molecules. Due to the relatively large contact area between a peptide and the nonpolar chromatographic surface in RPC, the binding is particularly sensitive to the concentration of the organic solvent in the eluent. For the reasons described above, conformational changes, including aggregation, may arise in either the isocratic or gradient elution modes. With inappropriate isocratic or gradient mobile-phase compositions, the residence-time effects may result in kinetically controlled conformational interconversions of larger peptides in the RPC environment, generating complex elution profiles under these non-optimized conditions.

7.1.4.2 Applications of Reversed-Phase Chromatography in Peptide Isolation

RPC procedures currently provide a greater range of complementary methods and more versatile opportunities for the separation of peptides than those offered by ion-exchange, size-exclusion, or the other chromatographic or electrophoretic procedures. As noted above, over 2000 citations were recorded in 1999 on the use of RPC procedures for the separation and purification of peptides, and this number has been increasing at the rate of about 10% per annum over the past decade. Thus, numerous examples of RPC separations of synthetic and naturally occurring peptides of importance to the pharmaceutical and biotechnology industries can be identified, including those detailed in Table 4. Typically, in all of these applications, wide-pore RPC sorbents of moderate hydrophobicity have been employed in conjunction with either 0.1% TFA/MeCN/H_2O or low-pH methanol, ethanol, propanol, or propan-2-ol/H_2O mobile-phase systems. The examples above, drawn randomly from the scientific literature, highlight the various uses of RPC procedures for the analysis and purification of peptides and lead to several conclusions. Firstly, the majority of the applications, to date, related to the high-resolution separations of peptides by RPC techniques on laboratory scale as well as on industrial level have involved the use of silica-based sorbents. Secondly, the evolution of suitable RPC protocols from the pioneering investigations of the late 1970s through the 1980s has now reached the point where synthetic and naturally occurring peptides can be separated at the 1 μg to kg level per injection; this is now achieved faster, with greater efficiency, and to a higher level of resolution and purity by RPC procedures than by any other chromatographic technique. Moreover, RPC procedures have proved very amenable to large-scale applications where GMP scale-up and rigorous analytical characterization is mandatory.

Table 4 Illustrative Examples of Synthetic and Naturally Occurring Peptides Isolated by RPC Procedures

Peptide	Ref
116-mer peptide containing the tandem cysteine-rich C1 domains of protein kinase C gamma	[214]
antibacterial acaloleptins A inducible peptides from larvae of the beetle *Acalolepta luxuriosa*	[215]
adenylate cyclase activating polypeptide (PACAP) from the brain of an elasmobranch stingray *Dasyatis akajei*	[216]
α-conotoxins isolated from the venom of *Conus consors*	[217]
amyloid polypeptide expression in human insulinomas	[218]
angiotensin and endothelin peptides	[219]
angiotensin I converting enzyme-inhibitory peptides from wine	[220]
antimicrobial defensin I peptide	[221]
antihypertensive peptides from a yogurt-like product fermented by *Lactobacillus helveticus CPN4*	[222]
bradykinin analogues	[223]
buforin I, a peptide inhibitor of botulinum neurotoxin B activity	[224]
bursicon, a cuticle sclerotizing neuropeptide from *Tenebrio molitor*	[225]
carcinoembryonic antigen peptides	[226]
casomorphin and casomorphin-like peptides	[227]
cholecystokinin-related peptides	[228]
chloroplast targeting peptides	[229]
diuretic myokinin neuropeptides from the housefly, *Musca domestica*	[230]
D-leucine-containing *Conus* peptides	[231]
double cysteine-rich motif 116-mer peptide of protein kinase C gamma	[232]
eledoisin peptides	[233]
glucagon-like peptide-1(GLP-1^{7-36}) amide, a potent insulin-releasing hormone	[234]
HLA-presented Epstein–Barr virus peptides recognized by CD4(+) or CD8(+) T lymphocytes from human blood	[235]
honey bee venom polypeptides	[236]
hydrophobic microcystins (peptide cyanotoxins)	[237]
hydrophobic peptides LH2 from *Rps. Palustris*	[238]
lichenysins G, lipopeptide biosurfactants from *Bacillus licheniformis IM 1307*	[239]
macrophage migration inhibitory factor	[240]

for references see p 622

Table 4 (cont.)

Peptide	Ref
neuropeptide Y analogues	[241]
neuroparsin A, a polytropic neurohormone of the locust *Locusta migratoria*	[242]
osteocalcin analogues	[243]
paralytic peptides from the hemolymph of the silkworm, *Bombyx mori*	[244]
Phe8 caerulein and the antibiotic peptide caerin 1.10 from the Australian tree frog *Litoria splendida*	[245]
polymyxin B1	[246]
POMC peptides derived from rat pituitary extracts	[247]
prolactin-releasing peptides in the bovine hypothalamic extract	[248]
pseudechetoxin, a peptide blocker of cyclic nucleotide-gated ion channels	[249]
pseudo-peptide analogues of thiol proteinase inhibitors	[250]
relaxin-like molecule from the male atlantic stingray, *Dasyatis sabina*	[251]
Respiratory syncytial virus protein G related peptides containing two multiple disulfide bridges	[252]
rice bran peptides	[253]
RGD peptides	[254]
superagonist peptides for a self-K-ras-reactive CD4(+) T cell clone	[255]
tachykinin-related peptides	[256]
TRH analogues	[257]
TRH-like peptides in rat thyroid	[258]
urocortin	[259]
urotensin I-like peptide in the pituitary of the lungfish *Protopterus annectens*	[260]
viscosinamide, a new cyclic depsipeptide with surfactant and antifungal properties produced by *Pseudomonas fluorescens DR54*	[261]

A further extensive field of application of RPC techniques relates to the structural characterization of proteins. Since their introduction[21,58,159,222] in the late 1970s, the separation of enzymatic maps of proteins by RPC procedures has become the dominant experimental tool for such structural studies. Differences in the peptide maps for different proteins digested with the same enzyme can be used to reveal compositional differences, whilst differences in the peptide maps that arise when different enzymes are used for the digestion can be employed to further identify and characterize the origins of these microheterogeneities. Moreover, the effects of amino acid substitution in small peptide fragments are attenuated in peptide maps as significant retention time changes, with the separation of peptides with structural/sequential variations relatively straightforward by these RPC procedures. Since some peptides within a complex mixture may co-elute under a specified RPC regime, the use of alternative mobile-phase conditions and appropriate computer-assisted optimization procedures should always be examined. In the case of peptide "maps" derived from the proteolytic cleavage of proteins, the sequence sites for fragmentation are usually known, or can be predicted with reasonable precision. Application of empirical rules can be employed then to facilitate the identification of the fragments following RPC separation from algorithmic analysis of the retention times.[32,34,35,223] Moreover, changes in intrinsic hydrophobicities of peptides can often be readily exploited either from knowledge of the hydrophobic fragmental coefficients of the amino acids within the sequence or following chemical modification of the peptide. Thus, generation of oxidation products such as methionine sulfoxide or cysteic acid often results in a significant (predicted) decrease in retention, whilst deamidation of a peptide often is associated with a slight increase in retention time in RPC under low pH elution conditions. Integration of this knowledge provides a particularly valuable basis to confirm the identity of a recombinant protein manufactured in a novel cell system by genetically engineered approaches compared to the authentic wild-type protein. For example, RPC separation and analysis of peptides derived from the enzymatic proteolysis of native and mis-posttranslationally modified proteins are now routine procedures. Such approaches have been employed to characterize Met from Nle variants of recombinant proteins, as well as to identify the molecular differences between the secreted and stored

forms of isoforms of glycoprotein hormones and other recombinant glycoproteins. Even with very complex mixtures, RPC methods offer a powerful method for characterizing glycopeptides, such as the tryptic peptides derived from the β-subunit of bovine thyrotropin, where 10 of the expected 11 peptides could be resolved,[224] in a single RPC experiment or recombinant tissue plasminogen activator (rt-PA), where 49 of the expected 51 peptides[225] could be identified.

The very high resolving power of RPC enzymatic mapping procedures in terms of their simplicity and high information content has been amply demonstrated, including in studies with recombinant t-PA,[226] seminal plasma cystine rich secretory protein,[227] hGH,[228,229] basic fibroblast growth factor,[230] endomorphins,[231] interleukin-2,[232] brain-derived neurotropic factor,[233] insulin-like growth factor 1,[234] keratinocyte growth factor,[235] therapeutic anti-CD4 IgG1 monoclonal antibody,[236] and in tracking the two-dimensional separation and proteolytic digestion of SMPT-modified RNase A with ESI/FTICR-MS and HPLC/ESI/QIT-MS.[193] As a consequence, peptide mapping by RPC and enzymatic procedures has become an essential requirement for the characterization of all naturally occurring proteins, rDNA-derived proteins and in the quality control of therapeutic proteins as part of Drug MasterFile registration. Since automated proteolytic mapping/RPC procedures are now available,[277] these on-line techniques when integrated with ESI-MS methods greatly streamline such characterizations.[237] In addition, algorithmic scoring techniques, which include spectral as well as retention time and peak area analysis by principal component analysis, allow computer-assisted comparisons of enzymatic maps of different batches of a given recombinant protein.[228,238] A challenge still facing the use of analytical RPC methods is the resolution of structural variants or contaminating species with the same relative hydrophobicity as the parent peptide but present at low levels (typically less than 2%) of abundance. At this concentration, the parent peptide effectively acts as displacer, masking the detection or separation of the contaminant. Fortunately, other high-performance separation procedures are available to address this constraint. In the remaining part of this section, some of the features of these alternative procedures will be discussed.

7.1.5 Achievement of Other Types of Chromatographic Selectivity of Peptide Separation and Purification

7.1.5.1 High-Performance Size-Exclusion Chromatographic Separations

Compared to hydrocarbonaceous silica RPC sorbents, not as much commitment has been made to the development of bonded, polar-phase sorbents suitable for the high-performance chromatographic separation of peptides. Due to polar, notably hydrogen bonding, interactions between the peptide and the hydrophilic surface of the sorbent useful selectivity effects can, however, be achieved. In fact, at least two types of separation mechanisms can be identified with bonded polar-phase sorbents. In the first mode, the peptides do not interact per se with the bonded polar-phase sorbent but, rather, are separated on the basis of their ability to permeate into the pores and elute in order of their hydrodynamic volume. In this mode, peptides are separated by steric exclusion effects, with the retention (in terms of elution volume, V_e) of a partial retained peptide, P_i, described by the following relationships:

$$V_{e,i} = V_o + K_{p,i}V_i = A + n \ln (M_r)_{p_i} \quad \text{eq 37}$$

where V_o is the void volume and V_i is the total pore volume of column, M_r is the molecular weight of the peptide(s), and A and n are constants characteristic of the HP-SEC system. The term $K_{p,i}$ is called the permeation coefficient, which is assumed in ideal cases to be $0 \leq K_{p,i} \leq 1$. However, when hydrogen bonding or electrostatic effects occur between the

for references see p 622

peptide and the HP-SEC sorbent, then non-ideal dependencies arise with $K_{p,i} > 1$. Because $V_{e,i}$ is related empirically to the logarithm of the molecular weight of a peptide [i.e., to $\ln(M_r)$] for peptides encompassing a suitable range of molecular weights, the selectivity in HPSEC between two peptides P_i and P_j will be controlled by the slope, $n_{i,j}$, expressed as

$$n_{i,j} = \frac{\Delta \ln MW}{\Delta V_{e,i,j}} \quad \text{eq 38}$$

Alternative expressions linking $V_{e,i}$ to M_r have been developed[239] for peptides, based on the relationship between the radius of gyration, R_g, and the molecular mass, M_r, i.e. $R_g \propto M_r^{\alpha}$ where $\alpha = 1$ for prolate or rod-like peptides, $\alpha = 1/2$ for flexible coiled-coil peptides, and $\alpha = 1/3$ for spherical peptides. For compact coiled-coil peptides, the hydrodynamic volume, V_h, can be calculated as follows:

$$V_h = \frac{\eta M_r}{\upsilon N_A} \quad \text{eq 39}$$

where η is the intrinsic viscosity, expressed as the volume of molecules per unit mass, N_A is Avogadro's number, and υ is the Simhas factor. For spherical globular peptides, $\upsilon = 2.5$, whilst for ellipsoidal peptides $\upsilon > 2.5$. Application of the modified Himmel–Squire dependency[240] and related relationships[239] permits a useful expression to be employed for the characterization of the HPSEC selectivities of peptides:

$$-b = \frac{dK_{p,i}}{d \ln R} = \frac{dK_{p,i}}{d \ln M_r^{0.33}} \quad \text{eq 40}$$

where

$$K_{p,i} \approx a - b \ln M_r \quad \text{eq 41}$$

and a and b are related to the pore size distribution of the HP-SEC sorbent, whilst the slope of the selectivity curve, b, depends on the pore size distribution. Peptides with molecular masses encompassing the range up to 7×10^3 daltons can be successfully chromatographed under size-exclusion conditions on HP-SEC sorbents, such as the TSK G2000 SPW or the Pharmacia Superose 12 materials, with pore sizes of 130 Å to larger than 500 Å, respectively. With such systems, a variety of isocratic mobile-phase conditions can be employed, including 50 mM KH_2PO_4/100 mM KCl, with pH values in the range of pH 3.0–6.5. To achieve adequate resolution of adjacent peak zones with currently available HP-SEC sorbents, molecular weight differences corresponding to (at least) a doubling in the size or sequence length of each of the polypeptides is required. A major limitation of most HP-SEC sorbents currently available is that they do not consequently have appropriate pore size characteristics to permit the routine resolution of peptide mixtures produced by SPPS methods. Many polypeptides may also deviate from the ideal size-exclusion behavior due to interactions with the surface of the HP-SEC sorbent or, alternatively, due to conformational effects. In addition, most of the HP-SEC sorbents currently available exhibit some electrostatic characteristics, necessitating the use of mobile phases of relatively high ionic strengths and/or containing salts such as sodium chloride. Recovery of the separated peptides then requires a further step, usually an RPC procedure. Because hydrogen bonding and hydrophobic interaction effects can also give rise to anomalous retention behavior for peptides with HPSEC sorbents, a common practice is also to add a small amount of a suitable organic solvent to the mobile phase in these cases. For these reasons alone, HP-SEC methods have been relegated to the final stage of purification of an occasional peptide or for some specialized analytical studies where quantitation of the presence of a peptide monomer, dimer, etc., may be required.

7.1.5.2 High-Performance Hydrophilic Interaction Chromatographic Separation of Peptides

High-performance hydrophilic interaction (HP-HILIC) and high-performance polar or normal phase (NP-HPLC) chromatography represent the second mode of HPLC separation of peptides with bonded polar-phase sorbents where the selectivity is largely based on hydrogen-bonding effects. In both of these modes, the separation selectivity can be significantly modulated through the use of solvents that strongly hydrogen bond with the sorbent and the peptide, such as methanol. Thus, basic or amphipathic polypeptides with a propensity for self-association can be resolved with good recovery from hydrophilic HP-HILIC sorbents, such as the PolySulfoethyl A sorbent.[241,242] With silica-based sorbents, the participation of type I and type II silanols results in significant hydrogen bond and zeta potential contributions, particularly when a low density of an introduced ligand is present. For this reason, these types of sorbents are better suited to HP-CEC applications rather than in the chromatographic separation of peptides. In the extreme case of a naked silica surface, participation of these hydrogen bond and zeta potential effects can lead to irreversible binding of peptides. However, with protected synthetic peptides, or amphipathic polypeptides, both NP-HPLC and HP-HILIC procedures can be very useful, particularly when used[243] as "flash chromatographic" methods.

With bonded, NP-HPLC sorbents, such as the porous aminopropylsilica sorbents, the distribution constant, $K_{abs,i}$, can be equated with the solubility parameter, δ_i, which in turn is a measure of the intermolecular interaction energy per unit volume of the polypeptide in a pure liquid such that

$$\ln k'_i = \ln \Phi + V_i \frac{[(\delta_i - \delta_m)^2 - (\delta_i - \delta_s)^2]}{RT} \quad \text{eq 42}$$

where δ_i, δ_m, and δ_s are the solubility parameters of the peptide P_i, the mobile phase m, and the stationary phase s; and V_i is the molar volume of the peptide P_i. As the term $K_{abs,i}$ ($= \ln k'_i/\Phi$) becomes larger, the polarity of s becomes more similar to that of the peptide P_i, whilst when $K_{abs,i}$ becomes smaller, the polar characteristics of the peptide P_i, are more similar to those found for the mobile phase. Alternatively, $K_{abs,i}$ can be expressed in terms of the general relationship:

$$\ln K_{ads,i} = \ln V_{a,i} + \beta(S_o - A_i\varepsilon_0) \quad \text{eq 43}$$

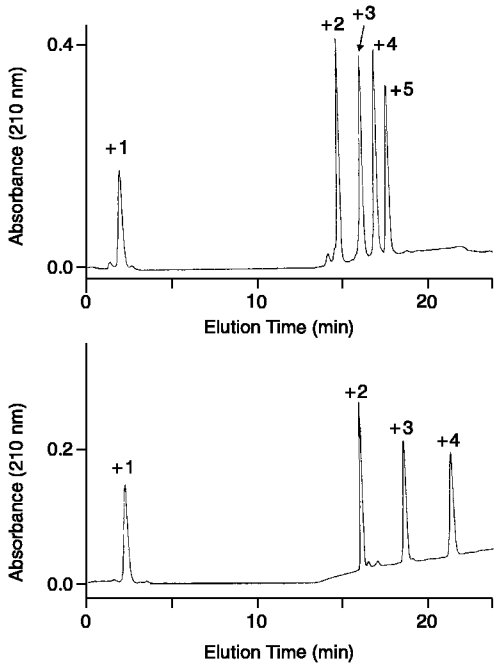
where $V_{a,i}$ is the volume of the mobile phase m, adsorbed to the sorbent; β is the activity coefficient (related to the zeta potential) of the sorbent; $K_{abs,i}$ is the ratio [mole P_i/g sorbent]/[mole P_i/mL mobile phase], ε_o is the solvent strength parameter (elutropicity) of the mobile phase, S_o is the adsorption energy of the peptide P_i for a standard sorbent where $\beta = 1$ for a standard mobile phase where $\varepsilon_o = 0$; and A_i is the molecular surface area of P_i. The variables $V_{a,i}$ and β represent characteristic properties of bonded polar phase sorbents that predominantly function via hydrogen bond effects. These properties will vary markedly with the amount of adsorbed water or the effective "hydroxylated" or "hydrophilated" content of the surface of the sorbents achieved by bonding the hydrophilic ligands. As the eluents employed with HP-HILIC and NP-HPLC sorbents typically contain a high ionic strength buffer together with a high content of a dipolar or hydrogen-bonding solvent, these techniques are better suited to the chromatographic resolution of side-chain-protected peptides, such as serine-protected analogues.

for references see p 622

7.1.5.3 High-Performance Ion-Exchange Chromatographic Separation of Peptides

Coulombic or electrostatic interactions are, like hydrogen bonds, of polar origin. The interactions occur between charged or ionizable groups accessible at the surface of the peptide and complementary charge groups on the HP-IEX sorbent. Many different types of electrostatic ligands are available for use in HP-IEX systems, ranging from the familiar $-NEt_2$, $-Me_3N^+$, $-CH_2SO_3H$, $-CH_2CO_2H$ moieties to the more exotic malachite green carbinol ligates, such as $(Me_2NC_6H_4)_2(H_2NHC_6H_4)COH$, which generate very strong anion exchangers as the corresponding immobilized carbenium species[244] at neutral or slightly acidic pH values. Because of the ascendancy of RPC procedures, HP-IEX methods today attract only limited interest for the separation of peptides, primarily due to difficulties in removing salts or other buffer components from the recovered peptide samples. Most studies related to the separation of peptides have employed cation-exchange sorbents such as sulfonopropylsilicas, e.g. PolySulfoethyl A or Synchropak S, or sulfone-polymers, such as Mono S, due to their greater stability, but this choice limits the selectivity options for peptides. With the PolySulfoethyl A and Mono S sorbents, gradient elution procedures based on a 5 mM KH_2PO_4 buffer containing 20–40% MeCN and 0–100 mM KCl gradients in the range of pH 3.0–6.5, have gained some popularity. Illustrative of the resolution that can be achieved with such cationic HP-IEX sorbents is the separation of various peptide analogues of different charge character shown in Figure 18.

Figure 18 Analytical Profiles of Several Synthetic Peptides Separated on the Strong Cation-Exchange Mono S Sorbent[347]a,b



peptides 1–5:
Ac-(Gly-Leu-Gly-Ala-Lys-Gly-Ala-Gly-Val-Gly)$_n$-$CONH_2$,
n = 1–5, q = +1 to +5

peptide 1:
Ac-Gly-Gly-Gly-Leu-Gly-Gly-Ala-Gly-Gly-Leu-Lys-$CONH_2$,
q = +1

peptide 2:
Ac-Lys-Tyr-Gly-Leu-Gly-Gly-Ala-Gly-Gly-Leu-Lys-$CONH_2$,
q = +2

peptide 3:
Ac-Gly-Gly-Ala-Leu-Lys-Ala-Leu-Lys-Gly-Leu-Lys-$CONH_2$,
q = +3

peptide 4:
Ac-Lys-Tyr-Ala-Leu-Lys-Ala-Leu-Lys-Gly-Leu-Lys-$CONH_2$,
q = +4

[a] Column: 50 × 5 mm i.d.; elution conditions involved a linear gradient from buffer A (5 mM KH_2PO_4, pH 6.5), following a 10-min isocratic elution with buffer A, to buffer B (buffer A plus 0.5 M NaCl), gradient 20 mM NaCl/min, both buffers contained 40% MeCN; flow rate: 1 mL · min^{-1}; T = 26 °C].

[b] Adapted from Burke, Mant, Black, and Hodges,[347] with permission.

Convergence or divergence in the linearity of the plots of retention time versus the logarithm of the number of residues, i.e. $k'_{iex,i}$ versus ln ℵ, can be employed to characterize the "regular" ion-exchange behavior of polypeptides in terms of the role of polypeptide chain length

and charge density. Empirically, these dependencies for a specified HP-IEX sorbent can be evaluated from the relationship between the equilibrium association constant, $K_{assoc,i}$, of the peptide P_i of charge state $\pm x$, and the concentration of the displacing ion can be described in terms of nonmechanistic, stoichiometric models, whereby

$$K_{assoc,i} = \frac{[P^{\pm x}]_s [D^{\pm y}]_m^{Z_t}}{[P^{\pm x}]_m [D^{\pm y}]_s^{Z_t}} \quad \text{eq 44}$$

where $D^{\pm y}$ is the displacing salt counterion, Z_t represents the ratio (x/y) of the effective charge on the peptide to that on the displacer counterion, and the subscripts m and s refer to the mobile and the stationary phase, respectively. Under near-equilibrium binding conditions, when T, P, V, Φ, and the flow rate, F, are constant, and the adsorption process approximates a linear (Langmuirean) isotherm, the dependency of the logarithm of the capacity factor $k'_{iex,i}$ on $[D^{\pm y}]_m$ can be expressed in a similar form to that employed for RPC separations of peptides:

$$\ln k'_{iex,i} = \alpha + \beta(\ln[1/C]) + \gamma(\ln[1/C])^2 + \delta(\ln[1/C])^3 + \cdots \quad \text{eq 45}$$

where C is the concentration of the displacing ion, i.e. $[D^{\pm y}]_m$, and α, β, γ, δ, etc. are coefficients dependent on the solubility parameter of the solute, δ_i the zeta potential, ζ, of the stationary phase the mobile-phase buffer composition, pH, polarizability, and dielectric properties. Over a narrow range of isocratic or gradient mobile-phase compositions, this relationship has often been approximated to a linear dependency, i.e. given in the familiar form

$$\ln k'_{iex,i} = \ln K_{dist,i} + Z_c(\ln[1/C]) \quad \text{eq 46}$$

where $K_{dist,i}$ is the distribution coefficient, a term that includes the value of $K_{assoc,i}$, the phase ratio, Φ, and the stationary phase electrostatic ligand concentration $[L^{\pm y}]$. Thus, under linear elution conditions, the slope coefficient, Z_c, and $\ln K_{dist,i}$ can be determined from the plot of $\ln k'_{iex,i}$ versus $\ln[1/C]$. However, it must be noted that Z_c reflects the apparent number of ionic charges associated with the adsorption of the peptide to the coulombic ligand surface, and is not identical in mathematical or physicochemical terms to the total number of charges, Z_t, present on the peptide for a defined pH value and buffer composition as determined by HP-CZE procedures or via the Henderson–Hasselbach relationship.[285] Moreover, curvilinear, rather than linear plots are more likely to occur in these $\ln k'_{iex,i}$ versus $\ln[1/C]$ plots due to the anisotropic nature of the charge distribution on the surface of peptides and the involvement of secondary binding processes, mediated for example by hydrophobic effects. For the isolation of peptides by HP-IEX methods, pyridine acetate gradients have been used because of their volatility and greater ease of removal by lyophilization.

Since peptides are amphoteric, Z_t and Z_c are expected to show nonlinear dependencies on pH. Similar behavior has been observed for various synthetic peptides separated on both strong anion and strong cation HP-IEX sorbents. As a consequence, the minima in the $\ln k'_{iex,i}$ versus pH plots at a defined concentration of displacing salt will not usually occur at the predicted pI value of the peptide, but rather at another pH value. Implementation of an optimized HP-IEX separation of peptides thus requires that the sequence microlocality and extent of ionization of the surface-accessible amino acid side chains, or the N- and C-terminal amino and carboxy groups, respectively, are taken into account.

In terms of optimizing the selectivity of the HPIEX system, various studies with synthetic peptides have shown that the slopes of the $\ln k'_{iex,i}$ versus $\ln[1/C]$ plots for different displacing buffers are not parallel but depend on the structure of the peptide. According to

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eqs 44–46, the magnitudes of the slope terms, i.e. the Z_c values, of the isocratically derived ln $k'_{iex,i}$ versus ln($[1/C_i]$) plots, or the case of gradient-derived data, the ln $k'_{iex,i}$ versus ln($1/[C_i]$) plots, are predicted to be dependent on the valency of the displacing salt. As a consequence, the use of high-molecular-weight charged displacers have potential for applications associated with the very large scale purification of synthetic peptides based on HP-IEX methods when used in the displacement chromatographic mode.

7.1.5.4 High-Performance Metal Ion Affinity Chromatographic Separation of Peptides

Many peptides bind to metal ions with relatively high affinity. For this reason, the use of high-performance immobilized metal ion affinity chromatography (HP-IMAC) also provides an alternative approach to purify peptides. In this case, the metal ions can be considered as Lewis acids, with the formation of the peptide-immobilized metal ion complexes rationalized in terms of Lewis acid–base interactions. This interaction can be visualized in terms of the ability of a metal ion, M^{n+}, to accept a pair of electrons (and thus act as a Lewis acid) from the peptide, P_i, typically the imidazolyl moiety of histidine residues but also from the amino terminal NH_2 group, the indole ring of tryptophan, or other electron-donor groups within the peptide, which act as an electron-donating base with an accessible lone pair of electrons. The π-bonding theory of Chatt[246] adequately rationalizes many of the characteristic features of the coordination interactions of metal ions with peptides. Thus, "hard" metal ions such as Fe^{3+}, Ca^{2+}, or Al^{3+} have a preference for "hard" donor groups within Asp, Glu, Lys, and Arg residues, as well as phospho-, sulfonato-, and sulfato-groups; borderline metal ions, e.g. Cu^{2+}, Zn^{2+}, or Ni^{2+}, have preferences for borderline donor groups associated with Tyr, Trp, or His residues; and "soft" metal ions, such as Hg^{2+}, have a preference for "soft" electron-donor groups on a peptide, such as SH or SMe of Cys or Met residues. Immobilized bi- to pentadentate metal ion complexes (IMICs) can be generated as chromatographic sorbents of the type $M^{n+} \Leftrightarrow CheL\ (H_2O)_m(O\bar{H})_j$ where $m \neq j$ with $0 \leq m \leq 4$ and $0 \leq j \leq 2$ by immobilizing suitable chelating compounds to a support material. If coordination sites of the metal ion are not taken up by the electron-donor groups of the chelate, they will then be available for binding to water, hydroxy groups, or other counterions X^{m-}, or alternatively to electron-donor groups accessible at the surface of the peptide P_i. The equilibrium association constant K_{Pi} for the complex formed following the coordinative interaction between the peptide P_i and the HP-IMAC sorbent can be represented by

$$M^{n+} \Leftrightarrow CheL_{(n-a)}X_{(a)} + P_{i,EDG(j)} \rightleftharpoons M^{n+} \Leftrightarrow CheL_{(n-a)}X_{(a-j)}P_{i,EDG(j-a)} + jX \quad \text{eq 47}$$

$$K_{P_i} = \frac{[M^{n+} \Leftrightarrow CheL_{(n-a)}X_{(a-j)}P_{i,EDG(j-a)}][X]^j}{[M^{n+} \Leftrightarrow CheL_{(n-a)}X_{(a)}][P_{i,EDG(j)}]} \quad \text{eq 48}$$

with the retention thus given by

$$k'_{imac,P_i} = K_{P_i} x \Phi \quad \text{eq 49}$$

In common with many tridentate ligands, significant leakage of borderline metal ions occurs under mild elution conditions in separation processes with *im*-M^{n+}-IDA complexes due to their relatively low metal ion stability constants (i.e., $7 \leq \log \beta_n \leq 10.5$).[247–250] Some synthetic peptides are, in addition, able to strip metal ions from *im*-M^{n+}-IDA complexes. To circumvent metal ion leakage, other chelating ligands have been investigated, including tris(carboxymethyl)ethylenediamine (TED),[251] a pentadentate ligand that coordinates metal ions via two nitrogen atoms of the secondary amino groups and three oxygen atoms from the three carboxy groups. When Ni^{2+} and other M^{2+} ions are chelated to the *im*-TED ligand, only

one vacant coordination site remains available within the coordination sphere of the metal ion for interaction with donor groups on a peptide, whilst three coordination sites are available with the corresponding *im*-Ni^{2+}-IDA complex. The *im*-Ni^{2+}-TED complex, therefore, has weaker affinity for peptides, i.e. a lower K_{Pi} value than the corresponding *im*-Ni^{2+}-IDA sorbent, although the propensity for metal ion leakage is reduced under comparable elution conditions because of the high $K_{Mn+} \rightarrow CheL$ value.

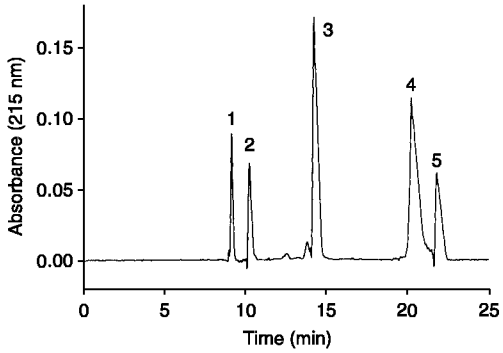
Other chelating ligands have been identified with lower metal ion leakage per se and more optimal association constants for peptide binding. Thus, other bidentate chelators [e.g., aminohydroxymate (AHM)[252] and 8-hydroxyquinoline[253]] or tridentate ligands [e.g., diaminomethylpyridine (DAMP),[254] *O*-phosphoserine (OPS),[255] 1,4,7-triazacyclononane (tacn),[256] dipicolylamine (DPA),[255] *cis*/*trans*-carboxymethylproline (CMP)[257]] or tetradentate ligands [e.g., nitrilotriacetic acid (NTA)[258,259] or carboxymethylaspartic acid (CM-ASP)[260]], and pentadentate ligands [e.g., tetraethylenepentamine (TEPA)[261]] have been examined for their potential use in IMAC applications with peptides and proteins. Typically, these immobilized metal ion chelating complexes (IMICs) have shown only modest increases in metal ion stability constants (i.e., in the range of $9 \leq \log \beta_n \leq 14$), the exception being the tacn-derived sorbents, although they nevertheless offer very useful, alternative separation options. Collectively, the selectivity of such IMAC systems for peptides has been found to be very dependent on the peptide sequence, and, more importantly, on the number of electron-donor amino acid residues of the peptide that are specific for the particular immobilized metal ion chelating complex. For this reason, such HP-IMAC sorbents provide class- or group-selective rather than high-resolution chromatographic opportunities with peptides. Acceptable resolution can be achieved with closely related peptides using gradient elution procedures with, for example, *im*-Cu^{2+}-IDA-HPLC sorbents as demonstrated[262] for the separation of 19 small peptides, whilst larger peptides of between 30–40 residues have been successfully resolved[263,264] with other types of HP-IMAC sorbents, utilizing pH gradients generated with 0–250 mM imidazole-based eluents. Nevertheless, the opportunities of HP-IMAC vis-à-vis RPC approaches still remain at the boutique level in terms of their versatility, and their use currently falls into the category of specialized biophysical investigations.

7.1.6 Application of Capillary Zone Electrophoresis for the Separation of Synthetic Peptides

The application of high-performance capillary zone electrophoresis (HP-CZE) in its various selectivity modes has become a very valuable adjunct to HPLC for the analysis of peptides. For synthetic peptides, in particular, both HPLC and HP-CZE now form essential components of the analytical characterization of these molecules. Increasingly, zonal, micellar, or (biospecific) affinity-based HP-CZE procedures with open tubular capillary systems are adapted to allow resolution with extremely high separation efficiencies (e.g., $>10^5$ plates per meter) of synthetic or naturally occurring peptides as part of the determination of their structural, biophysical, or functional properties. Illustrative of these capabilities are the results shown in Figure 19 for the separation of several peptides with different charge and Stokes radius characteristics by HP-CZE.

for references see p 622

Figure 19 Electrophoretograms Illustrating the Separation of Peptides[324]a,b



1 H_2N-LKRDQEPMDFHIWDDYLKRD-CO_2H, $q = 4.41$
2 H_2N-KNWDRAPQ-CO_2H, $q = 2.37$
3 H_2N-DDALYDDKNWDRAPQRCYYQ-CO_2H, $q = 2.41$
4 H_2N-CALEEADDDRS-CO_2H, $q = 0.88$
5 H_2N-TPYQQGTPCA-CO_2H, $q = 0.61$

[a] 40-cm capillary, 50 μm i.d.; 2 min pre-rinse with separation buffer, aliquots (7 s) of the peptide mixture (1 mg·mL^{-1} in 0.1 M H_3PO_4) with 0.02% DMF (4 s) as a marker were injected into the capillary; the electrophoretic separation was performed at 25 °C using 50 mM sodium phosphate buffer pH 3.0 at a potential of 15 kV; post-rinse cycle consisted of a wash with 0.1 M NaOH (3 min), H_2O (3 min), 0.1 M HCl (3 min), H_2O (3 min), and separation buffer (8 min) was utilized.
[b] Adapted from Sitaram, Keah, and Hearn,[324] with permission.

Under most operating pH values (i.e., at pH > 3) with fused silica capillaries, the silanol groups on the capillary walls are partially or fully ionized. Under the influence of an electric field, cations in the electrolyte are arranged as an ionic double layer around the capillary wall due to their negatively charged surfaces. The cations closest to the wall surface are constrained in a rigid plane, the so-called Stern layer, but solvated cations further away from the wall surface in this diffuse layer move towards the cathode and drag the inner bulk liquid along thus establishing an electro-osmotic flow (EOF) with a characteristic plug flow profile.[265,266] In HP-CZE, the electrophoretic mobility, μ_e, of a peptide is directly proportional to the applied field strength, E, the effective peptide charge, q, in the particular buffer, and the viscosity, η, of the buffer, and inversely proportional to its Stokes radius, R_S, i.e. the radius of the peptide as represented by the sphere swept out by the peptide in the solvent. Superimposed upon the electrophoretic mobility of the peptide is the electro-osmotic flow of ions within the buffer and the ensuing liquid flow, represented by the electro-osmotic mobility, μ_{eof}, or the electro-osmotic velocity, u_{eof}. Charged analytes, such as peptides, are transported through the open fused capillary due to the combined effects of their electrophoretic migration and this EOF effect. The apparent or observed electrophoretic mobility (μ_{obs}) of the peptide is thus the sum of intrinsic electrophoretic mobility of the peptide (μ_e) and the mobility of the EOF (μ_{eof}) such that

$$u_{eof} = \frac{\varepsilon_o \varepsilon_r \zeta E}{\eta} \quad \text{eq 50}$$

$$\mu_{eof} = \frac{\varepsilon_r \zeta}{\eta} \quad \text{eq 51}$$

$$\mu_e = \frac{q}{6\pi\eta r} \quad \text{eq 52}$$

$$\mu_{obs} = \mu_e + \mu_{eof} \quad \text{eq 53}$$

where ε_o is the permittivity in vacuo, ε_r is the relative permittivity (dielectric constant) of the mobile phase, ζ is the zeta potential of the liquid–solid interface, η is the viscosity of the mobile phase, E is the electric field strength (kV·m^{-1}), R_S is the ion radius or the Stokes radius of the peptide, and q is the intrinsic charge of the peptide. In HP-CZE, under ideal

conditions, peptide(s) are not retained by, or interact with, the capillary walls. The transit time(s) for the passage from the beginning of the capillary to the detector window can then be denoted as the migration time(s), t_e. The dominant band-broadening contribution in HP-CZE of peptides is associated with longitudinal diffusion, i.e. the B-term of the van Deemter equation. The interactive properties of the fused-silica capillary wall can be modified by different coatings, e.g. with polyvinyl alcohol, polyacrylamide, polyethylene glycol, etc., to diminish ionic or hydrophobic interactions with peptides that cause adsorption and tailed peaks. By minimizing peak dispersion, efficiencies with coated capillaries can be increased significantly. Coated capillaries also have a very low or negligible EOF and therefore the separation is almost totally due to the intrinsic electrophoretic mobility of the peptide(s). In open tubular HP-CZE, the peak shapes of peptides, separated at a low pH, can often be improved by increasing the buffer ionic strength. Joule heating is, however, a limiting factor when buffers of very high molarity are employed.

Under the selected pH conditions of the buffer in HP-CZE, the protonic dissociation of a generic peptide (with a charge state given by X^{charge}) of n amino acid residues with j basic ionogenic groups (N-terminus; histidine, lysine, arginine) and i acid ionogenic groups (C-terminus; aspartic and glutamic acid; cysteine, tyrosine) can be described by

$$X^{j} \xrightarrow[K_1]{} X^{j-1} \xrightarrow[K_2]{} \cdots\cdots \xrightarrow[K_{j+i}]{} X^{-i} \quad \text{eq 54}$$

In common with all protonic dissociation, this reaction pathway assumes that all k species of the same charge ($k=j+i+1$) are equivalent and takes advantage of the assumed independence of the macroscopic dissociation constants K_1, K_2, ...K_{j+i}. Moreover, this dissociation pathway assumes that the proton exchange kinetics as well as conformational interconversions of the peptide(s) are fast compared to the electrophoretic migration times. Because peptides fall into the class of compounds that have variable charge as a function of the pH of the surrounding solvent, changes in the proton activity of the buffer can lead to variation in the Stokes radius, R_S, of the peptide(s). The observed electrophoretic mobility (μ_{obs}) of a peptide as a function of pH can thus be considered as weighted mean mobility of the different protonated species such that

$$\mu_{obs} = \mu_j\chi_j + \mu_{j-1}\chi_{j-1} + \mu_{j-2}\chi_{j-2} + \cdots\cdots + \mu_{-i}\chi_{-i} = \sum_{k=j}^{k=-i} \mu_k\chi_k \quad \text{eq 55}$$

where μ_j, μ_{j-1}, μ_{j-2}, ...μ_{-i} are the electrophoretic mobilities of the individually ionized species and χ_j, χ_{j-1}, χ_{j-2}, ... χ_{-i} are the molar fractions of the different species. Because the Stokes radius, R_S, of the peptide will not be constant as a function of the buffer pH, eq 56 can be written as follows:

$$\mu_{obs} = \frac{q}{6\pi\eta} \sum_{k=j}^{k=-i} \frac{k}{R_{s,k}} \chi_k \quad \text{eq 56}$$

where χ_k is the molar fraction of any species j; q is the electron charge (1.60×10^{-19} C), η is the solution viscosity (water: 8.95 10^{-4} N·s·m^{-2} at 298 K), $R_{S,k}$ is the Stokes radius (i.e., the radius of the sphere equivalent to hydrated polypeptide) of any species j; and k is the dimensionless value of the species integer charge.

The molar fraction, χ, of any ionized species of the peptide can be expressed as a function of proton activity and dissociation constants, as follows:

$$\chi_{j-n} = \frac{[\chi^{j-n}]}{[[\chi^{j}] + [\chi^{j-1}] + [\chi^{j-2} + \cdots\cdots + [\chi^{-i}]]} = \frac{\beta_n/[H^+]^n}{\Re} \quad \text{eq 57}$$

for references see p 622

where $n = 0, 1, 2, \ldots\ldots, j+1$, and $\Re$ is the protonic ionization polynomial that links the respective dissociation constants to the hydrogen ion concentration and is given by

$$\Re = \beta_o + \beta_1/[H^+] + \beta_2/[H^+]^2 + \cdots\cdots + \beta_{j+1}/[H^+]^{j+1} \quad \text{eq 58}$$

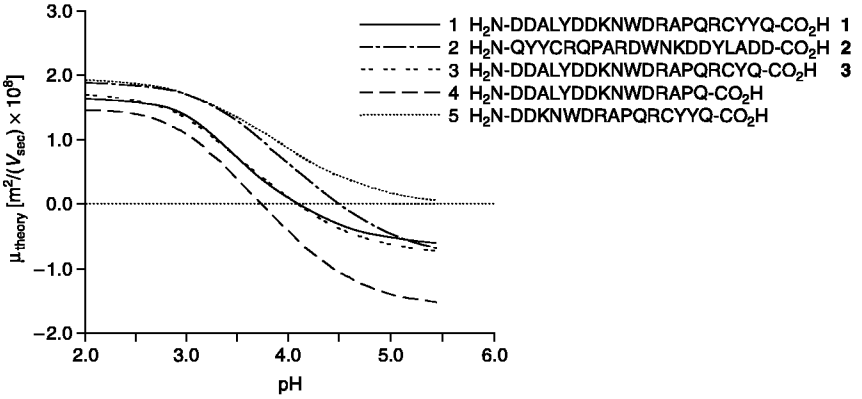
and β_n is given by

$$\beta_o = 1;\ \beta_1 = K_1;\ \beta_2 = K_1K_2;\cdots;\ \beta_{j+1} = K_1K_2K_3\cdots_{j+1} \quad \text{eq 59}$$

Based on these relationships, the respective mole fractions of the different charged species that contribute to the electromigration of each peptide in a mixture under a particular set of HP-CZE conditions can be derived, as well as the corresponding Stokes radii and effective charges. This information can then be employed to facilitate the development of an optimal pH, buffer condition, and electric field strength for the resolution of the mixture of peptides. Results reported for peptides related to the gp120 envelope protein of HIV-1,[267] the thrombin receptor inhibitors[268] as well as an array of other bioactive peptides have documented the advantages of these procedures for the isolation and characterization of synthetic peptides. Illustrative of the changes in electromobility, and thus selectivity, that can be achieved as the pH is varied in HPCZE are the plots for the all-L-α-polypeptide, H_2N-DDALYDDKNWDRAPQRCYYQ-CO_2H, (**1**), the *retro-inverso* D-α-isomer, H_2N-QYYCRQPARDWNKDDYLADD-CO_2H (2), and several analogues shown in Figure 20, whilst in Figure 21 and 22 are shown the corresponding fractional and total electromobilties, μ_{obs}, and mole fraction, χ, of each charge species for the polypeptides **1** and **2**. As evident from Figure 20, the resolution of the all-L-α-polypeptide, H_2N-DDALYDDKNWDRAPQRCYYQ-CO_2H (**1**) from its des-Tyr[19] analogue H_2N-DDALYDDKNWDRAPQRCYQ-CO_2H (**3**) represents a formidable task under these HP-CZE, despite the difference in effective charge, q, that arises as a consequence of the tyrosine deletion. However, these peptides can be resolved by HP-RPC methods. With regard to the all-L-α-polypeptide **1** and its *retro-inverso* D-α-isomer **2**, although these peptides have the same effective charge, q, and composition, it is apparent that the polarizability, dipole moments, and Stokes radii of these two very similar polypeptides are different, i.e. the calculated Stokes radii of the polypeptides **1** and **2** in their −2 charge state are 30 ±0.8 Å and 24 ±0.7 Å, respectively. In addition, as apparent from Figure 21 and 22, the individual contributions to the overall electrophoretic mobilities of polypeptides **1** and **2** from the electrophoretic mobilities of the individually ionized species in charge states +4, +3, ...−2, are also slightly different, partially due to the relative differences in the molar fraction of each ionized species present under the HP-CZE experimental conditions. Analogous observations can be made with many polypeptides, related to each other as diastereomers, or as sequence analogues. For these reasons, HP-CZE methods provide very powerful procedures to characterize the origins of differences between closely related polypeptides, in addition to enabling rapid and sensitive assessment of their purity.

Since HP-CZE separations can be achieved with picomole to attomole quantities of peptides at nanoliter/min flow rates using relatively simple aqueous or aqueous/organic solvent buffer systems,[269–271] these methods represent very economical approaches to acquire essential information on the molecular properties of peptides under various bulk solution and ligand-interaction conditions. Moreover, successes in interfacing HP-CZE with electrospray-mass spectrometry (ES-MS) and single-cell biosensors has potentially provided an ideal system to permit studies of the structure–activity relationships of biologically active peptides under conditions that allow a very high level of discrimination of their compositional and functional integrity.[272–276]

Figure 20 Plots of the Experimentally Observed Electromobilities, μ_{obs}, for L-α-Polypeptide **1**, its *retro-inverso* D-α-Isomer **2**, and the Synthetic Analogues **3–5** over the pH Range $2.0 \leq pH \leq 6.0$[307]a,b



[a] The curves were obtained by fitting the experimental data to eqs 55–57 and computed by utilizing parametric fitting procedures based on Marquardts algorithm. The CE apparatus employed was a Beckman P/ACE 2100 instrument, equipped with the Beckman Gold Series 711 system for automated apparatus control and data acquisition. Coated fused silica capillaries were used with no appreciable electroendosmosis occurring at low pH values, as evident from the migration behavior of the marker, acetanilide. The capillary dimensions were 56.5 cm (50 cm at the detection window) × 75 μm. A set of phosphate, formate, and acetate solutions was used for the electrophoretic buffers, characterized by constant ionic strength in the range of pH 2.2 to pH 5.5. The synthetic polypeptides were dissolved in H_2O (conductivity <18 μmho) at a concentration of 2.0 mg·mL^{-1}. Electrophoretic separations were carried out at a constant voltage of 25 kV at 298 K. The anode electrode was coincident with the capillary injection terminal, with injections performed by the pressure fill method for 1 or 2 s, corresponding to an injected volume of 5 or 10 nL. Polypeptides were detected at 200 nm.

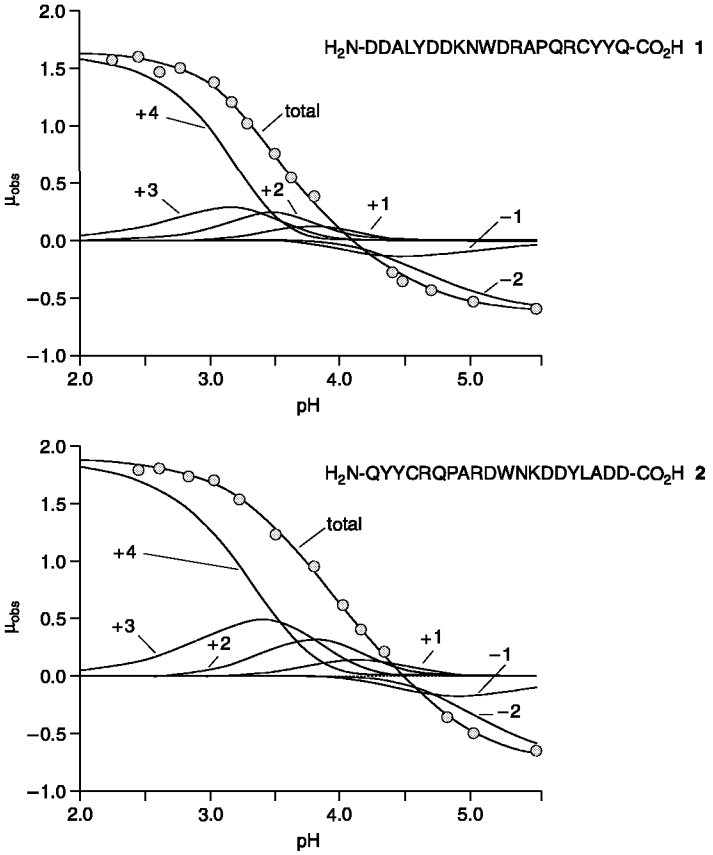
[b] Adapted from Hearn, Keah, Boysen, Cassiano, Messana, Rossetti, and Castagnola,[307] with permission.

When operated under these zonal modes, or alternatively under affinity interaction modes with buffers containing lipophilic, metal ion chelated ligands, or bioselective ligands, at low and/or high temperatures, the opportunity also exists to explore the conformational properties that peptides can manifest in different solution milieus.[17,278] Work from this and other laboratories has addressed several of these issues with the aim to both optimize resolution and separation efficiency as part of nano-isolation studies, whilst at the same time examining the relationships between electrophoretic mobility and the primary and/or secondary structures of peptides and peptidomimetics of interest as drug candidates in structure–activity studies related to peptide-based vaccines, cystine knot glycoprotein and growth factor antagonists, sperm tail and other gonadal proteins, and hypoglycaemic peptide agonists.

Arising from these investigations has been the opportunity to examine the physicochemical relationships that link $\mu_{e,i}$ with the effective charge, q, the molecular weight, M_r, and the frictional size to mass ratio parameter, ζ, of any peptide in a mixture. In addition, these studies permit the contribution from secondary structure features, e.g. α-helical, β-sheet, or random coil features, to the electrophoretic mobility of peptides in HP-CZE systems to be clarified. One important conclusion arising from these investigations with mixtures of synthetic or naturally occurring peptides that directly has bearing on the isolation of peptides by HP-CZE methods prior to, for example, subsequent ESI-MS analysis, is that the incremental contributions of the fractional charges of the individual amino acid residues of a peptide are not the sole determinant of $\mu_{e,i}$. Factors other than the summated fractional charge, q, calculated according to the Henderson–Hasselbach equation[245] influence the $\mu_{e,i}$ values of peptides. In addition to the contribution from q, the contribution of molecular frictional drag and the influence of the sequential arrangement of the amino acid residues within a particular peptide can be identified as two significant factors that affect the electromobility of peptide(s) in HP-CZE systems.

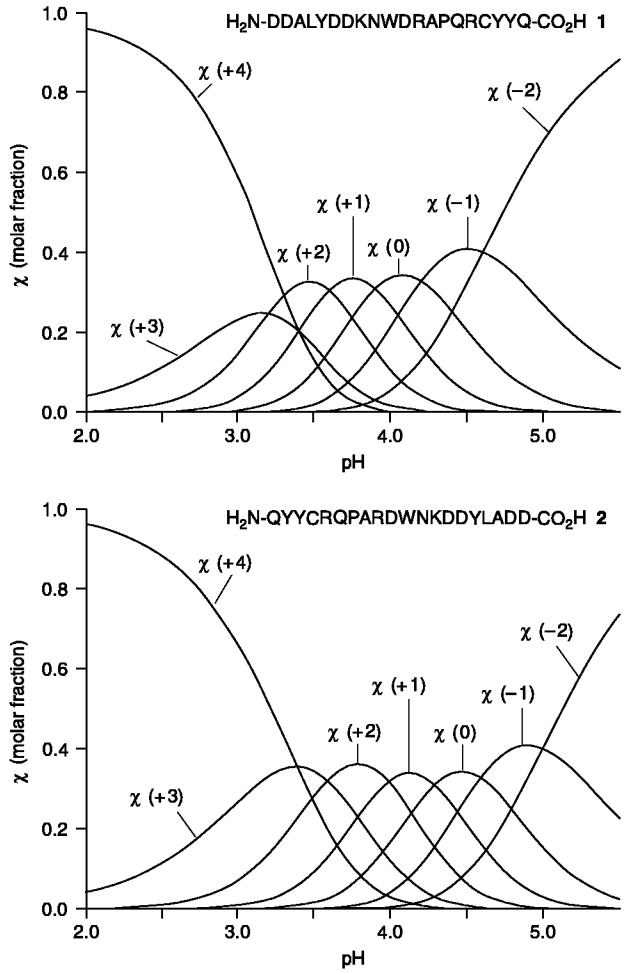
for references see p 622

Figure 21 Intrinsic Electrophoretic Mobility Contributions to the Total Mobility, μ_{total}, of Each Charged Form of the L-α-polypeptide **1** and Its *retro-inverso* D-α-Isomer **2**, Computed According to the Derived Values of the Molar Fraction, χ, of the Different Charged Forms Corresponding to the +4, +3, +2,...,−2 Species and Equations 55–59 (adapted from Hearn, Keah, Boysen, Cassiano, Messana, Rossetti, and Castagnola,[307] with permission)



According to Cifuentes and Poppe,[278,279] an optimum fit of the dependency that links $\mu_{e,i}$ to the respective charge-size parameters is observed when $\mu_{e,i}$ is plotted against ζ, where $\zeta = q/M_r^{2/3}$. Although this relationship has been widely used, other investigators[280–283] have proposed that plots of $\mu_{e,i}$ versus $q/M_r^{1/3}$, $\mu_{e,i}$ versus $q/M_r^{1/2}$, or $\mu_{e,i}$ versus $\ln(q+1)/N^{0.43}$, where N is the number of amino acid residues in the sequence, should be employed in order to achieve optimal discrimination of peptides in these HP-CZE systems. Illustrative of the effects of molecular frictional drag and amino acid sequence for peptides of notionally similar effective charge are the data[284] shown in Figure 23 for 18 HIV-1 peptide mimetics, where $\mu_{e,i}$ has been plotted against q and against ζ ($=q/M_r^{2/3}$) respectively. Inspection of these HP-CZE data for peptides 1–18 indicated that in all cases where the $\mu_{e,i}$ values experimentally derived from these HP-CZE measurements represented outlying data with lower correlation compared to the corresponding values predicted on the basis of a linear dependency of $\mu_{e,i}$ on q, or $\mu_{e,i}$ on ζ, these peptides were found by CD or ^{1}H NMR techniques to adopt under low pH conditions stable α-helical structures as a direct consequence of the particular amino acid arrangement within their sequences. When this subgroup of conformationally more constrained peptides were removed from the mixture of peptides **1–18**, the remaining peptides showed improved correlation for a linear dependency of μ_{em} on q or μ_{em} on ζ under the low pH conditions typically employed with HP-CZE systems. Such structural effects need to be factored into HP-CZE separations with low pH buffer systems as well as those of higher pH values.

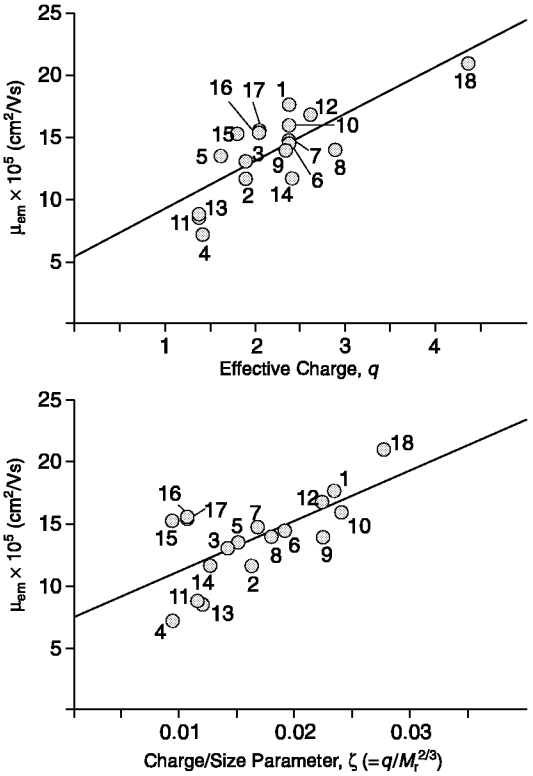
Figure 22 The pH Dependency of the Molar Fraction, χ, of the Different Charged Forms of L-α-Polypeptide **1** and Its *retro-inverso* D-α-Isomer **2** Obtained from the Application of Calculated Dissociation Constants for the C-Terminus and Side Chains and Computed According to Equations 55–59 (adapted from Hearn, Keah, Boysen, Cassiano, Messana, Rossetti, and Castagnola,[307] with permission)



As a nanotechnology, HP-CZE, as well as the micellar electrokinetic chromatographic extension (HP-MEKC) of the technique, provide important avenues for the rapid analysis of peptides. As such, the techniques offer valuable orthogonal capabilities to the RPC procedures, which form the mainstay of peptide separation and purification.

for references see p 622

Figure 23 Fit of Effective Electrophoretic Mobilities, μ_{em}, versus ζ (Calculated According to the Relationship $\zeta = q/M_r^{2/3}$) for Peptides Separated Using a 50 mM Sodium Phosphate Buffer pH 3.0[324][a,b]



1 H_2N-KNWDRAPQ-CO_2H
2 H_2N-DDKNWDRAPQ-CO_2H
3 H_2N-LYDDKNWDRAPQ-CO_2H
4 H_2N-DDALYDDKNWDRAPQ-CO_2H
5 H_2N-APQRCYYQ-CO_2H
6 H_2N-DRAPQRCYYQ-CO_2H
7 H_2N-DRAPQRCYYQ-CO_2H
8 H_2N-DDKNWDRAPQRCYYQ-CO_2H
9 H_2N-DRAPQRCY-CO_2H
10 H_2N-DRAPQRAY-CO_2H
11 H_2N-DAPQRCYYQ-CO_2H
12 H_2N-RAPQRCYYQ-CO_2H
13 H_2N-DAAPQRCYYQ-CO_2H
14 H_2N-DDALYDDKNWDRAPQRCYYQ-CO_2H
15 H_2N-DDALYDDKNWDRAPQRCYYQ-$CONH_2$
16 H_2N-DLALYDDKNWDRAPQRCYYQ-$CONH_2$
17 H_2N-DDALYLDKNWDRAPQRCYYQ-$CONH_2$
18 H_2N-CKRIHIGPGRAFYTTC-CO_2H

[a] Separation of the synthetic peptides was achieved using eCAP plain fused silica capillaries without coating. The capillaries were 50 µm i.d. and 375 µm o.d., and were of a total length of 47 cm, with length of 40 cm to the detector. The following method was routinely used for analytical separations with these capillaries: (i) rinse with separation buffer (2 min); (ii) pressure injection of a peptide solution 1 mg·mL^{-1} in 0.1 M H_3PO_4 (7 s), followed by a pressure injection of neutral marker DMF, (0.02% w/w in H_2O) (4 s); (iii) separation at 15 kV (constant voltage with the inlet as the anode and the outlet as cathode); and (iv) wash with 0.1 M NaOH (3 min), H_2O (3 min), 0.1 M HCl (3 min), H_2O (3 min), and separation buffer (8 min). All electrophoretic separations were carried out at a constant voltage of 25 kV and at a capillary temperature of 25 °C. Detection was performed by UV absorbance spectroscopy at 200 nm. Electro-osmotic flow was estimated from the migration time of DMF. The charge, q, associated with each peptide sequence, assuming an average random coil conformation for the species under the pH conditions of the separation, was calculated using the Henderson–Hasselbach equation.[285] The charge, q (or notional valence), can be defined as the sum of the fractional charges carried by each amino acid residue in the polypeptide sequence at pH 3.0. The pK_a values for the N-terminal, C-terminal, and side-chain functional groups of the individual amino acid residues incorporated in the peptide sequences studied were obtained from Rickard et al.[320] Calculations based on the pK_a values for the same functional groups in the free amino acids were also utilized. The pK_a values for the free amino acids were obtained from Dawson et al.[348]

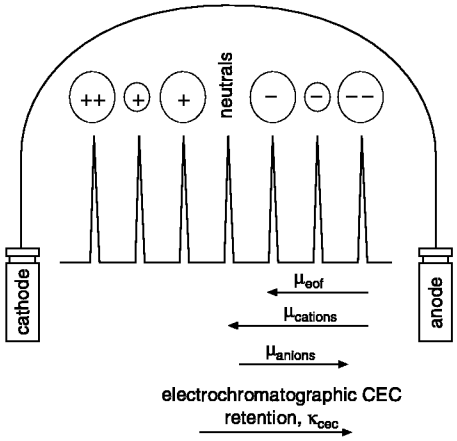
[b] Adapted from Sitaram, Keah, and Hearn,[324] with permission.

7.1.7 Application of Capillary Electrochromatography for the Separation of Synthetic Peptides

Capillary electrochromatography (CEC) is a rapidly emerging analytical separation technique, with several different instrumental formats and prepacked CEC capillary columns now available.[315–323] As an advanced nanoseparation technology, CEC represents an orthogonal hybrid of HPLC and HP-CZE. As a consequence, resolution can be achieved

through the migration of the peptides due to their relative electrophoretic mobility simultaneously with their specific interaction with ligands immobilized onto the surface of a chromatographic sorbent. Currently, most CEC analyses are performed with in-house built equipment or with modified commercial HP-CZE instruments.[325–336] Usually packed CEC (pCEC) capillaries of 50–100 μm i.d. containing *n*-alkylsilica particles are used for analytical separations, but open-tubular (OT-CEC) fused silica capillaries with their inner walls etched to increase the surface area and subsequently chemically modified via hydrosilanation methods have been employed in CEC modes of peptide separation and nano-isolation. The principle of and experimental layout for the separation of basic, neutral, and acidic peptides by CEC procedures is shown schematically in Figure 24.

Figure 24 Schematic Illustration of the Migration Order of Positively Charged, Negatively Charged, and Uncharged (Neutral) Peptides Separated by Capillary Electrochromatography (CEC) with H_2O/Organic Mobile Phases and Sorbents of Appropriate EOF Properties



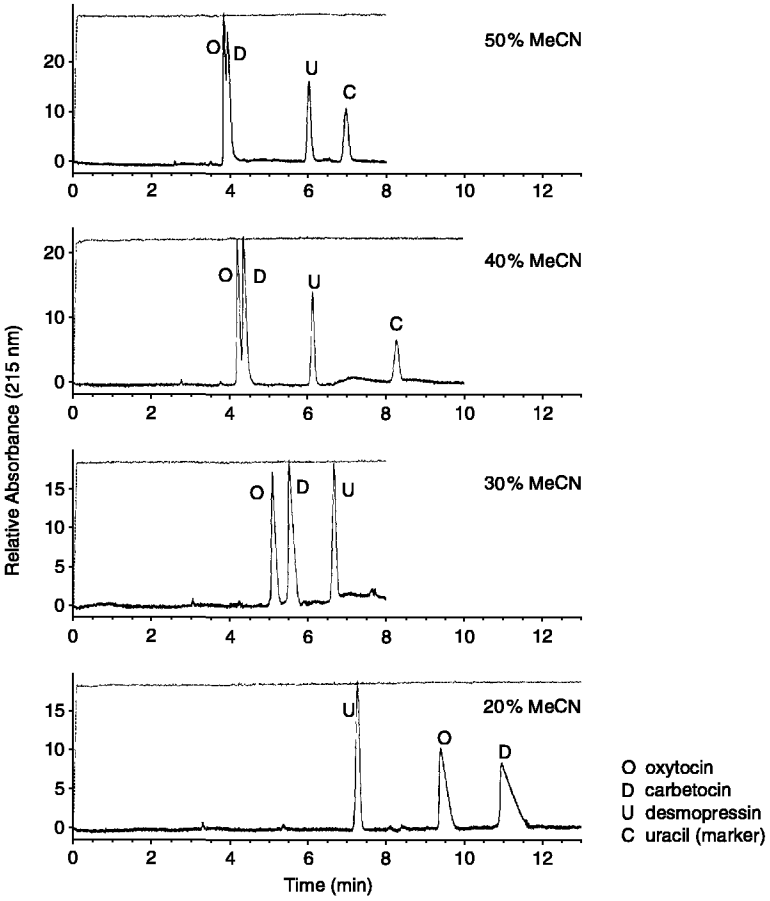
In both pCEC and OT-CEC systems, the mobile phase is transported through the capillary again by electro-osmotic flow (EOF), which originates from the migration of solvated ions when an electric field is applied across the solid–liquid interfaces.[337–339] In the case of pCEC, the main contribution to the EOF is due to the particles themselves, with only a minor effect arising from the capillary wall.[285] With OT-CEC systems, a similar effect is generated due to the greatly enhanced surface area (typically by a factor >1000) that is achieved by the etching process. When *n*-alkylsilica sorbents are used with peptides, CEC resolution is achieved from the combined effects due to hydrophobic as well as silanophilic interactions between the peptides and the RPC sorbent in the packed capillaries, or with the modified surface of the capillary wall in OT-CEC systems, superimposed with the intrinsic electrophoretic migration of the peptide due to the electro-osmotic velocity of the system. In CEC, the EOF is very sensitive to changes in the charged density of the sorbent surface. In turn, the charged density depends on the physical characteristics of the electrolyte used for the separation with respect to its pH, ionic strength, temperature, content of organic modifier, as well as the magnitude of the electric field strength applied to the packed or open tubular capillary.[286–288] Depending on the composition of the eluent, the CEC migration of peptides will thus reflect the interplay of both separation processes, with the dominance of one separation mechanism over the other achieved simply by changing the content of an organic solvent in the eluent. Similarly, the participation of electrostatic interactions between peptide and sorbent can be decreased or suppressed by increasing the molarity of the electrolyte, which at the same time will affect the magnitude of the EOF and thus the selectivity.

Separation of fully protected, neutral peptides by CEC occurs predominantly by chromatographic interaction with the RPC sorbent, with the observed retention coefficient,

for references see p 622

k_{cec}, having an analogous significance as the capacity factor, k', in RP-HPLC, systems, with the eluent delivered either by electrically or pressure-driven processes. In contrast, for unprotected or charged peptides in CEC, the retention coefficient, κ_{cec}, involves contributions from both chromatographic and electrophoretic effects, and thus incorporates retention contributions due to peptide–nonpolar ligand interaction and the EOF.[130] At low pH values with *n*-alkylsilicas, i.e. at pH <3.0, the EOF is reduced, since a higher percentage of the silanol groups become protonated. One way to obtain a high EOF at pH <3.0 is to use mixed mode sorbents containing ligands with both hydrophobic and strong ion-exchange characteristics, i.e. ionized alkylsulfonic acid groups,[289] in the normal, positive potential separation mode, or quaternary alkylammonium groups, for situations when a negative potential is applied. Modulation of the ligand density as mixed-mode sorbents or reduced endcapped RPC sorbents provides alternative opportunities for peptide separation in the CEC mode. Illustrative of the power of HP-CEC procedures for nano-scale separations of peptides are the results shown in Figure 25 for several oxytocin-related peptides. Progress in the field has shown that such systems can be interfaced with ESI-MS instrumentation, permitting rapid on-line analysis and characterization of synthetic as well as naturally occurring peptide mixtures.[340,341]

Figure 25 Elution Profiles for Oxytocin (O), Carbetocin (C), and Desmopressin (D) Separated by Capillary Electrochromatography[130]a,b

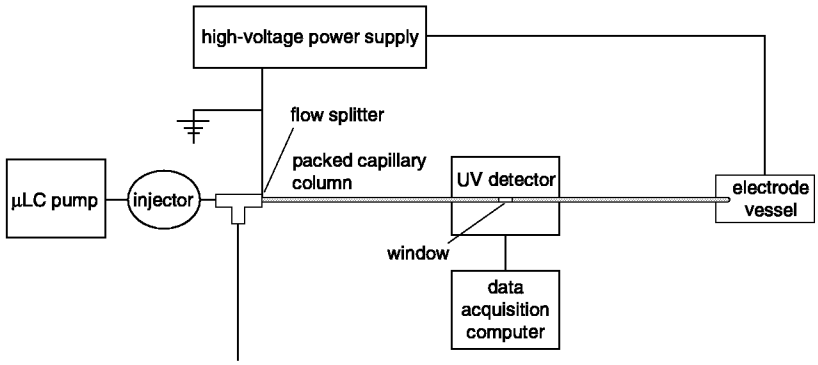


[a] At 25 kV with a Hypersil C_8 CEC column at different concentrations of MeCN. The EOF marker was uracil (U) and the eluents comprised X mL MeCN added to 2 mL 38 mM TEAP at pH 3.0, and (10–2-X) mL H_2O added to a final volume of 10 mL. The stability of the electric field, as monitored by the current profile throughout the CEC separation, is indicated by the dashed line.

[b] Adapted from Walhagen, Unger, Olsson, and Hearn,[130] with permission.

Several different analytical and ultra-micropreparative CEC approaches have been described for such peptide separations. For example, open tubular (OT-CEC) methods have been used[290–294] with etched fused silicas to increase the surface area with diols or octadecyl chains then bonded to the surface.[330,331] With such OT-CEC systems, the peptide–ligand interactions of, for example, angiotensin I-III increased with increasing hydrophobicity of the bonded phase on the capillary wall. Porous layer open tubular (PLOT) capillaries coated with anionic polymers[295] or poly(aspartic acid)[296] have also been employed[297] to separate basic peptides on the inner wall of fused silica capillaries of 20 μm i.d. When the same eluent conditions were employed, superior performance was observed for these PLOT capillaries compared to the corresponding capillary zone electrophoresis (HP-CZE) separation. Peptide mixtures can be analyzed[298–300] with OT-CEC systems based on octyl-bonded fused silica capillaries that have been coated with (3-aminopropyl)trimethoxysilane (APS), as well as with pressurized CEC (pCEC) packed with particles of similar surface chemistry, to decrease the electrostatic interactions between the solute and the surface, coupled to a mass spectrometer (MS). In the pressurized flow version of electrochromatography, a μLC pump is also employed (Figure 26) to facilitate liquid flow, reduce bubble formation, and to fine-tune the selectivity of the separation of the peptide mixture.

Figure 26 Instrumentation Design for Pressurized Flow Electrochromatography of Peptides



MS-detection procedures have similarly been employed with capillaries packed with 1.5-μm RPC sorbents to examine[301] the impact on the selectivity with peptides when either pressure or electrically driven eluent flow was used.[302,303] Modified continuous beds using normal flow and counter-flow gradients for both capillaries as well as microchips have been proposed for separating larger polypeptides.[304–307] The retention behavior of small linear peptides with free N- and C-terminal groups have also been investigated[308] with the strong cation-exchanger, Spherisorb-SCX, documenting that strongly electrostatic interactions between the sorbent and positively charged peptides could be modulated with eluents of low pH, i.e. pH 3, with 60% MeCN. Work undertaken in this laboratory extended our earlier findings[130–133] on the use of chemically modified porous and nonporous sorbents of various ligand characteristics with particle sizes below 2 μm. These studies have provided a basis to rationalize the retention behavior of peptides in CEC with respect to their dual chromatographic retention and electrophoretic migration behavior.[309] These studies also provided a linkage to the underlying linear free energy dependencies, as well as new procedures to optimize CEC separations of peptides through temperature modulation, solvent composition effects and particle properties. Arising from these investigations the following conclusions can now be drawn.

(1) To obtain high EOF values in HP-CEC with RPC sorbents, such as C_8- or C_{18}-bonded silica, and thus to obtain enhanced selectivity and increased separation speed with peptides, it is necessary to use non-endcapped sorbents.[130,285] With *n*-alkylsilicas, a high

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surface density of ionized silanol groups on the particle increases the magnitude of the EOF, but, on the other hand, it also increases the silanophilic interaction with the peptide(s). At pH ≥8, the type I and type II silanols are fully ionized and therefore a high EOF will be obtained when the pH of the eluent/buffer is near this value. In contrast, at low pH values, i.e. pH <3.0, the silanol groups are partially protonated and a significant decrease (or lack) of the EOF effect can result with the currently available RPC sorbents. Alternatively, hydrolytically stabilized amino- and diol-phase coated silica materials[290,310] or macroporous, polyacrylamide/polyethylene glycol matrices[311] can be used for manipulation of the electroosmotic flow. Another way to establish a high EOF effect at a low pH is to use mixed-mode phases. For example, *n*-alkyl-bonded silica materials containing sulfonic acid groups give a high EOF even at low pH.

(2) Because the u_{eo} term is inversely related to the viscosity of the eluent, but proportional to $\sqrt{T}$, these dependencies will result in nonlinear retention behavior for peptides in terms of the overall effect of the EOF and the chromatographic interaction process when unexpected or for that matter deliberately induced changes in temperature occur in HP-CEC systems. As a consequence, even more complex nonlinear van't Hoff behavior, expressed as a function of ln κ_{cec} versus $1/T$, can be anticipated for the HP-CEC of peptides compared to that experienced in RPC.

(3) With pressurized HP-CEC, the electrophoretic migration rate and the mobile-phase flow rate can be optimized independently for peptide separations. Under such conditions, the retention factor, κ_{cec}, of a peptide(s) is dependent on both the electric field and the applied pressure[299] according to the following:

$$\kappa_{cec} = \frac{(a-c)E + dk_{LC}\Delta P}{cE + d\Delta P} \quad \text{eq 60}$$

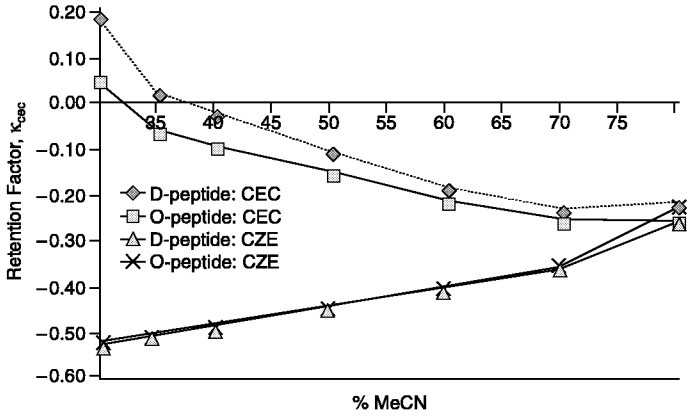
where a, c, and d are molecular descriptor constants of the peptide for a specified capillary and solvent system, E is the field strength (in $kV \cdot m^{-1}$), k'_{LC} is the capacity factor of the peptide in the HPLC mode, ΔP is the supplementary pressure drop. If no pressure is applied, then the retention factor, κ_{cec}, of the peptide(s) is, in theory, independent of the applied electric field. When an additional pressure is used, both the applied electric field and the pressure can be used to tune the κ_{cec} of the peptide and thereby increase selectivity in the separation.

(4) The tailing of basic peptides in HP-CEC can be suppressed by adding an organic base, such as triethylamine, triethanolamine, or hexylamine, to the buffer[130,288] or alternatively through the use of ion-pairing reagents.[21,130,312,313]

(5) Because of the combined effects from EOF and chromatographic interaction, the content of the organic solvent in the buffer is often higher than that used in RPC separation of peptides. This feature facilitates resolution of hydrophobic peptides, whilst at the same time permitting the fractions to be presented in a state compatible for direct ESI-MS analysis.

(6) As evident from Figure 27, which shows the CEC separations on a Hypersil C_8 sorbent of two desmopressin-related peptides versus their electrophoretic migration (HP-CZE), the chromatographic effect becomes small at higher concentrations of acetonitrile but increased as the content of acetonitrile was decreased. These results can be contrasted to the findings with these peptides when separated[130] on a Hypersil mixed-mode (C_{18}/SCX) CEC column and also on the Spherisorb C_{18}/SCX material. With the mixed-mode CEC sorbents the elution order of these peptides was different from that obtained with the Hypersil C_8 (or Hypersil C_{18}) sorbent although a common eluent and capillary configuration was used. Such selectivity reversals provide significant scope in nanoseparation and isolation studies.

Figure 27 Overlay of the Plots of the Retention Factors κ_{cec} and k_{cze} of Desmopressin (D) and Oxytocin (O) versus the Concentration of MeCN[130,133][a,b]



[a] Using the CEC Hypersil C_8 and CZE-opened fused silica capillary systems operated at 20 kV, with the same buffer/eluent containing X mL MeCN added to 2 mL 25 mM TEAP at pH 3.0, and (10–2-X) mL H_2O added to a final volume of 10 mL.

[b] Adapted from Walhagen, Unger, Olsson, and Hearn,[130,133] with permission.

(7) One of the major differences between HP-CEC and μ-HPLC with RPC sorbents is that the peak dispersion of peptide(s) is larger in μ-HPLC than in HP-CEC. As noted above, the dominating term controlling the total plate height based on the van Deemter equation in HP-CEC is the longitudinal diffusion of the peptide(s) (B/v), whereas in μ-HPLC the eddy diffusion ($Av^{1/3}$) and the mass transfer (Cv) terms dominate the peak dispersion of peptide analytes.[314] Since the B term is proportional to the diffusion coefficient of the peptide (i.e., $B = 2\gamma D_m$) the molecular properties of the peptide(s) per se rather than the physical properties of the sorbent provide the dominant contribution to the peak dispersion in HP-CEC. Some major differences between the three separation methods CZE, μ-HPLC, and CEC are illustrated in Table 5. Since the separation mechanisms differ significantly between these three analytical methods, it is necessary to distinguish between the retention of analyte in terms of their definition, i.e. for μ-HPLC: capacity factor k' (or chromatographic retention factor), for HP-CZE: electrophoretic velocity factor u_e and for HP-CEC: electro-chromatographic retention factor κ_{cec}.

In view of the increasing level of research activities associated with the application of HP-CEC for peptide analysis, it can be confidently expected that new miniaturized, high-throughput separation systems will become available over the next several years, fully integrated with MS/MS instrumentation and other types of spectrometers capable of determining structural characteristics of the resolved peptides at the low nanomole level.

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Table 5 A Comparison of CZE, μ-HPLC, and CEC Procedures for the Analysis of Peptides

Properties	HP-CZE	μ-HPLC	HP-CEC
Velocity of eluent (v)	electrically driven	pressure driven	electrically driven
Column i.d. (μm)	25–100	100–500	50–100
Separation by	differential electrophoretic mobilities, charge state, and Stokes radii of peptides	chromatographic interaction of peptides with the stationary phase	both chromatographic interactions and electrophoretic mobility of the peptides
Capillary	open tubular fused silica capillary with/without wall modifications	capillary packed with reversed-phase or mixed-mode sorbent materials	capillary packed with reversed-phase or mixed-mode sorbent materials
Flow profile	plug flow profile front is flat, peak dispersion dominated by the longitudinal diffusion term $H = B/v$	parabolic flow between particles in well-packed capillaries; peak dispersion can be approximated to pseudo-plug flow shape under some optimized v conditions but overall plate height, H, is given by van Deemter dependency, i.e. $H = Av^{1/3} + B/v + Cv$, and dominated by the eddy diffusion, $Av^{1/3}$, and the mass transfer, Cv, terms as v increases, hence, H increases at high v values	plug flow between particles in well-packed capillaries; overall profile front is also plug flow at high v; peak dispersion, H, has very negligible contributions from the eddy diffusion and mass transfer terms of the van Deemter dependency, i.e. both $Av^{1/3}$ and $Cv \approx 0$, with H dominated by the longitudinal diffusion, B/v, term, hence, H decreases as v increases
Dispersion of zones relatively to each other	small	large	small–medium

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7.2 Chromatographic Methods

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When reading the peptide literature, such descriptive words as "pure", "highly purified", and "homogeneous" are encountered. These words are generally used to qualify the extent of effort devoted to the purification of such peptides rather than to specify the actual degree of purity by multiple and orthogonal analytical techniques. My personal conclusion, from reading some of these articles, is that peptides are typically reported to be "pure" after being subjected to one preparative reversed-phase HPLC step (generally 0.1% TFA/MeCN on a C_{18} column/cartridge) and analyzed using the same eluent on an analytical column of the same silica gel. A peptide is "highly purified" when, after the above purification step, an analytical run in the same system as the one used for purification yields a significant main absorbance with some closely related minor impurities that are rarely quantitated. Last but not least, a peptide is said to be "homogeneous" when in the same sentence both "pure" and "highly purified" descriptors have already been used. Clearly, this is a cynical point of view and the use of these terms in the scientific literature is unsatisfactory and should not be accepted.

Therefore, in this chapter and cynicism aside, those analytical techniques that were found most useful and reproducible for the successful detection, quantitation, and characterization of peptides are described. It is certainly far from being all-encompassing and readers should be encouraged to experiment with systems other than those described here when they encounter unsatisfactory elution patterns (tailing spots on TLC or tailing absorbances in liquid chromatography or CE) or what may seem to be artifactual. Abundant relevant literature is cited for reference.

It should be noted that no single technique will give an absolute appreciation of the degree of purity of synthetic peptides, regardless of the strategy used for their synthesis (stepwise assembly, enzymatic synthesis, fragment condensation, or in vitro expression) or their size. In order to be accurate, any method of analysis must detect all components of a mixture to the same extent without selective loss of any of the components of the mixture on the chromatographic support. Ideally, as is the case when peptides are synthesized and characterized under current Good Manufacturing Procedures, chromatographic methods must be validated. The reader interested in this regulatory aspect may consult the specialized literature. See, for example, the article by Krull and Swartz on "System-suitability samples – what they are, why we use them, and what they should do".[1]

The fact is that a synthetic peptide will never be homogeneous (100% pure) and the question remains as to what level of purity is acceptable for a particular investigation. Most scientific journals have their own policy in this regard; this question will therefore remain unanswered, although it is clear that for sequence determination (Edman degradation) and structural studies, peptides and proteins should be as pure as possible. Similarly, peptides should meet the Food and Drug Administration's guidelines before they can be used for clinical investigations. However, the total synthesis of a peptide cannot be claimed until the peptide has been fully characterized using all chromatographic and structural tools available and applicable. Ideally, evidence should also be given that the synthetic and purification protocols can be reproduced on a preparative scale.

Some of the technical details associated with those chromatographic techniques that can be used to characterize peptides are presented. These include the use of TLC, HPLC, and CE. Partition chromatography, gel permeation, and ion-exchange chromatographies on soft gels, and countercurrent distribution, all formerly used for analytical and preparative purposes, are now superseded by normal-phase, reversed-phase (RP), exclusion, and ion-exchange chromatographies on noncompressible supports operated under high pressure. Similarly, gas chromatography of derivatized peptides[2] has been supplanted by RP-HPLC of

the underivatized analytes. Additionally, paper chromatography[3] and paper electrophoresis, used extensively in the 1940s to 1970s, are now obsolete and have been replaced by CE.

7.2.1 Thin-Layer Chromatography and Coloring Reagents

In the last 20 years, analytical methods (HPLC and CE) that are more resolutive and quantitative than TLC have been developed for the analysis and quantitation of peptides. As a result, only a limited number of applications of TLC that remain of current interest will be addressed.

Amino acids and peptides can be separated and their purity estimated rapidly and economically using TLC. In fact, until not long ago, TLC was to the peptide chemist what the hammer was to the carpenter, and extensive literature exists that describes all aspects of TLC including supports and solvent systems as well as the theory behind the separations.[4–14] TLC supports and solvent systems were developed since the beginning of this century to assess the purity of protected and unprotected amino acids and peptides reproducibly and quantitatively.

Today, TLC has lost its popularity as an analytical tool for peptides since it is much less resolutive and quantitative than HPLC. This evolution came at great cost: expense of the instrument, increased time consumption, and loss of some information on the composition of the analyte. An example of the last case is the use of color reagents that are specific for free amino functions (ninhydrin) and an imidazole ring (Pauly reagent). This is why TLC still plays a critical diagnostic, rather than analytical and quantitative, role in a peptide laboratory. As shown in Sections 7.2.1.1–7.2.1.4, the development of chiral and reversed-phase silica gel coated plates and a variety of detection methods has broadened the field of use for this simple and affordable technique. In fact, there are several major competing suppliers of TLC products, including Aldrich, Alltech, Analtech, J.T. Baker, EM Science, Fisher, and Whatman. The literature that these companies provide to promote their products is the most updated source of information. Additionally, they offer a number of reference books. Stahl[9] describes the science and art of TLC, discussing history, specific TLC methods, and formulations. Chapters by Brenner et al. on amino acids and derivatives[15] and by Waldi on spray reagents[16] for TLC are particularly useful. Touchstone[11] and Bauer et al.[12] describe in detail the basic operations necessary for successful TLC. Kirchner also covers all aspects of modern TLC in two parts: general aspects and chromatography of specific product groups.[10] The contribution by Jork et al.[14] is an all-encompassing book covering an introduction to the method of TLC, references, and the description of laboratory experiments and reagents. Dallas et al.[7] present in their monograph a detailed review of the most recent developments in TLC, inclusive of sophisticated instrumentation for qualitative and quantitative detection, sorbents, modifiers, and applications. Sherma and Fried present the latest techniques of TLC inclusive of instrumentation and applications of overpressurized, rotational, and high-performance qualitative TLC.[13,17]

In summary, most manufacturers offer a choice of adsorbents of different thicknesses (analytical and preparative) with varying composition, pore size and particle size and distribution, a choice of composition of the inert binder, and the optional addition of a fluorescent indicator for different types of chromatography, see Sections 7.2.1.5–7.2.1.7. Additionally, these adsorbents are spread over a choice of backings (glass, plastic, and aluminum sheets of different sizes). Aluminum and glass are preferred to plastic supports, that will melt at the elevated temperatures often reached with charring and certain sprays. An advantage of plastic and aluminum backing is that they can be more easily cut to the desired size than glass supports.

for references see p 649

For maximal resolution, the product in solution should be carefully applied in order to avoid under- or over-loading. This is best determined experimentally depending on the reagent(s) used for detection. When using high performance TLC plates, characterized by a homogeneously small particle size (4.5–10.0 μm, depending on the manufacturer), a sample applicator is recommended. Additionally, some manufacturers offer TLC plates with a preadsorbent strip to maximize sample volume and ease of sample application. To avoid artifacts and achieve a certain degree of reproducibility, it is also important to saturate the developing chamber with the solvent vapors. This is best achieved with filter paper lining the sides of the tank or using a channeled developing chamber. TLC is ideal for the rapid screening of reaction mixtures as it often takes just a few minutes to achieve separation of starting material, desired product, and byproducts.

7.2.1.1 Adsorption Thin-Layer Chromatography

Adsorption TLC is best carried out on silica gel plates. For example, SilicaGel 60 F_{254} on aluminum sheets from EM Science is used to separate weakly to strongly polar substances. Some solvent systems used for development are described in Tables 1 and 2. It should be noted that these solvents are acidic, neutral, or basic. Chromatographic results (R_f and spot shape; R_f is ratio of the distance of the compound from the origin over the distance of the solvent front from the origin expressed in cm) will, therefore, depend on the support, the eluent, the isoelectric point of the peptide, and quite often the peptide's molecular weight. A poor or unfortunate choice of support or eluent will result in spots that remain at the origin, streak, or migrate with the solvent front. One of the advantages of adsorption TLC is the almost unlimited choice of solvent systems with unique selectivities that can be easily adapted to particular peptide families. It is, therefore, a very versatile technique.

Table 1 Solvent Systems for TLC of Protected Amino Acids and Peptides[19]

Solvent Systems[a]	Adsorption	Reversed Phase	Ref
$CHCl_3$ (70–95), MeOH (28–3), AcOH (2)	×		[19]
acetone (98), AcOH (2)	×		[19]
$CHCl_3$ (5), acetone (1)	×		[19]
PrOH (84), concd NH_4OH (37)	×		[19]
TFA (0.1%), H_2O (40–60), MeCN (60–40)		×	[19]

[a] Numbers in parentheses are (v/v).

7.2.1.2 Reversed-Phase Thin-Layer Chromatography

Reversed-phase chromatography is carried out on derivatized silica gel such as RP-2, RP-8, RP-18, cyanopropyl, or diphenyl. Whereas RP refers to reversed phase, the numbers 2, 8, and 18 refer to the length of the hydrocarbon chains coating the silica gel. Other coatings of silica gel include the cyanopropyl and diphenyl groups. As these derivatized silica gels resemble those found in RP-HPLC columns, the TLC plates coated with these materials are often used to determine the solvent strength (MeCN concentration in this case) best suited for column chromatography in the reversed-phase mode (see Section 7.2.2.3 for more details).

In adsorption TLC, selectivity is mostly determined by the developing solvent system, whereas in RP-TLC the selectivity is determined primarily by the reversed-phase support and degree of carbon loading on the silica gel base. It should be kept in mind that the coating material of RP-TLC plates will always contain a certain amount of underivatized silanols that will confer to the plates some adsorptive and ionic characteristics that affect selectivity. It is, therefore, erroneous to think that all RP-C_2, C_8, C_{18}, cyanopropyl, and diphenyl materials will behave similarly and it is strongly suggested that, in an optimization process, an investigator should experiment with plates from different suppliers.

Common solvent systems are shown in Table 2. Although there is a direct correlation between increasing R_f values and increasing the eluotropic strength of the organic solvent, Wilson pointed out that this correlation exists over a very wide range, contrary to what is seen on RP-HPLC. Additionally, R_f values are not particularly sensitive to pH, which makes RP-TLC a particularly robust and simple technique to use.

Table 2 Solvent Systems for TLC of Unprotected Peptides

Solvent System[a]	Adsorption	Reversed Phase	Partition	Chiral	Ref
BuOH (4), AcOH (1), H_2O (5) UP	×		×		[19]
PhOH (88%, liq) (3), H_2O (1)	×		×		[19]
BuOH (1), EtOAc (1), 0.2–1 M NH_3 (2) UP	×				[19]
BuOH (1), EtOAc (1), AcOH (1), H_2O (1)	×				[19]
BuOH (1), iPrOH (1), 1 M NH_3 (2.5), EtOAc (1)	×				[19]
iPrOH (2), 1 M AcOH (1)	×				[19]
iPrOH (2), 1 M NH_3 (1)	×				[19]
BuOH (5), Py (3), 0.1% AcOH (11) UP	×		×		[19]
isopentyl alcohol (7), Py (7), H_2O (6)	×				[19]
EtOAc (5), Py (5), AcOH (1), H_2O (3)	×				[19]
Py (50), AcOH (30), H_2O (15)	×		×		[19]
TFA (0.1%), H_2O (20–60), MeCN (80–40)		×			[19]
MeOH (1), H_2O (1), MeCN (4)				×	[19]
MeOH (5), H_2O (5), MeCN (3)				×	[19]
MeOH (1), H_2O (8)				×	[19]

[a] UP is upper phase; numbers in parentheses are (v/v).

7.2.1.3 Partition Chromatography

Partition chromatography is best carried out on cellulose plates to separate weakly to strongly polar substances. Solvent systems are generally similar to those used for silica gel plates.

7.2.1.4 Chiral Thin-Layer Chromatography

TLC plates based on ligand exchange were developed in the mid-1980s by Günther et al.[18] and are now commercially available. The plates are coated with C_{18} silica gel containing Cu^{2+} and a chiral reagent (Chiralplate). The separation mechanism is based on ligand exchange whereby diastereomeric complexes are formed between the Cu^{2+} ion, the chiral selector, and the substrate. Enantiomeric resolution is possible if the antipodes of the chiral substrate form complexes of different stability. Complex formation with the unprotected N- and C-termini of amino acids is a prerequisite for successful separations. TLC plates have been developed using solvent systems similar to those used for reversed-phase separations described in Tables 1 and 2. The eluting system can vary considerably in terms of organic strength, depending on the hydrophobicity of the substrate. The examples given in Tables 1 and 2 are those described by Rausch for the separation of several amino acids and amino acid dipeptides.[19] EM Science 14101-1 HPLC precoated plates CHIR and Aldrich Z14,87 (5×20 cm) chiral silica gel plates and the MeOH/water/MeCN (1:1:4) solvent system were used successfully to follow the enzymatic resolution of Trp(6-Br).[20,21]

7.2.1.5 Nondestructive Detection Methods: UV Light, Iodine Vapors, and Water

7.2.1.5.1 Exposure to UV Light

The coatings of many TLC plates contain a manganese-activated zinc silicate that glows bright green under fluorescent light (254 nm). Samples that absorb UV light at that wave-

for references see p 649

length are detected owing to fluorescence quenching and appear as dark spots. Hand-held UV lamps or UV-viewing cabinets are commercially available that allow visualization of most peptides on these TLC plates. UV-positive spots should be marked lightly with a pencil and labeled. Because this mode of detection does not alter the characteristics of the plate or the products, this visualization should be done first. **CAUTION:** *Glasses that absorb UV light should be worn to prevent eye damage.*

7.2.1.5.2 Iodine Vapors

Upon exposure to iodine vapors, peptides and many other organic molecules on TLC plates will turn different shades of brown, depending on the duration of exposure. Iodine crystals in a wide-mouth jar with a tight cover will readily saturate the atmosphere, into which dried TLC plates are quickly introduced. After a few minutes, brown spots appear where peptides have migrated. These spots should be marked lightly with a pencil as the reaction is reversible and the color will disappear with time. As this reaction is reversible, in most cases (see Jork et al.[14] for exceptions), it does not interfere with other color tests and can, therefore, be used after UV viewing and prior to the use of any color reagent that involves a specific chemical reaction. **CAUTION:** *Iodine is a skin and mucous membrane irritant. Appropriate safety precautions and procedures should be adopted when handling this reagent. This procedure should be carried out in a fume hood.*

7.2.1.5.3 Water

Another nondestructive detection method takes advantage of the low solubility of certain peptides or protected peptides. Differential wetting of the TLC plate's surface can be observed (especially against the light) after quickly dipping the plate in water or lightly spraying with water. Upon drying, translucent spots appear where hydrophobic substances are present.

7.2.1.6 Detection Methods: Coloring Reagents

7.2.1.6.1 Ninhydrin Test

Ninhydrin will react with free amino functions to yield a distinctive purple color. Of all the color tests, this is the most versatile and easiest to perform.[22]

To perform the ninhydrin test a solution of ninhydrin in ethanol or acetone (0.2% w/v) is sprayed lightly over the plate. Brown to purple colors develop slowly (within 1–2 min) upon heating to 60–90 °C. The spray is stable for several weeks.

It should be noted that before developing a plate with ninhydrin, a reliable and fast developing positive control can be obtained through application of a wet thumb print on the upper corner of the plate. Whereas the most classic way to develop a plate is to leave it in an oven (ca. 90 °C) for a few minutes, heating the plate with a hot air gun is preferred. In this manner, a rough appreciation of how much material is present on the plate is obtained by comparing how fast the spots and the thumb print develop. Different shades of color that distinguish substances that may have similar R_f but different reactivities to ninhydrin can be distinguished in this manner. For example, primary amines are more reactive and give a brighter color than secondary amines.

7.2.1.6.2 Fluorescamine Spray

Fluorescamine will react with free primary amino groups to yield spots that are easily distinguishable under UV light.[23] Secondary amino groups will not react.

In a first step of the detection method using fluorescamine spray, a solution of TEA (10% v/v in CH_2Cl_2) is lightly sprayed over the plate and allowed to dry for a few seconds. The fluorescamine spray is then applied (0.1% w/v in acetone) followed by a second application of the TEA spray. The plate is viewed under a long wavelength UV light (366 nm).

7.2.1.6.3 Pauly Reagent

The Pauly reagent will react with histidine, tyrosine, thyroxine, and dihydroxyphenylalanine to give a bright yellow and red color. Diiodotyrosine and creatinine give much weaker reactions.

In order to prepare the Pauly reagent two solutions, solution I (0.4 M sodium sulfanilate in water) and solution II (0.4 M sodium nitrite in water) are made separately. Just before use, solution I with solution II, 0.25 M HCl, and 2 M NaOH (1:1:8:10) are mixed successively. This solution, which is not stable, should be applied as a spray within minutes and discarded.

The ninhydrin test will not interfere with the Pauly reagent.

7.2.1.6.4 Ehrlich Reagent

The Ehrlich reagent will react with tryptophan and ureas to give a purple and yellow color, respectively.

To prepare the Ehrlich reagent, solution I (10% 4-(dimethylamino)benzaldehyde in concd HCl) is first made. Just before use, solution I and acetone (1:4) are mixed in a fume hood and used as a spray.

The ninhydrin test will not interfere with the Ehrlich reagent. Therefore, a plate can be examined under UV light, immersed in iodine vapors, treated with ninhydrin, and the Pauly or Ehrlich reagents successively.

7.2.1.6.5 Nitroprusside – Cyanide Reagent

The nitroprusside – cyanide reagent will react with thiol and disulfide groups to give a bright red color.

For the nitroprusside – cyanide reagent two solutions are prepared. For solution I, sodium nitroferricyanide (sodium nitroprusside, 1.5 g) is dissolved in 2 M H_2SO_4 (5 mL) to which MeOH (95 mL) and NH_4OH (10 mL) is added. A precipitate forms that is discarded. The clear, red solution is refrigerated. For solution II, sodium cyanide (2 g) is dissolved in H_2O (5 mL) and diluted with MeOH (95 mL), this is filtered if necessary. **CAUTION:** *Sodium cyanide can be absorbed through the skin and is extremely toxic. Appropriate safety precautions and first aid procedures should be adopted when handling sodium cyanide.*

Thiol-containing substances on the TLC plates will stain bright red after exposure (several min) with solution I. Disulfide-containing substances will turn red more slowly upon administration of solution II. Maximum intensity is reached after about 10 minutes.

7.2.1.6.6 Chlorine Peptide Spray

Chlorine spray is used to identify peptides and amino acid derivatives.[23] A blue color is given by almost all compounds that contain NH groups. Alcohols and acids interfere with the test and should be removed prior to application of the spray. The test fails when dioxane is used to develop the plate or when certain fluorescent indicators are present. Several protocols have been described for this test.[4] These protocols vary in the reagents that are used to generate the reactive chlorine and the iodine–starch reagent that is used for visualization.

for references see p 649

To detect peptides and amino acid derivatives using the chlorine peptide spray the following procedure is carried out. Solution I (1% *tert*-butyl hypochlorite in cyclohexane) and solution II (1% soluble starch and 1% KI in water) are prepared. To prepare solution II starch is dissolved in boiling water first and potassium iodide is added to the cold solution. A small amount of chloroform is added to inhibit bacterial growth.

In the first step, the thoroughly dried TLC plate is sprayed with solution I and dried in a ventilated area. In the second step, the TLC plate is sprayed with solution II. Blue spots that appear will fade rapidly. This test needs careful optimization of the first step. Indeed, not enough drying at step I results in the appearance of a blue colored plate with no distinction of spots, while too much drying results in either a faint color or no color at all.

7.2.1.6.7 Charring

Another test that may be considered is charring at 150 °C after a spray of 5% sulfuric acid in ethanol. Of all tests, it is the most dramatic in that it will not allow any further analysis. Charring will leave dark spots for all organic materials on glass, aluminum, and polyester plates/sheets. Higher temperatures will melt the polyester sheets.

7.2.1.7 Conclusion

A very limited number of spray reagents with advantages of sensitivity, selectivity, and ease of application have been described. Additional tests have been reported that can be applied to more specific situations.[5,9]

7.2.2 High-Performance Liquid Chromatography

7.2.2.1 General Considerations

7.2.2.1.1 Historical

With the advent of HPLC, a new tool was born that revolutionized peptide/protein chemistry as a whole as it allowed not only purification of biologically active peptides and proteins from complex mixtures of tissue or plant extracts,[23] but also allowed purification of synthetic unprotected peptide mixtures analytically and preparatively.[24–26] This was particularly significant in that purity could be assessed of peptide intermediates made by the classical solution-phase methodology that promoted characterization of all intermediates, as well as the purity of final products made by the solid-phase approach of Merrifield.[27]

Whereas HPLC had been first used for protected amino acids and small peptides using normal-phase silica gel based supports,[28] the years 1976–1977 saw the first publications describing the use of RP-HPLC for unprotected peptides of the size accessible by both classical and solid-phase approaches, using carbonaceous phases on silica gel supports in a reversed-phase mode.[29,30] Since then, thousands of individual reports and reviews in journals, books, and magazines as well as contributions presented at yearly scientific meetings have been published that cover all aspects of theoretical and practical chromatography.[31–44] It is impossible in a few pages to summarize the state of the art advances of these techniques. The reader is encouraged to consult such literature as "LC-GC – The Magazine of Separation Science" or companies' bulletins as a valuable resource to find the latest developments in analytical methods inclusive of LC and CZE.[45] This section will illustrate, with specific examples, a limited number of practical aspects of RP-HPLC that are useful to the analytical and synthetic peptide chemist. The main objective is to rapidly obtain a separation of the components of peptide mixtures with good sensitivity and high recovery. The variables

are numerous and are related to the nature of the sample, the equipment, the column, the experimental conditions, the mobile phase, and the temperature.[46] Only a few of these parameters will be addressed here.

7.2.2.1.2 Equipment

An HPLC system is composed of several elements: controller, pump(s), mixer, injector (manual or automatic), column, on-line detector(s), optional fraction collector, recorder/integrator, and some dead volumes such as connecting lines and mixers. It is critical that each element of an HPLC system be functioning optimally. The controller and the pumps should generate constant flow rates and reproducible gradients. Such an integrated system will periodically be subject to problems. Changes in back pressure resulting from pump (check valve, seals, and other parts) malfunction will result in an unsteady baseline. Similarly, a faulty mixer will result in a wavy baseline and distorted absorbance shapes. A dirty manual injector is a source of contamination from run to run and should be both flushed and backflushed with eluent between runs. Problems with the column are caused from it becoming partially plugged, developing a void volume at its top, or being incompatible with the product to be eluted. Bubbles that are generated after decompression of an undegassed solvent will create havoc with the detection system. Finally, the connections between the different elements of the system must fit precisely to eliminate dead volumes and leaks. Despite that effort, every system has a certain amount of dead volume, inclusive of a static or dynamic mixer, the injector, and the connecting tubings. It is important to know the size of this dead volume when running gradients because the composition of the buffer system at the top of the column will be shifted from that shown on the controller.

7.2.2.1.3 General Maintenance

For best operation, HPLC instruments should be kept in a dust-free, temperature-controlled environment. An HPLC system should be kept clean and used regularly to avoid the drying of seals or the accumulation of salts. The system should be flushed thoroughly with water/methanol (50:50) or water/acetonitrile (50:50) at the end of every day and not be allowed to dry out. The disadvantage of methanol over acetonitrile is that it generates significantly higher back pressures on silica gel based supports, although it is less expensive than acetonitrile. Both solvents are UV transparent at the wavelengths used for the detection of amide bond containing compounds (210–225 nm). Yet, acetonitrile is generally the organic solvent of choice and is used as the organic modifier in RP-HPLC or as a cosolvent that enhances peptide solubility in ion-exchange and exclusion chromatographies (see Sections 7.2.2.4 and 7.2.2.5).

7.2.2.2 Normal Phase

The stationary phase for normal-phase chromatography is silica gel that is more uniform in particle size (20–40 μm), has a pore size of 80–300 Å, and spherical or irregular particle shape. In normal phase, peptides elute in order of decreased hydrophobicity, with the most hydrophilic peptide eluting last at the highest concentration of the hydrophilic solvent. This would be the aqueous or alcoholic component of the buffer system, which is also often acidic.

The advantage of normal phase over the reversed-phase mode of liquid chromatography is that the column load can be 10-fold higher in the former. The main limitation of normal phase is that only those peptides that are soluble in organic solvents, such as chloroform and methanol or ethanol mixtures, will yield some separation. As a result, RP-HPLC is preferred for the separation of unprotected peptides as well as of protected or partially protected derivatives.

for references see p 649

7.2.2.3 Reversed-Phase High-Performance Liquid Chromatography

Peptides interact with the carbonaceous phase of RP-HPLC supports and are eluted by increasing the strength (organic phase, most often acetonitrile) of the buffer.

7.2.2.3.1 Solid Support

Solid supports for RP-HPLC are, in most cases, silica gel derivatized with cyano, C_4, C_8, C_{18}, or diphenyl groups and, therefore, have limited stability at pH values <2.0 or >8.0. Less commonly used supports are polymer based that will withstand high pH. Besides their carbon load, silica gel particles can be of different sizes, shapes, and with pore sizes ranging from 80 to 500 Å. Literally hundreds of different columns are available.[47,48] A commonly used particle is spherical, 5 μm in diameter, and with 300-Å pore size. Whereas C_4 and C_8 columns are preferred for proteins or very hydrophobic peptides, C_{18} is most commonly used for a wide variety of peptides.

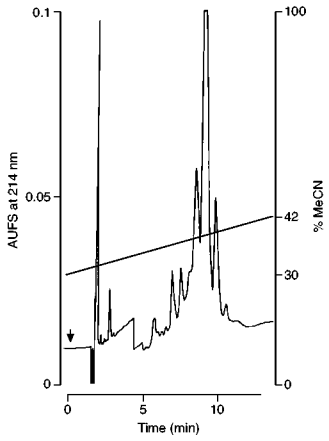
7.2.2.3.2 Solvent Systems

Owing to the instability of silica gel based columns at pH values <2.0 or >8.0, mobile phases must have a pH range of 2.1–7.5. Within this range, a large number of solvent systems have been described. Solvent systems are selected on the basis of several criteria: optimal resolution, volatility, UV transparency, biocompatibility, selectivity, viscosity, and cost.

7.2.2.3.2.1 Volatile Solvent Systems

The most commonly used volatile aqueous solvent is water containing 0.1% TFA.[49] To generate the acetate salt instead, 0.2–0.5% acetic acid is often used preparatively. An advantage of the triethylammonium formate (TEAF) buffer is that the pH can be adjusted between 3.0 and 7.5 by increasing the concentration of the TEAF.[24] Each one of these solvents containing similar concentrations of acetonitrile will generate different profiles of the same peptide mixture as the result of their individual properties and resulting selectivities. This can be taken advantage of preparatively or in the isolation of native molecules.[23,26,50] Perfluorinated carboxylic acids have also been used successfully.[51] A particularly striking example of the use of heptafluorobutyric acid (HFBA) is illustrated in Figure 1, where synthetic inhibin-31-OH (H-His-Asn-Lys-Gln-Glu-Gly-Arg-Asp-His-Asp-Lys-Ser-Lys-Gly-His-Phe-His-Arg-Val-Val-Ile-His-His-Lys-Gly-Gly-Lys-Ala-His-Arg-Gly-OH, MW = 3591) was obtained as a single symmetrical absorbance after purification in two solvent systems [triethylammonium phosphate (TEAP, pH 2.25) and 0.1% TFA]. An unacceptable amino acid analysis (poor ratios for basic amino acids) suggested that the compound could not be pure. It was rationalized that if the peptide was impure because of the deletion of one or several arginines, histidines, or lysines, a strong cation pairing reagent would most likely help discriminate between these different entities. Indeed, Figure 1 shows the presence of several major impurities as the result of chromatography using HFBA; the major eluting absorbance (<50% of total) was that of the desired product, as shown by MS.

Figure 1 HPLC Analysis of Inhibin-31-OH[26]a, b



[a] Column: Vydac C_{18} (300 Å, 5 μm, 0.46×25 cm); solvent: 0.1% HFBA; flow rate: 2.0 mL/min.
[b] Load, ca. 20 μg.

7.2.2.3.2.2 Nonvolatile Solvent Systems

If UV transparency is an issue for sensitive detection, nonvolatile solvents have been used. These include phosphoric acid,[52,53] and TEAP,[24] among others. The latter is preferred as it allows for pH adjustment (2.0–8) and is biocompatible in most in vitro biological systems. It has been proposed that the role of the added triethylamine (to phosphoric acid, formic acid, TFA, and acetic acid) is to cap the unreacted silanols present on the silica gel. This issue has been recently addressed in a short review.[54] The TEAP buffer has been used extensively for the isolation of natural products[23] and synthetic peptides.[26] Preparatively, the TEAP buffers were found to be good first solvents to use as they are highly resolutive with a selectivity that is often different from that conferred by 0.1% TFA. Since excess TFA will be eliminated by lyophilization, it is used as the last purification and desalting step.

7.2.2.3.3 Solvent Preparation

Typically, two solvents are prepared that are mixed in a controlled manner by the pumping system according to a preset gradient. The content of the organic modifier is increased linearly, stepwise, or exponentially in order to displace the sample. There is, therefore, the need for two solvent systems referred to as A (which is the aqueous component) and B (which contains the organic modifier HPLC grade, UV transparent).

0.1% TFA System

Component A: Distilled TFA (1.0 mL) is added to HPLC grade water (deionized, 1 L).

Component B: Distilled TFA (0.9 mL) is added to HPLC grade water/MeCN (40:60, 1 L).

As the gradient develops, a shift of the baseline may be observed resulting from the different absorbance of the A and B components. This can be adjusted experimentally by increasing or decreasing the amount of added TFA to component A or B.

TEAP 2.25 System

Component A: 85% H_3PO_4 (AR grade, 1 mL) is added to HPLC grade water (1 L). To this solution distilled TEA is added under stirring to adjust pH to desired value.

Component B: MeCN is added to component A (60:40).

TEAP 7.0 Buffer System

Component A: 85% H_3PO_4 (AR grade, 0.5 mL) is added to HPLC grade water (1 L). To this solution distilled TEA is added under stirring to adjust pH to desired value.

for references see p 649

Component B: MeCN is added to component A (60:40).

Both H_3PO_4 and TEA should be free of UV absorbing materials. If they are not, a convenient procedure to eliminate all UV absorbing materials is to apply component A to a preparative column (high or low pressure) of C_{18} material that will absorb all hydrophobic UV absorbing impurities. Typically, the concentrations of H_3PO_4 and TEA are increased five- to eightfold for preparative scale uses at acidic pH and two- to fourfold at neutral pH.

Ammonium Acetate System

Component A: Ammonium acetate (Fluka, puriss. p.a., 7.7 g) is dissolved in HPLC grade water (1 L).

Component B: UV transparent (HPLC grade) MeCN is mixed with component A (60:40).

1% Acetic Acid Buffer System

Component A: Glacial AcOH (5 mL) is added to HPLC grade water (1 L).

Component B: Glacial AcOH (9.9 mL) is added to HPLC grade water/MeCN (40:60, 1 L).

This system will absorb at 210–220 nm. Detection at 280 nm is possible.

The ammonium acetate and acetic acid systems are less resolutive and practical (not UV transparent) than the TFA and TEAP based systems. The former are generally used to generate the acetate salt of peptides. For example, a peptide fraction partially purified with the TEAP system will be reapplied onto an HPLC column after dilution with water (1:1). The column is washed with 1% AcOH and the peptide is eluted by increasing the concentration of MeCN. As another example, the peptide is partially purified using 0.1% TFA, then reapplied onto an HPLC column after dilution with water (1:1). The column is washed first with an ammonium acetate solution to eliminate all TFA, followed by the 0.5% AcOH system to eliminate the ammonium counterion. The peptide is displaced with increasing concentrations of MeCN.

It is recommended that all HPLC solvents be passed through a Millipore filter to eliminate any solid particles that might plug the columns, and degassed to avoid the possibility of bubble formation during decompression of the solvent. Bubbles will interfere with the detection system. Components A and B can be kept refrigerated for a few weeks. **CAUTION**: *The aqueous component of the TEAP system will grow mold/bacteria upon standing at room temperature.*

The B components as described above all contain a certain amount of aqueous phase to safeguard the B pump seals in two-pump systems. If the pumping system uses one pump and partitioning valves, the B component as well as additional ones can be pure solvents. Indeed, with very hydrophobic peptides that will not elute with 60% MeCN, the concentration of the B component in MeCN must be increased to 80 or even 100%.

7.2.2.3.4 Basic Operating Instructions

Because column performance decreases with age, it is important to keep good records of its performance. Entries useful to keep track of are: chromatogram number, sample identification number, amount injected, volume injected, solvents used (A and B components), column characteristics (ID number, packing, size), back pressure, flow rate, detector setting (UV, absorbance units full scale at given wavelength), gradient conditions, results (retention time of desired product), remarks (baseline shifts, unusual wash profile, etc.), and date.

Prior to any analytical run, the column must be washed with full strength component B in order to free the column of any impurities remaining from earlier run(s). In general, a 10 minute wash at the flow rates shown in Table 3 will be sufficient. The column is then equilibrated with the starting composition of the solvent to be used for as long as it takes to obtain a steady baseline (2–4 column volumes). The material is injected and the desired gradient is run followed by a wash (2–4 column volumes) at the highest concentration of component B before reequilibration to initial conditions.

Table 3 Suggested Flow Rates, Loads, and Injection Volumes for a Given Column Size

Column/Cartridge Size (cm)	Flow Rate ($mL \cdot min^{-1}$)	Load (g)	Injection Volume (mL)
2.1×150 or 250×10^{-3} analytical	1.0	1×10^{-6}–2×10^{-6}	5×10^{-3}
4.5×150 or 250×10^{-3} analytical	2.0	5×10^{-6}–10×10^{-6}	5×10^{-3}–10×10^{-3}
$10 \times 250 \times 10^{-3}$ semipreparative	4.0–8.0	1.0×10^{-3}–5.0×10^{-3}	1.0
2.3×25 preparative	8–20	20×10^{-3}–50×10^{-3}	1.0×10^{-3}–5.0
4.5×300 preparative	70–100	1.0–2.0 crude $<500 \times 10^{-3}$ partially purified peptide	>60

Always flush the HPLC system for 20–30 min at flow rates suggested in Table 3. This is inclusive of pumps, injector, columns, and detector with MeCN/water (50:50) when not in use.

Samples should not be applied to columns in a solvent containing higher amounts of organic modifier than that of the starting conditions. It may happen that, under those conditions, the sample is not completely soluble. In this case, two options are available. The concentration of organic modifier is increased and the volume of the injection is limited to a few microliters (<10 μL for a 4.5×250 mm column; see Table 3 for directions). Alternatively, the sample is filtered. Large injection volumes containing significant quantities of organic modifier may result in partial elution of the material in the void volume.

7.2.2.3.5 Illustration of a Coelution Experiment

The aim of a coelution experiment is to demonstrate that two different preparations of a particular peptide, coinjected in equal amounts, give a single and symmetrical absorbance on HPLC and are, therefore, likely to have identical structures. First, it should be shown that the system used is highly discriminatory in that it will separate very closely related compounds. For example, two peptides that differ by a single amino acid, one methyl group, or an oxygen atom will have different retention times in that analytical system. A coelution experiment for native rat growth hormone releasing factor (GRF) and its synthetic replica (H-His-Ala-Asp-Ala-Ile-Phe-Thr-Ser-Ser-Tyr-Arg-Arg-Ile-Leu-Gly-Gln-Leu-Tyr-Ala-Arg-Lys-Leu-Leu-His-Glu-Ile-Met-Asn-Arg-Gln-Gln-Gly-Glu-Arg-Asn-Gln-Glu-Gln-Arg-Ser-Arg-Phe-Asn-OH) illustrates an important protocol.[55]

The protocol for coelution of a native and synthetic peptide is as follows. Plan on having enough solvent of the same composition (solvents A and B) for a minimum of six runs.

First run: Inject 1.0 mL of water and run a gradient optimized earlier with synthetic material that would show good separation of the desired peptide and that of a closely related impurity. Collect 3 mL of the eluent around the time when the desired peptide is supposed to elute.

Second run: Inject 0.5 mL of collected eluent from run 1, diluted with water (1:1) and run the same gradient as above. This will be the control experiment to show no absorbance of the solvent in the area of the chromatogram where the desired product is expected to elute.

Third run: Inject 2–4 μg of synthetic peptide dissolved in 0.5 mL of collected eluent from run 1 diluted with water (1:1) to 1 mL total volume, and run the same gradient as above. Collect the eluent containing the desired peptide (ca. 250 μL).

Fourth run: Repeat the second run to ascertain that the baseline is clean with no memory effect from synthetic peptide.

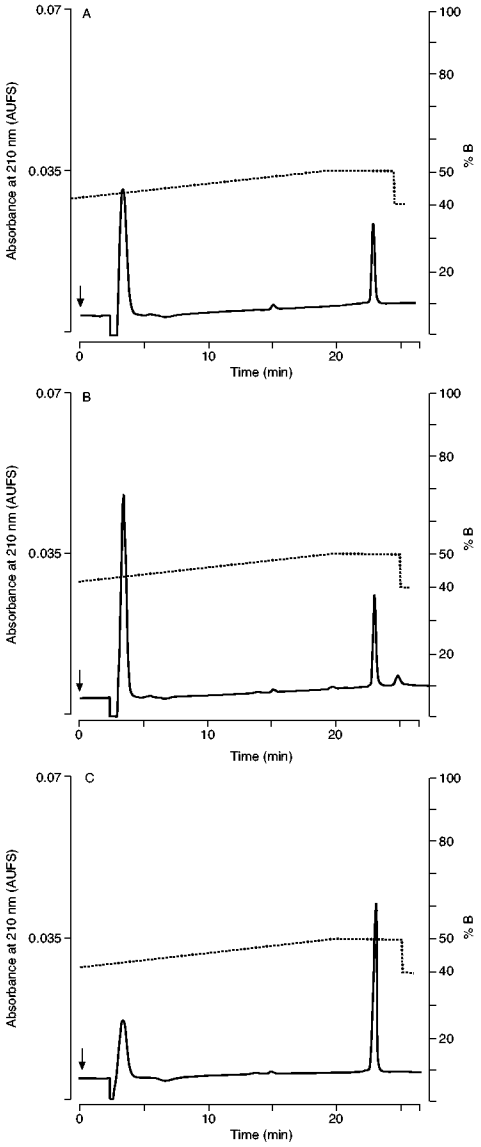
for references see p 649

Fifth run: Inject 2–4 μg of the native peptide dissolved in 0.5 mL of collected eluent from run 1, diluted with water (1:1) to 1 mL total volume, and run the same gradient as above. Collect the eluent containing the desired peptide (ca. 250 μL).

Sixth run: Inject fractions (250 μL each) from runs 3 and 5, diluted with water (1:1) to 1 mL, and run the same gradient as above.

In this manner, all injections are of the same volume and of the same composition (within a factor of 2); the same amount of peptide are injected in runs 3 and 5 and absorbances of synthetic and native peptides should be about the same. Coelution run 6 will contain twice as much material as runs 3 and 5 and absorbance should be twice as intense as that seen in these runs. Such an experiment is illustrated in Figure 2.

Figure 2 Coelution Experiment: A: Synthetic Rat Growth Hormone Releasing Factor (ca. 0.7 μg); B: Native Rat Growth Hormone Releasing Factor (ca. 0.7 μg); C: Coelution of Collected Fractions in A and B[23][a]



[a] Column: Vydac Phenyl (4.5×250 mm); solvent A: 0.1% TFA in H_2O; solvent B: 0.1% TFA in H_2O/MeCN.

In addition to the precautions already mentioned, the time of column equilibration between each run should be about the same. Ideally the profile of one of the two peptides (native or synthetic) should give a symmetrical absorbance as well as show the presence of a small amount of an impurity. Under such circumstances, the coelution run should show a symmetrical absorbance of the size of that of the synthetic and native molecules. In addition, the absorbance of the impurity should now be half the size of that seen in the chromatogram of the contaminated peptide. All other absorbances such as injection artifacts should be identical in all runs 2–6. **CAUTION**: *Two peptides that show the same retention times after being run separately under the same conditions cannot be said to have coeluted!*

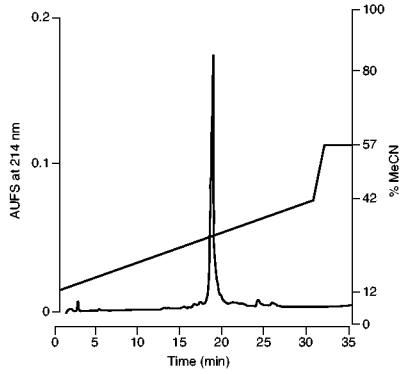
Fraction collection from analytical systems: When collecting fractions, whether manually or automatically, be sure that the leads from the detector to the fraction collector are narrow-bore metal tubing rather than plastic as the latter may induce peak broadening and potential loss of material.

7.2.2.4 Ion-Exchange High-Performance Liquid Chromatography

The theory behind ion-exchange HPLC (IE-HPLC) is the same as that of normal ion-exchange chromatography as described in several publications.[34–36,56,57] Solid supports for IE-HPLC can be either derivatized silica gel, where the bonded phase is trimethylpropylamine for anion exchange and propylsulfonic acid for cation exchange, or polymer-based, such as MonoBeads of Pharmacia Biotech (Mono S and Mono Q are strong cation and anion exchangers, respectively). Each of a large number of manufacturers offers a variety of proprietary materials in different particle and column sizes. Literature is available from these manufacturers describing how to optimally use their products.

As shown in the purification of rat histone $H2A_{1-53}$ amide synthesized using the FMOC strategy, remarkable results using a Pharmacia Biotech Mono S column were obtained for the purification of complex synthetic mixtures (Figures 3 and 4).[58] This system has been used with similar success with several other peptides. With a complex mixture of peptide/proteins, ion-exchange chromatography was found to be an important first step for preliminary fractionation to be further purified by RP-HPLC.

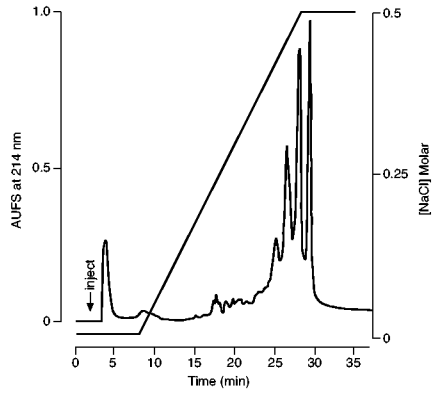
Figure 3 Analytical HPLC Profile of Crude Rat Histone $H2A_{1-53}$-NH_2 (15 μg in 10 μL)[58a]



[a] Column: Vydac C_{18} (5 μm, 0.46×25 cm); solvent A: 0.1% TFA in H_2O; flow rate: 2.0 mL/min.

for references see p 649

Figure 4 Crude Rat Histone $H2A_{1-53}$-NH_2 (60 μg in 50 μL)[58]a



[a] Column: Pharmacia Mono S Precision, 1.6/5 (5.0×0.16 cm i.d.); solvent A: boric acid (MeCN/H_2O 35:65) pH 9.7; solvent B: 0.5 M NaCl in A, gradient from 0–100% B in 20 min; flow rate: 0.1 mL/min.

The conclusion drawn from this work is that the desired peptide present in a relatively small amount in the synthetic mixture could never have been isolated in a highly purified form if it had not been for the discriminating first ion-exchange chromatography step.

Ion-exchange chromatography is the method of choice for the separation of underivatized amino acids and dipeptides and as a result is also the most widely used HPLC method with reverse phase, depending on whether the amino acids are, or are not, pre-column derivatized.[59–61] In 1999, a development[62] whereby silica gel based mix-beds of ion exchanger and C_{18} are used for the separation of underivatized amino acids coupled to an evaporative light-scattering detector is worthwhile mentioning. With this separation technique sensitivity is high and the instrumentation simple.

Purification of Rat Histone $H2A_{1-53}$-NH_2 Synthesized Using the Fmoc Strategy:[58]
Rat histone $H2A_{1-53}$-NH_2 (Ac-Ser-Gly-Arg-Gly-Lys-Gln-Gly-Gly-Lys-Ala-Arg-Ala-Lys-Ala-Lys-Ser-Arg-Ser-Ser-Arg-Ala-Gly-Leu-Gln-Phe-Pro-Val-Gly-Arg-Val-His-Arg-Leu-Leu-Arg-Lys-Gly-Asn-Tyr-Ala-Glu-Arg-Val-Gly-Ala-Gly-Ala-Pro-Val-Tyr-Leu-Ala-Ala-NH_2, MW = 5605) was assembled manually on a 2,4-dimethoxybenzhydrylamine resin. After cleavage using TFA, a crude product was obtained that appeared suspiciously pure using RP-HPLC (Figure 3) with conditions shown in the footnote. Indeed, cation-exchange chromatography on a Pharmacia Mono S Precision column (1.6×50 mm) gave the profile presented in Figure 4 using the conditions shown in the footnote. The last eluting absorbance contained the desired product, as shown by MS. It was isolated with a purity estimated to be greater than 90% using an additional RP-HPLC step.[58]

7.2.2.5 Size-Exclusion High-Performance Liquid Chromatography

Fractionation of proteins according to size utilizing cross-linked dextran or polyacrylamide gel columns was first demonstrated by Porath and Flodin[63] in 1959. This technique has become the most widely accepted method for separation and molecular weight determination of hydrophilic and some hydrophobic macromolecules using aqueous buffers with or without organic modifier. While this technique might not be unique in its ability to resolve and separate proteins, it is one additional simple and effective tool in the chemist's armamentarium. The theories behind size-exclusion HPLC and size-exclusion chromatography at low pressure are identical and are described in several publications.[31,34,36,39,44,64,65]

Derivatized silica gel that is used for size-exclusion HPLC contains a proprietary bonded phase that is defined as hydrophilic to minimize nonspecific hydrophobic and ionic interactions. Polymeric supports consist of highly cross-linked agarose beads, with or without bonded dextran or cross-linked copolymers of allyl dextran and *N,N'*-methylenebisacrylamide. All supports are available with a variety of particle and pore sizes and distribution.

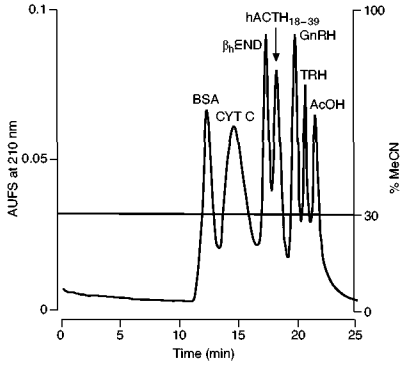
Each of a large number of manufacturers offer such materials in different particle, pore, and column sizes to be selected on the basis of the peptide/protein sizes to be separated. Manufacturers generally provide detailed literature on how to use their products for optimal results.

Although dated, the following example illustrates the fact that size-exclusion chromatography can be conducted efficiently and reliably under high pressure using a less conventional solvent system (TEAP) than the acetate-, phosphate-, and borate-based buffers described in the manufacturers' catalogues.[66]

Note that proteins are now synthesized routinely by stepwise assembly or fragment condensation (see Vol. E 22a, Section 4.1)[67,68] and chances of product contamination with shorter fragments that may have similar retention times under RP-HPLC should be considered.[69] Additionally, size-exclusion HPLC was found to be particularly useful in distinguishing linear from cyclic peptides.

Figure 5 illustrates the separation of bovine serum albumin (BSA, MW = ca. 44,000), cytochrome *c* (CYT C, MW = ca. 12,200), β-endorphin (β_hEND, H-Tyr-Gly-Gly-Phe-Met-Thr-Ser-Glu-Lys-Ser-Gln-Thr-Pro-Leu-Val-Thr-Leu-Phe-Lys-Asn-Ala-Ile-Ile-Lys-Asn-Ala-Tyr-Lys-Lys-Gly-Glu-OH, MW = 3461), hACTH(18–39) (H-Arg-Pro-Val-Lys-Val-Tyr-Pro-Asn-Gly-Ala-Glu-Asp-Glu-Ser-Ala-Glu-Ala-Phe-Pro-Leu-Glu-Phe-OH, MW = 2467), GnRH (pGlu-His-Trp-Ser-Tyr-Gly-Leu-Arg-Pro-Gly-NH_2, MW = 1182), TRF (H-pGlu-His-Pro-NH_2, MW = 362), and acetic acid (MW = 60) on a Waters PAC I-125 column using the TEAP solvent (pH 2.25) in the presence of 30% MeCN at 50 °C.

Figure 5 Separation of BSA, 729; Cytochrome c, 861; β-Endorphin (Human), 1011; hACTH(18–39), 1075; GnRH, 1167; TRH, 1221; Acetic Acid, 1273[66]a



[a] Loading: 50 μL protein-peptide mixture; solvent: TEAP, pH 2.25; numbers are the retention times for the different components.

In the range of 15–30% acetonitrile, the different components of the mixture were separated according to size, in a size-exclusion mode, with very little nonspecific adsorption and good peak symmetry. It was believed that the addition of TEA to the solvent contributed to these desirable effects by neutralizing residual silanol groups. Present supports should not suffer from the presence of such uncapped silanol groups.

for references see p 649

7.2.3 Capillary Electrophoresis

7.2.3.1 General Considerations

Capillary zone electrophoresis (CZE) is now routinely used in many laboratories for the analysis of native and synthetic peptides and proteins as a complementary method to HPLC analysis. The high sensitivity, high automated throughput, high resolution, and low solvent consumption of CZE make it an invaluable technique. As the principle of separation relies mostly on charges of analyte as opposed to hydrophobic interactions for RP-HPLC, these two techniques are complementary or orthogonal. Any one of a large number of books will introduce and describe the principles and applications of CZE,[70–78] including approaches for the optimization of experimental parameters.[79] Various booklets are also available from CZE instrument suppliers; these will supply information on general as well as more specialized applications, such as chiral separations.

As pointed out by Whatley in an article in 1999,[80] CZE is a low-volume technique and, as such, any disturbance of this volume will affect the reproducibility or the quality of the separation. The pump is inside the separation zone in that the motive force that drives the analytes through the column toward the detector comes from the electrical field and the electroosmotic flow. The capillary, because of its composition, is a reagent and is an integral part of the separation process. Changes in the chemistry of the capillary wall will affect the electroosmotic flow as well as the separation, in that it may introduce other types of interactions (ionic or hydrophobic). As shown in Section 7.2.3.2, these effects can be taken advantage of to achieve separations that would not be possible on the basis of charges only. Last but not least, composition and pH of the solvent is critical in CZE, not only to achieve reproducible results but also to optimize separations. Relating to the above, Altria and Campi have identified 10 ways to spoil a CZE separation;[81,82] among them are broken or cracked capillaries, misassembled capillary cartridges, distorted capillary tips, wrong capillary, siphoning effects, carry over, high-voltage leakage, and plugged capillaries as a result of improper storage.

7.2.3.2 Capillaries

Capillaries used for CZE are fused silica gel externally coated with polyimide. A small area of polyimide is removed to form a window for on-line detection. Bare fused silica gel and coated capillaries have been used. A capillary coated with a neutral hydrophilic bonded phase (Supelco CElect P175) has been extremely useful in work with hormonal peptides,[23,25,26] which have isoelectric points >7. Basic peptides analyzed using a bare fused silica gel capillary often separate as broad peaks. The hydrophilic coating seems to minimize interactions with the capillary wall. One example of this is the separation of a mixture of [D- and L-Agl(Me,Ac)^7]-Acyline [Ac-D-Nal-D-Cpa-D-Pal-Ser-Lys(Ac)-D-Lys(Ac)-D- and L-Agl(Me,Ac)- L-Lys-Pro-D-Ala-NH_2] using a fused silica gel capillary or CElect P175. All other conditions of the separation remained the same and no separation of the D,L-mixture was observed using the fused silica gel capillary while the same mixture was well separated (allowing quantitation 54:46) using the CElect P175 capillary.

Acidic phosphate-containing solvents are useful for CZE separations of most peptides except some negatively charged ones. For those cases, a borate buffer at pH 8.5 is used. In either case, the peptide must be positively charged or there must be enough electroosmotic flow to cause net movement of the peptide towards the cathode.

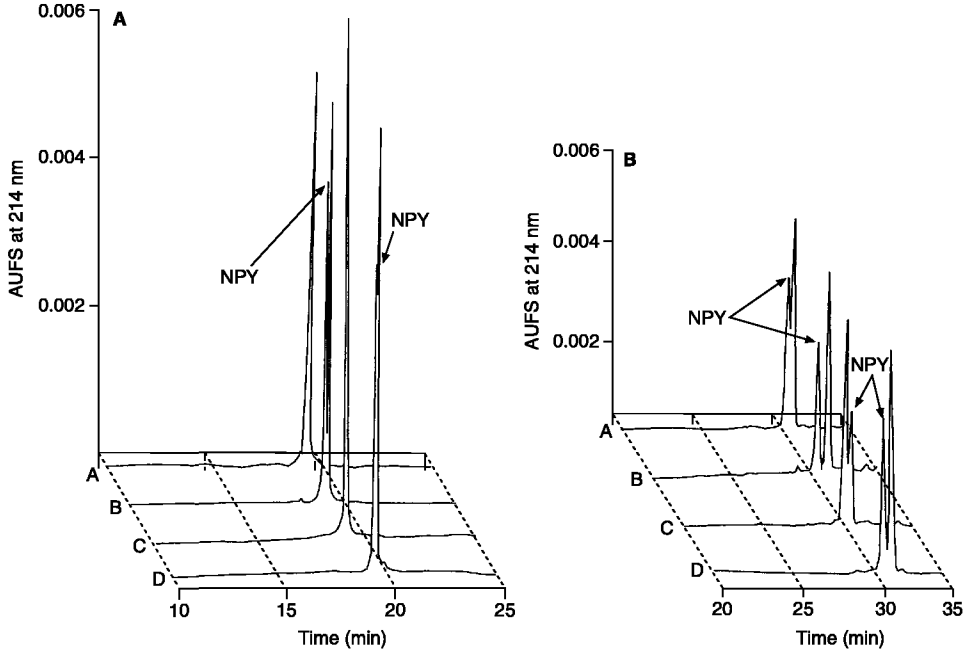
7.2.3.3 Useful Buffer Systems

Since phosphate and borate buffers are well described in the literature, the power of CZE in peptide chemistry will be illustrated with two examples that emphasize less conventional conditions than those offered by the manufacturers, such as using TEAP solvent systems or standard buffers in the presence of MeCN. These were utilized for some unusual separations.

Separation of Neuropeptide Y Diastereomers

The use of racemic mixtures of amino acids or epimerization at any one of the many amino acid α-carbons during synthesis can lead to the undesired incorporation of D-isomer impurities. Such impurities represent a significant challenge to chromatographic detection systems, since they differ by only a single inversion of a peptide backbone α-carbon. In Figure 6 the separation of four NPY diastereomers ([D-Tyr^1]NPY, [D-Ser^3]NPY, [D-Lys^4]-NPY, and [D-Tyr^{36}]NPY; NPY = H-Tyr-Pro-Ser-Lys-Pro-Asp-Asn-Pro-Gly-Glu-Asp-Ala-Pro-Ala-Glu-Asp-Leu-Ala-Arg-Tyr-Tyr-Ser-Ala-Leu-Arg-His-Tyr-Ile-Asn-Leu-Ile-Thr-Arg-Gln-Arg-Tyr-NH_2) synthesized on a 4-methylbenzhydrylamine resin and purified by RP-HPLC, which could not be separated using RP-HPLC, is illustrated.[83,84] Conditions are shown in the footnote.

Figure 6 CZE Trace of NPY and Diastereomers: A, [D-Tyr^1]NPY; B, [D-Ser^3]NPY; C, [D-Lys^4]NPY; and D, [D-Tyr^{36}]NPY[83,84][a]



[a] Beckman P/ACE 2050, Silica Gel Capillary (50 cm×75 μm), 30°C; A, voltage 12 kV, 62 μA, 0.1 M sodium phosphate, pH 2.5; B, voltage 20 kV, 100 μA, 0.1 M TEAP, pH 2.50.

Under standard CZE analysis conditions (sodium phosphate buffer, pH 2.5), [D-Ser^3]- and [D-Tyr^{36}]NPY were resolved from NPY (Figure 6A). The other isomers, [D-Tyr^1]NPY and [D-Lys^4]NPY, were not resolved under these conditions. All four isomers were finally resolved by changing the electrophoretic buffer to 0.1 M TEAP at pH 2.5 (see Figure 6B). Under these CZE conditions the entire series of 35 NPY diastereomers[84] was reevaluated and

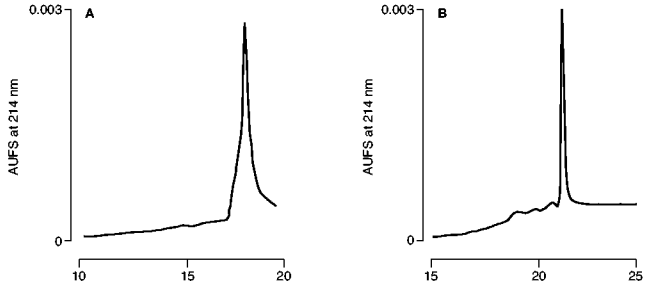
for references see p 649

resulted in the separation of 32 out of 35 isomers, emphasizing the need for multiple analytical techniques and systems for the characterization of peptides.

Analysis of a Hydrophobic Corticotropin Releasing Factor Antagonist Using Capillary Zone Electrophoresis in the Presence of Acetonitrile

As discussed in Section 7.2.2.5, the addition of MeCN to solvents used for ion-exchange and size-exclusion chromatographies was found to be extremely beneficial. MeCN was hypothesized to increase the solubility of the peptides and facilitate partitioning among phases (stationary and mobile). Since most separations have to be carried out in the presence of a support or of capillary walls in the case of CZE, it is extremely important that such interactions be minimized; certain capillary coatings will facilitate some separations. Figure 7A and Figure 7B illustrate the fact that the addition of MeCN to the buffer is very beneficial and allows better discrimination between the desired product and closely related impurities. More recent results suggest that the addition of trifluoroethanol (TFE), but not other common alcohols, may also dramatically improve the chromatographic process. TFE will induce a defined secondary structure that may be more amenable to rapid equilibrium between stationary and liquid phases.[86] Independently, Bhardwaj and Day have observed that TFE removes bound proteins from reversed-phase columns.[85]

Figure 7 CZE Trace of a CRF Antagonist[a]



[a] Beckman P/ACE 2050; Supelco P-175 Capillary (50 cm×75 μm); A, voltage 15 kV; buffer 0.1 M sodium phosphate, pH 2.5. B, voltage 12 kV, buffer 0.1 M sodium phosphate, pH 2.5, in MeCN (95:5).

7.2.3.4 Conclusion

Whereas HPLC continues to play a major role in the life of the peptide/protein chemist, it should not be relied upon as the only chromatographic approach for the purification and analysis of peptides greater than 40–50 residues in length. Selection of the column supports and solvents for peptide and protein separation is often critical. Preparative cation-exchange chromatography proved to be an excellent orthogonal technique for the purification of peptides. Narrow-bore cation-exchange chromatography and CZE complement HPLC for the determination of the purity of synthetic peptides and such techniques should be utilized in every laboratory dealing with bioorganic molecules. The addition of organic modifiers in ion-exchange and size-exclusion chromatographies and CZE should be encouraged. The examples of separations that have been achieved are illustrations of numerous years of experience and practice in the field. However, there is always room for further optimization for any given technique. There are limitations to this purely empirical approach when it comes to the development of a mechanistic understanding for the observed phenomena. When used critically and with discrimination, HPLC has been a magnificent tool both for the isolation of natural products and for the purification and characterization of peptides synthesized by Merrifield's solid-phase approach. What should be realized is that through these

chromatographic developments and constant improvements in synthetic methods we observe a consistent extension of the limits in size and complexity of synthetic peptides reported in the literature. The recent availability of other highly resolutive techniques ultimately broke the RP-HPLC monopoly and allows examination of peptide preparations from more than one angle, akin to what was taught originally by the pioneers of peptide chemistry.

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7.3 Amino Acid Analysis

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Among the analytical methods presently used for the characterization of natural and synthetic peptides and proteins, the primary value of amino acid analysis is the determination of absolute peptide and protein content in solids and solutions and the quantitation of their amino acid composition and stoichiometry. It involves two steps, i.e. complete hydrolysis of peptides and proteins, followed by photometric determination of the released amino acids. The steps are laborious and time-consuming, and there is a continuous need for improvement of the techniques to increase precision and sensitivity.

The original methodology for quantitative amino acid analysis, developed in the early 1950s by Moore and Stein,[1–3] relies on ion-exchange chromatography for separation of the amino acids and on their subsequent visualization by reaction with ninhydrin. With the advent of the high-performance liquid chromatography, new derivatization techniques provided alternative methods for amino acid analysis of increased speed and sensitivity. Nevertheless, both techniques, the ion-exchange chromatography with ninhydrin detection and the pre-column derivatization followed by HPLC resolution of the amino acid mixtures and photometric quantitation, are presently used.

Despite the high efficiency of the present-day chromatographic and electrophoretic techniques and the fast developments in mass spectrometry, amino acid analysis still represents a valuable analytical tool in peptide chemistry for characterization of the final products, but also for monitoring intermediate coupling steps in solution and on solid supports by comparing ratios of diagnostic amino acids.

7.3.1 Hydrolysis of Peptides and Proteins

The main concern in the sample preparation for amino acid analysis, particularly if performed at nano- to picomolar scale and with fluorescence detection, are contaminants readily introduced with chemicals and during handling of the samples. Therefore, glassware has to be heated in a muffle furnace for at least 12 h at 400 °C to destroy all possible amino acid contaminants and the chemicals used for hydrolysis, derivatization, and chromatography have to be of amino acid analysis grade. Depending upon the scope of the analysis, i.e. amino acid stoichiometry or absolute peptide and protein contents, the precision strongly depends on the accuracy in determining sample weights and volumes of solids and solutions, respectively. Whereby, non-amino acid contaminants such as buffer salts, detergents, or matrixes (resins in peptide chemistry) may interfere in the pre-column derivatization chemistries, but are less crucial in the post-column visualization techniques.

7.3.1.1 Acid Hydrolysis

Standard hydrolysis is performed with 6 M HCl at 110 °C for 24 h.[2] But the method can be varied in terms of acids used, temperature, and time, as well as in the mode of gas- or liquid-phase hydrolysis (Table 1). Time-dependent hydrolysis is required to determine the exact stoichiometry for unstable amino acids as well as for particular sequences difficult to hydrolyze. Tryptophan, methionine, cystine, tyrosine, serine, and threonine are known to be degraded at an extent of >50, >50, >30, >20, >10, and >5%, respectively.[4] Therefore, performing standard hydrolysis for 24, 48, and 72 h, and extrapolation to zero time of hydrolysis allows for better quantitation of these unstable amino acids, whereas extrapolation

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to 100 h time of hydrolysis is applied for amino acids which are slowly released such as valine, leucine, and isoleucine.

Table 1 General Procedures for Acid Hydrolysis of Peptides and Proteins[3–7]

Acids	Temperature (°C)	Time	Ref
6 M HCl	110	24 h	[3]
	145	4 h	[4]
	165	1.5 h	[5]
propionic acid/12 M HCl (1:1)	160	15 min	[6]
TFA/concd HCl (1:2)	166	25 min	[7]

For the hydrolysis of peptides linked to resins, the release of the C-terminal amino acid is strongly dependent upon the type of resin and amino acid residue. For this purpose the use of propanoic acid/12 M HCl (1:1)[6] is recommended. More studies, however, indicate hydrolysis with TFA/12 M HCl (1:3) at 160 °C is more appropriate since quantitative release of the most stable Phe residue from benzhydrylamine- (BHA-), 4-methylbenzhydrylamine- (pMeBHA-), and 4-(hydroxymethyl)phenylacetamidomethyl-resins occurs in 20, 6, and 5 h, respectively, compared to the 30, 18, and 18 h required with propanoic acid/12 M HCl (1:1).[8]

Since tryptophan and cysteine residues decompose under standard conditions of acid hydrolysis,[3] various alternative conditions have been proposed which are based on the addition of scavengers such as phenol (1–3%),[3,9] thioglycolic acid (0.1–4%),[10,11] β-ME (0.4%),[12] tryptamine (0.2%),[13,14] or sodium sulfite.[15] Whereby, degassing of the reaction mixture prior to sealing the tube or saturation with inert gas before closing is most important as tryptophan decomposes by a free radical auto-oxidation mechanism.[15,16] Alternatively, other acid mixtures are used for high recoveries of tryptophan and cysteine as listed in Table 2. A detailed analysis of the stability of tryptophan as a function of concentration of sulfanylethanesulfonic acid, time of hydrolysis, and temperature revealed as optimal conditions 2.5 M sulfanylethanesulfonic acid at 176 °C for 12.5 min.[20]

Under these reducing conditions of hydrolysis of tryptophan peptides, cystine is reduced to cysteine and its coelution with proline using standard buffer gradients, makes quantitation difficult. Thus, cysteine and cystine are generally derivatized prior to acid hydrolysis by oxidation to cysteic acid with performic acid[21] or alkylation, upon reduction in the case of cystine, with iodoacetic acid[21] or, more appropriately, with 4-vinylpyridine.[22,23,50] Conversion of cysteine into *S*-β-(4-pyridylethyl)cysteine bears the additional advantage of suppressing epimerization via the thiazoline intermediate, thus allowing for standardization of the acid-hydrolysis dependent racemization of cysteine in synthetic peptides.[24]

Due to the incompatibility of asparagine and glutamine with acid hydrolysis these amino acids can be converted into the corresponding diaminopropanoic and γ-diaminobutanoic acid by reaction with bis(1,1-trifluoroacetoxy)iodobenzene.[25] However, this procedure is generally not applied.

Table 2 Acid Hydrolysis of Peptides and Proteins Containing Unstable Amino Acid Residues[13,14,17–19]

Acid	Temperature (°C)	Time	Ref
3 M TosOH/0.2% tryptamine	110	24 h	[13]
4 M MsOH/0.2% tryptamine	110	24 h	[14]
3 M sulfanylethanesulfonic acid	110	24 h	[17]
3 M sulfanylethanesulfonic acid	160–170	15–30 min	[18,19]

Hydrolysis of Trp- and Cys-Containing Peptides; General Procedure:
Weighed amounts of peptide (100 nmol) were transferred into a hydrolysis tube into which freshly distilled 6 M HCl (0.8 mL) and thioglycolic acid (20 μL) was then added. The soln was frozen at −70 °C

and the tube evacuated at about 0.6 Pa and sealed with an oxygen flame. Hydrolysis was performed in a thermostated oven at 110 °C for 24, 48, and 72 h. The tube was opened and the solvent was removed from the mixture. The residue was reevaporated twice from H_2O (1 mL). The hydrolysate was dissolved in sample diluting buffer and applied to the amino acid analyzer in defined aliquots and concentrations depending upon the sensitivity of the instrumentation; recoveries of Trp and Cys by this procedure are usually >95%.

7.3.1.2 Alkaline Hydrolysis

Alkaline hydrolysis with barium, sodium, or lithium hydroxides (0.2–4 M) at 110 °C for 18–70 h[26–29] requires special reaction vessels and handling. Reaction mixtures are neutralized after hydrolysis and barium ions have to be removed by precipitation as their carbonate or sulfate salts prior to analysis which leads to loss of hydrolysate. Correspondingly, peptide contents are difficult to perform by this procedure. Preferred conditions for alkaline hydrolysis are 4 M LiOH at 145 °C for 4–8 h where >95% of tryptophan is recovered.[29] An additional inconvenience of the alkaline hydrolysis procedure is the dilution effect in the neutralization step and thus the difficult application to the analyzer if micro-scale analysis is to be performed. The main advantage is the good recovery of tryptophan and of acid-labile amino acid derivatives such as tyrosine-*O*-sulfate[26] (Section 6.6) as well as partial recovery of phosphoamino acids, particularly of threonine- and tyrosine-*O*-phosphate (Section 6.5).

7.3.1.3 Enzymatic Hydrolysis

For quantitative determination of tryptophan and particularly of glutamine and asparagine in peptides and proteins the most appropriate method of hydrolysis is the enzymatic digestion with exopeptidases or with the combined use of exo- and endopeptidases.[30] For peptides up to 35 residues in size, complete digestion is achieved with aminopeptidase M and prolidase, whereas for larger polypeptides an initial digestion with the unspecific pronase is required followed by aminopeptidase and prolidase. Pronase is a mixture of exo- and endoproteinases that is produced in the supernatant of *Streptomyces griseus* K-1, while aminopeptidase M (EC 3.4.11.2) is an exopeptidase that cleaves N-terminal residues from all peptides having a free α-amino or α-imino group. In peptides containing Xaa-Pro sequences where Xaa is glycine or a residue with bulky hydrophobic side chains, or in the case of an N-blocked amino acid, cleavage does not occur. Therefore, prolidase is used in combination with aminopeptidase M to achieve total hydrolysis of Pro-containing peptides. Prolidase (EC 3.4.13.9) cleaves from N-termini dipeptides with a prolyl or hydroxyprolyl residue in the Xaa+1 position. Finally, the pyroglutamate aminopeptidase (EC 3.4.19.3), a thiol exopeptidase that cleaves N-terminal pyroglutamyl residues, often present in naturally occurring polypeptides, is used in combination with the aminopeptidase M for total digestion of pyroglutamyl peptides. These enzymatic digestions are known to release quantitatively sulfated and phosphorylated amino acids and even some side-chain protected residues such as Cys(StBu), Cys(Acm) or Cys(CM).[31] Other side-chain derivatized amino acids, e.g. Thr(tBu), may act as inhibitors thus blocking further digestion. Similarly, digestion with aminopeptidase M stops at D-amino acid residues and this is exploited in synthetic peptides to detect racemization when it occurs to a large extent.

A promising new method of hydrolysis of peptides and proteins consists of a brief acid treatment with 6 M HCl for 15 min at 80–90 °C, followed by an enzymatic digestion with pronase for 12–16 h at 50 °C, and finally by an enzymatic hydrolysis with aminopeptidase M and peptidyl-D-amino acid hydrolase (EC 3.4.13.17) for 24 h. By this procedure the recovery averages 97–100%, and the amount of racemization derived from the hydrolysis step is less than 0.002%.[32,33] Correspondingly, racemization of single residues deriving from the synthetic procedures can be evaluated with high precision upon derivatization and gas-chromatographic enantiomeric resolution (see Section 7.4).

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Enzymatic Digestion of Proline-Containing Peptides; General Procedure:
Prolidase (0.05 mL of enzyme suspension, 200 U) was preincubated in degassed, argon-saturated 0.05 M Tris·HCl (pH 8.0, 0.45 mL) containing 0.2 M $MnCl_2$ at 37 °C for 45 min. Weighed samples of peptides (100 nmol) and aminopeptidase M at w/w ratio of 10:1 were added to the 0.5-mL prolidase soln. Digestion was allowed to proceed at 37 °C for 24 h. Then the mixture was acidified with 0.2 M lithium citrate buffer (pH 2.2)/1 M HCl (1:1, 0.5 mL) and aliquots of the soln were applied to the amino acid analyzer.

Enzymatic Digestion of Pyroglutamyl Peptides; General Procedure:
To weighed samples of pyroglutamyl peptides (100 nmol) in Tris·HCl (pH 8.0, 0.4 mL) was added pyroglutamate aminopeptidase at a w/w ratio of 10:1 and the digestion mixture was incubated at 37 °C for 24 h. Then aminopeptidase M (w/w of 10:1) was added and the mixture was incubated at 37 °C for an additional 24 h. The mixture was acidified with 0.2 M lithium citrate buffer (pH 2.2)/1 M HCl (1:1, 0.5 mL) and aliquots of the soln were applied to the amino acid analyzer.

7.3.2 Derivatization of the Hydrolysates

Amino acids are small and polar compounds that show no or weak absorbance in UV and fluorescence spectroscopy. For high-sensitivity detection the hydrolysates have, therefore, to be converted via a quantitative and reproducible reaction involving both primary and secondary amines under mild conditions to yield strongly UV-absorbing or fluorescent derivatives. This derivatization can be performed after separation of the amino acid mixture into the single components by ion-exchange chromatography in a post-column procedure or prior to separation where the derivatives, besides allowing sensitive detection, have to facilitate chromatographic separation.

7.3.2.1 Post-column Derivatization

The classical post-column derivatization, as developed by Moore and Stein,[3] uses ninhydrin and detection is achieved at 570 nm for primary amine derivatives and at 440 nm for secondary amine derivatives with detection limits that strongly depend upon the instrumentation used.

For analyses of synthetic peptides that are generally available in larger amounts than isolated natural peptides and proteins, the ninhydrin detection method is fully sufficient. The alternative detection method with orthophthaldialdehyde, which improves the sensitivity by a factor of five, was originally proposed for post-column derivatization,[34] but is presently used mainly in pre-column derivatization techniques.[35,36]

7.3.2.2 Pre-column Derivatization

Derivatization of amino acids with hydrophobic aromatic reagents alters their chromatographic behavior allowing the fast RP-HPLC to be used for separation. For this purpose, derivatizing procedures were developed that lead to detection limits in the femtomole range either by UV or by fluorescence on a routine basis. At this detection level, contaminating background levels of amino acids become a severe problem making the sample preparation the limiting factor rather than the sensitivity of the detection. The reagents generally used in these types of pre-column derivatization techniques are summarized in Table 3.

Table 3 Reagents for Pre-column Derivatization of Amino Acids[35–46]

Reagent	Amino Acid Derivative	Detection (nm)	Ref
phenylisothiocyanate	PTC derivatives	$\lambda_{max} = 254$	[37,38]
o-phthaldialdehyde/mercaptan	1-alkylthio-2-alkyl-isoindoles	$\lambda_{max} = 338$ $\lambda_{ex} = 335$ $\lambda_{em} = 455$	[35,36]
Fmoc-Cl	Fmoc derivatives	$\lambda_{max} = 260$ $\lambda_{ex} = 266$ $\lambda_{em} = 310$	[39–41]
DABSCl	Dabsyl derivatives	$\lambda_{max} = 436$	[42,43]
Dns-Cl	Dns derivatives	$\lambda_{max} = 330$ $\lambda_{ex} = 310$ $\lambda_{em} = 540$	[44]
aminoquinoyl-*N*-hydroxysuccinimidyl carbamate	AQC derivatives	$\lambda_{ex} = 245$ $\lambda_{em} = 395$	[45,46]

Advantages and limitations of the various derivatization procedures are discussed by Kellner et al.[47] Experimental protocols for optimized pre-column derivatization with the various reagents reported in Table 3 are reported by Kamp.[48]

7.3.3 Data Analysis

Whether post-column or pre-column derivatization is applied for visualization and correspondingly, the classical ion-exchange chromatography or RP-HPLC for separation of the amino acid mixtures, integration of the UV absorbance or fluorescence peak areas is used to quantify the single amino acid components. Thereby, precise recovery measurements performed on standard amino acid mixtures subjected to hydrolysis, derivatization, and chromatographic separation under identical conditions are elemental for determining the correction factors required for exact quantitation of the stoichiometry of the amino acid components as well as of the peptide and protein contents. For these computations protocols are available.[48,49] Limits in error of the quantitative determination of the amino acids are in the range of 2–3%.

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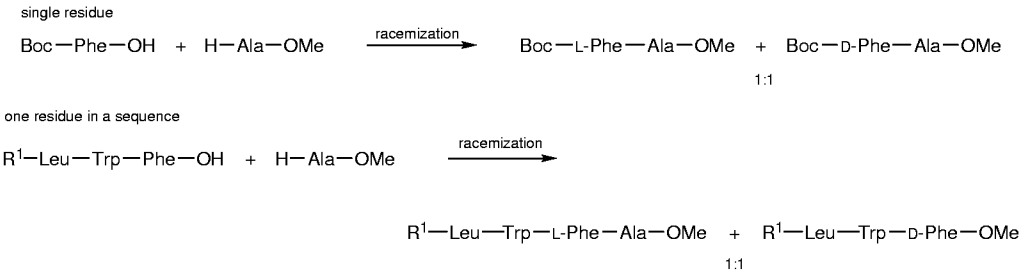
7.4 Racemization Assays

T. T. ROMOFF

A singular advantage for the use of peptides in drug discovery and other biological target-directed projects is the large pool of chiral building blocks (α-amino acids) from which they are constructed. This allows the rapid preparation of diverse molecules with well-defined stereochemistry. It is essential, therefore, that the methods chosen for peptide synthesis preserve the stereochemical integrity of the α-carbon atom of each amino acid residue. Numerous studies of racemization have been carried out over the 100 year history of peptide synthesis, but most of them were confined to a specific reaction or set of reactions. In this section, an overview of the parameters that control the rate of racemization will be presented, as well as a new generalized method for determining racemization rates.

Definitions: It is important to define precisely the stereochemical terms that will be employed in this discussion. The term *racemization* has often been used loosely by chemists to describe any situation in which a mixture of enantiomers or diastereomers is produced as a result of an amide-bond-forming reaction, without regard to the ratio of stereoisomers formed. For the purposes of this discussion though, the term *racemization* will be used to describe the situation leading toward the formation of an exact 1:1 mixture of stereoisomers. *Racemization*, therefore, is a process that occurs to a *collection* of molecules, and can happen to a single residue or to one residue in a peptide sequence (Scheme 1). This is a macroscopic event, as the result is detected subsequent to the amide bond formation.

Scheme 1 Examples of Racemization during a Peptide-Bond-Forming Reaction



On the other hand, a change in the configuration of a chiral center in a *single* molecule will be described by the term *epimerization*. This is a microscopic event that happens to a single molecule, and occurs prior to the amide bond formation (Scheme 2).

Scheme 2 Epimerization of an Amino Acid Residue



Put another way, epimerization is the mechanistic event, racemization is the observation. True racemization, the actual production of a racemic mixture, is rarely seen in peptide synthesis. Instead, it is the extent of epimerization that defines the stereochemical outcome of a peptide-bond-forming reaction. In order to assess the probability of epimerization under a given set of conditions, one must be aware of the mechanisms of epimerization, as well as the thermodynamic and kinetic factors that affect this process.

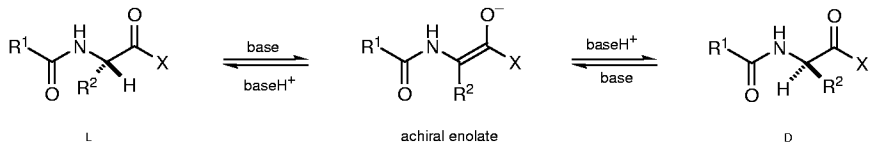
for references see p 668

Mechanisms of Epimerization

The central event in an epimerization reaction is the removal of a proton from the chiral α-carbon of an amino acid residue. This event generates an *enolate* with a trigonal planar carbon atom. Replacement of the proton on the face opposite from which it was abstracted results in the inversion of the configuration of the α-carbon. In theory, this event can occur at any stage of peptide synthesis. In practice, however, epimerization is observed almost exclusively during the amide-bond-formation step. This discussion will be confined to that type of epimerization.

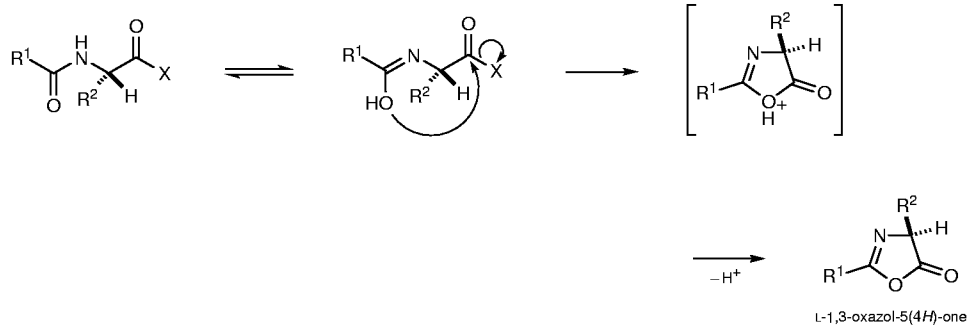
Removal of a proton from the α-carbon (the α-proton) requires the participation of a base. Epimerization is rarely observed on amino acid residues with unprotected carboxy groups, because the first equivalent of base generates the carboxylate. Formation of a second negative charge on the α-carbon in close proximity to the carboxylate anion is strongly disfavored. For typical peptide synthesis, however, the carboxy group is activated toward acceptance of a nucleophile by conversion into an acyl halide, or into an ester- or anhydride-type of functional group. The activating group is generally electron withdrawing, so that the electrophilicity of the carboxy carbon atom is increased. The same factors that serve to increase this electrophilicity, however, also serve to increase the acidity of the α-proton, rendering the molecule more susceptible to epimerization. When this proton abstraction and replacement process takes place on the unchanged activated intermediate, this is referred to as the direct abstraction mechanism (Scheme 3).

Scheme 3 The Direct Abstraction Mechanism of Epimerization



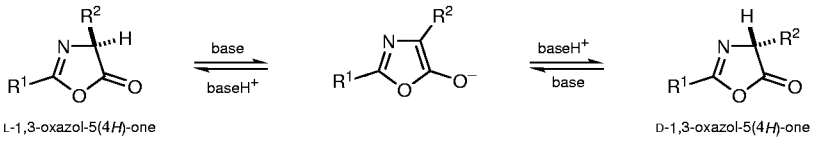
In parallel with the direct abstraction mechanism, a second important mechanism of epimerization can occur via an intermediate that can be produced from any type of activation. This common intermediate is the oxazol-5(4*H*)-one, produced by nucleophilic attack of the penultimate carbonyl oxygen atom attached to the amino group on the activated carboxy carbon atom (Scheme 4).

Scheme 4 Formation of Oxazol-5(4*H*)-ones from Activated Intermediates



It is evident from the structure of the oxazol-5(4*H*)-one, also sometimes called an azlactone or oxazolinone, why it is so susceptible to epimerization. Removal of the α-proton generates a five-membered ring with six π-electrons, an aromatic system according to the Hückel $4n+2$ rule (Scheme 5).

Scheme 5 Epimerization of Oxazol-5(4*H*)-ones

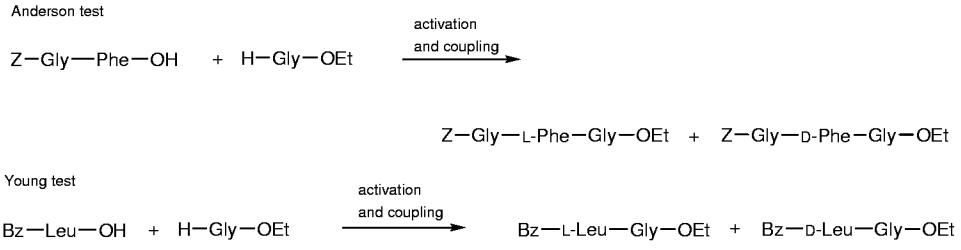


Oxazol-5(4*H*)-ones remain activated intermediates and can react with amines to form peptide bonds. The formation of the oxazol-5(4*H*)-one has been known and studied for many years. In 1928, Bergmann and Zervas reported that *N*-acyl protected amino acids racemized upon treatment with acetic anhydride in acetic acid.[1] They proposed that the mechanism responsible was acid-catalyzed enolization of the oxazol-5(4*H*)-one, which was formed subsequent to the (activating) formation of the mixed anhydride. Optically active oxazol-5(4*H*)-ones were first isolated in the 1940s by workers in the penicillin field,[2] and in 1964, Goodman and Levine reported the first synthesis of an optically active crystalline oxazol-5(4*H*)-one, derived from *N*-benzoyl-L-phenylalanine.[3] In 1966–1967, Goodman and McGahren[4,5] reported the synthesis and racemization studies[6] of the peptide oxazol-5(4*H*)-one derived from Z-Aib-Phe. In 1970, Goodman and Glaser published an extensive review of the reactions of oxazol-5(4*H*)-ones, including studies on the kinetics of racemization and aminolysis.[7] One of the most important findings was that formation of oxazol-5(4*H*)-ones from activated N-protected intermediates is significantly slower when the N-protecting group is changed from an amide to a carbamate (urethane), although urethane-protected oxazol-5(4*H*)-ones that do form are subject to decomposition to the unsubstituted α-amino acid *N*-carboxyanhydride (NCA). This result was predictable from a report by Scott[8] in which the nucleophilicities of the carbonyl oxygen of amide conjugate bases, neutral amides, and urethanes were compared in the course of a study of the neighboring group effects at alkyl carbons. The results of the peptide studies exactly parallel those found by Scott.

Classical Racemization Assays

The classical methods for detection and quantitation of racemization require analysis of the chiral purity of the product of a peptide-bond-forming reaction. For example, the Anderson test is used to explore a variety of reaction conditions for the coupling of Z-Gly-Phe-OH to H-Gly-OEt (Scheme 6).[9] The two possible enantiomeric tripeptides are separable by fractional crystallization, so that gravimetric analysis furnishes the racemization data. This procedure has a detection limit of 1–2% of the epimerized tripeptide. A modification by Kemp,[10] utilizing ^{14}C-labeled carboxy components, extends the detection limit by two to three orders of magnitude by an isotopic dilution procedure. The Young test[11] addresses the coupling of Bz-Leu-OH to H-Gly-OEt, and the extent of epimerization is determined by measurement of the specific rotation of the dipeptide product.

Scheme 6 The Anderson and Young Racemization Tests[9,11]



for references see p 668

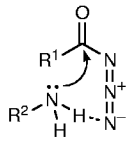
A method by Berger[12] utilized the stereoselective hydrolysis of diastereomeric peptides by leucine aminopeptidase. Z-L-Ala-D-Ala-OH was coupled to an all L-Ala peptide such as L-Ala-L-Ala-ONbz. Epimerization during coupling resulted in the formation of a small amount of L-Ala-L-Ala-L-Ala-L-Ala after deprotection, and since the peptidase has an absolute specificity for the all-L peptide, only the epimerized product was hydrolyzed. Quantification of the degradation products gave the extent of epimerization. These classical procedures, however, are specific to the particular coupling reaction under consideration and the results may not be fully applicable to all systems. Furthermore, they give no direct information about the rate of racemization.

Classical Approaches for Controlling Racemization

At the core of peptide synthesis are the techniques employed to form its central element, the peptide bond. By necessity, the carboxy component of the reaction mixture must be modified so that nucleophilic attack of the amine component can occur at the carboxy carbon. In living systems, this modification is an esterification of the amino acid to the 3′-hydroxy group of an adenosine nucleotide attached to transfer RNA. This esterification is sufficient to activate the carboxy group (in the presence of the appropriate enzyme) toward aminolysis of the ester by the incoming amino acid. In most cases, however, peptide chemists do not have the luxury of specific enzyme catalysis for their peptide-bond-forming reactions, and simple esterification of the carboxy component is insufficiently activating. A significant portion of the history of peptide synthesis involves the design of activating reagents.

As mentioned above, activation of the carboxy group of an α-amino acid generally occurs by attachment of an electron-withdrawing group. One of the first such moieties introduced to peptide synthesis was the azido group. The formation and use of acyl azides was first described in 1902 by Curtius.[13] Remarkably, this early method of peptide bond formation proved to be quite resistant to epimerization. Serendipity was at work here. It was suggested that the coupling rate was enhanced using acyl azides because of the ability of the activated intermediate to coordinate the incoming nucleophile via hydrogen bonding interactions (Scheme 7).[14] The theme of decreasing the observed epimerization by increasing the coupling rate is common and will be addressed again below, as will the theme of increasing the coupling rate by coordination of the incoming nucleophile.

Scheme 7 Coordination of the Incoming Nucleophile during Azide Coupling



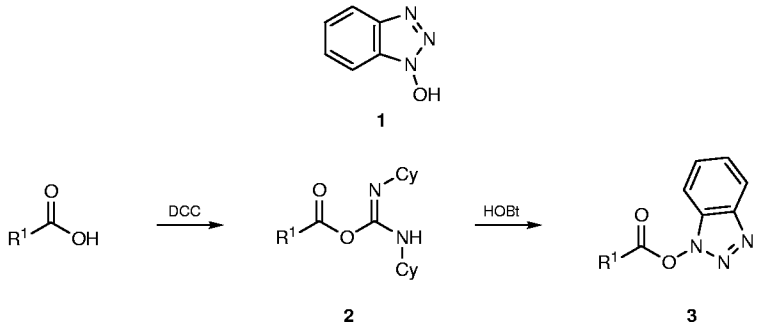
Indeed, there were those who described the azide coupling method as "racemization-free."[15] However, this viewpoint proved to be overly optimistic. In 1970, Sieber reported that during a synthesis of calcitonin M by the azide method, significant epimerization occurred during two of the segment condensation steps; in one of these reactions 40% of the epimerized product was observed.[16] There is a crucial detail in the experimental procedure here. The workers used *tert*-butyl nitrite to convert a peptide hydrazide into a peptide azide, but did not isolate the azide as was typical for research at that time. Instead, they neutralized the active intermediate in situ with DIPEA and added the amino segment for acylation. This demonstrates another important theme in the control of epimerization, the presence of a tertiary amine in the reaction mixture, even if only as a neutralization equivalent, can result in the formation of epimerized products. Indeed, most observations of racemization during

azide couplings occur when a base is present in the reaction mixture. The good reputation that the azide method earned was largely as a result of most workers isolating the peptide azide and carrying out the coupling reaction in the absence of added base.

The azide method, though useful in the first half of the 20th century, suffered from several drawbacks that motivated peptide chemists to find improved methods for peptide bond formation. One of the most significant discoveries was the carbodiimide method, reported in 1955 by Sheehan and Hess,[17] who utilized DCC for the coupling of Z-Gly-Phe-OH to H-Gly-OEt and reported no racemization. Again, this proved not to be a general observation. In 1958, Anderson and Callahan,[18] in a careful study of the same reaction, observed that epimerization did indeed occur to some extent and was dependent on temperature. For example, reaction in dichloromethane at room temperature gave 12% of the DL-product, but by carrying out the reaction at −5 °C only 0.5% of the DL-product was formed. Also in 1958, Hofmann[19] attempted the synthesis of the pentapeptide His-Phe-Arg-Trp-Gly using DCC and found that only 40% of the final product was the desired all-L form. The culprit in these cases was likely the intermediate *O*-acylisourea formed upon DCC activation, which can readily decompose to the readily epimerizable oxazol-5(4*H*)-one. In addition, reaction times using DCC alone were often on the order of several hours; the sluggishness of these reactions allowed additional time for epimerization to occur.

The utility of carbodiimide reagents was too important to consider abandoning them. Instead, it rapidly became obvious that carbodiimide activation could be used to prepare in situ active esters. Numerous nucleophilic additives were discovered and prepared for use in these reactions. The most important of these additives was 1,2,3-benzotriazol-1-ol (HOBt, **1**), first reported for use in peptide synthesis by König and Geiger in 1970.[20] As an example of a typical result, 1.2 equivalents of HOBt was added to the DCC coupling of Boc-Leu-Phe-OH to H-Val-OtBu in DMF as a solvent. Less than 1% of the L-D-L epimerized tripeptide was formed. When the reaction was carried out in the absence of HOBt, the amount of L-D-L product formed was 14.3%. Addition of HOBt to DCC reactions converts the intermediate *O*-acylisourea (**2**) (and any symmetrical anhydride) into the HOBt active ester **3** (Scheme 8).

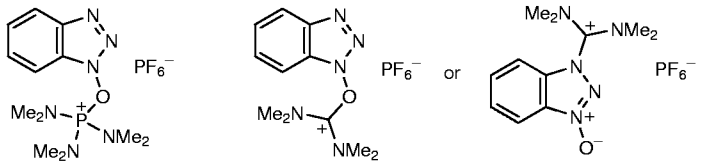
Scheme 8 Conversion of *O*-Acylisourea into HOBt Active Ester



This active ester is less prone to form oxazol-5(4*H*)-ones than is the *O*-acylisourea, and it is strongly activating, so that the coupling rate is increased. Both of these attributes serve to decrease the formation of epimerized product. This additive has been so useful that numerous reagents containing the HOBt moiety have been prepared, including those that generate the HOBt active ester in situ upon treatment with tertiary amine, for example the phosphonium reagent **4** (BOP)[21] and the uronium reagent **5** (HBTU).[22] The uronium salt is shown in its two tautomeric forms, based on crystal structure elucidation (Scheme 9).[23]

for references see p 668

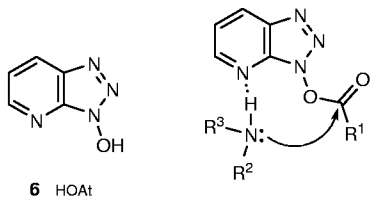
Scheme 9 HOBt Incorporated into Peptide Coupling Reagents[21,22]



4 BOP **5** HBTU

The utility of HOBt has been extended by preparation of its 7-aza derivative **6**, denoted HOAt.[24] In an illustrative study by Carpino,[25] the use of HOBt and HOAt were directly compared in a difficult solid-phase synthesis reaction. The dipeptide Fmoc-Phe-Ser(OtBu)-OH was coupled to H-Pro-PAL-PEG-PS resin. When DCC/HOBt was employed as the coupling reagent, 18% of the L-D-L epimerized product was formed. Upon substitution of HOAt for HOBt under the same conditions, the level of epimerized product was reduced to 6%. The HOAt moiety has also been incorporated into phosphonium and uronium reagents. Carpino has postulated that for HOAt esters, the nitrogen atom in the pyridine ring of HOAt is capable of coordinating the incoming nucleophile. This is analogous to the acyl azide situation, but the coordination is much more powerful, and the rate of amide bond formation is dramatically increased (Scheme 10).

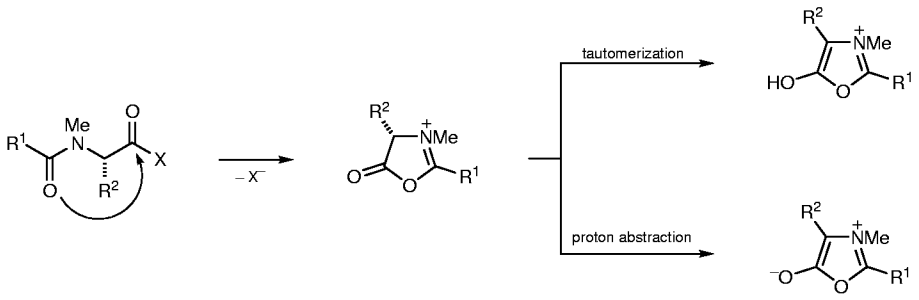
Scheme 10 HOAt Active Esters and Coordination of the Incoming Nucleophile[24]



6 HOAt

There are numerous studies of control of racemization for specific subsets of amino acids. For example, Benoiton has carried out extensive studies on racemization of *N*-methyl amino acids. In particular, McDermott and Benoiton[26] demonstrated that the presence of tertiary amine salts in coupling reactions had a profound effect on activated *N*-methyl amino acids in contrast to the nonmethylated form. In one example, when Z-Ala-MeLeu-OH was coupled to the tosylate salt of Gly-OBzl with ethyl 2-ethoxy-1,2-dihydroquinoline-1-carboxylate (EEDQ) in the presence of TEA, 15% of the L-D dipeptide was formed with a yield of 68%, compared to 0.5% for Z-Ala-Leu-OH with a yield of 78%. When the free base of Gly-OBzl was utilized in a DCC/HOSu coupling in the absence of tertiary amine, no L-D products were detected. The authors attributed this increased susceptibility to epimerization to an oxazolonium intermediate, which can epimerize by proton abstraction or merely by tautomerization (Scheme 11).

Scheme 11 Epimerization of Activated *N*-Methyl Amino Acids



This hypothetical intermediate is supported by the observation that *N*-acyl protected *N*-methyl amino acids have a higher propensity for epimerization than the corresponding *N*-carbamoyl protected analogues. Once again the theme of tertiary amine leading to epimerization is demonstrated.

The classical approaches to the control of racemization were generally empirical in nature, where peptide chemists found solutions to the particular epimerization problem at hand. Only recently have studies been undertaken that probe the general principles of racemization during peptide synthesis, isolating in turn the individual reaction parameters.

Measurement of the Intrinsic Rate of Racemization

The classical racemization assays are complicated by the fact that the rate of amide bond formation has a direct effect on the observed extent of epimerization. In every reaction there is a competition between the intrinsic rates of epimerization and amide bond formation. A reaction with a high coupling rate will generally show little or no epimerization, regardless of conditions, simply because the lifetime of the activated intermediate is too short for proton abstraction or oxazol-5(4*H*)-one formation to take place. Conversely, if the coupling rate is very slow, epimerization may be observed even under the most ideal conditions. Any racemization assay that relies on the stereochemical analysis of the product will therefore be dependent on the rate of amide bond formation. Since the goal of any method to assess the rate of racemization is the determination of the intrinsic factors that control epimerization, the ideal assay will be able to measure racemization rates independently of all other processes occurring during the reaction.

The ideal racemization assay has the following features:

(a) the epimerization process is separate from and independent of the amide bond formation process,

(b) epimerization takes place by a single mechanism,

(c) the activated intermediate is easily prepared and does not decompose over the lifetime of the experiment,

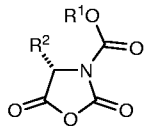
(d) the kinetics of the experiment are easily measured.

Reports in the literature that isolate any of these processes are rare and often require unusual conditions. For example, in addition to the oxazol-5(4*H*)-one studies described above, Kemp and Rebek[27] were able to use kinetic isotope effects to distinguish the enolization mechanism from oxazol-5(4*H*)-one formation in a simple peptide coupling experiment. α-^{2}H-Labeled Bz-L-Leu-OH and Z-Gly-Phe-OH were prepared and coupling reactions to H-Gly-OEt were carried out. In cases where oxazol-5(4*H*)-one formation is rate-determining, such as with Bz-L-Leu-OH, the isotope effect k_H/k_D is equal to 1 because the α-proton is not removed until after this rate-determining step. In contrast, enolization requires the direct removal of the α-proton, and the isotope effect measured for this mechanism was as high as 2.9 with Z-Gly-Phe-OH. Therefore, a measurement of the isotope

for references see p 668

effect for a given reaction provides evidence for the mechanism of epimerization. The urethane-protected α-amino acid *N*-carboxyanhydrides **7** (UNCAs) are a set of highly useful newer reagents for peptide synthesis (Scheme 12). The successful preparation and application of these relatively stable, crystalline, preactivated intermediates to peptide synthesis were first reported in 1990.[28] They react rapidly and quantitatively with most amines to form amides with the release of carbon dioxide as the only byproduct.

Scheme 12 Urethane-Protected α-Amino Acid *N*-Carboxyanhydride

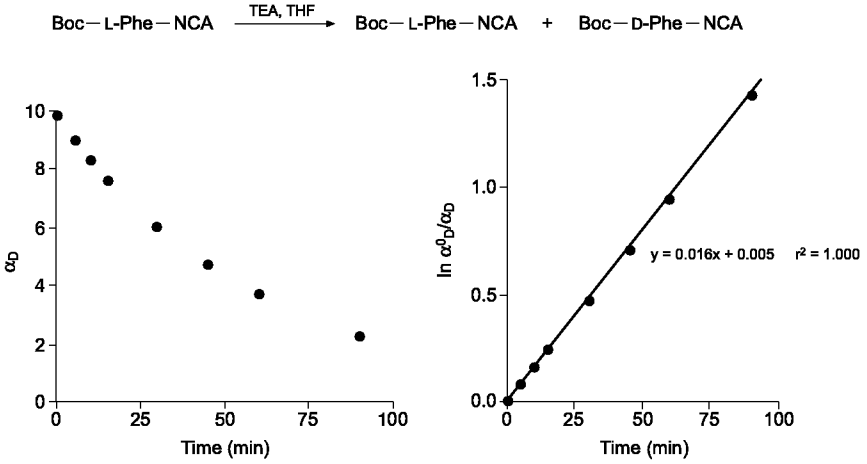


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The UNCAs are ideal activated intermediates for the study of intrinsic rates of racemization. Epimerization can occur only via the direct abstraction mechanism, because the five-membered ring structure precludes formation of an oxazol-5(4*H*)-one. In addition, the UNCA maintains its structural integrity during the epimerization process. A report by Romoff[29] describes two simple methods for the measurement of intrinsic rates of racemization under a wide variety of reaction conditions utilizing UNCAs as the prototypical activated intermediates.

The basic experiment consisted of simply treating an L-UNCA with a base and monitoring the formation of the D-UNCA. The most straightforward analysis is accomplished by polarimetry. For example, Boc-Phe-NCA was dissolved in THF at a concentration of 0.33 M, and 1.5 equivalents of TEA were added. The resulting solution was placed in a polarimeter cell and the optical rotation was monitored over time (Figure 1).

Figure 1 Racemization of Boc-Phe-NCA by TEA[29]



The optical rotation of the mixture approaches zero (a racemic mixture) over time, with apparent first-order kinetics. This observation was supported by the semi-log plot [$\ln(\alpha^{\circ}_D/\alpha_D)$ vs time], which is linear (Figure 1). It has been shown that this racemization process does in fact follow a true pseudo-first-order rate equation, the details of which have been described by Eliel.[30] Therefore, these processes can be described by the first-order rate constant associated with them, which reflects precisely the intrinsic rate of racemization. Comparison of the half-lives for racemization under conditions of varying amino acid side chain, base, and solvent is the basis for this new general method.

This polarimetric method was made even more general by utilizing chiral HPLC techniques. The L-UNCA was dissolved in the solvent at a concentration of 0.33 M at 20 °C. The tertiary amine (1.5 equiv) was added at time zero. The solution was allowed to stand for an experimentally determined delay time, during which the only process that can occur was epimerization, since there is no nucleophile present. The delay time was determined after carrying out several experiments with different delay times and chosen so as to fall within or just after the first half-life for racemization. At the end of the delay period, benzylamine was added. Benzylamine is a very powerful nucleophile that reacts virtually instantly (regardless of the type of activation) with the resulting mixture of L- and D-UNCAs to form the benzyl amides and quench the epimerization process. Thus, a "snapshot" of the ratio of L/D activated intermediates at the instant of benzylamine addition was obtained by measurement of the L/D ratio of the benzyl amide products.

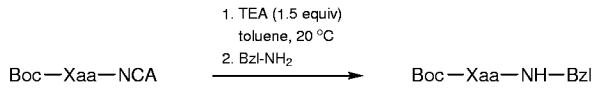
The ratio of the enantiomeric benzyl amide products was determined by analyzing a diluted aliquot of the quenched reaction mixture by HPLC using a chiral stationary phase column (Chiralcel OD, Daicel Chemical Co.). Since racemization is a pseudo-first-order kinetic process, these data (along with the time zero value) are sufficient for determination of the intrinsic rate of racemization k_R. The half-life for racemization $t^R_{1/2}$ can be directly calculated from the L/D ratio (or % enantiomeric excess, %ee) where t was the time of benzylamine addition (the delay time):

$$t^R_{1/2} = \frac{t}{ln[100/(\%ee)]} \cdot \ln 2$$

The Effect of Amino Acid Side Chain, Tertiary Amine, and Solvent on the Intrinsic Rate of Racemization

The UNCA methodology has been utilized to study the effect of the important parameters in peptide synthesis on the intrinsic rate of racemization.[29] The results of these experiments reflect both thermodynamic and kinetic influences on reaction rates and products. Table 1 shows the effect of varying the amino acid side chain, while maintaining the nature of the base and solvent.

Table 1 The Effect of the Amino Acid Side Chain[29]



Xaa	Delay Time	Ratio (L/D)[a]	$t^R_{1/2}$
Phe	30 min	65:35	16 min
Ser(OBzl)	3.0 min	65:35	1.7 min
Asp(OBzl)	11 min	64:36	6.0 min
Val	18 h	96:4	150 h

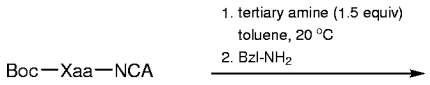
[a] Determined by chiral HPLC.

These results clearly show a correlation between the ease of formation of the α-deprotonated UNCA and the substituent(s) on the β-carbon. The presence of electron-withdrawing groups on the β-carbon stabilizes the α-carbanion, thus lowering the activation energy for its formation (decreased pK_a of the α-proton) and increasing the rate of racemization. This effect was shown previously by Kovacs,[31] who by the polarimetric method found high racemization rates for Z-Cys(Bzl) and Z-Asp(OMe) active esters, and a slower rate for Z-Glu(OMe) active esters. Conversely, the rate is decreased by branching at the β-carbon, because the increased steric bulk hinders approach of the amine to the α-proton.

for references see p 668

Nearly all peptide-bond-forming reactions employ tertiary amines in some manner, most commonly for neutralization of amine salts and/or as an adjunct to the activating agent. Table 2 shows the effect of tertiary amine structure on the rate of racemization of two UNCAs in toluene solvent.

Table 2 The Effect of the Tertiary Amine[29]



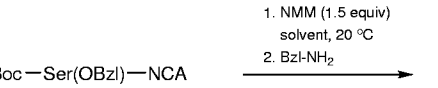
Xaa	Tertiary Amine	Delay Time	Ratio (L/D)[a]	$t^R_{1/2}$
Phe	TEA	30 min	65:35	16 min
Phe	DIPEA	2.0 h	87:13	4.6 h
Phe	4-methylmorpholine	14.2 h	98:2	240 h
Phe	4-ethylmorpholine	14.2 h	100:0	–
Ser(OBzl)	TEA	3.0 min	65:35	1.7 min
Ser(OBzl)	DIPEA	15 min	89:11	42 min
Ser(OBzl)	4-methylmorpholine	45 min	89:11	130 min
Ser(OBzl)	4-ethylmorpholine	2.3 h	85:15	260 min
Ser(OBzl)	4-cyclohexylmorpholine	2.3 hr	89:11	380 min

[a] Determined by chiral HPLC.

The relationship between the structure of the tertiary amine and the intrinsic rate of racemization was clear. This effect has been studied before by Williams,[32] among others. Two families of bases were compared: trialkylamines and 4-alkylmorpholines. The trends were most clearly expressed in the experiments using Boc-Ser(OBzl)-NCA. The rate was decreased within each family as the steric bulk of the amine was increased. This result was consistent with the direct enolization mechanism that requires a close approach of the tertiary amine for the abstraction of the α-proton. The rate was much lower with 4-alkylmorpholines than with trialkylamines because of the decreased basicity of the former (triethylamine and 4-ethylmorpholine have similar structures; TEA is actually more hindered). The most favorable results with respect to racemization were obtained when a weak base was combined with a sterically hindered substituent, as with 4-cyclohexylmorpholine. In the case of Boc-Phe-NCA, the same trends were seen, except that racemization by the 4-alkylmorpholines was so slow that the differences within that family were not significant.

The solvent also makes important contributions to the intrinsic rate of racemization. Table 3 shows the effect on varying solvent on racemization of Boc-Ser(OBzl)-NCA by 4-methylmorpholine.

Table 3 The Effect of the Solvent[29]



Solvent	Dielectric Constant (ε, 20 °C)	Amine Salts Soluble?	Delay Time (min)	Ratio (L/D)[a]	$t^R_{1/2}$ (min)
toluene	2.38	no	45	89:11	132
THF	7.58	no	45	91:9	157
CH_2Cl_2	9.08	yes	15	82:18	23
DMF	38.3	yes	15	73:27	13

[a] Determined by chiral HPLC.

The influence of the solvent on the rate of racemization has been studied for almost fifty years. One of the first reports on this topic was by Vaughan,[33] who studied peptide synthesis using the mixed carbonic anhydride approach and determined the extent of epimerization by product analysis. There are two important aspects to the influence of the solvent on the intrinsic rate of racemization. First, the rate of racemization is substantially increased in the polar solvent DMF. This is a result of increased solvation and thus stabilization of the activated intermediate. Second, a significant observation is that the rate in the nonpolar (as judged by the dielectric constant) solvent dichloromethane is comparable to the rate in the polar solvent DMF. Even though dichloromethane has a dielectric constant close to that of toluene and THF, it is commonly observed that hydrochloride salts of tertiary amines are soluble in dichloromethane and insoluble in the latter two solvents. This solvating ability of dichloromethane may lower the activation energy for protonation of the tertiary amine (and concurrent deprotonation of the activated intermediate), increasing the rate of racemization.

Summary – Thermodynamic and Kinetic Contributions to Epimerization

The parameters that control epimerization in a peptide-bond-forming reaction can be assessed in terms of their thermodynamic and kinetic components. Thermodynamic effects are those that stabilize the deprotonated activated intermediate or the protonated tertiary amine. Kinetic effects are expressed based on the degree of steric hindrance between the tertiary amine and activated intermediate. Table 4 summarizes these contributions and shows examples of high, moderate, and low propensities for contribution to the intrinsic rate of racemization among the various parameters.

Table 4 Summary of Contributions to the Intrinsic Rate of Racemization

	Propensities toward Epimerization, Examples			
Parameter	High	Moderate	Low	Types of Effects
amino acid	Ser, Cys, Asp	Phe, Ala	Val, Ile	thermodynamic: β-electron-withdrawing group on side chain increases propensity kinetic: β-branching on side chain decreases propensity
tertiary amine	TEA	DIPEA	4-ethylmorpholine, 2,4,6-collidine	thermodynamic: increasing base strength increases propensity kinetic: steric hindrance of base decreases propensity
solvent	–	DMF, CH_2Cl_2	THF, toluene	thermodynamic: stabilization of charged intermediates (e.g., amine salts) increases propensity

The trends among the parameters studied by the UNCA racemization assay have been borne out when other activation methods are used. By a modification of chiral HPLC procedure below, the assay can be used to study any method of activation (for example, uronium or phosphonium methods), keeping in mind that the activation must be rapid and that epimerization by multiple mechanisms becomes possible. These findings also hold for solid-phase synthesis reactions.

The stereochemical outcome of a peptide-bond-forming reaction is the result of the competition between the intrinisic rate of racemization of the activated intermediate and the rate of nucleophilic attack on that intermediate. It has been demonstrated that stereochemical integrity is often preserved by increasing the coupling rate. Significant progress has been made toward the minimization of epimerized products as a result of the development of reagents that increase the coupling rate, especially those that are able to coordinate the

for references see p 668

incoming nucleophile. At the same time, the presence or absence of tertiary amine and its structure are crucial elements. The majority of typical peptide synthesis reactions are generally free of epimerization products, mainly because the peptide-bond-forming rate is very high. However, the demands of modern drug discovery often require the synthesis of peptide fragments with sterically hindered structures and/or amino acids with a high propensity for epimerization, especially Cys. Moreover, the newest activating reagents often require the presence of large excesses of tertiary amine. In these cases, a judicious choice of conditions, based on the results described above, is crucial for minimizing the formation of epimerized products.

Polarimetric Determination of Intrinsic Racemization Rates; Typical Procedure:[29]
A soln of Boc-Phe-NCA (0.58 g, 2.0 mmol) in THF (6.0 mL) was stirred at 20 °C and TEA (0.42 mL, 3.0 mmol) was added. An aliquot of the mixture was immediately removed and placed in a 1.00-dm polarimeter cell. The total optical activity of the mixture at the sodium D line (589 nm) was monitored over time. After 90 min, the solvent was removed under reduced pressure from the remainder of the mixture to give a white solid. ^{1}H NMR analysis of this material showed it to be pure Boc-Phe-NCA, unchanged from starting material except for its optical activity. The material exhibited $[\alpha]_D^{20}$ +17 (*c* 1.1, THF), a loss of 86% of the original value for Boc-L-Phe-NCA of $[\alpha]_D^{20}$ +119 (*c* 1.1, THF).

Chiral HPLC Determination of Intrinsic Racemization Rates; Typical Procedure:[29]
To a soln of Boc-Ser(OBzl)-NCA (107 mg, 0.33 mmol) in toluene (1.0 mL) at 20 °C was added DIPEA (87 µL, 0.50 mmol). After stirring for 15 min, benzylamine (38 µL, 0.35 mmol) was added. After a further 5 min of stirring, a 10-µL aliquot of the mixture was diluted with a mixture of hexane and propan-2-ol (1:1, 1.0 mL). A 10-µL aliquot of this diluted soln was analyzed for the L/D ratio of the product Boc-Ser(OBzl)-NHBzl by injection onto a Chiralcel OD HPLC column (9:1 hexane/iPrOH, 1.5 mL/min, detection at 220 nm). The L/D product ratio was calculated by integration, based on previously synthesized standards.

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7.5 High Resolution NMR Spectroscopy

E. Locardi and H. Kessler

A few milligrams of a peptide, or less, are sufficient to run a routine ^{1}H NMR spectrum. The sample is dissolved in approximately 0.5 mL of water, DMSO, or other organic solvent and inserted into a very strong homogeneous magnetic field. Irradiation by radio frequency pulses results in excitation of magnetically active nuclei such as ^{1}H, ^{13}C, ^{15}N, or ^{19}F isotopes. For a given field strength each nucleus is excited only with a specific radio frequency, e.g. at a field strength of 11.7 Tesla protons require 500 MHz (^{13}C 125 MHz, ^{15}N 50 MHz, and ^{19}F 470 MHz). The excitation and its return to the initial state via "relaxation" can be detected and yields the NMR spectrum. The same nuclei in chemically different environments, such as protons in an aromatic ring or protons of a methyl group, will exhibit slightly different resonance frequencies, the "chemical shift". The range of chemical shifts of protons is about 10 ppm, i.e. up to 5000 Hz in the magnetic field given above. Neglecting accidental overlap, each chemically nonequivalent (except enantiotopic) nucleus exhibits separate signals in the NMR spectrum. Hence, counting the different signals can yield important information. However, each signal does not appear as a single line, but is split by through-bond coupling, "*J* coupling", caused by neighboring protons (or ^{13}C nuclei). Through-bond coupling is normally observed if the signals are only separated by one (^{13}C–H), two (e.g., ^{13}C–C–H, H^{α}–C–H^{β}), or three (e.g., H–C–C–H) bonds. The through-bond coupling decreases rapidly with an increasing number of bonds between the coupling nuclei. The identification of these couplings (e.g., in a COSY[1,2] or TOCSY[3,4]) spectra yields the connectivities of the different nuclei and, in total, the molecular constitution.

Excitation of nuclei results in nonequilibrium states that correspond to heating the nucleus. The return to the equilibrium can occur via heat flow from a "hot" nuclear spin state to a "cold" nucleus that is spatially close. These spins interact via their magnetic moments through space (dipolar coupling). Relaxation to spatially neighboring nuclei, equivalent to heat flow from a "hot" nucleus to a "cold" nucleus, causes changes in signal intensities (Nuclear Overhauser Effects, NOE), which are recorded in special experiments (ROESY,[5–8] NOESY[9–11]). As the rate of heat flow, the buildup rate of NOE, decays rapidly with increasing distance of dipolar coupling nuclei, the NOE (coupling of the nuclear dipoles causes the NOE) yields the spatial orientation (conformation, configuration) or can be used to connect different spin systems, which are not mutually coupled by *J* coupling.

Heteronuclei such as ^{13}C (this magnetically active nucleus has 1.1% natural abundant) and ^{15}N (0.3% natural abundance) are routinely measured with modern NMR spectrometers. Proton decoupled ^{13}C NMR spectra in natural abundance exhibit singlets for each specific carbon atom, which are easier to count than overlapping multiplet lines in ^{1}H NMR. 1D ^{13}C NMR can be used to investigate whether a peptide exhibits a single set of lines or a double (or more) set, which indicate conformational or configurational isomers (see Section 7.5.3). However, 1D ^{13}C NMR is rather insensitive and if there is not enough material or the solubility is low, more sensitive techniques have to be applied.

The best way to achieve this is to use the directly bound protons to excite the carbon. INEPT (insensitive nuclei enhanced by polarisation transfer) or DEPT (distortionless enhancement by polarisation transfer) techniques[12,13] are used for this purpose. The latter has the advantage of easy editing of spectra exhibiting "multiplicity selection", i.e. isolated spectra for CH, CH_2, and CH_3 groups. However, 2D dimensional versions, which yield the carbons in one dimension and their directly bound protons in the other, have only slightly lower intensity (by a factor of $1/\sqrt{2}$ if relaxation is neglected). As the level of information in 2D spectra is much higher, it is strongly recommended that they be used routinely. The common 2D ^{1}H–^{13}C correlation techniques are heteronuclear multiquantum correlation (HMQC)[14] and heteronuclear single-quantum correlation (HSQC).[15] Although quaternary

carbons are not visible in these spectra, the combined ^{1}H and ^{13}C information available and the high level of sensitivity render HMQC (for ^{13}C generally preferred over HSQC) the most efficient method for the collection of NMR data.

For the much less sensitive ^{15}N nucleus HSQC (preferred over HMQC for nitrogen) is often the only way to obtain spectra at natural abundance.

This section mainly discusses the use of NMR for the elucidation of the constitution of peptides, i.e. the identification of amino acids, presence or absence of protecting groups, and the assignment of the amino acid signals in the peptide sequence.

A few words about conformational analysis are given in Section 7.5.3. In Section 7.5.1 the NMR of peptides in general is discussed. Reviews of NMR spectroscopy of peptides can be found in the literature.[16–18]

7.5.1 NMR of Peptides

Peptides are amide-containing structures with unique sequences. *J* coupled correlated proton spectra (COSY, TOCSY) exhibit isolated spin systems for each amino acid. Coupling across the amide group is small and normally not observed. Some of amino acids have characteristic patterns and are readily identified (e.g., Gly, Pro, Val, Thr). Aromatic amino acids (e.g., Phe, His, Trp) have a typical AMXY spin system with A = NH proton, M = α-proton, X,Y = two β-protons. They can only be differentiated by connecting the aromatic protons with the AMXY, for example, by using NOE of the β-protons X and/or Y to some of the aromatic protons, or heteronuclear long-range couplings from X,Y and aromatic protons to a common ^{13}C atom, or by characteristic carbon shifts such as the typical low-field shift of the Trp β-carbon compared to the β-carbons of Phe and His (*vide infra*).

Linear peptides are very flexible. They have a number of conformations that undergo rapid interconversion on the NMR timescale. Hence, a mean spectrum is observed in which the chemical shift of the signals is mainly determined by the kind of amino acids and only to a lower extent by the sequence or conformation.

In general, the range of the chemical shifts of the signals increases when there is a lower number of populated conformations in equilibrium: the spread of signals for a more rigid head-to-tail cyclic peptide is larger than that of the linear precursor. An example is given in Figure 1 for the sequence Asn-Glu-His-Ala-Gly. The relatively good separation of the NH protons in the linear peptide is mainly induced by the effect of the two trityl and the *tert*-butyl residues as protecting groups. The dispersion of the NH resonances is very poor in a longer linear peptidic sequence (Figure 2).

for references see p 678

Figure 1 Comparison of a 1D ^{1}H NMR Spectrum (600 MHz, 300K, DMSO-d_6) of a Linear Peptide and a Cyclic Peptide of the Same Sequence

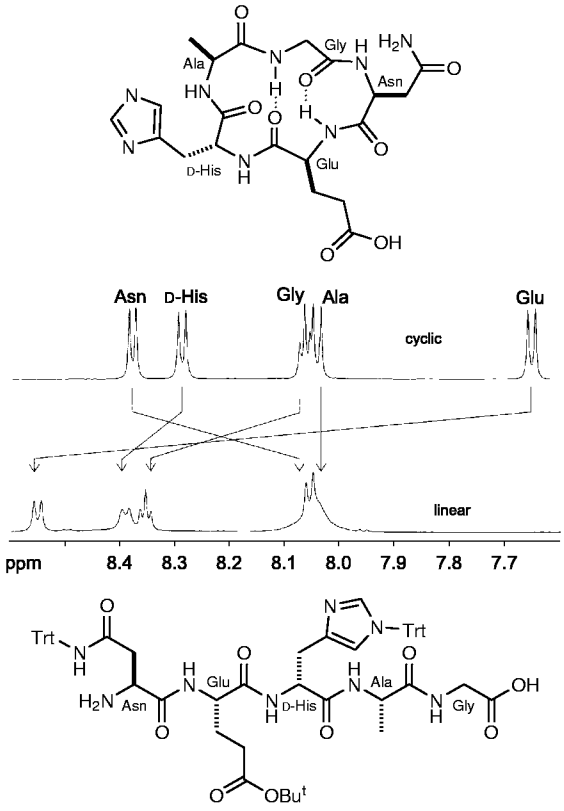
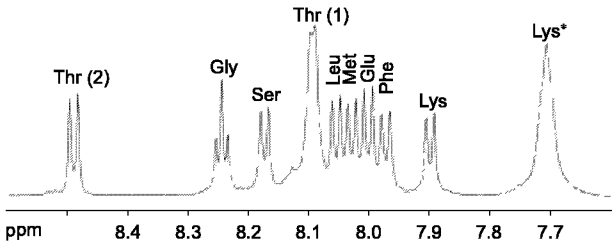


Figure 2 1D ^{1}H NMR (600 MHz, 300K, DMSO-d_6) for the Linear Nonapeptide H-Thr(1)-Thr(2)-Leu-Met-Lys-Phe-Ser-Glu-Gly-OH[a]



[a] Lys* denotes the ε-amino group of Lys and N-terminal amino group.

Furthermore, in flexible linear peptides the chemical shifts are typical of "random" structures similar to nonfolded proteins. Deviation from these random shifts sometimes identifies specific conformational preferences. NH-proton chemical shifts depend strongly on external influences (solvent, temperature, concentration, specific sequence). "Random coil" shifts for these protons correlate less well than chemical shifts of the α-protons or α-carbons.[19–26]

Not only are the shift differences of different heterotopic protons similar, but also those of diastereotopic β-protons. A preferred side-chain conformation is normally only found when there is also a preferred backbone conformation.

Coupling constants about bonds in which several conformations are in rapid dynamic equilibria also are averaged ($J_{HN-H\alpha}$ ~6–7 Hz, $J_{H\alpha-H\beta}$ ~7–8 Hz). In these cases, it is not recommended that much effort be expended on full assignment, in careful quantification of

NOE cross-peak intensity buildup, or quantitative extraction of coupling constants for conformational studies.[27] NMR spectroscopy is more recommended for proof of the constitution (Section 7.5.2).

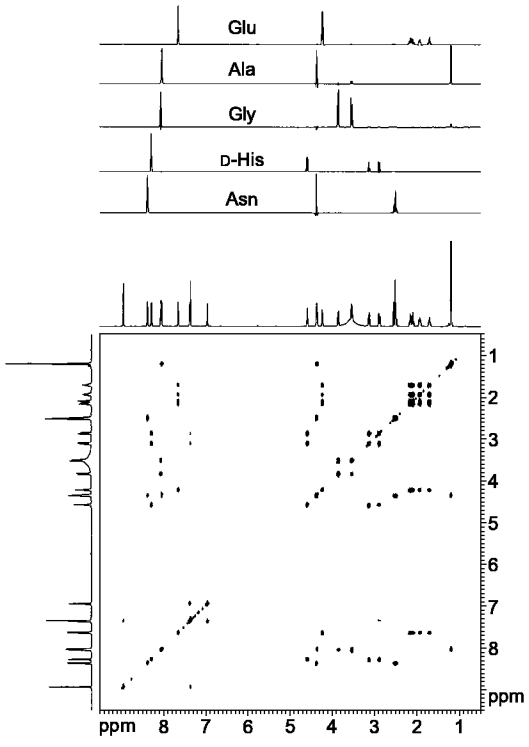
Assignment of diastereotopic groups (e.g., β-protons of aromatic amino acids, methyl groups in Val or Leu, both α-protons of Gly, pro-*R* or pro-*S*) is necessary for reliable conformational analysis but it is normally not important for proof of the constitution (Section 7.5.3).

7.5.2 Constitution

7.5.2.1 Identification of Amino Acids

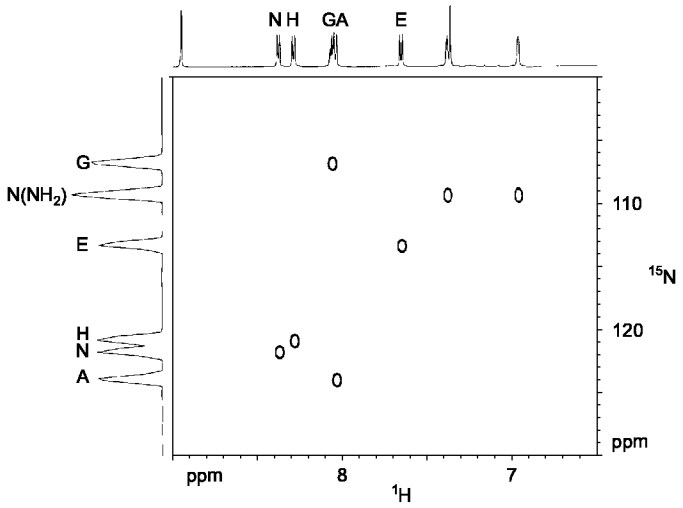
Numerous NMR techniques have been proposed for the assignment of NMR resonances to individual nuclei in peptides.[28,29] For each amino acid residue it is possible to identify a characteristic set of resonances that are connected by scalar spin–spin couplings. These resonances constitute a *spin system*. A TOCSY spectrum is the most efficient way to identify a proton spin system. Ideally, when all NH protons are resolved each amino acid spin system is completely established on cross-sections through its NH proton (Figure 3). A COSY spectrum can also be used for this purpose but then only neighboring protons are connected by a cross-peak as the appearance of a cross-peak requires a measurable coupling constant between two spin systems with different chemical shifts. This is normally only the case for geminal and vicinal protons. In the case of spectral overlap of the amide region a $^{1}H—^{15}N$ HSQC experiment can be helpful in the identification of all the NH groups (Figure 4).

Figure 3 TOCSY Spectra (600 MHz, 300K, DMSO-d_6) of the Cyclic Pentapeptide Sequence *c*[-Asn-Glu-D-His-Ala-Gly-]



for references see p 678

Figure 4 Natural Abundance ^{1}H–^{15}N HSQC Spectrum (600 MHz, 300K, DMSO-d_6, Total Measuring Time = 2 h) of the Cyclic Pentapeptide Sequence *c*[-Asn-Glu-D-His-Ala-Gly-][a]



[a] Signals are identified using their one-letter amino acid code.

7.5.2.2 Peptide Sequence

Sequential resonance assignment involves the assignment of the spin systems to their position along the peptide chain.

If the spin systems of the individual amino acids are identified, the connectivity in the sequence can be established via NOE or ROE across the amide bonds or via heteronuclear coupling of the carbonyl carbon to the NH and α-protons in the adjacent amino acids (Scheme 1). The NOEs from the NH to the C^{α} or C^{β} protons of the sequentially preceding residue are normally large and cross peaks connecting the spin systems of the amino acids are easy to observe (Figure 5).

Scheme 1 Peptide Sequence Sequential NOEs and Observable Through-Bond Couplings Useful for Sequential Assignment of the Peptide

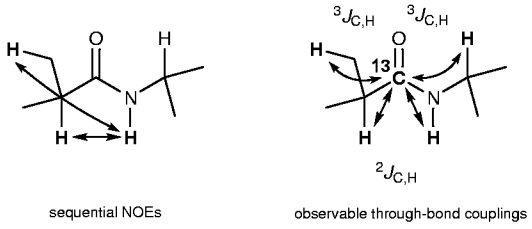
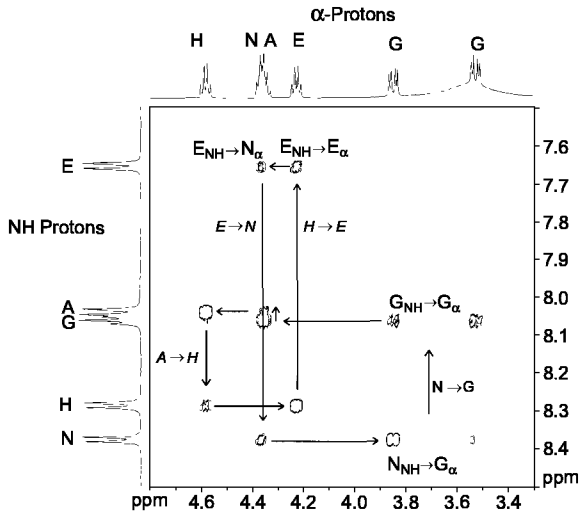


Figure 5 Finger Print of the NOESY Spectrum (600 MHz, 300K, DMSO-d_6) for the Cyclic Pentapeptide Sequence *c*[-Asn-Glu-D-His-Ala-Gly-] Showing the NH(i)–αH(i+1) Connectivities Valuable for the Sequential Assignment of the Resonances[a]



[a] Signals are identified using their one-letter amino acid code.

Heteronuclear long-range (two-bond) coupling from the carbonyl carbon to the C^{α}-proton of the same amino acid[30] and to the amide proton of the following amino acid unambiguously indicate the desired information about amino acid connectivity. For this purpose an HMBC sequence[31] is the most efficient technique.

7.5.2.3 Side-Chain Modification

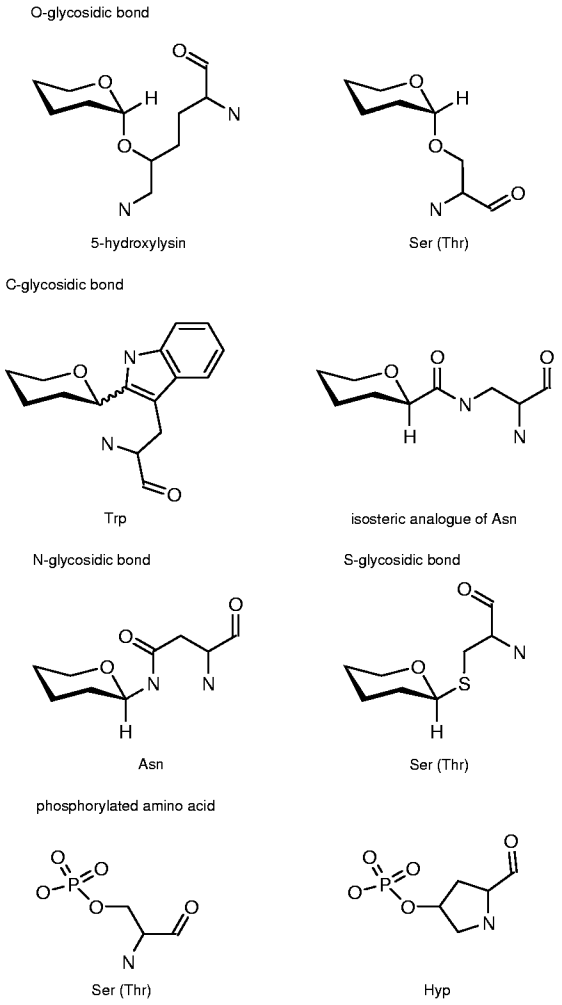
Amino acids with modified side-chains are frequently encountered in biological systems. Among all the modifications involving the side-chain of the amino acids, reversible glycosylation and phosphorylation play a central role in regulating the biological functionalities of numerous proteins.

Several types of glycosidic linkages in glycoproteins and unnatural products are shown in Scheme 2. The complete assignments of the resonances arising from the glycosylated amino acids can be carried out simply using a combination of COSY, ^{13}C–^{1}H shift correlation (HMQC), HMQC–TOCSY,[32] and ^{13}C–^{1}H long-range shift correlation (HMBC). For example, ^{1}H and ^{13}C NMR spectroscopic data have been used to establish the presence of a C-glycosidic linkage between a carbohydrate and a protein unequivocally.[33]

It has been shown that phosphorylation changes the local conformation of a protein and thereby affects the activity of the complete protein.[34] Phosphorylation of serine and threonine side chains often occurs (Scheme 2). Phosphoamino acids are readily characterized using ^{1}H and ^{31}P NMR experiments. The ^{1}H and ^{31}P NMR parameters are distinct for phosphorylated serine, threonine, and tyrosine and have also been used to identify both *cis*- and *trans-O*-phospho-4-hydroxy-L-proline.[35] Phosphorylation of Cys is rare, but it can be identified by NMR even in large proteins.[36]

for references see p 678

Scheme 2 Side-Chain Modified Amino Acids



7.5.2.4 NMR Methods for Monitoring Combinatorial Chemistry Libraries and Solid-Phase Synthesis

NMR spectroscopy has been proposed as a versatile tool for the analysis of complex mixtures, combinatorial libraries, and heterogeneous samples, such as peptides bound to resins during solid-phase synthesis.

The coupling of HPLC with NMR represents a powerful method for the high-throughput screening of peptides in mixtures run in stop-flow and continuous-flow modes. It is possible to obtain routine high-quality HPLC/NMR 1D NMR data with as little as 5 μg of compound in a chromatogram peak.[37] However, the HPLC/NMR technique cannot be favorably compared to mass spectrometry techniques (HPLC/MS) in terms of sensitivity and speed of analysis. To date, the majority of reports of the use of HPLC/NMR have been for drug metabolites.[38]

Developments in the editing of NMR spectra allows mixtures to be analyzed without chromatographic separation. The analysis of such unresolved mixtures is achieved using either diffusion-rate or relaxation-rate differences. Since compounds having different sizes and shapes tend to move and tumble at different rates, and the tumbling of a molecule can

influence its diffusion and relaxation rates, resonances arising from different compounds can be edited.[39–41] However, the application of NMR to the determination of structure in multicomponent organic systems prior to physical separation has not been widely applied and seems to be useful only in special cases.[42]

To overcome the problem of sample availability, smaller volume NMR probes, "nanoprobes", have been introduced for structure elucidations of samples that were previously considered too small for NMR analysis.[43]

NMR methods have also been introduced to follow directly reactions on solid support.[44] Optimization of chemical reactions on resin is normally done by cleavage from the resin and analysis of the separated component. High-resolution analysis by NMR on solid support has the advantage of yielding information on the bound species directly, e.g. these techniques are useful for structural studies on peptides that aggregate in solution after cleavage from the resin. For the preparation of the NMR sample, the resin is allowed to swell in a suitable solvent (CD_2Cl_2, $CDCl_3$), in a gel-phase system. High-resolution NMR spectra for the solid-phase synthesis resin-bound peptide is obtained by applying Magic Angle Spinning (MAS), a technique normally used for solid-state NMR.[45] The optimization of this technique has allowed samples bound to a single solid-phase synthesis bead to be detected, thus eliminating the restriction of sample availability.

7.5.3 Conformation and Dynamics

Molecular recognition between peptides and their receptors requires a natural surface fit and a specific orientation of the functional groups.[46] Hence, knowledge of the structural elements that characterize the peptides is fundamental to the understanding of their biological activity. However, peptides are flexible and adopt more than one conformation, and these conformations interconvert rapidly on an NMR timescale. If there is no strongly preferred structure in the equilibrium it is useless to attempt to determine a single structure that fulfils the experimental data. On the other hand, it might be that one or a few conformations are populated much more than others and those conformations can be elucidated by NMR. However, the flexibility of such structures allows an induced fit when bound to the receptor, e.g. the preferred conformation of the free molecule in solution may be different from the receptor-bound structure. The best methodology to obtain information in the bound state, when the structure of the complex cannot be determined directly, is to introduce conformational restrictions, which exclude parts of the conformational space available in the unconstrained linear peptide.[46] Cyclization via side chains or within the backbone can be used for this purpose. It is generally observed that in cyclic peptides conformational preferences are indicated by: (1) larger distribution of the NH chemical shifts; (2) stronger deviations from random chemical shift values for $C^{\alpha}H$ protons;[19–26] (3) larger chemical shift differences between diastereotopic protons such as the two α-protons of Gly or the two β-protons in His, Phe, Tyr, Asp, Glu, and also groups such as the two methyl groups in Val; (4) strong differentiation of temperature gradients of the NH shifts; and (5) deviations from mean coupling constants, which are about 7 Hz for $^3J_{C\alpha H-NH}$ or $^3J_{C\alpha H-C\beta H}$ when the conformations are equally populated.[16–18,46]

Conformational equilibria, which are slow on the NMR chemical shift timescale, exhibit a doubled signal set. These slow interconversions are observed when two amide bond rotamers are significantly populated, e.g. for N-alkylated amide bonds such as Xaa-Pro bonds. The barriers in these equilibria are in the order of 18 $kcal \cdot mol^{-1}$, leading to signal coalescence at 80–120 °C.[47] It is difficult to analyze each conformation in such equilibria in detail because frequently, but not always, even in cyclic peptides the peptide bond interconversion

for references see p 678

causes changes in both isomers only in the vicinity of the peptide bond. A rigorous analysis requires careful and tedious work, see as an example the study on the two conformations in thiocyclosporin.[48]

Also urethane bonds of protecting groups such as Boc often exhibit two rotamers, which interconvert with slightly lower barriers (~15 kcal·mol^{-1}).

It is beyond the scope of this section to describe the conformational analysis of peptides and the reader is directed to the literature.[16–18]

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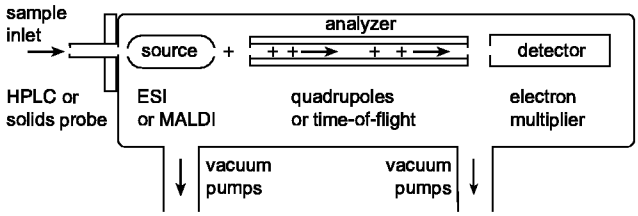
7.6 Mass Spectrometry

G. Barker, E. J. Want, J. Boydston, and G. Siuzdak

Mass spectrometry is an important tool in biochemical research. Scientists such as Fenn,[1] and Hillenkamp and Karas[2] have developed and established this technique into what it is today, a highly sensitive tool capable of analyzing small peptides as well as large proteins. Moreover, since adding electrospray ionization (ESI) and matrix-assisted laser desorption ionization (MALDI) to the repertoire of research tools, the demand for this instrumentation has literally exploded. As a result the commercial availability of these instruments that offer picomole to femtomole sensitivity and that enable the direct analysis of biological fluids with a minimum amount of sample preparation has made the analysis of proteins and peptides routine.[3] Peptide mapping can be performed directly on a protein digest by MALDI, or by compiling the mass data obtained from individual peaks in liquid chromatography.

A mass spectrometer determines the molecular weight of chemical compounds by separating molecular ions according to their mass-to-charge ratio (m/z). The ions are generated by inducing either the loss or the gain of a charge, for example through electron ejection, protonation, or deprotonation. Once the ions are formed they can be separated according to mass and finally detected. As a result molecular weight information as well as structural characterization can be obtained. The basic components of a mass spectrometer: the sample inlet, ionization source, mass analyzer, and ion detector are illustrated in Figure 1. Once the sample is introduced into the instrument through the sample inlet it undergoes ionization in the ionization source and the charged molecules are then electrostatically propelled into the mass analyzer, which separates the ions according to their mass-to-charge ratio.[3] The ion detector then transmits its signal to a computer where the information is stored and processed.

Figure 1 Components of a Mass Spectrometer and the Names of Those Commonly Used to Perform the Function of Ionization (Electrospray or MALDI), Analysis (Quadrupoles or Time-of-Flight), and Detection (Electron Multiplier)



7.6.1 Ionization and Ionization Sources

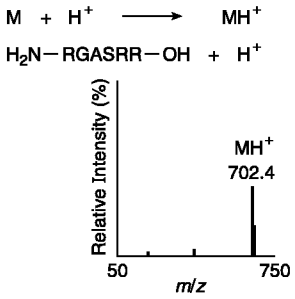
Electron ionization (EI) was the primary ionization source for mass analysis until the 1980s, limiting the chemist to the analysis of small molecules well below the mass range of common bioorganic compounds. This limitation motivated the development of the techniques commonly known as ESI,[1] MALDI,[2] and fast atom bombardment (FAB)[3,4] (Table 1). These ion sources allow for rapid and easy peptide analyses that previously required laborious sample preparation or were not possible with electron ionization. The mechanism of ionization these ion sources employ, which is somewhat responsible for their ability to generate stable molecular ions, is protonation and/or deprotonation.

Table 1 Ionization Techniques[1–4]

Ionization Technique	Acronym	Means of Ionization	Ref
electrospray ionization	ESI	evaporation of charged droplets	[1]
matrix-assisted laser desorption/ionization	MALDI	photon absorption/proton transfer	[2]
fast atom/ion bombardment	FAB	ion desorption/proton transfer	[3,4]

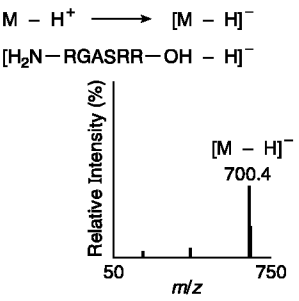
Protonation involves the addition of a proton to a molecule to produce a net positive charge of 1+ for every proton added (Figure 2). Positive charges tend to reside on the more basic residues of the molecule, such as amines, to form stable cations. A disadvantage of protonation is the covalent nature of the proton binding to the molecule which can initiate charge transfer and, in some cases, significant fragmentation.

Figure 2 The Mass Spectrum of a Peptide Obtained via Protonation



Deprotonation is the ejection of a proton from a molecule, resulting in a net negative charge of 1– for each proton ejected. This mechanism of ionization is very useful for acidic species, such as acidic peptides. With deprotonation, the net positive charge of 1– is achieved through the removal of a proton or even multiple negatively charged species (Figure 3).

Figure 3 The Mass Spectrum of a Peptide Obtained via Deprotonation



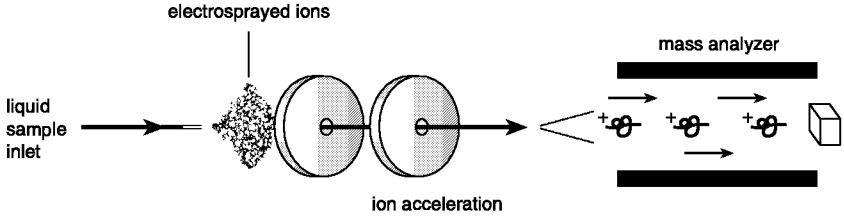
7.6.1.1 Electrospray Ionization

Electrospray ionization is a method used to produce gaseous ionized molecules from a liquid solution. This is done by creating a fine spray of highly charged droplets in the presence of a strong electric field. The sample solution is sprayed from a region of this field at the tip of a metal nozzle maintained at approximately 4000 V. The highly charged droplets are then electrostatically attracted to the mass spectrometer inlet where either dry gas, heat, or both are applied to the droplets before they enter the vacuum of the mass spectrometer, thus causing the solvent to evaporate from each droplet. As the droplet decreases in size, the electric field density on its surface increases. The mutual repulsion between like charges on this surface becomes so great that it exceeds the forces of surface tension, and ions begin to

for references see p 695

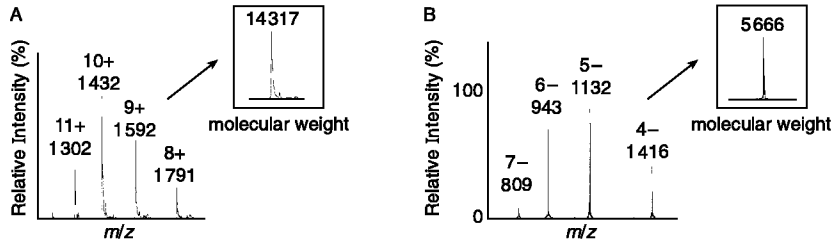
leave the droplet through what is known as a "Taylor cone". The ions are then directed into an orifice through electrostatic lenses leading to the mass analyzer (Figure 4). Since it is a continuous ionization method, ESI is suitable for using as an interface with modern separation techniques.

Figure 4 The Electrospray Ionization Source Generates Ions via Charged Droplet Evaporation Followed by Electrostatic Direction of the Ions into a Mass Analyzer



Electrospray ionization is conducive to the formation of singly-charged small molecules, but is also well-known for producing multiply-charged species of larger molecules. This is an important feature since the mass spectrometer measures the *m/z*, making it possible to observe very large molecules with an instrument having a relatively small mass range. Figure 5 illustrates how the ions from proteins and large peptides are observed through the multiple-charging phenomenon, where each of the peaks can be associated with different charge states of the molecular ion. Fortunately, electrospray mass spectrometers are equipped with computer programs that facilitate these molecular weight calculations.

Figure 5 Examples of Data Generated on an Electrospray Ionization Mass Spectrometer. (a) Proteins Typically Produce Positive, Multiply Charged Ions and (b) Oligonucleotides Generate Negative, Multiply Charged Ions. Inset are the Computer-Generated Molecular Weight Spectra

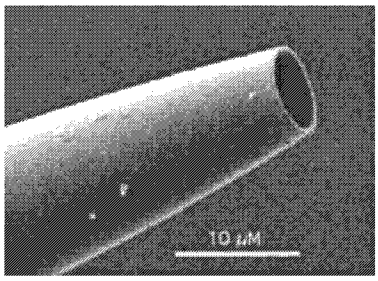


The extent of multiple charging that occurs in ESI is a unique characteristic of the technique that enables an analyte's mass to be determined with great precision, as masses can be independently calculated from several different charge states. The multiple charging in ESI also permits the measurement of high molecular weight analytes, using conventional mass analyzers that are normally limited to the detection of ions with relatively low *m/z* ratios (< *m/z* 3000). For example, a 50 kDa protein will typically retain on the order of 30 to 50 charges in ESI, yielding multiply charged species with *m/z* ratios between 1000 and 2000 that are easily detected with quadrupole mass analyzers. Another advantage of ESI-MS is its compatibility as an interface with liquid chromatography (LC).[4]

Nanoelectrospray ionization (nanoESI), also known as nanospray, nanoflow electrospray, and micro-electrospray, is a low flow/high sensitivity approach to ESI. NanoESI[5] is a slight variation on ESI such that the spray needle has been made very small and is positioned close to the entrance of the vacuum of the mass spectrometer and the mass analyzer (Figure 6). This greatly reduces required sample amounts allowing nanoliter flow rates and femtomole sample consumption. The end result is increased efficiency since the flow rates for

nanoESI sources are on the order of tens of nanoliters per minute. Another advantage of nanoESI is that the droplets formed are typically smaller than with normal ESI and thus nanoESI is more tolerant of salts and other impurities.

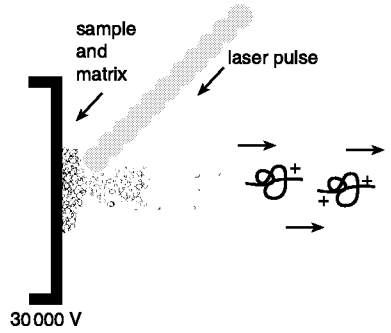
Figure 6 The Small Size Spray Needle for Nanoelectrospray



7.6.1.2 Matrix-Assisted Laser Desorption Ionization

Matrix-assisted laser desorption ionization mass spectrometry (MALDI-MS) permits the analysis of both low and high molecular weight compounds including carbohydrates,[6] lipids,[7] peptides,[8] proteins,[9] oligonucleotides,[10] as well as synthetic molecules and polymers[11] with high sensitivity. MALDI allows for the ionization and transfer of a sample from a condensed phase to the gas phase by first directing a pulsed laser beam to a sample suspended or dissolved in a matrix. This matrix plays a key role in the technique by absorbing the laser light energy and causing the matrix material to vaporize and the vaporized matrix carries some of the sample with it. Further, the matrix may play a role in the ionization of the analyte molecules once in the gas phase. Here, the charged molecules are directed by electrostatic lenses from the ionization source into the mass analyzer, while uncharged molecules will often react with the matrix or other molecules to produce charged species, transferred electrostatically into the mass analyzer. Once the molecules in the sample are vaporized, time-of-flight mass analysis is often used to separate the ions according to their mass-to-charge ratio (Figure 7).

Figure 7 Matrix-Assisted Laser Desorption/Ionization

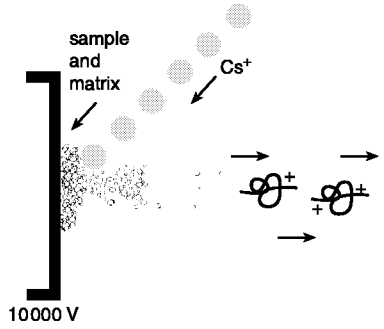


7.6.1.3 Fast Atom Bombardment

The fast atom bombardment ionization (FAB) technique is a soft ionization method, typically requiring the use of a direct insertion probe for sample introduction in which a high energy beam of Xe atoms, Cs^+ ions, or massive glycerol-NH_4^+ clusters sputter the sample and matrix from the probe surface (Figure 8).

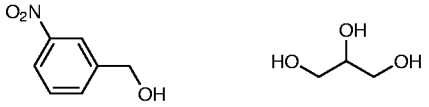
for references see p 695

Figure 8 Fast Atom Bombardment (FAB) Technique



The FAB matrix is essentially a nonvolatile liquid material, such as those illustrated in Scheme 1, that serves to constantly replenish the surface with new sample as the incident ion beam bombards the surface. The matrix also serves to minimize sample damage from the high-energy particle beam by absorbing most of the incident energy and is believed to facilitate the ionization process. The spectrum produced often includes matrix peaks along with some fragments and a peak for the protonated or cationized (i.e., $M+Na^+$) molecular ion.

Scheme 1 Two Commonly Used FAB Matrices



A general comparison of ionization sources is given in Table 2.

Table 2 General Comparison of Ionization Sources

Characteristics	Ionization Source			
	Electrospray Ionization (ESI)	NanoESI	MALDI	FAB
Typical Mass Range	70 000	70 000	300 000	8000
Matrix Interference	none	none	yes	yes
Degradation	none	none	possible photo-degradation & matrix reactions	matrix reactions & some thermal degradation
Ability to Analyze Complex Mixtures	somewhat limited	somewhat limited but better than ESI	excellent	Somewhat amenable
LC/MS Capability	excellent	OK, but low flow rates can present a problem	very limited	very limited
Sensitivity	high femtomole to low picomole	high zeptomole to low femtomole	low to high femtomole	nanomole
Salt Tolerance	low (low millimolar)	moderate (low–mid millimolar)	moderate (low millimolar)	high (to 0.01 M)
Other	multiple charging useful but significant suppression can occur with mixturessoft ionization = low fragmentation	multiple charging useful but significant suppression can occur with mixturessoft ionization = low fragmentation	matrix background can be problem for low mass ionssoft ionization = low fragmentation	solubility with matrix required soft ionization = low fragmentation

7.6.2 The Mass Analyzer

Once ions are formed the mass analyzer is primarily responsible for providing information on the mass of an ion. It accomplishes this by separating and measuring the ions according to their different mass-to-charge ratios. This concept is the most important thing to understand and remember about mass analyzers; they measure the *m/z* ratio, not the mass! This is often a point of confusion, especially in electrospray because if an ion is multiply charged, the *m/z* will be significantly less than the actual mass (e.g., a peptide ion, $C_{37}H_{68}N_{16}O_{14}^{2+}$, *m/z* 488.3, has a mass of 976.5 daltons). Since multiple charging is especially common with electrospray ionization that yields numerous peaks corresponding to the same species, one must keep in mind the importance of the *m/z* ratio.

The first mass analyzers, developed in the early 1900s, employed magnetic fields to separate ions according to their *m/z*. Modern analyzers, whose designs include variations on the early magnetic methods, have advanced to offer higher accuracy, sensitivity, mass range, and an ability to give structural information. Just as ionization techniques have evolved, mass analyzers have also changed to meet the demands for observing larger molecules with mass accuracy of the order of ±0.1 to 0.0001%. In addition, the sensitivity required (picomole to femtomole) for many biological applications has been obtained with electrospray and MALDI ionization sources, in part due to the high transmission efficiency of the analyzers.

The general characteristics of a mass analyzer are accuracy, resolving power, mass range, and tandem analysis capabilities.

Accuracy is the ability of the analyzer to provide accurate *m/z* information and is largely dependent on an instrument's stability. For example, an instrument with 0.01% accuracy can provide information on a 1000 Da peptide to ±0.1 Da or a 10 000 Da protein to ±1.0 Da. The accuracy will of course vary, sometimes dramatically, from analyzer to analyzer.

Resolving Power or Resolution is the ability of a mass spectrometer to distinguish between ions of different mass-to-charge ratios such that greater resolution corresponds directly to the increased ability to differentiate ions. For example, a mass spectrometer with a resolution of 500 can distinguish between ions of $m/z = 500$ and 501. The most common definition of resolution is given by the following equation:

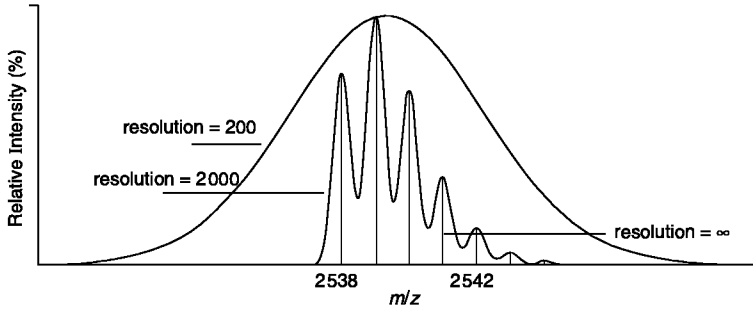
$$\text{Resolution} = M/\Delta M$$

where M is the *m/z* of an ion and ΔM is the full-width at half-maximum peak. For example, a peak with *m/z* 500 and a full-width at a half maximum of 1 has a resolution of $M/\Delta M = 500/1.0 = 500$.

The analyzer's resolving power does, to some extent, determine how much accuracy can be obtained with a particular instrument. The average mass of a molecule is calculated using the weighted average mass of all isotopes from each constituent element of the molecule. The monoisotopic mass is calculated by using the mass of the elemental isotope having the greatest abundance for each constituent element. Many instruments cannot resolve between the isotopes and will give only the average mass, yet sometimes the resolution is so low that it may be difficult to determine even the average mass with reasonable accuracy. Higher resolution allows separation of ions according to individual isotopes, or simply produces a very narrow peak allowing for greater accuracy. An example of how resolution affects the observed mass for a compound having a molecular formula $C_{101}H_{145}N_{34}O_{44}$ is shown in Figure 9.

for references see p 695

Figure 9 The Effect of Resolution upon Mass Accuracy. The Overlaid Spectra Were Calculated for the Same Molecular Formula $C_{101}H_{145}N_{34}O_{44}$, at Resolutions of 200, 2000, and Infinity[3][a]



[a] Notice that at higher resolution the accuracy is increased because the center of the peak can be more accurately identified.

Mass range is the *m/z* range of the mass analyzer. For example, quadrupole analyzers typically only scan up to *m/z* 3000, magnetic sector analyzers can go to *m/z* 10 000, and time-of-flight analyzers have virtually unlimited *m/z*.

Tandem analysis is the ability with which the analyzer can separate an ion from other ions of different *m/z*, perform collision-induced dissociation on that ion, and analyze the respective fragmentation ions. The process is called tandem analysis because subsequent analyses are performed either by a consecutive analyzer, such as a triple quadrupole mass analyzer or they are performed with the same analyzer, such as an ion trap which isolates the ion of interest, fragments it, and analyzes the fragment ions all within the same ion cell. Typically, tandem MS experiments are performed by generating the ion of interest and selecting it with the first analyzer. The ion is then collided with inert gas molecules such as argon or helium, and the fragments generated by the collision are analyzed. The information obtained can be used to sequence peptides, carbohydrates, small oligonucleotides, and lipids.

7.6.3 Ionization Method and Mass Platform Configurations

Having examined ionization sources and mass analyzers as separate components of a mass spectrometer in Sections 7.6.1–7.6.2, we can now turn to the most common ways these two components are interfaced.

ESI with Quadrupole and a Quadrupole Ion Trap

Quadrupole mass analyzers have shown new utility in their capacity to interface with electrospray ionization. This interface has three primary advantages. First, quadrupoles are tolerant of relatively high pressures ($\sim 5 \times 10^{-5}$ Torr), which makes them well suited to electrospray ionization since the ions are produced under atmospheric pressure conditions. Secondly, quadrupoles are now capable of routinely analyzing up to an *m/z* of 3000, which is useful because electrospray ionization of proteins and other biomolecules commonly produces a charge distribution in that range. Finally, the relatively low cost of quadrupole mass spectrometers makes them attractive as electrospray analyzers. Considering these mutually beneficial features of electrospray and quadrupoles, it is not surprising that most of the successful commercial electrospray instruments thus far have been coupled with quadrupole mass analyzers.

The quadrupole ion trap mass analyzer was developed at the same time as the quadrupole mass analyzer. The physics behind both of these analyzers are very similar. Ion traps also allow one to perform multiple collision-induced dissociation (CID) experiments (also known as tandem mass analysis events) which represent a valuable tool for obtaining sequence information. Quadrupole ion traps, primarily utilized with the electrospray ionization of biomolecules, typically consist of a ring electrode and two hyperbolic endcap electrodes. After applying radio frequency and direct current voltages, the motion of the ions trapped by the electric field allows them to be trapped or ejected from the ion trap. In the normal mode the radio frequency is scanned to higher voltages, and the trapped ions with the lowest *m/z* are ejected to a detector through small holes in the endcap. As the radio frequency is scanned further, higher *m/z* ratios are ejected and detected.

MALDI with Time-of-Flight

A time-of-flight (TOF) analyzer[12] is one of the simplest mass analyzing devices and is commonly used with MALDI ionization. Time-of-flight analysis is based on accelerating a set of ions to a detector with the same amount of energy. As the ions have the same energy, yet a different mass, they reach the detector at different times. The arrival time of an ion at the detector is dependent upon the mass, charge, and kinetic energy of the ion. Since kinetic energy (KE) is equal to $^1/_2 mv^2$, or velocity $v = (2KE/m)^{1/2}$, ions will travel a given distance, d, within a time, t, where t is dependent on their *m/z*. The process is analogous to a pitcher throwing a golf ball and a basketball at a catcher with the same amount of energy. The golf ball will reach the catcher faster because it has a smaller mass and therefore a greater velocity. In mass spectrometry, the smaller ions reach the detector first because of their greater velocity and the larger ions take longer, thus the analyzer is called time-of-flight. Due to the greatly improved resolving power that has been developed for TOF analyzers over the past decade, along with the advantage of having virtually no upper mass limitation, the MALDI-TOF interface now offers good accuracy, excellent mass range, as well as its inherent high sensitivity.

7.6.4 Peptide and Protein Analysis

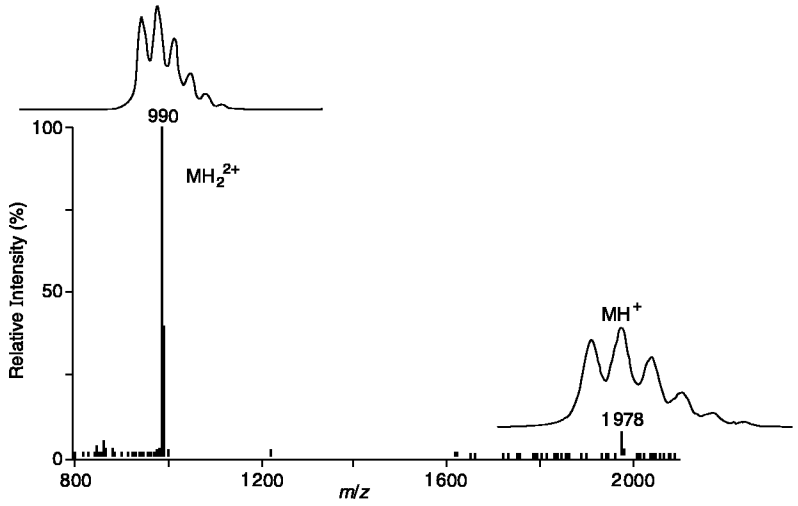
7.6.4.1 By Electrospray

The electrospray ionization of peptides and proteins involves the addition of a proton or multiple protons. Sample preparation is achieved by dissolving the sample in a protic volatile solvent system (H_2O, MeOH, and $MeCN/H_2O$) that is relatively homogeneous and less than 5 mM of salt. An important feature of most electrospray mass spectrometers is their ability to generate and observe multiply charged species. This multiple charging makes it possible to observe large proteins with mass analyzers that have a relatively small mass range (such as quadrupoles). In addition, observing multiple peaks for the same peptide allows one to extract multiple molecular weight calculations from a single spectrum. Thus, one can average these values and obtain a very accurate molecular weight.

The question often arises, especially concerning the analysis of peptides and proteins, how is the charge state of the observed ion determined? The answer is straightforward. The spectrum in Figure 10 below shows two peaks, one at *m/z* 1978 and another at *m/z* 990.

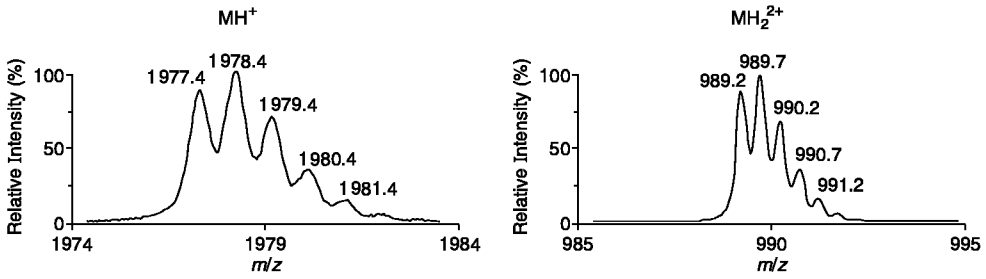
for references see p 695

Figure 10 Electrospray Mass Spectrum of a Peptide Having a Molecular Weight of 1977 Daltons[3]



How does the investigator know there are not two separate peptides? Since most electrospray spectrometers have good resolution, it is often possible to look at the isotopic distribution of singly and doubly charged ions. By looking at the isotopic distribution of the ions MH^+ and MH^{2+} (Figure 11) we can readily see that the isotopes are separated by one mass unit at *m/z* 1978 and $^1/_2$ mass unit at *m/z* 990. Each pattern therefore corresponds to the 1+ and 2+ charge states, respectively. This is due to the fact that the mass spectrometer measures the mass-to-charge ratio. Therefore, as the calculation demonstrates, for the 2+ charge state the isotopes are separated by $^1/_2$ mass unit, at 3+ the isotopes are separated by $^1/_3$ mass unit, and at 4+ the isotopes are separated by $^1/_4$ mass unit. Unfortunately, the resolving power of the common quadrupole electrospray mass spectrometers only allow for distinguishing between singly and doubly charged species.

Figure 11 The Isotopic Pattern of a Peptide with Singly and Doubly Charged Molecular Ions Obtained from an Electrospray Quadrupole Mass Spectrometer[3]



(molecular mass + no. of protons)/charge = mass-to-charge ratio

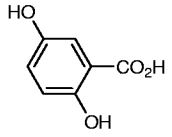
singly charged	doubly charged
(1976.4 + 1.0)/1 = 1977.4	(1976.4 + 2.0)/2 = 989.2
(1977.4 + 1.0)/1 = 1978.4	(1977.4 + 2.0)/2 = 989.7
(1978.4 + 1.0)/1 = 1979.4	(1978.4 + 2.0)/2 = 990.2
(1979.4 + 1.0)/1 = 1980.4	(1979.4 + 2.0)/2 = 990.7
(1980.4 + 1.0)/1 = 1981.4	(1980.4 + 2.0)/2 = 991.2

Fortunately, as larger peptides and proteins are analyzed a distribution of ions is obtained. Even though we cannot look at the individual charge states from the isotopic pattern (because the resolving power of the instruments are too low) we can deduce the charge state by looking at two consecutive peaks in the mass spectrum [m/z and $m/(z+1)$]. All commercial instruments now allow for these calculations to be carried out very easily.

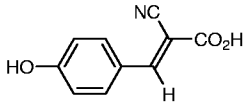
7.6.4.2 By Matrix-Assisted Laser Desorption

MALDI mass analysis of peptides and proteins is typically accomplished with a time-of-flight analyzer having resolving capabilities up to 2000. MALDI is a useful technique for these types of MS analysis for three reasons. The first is speed: the use of multi-sample probes (arranged on a single plate) allows one to prepare and run many samples simultaneously. Secondly, MALDI is very sensitive. Typical accuracies range from ±0.5 to 0.01% depending upon the presence of an internal standard, the type of instrument being used, and the selection of matrix material (Scheme 2). Lastly, MALDI is relatively tolerant of heterogeneous samples. Included here is the ability to measure peptides and other compounds in the presence of salt and the ability to analyze complex mixtures. MALDI can also be applied to the analysis of peptides generated from digests of a protein, a procedure used for peptide mapping.

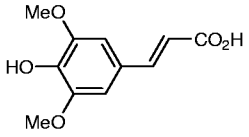
Scheme 2 Commonly Used MALDI Matrices



2,5-dihydroxybenzoic acid
peptides and small proteins



2-cyano-3-(4-hydroxyphenyl)acrylic acid
peptides and glycopeptides



3-(4-hydroxy-3,5-dimethoxyphenyl)acrylic acid
(sinapinic acid)
peptides and proteins

Many of the advantages that MALDI offers for peptide analysis are equally applicable to proteins. Protein analysis is similar to peptide analysis, in which ionization usually occurs through the addition of one, two, or three protons. However, since proteins are significantly bigger than peptides, ion detection is typically less efficient. Therefore, while peptides are measured at the femtomole or even attomole level with MALDI, proteins are usually measured at the high femtomole to low picomole level.

Sample Preparation and Matrix Recommendations for MALDI; Typical Procedure:
Pipet 0.5 μL of sample to the sample plate. Preferable concentration ~ 1pmol·μL^{-1}. Mix the sample and matrix[a] by drawing in and out of the pipette. Allow to air dry and acquire spectra.
[a]For peptides, small proteins, and most other compounds: a sat. soln of 2-cyano-3-(4-hydroxyphenyl)acrylic acid in 0.1% TFA in H_2O/MeCN (1:1) is used. This matrix requires about half the laser energy of other matrices.
[a]For proteins and other large molecules: sinapinic acid is substituted for the matrix in the above procedure and sample concentration is increased if necessary.
[a]For less polar compounds or small molecules: a 2,5-dihydroxybenzoic acid sat. soln in H_2O/MeCN (1:1) is used. 2,5-Dihydroxybenzoic acid works well as a matrix because it produces only a minimal amount of interference in the low molecular weight range.
[a]For oligonucleotides: 2,3-dihydropyridine-2-carboxylic acid or 2,4,6-trihydroxyacetophenone matrix in H_2O/MeCN (1:1) is used. These must be combined in equal parts with about 30 mM ammonium citrate to work. Sample concentrations may need to be higher than those used for peptides and laser energy may have to be increased.

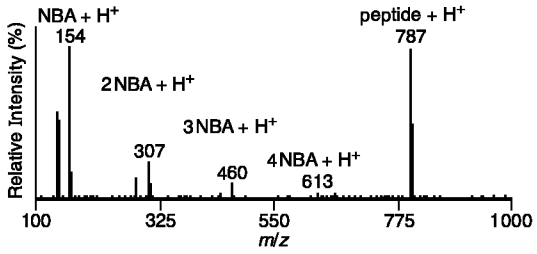
for references see p 695

7.6.4.3 By Fast Atom Bombardment

The application of FAB to peptide analysis has two general advantages: it typically generates accurate molecular weight information due to the mass analyzers to which it is interfaced, and it offers rapid analysis. Given that FAB is not commonly applied to the analysis of proteins due to its limited mass range, routinely of the order of 5000 to 8000 Da, and relatively poor sensitivity (typically requiring >500 pmol), we will only briefly focus on the sample preparation and analysis procedures for peptides.

FAB peptide preparation is accomplished by dissolving a relatively pure (>70%) peptide sample into a 3-nitrobenzyl alcohol (NBA) matrix. There are many matrices used in FAB analyses, however, 3-nitrobenzyl alcohol has become a standard for FAB peptide analysis (Figure 12). The peaks associated with the NBA matrix can make the spectra more cumbersome to interpret. However, to the experienced mass spectroscopist the matrix ions can act as reference ions and are a useful indicator of accuracy. Rarely are the NBA matrix ions observed above *m/z* 1000.

Figure 12 The FAB Mass Spectrum of a Peptide Analyzed in the NBA Matrix



The sample can be prepared in a cosolvent, such as 5–50% acetic acid in water, acetonitrile/water, chloroform, methanol, and trifluoroacetic acid (0.1 to 25%), at a concentration of approximately 1 to 5 mg·mL^{-1}. Sample solubility in this cosolvent is crucial because it acts as a medium between the sample and the matrix. The peptide solution (1–4 μL) is dissolved directly into the matrix on the probe tip (a flat round surface approximately 2–5 mm in diameter). When large amounts of sample (hundreds of micrograms) are available, the peptide can be added directly into the matrix, followed by the addition of a cosolvent such as trifluoroacetic acid (TFA) which, in addition to sample matrix solubility, can also facilitate protonation.

FAB – Sample Preparation; Typical Procedure:
Place 2–5 μL of NBA directly on the probe. Dissolve sample (~5 μg·μL^{-1}) in 0.1% TFA in H_2O/MeCN (1:1). Add 1–4 μL of sample soln to the matrix. Wait ca. 1 min for solvent to evaporate. Acquire spectra.

7.6.5 Sequencing of Peptides and Proteins

Mass spectrometry has developed a symbiotic relationship with other technologies, such as the widely used Edman degradation techniques, in determining protein structure. Having played a central role in the discovery of unique protein modifications, mass determination is also becoming more important in obtaining total sequence information. This section discusses the different methods for obtaining sequence information on peptides and proteins, including tandem mass spectrometry and protein ladder sequencing.

Sequencing with MS/MS

In general, mass spectrometry yields structural information on peptides and proteins by collision-induced dissociation, in which precursor (or parent) ions are subjected to collisions and the *m/z* of the resulting ions is measured. In order to obtain sequence information, fragments of an ion must be produced that reflect structural features of the original compound. Fortunately, most peptides are linear molecules, which allows for relatively straightforward interpretation of the fragmentation data. The process is initiated by converting some of the kinetic energy from the peptide ion into vibrational energy. This is achieved by introducing the selected ion, usually an $(M+H)^+$ or $(M+nH)^{n+}$ ion, into a collision cell where it collides with neutral Ar, Xe, or He molecules. The resulting fragment ions are then mass analyzed. More specifically, this application of tandem mass spectrometry allows for a heterogeneous solution of peptides to be analyzed, and then by filtering the ion of interest into the collision cell, structural information can be derived on each peptide from a complex mixture.

Sequence information can be obtained for peptides with molecular weights up to 2500 Da. Collision-induced dissociation of larger peptides reveal at least partial sequence information that will often suffice to solve a particular problem. The collision-induced dissociation method has been particularly useful on peptides from proteolytic digests, from which MS/MS data on different peptides can help identify the structure of a digested protein.

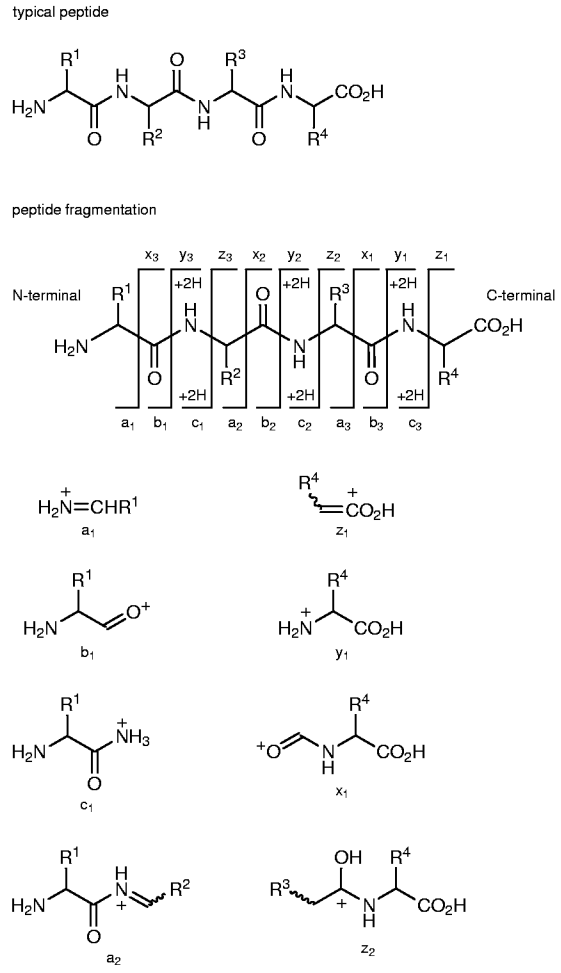
Collision-induced dissociation can be accomplished with a variety of instruments, including tandem double-focusing magnetic sectors, triple quadrupoles, quadrupole ion traps, and Fourier transform-ion cyclotron resonance. The quadrupole ion trap combined with electrospray is currently the most common means of generating structural data, being capable of high sensitivity, producing a reasonable amount of fragmentation information, and being relatively inexpensive.

From the discussion thus far, it is apparent that fragmentation is a vital step in sequencing peptides. The fragment ions produced in this process can be separated into two classes. One class retains the charge on the N-terminal while cleavage is observed from the C-terminal (Figure 13). This fragmentation can occur at three different positions, each of which is sequence-designated as types a_n, b_n, and c_n. The second class of fragment ions generated from the N-terminal retains the charge on the C-terminal, while cleavage is observed from the N-terminal. Like the first class, this fragmentation can occur at three different positions, types x_n, y_n, and z_n.

Certain limitations for complete sequence information with tandem mass analysis do exist. In determining the amino acid sequence of a peptide, a complete ion series could be used except that neither leucine and isoleucine, nor lysine and glutamine can be differentiated because they have the same mass. As a complete ion series is not usually observed, the information from both the N- and C-terminal ions must be used to help determine the entire sequence.

for references see p 695

Figure 13 Peptide Fragmentation Results in the Ions Shown. Collision-Induced Dissociation Spectra Often Result in the Dominant Fragmentation at the Amide Bonds in the Polyamide Backbone, Producing Ions of the Type b or y



It may seem difficult to identify a particular ion series in a spectrum, however there are several rules that can be used to determine the identity of peaks belonging to any ion of the backbone fragmentation (a_n, b_n, c_n, x_n, y_n, and z_n). First, backbone fragmentation can differ in *m/z* by only one of the 18 mass values associated with the amino acids and secondly, fragments a_n, b_n, and y_n generally form a continuous series.

Protein Ladder Sequencing

Protein ladder sequencing is a MALDI-based sequencing approach that consists of a two-step process. The first step is ladder-generating chemistry or enzymology; the controlled generation of sequence-defining peptide fragments, with each fragment differing from the next by one amino acid. The second step is MALDI mass analysis of the resulting protein-sequencing ladder. Each amino acid is identified from the mass difference between successive peaks, and the position in the data set defines the sequence of the original peptide chain. Protein ladder sequencing is currently being developed as a routine alternative to Edman sequencing and has been most useful for C-terminal sequencing when used in conjunction with carboxypeptidase enzymes.

7.6.6 Liquid Chromatography-Mass Spectrometry

Throughout the 1990s, tremendous progress has been made in the analysis of peptides and proteins by liquid chromatography-mass spectrometry (LC-MS). This two-dimensional analytical technique effectively couples the separation performance of liquid chromatography with the state-of-the-art detection, selectivity, and sensitivity of MS. The increased application of this technique for peptide and protein analysis is mostly a result of improvements in column chemistry, rendering a variety of commercially available novel columns and of technological advances achieved in interfacing the two analytical techniques, namely ESI and atmospheric pressure chemical ionization.[13,14] The utilization of ESI and atmospheric pressure chemical ionization as ionization sources has made the analysis of peptides and proteins routine at the low picomole levels with analytical columns, and sub picomole levels via small-bore columns. Mass spectrometry can serve as a general detector at full scan mode, a selective detector by monitoring single ions, and as a means to elucidate structure when utilizing tandem analysis. As a result of the inherent multicharging accompanied by atmospheric pressure ionization-mass spectrometry, the *m/z* charge ratios of large macromolecules such as proteins and proteins are well within the instrumental limits of commercially available quadrupoles platforms.

Chromatographic Conditions

The routine analysis of peptides and proteins using HPLC has been aided by advances in stationary phases particularly suited for the work. Most applications today are carried out via reversed-phase columns [e.g., octasilylated (C8) and octadecylsilylated (C18) materials] that have been deactivated (endcapped) to improve peak shape and overall sample throughput. The popularity of reversed-phase chromatography is primarily a result of its chromatographic characteristics, such as column efficiency and short analysis times, and compatible mobile phases consisting of volatile organic solvents specifically suited to currently available ionization interfaces. Other media typically used for routine analysis include size exclusion, ion exchange, and affinity column phases.

The mobile phase typically consists of an organic solvent such as methanol or acetonitrile diluted with aqueous buffer, normally applied via gradient-pump programming. Buffers are added to aid the chromatography, particularly as a means to increase solute retention and to improve peak shape. The use of volatile buffers that are compatible with LC-MS are also desirable as a means for enhancing atmospheric pressure chemical ionization. They include acetic acid, trifluoroacetic acid, formic acid, ammonium hydroxide, ammonium acetate, and triethylamine. The nonvolatile portion of the buffer must also be ionizable in the mode used (e.g., $H_2PO_4^-$ in the negative mode). Buffer concentrations should be optimized for chromatographic characteristics and detector sensitivity, and should be lower than 10 mM for ESI and 100 mM for atmospheric pressure chemical ionization. The gradient is typically programmed over the range of 0–90% organic phase, with the run time proportional to number of constituents in the sample matrix.

7.6.6.1 Liquid Chromatography-Electrospray Ionization-Mass Spectrometry

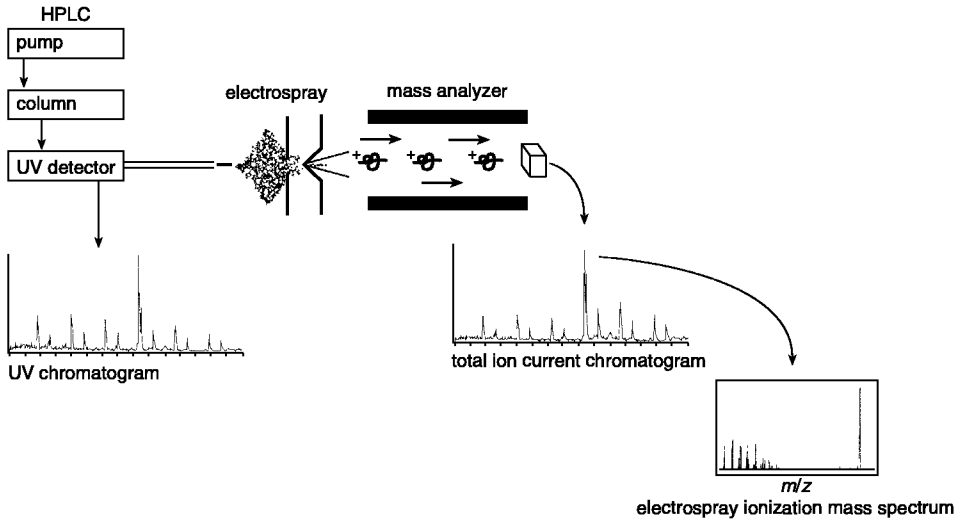
In order for mass spectrometry to be used as an effective tool in LC, the analyte must be ionized in the interface region (source) since MS can only detect charged species. Consequently, the role of the ionization source is twofold: (1) it is responsible for interfacing the chromatographic instrument with the detector and (2) it provides an effective and efficient means for sample ionization. It should be pointed out that source ionization efficiency plays a significant role in sample sensitivity for LC-MS analysis. The application of LC is ideally

for references see p 695

suited for the separation of nonvolatile and thermally labile compounds such as peptides and proteins. As a result, the ionization method should be efficient for the same compounds. Desorption ionization methods MALDI and FAB, while useful for peptides and proteins, are not suitable as a LC interface. However, since ESI and atmospheric pressure chemical ionization generate ions directly from a solution phase such as acetonitrile/water, they are the most successful LC interfaces used today.

Since the early 1990s, electrospray has been utilized routinely as a means for sample introduction in the field of mass spectrometry. An advantage of ESI is that ions are formed directly from solution which is usually an aqueous or aqueous/organic solvent system, a feature that has established the technique as a convenient mass detector for LC. While in the past, attempts to couple liquid chromatography with mass spectrometry resulted in limited success, ESI has made LC-MS routine. ESI also allows for MS analysis at relatively high LC flow rates ($1.0\,mL \cdot min^{-1}$) and high mass accuracy (±0.01%), adding a new dimension to the capabilities of LC characterization. The use of ESI as a detector for LC was one of its first obvious applications. Figure 14 illustrates how mass spectrometry is interfaced to liquid chromatography by ESI. Most instrument configurations utilize narrow-bore (ca. 0.005 inches) PEEK tubing to transfer HPLC effluent to the MS. For chromatographic methods that require high flow rates, such as rapid analysis of random libraries of peptides, a flow splitter is positioned between the LC and MS to divert a portion of the effluent directly to waste. This ESI ionization source is specifically tailored to molecules that are capable of supporting multiple charges, such as peptides and proteins. In addition, samples also compatible with the source include, but are not limited to, ions such as quaternary amines and sulfate conjugates, and heterocyclic compounds such as benzodiazepines. Nonpolar analytes are not very compatible with ESI and are more likely to ionize in atmospheric pressure chemical ionization sources.

Figure 14 Interfacing Liquid Chromatography with Electrospray Ionization Mass Spectrometry



Sample Preparation and Procedures for LC-MS; Typical Procedure:
Configure the LC instrument with a 5-μL injection loop (or autosampler); a precolumn (for endogenous samples); a column (ca. 2.1×30 mm, 3.5 μm particle size C18 column); two solvent reservoirs, Reservoir A: 0.05% TFA in H_2O and Reservoir B: 95% MeCN and 5% a soln of 0.05% TFA in H_2O; a mobile phase gradient (e.g., 5% A for 3 min, increase to 90% A over 8 min, hold at 90% A for 3 min, at $0.5\,mL \cdot min^{-1}$ flow rate); and an MS inlet flow appropriate for the mass spectrometer. Dissolve sample (preferably 1 $pmol \cdot \mu L^{-1}$) in 20% MeCN with 0.05% TFA. Inject 10 μL of sample into LC-MS and aquire the spectra.

7.6.6.2 Liquid Chromatography-Atmospheric Pressure-Chemical Ionization-Mass Spectrometry

The application of atmospheric pressure-chemical ionization has gained wide acceptance in bioanalysis. Similar to electrospray, the liquid effluent of atmospheric pressure chemical ionization is introduced directly into the ionization source. However, the similarity stops there; the atmospheric pressure chemical ionization source contains a heated vaporizer which facilitates rapid desolvation/vaporization of the droplets. Vaporized sample molecules are carried through an ion-molecule reaction region at atmospheric pressure. The ionization occurs through a corona discharge creating reagent ions from the solvent vapor. Chemical ionization of sample molecules is very efficient at atmospheric pressure due to the high collision frequency. Proton transfer (for protonation MH^+ reactions) occur in the positive mode, and either electron transfer or proton transfer (proton loss, $[M-H]^-$ in the negative mode). The moderating influence of the solvent clusters on the reagent ions, and of the high gas pressure, reduces fragmentation during ionization and results in primarily molecular ions. This ionization technique is more useful for a wider variety of samples (polar to nonpolar) than the ESI method.

7.6.6.3 Liquid Chromatography-Tandem Mass Spectrometry

Since LC-MS yields only one kind of ion (normally $[M+H]^+$), experiments with LC-MS/MS in the ion source must be carried out if increased detector specificity or structural information is needed. Furthermore, it is always possible in LC-MS to observe co-eluting compounds superimposed in the mass spectrum, making quantification and interpretation difficult. The technique of LC-MS/MS therefore offers distinct advantages over LC-MS by improving detection selectivity and sensitivity and by generating and acquiring ion fragmentation information to be used toward structural identification. For improved detector selectivity and sensitivity for peptides and proteins, typical protocols call for monitoring the parent ion (e.g., $[MH]^+$) signal simultaneously with a known ion fragment signal. This procedure significantly reduces the possibility of observing interference peaks present in complex mixtures.

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7.7 Spectroscopic Methods

7.7.1 Fluorescence Spectroscopy

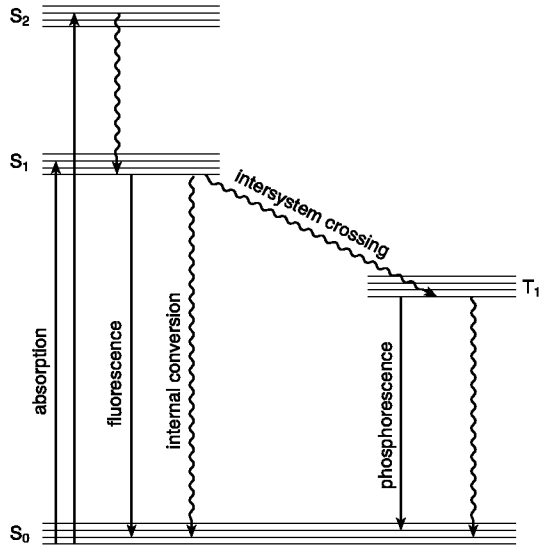
P. W. SCHILLER

Structural and rotational parameters of fluorophore-containing peptides and proteins have been determined by steady-state fluorescence spectroscopy for half a century. Steady-state fluorescence experiments are based on continuous excitation of the fluorescent sample. Great progress has been made in the development of methods permitting the measurement of time-resolved fluorescence. Both time domain (time-correlated single-photon counting) and frequency domain or phase modulation methods are now widely used to measure fluorescence lifetimes. Steady-state and time-resolved fluorescence decay measurements both provide information on three types of structural/dynamic features of the fluorophore-containing peptide: (i) determination of the fluorescence emission maximum and quantum yield of the fluorophore or measurement of its fluorescence lifetime permit the characterization of the fluorophore's immediate environment(s); (ii) fluorescence anisotropy measurements determine the rotational characteristics of the fluorophore and of the peptide to which it is attached; and (iii) the intramolecular distance between a donor fluorophore and an acceptor contained in a peptide can be estimated by a singlet–singlet resonance energy transfer experiment.

Fluorescence spectroscopy provides less structural detail than X-ray crystallography or NMR spectroscopy. However, fluorescence spectroscopic techniques also offer some unique advantages over other methods used in peptide and protein structure determination, including (i) high temporal resolution, permitting the monitoring of molecular motions taking place on the nanosecond and subnanosecond time scale; (ii) high sensitivity, allowing measurements at concentrations of 10^{-5} M or even lower and thus reducing the possibility of peptide self-aggregation often observed at concentrations greater than 10^{-4} M used in NMR experiments; and (iii) feasibility of fluorescence measurements in the presence of complex biological systems by use of appropriate fluorescent probes. A number of reviews and textbooks describing the scope and limitations of fluorescence spectroscopy in the study of peptide and protein conformations and dynamics have been published.[1–4] Another quite comprehensive review dealt specifically with the application of fluorescence techniques in studies of peptide conformations and interactions.[5]

The excited-state processes that give rise to the phenomena of fluorescence and phosphorescence are illustrated in Figure 1.

Figure 1 Energy Level Diagram of a luminophore[5][a,b]



[a] Processes involving emission or absorption of a photon are denoted by straight arrows; nonradiative processes are denoted by wavy arrows.

[b] Reprinted from (Schiller, *The Peptides: Analysis, Synthesis, Biology, Conformation in Biology and Drug Design*, Udenfriend; Meienhofer; Hruby, Eds.), Copyright (1985), p 115, with permission from Academic Press.

The luminophore may be excited either to an upper singlet state (e.g., S_2) or to a vibronic level of the lowest excited singlet state S_1 and then in either case rapidly returns to the lowest vibronic level of S_1 by nonradiative processes. From the latter level it may go to any vibronic level of the ground state S_0, either by fluorescence (photon emission) or by internal conversion (radiationless process), or via intersystem-crossing to the first excited triplet state T_1. From there the molecule may return to S_0 by phosphorescence (a radiative process) or by nonradiative processes. Because of the short lifetime of S_1 (typically a few nanoseconds), fluorescence can be observed in fluid solution at room temperature. In contrast, the lifetime of T_1 (milliseconds to seconds) is much longer and, therefore, it is susceptible to quenching processes. For this reason phosphorescence is usually only observed in rigid media (e.g., DMSO–ethyleneglycol mixtures at 77 K) and not in low viscosity solutions at room temperature. The fluorescence transition from the lowest vibronic level of S_1 to any vibronic level of S_0 gives rise to a spectral distribution of the fluorescence intensity that is characterized by an emission maximum. Upon excitation at a given wavelength (usually the wavelength corresponding to the maximum of the longest-wavelength absorption band) the fluorescence emission spectrum is obtained by recording the fluorescence intensity as a function of wavelength. Continuous variation of the excitation wavelength and concomitant recording of the fluorescence intensity at a given emission wavelength (usually the wavelength corresponding to the emission maximum) gives rise to the fluorescence excitation spectrum. Since in general the fluorescence intensity is proportional to the extinction coefficient, the corrected excitation spectrum is almost always identical with the absorption spectrum corresponding to the S_0–S_1 transition. More detailed descriptions of basic concepts of fluorescence spectroscopy can be found in a review[6] and in a textbook.[7]

for references see p 713

7.7.1.1 Fluorophores in Peptides

The determination of fluorescence parameters of peptides requires the presence of either natural fluorescent amino acid residues (intrinsic fluorescence) or of extrinsic fluorescent probes covalently attached to the peptide at appropriate sites. The use of extrinsic fluorescent probes is mandatory in cases where the conformational or rotational behavior of a peptide is examined in the presence of proteins that contain intrinsic fluorescent amino acids.

Tryptophan, tyrosine, and phenylalanine are the three natural amino acids that give rise to the intrinsic fluorescence of peptides in the ultraviolet region. Reliable, corrected fluorescence excitation and emission spectra of these aromatic amino acids were first published by Teale and Weber.[8] The fluorescence emission maxima of tryptophan, tyrosine, and phenylalanine in water are at 348, 303, and 282 nm, respectively. The photophysics and photochemistry of tryptophan and tyrosine have been comprehensively reviewed.[9,10]

In most cases peptides to be studied should not contain more than one of the fluorescent amino acids because the presence of multiple fluorophores often may not permit a clear-cut interpretation of the fluorescence parameters to be determined. This is indeed the case with several of the biologically active peptides that occur in nature. If several fluorescent amino acid residues are present in a peptide, it may be possible to replace one or several of them with a structurally similar non-fluorescent amino acid in order to arrive at a peptide analogue that contains only a single fluorophore. This will usually require the chemical synthesis of the desired analogue. Importantly, the analogue should display a biological activity profile similar to that of the native parent peptide. In cases where naturally occurring peptides do not contain a fluorescent amino acid residue, it may be feasible to incorporate one through substitution or addition at one of the termini by chemical synthesis of an analogue. Again, it is imperative that these structural modifications are not detrimental to biological activity.

Extrinsic fluorescent probes can be introduced into a peptide in a number of different ways. A relatively recent development has been the incorporation of artificial fluorescent amino acids into peptides. Such amino acids include 7-azatryptophan,[11] 3-(1-naphthyl)alanine, 3-(2-naphthyl)alanine, β-(9-anthryl)alanine, pyrenylalanine, etc. Obviously, this approach again requires the total chemical synthesis of a peptide analogue containing the artificial fluorescent amino acid at the desired position of the amino acid sequence. The second approach for introducing an extrinsic fluorescent probe into a peptide or protein is based on covalent labeling with a fluorescent reagent. Such reagents include sulfonyl chlorides, isothiocyanates, *N*-carboxyanhydrides, spirofurans (e.g., fluorescamine), iodoacetamides, maleimides, mercurials, or aziridines, which permit attachment of fluorescent probes to amino or thiol groups in peptides or proteins. A comprehensive handbook on fluorescent probes has been published by Haugland.[12] These fluorescent probes need to be attached at a specific site of the peptide such that biological activity is retained. Obviously, existing structure-activity relationships have to be carefully considered in the design of fluorescent peptide derivatives. Again, specific incorporation of a fluorescent probe often can only be achieved by total synthesis of the desired peptide derivative.

7.7.1.2 Fluorescence Quantum Yields and Lifetimes

An important parameter determined in steady-state fluorescence experiments is the fluorescence quantum yield φ, which is defined by eq 1

$$\varphi = k_e / \left(k_e + \sum_i k_i \right) \quad \text{eq 1}$$

where k_e represents the rate of fluorescence and $\Sigma_i k_i$ is the sum of all nonradiative processes depopulating the first excited singlet state S_1, such as internal conversion, solvent quenching, and intersystem crossing.

Methods for the measurement of absolute quantum yields have been reviewed.[6] In most steady-state fluorescence experiments relative fluorescence quantum yields are determined by comparison with appropriate standards (see Section 7.7.1.2.1).

The principal parameter determined in fluorescence decay experiments is the fluorescence lifetime τ which is defined as the reciprocal of the sum of the rates of all processes deactivating S_1 and is defined by the eq 2.

$$\tau = \left(k_e + \sum_i k_i\right)^{-1} \quad \text{eq 2}$$

The natural fluorescence lifetime, τ_o, is defined as the fluorescence lifetime that would be observed in the absence of any nonradiative quenching processes ($\tau_o = k_e^{-1}$).

It then follows that quantum yield and fluorescence lifetime are related by the eq 3.

$$\varphi = k_e\tau \quad \text{eq 3}$$

In the simplest case the decay of the fluorescence intensity F after excitation is described by eq 4

$$F(t) = F_o e^{-t/\tau} \quad \text{eq 4}$$

where F_o is the fluorescence intensity at the instance of excitation. The fluorescence lifetime τ can be measured either by time domain or frequency domain (phase modulation) methods.

There have been major advances in the development of both methods in recent years and fluorescence lifetime measurements on the picosecond time scale are now quite feasible. Precise analysis of the fluorescence decay revealed that even in the case of a peptide containing a single fluorophore multiexponential decay rather than single-exponential decay is often observed, as described by eq 5

$$F(t) = \sum_i \alpha_i e^{-t/\tau_i} \quad \text{eq 5}$$

where α_i is the relative amplitude factor ($\Sigma\alpha_i = 1$) and τ_i is the decay constant of the ith component. Basic principles of fluorescence decay measurements have been reviewed.[13,14]

The fluorescence emission maximum, quantum yield, and lifetime of a fluorophore are very sensitive to its immediate environment. A blue shift in the emission maximum and an increase in the fluorescence quantum yield or lifetime is generally observed when a fluorophore is transferred form a polar solvent to a nonpolar one or when it binds to a hydrophobic protein site. Furthermore, fluorescence quenching or enhancement may result from interactions of the fluorophore with various structural elements in its vicinity.

7.7.1.2.1 Determination of Fluorescence Quantum Yields and Emission Maxima

Fluorescence quantum yields of 0.13, 0.14, and 0.024 have been determined[15] for tryptophan, tyrosine, and phenylalanine in water, respectively, and these values have found general acceptance. Dissociation of the phenolic group of tyrosine at high pH results in strong fluorescence quenching.[16] The fluorescence quantum yield of a Trp, Tyr, or Phe residue contained in a peptide (φ_{PP}) can be determined by comparison with the corresponding amino acid in H_2O as standard on the basis of eq 6

$$\varphi_{PP} = \varphi_{AA}\,(I_{PP}A_{AA}/I_{AA}A_{PP}) \quad \text{eq 6}$$

for references see p 713

where I_{PP} and I_{AA} are the fluorescence intensities and A_{PP} and A_{AA} are the absorbances of the peptide fluorophore and standard amino acid, respectively.

Analogously, the fluorescence quantum yield of an extrinsic fluorescent probe contained in a peptide can be measured by comparison with an appropriate standard. If the fluorescent peptide exists in a conformational equilibrium, the fluorophore may be located in a number of different environments and may have a distinct quantum yield (φ_i) in each environment. In this case the determined fluorescence quantum yield represents a population-weighted average of the individual φ_i, as described by eq 7

$$\varphi = \sum_i f_i \varphi_i \quad \text{eq 7}$$

where f_i is the fraction of fluorophores in a distinct environment.

7.7.1.2.1.1 Studies with Model Compounds

Studies with tryptophan and tyrosine derivatives and with tryptophan- or tyrosine-containing dipeptides or piperazine-2,5-diones (diketopiperazines, DKP) revealed structural moieties that affect the fluorescence quantum yield of these aromatic amino acids (for a review, see ref[5]).

Tryptophan fluorescence is quenched by the following structural elements present in peptides:

(i) Peptide bond, the amide carbonyl group.
(ii) Protonated form of the N-terminal amino group.
(iii) Neutral carboxy group.
(iv) Histidinium moiety.
(v) Disulfide bridge and thiol group.

The tryptophan fluorescence emission maximum of nearly all model compounds in H_2O was located at 350 nm, indicating that the indole ring was completely exposed to the aqueous environment.

Tyrosine fluorescence quenchers are the following:

(i) Peptide bond (interaction between carbonyl group and phenyl ring).
(ii) Protonated α-amino group and unprotonated side-chain amino groups.
(iii) Protonated α-CO_2H group and deprotonated side-chain carboxylate groups.
(iv) Vicinal disulfide links.

In contrast to the variability of the fluorescence emission maximum observed with tryptophan, the tyrosine fluorescence emission maximum is always located at 303 nm and is not environment dependent.

Because of the low extinction coefficient and low fluorescence quantum yield of phenylalanine, phenylalanine fluorescence is not easily observed in tryptophan- or tyrosine-containing peptides and only very few fluorescence studies on Phe-containing peptides have been carried out.

7.7.1.2.1.2 Studies with Fluorophore-Containing, Biologically Active Peptides

The fluorescence quantum yields (φ^{Trp}) of selected, biologically active peptides and peptide analogues containing a single tryptophan residue are presented in Table 1.

Table 1 Trp Fluorescence Quantum Yields (φ^{Trp}) of Biologically Active Peptides[a]

Peptide or Peptide Analogue	φ^{Trp}	Ref
ACTH	0.10	[17]
Glucagon	0.12	[18]
c[-S-$(CH_2)_2$-CO-Lys-Asn-Tyr-Phe-Trp-Lys-Thr-Phe-Thr-Ser-NH-$(CH_2)_2$-S-]	0.023	[19]
c[-S-$(CH_2)_2$-CO-Lys-Asn-Tyr-Phe-D-Trp-Lys-Thr-Phe-Thr-Ser-NH-$(CH_2)_2$-S-]	0.071	[19]
[Trp^4,Met^5]enkephalin (Tyr-Gly-Gly-Trp-Met)	0.058	[21]
[D-Trp^4,Met^5]enkephalin (Tyr-Gly-Gly-D-Trp-Met)	0.088	[21]
[D-Cys^2,Trp^4,D-Cys^5]enkephalinamide (Tyr-D-Cys-Gly-Trp-D-Cys-NH_2)	0.036	[22]
[Trp^8]oxytocin (Cys-Tyr-Ile-Gln-Asn-Cys-Pro-Trp-Gly-NH_2)	0.038	[23]

[a] Determined in H_2O.

The listed φ^{Trp} values range from 0.12 to 0.023, indicating that they very much depend on the immediate environment of the indole moiety in the peptide. The tryptophan fluorescence emission maxima of all peptides are located in the narrow range of 348–350 nm, demonstrating that their indole moieties are located in a predominantly aqueous environment. Interestingly, inversion of the configuration of the Trp residue in position 8 of the somatostatin fragment *c*[-S-$(CH_2)_2$-CO-Lys-Asn-Tyr-Phe-Trp^8-Lys-Thr-Phe-Thr-Ser-NH-$(CH_2)_2$-S-] from L to D produced a threefold increase in the tryptophan fluorescence quantum yield (Table 1) accompanied by a 1.5-nm blue shift in the fluorescence emission maximum from 350 to 348.5 nm.[19] This result indicates that, in comparison with the L-isomer, the indole ring in the D-isomer is located in a slightly more hydrophobic environment due to interaction with the side chain of Lys^9. This interaction had also been indicated by an upfield-shift of the γ-methylene protons of Lys^9 that had been detected in the NMR spectra of D-Trp^8 analogues of somatostatin as compared to those of L-Trp^8 analogues.[20] Similarly, the D-Trp^4-analogue of enkephalin showed a 60% increase in φ^{Trp} and a 2-nm blue shift in comparison with the corresponding L-Trp^4-analogue,[21] presumably due to an interaction of the indole ring with the Met^5 side chain occurring in the D-Trp^4-analogue but not in the L-Trp^4-analogue. The low tryptophan fluorescence quantum yields observed with the potent cyclic enkephalin analogue H-Tyr-*c*[-D-Cys-Gly-Trp-D-Cys]-NH_2[22] and with [Trp^8]oxytocin[23] are caused by the quenching effect of the disulfide bridge which is in proximity of the indole ring in these compounds. The low tyrosine fluorescence quantum yield ($\varphi^{Tyr} = 0.025$) determined for native oxytocin[23] is also indicative of an interaction of the Tyr^2 residue in this peptide hormone with the adjacent disulfide moiety which results in fluorescence quenching. Evidence for this interaction has also been obtained from a circular dichroism study.[24] On the other hand, the potent analogue [Asu^{1-6}]oxytocin (Asu = 2-aminosuberic acid), containing an ethylene group in place of the disulfide moiety present in the native hormone, showed a much higher φ^{Tyr} (0.066).[23] In this case, the interaction of the Tyr^2 side chain with the bridging element (-CH_2-CH_2-) connecting amino acid residues 1 and 6 produces a more hydrophobic environment for the phenol ring and, thereby, a fluorescence enhancement. The very low φ^{Tyr} (0.009) observed with the cyclic enkephalin analog H-Tyr-*c*[-D-Cys-Gly-Phe-D-Cys-]-NH_2 also reflects a close interaction between the phenol ring of Tyr^1 and the disulfide bridge.[22]

Fluorescence quantum yield and emission maximum determinations as a function of peptide concentration may also permit the detection of peptide self-aggregation at concentrations below 10^{-4} M, because the peptide fluorophore is likely to be located in a different environment in the peptide aggregate. For example, the concentration-dependent changes in the tryptophan fluorescence emission maximum of mellitin were monitored to determine the equilibrium dissociation constant and thermodynamic parameters of the monomer–tetramer self-association reaction of this peptide.[25] Similarly, measurement of the changes in the tryptophan fluorescence intensity of gramicidin A as a function of its concentration permitted the determination of an average monomer–dimer equilibrium constant.[26]

for references see p 713

7.7.1.2.2 Fluorescence Lifetime Measurements

A large number of fluorescence decay measurements have been performed with proteins.[27] Studies on the fluorescence decay of tyrosine and tryptophan and their derivatives, and on biologically active peptides containing intrinsic or extrinsic fluorophores have also been carried out and a few illustrative examples will be reviewed here.

7.7.1.2.2.1 Studies with Tryptophan, Tyrosine, and Model Compounds

The fluorescence decay of tryptophan has been analyzed in aqueous solution at various pH values. In early studies double-exponential decay kinetics with decay components of about 0.5 and 3.2 ns were observed for tryptophan in aqueous solution at neutral pH.[28,29] A more recent analysis of *N*-acetyltryptophanamide revealed triple-exponential fluorescence decay at pH 5.5.[30] Double- and triple-exponential fluorescence decay was also observed with a number of tryptophan-containing dipeptides[30,31] and piperazine-2,5-diones (diketopiperazines, DKP).[32,33] These results have been interpreted to indicate that the two or three different fluorescence decay times are due to the fact that the tryptophan side chain may exist in a conformational equilibrium involving different rotamers around the C^{α}–C^{β} bond (gauche$^+$, gauche$^-$, and *trans* configuration)[28,30,34] or around the C^{β}–C^{γ} bond.[35] The short decay components result from quenching by an adjacent carbonyl group or protonated amino group.

A number of studies on the fluorescence decay of tyrosine, tyrosine derivatives, and small tyrosyl peptides have been carried out.[36–38] Whereas the tyrosine zwitterion and tyrosine derivatives with an ionized α-carboxy group exhibited monoexponential fluorescence decay ($\tau = 3.26$–3.76 ns), double- or triple-exponential decay was observed in most other cases. As in the case of the tryptophan model compounds, the complex decay kinetics were again interpreted in terms of rotamer populations resulting from rotation around the C^{α}–C^{β} bond. There is evidence to indicate that the shorter fluorescence lifetimes may arise from rotamers in which the phenol ring is in close contact with a hydrated carbonyl group[36,37] and that a charge-transfer mechanism may be implicated in this quenching process.[39]

7.7.1.2.2.2 Studies with Biologically Active Peptides

There have been several studies on the fluorescence decay of biologically active peptides that contain a single tryptophanyl residue, including glucagon, ACTH (1–24), mellitin, luliberin, and [Trp^2]oxytocin. In all cases, the observed tryptophan fluorescence decay was multiexponential. In principle, multiexponential decay may result either from ground-state heterogeneity or from excited-state reactions or relaxation phenomena. Ground-state heterogeneity may be due to the existence of a conformational equilibrium involving two or more conformers or to the presence of an ionizable group close to the indole ring. Various types of excited-state reactions resulting in complex fluorescence decays have been reviewed.[40] A characteristic dependence of the decay parameters (α_i,τ_i) on the emission wavelength is observed in the case of excited-state reactions.

Depending on specific sample conditions, a short decay component in the range from 0.5–1.66 ns and a long decay component ranging from 3.3 to 5.04 ns have been determined for the tryptophanyl residue in position 25 of glucagon in aqueous solution.[29,41–43] The double-exponential decay was interpreted to arise from the existence of two energetically stable conformations of glucagon (helical and non-helical). Similar decay parameters were determined for various glucagon fragments, including glucagon(1–27), glucagon(1–26), and glucagon(22–29).[42] ACTH(1–24) also showed double-exponential decay in aqueous solution, in one study at pH 7.0 and at a temperature of 10 °C ($\alpha_1 = 0.57$, $\tau_1 = 2.03$ ns, $\alpha_2 = 0.43$, $\tau_2 = 5.13$ ns)[29] and in another at a temperature of 20 °C ($\alpha_1 = 0.53$, $\tau_1 = 1.14$ ns, $\alpha_2 = 0.47$,

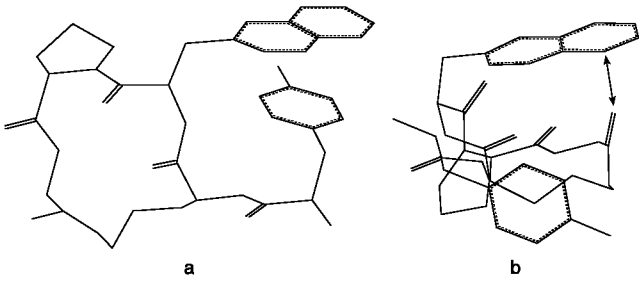
$\tau_2 = 3.62$ ns).[44] Double-exponential fluorescence decay of the tryptophanyl residue in mellitin was observed in Tris buffer at neutral pH in one study,[42] whereas in another study single-exponential decay was determined under the same conditions.[45] The authors of the latter study also observed double-exponential decay after addition of 0.15 M K_2HPO_4 and interpreted this result in terms of a different environment of the indole ring in the formed melittin tetramer as compared to the monomer. The single tryptophan residue in luliberin also showed double-exponential fluorescence decay.[42] In an interesting, more recent study the tryptophan fluorescence decay of the oxytocin analogue [Trp^2]oxytocin was determined.[46] The decay kinetics were best described by a sum of three exponential terms ($\alpha_1 = 0.39$, $\tau_1 = 0.32$ ns, $\alpha_2 = 0.42$, $\tau_2 = 1.07$ ns, $\alpha_3 = 0.19$, $\tau_3 = 2.62$ ns). These decay constants were found to be independent of the emission wavelength, thus indicating a situation of ground-state heterogeneity. Interestingly, the amplitude factors (α_i) correlated very well with the three χ_1 rotamer populations (gauche$^+$, gauche$^-$, *trans*) of the Trp side chain determined for this analogue by ^{1}H NMR spectroscopy and, therefore, the obtained fluorescence parameters were consistent with the χ_1 rotamer model.

Recent progress in instrumentation and data analysis methods permitted the determination of the fluorescence decay kinetics of a number of biologically active peptides containing a single tyrosine residue. For a review of work published before 1992, see ref[47]. Peptides studied included opioid peptides, oxytocin and vasopressin, and angiotensin II. [Leu^5]enkephalin in aqueous solution (pH 7, 25 °C) showed triple-exponential decay ($\alpha_1 = 0.12$, $\tau_1 = 72$ ps, $\alpha_2 = 0.40$, $\tau_2 = 321$ ps, $\alpha_3 = 0.48$, $\tau_3 = 1364$ ps).[48] Similar results were obtained in a subsequent study using 10-GHz frequency domain fluorometry,[49] whereas the decay parameters determined in a recent time-correlated single-photon counting study were somewhat different ($\alpha_1 = 0.55$, $\tau_1 = 240$ ps, $\alpha_2 = 0.35$, $\tau_2 = 1500$ ps, $\alpha_3 = 0.10$, $\tau_3 = 2500$ ps).[50] Triple-exponential decay was also observed with the δ receptor-selective, cyclic enkephalin analogue H-Tyr-*c*[-D-Pen-Gly-Phe-D-Pen-] in H_2O (pH 7, 25 °C) with the following decay parameters: $\alpha_1 = 0.69$, $\tau_1 = 50$ ps, $\alpha_2 = 0.20$, $\tau_2 = 812$ ps, $\alpha_3 = 0.11$, $\tau_3 = 3024$ ps.[50] The amplitude factors (α_i) were found to be in good agreement with the ground-state populations of rotamers of the Tyr^1 side chain of H-Tyr-*c*[-D-Pen-Gly-Phe-D-Pen-] determined by NMR spectroscopy. Furthermore, the average fluorescence lifetime of Tyr^1 in H-Tyr-*c*[-D-Pen-Gly-Phe-D-Pen-] was shorter than those determined for Tyr^1 in linear enkephalin peptides. This is due to quenching of the tyrosine fluorescence by the adjacent disulfide moiety. The same kind of tyrosine fluorescence quenching (low quantum yield) had previously been observed with the structurally similar cyclic enkephalin analogue H-Tyr-*c*[-D-Cys-Gly-Phe-D-Cys-]-NH_2[22] (see Section 7.7.1.2.1.2). The fluorescence decay of the Tyr^2 residue in oxytocin was determined by several groups. In one study carried out at pH 3 and 5 °C double-exponential fluorescence decay ($\alpha_1 = 0.75$, $\tau_1 = 0.69$ ns, $\alpha_2 = 0.25$, $\tau_2 = 1.89$ ns) was observed and explained by a ground-state rotamer model.[51] Comparison with NMR data led to the conclusion that the fluorescence of the Tyr^2 side chain in the gauche$^-$ configuration is statically quenched because of the close contact with the disulfide bridge. A three-component analysis of two dicarba analogues of oxytocin revealed that, as expected, this static quenching did not occur in these compounds, in confirmation of earlier results obtained from a Tyr^2 fluorescence quantum yield determination carried out with a dicarba analogue of oxytocin.[23] In another study using different conditions (pH 7, 25 °C), oxytocin showed triple-exponential fluorescence decay ($\alpha_1 = 0.29$, $\tau_1 = 80$ ps, $\alpha_2 = 0.28$, $\tau_2 = 359$ ps, $\alpha_3 = 0.43$, $\tau_3 = 927$ ps).[48] This different result can be explained by the fact that at pH 7 the N-terminal group is partially protonated. Under the same conditions (pH 7, 25 °C) lysine-vasopressin showed fluorescence decay parameters ($\alpha_1 = 0.59$, $\tau_1 = 170$ ps, $\alpha_2 = 0.30$, $\tau_2 = 750$ ps, $\alpha_3 = 0.11$, $\tau_3 = 1600$ ps) somewhat different from those of oxytocin.[52] These differences may reflect a different environment of the Tyr^2 side chain in the two hormones. Angiotensin II showed triple-exponential tyrosine fluorescence decay in aqueous buffer and double-exponential decay in alcoholic solvents.[53]

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Single-exponential fluorescence decays were determined for the artificial amino acids *N*-acetyl-1-naphthylalanine amide (Ac-1-Nal-NH_2) ($\tau = 30.6$ ns) and *N*-acetyl-2-naphthylalanine amide (Ac-2-Nal-NH_2) ($\tau = 34.0$ ns) in Tris buffer (pH 7.5, 20 °C).[54] The cyclic β-casomorphin analogue H-Tyr-*c*[-D-Orn-1-Nal-D-Pro-Gly-] (opioid μ agonist/δ agonist) also showed a single-exponential decay with a fluorescence lifetime (31.3 ns) similar to that of (Ac-1-Nal-NH_2), indicating that the naphthyl moiety in the peptide is fairly exposed to the aqueous environment. Double-exponential decay was observed with the cyclic analogue H-Tyr-*c*[-D-Orn-2-Nal-D-Pro-Gly-] (opioid μ agonist/δ antagonist). The major component ($\alpha_1 = 0.95$) was characterized by a fluorescence lifetime (τ_1) of 30.2 ns, similar to that of Ac-2-Nal-NH_2, reflecting again a mostly aqueous environment of the fluorophore. The minor component ($\alpha_2 = 0.05$) with a much shorter decay time ($\tau_2 = 6.9$ ns) indicated the existence of a small fraction of conformers in which drastic intramolecular quenching of the naphthyl fluorescence occurs. A molecular mechanics study (systematic grid search) resulted in a number of low energy conformers for both peptides in which the naphthyl moiety was exposed at the surface of the molecule (Figure 2a). This result is in agreement with the fluorescence data indicating a predominantly aqueous environment of the fluorophore in both peptides. However, whereas in none of the low-energy conformers of H-Tyr-*c*[-D-Orn-1-Nal-D-Pro-Gly-] the 1-Nal side chain is located close to a carbonyl group, the 2-Nal side chain in the third-lowest energy conformer of H-Tyr-*c*[-D-Orn-2-Nal-D-Pro-Gly-] (Figure 2b) is folded over the peptide ring with the naphthyl moiety being in close proximity to and on top of the Gly carbonyl group. Since the carbonyl group is a strong fluorescence quencher, this conformer may represent the structure which gives rise to the short fluorescence decay time (minor component) observed with the 2-Nal peptide.

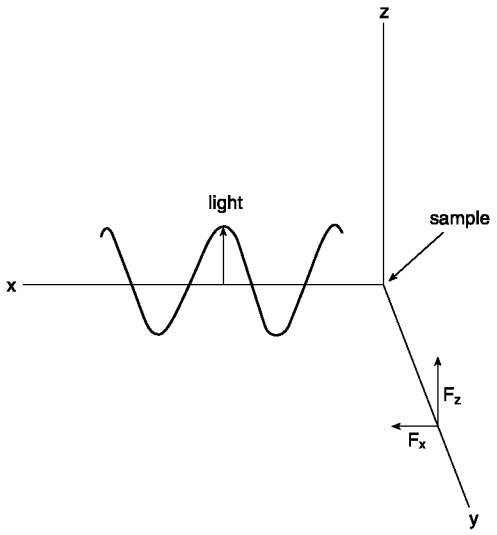
Figure 2 Low-Energy Conformers of H-Tyr-*c*[-D-Orn-2-Nal-D-Pro-Gly-]; (a) with Extended Nal^3 Side Chain, and (b) with the Nal^3 Side Chain in Proximity of the Gly^5 Carbonyl Group.



7.7.1.3 Fluorescence Polarization Studies

In a fluorescence polarization experiment the sample is excited by light traveling along the *x* axis and plane-polarized in the *xy* plane (Figure 3).

Figure 3 Principle of Fluorescence Polarization Experiment[a]



[a] The fluorescent sample is excited with light propagating along x and polarized along z. The emitted light is analyzed along y through a polarizer oriented either along z (I_z) or along x (I_x).

The emitted light is detected along y through a polarizer oriented either along z (F_z) or along x (F_x). In fluorescence polarization studies with continuous excitation (steady-state experiments), the emission anisotropy r and the emission polarization p are defined in eqs 8a and 8b.

$$r = \frac{I_z - I_x}{I_z + 2I_x} \quad \text{eq 8a}$$

$$p = \frac{I_z - I_x}{I_z + I_x} \quad \text{eq 8b}$$

The anisotropy r has been used more frequently in recent years because it depends only on rotational motion, whereas the polarization p depends on both the fluorescence lifetime and rotational motion. The so-called zero-point anisotropy r_o or the principal polarization p_o are observed in the absence of rotational motion during the fluorescence lifetime. The angle between the absorption and emission moments in the fluorophore determines the values of r_0 and p_0 which can vary within the following ranges:

$$-0.2 \leq r_0 \leq +0.4 \quad -0.33 \leq p_0 \leq +0.5$$

Values of +0.4 and +0.5 are theoretically expected for r_0 and p_0, respectively, if the absorption and emission transition moments are in the same direction, as is often the case with excitation at the longest-wavelength absorption maximum. However, due to rapid internal rotation of the emission transition moments immediately after excitation, the experimentally determined values of r_0 and p_0 are always lower than the maximal values. Thus, the highest value ever observed for r_0 is +0.35. In the common case where the fluorophore undergoes rotational motion during the excited-state lifetime, values of r or p closer to zero are observed depending on the extent of depolarization, and in the case of complete depolarization these parameters assume values of zero. The dependence of the anisotropy on rotational motion is described by eq 9[55]

$$r_0 = 1 + \tau/\theta \quad \text{eq 9}$$

for references see p 713

where θ is the rotational correlation time which depends on the solvent viscosity η according to eq 10

$$\theta = V\eta/kT \quad \text{eq 10}$$

where V is the hydrated volume of the molecule and k the Boltzmann constant. Combination of eqs 9 and 10 leads to eq 11.

$$\frac{1}{r} = \frac{1}{r_o} + \frac{1}{r_o}\frac{\tau kT}{V\eta} \quad \text{eq 11}$$

According to eq 11, r_o can be determined by plotting $1/r$ versus T/η (Perrin plots). The rotational correlation time θ can then be calculated if τ is known. In these experiments viscosity changes are usually achieved by varying the sample temperature or by adding sucrose or glycerol. This represents a disadvantage of the steady-state fluorescence polarization method, since the viscosity or temperature changes may alter the peptide conformation(s).

This problem does not exist with time-dependent fluorescence polarization measurements where the decay of the emission anisotropy $r(t)$ is obtained by determining the decay of I_z and I_x according to eq 12.

$$r(t) = \frac{I_z(t) - I_x(t)}{I_z(t) + 2I_x(t)} \quad \text{eq 12}$$

The decay of the anisotropy of a rigid spherical molecule is described by eq 13.

$$r(t) = r_o e^{-t/\theta} \quad \text{eq 13}$$

If the molecule is nonspherical or in the case of internal rotational motions of the fluorophore independent of the overall rotational motion of the peptide, the anisotropy decays can be described as a sum of exponentials as in eq 14

$$r(t) = r_o \sum_i g_i e^{-t/\theta_i} \quad \text{eq 14}$$

where $r_o g_i$ are the amplitudes and θ_i are the rotational correlation times associated with each component of the decay.

Theoretical and experimental aspects of time-resolved fluorescence anisotropy measurements have been described in detail in the literature.[7,40]

7.7.1.3.1 Determination of Rotational Correlation Times of Fluorophores in Peptides

Steady-state fluorescence polarization studies have been carried out with a number of peptides, including model peptides, ACTH, glucagon, melittin, and thyrocalcitonin. This work has been reviewed[5] and will not be discussed in the present article. More recently, interesting information on the rotational behavior and structural flexibility of various peptides has been obtained from fluorescence anisotropy decay measurements.

The anisotropy decay of the tryptophan fluorescence of both model peptides and biologically active peptides containing a single tryptophan residue has been determined in various studies. Even in the case of the tripeptide H-Gly-Trp-Gly-OH quenched by acrylamide the anisotropy decay displayed two correlation times with values of 39 and 135 ps.[44] The shorter correlation time was thought to be due to motions of the indole ring relative to the tripeptide. In the case of ACTH(1–24) the fluorescence anisotropy decay of the single tryptophan residue in position 9 of the peptide sequence obtained in phosphate buffer (pH 7, 3.5 °C) was also double-exponential.[29] The shorter rotational correlation time ($\theta_1 = 92$ ps)

determined may reflect the internal rotation of the indole ring, whereas the longer one ($\theta_2 = 4.47$ ns) is likely to be due to the overall motion of the entire peptide molecule. In a subsequent study, double-exponential decay of the tryptophan fluorescence anisotropy of ACTH(1–24) in phosphate buffer (pH 7, 20 °C) was also observed, albeit the determined rotational correlation times were somewhat different ($\theta_1 = 200$ ps, $\theta_2 = 1.80$ ns).[44] The anisotropy decay of the tryptophan fluorescence determined with glucagon was also found to be double-exponential.[56] Again, the shorter rotational correlation time (~0.5 ns) was attributed to the Trp^9 side-chain flexibility and the longer one was thought to reflect the rotational motion of entire glucagon molecule. In the case of melittin, nanosecond emission anisotropy measurements resulted in rotational correlation times of 1.1 and 3.9 ns for the monomer and tetramer, respectively.[45] Frequency domain fluorescence anisotropy measurements were performed with two zinc finger peptides (28- and 29-residue peptides) that contain a single tryptophan residue in the middle of the peptide sequence.[57] Both in the metal-free and in the metal-bound form two rotational correlation times were obtained for both peptides: a longer one (θ_1 ~2 ns) reflecting the global motion of the zinc finger molecule and a shorter one ($\theta_2 = 49$–463 ps) associated with the Trp side-chain rotation. However, whereas in the zinc-free peptides the amplitude $r_0 g_i$ of both components were nearly equal, the amplitudes of the longer component in both zinc-bound peptides were four- to fivefold longer than those of the shorter components. The latter result was interpreted to indicate that the metal-bound peptides assume a compact structure, whereas the metal-free peptides are relatively more unfolded and less compact.

Progress in instrumentation and data analysis has permitted the determination of the tyrosine fluorescence anisotrope decay in peptides. Biologically active peptides with a single Tyr residue that were studied included [Leu^5]enkephalin, H-Tyr-*c*[-D-Pen-Gly-Phe-D-Pen-], oxytocin, vasopressin, and a zinc finger peptide. In an initial study, [Leu^5]enkephalin in Tris buffer (pH 7.0) displayed a complex fluorescence anisotropy decay, with correlation times of 68 and 247 ps.[48] Similar values were obtained in two subsequent studies.[49,50] Again, the shorter correlation time was attributed to the local motions of the tyrosine side chain and the longer one to the overall rotational diffusion of the entire peptide. The rotational correlation times determined for the cyclic enkephalin analogue H-Tyr-*c*[-D-Pen-Gly-Phe-D-Pen-] ($\theta_1 = 36$ ps, $\theta_2 = 300$ ps) were quite similar to those obtained for the linear [Leu^5]enkephalin peptide.[50] Oxytocin also showed double-exponential tyrosine fluorescence anisotropy decay ($\theta_1 = 29$ ps, $\theta_2 = 454$ ps), reflecting the torsional motions of the phenol ring and the overall rotational diffusion of the peptide, respectively.[48] Two-component analysis of the anisotropy decay of lysine-vasopressin resulted in about the same value for θ_1 (37 ns) as in the case of oxytocin but in a significantly larger value for θ_2 (570 ps).[49] This result, together with the observed higher amplitude associated with the shorter correlation time and lower amplitude associated with the longer correlation time, as compared to oxytocin, suggests that the structure of lysine-vasopressin is less compact and more flexible than that of oxytocin. The fluorescence anisotropy decay of the 25 amino acid zinc finger peptide, Xfin-31, containing a single tyrosine residue was determined.[58] Double-exponential decay was observed with the zinc-free peptide ($\theta_1 = 590$ ps, $r_0 g_1 = 0.13$, $\theta_2 = 47$ ps, $r_0 g_2 = 0.18$) and with the zinc-bound peptide ($\theta_1 = 1420$ ps, $r_0 g_1 = 0.23$, $\theta_2 = 35$ ps, $r_0 g_2 = 0.08$). These data indicate that the overall rotational correlation time of the zinc-bound peptide is larger than that of the metal-free peptide and that the amplitude of the longer anisotropy decay time is greater than that of the shorter one in the case of the zinc-bound peptide. Taken together, these results suggest that the zinc-bound peptide is more compact and that the metal-free peptide is structurally more flexible.

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7.7.1.4 Measurement of Intramolecular Distances in Peptides by Singlet-Singlet Energy Transfer Experiments

The condition for transfer of singlet excitation energy from a fluorophore (donor) to another chromophore (acceptor) to occur is spectral overlap of the donor's fluorescence emission spectrum with the acceptor's absorption spectrum. The efficiency of energy transfer E is defined in eq 15

$$E = \frac{k_t}{k_e + \sum_i k_i + k_t} \quad \text{eq 15}$$

where k_e is the rate of donor fluorescence, k_t is the rate of energy transfer and $\Sigma_i k_i$ is the sum of the rates of all other quenching processes depopulating the first excited singlet state of the donor.

Energy transfer experiments permit the measurement of distances because the rate of transfer is proportional to the inverse sixth power of the distance between donor and acceptor. According to the theory of Förster the donor–acceptor distance r is related to the transfer efficiency E by eq 16

$$r = (E^{-1} - 1)^{1/6} R_o \quad \text{eq 16}$$

where R_o, the Förster critical distance, is the donor-acceptor distance at which 50% of the donor excitation energy is transferred to the acceptor.[59]

The experimental determination of R_o, measured in cm, is based on eq 17

$$R_o = [(8.79 \times 10^{-25})\ (\kappa^2/n^4)\ \varphi_D^\circ\ J_{AD}]^{1/6} \quad \text{eq 17}$$

where κ^2 is the dipole-dipole orientation factor, n the refractive index, φ_D° the donor fluorescence quantum yield in the absence of transfer, $J_{AD} = \int_0^\infty F_D(\lambda)\varepsilon_A(\lambda)\lambda^4 d\lambda$, which is the spectral overlap integral between the molar decadic absorption coefficient of the acceptor (ε_A) and the spectral distribution of the donor fluorescence, normalized to unity (F_D), modified by the wavelength factor λ^4. E can be experimentally determined from the donor fluorescence quenching by measuring either φ_D and φ_D°, the donor fluorescence quantum yield in the presence and absence of energy transfer, or τ_D and τ_D°, the donor fluorescence lifetime in the presence and absence of the acceptor, respectively, according to the eqs 18 and 19.

$$E = 1 - \varphi_D/\varphi_D^o \quad \text{eq 18}$$

$$E = 1 - \tau_D/\tau_D^o \quad \text{eq 19}$$

It is important to point out that for the measurement of an intramolecular donor–acceptor distance in a peptide φ_D° or τ_D° should be determined with a peptide analogue in which the acceptor has been replaced with a different amino acid residue with similar structural characteristics but unable to act as an acceptor. In such an analogue the microenvironment of the donor fluorophore is likely to be very similar to that in the peptide under study.

Alternatively, E can be determined from the relative increase in acceptor fluorescence on the basis of the eq 20[60]

$$E = \frac{A_A(\lambda)}{A_D(\lambda)} \left[\frac{I(\lambda)}{I_A(\lambda)} - 1 \right] \quad \text{eq 20}$$

where $A_A(\lambda)$ and $A_D(\lambda)$ are the absorbances at wavelength λ of the acceptor and the donor, respectively, and $I(\lambda)$ and $I_A(\lambda)$ are the intensities of the acceptor emission in the

presence and absence of the donor, respectively, with excitation at λ (fluorescence excitation spectra).

The eq 20 is simplified after normalization of the absorption and fluorescence excitation spectrum of the acceptor $[A_A(\lambda) = I_A(\lambda)]$ to give the eq 21.

$$E = [I(\lambda) - I_A(\lambda)]/A_D(\lambda) \quad \text{eq 21}$$

It thus follows that E can be determined by comparison of the fluorescence excitation spectra of the acceptor in the presence and absence of the donor with the absorption spectrum of the donor–acceptor pair after the three spectra have been normalized at a wavelength where the donor does not absorb.[61] It is best to obtain the fluorescence excitation spectrum of the acceptor in the absence of transfer with a peptide analogue in which the donor is either lacking or has been replaced with a structurally similar moiety that does not engage in energy transfer.[17] For best accuracy, E should be determined at an excitation wavelength where the donor/acceptor absorbance ratio is as high as possible. It is recommended that E be determined from both donor fluorescence quenching and acceptor fluorescence enhancement, because in some cases the observed donor fluorescence quenching may not exclusively be due to energy transfer but may also be partially caused by additional quenching mechanisms as a result of the incorporation of the acceptor into the peptide. If such additional quenching occurs, a value of φ_D° or τ_D° corrected for the additional quenching has to be used in the calculation of R_o.[5]

An important parameter required for the calculation of R_o is the orientation factor κ^2 which takes into account the angular dependence of dipole–dipole energy transfer, as described by eq 22

$$\kappa^2 = (\cos\alpha - 3\cos\beta\cos\gamma)^2 \quad \text{eq 22}$$

where α is the angle between the donor and acceptor dipoles, and β and γ are the angles between the donor and acceptor dipoles, respectively, and the line joining the centers of the donor and the acceptor.

Theoretically values of κ^2 can range from 0 to 4. In a situation of complete dynamic randomization of donor–acceptor orientation, κ^2 assumes a value of 2/3. If rotational mobility for both the donor and the acceptor can be established by a fluorescence polarization experiment and if the donor and acceptor electronic transitions are polarized along more than one direction, as is the case with a number of fluorophores, then the error in the measured distance is usually less than 20% if a value of $\kappa^2 = 2/3$ is used in the calculation of R_o. The value of the overlap integral J_{AD} is quite solvent dependent because the spectral distribution of the donor fluorescence varies as a function of solvent polarity. The range of measurable distances depends on R_o and, therefore, on the choice of the donor–acceptor pair. Distance measurements in the range 8–16 Å are possible with the Tyr-Trp pair, whereas distances ranging from 15 to 35 Å can be determined with the Trp-Dns donor–acceptor pair. An extensive list of R_o values calculated for numerous donor–acceptor pairs has been published.[62] Distances of less than ~8 Å cannot be determined by singlet–singlet energy transfer measurements because multipole and exchange interactions occur at very short donor–acceptor separations. Theoretical and experimental aspects of energy transfer distance measurements and their application in the field of biopolymers have been reviewed.[5,62–65]

The validity of the Förster theory was tested and confirmed in a number of model studies with compounds that contained a donor and an acceptor separated by well-defined rigid spacers. This work has been reviewed.[5] In a classical study, a naphthyl group (donor) was attached to the C-terminal and a dansyl group (acceptor) to the N-terminal of poly-L-proline oligomers (1–12 proline residues).[61] These proline oligomers assume a trans helical conformation in ethanol and thus represent spacers of well-defined length (12–46 Å). A continuous decrease in the transfer efficiency from 100% at a donor–acceptor separation of

for references see p 713

12 Å to 16% at a distance of 46 Å was observed by evaluation of the relative enhancement of the acceptor fluorescence. The experimentally determined transfer efficiency was found to be almost exactly proportional to the inverse sixth power of the donor-acceptor distance as expected on the basis of eq 16. In another important model study, the dependence of the rate of energy transfer on the magnitude of the overlap integral was examined by determining energy transfer between an *N*-methylindole moiety (donor) and a ketone group acceptor contained in a rigid steroid.[66] The value of *J* could be varied over a 40-fold range by changing the solvent. In agreement with Förster's equation, the measured transfer rate was found to be proportional to the magnitude of *J*. Clearly, the results of these two studies confirmed the validity of Förster's theory of energy transfer by weak dipole–dipole coupling and demonstrated the feasibility of intramolecular distance measurements in biopolymers by energy transfer experiments.

7.7.1.4.1 Measurement of Intramolecular Distances in Biologically Active Peptides

Whereas donor–acceptor pairs such as Tyr-Trp are already contained in some biologically active peptides, it will be necessary in many cases to incorporate a donor and/or acceptor group at appropriate sites of the peptide. Such peptide analogues should be designed under careful consideration of existing structure–activity relationships and often require the total synthesis of the desired compound. Intramolecular distances have been determined in a number of biologically active peptides, including ACTH, opioid peptides, angiotensins, a cholecystokinin fragment, luliberin, somatostatin fragments, oxytocin and vasopressin analogues, bradykinin analogues, insulin analogues, *Naja nigricollis* toxin, and zinc finger peptides. Many of these studies have been reviewed[5] and only a few representative examples will be presented here.

Early energy transfer distance measurements were performed with the N-terminal 24-peptide fragment of corticotropin [ACTH(1–24)]: H-Ser-Tyr-Ser-Met-Glu-His-Phe-Arg-Trp-Gly-Lys-Pro-Val-Gly-Lys-Lys-Arg-Arg-Pro-Val-Lys-Val-Tyr-Pro-OH. In a first study, efficiencies of energy transfer from the two tyrosines (Tyr^2, Tyr^{23}) to tryptophan (Trp^9) were determined with the entire 24-peptide and with the two shorter fragments ACTH(1–16) and ACTH(4–24) that contain only a single tyrosine residue.[67] An average intramolecular distance of 10 ± 1.5 Å between Tyr^2 and Trp^9 was determined, whereas the Trp^9-Tyr^{23} distance was found to be larger (>19 Å). The results were interpreted to indicate that folded conformations of the segment encompassing residues 2–9 must exist. In another study, the biologically active analogue N^{ε}-dansyllysine^{21}-ACTH(1–24), [Lys(Dns)^{21}]ACTH(1–24), was synthesized and energy transfer from the indole ring of Trp^9 (donor) to the dansyl group (acceptor) attached to the ε-amino group of Lys^{21} was determined.[17] The transfer efficiency was determined from donor fluorescence quenching, whereby φ_D° was measured with ACTH(1–24) which does not carry the dansyl acceptor. *E* was also determined from the relative enhancement of the acceptor fluorescence, using the fragment [Lys(Dns)^{21}]-ACTH(11–24), as the reference compound lacking the donor. Respective transfer efficiencies of 0.40 and 0.45 were obtained by the two methods in the initial study carried out in phosphate buffer (pH 7, 25 °C). Since these two values were very similar, it was concluded that the incorporation of the acceptor into the peptide did not result in any additional quenching of the donor fluorescence aside from energy transfer. The results of fluorescence anisotropy measurements indicated rotational motion of both the donor and the acceptor moiety relative to the peptide and thus justified the use of 2/3 as the value for κ^2. An average intramolecular distance between Trp^9 and Lys(Dns)^{21} of 24.5 Å was calculated based on Förster's equations. Similar average distances of 23.6 and 26.0 Å were determined under conformation-breaking conditions in 8 M urea and 6 M Gn-Cl, respectively. These results were interpreted to indicate that the segment encompassing residues 9–21 in ACTH(1–24)

had a "random coil" structure. Conformational heterogeneity in the case of both ACTH(1–24) and [Lys(Dns)^{21}]ACTH(1–24) was also indicated by the results of a theoretical conformational analysis.[68] In both cases an average transfer efficiency was calculated on the basis of the obtained donor–acceptor distance distribution function and under the assumption of a static regime of the peptide backbone conformations with rotational motion of the donor and acceptor moieties. Since the calculated values of E were in good agreement with the experimentally determined ones, it was concluded that these ACTH peptides are structurally flexible and exist in a conformational equilibrium.

In a number of studies, intramolecular distances in various opioid peptides were determined by fluorescence energy transfer experiments. For [Trp^4,Met^5]enkephalin (H-Tyr-Gly-Gly-Trp-Met-OH) in H_2O an average intramolecular distance of 9.3±0.2 Å between the phenol ring of Tyr^1 and the indole moiety of Trp^4 was obtained.[21] This result indicated the existence of folded structures, since in a fully extended conformation the phenol–indole separation would be 14–18 Å, depending on side-chain orientations. Again, the determined transfer efficiency was found to be in good agreement with the average value of E computed for a distribution of donor–acceptor distances resulting from a theoretical conformational study, the results of which suggested the existence of a conformational equilibrium of conformers with Tyr^1-Trp^4 distances ranging from 5 to 15 Å.[69] Intramolecular Tyr^1-Trp^4 distances were also determined with the four enkephalin analogues H-Tyr-Gly-Gly-Trp(N$^{\alpha}$Me)-Met-OH, H-Tyr-Gly-Gly-Trp-Leu(N$^{\alpha}$Me)-OH, H-Tyr-Gly-Gly-Trp-Pro-OH, and H-Tyr-(OMe)-Gly-Gly-Trp-Met-OH, which had been designed to prevent the formation of various proposed hydrogen bonds.[70] The distances obtained for these compounds ranged from 8.8 to 10.7 Å and, thus, were similar to the Tyr^1-Trp^4 distance (9.3 Å) determined with [Trp^4,Met^5]enkephalin. These results suggested that intramolecular 4→1 or 5→2 hydrogen bonds between amino and carbonyl groups of the peptide backbone or a hydrogen bond between the tyrosyl hydroxy group and a backbone carbonyl group are not required for the stabilization of folded conformations of enkephalin in aqueous solution. Evaluation of energy transfer in the potent cyclic enkephalin analogue H-Tyr-c[-D-Cys-Gly-Trp-D-Cys-]-NH_2 resulted in an average intramolecular Tyr^1-Trp^4 distance of 9.7 Å, similar to that determined with its linear correlate H-Tyr-D-Ala-Gly-Trp-Met-OH (9.5 Å), in spite of the conformational constraints existing in the cyclic peptide.[22] The average Tyr^1-Trp^4 distance (~15 Å) determined with [Trp^4]dynorphin(1–13) (H-Tyr-Gly-Gly-Trp-Leu-Arg-Arg-Ile-Arg-Pro-Lys-Leu-Lys-OH) at pH 8.0 was considerably longer than the corresponding distance (10.0 Å) obtained with [Trp^4,Leu^5]enkephalin (H-Tyr-Gly-Gly-Trp-Leu-OH) under the same conditions.[71] This result indicated that the predominant conformation of the N-terminal tetrapeptide segment in [Trp^4]dynorphin(1–13) is almost completely extended, in contrast to the folded conformation(s) of that same segment observed with [Trp^4,Leu^5]enkephalin. This conformational difference between the two peptides may be of relevance to their different opioid receptor selectivity profiles. Since a Tyr^1-Trp^{14} distance of greater than 22 Å was determined with dynorphin(1–17) (H-Tyr-Gly-Gly-Phe-Leu-Arg-Arg-Ile-Arg-Pro-Lys-Leu-Lys-Trp-Asp-Asn-Gln-OH), there seems to be no interaction between the N- and C-terminal segments of this peptide.[71] For the unsulfated and partially protected 7-peptide of cholecystokinin (Boc-Tyr-Met-Gly-Trp-Met-Asp-Phe-NH_2) a Tyr^1-Trp^4 distance (10.0 Å) similar to the Tyr^1-Trp^4 distance in the Trp^4-analogues of enkephalins was determined.[72] This conformational similarity is of interest because the cholecystokinin 7-residue peptide with a free N-terminal amino group is a μ-opioid receptor ligand.

Conformational properties of zinc finger peptides have also been investigated by energy transfer distance measurements and a review is available.[73] In one study, the evaluation of energy transfer between Tyr^{28} and Trp^{35} in the "zinc finger motif"-containing nucleocapsid protein of Moloney murine leukemia virus resulted in an average intramolecular distance of 10.8 Å.[74] On binding of zinc to the protein this intrafinger distance increased to 12.8 Å as a consequence of the metal-induced conformational change. An interfinger Tyr^{16}-Trp^{37} dis-

for references see p 713

tance in the range of 7–12 Å was determined in an energy-transfer study with the mutant peptide Tyr^{16}(12–53) of the HIV nucleocapsid protein and was interpreted to indicate a close proximity of the two zinc finger motifs.[75]

Whereas, the steady-state fluorescence resonance energy-transfer experiments described above permitted the measurement of discrete donor–acceptor distances, distributions of donor–acceptor distances can be determined with time-resolved fluorescence energy transfer experiments. In these measurements, distance distributions $P(r)$ are assumed to be Gaussian and are characterized by a most probable distance (r_{av}) and the width of the distribution at half-maximum probability (hw).[76] Furthermore, the mutual donor-to-acceptor diffusion coefficient (D) is usually also determined in these experiments. A distance distribution determination performed with a 29-residue single domain CCHH-type zinc finger peptide[57] serves as an illustrative example. In this study, the single Trp residue (Trp^{15}) in the middle of the sequence was the donor and one acceptor (Dns) was attached to the N-terminal amino group. In a second peptide derivative the acceptor 7-amino-4-methylcoumarin-3-acetyl (AMCA) was coupled to the ε-amino group of the C-terminal lysine residue. For the metal-free peptide the distance distribution parameters for the Trp-Dns pair were as follows: $r_{av} = 20$ Å, $hw = 15$ Å, $D = 12$ Å^2/ns, and those for the Trp-AMCA pair were as follows: $r_{av} = 15$ Å, $hw = 28$ Å and $D = 20$ Å^2/ns. In the zinc-bound state the distributions of both donor–acceptor pairs covered a narrower range and the donor-receptor diffusion was much more limited (Trp-Dns, $r_{av} = 12$ Å, $hw = 3$ Å, $D < 3$ Å^2/ns; Trp-AMCA, $r_{av} = 21.5$ Å, $hw = 10$ Å, $D < 3$ Å^2/ns). Furthermore, r_{av} for the Trp-Dns pair was 8 Å shorter in the zinc-bound state as compared to the metal-free state. These data were interpreted to indicate that in the zinc-bound state the peptide assumes a more compact and less flexible structure than in the absence of the metal.

7.7.1.5 Studies of Peptide Interactions with Macromolecular Systems by Fluorescence Techniques

The possibility to carry out conformational studies of peptides at low concentrations and in the presence of complex biological systems represents a major advantage of fluorescence spectroscopy over other techniques. Fluorescence quantum yield or lifetime determinations, anisotropy measurements and singlet–singlet resonance energy transfer experiments can be used to study the interaction of peptides with lipid micelles, membranes, proteins, or receptors. These fluorescence techniques can be used to determine binding parameters and to elucidate conformational aspects of the interaction of the peptide with a particular macromolecular system. The limited scope of this chapter does not permit a comprehensive review of the numerous studies of this kind that have been carried and only a few general aspects are briefly discussed here. Fluorescence studies of peptide interactions with macromolecular systems published prior to 1984 have been reviewed.[5]

Fluorescence studies of the interaction of peptides with lipid micelles or vesicles generally provided information on lipid binding parameters and on changes in the conformation and rotational motion of the lipid-bound peptide. Blue shifts in the fluorescence emission spectrum of the peptide fluorophore are often observed due to the hydrophobic lipid environment. Depending on the type of lipid, these blue shifts are accompanied by either an increase or decrease in fluorescence intensity. The changes in these parameters often permit the construction of a binding curve and the determination of the peptide-micelle (or vesicle) affinity constant and the number of peptide molecules bound per micelle or vesicle. As expected, the rotational correlation time of the peptide fluorophore always increases upon binding of the peptide to the lipid system. The orientation of the peptide in the lipid can be examined by determining the depth of insertion of its fluorophore into the lipid bilayer via fluorescence quenching with spin-labeled lipids.

For studies of peptide–protein interactions an extrinsic fluorescent probe usually has to be incorporated into the peptide and/or into the protein or receptor. The binding parameters of a fluorescent peptide interacting with a protein or receptor can be determined by fluorescence quantum yield/lifetime determinations or fluorescence anisotropy measurements. Such studies may also permit the determination of the binding kinetics (k_{on} and k_{off} rates) and thereby provide information on the binding process. Fluorescence energy transfer between a fluorophore contained in a protein such as an enzyme and a fluorophore in a peptide ligand (e.g., an inhibitor) binding to the protein can be used to measure intermolecular peptide ligand–protein distances. In recent years, specific incorporation of a single extrinsic fluorescent probe into a receptor has been achieved by chemical or recombinant DNA techniques. Measurements of the fluorescence quantum yield or lifetime of the probe in the reconstituted receptor may reveal ligand-induced conformational changes in the receptor. Furthermore such fluorescent-labeled receptors may permit energy transfer distance determinations between a fluorophore contained in the peptide ligand and the fluorescent probe in the receptor, thereby providing information on the location of the ligand-binding site in the receptor.

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7.7.2 Conformational Studies Using Infrared Techniques

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Shortly after the initial syntheses of poly-α-amino acids (homopolypeptides), infrared (IR) absorption was established as an important tool for characterizing the secondary structure of the polymer fold.[1–10] IR frequencies were known for the vibrational transitions associated with the amide group (peptide link) and empirical correlation of these frequencies showed them to shift in characteristic patterns depending on secondary structure.[5,7,11,12] In hindsight, the advantage of IR was its ability to probe both solid and solution samples of widely varying quality (mulls, powders, films, gels, scattering solutions, etc.)[1,3,5,6,9,13,14] and still yield usable data. Thus, IR could bridge the emerging structural insight available from X-ray diffraction for solid-state (crystalline) structures to the more broadly prepared samples (e.g., solutions, powders, films) accessible with spectroscopic techniques. In addition, the ability to study oriented samples (e.g., crystals, fibers, or stretched films) allowed polarization to become an important tool for interpreting the observed IR band shapes and frequencies.[1,5,6]

The most studied band in the IR of peptides is designated as the amide I, primarily composed of the C=O stretch on the amide linkage, which appears between 1600 and 1700 cm^{-1}, and typically has frequencies of 1650–1660 cm^{-1} for α-helices and 1620–1640 cm^{-1} for β-sheets. This band is most useful for the typical analytical reasons of its being in a relatively isolated region of the spectrum, which leads to less interference by other bands, and due to its high dipole intensity which makes it easier to study small samples and still identify this band even under conditions of spectral interference from the solvent. Due to the well-defined orientation of the C=O group with respect to the helix axis in α-helices and β-sheets, the amide I is polarized. Thus, IR polarization ratios provide an added diagnostic for oriented samples[1,2,5,6] and a tool to discriminate against frequency shifts induced by environmental perturbations or side-chain effects (see Section 7.7.2.3).

In addition, for solid samples or peptides in nonaqueous solvents, the amide II (primarily in-plane NH deformation mixed with C–N stretch, ~1500–1530 cm^{-1}) and the amide A (NH stretch, ~3300 cm^{-1} but quite broad) bands are also useful added diagnostics of secondary structure.[5,15–17] Due to their relatively broader profiles and complicated by their somewhat weaker intensities, the frequency shifts of these two bands with change in secondary structure are less dramatic than for the amide I; yet for oriented samples their polarization properties remain useful.[18] Additionally, the amide A and amide II bands are highly sensitive to deuteration effects. Thus, they can be diagnostic of the degree of exchange for a peptide and consequently act as a measure of protected or buried residues as compared to those fully exposed to solvent.[9,19,20] Amide A measurements are not useful in aqueous solution due to overlap with very intense water transitions, but amide II measurements can usefully be measured under such conditions.[5,19,20] The amide III (opposite-phase NH deformation plus C–N stretch combination) is very weak in the IR and is mixed with other local modes, but has nonetheless been the focus of a few protein-based studies.[5,21–26] Finally, other amide modes (IV–VII) have been identified at lower frequencies, but have been the subject of relatively few studies in peptides.[5–8,18,27,28]

7.7.2.1 Complementary Raman Method

Due to limitations in signal-to-noise ratio available for the then common dispersive IR instruments, peptide and protein vibrational spectroscopic studies shifted to emphasize Raman measurements in the 1970s.[29–32] Qualitatively the same sorts of empirical correlations as discussed above have been found between frequencies of amide bands in the Raman and secondary structure. However, due to the complementary selection rules for Raman as compared to IR and to the multi-component nature of these polymeric spectral bands, the

for references see p 733

dominant frequencies observed in Raman spectra differ from those in IR.[30,33–36] In Raman, the most useful and intense amide bands have proven to be the amide I and III bands.

As Raman intensity depends on the polarizability change in the vibration, Raman has unique sensitivity to some side-chain modes such as the ring-expansion modes of aromatics and stretches of disulfide (-S–S-) linkages, the latter of which has proven to be useful in defining peptide conformations for sequences containing two or more cysteins.[37] Aromatic-based transitions have also been the subject of a series of resonance Raman-based studies.[38–42] These could provide more site-specific information about the residues. UV Resonance Raman (UVRR) studies of the amide modes have also appeared and have different intensity profiles than the nonresonant spectra.[42–46] Recent studies have shown these UVRR spectra of the amide III (frequency and intensity) to be affected by its mixing with (C) $C_\alpha H$ deformation leading to correlation with local torsional conformations at the C_α carbon.[232,233]

7.7.2.2 Fourier-Transform Infrared Impact

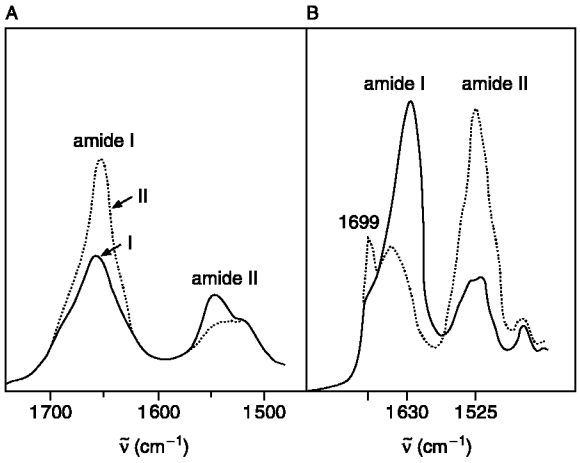
The switch in emphasis back to IR for peptide conformational studies using vibrational spectroscopy occurred in the 1980s with the widespread availability of inexpensive, high-quality Fourier-transform infrared (FTIR) spectrometers[47] and the development of sampling methods to allow very high signal-to-noise ratio measurements as well as unique polarization modulation and very small or dilute sampling experiments.[48–53] Raman remains a contributor to protein and peptide secondary structure analysis, but is more utilized for applications which can better utilize its unique strengths. Hence, all the discussion to follow will be on IR techniques or combinations of IR with other techniques. Developments in technology have made FTIR analyses of peptide structure quite routine such that they have become a standard component of some areas of peptide structure study.

This section focuses on the use of IR spectroscopy for determining characteristic secondary structure properties of peptides, primarily dealing with solution behavior but encompassing other environments. Numerous reviews and papers of IR applications to protein secondary structure analysis are available.[49,50,52–55] Consistent with the topic of this work, we here take a narrower focus on synthetic and smaller peptides and what can be learned from IR studies. Haris and Chapman[51] have published a review focused on IR studies of peptides, and Haris[234] has recently critically reviewed peptide FTIR. Freedman et al.[56] have similarly covered peptide applications of vibrational circular dichroism (VCD). An important aspect/limiting structural application of IR studies of peptides (and any polymer species) is the frequency overlap of vibrational transitions arising from repeating functional groups (such as the most important feature, the amide link). This overlap, coupled with the natural width of an IR or Raman band, normally prevents use of IR spectroscopy for site-specific studies such as those that have made NMR and X-ray crystallography so valuable. (An exception are the site-specific, isotopically labeled peptide studies.[57–60,235]) IR transitions are best used to characterize repeating units of the same conformation and hence are useful for secondary structure analyses in terms of average composition. Some techniques have been developed to circumvent these resolution limitations, at least in part. To better identify overlapped spectral contributions arising from different secondary structural types, the use of Fourier self-deconvolution with band fitting[48,61–66] or second-derivative[65,67] techniques to identify underlying band components in a heterogenously broadened IR feature have become very common. Band shape based techniques have also been employed to quantify secondary structure in proteins.[68–73] Some applications of these and other methods have appeared for peptide IR.[12,74,75] These enhancements of spectral analysis are made possible by the very high signal-to-noise ratio available with FTIR techniques.[47,52]

7.7.2.3 Polarization Advantages

The above analytical approaches still only address the frequency (or eigen value) aspect of the vibrational transition. Other techniques using polarization properties (intensity components) probe the nature of the vibrational state (or eigen vector) and are partially immune to frequency variations. The simplest of these is linear dichroism (LD) whereby the molecule is constrained to a known orientation, e.g. in a crystal, oriented film, or as part of a membrane, and probed with linearly polarized light.[76–81] If the polarization is parallel to the transition moment, the absorbance will be maximized. In this regard, early peptide studies used polarized IR studies of oriented helices and sheets in fibers and stretched or combed films to characterize structure.[1,5] In α-helices, the amide I is polarized parallel to the helix axis and the amide II perpendicular to it (Figure 1, A). This follows directly from the hydrogen-bonding pattern being from one amide to the one above it in the helix (four residues away). In β-sheets the opposite polarization pattern is found (Figure 1, B), due to interstrand hydrogen bonding. In addition, the parallel and perpendicular amide I components are resolved providing a characteristic band shape to distinguish the sheet-based transitions from those that might arise from a misoriented helix or a coil form. Such discrimination between peptide secondary structure was evident in early oriented film IR studies of wool keratin, porcupine quill (α-helical), and *Bombyx mori* silk thread (β-sheet) studies.[82–84]

Figure 1 Polarized IR Spectra for (A) α-Helical Wool Keratin and (B) β-Sheet Stranded *Bombyx mori* Silk Thread[18]

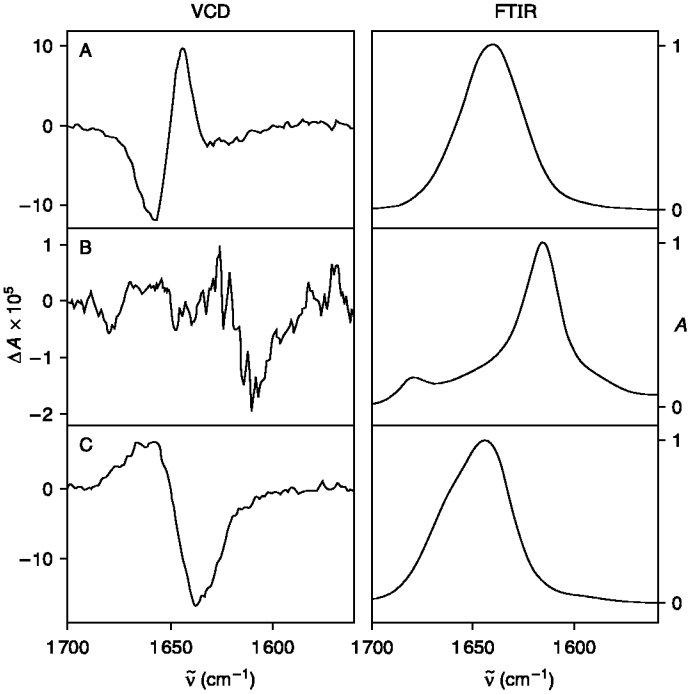


An alternative means of probing the character of the excited state with polarized light is circular dichroism (CD), the measure of the differential absorption of left and right circularly polarized light ($\Delta A = A_l - A_r$).[85,86] While most CD studies of peptides are done on electronic transitions (ECD) in the UV region,[87–90] studies of vibrational circular dichroism (VCD) in the IR are well-established and provide a very useful means of structurally characterizing peptide conformations.[55,56,91–96] VCD is a measure of chirality-based optical activity that arises from the rotational strength [$R = \mathrm{Im}(\mu \times m)$] of a transition which in turn depends on the interference of the electric (μ) and magnetic (m) dipole moments. Thus, it is a polarization measurement that senses the intramolecular interference of two components of the transition and thus can be measured for isotropic systems such as solution samples. VCD is exceptionally sensitive to conformational variation since much of the magnetic moment in a peptide arises from coupling of repeated oscillators in the polymeric chain. Unique amide I and amide II patterns for α-helices and β-sheets as well as for 3_{10}-helices, Pro II helices (which relate to coils), and some β-turns are found with VCD.[19,20,56,92,97–109] Some

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typical results for α-helix, β-sheet, and random coil peptides are illustrated in Figure 2. This enhancement in structural sensitivity comes at some cost in sampling flexibility (the usable concentration range becomes more restricted and solvent interference can be more difficult) and in the achievable signal-to-noise ratio (since ΔA is typically 10^{-4} to 10^{-5} of the absorbance, A).

Figure 2 Typical VCD and FTIR Solution-Phase Spectra of Helix, Sheet, and Coil Conformations for Polypeptides: (A) Alternate LKKL, (B) Alternate LK (high salt), and (C) Poly-L-Lys (neutral pH)[230][a]



[a] Reprinted from (Keiderling, *Infrared and Raman Spectroscopy of Biological Materials*, Gremlich; Yan, Eds.), Copyright (2001), p 68, with permission from Marcel Dekker.

Of course, as with any spectroscopic technique, FTIR should not be used in isolation but always in conjunction with other techniques. FTIR is most usefully paired with circular dichroism in both the UV (i.e., ECD) and, if possible, also in the IR (i.e., VCD).[64,70,73,110–112] Coupling IR to NMR studies can usefully aid interpretation of spectra from fluctional structures and can lead to some site-specific insight.[113–115] Even the generality of X-ray diffraction results (especially for linear oligopeptides whose conformation can be strongly influenced by intermolecular interactions in the crystal) can be tested by comparison of solid-state and solution-phase IR data with solution-phase CD data.[116,117]

7.7.2.4 Experimental Considerations

Fourier-transform infrared (FTIR) spectrometers encode infrared wavenumbers by moving a mirror in a Michelson interferometer which results in a unique, path-dependent pattern of interference for each light wavelength in the IR beam. FTIRs have come to totally dominate the IR market and are the means by which most of the work described in this review was accomplished. Only for some special applications (modulation spectra and time-dependence studies) are dispersive-based (scanning monochromator or tuned laser) spectrometers still used. The advantages of the FTIR approach are that the entire spectral region of interest can

be measured simultaneously (multiplex advantage), energy losses that occur due to narrow slits on a conventional scanning monochromator are minimized by use of a circular aperture (throughput advantage), and all spectra are referenced in terms of frequency to a stable helium–neon laser signal which allows highly reproducible subtraction and co-addition.[47] The end result provides the potential for obtaining exceptionally high (> 3000:1) signal-to-noise ratio spectra of normal samples and the possibility of measuring the IR spectra for samples of very low absorbance ($A \leq 0.001$ are feasible) or the subtraction of interfering solvent bands with high absorbances (e.g., $A \sim 1$ at $1650\,cm^{-1}$ with H_2O for a 6-μm pathlength cell) from sample spectra.

7.7.2.4.1 Sampling Techniques

The high instrumental stability and signal-to-noise ratio available with modern FTIR spectrometers has prompted the growth of alternative sampling techniques. Solution samples are normally contained between crystal plates (KBr for organic solvents; CaF_2, BaF_2, or ZnSe for aqueous samples) separated by a thin spacer (3.5–6 μm for H_2O, 25–100 μm for D_2O, and up to 1 mm for "IR-compatible solvents" such as CCl_4 or CS_2). These cells are widely available commercially or easily constructed in house for specific applications and can be refilled via external ports or demounted, cleaned, and refilled as best fits the experimental constraints. In these cells, aqueous samples of as low as $1\,mg \cdot mL^{-1}$ can be studied under the most ideal circumstances (more typically $\sim 10\,mg \cdot mL^{-1}$ is a minimum in H_2O) using sample sizes of ≤20 μL. Thus analyses of quantities of the order of 20–200 μg are accessible with ordinary transmission techniques. In addition, samples in solution can be dried to form films on a single plate and their solid-phase spectra can be studied. Older techniques commonly used for solids, such as grinding the sample to form a Nujol mull or a KBr pellet, are less useful for peptides and can broaden the spectral bands. In all these cases, linear peptides can, of course, change conformation when in the solid state and can interact with (adhere to) a salt plate that could result in further conformational effects. Thus results from such experiments must be interpreted with care.

A popular sampling technique for scarce and difficult to dissolve samples is attenuated total reflectance (ATR) whereby a film or solution is coated on the surface a thin zinc selenide or germanium plate into which the IR beam is directed through an edge to propagate down the long crystal axis.[79–81,118–122] As the beam transverses in the crystal it undergoes multiple internal reflections at the coated surface. On each reflection the beam develops an evanescent wave that penetrates the coating and thereby samples the absorbance of whatever is in contact with the surface, such as a solution or dried film. Since this phenomenon damps off quickly, ATR spectra will have a preferential sensitivity for those molecules closest to the surface. ATR is a good way to minimize solvent effects, but, since the peptide may denature on the surface, it poses a conformational ambiguity. Perhaps the most significant impact of ATR is one of sensitivity enhancement. Reports of obtaining IR spectra for samples as small as a few micrograms have appeared recently.[78,79,123] Another significant advantage of ATR is its natural polarization sensitivity. One can determine the alignment of transition dipoles from the relative linear polarization of the absorbance bands corresponding to oriented peptides.[76–81,124–126] If a peptide binds to the plate with a preferential orientation, the amide I and amide II bands will be polarized and that polarization will be a function of the orientation and conformation. Thus the polarization ratio can be used to determine if the peptide is parallel or perpendicular to the ATR plate surface, provided the conformation can be determined independently from, for example, the frequency. Such studies are most useful if a layer of membrane molecules is coated on the ATR crystal plate and the peptide is inserted in or bound to this layer. The relative orientation of the peptide and membrane can then be determined by correlating polarization ratios of the lipid CH-modes with the peptide ones.[78,120,127–129]

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7.7.2.4.2 Analysis Methods

Fourier self-deconvolution (FSD) is a computational method of enhancing the representation of underlying components in an overlapped transition.[61,62] It is sometimes referred to as resolution enhancement, but that is technically incorrect since the spectral resolution is set by the original experiment. (In FTIR this is ultimately controlled by the maximum displacement of the moving mirror in the interferometer during the scan.) The FSD method involves computational manipulation of the interferogram using a weighting scheme such that the higher-resolution components (corresponding to larger mirror displacements) are emphasized while the lower-resolution parts (near the center burst) are de-emphasized. Consequently, in a broadened band composed of multiple components, the resulting shoulders and inflection points are enhanced to appear as identifiable peaks. All of this is not without a penalty. In this case, the noise and, particularly, any residual water vapor bands are enhanced even more, since they are intrinsically narrow. These water bands, and even the commonly found band at 1673 cm^{-1} arising from TFA residuals,[50,74,130,131,234] which can result from the HPLC processing of peptides obtained via solid-phase synthesis, have been mistakenly assigned as due to secondary structure by some researchers. It is an unfortunate side effect of making the FSD-FTIR techniques easy to use that some users will apply them without proper care.

Second derivatives are an alternative way of identifying underlying components, ideally yielding a peak for each inflection point in the band shape.[65,67] However, since a second derivative must have oppositely signed wings, the resultant shapes are quite complex. Often the second-derivative peak positions are used to constrain the center frequencies of component bands used to fit the original band shape.[48] Such constraints are useful since the IR band of a peptide is normally so smooth as to permit multiple solutions for any band fitting exercise. It is important not to overfit an IR band profile, as reliable interpretability can be lost.

7.7.2.4.3 Polarization Methods

Most linear dichroism (LD) experiments with an FTIR can be done by simple subtraction of spectra obtained using two polarizations. Polarizers are generally commercially available, composed of an aluminum or gold grid deposited on an IR transparent substrate (e.g., BaF_2 or ZnSe), which can be mounted on a rotating holder to access various polarization orientations. Alternatively, one can modulate the polarization by physical manipulation of these elements or in conjunction with a polarization modulator [e.g., with a photoelastic modulator (PEM) in half-wave mode or with a Faraday rotator] to create a dynamic polarization-dependent signal that can be detected using phase-sensitive techniques.[126] Such experiments have become popular for two-dimensional polymer stretching analyses so that commercial manufacturers now provide all the needed components.[132,133] It is important to recognize that many components in an IR spectrometer will be polarization sensitive and that these may give rise to features that could be misinterpreted as spectral bands. Thus calibration of the instrumental response and care in establishing a baseline is vital to success in such LD measurements. LD samples can be crystals or oriented films of long polymers created by stroking or otherwise ordering the molecules during drying of the film.[5,6,134,135] For longer, more-rigid molecules, laminar flow can also be used to partially orient even solution samples. Alternatively LD can be carried out using ATR techniques, as noted above, to determine the sample orientation with regard to the ATR cell. This then becomes a function of the binding of the peptide to the cell surface unless some self-assembled monolayer, such as a lipid, is first coated on the plate and the interaction of this layer with the peptide is used to orient the sample. This latter capability is a great strength of ATR for membrane-active peptide studies. More recent studies have used the absorbance associated

with a single reflection at the air–water interface to study peptides and to use LD to determine their orientation.[136,137]

VCD experiments have been done on both dispersive- (scanning monochromator based) and FTIR-VCD instruments.[93,138–144] Both incorporate a linear polarizer combined with a PEM, operating in quarter-wave mode, to create a polarization state oscillating between effectively left and right circular states (true circular polarization is difficult to achieve dynamically and, for any set of experimental parameters, would apply to only a single wavelength, even if successful). The sample differential absorption ($\Delta A = A_l - A_r$) converts the polarization modulation into a differential intensity modulation in the beam, which is detected by a fast semiconductor diode detector, such as a mercury cadmium telluride. The modulation signal is recovered from the noise using a lock-in amplifier, with phase-sensitive detection techniques (much as in ordinary ECD spectrometers operating in the UV). The result is then processed by the instrumental electronics and an associated computer to generate the VCD spectrum. (This same approach can be used for high-sensitivity LD experiments if half-wave modulation is used and detection is at twice the modulation frequency.) Until recently all VCD measurements were made on instruments constructed by research groups,[93,138,139,145,146] but commercial versions are now available both as complete, stand-alone VCD instruments or as component systems (packaged as FTIR accessories that need to be assembled and optimized by the user). As with ECD, VCD spectra are subject to baseline artifacts, many of which arise from the non-ideal properties of the sample cell. However, these can be more severe for VCD than ECD due to the smaller signals generally measured and the need to make the cells from IR-compatible materials that are more likely to be strained than are the quartz cells normally used in the ultraviolet ECD experiments. Such artifacts can be reduced by careful optical design, but have not yet been fully eliminated in practice.[93,236–238] As a consequence, care must be taken in reproducibly assembling the sample cell, and a baseline scan of the cell plus solvent must be obtained under identical conditions as used for the sample spectrum. Unlike LD, which requires an oriented (hence solid-state or otherwise ordered) sample, VCD demands an isotropic (totally unoriented) sample and is thus best done on solution-phase peptides. Some VCD studies of films have been done by taking great care to prepare an unoriented film,[100,147,148] but these have had limited success in terms of interpretation due to interaction between peptides in the film which can lead to spectral consequences that are not molecular in origin. More interpretable results are possible with film samples of globular proteins, but these samples are often more easily studied in solution. It is perhaps useful to note that ATR techniques are incompatible with VCD as the circular polarization is altered by the multiple reflections.

7.7.2.4.4 Time-Dependent Methods

A more recent development in IR technology has been the capability of obtaining time-resolved spectral data for relatively large parts of the spectrum.[149–153] Early studies monitored the change in absorbancy at a single frequency but were often limited by the slow detector response. More recently with the development of rapid-response diode detectors and tunable IR laser light sources, time-dependent absorbance changes can be differentiated down to at least the nanosecond level.[154–158] The spectral response variation with time can be followed at single frequencies, and a pseudo spectrum can be assembled by tuning the laser through the band. Processes to be studied in this manner are often initiated photochemically or by temperature jump, which is usually obtained by means of a pulsed laser excitation.

Such time-dependent studies can also be carried out using an FTIR to obtain full, or at least broad, spectral coverage as a function of time.[153] In this case it is imperative that the time-dependent process to be studied has a cyclic character or be virtually indefinitely repeatable (e.g., T-jump or reversible photo excitation). The simplest way of doing such

for references see p 733

studies is to step-scan the interferometer and collect a decay curve (temporal response of the signal) at each step. When enough steps are accumulated to yield the desired spectral resolution, a series of time slices (using the initiating pulse as $t=0$) is taken through all the steps and from these resulting inteferograms, a series of time-dependent spectra are constructed. In this approach the time-dependent behavior of single frequencies, isolated bands, correlated bands, or the entire spectral shape can be analyzed. The signal-to-noise ratio of the kinetic profile will greatly improve with inclusion of more spectral points that have the same time dependence in any given analysis. Thus, where possible, this is the optimal technique for rapid, IR-detected, time-dependent processes and is growing in popularity.

7.7.2.5 Applications

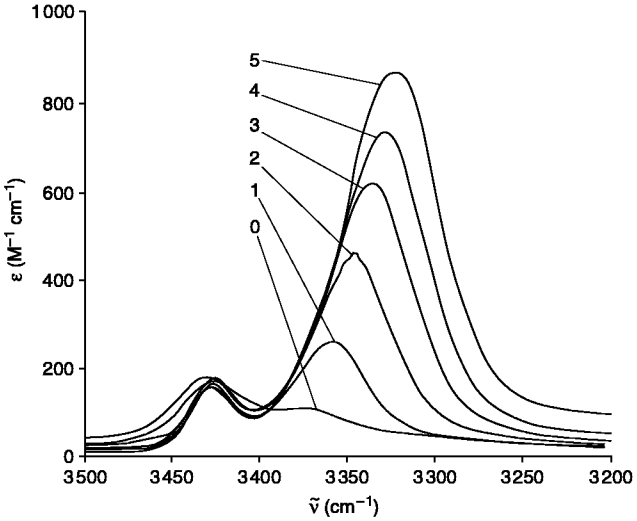
Due to the huge number and variety of peptide studies carried out using IR techniques, it is impossible to provide a comprehensive review, particularly in this format. Thus a few selected examples will be provided to give the reader just a flavor of the types of experiments done, qualities of systems studied, and insights into structure obtained using IR techniques.

7.7.2.5.1 Polypeptide Studies

Early peptide IR studies that established typical polarization and frequency characteristics for α-helix and β-sheets were actually done on proteins,[82–84,159] (see Figure 1). These structural protein systems had the advantage of being naturally oriented for polarization studies as well as having a dominant secondary structure type. Early syntheses of polypeptides used these insights to demonstrate the properties of the new polymers that were prepared. As X-ray diffraction methods developed an understanding of the solid-state structures of various polypeptides, correlation of IR bands and structural types continued to develop, indicating that the amide group indeed had characteristic frequencies that were relatively independent of the amino acids involved. It is this observation that is key to extended use of IR today, yet it is also this thinking that has led to abuse of IR analyses in peptides and protein conformational analyses.

Systematic synthesis of peptides with controlled amino acid composition and length led to some refinements of these observations and to development of alternate applications of IR spectra for peptides. For example, in aqueous solutions the amide A (N—H stretch) is not very useful, due to overlap with solvent bands, but in nonpolar environments it can be used as a probe of hydrogen bonding to the amide.[16,160,161] Amide NH groups that remain exposed only to solvent will yield a relatively sharp band near 3450 cm^{-1}. However, if the peptide forms α- or 3_{10}-helices with hydrogen bonds between residues displaced along the helix or forms β-sheets with intrastrand hydrogen bonds, or even the $i \rightarrow i+3$ hydrogen bond forming a β-turn, the amide A will dramatically broaden and gain intensity while shifting down in wavenumber to 3300–3200 cm^{-1}. This is shown in Figure 3 for a series of Aib-containing oligomers; as the peptide gets longer, a 3_{10}-helical conformation becomes stabilized forming more internal hydrogen bonds as reflected by the growing feature from 3375–3300 cm^{-1}. The small peak at 3425 cm^{-1} corresponds to the free, non-hydrogen-bonded NH. The changes in bands corresponding to these two states consequently provide a means of monitoring conformational change in nonaqueous environments. In such environments, the amide A can also be used as a monitor of the degree of H/D exchange which can be indicative of stable secondary structure formation, but this is often more practical with the amide II band, as most exchange media also contain OH or NH stretching bands which would strongly interfere in such an amide-A-based study. In addition, the amide A is overlapped and mixed with the amide B mode, which has some dependence on the amide II overtone.

Figure 3 IR Absorption in the Amide A Region for a Series of Oligopeptides, $(Aib)_n$-(L-Leu)-Aib_2 (Where n = 0, 1, 2, 3, 4, 5), in Chloroform-d_1 ($CDCl_3$) Solution[181][a]



[a] Adapted from Yasui; Keiderling; Formaggio; Bonora; Toniolo,[181] with permission.

7.7.2.5.2 Theoretical Basis

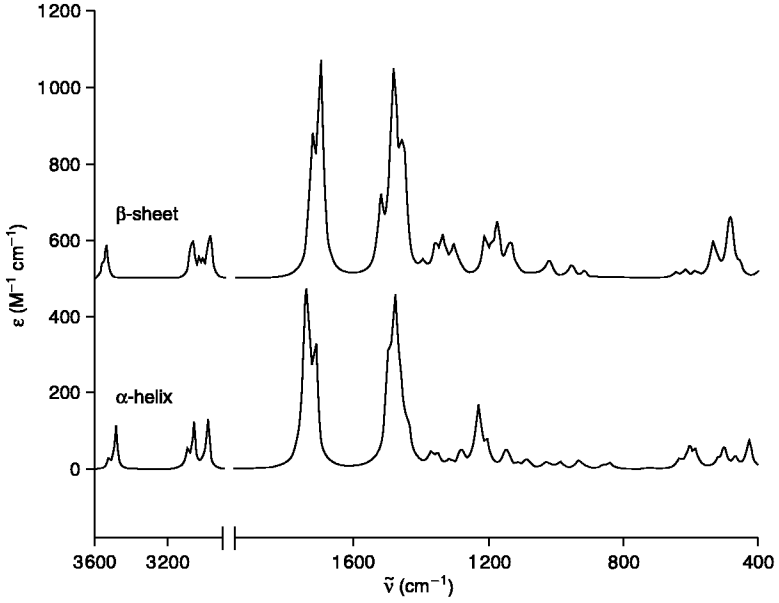
Characterization of amide vibrational modes as seen in IR and Raman spectra has developed from a series of theoretical analyses of empirical data. The designation of amide A, B, I, II, etc., modes stem from several early studies of the *N*-methyl acetamide (NMA) molecule vibrational spectra which continues to be a target of theoretical analysis.[5,27,34,162–166,239] Experimental frequencies were originally fitted to a valence force field using standard vibrational analysis techniques and subsequently were compared to ab initio quantum mechanical force field results.

It was then shown that these amide modes persisted in polypeptides, which is a result of the relative localization of vibrational motions and the repeating nature of the peptide chain in polypeptides characterized by a single secondary structure type. In addition, several amide modes exhibit reasonable independence from the nature of the residue side chains. Empirical polypeptide force-field studies, historically dominated by the work of Krimm and coworkers,[34,35,167–171] led to an improved understanding of the conformational dependence of the observed amide mode frequencies. Refinements of this approach to include dipole coupling effects sharpened the interpretations of characteristic frequencies and their variations.[35,172] More recent studies have used transferred force fields from *N*-methyl acetamide or similar small molecules that are obtained from ab initio quantum mechanical calculations.[171,173–175] These have the advantage of reducing the rather arbitrary selection of off-diagonal force constants used in traditional empirical force-field studies and also are appropriate for direct calculation of vibrational spectra for small oligopeptides constrained to a selected conformation using modern, fast, high capacity computers (work stations) with optimized quantum chemical codes. Comparison of these direct results with those obtained by transfer of force field from smaller segments demonstrates the virtues and limitations of using simple small peptides as model systems for building larger peptide force fields. While detailed calculations, for example of tripeptides, are readily available by using density functional theory,[176] such small linear peptides rarely have well-defined structures, so a direct comparison for a single conformation with experiment is not possible (see Figure 4). Recent spectral results (Raman and IR) have been shown to indicate a high degree of structure for some tripeptides under specific pH conditions.[240] These limitations are not so

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severe for cyclic peptides. It is expected that detailed force-field calculations soon will be carried out for various conformations of such structures, and these analyses will lead to improved structural applications of IR data through comparison of calculated and experimental spectra for each conformer. Such studies will probably prove to be developmentally useful but are not expected to provide complete structural applications in themselves, since the NMR of cyclic peptides is likely to be a more efficient way of determining molecular structure in solution.

Figure 4 Comparison of Computed (Ab Initio DFT/B3LYP/6–31G* Level) IR Spectra for Ala_3 Tripeptides Constrained to α-Helical and β-Sheet φ, ψ Angles[231]

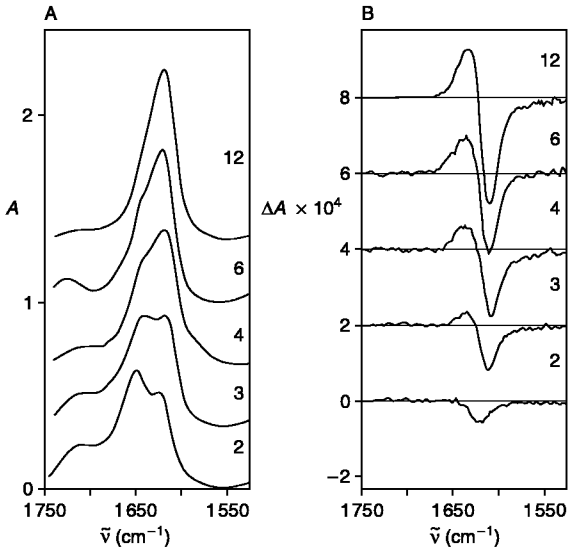


Since vibrational spectra of amide modes are not resolved, and as coupling of these modes leads to multiple components of different frequencies that yield the heterogeneous band widths observed,[172,177,241–243] and, most importantly, as even the best of these methods has considerable frequency error (~5%), such simple theoretical determination of frequencies is very limited in its application. The normal product of force-field analysis, frequencies, have primarily a relative value, even if the force-fields are scaled to correct for the error. Yet even these results make it clear that the relative frequencies of the amide I band in α-helices and β-sheets, for example, arise from the stereochemistry and associated local dipole coupling and not just from hydrogen bond strength differences. Hydrogen bond differences are undoubtedly important and may influence underlying interactions leading to characteristic antiparallel sheet frequencies seen in aggregates as opposed to those characteristic of globular proteins.[53,242] Furthermore, these bands result from a distribution of transitions whose intensities and properties reflect the coupling of near degenerate modes in the polymer. (If the polymer is viewed as a one-dimensional crystal, these transitions form an exciton band spread over a range of frequencies determined by the strength of the interaction between monomer transitions.[172,177,242,243])

Once the normal modes are computed, the nature or properties of the excited vibrational states can also be of use for structural studies. The most obvious of these is the dipole strength, directly proportional to the IR absorption intensity, which will depend on the environment and stereochemistry of the amide group. A key element in utilizing these calculational results is the use of intensities, through spectral simulation, to compare pre-

dicted and experimental bandshapes.[241–243] Natural extensions of the transition dipole moment are the polarization properties, should the peptide be oriented. Alternatively, the VCD intensities can also be computed using ab initio methods,[174,175,178,179,241–244] given that a reasonably good ground state wave function was initially determined. We have experimentally demonstrated that peptide VCD band shapes (intensity dependent) are dominated by near neighbor interactions, as illustrated in Figure 5 for $(Pro)_n$ oligomers.[98,111,180,181] [Note how as a function of length the multiple components of the amide I band coalesce into a single intense IR band (with a weak shoulder to a higher frequency). The VCD band shape is seen to be almost constant from the tetramer, indicating it is dominated by local interactions (Figure 5).] This short-range property is well-correlated with the theoretical result that VCD patterns computed for even a dipeptide reflect the dominant components of what is seen for the corresponding oligopeptides.[174,241]

Figure 5 (A) IR and (B) VCD Spectra of $(Pro)_n$ (Where n = 2, 3, 4, 6, 12)[98][a]



[a] Reprinted from (Dukor; Keiderling, *Biopolymers*), Copyright (1991), p 1752, with permission from John Wiley and Sons, Inc.

Subsequent theoretical studies have shown that when these smaller oligopeptide results are propagated onto longer peptides using a transfer function, this relationship is qualitatively maintained and has the most reliability for α-helix and Pro II helix amide I and amide II bands.[175,182,206,241–243] Hydrogen bonds can also be added to such model calculations, but preliminary results[259] indicate that they will not change the qualitative picture for the helical species beyond shifting the frequencies to more realistic values.[176,182,245,246] By contrast, such computational enhancements may be more important in distinguishing parallel and antiparallel sheets which are not distinguished in such a single peptide chain based approach.[242,260]

7.7.2.5.3 Alanine-Rich Peptides

The α-helix is perhaps the most important structural unit in the folding of proteins. However, it is not very stable as an isolated unit in aqueous solution due to the competition for hydrogen bonding with water. Thus, in globular proteins helices often pack together or with other structural elements, such as sheets, and form solvent-excluded (hydrophobic) regions.

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The corollary to this behavior is that addition of various organic solvents, e.g. trifluroethanol (TFE), acts as a helix-promoting perturbation, probably by dehydrating the peptide. There are exceptions to this behavior pattern, the most studied example of which for linear peptides (of the order of 20 residues), has been those with a very high alanine content.[183–189] This work originally stemmed from noting that the *S*-peptide (residues 1–20 of the N-terminus) of ribonuclease A had a significant amount of helical structure, even when isolated in solution.[190,191] Following that, a number of peptides, typically containing a repeated sequence of (AAXAA) where X is a charged residue, such as Lys, Glu, Arg, were synthesized in various lengths and with various termini to study helix stability.

IR studies impacted this field in a few different ways. Millhauser et al.[192–196] observed that these peptides had amide I′ (*N*-deuterated) frequencies that were abnormally low (~1637 cm^{-1} in D_2O).[192] By reference to some earlier papers that tried to draw correlations between FSD-FTIR spectra and protein crystal structures, as well as their own systematic analysis of ESR spectra for spin-labeled variants of these peptides; this was originally proposed as evidence for 3_{10}-helical formation in whole or in part, with 3_{10}-helical structures being more important for shorter variants and α-helices for longer peptides.[193–196] Later analyses recognized this low frequency as characteristic of a solvated α-helical structure, and viewed the 3_{10}-helix as a probable intermediate in the helix-coil transition.[74,158,206,247,248] Subsequent IR and VCD studies demonstrated that the longer peptides (22-mer and 17-mer) were indeed dominantly α-helical at the low temperatures and provided graphic evidence of the dangers of using frequencies alone for secondary structure determination, particularly for solvated linear peptides.[74] Comparison of data for different concentrations also implied a helix stabilization through interactions at higher concentrations. A similar effect was seen for 3_{10}-helical forming peptides in nonaqueous environments.[111] Furthermore these IR and VCD studies demonstrated an ability to detect the junction between the helix and coil segments as they unfolded with increasing temperature. Isotope labelling experiments (see Section 7.7.2.5.4) confirmed that these peptides unfold from the ends resulting in an ensemble of intermediate structures.[206] The same interpretation could be consistently applied to the whole amide I′ bandshape variations of both the FTIR and VCD spectra with temperature. Such a behavior emphasizes the short-range sensitivity of vibrational spectra to conformation. Longer-range spectral interactions, as seen in conventional CD, show only two-state behavior with no evidence of the intermediate junction structure.

In a kinetically oriented approach, Williams et al.[158] have undertaken temperature-jump (T-jump) studies of analogous alanine-rich peptides and studied their rates of relaxation to a new equilibrium structure. These studies projected a very fast (180 ns) rate for folding based on a two-state model of the relaxation to equilibrium. This conformational conversion (melting) was monitored by transient IR studies at selected frequencies using a tunable diode laser probe. These studies, and analogous ones using fluorescence probes,[197–200] demonstrate the very fast intrinsic folding kinetics for this most important (for protein folding) peptide conformation in water. Such a model discriminates between steps in the protein folding process, such as those involving diffusion of segments together to form contacts that may be required in the formation of a β-hairpin[201] or coiled-coil helices.[202] These latter processes are also important for aggregation phenomena and kinetically critical steps in global protein folding (hydrophobic collapse).

7.7.2.5.4 Isotopic Editing

IR spectra are naturally of moderate resolution; while vibrations of different types of bonds are spectrally resolved, contributions from different residues are not. Thus it is normally unrealistic to expect IR, or any vibrational spectroscopy, to yield site-specific information about peptide conformations. However, if one substitutes ^{13}C or ^{18}O for the C=O carbon or oxygen on specific residues, the amide I will show significant frequency shifts for those

labeled residues.[206,207,249] Ideally, the spectra of the whole and the isotope-edited portion of the molecule can then be studied independently.

In practice this is a bit more complex, but the spectral response of the labeled segments or residues is identifiable and can often be associated with a specific conformational type. In protein-based studies, focused on protein–protein and protein–peptide interactions, a useful approach is to uniformly label one species and not the other. Conformational changes that occur in each species upon interaction can then be separately modeled. This is analogous to labeling one type of lipid with deuterium in a mixed lipid membrane.[203] Preliminary demonstrations of the use of isotopically selected, edited FTIR for species differentiation in protein–protein interactions was made by Haris et al.[204] for two *E. coli* proteins from the phosphotransferase system. Similarly calmodulin binding to peptide sequences representing the binding domains of several proteins were studied with a similar isotopic discrimination scheme by Zhang et al.[205] These studies interpreted the binding domain to become helical while the calmodulin conformation remained unchanged. A previous study actually took this even further, to a level that can rightly be called isotopic editing, in which Braiman and Rothchild[57] labeled specific residues in bacteriorhodopsin to probe their function in the photocycle.

In a purely peptide-based study, Tadesse et al.[58] demonstrated that synthetic peptides with sequences of four residues labeled with ^{13}C on the amide C=O, could give distinct isotopically shifted bands that reflected definite conformer contributions. Subsequently, Decatur et al.[60] have prepared a series of alanine-rich peptides with specific positions isotopically labeled with ^{13}C (N-terminal, middle, C-terminal). As described in Section 7.7.2.5.3, these peptides have a high propensity for helix formation in water at low temperatures and undergo a helix–coil transition on heating. FTIR spectra of these peptides, as a function of temperature, can be interpreted to demonstrate the relative stability of the N- and C-termini to unwinding. Other studies[206] using VCD demonstrate that the substituted residues are actually helical at low temperatures and that they lose their helicity at different transition temperatures depending on the position of the isotopic substitution. These results illustrate that isotopic-editing can be developed into a true site-specific (or perhaps more realistically, segment-specific) conformational analysis tool when combined with FTIR and VCD spectroscopies.

In an alternative, creative approach to use isotope editing, Anderson et al.[207] studied linearly polarized IR spectra of cleromyrine [*c*-(GPLPGY)] crystals using an FTIR with a microscope attachment to isolate 10- to 20-μm long crystals for analysis. By ^{13}C labeling the C=O of different residues, individual peaks in the FTIR band profile could be associated with unique sites in this cyclic peptide which has two β-type hydrogen bonds between two pairs of residues. The frequencies seen showed an unusually low frequency, 1616 cm^{-1} for the ^{12}C species, for the hydrogen bonded C=O on the Leu residue. This acts as further evidence that single frequency assignments to a specific structural type can be influenced by the nature of nearby residues. LD analysis further allowed the authors to determine the orientation of each of the labeled C=O groups relative to that of the Leu residue. It was hoped that this isotope-editing–LD method could eventually be applied to the study of amyloid fibrils which are not approachable with X-ray or NMR techniques, but it may also be appropriate for peptides bound to lipids, as has already been shown for non-labeled sequences.

For an octapeptide sequence taken from the C-terminal residues of the Alzheimer's Aβ-peptide Lansbury et al. identified a large intensity enhancement for ^{13}C-labeled modes that was sequence-dependent and assessed as largely due to interstrand dipole-coupling.[207,235] Mendelsohn et al. found a similar effect by double labeling on alternate sites a peptide that formed a β-sheet-like structure in methanol.[250] Subsequent theoretical modeling showed this latter intensity enhancement to be a function of forming extended, flat, anti-parallel β-sheet structures,[251] and the overall effect to be highly position sensitive.[245]

for references see p 733

7.7.2.5.5 β-Hairpin Structures

After α-helices, β-sheets are the most prominent secondary structure feature in globular proteins. However, the most widely used tool for secondary analysis, ultraviolet ECD, is so dominated by its sensitivity to the α-helix that, at best, it can only poorly characterize β-sheet content. IR on the other hand has a good differential sensitivity to the β-sheet because of the large frequency shift (~30 cm^{-1}) from the helical band and because the extinction coefficient for model extended sheets is often higher than for other conformations. However, study of the β-sheet conformation with peptide models has long been hindered by their natural tendency to aggregate. Furthermore, no (or very few) peptide models of parallel sheets are available, to our knowledge.

Arrondo et al.[208] have studied the IR spectra of a model β-hairpin which had previously been well-characterized by NMR analyses and was known to be monomeric at 2 mM. Analysis by band fitting showed a pattern characteristic of extended antiparallel β-sheets with an intense band at 1620–23 cm^{-1} and a weak one at 1691–93 cm^{-1}. These frequencies are not typical of most globular proteins with β-sheets, but are, in fact, more like what is seen in aggregates. Differentiation from such an aggregate form was found in the lack of a deuteration shift for the 1690 cm^{-1} band, which for extended peptide aggregates shifts to ~1680 cm^{-1} in D_2O. The spectra also have contributions at ~1640 cm^{-1} which were attributed to the turn in the hairpin structure, but this region certainly must also have substantial contribution from the disordered residues resulting from the frayed ends of the hairpin. This is emphasized by the temperature dependence of the FTIR that collapses to a broad band at 1643 cm^{-1} for temperatures above 55 °C.

Blanco et al.[209] have studied the thermodynamic stability of another β-hairpin model sequence by use of NMR spectroscopy. This work was extended by Munoz et al.[201] by analysis of the T-jump-induced unfolding kinetics (with fluorescence monitoring) which revealed significantly longer time scales (μs) for bringing the two arms of the hairpin together than had previously been found for "zipping" the α-helix (180 ns).

In another approach, Nesloney and Kelly[210] have prepared a rigid biphenyl-based linker that acts as a β-sheet initiator. This rigid link lowers conformational flexibility to inhibit aggregation. In addition selected NH groups were methylated to disfavor intermolecular strand cross-linking. The result is that this peptide, while shown to be a β-hairpin with frayed ends by CD and NMR spectroscopy, gives IR absorption bands at 1635 and 1685 cm^{-1} in D_2O, much closer to the typical sort of bands seen in globular proteins. By study of different length peptides it was clear that the initiator linker element stabilized a β-form but the ends became disordered for the longer arms studied.

Stabilization of a β-hairpin structure can be achieved in two ways, promoting a stable (or restricted) turn structure (as done with mimetics) or linking the two arms either chemically, or, more naturally, by hydrophobic interactions. In an approach to utilizing both methods, a D-Pro-Gly linkage was used to stabilize a left-handed turn (type I′ or II′) and various charged and hydrophobic residues were used to stabilize the molecule and enhance the interaction between arms.[252–254] Examples of these peptides studied in nonaqueous solution by IR, VCD and NMR spectroscopy exhibit characteristics of well-formed hairpins.[255] Alternatively, in aqueous solution, IR, VCD, and ECD results for related peptides agree with the NMR interpretation of conformations characterized as hairpins stabilized at the turn and frayed at the ends.[256] These latter results also have a qualitative match with theoretical simulations. Recently, examples of hydrophobically stabilized hairpins studied by NMR spectroscopy have avoided use of a nonnatural amino acid.[257,258]

7.7.2.5.6 Amyloid Peptides

Alzheimer's disease has been attributed to aggregation, in a primarily β-sheet conformation, of a (39–42)-residue long peptide (Aβ) that is derived from proteolysis of the amyloid precusor protein, βAPP. Fraser et al.[211] studied various sequences from the supposedly water-soluble N-terminal residue segment Aβ(1–28) to determine which components promoted the aggregated conformation. IR studies showed that β-structure formation was pH dependent for the longer peptides studied, being most stable at medium pH values and least stable at high (10) and low (2) pHs. These secondary structure formations correlated with fibril formation.

Fraser et al.[212] then studied, with FTIR and electron microscopy (EM), two Aβ sequences derived from primates contrasted with that from rodents (3 residues different), to determine species specificity, and two others, to determine effects of sequence reversal and substantial substitution. The reversed and highly substituted peptide sequences did not yield peptides that form fibrils, whereas the "Dutch" variant of the human amyloid peptide which clinically correlated to a more severe disease condition had the most stable fibril formation, being observable even at high pH values where the native primate sequence does not seem to form fibrils. FTIR was used to follow the time course of aggregation with the spectra showing antiparallel β-sheet character growing with time for samples at low pH values. Normally this process took hours to days, but the rodent species exhibited such FTIR spectra after very short times. Such species differences were not observable using X-ray and EM studies. The potential importance of this is that the rodent species does not develop plaques in the same way as seen for primates and Alzheimer's patients. Fabian et al.[213] compared the "normal" and the "Dutch" mutation of the (1–42) peptide and found both of these to form β-sheet structures that evidenced FTIR spectra that were similar to those spectra of fibrils taken from an Alzheimer's patient.

Prion diseases are thought to involve a similar β-aggregation process that is instigated by the scrapie isoform (PrP^{Sc}) of the prion protein (PrP). Gasset et al.[214] synthesized peptides homologous to segments of the Syrian hamster prion protein and studied them by ATR-FTIR. While these peptides were chosen for their sequences that indicated an α-helix propensity, three of the four segments studied showed β-structure FTIR spectra and had other amyloid properties. This FTIR peptide study was instrumental in suggesting that the conversion of PrP into PrP^{Sc} involved unfolding of a helix and reformation as a sheet or aggregate structure.

Much of the work to date on amyloid-like processes has focused on study of specific sequences or on peptide fragments from proteins that have been implicated in amyloid-related diseases. Chiti et al.[215] have shown that in a combined FTIR, electron microscopy, and fluorescence study that a small α/β-protein could be induced to form amyloid-like fibrils by partial destabilization of the native fold [in their case induced by adding TFE(25%)] and by allowing sufficient time (2 weeks) for fibrils to form in a quasi-equilibrated process. This same process could be important in low pH environments such as lipozomes. The important result is nonspecificity, suggesting that many proteins may undergo amyloid fibril formation under the right conditions, but it may also support the importance of an α-proteolysis step to expose sequence residues from a longer, stable protein to the destabilizing environment.

7.7.2.5.7 β-Turns and Related Structures

A vital component of globular protein structure, yet one that is very hard to characterize, is the β-turn. These have importance in folding the protein and are often recognition sites for interactions, but have many detailed structural variations that lead to a variety of spectral signatures. Several researchers have attempted to categorize turns by detection of bands at

for references see p 733

unique frequency positions within the amide I region of the IR in protein spectra, and some others have been based on peptide model studies.

Gallo and Gellman[216] have modeled various turn configurations with depsipeptides and monitored their thermodynamic stability in chloroform with the amide A (equivalent to NH stretch) IR bands. This allows thermodynamic evaluation of the propensity of the amide to form a hydrogen bond to some part of the rest of the molecule. In this study, depsipeptides were constructed to favor α-helical (13-membered ring) hydrogen bonds yet significant β-turn (10-membered ring) formation was found as well. The relative stability study indicated β-turns to be favored over α-helical hydrogen-bond closure that is consistent with the occurrence of 3_{10}-helical turns being sometimes observed at the ends of α-helical segments in proteins. The work also demonstrated the propensity of Pro to be a helix initiator if located at the N-terminus of such a segment. This latter property may follow from a reduction in conformational flexibility for bonds to the Pro residue.

Hollosi et al.[64] have studied linear sequences with a Pro-Gly sequence to act as a potential type-II β-turn motif for several peptides in TFE, $CHCl_3$, and DMSO using FTIR and CD spectroscopy. These authors found very complex amide I FTIR spectra and attributed the presence of a band between 1640–33 cm^{-1} in the Pro-Gly containing peptides to be evidence of a type-II turn. This is based on CD results which they interpret as being consistent with a turn conformation and on structural interpretations in previous FTIR studies of cyclic hexapeptide analogues[217] which in turn are based on an assignment for 3_{10}-helix which, however, is a type-III turn. Although isolated turns may well have bands between 1640 and 1633 cm^{-1}, as noted above, assignments of 3_{10}-helical features to low frequencies are probably incorrect[17,218] as they certainly are for the alanine-rich peptides[74] and have clearly been demonstrated to be for well-defined 3_{10}-helices in these same sort of nonaqueous solvents.[101,111,181] The observed spectra do include bands at ~1670 cm^{-1} which may well be due to some components of the β-turn that was sought. Subsequent studies[112] of cyclic penta- and hexapeptides used that same assignment for β-turns to next argue that γ-turns absorb at 1620 cm^{-1}. Given the expected weak strength of the γ-turn hydrogen bond, this would be a surprising value for the amide I frequency. Again these spectra all are very complex and have evidence of high-frequency bands (1670–1685 cm^{-1}) which must also result from these turn conformations since the cyclic models have less conformational freedom. There is no doubt that the nonuniform conformation (ψ, φ angles) and hydrogen bonding pattern for the amide C=O groups in a β-turn will lead to a variety of bond strengths and thus a variety of amide I frequencies. This makes it difficult, if not impossible, to formulate a single "turn" frequency and makes it even more dangerous to use such concepts when the solvent environment changes. Nonetheless, these papers provide important baseline studies that help understand the limitations of applying FTIR to complete secondary structure analyses of proteins.

7.7.2.5.8 Membrane Peptides

One of the strengths of infrared spectroscopy is its ability to study molecules in highly scattering environments, which can cause major difficulties for ultraviolet ECD measurements, due to the longer wavelengths and alternative sampling techniques available with FTIR. This includes the ability to study peptides in membranes, attached to micelles and vesicles, and on surfaces. Several such studies have appeared and some have been very nicely reviewed by Haris and Chapman.[51] Dluhy et al.[219] have also reviewed various IR studies of monolayers, including lipid systems. The reader is referred to both of these reviews for more extensive discussions. One very important limitation of such studies is the heterogeneity in terms of the impact of solvation effects for a single sample that is intrinsic to such membrane-like environments. While the membrane surface is highly polar, or even charged, the

interior is highly hydrophobic, which of course leads to stable formation of micelles and vesicles. Thus great care must be taken in interpreting frequency shifts as they may reflect environmental and not conformational variation.

Early investigations of peptides in membrane model systems included studies of melletin[124,125,220,221] spectra and polarization properties. This water-soluble peptide is found to be structureless in solution at neutral pH but was sensitive to environmental change. The undecapeptide hormone, substance P, a member of the tackykinin family, was also found by Choo et al.[222] to be unstructured in solution at physiological pH and to aggregate at high pH or on interaction with charged lipids. These data were used as counter-evidence to a hypothesis that the membrane surface structured the peptide to facilitate interaction with the receptor.

Lins et al.[127] have studied five peptide sequences derived from the ApoB-100 protein, which is the protein moiety in low-density lipoproteins (LDC) that transport cholesterol. ApoB-100 is insoluble and binds to the surface of the LDC particle, and these selected sequences for this study have been implicated as being important in the lipid binding. ATR-FTIR studies showed the one core and three C-terminal originating sequences were mostly sheet-like in the presence of unilamellar vesicles but the N-terminal one was different, probably representing a complex mixture of conformers with some helical component. Furthermore, these workers were able to carry out ATR-LD measurements and determine the orientation of the peptide as being "oblique" to the membrane. These studies are in contrast to ultraviolet ECD results which were adversely affected by scattering artifacts.

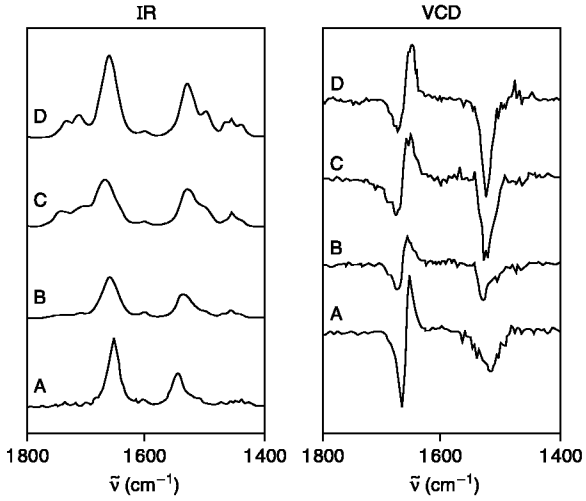
The magainin family of antimicrobial peptides are relatively small (21- to 35-residues), positively charged oligomers. The sequence suggests an amphipathic helix that might act as a membrane pore and data show they can form anion channels. Jackson et al.[223] have shown magainin-2 and two related peptides to be structureless in water at neutral pH and to become helical in basic media (pH ~10) using FTIR studies. With negatively charged liposomes they were helical but with zwitterionic phospholipids they were disordered. Cholesterol was found to induce β-sheet formation in some of them. Difficulties due to aggregation had led an earlier Raman experiment to overinterpret the influence of sheet formation on the structure. From this FTIR data it was then possible to propose a mechanism for antimicrobial action.

Angiotensin II is a linear octapeptide of importance in blood pressure control. Its active structure and mechanism have been the target of many studies. Surewicz and Mantsch[224] have studied the FTIR of angiotensin II and its Asp-deleted heptamer in aqueous and lipid-containing solutions. While disordered in water, when the peptide interacts with an acidic phospholipid, an ordered β-strand and β-turn are formed for the octopeptide, but no β-strand is evident for the heptamer.

Alamethecin is a 19-residue antibiotic that has been widely studied as an example peptaibol with membrane channel forming activity. Crystal structures have shown the pure peptide to be a bent helix (due to proline) with some 3_{10}-helical content.[225] FTIR studies[226] have found a dominant 1662 cm^{-1} band for the peptide in a membrane that was attributed to partial 3_{10}-helix formation. This was done by analogy to studies on Aib-containing peptides[17] such as $(Aib\text{-}Ala)_n$ oligomers dissolved in $CDCl_3$ and in phospholipid vesicles which were used to implicate a difference in α- and 3_{10}-helical content in the two environments. However, since such studies depend on frequency assignments for two different media, they must be interpreted with caution. VCD studies on these same peptides in $CDCl_3$ show that the helical content changes with the length of the peptide. As for similar length α-methylated peptides, the heptamer has a predominantly 3_{10}-conformation, while the dodecamer has a mixed α- and 3_{10}-helical structure[227] as is illustrated in Figure 6 by the comparison of IR and VCD spectra for these peptides with those of α-helical and 3_{10}-helical model peptides.

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Figure 6 Comparison of IR and VCD for (A) α-Helical, (L-Met_2-L-Leu)$_8$, (B) Mixed Structure, (Aib-Ala)$_6$, (C) 3_{10}-Helical, Ala(Aib-Ala)$_3$, and (D) 3_{10}-Helical, Aib_5-L-Leu-Aib_2, in Chloroform-d_1 Solution[230],[a]



[a] Reprinted from (Keiderling, *Infrared and Raman Spectroscopy of Biological Materials*, Gremlich; Yan, Eds.), Copyright (2001), p 70, with permission from Marcel Dekker.

Most of the above membrane-oriented studies were carried out for peptides in multilayer systems that were collapsed or transferred onto a sample cell surface. An alternative and very interesting way to study membrane systems is by IRRAS (infrared reflection absorption spectroscopy) at the air-water interface. In this way, unilamellar systems can be studied as a function of surface pressure and under the influence of various membrane proteins and peptides added. Mendelsohn et al.[136] have studied a model series of peptides, [$K_2(LA)_n$] (n = 6, 8, 10, 12), in nonaqueous (solution), multilamellar (lipid), and unilamellar (peptide-IRRAS) conditions. In the multilamellar vesicles these peptides are predominantly helical in conformation, but as peptide only monolayers on a D_2O subphase the conformation is β-sheet like, at least initially. For different lengths, the peptides show variable surface pressure sensitivity to development of some helical component. These authors further use their IR data to hypothesize the existence of the less-usual parallel β-sheet conformation in these peptides. A critical comparison is available for different secondary structures as detected using the IRRAS data for peptides on H_2O and D_2O subphases.[137]

Unilamellar lipid layers provide a good model of the lipid-layer lining of the alveoli in the lungs. Hydrophobic surfactant proteins are present in these single-layer membranes and provide a needed modification of the surface tension. These proteins are essentially peptides, being of relatively low molecular weight. Flach et al.[228] have shown one of these surfactant proteins, dSP-C, to be substantially helical in the monolayer suggesting that it inserts into the membrane. This is consistent with the increased angle of tilt for the helix with respect to the lipid hydrocarbon chain in the monolayer as compared to a lipid bilayer system as was determined from the LD measured for these state-of-the-art IRRAS lipid–peptide spectra.[229]

7.7.2.6 Conclusions

Infrared spectroscopy has been an important part of peptide structural analysis for ~50 years now. From a rather basic beginning, applications have blossomed to encompass secondary structure analysis, polarization phenomena, membrane conformation, and orientation, and have extended to time-dependent conformational folding mechanisms. Questions have evolved from basic polymer chemistry to issues centered on peptides involved in socially

important human disease issues. All these have had IR-based studies in the background providing a conformational foundation to the structure interpretation. Over this period, it has been the sampling flexibility in terms of state of matter, time scale, and sensitivity to molecular structure that has permitted IR to have this important supplemental role in peptide conformational analysis. Continued technological developments assure that this role will continue to develop in the years to come.

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7.7.3 Circular Dichroism

R. W. Woody

Circular dichroism (CD) is a spectroscopic method that is uniquely sensitive to conformation. CD has therefore been widely applied to the conformational analysis of peptides.[1,2]

CD is closely related to optical rotation and its wavelength dependence, optical rotatory dispersion (ORD). Linearly polarized light consists of two circularly polarized beams of equal intensity, i.e. right- and left-circularly polarized light, denoted rcp and lcp, respectively.[3] In circularly polarized light, the electric vector executes a full rotation about the direction of propagation over the distance of one wavelength, λ, or the time of one period (λ/c, where c is the velocity of light). In rcp, the electric vector rotates clockwise as seen by a fixed observer looking toward the source. In a snapshot, the tip of the electric vector of rcp follows a right-handed helical path. For lcp, an observer looking toward the source sees the electric vector rotate counterclockwise, and a snapshot of lcp reveals a left-handed helical path for the electric vector.

Circularly polarized light is chiral and therefore rcp and lcp will interact to differing extent with chiral objects, i.e. a chiral medium will exhibit differing optical properties for rcp and lcp. CD is the difference in absorbance (A) of rcp and lcp:

$$\Delta A = A_l - A_r = (\varepsilon_l - \varepsilon_r) \cdot C \cdot l = \Delta\varepsilon \cdot C \cdot l \quad \text{eq 1}$$

where $\Delta\varepsilon = \varepsilon_l - \varepsilon_r$ is the molar CD, the difference in molar extinction coefficients for lcp and rcp; l is the path length in cm; and C is the molar concentration. CD is often reported as molar ellipticity, referring to the way in which CD was originally measured, as the ellipticity induced in a linearly polarized light beam after passing through an absorbing chiral sample. The molar ellipticity, $[\theta]$, is directly proportional to the molar CD:

$$[\theta] = 3298\Delta\varepsilon \quad \text{eq 2}$$

In reporting CD spectra of peptides and proteins, especially in the far UV where the peptide groups dominate the spectra, it is customary to give the CD or ellipticity per mole of amino acid residues or, in some cases, peptide groups. In the near UV, data are sometimes reported on a per residue (or peptide) basis, and sometimes per mole of peptide or protein. It is important for authors to clearly specify the basis for reported CD spectra, and the reader must be careful to note this.

Throughout this section, amino acid names without a configurational designation are to be understood as referring to the L-isomer.

7.7.3.1 Measurement of Circular Dichroism

Most modern CD instruments[4] use a modulation technique to measure the very small difference between ε_l and ε_r. (Typically, $\Delta\varepsilon/\varepsilon \approx 10^{-3} - 10^{-5}$.) A 150- or 450-W xenon arc is used as the light source, providing a continuous spectrum from the edge of the vacuum UV into the IR. The light passes through a double monochromator and a polarizing prism. The plane-polarized light is then incident upon a photoelastic modulator,[5] which is a transparent plate rendered birefringent by stress induced by mechanical coupling to a piezoelectric quartz crystal. The amplitude of the quartz crystals vibration is determined by an applied alternating current (AC) field (ca. 50 KHz) adjusted in magnitude to make the modulating plate a quarter-wave plate ($\lambda/4$ plate) at the extremes. The light emerging from the modulator is generally elliptically polarized, but at one extremum of the applied field, where the $\lambda/4$ condition is met, the light is rcp and at the next extremum it is lcp. The light then passes

for references see p 761

through the sample and on to the detector, a photomultiplier tube. If the sample exhibits CD, the light intensity reaching the photomultiplier will vary sinusoidally with a non-zero mean value, and the photomultiplier signal will consist of an AC component and a direct current (DC) component. The AC and DC components are separated, and the AC component is amplified by a lockin amplifier tuned to the modulator frequency. The AC/DC ratio is directly proportional to ΔA, the CD of the sample.

As implied by this description, CD measurements by the modulation method are not absolute. The instrument must be calibrated using a standard well-known CD.[2,6–8] Due to slow changes in the amplifier and other components, CD instruments must be calibrated regularly.

Currently, CD spectrophotometers are produced by four companies: Protein Solutions, formerly Aviv (Lakewood, NJ, USA); JASCO (Tokyo, Japan and Easton, MD, USA); Jobin–Yvon (Longjumeau, France); and OLIS (Bogart, GA, USA). In addition, CD accessories for stopped-flow instruments are available from Applied Photophysics Ltd. (Leatherhead, UK) and Biologic (Grenoble, France). All of these instruments use the modulation method, except for the OLIS. The OLIS instrument uses a method called the digital subtraction method. Both of the two diverging orthogonally polarized beams emerging from the linear polarizer pass through the photoelastic modulator, the sample, and then on to separate photomultipliers. The digitized signals from the two photomultiplier tubes are combined in such a way as to yield the CD signal directly and to cancel the contributions of noise. In this system, absolute CD is measured, and no calibration is required.

Those contemplating the purchase of a CD instrument should compare the various instruments by submitting representative samples to be run. It is best to visit as many as possible of the vendors or to go to laboratories with well-maintained instruments to see them in operation and participate in running samples.

The amount of material required to acquire a CD spectrum is quite modest, comparable to the amount needed to run an absorption spectrum. The total absorbance of the sample (solute, solvent, and cell windows) should not exceed 1.0 throughout the spectral region to be studied. A maximum absorbance of 0.87 gives the optimal signal-to-noise ratio.[2] Water is an excellent solvent for CD measurements because of its excellent UV transparency, permitting measurements down to about 185 nm in a 1 mm path length cell. Fluorinated alcohols such as TFE and HFIP are even better. Among the other common organic solvents used for peptides, MeCN is best. Some common salts and buffer components seriously interfere with measurements in the far UV. Phosphate buffers are best, contributing little absorbance in the region of interest. Tris and related buffers should be kept at 10 mM or less. Likewise, chloride salts should be minimized or replaced by fluorides, perchlorates, or sulfates, if possible. Tabulations of the lower limits of wavelength achievable with various solvents and buffer components are available.[2,8,9]

To acquire CD data over a substantial wavelength region, it is generally necessary to use cells with different path lengths. Cells with 10- and 1-mm path lengths are convenient and easy to fill and clean, but can not generally be used below 200 and 190 nm, respectively. For shorter wavelengths, and for more strongly absorbing solvents at longer wavelengths, cells with path lengths of 0.1 or 0.05 mm are necessary. With such cells, it is necessary to use higher solute concentrations, but the total volume needed is smaller.

More details on CD measurements and on maintenance of CD spectrophotometers can be found in refs[1,2,4,8,10].

7.7.3.2 Chromophores

7.7.3.2.1 Peptides

The most abundant chromophore in peptides is the amide group. The excited states of the amide group are now well-characterized, both experimentally[11–14] and theoretically.[15–22] The two low-energy amide excited states that are important for interpreting the CD spectra of peptides are the nπ* and the ππ* (NV_1). The nπ* transition is weak in absorption ($\varepsilon \approx 100\,M^{-1}\cdot cm^{-1}$), but has a large magnetic dipole transition moment and therefore can give rise to strong CD bands. The wavelength of the nπ* transition typically ranges from 210 to 230 nm, depending on the extent and strength of hydrogen bonding to the carbonyl oxygen, with the most strongly hydrogen-bonded amides having $\lambda_{n\pi^*} \approx 210$ nm. The ππ* transition is strong in absorption ($\varepsilon_{max} \approx 5\text{–}10 \times 10^3\ M^{-1}\cdot cm^{-1}$) and occurs at ~190 nm for secondary amides and at ~200 nm for tertiary amides, e.g. in Xaa-Pro peptides. Hydrogen bonding has a relatively small effect on the NV_1 transition wavelength.

The nπ* transition acquires CD intensity (rotational strength) through two mechanisms, both involving mixing with the allowed ππ* transition. The local electrostatic field at the amide group causes the nπ* and ππ* excited states within the amide group to mix. This is called the one-electron effect or static-coupling mechanism.[23] The nπ* transition can also mix with the ππ* transitions in other nearby amide groups, or with allowed transitions in other types of chromophores, such as aromatic side chains. This is called the μ-m mechanism.[24] In all cases, the transition with which the nπ* transition mixes acquires CD intensity of equal magnitude but opposite sign.

The ππ* transition in an amide acquires rotational strength from nπ* transitions through the one-electron and the μ-m mechanisms, as just described. It also can interact with ππ* transitions in other groups in the coupled-oscillator or dynamic-coupling mechanism.[25] A special case of the coupled-oscillator mechanism arises from mixing of ππ* transitions in equivalent peptide groups. This is called the exciton mechanism[26] and leads to a splitting of the ππ* band (exciton splitting), producing both positive and negative CD components.

Methods for predicting the CD of molecules with a known (or hypothesized) geometry are available.[27–30] Several reviews[24,31–35] provide further discussion of the various mechanisms and other theoretical aspects.

7.7.3.2.2 Aromatic Side Chains

The three aromatic amino acids (Phe, Tyr, Trp) have side-chain groups corresponding to the benzene, phenol, and indole chromophores, respectively. The spectroscopic properties of the ππ* transitions in these chromophores have been reviewed.[36–38] Coupling of aromatic transitions among themselves and with peptide transitions can give rise to CD bands in the near and far ($\lambda < 250$ nm) UV. Near-UV CD bands are useful indicators of the environment of the aromatic chromophores and can frequently be assigned to specific types of side chain, based upon band position, presence of vibrational fine structure, etc. Far-UV CD bands due to aromatic side chains, except for the L_a band of Tyr (~230 nm) and the B_b band of Trp (~225 nm), are generally difficult to resolve from peptide CD bands and can complicate the conformational analysis of peptides.

7.7.3.2.3 Disulfides

The disulfide group has two nσ* bands in the near UV that are weak in absorption but can give rise to substantial CD bands. The two n orbitals arise from lone pairs on the sulfurs, and the σ* orbital is the antibonding orbital formed from the same σ orbitals responsible for the S–S bond. When the disulfide dihedral angle $\chi_{ss} \approx 90°$, the preferred dihedral angle for an

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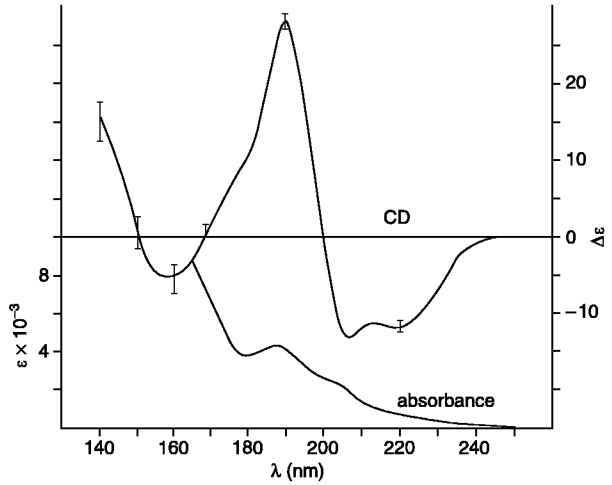
unstrained disulfide, the two nσ* excited states are degenerate, and therefore the two nσ* transitions appear as a single broad band near 260 nm. As the dihedral angle is increased or decreased from ~90°, the two transitions move apart in energy and two separate bands are observed. The disulfide group is an example of an inherently chiral chromophore. The preferred *gauche* conformation has no plane or center of symmetry, so disulfide transitions give rise to CD even in the absence of perturbing chromophores. The two nσ* transitions have CD of opposite sign and approximately equal magnitude. Thus, for $\chi_{ss} \approx 90°$, they cancel. For smaller or larger dihedral angles one observes either a positive or a negative couplet centered near 260 nm. (A couplet[24,31] is a pair of closely spaced CD bands of opposite sign. The sign of the couplet is that of the long-wavelength component.) The sign of the couplet is positive for a right-handed disulfide (*P* chirality) and negative for a left-handed disulfide (*M* chirality), if $\chi_{ss} < 90°$.[39–41] The reverse is the case for $\chi_{ss} > 90°$. Thus, the absolute configuration of a disulfide can be assigned from the sign of the 260 nm couplet, if χ_{ss} is known approximately. However, cases are known[42] in which coupling of the disulfide nσ* transitions with other chromophore in the molecule reverses the sign of the long-wavelength band, which would lead to an erroneous assignment of chirality.

7.7.3.3 Secondary Structural Characterization

7.7.3.3.1 α-Helix

The α-helix is the best-defined and most thoroughly investigated type of peptide secondary structure. It also has the most intense and characteristic CD spectrum, shown in Figure 1.[46] The long-wavelength negative band ($\lambda_{max} \approx 220$ nm) is due to the amide nπ* transition.[43,44] The negative band at ~207 nm and the positive band at ~190 nm are due to exciton splitting of the amide ππ* transition.[26,44,45]

Figure 1 Absorption and CD Spectra of α-Helical Poly(GluOMe) in HFIP[46a]



[a] Reprinted from (Johnson; Tinoco, *Journal of the American Chemical Society*, Vol. 94), Copyright (1972), p 4389, with permission from The American Chemical Society.

The α-helix CD spectrum is remarkably insensitive to solvent. One apparent exception is poly(Ala) in HFIP.[47] The CD amplitude for poly(Ala) in HFIP is only about half that of typical α-helical polypeptides, the CD bands are shifted several nm to the blue, and the long-wavelength nπ* band is observed as a shoulder rather than a discrete band. These features are unique to poly(Ala), as other helix-forming polypeptides exhibit normal CD spectra in

HFIP. These changes in CD as well as alterations in IR and UV absorption spectra are explained[47] by a modified α-helix conformation in which the amide planes, parallel to the helix axis in the standard Pauling and Corey geometry,[48] are tilted significantly to permit the carbonyl oxygens to accept hydrogen bonds from the solvent HFIP as well as the NH group of the helix. Poly(Ala) is unique in assuming this conformation because of its small side chain.

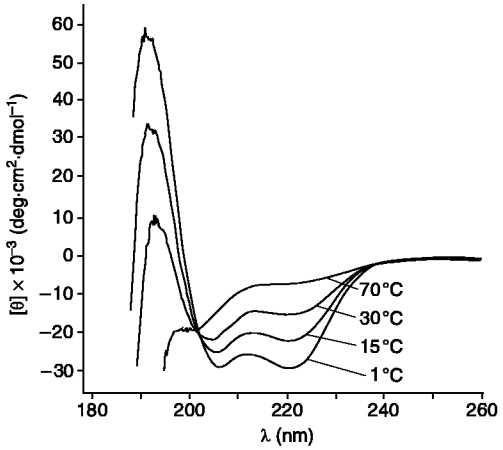
The α-helix CD spectrum is also largely unaffected by the nature and sequence of the amino acid residues, except for those with aromatic groups. α-Helical poly(Tyr)[49,50] and poly(Trp),[51] for example, have CD spectra that are not recognizable as α-helical. α-Helices with 5–10% aromatic residues, typical for globular proteins, are expected to show at most minor perturbations of the α-helix spectrum. N-Terminal Tyr and Trp residues on an Ala-rich heptadecapeptide under conditions favoring α-helix formation have $\Delta[\theta]_{220}$ of +5000 $deg \cdot cm^2 \cdot dmol^{-1}$ and −3000 $deg \cdot cm^2 \cdot dmol^{-1}$, respectively, relative to the reference peptide with N-terminal Ala.[52] The same type of peptide with a C-terminal peptide group gave a similar difference relative to the appropriate reference peptide. Curiously, the peptide with Tyr at both the N- and C-termini gave the same $\Delta[\theta]_{220}$ value as the peptide with only the C-terminal Tyr.

A method has been proposed for correcting for side-chain contributions to the CD spectra of synthetic peptides that should be useful in studies of helix-unordered equilibria.[53] The method consists of measuring a series of spectra of homologous peptides with and without the side-chain chromophore, at various stages in the helix-unordered transition. The data for peptides lacking the side-chain chromophore are analyzed by singular value decomposition,[54] a procedure that generates a series of basis spectra, a small subset of which can be used to regenerate the full set of spectra within experimental error. These basis spectra are fit to a region of the CD spectrum of the peptide with the chromophoric side chain that is not expected to be affected by the chromophore, the 195–210-nm region. The fitting coefficients are then used with the basis spectra to generate a spectrum of the chromophoric peptide corrected for the side-chain contributions. The family of spectra thus generated can then be analyzed by the usual procedures, but the results should not be influenced by contributions from the side-chain chromophores. The effect of using just one model peptide with an aliphatic amino acid replacing the aromatic or sulfur-containing amino acid(s) was examined and the results obtained are nearly identical with those using all 14 nonchromophoric amino acids as replacements. Although one can argue that the 195–210 nm region is not really devoid of aromatic contributions,[38] this region seems to be the only one suitable for curve-fitting of the basis spectra, and the aromatic contributions in the 210–230 nm region seem to be more significant.

Accurate estimation of α-helix content in peptides is important in many applications, especially studies of α-helix stability as measured in helix–coil equilibria.[55–58] Due to the large difference in CD of the α-helix and the unordered ("random coil") conformation (see Section 7.7.3.3.7), CD is the method of choice for monitoring the equilibrium between the helical and the unordered form of peptides as a function of temperature, pH, and solvent composition. As shown in Figure 2 for a thermal transition, interconversion of helix and unordered conformations generates a family of CD spectra with an isodichroic point near 203 nm, characteristic of a two-state equilibrium. The difference between the α-helix and unordered conformation is generally maximal near 222 nm, corresponding to the long-wavelength negative band of the α-helix. Therefore, this wavelength is commonly used to follow the equilibrium.

for references see p 761

Figure 2 CD Spectra of Ac-A(EAAAK)$_3$A-NH$_2$ (0.01 M NaF, pH 7) at Four Temperatures[59][a]



[a] Reprinted from (Marqusee; Baldwin, *Proceedings of the National Academy of Sciences of the United States of America*, Vol. 84), Copyright (1987), p 8898, with permission from the authors.

If the helix and unordered forms are the only forms present, as indicated by an isodichroic point, the fraction of helix, f_α is given by:

$$f_\alpha = ([\theta]_{222,obs} - [\theta]_{222,u})/([\theta]_{222,\alpha} - [\theta]_{222,u}) \quad \text{eq 3}$$

where $[\theta]_{222,obs}$ is the measured ellipticity at 222 nm, and $[\theta]_{222,\alpha}$ and $[\theta]_{222,u}$ are, respectively, the ellipticity of the α-helix form and unordered form. Typical numerical values have been proposed[60] for the fully helical and fully unordered conformations of −32640 and −2340, respectively, giving the equation:

$$f_\alpha = -([\theta]_{222} + 2340)/30300 \quad \text{eq 4}$$

This simple equation still provides a useful way of estimating helix content of a peptide, but more exact parameters are needed for precise work.

In working with peptides, it is important to take into account the dependence of the α-helix CD upon the length of the helix, i.e. upon the number of peptide groups, r. This dependence results from end effects: amide groups at each end of the helix have a different environment and therefore a different CD from those in the middle of the helix. An empirical equation has been proposed[61,62] that can be written in the form:

$$[\theta_r]_\lambda = [\theta_r]_\lambda^\infty (r - k_r)/r \quad \text{eq 5}$$

where $[\theta_r]_\lambda$ is the molar ellipticity per amide group of a helix containing r amides at wavelength λ; $[\theta_r]_\lambda^\infty$ is the molar ellipticity per amide of an infinite α-helix at the same wavelength; and k_r is an empirical constant that corrects for end effects. It can be interpreted as the number of amides that are "missing" because of end effects, and its value depends on the CD band that is used. It has been shown[62] that equation 5 fits predicted ORD[44] and CD[63] data as a function of helix length. From fitting CD spectra of several proteins of known structure, values of $k_r = 3.5$ and $[\theta_r]_{222}^\infty = -37300$ deg · cm^{2}dmol^{-1} have been proposed.

It should be noted that a different formulation of equation 5 has been used[62] framed in terms of the number of amino acid residues, n, rather than the number of amides:

$$[\theta]_{222} = [\theta]_{222}^\infty (n - k_n)/n \quad \text{eq 6}$$

Here, the molar ellipticities are calculated per residue, which is the usual practice in peptide and protein work, and k_n is the number of "missing" residues. For a peptide without blocking groups, the number of amides is one less than the number of residues, $r = n - 1$, and $k_r = k_n - 1$. If one terminus is blocked to form an additional amide, e.g. N-acylation or C-carboxyamidation, r and n are equal and so are k_r and k_n. If both termini are converted into amides, $r = n + 1$ and $k_r = k_n + 1$.

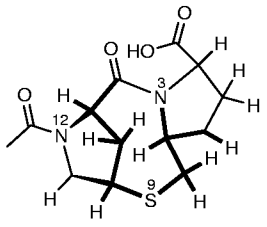
Using[64] the calculations of Manning and Woody[65] values of $k_r = 4.6$ were derived for the nπ* rotational strength. From experimental data on short peptides which are expected to be nearly 100% helical, $k_r = 4.7$ and $[\theta]_{222}^{\infty} = -40\,000$ deg·cm^2·dmol^{-1}. A value of $[\theta]_{222}^{\infty} = -42\,500 + 125t$ deg·cm^2·dmol^{-1} has been used,[57,66] where t is the temperature (°C), for peptides with the repeating sequence (AEAAKA), and a slightly different value of $[\theta]_{222}^{\infty}$ is appropriate for peptides based upon the repeating sequence (AAKAA).[67] For k_r a value of 3 was recommended, corresponding to the number of peptide carbonyls in a complete helix that are not engaged in hydrogen bonds within the helix. The values for $[\theta]_{222}^{\infty}$ are near $-40\,000$ deg·cm^2·dmol^{-1}, which agrees well with the body of data from studies of homopolypeptides in a variety of solvents.[33]

CD spectra have been reported[68] for a series of peptides with the sequence AcHel-$(AAAAK)_n$AA-NH_2 with n = 1–5. AcHelOH = (1*S*,4*S*,7*S*,10*S*)-12-acetyl-2-oxo-9-thia-3,12-diazatricyclo[8.2.1.0^{3,7}]tridecane-4-carboxylic acid (**1**, Scheme 1) is a template developed by Kemp's group[69,70] that serves as a nucleus for α-helix formation because in one of its limited number of conformations, three amide carbonyls have the same relative geometry as three neighboring peptide carbonyls in an α-helix. In a 16 mol% ethylene glycol/water mixture, pH 1, 0.2 M sodium perchlorate, $[\theta]_{222}$ values of about $-50\,000$ deg·cm^2·dmol^{-1} at 0 °C were observed and nearly $-60\,000$ deg·cm^2·dmol^{-1} at −20 °C. Interestingly, the 208 nm band does not undergo the intensity enhancement observed for the nπ* band. The CD data do not extend below 200 nm, and therefore the behavior of the 190 nm positive band is not known. It was reported that similar spectra are observed for this family of peptides in 5 M NaCl, pH 1 aqueous solution and in water/TFE mixtures (composition unspecified) at pH <2. It is possible that the remarkable enhancement of the 222 nm band in this system results from unusually strong static-field mixing of the nπ* and ππ* transitions in the peptide groups produced by folding of the lysine side chains against the barrel of the helix. It has been proposed[71] that in N-template-nucleated peptides with the sequence $(Ala)_n$-Lys-$(Ala)_n$, the lysyl ammonium group at position i interacts with the carbonyl oxygen at residue $i - 4$, and this interaction is reinforced by hydrophobic interactions between methylene groups in the Lys side chain and the methyl group of the Ala residue at $i - 3$, for which there is support from observed NOEs. The small size of the Ala side chain is probably essential for the R^1-NH_3^+–carbonyl oxygen interaction, and such an effect would be precluded by the presence of larger side chains.

An alternative explanation for the unusually large magnitude of $[\theta]_{220}$ reported for the template-nucleated peptides AcHel-$(AAAAK)_n$AA-NH_2[68] has been proposed. Hirst has suggested that the large intensity results from unusually short hydrogen bonds in the α-helix, induced at low temperatures by the co-solvents ethylene glycol and TFE.[217] Calculations of $[\theta]_{220}$ as a function of hydrogen bond length gave values near -4×10^4 deg·cm^2·dmol^{-1} for O–N distances near the normal value of 3.0 Å, but the ellipticity became more negative for shorter hydrogen bonds, with values near -5×10^4 deg·cm^2·dmol^{-1} at an O–N distance of 2.7 Å. Significant strengthening of interpeptide hydrogen bonds in the presence of polar solvent components has been predicted by ab initio calculations.[218]

for references see p 761

Scheme 1 (1*S*,4*S*,7*S*,10*S*)-12-Acetyl-2-oxo-9-thia-3,12-diazatricyclo[8.2.1.$0^{3,7}$]tridecane-4-carboxylic Acid



1

Bierzynski et al. have shown that the four peptide groups at the C-terminus of a 12-residue peptide based on a Ca^{2+}-binding loop from calmodulin adopts an α-helical conformation when the peptide binds La^{3+}.[219] This observation provides a different approach to nucleation of short, stable α-helices. Chin et al. have determined the CD spectra of α-helices with 4, 8, and 11 peptides based on the La^{3+} complexes of the Ca^{2+}-binding peptides with C-terminal helices of 3, 7, and 10 residues.[220] The most striking result is that the α-helix CD pattern, including the double minimum, persists even with four amides, which is barely more than one turn of the helix. This contradicts earlier theoretical predictions[44,45,63,65] that the CD of short α-helices should differ qualitatively from the spectra of long helices. However, calculations[220,221] using the more recently determined[14] ππ*-transition moment direction of the peptide group are consistent with these newer results. Earlier calculations used the transition moment direction for primary amides,[222] which differs from that for secondary amides by ~15°.

7.7.3.3.2 3_{10}-Helix

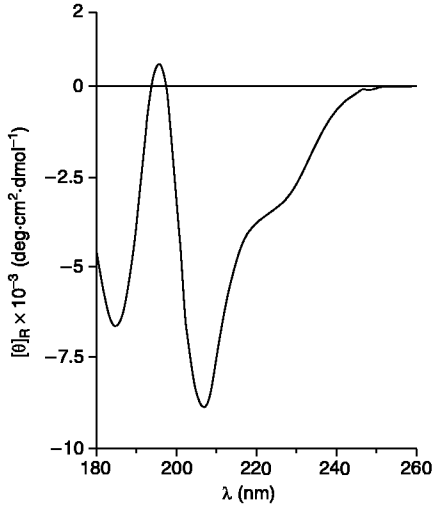
The 3_{10}-helix is an alternative helical conformation that has been observed in many peptide and protein structures. Whether the α- and 3_{10}-helix can be distinguished by CD has been a long-standing question.[44] In one study, NMR was used to determine 3_{10}-helix character in a number of Aib-containing peptides, based on the number of NH bonds involved in intramolecular hydrogen bonds.[72] CD spectra for these peptides showed negative bands at about 220 and 205 nm. The $[\theta]_{220}/[\theta]_{205}$ ratio ranged from about 0.3 to about 0.85. (In the remainder of this section, the ratio will be referred to as $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ because various wavelengths ranging from 205 to 208 nm have been used for the parallel exciton component of the ππ* transition and the wavelength of the nπ* transition has been taken to be either 220 or 222 nm.) No correlation was found, however, between the $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ ratio and the 3_{10}-helix content assessed by NMR, leading to the conclusion that CD cannot be used to distinguish between α- and 3_{10}-helix. It should be noted that the NMR spectra of the peptides were measured in $CDCl_3$ and DMSO, whereas the CD spectra were determined in TFE and MeOH. Since the balance between α- and 3_{10}-helix in Aib-rich peptides appears to be a delicate one, solvent-induced conformational changes might be responsible for the poor correlation. In addition, the $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ ratio can be low because of incomplete helix formation.

The CD spectra of 3_{10}-helices have been predicted using three different geometries.[65] For all three geometries, the nπ* band and the positive 190 nm band were predicted to be substantially weaker than those for the α-helix. In two of the three cases, the 205 nm band was predicted to be stronger than in the α-helix case. Thus, these calculations predict that the $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ ratio should be less than one.

The CD spectrum of an authentic 3_{10}-helix has been reported for Ac(α-MeVal)$_8$-OtBu in TFE (Figure 3).[73] This shows a ratio of $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ of 0.4 and a very weak positive band near 190 nm, both in agreement with theory.[65] Therefore, a criterion based upon the ratio of $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ appears to be reasonable for distinguishing between 3_{10}- and α-helix: a ratio near

0.4 is characteristic of 3_{10}-helix, whereas a ratio near 1 is indicative of an α-helix. However, this criterion must be used with caution,[74] since mixtures of α-helix and unordered conformations have $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ ratio less than one. Variation of temperature, pH, or solvent may be useful in distinguishing the two sources of low $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$. If an isodichroic point is observed near 203 nm, a ratio much less than one should be attributed to a helix–coil equilibrium rather than an α-helix–3_{10}-helix equilibrium.

Figure 3 CD Spectrum of Ac-(αMeVal)$_8$-OtBu in TFE Solution[73]a



a Reprinted from (Toniolo; Polese; Formaggio; Crisma; Kamphuis, *Journal of the American Chemical Society*, Vol. 118), Copyright (1996), p 2744, with permission from The American Chemical Society.

The question of CD criteria for distinguishing α- and 3_{10}-helix acquired additional urgency with the report[75–77] that Ala-rich peptides of the type used extensively by the Baldwin group and others for studying α-helix stability may form predominantly 3_{10}-helices. This conclusion was based upon studies of line broadening in doubly spin-labeled Ala-rich peptides. CD spectra of the spin-labeled peptides had the typical double minimum characteristic of α-helices with $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*} \approx 0.76$–$0.94$. Since conditions had not been varied, the low ratios in some cases may have been due to incomplete helix formation. In fact, the helix content for the peptide that gave the lowest ratio calculated using equation 5 and the parameters of Rohl and Baldwin,[57] was only 50%, and that for the peptide giving the highest ratio was 72%. It is therefore likely that the low $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ ratios are due primarily (or solely) to incomplete helix formation.

An EPR study[78] has been carried out on doubly spin-labeled Ala-rich peptides, using much less flexible spin labels; this study concluded that these peptides are predominantly α-helical in character and argued that a previous study[75] was misled by using spin labels that were too flexible. The CD spectra of the peptides in this later work[78] are of somewhat higher amplitude and have $[\theta]_{n\pi^*}/[\theta]_{\pi\pi^*}$ slightly greater than one. Using equation 5 with $[\theta]_{222}^{\infty} = -40\,000$ deg·cm^2·dmol^{-1} and $k_n = k_r = 2.5$, the helix content was found to be 88–98%.

Monte Carlo and molecular dynamics calculations predict that, in aqueous solutions, the α-helix is substantially more stable than the 3_{10}-helix for $(Ala)_{11}$[79] and even for a decamer of Aib.[80] Thus, it seems very unlikely that Ala-rich peptides in water have any significant level of molecules that are wholly or largely in the 3_{10}-helix. However, isolated 3_{10}-helix hydrogen bonds or perhaps short stretches of 3_{10}-helix near the termini are possible. High-resolution NMR of the peptides Ac-(A_4K)A-NH_2 and Ac-AMAAKAWAAKAAAARA-NH_2[81] have led to estimates of about 50% 3_{10}-helix population at the termini and 25% in the interior. The CD spectrum for a mixed α–3_{10}-helix has not been determined.

for references see p 761

7.7.3.3.3 Comparison of Circular Dichroism and NMR

Significant discrepancies have been reported[82] between helix contents determined by CD and NMR for several synthetic polypeptides, with NMR-derived α-helix fractions substantially higher than CD-derived values (~70 vs ~30% in one case). Several possible sources of error in the CD analysis have been discussed, especially uncertainty in the limiting $[\theta]_{222}$ for 100% helix and aromatic side-chain contributions, and more credence was given to the NMR-based methods. The fraction of helix was deduced[82] from measured $^3J_{\alpha N}$ coupling constants and from integrated intensities of crosspeaks in NOESY spectra. However, purely theoretical values were used for the values of helix and unordered forms, and some of these values could be significantly different from empirical values. Methods for quantitating secondary structure based upon chemical shifts were later developed.[83–85] The present author has applied this method to one of these data sets, and a value of 25% for the helix content was obtained for the above-mentioned peptide, comparable to the CD value (31%).

The helix contents of five peptide fragments from the protein thermolysin have been determined by CD and NMR in both water and 30% TFE.[85] The helix content was obtained from CD by the method of Chen et al.[62] and the NMR method utilized chemical shifts.[84] Four of the five peptides correspond to helical regions in the intact protein, and one corresponds to an Ω-loop.[86] The rms difference between CD and NMR helix contents for the five peptides under the two conditions is 7.5%. One peptide shows the largest deviations (0 vs 13% in water, 45 vs 62% in 30% TFE). If it is excluded, the rms deviation decreases to 4%. The peptide showing the largest deviation, residues 258–276 from the thermolysin sequence, has two Tyr and one Phe (with the Phe adjacent to one of the Tyr in the sequence), and therefore it has an above-average content of aromatic amino acids, which can perturb both the CD spectrum and NMR chemical shifts. Of the other four peptides, three have a single aromatic residue and the fourth has two aromatics.

A comparative study of NMR and CD estimates of helix content in more than 50 peptides has been reported[87] using literature data for peptides for which both CD data and NMR $C_\alpha H$ proton chemical shifts were reported. If the original paper included a helix content calculated from CD, it was used. Otherwise, the helix content was calculated from the $[\theta]_{222}$ reported or read from the published spectrum. The NMR-based helix contents were calculated from the published chemical shifts using the method of Wishart et al.[84] The two estimates showed a good correlation ($r = 0.86$), when those samples in which the temperature for the NMR and CD measurements differed by more than 5 °C were rejected ($r = 0.71$ for the full data set). Of this smaller set (50 peptides), 50% showed differences of 5% or less in the two measures of helix content, 68% were within 10%, 82% within 15%, 90% within 20%, and 100% within 30%. Given that these samples were drawn from the general literature, no corrections were made to the chemical shift differences due to solvents, there were frequently differences in pH between the NMR and CD measurements, and there was always a peptide concentration difference of ~10^3, the level of agreement is encouraging.

Exchange rates for NH protons provide another basis for quantitating secondary structure. A peptide NH engaged in a hydrogen bond in an α-helix or β-sheet cannot undergo exchange. Secondary structure formation shields NH groups from exchange and, under appropriate conditions, measurement of the exchange rate of a specific NH group relative to that of an exposed peptide NH gives a measure of the fraction of time the NH group is not in a helix. Comparison of hydrogen exchange, followed by NMR at individual NH groups, and CD have been used as measures of helix content and for determining helix–coil transition parameters.[88] Both data sets could be fit by helix–coil transition theory, but the exchange rates indicated a higher helix content than given by CD. Using the helix–coil transition parameters, determined from exchange rates to fit the measured CD, led to a 4% decrease in $[\theta]_{222}^{\infty}$, but also caused the rms deviations between calculated and observed ellipticities to increase by more than 50%. As an alternative, the CD-derived

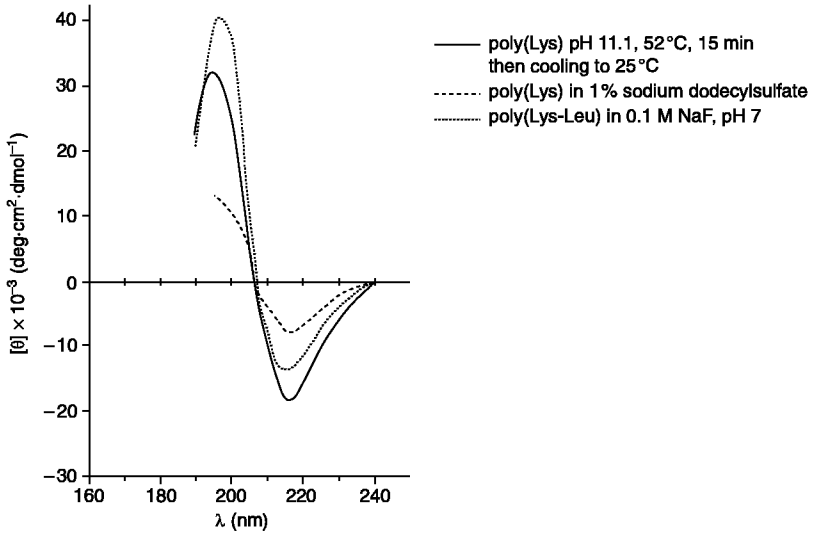
helix–coil transition parameters were used to fit the exchange rate data, k_{int}, allowing the intrinsic exchange rate constant to vary.

The exchange data could be fit just as well by the CD-derived parameters with a k_{int} about 17% lower than the value used in the fit to exchange rate data alone. In the latter case, k_{int} was taken to be the same as the rate of exchange observed for an NH group in an unstructured peptide. It is quite plausible that a peptide NH in a partially helical peptide may differ from that in a completely unstructured peptide. Thus, these results indicate that if the CD parameters are carefully determined for the helical and the unordered conformations, the helix content of peptides can be determined accurately, in agreement with alternative measurements by exchange kinetics.

7.7.3.3.4 β-Sheets

The characteristic features of the β-sheet CD spectrum[89–92] [shown in Figure 4 for poly(Lys) at pH 11.1[94] and in 1% sodium dodecylsulfate[95] and poly(Lys-Leu)[92]] are a negative band near 217 nm, a positive band near 195 nm, and a negative band near 175 nm. The 217 nm band is due to the nπ* transition, whereas the 195 and 175 nm bands are attributable to exciton components of the first ππ* transition. Unlike the α-helix, the absolute and relative magnitudes of the β-sheet CD bands show a marked variability. β-Sheets themselves are quite variable: they can be antiparallel, parallel, or mixed; they vary in strand length and number of strands; and they vary widely in the degree of twist,[93] from nearly planar to strongly twisted. It is therefore, not surprising that the CD of β-sheets is variable.

Figure 4 CD Spectra of Polypeptide Models in the β-Sheet Conformation[1,92,94,95][a]



[a] Reprinted from (Woody, In *The Peptides: Analysis, Synthesis, Biology*, Hruby, V. J., Ed.; Vol. 7), Copyright (1985), p 15, with permission from Academic Press.

Can parallel and antiparallel β-sheets be distinguished by CD? It has been suggested that a difference in the crossover wavelength, the wavelength at which the CD spectrum passes through zero on going from the 195 nm positive maximum to the 175 nm negative maximum, can be used to distinguish these two forms of β-sheet.[96] Films of Boc-Xaa^1-OMe, where Xaa^1 = Ala and Abu, were shown to be antiparallel β-sheets by IR[97] and showed a crossover wavelength of ~178 nm. Films of blocked heptapeptides with Xaa^1 = Val or Phe were shown to be parallel β-sheets by IR criteria and had crossover wavelengths of ~192 nm. When

for references see p 761

Xaa[1] = Leu, Cha, Nva, Nle, the crossover wavelengths were found to have intermediate crossover wavelengths and were assigned as mixed β-sheets. However, it has also been predicted that residues with small or linear side chains (Ala, Abu) form β-sheets that have a small twist,[98] whereas residues with branching at the β-carbon (Val)[99] form sheets that are strongly twisted. Theoretical calculations on slightly and strongly twisted sheets, both parallel and antiparallel, predict that the CD spectrum is much more sensitive to the extent of twist than it is to parallel vs antiparallel chain orientations.[100] Therefore, the crossover wavelength is probably sensing the degree of twist of the sheets rather than their parallel or antiparallel nature.

Theoretical calculations indicate that the CD of parallel and antiparallel β-sheets are quite similar. This conclusion is supported by the CD spectrum of pelC,[101,102] a protein that has a parallel β-helix motif,[103] and therefore is more than 30% parallel β-sheet with no antiparallel β-sheet. Moreover, the parallel β-sheet has a rather small degree of twist. The band positions and relative amplitudes in the CD spectrum of this protein resemble those for poly(Lys) in the β-sheet form, which is an antiparallel sheet that is expected to be only slightly twisted because of the linear side chains.

Model peptides that can adopt the β-sheet conformation have until recently been confined to ones that form intermolecular β-sheets.[96,104,105] Peptides that can form intramolecular β-sheets have been avidly sought because the coil-β transition[106] in such peptides would provide thermodynamic data on the effects of sequence and individual residues on β-sheet stability like those obtainable from model α-helical peptides. Two types of models have been developed:[107] β-hairpins, i.e. two-stranded β-sheets, and three-stranded β-sheets.

In several systems, significant levels of β-hairpin conformers have been indicated by NMR, but no clear evidence for such conformers was found by CD.[108–110] To some extent, this may be attributable to complications in the CD spectrum due to aromatic side chains. The 16-residue β-hairpin of the B1 domain of protein G,[111] residues 41–56,[108] contains a Trp, a Tyr, and a Phe, with the Trp and Tyr separated by only one amino acid residue in the sequence. Three of the β-hairpin peptides from the α-spectrin SH3 domain[109] contain two adjacent Trp residues, and these show clear signs of aromatic contributions in their CD spectra (negative bands near 225 nm). The 12-residue peptide designed by Sieber and Moe[110] shows an unordered type spectrum in its reduced form. In the oxidized form, the spectrum has two weak negative bands with maxima at 220 and 200 nm. This spectrum may reflect contributions from unordered and type I β-turn conformations, as well as the disulfide and Tyr groups, but there is no clear evidence for β-sheet. Although NOEs indicative of proximity between the strands of the putative β-hairpin were observed, no NMR evidence was adduced for a β-sheet-like hydrogen-bonding network. It may be that the peptides discussed here are "nascent[112] β-hairpins", with transient structures having some of the features of β-hairpins detectable by NOEs, but not forming the full hairpin structure that would be detected by CD.

Several peptides have been reported[108,113,114] that show distinct negative CD bands in the 215–220 nm region, pointing clearly to β-hairpin formation. For the peptides exhibiting an identifiable β-sheet feature, estimates of the β-hairpin content have been made from CD and NMR that are in reasonable agreement. For example, the β-hairpin[114] content of a model hexadecapeptide in water was found to be 55% by CD ($[\theta]_{216}$), 41–47% by NOE intensities, and 47% by chemical shifts. In 20% MeOH, these percentages were 80, 70–78, and 66%, respectively.

NMR evidence[115] demonstrated that a designed 20-residue peptide (betanova) adopts a three-stranded β-sheet in water; CD was also measured but the spectrum was not shown. The CD at 217 nm is negative and was used to monitor the thermal unfolding of the peptide. CD and fluorescence indicate a cooperative thermal transition. These CD and fluorescence data can be interpreted as evidence for a two-state cooperative thermal transition. However, a

UV resonance Raman study of betanova, while consistent with a β-sheet secondary structure, provides no evidence for a cooperative thermal transition.[223] The CD spectra reported by Boyden and Asher[223] showed no indication of β-sheet features, even at 0 °C, and were indicative of an unordered conformation. It has been suggested[115,223] that the CD in the 220 nm region is dominated by aromatic side-chain contributions, yet the resonance Raman data suggest a molten globule-like structure, which appears inconsistent with substantial CD contributions from aromatic side chains.[224]

There is no generally accepted way to quantitate β-sheet content in peptides, in contrast to α-helix. It is proposed[116] that β-sheet content could be calculated from the difference between the CD at the negative 217 nm maximum and the positive 195 nm maximum, using $\Delta\varepsilon_{195} - \Delta\varepsilon_{217} = 15 - 16.5\,M^{-1}\cdot cm^{-1}$ ($50 - 55 \times 10^3$ deg·cm²·dmol⁻¹) for 100% sheet. This is a reasonable procedure as long as the sheet is not strongly twisted. It agrees, for example, with the data for poly(Lys)[94] and for poly(Lys-Leu).[92] However, CD data for Boc-Xaa_7-OMe in TFE do not agree with the proposed standard value: for Xaa = Ala, $[\theta]_{217} - [\theta]_{197} \approx 70 \times 10^3$ deg·cm²·dmol⁻¹,[117] and for Xaa = Val, $[\theta]_{217} - [\theta]_{197} \approx 104 \times 10^3$ deg·cm²·dmol⁻¹.[118] For Xaa = Val, this could be attributed to the large twist resulting from the β-branched side chains,[99,100] but this cannot be the explanation for Xaa = Ala.

Solubility is a major problem with peptides that form β-sheet. Polyethylene glycols have been attached to the N- or C-termini of peptides to improve solubility.[119] Decapeptides with the sequences Val_5-Ala_2-Val_3-NH-PEGM and Ala_5-Val_3-Ala_2-NH-PEGM, where PEGM is polyethylene glycol monomethyl ether ($M_r \approx 5 \times 10^3$) have been prepared and studied by CD in water and TFE.[120] In both solvents, Ala_5-Val_3-Ala_2-NH-PEGM gave CD spectra with a characteristic β-sheet pattern (negative maxima at 216 nm in water and 215 nm in TFE; positive maximum at ~190 nm in water). By contrast, Val_5-Ala_2-Val_3-NH-PEGM gave a CD spectrum consistent with an unordered conformation in water and a mixture of unordered and α-helix conformations in TFE.

The critical length for forming a β-sheet has been investigated in peptides of the type Val-Xaa^1-Val_3-NH-PEGM, where Xaa^1 = Val, Leu, Ile[120] and $(Xaa^1\text{-}Xaa^2)_y$-Xaa^3-NH-PEGM,[121] where Xaa^1 is a hydrophilic amino acid and Xaa^2 is a hydrophobic amino acid, or vice versa, and Xaa^3 is Gly or Ser. At or above the critical length, the CD spectra are characteristic of β-sheets. The critical chain lengths ranged from six to more than eleven residues, depending on the sequence and the solvent. Generally, the bulkier the side chain, the shorter the critical chain length. Interestingly, IR criteria indicate that all of the peptides with the host–guest block sequence adopt an antiparallel sheet conformation, in contrast to the demonstration[96,97] that Val and Ile homooligomers assume a parallel β-sheet conformation beyond the critical chain length. The preference[120] for antiparallel sheets in the PEGM-protected peptides was attributed to avoidance of steric clashes that would arise between the protecting groups in a parallel sheet.

7.7.3.3.5 β- and γ-Turns

β-Turns (also called reverse turns, β-bends, 3_{10}-turns) occur at sites where a peptide chain undergoes a reversal in direction within the span of three peptide groups or two α-carbons. In γ-turns, a large change in direction occurs over two amides and a single α-carbon. β-Turns are far more abundant in peptides than γ-turns. Both types of turns have been discussed in several excellent reviews,[122–124] one of which[124] is largely devoted to the CD spectra of turns.

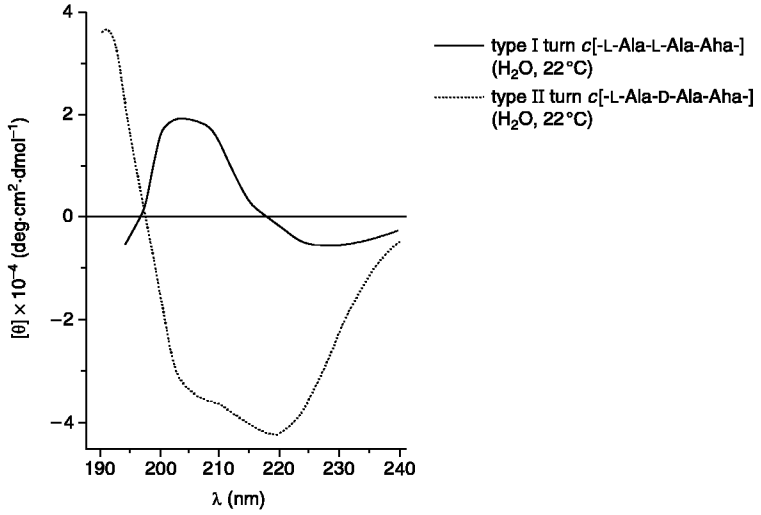
Due to the variety of β-turn types, and the range of φ,ψ values at each of the α-carbons involved in the turn, there is no unique CD signature that characterizes β-turns. Theoretical calculations[125] on a wide range of β-turn conformations[126] led to the prediction that a variety of CD spectral types could be associated with β-turns, but one type of spectrum was most prevalent. This type (class B) has a weak negative band between 220 and 230 nm, a

for references see p 761

strong positive band between 200 and 210 nm, and a strong negative band between 180 and 190 nm. This spectrum is qualitatively like that of a β-sheet, with a red shift of ~10 nm. Certain types of β-turns were predicted to have distinctively different CD spectra. For example, an α-helix-like (class C) pattern was predicted for type II′ β-turns which are adopted by the sequence D-Xaa^1-L-Pro. (The prime indicates a mirror image, and therefore a type II′ turn has Ramachandran angles that are reversed in sign relative to those for the type II turn.)

The CD of β-turns has been characterized[127] by using the peptides *c*(-L-Ala-L-Ala-Aha-) as a model for a type I turn and *c*(-L-Ala-D-Ala-Aha-) for a type II turn. The CD spectra for these two peptides are shown in Figure 5. The spectrum for the type I turn is of class C and that for the type II turn is of class B. The latter result is consistent with the calculations of Woody,[125] but the former result is not. Moreover, explicit calculations for the low-energy conformers[127] predicted a class B spectrum for type I and a class C spectrum for type II. However, using a classical dipole interaction model[128,129] Sathyanarayanan and Applequist[130] obtained qualitative agreement with experiment for these two model systems.

Figure 5 CD Spectra of Two Model β-Turns[127][a]



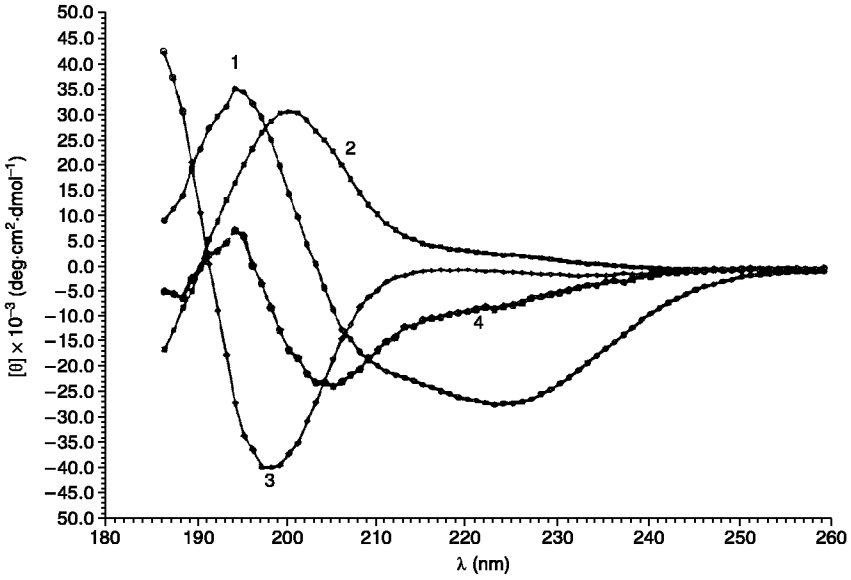
[a] Reprinted from (Woody, *Methods in Enzymology*, Vol. 246), Copyright (1996), p 71, with permission from Academic Press.

Further support for this assignment of CD spectra for the two most common types of β-turns has been provided by studies of cyclic hexapeptides.[92,131,132] The prediction of a class C spectrum for the type II′ turn was verified by studies of cyclic peptides with the D-Xaa^1-L-Pro sequence.[131–133] A peptide with the sequence L-Pro-D-Ala, expected to have a variant of a type II turn,[126] was found[134] to have a class C′ spectrum (mirror image of a class C spectrum), in accordance with predictions.[125] A type VI turn, with a *cis*-Xaa^1-Pro bond has been studied in *c*(-L-Phe-L-Pro-Aha-) and found to have a strong negative band at ~212 nm,[133] but data below 200 nm were not reported. Of course, there may be significant aromatic contributions in the spectrum of this peptide.

A series of 14 linear and cyclic peptides have been studied[135] by CD, NMR, and MD simulations. Quantitative NOE measurements and MD were used to determine the fractions of type I and II β-turns in each peptide. The CD spectra for these same peptides were analyzed by convex constraint analysis[136] to derive a set of CD curves for the two types of β-turns (Figure 6). These basis spectra gave fractional β-turn contents in good agreement with NMR measurements. Two basis spectra (components 1 and 4) were obtained for the type I

turn, both of which are described as class C spectra. Component 1 is qualitatively similar to the α-helix-like spectrum generally assigned to type I β-turns. Component 4 has the same sign pattern but is very different in appearance. It has been suggested[135] that component 1 corresponds to a type I turn in a strained conformation, whereas component 4 arises from unstrained type I turns. However, it should be noted that type I β-turns in cyclic hexapeptides, which should be unstrained, have CD spectra resembling component 1. Component 2 is related to type II turns, and the type II′ β-turn is associated with component 3 in these results. This is a class C′ spectrum and differs from the spectrum generally accepted for type II turns[92,127,131,132] in having a very weak positive band in the nπ* region, rather than a negative band. This basis spectrum is primarily determined by the CD of two linear peptides, Boc-Pro-D-Ser-NHMe and Boc-Val-D-Ser-NHMe, which are the only peptides in the sample having predominantly type II turns. The other peptides have at most about 20% type II turns and more than half of them have less than 10%. As noted before, a Pro-D-Ala peptide has been shown to give rise to a class C′ spectrum,[134] and such a spectrum is predicted[125] for conformations near those of the canonical type II turns.

Figure 6 The Four Component CD Curves from Convex Constraint Analysis of CD Spectra of Cyclic β-Turn Models[135]a,b



[a] Reprinted from (Perczel; Fasman, *Protein Science*, Vol. 1), Copyright (1992), p 378, with permission from Cambridge University Press.

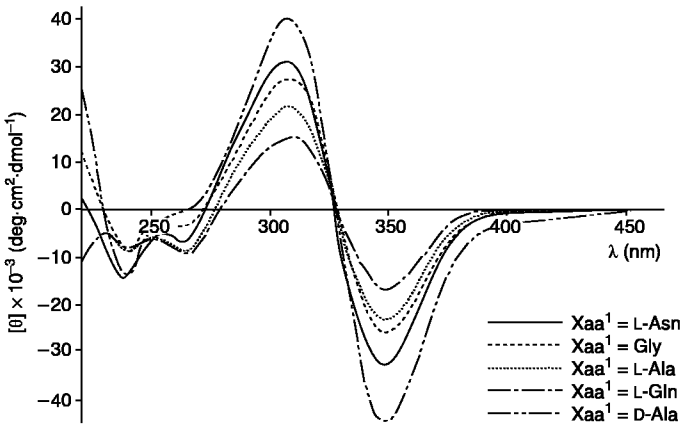
[b] Components 1 and 4 are the two forms of type I turns, component 2 is related to type II turns, and component 3 to open non-β-turn conformations.

The presence of β-turns[137] in small linear peptides has been inferred from CD studies of modified peptides using the exciton chirality method.[138] A 2,4-dinitrophenyl group is attached to the N-terminal amino group and a 4-nitroanilide group is used to amidate the C-terminus. If the two termini are in close proximity, as in a β-turn, exciton coupling between the two chromophores gives a strong exciton couplet. Figure 7 shows spectra for a series of tetrapeptides with the general structure Dnp-Gly-Pro-Xaa^1-Gly-pNA. A strong negative couplet is observed, centered near 330 nm, with amplitudes of up to 80×10^3 deg·cm^2·dmol^{-1}. The amplitude of the couplet decreases in the order Asn > Gly > Ala > Gln, roughly in the order of decreasing β-turn propensities at the third (Xaa^1) position in a β-turn.[139] The highest amplitude is observed when Xaa^1 = D-Ala. From the data in Figure 7,

for references see p 761

one can also infer that the sign of the couplet does not permit one to distinguish between type I and type II turns. When Xaa^1 = D-Ala, the turn must be of type II, whereas for Xaa^1 = Asn, Ala, Gln, it must be of type I, and yet the sign of the couplet is negative for all four peptides. However, for the sequence Dnp-Gly-D-Ala-Pro-Gly-pNA, a positive couplet is observed, nearly a mirror image of the curve for Xaa^1 = Asn in Figure 7. Thus, the sign of the couplet does permit one to distinguish between type I and II turns (negative couplet) and type I′ and II′ β-turns (positive couplet). The exciton chirality method has been used to explore effects of amino acids at various positions on β-turn stability and conformation, e. g. the effects of positions 1 and 4 of the β-turn.[140]

Figure 7 CD Spectra (Methanol) of Five Peptides with the General Structure of Dnp-Gly-Pro-Xaa^1-Gly-pNA[137]a



[a] Reprinted from (Sato; Kawai; Nagai, *Biopolymers*, Vol. 20), Copyright (1981), p 1921, with permission from John Wiley & Sons, Inc.

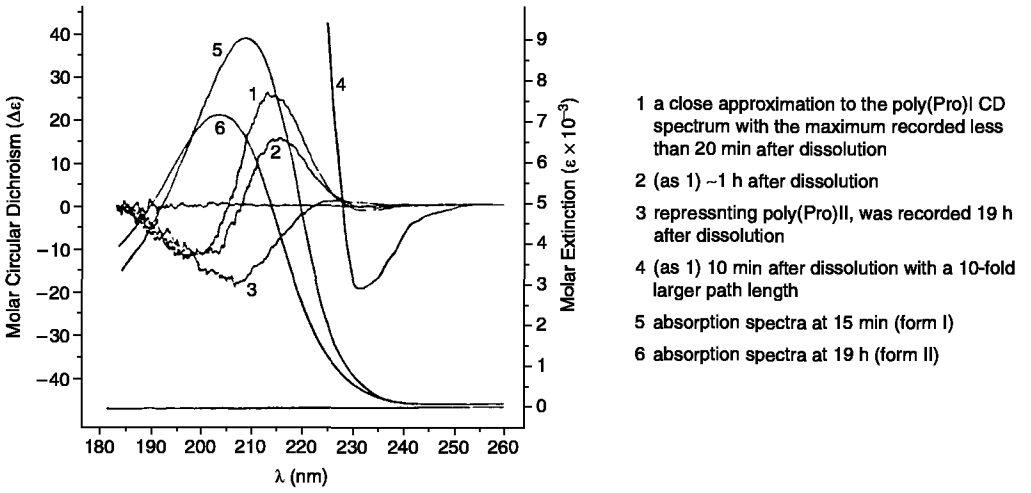
The CD of γ-turns has not been well-characterized. In the original definition of the γ-turn,[141] the Ramachandran angles at the α-carbon located at the turn are (+68, −61), which is near the C_7^{ax} conformation, in a region that is unfavorable for L-amino acid residues and excluded for L-Pro. The inverse γ-turn, with Ramachandran angles of (−68, +61) is near the C_7^{eq} conformation and is accessible to all L-amino acid residues. Therefore, the inverse γ-turn is the only form that is likely to be encountered in most peptides, except at Gly or D-amino acid residues. It has been reported[142] that the inverse γ-turn is a dominant conformer in Ac-Pro-NHMe and Ac-Ala-NHMe in carbon tetrachloride and is populated to a significant extent in chloroform. CD measurements can be performed in the latter solvent down to 220 nm and reveal a strong negative band in the nπ* region ($[\theta]_{228} \sim -5 \times 10^4$ deg·cm^2·dmol^{-1} for Ac-Pro-NHMe and $[\theta]_{220} \sim -2 \times 10^4$ deg·cm^2·dmol^{-1} for Ac-Ala-NHMe). Thus an inverse γ-turn is expected to give rise to a strong negative CD band in the 220–230 nm region and a standard γ-turn should give rise to a large positive nπ* band.

7.7.3.3.6 Poly(Proline) Helices

Poly(Pro) can assume two very different conformations: poly(Pro)I, a right-handed helix of all *cis*-peptides[143] and poly(Pro)II, a left-handed helix of all *trans*-peptides.[144] Poly(Pro)I is found only in the solid state and in solution in solvents of low polarity, e.g. higher alcohols. Poly(Pro)II is the conformation found in water, TFE, and other polar solvents, and the focus will be on it in this section. The CD spectra of both forms are shown in Figure 8. The poly(Pro)II CD spectrum is characterized by a weak positive nπ* maximum at 226 nm and a

strong negative ππ* maximum at 206 nm.[145,146] A second, weaker negative band is observed at ~165 nm.[146] In contrast to peptides forming α-helices and β-sheets, $(Pro)_n$ oligomers exhibit the qualitative CD features of poly(Pro)II even for n = 3, although with substantially reduced amplitude and shifted wavelengths.[147,148]

Figure 8 CD and Absorption Spectra of Poly(Proline) in TFE[145]a



[a] Reprinted from (Bovey; Hood, *Biopolymers*, Vol 5), Copyright (1967), p 915, with permission from John Wiley & Sons, Inc.

Each of the three strands of the collagen triple helix is based upon the poly(Pro)II helix. Numerous studies of peptides capable of triple helix formation have been performed, and these have been reviewed.[149] The CD of the collagen triple helix[146] is qualitatively like that of the poly(Pro)II, but the position of the maximum depends on the content of secondary vs tertiary peptide bonds. In natural collagen, the positive band is at about 220 nm and the negative band at 197 nm.

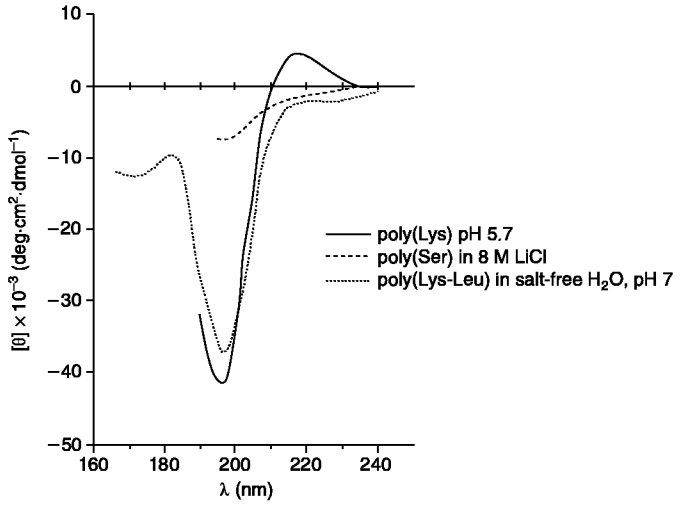
7.7.3.3.7 Unordered Peptides

Many peptides have no well-defined conformation, but exist as an ensemble of conformers, especially in aqueous solution. Unordered peptides have a characteristic strong negative CD band in the 195–200 nm region and a weaker feature in the 220 nm region that may be either positive or negative [shown in Figure 9 for poly(Lys),[94] poly(Ser),[150] and poly(Lys-Leu)[92]].

The distribution of conformers in the unordered conformation depends on the amino acid sequence and on the solvent and temperature. However, there is a substantial body of evidence[151] that poly(Pro)II-like conformers are present at significant levels in unordered peptides at room temperature, and become increasingly important as the temperature decreases. This was first pointed out by Tiffany and Krimm,[152] who noted the striking similarity of the CD spectra of ionized poly(Lys) and poly(Glu) to that of poly(Pro)II. The spectra are indeed very similar, if one allows for a red shift of ~10 nm in the poly(Pro)II spectrum, attributable to the difference between tertiary and secondary amides. The case for poly(Pro)-II-like conformers in unordered peptides is greatly strengthened by observations that the vibrational CD spectrum of unordered peptides is qualitatively like that of poly(Pro)II.[153,154]

for references see p 761

Figure 9 CD Spectra of Unordered Polypeptides[1,92,94,150][a]



[a] Reprinted from (Woody, In *The Peptides: Analysis, Synthesis, Biology*, Hruby, V. J., Ed.; Vol 7) Copyright (1985), p 15, with permission from Academic Press.

A number of studies have shown that unordered peptides exhibit an isodichroic point over a broad temperature range.[155–158] This implies an equilibrium between two forms. The CD spectrum of the low-temperature form is characteristic of a poly(Pro)II helix, whereas that of the high-temperature form has a weaker negative band near 200 nm and a negative shoulder in the 220 nm region. This high-temperature form is an ensemble with significant populations of nearest neighbor pairs in the right-handed α-helix and β-sheet region, as well as the poly(Pro)II region of the Ramachandran map. The importance of the poly(Pro)II conformation in peptides has been reviewed.[159,225]

The contribution of the poly(Pro)II conformation to the ensemble of unordered peptides has been considered.[158] The temperature dependence of $[\theta]_{222}$ for the peptide Ac-YEAAAKEAPAKEAAAKA-NH_2 in 8 M guanidinium chloride and of poly(Lys) in water and in ethylene glycol/water (2:1) mixtures[156] was fitted to a two-state equation for a poly(Pro)II-unordered equilibrium with a temperature-independent ΔH and temperature-independent molar ellipticities for the two components. The peptide with a Pro at the central position is an unordered peptide, the spectrum of which has pronounced poly(Pro)II-like features at low temperatures. This fit yielded $[\theta]_{222} = +9580$ deg·cm^2·dmol^{-1} for the poly (Pro)II component and −5560 deg·cm^2·dmol^{-1} for the unordered component. These values provide a method for roughly estimating the poly(Pro)II content, f_{PII}, of an unordered peptide from $[\theta]_{222}$:

$$f_{PII} = ([\theta]_{222} + 5560)/15140 \quad \text{eq 7}$$

The CD of the unordered form at 222 nm is an important parameter in the determination of α-helix content in peptides. The CD of short peptides (5–7 residues) has been used to evaluate the CD of the unordered form in a homologous series of peptides, fitting the temperature dependence to a linear function of temperature. For example, equation 8 has been used:[160]

$$[\theta]_c = +640 - 45t \quad \text{eq 8}$$

where $[\theta]_c$ is the ellipticity per residue of unordered conformation, in $deg \cdot cm^2 \cdot dmol^{-1}$, and t is the temperature (°C). The intercept and slope need to be evaluated for each type of peptide and for each solvent composition.

The effect of the poly(Pro)II conformation on the estimation of α-helix content in helix-unordered transitions has been discussed.[158] It was pointed out that, in general, we are dealing with a three-state equilibrium, so the ellipticity has contributions from α-helix, poly(Pro)II helix, and truly unordered conformers. In the usual method of analyzing α-helix-unordered equilibria, the poly(Pro)II contribution is taken into account through the temperature dependence of the ellipticity for the unordered form, as described *vide supra*. It was suggest that the unordered form be represented by a modification of the peptide under consideration with a Pro substituted for the central residue. In this approach, the 222 nm ellipticity of the Pro-modified peptide under the solvent conditions would be used as the model for the unordered form. An analysis of synthetic data indicates a distinct improvement over the assumption that the unordered conformation has $[\theta]_{222} = 0$. It is not clear, however, that the proposed method is superior to the use of a short fragment of the peptide under study as the model for the unordered form, assuming a linear temperature dependence of $[\theta]_{222}$.[57,160]

7.7.3.4 Analysis of Secondary Structure in Peptides

In globular proteins, methods have been developed to determine the fraction of various secondary structural types and these are widely used.[161–165] These methods utilize information from the CD spectra of globular proteins of known structure, thus achieving higher accuracy than methods that use model peptides[94] or basis spectra derived from the spectra of a set of proteins without utilizing the information from crystal structures.[136] The methods that work well for globular proteins are generally less suitable for the analysis of peptides, however, because of the different types of distortions and constraints that modify the CD of secondary structures in compact globular proteins as compared to those operating in the open and generally flexible conformations of peptides in solution.

In Section 7.7.3.3, methods for quantitating α-helix and other secondary structural types in peptides were described. These are generally applicable to a series of peptides in which a regular conformation [α-helix, β-sheet, poly(Pro)II] is in equilibrium with an ensemble of unordered conformations, as evidenced by an isodichroic point observed over a range of temperature, pH, or solvent composition.

A method has been described[166] that uses a set of CD spectra for various secondary structures to fit the CD spectrum of a peptide of unknown structure. The weighting coefficients from the least-squares fit are interpreted as the fractions of the respective secondary structures in the peptides. The basis spectra were obtained by various means. Those for the α-helix, β-sheet, and unordered conformation were derived by averaging over the CD spectra for a number of spectra for homo- and copolymers adopting the given conformation, taken from the literature. In constructing the basis spectra, the spectrum for each homo- or copolymer was weighted by a factor proportional to the abundance of the residue(s) in a general protein. The objective of this averaging was to avoid a bias in using any single homo- or copolymer. β-Turns were modeled by spectra for types I, II, and III from Crisma et al.,[167] as was the γ-turn. A general β-turn spectrum[168] and a spectrum for the unordered conformation[167] were also introduced for improving the fit in special cases. Validation of this approach is very difficult because of the absence of well-defined peptide solution conformations determined by the independent methods of X-ray diffraction and NMR, in contrast to the case of globular proteins. The authors cite several cases where their analysis was subsequently supported by NMR evidence in solution or by determination of the conformation of the peptide bound to a protein.

for references see p 761

7.7.3.5 Discussion of Examples

7.7.3.5.1 Cyclic Peptides

The conformational analysis of cyclic peptides is generally more straightforward than that of a linear peptide with the same number of residues. This is a result of the constraints imposed by cyclization.

The simplest cyclic peptides are cyclic dipeptides, called piperazine-2,5-diones (diketopiperazines, DKPs). DKPs have been extensively studied by CD, in addition to NMR, X-ray diffraction, and molecular mechanics. The CD and absorption spectra of *c*(-Ala-Ala-) and *c*(-Pro-Pro-) in several solvents are shown in Figure 10.[169] The CD spectrum of *c*(-Ala-Ala-) has a negative couplet with a crossover near the absorption maximum of the NV_1 transition at 187 nm in water. This couplet is due to exciton splitting of the NV_1 transition. There is also a positive band near 140 nm which can be assigned to the NV_2 transition.[14] A very weak positive band has also been reported near 240 nm in water, and in triethyl phosphate a negative shoulder was observed at 220 nm, as well as the very weak positive band at long wavelengths.[170] It is likely that in water, *c*(-Ala-Ala-) gives a positive $n\pi^*$ couplet, but because the $n\pi^*$ transition is blue-shifted, the negative lobe of the $n\pi^*$ couplet is not resolved from the negative lobe of the $\pi\pi^*$ couplet. The spectra in water and in HFIP are very similar except for small wavelength shifts in the extrema of the couplet.

The CD spectrum of *c*(-Pro-Pro-)[169] shows a negative $n\pi^*$ band at 219 nm in water and HFIP (224 nm in MeCN) and a distinctly asymmetric positive $\pi\pi^*$ couplet in the NV_1 region. A nearly symmetric positive couplet is observed at shorter wavelengths, probably attributable to the NV_2 transition.

It has been proposed by Hooker et al. that the sign of the NV_1 couplet is correlated with the folding angle, β, of the DKP ring.[170] The angle β is the dihedral angle between the $C_1^{\alpha}C_1'N_2C_2^{\alpha}$ plane and the plane formed by the same atoms in the planar DKP conformation. The angle is positive if the pro-*S* substituents on the α-carbons are in the equatorial positions and negative if they are in the axial positions. This correlation states that the NV_1 couplet has the same sign as the angle β.

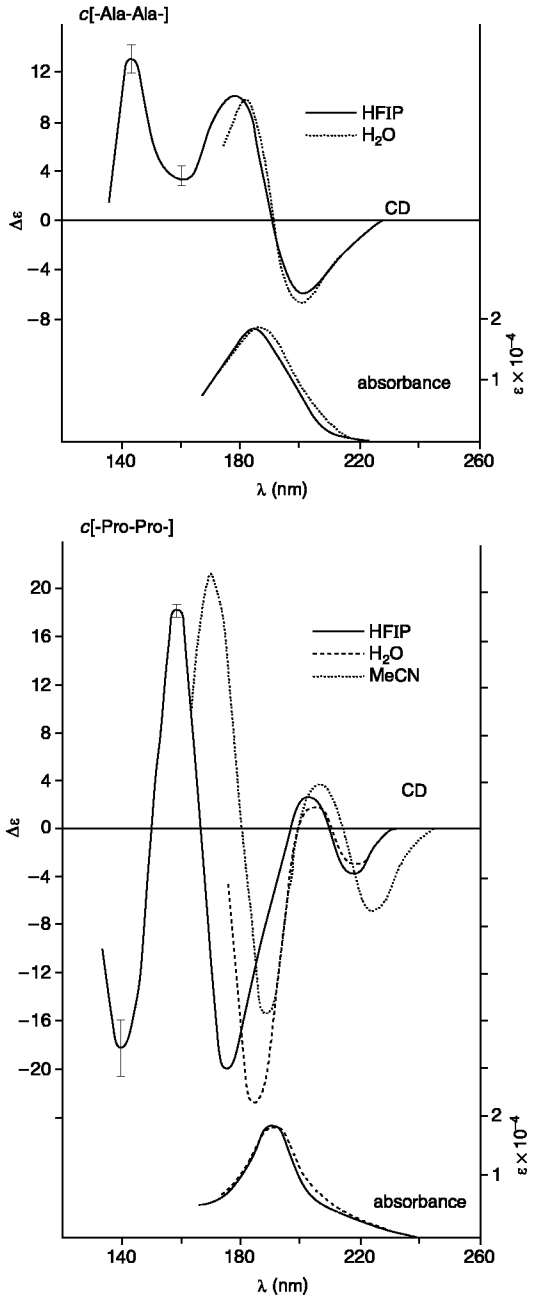
There are two apparent exceptions to the rule of Hooker et al.[1,170] These are *c*(-Ala-Ala-) and *c*[-(Me)Ala-(Me)Ala-]. The X-ray structure of *c*(-Ala-Ala-)[171,172] gives a positive β, but the NV_1 couplet is observed to be negative. However, molecular mechanics calculations[173,174] predict a negative β for *c*(-Ala-Ala-) and its *N*-methyl analogue, in agreement with the CD assignment, so crystal packing effects are responsible for the apparent exception.

A correlation for the $n\pi^*$ transition was also proposed[170] based on theoretical calculations: the sign of the $n\pi^*$ band (or its dominant component for a couplet) is opposite to that of β, except for small positive values of β, where the $n\pi^*$ band has the same sign as β. However, there are numerous exceptions[1] to this correlation, and therefore only the $\pi\pi^*$ correlation should be used.

Cyclic hexapeptides, and other cyclic peptides with $2(2j+1)$ residues, where *j* is integral, have a strong tendency to form structures with two β-turns separated by a central β-sheet segment, as first proposed by Hodgkin and Oughton[175] and by Schwyzer et al.[176] Cyclic hexapeptides lacking chromophoric side chains have been studied extensively by CD.[131,132,177–179] Those with a single chiral amino acid, e.g. *c*(-Ala-Gly_5-)[177,178] and *c*(-Leu-Gly_5-),[180] have very weak CD reflecting a high degree of flexibility. With two or more chiral amino acids, stronger CD bands are generally observed, consistent with a more restricted conformation.

The cyclic hexapeptide *c*(-Pro-Gly-)$_3$ has been extensively investigated.[181–184] CD, NMR, and molecular mechanics indicate a C_3-symmetric conformation with three inverse γ-turns in nonpolar solvents. In polar solvents, a nonsymmetrical conformation is dominant, but in the presence of divalent cations, 1:1 and 2:1 complexes with C_3 symmetry are formed.

Figure 10 CD and Absorption Spectra of *c*(-Ala-Ala-) and *c*(-Pro-Pro-)[169]a



[a] Reprinted from (Bowman; Kellerman; Johnson, Jr., *Biopolymers*, Vol. 22) Copyright (1983), p 1045, with permission from John Wiley & Sons, Inc.

Gramicidin S is a cyclic decapeptide with a characteristic CD spectrum that qualitatively resembles that of an α-helix,[185–189] but numerous studies described below strongly support the structure predicted by Hodgkin and Oughton[175] and by Schwyzer et al.[176] in which β-turns (type II′) at the D-Phe-L-Pro sequences link two extended tripeptide sequences (Val-Orn-Leu) that form an antiparallel β-structure.

for references see p 761

Based upon theoretical calculations, it was[31,190] proposed that the negative nπ* band is largely due to the antiparallel β-sheet region, whereas the D-Pro residues are responsible for the negative 207 nm band. It has also been proposed[125] that the two type II′ β-turns are primarily responsible for the entire α-helix-like spectrum of gramicidin S.

A series of cyclic peptides with an even number (n) of residues, ranging from six to sixteen, have been studied by CD and NMR.[189] In each case, two D-Tyr-L-Pro sequences are present, starting at sites 1 and $(n/2)+1$, analogous to gramicidin S, for which $n=10$ and D-Phe replaces D-Tyr. For $n=6$, 10, 14, well-defined structures were obtained by NMR, corresponding to two type II′ β-turns linked by one, three, and five extended residues, respectively. The average β-sheet content was estimated by averaging values based upon chemical shifts and $^3J_{N\alpha}$ as 50% for $n=6$ and >90% for $n=10$ and 14. The CD spectra for these three cyclic peptides were all very similar in shape and amplitude, resembling the α-helix spectrum at wavelengths above 195 nm. By contrast, for $n=8$ and 12, NMR did not give a well-defined structure, and the β-sheet content was estimated as 5%. The CD spectra for these three peptides and for $n=16$ were indicative of an unordered conformation. Thus, this study strongly supports the models of Hodgkin and Oughton[175] and of Schwyzer et al.[176] for cyclic peptides with $2(2j+1)$ residues (j integral). It has also been pointed out that only sequences giving rise to type II′ or I′ β-turns can occur in Schwyzer-type structures.[189] Thus (D-Xaa^1)-(L-Pro) or (D-Xaa^1)-(L-Xaa^2) sequences for the turns in a background of L-residues are required. Peptides with $4j$ (or with an odd number of) residues cannot adopt a stable β-sheet + 2β-turn structure and will have an unordered conformation.

The observation[189] that the series of $2(2j+1)$ cyclic peptides has a CD amplitude nearly independent of j raises problems for both proposed explanations[31,125] of the gramicidin S CD spectrum. According to one proposal,[125] the amplitude per residue should decrease with increasing j since the β-turns represent a decreasing fraction of the residues. The other model[31] implies that the 220 nm band should increase relative to the 208 nm band as j increases.

7.7.3.5.2 Linear Peptides

Gramicidin A is a linear peptide with 15 residues that acts as an antibiotic because it can form channels across the lipid bilayer that lead to a breakdown of the transmembrane electrochemical gradient that drives ATP synthesis. The sequence is For-Val-Gly-Ala-D-Leu-Ala-D-Val-Val-D-Val-Trp-D-Leu-Trp-D-Leu-Trp-D-Leu-Trp-NH(CH)$_2$OH, i.e. it consists of alternating L- and D-amino acids, with Gly occupying one of the D-sites. A π-helix[191] conformation was proposed by Urry,[192] who noted that such helices can accommodate the alternating L–D-sequence. An alternating L–D-sequence in such a π-helix has carbonyl groups pointing alternately up and down, and is able to form hydrogen bonds analogous to those in parallel β-sheets. Such structures are often called β-helices. Urry also proposed that two such helices can form head-to-head dimers and such a dimer would both span the lipid bilayer and provide a pore large enough to pass a partially unsolvated potassium ion. Double-stranded β-helices in which two-stranded parallel or antiparallel β-sheets are wrapped around a cylindrical surface have also been proposed for gramicidin A.[193–195]

CD has been used to characterize four chromatographically distinct gramicidin A species in organic solvents.[193,196] In combination with NMR and IR, conformational assignments have been made for these species. In organic solvents, the dominant forms are double-stranded β-helices.[197–199] The CD spectrum of gramicidin A in phospholipid bilayers is distinct from that in any homogeneous organic solvent.[200] NMR[201,202] and X-ray diffraction[203,204] studies strongly support a single-stranded β-helix in a head-to-head dimer as the form present in lipid bilayers and as the biologically active form of gramicidin A. The results of an X-ray study[205] have been interpreted[206,207] as support for the double-stranded β-helix as the active ionophore. However, although the structure determined was a Cs^+

complex of gramicidin A, the two forms studied were crystallized from media (CsCl-saturated MeOH and glacial AcOH) that resemble simple organic solvents much more than a phospholipid bilayer. The single-stranded dimer structure has been supported in two communications.[208,209] Moreover, a refinement of the structure of gramicidin A in a liquid crystalline lipid bilayer by solid-state NMR has provided a detailed structure[210] and verified the single-stranded character of the dimer in the lipid phase.

CD has been used extensively in studies of peptides related to Alzheimers disease, although studies of the amyloid plaques associated with the disease are precluded by their insolubility.[211] A peptide consisting of 39–43 residues is the principal component of amyloid deposits that are found in the brains of Alzheimers patients. This peptide, called A4, β-peptide, or βAP for β-amyloid peptide, is derived by proteolytic cleavage of a protein called amyloid protein (APP). The mechanism of aggregation of βAP is clearly of great interest.

A study of four peptides, corresponding to residues 1–42, 1–39, 1–28, and 29–42 of βAP, found that in water at neutral pH three of these peptides are predominately β-sheet, whereas βAP(1–28) has essentially no β-sheet.[212] Addition of TFE or HFIP leads to an increase in α-helix and a loss in β-sheet and unordered conformations. Variation in peptide concentration showed that β-sheet formation is intermolecular, whereas α-helices are monomeric. The rate of aggregate formation was monitored by CD and appears to involve an unordered intermediate between the α-helix and β-sheet conformations. The C-terminal fragment, βAP(29–42) is 100% β-sheet in water, regardless of temperature or pH, in keeping with its hydrophobic character. It was suggested that the C-terminal region may serve as the nucleus for aggregation, and the conclusion was that βAP is not intrinsically insoluble, but that fluctuations in concentration or other variables may trigger amyloid formation.

The aggregation of βAP in the presence and absence of negatively charged lipid vesicles has been investigated using CD.[213] In the absence of added lipids, a family of CD curves was observed at various concentrations, with an isodichroic point at 212 nm, reflecting an equilibrium between unordered and β-sheet conformations. At 25 μM peptide, which exhibits an unordered CD pattern in aqueous buffer, addition of increasing concentrations of lipid vesicles leads to a family of CD curves with the 212 nm isodichroic point characteristic of an unordered–β-sheet equilibrium. Thus, lipid vesicles promote β-sheet formation. Other studies have also reported that phospholipid vesicles[214] and glycolipid-containing membranes[215] promote β-sheet formation in βAP peptides.

Normal human βAP(1–42) peptide has been compared[216] with the corresponding peptide from a mutant from patients with Dutch-type hereditary cerebral hemorrhage with amyloidosis. In these patients, Glu 22 of βAP is mutated to Gln. Both CD and FTIR showed a higher β-sheet content for the mutant sequence, relative to the wild type.

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7.8 Quality Standards and Comparative Syntheses

L. F. Bonewald

In this section, the most practical and useful procedures for analyzing synthetic peptides will be defined and outlined as determined by studies performed by the Peptide Synthesis Research Group as a component of the Association of Biomolecular Resource Facilities. The goal of this unique society is to support resource facilities and related research laboratories. This is the only known society that utilizes research groups composed of members of the society to perform and publish studies in specific areas of technology. The purpose of the Peptide Synthesis Research Group (PSRG) when formed was to evaluate the quality of the synthetic methods utilized by its member laboratories for peptide synthesis. Over the years, 1991–1998, this was modified to allow introduction and evaluation of new chemistries.[1] Over these eight years, the PSRG performed eight studies and was able to compare a variety of analytical techniques, side-by-side, within each study. These techniques included reversed-phase high-performance liquid chromatography (RP-HPLC), amino acid analysis, capillary electrophoresis, sequencing using Edman degradation chemistry, mass determination by mass spectrometry, and sequence analysis using mass spectrometry. Each technique was found to have its own advantages and its limitations. These types of comparative studies had not been done before and allowed the identification and establishment of optimal analytical methods for peptide analysis.

When these studies were first initiated in 1991, a misconception existed that "routine" or "standard" automated peptide synthesis almost always resulted in the correct or expected product. This misconception was probably founded by overzealous salespersons and then promulgated by researchers anxious to obtain the technology for their institutions. It quickly became apparent in the first two studies that this was not the case and that protocols and chemistries required optimization.[2] Side reactions during post-assembly, i.e. cleavage and deprotection, were more significant than actual peptide assembly. Other findings of the PSRG include a documented shift from Boc to Fmoc chemistry for automated peptide synthesis over this time period, the importance of mass spectrometry in peptide analysis, and that peptide quality depends the same or to a greater extent on the experience and skill of personnel who carry out the synthesis than on the instrument or method used.

7.8.1 Importance of Peptide Analysis

One approach to producing biologically active peptides is to first design and synthesize the peptide followed by characterization before testing in a chemical or biological system. An alternate approach is to test peptide mixtures or libraries first, then characterize the peptide(s) that gave a biological or other desired effect. Once a peptide with the desired activity is obtained, a reiterative process is performed to optimize and enhance or stabilize activity. As many peptides are used clinically and many have potential clinical application, it is important to be able to modify peptides to enhance potency and/or delivery. Often the goal is to develop a peptidomimetic, a compound that mimics the side chains of the amino acids, but lacks the peptide backbone.[3,4] However, modifications that give optimal activity in vitro may have no effect in vivo. Therefore, it is important that the peptide be accurately characterized at each step with each modification.

When the peptide chemist collaborates with the biologist, it becomes obvious to both that a correct correlation must be made between composition and purity of the peptide and its function or biological effect. Should the peptide not be correct or the true nature of the peptide not known, then function studies are misinterpreted. Misleading results can consume

a considerable amount of time and resources before being caught and/or corrected. Misleading results can be especially costly if not caught until the stage of in vivo analysis. Often an impurity and not the designed peptide can be responsible for a biological effect. A simple modification of the peptide such as deamidation can influence biological activity but can be easily missed during the characterization process (*vide infra*). The complexity of biological systems, the use of peptides to investigate biological mechanisms, and the potential use of peptides in drug development emphasizes the importance of accurately analyzing a peptide preparation. A full or stringent characterization for complete confidence requires amino acid analysis, various high-pressure chromatography separations, possibly capillary electrophoresis, mass spectrometric analysis using different ionization methods, two-dimensional nuclear magnetic resonance, and X-ray crystallography. However, in this section, only initial or routine analysis of peptide synthesis will be addressed.

7.8.2 Routine Analysis of Peptide Synthesis

It should never be automatically assumed that a correct peptide has been made using either manual or automated peptide synthesis.[1] In the PSRG study of 1991, 17% of the submitted samples did not contain any of the correct peptide, 28% of the crude preparations contained greater than 75% correct peptide and only 68% of the purified samples contained greater than 75% correct product.[5] In the PSRG study of 1992, 48% of the crude preparations and 21% of the purified samples contained less than 75% of the desired product.[6] However, considerable improvement was observed in the 1994 study[7] and the 1996 study.[8] In the 1996 study, of the 53 peptides submitted, only two had less than 50% of the desired product and only three others had less than 70% desired product. This improvement in purity was probably due to more reliable chemistries, improved instrument design, and, perhaps, a heightened awareness of quality control and increased use of mass spectrometry in core facilities.

7.8.2.1 HPLC Analysis and Purification

HPLC separation is recommended in place of desalting as greater purity can be achieved due to enhanced separation of the correct peptide from deletions, truncations, or unblocked peptides. Today, automated systems are available which will automatically inject and collect all or designated absorbance peaks from a chromatogram, thereby greatly speeding up the process of purifying peptides from multiple peptide syntheses. These instruments in combination with large preparative columns allow purification in one HPLC run (see also Sections 7.1 and 7.2).

For routine analysis, it is suggested that mass determination of an HPLC purified product is sufficient to be relatively assured that the correct product has been made. Of course, this approach will neither determine if the correct amino acids have been incorporated into an incorrect sequence nor determine if a substitution of amino acids of identical mass has occurred. However, the numerous checkpoints for automated peptide synthesis (bar codes, printouts, etc.) should greatly reduce the probability of this occurring. Should access to mass spectrometric analysis not be available, amino acid analysis is preferable.

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7.8.2.2 Amino Acid Analysis

Some laboratories do not have access to mass spectrometric analysis, but the number is fewer as the cost for this type of instrumentation is decreasing. It is suggested that these laboratories utilize amino acid analysis due to reduced cost and rapid turnaround. Peptide composition and stoichiometry can be determined, the technique is highly reproducible, and can be used to monitor cycle-to-cycle coupling efficiency.[9] However, not all amino acids are recovered quantitatively. Cys and Trp are totally destroyed and must be quantitated using distinctly different hydrolysis procedures. Ser and Thr can be partially destroyed. Some laboratories perform amino acid analysis in addition to mass spectrometric analysis in order to assure peptide composition, stoichiometry, and quantity (see also Sections 7.3, 7.3.1 and 7.3.2).

7.8.2.2.1 Mass Spectrometry

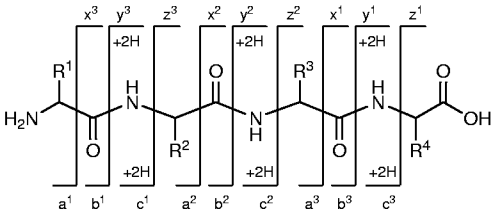
Accuracy of mass spectrometric analysis with modern instrumentation is within tenths of a mass unit. Mass spectrometric analysis is rapid, accurate, and sensitive and gives direct evidence of the presence or absence of the correct product. The chemistries used for amino acid analysis and Edman sequencing can cleave residual protecting groups, thereby making a poor synthesis appear correct. Should the mass of a synthetic peptide be different from the expected mass by more than a mass unit, it should be assumed that the correct peptide has not been made. At this point, some peptide chemists choose to repeat the synthesis in hopes of synthesizing the correct peptide, while others choose to analyze the peptide in hopes of understanding what happened during the synthesis.[10] The first method of choice for analysis of problem peptides is the Edman sequence analysis as is discussed in Section 7.8.3.1.

7.8.3 Analysis of Problem Peptides

7.8.3.1 Edman Sequence Analysis

Edman sequence analysis can be particularly useful in detecting amino acid deletions and can be used to monitor progress of a problem synthesis as this analysis technique can be performed before and after cleavage from the resin.[11] Most peptides can be sequenced to the penultimate carboxy terminal amino acid and some can be completely sequenced. This technique will give the correct amino acids and their correct sequence. However, Cys can only be detected by Edman sequence analysis if modified and the technique is not quantitative; therefore, the degree of adduct formation cannot be determined. Another approach to solving peptide problems is peptide sequencing by mass spectrometry upon fragmentation of the peptide[12] (see Scheme 1). This analytical technique has become particularly useful in solving peptide problems and for thorough characterization of a synthetic peptide. This approach is more time consuming in analysis of the data, but can yield useful information (see also Section 7.3).

Scheme 1 The Expected Fragmentation Pattern of Protonated Peptide Ions and the Nomenclature of the Amino Acid Sequence Fragment Ions for Determining Amino Acid Sequences



7.8.3.2 Sequencing by Mass Spectrometry

In the 1980s fast atom bombardment (FAB) was used for peptide analysis.[11] Newer ionization techniques such as electrospray (ES) or matrix-assisted laser desorption ionization (MALDI) have replaced FAB in peptide analysis. In previous studies by the PSRG, all three, FAB, ES, and MALDI were found to give similar results with respect to mass determinations[13] but each has its advantages and disadvantages.[13,14] A major advantage of ES is that it can be interfaced with liquid chromatography however, detergents and salts are detrimental and interfere with ionization. MALDI has the advantage of a very wide mass range (>200,000) and can tolerate most detergents and salts. These techniques can be used to determine peptide sequence after fragmentation. When a peptide is fragmented by cleavage of the bonds in the peptide backbone, charge is retained on either the amino (N) or carboxy (C) terminus. The ions on the N-terminus are designated a_n, b_n, and c_n (n = numerical sequence of amino acids from the N-terminus) and the C-terminus are designated x_n, y_n, and z_n (n = numerical sequence of amino acids from the C-terminus) (see Scheme 1). The ions of the same type with adjacent n values differ by the mass of the amino acid residue. The amino acid sequence can be deduced from a complete series of fragment ions (see also Section 7.6).

Sequencing by both Edman sequence analysis and mass spectrometric analysis can be complimentary and where one may fail in yielding information concerning the synthesis problem, the other may succeed. This is because every peptide has different properties and each technique has advantages and disadvantages. However, sequence analysis is expensive and time consuming, which must be taken into consideration when solving the problems of peptide synthesis, especially those of routine peptide synthesis.

7.8.4 Examples of Comparative Peptide Analysis

The last four studies performed by the PSRG have been selected as examples of comparative peptide analysis for this section on the analytics of synthetic peptides. These include techniques for the synthesis and production of cyclic peptides and phosphopeptides and include studies on the potential problems of racemization and deamidation[10,15,16,19] These studies were selected for the following reasons: first, the cyclized and phosphorylated forms of peptides are synthesized and used by many scientists; secondly, these studies also address potential problems that can arise using these techniques; and thirdly, these studies are provided as references in which readers might find useful methods for each type of analysis.

7.8.4.1 Cyclization

For the PSRG study of 1995, members were asked to prepare an octreotide-like peptide with intact disulfide bonds.[15] At that time few cyclized peptides were being produced on a routine basis, and the PSRG sought to introduce this technique to its member laboratories. Cyclized peptides are more stable and less susceptible to proteolysis than their linear counterparts and are easily translated into peptidomimetics. Member laboratories were asked to select from either two on-resin and/or two off-resin protocols. As with all studies performed by the PSRG, the test peptides are first made by the committee and characterized by a variety of techniques always including HPLC, amino acid analysis, Edman sequence analysis, and mass spectrometric analysis. Depending on the nature of the study, other analysis techniques are added and in this study, the use of Ellman's reagent to detect free thiol groups was also used in the analysis. HPLC was found to readily separate linear, cyclized, dimeric, deleted, and incompletely deprotected peptides or peptides containing

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mercury adducts. Mass spectrometry was very useful in resolving the two mass unit differences between linear and cyclized peptides. However, it is still best to use a combination of analytical techniques that complement each other in detecting the correct product. In this study the use of Ellman's reagent in combination with mass spectrometric analysis was useful to record the loss of free thiol groups, but alone could not discriminate between non-deprotected cysteines, mercury adducts, or disulfide bonds in dimers. The mercuric acetate methods did not yield correct product. This study showed that the less expensive post-cleavage techniques provide reasonable yields of the desired product, whereas the more expensive on-resin techniques produce greater yields (see also Section 6.8).

7.8.4.2 Racemization

Although the biological importance of racemization is well recognized, it was not clear whether racemization is a problem during routine peptide synthesis. The racemized form of a peptide may be significantly different in biological activity and in antigenicity. Racemization may explain why positive results from a peptide library cannot be repeated with a newly synthesized peptide. The extent of racemization during routine peptide synthesis was addressed by the PSRG study in 1996.[16] A peptide was designed containing an amino acid with potential to racemize, His (see Table 1). Racemization was judged by three methods; amino acid analysis followed by reaction with Marfey's reagent, HPLC, and MALDI-MS of products produced by digestion with carboxypeptidase A. With the 53 samples that were submitted that year, racemization was found not to be a major problem. The percent racemization ranged from approximately 2–8%, but only peptides from one laboratory possessed the higher level of racemization. It was also learned from this study that the designed, racemized peptides could be separated by HPLC, however this may not be the case for all racemized peptides. Therefore, if two peaks are found to have identical masses, one may be the stereoisomer. This can be determined by performing amino acid analysis followed by analysis using Marfey's reagent. Another novel potentially applicable approach which worked well in this study was to treat the peptide with carboxypeptidase A which will cleave the L-form, but not the D-form of His, followed by analysis by mass spectrometry. Determining the extent of racemization is not recommended for routine synthesis, but should be analyzed for important peptides more advanced in their development (see also Section 7.4).

Table 1 Test Peptides Synthesized for PSRG Studies

Organization, Year	Peptide
ABRF, 1995	Phe-*c*[-Cys-Phe-Trp-Lys-Thr-Cys-]-Thr-NH_2
ABRF, 1996	Arg-Glu-Arg-His-Ala-Tyr-OH
ABRF, 1997	H-Glu-Asp-Tyr-Glu-Tyr-Thr-Ala-Arg-Phe-NH_2 (nonphosphorylated, $Y^{3,5}$)
	H-Glu-Asp-Tyr-Glu-pTyr-Thr-Ala-Arg-Phe-NH_2 (pY^5)[a]
	H-Glu-Asp-pTyr-Glu-Tyr-Thr-Ala-Arg-Phe-NH_2 (pY^3)[a]
	H-Glu-Asp-pTyr-Glu-pTyr-Thr-Ala-Arg-Phe-NH_2 ($pY^{3,5}$)[a]
ABRF, 1998	"requested" peptide sequence: H-Ser-Arg-Pro-Ala-Met-His-Phe-NH_2
	peptide sent to facilities: 70% H-Ser-Arg-Pro-Ala-Met-His-Phe-NH_2
	30% H-Ser-Arg-Pro-Ala-Met(O)-His-Phe-NH_2
	"requested" peptide sequence: H-Asp-Glu-Gln-Glu-Ala-Leu-Asn-Arg-Ser-NH_2
	peptide sent to facilities: H-Thr-Arg-Asn-Leu-Ala-Asp-Gln-Glu-Asp-NH_2

[a] p = phosphorylated.

7.8.4.3 Phosphopeptides

Phosphorylation and dephosphorylation are key events in receptor-mediated and post-receptor-mediated signal transduction.[17,18] Synthetic phosphopeptides have been shown to have dramatic agonist or antagonist effects in these signaling pathways. For its 1997 study, member laboratories were asked to synthesize a specific peptide containing a tyrosine and a phosphotyrosine.[19] As with all previous studies, the PSRG synthesized and characterized the test peptides; the nonphosphorylated form of the peptide, the doubly phosphorylated form and the peptides singly phosphorylated on either the first or the second tyrosine (see Table 1). These peptide standards were found to be easily separated by HPLC and capillary electrophoresis and the phosphotyrosine was readily detected by Edman degradation sequence analysis. No differences between the compositions of the peptides were seen by amino acid analysis. The expected masses were observed by mass spectrometry. The two singly phosphorylated peptides were easily distinguished by MALDI-PSD (PSD = post-source delay).

Analysis of the phosphotyrosine containing peptides submitted from member facilities revealed that all but four of the 33 samples contained the correct product as determined by HPLC and mass spectrometry. HPLC analysis indicated that 20 of the 33 submitted samples contained greater than 75% correct product, five contained less than 50% correct product and four did not contain any correct product. Truncated peptide not containing the phosphotyrosine was present in six of the samples, suggesting problems with phosphotyrosine addition. No correlation was found to exist between coupling time and percent correct product. A correlation may exist between a greater percentage of correct product with the use of nonprotected phosphotyrosine as 15 of the 20 samples with greater than 75% correct product used nonprotected phosphotyrosine.

7.8.4.4 Deamidation

The purpose for the 1998 study was to assess the ability of and determine the means whereby member facilities identify and solve problems in peptide synthesis.[10] The potential for oxidation of amino acids such as methionine is always a concern for peptide chemists and biomedical researchers. A peptide mixture containing 70% correct peptide and 30% oxidized peptide was prepared and sent to member facilities to determine if the oxidized methionine would be detected (see Table 1). In addition to the oxidized peptide, a reverse synthesized peptide was sent to the participants. In previous studies, peptides had been submitted which had been synthesized in the reverse order and if only HPLC and mass spectrometric analyses was performed, the reverse synthesis would not be identified. Therefore, two peptides were designed with the second in the reverse order with two substitutions to equal the mass of the first peptide. These two peptides were readily separated by HPLC. The second peptide was sent to the laboratories, but the laboratories were given the first sequence and asked if the correct peptide had been made. Out of 20 participating laboratories:

Seven identified the oxidized methionine as being present in the first peptide.

Seven out of the 20 laboratories assumed that the reverse peptide was correctly made.

The 10 laboratories that correctly identified the error in the reverse peptide used Edman sequencing.

Of the 20 respondents, 19 use mass spectrometry for routine analysis of peptides, 16 use HPLC, and seven use amino acid analysis.

Of the 20 respondents, 11 use Edman sequencing for problem solving, three use amino acid analysis, three use fragmentation sequencing, two use CID, and five just resynthesize the peptide.

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Although the 1998 study was aimed at determining how laboratories solve problems with peptide synthesis, other important observations were made concerning rates of deamidation.[10] As the study proceeded, additional HPLC peaks for the reverse peptide were reported. More extensive stability studies were then performed. The peptide mixture containing the oxidized methionine in diluted 0.1% TFA was found to be stable for at least 2 weeks at room temperature. The peptide made in the correct order was also found to be stable, but the peptide made in the reverse order was found to be relatively unstable in either water or 0.1% TFA when stored at 4 °C and was especially unstable at room temperature. All of the peptides sent to the facilities were amidated on the carboxy terminus (see Table 1). It was the reverse peptide with the C-terminal amino acid, Asp, that was highly susceptible to deamidation on the carboxy terminus. The Gln in the third position from the C-terminus was also susceptible to deamidation, but less than the Asp. This same amino acid Gln, when in the third position from the amino terminus (in the requested peptide) did not deamidate. This deamidation was not readily detectable by MALDI-TOF/MS or by ESI/MS using an SSQ700 quadrupole mass spectrometer. It was found that electrospray ionization mass spectrometry with enhanced resolution using ion trap analyses could be used to identify deamidation occurring in a synthetic peptide. Therefore, it is suggested that this technique be used if deaminidation is suspected or would interfere with peptide activity.

7.8.5 Summary

A large number of wanted and unwanted products can be generated using solid-phase peptide synthesis. To insure quality peptides, it is important to characterize synthesized products with respect to quantity and purity. An array of analytical techniques is needed for peptide analysis; one is not enough. Studies performed by the Peptide Synthesis Research Group have shown improved quality of submitted peptides over eight years and have shown the usefulness and necessity of mass spectrometric analysis of peptides. This enhanced capability to characterize wanted and unwanted products should encourage peptide scientists to undertake more complicated synthesis.

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